

Analysis 3000 Papers

2026-09-20

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1 3000 篇 AI、具身智能与系统论文快速分析汇总

本报告基于已锁定的 3000 篇论文清单生成。逐篇内容是快速理解入口，不能替代全文精读。

1.1 1. 阅读范围与证据边界

- 论文总数：3000篇。
- AI: 2500 篇；系统: 500 篇。
- 证据层级：索引摘要 1123 篇，本地 PDF 前页 1821 篇，标题/主题推断 56 篇。
- 摘要抽取保留一至两句，方法、指标、数据集和局限性需要在后续全文阶段复核。
- 主题来自清单筛选器的可复核标签；一篇论文可以属于多个主题。

1.2 2. 全局主题分布

主题	论文数	主题含义
Supporting Perception	1199	3D、视频、多模态视觉或具身感知支撑能力
Agent RL	834	强化学习、模仿学习、规划、奖励或多智能体决策
AI Infra	713	训练、推理、服务、编译、调度、通信或 AI 加速
Systems	596	操作系统、分布式系统、网络、存储、架构或并行运行时
Embodied/WBC	275	具身任务中的感知、动作生成、全身协调或运动控制
Robot Learning	139	机器人数据、策略学习、操作、触觉或 sim-to-real
World Model	128	环境动力学、视频预测、潜变量状态或模型式决策
VLA	87	视觉、语言条件与机器人动作策略之间的对齐
Teleoperation	32	人类示范、远程操作、共享自治或动作重定向

1.3 3. 按会议分析

1.3.1 AI · ACL

本会议纳入 23 篇。自动主题统计为：Supporting Perception (13)、Agent RL (12)、AI Infra (9)、Systems (9)、VLA (3)、Embodied/WBC (1)。这些统计用于组织阅读顺序，不能单独证明论文结论或实验优越性。

优先阅读候选：- MARS-RA: Rank Aggregation for Credit Assignment via Multimodal Comparisons in Embodied Multi-Agent Cooperation (优先级分数 87.5) - DORA: A Dual-Objective Reinforcement Learning Framework for Effective and Efficient Multimodal Agentic Search (优先级分数 60.5) - EchoVLM: Dynamic Mixture-of-Experts Vision-Language Model for Universal Ultrasound Intelligence (优先级分数 60.5) - HeteroCache: A Dynamic Retrieval Approach to Heterogeneous KV Cache Compression for Long-Context LLM Inference (优先级分数 60.5) - HybridKV: Hybrid KV Cache Compression for Efficient Multimodal Large Language Model Inference (优先级分数 60.5)

1.3.1.1 MARS-RA: Rank Aggregation for Credit Assignment via Multimodal Comparisons in Embodied Multi-Agent Cooperation

- 作者：作者信息未提供
- 年份：2026；主题：Embodied/WBC; Agent RL; Supporting Perception; 优先级：核心
- [来源](#)

论文：MARS-RA: Rank Aggregation for Credit Assignment via Multimodal Comparisons in Embodied Multi-Agent Cooperation

快速理解：Credit assignment is a fundamental challenge in cooperative multi-agent reinforcement learning, particularly in embodied AI settings characterized by limited and delayed feedback as well as dynamically changing numbers of active agents. We propose MARS-RA, a framework that reformulates credit assignment as a rank aggregation problem using contributionbased pairwise comparisons among agents generated by large multimodal models.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；强化学习、模仿学习、规划、奖励或多智能体决策；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.1.2 DORA: A Dual-Objective Reinforcement Learning Framework for Effective and Efficient Multimodal Agentic Search

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL; Supporting Perception; 优先级：重要
- [来源](#)

论文：DORA: A Dual-Objective Reinforcement Learning Framework for Effective and Efficient Multimodal Agentic Search

快速理解：The most recent research uses reinforcement learning (RL) to post-train Multi-modal Large Language Models (MLLMs) such that these models are able to iteratively call search engines to dynamically access external knowledge when handling complex Visual Question Answering (VQA) tasks. However, existing methods face two major limitations in effectiveness and efficiency: i) For effectiveness, the objective of these methods, which only considers the correctness of the generated final response, overlooks the quality of intermediate search results, thus leading to suboptimal search strategies. ii) For efficiency, existing methods often unnecessarily invoke search calls during reasoning, making the inference inefficient.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.1.3 EchoVLM: Dynamic Mixture-of-Experts Vision-Language Model for Universal Ultrasound Intelligence

- 作者：作者信息未提供
- 年份：2026；主题：AI Infra; Supporting Perception; 优先级：重要
- [来源](#)

论文：EchoVLM: Dynamic Mixture-of-Experts Vision-Language Model for Universal Ultrasound Intelligence

快速理解：Ultrasound is the preferred early cancer screening modality due to non-ionizing radiation, cost-effectiveness, and real-time imaging, yet conventional diagnosis relies heavily on physician expertise, causing significant subjectivity and limited efficiency. Vision-Language Models (VLMs) show

promise but lack ultrasound-specific knowledge and multi-organ generalization.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.1.4 HeteroCache: A Dynamic Retrieval Approach to Heterogeneous KV Cache Compression for Long-Context LLM Inference

- 作者：作者信息未提供
- 年份：2026；主题：AI Infra; Systems; 优先级：重要
- [来源](#)

论文：HeteroCache: A Dynamic Retrieval Approach to Heterogeneous KV Cache Compression for Long-Context LLM Inference

快速理解：The linear memory growth of the KV cache poses a significant bottleneck for LLM inference in long-context tasks. Existing static compression methods often fail to preserve globally important information.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.1.5 HybridKV: Hybrid KV Cache Compression for Efficient Multimodal Large Language Model Inference

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception; Systems; 优先级：重要
- [来源](#)

论文：HybridKV: Hybrid KV Cache Compression for Efficient Multimodal Large Language Model Inference

快速理解：Multimodal Large Language Models (MLLMs) have advanced unified reasoning over text, images, and videos, but their inference is hindered by the rapid growth of key-value (KV) caches. Each visual input expands into thousands of tokens, causing caches to scale linearly with context length and remain resident in GPU memory throughout decoding, which leads to prohibitive memory overhead and latency even on high-end GPUs.

关注点：3D、视频、多模态视觉或具身感知支撑能力；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.1.6 LLM-as-Scheduler: Agentic Workflow Dynamic Scheduling

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL; Systems; 优先级：重要
- [来源](#)

论文：LLM-as-Scheduler: Agentic Workflow Dynamic Scheduling

快速理解：As large language models (LLMs) become more capable, many applications are shifting from a single LLM call to multi-agent systems. Manually designed or automatically optimized workflows often include multiple verification and testing stages.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.1.7 Learning What Matters: Dynamic Dimension Selection and Aggregation for Interpretable Vision-Language Reward Modeling

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL; Supporting Perception；优先级：重要
- [来源](#)

论文：Learning What Matters: Dynamic Dimension Selection and Aggregation for Interpretable Vision-Language Reward Modeling

快速理解：Vision-language reward modeling faces a dilemma: generative approaches are interpretable but slow, while discriminative ones are efficient but act as opaque “black boxes.” To bridge this gap, we propose VL-MDR (Vision-Language Multi-Dimensional Reward), a framework that dynamically decomposes evaluation into granular, interpretable dimensions. Instead of outputting a monolithic scalar, VL-MDR employs a visual-aware gating mechanism to identify relevant dimensions and adaptively weight them (e.g., Hallucination, Reasoning) for each specific input.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.1.8 MIRTH: Mutual-Information Reasoning with Temporal Hubs for Vision-Language-Action Agents

- 作者：作者信息未提供
- 年份：2026；主题：VLA; Supporting Perception；优先级：重要
- [来源](#)

论文：MIRTH: Mutual-Information Reasoning with Temporal Hubs for Vision-Language-Action Agents

快速理解：VLA models have emerged as a powerful paradigm for transferring semantic knowledge from web-scale data to physical robotic control. However, current single-frame architectures suffer from intrinsic limitations: temporal myopia that discards historical dynamics, reasoning gaps between high-level instructions and low-level motor commands, and inference inefficiency due to autoregressive scalar decoding.

关注点：视觉、语言条件与机器人动作策略之间的对齐；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.1.9 MONETA: Multimodal Industry Classification through Geographic Information with Multi Agent Systems

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL; Supporting Perception；优先级：重要
- [来源](#)

论文：MONETA: Multimodal Industry Classification through Geographic Information with Multi Agent Systems

快速理解: Industry classification schemes are integral parts of public and corporate databases as they classify businesses based on economic activity. Due to the size of the company registers, manual annotation is costly, and fine-tuning models with every update in industry classification schemes requires significant data collection.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.1.10 MixKVQ: Query-Aware Mixed-Precision KV Cache Quantization for Long-Context Reasoning

- 作者: 作者信息未提供
- 年份: 2026; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: MixKVQ: Query-Aware Mixed-Precision KV Cache Quantization for Long-Context Reasoning

快速理解: Long Chain-of-Thought (CoT) reasoning has significantly advanced the capabilities of Large Language Models (LLMs), but this progress is accompanied by substantial memory and latency overhead from the extensive Key-Value (KV) cache. Although KV cache quantization is a promising compression technique, existing low-bit quantization methods often exhibit severe performance degradation on complex reasoning tasks.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.1.11 On-the-Fly VLA Adaptation via Test-Time Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2026; 主题: VLA; Agent RL; 优先级: 重要
- [来源](#)

论文: On-the-Fly VLA Adaptation via Test-Time Reinforcement Learning

快速理解: Vision-Language-Action (VLA) models have recently emerged as a powerful paradigm for general-purpose robot learning, enabling agents to map visual observations and natural language instructions into executable robotic actions. Though popular, they are primarily trained via supervised fine-tuning or training time reinforcement learning, requiring explicit fine-tuning phases, human interventions, or controlled data collection.

关注点: 视觉、语言条件与机器人动作策略之间的对齐; 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.1.12 ProActor: Timing-Aware Reinforcement Learning for Proactive Task Scheduling Agents

- 作者: 作者信息未提供
- 年份: 2026; 主题: Agent RL; Systems; 优先级: 重要
- [来源](#)

论文: ProActor: Timing-Aware Reinforcement Learning for Proactive Task Scheduling Agents

快速理解: Proactive task-oriented agents must autonomously anticipate user needs, identify actionable

opportunities, and trigger software actions at appropriate moments—fundamentally shifting from reactive systems that await explicit instructions. However, existing approaches lack generalizable end-to-end solutions for measuring and optimizing such anticipatory behaviors.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.1.13 QA-MoE: Towards a Continuous Reliability Spectrum with Quality-Aware Mixture of Experts for Robust Multimodal Sentiment Analysis

- 作者：作者信息未提供
- 年份：2026；主题：AI Infra; Supporting Perception; 优先级：重要
- [来源](#)

论文：QA-MoE: Towards a Continuous Reliability Spectrum with Quality-Aware Mixture of Experts for Robust Multimodal Sentiment Analysis

快速理解：Multimodal Sentiment Analysis (MSA) aims to infer human sentiment from textual, acoustic, and visual signals. In real-world scenarios, however, multimodal inputs are often compromised by dynamic noise or modality missingness.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.1.14 ReLook: Vision-Grounded RL with a Multimodal LLM Critic for Agentic Web Coding

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL; Supporting Perception; 优先级：重要
- [来源](#)

论文：ReLook: Vision-Grounded RL with a Multimodal LLM Critic for Agentic Web Coding

快速理解：While Large Language Models (LLMs) excel at algorithmic code generation, they struggle with front-end development, where correctness is judged on rendered pixels and interaction. We present ReLook, an agentic, vision-grounded reinforcement learning framework that empowers an agent to close a robust generate-diagnose-refine loop by invoking a multimodal LLM (MLLM) as a tool.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.1.15 Resonating with RoPE: Spectral Quantization for High-Fidelity Key Cache Compression

- 作者：作者信息未提供
- 年份：2026；主题：AI Infra; Systems; 优先级：重要
- [来源](#)

论文：Resonating with RoPE: Spectral Quantization for High-Fidelity Key Cache Compression

快速理解：The linear growth of KV cache bottlenecks long-context LLMs, yet RoPE-induced oscillations complicate Key cache quantization. To address this issue, we propose SpectrumQuant, a

frequency-domain framework that utilizes the Discrete Cosine Transform (DCT) to convert these oscillations into sparse spectral representations.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.1.16 Seeing but Not Thinking: Routing Distraction in Multimodal Mixture-of-Experts

- 作者：作者信息未提供
- 年份：2026；主题：AI Infra; Supporting Perception; 优先级：重要
- [来源](#)

论文：Seeing but Not Thinking: Routing Distraction in Multimodal Mixture-of-Experts

快速理解：Multimodal Mixture-of-Experts (MoE) models have achieved remarkable performance on vision-language tasks. However, we identify a puzzling phenomenon termed Seeing but Not Thinking: models accurately perceive image content yet fail in subsequent reasoning, while correctly solving identical problems presented as pure text.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.1.17 SpiderFlow: Efficient Topology-Aware Scheduling for LLM Training Across Decentralized GPU Clusters

- 作者：作者信息未提供
- 年份：2026；主题：AI Infra; Systems; 优先级：重要
- [来源](#)

论文：SpiderFlow: Efficient Topology-Aware Scheduling for LLM Training Across Decentralized GPU Clusters

快速理解：In response to increasing demands for largescale machine learning training jobs, many organizations have deployed GPU clusters across geographically distributed regions. However, existing Integer Linear Programming (ILP)- or genetic-based cross-cluster training approaches largely overlook the topology of decentralized clusters, lacking both topology-aware task scheduling mechanisms and automated model parallelization strategies.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.1.18 Towards Stable and Effective Reinforcement Learning for Mixture-of-Experts

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL; AI Infra; 优先级：重要
- [来源](#)

论文：Towards Stable and Effective Reinforcement Learning for Mixture-of-Experts

快速理解：Reinforcement learning with verifiable rewards (RLVR) has emerged as a powerful paradigm for improving reasoning capabilities. However, training RLVR with Mixture-of-Experts (MoE) policies remains fragile and is often prone to reward collapse.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.1.19 VecInfer: Efficient LLM Inference with Low-Bit KV Cache via Outlier-Suppressed Vector Quantization

- 作者：作者信息未提供
- 年份：2026；主题：AI Infra; Systems；优先级：重要
- [来源](#)

论文：VecInfer: Efficient LLM Inference with Low-Bit KV Cache via Outlier-Suppressed Vector Quantization

快速理解：The Key-Value (KV) cache introduces substantial memory overhead during large language model (LLM) inference. Although existing vector quantization (VQ) methods reduce KV cache usage and provide flexible representational capacity across bit-widths, they suffer severe performance degradation at ultra-low bit-widths due to key cache outliers that hinder effective codebook utilization.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.1.20 Visual Inception: Compromising Long-term Planning in Agentic Recommenders via Multimodal Memory Poisoning

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL; Supporting Perception；优先级：重要
- [来源](#)

论文：Visual Inception: Compromising Long-term Planning in Agentic Recommenders via Multimodal Memory Poisoning

快速理解：The evolution from static ranking models to Agentic Recommender Systems (Agentic Rec- Sys) empowers AI agents to maintain longterm user profiles and autonomously plan service tasks. While this paradigm shift enhances personalization, it introduces a vulnerability: reliance on Long-term Memory (LTM).

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.1.21 Visually-Guided Policy Optimization for Multimodal Reasoning

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL; Supporting Perception；优先级：重要
- [来源](#)

论文：Visually-Guided Policy Optimization for Multimodal Reasoning

快速理解：Reinforcement learning with verifiable rewards (RLVR) has significantly advanced the reasoning ability of vision-language models (VLMs). However, the inherent text-dominated nature of VLMs often leads to insufficient visual faithfulness, characterized by sparse attention activation to visual tokens.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.1.22 When Does Language Matter? Multilingual Instructions Reveal Step-wise Language Sensitivity in Vision-Language-Action Models

- 作者：作者信息未提供
- 年份：2026；主题：VLA; Supporting Perception；优先级：重要
- [来源](#)

论文：When Does Language Matter? Multilingual Instructions Reveal Step-wise Language Sensitivity in Vision-Language-Action Models

快速理解：Vision-Language-Action (VLA) models have shown strong performance in language-conditioned robotic manipulation, yet their robustness to linguistic variation remains poorly understood. In this work, we present the first systematic multilingual evaluation of VLA models by translating the LIBERO benchmark into ten languages, revealing severe performance degradation under non-English instructions, with success rates dropping by 30-50%.

关注点：视觉、语言条件与机器人动作策略之间的对齐；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.1.23 When KV Cache Reuse Fails in Multi-Agent Systems: Cross-Candidate Interaction is Crucial for LLM Judges

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL; Systems；优先级：重要
- [来源](#)

论文：When KV Cache Reuse Fails in Multi-Agent Systems: Cross-Candidate Interaction is Crucial for LLM Judges

快速理解：Multi-agent LLM systems routinely generate multiple candidate responses that are aggregated by an LLM judge. To reduce the dominant prefill cost in such pipelines, recent work advocates KV cache reuse across partially shared contexts and reports substantial speedups for generation agents.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.2 AI · CVPR

本次会议纳入 **699** 篇。自动主题统计为：Supporting Perception (488)、Agent RL (97)、Systems (48)、Embodied/WBC (47)、VLA (26)、AI Infra (26)、World Model (21)、Robot Learning (15)、Teleoperation (1)。这些统计用于组织阅读顺序，不能单独证明论文结论或实验优越性。

优先阅读候选：- ActiveVLA: Injecting Active Perception into Vision-Language-Action Models for Precise 3D Robotic Manipulation (优先级分数 92.5) - DriveMoE: Mixture-of-Experts for Vision-Language-Action Model in End-to-End Autonomous Driving(优先级分数 92.5)- Explore with Long-term Memory: A Benchmark and Multimodal LLM-based Reinforcement Learning Framework for Embodied Exploration (优先级分数 92.5) - Global Prior Meets Local Consistency: Dual-Memory Augmented Vision-Language-Action Model for Efficient Robotic Manipulation (优先级分数 92.5)- Localizing, Structuring,

and Rendering: Bridging 3D and 2D Vision-Language-Action Models for Robotic Manipulation (优先级分数 92.5)

1.3.2.1 ActiveVLA: Injecting Active Perception into Vision-Language-Action Models for Precise 3D Robotic Manipulation

- 作者：作者信息未提供
- 年份：2026；主题：Embodied/WBC; VLA; Supporting Perception; 优先级：核心
- [来源](#)

论文：ActiveVLA: Injecting Active Perception into Vision-Language-Action Models for Precise 3D Robotic Manipulation

快速理解：Recent advances in robot manipulation have leveraged pretrained vision-language models (VLMs) and explored integrating 3D spatial signals into these models for effective action prediction, giving rise to the promising visionlanguage-action (VLA) paradigm. However, most existing approaches overlook the importance of active perception: they typically rely on static, wrist-mounted cameras that provide an end-effector-centric viewpoint.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；视觉、语言条件与机器人动作策略之间的对齐；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.2 DriveMoE: Mixture-of-Experts for Vision-Language-Action Model in End-to-End Autonomous Driving

- 作者：作者信息未提供
- 年份：2026；主题：VLA; AI Infra; Supporting Perception; 优先级：核心
- [来源](#)

论文：DriveMoE: Mixture-of-Experts for Vision-Language-Action Model in End-to-End Autonomous Driving

快速理解：End-to-end autonomous driving (E2E-AD) demands effective processing of multi-view sensor data and robust handling of diverse and complex driving scenarios, particularly rare maneuvers such as aggressive turns. The recent success of the Mixture-of-Experts (MoE) architectures in Large Language Models (LLMs) demonstrates that expert specialization enables strong scalability.

关注点：视觉、语言条件与机器人动作策略之间的对齐；训练、推理、服务、编译、调度、通信或 AI 加速；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.3 Explore with Long-term Memory: A Benchmark and Multimodal LLM-based Reinforcement Learning Framework for Embodied Exploration

- 作者：作者信息未提供
- 年份：2026；主题：Embodied/WBC; Agent RL; Supporting Perception; 优先级：核心
- [来源](#)

论文：Explore with Long-term Memory: A Benchmark and Multimodal LLM-based Reinforcement Learning Framework for Embodied Exploration

快速理解: An ideal embodied agent should possess lifelong learning capabilities to handle long-horizon and complex tasks, enabling continuous operation in general environments. This not only requires the agent to accurately accomplish given tasks but also to leverage long-term episodic memory to optimize decision-making.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 强化学习、模仿学习、规划、奖励或多智能体决策; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.4 Global Prior Meets Local Consistency: Dual-Memory Augmented Vision-Language-Action Model for Efficient Robotic Manipulation

- 作者: 作者信息未提供
- 年份: 2026; 主题: Embodied/WBC; VLA; Supporting Perception; 优先级: 核心
- [来源](#)

论文: Global Prior Meets Local Consistency: Dual-Memory Augmented Vision-Language-Action Model for Efficient Robotic Manipulation

快速理解: Hierarchical Vision-Language-Action (VLA) models have rapidly become a dominant paradigm for robotic manipulation. It typically comprising a Vision-Language backbone for perception and understanding, together with a generative policy for action generation.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 视觉、语言条件与机器人动作策略之间的对齐; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.5 Localizing, Structuring, and Rendering: Bridging 3D and 2D Vision-Language-Action Models for Robotic Manipulation

- 作者: 作者信息未提供
- 年份: 2026; 主题: Embodied/WBC; VLA; Supporting Perception; 优先级: 核心
- [来源](#)

论文: Localizing, Structuring, and Rendering: Bridging 3D and 2D Vision-Language-Action Models for Robotic Manipulation

快速理解: Robotic manipulation in complex 3D environments requires unifying spatial reasoning with intuitive visual perception, which is a capability that current Vision-Language-Action paradigms address separately. While 3D VLAs excel in geometric and physical reasoning, they lack intuitive, imagelevel understanding and dense visual semantics; conversely, 2D VLAs (even with depth image) provide rich visual intuition and semantic continuity but miss explicit spatial global grounding.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 视觉、语言条件与机器人动作策略之间的对齐; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.6 MoE-GRPO: Optimizing Mixture-of-Experts via Reinforcement Learning in Vision-Language Models

- 作者: 作者信息未提供
- 年份: 2026; 主题: Agent RL; AI Infra; Supporting Perception; 优先级: 核心

- [来源](#)

论文: MoE-GRPO: Optimizing Mixture-of-Experts via Reinforcement Learning in Vision-Language Models

快速理解: Mixture-of-Experts (MoE) has emerged as an effective approach to reduce the computational overhead of Transformer architectures by sparsely activating a subset of parameters for each token while preserving high model capacity. This paradigm has recently been extended to Vision-Language Models (VLMs), enabling scalable multi-modal understanding with reduced computational cost.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 训练、推理、服务、编译、调度、通信或 AI 加速; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.7 QuantVLA: Scale-Calibrated Post-Training Quantization for Vision-Language-Action Models

- 作者: 作者信息未提供
- 年份: 2026; 主题: VLA; AI Infra; Supporting Perception; 优先级: 核心
- [来源](#)

论文: QuantVLA: Scale-Calibrated Post-Training Quantization for Vision-Language-Action Models

快速理解: Vision-language-action (VLA) models unify perception, language, and control for embodied agents but face significant challenges in practical deployment due to rapidly increasing compute and memory demands, especially as models scale to longer horizons and larger backbones. To address these bottlenecks, we introduce QuantVLA, a trainingfree post-training quantization (PTQ) framework that, to our knowledge, is the first PTQ approach for VLA systems and the first to successfully quantize a diffusion transformer (DiT) action head.

关注点: 视觉、语言条件与机器人动作策略之间的对齐; 训练、推理、服务、编译、调度、通信或 AI 加速; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.8 SAGE: Scalable Agentic 3D Scene Generation for Embodied AI

- 作者: 作者信息未提供
- 年份: 2026; 主题: Embodied/WBC; Agent RL; Supporting Perception; 优先级: 核心
- [来源](#)

论文: SAGE: Scalable Agentic 3D Scene Generation for Embodied AI

快速理解: Real-world data collection for embodied agents remains costly and unsafe, calling for scalable, realistic, and simulator-ready 3D environments. However, existing scenegeneration systems often rely on rule-based or task-specific pipelines, yielding artifacts and physically invalid scenes.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 强化学习、模仿学习、规划、奖励或多智能体决策; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.9 SRPO: Self-Referential Policy Optimization for Vision-Language-Action Models

- 作者：作者信息未提供
- 年份：2026；主题：VLA; Agent RL; Supporting Perception; 优先级：核心
- [来源](#)

论文：SRPO: Self-Referential Policy Optimization for Vision-Language-Action Models

快速理解：Vision-Language-Action (VLA) models excel in robotic manipulation but are constrained by their heavy reliance on expert demonstrations, leading to demonstration bias and limiting performance. Reinforcement learning (RL) is a vital post-training strategy to overcome these limits, yet current VLA-RL methods, including group-based optimization approaches, are crippled by severe reward sparsity.

关注点：视觉、语言条件与机器人动作策略之间的对齐；强化学习、模仿学习、规划、奖励或多智能体决策；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.10 ARM-Thinker: Reinforcing Multimodal Generative Reward Models with Agentic Tool Use and Visual Reasoning

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL; Supporting Perception; 优先级：核心
- [来源](#)

论文：ARM-Thinker: Reinforcing Multimodal Generative Reward Models with Agentic Tool Use and Visual Reasoning

快速理解：Reward models are critical for aligning vision-language systems with human preferences, yet current approaches suffer from hallucination, weak visual grounding, and an inability to use tools for verification, limiting their reliability on complex multimodal reasoning tasks. We present ARM-Thinker, an Agentic multimodal Reward Model that autonomously invokes external tools (e.g., image cropping, doc page retrieval) to ground judgments in verifiable evidence, replacing static, non-interactive reward scoring.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.11 AVA-VLA: Improving Vision-Language-Action models with Active Visual Attention

- 作者：作者信息未提供
- 年份：2026；主题：VLA; Supporting Perception; 优先级：核心
- [来源](#)

论文：AVA-VLA: Improving Vision-Language-Action models with Active Visual Attention

快速理解：Vision-Language-Action (VLA) models have shown remarkable progress in embodied tasks recently, but most methods process visual observations independently at each timestep. This history-agnostic design treats robot manipulation as a Markov Decision Process, even though realworld robotic control is inherently partially observable and requires reasoning over past interactions.

关注点：视觉、语言条件与机器人动作策略之间的对齐；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.12 Adapting Point Cloud Analysis via Multimodal Bayesian Distribution Learning

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception; Systems; 优先级：核心
- [来源](#)

论文：Adapting Point Cloud Analysis via Multimodal Bayesian Distribution Learning

快速理解：Large multimodal 3D vision-language models show strong generalization across diverse 3D tasks, but their performance still degrades notably under domain shifts. This has motivated recent studies on test-time adaptation (TTA), which enables models to adapt online using test-time data.

关注点：3D、视频、多模态视觉或具身感知支撑能力；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.2.13 Adaptive Action Chunking at Inference-time for Vision-Language-Action Models

- 作者：作者信息未提供
- 年份：2026；主题：VLA; Supporting Perception; 优先级：核心
- [来源](#)

论文：Adaptive Action Chunking at Inference-time for Vision-Language-Action Models

快速理解：In Vision-Language-Action (VLA) models, action chunking (i.e., executing a sequence of actions without intermediate replanning) is a key technique to improve robotic manipulation abilities. However, a large chunk size reduces the model's responsiveness to new information, while a small one increases the likelihood of mode-jumping, jerky behavior resulting from discontinuities between chunks.

关注点：视觉、语言条件与机器人动作策略之间的对齐；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.14 Boosting Vision-Language-Action Finetuning with Feasible Action Neighborhood Prior

- 作者：作者信息未提供
- 年份：2026；主题：VLA; Supporting Perception; 优先级：核心
- [来源](#)

论文：Boosting Vision-Language-Action Finetuning with Feasible Action Neighborhood Prior

快速理解：In real-world robotic manipulation, states typically admit a neighborhood of near-equivalent actions. That is, for each state, there exist a feasible action neighborhood (FAN) rather than a single correct action, within which motions yield indistinguishable progress.

关注点：视觉、语言条件与机器人动作策略之间的对齐；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.15 Cross-Hand Latent Representation for Vision-Language-Action Models

- 作者：作者信息未提供
- 年份：2026；主题：VLA；Supporting Perception；优先级：核心
- [来源](#)

论文：Cross-Hand Latent Representation for Vision-Language-Action Models

快速理解：Dexterous manipulation is essential for real-world robot autonomy, mirroring the central role of human hand coordination in daily activity. Humans rely on rich multimodal perception—vision, sound, and language-guided intent—to perform dexterous actions, motivating vision-based, language-conditioned manipulation systems for robots.

关注点：视觉、语言条件与机器人动作策略之间的对齐；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.16 D3D-VLP: Dynamic 3D Vision-Language-Planning Model for Embodied Grounding and Navigation

- 作者：作者信息未提供
- 年份：2026；主题：Embodied/WBC；Supporting Perception；优先级：核心
- [来源](#)

论文：D3D-VLP: Dynamic 3D Vision-Language-Planning Model for Embodied Grounding and Navigation

快速理解：Embodied agents face a critical dilemma that end-to-end models lack interpretability and explicit 3D reasoning, while modular systems ignore cross-component interdependencies and synergies. To bridge this gap, we propose the Dynamic 3D Vision-Language-Planning Model (D3D-VLP).

关注点：具身任务中的感知、动作生成、全身协调或运动控制；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.17 Diverse Video Generation with Determinantal Point Process-Guided Policy Optimization

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL；Supporting Perception；优先级：核心
- [来源](#)

论文：Diverse Video Generation with Determinantal Point Process-Guided Policy Optimization

快速理解：While recent text-to-video (T2V) diffusion models have achieved impressive quality and prompt alignment, they often produce low-diversity outputs when sampling multiple videos from a single text prompt. We tackle this challenge by formulating it as a set-level policy optimization problem, with the goal of training a policy that can cover the diverse range of plausible outcomes for a given prompt.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.18 EE-RL: Vision Language Guided Reinforcement Learning with Explorer and Expert model for End-to-End Autonomous Driving

- 作者: 作者信息未提供
- 年份: 2026; 主题: Agent RL; Supporting Perception; 优先级: 核心
- [来源](#)

论文: EE-RL: Vision Language Guided Reinforcement Learning with Explorer and Expert model for End-to-End Autonomous Driving

快速理解: End-to-end driving frameworks, which directly map raw sensor data to vehicle control commands, have shown remarkable potential. However, their performance often deteriorates in sparse-critical scenarios, where rare but safety-sensitive events occur.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.19 Efficient Hybrid SE(3)-Equivariant Visuomotor Flow Policy via Spherical Harmonics for Robot Manipulation

- 作者: 作者信息未提供
- 年份: 2026; 主题: Embodied/WBC; Robot Learning; 优先级: 核心
- [来源](#)

论文: Efficient Hybrid SE(3)-Equivariant Visuomotor Flow Policy via Spherical Harmonics for Robot Manipulation

快速理解: While existing equivariant methods enhance data efficiency, they suffer from high computational intensity, reliance on single-modality inputs, and instability when combined with fast-sampling methods. In this work, we propose E3Flow, a novel framework that addresses the critical limitations of equivariant diffusion policies.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 本地 PDF 前页。

1.3.2.20 Evo-1: Lightweight Vision-Language-Action Model with Preserved Semantic Alignment

- 作者: 作者信息未提供
- 年份: 2026; 主题: VLA; Supporting Perception; 优先级: 核心
- [来源](#)

论文: Evo-1: Lightweight Vision-Language-Action Model with Preserved Semantic Alignment

快速理解: Vision-Language-Action (VLA) models have emerged as a powerful framework that unifies perception, language, and control, enabling robots to perform diverse tasks through multimodal understanding. However, current VLA models typically contain massive parameters and rely heavily on large-scale robot data pretraining, leading to high computational costs during training, as well as limited deployability for real-time inference.

关注点: 视觉、语言条件与机器人动作策略之间的对齐; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.21 EvoGraph-R1: Self-Evolving Multimodal Knowledge Hypergraphs for Agentic Retrieval

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL; Supporting Perception；优先级：核心
- [来源](#)

论文：EvoGraph-R1: Self-Evolving Multimodal Knowledge Hypergraphs for Agentic Retrieval

快速理解：Retrieval-augmented generation (RAG) has emerged as a critical paradigm for grounding Multimodal Large Language Models (MLLMs) in external knowledge. Recent GraphRAG methods introduce structured entity-relation graphs to improve retrieval and reasoning.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.22 Fast-ThinkAct: Efficient Vision-Language-Action Reasoning via Verbalizable Latent Planning

- 作者：作者信息未提供
- 年份：2026；主题：VLA; Supporting Perception；优先级：核心
- [来源](#)

论文：Fast-ThinkAct: Efficient Vision-Language-Action Reasoning via Verbalizable Latent Planning

快速理解：Vision-Language-Action (VLA) tasks require reasoning over complex visual scenes and executing adaptive actions in dynamic environments. While recent studies on reasoning VLAs show that explicit chain-of-thought (CoT) can improve generalization, they suffer from high inference latency due to lengthy reasoning traces.

关注点：视觉、语言条件与机器人动作策略之间的对齐；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.23 Fine-Grained Post-Training Quantization for Large Vision Language Models with Quantization-Aware Integrated Gradients

- 作者：作者信息未提供
- 年份：2026；主题：AI Infra; Supporting Perception；优先级：核心
- [来源](#)

论文：Fine-Grained Post-Training Quantization for Large Vision Language Models with Quantization-Aware Integrated Gradients

快速理解：Large Vision Language Models (LVLMs) have achieved remarkable success in a range of downstream tasks that require multimodal interaction, but their capabilities come with substantial computational and memory overhead, which hinders practical deployment. Among numerous acceleration techniques, post-training quantization is a popular and effective strategy for reducing memory cost and accelerating inference.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.24 FlowHijack: A Dynamics-Aware Backdoor Attack on Flow-Matching Vision-Language-Action Models

- 作者：作者信息未提供
- 年份：2026；主题：VLA; Supporting Perception；优先级：核心
- [来源](#)

论文：FlowHijack: A Dynamics-Aware Backdoor Attack on Flow-Matching Vision-Language-Action Models

快速理解：Vision-Language-Action (VLA) models are emerging as a cornerstone for robotics, with flow-matching policies like 0 showing great promise in generating smooth, continuous actions. As these models advance, their unique action generation mechanism—the vector field dynamics—presents a critical yet unexplored security vulnerability, particularly backdoor vulnerabilities.

关注点：视觉、语言条件与机器人动作策略之间的对齐；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.25 GTR-Turbo: Merged Checkpoint is Secretly a Free Teacher for Agentic VLM Training

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL; AI Infra；优先级：核心
- [来源](#)

论文：GTR-Turbo: Merged Checkpoint is Secretly a Free Teacher for Agentic VLM Training

快速理解：Multi-turn reinforcement learning (RL) for multi-modal agents built upon vision-language models (VLMs) is hampered by sparse rewards and long-horizon credit assignment. Recent methods densify the reward by querying a teacher that provides step-level feedback, e.g., Guided Thought Reinforcement (GTR) [50] and On-Policy Distillation [25], but rely on costly, often privileged models as the teacher, limiting practicality and reproducibility.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.2.26 GaussianDWM: 3D Gaussian Driving World Model for Unified Scene Understanding and Multi-Modal Generation

- 作者：作者信息未提供
- 年份：2026；主题：World Model; Supporting Perception；优先级：核心
- [来源](#)

论文：GaussianDWM: 3D Gaussian Driving World Model for Unified Scene Understanding and Multi-Modal Generation

快速理解：Driving World Models (DWMs) have been developing rapidly with the advances of generative models. However, existing DWMs lack 3D scene understanding capabilities and can only generate content conditioned on input data, without the ability to interpret or reason about the driving environment.

关注点：环境动力学、视频预测、潜变量状态或模型式决策；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.27 GenieDrive: Towards Physics-Aware Driving World Model with 4D Occupancy Guided Video Generation

- 作者：作者信息未提供
- 年份：2026；主题：World Model; Supporting Perception; 优先级：核心
- [来源](#)

论文：GenieDrive: Towards Physics-Aware Driving World Model with 4D Occupancy Guided Video Generation

快速理解：Physics-aware driving world model is essential for drive planning, out-of-distribution data synthesis, and closedloop evaluation. However, existing methods often rely on a single diffusion model to directly map driving actions to videos, which makes learning difficult and leads to physically inconsistent outputs.

关注点：环境动力学、视频预测、潜变量状态或模型式决策；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.28 GeoDexGrasp: Geometry-aware Generation for Data-efficient and Physics-plausible Dexterous Grasping

- 作者：作者信息未提供
- 年份：2026；主题：Embodied/WBC; Robot Learning; 优先级：核心
- [来源](#)

论文：GeoDexGrasp: Geometry-aware Generation for Data-efficient and Physics-plausible Dexterous Grasping

快速理解：Achieving dexterous grasping remains a key challenge in robotics. Recent generative approaches enable diverse grasps through large-scale data-driven training, yet they often neglect object geometric priors, leading to low data efficiency and poor physical plausibility.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：本地 PDF 前页。

1.3.2.29 HiF-VLA: Hindsight, Insight and Foresight through Motion Representation for Vision-Language-Action Models

- 作者：作者信息未提供
- 年份：2026；主题：VLA; Supporting Perception; 优先级：核心
- [来源](#)

论文：HiF-VLA: Hindsight, Insight and Foresight through Motion Representation for Vision-Language-Action Models

快速理解：Vision-Language-Action (VLA) models have recently enabled robotic manipulation by grounding visual and linguistic cues into actions. However, most VLAs assume the Markov property, relying only on the current observation and thus suffering from temporal myopia that degrades long-horizon coherence.

关注点：视觉、语言条件与机器人动作策略之间的对齐；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.30 HybridDriveVLA: Vision-Language-Action Model with Visual CoT reasoning and ToT Evaluation for Autonomous Driving

- 作者：作者信息未提供
- 年份：2026；主题：VLA; Supporting Perception; 优先级：核心
- [来源](#)

论文：HybridDriveVLA: Vision-Language-Action Model with Visual CoT reasoning and ToT Evaluation for Autonomous Driving

快速理解：Vision-Language-Action (VLA) models are emerging as an important technology in autonomous driving, recognized for their sophisticated reasoning and interpretability. How- †Corresponding author.

关注点：视觉、语言条件与机器人动作策略之间的对齐；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.31 Image-to-Point Cloud Feature Back-Projection for Multimodal Training of 3D Semantic Segmentation

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception; Systems; 优先级：核心
- [来源](#)

论文：Image-to-Point Cloud Feature Back-Projection for Multimodal Training of 3D Semantic Segmentation

快速理解：The effective integration and utilization of multimodal data acquired from image cameras and LiDAR is of paramount importance for perception systems. This paper proposes Image-to-Point Cloud Feature Back-Projection (IPFP), a novel method for training multimodal fusion networks that back-projects aggregated image-feature centers (from nonprojection-aligned image pixels) into the point-cloud feature set via the estimated depth map.

关注点：3D、视频、多模态视觉或具身感知支撑能力；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.2.32 Improving Vision-language Models with Perception-centric Process Reward Models

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL; Supporting Perception; 优先级：核心
- [来源](#)

论文：Improving Vision-language Models with Perception-centric Process Reward Models

快速理解：Recent advancements in reinforcement learning with verifiable rewards (RLVR) have significantly improved the complex reasoning ability of vision-language models (VLMs). However, its outcome-level supervision is too coarse to diagnose and correct errors within the reasoning chain.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.33 Lifelong Imitation Learning with Multimodal Latent Replay and Incremental Adjustment

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL; Supporting Perception；优先级：核心
- [来源](#)

论文：Lifelong Imitation Learning with Multimodal Latent Replay and Incremental Adjustment

快速理解：We introduce a lifelong imitation learning framework that enables continual policy refinement across sequential tasks under realistic memory and data constraints. Our approach departs from conventional experience replay by operating entirely in a multimodal latent space, where compact representations of visual, linguistic, and robot's state information are stored and reused to support future learning.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.34 MASQuant: Modality-Aware Smoothing Quantization for Multimodal Large Language Models

- 作者：作者信息未提供
- 年份：2026；主题：AI Infra; Supporting Perception；优先级：核心
- [来源](#)

论文：MASQuant: Modality-Aware Smoothing Quantization for Multimodal Large Language Models

快速理解：Post-training quantization (PTQ) with computational invariance for Large Language Models (LLMs) have demonstrated remarkable advances, however, their application to Multimodal Large Language Models (MLLMs) presents substantial challenges. In this paper, we analyze SmoothQuant as a case study and identify two critical issues: Smoothing Misalignment and Cross-Modal Computational Invariance.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.35 MSGNav: Unleashing the Power of Multi-modal 3D Scene Graph for Zero-Shot Embodied Navigation

- 作者：作者信息未提供
- 年份：2026；主题：Embodied/WBC; Supporting Perception；优先级：核心
- [来源](#)

论文：MSGNav: Unleashing the Power of Multi-modal 3D Scene Graph for Zero-Shot Embodied Navigation

快速理解：Embodied navigation is a fundamental capability for robotic agents operating. Real-world deployment requires open vocabulary generalization and low training overhead, motivating zero-shot methods rather than task-specific RL training.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.36 MSRL: Scaling Generative Multimodal Reward Modeling via Multi-Stage Reinforcement Learning

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL; Supporting Perception；优先级：核心
- [来源](#)

论文: MSRL: Scaling Generative Multimodal Reward Modeling via Multi-Stage Reinforcement Learning

快速理解: Recent advances in multimodal reward modeling have been largely driven by a paradigm shift from discriminative to generative approaches. Building on this progress, recent studies have further employed reinforcement learning from verifiable rewards (RLVR) to enhance multimodal reward models (MRMs).

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策；3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.37 MaskDexGrasp: Generative Masked Modeling for Part-Aware Dexterous Grasp Synthesis

- 作者：作者信息未提供
- 年份：2026；主题：Embodied/WBC; Robot Learning；优先级：核心
- [来源](#)

论文: MaskDexGrasp: Generative Masked Modeling for Part-Aware Dexterous Grasp Synthesis

快速理解: Dexterous grasp generation is a predominant task that enables robots to perform human-level manipulation. However, a dexterous hand always maintains high-dimensional DoF and actuation space, making existing approaches that rely on holistic latent representations difficult to produce high-quality and semantically aligned grasps.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制；机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 本地 PDF 前页。

1.3.2.38 MergeVLA: Cross-Skill Model Merging Toward a Generalist Vision-Language-Action Agent

- 作者：作者信息未提供
- 年份：2026；主题：VLA; Supporting Perception；优先级：核心
- [来源](#)

论文: MergeVLA: Cross-Skill Model Merging Toward a Generalist Vision-Language-Action Agent

快速理解: Recent Vision-Language-Action (VLA) models reformulate vision-language models by tuning them with millions of robotic demonstrations. While they perform well when fine-tuned for a single embodiment or task family, extending them to multi-skill settings remains challenging: directly merging VLA experts trained on different tasks results in near-zero success rates.

关注点: 视觉、语言条件与机器人动作策略之间的对齐；3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.39 MoDES: Accelerating Mixture-of-Experts Multimodal Large Language Models via Dynamic Expert Skipping

- 作者：作者信息未提供
- 年份：2026；主题：AI Infra; Supporting Perception; 优先级：核心
- [来源](#)

论文：MoDES: Accelerating Mixture-of-Experts Multimodal Large Language Models via Dynamic Expert Skipping

快速理解：Mixture-of-Experts (MoE) Multimodal large language models (MLLMs) excel at vision-language tasks, but they suffer from high computational inefficiency. To reduce inference overhead, expert skipping methods have been proposed to deactivate redundant experts based on the current input tokens.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.40 MoEActok: A MoE-based Action Tokenizer for Vision-Language-Action Models

- 作者：作者信息未提供
- 年份：2026；主题：VLA; Supporting Perception; 优先级：核心
- [来源](#)

论文：MoEActok: A MoE-based Action Tokenizer for Vision-Language-Action Models

快速理解：Recent works on vision-language-action (VLA) models have made great progress in exploring action tokenizers that convert continuous control signals into discrete tokens to align with LLM/VLM training paradigms. These approaches typically train a single tokenizer over entire manipulation trajectories, which often comprise multiple distinct skills and thus pose a challenging optimization trade-off.

关注点：视觉、语言条件与机器人动作策略之间的对齐；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.41 ModularAgent: A Task-Aware Modular Framework for Joint Optimization of Multimodal Large Language Models and World Models

- 作者：作者信息未提供
- 年份：2026；主题：World Model; Supporting Perception; 优先级：核心
- [来源](#)

论文：ModularAgent: A Task-Aware Modular Framework for Joint Optimization of Multimodal Large Language Models and World Models

快速理解：Building generalist embodied agents requires a unified system that can interpret multimodal goals, model environment dynamics, and execute reliable actions across diverse realworld tasks. Multimodal large language models (MLLMs) offer strong semantic priors and cross-modal generalization, while world models (WMs) provide actionable latent dynamics for prediction and control.

关注点：环境动力学、视频预测、潜变量状态或模型式决策；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.42 Multimodal RewardBench 2: Evaluating Omni Reward Models for Interleaved Text and Image

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL; Supporting Perception; 优先级：核心
- [来源](#)

论文：Multimodal RewardBench 2: Evaluating Omni Reward Models for Interleaved Text and Image

快速理解：Reward models (RMs) are essential for training large language models (LLMs), but remain underexplored for omni models that handle interleaved image and text sequences. We introduce Multimodal RewardBench 2 (MMRB2), the first comprehensive benchmark for reward models on multimodal understanding and (interleaved) generation.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.43 PAM: A Pose-Appearance-Motion Engine for Sim-to-Real HOI Video Generation

- 作者：作者信息未提供
- 年份：2026；主题：Robot Learning; Supporting Perception; 优先级：核心
- [来源](#)

论文：PAM: A Pose-Appearance-Motion Engine for Sim-to-Real HOI Video Generation

快速理解：Hand-object interaction (HOI) reconstruction and synthesis are becoming central to embodied AI and AR/VR. Yet, despite rapid progress, existing HOI generation research remains fragmented across three disjoint tracks: (1) pose-only synthesis that predicts MANO trajectories without producing pixels; (2) single-image HOI generation that hallucinates appearance from masks or 2D cues but lacks dynamics; and (3) video generation methods that require both the entire pose sequence and the ground-truth first frame as inputs, preventing true sim-to-real deployment.

关注点：机器人数据、策略学习、操作、触觉或 sim-to-real；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.44 PointWorld: Scaling 3D World Models for In-The-Wild Robotic Manipulation

- 作者：作者信息未提供
- 年份：2026；主题：Embodied/WBC; World Model; 优先级：核心
- [来源](#)

论文：PointWorld: Scaling 3D World Models for In-The-Wild Robotic Manipulation

快速理解：Humans anticipate, from a glance and a contemplated action of their bodies, how the 3D world will respond, a capability that is equally vital for robotic manipulation. We introduce POINTWORLD, a large pre-trained 3D world model that unifies state and action in a shared 3D space as 3D point flows: given one or few RGB-D images and a sequence of low-level robot action commands, POINTWORLD forecasts per-pixel displacements in 3D that respond to the given actions.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；环境动力学、视频预测、潜变量状态或模型式决策。

证据层级：本地 PDF 前页。

1.3.2.45 PvP: Data-Efficient Humanoid Robot Learning with Proprioceptive-Privileged Contrastive Representations

- 作者：作者信息未提供
- 年份：2026；主题：Embodied/WBC; Robot Learning; 优先级：核心
- [来源](#)

论文：PvP: Data-Efficient Humanoid Robot Learning with Proprioceptive-Privileged Contrastive Representations

快速理解：Achieving efficient and robust whole-body control (WBC) is essential for enabling humanoid robots to perform complex tasks in dynamic environments. Despite the success of reinforcement learning (RL) in this domain, its sample inefficiency remains a significant challenge due to the intricate dynamics and partial observability of humanoid robots.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：本地 PDF 前页。

1.3.2.46 Quant Experts: Token-aware Adaptive Error Reconstruction with Mixture of Experts for Large Vision-Language Models Quantization

- 作者：作者信息未提供
- 年份：2026；主题：AI Infra; Supporting Perception; 优先级：核心
- [来源](#)

论文：Quant Experts: Token-aware Adaptive Error Reconstruction with Mixture of Experts for Large Vision-Language Models Quantization

快速理解：Post-Training Quantization (PTQ) has emerged as an effective technique for alleviating the substantial computational and memory overheads of Vision-Language Models (VLMs) by compressing both weights and activations without retraining the full model. Existing PTQ methods primarily rely on static identification and global compensation of sensitive or outlier channels, yet they often overlook the distributional differences of these important channels across inputs, leading to unsatisfactory quantization.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.47 Recurrent Reasoning with Vision-Language Models for Estimating Long-Horizon Embodied Task Progress

- 作者：作者信息未提供
- 年份：2026；主题：Embodied/WBC; Supporting Perception; 优先级：核心
- [来源](#)

论文：Recurrent Reasoning with Vision-Language Models for Estimating Long-Horizon Embodied Task Progress

快速理解：Accurately estimating task progress is critical for embodied agents to plan and execute long-

horizon, multistep tasks. Despite promising advances, existing Vision- Language Models (VLMs) based methods primarily leverage their video understanding capabilities, while neglecting their complex reasoning potential.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.48 SenseSearch: Empowering Vision-Language Models with High-Resolution Agentic Search-Reasoning via Reinforcement Learning

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL; Supporting Perception；优先级：核心
- [来源](#)

论文：SenseSearch: Empowering Vision-Language Models with High-Resolution Agentic Search-Reasoning via Reinforcement Learning

快速理解：Vision-Language Models (VLMs) are limited by static knowledge and insufficient fine-grained visual analysis, hindering their performance on knowledge-intensive and visually complex tasks. While recent research has explored VLMs that employ external tools like search or cropping to enhance model performance, they typically employ tools in isolation and lack the ability to coordinate multiple tools effectively.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.49 Stereo World Model: Camera-Guided Stereo Video Generation

- 作者：作者信息未提供
- 年份：2026；主题：World Model; Supporting Perception；优先级：核心
- [来源](#)

论文：Stereo World Model: Camera-Guided Stereo Video Generation

快速理解：We present StereoWorld, a camera-conditioned stereo world model that jointly learns appearance and binocular geometry for end-to-end stereo video generation. Unlike monocular RGB or RGBD approaches, StereoWorld operates exclusively within the RGB modality, while simultaneously grounding geometry directly from disparity. To efficiently achieve consistent stereo generation, our approach introduces two key designs: (1) a unified camera-frame RoPE that augments latent tokens with camera-aware rotary positional encoding, enabling relative, view and time consistent conditioning while preserving pretrained video priors via a stable attention initialization; and (2) a stereo-aware attention decomposition that factors full 4D attention into 3D intra-view attention plus horizontal row attention, leveraging the epipolar prior to capture disparity-aligned * Project Lead, †Corresponding Author correspondences with substantially lower compute.

关注点：环境动力学、视频预测、潜变量状态或模型式决策；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.50 TRM-VLA: Temporal-Aware Chain-of-Thought Reasoning and Memorization for Vision-Language-Action Models

- 作者：作者信息未提供

- 年份: 2026; 主题: VLA; Supporting Perception; 优先级: 核心
- [来源](#)

论文: TRM-VLA: Temporal-Aware Chain-of-Thought Reasoning and Memorization for Vision-Language-Action Models

快速理解: Vision-Language-Action (VLA) models have emerged as a powerful paradigm for general robotic manipulation. However, existing approaches typically omit intermediate reasoning steps and directly regress actions, limiting reasoning interpretability and performance in long-horizon or compositional tasks.

关注点: 视觉、语言条件与机器人动作策略之间的对齐; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.51 TTRV: Test-Time Reinforcement Learning for Vision Language Models

- 作者: 作者信息未提供
- 年份: 2026; 主题: Agent RL; Supporting Perception; 优先级: 核心
- [来源](#)

论文: TTRV: Test-Time Reinforcement Learning for Vision Language Models

快速理解: Existing methods for extracting reward signals in Reinforcement Learning typically rely on labeled data and dedicated training splits, a setup that contrasts with how humans learn directly from their environment. In this work, we propose TTRV to enhance vision-language understanding by adapting the model on-the-fly at inference time, without the need for any labeled data.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.52 Test-Time Perturbation Tuning with Delayed Feedback for Vision-Language-Action Models

- 作者: 作者信息未提供
- 年份: 2026; 主题: VLA; Supporting Perception; 优先级: 核心
- [来源](#)

论文: Test-Time Perturbation Tuning with Delayed Feedback for Vision-Language-Action Models

快速理解: Vision-Language-Action models (VLAs) achieve remarkable performance in sequential decision-making but remain fragile to subtle environmental shifts, such as small changes in object pose. We attribute this brittleness to trajectory overfitting, where VLAs over-attend to the spurious correlation between actions and entities, then reproduce memorized action patterns.

关注点: 视觉、语言条件与机器人动作策略之间的对齐; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.53 Thinking With Videos: Multimodal Tool-Augmented Reinforcement Learning for Long Video Reasoning

- 作者: 作者信息未提供
- 年份: 2026; 主题: Agent RL; Supporting Perception; 优先级: 核心
- [来源](#)

论文: Thinking With Videos: Multimodal Tool-Augmented Reinforcement Learning for Long Video Reasoning

快速理解: The video reasoning ability of multimodal large language models (MLLMs) is crucial for downstream tasks like video question answering and temporal grounding. While recent approaches have explored text-based chain-of-thought (CoT) reasoning for MLLMs, these methods often suffer from limited cross-modal interaction and increased hallucination, especially with longer videos or reasoning chains.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.54 TopoMA: Topology-Guided Multi-Agent Dense RGB 3D Reconstruction via Distributed Inference

- 作者: 作者信息未提供
- 年份: 2026; 主题: Agent RL; Supporting Perception; 优先级: 核心
- [来源](#)

论文: TopoMA: Topology-Guided Multi-Agent Dense RGB 3D Reconstruction via Distributed Inference

快速理解: Multi-agent 3D reconstruction, as a key technology for large-scale VR/AR, robot swarms, and digital twins, has attracted growing attention. Recent end-to-end 3D reconstruction methods achieve strong performance in singleagent scenarios, but they are difficult to directly extend to multi-agent collaborative settings, where they often suffer from unstable tracking, excessive memory consumption, and frequent loop-closure failures, thus failing to meet realtime and large-scale deployment requirements.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.55 Towards Foundation Models for 3D Scene Understanding: Instance-Aware Self-Supervised Learning for Point Clouds

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; Systems; 优先级: 核心
- [来源](#)

论文: Towards Foundation Models for 3D Scene Understanding: Instance-Aware Self-Supervised Learning for Point Clouds

快速理解: Recent advances in self-supervised learning (SSL) for point clouds have substantially improved 3D scene understanding without human annotations. Existing approaches emphasize semantic awareness by enforcing feature consistency across augmented views or by masked scene modeling.

关注点: 3D、视频、多模态视觉或具身感知支撑能力; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.2.56 Training High-Level Schedulers with Execution-Feedback Reinforcement Learning for Long-Horizon GUI Automation

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL; Systems; 优先级：核心
- [来源](#)

论文：Training High-Level Schedulers with Execution-Feedback Reinforcement Learning for Long-Horizon GUI Automation

快速理解：The rapid development of large vision-language model (VLM) has greatly promoted the research of GUI agent. However, GUI agents still face significant challenges in handling long-horizon tasks.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.2.57 VLM-Loc: Localization in Point Cloud Maps via Vision-Language Models

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception; Systems; 优先级：核心
- [来源](#)

论文：VLM-Loc: Localization in Point Cloud Maps via Vision-Language Models

快速理解：Text-to-point-cloud (T2P) localization aims to infer precise spatial positions within 3D point cloud maps from natural language descriptions, reflecting how humans perceive and communicate spatial layouts through language. However, existing methods largely rely on shallow text-point cloud correspondence without effective spatial reasoning, limiting their accuracy in complex environments.

关注点：3D、视频、多模态视觉或具身感知支撑能力；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.2.58 VLM-PTQ: Efficient Post-Training Quantization for Large Vision-Language Models

- 作者：作者信息未提供
- 年份：2026；主题：AI Infra; Supporting Perception; 优先级：核心
- [来源](#)

论文：VLM-PTQ: Efficient Post-Training Quantization for Large Vision-Language Models

快速理解：Post-training quantization (PTQ) serves as a vital technique for efficiently compressing large-scale models, with weight-compensation methods such as GPTQ (symmetric calibration) and GPTAQ (asymmetric calibration) showing remarkable success. However, directly applying these methods to Vision-Language Models (VLMs) exposes two notable shortcomings: 1) the standard rounding-to-nearest (RTN) method is suboptimal for the asymmetric objective, failing to account for residual-induced shifts in the optimal quantization target; and 2) all input channels are processed uniformly across modalities, overlooking the distinct information densities.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.59 VQRAE: Representation Quantization Autoencoders for Multimodal Understanding, Generation and Reconstruction

- 作者：作者信息未提供
- 年份：2026；主题：AI Infra; Supporting Perception; 优先级：核心
- [来源](#)

论文：VQRAE: Representation Quantization Autoencoders for Multimodal Understanding, Generation and Reconstruction

快速理解：Unifying multimodal understanding, generation and reconstruction representation in a single tokenizer remains a key challenge in building unified models. Previous research predominantly attempts to address this in a dual encoder paradigm, e.g., utilizing the separate encoders for understanding and generation respectively or balancing semantic representations and low-level features with contrastive loss.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.60 ViLoMem: Agentic Learner with Grow-and-Refine Multimodal Semantic Memory

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL; Supporting Perception; 优先级：核心
- [来源](#)

论文：ViLoMem: Agentic Learner with Grow-and-Refine Multimodal Semantic Memory

快速理解：MLLMs exhibit strong reasoning on isolated queries, yet they operate de novo—solving each problem independently and often repeating the same mistakes. Current memory agents mainly reuse past trajectories.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.61 VideoWeaver: Multimodal Multi-View Video-to-Video Transfer for Embodied Agents

- 作者：作者信息未提供
- 年份：2026；主题：Embodied/WBC; Supporting Perception; 优先级：核心
- [来源](#)

论文：VideoWeaver: Multimodal Multi-View Video-to-Video Transfer for Embodied Agents

快速理解：Recent progress in video-to-video (V2V) translation has enabled realistic resimulation of embodied AI demonstrations, a capability that allows pretrained robot policies to be transferable to new environments without additional data collection. However, prior works can only operate on a single view at a time, while embodied AI tasks are commonly captured from multiple synchronized cameras to support policy learning.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.62 phi-DPO: Fairness Direct Preference Optimization Approach to Continual Learning in Large Multimodal Models

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：phi-DPO: Fairness Direct Preference Optimization Approach to Continual Learning in Large Multimodal Models

快速理解：Fairness in Continual Learning for Large Multimodal Models (LMs) is an emerging yet underexplored challenge, particularly in the presence of imbalanced data distributions that can lead to biased model updates and suboptimal performance across tasks. While recent continual learning studies have made progress in addressing catastrophic forgetting, the problem of fairness caused by imbalanced data remains largely underexplored.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.63 3D sans 3D Scans: Scalable Pre-training from Video-Generated Point Clouds

- 作者：作者信息未提供
- 年份：2026；主题：Systems；优先级：扩展
- [来源](#)

论文：3D sans 3D Scans: Scalable Pre-training from Video-Generated Point Clouds

快速理解：Despite recent progress in 3D self-supervised learning, collecting large-scale 3D scene scans remains expensive and labor-intensive. In this work, we investigate whether 3D representations can be learned from unlabeled videos recorded without any real 3D sensors.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.2.64 3D-Aware Multi-Task Learning with Cross-View Correlations for Dense Scene Understanding

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：3D-Aware Multi-Task Learning with Cross-View Correlations for Dense Scene Understanding

快速理解：This paper addresses the challenge of training a single network to jointly perform multiple dense prediction tasks, such as segmentation and depth estimation, i.e., multi-task learning (MTL). Current approaches mainly capture crosstask relations in the 2D image space, often leading to unstructured features lacking 3D-awareness.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.65 3D-Fixer: Coarse-to-Fine In-place Completion for 3D Scenes from a Single Image

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展

- [来源](#)

论文: 3D-Fixer: Coarse-to-Fine In-place Completion for 3D Scenes from a Single Image

快速理解: Compositional 3D scene generation from a single view requires the simultaneous recovery of scene layout and 3D assets. Existing approaches mainly fall into two categories: feed-forward generation methods and per-instance generation methods.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.66 4D Local Modeling Toward Dynamic Global Perception for Ambiguity-free Rotation-Invariant Point Cloud Analysis

- 作者: 作者信息未提供
- 年份: 2026; 主题: Systems; 优先级: 扩展
- [来源](#)

论文: 4D Local Modeling Toward Dynamic Global Perception for Ambiguity-free Rotation-Invariant Point Cloud Analysis

快速理解: Rotation invariance remains a core challenge in point cloud analysis, where existing methods often struggle with structural ambiguities and insufficient global context. Most rotation-invariant (RI) representations are derived from local coordinate systems, which inherently suffer from point-pair ambiguities and fail to capture discriminative features in symmetric or repetitive structures, while discarding informative global pose cues.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.2.67 A Closed-Form Solution for Debiasing Vision-Language Models with Utility Guarantees Across Modalities and Tasks

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: A Closed-Form Solution for Debiasing Vision-Language Models with Utility Guarantees Across Modalities and Tasks

快速理解: While Vision-Language Models (VLMs) have achieved remarkable performance across diverse downstream tasks, recent studies have shown that they can inherit social biases from the training data and further propagate them into downstream applications. To address this issue, various debiasing approaches have been proposed, yet most of them aim to improve fairness without having a theoretical guarantee that the utility of the model is preserved.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.68 A Frame is Worth One Token: Efficient Generative World Modeling with Delta Tokens

- 作者: 作者信息未提供

- 年份: 2026; 主题: World Model; 优先级: 扩展
- [来源](#)

论文: A Frame is Worth One Token: Efficient Generative World Modeling with Delta Tokens

快速理解: Anticipating diverse future states is a central challenge in video world modeling. Discriminative world models produce deterministic predictions that implicitly average over possible futures, while existing generative world models remain computationally expensive.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策。

证据层级: 本地 PDF 前页。

1.3.2.69 A Multi-Agent Perception-Action Alliance for Efficient Long Video Reasoning

- 作者: 作者信息未提供
- 年份: 2026; 主题: Agent RL; 优先级: 扩展
- [来源](#)

论文: A Multi-Agent Perception-Action Alliance for Efficient Long Video Reasoning

快速理解: This paper presents a multi-agent perception-action exploration alliance, dubbed A4VL, for efficient long-video reasoning. A4VL operates in a multi-round perception-action exploration loop with a selection of VLM agents.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.2.70 AGENTS SAFE: Benchmarking the Safety of Embodied Agents on Hazardous Instructions

- 作者: 作者信息未提供
- 年份: 2026; 主题: Embodied/WBC; 优先级: 扩展
- [来源](#)

论文: AGENTS SAFE: Benchmarking the Safety of Embodied Agents on Hazardous Instructions

快速理解: The integration of vision-language models (VLMs) is driving a new generation of embodied agents capable of operating in human-centered environments. However, as deployment expands, these systems face growing safety risks, particularly when executing hazardous instructions.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制。

证据层级: 本地 PDF 前页。

1.3.2.71 AGFT: Alignment-Guided Fine-Tuning for Zero-Shot Adversarial Robustness of Vision-Language Models

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: AGFT: Alignment-Guided Fine-Tuning for Zero-Shot Adversarial Robustness of Vision-Language Models

快速理解: Pre-trained vision-language models (VLMs) exhibit strong zero-shot generalization but remain vulnerable to adversarial perturbations. Existing classification-guided adversarial fine-tuning methods often disrupt pre-trained crossmodal alignment, weakening visual-textual correspondence and degrading zero-shot performance.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.72 APPO: Attention-guided Perception Policy Optimization for Video Reasoning

- 作者: 作者信息未提供
- 年份: 2026; 主题: Agent RL; 优先级: 扩展
- [来源](#)

论文: APPO: Attention-guided Perception Policy Optimization for Video Reasoning

快速理解: Complex video reasoning, actually, relies excessively on fine-grained perception rather than on expert (e.g., Ph.D, Science)-level reasoning. Through extensive empirical observation, we have recognized the critical impact of perception.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.2.73 AREA3D: Active Reconstruction Agent with Unified Feed-Forward 3D Perception and Vision-Language Guidance

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: AREA3D: Active Reconstruction Agent with Unified Feed-Forward 3D Perception and Vision-Language Guidance

快速理解: Active 3D reconstruction enables an agent to autonomously select viewpoints to build accurate and complete scene geometry, rather than passively reconstructing from precollected datasets. Existing active reconstruction methods often rely on geometric heuristics, which may lead to redundant observations without improving reconstruction quality.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.74 ARGUS: Defending Against Multimodal Indirect Prompt Injection via Steering Instruction-Following Behavior

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: ARGUS: Defending Against Multimodal Indirect Prompt Injection via Steering Instruction-Following Behavior

快速理解: Multimodal Large Language Models (MLLMs) are increasingly vulnerable to multimodal

Indirect Prompt Injection (IPI) attacks, which embed malicious instructions in images, videos, or audio to hijack model behavior. Existing defenses, designed primarily for text-only LLMs, are unsuitable for countering these multimodal threats, as they are easily bypassed, modality-dependent, or generalize poorly.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.75 AURA: Multi-modal Shared Autonomy for Urban Navigation

- 作者：作者信息未提供
- 年份：2026；主题：Teleoperation；优先级：扩展
- [来源](#)

论文：AURA: Multi-modal Shared Autonomy for Urban Navigation

快速理解：Long-horizon navigation in complex urban environments relies heavily on continuous human operation, which leads to fatigue, reduced efficiency, and safety concerns. Shared autonomy, where a Vision-Language AI agent and a human operator collaborate on maneuvering the mobile machine, presents a promising solution to address these issues.

关注点：人类示范、远程操作、共享自治或动作重定向。

证据层级：本地 PDF 前页。

1.3.2.76 AVATAR: Reinforcement Learning to See, Hear, and Reason Over Video

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL；优先级：扩展
- [来源](#)

论文：AVATAR: Reinforcement Learning to See, Hear, and Reason Over Video

快速理解：Multimodal reasoning over long-horizon video is challenging due to the need for precise spatiotemporal fusion and alignment across modalities. While recent methods such as Group Relative Policy Optimization (GRPO) have shown promise in this domain, they suffer from three key limitations: (1) data inefficiency from their on-policy design, (2) a vanishing advantage problem, where identical or near-identical rewards within a group eliminate the learning signal by producing zero-valued advantages, and (3) uniform credit assignment that fails to emphasize critical reasoning steps.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.2.77 AVION: Aerial Vision-Language Instruction from Offline Teacher to Prompt-Tuned Network

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：AVION: Aerial Vision-Language Instruction from Offline Teacher to Prompt-Tuned Network

快速理解：Adapting vision-language models to remote sensing imagery remains challenging due to

two key factors: limited semantic coverage in textual representations and insufficient adaptability of visual features. These issues are particularly significant in aerial scenes, which involve various visual appearances and fine-grained object distinctions.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.78 AXG-Reasoner: Error Detection and Explanation in Long Task Videos with Vision-Language Models

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: AXG-Reasoner: Error Detection and Explanation in Long Task Videos with Vision-Language Models

快速理解: Virtual task assistants must recognize and explain users' mistakes to provide effective and corrective guidance. In this paper, we address the problem of error reasoning in long task videos, which is to detect and explain errors.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.79 Abstract 3D Perception for Spatial Intelligence in Vision-Language Models

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Abstract 3D Perception for Spatial Intelligence in Vision-Language Models

快速理解: 3D Perception for Spatial Intelligence in Vision-Language Models Yifan Liu^{1,2}, Fangneng Zhan^{2,3}, Kaichen Zhou³, Yilun Du², Paul Pu Liang³, Hanspeter Pfister^{2†} ¹Tsinghua University ²Harvard University ³Massachusetts Institute of Technology liuyifan22@mails.tsinghua.edu.cn, pfisher@seas.harvard.edu Abstract Vision-language models (VLMs) struggle with 3D-related tasks such as spatial cognition and physical understanding, which are crucial for real-world applications like robotics and embodied agents. We attribute this to a modality gap between the 3D tasks and the 2D training of VLM, which led to inefficient retrieval of 3D information from 2D input.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.80 Accelerating Autoregressive Video Diffusion via History-Guided Cache and Residual Correction

- 作者: 作者信息未提供
- 年份: 2026; 主题: Systems; 优先级: 扩展
- [来源](#)

论文: Accelerating Autoregressive Video Diffusion via History-Guided Cache and Residual Correction

快速理解: Caching-based acceleration methods have recently driven significant progress in efficient

video generation with diffusion models. However, we identify a critical limitation when directly applying these acceleration techniques to auto-regressive video diffusion models, which generate long videos by sequentially synthesizing segments conditioned on historical context.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.2.81 Accelerating Diffusion via Hybrid Data-Pipeline Parallelism Based on Conditional Guidance Scheduling

- 作者：作者信息未提供
- 年份：2026；主题：Systems；优先级：扩展
- [来源](#)

论文：Accelerating Diffusion via Hybrid Data-Pipeline Parallelism Based on Conditional Guidance Scheduling

快速理解：Diffusion models have achieved remarkable progress in high-fidelity image, video, and audio generation, yet inference remains computationally expensive. Nevertheless, current diffusion acceleration methods based on distributed parallelism suffer from noticeable generation artifacts and fail to achieve substantial acceleration proportional to the number of GPUs.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.2.82 Action-Sketcher: From Reasoning to Action via Visual Sketches for Robotic Manipulation

- 作者：作者信息未提供
- 年份：2026；主题：Embodied/WBC；优先级：扩展
- [来源](#)

论文：Action-Sketcher: From Reasoning to Action via Visual Sketches for Robotic Manipulation

快速理解：Long-horizon robotic manipulation is increasingly important for real-world deployment, requiring spatial disambiguation in complex layouts and temporal resilience under dynamic interaction. † Project leaders. B Corresponding author: shanghang@pku.edu.cn

关注点：具身任务中的感知、动作生成、全身协调或运动控制。

证据层级：本地 PDF 前页。

1.3.2.83 Activation Matters: Test-time Activated Negative Labels for OOD Detection with Vision-Language Models

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Activation Matters: Test-time Activated Negative Labels for OOD Detection with Vision-Language Models

快速理解：Out-of-distribution (OOD) detection aims to identify samples that deviate from in-distribution (ID). One popular pipeline addresses this by introducing negative labels distant from

ID classes and detecting OOD based on their distance to these labels.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.84 Active Intelligence in Video Avatars via Closed-loop World Modeling

- 作者：作者信息未提供
- 年份：2026；主题：World Model；优先级：扩展
- [来源](#)

论文：Active Intelligence in Video Avatars via Closed-loop World Modeling

快速理解：Current video avatar generation methods excel at identity preservation and motion alignment but lack genuine agency—they cannot autonomously pursue long-term goals through adaptive environmental interaction. We address this by introducing L-IVA (Long-horizon Interactive Visual Avatar), a task and benchmark for evaluating goaldirected planning in stochastic generative environments, and ORCA (Online Reasoning and Cognitive Architecture), the first framework enabling active intelligence in video avatars.

关注点：环境动力学、视频预测、潜变量状态或模型式决策。

证据层级：本地 PDF 前页。

1.3.2.85 Active Perceptual Inference: A Corticothalamic-Inspired Dynamic Nested Recurrent Network for Multimodal Sentiment Analysis with Incomplete Data

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Active Perceptual Inference: A Corticothalamic-Inspired Dynamic Nested Recurrent Network for Multimodal Sentiment Analysis with Incomplete Data

快速理解：Random frame-level data missing is a critical challenge in multimodal sentiment analysis. Existing methods are largely limited to passive completion via single-pass feedforward connections and static cross-modal fusion, which struggle to generate high-quality completed features.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.86 AdaCluster: Adaptive Query-Key Clustering for Sparse Attention in Video Generation

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：AdaCluster: Adaptive Query-Key Clustering for Sparse Attention in Video Generation

快速理解：Video diffusion transformers (DiTs) suffer from prohibitive inference latency due to quadratic attention complexity. Existing sparse attention methods either overlook semantic similarity, or fail to adapt to heterogeneous token distributions across layers, leading to model performance degradation.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.87 AdaSVD: Singular Value Decomposition with Adaptive Mechanisms for Large Multimodal Models

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：AdaSVD: Singular Value Decomposition with Adaptive Mechanisms for Large Multimodal Models

快速理解：Large Multimodal Models (LMMs) have attained impressive achievements in multimodal processing tasks, yet their massive memory demands pose major obstacles to deployment on resource-limited devices. Singular Value Decomposition (SVD) has emerged as a promising compression technique for LMMs, delivering substantial reductions in memory overhead.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.88 AdaptVision: Efficient Vision-Language Models via Adaptive Visual Acquisition

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：AdaptVision: Efficient Vision-Language Models via Adaptive Visual Acquisition

快速理解：Vision-Language Models (VLMs) have achieved remarkable success in visual question answering tasks, but their reliance on large numbers of visual tokens introduces significant computational overhead. While existing efficient VLM approaches reduce visual tokens through fixed-ratio compression, they operate passively and lack the ability to adapt to varying task requirements.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.89 Adaptive Confidence Regularization for Multimodal Failure Detection

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Adaptive Confidence Regularization for Multimodal Failure Detection

快速理解：The deployment of multimodal models in high-stakes domains, such as self-driving vehicles and medical diagnostics, demands not only strong predictive performance but also reliable mechanisms for detecting failures. In this work, we address the largely unexplored problem of failure detection in multimodal contexts.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.90 AffordGrasp: Cross-Modal Diffusion for Affordance-Aware Grasp Synthesis

- 作者：作者信息未提供
- 年份：2026；主题：Robot Learning；优先级：扩展
- [来源](#)

论文：AffordGrasp: Cross-Modal Diffusion for Affordance-Aware Grasp Synthesis

快速理解：Generating human grasping poses that accurately reflect both object geometry and user-specified interaction semantics is essential for natural hand-object interactions in AR/VR and embodied AI. However, existing semantic grasping approaches struggle with the large modality gap between 3D object representations and textual instructions, and often lack explicit spatial or semantic constraints, leading to physically invalid or semantically inconsistent grasps.

关注点：机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：本地 PDF 前页。

1.3.2.91 AffordMatcher: Affordance Learning in 3D Scenes from Visual Signifiers

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：AffordMatcher: Affordance Learning in 3D Scenes from Visual Signifiers

快速理解：Affordance learning is a complex challenge in many applications, where existing approaches primarily focus on the geometric structures, visual knowledge, and affordance labels of objects to determine interactable regions. However, extending this learning capability to a scene is significantly more complicated, as incorporating object and scene-level semantics is not straightforward.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.92 Affordance Field Intervention: Enabling VLAs to Escape Memory Traps in Robotic Manipulation

- 作者：作者信息未提供
- 年份：2026；主题：Embodied/WBC；优先级：扩展
- [来源](#)

论文：Affordance Field Intervention: Enabling VLAs to Escape Memory Traps in Robotic Manipulation

快速理解：Vision-Language-Action (VLA) models have shown great performance in robotic manipulation by mapping visual observations and language instructions directly to actions. However, they remain brittle under distribution shifts: when test scenarios change, VLAs often reproduce memorized trajectories instead of adapting to the updated scene, which is a failure mode we refer to as the “Memory Trap” .

关注点：具身任务中的感知、动作生成、全身协调或运动控制。

证据层级：本地 PDF 前页。

1.3.2.93 Agent4FaceForgery: Multi-Agent LLM Framework for Realistic Face Forgery Detection

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL；优先级：扩展
- [来源](#)

论文：Agent4FaceForgery: Multi-Agent LLM Framework for Realistic Face Forgery Detection

快速理解：Face forgery detection faces a critical challenge: a persistent gap between offline benchmarks and real-world efficacy, which we attribute to the ecological invalidity of training data. This work introduces Agent4FaceForgery to address two fundamental problems: (1) how to capture the diverse intents and iterative processes of human forgery creation, and (2) how to model the complex, often adversarial, text-image interactions that accompany forgeries in social media.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.2.94 AgentDet: A Shared-Blackboard Multi-Agent Framework for Zero-/Few-Shot Object Detection

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL；优先级：扩展
- [来源](#)

论文：AgentDet: A Shared-Blackboard Multi-Agent Framework for Zero-/Few-Shot Object Detection

快速理解：Large multimodal language models have made rapid progress on vision-language tasks, yet their potential for zero-/few-shot object detection (ZSOD/FSOD) under a closed set of target classes has yet to be fully explored. ZSOD/FSOD is hampered by data scarcity and catastrophic forgetting.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.2.95 AlcheMinT: Fine-grained Temporal Control for Multi-Reference Consistent Video Generation

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：AlcheMinT: Fine-grained Temporal Control for Multi-Reference Consistent Video Generation

快速理解：Recent advances in subject-driven video generation with large diffusion models have enabled personalized content synthesis conditioned on user-provided subjects. However, existing methods lack fine-grained temporal control over subject appearance and disappearance, which are essential for applications such as compositional video synthesis, storyboarding, and controllable animation.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.96 Aligning What Vision-Language Models See and Perceive with Adaptive Information Flow

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Aligning What Vision-Language Models See and Perceive with Adaptive Information Flow

快速理解：Vision-Language Models (VLMs) have demonstrated strong capability in a wide range of tasks such as visual recognition, document parsing, and visual grounding. Nevertheless, recent work shows that while VLMs often manage to capture the correct image region corresponding to the question, they do not necessarily produce the correct answers.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.97 An Efficient Token Compression Framework for Visual Object Tracking

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：An Efficient Token Compression Framework for Visual Object Tracking

快速理解：Refining visual representations by eliminating their internal feature-level redundancy is crucial for simultaneously optimizing the performance and computational cost of models in visual tracking. To enhance their performance, many contemporary Transformer-based trackers leverage a larger number of historical template frames to capture richer spatio-temporal cues.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.98 Anchor-Guided Gradient Alignment for Incomplete Multimodal Learning

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Anchor-Guided Gradient Alignment for Incomplete Multimodal Learning

快速理解：Vision-language pre-training (VLP) has achieved remarkable performance across diverse multimodal learning (MML) tasks. Recently, many efforts have focused on reconstructing missing modalities to improve the adaptability of VLP models in incomplete MML scenarios.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.99 Anchoring the Mind of Multimodal Reasoners: Cognitive Bias as a Vector for Jailbreak Attacks

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展

- [来源](#)

论文: Anchoring the Mind of Multimodal Reasoners: Cognitive Bias as a Vector for Jailbreak Attacks

快速理解: Multimodal Large Reasoning Models (MLRMs) exhibit remarkable performance on complex tasks by incorporating explicit multi-step reasoning. However, this capability also introduces new security vulnerabilities.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.100 Anti-I2V: Safeguarding your Photos from Malicious Image-to-video Generation

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Anti-I2V: Safeguarding your Photos from Malicious Image-to-video Generation

快速理解: Advances in diffusion-based video generation models, while significantly improving human animation, poses threats of misuse through the creation of fake videos from a specific person's photo and text prompts. Recent efforts have focused on adversarial attacks that introduce crafted perturbations to protect images from diffusion-based models.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.101 AnyPcc: Compressing Any Point Cloud with a Single Universal Model

- 作者: 作者信息未提供
- 年份: 2026; 主题: Systems; 优先级: 扩展
- [来源](#)

论文: AnyPcc: Compressing Any Point Cloud with a Single Universal Model

快速理解: Generalization remains a critical challenge in deep learning-based point cloud geometry compression. While existing methods perform well on standard benchmarks, their performance collapses in real-world scenarios due to two fundamental limitations: the lack of context models that are robust across diverse data densities, and the inability to efficiently adapt to out-of-distribution (OOD) data.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.2.102 Archon: A Unified Multimodal Model for Holistic Digital Human Generation

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Archon: A Unified Multimodal Model for Holistic Digital Human Generation

快速理解: Digital humans are fundamental to immersive interaction, yet creating a unified model for holistic modalities, including text, audio, motion, and visual content, remains an open challenge. In this paper, we present Archon, a fully pretrained, human-centric unified multimodal model for holistic avatar generation.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.103 Attention-aware Inference Optimizations for Large Vision-Language Models with Memory-efficient Decoding

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Attention-aware Inference Optimizations for Large Vision-Language Models with Memory-efficient Decoding

快速理解: Large Vision-Language Models (VLMs) have achieved remarkable success in multi-modal reasoning, but their inference time efficiency remains a significant challenge due to the memory overhead during decoding, especially when the query and answer of VLMs consist of long sequences of visual and text tokens. This paper presents AttentionPack, an adaptive and attention-aware optimization framework tailored for large vision-language models with improving memory-efficiency during decoding, focusing on addressing the challenges due to the increased high number of visual inputs and interactions, particularly in long-context tasks with multiple high-resolution images or videos.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.104 AudioAvatar: Personalized Audio-driven Whole-body Talking Avatars

- 作者: 作者信息未提供
- 年份: 2026; 主题: Embodied/WBC; 优先级: 扩展
- [来源](#)

论文: AudioAvatar: Personalized Audio-driven Whole-body Talking Avatars

快速理解: Prior expressive whole-body conversational avatar systems map audio to parametric poses and then render, creating a lossy bottleneck where quantization, retargeting, and tracking errors accumulate. This degrades audio-motion synchronization and suppresses micro-articulations critical for realism—such as bilabial closures, cheek inflation, nasolabial motion, blinks, and fine hand gestures—especially under single-image personalization.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制。

证据层级: 本地 PDF 前页。

1.3.2.105 AutoTraces: Autoregressive Trajectory Forecasting via Multimodal Large Language Models

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展

- [来源](#)

论文: AutoTraces: Autoregressive Trajectory Forecasting via Multimodal Large Language Models

快速理解: We present AutoTraces, an autoregressive vision-language trajectory model for robot trajectory forecasting in human-populated environments, which harnesses the inherent reasoning capabilities of large language models (LLMs) to model complex human behaviors. In contrast to prior works that rely solely on textual representations, our key innovation lies in a novel trajectory tokenization scheme, which represents waypoints with tokens as categorical and positional markers while encoding waypoint numerical values as corresponding point embeddings, seamlessly integrated into the LLM's space through a lightweight encoder-decoder architecture.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.106 AviaSafe: A Physics-Informed Data-Driven Model for Aviation Safety-Critical Cloud Forecasts

- 作者: 作者信息未提供
- 年份: 2026; 主题: Systems; 优先级: 扩展
- [来源](#)

论文: AviaSafe: A Physics-Informed Data-Driven Model for Aviation Safety-Critical Cloud Forecasts

快速理解: Current AI weather forecasting models predict conventional atmospheric variables but cannot distinguish between cloud microphysical species critical for aviation safety. We introduce AviaSafe, a hierarchical, physics-informed neural forecaster that produces global, six-hourly predictions of these four hydrometeor species for lead times up to 7 days.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.2.107 AwareVLN: Reasoning with Self-awareness for Vision-Language Navigation

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: AwareVLN: Reasoning with Self-awareness for Vision-Language Navigation

快速理解: Vision-and-Language Navigation (VLN) requires an agent to ground language instructions to its own movement within a visual environment. While state-of-the-art methods leverage the reasoning capabilities of Vision-Language Models (VLMs) for end-to-end action prediction, they often lack an explicit and explainable understanding of the relationships between the agent, the instruction, and the scene.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.108 BA-GS: Bayesian Adaptive Gaussian Splatting for SFM-Free 3D Reconstruction

- 作者: 作者信息未提供

- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: BA-GS: Bayesian Adaptive Gaussian Splatting for SFM-Free 3D Reconstruction

快速理解: 3D Gaussian Splatting (3DGS) has demonstrated exceptional performance in reconstruction and novel view synthesis tasks. However, its reliance on Structure-from-Motion preprocessing may lead to degraded performance under sparse-view scenarios.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.109 Beyond Layer-Wise Merging: Chain-of-Merging for Vision-Language Models

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Beyond Layer-Wise Merging: Chain-of-Merging for Vision-Language Models

快速理解: (a) Vision-Language model Question Image Answer Visual Encoder Language Model Projection Merge (b) Layer-Wise Merging VLM LLM Merging VLM Merging Plan VLM-L2 LLM-L4 (2%), ..., VLM-L10 LLM-L9(6%), ..., VLM-L23 LLM-L23(4%) Merge (c) Chain-of-Merging LLM Merging VLM For each (Image, Question) This CVPR paper is the Open Access version, provided by the Computer Vision Foundation. Except for this watermark, it is identical to the accepted version; the final published version of the proceedings is available on IEEE Xplore.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.110 Beyond Mimicry: Learning Whole-Body Human-Humanoid Interaction from Human-Human Demonstrations

- 作者: 作者信息未提供
- 年份: 2026; 主题: Embodied/WBC; 优先级: 扩展
- [来源](#)

论文: Beyond Mimicry: Learning Whole-Body Human-Humanoid Interaction from Human-Human Demonstrations

快速理解: Enabling humanoid robots to physically interact with humans is a critical frontier, but progress is hindered by the scarcity of high-quality Human-Humanoid Interaction (HHoI) data. While leveraging abundant Human-Human Interaction (HHI) data presents a scalable alternative, we first demonstrate that standard retargeting fails by breaking the essential contacts.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制。

证据层级: 本地 PDF 前页。

1.3.2.111 Beyond Single Images: A Comprehensive Benchmark for Album-Level Vision-Language Understanding

- 作者: 作者信息未提供

- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Beyond Single Images: A Comprehensive Benchmark for Album-Level Vision-Language Understanding

快速理解: Automatic album organization has been studied extensively over the past decades due to significant progress in digital photography. Recent Vision-Language Models (VLMs) have shown strong performance on multi-image understanding, making them natural candidates for automating album organization workflows.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.112 Beyond Success: Refining Elegant Robot Manipulation from Mixed-Quality Data via Just-in-Time Intervention

- 作者: 作者信息未提供
- 年份: 2026; 主题: Embodied/WBC; 优先级: 扩展
- [来源](#)

论文: Beyond Success: Refining Elegant Robot Manipulation from Mixed-Quality Data via Just-in-Time Intervention

快速理解: Vision-Language-Action (VLA) models have enabled notable progress in general-purpose robotic manipulation, yet their learned policies often exhibit variable execution quality. We attribute this variability to the mixed-quality nature of human demonstrations, where the implicit principles that govern how actions should be carried out are only partially satisfied.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制。

证据层级: 本地 PDF 前页。

1.3.2.113 BiGMINT: Biologically-guided Hierarchical Multimodal Integration for Modeling Multiple Compound Activities in Drug Discovery

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: BiGMINT: Biologically-guided Hierarchical Multimodal Integration for Modeling Multiple Compound Activities in Drug Discovery

快速理解: Compound activity modeling is critical for drug discovery, where accurate in silico predictions can significantly reduce reliance on expensive, time-consuming targetspecific experimental assays. Traditional machine learning approaches for compound activity modeling typically rely on either chemoproteomics-centric molecular data or phenotype-centric imaging screens, limiting their ability to capture complementary biological signals.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.114 Bidirectional Multimodal Prompt Learning with Scale-Aware Training for Few-Shot Multi-Class Anomaly Detection

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Bidirectional Multimodal Prompt Learning with Scale-Aware Training for Few-Shot Multi-Class Anomaly Detection

快速理解：Few-shot multi-class anomaly detection is crucial in real industrial settings, where only a few normal samples are available while numerous object types must be inspected. This setting is challenging as defect patterns vary widely across categories while normal samples remain scarce.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.115 BiomedCCPL: Causal Conditional Prompt Learning for Biomedical Vision-Language Models

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：BiomedCCPL: Causal Conditional Prompt Learning for Biomedical Vision-Language Models

快速理解：Vision-language models (VLMs) have demonstrated strong potential for adapting to downstream biomedical tasks with limited training samples. However, their generalization to unseen classes within the same dataset remains limited, as the image-text alignment semantics often rely on spurious cues present in seen classes that do not transfer.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.116 Blink: Dynamic Visual Token Resolution for Enhanced Multimodal Understanding

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Blink: Dynamic Visual Token Resolution for Enhanced Multimodal Understanding

快速理解：Multimodal large language models (MLLMs) have achieved remarkable progress on various vision-language tasks, yet their visual perception remains limited. Humans, in comparison, perceive complex scenes efficiently by dynamically scanning and focusing on salient regions in a sequential “blink-like” process.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.117 Boosting Reasoning in Large Multimodal Models via Activation Replay

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Boosting Reasoning in Large Multimodal Models via Activation Replay

快速理解：Recently, Reinforcement Learning with Verifiable Rewards (RLVR) has emerged as an effective approach to incentivizing reasoning capability in Large Multimodal Models (LMMs), while the underlying mechanisms behind this post-training paradigm are poorly understood. We begin by exploring how input activations are affected by RLVR through the perspective of logit lens.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.118 Boosting Vision-Language Models Towards Cross-Domain Incremental Object Detection

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Boosting Vision-Language Models Towards Cross-Domain Incremental Object Detection

快速理解：Incremental Object Detection (IOD) aims to equip detectors with the ability to handle dynamic environments and emerging object categories, and the rise of vision-language models has substantially advanced this goal. However, existing studies often oversimplify real-world scenarios by assuming the incremental tasks come from a single general domain.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.119 Bootstrap Dynamic-Aware 3D Visual Representation for Scalable Robot Learning

- 作者：作者信息未提供
- 年份：2026；主题：Robot Learning；优先级：扩展
- [来源](#)

论文：Bootstrap Dynamic-Aware 3D Visual Representation for Scalable Robot Learning

快速理解：Despite strong results on recognition and segmentation, current 3D visual pre-training methods often underperform on robotic manipulation. We attribute this gap to two factors: the lack of state-action-state dynamics modeling and the unnecessary redundancy of explicit geometric reconstruction.

关注点：机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：本地 PDF 前页。

1.3.2.120 Breaking Multimodal LLM Safety via Video-Driven Prompting

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文: Breaking Multimodal LLM Safety via Video-Driven Prompting

快速理解: Multimodal Large Language Models (MLLMs) have achieved remarkable progress in visual reasoning tasks, serving as the core perception engines for emerging AI agents like OpenClaw. While recent studies have introduced several effective image-based jailbreak methods, the vulnerabilities inherent in the video modality remain a largely unexplored frontier.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.121 Breaking Smooth-Motion Assumptions: A UAV Benchmark for Multi-Object Tracking in Complex and Adverse Conditions

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Breaking Smooth-Motion Assumptions: A UAV Benchmark for Multi-Object Tracking in Complex and Adverse Conditions

快速理解: The rapid movements and agile maneuvers of unmanned aerial vehicles (UAVs) induce significant observational challenges for multi-object tracking (MOT). However, existing UAV-perspective MOT benchmarks often lack these complexities, featuring predominantly predictable camera dynamics and linear motion patterns.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.122 Breaking the 3D Dataset Bottleneck: Fast Scalable Generation of Aligned 3D Assets from Scratch for Category 6D Pose Estimation and Robotic Grasping

- 作者: 作者信息未提供
- 年份: 2026; 主题: Robot Learning; 优先级: 扩展
- [来源](#)

论文: Breaking the 3D Dataset Bottleneck: Fast Scalable Generation of Aligned 3D Assets from Scratch for Category 6D Pose Estimation and Robotic Grasping

快速理解: While 2D vision has been revolutionized by large-scale datasets like ImageNet, 3D vision remains constrained by the scarcity of high-quality, canonically aligned data. We introduce the first scalable, automated framework that generates complete category-level 6D pose datasets directly from text prompts, bypassing the need for existing 3D assets.

关注点: 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 本地 PDF 前页。

1.3.2.123 Breaking the Illusion: When Positive Meets Negative in Multimodal Decoding

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Breaking the Illusion: When Positive Meets Negative in Multimodal Decoding

快速理解: Vision-Language Models (VLMs) are frequently undermined by object hallucination—generating content that contradicts visual reality—due to an over-reliance on linguistic priors. We introduce Positive-and-Negative Decoding (PND), a training-free inference framework that intervenes directly in the decoding process to enforce visual fidelity.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.124 Breaking the Regional Perception Bottleneck of Multimodal Large Language Models via External Reasoning Framework

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Breaking the Regional Perception Bottleneck of Multimodal Large Language Models via External Reasoning Framework

快速理解: High-quality pixel-level responses remain a major bottleneck for multimodal large language models (MLLMs) in regional perception. Existing approaches generally attach regression decoders to MLLM features, achieving strong grounding performance but compromising end-to-end design and increasing training costs.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.125 Bridging Pixels and Words: Mask-Aware Local Semantic Fusion for Multimodal Media Verification

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Bridging Pixels and Words: Mask-Aware Local Semantic Fusion for Multimodal Media Verification

快速理解: As multimodal misinformation becomes more sophisticated, its detection and grounding are crucial. However, current multimodal verification methods, relying on passive holistic fusion, struggle with sophisticated misinformation.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.126 BuildAnyPoint: 3D Building Structured Abstraction from Diverse Point Clouds

- 作者: 作者信息未提供
- 年份: 2026; 主题: Systems; 优先级: 扩展
- [来源](#)

论文: BuildAnyPoint: 3D Building Structured Abstraction from Diverse Point Clouds

快速理解: We introduce BuildAnyPoint, a novel generative framework for structured 3D building reconstruction from point clouds with diverse distributions, such as those captured by airborne LiDAR and Structure-from-Motion. To recover artist-created building abstraction in this highly underconstrained setting, we capitalize on the role of explicit 3D generative priors in autoregressive mesh generation.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.2.127 BuildingGPT: Auto-Regressive Building Wireframe Reconstruction Model with Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2026; 主题: Agent RL; 优先级: 扩展
- [来源](#)

论文: BuildingGPT: Auto-Regressive Building Wireframe Reconstruction Model with Reinforcement Learning

快速理解: In this paper, we propose BuildingGPT, a novel autoregressive model for building wireframe reconstruction from point clouds with reinforcement learning. Unlike prior works based on detection or diffusion models, BuildingGPT reformulates the building wireframe reconstruction task into a sequence prediction problem.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.2.128 BulletTime: Decoupled Control of Time and Camera Pose for Video Generation

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: BulletTime: Decoupled Control of Time and Camera Pose for Video Generation

快速理解: Emerging video diffusion models achieve high visual fidelity but fundamentally couple scene dynamics with camera motion, limiting their ability to provide precise spatial and temporal control. We introduce a 4D-controllable video diffusion framework that explicitly decouples scene dynamics from camera pose, enabling fine-grained manipulation of both scene dynamics and camera viewpoint.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.129 C-GenReg: Training-Free 3D Point Cloud Registration by Multi-View-Consistent Geometry-to-Image Generation with Probabilistic Modalities Fusion

- 作者: 作者信息未提供
- 年份: 2026; 主题: Systems; 优先级: 扩展
- [来源](#)

论文: C-GenReg: Training-Free 3D Point Cloud Registration by Multi-View-Consistent Geometry-to-Image Generation with Probabilistic Modalities Fusion

快速理解: We introduce C-GenReg, a training-free framework for 3D point cloud registration that leverages the complementary strengths of world-scale generative priors and registration-oriented Vision Foundation Models (VFMs). Current learning-based 3D point cloud registration methods struggle to generalize across sensing modalities, sampling differences, and environments.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.2.130 CAPT: Confusion-Aware Prompt Tuning for Reducing Vision-Language Misalignment

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: CAPT: Confusion-Aware Prompt Tuning for Reducing Vision-Language Misalignment

快速理解: Vision-language models like CLIP have achieved remarkable progress in cross-modal representation learning, yet suffer from systematic misalignment among visually and semantically similar categories. We observe that such confusion patterns are not random but persistently occur between specific category pairs, revealing the model's intrinsic bias and limited fine-grained discriminative ability.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.131 CAR-SAM: Cross-Attention Reconstruction for Post-Training Quantization of the Segment Anything Model

- 作者: 作者信息未提供
- 年份: 2026; 主题: AI Infra; 优先级: 扩展
- [来源](#)

论文: CAR-SAM: Cross-Attention Reconstruction for Post-Training Quantization of the Segment Anything Model

快速理解: Segment Anything Models (SAMs) are extensively used in computer vision for universal image segmentation, but deploying them on resource-constrained devices is challenging due to their high computational and memory demands. Post-Training Quantization (PTQ) is a widely used technique for model compression and acceleration.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 本地 PDF 前页。

1.3.2.132 CARD: A Multi-Modal Automotive Dataset for Dense 3D Reconstruction in Challenging Road Topography

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展

- [来源](#)

论文: CARD: A Multi-Modal Automotive Dataset for Dense 3D Reconstruction in Challenging Road Topography

快速理解: Autonomous driving must operate across diverse surfaces to enable safe mobility. However, most driving datasets are captured on well-paved flat roads.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.133 CCCaption: Dual-Reward Reinforcement Learning for Complete and Correct Image Captioning

- 作者: 作者信息未提供
- 年份: 2026; 主题: Agent RL; 优先级: 扩展
- [来源](#)

论文: CCCaption: Dual-Reward Reinforcement Learning for Complete and Correct Image Captioning

快速理解: Image captioning remains a fundamental task for vision-language understanding, yet ground-truth supervision still relies predominantly on human-annotated references. Because human annotations reflect subjective preferences and expertise, ground-truth captions are often incomplete or even incorrect, which in turn limits caption models.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.2.134 CF-IPT: Cross-Modal Fusion Interactive Prompt Tuning of Vision-Language Pre-Trained Model for Multisource Remote Sensing Data Classification

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: CF-IPT: Cross-Modal Fusion Interactive Prompt Tuning of Vision-Language Pre-Trained Model for Multisource Remote Sensing Data Classification

快速理解: Fine-tuning Vision-Language Models (VLMs) trained on large-scale datasets of natural image-text pairs has demonstrated impressive performance for various downstream tasks. However, their fine-tuning for remote sensing (RS) tasks faces dual barriers: (1) Data-level barrier caused by the fundamental modality gap between natural imagery and RS data, and (2) Task-level barrier stemming from the requirement for multi-source interaction modeling capabilities.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.135 CICA: Coupling Confidence-Aware Pretraining with Confidence-Informed Attention for Robust Multimodal Sentiment Analysis

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: CICA: Coupling Confidence-Aware Pretraining with Confidence-Informed Attention for Robust Multimodal Sentiment Analysis

快速理解: Multimodal sentiment analysis requires integrating language, visual, and acoustic cues, yet these modalities are often noisy, incomplete, or contradictory, making fusion unreliable. Most existing methods assume uniformly trustworthy modalities and thus degrade when signals conflict.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.136 CLAY: Conditional Visual Similarity Modulation in Vision-Language Embedding Space

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: CLAY: Conditional Visual Similarity Modulation in Vision-Language Embedding Space

快速理解: Human perception of visual similarity is inherently adaptive and subjective, depending on the users' interests and focus. However, most image retrieval systems fail to reflect this flexibility, relying on a fixed, monolithic metric that cannot incorporate multiple conditions simultaneously.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.137 CLIPoint3D: Language-Grounded Few-Shot Unsupervised 3D Point Cloud Domain Adaptation

- 作者: 作者信息未提供
- 年份: 2026; 主题: Systems; 优先级: 扩展
- [来源](#)

论文: CLIPoint3D: Language-Grounded Few-Shot Unsupervised 3D Point Cloud Domain Adaptation

快速理解: Recent vision-language models (VLMs) such as CLIP demonstrate impressive cross-modal reasoning, extending beyond images to 3D perception. Yet, these models remain fragile under domain shifts, especially when adapting from synthetic to real-world point clouds.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.2.138 CLaD: Planning with Grounded Foresight via Cross-Modal Latent Dynamics

- 作者: 作者信息未提供
- 年份: 2026; 主题: World Model; 优先级: 扩展
- [来源](#)

论文: CLaD: Planning with Grounded Foresight via Cross-Modal Latent Dynamics

快速理解: Robotic manipulation involves kinematic and semantic transitions that are inherently coupled via underlying actions. However, existing approaches plan within either semantic or latent

space without explicitly aligning these cross-modal transitions.

关注点：环境动力学、视频预测、潜变量状态或模型式决策。

证据层级：本地 PDF 前页。

1.3.2.139 CLiViS: Unleashing Cognitive Map through Linguistic-Visual Synergy for Embodied Visual Reasoning

- 作者：作者信息未提供
- 年份：2026；主题：Embodied/WBC；优先级：扩展
- [来源](#)

论文：CLiViS: Unleashing Cognitive Map through Linguistic-Visual Synergy for Embodied Visual Reasoning

快速理解：Embodied Visual Reasoning (EVR) seeks to follow complex, free-form instructions based on egocentric video, enabling semantic understanding and spatiotemporal reasoning in dynamic environments. Despite its promising potential, EVR encounters significant challenges stemming from the diversity of complex instructions and the intricate spatiotemporal dynamics in long-term egocentric videos.

关注点：具身任务中的感知、动作生成、全身协调或运动控制。

证据层级：本地 PDF 前页。

1.3.2.140 CME-CAD: Heterogeneous Collaborative Multi-Expert Reinforcement Learning for CAD Code Generation

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL；优先级：扩展
- [来源](#)

论文：CME-CAD: Heterogeneous Collaborative Multi-Expert Reinforcement Learning for CAD Code Generation

快速理解：Computer-Aided Design (CAD) is essential in industrial design, but the complexity of traditional CAD modeling and workflows presents significant challenges for automating the generation of high-precision, editable CAD models. Existing methods that reconstruct 3D models from sketches often produce non-editable and approximate models that fall short of meeting the stringent requirements for precision and editability in industrial design.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.2.141 COPO: Causal-Oriented Policy Optimization for Hallucinations of MLLMs

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL；优先级：扩展
- [来源](#)

论文：COPO: Causal-Oriented Policy Optimization for Hallucinations of MLLMs

快速理解：Despite Multimodal Large Language Models (MLLMs) having shown impressive capabilities, they may suffer from hallucinations. Empirically, we find that MLLMs attend disproportionately to

task-irrelevant background regions compared with text-only LLMs, implying spurious background-answer correlations.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.2.142 CREward: A Type-Specific Creativity Reward Model

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL；优先级：扩展
- [来源](#)

论文：CREward: A Type-Specific Creativity Reward Model

快速理解：Creativity is a complex phenomenon. When it comes to representing and assessing creativity, treating it as a single undifferentiated quantity would appear naive and underwhelming.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.2.143 CUPID: Generative 3D Reconstruction via Joint Object and Pose Modeling

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：CUPID: Generative 3D Reconstruction via Joint Object and Pose Modeling

快速理解：We introduce CUPID, a generative 3D reconstruction framework that jointly models the full distribution over both canonical objects and camera poses. Our two-stage flowbased model first generates a coarse 3D structure and 2D-3D correspondences to estimate the camera pose robustly.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.144 CaT-GS: Efficient 3DGS Rendering for Large-Scale Scenes with Inter-frame Caching and Tile Scheduling

- 作者：作者信息未提供
- 年份：2026；主题：Systems；优先级：扩展
- [来源](#)

论文：CaT-GS: Efficient 3DGS Rendering for Large-Scale Scenes with Inter-frame Caching and Tile Scheduling

快速理解：Recent breakthroughs in 3D Gaussian Splatting (3DGS) have advanced neural rendering with high fidelity and efficiency. However, its performance degrades severely in large-scale scenes due to the increasing computational workload of tile-based rasterization.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.2.145 CapNav: Benchmarking Vision Language Models on Capability-conditioned Indoor Navigation

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：CapNav: Benchmarking Vision Language Models on Capability-conditioned Indoor Navigation

快速理解：Vision-Language Models (VLMs) have shown remarkable progress in Vision-Language Navigation (VLN), offering new possibilities for navigation decision-making that could benefit both robotic platforms and human users. However, real-world navigation is inherently conditioned by the agent's mobility constraints.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.146 CausalLens: Sensitivity-Guided Multi-Head Causal Intervention for Hallucination Mitigation in Large Vision-Language Models

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：CausalLens: Sensitivity-Guided Multi-Head Causal Intervention for Hallucination Mitigation in Large Vision-Language Models

快速理解：Recent Large Vision-Language Models (LVLMs) have shown impressive capabilities in multimodal understanding and generation. Despite this progress, they remain prone to hallucination, where model outputs conflict with the visual input due to an over-reliance on textual priors.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.147 CineBrain: A Large-Scale Multi-Modal Audiovisual Brain Dataset for Brain-Conditioned Video Generation

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：CineBrain: A Large-Scale Multi-Modal Audiovisual Brain Dataset for Brain-Conditioned Video Generation

快速理解：Most research decoding brain signals into images, often using them as priors for generative models, has focused only on visual content. This overlooks the brain's natural ability to integrate auditory and visual information, for instance, sound strongly influences how we perceive visual scenes.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.148 CineScene: Implicit 3D as Effective Scene Representation for Cinematic Video Generation

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：CineScene: Implicit 3D as Effective Scene Representation for Cinematic Video Generation

快速理解：Cinematic video production requires control over scenesubject composition and camera movement, but live-action shooting remains costly due to the need for constructing physical sets. To address this, we introduce the task of cinematic video generation with decoupled scene context: given multiple images of a static environment, the goal is to synthesize high-quality videos featuring dynamic subject while preserving the underlying scene consistency and following a user-specified camera trajectory.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.149 Cloning Deterministic Worlds: The Critical Role of Latent Geometry in Long-Horizon World Models

- 作者：作者信息未提供
- 年份：2026；主题：World Model；优先级：扩展
- [来源](#)

论文：Cloning Deterministic Worlds: The Critical Role of Latent Geometry in Long-Horizon World Models

快速理解：A world model is an internal model that simulates how the world evolves. Given past observations and actions, it predicts the future physical state of both the embodied agent and its environment.

关注点：环境动力学、视频预测、潜变量状态或模型式决策。

证据层级：本地 PDF 前页。

1.3.2.150 Cluster-Aware Neural Collapse Prompt Tuning for Long-Tailed Generalization of Vision-Language Models

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Cluster-Aware Neural Collapse Prompt Tuning for Long-Tailed Generalization of Vision-Language Models

快速理解：Prompt learning has emerged as an efficient alternative to fine-tuning pre-trained vision-language models (VLMs). Despite its promise, current methods still struggle to maintain tail-class discriminability when adapting to classimbalanced datasets.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.151 CoIn: Coverage and Informativeness-Guided Token Reduction for Efficient Large Multimodal Models

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：CoIn: Coverage and Informativeness-Guided Token Reduction for Efficient Large Multimodal Models

快速理解：Large Multimodal Models (LMMs) have shown remarkable success in visual understanding tasks. LMMs encode visual and textual inputs into tokens, which are then processed by Large Language Models (LLMs).

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.152 CoRiM: Conflict-driven Risk Minimization for Dynamic Multimodal Fusion

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：CoRiM: Conflict-driven Risk Minimization for Dynamic Multimodal Fusion

快速理解：Dynamic multimodal fusion methods lack robust theoretical guidance for handling modal conflicts and inconsistent data quality. While recent theoretical work has associated modal weights with scalar quantities such as loss or confidence, this paradigm struggles to comprehensively address the risks caused by inconsistent predictive distributions across modalities.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.153 CoVFT: Context-aware Visual Fine-tuning for Multimodal Large Language Models

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：CoVFT: Context-aware Visual Fine-tuning for Multimodal Large Language Models

快速理解：Multimodal large language models (MLLMs) achieve remarkable progress in cross-modal perception and reasoning, yet a fundamental question remains unresolved: should the vision encoder be fine-tuned or frozen? Despite the success of models such as LLaVA and Qwen-VL, inconsistent design choices and heterogeneous training setups hinder a unified understanding of visual fine-tuning (VFT) in MLLMs.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.154 CodeV: Code with Images for Faithful Visual Reasoning via Tool-Aware Policy Optimization

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL；优先级：扩展
- [来源](#)

论文：CodeV: Code with Images for Faithful Visual Reasoning via Tool-Aware Policy Optimization

快速理解：Agentic vision-language models are increasingly trained to “think with images” by calling image operations. However, we show that high final-answer accuracy often hides unfaithful visual reasoning: models may invoke tools on irrelevant regions or ignore tool outputs entirely, yet still guess the correct answer.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.2.155 Complementary Prototype Mapping for Efficient Multimodal Anomaly Detection

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Complementary Prototype Mapping for Efficient Multimodal Anomaly Detection

快速理解：Multimodal unsupervised anomaly detection has garnered increasing attention for robust defect localization. Recent approaches rely on establishing cross-modal matching relationships under normal conditions without explicit guidance.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.156 Conflict-Aware Adaptive Cross-Reconstruction for Multimodal Sentiment Analysis

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Conflict-Aware Adaptive Cross-Reconstruction for Multimodal Sentiment Analysis

快速理解：Disentanglement-based methods for learning shared representations are widely used in multimodal sentiment analysis. However, existing methods often overlook potential emotional conflicts across modalities within the same sample.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.157 ConsisVLA-4D: Advancing Spatiotemporal Consistency in Efficient 3D-Perception and 4D-Reasoning for Robotic Manipulation

- 作者：作者信息未提供
- 年份：2026；主题：Embodied/WBC；优先级：扩展
- [来源](#)

论文: ConsisVLA-4D: Advancing Spatiotemporal Consistency in Efficient 3D-Perception and 4D-Reasoning for Robotic Manipulation

快速理解: Current Vision-Language-Action (VLA) models primarily focus on mapping 2D observations to actions but exhibit notable limitations in spatiotemporal perception and reasoning: 1) spatial representations often rely on additional sensors, introducing substantial computational overhead; 2) visual reasoning is typically limited to future-frame prediction, lacking alignment with the instruction-grounded scene and thus compromising spatiotemporal consistency. To address these challenges, we propose ConsisVLA-4D, a unified and efficient framework that enhances spatiotemporal consistency in 3D-Perception and 4D-Reasoning.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制。

证据层级: 本地 PDF 前页。

1.3.2.158 Consistent Instance Field for Dynamic Scene Understanding

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Consistent Instance Field for Dynamic Scene Understanding

快速理解: We introduce Consistent Instance Field, a continuous and probabilistic spatio-temporal representation for dynamic scene understanding. Unlike prior methods that rely on discrete tracking or view-dependent features, our approach disentangles visibility from persistent object identity by modeling each space-time point with an occupancy probability and a conditional instance distribution.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.159 Copy-Transform-Paste: Zero-Shot Object-Object Alignment Guided by Vision-Language and Geometric Constraints

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Copy-Transform-Paste: Zero-Shot Object-Object Alignment Guided by Vision-Language and Geometric Constraints

快速理解: We study zero-shot 3D alignment of two given meshes, using a text prompt describing their spatial relation—an essential capability for content creation and scene assembly. Earlier approaches primarily rely on geometric alignment procedures, while recent work leverages pretrained 2D diffusion models to model language-conditioned object-object spatial relationships.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.160 Cross-Architecture Adaptation: Cloud-Edge Continual Test-Time Adaptation with Dynamic Sampling and Heterogeneous Distillation

- 作者：作者信息未提供
- 年份：2026；主题：Systems；优先级：扩展
- [来源](#)

论文：Cross-Architecture Adaptation: Cloud-Edge Continual Test-Time Adaptation with Dynamic Sampling and Heterogeneous Distillation

快速理解：Cloud-Edge Continual Test-Time Adaptation (CTTA)—with edge devices processing real-time data and the cloud offering strong computing power—is a critical paradigm for models that adapt to dynamic data distributions in realworld scenarios. However, most existing frameworks assume architectural homogeneity between cloud and edge CNNs, which poses a significant performance bottleneck, particularly given the rapid emergence of Transformerbased models.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.2.161 Cross-Modal Guided Visual Synthesis for Data-Efficient Multimodal Depression Recognition

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Cross-Modal Guided Visual Synthesis for Data-Efficient Multimodal Depression Recognition

快速理解：The performance of multimodal learning systems, particularly in high-stakes domains like automated depression recognition, is fundamentally constrained by the challenge of learning robust visual representations from limited and complex clinical data. To overcome this, we introduce Cross-Modal Guided Visual Synthesis (CMG-VS), a novel training framework that internally enhances the learning process by synthesizing new, task-relevant visual features.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.162 Cross-modal Identity Mapping: Minimizing Information Loss in Modality Conversion via Reinforcement Learning

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL；优先级：扩展
- [来源](#)

论文：Cross-modal Identity Mapping: Minimizing Information Loss in Modality Conversion via Reinforcement Learning

快速理解：Large Vision-Language Models (LVLMs) often omit or misrepresent critical visual content in generated image captions. Minimizing such information loss will force LVLMs to focus on image details to generate precise descriptions.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.2.163 CrossHOI-Bench: A Unified Benchmark for HOI Evaluation across Vision-Language Models and HOI-Specific Methods

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：CrossHOI-Bench: A Unified Benchmark for HOI Evaluation across Vision-Language Models and HOI-Specific Methods

快速理解：HOI detection has long been dominated by task-specific models, sometimes with early vision-language backbones such as CLIP. With the rise of large generative VLMs, a key question is whether standalone VLMs can perform HOI detection competitively against specialized HOI methods.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.164 CrossVL: Complexity-Aware Feature Routing and Paired Curriculum for Cross-View Vision-Language Detection

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：CrossVL: Complexity-Aware Feature Routing and Paired Curriculum for Cross-View Vision-Language Detection

快速理解：Vision-language models (VLMs) enable text-guided object detection but degrade severely under cross-view scenarios where ground and aerial viewpoints differ in altitude, scale, and spatial layout. These geometric changes introduce systematic complexity variations between viewpoints, e.g., ground view images contain dense and highly occluded structures, while aerial images are sparse and globally organized.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.165 CubeComposer: Spatio-Temporal Autoregressive 4K 360deg Video Generation from Perspective Video

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：CubeComposer: Spatio-Temporal Autoregressive 4K 360deg Video Generation from Perspective Video

快速理解：Generating high-quality 360° panoramic videos from perspective input is one of the crucial applications for virtual reality (VR), whereby high-resolution videos are especially important for immersive experience. Existing methods are constrained by computational limitations of vanilla diffusion models, only supporting 1K resolution native generation and relying on suboptimal post-hoc super-resolution.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.166 Curvature-Aware Captioning: Leveraging Geodesic Attention for 3D Scene Understanding

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Curvature-Aware Captioning: Leveraging Geodesic Attention for 3D Scene Understanding

快速理解：Accurate 3D scene description is fundamental to robotic navigation and augmented reality, yet current dense captioning methods face significant limitations in processing sparse point cloud data. Existing approaches that apply Euclidean embedding spaces struggle to simultaneously preserve fine-grained local geometric details and model exponentially growing global semantic hierarchies, leading to either inaccurate localization or disjointed, shallow scene descriptions.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.167 Cut to the Chase: Training-free Multimodal Summarization via Chain-of-Events

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Cut to the Chase: Training-free Multimodal Summarization via Chain-of-Events

快速理解：Multimodal Summarization (MMS) aims to generate concise textual summaries by understanding and integrating information across videos, transcripts, and images. However, existing approaches still suffer from three main challenges: (1) reliance on domain-specific supervision, (2) implicit fusion with weak cross-modal grounding, and (3) flat temporal modeling without event transitions.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.168 D2Cache: Second-Order Delta Caching for Higher Video Diffusion Acceleration

- 作者：作者信息未提供
- 年份：2026；主题：Systems；优先级：扩展
- [来源](#)

论文：D2Cache: Second-Order Delta Caching for Higher Video Diffusion Acceleration

快速理解：Video diffusion models achieve impressive visual fidelity but remain computationally prohibitive for real-time or interactive generation due to their sequential denoising process. Recent caching methods accelerate inference by reusing outputs across timesteps, typically estimating each new output from the first-order residual, which is the difference between adjacent model predictions.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.2.169 DDiT: Dynamic Patch Scheduling for Efficient Diffusion Transformers

- 作者：作者信息未提供
- 年份：2026；主题：Systems；优先级：扩展
- [来源](#)

论文：DDiT: Dynamic Patch Scheduling for Efficient Diffusion Transformers

快速理解: Diffusion Transformers (DiTs) have achieved state-of-the-art performance in image and video generation, but their success comes at the cost of heavy computation. This inefficiency is largely due to the fixed tokenization process, which uses constant-sized patches throughout the entire denoising phase, regardless of the content's complexity.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.2.170 DEVA: Fine-tuning Multimodal Large Language Models for Visual Perception Tasks

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：DEVA: Fine-tuning Multimodal Large Language Models for Visual Perception Tasks

快速理解: Fine-tuning large language models (LLMs) using reinforcement learning (RL) objectives has gained traction, especially in scenarios where labeled data is limited. Building on its success in the language domain, recent efforts have extended RL-based fine-tuning to multimodal tasks.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.171 DLWM: Dual Latent World Models enable Holistic Gaussian-centric Pre-training in Autonomous Driving

- 作者：作者信息未提供
- 年份：2026；主题：World Model；优先级：扩展
- [来源](#)

论文：DLWM: Dual Latent World Models enable Holistic Gaussian-centric Pre-training in Autonomous Driving

快速理解: Vision-based autonomous driving has gained much attention due to its low costs and excellent performance. Compared with dense BEV (Bird's Eye View) or sparse query models, Gaussian-centric method is a comprehensive yet sparse representation by describing scene with 3D semantic Gaussians.

关注点：环境动力学、视频预测、潜变量状态或模型式决策。

证据层级：本地 PDF 前页。

1.3.2.172 DPL: Decoupled Prototype Learning for Enhancing Robustness of Vision-Language Transformers to Missing Modalities

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：DPL: Decoupled Prototype Learning for Enhancing Robustness of Vision-Language Transformers to Missing Modalities

快速理解：The performance of Vision-Language Transformers drops sharply when an input modality (e.g., image) is missing, because the model is forced to make predictions using incomplete information. Existing missing-aware prompt methods help reduce this degradation, but they still rely on conventional prediction heads (e.g., a Fully-Connected layer) that compute class scores in the same way regardless of which modality is present or absent.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.173 DRM: Diffusion-based Reward Model With Step-wise Guidance

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL；优先级：扩展
- [来源](#)

论文：DRM: Diffusion-based Reward Model With Step-wise Guidance

快速理解：Current mainstream methods of aligning diffusion models with human preferences typically employ VLM-based reward models. However, these reward models, pre-trained for semantic alignment, struggle to capture the essential perceptual qualities—such as aesthetics, composition, and visual harmony.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.2.174 DarkAct: A RGB-Thermal Dataset and Fusion Framework for Multimodal Low-Light Action Recognition

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：DarkAct: A RGB-Thermal Dataset and Fusion Framework for Multimodal Low-Light Action Recognition

快速理解：Human action recognition (HAR) in low-light environments remains challenging due to degraded visibility, illumination variance, and loss of appearance cues. We introduce DarkAct, a large-scale and high-quality RGB-thermal video dataset purpose-built for multimodal action recognition under low illumination.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.175 DeRVOS: Decoupling Consistent Trajectory Generation and Multimodal Understanding for Referring Video Object Segmentation

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：DeRVOS: Decoupling Consistent Trajectory Generation and Multimodal Understanding for Referring Video Object Segmentation

快速理解：Referring video object segmentation (RVOS) aims to segment objects within a video according to natural language expressions. Unlike earlier works focusing on static singleobject scenarios, recent studies address more complex motion scenes.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.176 DeepScan: A Training-Free Framework for Visually Grounded Reasoning in Large Vision-Language Models

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：DeepScan: A Training-Free Framework for Visually Grounded Reasoning in Large Vision-Language Models

快速理解：Humans can robustly localize visual evidence and provide grounded answers even in noisy environments by identifying critical cues and then relating them to the full context in a bottom-up manner. Inspired by this, we propose Deep- Scan, a training-free framework that combines Hierarchical Scanning, Refocusing, and Evidence-Enhanced Reasoning for visually grounded reasoning in Large Vision-Language Models (LVLMs).

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.177 Deeper Thought, Weaker Aim: Understanding and Mitigating Perceptual Impairment during Reasoning in Multimodal Large Language Models

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Deeper Thought, Weaker Aim: Understanding and Mitigating Perceptual Impairment during Reasoning in Multimodal Large Language Models

快速理解：Multimodal large language models (MLLMs) often suffer from perceptual impairments under extended reasoning modes, particularly in visual question answering (VQA) tasks. We identify attention dispersion as the underlying cause: during multi-step reasoning, model’ s visual attention becomes scattered and drifts away from questionrelevant regions, effectively “losing focus” on the visual input.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.178 Deformation-based In-Context Learning for Point Cloud Understanding

- 作者：作者信息未提供
- 年份：2026；主题：Systems；优先级：扩展
- [来源](#)

论文：Deformation-based In-Context Learning for Point Cloud Understanding

快速理解：Recent advances in point cloud In-Context Learning (ICL) have demonstrated strong multitask capabilities. Existing approaches typically adopt a Masked Point Modeling (MPM)-based paradigm for point cloud ICL.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.2.179 Dejavu: Towards Experience Feedback Learning for Embodied Intelligence

- 作者：作者信息未提供
- 年份：2026；主题：Embodied/WBC；优先级：扩展
- [来源](#)

论文：Dejavu: Towards Experience Feedback Learning for Embodied Intelligence

快速理解：Embodied agents face a fundamental limitation: once deployed in real-world environments, they cannot easily acquire new knowledge to improve task performance. In this paper, we propose Dejavu, a general postdeployment learning framework that augments a frozen Vision-Language-Action (VLA) policy with retrieved execution memories through an Experience Feedback Network (EFN).

关注点：具身任务中的感知、动作生成、全身协调或运动控制。

证据层级：本地 PDF 前页。

1.3.2.180 Denoising as Path Planning: Training-Free Acceleration of Diffusion Models with DPCache

- 作者：作者信息未提供
- 年份：2026；主题：Systems；优先级：扩展
- [来源](#)

论文：Denoising as Path Planning: Training-Free Acceleration of Diffusion Models with DPCache

快速理解：Diffusion models have demonstrated remarkable success in image and video generation, yet their practical deployment remains hindered by the substantial computational overhead of multi-step iterative sampling. Among acceleration strategies, caching-based methods offer a training-free and effective solution by reusing or predicting features across timesteps.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.2.181 Depth Any Endoscopy: Towards Self-Supervised Generalizable Depth Estimation in Monocular Endoscopy

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文: Depth Any Endoscopy: Towards Self-Supervised Generalizable Depth Estimation in Monocular Endoscopy

快速理解: Monocular depth estimation serves as a core technique in endoscopic applications such as 3D reconstruction and localization. However, most existing methods focus primarily on in-domain depth estimation, which limits their robustness and prevents them from delivering impressive crossdomain performance, due to variations in depth distributions, illumination conditions, and texture patterns.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.182 Depth Any Panoramas: A Foundation Model for Panoramic Depth Estimation

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文: Depth Any Panoramas: A Foundation Model for Panoramic Depth Estimation

快速理解: In this work, we present a panoramic metric depth foundation model that generalizes across diverse scene distances. We explore a data-in-the-loop paradigm from the view of both data construction and framework design.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.183 Depth Hypothesis Guided Iterative Refinement for Event-Image Monocular Depth Estimation

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文: Depth Hypothesis Guided Iterative Refinement for Event-Image Monocular Depth Estimation

快速理解: Event cameras hold excellent dynamic properties, showing great potential for monocular depth estimation (MDE). However, existing methods mainly improve performance by optimizing contextual features, but still struggle with the illposed and nonlinear nature of direct full-depth regression.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.184 DiG: Differential Grounding for Enhancing Fine-Grained Perception in Multimodal Large Language Models

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展

- [来源](#)

论文: DiG: Differential Grounding for Enhancing Fine-Grained Perception in Multimodal Large Language Models

快速理解: Multimodal Large Language Models have achieved impressive performance on a variety of vision-language tasks, yet their fine-grained visual perception and precise spatial reasoning remain limited. In this work, we introduce DiG (Differential Grounding), a novel proxy task framework where MLLMs learn fine-grained perception by identifying and localizing all differences between similar image pairs without prior knowledge of their number.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.185 Diagnosing and Repairing Unsafe Channels in Vision-Language Models via Causal Discovery and Dual-Modal Safety Subspace Projection

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Diagnosing and Repairing Unsafe Channels in Vision-Language Models via Causal Discovery and Dual-Modal Safety Subspace Projection

快速理解: Large Vision-Language Models (LVLMs) have achieved impressive performance across multimodal understanding and reasoning tasks, yet their internal safety mechanisms remain opaque and poorly controlled. In this work, we present a comprehensive framework for diagnosing and repairing unsafe channels within LVLMs (CARE).

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.186 Dictionary-Aligned Concept Control for Safeguarding Multimodal LLMs

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Dictionary-Aligned Concept Control for Safeguarding Multimodal LLMs

快速理解: Multimodal Large Language Models (MLLMs) have been shown to be vulnerable to malicious queries that can elicit unsafe responses. Recent work uses prompt engineering, response classification, or finetuning to improve MLLM safety.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.187 DiffuView: Multi-View Diffusion Pretraining for 3D Aware Robotic Manipulation

- 作者: 作者信息未提供
- 年份: 2026; 主题: Embodied/WBC; 优先级: 扩展
- [来源](#)

论文: DiffuView: Multi-View Diffusion Pretraining for 3D Aware Robotic Manipulation

快速理解: Robotic manipulation from visual observations remains challenging due to the lack of 3D consistent representations that can generalize across diverse viewpoints and sensor configurations. Existing methods, primarily based on masked autoencoders or neural scene representations, struggle to capture robust view correspondences due to a lack of global 3D consistency or the requirement for scene specific optimization.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制。

证据层级: 本地 PDF 前页。

1.3.2.188 Direction-aware 3D Large Multimodal Models

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Direction-aware 3D Large Multimodal Models

快速理解: 3D large multimodal models (3D LMMs) rely heavily on ego poses for enabling directional question-answering and spatial reasoning. However, most existing point cloud benchmarks contain rich directional queries but lack the corresponding ego poses, making them inherently ill-posed in 3D large multimodal modelling.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.189 Disentangle-then-Align: Non-Iterative Hybrid Multimodal Image Registration via Cross-Scale Feature Disentanglement

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Disentangle-then-Align: Non-Iterative Hybrid Multimodal Image Registration via Cross-Scale Feature Disentanglement

快速理解: Multimodal image registration is a fundamental task and a prerequisite for downstream cross-modal analysis. Despite recent progress in shared feature extraction and multi-scale architectures, two key limitations remain.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.190 Distribution-Aligned Multimodal Fusion for Robust Object Detection

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Distribution-Aligned Multimodal Fusion for Robust Object Detection

快速理解: Cross-degradation generalization remains a critical challenge for RGB-infrared multimodal

object detection, especially when training data covers limited degradation types. This paper presents a distribution alignment framework with a key insight: aligning fused features to the pretrained distribution where the frozen detector performs optimally, rather than adapting to training-specific degradations.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.191 Do Vision-Language Models Leak What They Learn? Adaptive Token-Weighted Model Inversion Attacks

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Do Vision-Language Models Leak What They Learn? Adaptive Token-Weighted Model Inversion Attacks

快速理解: Model inversion (MI) attacks pose significant privacy risks by reconstructing private training data from trained neural networks. While prior studies have primarily examined unimodal deep networks, the vulnerability of visionlanguage models (VLMs) remains largely unexplored.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.192 Do Vision-Language Models Measure Up? Benchmarking Visual Measurement Reading with MeasureBench

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Do Vision-Language Models Measure Up? Benchmarking Visual Measurement Reading with MeasureBench

快速理解: Reading measurement instruments is effortless for humans and requires relatively little domain expertise, yet it remains surprisingly challenging for current vision-language models (VLMs) as we find in preliminary evaluation. In this work, we introduce MeasureBench, a benchmark on visual measurement reading covering both real-world and synthesized images of various types of measurements, along with an extensible pipeline for data synthesis.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.193 DreamOmni2: Multimodal Instruction-based Generation and Editing

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: DreamOmni2: Multimodal Instruction-based Generation and Editing

快速理解: attributions. Abstract Recent advancements in instruction-based image editing and subject-

driven generation have garnered significant attention, yet both tasks still face limitations in meeting practical user needs.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.194 DriveLaW: Unifying Planning and Video Generation in a Latent Driving World

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：DriveLaW: Unifying Planning and Video Generation in a Latent Driving World

快速理解：World models have become crucial for autonomous driving, as they learn how scenarios evolve over time to address the long-tail challenges of the real world. However, current approaches relegate world models to limited roles: they operate within ostensibly unified architectures that still keep world prediction and motion planning as decoupled processes.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.195 Dual-Granularity Memory for Efficient Video Generation

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Dual-Granularity Memory for Efficient Video Generation

快速理解：Recurrent architectures offer significant efficiency advantages over attention-based transformers for video generation, particularly for long sequences. However, chunked processing in these models creates temporal discontinuities that degrade long-range consistency.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.196 DualPrim: Compact 3D Reconstruction with Positive and Negative Primitives

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：DualPrim: Compact 3D Reconstruction with Positive and Negative Primitives

快速理解：Neural reconstructions often trade structure for fidelity, yielding dense and unstructured meshes with irregular topology and weak part boundaries that hinder editing, animation, and downstream asset reuse. We present Dual-Prim, a compact and structured 3D reconstruction framework.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.197 DuetSVG: Unified Multimodal SVG Generation with Internal Visual Guidance

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：DuetSVG: Unified Multimodal SVG Generation with Internal Visual Guidance

快速理解：Recent vision-language model (VLM)-based approaches have achieved impressive results on SVG generation. However, because they generate only text and lack visual signals during decoding, they often struggle with complex semantics and fail to produce visually appealing or geometrically coherent SVGs.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.198 DyFCLT: Dynamic Frequency-Decoupled Cross-Modal Learning Transformer for Multimodal Tiny Object Detection

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：DyFCLT: Dynamic Frequency-Decoupled Cross-Modal Learning Transformer for Multimodal Tiny Object Detection

快速理解：Multimodal tiny object detection plays a critical role in real-world applications, yet remains highly challenging due to weak target representations and complex cross-modal interference. Existing frequency-domain methods for tiny object detection are still largely limited to the visible modality and overlook complementary cross-modal frequency cues in multimodal scenes.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.199 DynBridge: Bridging Imagination and Control through Interaction Dynamics for Robot Manipulation

- 作者：作者信息未提供
- 年份：2026；主题：Embodied/WBC；优先级：扩展
- [来源](#)

论文：DynBridge: Bridging Imagination and Control through Interaction Dynamics for Robot Manipulation

快速理解：Recent generative models allow robots to generate future visual outcomes for action guidance, yet most still address imagination and control independently, resulting in visually coherent rollouts but physically inconsistent behaviors. While structural priors enhance spatial grounding, these methods remain visually correlation-driven rather than causally informed, overlooking the bidirectional coupling between robot actions and the evolving environment.

关注点：具身任务中的感知、动作生成、全身协调或运动控制。

证据层级：本地 PDF 前页。

1.3.2.200 Dynamic Token Reweighting for Robust Vision-Language Models

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Dynamic Token Reweighting for Robust Vision-Language Models

快速理解：Large vision-language models (VLMs) are highly vulnerable to multimodal jailbreak attacks that exploit visual-textual interactions to bypass safety guardrails. In this paper, we present DTR, a novel inference-time defense that mitigates multimodal jailbreak attacks through optimizing the model's key-value (KV) caches.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.201 Dynamic-eDiTor: Training-Free Text-Driven 4D Scene Editing with Multimodal Diffusion Transformer

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Dynamic-eDiTor: Training-Free Text-Driven 4D Scene Editing with Multimodal Diffusion Transformer

快速理解：Recent progress in 4D representations, such as Dynamic NeRF and 4D Gaussian Splatting (4DGS), has enabled dynamic 4D scene reconstruction. However, textdriven 4D scene editing remains under-explored due to the challenge of ensuring both multi-view and temporal consistency across space and time during editing.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.202 Dynamics-Aware Preference Optimization for Vision-Language Models

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Dynamics-Aware Preference Optimization for Vision-Language Models

快速理解：Preference-based finetuning of vision-language models (VLMs) is notoriously unstable, i.e., trivially wrong negatives inject uninformative gradients that distort optimization and degrade calibration. This work revisits this issue through the lens of learning dynamics and identifies a core pathology, i.e., the squeezing effect, where easy negatives retain large, misaligned gradients despite negligible loss.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.203 E²-SCI: Elastic Edge-Cloud Speculative Decoding via Credit Inertia

- 作者：作者信息未提供
- 年份：2026；主题：Systems；优先级：扩展
- [来源](#)

论文：E²-SCI: Elastic Edge-Cloud Speculative Decoding via Credit Inertia

快速理解：In edge-cloud environments, efficiency of speculative decoding is heavily constrained by uplink transmission and cloud-side verification. In this work, we identify a phenomenon we term credit inertia, where acceptance rates of adjacent token windows exhibit strong temporal consistency.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.2.204 E2EGS: Event-to-Edge Gaussian Splatting for Pose-Free 3D Reconstruction

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：E2EGS: Event-to-Edge Gaussian Splatting for Pose-Free 3D Reconstruction

快速理解：The emergence of neural radiance fields (NeRF) and 3D Gaussian splatting (3DGS) has advanced novel view synthesis (NVS). These methods, however, require high-quality RGB inputs and accurate corresponding poses, limiting robustness under real-world conditions such as fast camera motion or adverse lighting.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.205 EMAD: Evidence-Centric Grounded Multimodal Diagnosis for Alzheimer' s Disease

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：EMAD: Evidence-Centric Grounded Multimodal Diagnosis for Alzheimer' s Disease

快速理解：Deep learning models for medical image analysis often act as “black boxes,” seldom aligning with clinical guidelines or explicitly linking decisions to supporting evidence. This is especially critical in Alzheimer' s disease (AD), where predictions should be grounded in both anatomical and clinical findings.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.206 EMMA: Extracting Multiple physical parameters from Multimodal Data

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：EMMA: Extracting Multiple physical parameters from Multimodal Data

快速理解: We introduce EMMA, a physics-informed multimodal framework that recovers all identifiable dynamical parameters of a system directly from raw video, audio, and image-based time-series observations. Unlike prior videoonly approaches that struggle with occluded states, hidden actuation inputs, or assumptions about known initial conditions and coordinate frames, EMMA performs joint inference of explicit parameters, implicit dynamical components, and calibration invariants within a unified continuous-time model.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.207 ENC-Bench: A Benchmark for Evaluating Multimodal Large Language Models in Electronic Navigational Chart Understanding

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: ENC-Bench: A Benchmark for Evaluating Multimodal Large Language Models in Electronic Navigational Chart Understanding

快速理解: Electronic Navigational Charts (ENCs) are the safetycritical backbone of modern maritime navigation, yet it remains unclear whether multimodal large language models (MLLMs) can reliably interpret them. Unlike natural images or conventional charts, ENCs encode regulations, bathymetry, and route constraints via standardized vector symbols, scale-dependent rendering, and precise geometric structure—requiring specialized maritime expertise for interpretation.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.208 EVA: Efficient Reinforcement Learning for End-to-End Video Agent

- 作者: 作者信息未提供
- 年份: 2026; 主题: Agent RL; 优先级: 扩展
- [来源](#)

论文: EVA: Efficient Reinforcement Learning for End-to-End Video Agent

快速理解: Video understanding with multimodal large language models (MLLMs) remains challenging due to the long token sequences of videos, which contain extensive temporal dependencies and redundant frames. Existing approaches typically treat MLLMs as passive recognizers, processing entire videos or uniformly sampled frames without adaptive reasoning.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.2.209 EVLF: Early Vision-Language Fusion for Generative Dataset Distillation

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: EVLF: Early Vision-Language Fusion for Generative Dataset Distillation

快速理解: Dataset distillation (DD) aims to synthesize compact training sets that enable models to achieve high accuracy with significantly fewer samples. Recent diffusion-based DD methods commonly introduce semantic guidance through late-stage cross-attention, where textual prompts tend to dominate the generative process.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.210 Edges Compete for Trust: Group Relative Edge Optimization for Building Reconstruction from Point Clouds

- 作者: 作者信息未提供
- 年份: 2026; 主题: Systems; 优先级: 扩展
- [来源](#)

论文: Edges Compete for Trust: Group Relative Edge Optimization for Building Reconstruction from Point Clouds

快速理解: Building reconstruction aims to extract compact wireframes from point clouds. Recent edge-based methods achieve impressive results but suffer from sparse supervision from one-to-one matching, which leaves most edge proposals under-optimized.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.2.211 EduDiag: A Benchmark for Educational Diagnostic Reasoning with Error Tracing and Correction on Large Multimodal Models

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: EduDiag: A Benchmark for Educational Diagnostic Reasoning with Error Tracing and Correction on Large Multimodal Models

快速理解: Large multimodal models (LMMs) have achieved impressive performance on multimodal reasoning, becoming crucial technology for the advancement of intelligent questionanswering systems. In real-world educational scenarios, effective teaching extends far beyond providing answers.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.212 Efficient Encoder-Free Fourier-based 3D Large Multimodal Model

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Efficient Encoder-Free Fourier-based 3D Large Multimodal Model

快速理解: Large Multimodal Models (LMMs) that process 3D data typically rely on heavy, pre-trained visual encoders to extract geometric features. While recent 2D LMMs have begun to eliminate such encoders for efficiency and scalability, extending this paradigm to 3D remains challenging due to

the unordered and large-scale nature of point clouds.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.213 Efficient Frame Selection for Long Video Understanding via Reinforcement Learning

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL；优先级：扩展
- [来源](#)

论文：Efficient Frame Selection for Long Video Understanding via Reinforcement Learning

快速理解：Recent advances in Multimodal Large Language Models (MLLMs) have led to significant progress in video understanding. Due to limited context windows and computational overhead, most MLLMs adopt uniform frame sampling.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.2.214 EgoControl: Controllable Egocentric Video Generation via 3D Full-Body Poses

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：EgoControl: Controllable Egocentric Video Generation via 3D Full-Body Poses

快速理解：Egocentric video generation with fine-grained control through body motion is a key requirement towards embodied AI agents that can simulate, predict, and plan actions. In this work, we propose EgoControl, a pose-controllable video diffusion model trained on egocentric data.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.215 EmbodMocap: In-the-Wild 4D Human-Scene Reconstruction for Embodied Agents

- 作者：作者信息未提供
- 年份：2026；主题：Embodied/WBC；优先级：扩展
- [来源](#)

论文：EmbodMocap: In-the-Wild 4D Human-Scene Reconstruction for Embodied Agents

快速理解：Human behaviors in the real world naturally encode rich, long-term contextual information that can be leveraged to train embodied agents for perception, understanding, and acting. However, existing capture systems typically rely on costly studio setups and wearable devices, limiting the large-scale collection of scene-conditioned human motion data in the wild.

关注点：具身任务中的感知、动作生成、全身协调或运动控制。

证据层级：本地 PDF 前页。

1.3.2.216 End-to-End Language-Action Model for Humanoid Whole Body Control

- 作者：作者信息未提供
- 年份：2026；主题：Embodied/WBC；优先级：扩展
- [来源](#)

论文：End-to-End Language-Action Model for Humanoid Whole Body Control

快速理解：Existing humanoid control systems often rely on teleoperation or modular generation pipelines that separate language understanding from physical execution. However, the former is entirely human-driven, and the latter lacks tight alignment between language commands and physical behaviors.

关注点：具身任务中的感知、动作生成、全身协调或运动控制。

证据层级：本地 PDF 前页。

1.3.2.217 Enhance-then-Balance Modality Collaboration for Robust Multimodal Sentiment Analysis

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Enhance-then-Balance Modality Collaboration for Robust Multimodal Sentiment Analysis

快速理解：Multimodal sentiment analysis (MSA) integrates heterogeneous text, audio, and visual signals to infer human emotions. While recent approaches leverage crossmodal complementarity, they often struggle to fully utilize weaker modalities.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.218 Enhancing Hands in 3D Whole-Body Pose Estimation with Conditional Hands Modulator

- 作者：作者信息未提供
- 年份：2026；主题：Embodied/WBC；优先级：扩展
- [来源](#)

论文：Enhancing Hands in 3D Whole-Body Pose Estimation with Conditional Hands Modulator

快速理解：Accurately recovering hand poses within the body context remains a major challenge in 3D whole-body pose estimation. This difficulty arises from a fundamental supervision gap: whole-body pose estimators are trained on full-body datasets with limited hand diversity, while hand-only estimators, trained on hand-centric datasets, excel at detailed finger articulation but lack global body awareness.

关注点：具身任务中的感知、动作生成、全身协调或运动控制。

证据层级：本地 PDF 前页。

1.3.2.219 Enhancing Mixture-of-Experts Specialization via Cluster-Aware Upcycling

- 作者：作者信息未提供

- 年份: 2026; 主题: AI Infra; 优先级: 扩展
- [来源](#)

论文: Enhancing Mixture-of-Experts Specialization via Cluster-Aware Upcycling

快速理解: Sparse Upcycling provides an efficient way to initialize a Mixture-of-Experts (MoE) model from pretrained dense weights instead of training from scratch. However, since all experts start from identical weights and the router is randomly initialized, the model suffers from expert symmetry and limited early specialization.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 本地 PDF 前页。

1.3.2.220 Envision, Attend, Then Respond: Counterfactual Hallucination Mitigation in Large Vision-Language Models

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Envision, Attend, Then Respond: Counterfactual Hallucination Mitigation in Large Vision-Language Models

快速理解: Large Vision-Language Models (LVLMs) often hallucinate when visual evidence conflicts with world knowledge, i.e., in counterfactual scenarios. We propose Envision- Attend-Respond (EnAR), a training-free framework that leverages visual priors to steer the model's attention toward counterfactual elements in the image.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.221 Evo-Retriever: LLM-Guided Curriculum Evolution with Viewpoint-Pathway Collaboration for Multimodal Document Retrieval

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Evo-Retriever: LLM-Guided Curriculum Evolution with Viewpoint-Pathway Collaboration for Multimodal Document Retrieval

快速理解: Visual-language models (VLMs) excel at data mappings, but real-world document heterogeneity and unstructuredness disrupt the consistency of cross-modal embeddings. Recent late-interaction methods enhance image-text alignment through multi-vector representations, yet traditional training with limited samples and static strategies cannot adapt to the model's dynamic evolution, causing crossmodal retrieval confusion.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.222 Evolutionary Multimodal Reasoning via Hierarchical Semantic Representation for Intent Recognition

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文: Evolutionary Multimodal Reasoning via Hierarchical Semantic Representation for Intent Recognition

快速理解: Multimodal intent recognition aims to infer human intents by jointly modeling various modalities, playing a pivotal role in real-world dialogue systems. However, current methods struggle to model hierarchical semantics underlying complex intents and lack the capacity for selfevolving reasoning over multimodal representations.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.223 ExPose: Reinforcing Video Generation Models for Extreme Pose Estimation

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文: ExPose: Reinforcing Video Generation Models for Extreme Pose Estimation

快速理解: Pose estimation remains challenging under sparse views, especially when visual overlap across images is extremely limited. Recent advances in video generation models offer a promising solution by enabling keyframe interpolation, which can enrich contextual cues and improve pose estimation performance.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.224 Experience Transfer for Multimodal LLM Agents in Minecraft Game

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文: Experience Transfer for Multimodal LLM Agents in Minecraft Game

快速理解: Multimodal LLM agents operating in complex game environments must continually reuse past experience to solve new tasks efficiently. In this work, we propose Echo, a transfer-oriented memory framework that enables agents to derive actionable knowledge from prior interactions rather than treating memory as a passive repository of static records.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.225 Extending Embodied Question Answering from Perception to Decision

- 作者：作者信息未提供
- 年份：2026；主题：Embodied/WBC；优先级：扩展
- [来源](#)

论文: Extending Embodied Question Answering from Perception to Decision

快速理解: Embodied Question Answering (EQA) connects perception, reasoning, and interaction within embodied environments. However, existing datasets and benchmarks remain fragmented, each focusing on a limited subset of reasoning skills such as spatial understanding or procedural reasoning, without offering a unified large-scale framework for comprehensive evaluation.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制。

证据层级: 本地 PDF 前页。

1.3.2.226 FAVE: A Structured Benchmark for Fine-Grained Audio-Visual Temporal Evaluation in Multimodal LLMs

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: FAVE: A Structured Benchmark for Fine-Grained Audio-Visual Temporal Evaluation in Multimodal LLMs

快速理解: Audio-visual large language models (AVLLMs) have achieved strong performance in multimodal understanding, yet their ability to model fine-grained temporal relationships between audio and visual streams remains underexplored. We introduce FAVE (Fine-grained Audio-Visual Temporal Evaluation), a benchmark designed to evaluate three aspects of temporal perception: cross-modal temporal alignment (FAVE-align), event-level temporal relationships (FAVE-low), and moment-level captioning (FAVEhigh).

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.227 FBTA: Enabling Single-GPU End-to-End Gigapixel WSI Classification with Feature Bridging and Translation Alignment

- 作者: 作者信息未提供
- 年份: 2026; 主题: AI Infra; 优先级: 扩展
- [来源](#)

论文: FBTA: Enabling Single-GPU End-to-End Gigapixel WSI Classification with Feature Bridging and Translation Alignment

快速理解: Whole-slide images (WSIs) in computational pathology contain billions of pixels, making end-to-end training of feature extractors and multi-instance learning (MIL) networks infeasible on a single commodity GPU. Existing methods often freeze the feature extractor and train MIL networks on the resulting frozen features, which introduces a semantic gap that limits downstream performance.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 本地 PDF 前页。

1.3.2.228 FLARE: A Failure-Aware Framework for Autonomous Correction and Recovery in Visual-Language Robotic Manipulation

- 作者：作者信息未提供
- 年份：2026；主题：Embodied/WBC；优先级：扩展
- [来源](#)

论文：FLARE: A Failure-Aware Framework for Autonomous Correction and Recovery in Visual-Language Robotic Manipulation

快速理解：Vision-Language-Action Models (VLAs) have demonstrated significant promise in generalizing to complex, long-horizon robotic manipulation tasks. However, their performance remains brittle, as they are typically trained on trajectory-monotonic, failure-free demonstrations.

关注点：具身任务中的感知、动作生成、全身协调或运动控制。

证据层级：本地 PDF 前页。

1.3.2.229 FUN REC * Reconstructing Functional 3D Scenes from Egocentric Interaction Videos

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：FUN REC * Reconstructing Functional 3D Scenes from Egocentric Interaction Videos

快速理解：We present FunREC, a method for reconstructing functional 3D digital twins of indoor scenes directly from egocentric RGB-D interaction videos. Unlike existing methods on articulated reconstruction, which rely on controlled setups, multi-state captures, or CAD priors, FunREC operates directly on in-the-wild human interaction sequences to recover interactable 3D scenes.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.230 FantasyVLN: Unified Multimodal Chain-of-Thought Reasoning for Vision-and-Language Navigation

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：FantasyVLN: Unified Multimodal Chain-of-Thought Reasoning for Vision-and-Language Navigation

快速理解：Achieving human-level performance in Vision-and- Language Navigation (VLN) requires an embodied agent to understand textual instructions, perceive visual observations, and reason over long action sequences. Recent works, such as NavCoT and NavGPT-2, demonstrate the potential of Chain-of-Thought (CoT) reasoning for improving interpretability and long-horizon planning.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.231 Fast3Dcache: Training-free 3D Geometry Synthesis Acceleration

- 作者：作者信息未提供
- 年份：2026；主题：Systems；优先级：扩展

- [来源](#)

论文: Fast3Dcache: Training-free 3D Geometry Synthesis Acceleration

快速理解: Diffusion models have achieved impressive generative quality across modalities like 2D images, videos, and 3D shapes, but their inference remains computationally expensive due to the iterative denoising process. While recent caching-based methods effectively reuse redundant computations to speed up 2D and video generation, directly applying these techniques to 3D diffusion models can severely disrupt geometric consistency.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.2.232 FedAFD: Multimodal Federated Learning via Adversarial Fusion and Distillation

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: FedAFD: Multimodal Federated Learning via Adversarial Fusion and Distillation

快速理解: Multimodal Federated Learning (MFL) enables clients with heterogeneous data modalities to collaboratively train models without sharing raw data, offering a privacy-preserving framework that leverages complementary cross-modal information. However, existing methods often overlook personalized client performance and struggle with modality/- task discrepancies, as well as model heterogeneity.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.233 FedMPT: Federated Multi-Label Prompt Tuning of Vision-Language Models

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: FedMPT: Federated Multi-Label Prompt Tuning of Vision-Language Models

快速理解: Multi-Label Recognition (MLR) based on Vision-Language Models (VLMs) aims to leverage their pre-trained knowledge to better adapt complex recognition scenarios, thereby enhancing model robustness. However, for realistic decentralized applications requiring federated learning, adapting VLMs to each client that possesses private and heterogeneous data can cause the model to overfit spurious label correlations, consequently triggering irrelevant categories when encountering new samples.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.234 Few-shot Acoustic Synthesis with Multimodal Flow Matching

- 作者: 作者信息未提供

- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Few-shot Acoustic Synthesis with Multimodal Flow Matching

快速理解: Generating audio that is acoustically consistent with a scene is essential for immersive virtual environments. Recent neural acoustic field methods enable spatially continuous sound rendering but remain scene-specific, requiring dense audio measurements and costly training for each environment.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.235 FinPercep-RM: A Fine-grained Reward Model and Co-evolutionary Curriculum for RL-based Real-world Super-Resolution

- 作者: 作者信息未提供
- 年份: 2026; 主题: Agent RL; 优先级: 扩展
- [来源](#)

论文: FinPercep-RM: A Fine-grained Reward Model and Co-evolutionary Curriculum for RL-based Real-world Super-Resolution

快速理解: Inspired by the success of Reinforcement Learning with Human Feedback (RLHF) in image generation, recent work has adapted reward-based learning to image superresolution (ISR) by using Image Quality Assessment (IQA) models as rewards. However, existing IQA models typically output only a single global score and are insensitive to local, fine-grained distortions, allowing perceptually undesirable artifacts to obtain spuriously high rewards and leading to reward hacking.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.2.236 First Logit Boosting: Visual Grounding Method to Mitigate Object Hallucination in Large Vision-Language Models

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: First Logit Boosting: Visual Grounding Method to Mitigate Object Hallucination in Large Vision-Language Models

快速理解: Recent Large Vision-Language Models (LVLMs) have demonstrated remarkable performance across various multimodal tasks that require understanding both visual and linguistic inputs. However, object hallucination —the generation of nonexistent objects in answers —remains a persistent challenge.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.237 Flash-DMD: Towards High-Fidelity Few-Step Image Generation with Efficient Distillation and Joint Reinforcement Learning

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL；优先级：扩展
- [来源](#)

论文: Flash-DMD: Towards High-Fidelity Few-Step Image Generation with Efficient Distillation and Joint Reinforcement Learning

快速理解: Diffusion Models have emerged as a leading class of generative models, yet their iterative sampling process remains computationally expensive. Timestep distillation is a promising technique to accelerate generation, but it often requires extensive training and leads to image quality degradation.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.2.238 FlexTraj: Image-to-Video Generation with Flexible Point Trajectory Control

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文: FlexTraj: Image-to-Video Generation with Flexible Point Trajectory Control

快速理解: We present FlexTraj, a framework for image-to-video generation with flexible point trajectory control. FlexTraj introduces a unified point-based representation that encodes † Corresponding Author: jingliao@cityu.edu.hk point with a temporally consistent trajectory ID, a segmentation ID, and an optional color channel for appearance cues, enabling both dense and sparse trajectory control.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.239 Flow Matching for Multimodal Distributions

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文: Flow Matching for Multimodal Distributions

快速理解: Recently, vision foundation models have been shown to boost the efficiency of flow-based generative models by revealing the intrinsic union-of-manifold structures and lowering the complexity of the latent/target distribution. In this paper, we exploit the multimodality aspect of the union-of-manifold structures, and aim to further improve the learning and inference efficiency for flow-matching models.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.240 Flowception: Temporally Expansive Flow Matching for Video Generation

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展

- [来源](#)

论文: Flowception: Temporally Expansive Flow Matching for Video Generation

快速理解: We present Flowception, a novel non-autoregressive and variable-length video generation framework. Flowception learns a probability path that interleaves discrete frame insertions with continuous frame denoising.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.241 FluidGaussian: Propagating Simulation-Based Uncertainty Toward Functionally-Intelligent 3D Reconstruction

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: FluidGaussian: Propagating Simulation-Based Uncertainty Toward Functionally-Intelligent 3D Reconstruction

快速理解: Real objects that inhabit the physical world follow physical laws and thus behave plausibly during interaction with other physical objects. However, current methods that perform 3D reconstructions of real-world scenes from multi-view 2D images optimize primarily for visual fidelity, i.e., they train with photometric losses and reason about uncertainty in the image or representation space.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.242 ForeAct: Steering Your VLA with Efficient Visual Foresight Planning

- 作者: 作者信息未提供
- 年份: 2026; 主题: VLA; 优先级: 扩展
- [来源](#)

论文: ForeAct: Steering Your VLA with Efficient Visual Foresight Planning

快速理解: Vision-Language-Action (VLA) models convert high-level language instructions into concrete, executable actions, a task that is especially challenging in open-world environments. We present Visual Foresight Planning (ForeAct), a general and efficient planner that guides a VLA step-by-step using imagined future observations and subtask descriptions.

关注点: 视觉、语言条件与机器人动作策略之间的对齐。

证据层级: 本地 PDF 前页。

1.3.2.243 Free-Lunch Long Video Generation via Layer-Adaptive O.O.D Correction

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Free-Lunch Long Video Generation via Layer-Adaptive O.O.D Correction

快速理解: Generating long videos using pre-trained video diffusion models, which are typically trained on short clips, presents a significant challenge. Directly applying these models for long-video inference often leads to a notable degradation in visual quality.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.244 FreeScale: Scaling 3D Scenes via Certainty-Aware Free-View Generation

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: FreeScale: Scaling 3D Scenes via Certainty-Aware Free-View Generation

快速理解: The development of generalizable Novel View Synthesis (NVS) models is critically limited by the scarcity of largescale training data featuring diverse and precise camera trajectories. While real-world captures are photorealistic, they are typically sparse and discrete.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.245 From 3D Pose to Prose: Biomechanics-Grounded Vision-Language Coaching

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: From 3D Pose to Prose: Biomechanics-Grounded Vision-Language Coaching

快速理解: We present BioCoach, a biomechanics-grounded vision-language framework for fitness coaching from streaming video. BioCoach fuses visual appearance and 3D skeletal kinematics, through a novel three-stage pipeline: an exercise-specific degree-of-freedom selector that focuses analysis on salient joints; a structured biomechanical context that pairs individualized morphometrics with cycle and constraint analysis; and a vision-biomechanics conditioned feedback module that applies cross-attention to generate precise, actionable text.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.246 From None to All: Self-Supervised 3D Reconstruction via Novel View Synthesis

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: From None to All: Self-Supervised 3D Reconstruction via Novel View Synthesis

快速理解: In this paper, we introduce NAS3R, a self-supervised feed-forward framework that jointly learns explicit 3D geometry and camera parameters with no ground-truth annotations and no pre-trained priors. During training, NAS3R reconstructs 3D Gaussians from uncalibrated and unposed

context views and renders target views using its self-predicted camera parameters, enabling self-supervised training from 2D photometric supervision.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.247 From Observation to Action: Latent Action-based Primitive Segmentation for VLA Pre-training in Industrial Settings

- 作者：作者信息未提供
- 年份：2026；主题：VLA；优先级：扩展
- [来源](#)

论文: From Observation to Action: Latent Action-based Primitive Segmentation for VLA Pre-training in Industrial Settings

快速理解: We present a novel unsupervised framework to unlock vast unlabeled human demonstration data from continuous industrial video streams for Vision-Language-Action (VLA) model pre-training. Our method first trains a lightweight motion tokenizer to encode motion dynamics, then employs an unsupervised action segmenter leveraging a novel "Latent Action Energy" metric to discover and segment semantically coherent action primitives.

关注点：视觉、语言条件与机器人动作策略之间的对齐。

证据层级：本地 PDF 前页。

1.3.2.248 From Panel to Pixel: Zoom-In Vision-Language Pretraining from Biomedical Scientific Literature

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文: From Panel to Pixel: Zoom-In Vision-Language Pretraining from Biomedical Scientific Literature

快速理解: There is a growing interest in developing strong biomedical vision-language models. A popular approach to achieve robust representations is to use web-scale scientific data.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.249 From Selection to Scheduling: Federated Geometry-Aware Correction Makes Exemplar Replay Work Better under Continual Dynamic Heterogeneity

- 作者：作者信息未提供
- 年份：2026；主题：Systems；优先级：扩展
- [来源](#)

论文: From Selection to Scheduling: Federated Geometry-Aware Correction Makes Exemplar Replay Work Better under Continual Dynamic Heterogeneity

快速理解: Exemplar replay has become an effective strategy for mitigating catastrophic forgetting in federated continual learning (FCL) by retaining representative samples from past tasks. Existing

studies focus on designing sample importance estimation mechanisms to identify information-rich samples.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.2.250 From Where Things Are to What They Are For: Benchmarking Spatial-Functional Intelligence in Multimodal LLMs

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：From Where Things Are to What They Are For: Benchmarking Spatial-Functional Intelligence in Multimodal LLMs

快速理解：Human-level agentic intelligence extends beyond low-level geometric perception, evolving from recognizing where things are to understanding what they are for. While existing benchmarks effectively evaluate the geometric perception capabilities of multimodal large language models (MLLMs), they fall short of probing the higher-order cognitive abilities required for grounded intelligence.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.251 GT-SVJ: Generative-Transformer-Based Self-Supervised Video Judge For Efficient Video Reward Modeling

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL；优先级：扩展
- [来源](#)

论文：GT-SVJ: Generative-Transformer-Based Self-Supervised Video Judge For Efficient Video Reward Modeling

快速理解：Aligning video generative models with human preferences remains challenging; current approaches rely on Vision-Language Models (VLMs) for reward modeling, but these models struggle to capture subtle temporal dynamics. We propose a fundamentally different approach: repurposing video generative models, which are inherently designed to model temporal structure, as reward models.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.2.252 GThinker: Towards General Multimodal Reasoning via Cue-Guided Rethinking

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：GThinker: Towards General Multimodal Reasoning via Cue-Guided Rethinking

快速理解: Despite recent advances in multimodal reasoning, Multimodal Large Language Models (MLLMs) still struggle on complex tasks where initial visual perceptions can be misleading. This performance gap stems from a critical reasoning flaw we term Visual Inertia: while MLLMs excel at iterative reflection in textual contexts, they tend to uncritically commit to their initial visual interpretations and rarely revise them.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.253 Gallant: Voxel Grid-based Humanoid Locomotion and Local-navigation across 3-D Constrained Terrains

- 作者: 作者信息未提供
- 年份: 2026; 主题: Embodied/WBC; 优先级: 扩展
- [来源](#)

论文: Gallant: Voxel Grid-based Humanoid Locomotion and Local-navigation across 3-D Constrained Terrains

快速理解: Robust humanoid locomotion requires accurate and globally consistent perception of the surrounding 3D environment. However, existing perception modules, mainly based on depth images or elevation maps, offer only partial and locally flattened views of the environment, failing to capture the full 3D structure.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制。

证据层级: 本地 PDF 前页。

1.3.2.254 Gastric-X: A Multimodal Multi-Phase Benchmark Dataset for Advancing Vision-Language Models in Gastric Cancer Analysis

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Gastric-X: A Multimodal Multi-Phase Benchmark Dataset for Advancing Vision-Language Models in Gastric Cancer Analysis

快速理解: Recent vision-language models (VLMs) have shown strong generalization and multimodal reasoning abilities in natural domains. However, their application to medical diagnosis remains limited by the lack of comprehensive and structured datasets that capture real clinical workflows.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.255 Gated Condition Injection without Multimodal Attention: Towards Controllable Linear-Attention Transformers

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Gated Condition Injection without Multimodal Attention: Towards Controllable Linear-

Attention Transformers

快速理解: Recent advances in diffusion-based controllable visual generation have led to remarkable improvements in image quality. However, these powerful models are typically deployed on cloud servers due to their large computational demands, raising serious concerns about user data privacy.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.256 GaussFusion: Improving 3D Reconstruction in the Wild with A Geometry-Informed Video Generator

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: GaussFusion: Improving 3D Reconstruction in the Wild with A Geometry-Informed Video Generator

快速理解: We present GaussFusion, a novel approach for improving 3D Gaussian splatting (3DGS) reconstructions in the wild through geometry-informed video generation. GaussFusion mitigates common 3DGS artifacts, including floaters, flickering, and blur caused by camera pose errors, incomplete coverage, and noisy geometry initialization.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.257 GaussianGrow: Geometry-aware Gaussian Growing from 3D Point Clouds with Text Guidance

- 作者: 作者信息未提供
- 年份: 2026; 主题: Systems; 优先级: 扩展
- [来源](#)

论文: GaussianGrow: Geometry-aware Gaussian Growing from 3D Point Clouds with Text Guidance

快速理解: 3D Gaussian Splatting has demonstrated superior performance in rendering efficiency and quality, yet the generation of 3D Gaussians still remains a challenge without proper geometric priors. Existing methods have explored predicting point maps as geometric references for inferring Gaussian primitives, while the unreliable estimated geometries may lead to poor generations.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.2.258 GaussianVision: Vision-Language Alignment from Compressed Image Representations using 2D Gaussian Splatting

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: GaussianVision: Vision-Language Alignment from Compressed Image Representations using 2D Gaussian Splatting

快速理解: Modern vision-language pipelines are driven by RGB vision encoders trained on massive image-text corpora. While these pipelines have enabled impressive zero-shot capabilities and strong transfer across tasks, they still inherit two structural inefficiencies from the pixel domain: (i) transmitting dense RGB images from edge devices to the cloud is energy-intensive and costly, and (ii) patch-based tokenization explodes sequence length, stressing attention budgets and context limits.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.259 GeCo-SRT: Geometry-aware Continual Adaptation for Cross-Task Sim-to-Real Transfer

- 作者: 作者信息未提供
- 年份: 2026; 主题: Robot Learning; 优先级: 扩展
- [来源](#)

论文: GeCo-SRT: Geometry-aware Continual Adaptation for Cross-Task Sim-to-Real Transfer

快速理解: Bridging the sim-to-real gap is important for applying low-cost simulation data to real-world robotic systems. However, previous methods are severely limited by treating each transfer as an isolated endeavor, demanding repeated, costly tuning and wasting prior transfer experience.

关注点: 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 本地 PDF 前页。

1.3.2.260 Gen3R: 3D Scene Generation Meets Feed-Forward Reconstruction

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Gen3R: 3D Scene Generation Meets Feed-Forward Reconstruction

快速理解: We present Gen3R, a method that bridges the strong priors of foundational reconstruction models and video diffusion models for scene-level 3D generation. We repurpose the VGGT reconstruction model to produce geometric latents by training an adapter on its tokens, which are regularized to align with the appearance latents of pre-trained video diffusion models.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.261 Generalizable Sparse-View 3D Reconstruction from Unconstrained Images

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Generalizable Sparse-View 3D Reconstruction from Unconstrained Images

快速理解: Reconstructing 3D scenes from sparse, unposed images remains challenging under real-world conditions with varying illumination and transient occlusions. Existing methods rely on scene-specific optimization using appearance embeddings or dynamic masks, which requires extensive per-scene training and fails under sparse views.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.262 Generalizable Structure-Aware Keypoint Correspondence for Category-Unified 3D Single Object Tracking

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Generalizable Structure-Aware Keypoint Correspondence for Category-Unified 3D Single Object Tracking

快速理解：3D single object tracking (SOT) in point clouds is essential for real-world 3D perception, yet it remains challenging due to data sparsity and large variations in scale and structure across diverse object categories. Most existing methods rely on a category-specific paradigm that trains separate models for each class, severely limiting scalability and generalization in real deployment.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.263 Generalized-CVO: Fast and Correspondence-Free Local Point Cloud Registration with Second Order Riemannian Optimization

- 作者：作者信息未提供
- 年份：2026；主题：Systems；优先级：扩展
- [来源](#)

论文：Generalized-CVO: Fast and Correspondence-Free Local Point Cloud Registration with Second Order Riemannian Optimization

快速理解：We propose a fast and correspondence-free local point cloud registration method that leverages geometric surface structure and reproducing kernel Hilbert space (RKHS) embeddings. The method represents point clouds as continuous functions with point-wise anisotropic kernels that encode local geometry.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.2.264 GeoDiT: A Diffusion-based Vision-Language Model for Geospatial Understanding

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：GeoDiT: A Diffusion-based Vision-Language Model for Geospatial Understanding

快速理解：Autoregressive models are structurally misaligned with the inherently parallel nature of geospatial understanding, forcing a rigid sequential narrative onto scenes and fundamentally hindering the generation of structured and coherent outputs. We challenge this paradigm by reframing geospatial generation as a parallel refinement process, enabling a holistic, coarse-to-fine synthesis

that resolves all semantic elements simultaneously.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.265 GeoFree-CoSeg: Unsupervised Point Cloud-Image Cross-Modal Co-Segmentation Without Geometric Alignment

- 作者：作者信息未提供
- 年份：2026；主题：Systems；优先级：扩展
- [来源](#)

论文：GeoFree-CoSeg: Unsupervised Point Cloud-Image Cross-Modal Co-Segmentation Without Geometric Alignment

快速理解：Co-segmentation aims to identify and segment common objects across a set of point clouds or images. Existing methods focus on single-modal co-segmentation.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.2.266 GeoMMBench and GeoMMAgent: Toward Expert-Level Multimodal Intelligence in Geoscience and Remote Sensing

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：GeoMMBench and GeoMMAgent: Toward Expert-Level Multimodal Intelligence in Geoscience and Remote Sensing

快速理解：Recent advances in multimodal large language models (MLLMs) have accelerated progress in domain-oriented AI, yet their development in geoscience and remote sensing (RS) remains constrained by distinctive challenges: wide-ranging disciplinary knowledge, heterogeneous sensor modalities, and a fragmented spectrum of tasks. To bridge these gaps, we introduce GeoMMBench, a comprehensive multimodal question-answering benchmark covering diverse RS disciplines, sensors, and tasks, enabling broader and more rigorous evaluation than prior benchmarks.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.267 GeoTikzBridge: Advancing Multimodal Code Generation for Geometric Perception and Reasoning

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：GeoTikzBridge: Advancing Multimodal Code Generation for Geometric Perception and Reasoning

快速理解：Multimodal Large Language Models (MLLMs) have recently demonstrated remarkable perceptual and reasoning abilities. However, they struggle to perceive fine-grained geometric structures,

constraining their ability of geometric understanding and visual reasoning.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.268 GeoWorld: Geometric World Models

- 作者：作者信息未提供
- 年份：2026；主题：World Model；优先级：扩展
- [来源](#)

论文：GeoWorld: Geometric World Models

快速理解：Energy-based predictive world models provide a powerful approach for multi-step visual planning by reasoning over latent energy landscapes rather than generating pixels. However, existing approaches face two major challenges: (i) their latent representations are typically learned in Euclidean space, neglecting the underlying geometric and hierarchical structure among states, and (ii) they struggle with long-horizon prediction, which leads to rapid degradation across extended rollouts.

关注点：环境动力学、视频预测、潜变量状态或模型式决策。

证据层级：本地 PDF 前页。

1.3.2.269 Geoint-R1: Formalizing Multimodal Geometric Reasoning with Dynamic Auxiliary Constructions

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Geoint-R1: Formalizing Multimodal Geometric Reasoning with Dynamic Auxiliary Constructions

快速理解：Mathematical geometric reasoning is essential for scientific discovery and educational development, requiring precise logic and rigorous formal verification. While recent advances in Multimodal Large Language Models (MLLMs) have improved reasoning tasks, existing models typically struggle with formal geometric reasoning, particularly when dynamically constructing and verifying auxiliary geometric elements.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.270 Ghosts in the Point Clouds: De-glaring LiDAR in the Transient Domain

- 作者：作者信息未提供
- 年份：2026；主题：Systems；优先级：扩展
- [来源](#)

论文：Ghosts in the Point Clouds: De-glaring LiDAR in the Transient Domain

快速理解：Ghosts in the Point Clouds: De-glaring LiDAR in the Transient Domain Avery Gump* Connor Henley* Sungjin Cheong Akarsh Prabhakara Mohit Gupta University of Wisconsin-Madison, Madison, WI, USA {gump, connor.henley, sungjin.cheong, aprabhakara, mgupta37}@wisc.edu

wisionlab.com/project/deglaring-LiDAR Abstract Modern LiDARs are rapidly transitioning from bulky, mechanically scanned systems to ultra-compact, lowcost, solid-state arrays. This miniaturization—while enabling scalability, affordability, and camera-like data structures—introduces a new and severe failure mode: internal multipath glare.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.2.271 Gloria: Consistent Character Video Generation via Content Anchors

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Gloria: Consistent Character Video Generation via Content Anchors

快速理解：Digital characters are central to modern media, yet generating character videos with long-duration, consistent multiview appearance and expressive identity remains challenging. Existing approaches either provide insufficient context to preserve identity or leverage non-character-centric information as the “memory”, leading to suboptimal consistency.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.272 Goal-Driven Reward by Video Diffusion Models for Reinforcement Learning

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL；优先级：扩展
- [来源](#)

论文：Goal-Driven Reward by Video Diffusion Models for Reinforcement Learning

快速理解：Reinforcement Learning (RL) has achieved remarkable success in various domains, yet it often relies on carefully designed programmatic reward functions to guide agent behavior. Designing such reward functions can be challenging and may not generalize well across different tasks.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.2.273 GraPHFormer: A Multimodal Graph Persistent Homology Transformer for the Analysis of Neuroscience Morphologies

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：GraPHFormer: A Multimodal Graph Persistent Homology Transformer for the Analysis of Neuroscience Morphologies

快速理解：Neuronal morphology encodes critical information about circuit function, development, and disease, yet current methods analyze topology or graph structure in isolation. We introduce GraPHFormer, a multimodal architecture that unifies these complementary views through CLIP-style contrastive learning.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.274 Gradient Knows Best: Mixed-Precision Quantization via Gradient-Guided Bit Allocation for Super-Resolution

- 作者：作者信息未提供
- 年份：2026；主题：AI Infra；优先级：扩展
- [来源](#)

论文：Gradient Knows Best: Mixed-Precision Quantization via Gradient-Guided Bit Allocation for Super-Resolution

快速理解：Although deep learning-based image super-resolution (SR) models have achieved remarkable progress in reconstruction quality, their high computational and memory demands make them unsuitable for lightweight platforms. To address this issue, various quantization techniques have been introduced.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.2.275 GraphVLM: Benchmarking Vision Language Models for Multimodal Graph Learning

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：GraphVLM: Benchmarking Vision Language Models for Multimodal Graph Learning

快速理解：Vision-Language Models (VLMs) have demonstrated remarkable capabilities in aligning and understanding multimodal signals, yet their potential to reason over structured data, where multimodal entities are connected through explicit relational graphs, remains largely underexplored. Unlocking this capability is crucial for real-world applications such as social networks, recommendation systems, and scientific discovery, where multimodal information is inherently structured.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.276 GraspALL: Adaptive Structural Compensation from Illumination Variation for Robotic Garment Grasping in Any Low-Light Conditions

- 作者：作者信息未提供
- 年份：2026；主题：Robot Learning；优先级：扩展
- [来源](#)

论文：GraspALL: Adaptive Structural Compensation from Illumination Variation for Robotic Garment Grasping in Any Low-Light Conditions

快速理解：Achieving accurate garment grasping under dynamically changing illumination is crucial for all-day operation of service robots. However, the reduced illumination in lowlight scenes severely degrades garment structural features, leading to a significant drop in grasping robustness.

关注点：机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：本地 PDF 前页。

1.3.2.277 GraspLDP: Towards Generalizable Grasping Policy via Latent Diffusion

- 作者：作者信息未提供
- 年份：2026；主题：Robot Learning；优先级：扩展
- [来源](#)

论文：GraspLDP: Towards Generalizable Grasping Policy via Latent Diffusion

快速理解：This paper focuses on enhancing the grasping precision and generalization of manipulation policies learned via imitation learning. Diffusion-based policy learning methods have recently become the mainstream approach for robotic manipulation tasks.

关注点：机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：本地 PDF 前页。

1.3.2.278 GroundVTS: Visual Token Sampling in Multimodal Large Language Models for Video Temporal Grounding

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：GroundVTS: Visual Token Sampling in Multimodal Large Language Models for Video Temporal Grounding

快速理解：Video temporal grounding (VTG) is a critical task in video understanding and a key capability for extending video large language models (Vid-LLMs) to broader applications. However, existing Vid-LLMs rely on uniform frame sampling to extract video information, resulting in a sparse distribution of key frames and the loss of crucial temporal cues.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.279 Grounded 3D-Aware Spatial Vision-Language Modeling

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Grounded 3D-Aware Spatial Vision-Language Modeling

快速理解：We present GR3D, a spatial vision language model equipped with three complementary grounding capabilities—explicit 2D grounding, implicit 2D grounding, and monocular 3D grounding—within a single framework. GR3D introduces an implicit grounding mechanism that identifies entity mentions during generation and inserts the corresponding region tokens into the text stream, allowing the model to reference visual evidence on the fly when producing spatial chain-of-thought responses.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.280 Grounded Chain-of-Thought for Multimodal Large Language Models

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Grounded Chain-of-Thought for Multimodal Large Language Models

快速理解：Despite great progress, existing multimodal large language models (MLLMs) are still inferior in visual-spatial reasoning and also suffer from visual hallucination, which greatly impedes their trustworthy applications, such as Embodied AI. To investigate this problem, we propose a new task for MLLMs in this paper, called Grounded Chain-of-Thought (GCoT).

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.281 Grounding Everything in Tokens for Multimodal Large Language Models

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Grounding Everything in Tokens for Multimodal Large Language Models

快速理解：Multimodal large language models (MLLMs) have made significant advancements in vision understanding and reasoning. However, the autoregressive Transformer architecture used by MLLMs requires tokenization on input images, which limits their ability to accurately ground objects within the 2D image space.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.282 HAD: Hallucination-Aware Diffusion Priors for 3D Reconstruction

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：HAD: Hallucination-Aware Diffusion Priors for 3D Reconstruction

快速理解：Diffusion priors have recently demonstrated strong capability in enhancing the quality of sparse-view 3D reconstruction by augmenting training views at novel viewpoints, but they inevitably introduce hallucinated content -artifacts inconsistent with the input views -into the final 3D model. To address this challenge, we propose Hallucination-Aware Diffusion prior (HAD), which estimates pixel-wise hallucination score maps for augmented images by leveraging multi-view reasoning capabilities from a feedforward novel view synthesis (NVS) network pre-trained on large-scale 3D data.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.283 HBridge: H-Shape Bridging of Heterogeneous Experts for Unified Multimodal Understanding and Generation

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：HBridge: H-Shape Bridging of Heterogeneous Experts for Unified Multimodal Understanding and Generation

快速理解：Recent unified models integrate understanding experts (e.g., LLMs) with generative experts (e.g., diffusion models), achieving strong multimodal performance. However, recent advanced methods such as BAGEL and LMFusion follow the Mixture-of-Transformers (MoT) paradigm, † Corresponding author. adopting a symmetric design that mirrors one expert to another for convenient initialization and fusion, which remains suboptimal due to inherent modality discrepancies.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.284 HOG-Layout: Hierarchical 3D Scene Generation, Optimization and Editing via Vision-Language Models

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：HOG-Layout: Hierarchical 3D Scene Generation, Optimization and Editing via Vision-Language Models

快速理解：3D layout generation and editing play a crucial role in Embodied AI and immersive VR interaction. However, manual creation requires tedious labor, while data-driven generation often lacks diversity.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.285 HUMORCHAIN: Theory-Guided Multi-Stage Reasoning for Interpretable Multimodal Humor Generation

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：HUMORCHAIN: Theory-Guided Multi-Stage Reasoning for Interpretable Multimodal Humor Generation

快速理解：Humor, as both a creative human activity and a social binding mechanism, has long posed a major challenge for AI generation. Although producing humor requires complex cognitive reasoning and social understanding, theories of humor suggest that it follows learnable patterns and structures, making it theoretically possible for generative models to acquire them implicitly.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.286 Haptic Neural Fields: Bringing Tactile Interactions to 3D Rendered Scenes

- 作者：作者信息未提供
- 年份：2026；主题：Robot Learning；优先级：扩展
- [来源](#)

论文：Haptic Neural Fields: Bringing Tactile Interactions to 3D Rendered Scenes

快速理解：We address the problem of making 3D scenes interactive by asking: what would objects feel like if touched in a virtual environment? While neural scene representations achieve high visual realism, they fall short in modeling physical interactions, such as the vibration patterns produced when surfaces are explored with different actions.

关注点：机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：本地 PDF 前页。

1.3.2.287 Harmony: Harmonizing Audio and Video Generation through Cross-Task Synergy

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Harmony: Harmonizing Audio and Video Generation through Cross-Task Synergy

快速理解：The synthesis of synchronized audio-visual content is a key challenge in generative AI, with open-source models facing challenges in robust audio-video alignment. Our analysis reveals that this issue is rooted in three fundamental challenges of the joint diffusion process: (1) Correspondence Drift, where concurrently evolving noisy latents impede stable learning of alignment; (2) inefficient global attention mechanisms that fail to capture fine-grained temporal cues; and (3) the intra-modal bias of conventional Classifier-Free Guidance (CFG), which enhances conditionality but not cross-modal synchronization.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.288 Harnessing Chain-of-Thought Reasoning in Multimodal Large Language Models for Face Anti-Spoofing

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Harnessing Chain-of-Thought Reasoning in Multimodal Large Language Models for Face Anti-Spoofing

快速理解：Face Anti-Spoofing (FAS) typically depends on a single visual modality when defending against presentation attacks such as print attacks, screen replays, and 3D masks, resulting in limited generalization across devices, environments, and attack types. Meanwhile, Multimodal Large Language Models (MLLMs) have recently achieved breakthroughs in image-text understanding and semantic reasoning, suggesting that integrating visual and linguistic coinference into FAS can substantially improve both robustness and interpretability.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.289 Hg-I2P: Bridging Modalities for Generalizable Image-to-Point-Cloud Registration via Heterogeneous Graphs

- 作者：作者信息未提供
- 年份：2026；主题：Systems；优先级：扩展
- [来源](#)

论文：Hg-I2P: Bridging Modalities for Generalizable Image-to-Point-Cloud Registration via Heterogeneous Graphs

快速理解：Image-to-point-cloud (I2P) registration aims to align 2D images with 3D point clouds by establishing reliable 2D-3D correspondences. The drastic modality gap between images and point clouds makes it challenging to learn features that are both discriminative and generalizable, leading to severe performance drops in unseen scenarios.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.2.290 HiCoGen: Hierarchical Compositional Text-to-Image Generation in Diffusion Models via Reinforcement Learning

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL；优先级：扩展
- [来源](#)

论文：HiCoGen: Hierarchical Compositional Text-to-Image Generation in Diffusion Models via Reinforcement Learning

快速理解：Recent advances in diffusion models have demonstrated impressive capability in generating high-quality images for simple prompts. However, when confronted with complex prompts involving multiple objects and hierarchical structures, existing models struggle to accurately follow instructions, leading to issues such as concept omission, confusion, and poor compositionality.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.2.291 HiFICL: High-Fidelity In-Context Learning for Multimodal Tasks

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：HiFICL: High-Fidelity In-Context Learning for Multimodal Tasks

快速理解：In-Context Learning (ICL) is a significant paradigm for Large Multimodal Models (LMMs), using a few in-context demonstrations (ICDs) for new task adaptation. However, its performance is sensitive to demonstration configurations and computationally expensive.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.292 HiSpatial: Taming Hierarchical 3D Spatial Understanding in Vision-Language Models

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：HiSpatial: Taming Hierarchical 3D Spatial Understanding in Vision-Language Models

快速理解：Spatial Reasoning Spatial Understanding Performance Level 2 Accuracy Level 3 Accuracy Figure 1. Trained on our large-scale spatial VQA data, our model develops hierarchical 3D spatial intelligence from geometric perception to abstract reasoning (left), and achieves state-of-the-art results on multiple spatial benchmarks (top-right).

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.293 HiconAgent: History Context-aware Policy Optimization for GUI Agents

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL；优先级：扩展
- [来源](#)

论文：HiconAgent: History Context-aware Policy Optimization for GUI Agents

快速理解：Graphical User Interface (GUI) agents require effective utilization of historical context to perform sequential navigation tasks. While incorporating past actions and observations can significantly improve decision-making, naively using full history leads to excessive computational overhead and potential distraction from irrelevant information.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.2.294 Hierarchical Long Video Understanding with Audiovisual Entity Cohesion and Agentic Search

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL；优先级：扩展
- [来源](#)

论文：Hierarchical Long Video Understanding with Audiovisual Entity Cohesion and Agentic Search

快速理解：Long video understanding presents significant challenges for vision-language models due to extremely long context windows. Existing solutions relying on naive chunking strategies with retrieval-augmented generation, typically suffer from information fragmentation and a loss of global coherence.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.2.295 Hierarchically Robust Zero-shot Vision-language Models

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Hierarchically Robust Zero-shot Vision-language Models

快速理解：Vision-Language Models (VLMs) can perform zero-shot classification but are susceptible to adversarial attacks. While robust fine-tuning improves their robustness, existing approaches align fixed text embeddings with an image embedding, sacrificing natural performance and robustness.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.296 HoneyBee: Data Recipes for Vision-Language Reasoners

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：HoneyBee: Data Recipes for Vision-Language Reasoners

快速理解：Recent advances in vision-language models (VLMs) have made them highly effective at reasoning tasks. However, the principles underlying the construction of performant VL reasoning training datasets remain poorly understood.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.297 HulluEdit: Single-Pass Evidence-Consistent Subspace Editing for Mitigating Hallucinations in Large Vision-Language Models

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：HulluEdit: Single-Pass Evidence-Consistent Subspace Editing for Mitigating Hallucinations in Large Vision-Language Models

快速理解：Object hallucination in Large Vision-Language Models (LVLMs) significantly hinders their reliable deployment. Existing methods struggle to balance efficiency and accuracy: they often require expensive reference models and multiple forward passes, or apply static edits that risk suppressing genuine visual evidence.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.298 Human Interaction-Aware 3D Reconstruction from a Single Image

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Human Interaction-Aware 3D Reconstruction from a Single Image

快速理解: Reconstructing textured 3D human models from a single image is fundamental for AR/VR and digital human applications. However, existing methods mostly focus on single individuals and thus fail in multi-human scenes, where naive composition of individual reconstructions often leads to artifacts such as unrealistic overlaps, missing geometry in occluded regions, and distorted interactions.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.299 Humanoid Generative Pre-Training for Zero-Shot Motion Tracking

- 作者: 作者信息未提供
- 年份: 2026; 主题: Embodied/WBC; 优先级: 扩展
- [来源](#)

论文: Humanoid Generative Pre-Training for Zero-Shot Motion Tracking

快速理解: We introduce Humanoid-GPT, a GPT-style Transformer with causal attention trained on a billion-scale motion corpus for whole-body control. Unlike prior shallow MLP trackers constrained by scarce data and an agility-generalization trade-off, Humanoid-GPT is pretrained on a 2B-frame retargeted corpus that unifies all major mocap datasets with large-scale in-house recordings.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制。

证据层级: 本地 PDF 前页。

1.3.2.300 Hybrid Token Compression for Vision-Language Models

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Hybrid Token Compression for Vision-Language Models

快速理解: Vision-language models (VLMs) have transformed multimodal reasoning, but feeding hundreds of visual patch tokens to LLMs incurs quadratic computational costs, straining memory and context windows. Traditional approaches face a trade-off: continuous compression dilutes high-level semantics like object identities, while discrete quantization loses granular details such as textures.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.301 Hyper-PCN: Hypergraph-Based Point Cloud Completion via High-Order Correlation Modeling

- 作者: 作者信息未提供
- 年份: 2026; 主题: Systems; 优先级: 扩展
- [来源](#)

论文: Hyper-PCN: Hypergraph-Based Point Cloud Completion via High-Order Correlation Modeling

快速理解: Point cloud completion is an important yet challenging problem in 3D computer vision, which aims to reconstruct complete and dense 3D shapes from partial point clouds. Although

transformer-based and geometry-based approaches have made significant progress, they often struggle to capture the complex, high-order correlations inherent in point clouds.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.2.302 Hyperbolic Gramian Volumes for Multimodal Alignment

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Hyperbolic Gramian Volumes for Multimodal Alignment

快速理解：Multimodal contrastive learning typically relies on pairwise similarities for alignment, but recent work has shown that Gramian volumes can capture higher-order correlations across modalities. However, Euclidean Gramian volumes suffer from volume collapse under L2 normalization, concentrating near unity with minimal discriminative variance.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.303 Hypergraph-State Collaborative Reasoning for Multi-Object Tracking

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Hypergraph-State Collaborative Reasoning for Multi-Object Tracking

快速理解：Motion reasoning serves as the cornerstone of multiobject tracking (MOT), as it enables consistent association of targets across frames. However, existing motion estimation approaches face two major limitations: (1) instability caused by noisy or probabilistic predictions, and (2) vulnerability under occlusion, where trajectories often fragment once visual cues disappear.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.304 ID-Crafter: VLM-Grounded Online RL for Compositional Multi-Subject Video Generation

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：ID-Crafter: VLM-Grounded Online RL for Compositional Multi-Subject Video Generation

快速理解：Significant progress has been achieved in high-fidelity video synthesis, yet current paradigms often fall short in effectively integrating identity information from multiple subjects. This leads to semantic conflicts and suboptimal performance in preserving identities and interactions, limiting controllability and applicability.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.305 IF-Prune: Information-Flow Guided Token Pruning for Efficient Vision-Language Models

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：IF-Prune: Information-Flow Guided Token Pruning for Efficient Vision-Language Models

快速理解：Vision-language models (VLMs) with dynamic-resolution vision encoders achieve strong performance but face significant efficiency challenges due to long input sequences. A common approach is to assess the importance of tokens and prune those that are less informative.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.306 IGen: Scalable Data Generation for Robot Learning from Open-World Images

- 作者：作者信息未提供
- 年份：2026；主题：Robot Learning；优先级：扩展
- [来源](#)

论文：IGen: Scalable Data Generation for Robot Learning from Open-World Images

快速理解：The rise of generalist robotic policies has created an exponential demand for large-scale training data. However, on-robot data collection is labor-intensive and often limited to specific environments.

关注点：机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：本地 PDF 前页。

1.3.2.307 Identity-Preserving Image-to-Video Generation via Reward-Guided Optimization

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Identity-Preserving Image-to-Video Generation via Reward-Guided Optimization

快速理解：Recent advances in image-to-video (I2V) generation have achieved remarkable progress in synthesizing high-quality, *Equal contribution. †Corresponding authors. temporally coherent videos from static images. Among all the applications of I2V, human-centric video generation includes a large portion.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.308 Improving Calibration in Test-Time Prompt Tuning for Vision-Language Models via Data-Free Flatness-Aware Prompt Pretraining

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文: Improving Calibration in Test-Time Prompt Tuning for Vision-Language Models via Data-Free Flatness-Aware Prompt Pretraining

快速理解: Test-time prompt tuning (TPT) has emerged as a promising technique for enhancing the adaptability of vision-language models by optimizing textual prompts using unlabeled test data. However, prior studies have observed that TPT often produces poorly calibrated models, raising concerns about the reliability of their predictions.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.309 Incentivizing Generative Zero-Shot Learning via Outcome-Reward Reinforcement Learning with Visual Cues

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL；优先级：扩展
- [来源](#)

论文: Incentivizing Generative Zero-Shot Learning via Outcome-Reward Reinforcement Learning with Visual Cues

快速理解: Recent advances in zero-shot learning (ZSL) have demonstrated the potential of generative models. Typically, generative ZSL synthesizes visual features conditioned on semantic prototypes to model the data distribution of unseen classes, followed by training a classifier on the synthesized data.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.2.310 Inconsistency-aware Multimodal Schrodinger Bridge for Deepfake Localization

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文: Inconsistency-aware Multimodal Schrodinger Bridge for Deepfake Localization

快速理解: Audio-visual deepfake localization demands interval-level outputs that serve as temporal evidence. Despite recent progress, symmetric fusion under single-sided or asynchronous forgeries propagates cross-modal noise, degrading high-precision localization.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.311 Infinity-RoPE: Action-Controllable Infinite Video Generation Emerges From Autoregressive Self-Rollout

- 作者：作者信息未提供

- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Infinity-RoPE: Action-Controllable Infinite Video Generation Emerges From Autoregressive Self-Rollout

快速理解: Current autoregressive video diffusion models are constrained by three core bottlenecks: (i) the finite temporal horizon imposed by the base model's 3D Rotary Positional Embedding (3D-RoPE), (ii) slow prompt responsiveness in maintaining fine-grained action control during long-form rollouts, and (iii) the inability to realize discontinuous cinematic transitions within a single generation stream. We introduce $\rightarrow$ -RoPE, a unified inference-time framework that addresses all three limitations through three interconnected components: Block-Relativistic RoPE, KV Flush, and RoPE Cut.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.312 InstAP: Instance-Aware Vision-Language Pre-Train for Spatial-Temporal Understanding

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: InstAP: Instance-Aware Vision-Language Pre-Train for Spatial-Temporal Understanding

快速理解: Current vision-language pre-training (VLP) paradigms excel at global scene understanding but struggle with instance-level reasoning due to global-only supervision. We introduce InstAP, an Instance-Aware Pre-training framework that jointly optimizes global vision-text alignment and fine-grained, instance-level contrastive alignment by grounding textual mentions to specific spatial-temporal regions.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.313 InterAgent: Physics-based Multi-agent Command Execution via Diffusion on Interaction Graphs

- 作者: 作者信息未提供
- 年份: 2026; 主题: Agent RL; 优先级: 扩展
- [来源](#)

论文: InterAgent: Physics-based Multi-agent Command Execution via Diffusion on Interaction Graphs

快速理解: Humanoid agents are expected to emulate the complex coordination inherent in human social behaviors. However, existing methods are largely confined to single-agent scenarios, overlooking the physically plausible interplay essential for multi-agent interactions.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.2.314 Interpretable Debiasing of Vision-Language Models for Social Fairness

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Interpretable Debiasing of Vision-Language Models for Social Fairness

快速理解：The rapid advancement of Vision-Language models (VLMs) has raised growing concerns that their blackbox reasoning processes could lead to unintended forms of social bias. Current debiasing approaches focus on mitigating surface-level bias signals through post-hoc learning or test-time algorithms, while leaving the internal dynamics of the model largely unexplored.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.315 Iris: Bringing Real-World Priors into Diffusion Model for Monocular Depth Estimation

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Iris: Bringing Real-World Priors into Diffusion Model for Monocular Depth Estimation

快速理解：In this paper, we propose Iris, a deterministic framework for Monocular Depth Estimation (MDE) that integrates realworld priors into the diffusion model. Conventional feedforward methods rely on massive training data, yet still miss details.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.316 Is Bin Generation Indispensable? A Bin-Generation-Free Dataset Quantization via Semantic Perspective

- 作者：作者信息未提供
- 年份：2026；主题：AI Infra；优先级：扩展
- [来源](#)

论文：Is Bin Generation Indispensable? A Bin-Generation-Free Dataset Quantization via Semantic Perspective

快速理解：Dataset quantization has recently emerged as a promising solution for mitigating the computational and memory challenges of large-scale datasets. However, existing approaches rely on a bin generation step that is computationally expensive and inefficient for large-scale datasets.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.2.317 Jailbreaking Vision-Language Models via Dissonance-Guided Suffix Optimization and Image-Phrase Injection

- 作者：作者信息未提供

- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Jailbreaking Vision-Language Models via Dissonance-Guided Suffix Optimization and Image-Phrase Injection

快速理解: The integration of vision and language in Vision-Language Models (VLMs), while enabling multimodal capabilities, inherently expands their attack surface. Among existing white-box jailbreak methods, su!x-optimization-based approaches often rely on gradient approximations over discrete token spaces, yielding insu!cient guidance and causing optimization to stagnate in local optima, while imageperturbation-based ones frequently exhibit poor crossmodel transferability.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.318 JoPPO: Hierarchical Photography Assessment via Contrastive Joint Conditional Probabilistic Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2026; 主题: Agent RL; 优先级: 扩展
- [来源](#)

论文: JoPPO: Hierarchical Photography Assessment via Contrastive Joint Conditional Probabilistic Reinforcement Learning

快速理解: With the advancement of Vision-Language Models (VLMs), employing VLM-as-a-Judge for visual evaluation has become a widely adopted metric. However, existing VLM-asa-Judge approaches suffer from biased scoring with low discrimination and lack the capacity for unified multi-attribute compositional assessment.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.2.319 Joint-Aligned Latent Action: Towards Scalable VLA Pretraining in the Wild

- 作者: 作者信息未提供
- 年份: 2026; 主题: VLA; 优先级: 扩展
- [来源](#)

论文: Joint-Aligned Latent Action: Towards Scalable VLA Pretraining in the Wild

快速理解: Despite progress, Vision-Language-Action models (VLAs) are limited by a scarcity of large-scale, diverse robot data. While human manipulation videos offer a rich alternative, existing methods are forced to choose between small, preciselylabeled datasets and vast in-the-wild footage with unreliable hand tracking labels.

关注点: 视觉、语言条件与机器人动作策略之间的对齐。

证据层级: 本地 PDF 前页。

1.3.2.320 KAMP: Knowledge-Anchored Multimodal Pretraining Framework for Medical Image Representation

- 作者: 作者信息未提供

- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: KAMP: Knowledge-Anchored Multimodal Pretraining Framework for Medical Image Representation

快速理解: Cross-modal biomedical signals such as pathology and genomics can provide richer and more robust semantic guidance for medical image representation learning. However, the availability of such guidance remains limited, as privacy constraints and acquisition costs severely restrict access to medical images paired with other biomedical data.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.321 Keep it SymPL: Symbolic Projective Layout for Allocentric Spatial Reasoning in Vision-Language Models

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Keep it SymPL: Symbolic Projective Layout for Allocentric Spatial Reasoning in Vision-Language Models

快速理解: Perspective-aware spatial reasoning involves understanding spatial relationships from specific viewpoints—either egocentric (observer-centered) or allocentric (object-centered). While vision-language models (VLMs) perform well in egocentric settings, their performance deteriorates when reasoning from allocentric viewpoints, where spatial relations must be inferred from the perspective of objects within the scene.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.322 LLMind: Bio-inspired Training-free Adaptive Visual Representations for Vision-Language Models

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: LLMind: Bio-inspired Training-free Adaptive Visual Representations for Vision-Language Models

快速理解: Vision-Language Models (VLMs) typically assume a uniform spatial fidelity across the entire field of view of visual inputs, dedicating equal precision to even the uninformative regions. By contrast, human vision is neither uniform nor static.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.323 LOREAL: Mitigating Low-Resolution Challenges in Vision-Language Models with Attribute-driven Prompt Self-Distillation

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：LOREAL: Mitigating Low-Resolution Challenges in Vision-Language Models with Attribute-driven Prompt Self-Distillation

快速理解：Prompt Learning (PL) has emerged as a parameter-efficient technique for adapting Vision-Language Models (VLMs) to downstream tasks. However, almost all existing PL methods are primarily designed and evaluated on wellcurated datasets, overlooking a critical post-deployment phenomenon, i.e., the intrinsic connection between input resolution and storage-memory consumption.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.324 Language-Grounded Decoupled Action Representation for Robotic Manipulation

- 作者：作者信息未提供
- 年份：2026；主题：Embodied/WBC；优先级：扩展
- [来源](#)

论文：Language-Grounded Decoupled Action Representation for Robotic Manipulation

快速理解：The heterogeneity between high-level vision-language understanding and low-level action control remains a fundamental challenge in robotic manipulation. Although recent methods have advanced task-specific action alignment, they often struggle to generate robust and accurate actions for novel or semantically related tasks.

关注点：具身任务中的感知、动作生成、全身协调或运动控制。

证据层级：本地 PDF 前页。

1.3.2.325 Language-guided Frequency Modulation for Large Vision-Language Models

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Language-guided Frequency Modulation for Large Vision-Language Models

快速理解：Large Vision-Language Models (LVLMs) have demonstrated remarkable capabilities in visual reasoning across diverse tasks. These tasks place different demands on visual representations: some prioritize high-level global context, while others emphasize fine-grained local details.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.326 Learning 3D Reconstruction with Priors in Test Time

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文: Learning 3D Reconstruction with Priors in Test Time

快速理解: We introduce a test-time framework for multiview Transformers (MVTs) that incorporates priors (e.g., camera poses, intrinsics, and depth) to improve 3D tasks without retraining or modifying pre-trained image-only networks. Rather than feeding priors into the architecture, we cast them as constraints on the predictions and optimize the network at inference time.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.327 Learning Coordinate-based Convolutional Kernels for Continuous SE(3) Equivariant and Efficient Point Cloud Analysis

- 作者: 作者信息未提供
- 年份: 2026; 主题: Systems; 优先级: 扩展
- [来源](#)

论文: Learning Coordinate-based Convolutional Kernels for Continuous SE(3) Equivariant and Efficient Point Cloud Analysis

快速理解: A symmetry on rigid motion is one of the salient factors in efficient learning of 3D point cloud problems. Group convolution has been a representative method to extract equivariant features, but its realizations have struggled to retain both rigorous symmetry and scalability simultaneously.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.2.328 Learning from Itself: Mining Internal Knowledge from Vision Language Models for Continual Learning

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Learning from Itself: Mining Internal Knowledge from Vision Language Models for Continual Learning

快速理解: Vision-language models like CLIP excel at zero-shot recognition but struggle with continual learning due to two critical issues: (1) severe distribution gap between pretraining captions and post-training class names, and (2) performance mismatch between vision-only and dual-encoder approaches—vision-only methods achieve 20% higher accuracy on fine-grained tasks while CLIP dominates on natural images. We propose Learning from Itself (Lfi), which mines CLIP's internal knowledge to address both challenges.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.329 Learning to Assist: Physics-Grounded Human-Human Control via Multi-Agent Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2026; 主题: Agent RL; 优先级: 扩展

- [来源](#)

论文: Learning to Assist: Physics-Grounded Human-Human Control via Multi-Agent Reinforcement Learning

快速理解: Humanoid robotics has strong potential to transform daily service and caregiving applications. Although recent advances in general motion tracking within physics engines (GMT) have enabled virtual characters and humanoid robots to reproduce a broad range of human motions, these behaviors are primarily limited to contact-less social interactions or isolated movements.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.2.330 Learning to Focus and Precise Cropping:A Reinforcement Learning Framework with Information Gaps and Grounding Loss for MLLMs

- 作者: 作者信息未提供
- 年份: 2026; 主题: Agent RL; 优先级: 扩展
- [来源](#)

论文: Learning to Focus and Precise Cropping:A Reinforcement Learning Framework with Information Gaps and Grounding Loss for MLLMs

快速理解: To enhance the perception and reasoning capabilities of multimodal large language models in complex visual scenes, recent research has introduced agent-based workflows. In these works, MLLMs autonomously utilize image cropping tool to analyze regions of interest for question answering.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.2.331 Learning to Reason in 4D: Dynamic Spatial Understanding for Vision Language Models

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Learning to Reason in 4D: Dynamic Spatial Understanding for Vision Language Models

快速理解: Vision-language models (VLM) excel at general understanding yet remain weak at dynamic spatial reasoning (DSR), i.e., reasoning about the evolvement of object geometry and relationship in 3D space over time, largely due to the scarcity of scalable 4D-aware training resources. To bridge this gap across aspects of dataset, benchmark and model, we introduce DSR Suite.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.332 LensWalk: Agentic Video Understanding by Planning How You See in Videos

- 作者: 作者信息未提供
- 年份: 2026; 主题: Agent RL; 优先级: 扩展

- [来源](#)

论文: LensWalk: Agentic Video Understanding by Planning How You See in Videos

快速理解: The dense, temporal nature of video presents a profound challenge for automated analysis. Despite the use of powerful Vision-Language Models, prevailing methods for video understanding are limited by the inherent disconnect between reasoning and perception: they rely on static, pre-processed information and cannot actively seek raw evidence from video as their understanding evolves.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.2.333 Lenses: Toward Polysemous Vision-Language Understanding

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Lenses: Toward Polysemous Vision-Language Understanding

快速理解: Emotional Background A) Multi-Lens Views B) Category-Aligned Set Matching A woman's face emerges from a vibrant mural of fruits, her gaze fixed on the colorful display She blends into the harvest like a shadow in a cornucopia, her identity woven into the produce A collision of humanity and harvest: the face in the fruit becomes a metaphor for belonging Her eyes, painted in muted gold, hold the quiet mystery of a hidden observer in a world of color Weathered bricks cradle the mural, their rough texture contrasting with the smooth, painted fruit Literal Figurative Abstract Emotional Background 2 3 1 4 Multi-Perspective Views & Alignment 5 Figure 1. Overview of Lenses. (Left) Standard datasets treat each image as having one literal meaning. (Middle) Lenses decomposes images into five interpretive lenses: Literal, Figurative, Abstract, Background, and Emotional, with diverse captions for each perspective.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.334 Less is More: Data-Efficient Adaptation for Controllable Text-to-Video Generation

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Less is More: Data-Efficient Adaptation for Controllable Text-to-Video Generation

快速理解: Fine-tuning large-scale text-to-video diffusion models to add new generative controls, such as those over physical camera parameters (e.g., shutter speed or aperture), typically requires vast, high-fidelity datasets that are difficult to acquire. In this work, we propose a data-efficient finetuning strategy that learns these controls from sparse, lowquality synthetic data.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.335 Let VLMs Grade Their Own Thoughts: A Self-Quantification Approach to Reasoning-Aware Reward Modeling

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL；优先级：扩展
- [来源](#)

论文：Let VLMs Grade Their Own Thoughts: A Self-Quantification Approach to Reasoning-Aware Reward Modeling

快速理解：Chain-of-Thought (CoT) is a key technique for enhancing the reasoning capabilities of Vision Language Models (VLMs). Existing methods often employ Reinforcement Learning (RL) with external constraints to align the model's reasoning process with human cognitive patterns.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.2.336 Leveraging Verifier-Based Reinforcement Learning in Image Editing

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL；优先级：扩展
- [来源](#)

论文：Leveraging Verifier-Based Reinforcement Learning in Image Editing

快速理解：While Reinforcement Learning from Human Feedback (RLHF) has become a pivotal paradigm for text-to-image generation, its application to image editing remains largely unexplored. A key bottleneck is the lack of a robust general reward model for all editing tasks.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.2.337 LifeEval: A Multimodal Benchmark for Assistive AI in Egocentric Daily Life Tasks

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：LifeEval: A Multimodal Benchmark for Assistive AI in Egocentric Daily Life Tasks

快速理解：The rapid progress of Multimodal Large Language Models (MLLMs) marks a significant step toward artificial general intelligence, offering great potential for augmenting human capabilities. However, their ability to provide effective assistance in dynamic, real-world environments remains largely underexplored.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.338 Lifting Unlabeled Internet-level Data for 3D Scene Understanding

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展

- [来源](#)

论文: Lifting Unlabeled Internet-level Data for 3D Scene Understanding

快速理解: Annotated 3D scene data is scarce and expensive to acquire, while abundant unlabeled videos are readily available on the internet. In this paper, we demonstrate that carefully designed data engines can leverage web-curated, unlabeled videos to automatically generate training data, to facilitate end-to-end models in 3D scene understanding alongside human-annotated datasets.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.339 LinVideo: A Post-Training Framework towards $O(n)$ Attention in Efficient Video Generation

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: LinVideo: A Post-Training Framework towards $O(n)$ Attention in Efficient Video Generation

快速理解: Video diffusion models (DMs) have enabled high-quality video synthesis, but their computation costs scale quadratically with sequence length due to the nature of selfattention. While linear attention offers a more efficient alternative, fully replacing quadratic attention demands costly pretraining.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.340 LiteSense: Lifting Lightweight ToF with RGB for High-Resolution Metric Depth Estimation

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: LiteSense: Lifting Lightweight ToF with RGB for High-Resolution Metric Depth Estimation

快速理解: Metric depth estimation aims to recover depth maps with absolute scale, high resolution, and cross-scene consistency from visual observations. Existing approaches either rely on large-scale models or costly sensors to preserve metric accuracy and generalization, both ill-suited to resource-constrained deployment.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.341 LiteVGGT: Boosting Vanilla VGGT via Geometry-aware Cached Token Merging

- 作者: 作者信息未提供
- 年份: 2026; 主题: Systems; 优先级: 扩展
- [来源](#)

论文: LiteVGGT: Boosting Vanilla VGGT via Geometry-aware Cached Token Merging

快速理解: 3D vision foundation models like Visual Geometry Grounded Transformer (VGGT) have advanced greatly in geometric perception. However it is time-consuming and memory-intensive for long sequences, limiting application to large-scale scenes beyond hundreds of images.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.2.342 LoL: Longer than Longer, Scaling Video Generation to Hour

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: LoL: Longer than Longer, Scaling Video Generation to Hour

快速理解: Recent research in long-form video generation has shifted from bidirectional to autoregressive models, yet these methods commonly suffer from error accumulation and a loss of long-term coherence. While attention sink frames have been introduced to mitigate this performance decay, they often induce a critical failure mode we term sink-collapse: the generated content repeatedly reverts to the sink frame, resulting in abrupt scene resets and cyclic motion patterns.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.343 Local Motion Matters: A Deconstruct-Recompose Paradigm for Reinforcement Learning Pre-training from Videos

- 作者: 作者信息未提供
- 年份: 2026; 主题: Agent RL; 优先级: 扩展
- [来源](#)

论文: Local Motion Matters: A Deconstruct-Recompose Paradigm for Reinforcement Learning Pre-training from Videos

快速理解: Pre-training on large-scale videos to improve reinforcement learning efficiency is promising yet remains challenging. Existing methods typically treat the agent as an indivisible entity, modeling motion patterns globally.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.2.344 Local Precise Refinement: A Dual-Gated Mixture-of-Experts for Enhancing Foundation Model Generalization against Spectral Shifts

- 作者: 作者信息未提供
- 年份: 2026; 主题: AI Infra; 优先级: 扩展
- [来源](#)

论文: Local Precise Refinement: A Dual-Gated Mixture-of-Experts for Enhancing Foundation Model Generalization against Spectral Shifts

快速理解: Domain Generalization Semantic Segmentation (DGSS) in spectral remote sensing is severely challenged by spectral shifts across diverse acquisition conditions, which cause significant performance degradation for models deployed in unseen domains. While fine-tuning foundation models is a promising direction, existing methods employ global, homogeneous adjustments.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 本地 PDF 前页。

1.3.2.345 LocateAnything3D: Vision-Language 3D Detection with Chain-of-Sight

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: LocateAnything3D: Vision-Language 3D Detection with Chain-of-Sight

快速理解: To act in the world, a model must name what it sees and know where it is in 3D. Today's vision-language models (VLMs) excel at open-ended 2D description and grounding, yet multi-object 3D detection remains largely missing from the VLM toolbox.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.346 Low-Rank Test-Time Training for Pre-Trained Point Cloud Models

- 作者: 作者信息未提供
- 年份: 2026; 主题: Systems; 优先级: 扩展
- [来源](#)

论文: Low-Rank Test-Time Training for Pre-Trained Point Cloud Models

快速理解: Test-time training (TTT) enhances the robustness of pretrained models to out-of-distribution (OOD) data through auxiliary self-supervised tasks, without requiring labeled samples. However, existing TTT methods predominantly rely on decoder-based auxiliary objectives, which suffer from inefficient adaptation and weak coupling with the primary task.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.2.347 Lynx: Towards High-Fidelity Personalized Video Generation

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Lynx: Towards High-Fidelity Personalized Video Generation

快速理解: We present Lynx, a high-fidelity model for personalized video synthesis from a single input image. Built on an opensource Diffusion Transformer (DiT) foundation model, Lynx introduces two lightweight adapters to ensure identity fidelity.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.348 M3DocDep: Multi-modal, Multi-page, Multi-document Dependency Chunking with Large Vision-Language Models

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：M3DocDep: Multi-modal, Multi-page, Multi-document Dependency Chunking with Large Vision-Language Models

快速理解：In long, multi-page industrial documents, retrieval-augmented generation (RAG) depends heavily on whether chunk boundaries follow the document's true structure. Existing text-centric chunkers and generative hierarchy parsers often miss cross-page parent-child relations, figure/table-caption bindings, and boundary cues, which leads to fragmented or redundant chunks and degrades both retrieval and answer quality.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.349 M4-RAG: A Massive-Scale Multilingual Multi-Cultural Multimodal RAG

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：M4-RAG: A Massive-Scale Multilingual Multi-Cultural Multimodal RAG

快速理解：Vision-language models (VLMs) have achieved strong performance in visual question answering (VQA), yet they remain constrained by static training data. Retrieval-Augmented Generation (RAG) mitigates this limitation by enabling access to up-to-date, culturally grounded, and multilingual information; however, multilingual multimodal RAG remains largely underexplored.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.350 M4Human: A Large-Scale Multimodal mmWave Radar Benchmark for Human Mesh Reconstruction

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：M4Human: A Large-Scale Multimodal mmWave Radar Benchmark for Human Mesh Reconstruction

快速理解：Human mesh reconstruction (HMR) provides direct insights into body-environment interaction, enabling various immersive applications. However, existing large-scale HMR benchmarks largely rely on line-of-sight RGB sensing, causing HMR systems to inherit the limitations of vision-based systems, including sensitivity to occlusion, lighting variation, and privacy concerns.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.351 MAD: Modality-Adaptive Decoding for Mitigating Cross-Modal Hallucinations in Multimodal Large Language Models

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：MAD: Modality-Adaptive Decoding for Mitigating Cross-Modal Hallucinations in Multimodal Large Language Models

快速理解：Multimodal Large Language Models (MLLMs) suffer from cross-modal hallucinations, where one modality inappropriately influences generation about another, leading to fabricated output. This exposes a more fundamental deficiency in modality-interaction control.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.352 MANSION: Multi-floor lANguage-to-3D Scene generatIOOn for loNg-horizon tasks

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：MANSION: Multi-floor lANguage-to-3D Scene generatIOOn for loNg-horizon tasks

快速理解：Real-world robotic tasks are long-horizon and often span multiple floors, demanding rich spatial reasoning. However, existing embodied benchmarks are largely confined to single-floor in-house environments, failing to reflect the complexity of real-world tasks.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.353 MD2E: Modeling Depth-to-Edge Cues for Monocular Metric Depth Estimation

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：MD2E: Modeling Depth-to-Edge Cues for Monocular Metric Depth Estimation

快速理解：We study monocular metric depth estimation (MMDE) without camera intrinsics at training or inference. When focal length and scene depth vary together, depth changes are difficult to perceive in the image, yet the edge-frequency statistics exhibit systematic, scale-correlated shifts.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.354 MHopReg: Efficient Hierarchical Multi-Hop Graph Search for Point Cloud Registration

- 作者：作者信息未提供

- 年份: 2026; 主题: Systems; 优先级: 扩展
- [来源](#)

论文: MHopReg: Efficient Hierarchical Multi-Hop Graph Search for Point Cloud Registration

快速理解: Outlier rejection for correspondence-based point cloud registration confronts two fundamental challenges in realworld scenarios. First, low-overlap regions yield sparse and fragmented inlier distributions that are difficult to discover using conventional one-step global search strategies.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.2.355 MM-ACT: Learn from Multimodal Parallel Generation to Act

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: MM-ACT: Learn from Multimodal Parallel Generation to Act

快速理解: A generalist robotic policy needs both semantic understanding for task planning and the ability to interact with the environment through predictive capabilities. To tackle this, we present MM-ACT, a unified Vision- Language-Action (VLA) model that integrates text, image, and action in shared token space and performs generation across all three modalities.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.356 MM-OVSeg: Multimodal Optical-SAR Fusion for Open-Vocabulary Segmentation in Remote Sensing

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: MM-OVSeg: Multimodal Optical-SAR Fusion for Open-Vocabulary Segmentation in Remote Sensing

快速理解: Open-vocabulary segmentation enables pixel-level recognition from an open set of textual categories, allowing generalization beyond fixed classes. Despite great potential in remote sensing, progress in this area remains largely limited to clear-sky optical data and struggles under cloudy or haze-contaminated conditions.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.357 MMDIR: Multimodal Instruction-Driven Framework for Mixed-Degradation Document Image Restoration

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: MMDIR: Multimodal Instruction-Driven Framework for Mixed-Degradation Document Image Restoration

快速理解: Restoring degraded document image is essential for both improving visual quality and optimizing performance in downstream document analysis tasks. Although existing methods have demonstrated substantial improvements in restoration outcomes, they primarily address single-type degradation scenarios.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.358 MMR-AD: A Large-Scale Multimodal Dataset for Benchmarking General Anomaly Detection with Multimodal Large Language Models

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: MMR-AD: A Large-Scale Multimodal Dataset for Benchmarking General Anomaly Detection with Multimodal Large Language Models

快速理解: In the progress of industrial anomaly detection, general anomaly detection (GAD) is an emerging trend and also the ultimate goal. Unlike the conventional singleand multiclass AD, general AD aims to train a general AD model that can directly detect anomalies in diverse novel classes without any retraining or fine-tuning on the target data.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.359 MMSD3.0: A Multi-Image Benchmark for Real-World Multimodal Sarcasm Detection

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: MMSD3.0: A Multi-Image Benchmark for Real-World Multimodal Sarcasm Detection

快速理解: Despite progress in multimodal sarcasm detection, existing datasets and methods predominantly focus on singleimage scenarios, overlooking potential semantic and affective relations across multiple images. This leaves a gap in modeling cases where sarcasm is triggered by multiimage cues in real-world settings.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.360 MODIX: A Training-Free Multimodal Information-Driven Positional Index Scaling for Vision-Language Models

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: MODIX: A Training-Free Multimodal Information-Driven Positional Index Scaling for Vision-Language Models

快速理解: Vision-Language Models (VLMs) have achieved remarkable progress in multimodal understanding, yet their positional encoding mechanisms remain suboptimal. Existing approaches uniformly assign positional indices to all tokens, overlooking variations in information density within and across modalities, which leads to inefficient attention allocation where redundant visual regions dominate while informative content is underrepresented.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.361 MR-RAG: Multimodal Relevance-Aware Retrieval-Augmented Generation for Medical Visual Question Answering

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: MR-RAG: Multimodal Relevance-Aware Retrieval-Augmented Generation for Medical Visual Question Answering

快速理解: Large Vision Language Models (LVLMs) with retrieval-augmented generation (RAG) are emerging as a main paradigm for processing vision-language medical tasks due to their promising achievements. However, existing approaches exhibit two significant limitations in both retrieval and generation stage: First, during the retrieval stage, most methods typically rely on a single similarity signal to estimate document relevance, ignoring the rich information available in multimodal data, which may fail to accurately retrieve matching content.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.362 MRI Contrast Enhancement Kinetics World Model

- 作者: 作者信息未提供
- 年份: 2026; 主题: World Model; 优先级: 扩展
- [来源](#)

论文: MRI Contrast Enhancement Kinetics World Model

快速理解: Clinical MRI contrast acquisition suffers from inefficient information yield, which presents as a mismatch between the risky and costly acquisition protocol and the fixed and sparse acquisition sequence. Applying world models to simulate the contrast enhancement kinetics in the human body enables continuous contrast-free dynamics.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策。

证据层级: 本地 PDF 前页。

1.3.2.363 MTA: Multimodal Task Alignment for BEV Perception and Captioning

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展

- [来源](#)

论文: MTA: Multimodal Task Alignment for BEV Perception and Captioning

快速理解: Bird' s eye view (BEV)-based 3D perception plays a crucial role in autonomous driving applications. The rise of large language models has spurred interest in BEV-based captioning to understand object behavior in the surrounding environment.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.364 MVGGT: Multimodal Visual Geometry Grounded Transformer for Multi-view 3D Referring Expression Segmentation

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: MVGGT: Multimodal Visual Geometry Grounded Transformer for Multiview 3D Referring Expression Segmentation

快速理解: Most existing 3D referring expression segmentation (3DRES) methods rely on dense, high-quality point clouds, while real-world agents such as robots and mobile phones †Equal Contribution. *Corresponding Author. operate with only a few sparse RGB views and strict latency constraints. We introduce Multi-view 3D Referring Expression Segmentation (MV-3DRES), where the model must recover scene structure and segment the referred object directly from sparse multi-view images.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.365 MVLM: Template-Free Tracking via Vision-Language Margin Confidence and Memory-Gated Tracking

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: MVLM: Template-Free Tracking via Vision-Language Margin Confidence and Memory-Gated Tracking

快速理解: We introduce a new template-free tracking paradigm based solely on natural language, capable of tracking an arbitrary object and seamlessly switching to a new target without box initialization. Our key idea is to localize an object via vision-language (VL) correlation.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.366 M^3KG-RAG: Multi-hop Multimodal Knowledge Graph-enhanced Retrieval-Augmented Generation

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: M³KG-RAG: Multi-hop Multimodal Knowledge Graph-enhanced Retrieval-Augmented Generation

快速理解: Retrieval-Augmented Generation (RAG) has recently been extended to multimodal settings, connecting multimodal large language models (MLLMs) with vast corpora of external knowledge such as multimodal knowledge graphs (MMKGs). Despite their recent success, multimodal RAG in the audio-visual domain remains challenging due to 1) limited modality coverage and multi-hop connectivity of existing MMKGs, and 2) retrieval based solely on similarity in a shared multimodal embedding space, which fails to filter out off-topic or redundant knowledge.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.367 Machine Mental Imagery: Empower Multimodal Reasoning with Latent Visual Tokens

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Machine Mental Imagery: Empower Multimodal Reasoning with Latent Visual Tokens

快速理解: Vision-language models (VLMs) excel at multimodal understanding, yet their text-only decoding forces them to verbalize visual reasoning, limiting performance on tasks that demand visual imagination. Recent attempts train VLMs to render explicit images, but the heavy image-generation pretraining often hinders the reasoning ability.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.368 Mamba Learns in Context: Structure-Aware Domain Generalization for Multi-Task Point Cloud Understanding

- 作者: 作者信息未提供
- 年份: 2026; 主题: Systems; 优先级: 扩展
- [来源](#)

论文: Mamba Learns in Context: Structure-Aware Domain Generalization for Multi-Task Point Cloud Understanding

快速理解: While recent Transformer and Mamba architectures have advanced point cloud representation learning, they are typically developed for single-task or single-domain settings. Directly applying them to multi-task domain generalization (DG) leads to degraded performance.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.2.369 MangoBench: A Benchmark for Multi-Agent Goal-Conditioned Offline Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2026; 主题: Agent RL; 优先级: 扩展

- [来源](#)

论文: MangoBench: A Benchmark for Multi-Agent Goal-Conditioned Offline Reinforcement Learning

快速理解: Offline Multi-Agent Reinforcement Learning (MARL) is critical for coordinating multiple agents in costly and unsafe environments, yet existing methods struggle with high sensitivity to reward functions and weak generalization to new goals, limiting its practical impact. Inspired by singleagent Offline Goal-Conditioned RL (OGCRL), we propose the first goal-conditioned offline MARL framework, extending OGCRL to multi-agent settings under both fully decentralized and centralized training with decentralized execution (CTDE) paradigms.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.2.370 Mario: Multimodal Graph Reasoning with Large Language Models

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Mario: Multimodal Graph Reasoning with Large Language Models

快速理解: Recent advances in large language models (LLMs) have opened new avenues for multimodal reasoning. Yet, most existing methods still rely on pretrained vision-language models (VLMs) to encode image-text pairs in isolation, ignoring the relational structure that real-world multimodal data naturally form.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.371 MarkushGrapher-2: End-to-end Multimodal Recognition of Chemical Structures

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: MarkushGrapher-2: End-to-end Multimodal Recognition of Chemical Structures

快速理解: Automatically extracting chemical structures from documents is essential for the large-scale analysis of the literature in chemistry. Automatic pipelines have been developed to recognize molecules represented either in figures or in text independently.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.372 Masking Matters: Unlocking the Spatial Reasoning Capabilities of LLMs for 3D Scene-Language Understanding

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Masking Matters: Unlocking the Spatial Reasoning Capabilities of LLMs for 3D Scene-Language Understanding

快速理解: Recent advances in 3D scene-language understanding have leveraged Large Language Models (LLMs) for 3D reasoning by transferring their general reasoning ability to 3D multi-modal contexts. However, existing methods typically adopt standard decoders from language modeling, which rely on a causal attention mask.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.373 Med-CMR: A Fine-Grained Benchmark Integrating Visual Evidence and Clinical Logic for Medical Complex Multimodal Reasoning

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Med-CMR: A Fine-Grained Benchmark Integrating Visual Evidence and Clinical Logic for Medical Complex Multimodal Reasoning

快速理解: MLLMs are beginning to appear in clinical workflows, but their ability to perform complex medical reasoning remains unclear. We present Med-CMR, a fine-grained Medical Complex Multimodal Reasoning benchmark.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.374 MedCLIPSeg: Probabilistic Vision-Language Adaptation for Data-Efficient and Generalizable Medical Image Segmentation

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: MedCLIPSeg: Probabilistic Vision-Language Adaptation for Data-Efficient and Generalizable Medical Image Segmentation

快速理解: Medical image segmentation remains challenging due to limited annotations for training, ambiguous anatomical features, and domain shifts. While vision-language models such as CLIP offer strong cross-modal representations, their potential for dense, text-guided medical image segmentation remains underexplored.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.375 MedGRPO: Multi-Task Reinforcement Learning for Heterogeneous Medical Video Understanding

- 作者: 作者信息未提供
- 年份: 2026; 主题: Agent RL; 优先级: 扩展
- [来源](#)

论文: MedGRPO: Multi-Task Reinforcement Learning for Heterogeneous Medical Video Understanding

快速理解: Large vision-language models struggle with medical video understanding, where spatial precision, temporal reasoning, and clinical semantics are critical. To address this, we first introduce MedVidBench, a large-scale benchmark of 531,850 video-instruction pairs across 8 medical sources spanning video, segment, and frame-level tasks, curated through a rigorous quality assurance pipeline with expert-guided prompting and dual-model validation.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.2.376 MedLoc-R1: Performance-Aware Curriculum Reward Scheduling for GRPO-Based Medical Visual Grounding

- 作者: 作者信息未提供
- 年份: 2026; 主题: Systems; 优先级: 扩展
- [来源](#)

论文: MedLoc-R1: Performance-Aware Curriculum Reward Scheduling for GRPO-Based Medical Visual Grounding

快速理解: Medical visual grounding serves as a crucial foundation for fine-grained multimodal reasoning and interpretable clinical decision support. Despite recent advances in reinforcement learning (RL) for grounding tasks, existing approaches such as Group Relative Policy Optimization (GRPO) suffer from severe reward sparsity when directly applied to medical images, primarily due to the inherent difficulty of localizing small or ambiguous regions of interest, which is further exacerbated by the rigid and suboptimal nature of fixed IoU-based reward schemes in RL.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.2.377 MedMO: Grounding and Understanding Multimodal Large Language Model for Medical Images

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: MedMO: Grounding and Understanding Multimodal Large Language Model for Medical Images

快速理解: Multimodal large language models (MLLMs) have rapidly advanced, yet their adoption in medicine remains limited by gaps in domain coverage, modality alignment, and grounded reasoning. In this work, we introduce MedMO, a medical foundation model built upon a generalized MLLM architecture and trained exclusively on large-scale, domain-specific data.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.378 Medic-AD: Towards Medical Vision-Language Model's Clinical Intelligence

- 作者: 作者信息未提供

- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Medic-AD: Towards Medical Vision-Language Model' s Clinical Intelligence

快速理解: Lesion detection, symptom tracking, and visual explainability are central to real-world medical image analysis, yet current medical Vision-Language Models (VLMs) still lack mechanisms that translate their broad knowledge into clinically actionable outputs. To bridge this gap, we present MEDIC-AD, a clinically oriented VLM that strengthens these three capabilities through a stage-wise framework.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.379 Memory-Augmented Scene Understanding and Exploration for Open-World Aerial Object-Goal Navigation

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Memory-Augmented Scene Understanding and Exploration for Open-World Aerial Object-Goal Navigation

快速理解: Aerial object-goal navigation (Aerial ObjectNav) requires an Unmanned Aerial Vehicle (UAV) to navigate to target objects in large-scale outdoor environments using only visual observations and high-level object descriptions, without detailed step-by-step instructions. Existing approaches rely on local observations or short-term history, lacking comprehensive scene understanding and effective spatial exploration strategies, which constrains their navigation *Equal contribution. †Corresponding author. capability in complex aerial scenarios.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.380 MimicTalker: A Multimodal Interactive and Memory-Enhanced Framework for Real-Time Dyadic 3D Head Generation

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: MimicTalker: A Multimodal Interactive and Memory-Enhanced Framework for Real-Time Dyadic 3D Head Generation

快速理解: Dyadic interactive head generation aims to synthesize realistic head motions that respond both verbally and nonverbally to an interlocutor in real-time conversation. The existing works often focus on offline scenarios, and struggle with a shallow understanding of the multimodal conversational context while also lacking long-term coherence.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.381 MindDriver: Introducing Progressive Multimodal Reasoning for Autonomous Driving

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：MindDriver: Introducing Progressive Multimodal Reasoning for Autonomous Driving

快速理解：Vision-Language Models (VLM) exhibit strong reasoning capabilities, showing promise for end-to-end autonomous driving systems. Chain-of-Thought (CoT), as VLM's widely used reasoning strategy, is facing critical challenges.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.382 Mining Instance-Centric Vision-Language Contexts for Human-Object Interaction Detection

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Mining Instance-Centric Vision-Language Contexts for Human-Object Interaction Detection

快速理解：Human-Object Interaction (HOI) detection aims to localize human-object pairs and classify their interactions from a single image, a task that demands strong visual understanding and nuanced contextual reasoning. Recent approaches have leveraged Vision-Language Models (VLMs) to introduce semantic priors, significantly improving HOI detection performance.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.383 Mixture of States: Routing Token-Level Dynamics for Multimodal Generation

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Mixture of States: Routing Token-Level Dynamics for Multimodal Generation

快速理解：We introduce MoS (Mixture of States), a novel fusion paradigm for multimodal diffusion models that merges modalities using flexible, state-based interactions. The core of MoS is a learnable, token-wise router that creates denoising timestep and input-dependent interactions between modalities' hidden states, precisely aligning token level features with the diffusion trajectory.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.384 Mixture-of-Experts based Feature Decoupling for Open Vocabulary Scene Graph Generation

- 作者：作者信息未提供
- 年份：2026；主题：AI Infra；优先级：扩展
- [来源](#)

论文：Mixture-of-Experts based Feature Decoupling for Open Vocabulary Scene Graph Generation

快速理解：In recent years, while Scene Graph Generation has advanced significantly, mainstream methods remain constrained by predefined object and relationship categories, limiting generalization to open real-world scenarios. Inspired by open vocabulary object detection, recent efforts have expanded SGG to the open vocabulary domain.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.2.385 MoRGS: Efficient Per-Gaussian Motion Reasoning for Streamable Dynamic 3D Scenes

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：MoRGS: Efficient Per-Gaussian Motion Reasoning for Streamable Dynamic 3D Scenes

快速理解：Online reconstruction of dynamic scenes aims to learn from streaming multi-view inputs under low-latency constraints. The fast training and real-time rendering capabilities of 3D Gaussian Splatting have made on-the-fly reconstruction practically feasible, enabling online 4D reconstruction.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.386 MoReGen: Multi-Agent Motion-Reasoning Engine for Code-based Text-to-Video Synthesis

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL；优先级：扩展
- [来源](#)

论文：MoReGen: Multi-Agent Motion-Reasoning Engine for Code-based Text-to-Video Synthesis

快速理解：While text-to-video (T2V) generation has achieved remarkable progress in photorealism, generating intent-aligned videos that faithfully obey physics principles remains a core challenge. In this work, we systematically study Newtonian motion-controlled text-to-video generation and evaluation, emphasizing physical precision and motion coherence.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.2.387 Molmo2: Open Weights and Data for Vision-Language Models with Video Understanding and Grounding

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文: Molmo2: Open Weights and Data for Vision-Language Models with Video Understanding and Grounding

快速理解: Today's strongest video-language models (VLMs) remain proprietary, and the strongest open-weight models often rely on synthetic data from proprietary VLMs and do not disclose their training data or recipe. As a result, the opensource community lacks the foundations needed to improve on the state-of-the-art video (and image) language models.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.388 MonoVLM: Monocular 3D Visual Grounding with Vision Language Models

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: MonoVLM: Monocular 3D Visual Grounding with Vision Language Models

快速理解: Vision-Language Models (VLMs) have demonstrated remarkable capabilities in instruction following and 2D visual understanding. However, state-of-the-art VLMs, including GPT-5, still struggle with 3D perception, particularly in tasks such as monocular 3D visual grounding.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.389 MorphSeek: Fine-grained Latent Representation-Level Policy Optimization for Deformable Image Registration

- 作者: 作者信息未提供
- 年份: 2026; 主题: Agent RL; 优先级: 扩展
- [来源](#)

论文: MorphSeek: Fine-grained Latent Representation-Level Policy Optimization for Deformable Image Registration

快速理解: Deformable image registration (DIR) remains a fundamental yet challenging problem in medical image analysis, largely due to the prohibitively high-dimensional deformation space of dense displacement fields and the scarcity of voxel-level supervision. Existing reinforcement learning frameworks often project this space into coarse, low-dimensional representations, limiting their ability to capture spatially variant deformations.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.2.390 Mostly Text, Smart Visuals: Asymmetric Text-Visual Pruning for Large Vision-Language Models

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Mostly Text, Smart Visuals: Asymmetric Text-Visual Pruning for Large Vision-Language

Models

快速理解: Network pruning is an effective technique for enabling lightweight Large Vision-Language Models (LVLMs), which primarily incorporates both weights and activations into the importance metric. However, existing efforts typically process calibration data from different modalities in a unified manner, overlooking modality-specific behaviors.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.391 MotionEnhancer: Leveraging Video Diffusion for Motion-Enhanced Vision-Language Models

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: MotionEnhancer: Leveraging Video Diffusion for Motion-Enhanced Vision-Language Models

快速理解: The new era has witnessed a remarkable capability to extend Vision-Language Models (VLMs) for tackling tasks of video understanding. While current VLMs excel at event or story-level understanding, their ability to capture fine-grained motion details remains limited, primarily due to their focus on high-level static semantic structures and macro-event logic.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.392 Motus: A Unified Latent Action World Model

- 作者: 作者信息未提供
- 年份: 2026; 主题: World Model; 优先级: 扩展
- [来源](#)

论文: Motus: A Unified Latent Action World Model

快速理解: While a general embodied agent must function as a unified system, current methods are built on isolated models for understanding, world modeling, and control. This fragmentation prevents unifying multimodal generative capabilities and hinders learning from large-scale, heterogeneous data.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策。

证据层级: 本地 PDF 前页。

1.3.2.393 MovieRecapsQA: A Multimodal Open-Ended Video Question-Answering Benchmark

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: MovieRecapsQA: A Multimodal Open-Ended Video Question-Answering Benchmark

快速理解: Understanding real-world videos such as movies requires integrating visual and dialogue

cues. Yet existing VideoQA benchmarks struggle to capture this multimodal reasoning and, given the difficulty of evaluating free-form answers, largely resort to simple multiple choice questions.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.394 MuCo: Multi-turn Contrastive Learning for Multimodal Embedding Model

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：MuCo: Multi-turn Contrastive Learning for Multimodal Embedding Model

快速理解：Universal Multimodal embedding models built on Multimodal Large Language Models (MLLMs) have traditionally employed contrastive learning, which aligns representations of query-target pairs across different modalities. Yet, despite its empirical success, they are primarily built on a “single-turn” formulation where each query-target pair is treated as an independent data point.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.395 MuKV: Multi-Grained KV Cache Compression for Long Streaming Video Question-Answering

- 作者：作者信息未提供
- 年份：2026；主题：Systems；优先级：扩展
- [来源](#)

论文：MuKV: Multi-Grained KV Cache Compression for Long Streaming Video Question-Answering

快速理解：Long streaming video QA remains challenging due to growing visual tokens and limited reasoning length of large language models (LLMs). KV-caching stores the Key-Value (KV) of the historical tokens via LLM prefill and enables more efficient streaming QA.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.2.396 Multi-Metric Representation Learning Strategy Based on Clustering for Fine-Grained Multimodal Sentiment Analysis

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Multi-Metric Representation Learning Strategy Based on Clustering for Fine-Grained Multimodal Sentiment Analysis

快速理解：Multimodal sentiment analysis (MSA) aims to identify human emotions through multimodal data. Despite considerable advances in MSA, we find that emotional class centers often overlap when integrating data from different modalities into the same representation space.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.397 Multi-Scale Gaussian-Language Map for Zero-shot Embodied Navigation and Reasoning

- 作者：作者信息未提供
- 年份：2026；主题：Embodied/WBC；优先级：扩展
- [来源](#)

论文：Multi-Scale Gaussian-Language Map for Zero-shot Embodied Navigation and Reasoning

快速理解：Understanding the geometric and semantic structure of environments is essential for embodied navigation and reasoning. Existing semantic mapping methods trade off between explicit geometry and multi-scale semantics, and lack a native interface for large models, thus requiring additional training of feature projection for semantic alignment.

关注点：具身任务中的感知、动作生成、全身协调或运动控制。

证据层级：本地 PDF 前页。

1.3.2.398 Multi-speaker Attention Alignment for Multimodal Social Interaction

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Multi-speaker Attention Alignment for Multimodal Social Interaction

快速理解：Understanding social interaction in video requires reasoning over a dynamic interplay of verbal and non-verbal cues: who is speaking, to whom, and with what gaze or gestures. While Multimodal Large Language Models (MLLMs) are natural candidates, simply adding visual inputs yields surprisingly inconsistent gains on social tasks.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.399 MultiModalPFN: Extending Prior-Data Fitted Networks for Multimodal Tabular Learning

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：MultiModalPFN: Extending Prior-Data Fitted Networks for Multimodal Tabular Learning

快速理解：Recently, TabPFN has gained attention as a foundation model for tabular data. However, it struggles to integrate heterogeneous modalities such as images and text, which are common in domains like healthcare and marketing, thereby limiting its applicability.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.400 MultiShotMaster: A Controllable Multi-Shot Video Generation Framework

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：MultiShotMaster: A Controllable Multi-Shot Video Generation Framework

快速理解：Current video generation techniques excel at single-shot clips but struggle to produce narrative multi-shot videos, which require flexible shot arrangement, coherent narrative, and controllability beyond text prompts. To tackle these challenges, we propose MultiShotMaster, a framework for highly controllable multi-shot video generation.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.401 Multimodal Causality-Driven Representation Learning for Generalizable Medical Image Segmentation

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Multimodal Causality-Driven Representation Learning for Generalizable Medical Image Segmentation

快速理解：Vision-Language Models (VLMs), such as CLIP, have demonstrated remarkable zero-shot capabilities in various computer vision tasks. However, their application to medical imaging remains challenging due to the high variability and complexity of medical data.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.402 Multimodal Distribution Matching for Vision-Language Dataset Distillation

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Multimodal Distribution Matching for Vision-Language Dataset Distillation

快速理解：Dataset distillation compresses large training sets into compact synthetic datasets while preserving downstream performance. As modern systems increasingly operate on paired vision-language inputs, multimodal distillation must preserve representation quality and cross-modal alignment under tight compute and memory budgets, yet prior methods often require heavy computes and overlook their correlations.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.403 Multimodal Protein Language Models for Enzyme Kinetic Parameters: From Substrate Recognition to Conformational Adaptation

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展

- [来源](#)

论文: Multimodal Protein Language Models for Enzyme Kinetic Parameters: From Substrate Recognition to Conformational Adaptation

快速理解: Predicting enzyme kinetic parameters quantifies how efficiently an enzyme catalyzes a specific substrate under defined biochemical conditions. Canonical parameters such as the turnover number (k_{cat}), Michaelis constant (K_m), and inhibition constant (K_i) depend jointly on the enzyme sequence, the substrate chemistry, and the conformational adaptation of the active site during binding.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.404 Multimodal Semantic Bias Mitigation for Diverse Text-To-3D Generation

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Multimodal Semantic Bias Mitigation for Diverse Text-To-3D Generation

快速理解: The latest progress in text-to-3D generative models makes it possible to generate high-quality 3D content. Recent text-to-3D large model have achieved remarkable breakthroughs in multi-view consistency.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.405 NERFIFY: A Multi-Agent Framework for Turning NeRF Papers into Code

- 作者: 作者信息未提供
- 年份: 2026; 主题: Agent RL; 优先级: 扩展
- [来源](#)

论文: NERFIFY: A Multi-Agent Framework for Turning NeRF Papers into Code

快速理解: The proliferation of neural radiance field (NeRF) research requires significant efforts to reimplement papers before building upon them. We introduce NERFIFY, a multiagent framework that reliably converts NeRF research papers into trainable Nerfstudio plugins, in contrast to generic paper-to-code methods and frontier models like GPT-5 that usually fail to produce runnable code.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.2.406 NIL: No-data Imitation Learning

- 作者: 作者信息未提供
- 年份: 2026; 主题: Agent RL; 优先级: 扩展
- [来源](#)

论文: NIL: No-data Imitation Learning

快速理解: Acquiring physically plausible motor skills across diverse and unconventional embodiments,

including humanoids and quadrupeds, is essential for advancing character simulation and robotics. Traditional methods, such as reinforcement learning (RL), require extensive reward function engineering.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.2.407 Nano-EmoX: Unifying Multimodal Emotional Intelligence from Perception to Empathy

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Nano-EmoX: Unifying Multimodal Emotional Intelligence from Perception to Empathy

快速理解：The development of affective multimodal language models (MLMs) has long been constrained by a gap between lowlevel perception and high-level interaction, leading to fragmented affective capabilities and limited generalization. To bridge this gap, we propose a cognitively inspired threelevel hierarchy that organizes affective tasks according to their cognitive depth—perception, understanding, and interaction—and provides a unified conceptual foundation for advancing affective modeling.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.408 Neighbor GRPO: Contrastive ODE Policy Optimization Aligns Flow Models

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL；优先级：扩展
- [来源](#)

论文：Neighbor GRPO: Contrastive ODE Policy Optimization Aligns Flow Models

快速理解：Group Relative Policy Optimization (GRPO) has shown promise in aligning image and video generative models with human preferences. However, applying it to modern flow matching models is challenging because of its deterministic sampling paradigm.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.2.409 Neuro-Cognitive Reward Modeling for Human-Centered Autonomous Vehicle Control

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL；优先级：扩展
- [来源](#)

论文：Neuro-Cognitive Reward Modeling for Human-Centered Autonomous Vehicle Control

快速理解：Recent advancements in computer vision have accelerated the development of autonomous driving. Despite these advancements, training machines to drive in a way that aligns with human expectations remains a significant challenge.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.2.410 NimbusGS: Unified 3D Scene Reconstruction under Hybrid Weather

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：NimbusGS: Unified 3D Scene Reconstruction under Hybrid Weather

快速理解：We present NimbusGS, a unified framework for reconstructing high-quality 3D scenes from degraded multi-view inputs captured under diverse and mixed adverse weather conditions. Unlike existing methods that target specific weather types, NimbusGS addresses the broader challenge of generalization by modeling the dual nature of weather: a continuous, view-consistent medium that attenuates light, and dynamic, view-dependent particles that cause scattering and occlusion.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.411 Nonparametric Deep Fine-grained Clustering with Low-Rank Guided Vision-Language Model

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Nonparametric Deep Fine-grained Clustering with Low-Rank Guided Vision-Language Model

快速理解：The scarcity of labeled fine-grained data presents a significant challenge for deep clustering. Vision-Language Models (VLMs) on existing coarse-grained datasets (characterized by high inter-class and low intra-class variance) struggle to capture the subtle distinctions essential for finegrained categorization, leading to suboptimal clustering performance.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.412 ORIC: Benchmarking Object Recognition under Contextual Incongruity in Large Vision-Language Models

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：ORIC: Benchmarking Object Recognition under Contextual Incongruity in Large Vision-Language Models

快速理解：Large Vision-Language Models (LVLMs) excel at captioning, visual question answering, and robotics by combining vision and language, yet they often miss obvious objects or hallucinate nonexistent ones in atypical scenes. We examine these failures through the lens of uncertainty, focusing on contextual incongruity, where objects appear unexpectedly or fail to appear in expected contexts, and show that such cases increase recognition difficulty for state-of-the-art LVLMs.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.413 ORV: 4D Occupancy-centric Robot Video Generation

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：ORV: 4D Occupancy-centric Robot Video Generation

快速理解：Recent embodied intelligence suffers from data scarcity, while conventional simulators lack visual realism. Controllable video generation is emerging as a promising data engine, yet current action-conditioned methods still fall short: generated videos are limited in fidelity and temporal consistency, poorly aligned with controls, and often constrained to singleview settings.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.414 OctoMed: Data Recipes for State-of-the-Art Multimodal Medical Reasoning

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：OctoMed: Data Recipes for State-of-the-Art Multimodal Medical Reasoning

快速理解：High-quality and carefully curated data is a cornerstone of training medical large language models, as it directly impacts both generalization and robustness to unseen clinical tasks. We investigate strategies for training and data curation to develop a robust multimodal reasoning model in the medical domain.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.415 OctoT2I: A Self-Evolving Agentic Text-to-Image Router

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL；优先级：扩展
- [来源](#)

论文：OctoT2I: A Self-Evolving Agentic Text-to-Image Router

快速理解：The explosive growth of Text-to-Image (T2I) models, from large-scale versions to lightweight, real-time ones, now faces diminishing marginal returns from single-model scaling. Agentic T2I methods emerged to alleviate this bottleneck by using multiple models.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.2.416 Octopus: History-Free Gradient Orthogonalization for Continual Learning in Multimodal Large Language Models

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文: Octopus: History-Free Gradient Orthogonalization for Continual Learning in Multimodal Large Language Models

快速理解: Continual learning in multimodal large language models (MLLMs) aims to sequentially acquire knowledge while mitigating catastrophic forgetting, yet existing methods face inherent limitations: architecture-based approaches incur additional computational overhead and often generalize poorly to new tasks, rehearsal-based methods rely on storing historical data, raising privacy and storage concerns, and conventional regularization-based strategies alone are insufficient to fully prevent parameter interference. We propose Octopus, a two-stage continual learning framework based on History-Free Gradient Orthogonalization (HiFGO), which enforces gradient-level orthogonality without historical task data.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.417 OlmoEarth: Stable Latent Image Modeling for Multimodal Earth Observation

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文: OlmoEarth: Stable Latent Image Modeling for Multimodal Earth Observation

快速理解: Earth observation data presents a unique challenge: it is spatial like images, sequential like video or text, and highly multimodal. We present OlmoEarth: a spatiotemporal, multimodal foundation model that employs a novel self-supervised learning formulation, masking strategy, and loss, which are all designed for Earth observation.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.418 Omni-Attack: Adversarial Attacks on Open-Ended VQA in Black-Box Multimodal LLMs

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文: Omni-Attack: Adversarial Attacks on Open-Ended VQA in Black-Box Multimodal LLMs

快速理解: Multimodal large language models (MLLMs) have achieved remarkable success across diverse applications, from autonomous driving to document understanding. As these models are deployed in safety-critical contexts, understanding their adversarial robustness becomes crucial.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.419 Omni-Fake: Benchmarking Unified Multimodal Social Media Deepfake Detection

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Omni-Fake: Benchmarking Unified Multimodal Social Media Deepfake Detection

快速理解：Multimodal deepfakes are proliferating on social media and threaten authenticity, information integrity, and digital forensics. Existing benchmarks are constrained by their single-modality scope, simplified manipulations, or unrealistic distributions, which limit their ability to assess real-world robustness.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.420 OmniBrainBench: A Comprehensive Multimodal Benchmark for Brain Imaging Analysis Across Multi-stage Clinical Tasks

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：OmniBrainBench: A Comprehensive Multimodal Benchmark for Brain Imaging Analysis Across Multi-stage Clinical Tasks

快速理解：Brain imaging analysis is crucial for diagnosing and treating brain disorders, and multimodal large language models (MLLMs) are increasingly supporting it. However, current brain imaging visual question-answering (VQA) benchmarks either cover a limited number of imaging modalities or are restricted to coarse-grained pathological descriptions, hindering a comprehensive assessment of MLLMs across the full clinical continuum.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.421 OmniGen2: Towards Instruction-Aligned Multimodal Generation

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：OmniGen2: Towards Instruction-Aligned Multimodal Generation

快速理解：Multimodal generative models can process instructions in various modalities and demonstrate outstanding performance across a wide range of image generation tasks. However, their robustness in complex real-world scenarios remains limited due to insufficient generalized instruction alignment.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.422 OneOcc: Semantic Occupancy Prediction for Legged Robots with a Single Panoramic Camera

- 作者：作者信息未提供
- 年份：2026；主题：Embodied/WBC；优先级：扩展
- [来源](#)

论文：OneOcc: Semantic Occupancy Prediction for Legged Robots with a Single Panoramic Camera

快速理解：Robust 3D semantic occupancy is crucial for legged/humanoid robots, yet most semantic scene completion (SSC) systems target wheeled platforms with forward-facing sensors. We present OneOcc, a vision-only panoramic SSC framework designed for gait-introduced body jitter and 360° continuity.

关注点：具身任务中的感知、动作生成、全身协调或运动控制。

证据层级：本地 PDF 前页。

1.3.2.423 Open-world Hand-Object Interaction Video Generation Based on Structure and Contact-aware Representation

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Open-world Hand-Object Interaction Video Generation Based on Structure and Contact-aware Representation

快速理解：Generating realistic hand-object interactions (HOI) videos is a significant challenge due to the difficulty of modeling physical constraints (e.g., contact and occlusion between hands and manipulated objects). Current methods utilize HOI representation as an auxiliary generative objective to guide video synthesis.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.424 OpenDance: Multimodal Controllable 3D Dance Generation with Large-scale Internet Data

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：OpenDance: Multimodal Controllable 3D Dance Generation with Large-scale Internet Data

快速理解：Music-driven 3D dance generation offers significant creative potential, yet practical applications demand versatile and multimodal control. Given the highly dynamic and complex human motion covering various styles and genres, dance generation requires satisfying diverse conditions beyond just music (e.g., spatial trajectories, keyframe gestures, or style descriptions).

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.425 OpenMMReasoner: Pushing the Frontiers in Multimodal Reasoning with an Open and General Recipe

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：OpenMMReasoner: Pushing the Frontiers in Multimodal Reasoning with an Open and General Recipe

快速理解：Recent advancements in large reasoning models have fueled growing interest in extending such capabilities to multimodal domains. However, despite notable progress in visual reasoning, the lack of transparent and reproducible data curation and training strategies remains a major barrier to scalable research.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.426 OpenMarcie: Dataset for Multimodal Action Recognition in Industrial Environments

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：OpenMarcie: Dataset for Multimodal Action Recognition in Industrial Environments

快速理解：Smart factories use advanced technologies to optimize production and increase efficiency. To this end, the recognition of worker activity allows for accurate quantification of performance metrics, improving efficiency holistically while contributing to worker safety.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.427 OpenVision 2: A Family of Generative Pretrained Visual Encoders for Multimodal Learning

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：OpenVision 2: A Family of Generative Pretrained Visual Encoders for Multimodal Learning

快速理解：This paper provides a simplification on OpenVision' s architecture and loss design for enhancing its training efficiency. Following the prior vision-language pretraining works CapPa and AIMv2, as well as modern multimodal designs like LLaVA, our changes are straightforward: we remove the text encoder—and therefore the contrastive loss—retaining only the captioning loss as a purely generative training signal.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.428 OpenVoxel: Training-Free Grouping and Captioning Voxels for Open-Vocabulary 3D Scene Understanding

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：OpenVoxel: Training-Free Grouping and Captioning Voxels for Open-Vocabulary 3D Scene Understanding

快速理解：We propose OpenVoxel, a training-free algorithm for grouping and captioning sparse voxels for the openvocabulary 3D scene understanding tasks. Given the sparse voxel rasterization (SVR) model obtained from multi-view images of a 3D scene, our OpenVoxel is able to produce meaningful groups that describe different objects in the scene.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.429 OralGPT-Omni: A Versatile Dental Multimodal Large Language Model

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：OralGPT-Omni: A Versatile Dental Multimodal Large Language Model

快速理解：Multimodal Large Language Models (MLLMs) have exhibited immense potential across numerous medical specialties; yet, dentistry remains underexplored, in part due to limited domain-specific data, scarce dental expert annotations, insufficient modality-specific modeling, and challenges in reliability. In this paper, we present OralGPT- Omni, the first dental-specialized MLLM designed for comprehensive and trustworthy analysis across diverse dental imaging modalities and clinical tasks.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.430 Otil: Accelerating Diffusion Model Inference via Communication-Efficient Multi-GPU Parallelism

- 作者：作者信息未提供
- 年份：2026；主题：AI Infra；优先级：扩展
- [来源](#)

论文：Otil: Accelerating Diffusion Model Inference via Communication-Efficient Multi-GPU Parallelism

快速理解：Diffusion models (DMs) have recently achieved remarkable success across diverse modalities, including high-fidelity image and video synthesis. However, their inherent step sequential denoising process introduces substantial cumulative latency, which significantly degrades user experience.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.2.431 Ov3R: Open-Vocabulary Semantic 3D Reconstruction from RGB Videos

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Ov3R: Open-Vocabulary Semantic 3D Reconstruction from RGB Videos

快速理解：We present Ov3R, a novel framework for open-vocabulary semantic 3D reconstruction from RGB video streams, designed to advance Spatial AI. The system features two key components: CLIP3R, a CLIP-informed 3D reconstruction module that predicts dense point maps from overlapping clips alongside object-level semantics; and 2D-3D OVS, a 2D-3D open-vocabulary semantic module that lifts 2D features into 3D by learning fused descriptors integrating spatial, geometric, and semantic cues.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.432 PACT: Phase-Like Transition Constraints in Adapter-Based Continual Learning of Vision-Language Models

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：PACT: Phase-Like Transition Constraints in Adapter-Based Continual Learning of Vision-Language Models

快速理解：Continual Learning (CL) enables Vision-Language Models (VLMs) to acquire new capabilities while retaining prior knowledge, for example, by employing task-specific adapters. Existing CL approaches typically optimize these adapters to convergence, often with (near-)orthogonality constraints to reduce interference; however, isolating adapters in orthogonal subspaces can suppress cross-task transfer and sharing.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.433 PALM: Progress-Aware Policy Learning via Affordance Reasoning for Long-Horizon Robotic Manipulation

- 作者：作者信息未提供
- 年份：2026；主题：Embodied/WBC；优先级：扩展
- [来源](#)

论文：PALM: Progress-Aware Policy Learning via Affordance Reasoning for Long-Horizon Robotic Manipulation

快速理解：Recent advancements in vision-language-action (VLA) models have shown promise in robotic manipulation, yet they continue to struggle with long-horizon, multi-step tasks. Existing methods lack internal reasoning mechanisms that can identify task-relevant interaction cues or track progress within a subtask, leading to critical execution errors such as repeated actions, missed steps, and premature termination.

关注点：具身任务中的感知、动作生成、全身协调或运动控制。

证据层级：本地 PDF 前页。

1.3.2.434 PAS: Prelim Attention Score for Detecting Object Hallucinations in Large Vision-Language Models

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：PAS: Prelim Attention Score for Detecting Object Hallucinations in Large Vision-Language Models

快速理解：Large vision-language models (LVLMs) are powerful, yet they remain unreliable due to object hallucinations. In this work, we show that in many hallucinatory predictions the LVLm effectively ignores the image and instead relies on previously generated output (“prelim”) tokens to infer new objects.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.435 PDCR: Perception-Decomposed Confidence Reward for Vision-Language Reasoning

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：PDCR: Perception-Decomposed Confidence Reward for Vision-Language Reasoning

快速理解：Reinforcement Learning with Verifiable Rewards (RLVR) traditionally relies on a sparse, outcome-based signal. Recent work shows that providing a fine-grained, model-intrinsic signal—rewarding the confidence growth in the ground-truth answer—effectively improves language reasoning training by providing step-level guidance without costly external models.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.436 PE3R: Perception-Efficient 3D Reconstruction

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：PE3R: Perception-Efficient 3D Reconstruction

快速理解：Recent advances in 2D-to-3D perception have enabled the recovery of 3D scene semantics from unposed images. However, prevailing methods often suffer from limited generalization, reliance on per-scene optimization, and semantic inconsistencies across viewpoints.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.437 PQDT: Pseudo-Query Dual Transformer for Robust Point Cloud Restoration

- 作者：作者信息未提供
- 年份：2026；主题：Systems；优先级：扩展
- [来源](#)

论文：PQDT: Pseudo-Query Dual Transformer for Robust Point Cloud Restoration

快速理解：Point clouds are a fundamental 3D representation in computer vision, enabling a wide range of perception tasks. However, real-world point clouds often suffer from degradations such as incompleteness, noise, outliers, and irregular density, caused by sensor limitations or occlusions.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.2.438 PTC-Depth: Pose-Refined Monocular Depth Estimation with Temporal Consistency

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：PTC-Depth: Pose-Refined Monocular Depth Estimation with Temporal Consistency

快速理解：Monocular depth estimation (MDE) has been widely adopted in the perception systems of autonomous vehicles and mobile robots. However, existing approaches often struggle to maintain temporal consistency in depth estimation across consecutive frames.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.439 PanDA: Unsupervised Domain Adaptation for Multimodal 3D Panoptic Segmentation in Autonomous Driving

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：PanDA: Unsupervised Domain Adaptation for Multimodal 3D Panoptic Segmentation in Autonomous Driving

快速理解：This paper presents the first study on Unsupervised Domain Adaptation (UDA) for multimodal 3D panoptic segmentation (mm-3DPS), aiming to improve generalization under domain shifts commonly encountered in real-world autonomous driving. A straightforward solution is to employ a pseudo-labeling strategy, which is widely used in UDA to generate supervision for unlabeled target data, combined with an mm-3DPS backbone.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.440 Pano3DComposer: Feed-Forward Compositional 3D Scene Generation from Single Panoramic Image

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Pano3DComposer: Feed-Forward Compositional 3D Scene Generation from Single Panoramic Image

快速理解：Current compositional image-to-3D scene generation approaches construct 3D scenes by time-consuming iterative layout optimization or inflexible joint object-layout generation. Moreover, most methods rely on limited field-of-view perspective images, hindering the creation of complete 360° environments.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.441 PanoEnv: Exploring 3D Spatial Intelligence in Panoramic Environments with Reinforcement Learning

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL；优先级：扩展
- [来源](#)

论文：PanoEnv: Exploring 3D Spatial Intelligence in Panoramic Environments with Reinforcement Learning

快速理解：360° panoramic images are increasingly used in VR, autonomous driving, and robotics for holistic scene understanding. However, current Vision-Language Models (VLMs) struggle with 3D spatial reasoning on Equirectangular Projection (ERP) images due to geometric distortion and limited 3D supervision.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.2.442 Paper2Figure: A Multi-Agent Collaborative System for Figure Generation Towards Academic Research Paper

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL；优先级：扩展
- [来源](#)

论文：Paper2Figure: A Multi-Agent Collaborative System for Figure Generation Towards Academic Research Paper

快速理解：Automatically generating clear and accurate figures for research papers remains challenging, as it requires semantic understanding, precise structure, and visual aesthetics. Existing approaches struggle to balance fidelity and quality: large language model (LLM) code-based methods (e.g., SVG, Mermaid) are structured but inflexible, while image generation models (e.g., GPT-Image-1, Nano Banana) produce hard-to-edit and often inaccurate figures.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.2.443 ParaUni: Enhance Generation in Unified Multimodal Model with Reinforcement-driven Hierarchical Parallel Information Interaction

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：ParaUni: Enhance Generation in Unified Multimodal Model with Reinforcement-driven Hierarchical Parallel Information Interaction

快速理解：Unified multimodal models significantly improve visual generation by combining vision-language models (VLMs) with diffusion models. However, existing methods struggle to fully balance sufficient interaction and flexible implementation due to vast representation difference.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.444 PersonaVLM: Long-Term Personalized Multimodal LLMs

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：PersonaVLM: Long-Term Personalized Multimodal LLMs

快速理解：Multimodal Large Language Models (MLLMs) serve as daily assistants for millions. However, their ability to generate responses aligned with individual preferences remains limited.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.445 PhyCritic: Multimodal Critic Models for Physical AI

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：PhyCritic: Multimodal Critic Models for Physical AI

快速理解：With the rapid development of large multimodal models, reliable judge and critic models have become essential for open-ended evaluation and preference alignment, providing pairwise preferences, numerical scores, and explanatory justifications for assessing model-generated responses. However, existing critics are primarily trained in general visual domains such as captioning or image question answering, leaving physical AI tasks involving perception, causal reasoning, and planning largely underexplored.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.446 Physical Object Understanding with a Physically Controllable World Model

- 作者：作者信息未提供
- 年份：2026；主题：World Model；优先级：扩展

- [来源](#)

论文: Physical Object Understanding with a Physically Controllable World Model

快速理解: A central challenge in visual intelligence is learning the physical structure of scenes from raw videos: how regions form objects and the laws that govern their interactions. Solving these tasks requires world models capable of inferring distributional states of the world from partial observations –capabilities that current architectures do not provide.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策。

证据层级: 本地 PDF 前页。

1.3.2.447 Physical Simulator In-the-Loop Video Generation

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Physical Simulator In-the-Loop Video Generation

快速理解: Recent advances in diffusion-based video generation have achieved remarkable visual realism but still struggle to obey basic physical laws such as gravity, inertia, and collision. Generated objects often move inconsistently across frames, exhibit implausible dynamics, or violate physical constraints, limiting the realism and reliability of AI-generated videos.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.448 Planning in 8 Tokens: A Compact Discrete Tokenizer for Latent World Model

- 作者: 作者信息未提供
- 年份: 2026; 主题: World Model; 优先级: 扩展
- [来源](#)

论文: Planning in 8 Tokens: A Compact Discrete Tokenizer for Latent World Model

快速理解: World models provide a powerful framework for simulating environment dynamics conditioned on actions or instructions, enabling downstream tasks such as action planning or policy learning. Recent approaches leverage world models as learned simulators, but its application to decision-time planning remains computationally prohibitive for real-time control.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策。

证据层级: 本地 PDF 前页。

1.3.2.449 PoInit-of-View: Poisoning Initialization of Views Transfers Across Multiple 3D Reconstruction Systems

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: PoInit-of-View: Poisoning Initialization of Views Transfers Across Multiple 3D Reconstruction

Systems

快速理解: Poisoning input views of 3D reconstruction systems has been recently studied. However, we identify that existing studies simply backpropagate adversarial gradients through the 3D reconstruction pipeline as a whole, without uncovering the new vulnerability rooted in specific modules of the 3D reconstruction pipeline.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.450 PointCSP: Cross-Sample Semantic Propagation and Stability Preservation in Self-Supervised Point Cloud Learning

- 作者: 作者信息未提供
- 年份: 2026; 主题: Systems; 优先级: 扩展
- [来源](#)

论文: PointCSP: Cross-Sample Semantic Propagation and Stability Preservation in Self-Supervised Point Cloud Learning

快速理解: Scene-level point cloud self-supervised learning (PC-SSL) has demonstrated potential in enhancing the generalization capability of 3D vision models. Despite the advances in the field through existing methods, the sample-independent modeling paradigm still poses significant limitations in terms of maintaining consistent semantic representations across scenes.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.2.451 PointThinker: Point-Incentivized Parallel Thinking for Multimodal Large Language Model

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: PointThinker: Point-Incentivized Parallel Thinking for Multimodal Large Language Model

快速理解: This paper explores parallel thinking for Multi-modal Large Language Models (MLLMs), aiming to improve Chain-of-Thought (CoT) through multiple diverse reasoning paths. We guide the model to list multiple visual key points and develop an independent reasoning path for each.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.452 PoseAnything: General Pose-guided Video Generation with Part-aware Temporal Coherence

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: PoseAnything: General Pose-guided Video Generation with Part-aware Temporal Coherence

快速理解: Pose-guided video generation refers to controlling the motion of subjects in generated video through a sequence of poses. It enables precise control over subject motion and has important applications in animation.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.453 Predict Before You Explore: Predictive Planning with Specialized Memory for Embodied Question Answering

- 作者: 作者信息未提供
- 年份: 2026; 主题: Embodied/WBC; 优先级: 扩展
- [来源](#)

论文: Predict Before You Explore: Predictive Planning with Specialized Memory for Embodied Question Answering

快速理解: Embodied Question Answering (EQA) requires agents to navigate 3D environments, accumulate visual evidence, and reason over partial observations to answer questions. However, current agents struggle to maintain coherent, longhorizon behavior: planning remains reactive, causing inconsistent actions, while monolithic memories entangle all observations, hindering retrieval of the sparse but crucial evidence.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制。

证据层级: 本地 PDF 前页。

1.3.2.454 ProM3E: Probabilistic Masked MultiModal Embedding Model for Ecology

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: ProM3E: Probabilistic Masked MultiModal Embedding Model for Ecology

快速理解: We introduce ProM3E, a probabilistic masked multimodal embedding model for any-to-any generation of multimodal representations for ecology. ProM3E is based on masked modality reconstruction in the embedding space, learning to infer missing modalities given a few context modalities.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.455 Probabilistic Concept Graph Reasoning for Multimodal Misinformation Detection

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Probabilistic Concept Graph Reasoning for Multimodal Misinformation Detection

快速理解: Multimodal misinformation poses an escalating challenge that often evades traditional detectors, which are opaque black boxes and fragile against new manipulation tactics. We present

Probabilistic Concept Graph Reasoning (PCGR), an interpretable and evolvable framework that reframes multimodal misinformation detection (MMD) as structured and concept-based reasoning.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.456 ProgTrack: A Multi-Object Tracking Algorithm with Progressive Matching Strategy

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：ProgTrack: A Multi-Object Tracking Algorithm with Progressive Matching Strategy

快速理解：Multi-object tracking (MOT) based on unmanned aerial vehicle (UAV) aims to identify and continuously track the positions of multiple ground targets during UAV flight. Current mainstream methods utilize appearance matching and motion matching to match targets in consecutive frames.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.457 ProjFlow: Projection Sampling with Flow Matching for Zero-Shot Exact Spatial Motion Control

- 作者：作者信息未提供
- 年份：2026；主题：Embodied/WBC；优先级：扩展
- [来源](#)

论文：ProjFlow: Projection Sampling with Flow Matching for Zero-Shot Exact Spatial Motion Control

快速理解：Generating human motion with precise spatial control is a challenging problem. Existing approaches often require task-specific training or slow optimization, and enforcing hard constraints frequently disrupts motion naturalness.

关注点：具身任务中的感知、动作生成、全身协调或运动控制。

证据层级：本地 PDF 前页。

1.3.2.458 PromptDepth: Efficient and Promptable Geometric 3D Vision Model for Embodied Intelligence

- 作者：作者信息未提供
- 年份：2026；主题：Embodied/WBC；优先级：扩展
- [来源](#)

论文：PromptDepth: Efficient and Promptable Geometric 3D Vision Model for Embodied Intelligence

快速理解：Vision models for embodied intelligence require efficient 3D comprehension and interaction with objects within the scene. Existing 3D reconstruction models either overlook instance-level perception or rely on time-consuming offline reasoning, showing a less adaptability in real-time embodied scenario.

关注点：具身任务中的感知、动作生成、全身协调或运动控制。

证据层级：本地 PDF 前页。

1.3.2.459 PromptMoE: A Segmentation Refinement Framework Leveraging Mixture of Experts for Improved Prompting

- 作者：作者信息未提供
- 年份：2026；主题：AI Infra；优先级：扩展
- [来源](#)

论文：PromptMoE: A Segmentation Refinement Framework Leveraging Mixture of Experts for Improved Prompting

快速理解: High-quality segmentations are critical in vision tasks where boundary accuracy is important (e.g., medical diagnostics, quality control, etc.). Recently, promptable vision models have emerged as effective backbones for segmentation refinement frameworks.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.2.460 Proof-of-Perception: Certified Tool-Using Multimodal Reasoning with Compositional Conformal Guarantees

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Proof-of-Perception: Certified Tool-Using Multimodal Reasoning with Compositional Conformal Guarantees

快速理解: We present Proof-of-Perception (PoP), a tool-using framework that casts multimodal reasoning as an executable graph with explicit reliability guarantees. Each perception or logic node outputs a conformal set $\Gamma(t)(x)$, yielding calibrated, stepwise uncertainty; a lightweight controller uses these certificates to allocate compute under a budget—expanding with extra tool calls only when needed and stopping early otherwise.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.461 Proxy-Tuning: Tailoring Multimodal Autoregressive Models for Subject-Driven Image Generation

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Proxy-Tuning: Tailoring Multimodal Autoregressive Models for Subject-Driven Image Generation

快速理解: Multimodal autoregressive (AR) models, based on nexttoken prediction and transformer architecture, have demonstrated remarkable capabilities in various multimodal tasks including text-to-image (T2I) generation. Despite their strong performance in general T2I tasks, our research reveals that these models initially struggle with subject-driven image generation compared to dom-

inant diffusion models.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.462 Proxy3D: Efficient 3D Representations for Vision-Language Models via Semantic Clustering and Alignment

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Proxy3D: Efficient 3D Representations for Vision-Language Models via Semantic Clustering and Alignment

快速理解：Spatial intelligence in vision-language models (VLMs) attracts research interest with the practical demand to reason in the 3D world. Despite promising results, most existing methods follow the conventional 2D pipeline in VLMs and use pixel-aligned representations for the vision modality.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.463 Prune2Drive: A Plug-and-Play Framework for Accelerating Vision-Language Models in Autonomous Driving

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Prune2Drive: A Plug-and-Play Framework for Accelerating Vision-Language Models in Autonomous Driving

快速理解：Vision-Language Models (VLMs) have emerged as a promising paradigm in autonomous driving (AD), providing a unified framework for perception and decisionmaking. However, their real-world deployment is hindered by significant computational overhead when processing high-resolution, multi-view images.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.464 QD-PCQA: Quality-Aware Domain Adaptation for Point Cloud Quality Assessment

- 作者：作者信息未提供
- 年份：2026；主题：Systems；优先级：扩展
- [来源](#)

论文：QD-PCQA: Quality-Aware Domain Adaptation for Point Cloud Quality Assessment

快速理解：No-Reference Point Cloud Quality Assessment (NR-PCQA) still struggles with generalization, primarily due to the scarcity of annotated point cloud datasets. Since the Human Visual System (HVS) drives perceptual quality assessment independently of media types, prior knowledge on quality learned from images can be repurposed for point clouds.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.2.465 QUANTIPHY: A Quantitative Benchmark Evaluating Physical Reasoning Abilities of Vision-Language Models

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：QUANTIPHY: A Quantitative Benchmark Evaluating Physical Reasoning Abilities of Vision-Language Models

快速理解：Understanding the physical world is essential for generalist AI agents. However, it remains unclear whether state-of-the-art vision perception models (e.g., large VLMs) can reason physical properties quantitatively.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.466 Quantized Residuals to Continuous Prompts for Few-Shot Class Incremental Learning in Vision-Language Models

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Quantized Residuals to Continuous Prompts for Few-Shot Class Incremental Learning in Vision-Language Models

快速理解：Few-shot Class-Incremental Learning (FSCIL) requires learning new classes from very limited data while preventing catastrophic forgetting. Existing methods rely mainly on visual features and are prone to overfitting, while recent vision-language models (VLMs) offer better transferability but suppress fine-grained information due to contrastive feature decorrelation.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.467 QueryMe: Query-Driven Open-Vocabulary 3D Object Affordances Grounding from Multimodal Evidence

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：QueryMe: Query-Driven Open-Vocabulary 3D Object Affordances Grounding from Multimodal Evidence

快速理解：Open-vocabulary 3D object affordance grounding aims to identify functional regions of objects given arbitrary semantic descriptions. However, existing methods often rely on fixed training categories and geometric priors, lacking geometric invariance and analogical reasoning capabilities.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.468 QuietPrune: Query-Guided Early Token Pruning for Vision-Language Models

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：QuietPrune: Query-Guided Early Token Pruning for Vision-Language Models

快速理解：Vision-language models (VLMs) demonstrate powerful capabilities in multimodal tasks. However, the large number of visual tokens imposes a significant computational cost.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.469 R3-PCQA: Ray-Reprojection-Reinforcement for No-Reference 3D Point Cloud Quality Assessment

- 作者：作者信息未提供
- 年份：2026；主题：Systems；优先级：扩展
- [来源](#)

论文：R3-PCQA: Ray-Reprojection-Reinforcement for No-Reference 3D Point Cloud Quality Assessment

快速理解：Prevailing no-reference 3D point cloud quality assessment methods predominantly treat 2D projections and 3D point clouds as independent modalities and rely on simplistic feature fusion, thereby neglecting fundamental mechanisms underlying human 3D perception. To address this limitation, we introduce R3-PCQA (Ray-Reprojection- Reinforcement 3D Point Cloud Quality Assessor), a novel and principled framework that explicitly encodes perceptual priors into the assessment pipeline: A geometricaware ray-based reprojection pipeline simulates viewpointdependent observation of 3D structure.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.2.470 R4-CGQA: Retrieval-based Vision Language Models for Computer Graphics Image Quality Assessment

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：R4-CGQA: Retrieval-based Vision Language Models for Computer Graphics Image Quality Assessment

快速理解：Immersive Computer Graphics (CGs) rendering has become ubiquitous in modern daily life. However, comprehensively evaluating CG quality remains challenging for two reasons: (1) existing CG datasets lack systematic descriptions of rendering quality; and (2) existing CG quality assessment methods cannot provide reasonable text-based explanations.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.471 R4: Retrieval-Augmented Reasoning for Vision-Language Models in 4D Spatio-Temporal Space

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：R4: Retrieval-Augmented Reasoning for Vision-Language Models in 4D Spatio-Temporal Space

快速理解：Humans perceive and reason about their surroundings in four dimensions—three spatial and one temporal axis—by building persistent, structured internal representations that encode semantic meaning, spatial layout, and temporal dynamics. These multimodal memories enable them to recall past events, infer unobserved states, and integrate new information into context-dependent reasoning.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.472 RAAS: LLM Agentic System Architecture Search with GRPO

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL；优先级：扩展
- [来源](#)

论文：RAAS: LLM Agentic System Architecture Search with GRPO

快速理解：Large Language Model (LLM) agentic systems solve complex tasks through coordinated workflows, but designing them remains labor-intensive. The Agentic Supernet paradigm automates this by optimizing a probabilistic architecture space, yet suffers from critical evaluation instabilities: absolute performance scores entangle architectural merit with query difficulty, while single-execution protocols capture execution randomness rather than true capability.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.2.473 RAMEN: Resolution-Adjustable Multimodal Encoder for Earth Observation

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：RAMEN: Resolution-Adjustable Multimodal Encoder for Earth Observation

快速理解：Earth observation (EO) data spans a wide range of spatial, spectral, and temporal resolutions, from high-resolution optical imagery to low resolution multispectral products or radar time series. While recent foundation models have improved multimodal integration for learning meaningful representations, they often expect fixed input resolutions or are based on sensor-specific encoders limiting generalization across heterogeneous EO modalities.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.474 RAYNOVA: Scale-Temporal Autoregressive World Modeling in Ray Space

- 作者：作者信息未提供
- 年份：2026；主题：World Model；优先级：扩展
- [来源](#)

论文：RAYNOVA: Scale-Temporal Autoregressive World Modeling in Ray Space

快速理解：World foundation models aim to simulate the evolution of the real world with physically plausible behavior. Unlike prior methods that handle spatial and temporal correlations separately, we propose RAYNOVA, a geometryagonistic multiview world model for driving scenarios that employs a dual-causal autoregressive framework.

关注点：环境动力学、视频预测、潜变量状态或模型式决策。

证据层级：本地 PDF 前页。

1.3.2.475 RE-VLM: Event-Augmented Vision-Language Model for Scene Understanding

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：RE-VLM: Event-Augmented Vision-Language Model for Scene Understanding

快速理解：Conventional vision-language models (VLMs) struggle to interpret scenes captured under adverse conditions (e.g., low light, high dynamic range, or fast motion) because standard RGB images degrade in such environments. Event cameras provide a complementary modality: they asynchronously record per-pixel brightness changes with high temporal resolution and wide dynamic range, preserving motion cues where frames fail.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.476 REVISOR: Beyond Textual Reflection, Towards Multimodal Introspective Reasoning in Long-Form Video Understanding

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：REVISOR: Beyond Textual Reflection, Towards Multimodal Introspective Reasoning in Long-Form Video Understanding

快速理解：Self-reflection mechanisms that rely on purely text-based rethinking processes perform well in most multimodal tasks. However, when directly applied to long-form video understanding scenarios, they exhibit clear limitations.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.477 RLFTSim: Realistic and Controllable Multi-Agent Traffic Simulation via Reinforcement Learning Fine-Tuning

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL；优先级：扩展
- [来源](#)

论文：RLFTSim: Realistic and Controllable Multi-Agent Traffic Simulation via Reinforcement Learning Fine-Tuning

快速理解：Supervised open-loop training has been widely adopted for training traffic simulation models; however, it fails to capture the inherently dynamic, multi-agent interactions common in complex driving scenarios. We introduce RLFT-Sim, a reinforcement-learning-based fine-tuning framework that enhances scenario realism by aligning simulator rollouts with real-world data distributions and provides a method for distilling goal-conditioned controllability in scenario generation.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.2.478 RMIR: A Benchmark Dataset for Reasoning-Intensive Multimodal Image Retrieval

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：RMIR: A Benchmark Dataset for Reasoning-Intensive Multimodal Image Retrieval

快速理解：Current multimodal image retrieval benchmarks focus on relatively simple queries where target images are either described directly or by simple composition with an input image. When retrieval requires complex reasoning to determine the target image, the task becomes significantly more challenging, yet standardized benchmarks for this setting do not exist.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.479 Radar-Guided Polynomial Fitting for Metric Depth Estimation

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Radar-Guided Polynomial Fitting for Metric Depth Estimation

快速理解：We propose POLAR, a novel radar-guided depth estimation method that introduces polynomial fitting to efficiently transform scaleless depth predictions from pretrained monocular depth estimation (MDE) models into metric depth maps. Unlike existing approaches that rely on complex architectures or expensive sensors, our method is grounded in a fundamental insight: although MDE models often infer reasonable local depth structure within each object or local region, they may misalign these regions relative to one another, making a linear scale and shift (affine) transformation

insufficient given three or more of these regions.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.480 Rascene: High-Fidelity 3D Scene Imaging with mmWave Communication Signals

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Rascene: High-Fidelity 3D Scene Imaging with mmWave Communication Signals

快速理解：Robust 3D environmental perception is critical for applications such as autonomous driving and robot navigation. However, optical sensors such as cameras and Li- DAR often fail under adverse conditions, including smoke, fog, and non-ideal lighting.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.481 Re-evaluating Continual VQA: Toward Fair and Robust Evaluation for Multimodal Continual Learning

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Re-evaluating Continual VQA: Toward Fair and Robust Evaluation for Multimodal Continual Learning

快速理解：Continual Visual Question Answering (Continual VQA) poses unique challenges for multimodal continual learning, requiring models to incrementally acquire new knowledge while preserving visual-semantic grounding across tasks. However, existing benchmarks hinder fair and robust evaluation of such capabilities, as they allow models to exploit dataset biases rather than demonstrate genuine continual reasoning.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.482 ReFact: Empowering Multimodal Web Agents with Visual and Context Focusing

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：ReFact: Empowering Multimodal Web Agents with Visual and Context Focusing

快速理解：Multimodal Web Search Agents demonstrate a practically valuable capability by fusing information from diverse modalities (e.g., text and vision), retrieved iteratively from the internet, to address complex user queries. However, the visual modality is prone to information overload, and the noise contained within it—such as irrelevant background details or complex structures—can

disrupt the model's attention, misdirecting its operational focus toward an erroneous path.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.483 ReMatch: Boosting Representation through Matching for Multimodal Retrieval

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：ReMatch: Boosting Representation through Matching for Multimodal Retrieval

快速理解：We present ReMatch, a framework that leverages the generative strength of MLLMs for multimodal retrieval. Previous approaches treated an MLLM as a simple encoder, ignoring its generative nature, and under-utilising its compositional reasoning and world knowledge.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.484 ReMoRa: Multimodal Large Language Model based on Refined Motion Representation for Long-Video Understanding

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：ReMoRa: Multimodal Large Language Model based on Refined Motion Representation for Long-Video Understanding

快速理解：While multimodal large language models (MLLMs) have shown remarkable success across a wide range of tasks, long-form video understanding remains a significant challenge. In this study, we focus on video understanding by MLLMs.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.485 ReMoT: Reinforcement Learning with Motion Contrast Triplets

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL；优先级：扩展
- [来源](#)

论文：ReMoT: Reinforcement Learning with Motion Contrast Triplets

快速理解：We present ReMoT, a unified training paradigm to systematically address the fundamental shortcomings of VLMs in spatio-temporal consistency—a critical failure point in navigation, robotics, and autonomous driving. ReMoT integrates two core components: (i) A rule-based automatic framework that generates ReMoT-16K, a large-scale (16.5K triplets) motion-contrast dataset derived from video meta-annotations, surpassing costly manual or modelbased generation. (ii) Group Relative Policy Optimization, which we empirically validate, yields optimal performance and data efficiency for learning this contrastive reasoning, far exceeding standard Supervised Fine-Tuning.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.2.486 ReScene4D: Temporally Consistent Semantic Instance Segmentation of Evolving Indoor 3D Scenes

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：ReScene4D: Temporally Consistent Semantic Instance Segmentation of Evolving Indoor 3D Scenes

快速理解：Indoor environments evolve as objects move, appear, or leave the scene. Capturing these dynamics requires maintaining temporally consistent instance identities across intermittently captured 3D scans, even when changes are unobserved.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.487 ReaGEN: Adaptive Generation of Structured Chains-of-Thought for Efficient Multimodal Reasoning

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：ReaGEN: Adaptive Generation of Structured Chains-of-Thought for Efficient Multimodal Reasoning

快速理解：Large Vision Language Models (LVLMs) exhibit strong perceptual and linguistic capabilities yet struggle with complex visual reasoning tasks that require structured, compositional, and adaptive inference. Existing approaches either rely on costly inference-time exploration—such as multipath or tree-based Chain-of-Thought (CoT) search—or on expensive post-training with large curated CoT datasets.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.488 Reading or Reasoning? Format Decoupled Reinforcement Learning for Document OCR

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL；优先级：扩展
- [来源](#)

论文：Reading or Reasoning? Format Decoupled Reinforcement Learning for Document OCR

快速理解：Reading text from images or scanned documents via OCR models has been a longstanding focus of researchers. Intuitively, text reading is perceived as a straightforward perceptual task, and existing work primarily focuses on constructing enriched data engineering to enhance SFT capabilities.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.2.489 Real-Time Multimodal Fingertip Contact Detection via Depth and Motion Fusion for Vision-Based Human-Computer Interaction

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Real-Time Multimodal Fingertip Contact Detection via Depth and Motion Fusion for Vision-Based Human-Computer Interaction

快速理解：Precise fingertip contact detection is a fundamental challenge for natural and immersive virtual reality (VR) interaction, yet existing vision-based methods suffer from insufficient accuracy, with typical depth errors (12–25 mm) far exceeding the <3 mm precision required to reliably distinguish hovering from true contact. While commercial motion-capture systems provide sub-millimeter accuracy, their prohibitive cost limits accessibility.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.490 Real2Sim2Real: RetinalDepth-64K for Depth Estimation in Posterior Segment Ophthalmic Surgery

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Real2Sim2Real: RetinalDepth-64K for Depth Estimation in Posterior Segment Ophthalmic Surgery

快速理解：Accurate depth estimation is crucial for 3D reconstruction and precise navigation in posterior segment ophthalmic surgery. However, acquiring annotated data remains challenging due to the impracticality of depth sensors under surgical microscopes.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.491 Reallocating Attention Across Layers to Reduce Multimodal Hallucination

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Reallocating Attention Across Layers to Reduce Multimodal Hallucination

快速理解：Multimodal large reasoning models (MLRMs) often suffer from hallucinations that stem not only from insufficient visual grounding but also from imbalanced allocation between perception and reasoning processes. Building upon recent interpretability findings suggesting a staged division of attention across layers, we analyze how this functional misalignment leads to two complementary failure modes: perceptual bias in shallow layers and reasoning drift in deeper layers.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.492 RecEdit-Drive: 3D Reconstruction-Guided Spatiotemporal Video Editing for Autonomous Driving Scenes

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文: RecEdit-Drive: 3D Reconstruction-Guided Spatiotemporal Video Editing for Autonomous Driving Scenes

快速理解: High-quality video editing and processing are crucial in domains such as filmmaking and autonomous driving, where accurate visual refinement and data preparation are essential. However, it is challenging to achieve precise control over dynamic objects while maintaining spatiotemporal consistency.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.493 Refining Few-Step Text-to-Multiview Diffusion via Reinforcement Learning

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL；优先级：扩展
- [来源](#)

论文: Refining Few-Step Text-to-Multiview Diffusion via Reinforcement Learning

快速理解: Text-to-multiview (T2MV) diffusion models have shown great promise in generating multiple views of a scene from a single text prompt. While few-step backbones enable realtime T2MV generation, they often compromise key aspects of generation quality, such as per-view fidelity and crossview consistency.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.2.494 Rejection Mixing: Fast Semantic Propagation of Mask Tokens for Efficient DLLM Inference

- 作者：作者信息未提供
- 年份：2026；主题：AI Infra；优先级：扩展
- [来源](#)

论文: Rejection Mixing: Fast Semantic Propagation of Mask Tokens for Efficient DLLM Inference

快速理解: Diffusion Large Language Models (DLLMs) promise fast non-autoregressive inference but suffer a severe qualityspeed trade-off in parallel decoding. This stems from the “combinatorial contradiction” phenomenon, where parallel tokens form semantically inconsistent combinations.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.2.495 Reliable Policy Transfer for Safety-Aware End-to-End Driving with Deep Reinforcement Learning

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL；优先级：扩展
- [来源](#)

论文: Reliable Policy Transfer for Safety-Aware End-to-End Driving with Deep Reinforcement Learning

快速理解: End-to-End (E2E) Reinforcement Learning (RL) for autonomous driving still struggles with safety and generalization under distribution shift, as perception-heavy encoders, sparse rewards, and ad hoc uncertainty handling yield brittle closed-loop behavior. This work introduces a unified Deep RL (DRL) framework built around a controller reliability interface where the same uncertainty signal informs relational attention, gates policy entropy, and regularizes transfer alignment.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.2.496 Reliev3R: Relieving Feed-forward 3D Reconstruction from Multi-View Geometric Annotations

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文: Reliev3R: Relieving Feed-forward 3D Reconstruction from Multi-View Geometric Annotations

快速理解: With recent advances, Feed-forward Reconstruction Models (FFRMs) have demonstrated great potential in reconstruction quality and adaptiveness to multiple downstream tasks. However, the excessive reliance on multi-view geometric annotations, e.g.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.497 Residual Decoding: Mitigating Hallucinations in Large Vision-Language Models via History-Aware Residual Guidance

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文: Residual Decoding: Mitigating Hallucinations in Large Vision-Language Models via History-Aware Residual Guidance

快速理解: Large Vision-Language Models (LVLMs) can reason from image-text inputs and perform well in various multimodal tasks. Despite this success, they are affected by language priors and often produce hallucinations.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.498 Resolving the Stability-Plasticity Dilemma in Reinforcement Learning via Complementary Continual Critics

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL；优先级：扩展
- [来源](#)

论文：Resolving the Stability-Plasticity Dilemma in Reinforcement Learning via Complementary Continual Critics

快速理解：This paper proposes the Continual Dual-Critic with Cross- Attention (CD-CCA) framework for visual reinforcement learning to address the plasticity-stability conflict. Our method introduces continual learning techniques into the visual RL architecture, constructing two complementary critics using Continual Backpropagation (CBP) and Elastic Weight Consolidation (EWC) -one for maintaining representational plasticity for rapid environmental adaptation, and the other for preserving knowledge stability to prevent catastrophic forgetting.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.2.499 RetFormer: Multimodal Retrieval for Enhancing Image Recognition

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：RetFormer: Multimodal Retrieval for Enhancing Image Recognition

快速理解：The rapid advancements in Transformers and the availability of high-quality multimodal datasets have enabled deep neural networks to achieve remarkable performance in vision and language tasks. However, catastrophic forgetting and the long-tailed distribution of real-world data make it non-trivial to apply these advances in real-world applications.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.500 Rethinking Asymmetric Quantization: Hidden Symmetry in Vision Model Weights

- 作者：作者信息未提供
- 年份：2026；主题：AI Infra；优先级：扩展
- [来源](#)

论文：Rethinking Asymmetric Quantization: Hidden Symmetry in Vision Model Weights

快速理解：Post-training quantization (PTQ) enables rapid deployment of deep pretrained models. In the low-bit regime, recent PTQ methods for vision models adopt asymmetric quantization (AsymQ), introducing zero-point offsets to mitigate quantization errors.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.2.501 Rethinking Intermediate Representation for VLM-based Robot Manipulation

- 作者：作者信息未提供
- 年份：2026；主题：Embodied/WBC；优先级：扩展
- [来源](#)

论文：Rethinking Intermediate Representation for VLM-based Robot Manipulation

快速理解：Vision-Language Model (VLM) is an important component to enable robust robot manipulation. Yet, using it to translate human instructions into an action-resolvable intermediate representation often needs a tradeoff between VLMcomprehensibility and generalizability.

关注点：具身任务中的感知、动作生成、全身协调或运动控制。

证据层级：本地 PDF 前页。

1.3.2.502 Rethinking Position Embedding as a Context Controller for Multi-Reference and Multi-Shot Video Generation

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Rethinking Position Embedding as a Context Controller for Multi-Reference and Multi-Shot Video Generation

快速理解：Recent proprietary models such as Sora2 demonstrate promising progress in generating multi-shot videos conditioned on multiple reference characters. However, academic research on this problem remains limited.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.503 Rethinking Token Reduction for Large Vision-Language Models

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Rethinking Token Reduction for Large Vision-Language Models

快速理解：Large Vision-Language Models (LVLMs) excel in visual understanding and reasoning, but the excessive visual tokens lead to high inference costs. Although recent token reduction methods mitigate this issue, they mainly target single-turn Visual Question Answering (VQA), leaving the more practical multi-turn VQA (MT-VQA) scenario largely unexplored.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.504 Rethinking Two-Stage Referring-by-Tracking in Referring Multi-Object Tracking: Make it Strong Again

- 作者：作者信息未提供

- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Rethinking Two-Stage Referring-by-Tracking in Referring Multi-Object Tracking: Make it Strong Again

快速理解: Referring Multi-Object Tracking (RMOT) aims to track multiple objects specified by natural language expressions in videos. With the recent significant progress of onestage methods, the two-stage Referring-by-Tracking (RBT) paradigm has gradually lost its popularity.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.505 Revisiting 3D Reconstruction Kernels as Low-Pass Filters

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Revisiting 3D Reconstruction Kernels as Low-Pass Filters

快速理解: 3D reconstruction is to recover 3D signals from the sampled discrete 2D pixels, with the goal to converge continuous 3D spaces. In this paper, we revisit 3D reconstruction from the perspective of signal processing, identifying the periodic spectral extension induced by discrete sampling as the fundamental challenge.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.506 RoadSceneBench: A Lightweight Benchmark for Mid-Level Road Scene Understanding

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: RoadSceneBench: A Lightweight Benchmark for Mid-Level Road Scene Understanding

快速理解: Understanding mid-level road semantics, which capture the structural and contextual cues that link low-level perception to high-level planning, is essential for reliable autonomous driving and digital map construction. However, existing benchmarks primarily target perception tasks such as detection or segmentation, overlooking the reasoning capabilities required to infer road topology and dynamic scene structure.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.507 RoboAgent: Chaining Basic Capabilities for Embodied Task Planning

- 作者: 作者信息未提供
- 年份: 2026; 主题: Embodied/WBC; 优先级: 扩展
- [来源](#)

论文: RoboAgent: Chaining Basic Capabilities for Embodied Task Planning

快速理解: This paper focuses on embodied task planning, where an agent acquires visual observations from the environment and executes atomic actions to accomplish a given task. Although recent Vision-Language Models (VLMs) have achieved impressive results in multimodal understanding and reasoning, their performance remains limited when applied to embodied planning that involves multi-turn interaction, long-horizon reasoning, and extended context analysis.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制。

证据层级: 本地 PDF 前页。

1.3.2.508 S²AM3D: Scale-controllable Part Segmentation of 3D Point Clouds

- 作者: 作者信息未提供
- 年份: 2026; 主题: Systems; 优先级: 扩展
- [来源](#)

论文: S²AM3D: Scale-controllable Part Segmentation of 3D Point Clouds

快速理解: Part-level point cloud segmentation has recently attracted significant attention in 3D computer vision. Nevertheless, existing research is constrained by two major challenges: native 3D models lack generalization due to data scarcity, while introducing 2D pre-trained knowledge often leads to inconsistent segmentation results across different views.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.2.509 S2D: Sparse to Dense Lifting for 3D Reconstruction with Minimal Inputs

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: S2D: Sparse to Dense Lifting for 3D Reconstruction with Minimal Inputs

快速理解: Explicit 3D representations have already become an essential medium for 3D simulation and understanding. However, the most commonly used point cloud and 3D Gaussian Splatting (3DGS) each suffer from non-photorealistic rendering and significant degradation under sparse inputs.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.510 SALMUBench: A Benchmark for Sensitive Association-Level Multimodal Unlearning

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: SALMUBench: A Benchmark for Sensitive Association-Level Multimodal Unlearning

快速理解: As multimodal models like CLIP become integral to downstream systems, the need to remove sensitive information is critical. However, machine unlearning for contrastively trained en-

coders remains underexplored, and existing evaluations fail to diagnose fine-grained, association-level forgetting.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.511 SCE-Depth: A Spherical Compound Eye Framework for Wide FOV Depth Estimation

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：SCE-Depth: A Spherical Compound Eye Framework for Wide FOV Depth Estimation

快速理解：Accurate depth estimation in wide field is highly desired in applications of autonomous driving, robot vision and drone controls. Biological compound eyes inspire wide Field of View (FOV) depth estimation, yet their artificial implementations face the challenge of modality misalignment.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.512 SCIEval: Evaluating and Benchmarking the Faithfulness of Scientific Image Generation and Interpretation with Large Multimodal Models

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：SCIEval: Evaluating and Benchmarking the Faithfulness of Scientific Image Generation and Interpretation with Large Multimodal Models

快速理解：Scientific images often require accurate numerical representations and correct object attributes. However, current faithfulness metrics are primarily tailored toward photorealistic, real-life imagery, rendering them ill-suited for scientific image evaluation.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.513 SCoRe: Saliency-Coverage Reduction for Vision Token Pruning in Vision-Language Models

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：SCoRe: Saliency-Coverage Reduction for Vision Token Pruning in Vision-Language Models

快速理解：The heavy computational burden of Large Vision-Language Models (LVLMs) stems primarily from the lengthy visual token sequences generated by their vision encoders. To mitigate this, recent work has shifted towards pruning tokens within the vision encoder.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.514 SDGS: Spatial Difference Guided Gaussian Splatting for Simultaneous Localization and 3D Reconstruction

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：SDGS: Spatial Difference Guided Gaussian Splatting for Simultaneous Localization and 3D Reconstruction

快速理解：3D Gaussian Splatting (3DGS) pioneers explicit scene representation, enabling photorealistic, real-time 3D reconstruction. Conventional pipelines require precomputed camera poses for Gaussian parameter optimization, which introduces latency between perception and reconstruction.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.515 SEATrack: Simple, Efficient, and Adaptive Multimodal Tracker

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：SEATrack: Simple, Efficient, and Adaptive Multimodal Tracker

快速理解：Parameter-efficient fine-tuning (PEFT) in multimodal tracking reveals a concerning trend where recent performance gains are often achieved at the cost of inflated parameter budgets, which fundamentally erodes PEFT's efficiency promise. In this work, we introduce SEATrack, a Simple, Efficient, and Adaptive two-stream multimodal tracker that tackles this performance-efficiency dilemma from two complementary perspectives.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.516 SEBA: Sample-Efficient Black-Box Attacks on Visual Reinforcement Learning

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL；优先级：扩展
- [来源](#)

论文：SEBA: Sample-Efficient Black-Box Attacks on Visual Reinforcement Learning

快速理解：Visual reinforcement learning has achieved remarkable progress in visual control and robotics, but its vulnerability to adversarial perturbations remains underexplored. Most existing black-box attacks focus on vector-based or discrete-action RL, and their effectiveness on image-based continuous control is limited by the large action space and excessive environment queries.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.2.517 SIF: Semantically In-Distribution Fingerprints for Large Vision-Language Models

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：SIF: Semantically In-Distribution Fingerprints for Large Vision-Language Models

快速理解：The public accessibility of Large Vision-Language Models (LVLMs) raises serious concerns about unauthorized model reuse and intellectual property infringement. Existing ownership verification approaches often rely on semantically abnormal queries or out-of-distribution responses as fingerprints, which are easily recognized and removed by adversaries.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.518 SIMPACT: Simulation-Enabled Action Planning using Vision-Language Models

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：SIMPACT: Simulation-Enabled Action Planning using Vision-Language Models

快速理解：Vision-Language Models (VLMs) exhibit remarkable common-sense and semantic reasoning capabilities. However, they lack a grounded understanding of physical dynamics.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.519 SLVMEval: Synthetic Meta Evaluation Benchmark for Text-to-Long Video Generation

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：SLVMEval: Synthetic Meta Evaluation Benchmark for Text-to-Long Video Generation

快速理解：This paper proposes the synthetic long-video metaevaluation (SLVMEval), a benchmark to perform metaevaluations of text-to-video (T2V) evaluation systems. The proposed SLVMEval benchmark focuses on assessing these systems on videos of up to 10,486 s (approximately 3 h).

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.520 SMAP: Semantic Route Planning with Map-Grounded Multimodal Alignment

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展

- [来源](#)

论文: SMAP: Semantic Route Planning with Map-Grounded Multimodal Alignment

快速理解: Semantic route planning involves generating itineraries that align with user intent while respecting real-world spatial constraints. However, text-only large language models (LLMs) often hallucinate geographically implausible routes due to poor spatial grounding.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.521 SMRABooth: Subject and Motion Representation Alignment for Customized Video Generation

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: SMRABooth: Subject and Motion Representation Alignment for Customized Video Generation

快速理解: Customized video generation aims to produce videos that faithfully preserve the subject's appearance from reference images while maintaining temporally consistent motion from reference videos. Existing methods struggle to ensure both subject appearance similarity and motion pattern consistency due to the lack of object-level guidance for subject and motion.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.522 SO-Bench: A Structural Output Evaluation of Multimodal LLM

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: SO-Bench: A Structural Output Evaluation of Multimodal LLM

快速理解: Multimodal large language models (MLLMs) are increasingly deployed in real-world, agentic settings where outputs must not only be correct, but also conform to pre-defined data schemas. Despite recent progress in structured generation in textual domain, there is still no benchmark that systematically evaluates schema-grounded information extraction and reasoning over visual inputs.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.523 SPEAR-1: Scaling Beyond Robot Demonstrations via 3D Understanding

- 作者: 作者信息未提供
- 年份: 2026; 主题: Robot Learning; 优先级: 扩展
- [来源](#)

论文: SPEAR-1: Scaling Beyond Robot Demonstrations via 3D Understanding

快速理解: Robotic Foundation Models (RFMs) hold great promise as generalist, end-to-end systems for robot control. Yet their ability to generalize across new environments, tasks, and embodiments

remains limited.

关注点：机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：本地 PDF 前页。

1.3.2.524 SR3R: Rethinking Super-Resolution 3D Reconstruction With Feed-Forward Gaussian Splatting

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：SR3R: Rethinking Super-Resolution 3D Reconstruction With Feed-Forward Gaussian Splatting

快速理解：3D super-resolution (3DSR) aims to reconstruct highresolution (HR) 3D scenes from low-resolution (LR) multiview images. Existing methods rely on dense LR inputs and per-scene optimization, which restricts the highfrequency priors for constructing HR 3D Gaussian Splatting (3DGS) to those inherited from pretrained 2D super-^{*}Equal contribution; [†] Project leader; [‡] Corresponding author. resolution (2DSR) models.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.525 STAR-R1: Multi-View Spatial TrAnsformation Reasoning by Reinforcing Multimodal LLMs

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：STAR-R1: Multi-View Spatial TrAnsformation Reasoning by Reinforcing Multimodal LLMs

快速理解：Multimodal Large Language Models (MLLMs) remain far from human-level performance in multi-view spatial reasoning, where models must establish object correspondences across views and infer coherent scene semantics. We analyze this limitation through the Transformation- Driven Visual Reasoning (TVR) task and find that Supervised Fine-Tuning (SFT) fails to capture cross-view consistency, whereas vanilla reinforcement learning (RL) often fails to reliably identify key referential objects.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.526 STAR: Test-Time Adaptation Can Enhance Universal Prompt Learning for Vision-Language Models

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：STAR: Test-Time Adaptation Can Enhance Universal Prompt Learning for Vision-Language Models

快速理解：This paper studies the problem of universal prompt learning at test-time for vision-language

models (VLMs), aiming to enhance the adaptation of pre-trained VLMs to noisy unlabeled target data comprising both in-distribution (ID) and out-of-distribution (OOD) instances. However, existing test-time adaptation approaches often rely on unreliable pseudo-labels due to inadequate uncertainty estimation and overlook class-specific diversity in the target domain, which may result in additional adaptation bias during test time.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.527 STRNet: Visual Navigation with Spatio-Temporal Representation through Dynamic Graph Aggregation

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：STRNet: Visual Navigation with Spatio-Temporal Representation through Dynamic Graph Aggregation

快速理解：Visual navigation requires the robot to reach a specified goal such as an image, based on a sequence of firstperson visual observations. While recent learning-based approaches have made significant progress, they often focus on improving policy heads or decision strategies while relying on simplistic feature encoders and temporal pooling to represent visual input.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.528 SURF: Signature-Retained Fast Video Generation

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：SURF: Signature-Retained Fast Video Generation

快速理解：The demand for high-resolution video generation is growing rapidly. However, the generation resolution is severely constrained by slow inference speeds.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.529 SVBench: Evaluation of Video Generation Models on Social Reasoning

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：SVBench: Evaluation of Video Generation Models on Social Reasoning

快速理解：Recent text-to-video generation models have made remarkable progress in visual realism, motion fidelity, and textvideo alignment, yet they still struggle to produce socially coherent behavior. Unlike humans, who readily infer intentions, beliefs, emotions, and social norms from brief visual cues, current models often generate literal scenes without capturing the underlying causal and

psychological dynamics.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.530 Saliency-Guided Representation with Consistency Policy Learning for Visual Unsupervised Reinforcement Learning

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL；优先级：扩展
- [来源](#)

论文：Saliency-Guided Representation with Consistency Policy Learning for Visual Unsupervised Reinforcement Learning

快速理解：Zero-shot unsupervised reinforcement learning (URL) offers a promising direction for building generalist agents capable of generalizing to unseen tasks without additional supervision. Among existing approaches, successor representations (SR) have emerged as a prominent paradigm due to their effectiveness in structured, low-dimensional settings.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.2.531 Saliency-R1: Enforcing Interpretable and Faithful Vision-language Reasoning via Saliency-map Alignment Reward

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Saliency-R1: Enforcing Interpretable and Faithful Vision-language Reasoning via Saliency-map Alignment Reward

快速理解：Vision-language models (VLMs) have achieved remarkable success across diverse tasks. However, concerns about their trustworthiness persist, particularly regarding tendencies to lean more on textual cues than visual evidence and the risk of producing ungrounded or fabricated responses.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.532 Same or Not? Enhancing Visual Perception in Vision-Language Models

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Same or Not? Enhancing Visual Perception in Vision-Language Models

快速理解：Vision-language models (VLMs) excel at broad visual understanding but remain coarse-grained, exhibit visual biases, and miss subtle visual details. Existing training corpora reinforce this limitation by emphasizing general recognition (“Is it a cat or a dog?”) over fine-grained perception.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.533 Scal3R: Scalable Test-Time Training for Large-Scale 3D Reconstruction

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Scal3R: Scalable Test-Time Training for Large-Scale 3D Reconstruction

快速理解：This paper addresses the task of large-scale 3D scene reconstruction from long video sequences. Recent feedforward reconstruction models have shown promising results by directly regressing 3D geometry from RGB images without explicit 3D priors or geometric constraints.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.534 Scaling Agentic Reinforcement Learning for Tool-Integrated Reasoning in VLMs

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL；优先级：扩展
- [来源](#)

论文：Scaling Agentic Reinforcement Learning for Tool-Integrated Reasoning in VLMs

快速理解：While recent vision-language models (VLMs) demonstrate strong image understanding, their ability to “think with images,” i.e., to reason through multi-step visual interactions, remains limited. We introduce VISTA-Gym, a scalable training environment for incentivizing tool-integrated visual reasoning capabilities in VLMs.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.2.535 Scaling Spatial Intelligence with Multimodal Foundation Models

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Scaling Spatial Intelligence with Multimodal Foundation Models

快速理解：Despite remarkable progress, multimodal foundation models still exhibit surprising deficiencies in spatial intelligence. In this work, we explore scaling up multimodal foundation models to cultivate spatial intelligence within the SenseNova-SI family¹, built upon established multimodal foundations including visual understanding models (i.e., Qwen3-VL and InternVL3) and unified understand-¹This paper is based on the v1.1 version of SenseNova-SI. ing and generation models (i.e., Bagel).

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.536 Scaling Test-Time Robustness of Vision-Language Models via Self-Critical Inference Framework

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文: Scaling Test-Time Robustness of Vision-Language Models via Self-Critical Inference Framework

快速理解: The emergence of Large Language Models (LLMs) has driven rapid progress in multi-modal learning, particularly in the development of Large Vision-Language Models (LVLMs). However, existing LVLM training paradigms place excessive reliance on the LLM component, giving rise to two critical robustness challenges: language bias and language sensitivity.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.537 Scaling the Long Video Understanding of Multimodal Large Language Models via Visual Memory Mechanism

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文: Scaling the Long Video Understanding of Multimodal Large Language Models via Visual Memory Mechanism

快速理解: Long video understanding is a key challenge that plagues the advancement of Multimodal Large language Models (MLLMs). In this paper, we study this problem from the perspective of visual memory mechanism, and proposed a novel and training-free approach, termed Flexible Memory (FlexMem).

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.538 SceneTok: A Compressed, Diffusable Token Space for 3D Scenes

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文: SceneTok: A Compressed, Diffusable Token Space for 3D Scenes

快速理解: We present SceneTok, a novel tokenizer for encoding view sets of scenes into a compressed and diffusable set of unstructured tokens. Existing approaches for 3D scene representation and generation commonly use 3D data structures or view-aligned fields.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.539 Scenes as Tokens: Multi-Scale Normal Distributions Transform Tokenizer for General 3D Vision-Language Understanding

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文: Scenes as Tokens: Multi-Scale Normal Distributions Transform Tokenizer for General 3D Vision-Language Understanding

快速理解: Recent advances in 3D vision-language models (VLMs) highlight a strong potential for 3D scene understanding and reasoning. However, effectively tokenizing 3D scenes into holistic scene tokens, and leveraging these tokens across diverse 3D understanding tasks, remain highly challenging.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.540 SeD-UD: An Influence-Driven and Hierarchically-Decoupled Information Bottleneck for Multimodal Intent Recognition

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文: SeD-UD: An Influence-Driven and Hierarchically-Decoupled Information Bottleneck for Multimodal Intent Recognition

快速理解: Multimodal intent recognition (MIR) is hindered by substantial redundancy and noise originating from text, speech, and visual inputs, which weakens feature distinctiveness and ultimately harms recognition performance. Although recent approaches based on the information bottleneck (IB) principle mitigate this issue via feature compression and reconstruction to obtain compact and noise-reduced representations, they still encounter two major drawbacks.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.541 SeaCache: Spectral-Evolution-Aware Cache for Accelerating Diffusion Models

- 作者：作者信息未提供
- 年份：2026；主题：Systems；优先级：扩展
- [来源](#)

论文: SeaCache: Spectral-Evolution-Aware Cache for Accelerating Diffusion Models

快速理解: Diffusion models are a strong backbone for visual generation, but their inherently sequential denoising process leads to slow inference. Previous methods accelerate sampling by caching and reusing intermediate outputs based on feature distances between adjacent timesteps.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.2.542 See Less, See Right: Bi-directional Perceptual Shaping For Multimodal Reasoning

- 作者：作者信息未提供

- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: See Less, See Right: Bi-directional Perceptual Shaping For Multimodal Reasoning

快速理解: Large vision-language models (VLMs) often benefit from intermediate visual cues, either injected via external tools or generated as latent visual tokens during reasoning, but these mechanisms still overlook fine-grained visual evidence (e.g., polylines in charts), generalize poorly across domains, and incur high inference-time cost. In this paper, we propose Bi-directional Perceptual Shaping (BiPS), which transforms question-conditioned masked views into bidirectional where-to-look signals that shape perception during training.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.543 See and Fix the Flaws: Enabling VLMs and Diffusion Models to Comprehend Visual Artifacts via Agentic Data Synthesis

- 作者: 作者信息未提供
- 年份: 2026; 主题: Agent RL; 优先级: 扩展
- [来源](#)

论文: See and Fix the Flaws: Enabling VLMs and Diffusion Models to Comprehend Visual Artifacts via Agentic Data Synthesis

快速理解: Despite recent advances in diffusion models, AI generated images still often contain visual artifacts that compromise realism. Although more thorough pre-training and bigger models might reduce artifacts, there is no assurance that they can be completely eliminated, which makes artifact mitigation a highly crucial area of study.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.2.544 See, Think, Act: Teaching Multimodal Agents to Effectively Interact with GUI by Identifying Toggles

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: See, Think, Act: Teaching Multimodal Agents to Effectively Interact with GUI by Identifying Toggles

快速理解: The advent of multimodal agents facilitates effective interaction within graphical user interface (GUI), especially in ubiquitous GUI control. However, their inability to reliably execute toggle control instructions remains a key bottleneck.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.545 SeeGroup: Multi-Layer Depth Estimation of Transparent Surfaces via Self-Determined Grouping

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文: SeeGroup: Multi-Layer Depth Estimation of Transparent Surfaces via Self-Determined Grouping

快速理解: Transparent objects are common in daily life, and understanding their multi-layer depth information, including both the transparent surface and the objects behind it, is crucial for real-world applications that interact with transparent materials. However, existing depth methods produce only a single depth map, which is inherently ambiguous for transparent surfaces.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.546 Seeing Clearly, Reasoning Confidently: Plug-and-Play Remedies for Vision Language Model Blindness

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文: Seeing Clearly, Reasoning Confidently: Plug-and-Play Remedies for Vision Language Model Blindness

快速理解: Vision language models (VLMs) have achieved remarkable success in broad visual understanding, yet they remain challenged by object-centric reasoning on rare objects due to the scarcity of such instances in pretraining data. While prior efforts alleviate this issue by retrieving additional data or introducing stronger vision encoders, these methods are still computationally intensive during finetuning VLMs and don't fully exploit the original training data.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.547 Seeing Depth Through Frequency and Motion: A Progressive Training Paradigm for Monocular Depth Estimation

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文: Seeing Depth Through Frequency and Motion: A Progressive Training Paradigm for Monocular Depth Estimation

快速理解: Self-supervised monocular depth estimation has achieved remarkable progress in recent years, yet frequency aliasing and the lack of fine-grained cross-frame motion modeling still lead to blurred depth boundaries and suboptimal camera motion estimation. To address these challenges, we propose a progressive self-supervised framework that integrates a Frequency-Guided Depth Network (FGDepth) and a PoseQuery Network (PQNet).

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.548 Seeing What Matters: A Training-Free Self-Guided Framework for Multimodal Detail Perception and Reasoning

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Seeing What Matters: A Training-Free Self-Guided Framework for Multimodal Detail Perception and Reasoning

快速理解：Multimodal large language models (MLLMs) have achieved remarkable success on diverse visual-language tasks. However, fixed-resolution models face challenges in perceiving fine-grained visual details, particularly due to distracted attention and blurry vision.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.549 Seeing What Matters: Visual Preference Policy Optimization for Visual Generation

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL；优先级：扩展
- [来源](#)

论文：Seeing What Matters: Visual Preference Policy Optimization for Visual Generation

快速理解：Reinforcement learning (RL) has become a powerful tool for post-training visual generative models, with Group Relative Policy Optimization (GRPO) increasingly used to align generators with human preferences. However, existing GRPO pipelines rely on a single scalar reward per sample, treating each image or video as a holistic entity and ignoring the rich spatial and temporal structure of visual content.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.2.550 SegMoTE: Token-Level Mixture of Experts for Medical Image Segmentation

- 作者：作者信息未提供
- 年份：2026；主题：AI Infra；优先级：扩展
- [来源](#)

论文：SegMoTE: Token-Level Mixture of Experts for Medical Image Segmentation

快速理解：Medical image segmentation is vital for clinical diagnosis and quantitative analysis, yet remains challenging due to the heterogeneity of imaging modalities and the high cost of pixel-level annotations. Although general interactive segmentation models like SAM have achieved remarkable progress, their transfer to medical imaging still faces two key bottlenecks: (i) the lack of adaptive mechanisms for modality and anatomy-specific tasks, which limits generalization in out-of-distribution medical scenarios; and (ii) current medical adaptation methods fine-tune on large, heterogeneous datasets without selection, leading to noisy supervision, higher cost, and negative transfer.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.2.551 Semantic Audio-Visual Navigation in Continuous Environments

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Semantic Audio-Visual Navigation in Continuous Environments

快速理解：Audio-visual navigation enables embodied agents to navigate toward sound-emitting targets by leveraging both auditory and visual cues. However, most existing approaches rely on precomputed room impulse responses (RIRs) for binaural audio rendering, restricting agents to discrete grid positions and leading to spatially discontinuous observations.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.552 SenCache: Accelerating Diffusion Model Inference via Sensitivity-Aware Caching

- 作者：作者信息未提供
- 年份：2026；主题：Systems；优先级：扩展
- [来源](#)

论文：SenCache: Accelerating Diffusion Model Inference via Sensitivity-Aware Caching

快速理解：Diffusion models achieve state-of-the-art video generation quality, but their inference remains expensive due to the large number of sequential denoising steps. This has motivated a growing line of research on accelerating diffusion inference.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.2.553 ShotDirector: Directorially Controllable Multi-Shot Video Generation with Cinematographic Transitions

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：ShotDirector: Directorially Controllable Multi-Shot Video Generation with Cinematographic Transitions

快速理解：Shot transitions play a pivotal role in multi-shot video generation, as they determine the overall narrative expression and the directorial design of visual storytelling. However, recent progress has primarily focused on low-level visual consistency across shots, neglecting how transitions are designed and how cinematographic language contributes to *Work done during an internship at Shanghai AI Laboratory. †Corresponding authors. coherent narrative expression.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.554 ShowUI-p: Flow-based Generative Models as GUI Dexterous Hands

- 作者：作者信息未提供
- 年份：2026；主题：Embodied/WBC；优先级：扩展
- [来源](#)

论文：ShowUI-p: Flow-based Generative Models as GUI Dexterous Hands

快速理解：Building intelligent agents capable of dexterous manipulation is essential for achieving human-like automation in both robotics and digital environments. However, existing GUI agents rely on discrete click predictions (x, y), which prohibits free-form, closed-loop trajectories (e.g., dragging a progress bar) that require continuous, on-the-fly perception and adjustment.

关注点：具身任务中的感知、动作生成、全身协调或运动控制。

证据层级：本地 PDF 前页。

1.3.2.555 Simple-ViLMedSAM: Simple Text Prompts Meet Vision-Language Models for Medical Image Segmentation

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Simple-ViLMedSAM: Simple Text Prompts Meet Vision-Language Models for Medical Image Segmentation

快速理解：Medical image segmentation is challenging due to limited annotated data, high labeling costs, and substantial image heterogeneity. Although large-scale vision foundation models (e.g., SAM) have shown great potential in this field, existing SAM-based methods typically rely on expert-defined geometric prompts or complex clinical text prompts, which limits their generalizability across diverse medical image segmentation tasks.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.556 SineProject: Machine Unlearning for Stable Vision-Language Alignment

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：SineProject: Machine Unlearning for Stable Vision-Language Alignment

快速理解：Multimodal Large Language Models (MLLMs) increasingly need to forget specific knowledge, such as unsafe or private information, without full retraining. However, existing unlearning methods often disrupt vision-language alignment, causing models to reject both harmful and benign queries simultaneously.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.557 SketchRevive: Fine-Grained Pixel-to-Vector Sketch Completion with Diffusion-Prior-Guided Multimodal LLMs

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：SketchRevive: Fine-Grained Pixel-to-Vector Sketch Completion with Diffusion-Prior-Guided Multimodal LLMs

快速理解：Transforming sparse, partial pixel sketches from diverse media into complete, editable vector drawings is essential yet underexplored in digital creation. Prior methods either generate from scratch or inpaint local gaps without predicting global structure, leading to coarse contours and limited detail.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.558 SocialNav: Training Human-Inspired Foundation Model for Socially-Aware Embodied Navigation

- 作者：作者信息未提供
- 年份：2026；主题：Embodied/WBC；优先级：扩展
- [来源](#)

论文：SocialNav: Training Human-Inspired Foundation Model for Socially-Aware Embodied Navigation

快速理解：Embodied navigation that adheres to social norms remains an open research challenge. Our SocialNav is a foundational model for socially-aware navigation with a hierarchical ” brain-action” architecture, capable of understanding high-level social norms and generating low-level, socially compliant trajectories.

关注点：具身任务中的感知、动作生成、全身协调或运动控制。

证据层级：本地 PDF 前页。

1.3.2.559 Socratic-Geo: Synthetic Data Generation and Cross-Modal Geometric Reasoning via Multi-Agent Interaction

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL；优先级：扩展
- [来源](#)

论文：Socratic-Geo: Synthetic Data Generation and Cross-Modal Geometric Reasoning via Multi-Agent Interaction

快速理解：Multimodal Large Language Models (MLLMs) have significantly advanced vision-language understanding. However, even state-of-the-art models struggle with geometric reasoning, revealing a critical bottleneck: the extreme scarcity of high-quality image-text pairs.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.2.560 SpaceMind: Camera-Guided Modality Fusion for Spatial Reasoning in Vision-Language Models

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文: SpaceMind: Camera-Guided Modality Fusion for Spatial Reasoning in Vision-Language Models

快速理解: object counting absolute distance object size room size relative distance relative direction route planning appearance order 73.9 61.5 77.6 74.8 67.7 88.6 46.9 70.7 This CVPR paper is the Open Access version, provided by the Computer Vision Foundation. Except for this watermark, it is identical to the accepted version; the final published version of the proceedings is available on IEEE Xplore.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.561 Sparse-LaViDa: Sparse Multimodal Discrete Diffusion Language Models

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文: Sparse-LaViDa: Sparse Multimodal Discrete Diffusion Language Models

快速理解: Masked Discrete Diffusion Models (MDMs) have achieved strong performance across a wide range of multimodal tasks, including image understanding, generation, and editing. However, their inference speed remains suboptimal due to the need to repeatedly process redundant masked tokens at every sampling step.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.562 Spatia: Video Generation with Updatable Spatial Memory

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文: Spatia: Video Generation with Updatable Spatial Memory

快速理解: Existing video generation models struggle to maintain longterm spatial and temporal consistency due to the dense, high-dimensional nature of video signals. To overcome this limitation, we propose Spatia, a spatial memory-aware video generation framework that explicitly preserves a 3D scene point cloud as persistent spatial memory.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.563 SpatiaLQA: A Benchmark for Evaluating Spatial Logical Reasoning in Vision-Language Models

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文: SpatiaLQA: A Benchmark for Evaluating Spatial Logical Reasoning in Vision-Language Models

快速理解: Vision-Language Models (VLMs) have been increasingly applied in real-world scenarios due to their outstanding understanding and reasoning capabilities. Although VLMs have already demonstrated impressive capabilities in common visual question answering and logical reasoning, they still lack the ability to make reasonable decisions in complex real-world environments.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.564 Spatial-Aware VLA Pretraining through Visual-Physical Alignment from Human Videos

- 作者: 作者信息未提供
- 年份: 2026; 主题: VLA; 优先级: 扩展
- [来源](#)

论文: Spatial-Aware VLA Pretraining through Visual-Physical Alignment from Human Videos

快速理解: Vision-Language-Action (VLA) models provide a promising paradigm for robot learning by integrating visual perception with language-guided policy learning. However, most existing approaches rely on 2D visual inputs to perform actions in 3D physical environments, creating a significant gap between perception and action grounding.

关注点: 视觉、语言条件与机器人动作策略之间的对齐。

证据层级: 本地 PDF 前页。

1.3.2.565 SpatialReward: Verifiable Spatial Reward Modeling for Fine-Grained Spatial Consistency in Text-to-Image Generation

- 作者: 作者信息未提供
- 年份: 2026; 主题: Agent RL; 优先级: 扩展
- [来源](#)

论文: SpatialReward: Verifiable Spatial Reward Modeling for Fine-Grained Spatial Consistency in Text-to-Image Generation

快速理解: Recent advances in text-to-image (T2I) generation via reinforcement learning (RL) have benefited from reward models that assess semantic alignment and visual quality. However, most existing reward models pay limited attention to finegrained spatial relationships, often producing images that appear plausible overall yet contain inaccuracies in object positioning.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.2.566 Specificity-aware reinforcement learning for fine-grained open-world classification

- 作者: 作者信息未提供
- 年份: 2026; 主题: Agent RL; 优先级: 扩展
- [来源](#)

论文: Specificity-aware reinforcement learning for fine-grained open-world classification

快速理解: Classifying fine-grained visual concepts under open-world settings, i.e., without a predefined label set, demands models to be both accurate and specific. Recent reasoning Large Multimodal Models (LMMs) exhibit strong visual understanding capability but tend to produce overly generic predictions when performing fine-grained image classification.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.2.567 Spectral Mixture-of-Experts for Continual Learning

- 作者: 作者信息未提供
- 年份: 2026; 主题: AI Infra; 优先级: 扩展
- [来源](#)

论文: Spectral Mixture-of-Experts for Continual Learning

快速理解: While Parameter-Efficient Fine-Tuning using Mixtureof-Experts (MoE) is a promising solution for continual learning (CL), it suffers from two critical failure modes: structural interference, where expert updates interfere, and compositional forgetting, where the model's routing policy drifts. To address these issues, we introduce Spectral MoE, a novel framework built for CL from three core components.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 本地 PDF 前页。

1.3.2.568 SpikeTrack: High-performance and Energy-efficient Event-Based Object Tracking with Spiking Neural Network

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: SpikeTrack: High-performance and Energy-efficient Event-Based Object Tracking with Spiking Neural Network

快速理解: Event cameras have attracted considerable attention for object tracking due to their microsecond-level temporal resolution and wide dynamic range, yet effectively harnessing spiking neural networks (SNNs) in this domain remains challenging. In this paper, we introduce SpikeTrack, a purely spike-driven framework for single-object tracking that addresses the shortcomings of RGB-based approaches in fast-motion or target appearance change.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.569 Stable and Efficient Single-Rollout RL for Multimodal Reasoning

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Stable and Efficient Single-Rollout RL for Multimodal Reasoning

快速理解: Reinforcement Learning with Verifiable Rewards (RLVR) has become a key paradigm

to improve the reasoning capabilities of Multimodal Large Language Models (MLLMs). However, prevalent group-based algorithms such as GRPO require multi-rollout sampling for each prompt.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.570 Stand-In: A Lightweight and Plug-and-Play Identity Control for Video Generation

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Stand-In: A Lightweight and Plug-and-Play Identity Control for Video Generation

快速理解：Generating high-fidelity human videos that match userspecified identities is important yet challenging in the field of generative AI. Existing methods often rely on an excessive number of training parameters and lack compatibility with other AIGC tools.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.571 StereoWorld: Geometry-Aware Monocular-to-Stereo Video Generation

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：StereoWorld: Geometry-Aware Monocular-to-Stereo Video Generation

快速理解：The growing adoption of XR devices has fueled strong demand for high-quality stereo video, yet its production remains costly and artifact-prone. To address this challenge, we present StereoWorld, an end-to-end framework that repurposes a pretrained video generator for high-fidelity monocular-to-stereo video generation.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.572 StreamDiT: Real-Time Streaming Text-to-Video Generation

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：StreamDiT: Real-Time Streaming Text-to-Video Generation

快速理解：Recently, great progress has been achieved in text-tovideo (T2V) generation by scaling transformer-based diffusion models to billions of parameters. However, existing models typically produce only short clips offline, restricting their use cases in interactive and real-time applications.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.573 StructXLIP: Enhancing Vision-language Models with Multimodal Structural Cues

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：StructXLIP: Enhancing Vision-language Models with Multimodal Structural Cues

快速理解：Edge-based representations are fundamental cues for visual understanding, a principle rooted in early vision research and still central today. We extend this principle to visionlanguage alignment, showing that isolating and aligning structural cues across modalities can greatly benefit finetuning on long, detail-rich captions, with a specific focus on improving cross modal retrieval.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.574 Style-GRPO: Semantic-Aware Preference Optimization for Image Style Transfer Guided by Reward Modeling

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL；优先级：扩展
- [来源](#)

论文：Style-GRPO: Semantic-Aware Preference Optimization for Image Style Transfer Guided by Reward Modeling

快速理解：We present a novel approach to bridging the gap between style transfer guided by reference styles and faithful content preservation in text-to-image generation. First, we introduce StyleReward-Dataset, an expert-annotated benchmark with over 300k adversarial image pairs spanning diverse real-world and virtual styles.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.2.575 StyleDoctor: Towards Specialist Reward Model for Style-centric Generation Tasks

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL；优先级：扩展
- [来源](#)

论文：StyleDoctor: Towards Specialist Reward Model for Style-centric Generation Tasks

快速理解：Style generation has made significant progress through diffusion models. Recent efforts have explored reinforcement learning with human-preference reward models to enhance diffusion models for general downstream applications.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.2.576 SuP: Sub-cloud Driven Point Cloud Registration

- 作者：作者信息未提供
- 年份：2026；主题：Systems；优先级：扩展
- [来源](#)

论文：SuP: Sub-cloud Driven Point Cloud Registration

快速理解：While existing point-cloud-registration methods can handle high-overlap scenarios of two point clouds well, they often struggle with low-overlap scenarios due to inevitable geometric/semantic ambiguities in the non-overlapping regions. In this paper, we introduce SuP, a novel framework that reformulates low-overlap registration as a high-overlap sub-cloud pairs (anchor pairs) mining problem.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.2.577 SwiftVLA: Unlocking Spatiotemporal Dynamics for Lightweight VLA Models at Minimal Overhead

- 作者：作者信息未提供
- 年份：2026；主题：VLA；优先级：扩展
- [来源](#)

论文：SwiftVLA: Unlocking Spatiotemporal Dynamics for Lightweight VLA Models at Minimal Overhead

快速理解：Vision-Language-Action (VLA) models built on pretrained Vision-Language Models (VLMs) show strong potential but are limited in practicality due to their large parameter counts. To mitigate this issue, using a lightweight VLM has been explored, but it compromises spatiotemporal reasoning.

关注点：视觉、语言条件与机器人动作策略之间的对齐。

证据层级：本地 PDF 前页。

1.3.2.578 SwitchCraft: Training-Free Multi-Event Video Generation with Attention Controls

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：SwitchCraft: Training-Free Multi-Event Video Generation with Attention Controls

快速理解：Recent advances in text-to-video diffusion models have enabled high-fidelity and temporally coherent video synthesis. However, current models are predominantly optimized for single-event generation.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.579 SymphoMotion: Joint Control of Camera Motion and Object Dynamics for Coherent Video Generation

- 作者：作者信息未提供

- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: SymphoMotion: Joint Control of Camera Motion and Object Dynamics for Coherent Video Generation

快速理解: Controlling both camera motion and object dynamics is essential for coherent and expressive video generation, yet current methods typically handle only one motion type or rely on ambiguous 2D cues that entangle camera-induced parallax with true object movement. We present SymphoMotion, a unified motion-control framework that jointly governs camera trajectories and object dynamics within a single model.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.580 Symphony: A Cognitively-Inspired Multi-Agent System for Long-Video Understanding

- 作者: 作者信息未提供
- 年份: 2026; 主题: Agent RL; 优先级: 扩展
- [来源](#)

论文: Symphony: A Cognitively-Inspired Multi-Agent System for Long-Video Understanding

快速理解: Despite rapid developments and widespread applications of MLLM agents, they still struggle with long-form video understanding (LVU) tasks, which are characterized by high information density and extended temporal spans. Recent research on LVU agents demonstrates that simple task decomposition and collaboration mechanisms are insufficient for long-chain reasoning tasks.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.2.581 SynMotion: Semantic-Visual Adaptation for Motion Customized Video Generation

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: SynMotion: Semantic-Visual Adaptation for Motion Customized Video Generation

快速理解: Diffusion-based video motion customization facilitates the acquisition of human motion from a few video samples, while achieving arbitrary subjects transfer through precise *Work done during internship at Ant Group. †Project lead. ‡Corresponding author. textual conditioning. Existing approaches often rely on semantic-level alignment, expecting the model to learn new motion concepts and combine them with other entities (e.g., “cats” or “dogs”) to produce visually appealing results.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.582 SyncMos: Scalable Motion Synchronisation for Multi-Agent Scene Interaction

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL；优先级：扩展
- [来源](#)

论文：SyncMos: Scalable Motion Synchronisation for Multi-Agent Scene Interaction

快速理解：Text-guided motion generation in 3D scenes has advanced the synthesis of human-scene interactions, contributing to embodied AI, scene understanding, and virtual agent simulation. While recent studies have begun exploring multi-agent scenarios, achieving temporally synchronised interactions among multiple agents remains an open challenge.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.2.583 Synthesizing Visual Concepts as Vision-Language Programs

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Synthesizing Visual Concepts as Vision-Language Programs

快速理解：Vision-Language models (VLMs) achieve strong performance on multimodal tasks but often fail at systematic visual reasoning, especially in inductive reasoning problems. Neuro-symbolic methods promise to address this by inducing interpretable logical programs from images, though they usually rely on rigid, domain-specific perception modules for this.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.584 TAG-MoE: Task-Aware Gating for Unified Generative Mixture-of-Experts

- 作者：作者信息未提供
- 年份：2026；主题：AI Infra；优先级：扩展
- [来源](#)

论文：TAG-MoE: Task-Aware Gating for Unified Generative Mixture-of-Experts

快速理解：Unified image generation and editing models suffer from severe task interference in dense diffusion transformers architectures, where a shared parameter space must compromise between conflicting objectives (e.g., local editing v.s. subject-driven generation). While the sparse Mixture-of-Experts (MoE) paradigm is a promising solution, its gating networks remain task-agnostic, operating based on local features, unaware of global task intent.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.2.585 TANGO: Text-Anchored Guided Optimization for Robust Fine-tuning Vision-Language Models under Label Noise

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文: TANGO: Text-Anchored Guided Optimization for Robust Fine-tuning Vision-Language Models under Label Noise

快速理解: Fine-tuning large-scale Vision-Language Models (VLMs) is crucial for specialized tasks, but their performance is often undermined by the label noise prevalent in real-world datasets. Traditional approaches to learning with noisy labels typically rely on a self-referential loop, using a model's own predictions to correct errors.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.586 TGT: Text-Grounded Trajectories for Locally Controlled Video Generation

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文: TGT: Text-Grounded Trajectories for Locally Controlled Video Generation

快速理解: Text-to-video generation has advanced rapidly in visual fidelity, whereas standard methods still have limited ability to control the subject composition of generated scenes. Prior work shows that adding localized text control signals, such as bounding boxes or segmentation masks, can help.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.587 TGTrack: Temporal Generative Learning for Unified Single Object Tracking

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文: TGTrack: Temporal Generative Learning for Unified Single Object Tracking

快速理解: Existing single object trackers typically treat temporal modeling superficially by passing limited inter-frame information, such as propagated tokens or template updates, without intrinsic temporal supervision learning. To address this limitation, we propose TGTrack, a new unified tracking framework that incorporates a temporally generative supervision task to guide the model in learning temporal dynamics.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.588 THE MORE, THE MERRIER: CONTRASTIVE FUSION FOR HIGHER-ORDER MULTIMODAL ALIGNMENT

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文: THE MORE, THE MERRIER: CONTRASTIVE FUSION FOR HIGHER-ORDER MULTI-MODAL ALIGNMENT

快速理解: Learning joint representations across multiple modalities remains a central challenge in multimodal machine learning. Prevailing approaches predominantly operate in pairwise settings, aligning two modalities at a time.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.589 TIPSv2: Advancing Vision-Language Pretraining with Enhanced Patch-Text Alignment

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: TIPSv2: Advancing Vision-Language Pretraining with Enhanced Patch-Text Alignment

快速理解: Recent progress in vision-language pretraining has enabled significant improvements to many downstream computer vision applications, such as classification, retrieval, segmentation and depth prediction. However, a fundamental capability that these models still struggle with is aligning dense patch representations with text embeddings of corresponding concepts.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.590 TSTM: Temporal Segmentation for Task-relevant Mask in Visual Reinforcement Learning Generalization

- 作者: 作者信息未提供
- 年份: 2026; 主题: Agent RL; 优先级: 扩展
- [来源](#)

论文: TSTM: Temporal Segmentation for Task-relevant Mask in Visual Reinforcement Learning Generalization

快速理解: Achieving strong policy generalization to unseen environments remains a core challenge in visual reinforcement learning, and segmenting task-relevant regions to mitigate the influence of irrelevant visual cues has emerged as a promising direction. However, existing methods rely solely on the current observation, lack temporal information, and fail to exploit preceding observations, leaving learned policies susceptible to task-irrelevant background variations and ultimately degrading policy performance.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.2.591 TTL: Test-time Textual Learning for OOD Detection with Pretrained Vision-Language Models

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展

- [来源](#)

论文: TTL: Test-time Textual Learning for OOD Detection with Pretrained Vision-Language Models

快速理解: Vision-language models (VLMs) such as CLIP exhibit strong Out-of-distribution (OOD) detection capabilities by aligning visual and textual representations. Recent CLIPbased test-time adaptation methods further improve detection performance by incorporating external OOD labels.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.592 TUNA: Taming Unified Visual Representations for Native Unified Multimodal Models

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: TUNA: Taming Unified Visual Representations for Native Unified Multimodal Models

快速理解: Unified multimodal models (UMMs) aim to jointly perform multimodal understanding and generation within a single framework. We present TUNA, a native UMM that builds a unified continuous visual representation by cascading a VAE encoder with a representation encoder.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.593 TV2TV: A Unified Framework for Interleaved Language and Video Generation

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: TV2TV: A Unified Framework for Interleaved Language and Video Generation

快速理解: Video generation models are rapidly advancing, but can still struggle with complex video outputs that require significant semantic branching or repeated high-level reasoning about what should happen next. We introduce a new class of omni video-text models that integrate ideas from recent LM reasoning advances to address this challenge.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.594 Tackling Model Bias via Game-theoretic Multi-agent Collaboration Framework for Hateful Meme Classification

- 作者: 作者信息未提供
- 年份: 2026; 主题: Agent RL; 优先级: 扩展
- [来源](#)

论文: Tackling Model Bias via Game-theoretic Multi-agent Collaboration Framework for Hateful Meme Classification

快速理解: Hateful meme classification aims to identify memes containing hateful content and has become increasingly important in the era of social media dominance. Large multimodal models (LMMs) have significantly enhanced the understanding of multimodal content, advancing this field.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.2.595 Talk2Move: Reinforcement Learning for Text-Instructed Object-Level Geometric Transformation in Scenes

- 作者: 作者信息未提供
- 年份: 2026; 主题: Agent RL; 优先级: 扩展
- [来源](#)

论文: Talk2Move: Reinforcement Learning for Text-Instructed Object-Level Geometric Transformation in Scenes

快速理解: We introduce Talk2Move, a reinforcement learning (RL) based diffusion framework for text-instructed spatial transformation of objects within scenes. Spatially manipulating objects in a scene through natural language poses a challenge for multimodal generation systems.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.2.596 Taming Preference Mode Collapse via Directional Decoupling Alignment in Diffusion Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2026; 主题: Agent RL; 优先级: 扩展
- [来源](#)

论文: Taming Preference Mode Collapse via Directional Decoupling Alignment in Diffusion Reinforcement Learning

快速理解: Recent studies have demonstrated significant progress in aligning text-to-image diffusion models with human preference via Reinforcement Learning from Human Feedback. However, while existing methods achieve high scores on automated reward metrics, they often lead to Preference Mode Collapse (PMC)—a specific form of reward hacking where models converge on narrow, high-scoring outputs (e.g., images with monolithic styles or pervasive overexposure), severely degrading generative diversity.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.2.597 TaskForce: Cooperative Multi-agent Reinforcement Learning for Multi-task Optimization

- 作者: 作者信息未提供
- 年份: 2026; 主题: Agent RL; 优先级: 扩展
- [来源](#)

论文: TaskForce: Cooperative Multi-agent Reinforcement Learning for Multi-task Optimization

快速理解: Multi-task learning (MTL) involves the simultaneous optimization of multiple task-specific losses, often leading to gradient conflicts and scale imbalances that result in negative transfer. While existing multi-task optimization methods attempt to mitigate these challenges, they either lack the stochasticity needed to escape poor local minima or fail to explicitly address conflicts at the gradient level.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.2.598 Taxonomy-Aware Representation Alignment for Hierarchical Visual Recognition with Large Multimodal Models

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Taxonomy-Aware Representation Alignment for Hierarchical Visual Recognition with Large Multimodal Models

快速理解: A high-performing, general-purpose visual understanding model should map visual inputs to a taxonomic tree of labels, identify novel categories beyond the training set for which few or no publicly available images exist. Large Multimodal Models (LMMs) have achieved remarkable progress in fine-grained visual recognition (FGVR) for known categories.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.599 Teacher-Guided Routing for Sparse Vision Mixture-of-Experts

- 作者: 作者信息未提供
- 年份: 2026; 主题: AI Infra; 优先级: 扩展
- [来源](#)

论文: Teacher-Guided Routing for Sparse Vision Mixture-of-Experts

快速理解: Recent progress in deep learning has been driven by increasingly large-scale models, but the resulting computational cost has become a critical bottleneck. Sparse Mixture of Experts (MoE) offers an effective solution by activating only a small subset of experts for each input, achieving high scalability without sacrificing inference speed.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 本地 PDF 前页。

1.3.2.600 TempR1: Improving Temporal Understanding of MLLMs via Temporal-Aware Multi-Task Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2026; 主题: Agent RL; 优先级: 扩展
- [来源](#)

论文: TempR1: Improving Temporal Understanding of MLLMs via Temporal-Aware Multi-Task Reinforcement Learning

快速理解: Enhancing temporal understanding of MLLMs is essential for long-form video analysis, supporting tasks such as temporal localization and time-sensitive question answering. While reinforcement learning (RL) has been explored for temporal reasoning, existing approaches are often limited to specific tasks or datasets, hindering generalization across diverse temporal scenarios.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.2.601 Test-Time Attention Purification for Backdoored Large Vision Language Models

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Test-Time Attention Purification for Backdoored Large Vision Language Models

快速理解: Despite the strong multimodal performance, large visionlanguage models (LVLMs) are vulnerable during fine-tuning to backdoor attacks, where adversaries insert triggerembedded samples into the training data to implant behaviors that can be maliciously activated at test time. Existing defenses typically rely on retraining backdoored parameters (e.g., adapters or LoRA modules) with clean data, which is computationally expensive and often degrades model performance.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.602 Text-Printed Image: Bridging the Image-Text Modality Gap for Text-centric Training of Large Vision-Language Models

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Text-Printed Image: Bridging the Image-Text Modality Gap for Text-centric Training of Large Vision-Language Models

快速理解: Recent large vision-language models (LVLMs) have been applied to diverse VQA tasks. However, achieving practical performance typically requires task-specific fine-tuning with large numbers of image-text pairs, which are costly to collect.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.603 The Geometry of Robustness: Optimizing Loss Landscape Curvature and Feature Manifold Alignment for Robust Finetuning of Vision-Language Models

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: The Geometry of Robustness: Optimizing Loss Landscape Curvature and Feature Manifold Alignment for Robust Finetuning of Vision-Language Models

快速理解: Fine-tuning approaches for Vision-Language Models (VLMs) face a critical three-way trade-off between In-Distribution (ID) accuracy, Out-of-Distribution (OOD) generalization, and adversarial robustness. Existing robust fine-tuning strategies resolve at most two axes of this trade-off.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.604 The Image as Its Own Reward: Reinforcement Learning with Adversarial Reward for Image Generation

- 作者: 作者信息未提供
- 年份: 2026; 主题: Agent RL; 优先级: 扩展
- [来源](#)

论文: The Image as Its Own Reward: Reinforcement Learning with Adversarial Reward for Image Generation

快速理解: A reliable reward function is essential for reinforcement learning (RL) in image generation. Most current RL approaches depend on pre-trained preference models that output scalar rewards to approximate human preferences.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.2.605 Think Visually, Reason Textually: Vision-Language Synergy in Abstract Reasoning

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Think Visually, Reason Textually: Vision-Language Synergy in Abstract Reasoning

快速理解: Reasoning Beichen Zhang^{1,2}, Yuhang Zang^{2B}, Xiaoyi Dong^{1,2}, Yuhang Cao² Haodong Duan², Dahua Lin^{1,2,4B}, Jiaqi Wang^{2,3B} ¹The Chinese University of Hong Kong ²Shanghai AI Laboratory ³Shanghai Innovation Institute ⁴CPII under InnoHK {zhangbeichen,zangyuhang}@pjlab.org.cn BCorresponding Authors. Abstract Abstract reasoning from minimal examples remains a core unsolved problem for frontier foundation models such as GPT-5.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.606 Thinking Diffusion: Penalize and Guide Visual-Grounded Reasoning in Diffusion Multimodal Language Models

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Thinking Diffusion: Penalize and Guide Visual-Grounded Reasoning in Diffusion Multimodal Language Models

快速理解: Diffusion large language models (dLLMs) are emerging as promising alternatives to autoregressive (AR) LLMs. Recently, this paradigm has been extended to multimodal tasks, leading to the development of diffusion multimodal large language models (dMLLMs).

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.607 Thinking in 360deg: Humanoid Visual Search in the Wild

- 作者: 作者信息未提供
- 年份: 2026; 主题: Embodied/WBC; 优先级: 扩展
- [来源](#)

论文: Thinking in 360deg: Humanoid Visual Search in the Wild

快速理解: Humans rely on the synergistic control of head (cephalomotor) and eye (oculomotor) to efficiently search for visual information in 360°. However, prior approaches to visual search are limited to a static image, neglecting the physical embodiment and its interaction with the 3D world.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制。

证据层级: 本地 PDF 前页。

1.3.2.608 Thinking with Frames: Generative Video Distortion Evaluation via Frame Reward Model

- 作者: 作者信息未提供
- 年份: 2026; 主题: Agent RL; 优先级: 扩展
- [来源](#)

论文: Thinking with Frames: Generative Video Distortion Evaluation via Frame Reward Model

快速理解: Recent advances in video reward models and post-training strategies have improved text-to-video (T2V) generation. While these models typically assess visual quality, motion quality, and text alignment, they often overlook key structural distortions, such as abnormal object appearances and interactions, which can degrade the overall quality of the generative video.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.2.609 Thinking with Video: Video Generation as a Promising Multimodal Reasoning Paradigm

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Thinking with Video: Video Generation as a Promising Multimodal Reasoning Paradigm

快速理解: The “Thinking with Text” and “Thinking with Images” paradigms significantly improve the reasoning abilities of large language models (LLMs) and Vision-Language Models (VLMs). However, these paradigms have inherent limitations. (1) Images capture only single moments and fail to represent dynamic processes or continuous changes, and (2) The separation of text and vision as distinct modalities, which hinders unified multimodal understanding and generation.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.610 TimeLens: Rethinking Video Temporal Grounding with Multimodal LLMs

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：TimeLens: Rethinking Video Temporal Grounding with Multimodal LLMs

快速理解：This paper does not introduce a novel method but instead establishes a straightforward, incremental, yet essential baseline for video temporal grounding (VTG), a core capability in video understanding. While multimodal large language models (MLLMs) excel at various video understanding tasks, the recipes for optimizing them for VTG remain under-explored.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.611 TimeViper: A Hybrid Mamba-Transformer Vision-Language Model for Efficient Long Video Understanding

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：TimeViper: A Hybrid Mamba-Transformer Vision-Language Model for Efficient Long Video Understanding

快速理解：We introduce TimeViper, a hybrid vision-language model designed to tackle challenges of long video understanding. Processing long videos demands both an efficient model architecture and an effective mechanism for handling extended temporal contexts.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.612 Tokenization Allows Multimodal Large Language Models to Understand, Generate and Edit Architectural Floor Plans

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Tokenization Allows Multimodal Large Language Models to Understand, Generate and Edit Architectural Floor Plans

快速理解：Architectural floor plan design demands joint reasoning over geometry, semantics, and spatial hierarchy, which remains a major challenge for current AI systems. Although recent diffusion and language models improve visual fidelity, they still struggle with coherent spatial reasoning and controllable generation.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.613 Topology-aware Feature Propagation for Unsupervised Non-rigid Point Cloud Correspondence

- 作者：作者信息未提供
- 年份：2026；主题：Systems；优先级：扩展
- [来源](#)

论文：Topology-aware Feature Propagation for Unsupervised Non-rigid Point Cloud Correspondence

快速理解：Unsupervised non-rigid point cloud correspondence aims to predict point-to-point correspondences without annotations. Existing methods leverage the spatial-relationbased feature propagation strategy that includes nonphysical connections, which are sensitive to non-rigid deformation.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.2.614 Towards Dynamic Modality Alignment in Multimodal Continual Learning

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Towards Dynamic Modality Alignment in Multimodal Continual Learning

快速理解：Multimodal Continual Learning (MMCL) aims to enable models to continuously accumulate knowledge across multiple tasks and modalities without forgetting prior information. MMCL presents more challenges than single-modal continual learning, as it requires effective cooperation and complementarity between modalities.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.615 Towards GUI Agents: Vision-Language Diffusion Models for GUI Grounding

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Towards GUI Agents: Vision-Language Diffusion Models for GUI Grounding

快速理解：Autoregressive (AR) vision-language models (VLMs) have long dominated multimodal understanding, reasoning, and graphical user interface (GUI) grounding. Recently, discrete diffusion vision-language models (DVLMS) have shown strong performance in multimodal reasoning, offering bidirectional attention, parallel token generation, and iterative refinement.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.616 Towards Generalized Multimodal Homography Estimation

- 作者：作者信息未提供

- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Towards Generalized Multimodal Homography Estimation

快速理解: Supervised and unsupervised homography estimation methods depend on image pairs tailored to specific modalities to achieve high accuracy. However, their performance deteriorates substantially when applied to unseen modalities.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.617 Towards Highly Transferable Vision-Language Attack via Semantic-Augmented Dynamic Contrastive Interaction

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Towards Highly Transferable Vision-Language Attack via Semantic-Augmented Dynamic Contrastive Interaction

快速理解: With the rapid advancement and widespread application of vision-language pre-training (VLP) models, their vulnerability to adversarial attacks has become a critical concern. In general, the adversarial examples can typically be designed to exhibit transferable power, attacking not only different models but also across diverse tasks.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.618 Towards Motion Turing Test: Evaluating Human-Likeness in Humanoid Robots

- 作者: 作者信息未提供
- 年份: 2026; 主题: Embodied/WBC; 优先级: 扩展
- [来源](#)

论文: Towards Motion Turing Test: Evaluating Human-Likeness in Humanoid Robots

快速理解: Humanoid robots have achieved significant progress in motion generation and control, exhibiting movements that appear increasingly natural and human-like. Inspired by the Turing Test, we propose the Motion Turing Test, a framework that evaluates whether human observers can discriminate between humanoid robot and human poses using only kinematic information.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制。

证据层级: 本地 PDF 前页。

1.3.2.619 Towards Open Environments and Instructions: General Vision-Language Navigation via Fast-Slow Interactive Reasoning

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Towards Open Environments and Instructions: General Vision-Language Navigation via Fast-Slow Interactive Reasoning

快速理解: Vision-Language Navigation (VLN) aims to enable agents to navigate to a target location based on language instructions. Traditional VLN often follows a close-set assumption, i.e., training and test data share the same style of the input images and instructions.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.620 Towards Open-Vocabulary Industrial Defect Understanding with a Large-Scale Multimodal Dataset

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Towards Open-Vocabulary Industrial Defect Understanding with a Large-Scale Multimodal Dataset

快速理解: We present IMDD-1M, the first large-scale Industrial Multimodal Defect Dataset comprising 1,000,000 aligned image-text pairs, designed to advance multimodal learning for manufacturing and quality inspection. IMDD- 1M contains high-resolution real-world defects spanning over 60 material categories and more than 400 defect types, each accompanied by expert-verified annotations and fine-grained textual descriptions detailing defect location, severity, and contextual attributes.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.621 Towards Reasoning-Preserving Unlearning in Multimodal Large Language Models

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Towards Reasoning-Preserving Unlearning in Multimodal Large Language Models

快速理解: Machine unlearning aims to erase requested data from trained models without full retraining. For Reasoning Multimodal Large Language Models (RMMLMs), this is uniquely challenging: intermediate chain-of-thought steps can still leak sensitive information even when final answers are forgotten, and overly aggressive interventions easily damage general reasoning ability.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.622 Towards Robust Multimodal Large Language Models Against Jailbreak Attacks

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Towards Robust Multimodal Large Language Models Against Jailbreak Attacks

快速理解: While multimodal large language models (MLLMs) have achieved remarkable success in recent advancements, their susceptibility to jailbreak attacks has come to light. In such attacks, adversaries exploit carefully crafted prompts to coerce models into generating harmful or undesirable content.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.623 Towards an Incremental Unified Multimodal Anomaly Detection: Augmenting Multimodal Denoising From an Information Bottleneck Perspective

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Towards an Incremental Unified Multimodal Anomaly Detection: Augmenting Multimodal Denoising From an Information Bottleneck Perspective

快速理解: The quest for incremental unified multimodal anomaly detection seeks to empower a single model with the ability to systematically detect anomalies across all categories and support incremental learning to accommodate emerging objects/categories. Central to this pursuit is resolving the catastrophic forgetting dilemma, which involves acquiring new knowledge while preserving prior learned knowledge.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.624 TraceGen: World Modeling in 3D Trace Space Enables Learning from Cross-Embodiment Videos

- 作者: 作者信息未提供
- 年份: 2026; 主题: World Model; 优先级: 扩展
- [来源](#)

论文: TraceGen: World Modeling in 3D Trace Space Enables Learning from Cross-Embodiment Videos

快速理解: Learning new robot tasks on new platforms and in new scenes from only a handful of demonstrations remains challenging. While videos of other embodiments—humans and different robots—are abundant, differences in embodiment, camera, and environment hinder their direct use.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策。

证据层级: 本地 PDF 前页。

1.3.2.625 Training One Model to Master Cross-Level Agentic Actions via Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2026; 主题: Agent RL; 优先级: 扩展
- [来源](#)

论文: Training One Model to Master Cross-Level Agentic Actions via Reinforcement Learning

快速理解: The paradigm of agentic AI is shifting from engineered complex workflows to post-training native models. However, existing agents are typically confined to static, predefined action spaces—such as exclusively using APIs, GUI events, or robotic commands.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.2.626 Training-free Motion Factorization for Compositional Video Generation

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Training-free Motion Factorization for Compositional Video Generation

快速理解: Compositional video generation aims to synthesize multiple instances with diverse appearance and motion. However, current approaches mainly focus on binding semantics, neglecting to understand diverse motion categories specified in prompts.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.627 Transform to Transfer: Boosting Adversarial Attack Transferability on Vision-Language Pre-training Models

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Transform to Transfer: Boosting Adversarial Attack Transferability on Vision-Language Pre-training Models

快速理解: Vision-Language Pre-training (VLP) models, while achieving state-of-the-art performance on various multimodal tasks, exhibit significant vulnerability to multimodal adversarial examples. In black-box attack scenarios of VLP models, a key challenge lies in the limited transferability of these adversarial examples.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.628 Transition Matching Distillation for Fast Video Generation

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Transition Matching Distillation for Fast Video Generation

快速理解: Large video diffusion and flow models have achieved remarkable success in high-quality video generation, but their use in real-time interactive applications remains limited due to their inefficient multi-step sampling process. In this work, we present Transition Matching Distillation

(TMD), a novel framework for distilling video diffusion models into efficient few-step generators.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.629 TreeTeaming: Autonomous Red-Teaming of Vision-Language Models via Hierarchical Strategy Exploration

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: TreeTeaming: Autonomous Red-Teaming of Vision-Language Models via Hierarchical Strategy Exploration

快速理解: The rapid advancement of Vision-Language Models (VLMs) has brought their safety vulnerabilities into sharp focus. However, existing red teaming methods are fundamentally constrained by an inherent linear exploration paradigm, confining them to optimizing within a predefined strategy set and preventing the discovery of novel, diverse exploits.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.630 Tri-Subspaces Disentanglement for Multimodal Sentiment Analysis

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Tri-Subspaces Disentanglement for Multimodal Sentiment Analysis

快速理解: Multimodal Sentiment Analysis (MSA) integrates language, visual, and acoustic modalities to infer human sentiment. Most existing methods either focus on globally shared representations or modality-specific features, while overlooking signals that are shared only by certain modality pairs.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.631 Tutor-Student Reinforcement Learning: A Dynamic Curriculum for Robust Deepfake Detection

- 作者: 作者信息未提供
- 年份: 2026; 主题: Agent RL; 优先级: 扩展
- [来源](#)

论文: Tutor-Student Reinforcement Learning: A Dynamic Curriculum for Robust Deepfake Detection

快速理解: Standard supervised training for deepfake detection treats all samples with uniform importance, which can be suboptimal for learning robust and generalizable features. In this work, we propose a novel Tutor-Student Reinforcement Learning (TSRL) framework to dynamically optimize the training curriculum.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.2.632 U4D: Uncertainty-Aware 4D World Modeling from LiDAR Sequences

- 作者：作者信息未提供
- 年份：2026；主题：World Model；优先级：扩展
- [来源](#)

论文：U4D: Uncertainty-Aware 4D World Modeling from LiDAR Sequences

快速理解：Modeling dynamic 3D environments from LiDAR sequences is central to building reliable 4D worlds for autonomous driving and embodied AI. Existing generative frameworks, however, often treat all spatial regions uniformly, overlooking the varying uncertainty across real-world scenes.

关注点：环境动力学、视频预测、潜变量状态或模型式决策。

证据层级：本地 PDF 前页。

1.3.2.633 UETTrack: A Unified and Efficient Framework for Single Object Tracking

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：UETTrack: A Unified and Efficient Framework for Single Object Tracking

快速理解：With growing real-world demands, efficient tracking has received increasing attention. However, most existing methods are limited to RGB inputs and struggle in multimodal scenarios.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.634 UNI-OOD: Unified Object- and Image-level Out-of-Distribution Detection via Cross-Context Attentive Vision-Language Modeling

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：UNI-OOD: Unified Object- and Image-level Out-of-Distribution Detection via Cross-Context Attentive Vision-Language Modeling

快速理解：Out-of-distribution (OOD) detection is a key requirement for reliable deployment in open-world environments, where a model must recognize inputs that fall outside the semantic scope of known concepts. While recent advances in visionlanguage models (VLMs) have achieved strong results in image-level OOD detection, most methods still assume that each image contains a single dominant object.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.635 UST-Hand: An Uncertainty-aware Spatiotemporal Point Cloud Interaction Network for 3D Self-supervised Hand Pose Estimation

- 作者：作者信息未提供
- 年份：2026；主题：Systems；优先级：扩展
- [来源](#)

论文：UST-Hand: An Uncertainty-aware Spatiotemporal Point Cloud Interaction Network for 3D Self-supervised Hand Pose Estimation

快速理解：Manually annotating accurate 3D hand poses is extremely time-consuming and labor-intensive. Existing selfsupervised hand pose estimation methods leverage the discrepancy between input images and rendered outputs, or multi-view consistency constraints, as the driving force to optimize networks and progressively refine pose accuracy.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.2.636 Uncertainty-Aware Knowledge Distillation for Multimodal Large Language Models

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Uncertainty-Aware Knowledge Distillation for Multimodal Large Language Models

快速理解：Knowledge distillation establishes a learning paradigm that learns from both data supervision and teacher guidance. However, the optimal balance between learning from data and learning from the teacher is hard to determine, as some samples are data-noisy while others are teacher-uncertain.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.637 Understanding Counting Mechanisms in Large Language and Vision-Language Models

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Understanding Counting Mechanisms in Large Language and Vision-Language Models

快速理解：Counting is one of the fundamental abilities of large language models (LLMs) and large vision-language models (LVLMs). This paper examines how these foundation models represent and compute numerical information in counting tasks.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.638 Understanding and Mitigating Hallucinations in Multimodal Chain-of-Thought Models

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展

- [来源](#)

论文: Understanding and Mitigating Hallucinations in Multimodal Chain-of-Thought Models

快速理解: Multimodal Chain-of-Thought (MCoT) models have demonstrated impressive capability in complex visual reasoning tasks. Unfortunately, recent studies reveal that they suffer from severe hallucination problems due to diminished visual attention during the generation process.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.639 Understanding the Role of Hallucination in Reinforcement Post-Training of Multimodal Reasoning Models

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Understanding the Role of Hallucination in Reinforcement Post-Training of Multimodal Reasoning Models

快速理解: The recent success of reinforcement learning (RL) in large reasoning models has inspired the growing adoption of RL for post-training Multimodal Large Language Models (MLLMs) to enhance their visual reasoning capabilities. Although many studies have reported improved performance, it remains unclear whether RL training truly enables models to learn from visual information.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.640 Uni3R: Unified 3D Reconstruction and Semantic Understanding via Generalizable Gaussian Splatting from Unposed Multi-View Images

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Uni3R: Unified 3D Reconstruction and Semantic Understanding via Generalizable Gaussian Splatting from Unposed Multi-View Images

快速理解: Reconstructing and semantically interpreting 3D scenes from sparse 2D views remains a fundamental challenge in computer vision. Conventional methods often decouple semantic understanding from reconstruction or necessitate costly per-scene optimization, thereby restricting their scalability and generalizability.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.641 UniAVGen: Unified Audio and Video Generation with Asymmetric Cross-Modal Interactions

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: UniAVGen: Unified Audio and Video Generation with Asymmetric Cross-Modal Interactions

快速理解: Due to the lack of effective cross-modal modeling, existing open-source audio-video generation methods often exhibit compromised lip synchronization and insufficient semantic consistency. To mitigate these drawbacks, we propose UniAVGen, a unified framework for human-centric joint audio and video generation.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.642 UniCompress: Token Compression for Unified Vision-Language Understanding and Generation

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: UniCompress: Token Compression for Unified Vision-Language Understanding and Generation

快速理解: Unified models aim to support both understanding and generation by encoding images into discrete tokens and processing them alongside text within a single autoregressive framework. This unified design offers architectural simplicity and cross-modal synergy, which facilitates shared parameterization, consistent training objectives, and seamless transfer between modalities.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.643 UniDAC: Universal Metric Depth Estimation for Any Camera

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: UniDAC: Universal Metric Depth Estimation for Any Camera

快速理解: Monocular metric depth estimation (MMDE) is a core challenge in computer vision, playing a pivotal role in realworld applications that demand accurate spatial understanding. Although prior works have shown promising zeroshot performance in MMDE, they often struggle with generalization across diverse camera types, such as fisheye and 360° cameras.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.644 UniDex: A Robot Foundation Suite for Universal Dexterous Hand Control from Egocentric Human Videos

- 作者: 作者信息未提供
- 年份: 2026; 主题: Embodied/WBC; 优先级: 扩展
- [来源](#)

论文: UniDex: A Robot Foundation Suite for Universal Dexterous Hand Control from Egocentric Human Videos

快速理解: Dexterous manipulation remains challenging due to the cost of collecting real-robot teleoperation data, the heterogeneity of hand embodiments, and the high dimensionality of control. We present UniDex, a robot foundation suite that couples a large-scale robot-centric dataset with a unified vision-language-action (VLA) policy and a practical human-data capture setup for universal dexterous hand control.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制。

证据层级: 本地 PDF 前页。

1.3.2.645 UniGame: Turning a Unified Multimodal Model Into Its Own Adversary

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: UniGame: Turning a Unified Multimodal Model Into Its Own Adversary

快速理解: Unified Multimodal Models (UMMs) have shown impressive performance in both understanding and generation with a single architecture. However, UMMs still exhibit a fundamental inconsistency: understanding favors compact embeddings, whereas generation favors reconstruction-rich representations.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.646 UniM: A Unified Any-to-Any Interleaved Multimodal Benchmark

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: UniM: A Unified Any-to-Any Interleaved Multimodal Benchmark

快速理解: In real-world multimodal applications, systems usually need to comprehend arbitrarily combined and interleaved multimodal inputs from users, while also generating outputs in any interleaved multimedia form. This capability defines the goal of any-to-any interleaved multimodal learning under a unified paradigm of understanding and generation, posing new challenges and opportunities for advancing Multimodal Large Language Models (MLLMs).

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.647 Unified Camera Positional Encoding for Controlled Video Generation

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Unified Camera Positional Encoding for Controlled Video Generation

快速理解: Transformers have emerged as a universal backbone across 3D perception, video generation, and world models for autonomous driving and embodied AI, where understanding camera geometry is essential for grounding visual observations in three-dimensional space. However, existing camera

encoding methods often rely on simplified pinhole assumptions, restricting generalization across the diverse intrinsics and lens distortions in real-world cameras.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.648 Unified Customized Generation by Disentangled Reward Modeling

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL；优先级：扩展
- [来源](#)

论文：Unified Customized Generation by Disentangled Reward Modeling

快速理解：Existing literature typically treats various customized generation tasks (e.g., subject-customized generation, style-customized generation) as distinct and disjoint problems, with each task focusing solely on customizing a specific aspect of the reference image. →Corresponding author. Email: huangmq@ustc.edu.cn.
† Project lead

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.2.649 Unified Multimodal Models as Auto-Encoders

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Unified Multimodal Models as Auto-Encoders

快速理解：Image-to-text (I2T) understanding and text-to-image (T2I) generation are two fundamental, important yet traditionally isolated multimodal tasks. Despite their intrinsic connection, existing approaches typically optimize them independently, missing the opportunity for mutual enhancement.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.650 UnityVideo: Unified Multi-Modal Multi-Task Learning for Enhancing World-Aware Video Generation

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：UnityVideo: Unified Multi-Modal Multi-Task Learning for Enhancing World-Aware Video Generation

快速理解：Recent video generation models demonstrate impressive synthesis capabilities but remain limited by single-modality conditioning, constraining their holistic world understanding. This stems from insufficient cross-modal interaction and limited modal diversity for comprehensive world knowledge representation.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.651 Unlearning without Forgetting: Securely Removing Targeted Concepts from Large-Scale Vision-Language Open-Vocabulary Detectors

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Unlearning without Forgetting: Securely Removing Targeted Concepts from Large-Scale Vision-Language Open-Vocabulary Detectors

快速理解：Open-vocabulary detectors (OvOD) inherit tightly coupled cross-modal knowledge from web-scale pretraining, creating privacy, copyright, and compliance risks. Existing machine unlearning methods face geometric entanglement interference in OvOD: forgetting updates inevitably distort preserved knowledge due to shared semantic factors in decomposable embeddings.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.652 Unleashing VLA Potentials in Autonomous Driving via Explicit Learning from Failures

- 作者：作者信息未提供
- 年份：2026；主题：VLA；优先级：扩展
- [来源](#)

论文：Unleashing VLA Potentials in Autonomous Driving via Explicit Learning from Failures

快速理解：Vision-Language-Action (VLA) models for autonomous driving often hit a performance plateau during Reinforcement Learning (RL) optimization. This stagnation arises from exploration capabilities constrained by previous Supervised Fine-Tuning (SFT), leading to “persistent failures” in long-tail scenarios.

关注点：视觉、语言条件与机器人动作策略之间的对齐。

证据层级：本地 PDF 前页。

1.3.2.653 Unleashing the Intrinsic Visual Representation Capability of Multimodal Large Language Models

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Unleashing the Intrinsic Visual Representation Capability of Multimodal Large Language Models

快速理解：Multimodal Large Language Models (MLLMs) have demonstrated remarkable proficiency in multimodal tasks. Despite their impressive performance, MLLMs suffer from the modality imbalance issue, where visual information is often underutilized compared to textual representations in deeper layers, leading to degraded visual performance or hallucinations.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.654 Unsupervised Multi-agent and Single-agent Perception from Cooperative Views

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL；优先级：扩展
- [来源](#)

论文：Unsupervised Multi-agent and Single-agent Perception from Cooperative Views

快速理解：The LiDAR-based multi-agent and single-agent perception has shown promising performance in environmental understanding for robots and automated vehicles. However, there is no existing method that simultaneously solves both multi-agent and single-agent perception in an unsupervised way.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.2.655 VGG-T³: Offline Feed-Forward 3D Reconstruction at Scale

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：VGG-T³: Offline Feed-Forward 3D Reconstruction at Scale

快速理解：We present a scalable 3D reconstruction model that addresses a critical limitation in offline feed-forward methods: their computational and memory requirements grow quadratically w.r.t. the number of input images. Our approach is built on the key insight that this bottleneck stems from the varying-length Key-Value (KV) space representation of scene geometry, which we distill into a fixed-size Multi-Layer Perceptron (MLP) via test-time training.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.656 VGGDrive: Empowering Vision-Language Models with Cross-View Geometric Grounding for Autonomous Driving

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：VGGDrive: Empowering Vision-Language Models with Cross-View Geometric Grounding for Autonomous Driving

快速理解：The significance of cross-view 3D geometric modeling capabilities for autonomous driving is self-evident, yet existing Vision-Language Models (VLMs) inherently lack this capability, resulting in their mediocre performance. While some promising approaches attempt to mitigate this by constructing Q&A data for auxiliary training, they still fail to fundamentally equip VLMs with the ability to comprehensively handle diverse evaluation protocols.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.657 VGGT-360: Geometry-Consistent Zero-Shot Panoramic Depth Estimation

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：VGGT-360: Geometry-Consistent Zero-Shot Panoramic Depth Estimation

快速理解：This paper presents VGGT-360, a novel training-free framework for zero-shot, geometry-consistent panoramic depth estimation. Unlike prior view-independent training-free approaches, VGGT-360 reformulates the task as panoramic reprojection over multi-view reconstructed 3D models by leveraging the intrinsic 3D consistency of VGGT-like foundation models, thereby unifying fragmented per-view reasoning into a coherent panoramic understanding.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.658 VKG-QA: Visual Knowledge Graph-based Question Answer for Large Multimodal Models

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：VKG-QA: Visual Knowledge Graph-based Question Answer for Large Multimodal Models

快速理解：Understanding and reasoning over structured knowledge is a fundamental capability for intelligent systems. While Large Language Models (LLMs) have leveraged textual Knowledge Graphs (KGs) for relational reasoning, linearizing graph structures into text leads to loss of higher-order relational cues.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.659 VL-RouterBench: A Benchmark for Vision-Language Model Routing

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：VL-RouterBench: A Benchmark for Vision-Language Model Routing

快速理解：Multi-model routing has evolved from an engineering technique into essential infrastructure, yet existing work lacks a systematic, reproducible benchmark for evaluating vision-language models (VLMs). We present VL- RouterBench to assess the overall capability of VLM routing systems systematically.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.660 VLA Models Are More Generalizable Than You Think: Revisiting Physical and Spatial Modeling

- 作者：作者信息未提供
- 年份：2026；主题：VLA；优先级：扩展
- [来源](#)

论文：VLA Models Are More Generalizable Than You Think: Revisiting Physical and Spatial Modeling

快速理解：Vision-language-action (VLA) models achieve strong indistribution performance but degrade sharply under novel camera viewpoints and visual perturbations. We show that this brittleness primarily arises from misalignment in Spatial Modeling, rather than Physical Modeling.

关注点：视觉、语言条件与机器人动作策略之间的对齐。

证据层级：本地 PDF 前页。

1.3.2.661 VLIC: Vision-Language Models As Perceptual Judges for Human-Aligned Image Compression

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：VLIC: Vision-Language Models As Perceptual Judges for Human-Aligned Image Compression

快速理解：Evaluations of image compression performance which include human preferences have generally found that naive distortion functions such as MSE are insufficiently aligned to human perception. In order to align compression models to human perception, prior work has employed differentiable perceptual losses consisting of neural networks calibrated on large-scale datasets of human psycho-visual judgments.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.662 VLM4RSDet: Collaborative Optimization with Vision-Language Model for Enhancing Remote Sensing Object Detection

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：VLM4RSDet: Collaborative Optimization with Vision-Language Model for Enhancing Remote Sensing Object Detection

快速理解：Closed-set object detection in remote sensing imagery has made significant progress, but achieving high detection accuracy remains challenging. Vision-Language Models (VLMs), which possess rich prior knowledge, offer a promising solution to this challenge.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.663 VOLD: Reasoning Transfer from LLMs to Vision-Language Models via On-Policy Distillation

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：VOLD: Reasoning Transfer from LLMs to Vision-Language Models via On-Policy Distillation

快速理解：Training vision-language models (VLMs) for complex reasoning remains a challenging task, i.a. due to the scarcity of high-quality image-text reasoning data. Conversely, text-based reasoning resources are abundant and scalable, but it is still an open question how to leveraging them for VLM reasoning.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.664 VS-Bench: Evaluating VLMs for Strategic Abilities in Multi-Agent Environments

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL；优先级：扩展
- [来源](#)

论文：VS-Bench: Evaluating VLMs for Strategic Abilities in Multi-Agent Environments

快速理解：Recent advancements in Vision Language Models (VLMs) have expanded their capabilities to interactive agent tasks, yet existing benchmarks remain limited to single-agent or text-only environments. In contrast, real-world scenarios often involve multiple agents interacting within rich visual and textual contexts, posing challenges with both multimodal observations and strategic interactions.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.2.665 VULCAN: Tool-Augmented Multi Agents for Iterative 3D Object Arrangement

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL；优先级：扩展
- [来源](#)

论文：VULCAN: Tool-Augmented Multi Agents for Iterative 3D Object Arrangement

快速理解：Despite the remarkable progress of Multimodal Large Language Models (MLLMs) in 2D vision-language tasks, their application to complex 3D scene manipulation remains underexplored. In this paper, we bridge this critical gap by tackling three key challenges in 3D object arrangement task using MLLMs.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.2.666 VerseCrafter: Dynamic Realistic Video World Model with 4D Geometric Control

- 作者：作者信息未提供
- 年份：2026；主题：World Model；优先级：扩展
- [来源](#)

论文：VerseCrafter: Dynamic Realistic Video World Model with 4D Geometric Control

快速理解：Video world models aim to simulate dynamic, real-world environments, yet existing methods struggle to provide unified and precise control over camera and multi-object motion, as videos inherently capture dynamics in the projected 2D image plane. To bridge this gap, we introduce VerseCrafter, a geometry-driven video world model that generates dynamic, realistic videos from a unified 4D geometric world state.

关注点：环境动力学、视频预测、潜变量状态或模型式决策。

证据层级：本地 PDF 前页。

1.3.2.667 VidPrism: Heterogeneous Mixture of Experts for Image-to-Video Transfer

- 作者：作者信息未提供
- 年份：2026；主题：AI Infra；优先级：扩展
- [来源](#)

论文：VidPrism: Heterogeneous Mixture of Experts for Image-to-Video Transfer

快速理解：With the rapid development of pre-training technologies, adapting large-scale Vision-Language Models (VLMs) for video understanding i.e. image-to-video transfer learning has become a dominant paradigm. To achieve superior performance, it raises as an effective strategy among recent advances to employ Mixture-of-Experts (MoE) to enhance VLMs' temporal modeling capabilities.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.2.668 Video Generation with Stable Transparency via Shiftable RGB-A Distribution Learner

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Video Generation with Stable Transparency via Shiftable RGB-A Distribution Learner

快速理解：Generating RGB-A videos, which include alpha channels for transparency, has wide applications. However, current methods often suffer from low quality due to confusion between RGB and alpha.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.669 Video-Only ToM: Enhancing Theory of Mind in Multimodal Large Language Models

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Video-Only ToM: Enhancing Theory of Mind in Multimodal Large Language Models

快速理解：As large language models (LLMs) continue to advance, there is increasing interest in their ability to infer human mental states and demonstrate a human-like Theory of Mind (ToM). Most existing ToM evaluations, however, are centered on text-based inputs, while scenarios relying solely on visual information receive far less attention.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.670 Video2Robo: 3DGS-based Synthetic Data from One Video Enables Scalable Robot Learning

- 作者：作者信息未提供
- 年份：2026；主题：Robot Learning；优先级：扩展
- [来源](#)

论文：Video2Robo: 3DGS-based Synthetic Data from One Video Enables Scalable Robot Learning

快速理解：Scalable robot learning is hindered by the high cost of acquiring diverse, high-quality embodied data. Existing data generation approaches partially mitigate this issue but typically depend on hard-to-access hardware and laborintensive manual effort, with limited generalization to diverse scene configurations.

关注点：机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：本地 PDF 前页。

1.3.2.671 VideoARM: Agentic Reasoning over Hierarchical Memory for Long-Form Video Understanding

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL；优先级：扩展
- [来源](#)

论文：VideoARM: Agentic Reasoning over Hierarchical Memory for Long-Form Video Understanding

快速理解：Long-form video understanding remains challenging due to the extended temporal structure and dense multimodal cues. Despite recent progress, many existing approaches still rely on hand-crafted reasoning pipelines or employ token-consuming video preprocessing to guide MLLMs in autonomous reasoning.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.2.672 VideoITG: Multimodal Video Understanding with Instructed Temporal Grounding

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展

- [来源](#)

论文: VideoITG: Multimodal Video Understanding with Instructed Temporal Grounding

快速理解: While Video Large Language Models (Video-LLMs) have shown significant potential in multimodal understanding and reasoning tasks, how to efficiently select the most informative frames from videos remains a critical challenge. Existing methods attempt to optimize frame sampling by reducing inter-frame redundancy or employing unsupervised event localization.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.673 VinQA: Visual Elements Interleaved Long-form Answer Generation for Real-World Multimodal Document QA

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: VinQA: Visual Elements Interleaved Long-form Answer Generation for Real-World Multimodal Document QA

快速理解: Real-world documents combine text with tables, charts, photographs, and diagrams arranged in diverse layouts, yet existing research on multimodal large language models (MLLMs) for document QA predominantly produces textonly responses, underutilizing these visual elements. We introduce VinQA, a dataset designed for long-form answer generation where cited visual elements are explicitly interleaved with their supporting text and grounded in relevant document pages.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.674 VisMem: Latent Vision Memory Unlocks Potential of Vision-Language Models

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: VisMem: Latent Vision Memory Unlocks Potential of Vision-Language Models

快速理解: Despite the remarkable success of Vision-Language Models (VLMs), their performance on a range of complex visual tasks is often hindered by a “visual processing bottleneck”: a propensity to lose grounding in visual evidence and exhibit a deficit in contextualized visual experience during prolonged generation. Drawing inspiration from human cognitive memory theory, which distinguishes short-term visually-dominant memory and longterm semantically-dominant memory, we propose VisMem, a cognitively-aligned framework that equips VLMs with dynamic latent vision memories, a short-term module for finegrained perceptual retention and a long-term module for abstract semantic consolidation.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.675 VisPlay: Self-Evolving Vision-Language Models

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：VisPlay: Self-Evolving Vision-Language Models

快速理解：Reinforcement learning (RL) provides a principled framework for improving Vision-Language Models (VLMs) on complex reasoning tasks. However, existing RL approaches often depend on human-annotated labels or task-specific heuristics to define verifiable rewards—both costly and limited in scalability.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.676 Vision-Language Model Guided Source-Free Domain Adaptation via Optimal Transport

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Vision-Language Model Guided Source-Free Domain Adaptation via Optimal Transport

快速理解：Unsupervised domain adaptation transfers knowledge from a labeled source domain to an unlabeled target domain. When source data cannot be accessed, source-free domain adaptation (SFDA) becomes a practical alternative.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.677 Vista4D: Video Reshooting with 4D Point Clouds

- 作者：作者信息未提供
- 年份：2026；主题：Systems；优先级：扩展
- [来源](#)

论文：Vista4D: Video Reshooting with 4D Point Clouds

快速理解：We present Vista4D, a robust and flexible video reshooting framework that grounds the input video and target cameras in a 4D point cloud. Specifically, given an input video, our method re-synthesizes the scene with the same dynamics from a different camera trajectory and viewpoint.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.2.678 Visual Document Understanding and Reasoning: A Multi-Agent Collaboration Framework with Agent-Wise Adaptive Test-Time Scaling

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL；优先级：扩展
- [来源](#)

论文: Visual Document Understanding and Reasoning: A Multi-Agent Collaboration Framework with Agent-Wise Adaptive Test-Time Scaling

快速理解: The dominant paradigm of monolithic scaling in Vision- Language Models (VLMs) is failing for understanding and reasoning in documents, yielding diminishing returns as it struggles with the inherent need of this domain for document-based procedural reasoning, cognitive complexity, and factual accuracy. To this end, we introduce MACT, a Multi-Agent Collaboration framework with agent-wise adaptive Test-time scaling that pioneers a paradigm shift to procedural scaling, adapting dynamically to the functional entities of visual documents understanding and reasoning.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.2.679 W2W: Language-Model-Based Trajectory Prediction with Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2026; 主题: Agent RL; 优先级: 扩展
- [来源](#)

论文: W2W: Language-Model-Based Trajectory Prediction with Reinforcement Learning

快速理解: Pedestrian trajectory prediction is crucial for applications such as autonomous driving and social robots. Recently, language model (LM)-based trajectory prediction has offered both prediction accuracy and interpretability.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.2.680 WalkGPT: Grounded Vision-Language Conversation with Depth-Aware Segmentation for Pedestrian Navigation

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: WalkGPT: Grounded Vision-Language Conversation with Depth-Aware Segmentation for Pedestrian Navigation

快速理解: Ensuring accessible pedestrian navigation requires reasoning about both semantic and spatial aspects of complex urban scenes, a challenge that existing Large Vision-Language Models (LVLMs) struggle to meet. Although these models can describe visual content, their lack of explicit grounding leads to object hallucinations and unreliable depth reasoning, limiting their usefulness for accessibility guidance.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.681 Wanderland: Geometrically Grounded Simulation for Open-World Embodied AI

- 作者: 作者信息未提供

- 年份: 2026; 主题: Embodied/WBC; 优先级: 扩展
- [来源](#)

论文: Wanderland: Geometrically Grounded Simulation for Open-World Embodied AI

快速理解: Reproducible closed-loop evaluation remains a major bottleneck in Embodied AI such as visual navigation. A promising path forward is high-fidelity simulation that combines photorealistic sensor rendering with geometrically grounded interaction in complex, open-world urban environments.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制。

证据层级: 本地 PDF 前页。

1.3.2.682 Wave-Former: Through-Occlusion 3D Reconstruction via Wireless Shape Completion

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Wave-Former: Through-Occlusion 3D Reconstruction via Wireless Shape Completion

快速理解: We present Wave-Former, a novel method capable of high-accuracy 3D shape reconstruction for completely occluded, diverse, everyday objects. This capability can open new applications spanning robotics, augmented reality, and logistics.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.683 WeMMU: Enhanced Bridging of Vision-Language Models and Diffusion Models via Noisy Query Tokens

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: WeMMU: Enhanced Bridging of Vision-Language Models and Diffusion Models via Noisy Query Tokens

快速理解: Recent progress in multimodal large language models (MLLMs) has highlighted the challenge of efficiently bridging pre-trained Vision-Language Models (VLMs) with Diffusion Models. While methods using a fixed number of learnable query tokens offer computational efficiency, they suffer from task generalization collapse, failing to adapt to new tasks that are distant from their pre-training tasks.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.684 What Are You Doing? A Closer Look at Controllable Human Video Generation

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展

- [来源](#)

论文: What Are You Doing? A Closer Look at Controllable Human Video Generation

快速理解: High-quality benchmarks are crucial for driving progress in machine learning research. However, despite the growing interest in video generation, there is no comprehensive dataset to evaluate human synthesis.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.685 What Do Visual Tokens Really Encode? Uncovering Sparsity and Redundancy in Multimodal Large Language Models

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: What Do Visual Tokens Really Encode? Uncovering Sparsity and Redundancy in Multimodal Large Language Models

快速理解: Multimodal large language models (MLLMs) project visual tokens into the embedding space of language models, yet the internal structuring and processing of visual semantics remain poorly understood. In this work, we introduce a two-fold analytical framework featuring a novel probing tool, EmbedLens, to conduct a fine-grained analysis.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.686 Which Concepts to Forget and How to Refuse? Decomposing Concepts for Continual Unlearning in Large Vision-Language Models

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Which Concepts to Forget and How to Refuse? Decomposing Concepts for Continual Unlearning in Large Vision-Language Models

快速理解: Continual unlearning poses the challenge of enabling large vision-language models to selectively refuse specific imageinstruction pairs in response to sequential deletion requests, while preserving general utility. However, sequential unlearning updates distort shared representations, creating spurious associations between vision-language pairs and refusal behaviors that hinder precise identification of refusal targets, resulting in inappropriate refusals.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.687 White-Balance First, Adjust Later: Cross-Camera Color Constancy via Vision-Language Evaluation

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展

- [来源](#)

论文: White-Balance First, Adjust Later: Cross-Camera Color Constancy via Vision-Language Evaluation

快速理解: Color constancy aims to keep object colors consistent under varying illumination. Cross-camera generalization in color constancy remains challenging because learningbased models often overfit to the color response characteristics of the training camera, resulting in degraded performance on images captured by other cameras.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.688 Will Multimodal Models Be Dazzled by Multi-Image Visual Puzzles?

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Will Multimodal Models Be Dazzled by Multi-Image Visual Puzzles?

快速理解: The rapid advancement of Multimodal Large Language Models (MLLMs) has revealed the limitations of existing benchmarks in evaluating complex reasoning over multiple images. To address this gap, we introduce MIRACLE, a novel benchmark for Multi-Image complex Reasoning And Comprehension Logic Evaluation, featuring 4,000 questions across diverse reasoning types such as visual comparison, temporal sequencing, and spatial relations.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.689 WorldLens: Full-Spectrum Evaluations of Driving World Models in Real World

- 作者: 作者信息未提供
- 年份: 2026; 主题: World Model; 优先级: 扩展
- [来源](#)

论文: WorldLens: Full-Spectrum Evaluations of Driving World Models in Real World

快速理解: Generative world models are reshaping embodied AI, enabling agents to synthesize realistic 4D driving environments that look convincing but often fail physically or behaviorally. Despite rapid progress, the field still lacks a unified way to assess whether generated worlds preserve geometry, obey physics, or support reliable control.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策。

证据层级: 本地 PDF 前页。

1.3.2.690 WorldReel: 4D Video Generation with Consistent Geometry and Motion Modeling

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: WorldReel: 4D Video Generation with Consistent Geometry and Motion Modeling

快速理解: Recent video generators achieve striking photorealism, yet remain fundamentally inconsistent in 3D. We present WorldReel, a 4D video generator that is natively spatiotemporally consistent.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.691 WorldStereo: Bridging Camera-Guided Video Generation and Scene Reconstruction via 3D Geometric Memories

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: WorldStereo: Bridging Camera-Guided Video Generation and Scene Reconstruction via 3D Geometric Memories

快速理解: Recent advances in foundational Video Diffusion Models (VDMs) have yielded significant progress. Yet, despite the remarkable visual quality of generated videos, reconstructing consistent 3D scenes from these outputs remains challenging, due to limited camera controllability and inconsistent generated content when viewed from distinct camera trajectories.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.2.692 X-WIN: Building Chest Radiograph World Model via Predictive Sensing

- 作者: 作者信息未提供
- 年份: 2026; 主题: World Model; 优先级: 扩展
- [来源](#)

论文: X-WIN: Building Chest Radiograph World Model via Predictive Sensing

快速理解: Chest X-ray radiography (CXR) is an essential medical imaging technique for disease diagnosis. However, as 2D projectional images, CXRs are limited by structural superposition and hence fail to capture 3D anatomies.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策。

证据层级: 本地 PDF 前页。

1.3.2.693 ZOO-Prune: Training-Free Token Pruning via Zeroth-Order Gradient Estimation in Vision-Language Models

- 作者: 作者信息未提供
- 年份: 2026; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: ZOO-Prune: Training-Free Token Pruning via Zeroth-Order Gradient Estimation in Vision-Language Models

快速理解: Large Vision-Language Models (VLMs) enable strong multimodal reasoning but incur heavy inference costs from redundant visual tokens. Token pruning alleviates this issue, yet existing

approaches face limitations.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.694 Zero-Shot Depth Completion with Vision-Language Model

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Zero-Shot Depth Completion with Vision-Language Model

快速理解：Vision language models (VLMs) have achieved remarkable success in semantic understanding tasks under language guidance, yet their potential for geometric perception remains largely underexplored. This paper introduces the first VLM-based depth completion framework.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.695 ZipMap: Linear-Time Stateful 3D Reconstruction via Test-Time Training

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：ZipMap: Linear-Time Stateful 3D Reconstruction via Test-Time Training

快速理解：Feed-forward transformer models have driven rapid progress in 3D vision, but state-of-the-art methods such as VGGT and 3 have a computational cost that scales quadratically with the number of input images, making them inefficient when applied to large image collections. Sequential-reconstruction approaches reduce this cost but sacrifice reconstruction quality.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.696 b-CLIP: Text-Conditioned Contrastive Learning for Multi-Granular Vision-Language Alignment

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：b-CLIP: Text-Conditioned Contrastive Learning for Multi-Granular Vision-Language Alignment

快速理解：CLIP achieves strong zero-shot image-text retrieval by aligning global vision and text representations, yet it falls behind on fine-grained tasks even when fine-tuned on long, detailed captions. In this work, we propose -CLIP, a multigranular text-conditioned contrastive learning framework designed to achieve hierarchical alignment between multiple textual granularities—from full captions to sentences and phrases—and their corresponding visual regions.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.697 iLRM: An Iterative Large 3D Reconstruction Model

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：iLRM: An Iterative Large 3D Reconstruction Model

快速理解：Feed-forward 3D modeling has emerged as a promising approach for rapid and high-quality 3D reconstruction. In particular, directly generating explicit 3D representations, such as 3D Gaussian splatting, has attracted significant attention due to its fast and high-quality rendering.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.698 mVLM: A Vision Language Model for mNPUs

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：mVLM: A Vision Language Model for mNPUs

快速理解：The proliferation of low-power intelligent processors with integrated Neural Processing Units (NPUs), called μ NPUs, has created new opportunities for on-device generative AI, benefitting end devices like smart wearables and small robots. However, deploying Vision-Language Models (VLMs) on μ NPUs is severely hindered by stringent memory constraints and limited operator support.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.2.699 tttLRM: Test-Time Training for Long Context and Autoregressive 3D Reconstruction

- 作者：作者信息未提供
- 年份：2026；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：tttLRM: Test-Time Training for Long Context and Autoregressive 3D Reconstruction

快速理解：We propose tttLRM, a novel large 3D reconstruction model that leverages a Test-Time Training (TTT) layer to enable long-context, autoregressive 3D reconstruction with linear computational complexity, further scaling the model's capability. Our framework efficiently compresses multiple image observations into the fast weights of the TTT layer, forming an implicit 3D representation in the latent space that can be decoded into various explicit formats, such as Gaussian Splats (GS) for downstream applications.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.3 AI · CoRL

本会议纳入 **227** 篇。自动主题统计为: Embodied/WBC (124)、Robot Learning (93)、Agent RL (89)、Supporting Perception (67)、VLA (25)、Teleoperation (25)、Systems (22)、World Model (17)、AI Infra (8)。这些统计用于组织阅读顺序, 不能单独证明论文结论或实验优越性。

优先阅读候选: - ClutterDexGrasp: A Sim-to-Real System for General Dexterous Grasping in Cluttered Scenes (优先级分数 147.25) - FLARE: Robot Learning with Implicit World Modeling (优先级分数 147.25) - AgentWorld: An Interactive Simulation Platform for Scene Construction and Mobile Robotic Manipulation (优先级分数 138.25) - ParticleFormer: A 3D Point Cloud World Model for Multi-Object, Multi-Material Robotic Manipulation (优先级分数 138.25) - ARCH: Hierarchical Hybrid Learning for Long-Horizon Contact-Rich Robotic Assembly (优先级分数 111.25)

1.3.3.1 ClutterDexGrasp: A Sim-to-Real System for General Dexterous Grasping in Cluttered Scenes

- 作者: Chen, Zeyuan; Yan, Qiyang; Chen, Yuanpei; Wu, Tianhao; Zhang, Jiyao; Ding, Zihan; Li, Jinzhou; Yang, Yaodong; Dong, Hao
- 年份: 2025; 主题: Embodied/WBC; VLA; Agent RL; Robot Learning; Supporting Perception; Systems; 优先级: 核心
- [来源](#)

论文: ClutterDexGrasp: A Sim-to-Real System for General Dexterous Grasping in Cluttered Scenes

快速理解: Dexterous grasping in cluttered scenes presents significant challenges due to diverse object geometries, occlusions, and potential collisions. Existing methods primarily focus on single-object grasping or grasp-pose prediction without interaction, which are insufficient for complex, cluttered scenes.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 视觉、语言条件与机器人动作策略之间的对齐; 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.3.2 FLARE: Robot Learning with Implicit World Modeling

- 作者: Zheng, Ruijie; Wang, Jing; Reed, Scott; Bjorck, Johan; Fang, Yu; Hu, Fengyuan; Jang, Joel; Kundalia, Kaushil; Lin, Zongyu; Magne, Loïc; Narayan, Avnish; Tan, You Liang; Wang, Guanzhi; Wang, Qi; Xiang, Jiannan; Xu, Yinzhen; Ye, Seonghyeon; Kautz, Jan; Huang, Furong; Zhu, Yuke; Fan, Linxi
- 年份: 2025; 主题: Embodied/WBC; VLA; World Model; Agent RL; Robot Learning; Supporting Perception; 优先级: 核心
- [来源](#)

论文: FLARE: Robot Learning with Implicit World Modeling

快速理解: We introduce **Future LATent Representation AlignmEnt (FLARE)**, a novel framework that integrates predictive world modeling into robot policy learning. By aligning features from a diffusion transformer with latent embeddings of future observations, **FLARE** enables a diffusion transformer policy to anticipate latent representations of future observations, allowing it to reason about long-term consequences while generating actions.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 视觉、语言条件与机器人动作策略之间的对齐; 环境动力学、视频预测、潜变量状态或模型式决策。

证据层级：索引摘要。

1.3.3.3 AgentWorld: An Interactive Simulation Platform for Scene Construction and Mobile Robotic Manipulation

- 作者: Zhang, Yizheng; Yu, Zhenjun; Lai, JiaXin; Lu, Cewu; Han, Lei
- 年份: 2025; 主题: Embodied/WBC; Teleoperation; VLA; Agent RL; Robot Learning; Supporting Perception; 优先级: 核心
- [来源](#)

论文: AgentWorld: An Interactive Simulation Platform for Scene Construction and Mobile Robotic Manipulation

快速理解: We introduce AgentWorld, an interactive simulation platform for developing household mobile manipulation capabilities. Our platform combines automated scene construction that encompasses layout generation, semantic asset placement, visual material configuration, and physics simulation, with a dual-mode teleoperation system supporting both wheeled bases and humanoid locomotion policies for data collection.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 人类示范、远程操作、共享自治或动作重定向; 视觉、语言条件与机器人动作策略之间的对齐。

证据层级：索引摘要。

1.3.3.4 ParticleFormer: A 3D Point Cloud World Model for Multi-Object, Multi-Material Robotic Manipulation

- 作者: Huang, Suning; Chen, Qianzhong; Zhang, Xiaohan; Sun, Jiankai; Schwager, Mac
- 年份: 2025; 主题: Embodied/WBC; World Model; Robot Learning; Supporting Perception; Systems; 优先级: 核心
- [来源](#)

论文: ParticleFormer: A 3D Point Cloud World Model for Multi-Object, Multi-Material Robotic Manipulation

快速理解: 3D world models (i.e., learning-based 3D dynamics models) offer a promising approach to generalizable robotic manipulation by capturing the underlying physics of environment evolution conditioned on robot actions. However, existing 3D world models are primarily limited to single-material dynamics using a particle-based Graph Neural Network model, and often require time-consuming 3D scene reconstruction to obtain 3D particle tracks for training.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 环境动力学、视频预测、潜变量状态或模型式决策; 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：索引摘要。

1.3.3.5 ARCH: Hierarchical Hybrid Learning for Long-Horizon Contact-Rich Robotic Assembly

- 作者: Sun, Jiankai; Curtis, Aidan; You, Yang; Xu, Yan; Koehle, Michael; Chen, Qianzhong; Huang, Suning; Guibas, Leonidas; Chitta, Sachin; Schwager, Mac; Li, Hui
- 年份: 2025; 主题: Embodied/WBC; Teleoperation; World Model; Agent RL; Robot Learning; 优先级: 核心

- [来源](#)

论文: ARCH: Hierarchical Hybrid Learning for Long-Horizon Contact-Rich Robotic Assembly

快速理解: Generalizable long-horizon robotic assembly requires reasoning at multiple levels of abstraction. While end-to-end imitation learning (IL) is a promising approach, it typically requires large amounts of expert demonstration data and often struggles to achieve the high precision demanded by assembly tasks.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 人类示范、远程操作、共享自治或动作重定向; 环境动力学、视频预测、潜变量状态或模型式决策。

证据层级: 索引摘要。

1.3.3.6 AutoEval: Autonomous Evaluation of Generalist Robot Manipulation Policies in the Real World

- 作者: Zhou, Zhiyuan; Atreya, Pranav; Tan, You Liang; Pertsch, Karl; Levine, Sergey
- 年份: 2025; 主题: Embodied/WBC; VLA; Robot Learning; Systems; 优先级: 核心
- [来源](#)

论文: AutoEval: Autonomous Evaluation of Generalist Robot Manipulation Policies in the Real World

快速理解: Scalable and reproducible policy evaluation has been a long-standing challenge in robot learning: evaluations are critical to assess progress and build better policies, but evaluation in the real world, especially at a scale that would provide statistically reliable results, is costly in terms of human time and hard to obtain. Evaluation of increasingly generalist robot policies requires an increasingly diverse repertoire of evaluation environments, making the evaluation bottleneck even more pronounced.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 视觉、语言条件与机器人动作策略之间的对齐; 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 索引摘要。

1.3.3.7 ControlVLA: Few-shot Object-centric Adaptation for Pre-trained Vision-Language-Action Models

- 作者: Li, Puhao; Wu, Yingying; Xi, Ziheng; Li, Wanlin; Huang, Yuzhe; Zhang, Zhiyuan; Chen, Yinghan; Wang, Jianan; Zhu, Song-Chun; Liu, Tengyu; Huang, Siyuan
- 年份: 2025; 主题: Embodied/WBC; VLA; Robot Learning; Supporting Perception; 优先级: 核心
- [来源](#)

论文: ControlVLA: Few-shot Object-centric Adaptation for Pre-trained Vision-Language-Action Models

快速理解: Learning real-world robotic manipulation is challenging, particularly when limited demonstrations are available. Existing methods for few-shot manipulation often rely on simulation-augmented data or pre-built modules like grasping and pose estimation, which struggle with sim-to-real gaps and lack extensibility.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 视觉、语言条件与机器人动作策略之间的对齐; 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：索引摘要。

1.3.3.8 DreamGen: Unlocking Generalization in Robot Learning through Video World Models

- 作者: Jang, Joel; Ye, Seonghyeon; Lin, Zongyu; Xiang, Jiannan; Bjorck, Johan; Fang, Yu; Hu, Fengyuan; Huang, Spencer; Kundalia, Kaushil; Lin, Yen-Chen; Magne, Loïc; Mandlekar, Ajay; Narayan, Avnish; Tan, You Liang; Wang, Guanzhi; Wang, Jing; Wang, Qi; Xu, Yinzhen; Zeng, Xiaohui; Zheng, Kaiyuan; Zheng, Ruijie; Liu, Ming-Yu; Zettlemoyer, Luke; Fox, Dieter; Kautz, Jan; Reed, Scott; Zhu, Yuke; Fan, Linxi
- 年份: 2025; 主题: Embodied/WBC; Teleoperation; World Model; Robot Learning; 优先级: 核心
- [来源](#)

论文: DreamGen: Unlocking Generalization in Robot Learning through Video World Models

快速理解: In this work, we unlock new capabilities in robot learning from neural trajectories, synthetic robot data generated from video world models. Our proposed recipe is simple, but powerful: we take the most recent state-of-the-art video generative models (world models), adapt them to the target robot embodiment, and generate new, synthetic robot data of the same task or even new behaviors.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 人类示范、远程操作、共享自治或动作重定向; 环境动力学、视频预测、潜变量状态或模型式决策。

证据层级：索引摘要。

1.3.3.9 Gen2Act: Human Video Generation in Novel Scenarios enables Generalizable Robot Manipulation

- 作者: Bharadhwaj, Homanga; Dwibedi, Debidatta; Gupta, Abhinav; Tulsiani, Shubham; Doersch, Carl; Xiao, Ted; Shah, Dhruv; Xia, Fei; Sadigh, Dorsa; Kirmani, Sean
- 年份: 2025; 主题: Embodied/WBC; World Model; Robot Learning; Supporting Perception; 优先级: 核心
- [来源](#)

论文: Gen2Act: Human Video Generation in Novel Scenarios enables Generalizable Robot Manipulation

快速理解: How can robot manipulation policies generalize to novel tasks involving unseen object types and new motions? In this paper, we provide a solution in terms of predicting motion information from web data through human video generation and conditioning a robot policy on the generated video.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 环境动力学、视频预测、潜变量状态或模型式决策; 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：索引摘要。

1.3.3.10 Generalist Robot Manipulation beyond Action Labeled Data

- 作者: Spiridonov, Alexander; Zaech, Jan-Nico; Nikolov, Nikolay; Gool, Luc Van; Paudel, Danda Pani
- 年份: 2025; 主题: Embodied/WBC; VLA; Robot Learning; Systems; 优先级: 核心
- [来源](#)

论文: Generalist Robot Manipulation beyond Action Labeled Data

快速理解: Recent advances in generalist robot manipulation leverage pre-trained Vision-Language Models (VLMs) and large-scale robot demonstrations to tackle diverse tasks in a zero-shot manner. A key challenge remains: scaling high-quality, action-labeled robot demonstration data, which existing methods rely on for robustness and generalization.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 视觉、语言条件与机器人动作策略之间的对齐; 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 索引摘要。

1.3.3.11 RICL: Adding In-Context Adaptability to Pre-Trained Vision-Language-Action Models

- 作者: Sridhar, Kaustubh; Dutta, Souradeep; Jayaraman, Dinesh; Lee, Insup
- 年份: 2025; 主题: VLA; Agent RL; Robot Learning; Supporting Perception; 优先级: 核心
- [来源](#)

论文: RICL: Adding In-Context Adaptability to Pre-Trained Vision-Language-Action Models

快速理解: Multi-task “vision-language-action” (VLA) models have recently demonstrated increasing promise as generalist foundation models for robotics, achieving non-trivial performance out of the box on new tasks in new environments. However, for such models to be truly useful, an end user must have easy means to teach them to improve.

关注点: 视觉、语言条件与机器人动作策略之间的对齐; 强化学习、模仿学习、规划、奖励或多智能体决策; 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 索引摘要。

1.3.3.12 Real2Render2Real: Scaling Robot Data Without Dynamics Simulation or Robot Hardware

- 作者: Yu, Justin; Fu, Letian; Huang, Huang; El-Refai, Karim; Ambrus, Rares Andrei; Cheng, Richard; Irshad, Muhammad Zubair; Goldberg, Ken
- 年份: 2025; 主题: Teleoperation; VLA; Agent RL; Robot Learning; Supporting Perception; 优先级: 核心
- [来源](#)

论文: Real2Render2Real: Scaling Robot Data Without Dynamics Simulation or Robot Hardware

快速理解: Scaling robot learning requires vast and diverse datasets. Yet the prevailing data collection paradigm—human teleoperation—remains costly and constrained by manual effort and physical robot access.

关注点: 人类示范、远程操作、共享自治或动作重定向; 视觉、语言条件与机器人动作策略之间的对齐; 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.3.13 RoboMonkey: Scaling Test-Time Sampling and Verification for Vision-Language-Action Models

- 作者: Kwok, Jacky; Agia, Christopher; Sinha, Rohan; Foutter, Matt; Li, Shulu; Stoica, Ion; Mirho-

seini, Azalia; Pavone, Marco

- 年份: 2025; 主题: VLA; Agent RL; Robot Learning; Supporting Perception; 优先级: 核心
- [来源](#)

论文: RoboMonkey: Scaling Test-Time Sampling and Verification for Vision-Language-Action Models

快速理解: Vision-Language-Action (VLA) models, pre-trained on large-scale imitation learning datasets, have demonstrated remarkable capabilities in visuomotor control. However, these models exhibit diverse failure modes in unstructured real-world environments, limiting the widespread adoption of VLAs in robotics.

关注点: 视觉、语言条件与机器人动作策略之间的对齐; 强化学习、模仿学习、规划、奖励或多智能体决策; 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 索引摘要。

1.3.3.14 WoMAP: World Models For Embodied Open-Vocabulary Object Localization

- 作者: Yin, Tenny; Mei, Zhiting; Sun, Tao; Sho, Ola; Majumdar, Anirudha; Zhou, Emily; Bao, Jeremy; Yamane, Miyu; Zha, Lihan
- 年份: 2025; 主题: Embodied/WBC; World Model; Agent RL; Robot Learning; 优先级: 核心
- [来源](#)

论文: WoMAP: World Models For Embodied Open-Vocabulary Object Localization

快速理解: Active object localization remains a critical challenge for robots, requiring efficient exploration of partially observable environments. However, state-of-the-art robot policies either struggle to generalize beyond demonstration datasets (e.g., imitation learning methods) or fail to generate physically grounded actions (e.g., VLMs).

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 环境动力学、视频预测、潜变量状态或模型式决策; 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.3.15 AirExo-2: Scaling up Generalizable Robotic Imitation Learning with Low-Cost Exoskeletons

- 作者: Fang, Hongjie; Wang, Chenxi; Wang, Yiming; Chen, Jingjing; Xia, Shangning; Lv, Jun; He, Zihao; Yi, Xiyang; Guo, Yunhan; Zhan, Xinyu; Yang, Lixin; Wang, Weiming; Lu, Cewu; Fang, Hao-Shu
- 年份: 2025; 主题: Embodied/WBC; Teleoperation; Agent RL; Robot Learning; 优先级: 核心
- [来源](#)

论文: AirExo-2: Scaling up Generalizable Robotic Imitation Learning with Low-Cost Exoskeletons

快速理解: Scaling up robotic imitation learning for real-world applications requires efficient and scalable demonstration collection methods. While teleoperation is effective, it depends on costly and inflexible robot platforms.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 人类示范、远程操作、共享自治或动作重定向; 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.3.16 Crossing the Human-Robot Embodiment Gap with Sim-to-Real RL using One Human Demonstration

- 作者: Lum, Tyler Ga Wei; Lee, Olivia Y.; Liu, Karen; Bohg, Jeannette
- 年份: 2025; 主题: Embodied/WBC; Teleoperation; Agent RL; Robot Learning; 优先级: 核心
- [来源](#)

论文: Crossing the Human-Robot Embodiment Gap with Sim-to-Real RL using One Human Demonstration

快速理解: Teaching robots dexterous manipulation skills often requires collecting hundreds of demonstrations using wearables or teleoperation, a process that is challenging to scale. Videos of human-object interactions are easier to collect and scale, but leveraging them directly for robot learning is difficult due to the lack of explicit action labels and human-robot embodiment differences.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 人类示范、远程操作、共享自治或动作重定向; 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.3.17 DexVLA: Vision-Language Model with Plug-In Diffusion Expert for General Robot Control

- 作者: Wen, Junjie; Zhu, Yichen; Li, Jinming; Tang, Zhibin; Shen, Chaomin; Feng, Feifei
- 年份: 2025; 主题: Embodied/WBC; VLA; Robot Learning; Supporting Perception; 优先级: 核心
- [来源](#)

论文: DexVLA: Vision-Language Model with Plug-In Diffusion Expert for General Robot Control

快速理解: Enabling robots to perform diverse tasks across varied environments is a central challenge in robot learning. While vision-language-action (VLA) models have shown promise for generalizable robot skills, realizing their full potential requires addressing limitations in action representation and efficient training.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 视觉、语言条件与机器人动作策略之间的对齐; 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 索引摘要。

1.3.3.18 Embrace Contacts: humanoid shadowing with full body ground contacts

- 作者: Zhuang, Ziwen; Zhao, Hang
- 年份: 2025; 主题: Embodied/WBC; Agent RL; Robot Learning; AI Infra; 优先级: 核心
- [来源](#)

论文: Embrace Contacts: humanoid shadowing with full body ground contacts

快速理解: Previous humanoid robot research works treat the robot as a bipedal mobile manipulation platform, where only the feet and hands contact the environment. However, we humans use all body parts to interact with the world, e.g., we sit in chairs, get up from the ground, or roll on the floor.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 强化学习、模仿学习、规划、奖励或多智能体决策; 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 索引摘要。

1.3.3.19 GraspVLA: a Grasping Foundation Model Pre-trained on Billion-scale Synthetic Action Data

- 作者: Deng, Shengliang; Yan, Mi; Wei, Songlin; Ma, Haixin; Yang, Yuxin; Chen, Jiayi; Zhang, Zhiqi; Yang, Taoyu; Zhang, Xuheng; Cui, Heming; Zhang, Zhizheng; Wang, He
- 年份: 2025; 主题: Embodied/WBC; VLA; Robot Learning; Supporting Perception; 优先级: 核心
- [来源](#)

论文: GraspVLA: a Grasping Foundation Model Pre-trained on Billion-scale Synthetic Action Data

快速理解: Embodied foundation models are gaining increasing attention for their zero-shot generalization, scalability, and adaptability to new tasks through few-shot post-training. However, existing models rely heavily on real-world data, which is costly and labor-intensive to collect.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 视觉、语言条件与机器人动作策略之间的对齐; 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 索引摘要。

1.3.3.20 Long-VLA: Unleashing Long-Horizon Capability of Vision Language Action Model for Robot Manipulation

- 作者: Fan, Yiguo; Bai, Shuanghao; Tong, Xinyang; Ding, Pengxiang; Zhu, Yuyang; Lu, Hongchao; Dai, Fengqi; Zhao, Wei; Liu, Yang; Huang, Siteng; Fan, Zhaoxin; Chen, Badong; Wang, Donglin
- 年份: 2025; 主题: Embodied/WBC; VLA; Supporting Perception; 优先级: 核心
- [来源](#)

论文: Long-VLA: Unleashing Long-Horizon Capability of Vision Language Action Model for Robot Manipulation

快速理解: Vision-Language-Action (VLA) models have become a cornerstone in robotic policy learning, leveraging large-scale multimodal data for robust and scalable control. However, existing VLA frameworks primarily address short-horizon tasks, and their effectiveness on long-horizon, multi-step robotic manipulation remains limited due to challenges in skill chaining and subtask dependencies.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 视觉、语言条件与机器人动作策略之间的对齐; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.3.21 Lucid-XR: An Extended-Reality Data Engine for Robotic Manipulation

- 作者: Ravan, Yajvan; Rashid, Adam; Yu, Alan; McClennen, Kai; Huh, Gio; Yang, Kevin; Yang, Zhutian; Yu, Qinxi; Wang, Xiaolong; Isola, Phillip; Yang, Ge
- 年份: 2025; 主题: Embodied/WBC; Teleoperation; Robot Learning; Supporting Perception; 优先级: 核心
- [来源](#)

论文: Lucid-XR: An Extended-Reality Data Engine for Robotic Manipulation

快速理解: We introduce Lucid-XR, a generative data engine for creating diverse and realistic-looking data to train real-world robot systems. At the core of Lucid-XR is vuer, a web-based physics simulation environment that runs directly on the XR headset, enabling internet-scale access to immersive, latency-free virtual interactions without requiring specialized equipment.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；人类示范、远程操作、共享自治或动作重定向；机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：索引摘要。

1.3.3.22 SAIL: Faster-than-Demonstration Execution of Imitation Learning Policies

- 作者：Arachchige, Nadun Ranawaka; Chen, Zhenyang; Jung, Wonsuhk; Shin, Woo Chul; Bansal, Rohan; Barroso, Pierre; He, Yu Hang; Lin, Yingyan Celine; Joffe, Benjamin; Kousik, Shreyas; Xu, Danfei
- 年份：2025；主题：Embodied/WBC; Agent RL; Robot Learning; Systems；优先级：核心
- [来源](#)

论文：SAIL: Faster-than-Demonstration Execution of Imitation Learning Policies

快速理解：Offline Imitation Learning (IL) methods such as Behavior Cloning are effective at acquiring complex robotic manipulation skills. However, existing IL-trained policies are confined to execute the task at the same speed as shown in demonstration data.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；强化学习、模仿学习、规划、奖励或多智能体决策；机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：索引摘要。

1.3.3.23 Sim-to-Real Reinforcement Learning for Vision-Based Dexterous Manipulation on Humanoids

- 作者：Lin, Toru; Sachdev, Kartik; Fan, Linxi; Malik, Jitendra; Zhu, Yuke
- 年份：2025；主题：Embodied/WBC; Agent RL; Robot Learning；优先级：核心
- [来源](#)

论文：Sim-to-Real Reinforcement Learning for Vision-Based Dexterous Manipulation on Humanoids

快速理解：Learning generalizable robot manipulation policies, especially for complex multi-fingered humanoids, remains a significant challenge. Existing approaches primarily rely on extensive data collection and imitation learning, which are expensive, labor-intensive, and difficult to scale.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；强化学习、模仿学习、规划、奖励或多智能体决策；机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：索引摘要。

1.3.3.24 X-Sim: Cross-Embodiment Learning via Real-to-Sim-to-Real

- 作者：Dan, Prithwish; Kedia, Kushal; Chao, Angela; Duan, Edward; Pace, Maximus Adrian; Ma, Wei-Chiu; Choudhury, Sanjiban
- 年份：2025；主题：Embodied/WBC; Teleoperation; Agent RL; Robot Learning；优先级：核心
- [来源](#)

论文：X-Sim: Cross-Embodiment Learning via Real-to-Sim-to-Real

快速理解：Human videos offer a scalable way to train robot manipulation policies, but lack the action labels needed by standard imitation learning algorithms. Existing cross-embodiment approaches try to map human motion to robot actions, but often fail when the embodiments differ significantly.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；人类示范、远程操作、共享自治或动作重定向；强化学习、模

仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.3.25 pi_0.5: a Vision-Language-Action Model with Open-World Generalization

- 作者: Black, Kevin; Brown, Noah; Darpinian, James; Dhabalia, Karan; Driess, Danny; Esmail, Adnan; Equi, Michael Robert; Finn, Chelsea; Fusai, Niccolo; Galliker, Manuel Y.; Ghosh, Dibya; Groom, Lachy; Hausman, Karol; ichter, brian; Jakubczak, Szymon; Jones, Tim; Ke, Liyiming; LeBlanc, Devin; Levine, Sergey; Li-Bell, Adrian; Mothukuri, Mohith; Nair, Suraj; Pertsch, Karl; Ren, Allen Z.; Shi, Lucy Xiaoyang; Smith, Laura; Springenberg, Jost Tobias; Stachowicz, Kyle; Tanner, James; Vuong, Quan; Walke, Homer; Walling, Anna; Wang, Haohuan; Yu, Lili; Zhilinsky, Ury
- 年份: 2025; 主题: Embodied/WBC; VLA; Supporting Perception; 优先级: 核心
- [来源](#)

论文: pi_0.5: a Vision-Language-Action Model with Open-World Generalization

快速理解: In order for robots to be useful, they must perform practically relevant tasks in the real world, outside of the lab. While vision-language-action (VLA) models have demonstrated impressive results for end-to-end robot control, it remains an open question how far such models can generalize in the wild.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 视觉、语言条件与机器人动作策略之间的对齐; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.3.26 CLONE: Closed-Loop Whole-Body Humanoid Teleoperation for Long-Horizon Tasks

- 作者: Li, Yixuan; Lin, Yutang; Cui, Jieming; Liu, Tengyu; Liang, Wei; Zhu, Yixin; Huang, Siyuan
- 年份: 2025; 主题: Embodied/WBC; Teleoperation; AI Infra; 优先级: 核心
- [来源](#)

论文: CLONE: Closed-Loop Whole-Body Humanoid Teleoperation for Long-Horizon Tasks

快速理解: Humanoid robot teleoperation plays a vital role in demonstrating and collecting data for complex interactions. Current methods suffer from two key limitations: (1) restricted controllability due to decoupled upper- and lower-body control, and (2) severe drift caused by open-loop execution.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 人类示范、远程操作、共享自治或动作重定向; 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.3.27 Disentangled Multi-Context Meta-Learning: Unlocking Robust and Generalized Task Learning

- 作者: Kim, Seonsoo; Kang, Jun-Gill; Kim, Taehong; Hong, Seongil
- 年份: 2025; 主题: Embodied/WBC; World Model; Agent RL; Robot Learning; 优先级: 核心
- [来源](#)

论文: Disentangled Multi-Context Meta-Learning: Unlocking Robust and Generalized Task Learning

快速理解: In meta-learning and its downstream tasks, many methods use implicit adaptation to represent task-specific variations. However, implicit approaches hinder interpretability and make it difficult to understand which task factors drive performance.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 环境动力学、视频预测、潜变量状态或模型式决策; 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.3.28 Elucidating the Design Space of Torque-aware Vision-Language-Action Models

- 作者: Zhang, Zongzheng; Xu, Haobo; Yang, Zhuo; Yue, Chenghao; Lin, Zehao; Gao, Huan-ang; Wang, Ziwei; Zhao, Hao
- 年份: 2025; 主题: Embodied/WBC; VLA; Supporting Perception; 优先级: 核心
- [来源](#)

论文: Elucidating the Design Space of Torque-aware Vision-Language-Action Models

快速理解: Many robotic manipulation tasks require sensing and responding to force signals such as torque to assess whether the task has been successfully completed and to enable closed-loop control. However, current Vision-Language-Action (VLA) models lack the ability to integrate such subtle physical feedback.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 视觉、语言条件与机器人动作策略之间的对齐; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.3.29 Ensuring Force Safety in Vision-Guided Robotic Manipulation via Implicit Tactile Calibration

- 作者: Wei, Lai; Ma, Jiahua; Hu, Yibo; Zhang, Ruimao
- 年份: 2025; 主题: Embodied/WBC; Robot Learning; Supporting Perception; 优先级: 核心
- [来源](#)

论文: Ensuring Force Safety in Vision-Guided Robotic Manipulation via Implicit Tactile Calibration

快速理解: In unstructured environments, robotic manipulation tasks involving objects with constrained motion trajectories—such as door opening—often experience discrepancies between the robot’s vision-guided end-effector trajectory and the object’s constrained motion path. Such discrepancies generate unintended harmful forces, which, if exacerbated, may lead to task failure and potential damage to the manipulated objects or the robot itself.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 机器人数据、策略学习、操作、触觉或 sim-to-real; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.3.30 FFHFlow: Diverse and Uncertainty-Aware Dexterous Grasp Generation via Flow Variational Inference

- 作者: Feng, Qian; Feng, Jianxiang; Chen, Zhaopeng; Triebel, Rudolph; Knoll, Alois
- 年份: 2025; 主题: Embodied/WBC; Robot Learning; Systems; 优先级: 核心
- [来源](#)

论文: FFHFlow: Diverse and Uncertainty-Aware Dexterous Grasp Generation via Flow Variational Inference

快速理解: Synthesizing diverse, uncertainty-aware grasps for multi-fingered hands from partial observations remains a critical challenge in robot learning. Prior generative methods struggle to model the intricate grasp distribution of dexterous hands and often fail to reason about shape uncertainty inherent in partial point clouds, leading to unreliable or overly conservative grasps.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 机器人数据、策略学习、操作、触觉或 sim-to-real; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.3.31 FLOWER: Democratizing Generalist Robot Policies with Efficient Vision-Language-Flow Models

- 作者: Reuss, Moritz; Zhou, Hongyi; R" {u}hle, Marcel; Yau{g}murlu, "{O}mer Erdinc{c}; Otto, Fabian; Lioutikov, Rudolf
- 年份: 2025; 主题: VLA; AI Infra; Supporting Perception; 优先级: 核心
- [来源](#)

论文: FLOWER: Democratizing Generalist Robot Policies with Efficient Vision-Language-Flow Models

快速理解: Developing efficient Vision-Language-Action (VLA) policies is crucial for practical robotics deployment, yet current approaches face prohibitive computational costs and resource requirements. Existing diffusion-based VLA policies require multi-billion-parameter models and massive datasets to achieve strong performance.

关注点: 视觉、语言条件与机器人动作策略之间的对齐; 训练、推理、服务、编译、调度、通信或 AI 加速; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.3.32 Focusing on What Matters: Object-Agent-centric Tokenization for Vision Language Action models

- 作者: Bendikas, Rokas; Dijkman, Daniel; Peschl, Markus; Haresh, Sanjay; Mazzaglia, Pietro
- 年份: 2025; 主题: Embodied/WBC; VLA; Supporting Perception; 优先级: 核心
- [来源](#)

论文: Focusing on What Matters: Object-Agent-centric Tokenization for Vision Language Action models

快速理解: Vision-Language-Action (VLA) models offer a pivotal approach to learning robotic manipulation at scale by repurposing large pre-trained Vision-Language-Models (VLM) to output robotic actions. However, adapting VLMs for robotic domains comes with an unnecessarily high computational cost, which we attribute to the tokenization scheme of visual inputs.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 视觉、语言条件与机器人动作策略之间的对齐; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.3.33 GraspQP: Differentiable Optimization of Force Closure for Diverse and Robust Dexterous Grasping

- 作者: Zurbrugg, Ren; Cramariuc, Andrei; Hutter, Marco
- 年份: 2025; 主题: Embodied/WBC; Robot Learning; Systems; 优先级: 核心
- [来源](#)

论文: GraspQP: Differentiable Optimization of Force Closure for Diverse and Robust Dexterous Grasping

快速理解: Dexterous robotic hands enable versatile interactions through the flexibility and adaptability of a multi-finger setup, allowing for a wide range of task-specific grasp configurations in diverse environments. However, access to diverse and high-quality grasp data is essential to fully exploit the capabilities of dexterous hands, be it to train grasp prediction models from point clouds, train manipulation policies, or to support high-level task planning with a broader range of action options.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 机器人数据、策略学习、操作、触觉或 sim-to-real; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.3.34 KineDex: Learning Tactile-Informed Visuomotor Policies via Kinesthetic Teaching for Dexterous Manipulation

- 作者: Zhang, Di; Yuan, Chengbo; Wen, Chuan; Zhang, Hai; Zhao, Junqiao; Gao, Yang
- 年份: 2025; 主题: Embodied/WBC; Teleoperation; Robot Learning; 优先级: 核心
- [来源](#)

论文: KineDex: Learning Tactile-Informed Visuomotor Policies via Kinesthetic Teaching for Dexterous Manipulation

快速理解: Collecting demonstrations enriched with fine-grained tactile information is critical for dexterous manipulation, particularly in contact-rich tasks that require precise force control and physical interaction. While prior works primarily focus on teleoperation or video-based retargeting, they often suffer from kinematic mismatches and the absence of real-time tactile feedback, hindering the acquisition of high-fidelity tactile data.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 人类示范、远程操作、共享自治或动作重定向; 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 索引摘要。

1.3.3.35 Mechanistic Interpretability for Steering Vision-Language-Action Models

- 作者: Hon, Bear; Stocking, Kaylene Caswell; Chuang, Ian; Tomlin, Claire
- 年份: 2025; 主题: Embodied/WBC; VLA; Supporting Perception; 优先级: 核心
- [来源](#)

论文: Mechanistic Interpretability for Steering Vision-Language-Action Models

快速理解: Vision-Language-Action (VLA) models are a promising path to realizing generalist embodied agents that can quickly adapt to new tasks, modalities, and environments. However, methods for interpreting and steering VLAs fall far short of classical robotics pipelines, which are grounded in explicit models of kinematics, dynamics, and control.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；视觉、语言条件与机器人动作策略之间的对齐；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.3.36 RobotxR1: Enabling Embodied Robotic Intelligence on Large Language Models through Closed-Loop Reinforcement Learning

- 作者：Boyle, Liam; Baumann, Nicolas; Sivasothilingam, Paviththiren; Magno, Michele; Benini, Luca
- 年份：2025；主题：Embodied/WBC; Agent RL; Systems；优先级：核心
- [来源](#)

论文: RobotxR1: Enabling Embodied Robotic Intelligence on Large Language Models through Closed-Loop Reinforcement Learning

快速理解: Future robotic systems operating in real-world environments require on-board embodied intelligence without continuous cloud connection, balancing capabilities with constraints on computational power and memory. This work presents an extension of the R1-zero approach, which enables the usage of small parameter-count Large Language Models (LLMs) in the robotic domain.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；强化学习、模仿学习、规划、奖励或多智能体决策；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.3.37 TWIST: Teleoperated Whole-Body Imitation System

- 作者：Ze, Yanjie; Chen, Zixuan; Araujo, Joao Pedro; Cao, Zi-ang; Peng, Xue Bin; Wu, Jiajun; Liu, Karen
- 年份：2025；主题：Embodied/WBC; Teleoperation; Agent RL；优先级：核心
- [来源](#)

论文: TWIST: Teleoperated Whole-Body Imitation System

快速理解: Teleoperating humanoid robots in a whole-body manner marks a fundamental step toward developing general-purpose robotic intelligence, with human motion providing an ideal interface for controlling all degrees of freedom. Yet, most current humanoid teleoperation systems fall short of enabling coordinated whole-body behavior, typically limiting themselves to isolated locomotion or manipulation tasks.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；人类示范、远程操作、共享自治或动作重定向；强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.3.38 Tactile Beyond Pixels: Multisensory Touch Representations for Robot Manipulation

- 作者：Higuera, Carolina; Sharma, Akash; Fan, Taosha; Bodduluri, Chaithanya Krishna; Boots, Byron; Kaess, Michael; Lambeta, Mike; Wu, Tingfan; Liu, Zixi; Hogan, Francois Robert; Mukadam, Mustafa
- 年份：2025；主题：Embodied/WBC; Agent RL; Robot Learning；优先级：核心
- [来源](#)

论文: Tactile Beyond Pixels: Multisensory Touch Representations for Robot Manipulation

快速理解: We present TacX, the first multisensory touch representations across four tactile modalities: image, audio, motion, and pressure. Trained on 1M contact-rich interactions collected with the Digit 360 sensor, TacX captures complementary touch signals at diverse temporal and spatial scales.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 强化学习、模仿学习、规划、奖励或多智能体决策; 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 索引摘要。

1.3.3.39 TypeTele: Releasing Dexterity in Teleoperation by Dexterous Manipulation Types

- 作者: Lin, Yuhao; Wei, Yi-Lin; Liao, Haoran; Lin, Mu; Xing, Chengyi; Li, Hao; Zhang, Dandan; Cutkosky, Mark; Zheng, Wei-Shi
- 年份: 2025; 主题: Embodied/WBC; Teleoperation; Agent RL; 优先级: 核心
- [来源](#)

论文: TypeTele: Releasing Dexterity in Teleoperation by Dexterous Manipulation Types

快速理解: Dexterous teleoperation plays a crucial role in robotic manipulation for real-world data collection and remote robot control. Previous dexterous teleoperation mostly relies on hand retargeting to closely mimic human hand postures.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 人类示范、远程操作、共享自治或动作重定向; 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.3.40 Vision in Action: Learning Active Perception from Human Demonstrations

- 作者: Xiong, Haoyu; Xu, Xiaomeng; Wu, Jimmy; Hou, Yifan; Bohg, Jeannette; Song, Shuran
- 年份: 2025; 主题: Embodied/WBC; Teleoperation; Robot Learning; Supporting Perception; 优先级: 核心
- [来源](#)

论文: Vision in Action: Learning Active Perception from Human Demonstrations

快速理解: We present Vision in Action (ViA), an active perception system for bimanual robot manipulation. ViA learns task-relevant active perceptual strategies (e.g., searching, tracking, and focusing) directly from human demonstrations.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 人类示范、远程操作、共享自治或动作重定向; 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 索引摘要。

1.3.3.41 Adapting by Analogy: OOD Generalization of Visuomotor Policies via Functional Correspondence

- 作者: Gupta, Pranay; Admoni, Henny; Bajcsy, Andrea
- 年份: 2025; 主题: Embodied/WBC; Agent RL; Robot Learning; 优先级: 核心
- [来源](#)

论文: Adapting by Analogy: OOD Generalization of Visuomotor Policies via Functional Correspondence

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快速理解: End-to-end visuomotor policies trained using behavior cloning have shown a remarkable ability to generate complex, multi-modal low-level robot behaviors. However, at deployment time, these policies still struggle to act reliably when faced with out-of-distribution (OOD) visuals induced by objects, backgrounds, or environment changes.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 强化学习、模仿学习、规划、奖励或多智能体决策; 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 索引摘要。

1.3.3.42 BEHAVIOR Robot Suite: Streamlining Real-World Whole-Body Manipulation for Everyday Household Activities

- 作者: Jiang, Yunfan; Zhang, Ruohan; Wong, Josiah; Wang, Chen; Ze, Yanjie; Yin, Hang; Gokmen, Cem; Song, Shuran; Wu, Jiajun; Fei-Fei, Li
- 年份: 2025; 主题: Embodied/WBC; Teleoperation; Robot Learning; 优先级: 核心
- [来源](#)

论文: BEHAVIOR Robot Suite: Streamlining Real-World Whole-Body Manipulation for Everyday Household Activities

快速理解: Real-world household tasks present significant challenges for mobile manipulation robots. An analysis of existing robotics benchmarks reveals that successful task performance hinges on three key whole-body control capabilities: bimanual coordination, stable and precise navigation, and extensive end-effector reachability.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 人类示范、远程操作、共享自治或动作重定向; 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 索引摘要。

1.3.3.43 DemoSpeedup: Accelerating Visuomotor Policies via Entropy-Guided Demonstration Acceleration

- 作者: Guo, Lingxiao; Xue, Zhengrong; Xu, Zijing; Xu, Huazhe
- 年份: 2025; 主题: Embodied/WBC; Agent RL; Robot Learning; 优先级: 核心
- [来源](#)

论文: DemoSpeedup: Accelerating Visuomotor Policies via Entropy-Guided Demonstration Acceleration

快速理解: Imitation learning has shown great promise in robotic manipulation, but the policy's execution is often unsatisfactorily slow due to commonly tardy demonstrations collected by human operators. In this work, we present DemoSpeedup, a self-supervised method to accelerate visuomotor policy execution via entropy-guided demonstration acceleration.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 强化学习、模仿学习、规划、奖励或多智能体决策; 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 索引摘要。

1.3.3.44 Dexplore: Scalable Neural Control for Dexterous Manipulation from Reference Scoped Exploration

- 作者: Xu, Sirui; Chao, Yu-Wei; Bian, Liuyu; Mousavian, Arsalan; Wang, Yu-Xiong; Gui, Liangyan; Yang, Wei
- 年份: 2025; 主题: Embodied/WBC; Teleoperation; Agent RL; 优先级: 核心
- [来源](#)

论文: Dexplore: Scalable Neural Control for Dexterous Manipulation from Reference Scoped Exploration

快速理解: Hand-object motion-capture (MoCap) repositories provide abundant, contact-rich human demonstrations for scaling dexterous manipulation on robots. Yet demonstration inaccuracy and embodiment gaps between human and robot hands challenge direct policy learning.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 人类示范、远程操作、共享自治或动作重定向; 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.3.45 DiWA: Diffusion Policy Adaptation with World Models

- 作者: Chandra, Akshay L; Nematollahi, Iman; Huang, Chenguang; Welschhold, Tim; Burgard, Wolfram; Valada, Abhinav
- 年份: 2025; 主题: World Model; Agent RL; Robot Learning; 优先级: 核心
- [来源](#)

论文: DiWA: Diffusion Policy Adaptation with World Models

快速理解: Fine-tuning diffusion policies with reinforcement learning (RL) presents significant challenges. The long denoising sequence for each action prediction impedes effective reward propagation.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策; 强化学习、模仿学习、规划、奖励或多智能体决策; 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 索引摘要。

1.3.3.46 HuB: Learning Extreme Humanoid Balance

- 作者: Zhang, Tong; Zheng, Boyuan; Nai, Ruiqian; Hu, Yingdong; Wang, Yen-Jen; Chen, Geng; Lin, Fanqi; Li, Jiongye; Hong, Chuye; Sreenath, Koushil; Gao, Yang
- 年份: 2025; 主题: Embodied/WBC; Agent RL; Robot Learning; 优先级: 核心
- [来源](#)

论文: HuB: Learning Extreme Humanoid Balance

快速理解: The human body demonstrates exceptional motor capabilities—such as standing steadily on one foot or performing a high kick with the leg raised over 1.5 meters—both requiring precise balance control. While recent research on humanoid control has leveraged reinforcement learning to track human motions for skill acquisition, applying this paradigm to balance-intensive tasks remains challenging.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 强化学习、模仿学习、规划、奖励或多智能体决策; 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 索引摘要。

1.3.3.47 Humanoid Policy Human Policy

- 作者: Qiu, Ri-Zhao; Yang, Shiqi; Cheng, Xuxin; Chawla, Chaitanya; Li, Jialong; He, Tairan; Yan, Ge; Yoon, David J.; Hoque, Ryan; Paulsen, Lars; Yang, Ge; Zhang, Jian; Yi, Sha; Shi, Guanya; Wang, Xiaolong
- 年份: 2025; 主题: Embodied/WBC; Teleoperation; Robot Learning; 优先级: 核心
- [来源](#)

论文: Humanoid Policy Human Policy

快速理解: Training manipulation policies for humanoid robots with diverse data enhances their robustness and generalization across tasks and platforms. However, learning solely from robot demonstrations is labor-intensive, requiring expensive tele-operated data collection, which is difficult to scale.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 人类示范、远程操作、共享自治或动作重定向; 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 索引摘要。

1.3.3.48 LaDi-WM: A Latent Diffusion-Based World Model for Predictive Manipulation

- 作者: Huang, Yuhang; Zhang, Jiazhao; Zou, Shilong; Liu, Xinwang; Hu, Ruizhen; Xu, Kai
- 年份: 2025; 主题: Embodied/WBC; World Model; Robot Learning; 优先级: 核心
- [来源](#)

论文: LaDi-WM: A Latent Diffusion-Based World Model for Predictive Manipulation

快速理解: Predictive manipulation has recently gained considerable attention in the Embodied AI community due to its potential to improve robot policy performance by leveraging predicted states. However, generating accurate future visual states of robot-object interactions from world models remains a well-known challenge, particularly in achieving high-quality pixel-level representations.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 环境动力学、视频预测、潜变量状态或模型式决策; 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 索引摘要。

1.3.3.49 Learning Impact-Rich Rotational Maneuvers via Centroidal Velocity Rewards and Sim-to-Real Techniques: A One-Leg Hopper Flip Case Study

- 作者: Kang, Dongyun; Kim, Gijeong; Choe, JongHun; Kim, Hajun; Park, Hae-Won
- 年份: 2025; 主题: Embodied/WBC; Agent RL; Robot Learning; 优先级: 核心
- [来源](#)

论文: Learning Impact-Rich Rotational Maneuvers via Centroidal Velocity Rewards and Sim-to-Real Techniques: A One-Leg Hopper Flip Case Study

快速理解: Dynamic rotational maneuvers, such as front flips, inherently involve large angular momentum generation and intense impact forces, presenting major challenges for reinforcement learning and sim-to-real transfer. In this work, we propose a general framework for learning and deploying impact-rich, rotation-intensive behaviors through centroidal velocity-based rewards and actuator-aware sim-to-real techniques.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 强化学习、模仿学习、规划、奖励或多智能体决策; 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：索引摘要。

1.3.3.50 Learning Long-Horizon Robot Manipulation Skills via Privileged Action

- 作者：Mao, Xiaofeng; XU, Yucheng; Sun, Zhaole; Miller, Elle; Layeghi, Daniel; Mistry, Michael
- 年份：2025; 主题：Embodied/WBC; Agent RL; Robot Learning; 优先级：核心
- [来源](#)

论文：Learning Long-Horizon Robot Manipulation Skills via Privileged Action

快速理解：Long-horizon contact-rich tasks are challenging to learn with reinforcement learning, due to ineffective exploration of high-dimensional state spaces with sparse rewards. The learning process often gets stuck in local optimum and demands task-specific reward fine-tuning for complex scenarios.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；强化学习、模仿学习、规划、奖励或多智能体决策；机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：索引摘要。

1.3.3.51 Learning a Unified Policy for Position and Force Control in Legged Loco-Manipulation

- 作者：Zhi, Peiyuan; Li, Peiyang; Yin, Jianqin; Jia, Baoxiong; Huang, Siyuan
- 年份：2025; 主题：Embodied/WBC; Agent RL; Robot Learning; 优先级：核心
- [来源](#)

论文：Learning a Unified Policy for Position and Force Control in Legged Loco-Manipulation

快速理解：Robotic loco-manipulation tasks often involve contact-rich interactions with the environment, requiring the joint modeling of contact force and robot position. However, recent visuomotor policies often focus solely on position or force control, overlooking their integration.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；强化学习、模仿学习、规划、奖励或多智能体决策；机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：索引摘要。

1.3.3.52 Omni-Perception: Omnidirectional Collision Avoidance of Legged Robots in Dynamic Environments

- 作者：Wang, Zifan; Ma, Teli; Jia, Yufei; Yang, Xun; Zhou, Jiaming; OUYANG, Wenlong; Zhang, Qiang; Liang, Junwei
- 年份：2025; 主题：Embodied/WBC; Robot Learning; Systems; 优先级：核心
- [来源](#)

论文：Omni-Perception: Omnidirectional Collision Avoidance of Legged Robots in Dynamic Environments

快速理解：Agile locomotion in complex 3D environments requires robust spatial awareness to safely avoid diverse obstacles such as aerial clutter, uneven terrain, and dynamic agents. Depth-based perception approaches often struggle with sensor noise, lighting variability, computational overhead from intermediate representations (e.g., elevation maps), and difficulties with non-planar obstacles, limiting performance in unstructured environments.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；机器人数据、策略学习、操作、触觉或 sim-to-real；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.3.53 Robot Trains Robot: Automatic Real-World Policy Adaptation and Learning for Humanoids

- 作者：Hu, Kaizhe; Shi, Haochen; He, Yao; Wang, Weizhuo; Liu, Karen; Song, Shuran
- 年份：2025；主题：Embodied/WBC; Agent RL; Robot Learning; 优先级：核心
- [来源](#)

论文：Robot Trains Robot: Automatic Real-World Policy Adaptation and Learning for Humanoids

快速理解：Simulation-based reinforcement learning (RL) has significantly advanced humanoid locomotion tasks, yet direct real-world RL from scratch or starting from pretrained policies remains rare, limiting the full potential of humanoid robots. Real-world training, despite being crucial for overcoming the sim-to-real gap, faces substantial challenges related to safety, reward design, and learning efficiency.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；强化学习、模仿学习、规划、奖励或多智能体决策；机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：索引摘要。

1.3.3.54 SLAC: Simulation-Pretrained Latent Action Space for Whole-Body Real-World RL

- 作者：Hu, Jiaheng; Stone, Peter; Mart' {i}n-Mart' {i}n, Roberto
- 年份：2025；主题：Embodied/WBC; Agent RL; Robot Learning; 优先级：核心
- [来源](#)

论文：SLAC: Simulation-Pretrained Latent Action Space for Whole-Body Real-World RL

快速理解：Building capable household and industrial robots requires mastering the control of versatile, high-degree-of-freedom (DoF) systems such as mobile manipulators. While reinforcement learning (RL) holds promise for autonomously acquiring robot control policies, scaling it to high-DoF embodiments remains challenging.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；强化学习、模仿学习、规划、奖励或多智能体决策；机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：索引摘要。

1.3.3.55 Text2Touch: Tactile In-Hand Manipulation with LLM-Designed Reward Functions

- 作者：Field, Harrison; Yang, Max; Lin, Yijiong; Psomopoulou, Efi; Barton, David A.W.; Lepora, Nathan F.
- 年份：2025；主题：Embodied/WBC; Robot Learning; Supporting Perception; 优先级：核心
- [来源](#)

论文：Text2Touch: Tactile In-Hand Manipulation with LLM-Designed Reward Functions

快速理解：Large language models (LLMs) are beginning to automate reward design for dexterous

manipulation. However, no prior work has considered tactile sensing, which is known to be critical for human-like dexterity.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；机器人数据、策略学习、操作、触觉或 sim-to-real；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.3.56 ToddlerBot: Open-Source ML-Compatible Humanoid Platform for Loco-Manipulation

- 作者：Shi, Haochen; Wang, Weizhuo; Song, Shuran; Liu, Karen
- 年份：2025；主题：Embodied/WBC; Teleoperation; Robot Learning；优先级：核心
- [来源](#)

论文：ToddlerBot: Open-Source ML-Compatible Humanoid Platform for Loco-Manipulation

快速理解：Learning-based robotics research driven by data demands a new approach to robot hardware design—one that serves as both a platform for policy execution and a tool for embodied data collection. We introduce ToddlerBot, a low-cost, open-source humanoid robot platform designed for robotics and AI research.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；人类示范、远程操作、共享自治或动作重定向；机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：索引摘要。

1.3.3.57 Towards Embodiment Scaling Laws in Robot Locomotion

- 作者：Ai, Bo; Dai, Liu; Bohlinger, Nico; Li, Dichen; Mu, Tongzhou; Wu, Zhanxin; Fay, K.; Christensen, Henrik I; Peters, Jan; Su, Hao
- 年份：2025；主题：Embodied/WBC; VLA; Agent RL；优先级：核心
- [来源](#)

论文：Towards Embodiment Scaling Laws in Robot Locomotion

快速理解：Developing generalist agents that operate across diverse tasks, environments, and robot embodiments is a grand challenge in robotics and artificial intelligence. While substantial progress has been made in cross-task and cross-environment generalization, achieving broad generalization to novel embodiments remains elusive.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；视觉、语言条件与机器人动作策略之间的对齐；强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.3.58 TrackVLA: Embodied Visual Tracking in the Wild

- 作者：Wang, Shaoan; Zhang, Jiazhao; Li, Minghan; Liu, Jiahang; Li, Anqi; Wu, Kui; Zhong, Fangwei; Yu, Junzhi; Zhang, Zhizheng; Wang, He
- 年份：2025；主题：Embodied/WBC; VLA; Supporting Perception；优先级：核心
- [来源](#)

论文：TrackVLA: Embodied Visual Tracking in the Wild

快速理解：Embodied visual tracking is a fundamental skill in Embodied AI, enabling an agent to

follow a specific target in dynamic environments using only egocentric vision. This task is inherently challenging as it requires both accurate target recognition and effective trajectory planning under conditions of severe occlusion and high scene dynamics.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；视觉、语言条件与机器人动作策略之间的对齐；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.3.59 Training Strategies for Efficient Embodied Reasoning

- 作者：Chen, William; Belkhale, Suneel; Mirchandani, Suvir; Pertsch, Karl; Driess, Danny; Mees, Oier; Levine, Sergey
- 年份：2025；主题：Embodied/WBC; VLA; Supporting Perception; 优先级：核心
- [来源](#)

论文：Training Strategies for Efficient Embodied Reasoning

快速理解：Robot chain-of-thought reasoning (CoT) -wherein a model predicts helpful intermediate representations before choosing actions -provides an effective method for improving the generalization and performance of robot policies, especially vision-language-action models (VLAs). While such approaches have been shown to improve performance and generalization, they suffer from core limitations, like needing specialized robot reasoning data and slow inference speeds.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；视觉、语言条件与机器人动作策略之间的对齐；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.3.60 VT-Refine: Learning Bimanual Assembly with Visuo-Tactile Feedback via Simulation Fine-Tuning

- 作者：Huang, Binghao; Xu, Jie; Akinola, Ireteayo; Yang, Wei; Sundaralingam, Balakumar; O'Flaherty, Rowland; Fox, Dieter; Wang, Xiaolong; Mousavian, Arsalan; Chao, Yu-Wei; Li, Yunzhu
- 年份：2025；主题：Agent RL; Robot Learning; AI Infra; 优先级：核心
- [来源](#)

论文：VT-Refine: Learning Bimanual Assembly with Visuo-Tactile Feedback via Simulation Fine-Tuning

快速理解：Humans excel at bimanual assembly tasks by adapting to rich tactile feedback—a capability that remains difficult to replicate in robots through behavioral cloning alone, due to the suboptimality and limited diversity of human demonstrations. In this work, we present VT-Refine, a visuo-tactile policy learning framework that combines real-world demonstrations, high-fidelity tactile simulation, and reinforcement learning to tackle precise, contact-rich bimanual assembly.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；机器人数据、策略学习、操作、触觉或 sim-to-real；训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.3.61 3DS-VLA: A 3D Spatial-Aware Vision Language Action Model for Robust Multi-Task Manipulation

- 作者：Li, Xiaoqi; Heng, Liang; Liu, Jiaming; Shen, Yan; Gu, Chenyang; Liu, Zhuoyang; Chen,

Hao; Han, Nuowei; Zhang, Renrui; Tang, Hao; Zhang, Shanghang; Dong, Hao

- 年份: 2025; 主题: VLA; Supporting Perception; 优先级: 核心
- [来源](#)

论文: 3DS-VLA: A 3D Spatial-Aware Vision Language Action Model for Robust Multi-Task Manipulation

快速理解: Recently, 2D vision-language-action (VLA) models have made significant strides in multi-task manipulation. However, these models struggle to reason about 3D spatial relationships from 2D image inputs.

关注点: 视觉、语言条件与机器人动作策略之间的对齐; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.3.62 Adapt3R: Adaptive 3D Scene Representation for Domain Transfer in Imitation Learning

- 作者: Wilcox, Albert; Ghanem, Mohamed; Moghani, Masoud; Barroso, Pierre; Joffe, Benjamin; Garg, Animesh
- 年份: 2025; 主题: Agent RL; Supporting Perception; 优先级: 核心
- [来源](#)

论文: Adapt3R: Adaptive 3D Scene Representation for Domain Transfer in Imitation Learning

快速理解: Imitation Learning can train robots to perform complex and diverse manipulation tasks, but learned policies are brittle with observations outside of the training distribution. 3D scene representations that incorporate observations from calibrated RGBD cameras have been proposed as a way to mitigate this, but in our evaluations with unseen embodiments and camera viewpoints they show only modest improvement.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.3.63 CASPER: Inferring Diverse Intents for Assistive Teleoperation with Vision Language Models

- 作者: Liu, Huihan; Shah, Rutav; Liu, Shuijing; Pittenger, Jack; Seo, Mingyo; Cui, Yuchen; Bisk, Yonatan; Mart' $\{i\}_n$ -Mart' $\{i\}_n$, Roberto; Zhu, Yuke
- 年份: 2025; 主题: Teleoperation; Supporting Perception; 优先级: 核心
- [来源](#)

论文: CASPER: Inferring Diverse Intents for Assistive Teleoperation with Vision Language Models

快速理解: Assistive teleoperation, where control is shared between a human and a robot, enables efficient and intuitive human-robot collaboration in diverse and unstructured environments. A central challenge in real-world assistive teleoperation is for the robot to infer a wide range of human intentions from user control inputs and to assist users with correct actions.

关注点: 人类示范、远程操作、共享自治或动作重定向; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.3.64 Contrastive Forward Prediction Reinforcement Learning for Adaptive Fault-Tolerant Legged Robots

- 作者: Fu, Yangqing; Zhang, Yang; Yang, Qiyue; Yan, Liyun; Cao, Zhanxiang; Gao, Yue
- 年份: 2025; 主题: Embodied/WBC; Agent RL; 优先级: 核心
- [来源](#)

论文: Contrastive Forward Prediction Reinforcement Learning for Adaptive Fault-Tolerant Legged Robots

快速理解: In complex environments, adaptive and fault-tolerant capabilities are essential for legged robot locomotion. To address this challenge, this study proposes a reinforcement learning framework that integrates contrastive learning with forward prediction to achieve fault-tolerant locomotion for legged robots.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.3.65 DexSkin: High-Coverage Conformable Robotic Skin for Learning Contact-Rich Manipulation

- 作者: Wistreich, Suzannah; Shi, Baiyu; Tian, Stephen; Clarke, Samuel; Nath, Michael; Xu, Chengyi; Bao, Zhenan; Wu, Jiajun
- 年份: 2025; 主题: Embodied/WBC; Agent RL; Robot Learning; 优先级: 核心
- [来源](#)

论文: DexSkin: High-Coverage Conformable Robotic Skin for Learning Contact-Rich Manipulation

快速理解: Human skin provides a rich tactile sensing stream, localizing intentional and unintentional contact events over a large and contoured region. Replicating these tactile sensing capabilities for dexterous robotic manipulation systems remains a longstanding challenge.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 强化学习、模仿学习、规划、奖励或多智能体决策; 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 索引摘要。

1.3.3.66 EndoVLA: Dual-Phase Vision-Language-Action for Precise Autonomous Tracking in Endoscopy

- 作者: KIT, NG CHI; Bai, Long; Wang, Guankun; Wang, Yupeng; Gao, Huxin; yuan, Kun; Jin, Chenhan; Zeng, Tieyong; Ren, Hongliang
- 年份: 2025; 主题: VLA; Supporting Perception; 优先级: 核心
- [来源](#)

论文: EndoVLA: Dual-Phase Vision-Language-Action for Precise Autonomous Tracking in Endoscopy

快速理解: In endoscopic procedures, autonomous tracking of abnormal regions and following of circumferential cutting markers can significantly reduce the cognitive burden on endoscopists. However, conventional model-based pipelines are fragile—each component (e.g., detection, motion planning) requires manual tuning and struggles to incorporate high-level endoscopic intent, resulting in poor generalization across variable scenes.

关注点: 视觉、语言条件与机器人动作策略之间的对齐; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.3.67 First Order Model-Based RL through Decoupled Backpropagation

- 作者：Amigo, Joseph; Khorrambakht, Rooholla; Chane-Sane, Elliot; Mansard, Nicolas; Righetti, Ludovic
- 年份：2025；主题：Embodied/WBC; World Model; Agent RL; 优先级：核心
- [来源](#)

论文：First Order Model-Based RL through Decoupled Backpropagation

快速理解：There is growing interest in reinforcement learning (RL) methods that leverage the simulator's derivatives to improve learning efficiency. While early gradient-based approaches have demonstrated superior performance compared to derivative-free methods, accessing simulator gradients is often impractical due to their implementation cost or unavailability.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；环境动力学、视频预测、潜变量状态或模型式决策；强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.3.68 ImMimic: Cross-Domain Imitation from Human Videos via Mapping and Interpolation

- 作者：Liu, Yangcen; Shin, Woo Chul; Han, Yunhai; Chen, Zhenyang; Ravichandar, Harish; Xu, Danfei
- 年份：2025；主题：Embodied/WBC; Teleoperation; Robot Learning; 优先级：核心
- [来源](#)

论文：ImMimic: Cross-Domain Imitation from Human Videos via Mapping and Interpolation

快速理解：Learning robot manipulation from abundant human videos offers a scalable alternative to costly robot-specific data collection. However, domain gaps across visual, morphological, and physical aspects hinder direct imitation.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；人类示范、远程操作、共享自治或动作重定向；机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：索引摘要。

1.3.3.69 Imagine, Verify, Execute: Memory-guided Agentic Exploration with Vision-Language Models

- 作者：Lee, Seungjae; Ekpo, Daniel; Liu, Haowen; Huang, Furong; Shrivastava, Abhinav; Huang, Jia-Bin
- 年份：2025；主题：Agent RL; Supporting Perception; 优先级：核心
- [来源](#)

论文：Imagine, Verify, Execute: Memory-guided Agentic Exploration with Vision-Language Models

快速理解：Exploration is key for general-purpose robotic learning, particularly in open-ended environments where explicit guidance or task-specific feedback is limited. Vision-language models (VLMs), which can reason about object semantics, spatial relations, and potential outcomes, offer a promising foundation for guiding exploratory behavior by generating high-level goals or transitions.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.3.70 JaxRobotarium: Training and Deploying Multi-Robot Policies in 10 Minutes

- 作者：Jain, Shalin; Liu, Jiazhen; Kailas, Siva; Ravichandar, Harish
- 年份：2025；主题：Agent RL; Robot Learning; AI Infra; 优先级：核心
- [来源](#)

论文：JaxRobotarium: Training and Deploying Multi-Robot Policies in 10 Minutes

快速理解：Multi-agent reinforcement learning (MARL) has emerged as a promising solution for learning complex and scalable coordination behaviors in multi-robot systems. However, established MARL platforms (e.g., SMAC and MPE) lack robotics relevance and hardware deployment, leaving multi-robot learning researchers to develop bespoke environments and hardware testbeds dedicated to the development and evaluation of their individual contributions.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；机器人数据、策略学习、操作、触觉或 sim-to-real；训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.3.71 KDPE: A Kernel Density Estimation Strategy for Diffusion Policy Trajectory Selection

- 作者：Rosasco, Andrea; Ceola, Federico; Pasquale, Giulia; Natale, Lorenzo
- 年份：2025；主题：Agent RL; Robot Learning; Supporting Perception; 优先级：核心
- [来源](#)

论文：KDPE: A Kernel Density Estimation Strategy for Diffusion Policy Trajectory Selection

快速理解：Learning robot policies that capture multimodality in the training data has been a long-standing open challenge for behavior cloning. Recent approaches tackle the problem by modeling the conditional action distribution with generative models.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；机器人数据、策略学习、操作、触觉或 sim-to-real；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.3.72 LocoTouch: Learning Dynamic Quadrupedal Transport with Tactile Sensing

- 作者：Lin, Changyi; Song, Yuxin Ray; Huo, Boda; Yu, Mingyang; Wang, Yikai; Liu, Shiqi; Yang, Yuxiang; Yu, Wenhao; Zhang, Tingnan; Tan, Jie; Luo, Yiyue; Zhao, Ding
- 年份：2025；主题：Embodied/WBC; Robot Learning; 优先级：核心
- [来源](#)

论文：LocoTouch: Learning Dynamic Quadrupedal Transport with Tactile Sensing

快速理解：Quadrupedal robots have demonstrated remarkable agility and robustness in traversing complex terrains. However, they struggle with dynamic object interactions, where contact must be precisely sensed and controlled.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：索引摘要。

1.3.3.73 ManiFlow: A General Robot Manipulation Policy via Consistency Flow Training

- 作者: Yan, Ge; Zhu, Jiyue; Deng, Yuquan; Yang, Shiqi; Qiu, Ri-Zhao; Cheng, Xuxin; Memmel, Marius; Krishna, Ranjay; Goyal, Ankit; Wang, Xiaolong; Fox, Dieter
- 年份: 2025; 主题: Embodied/WBC; Robot Learning; 优先级: 核心
- [来源](#)

论文: ManiFlow: A General Robot Manipulation Policy via Consistency Flow Training

快速理解: Generative models based on flow matching offer significant potential for learning robot policies, particularly in generating high-dimensional, dexterous behaviors that are conditioned on diverse observations. In this work, we introduce ManiFlow, an advanced flow matching model specifically designed to support dexterous manipulation tasks.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：索引摘要。

1.3.3.74 ManipBench: Benchmarking Vision-Language Models for Low-Level Robot Manipulation

- 作者: Zhao, Enyu; Raval, Vedant; Zhang, Hejia; Mao, Jiageng; Shangguan, Zeyu; Nikolaidis, Stefanos; Wang, Yue; Seita, Daniel
- 年份: 2025; 主题: Embodied/WBC; Supporting Perception; 优先级: 核心
- [来源](#)

论文: ManipBench: Benchmarking Vision-Language Models for Low-Level Robot Manipulation

快速理解: Vision-Language Models (VLMs) have revolutionized artificial intelligence and robotics due to their commonsense reasoning capabilities. In robotic manipulation, VLMs are used primarily as high-level planners, but recent work has also studied their lower-level reasoning ability, which refers to making decisions about precise robot movements.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.3.75 Merging and Disentangling Views in Visual Reinforcement Learning for Robotic Manipulation

- 作者: Almuzairee, Abdulaziz; Patil, Rohan Prashant; Bhatt, Dwait; Christensen, Henrik I
- 年份: 2025; 主题: Embodied/WBC; Agent RL; 优先级: 核心
- [来源](#)

论文: Merging and Disentangling Views in Visual Reinforcement Learning for Robotic Manipulation

快速理解: Vision is well-known for its use in manipulation, especially using visual servoing. To make it robust, multiple cameras are needed to expand the field of view.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.3.76 Multi-Loco: Unifying Multi-Embodiment Legged Locomotion via Reinforcement Learning Augmented Diffusion

- 作者: Yang, Shunpeng; Fu, Zhen; Cao, Zhefeng; Junde, Guo; Wensing, Patrick; Zhang, Wei; Chen, Hua
- 年份: 2025; 主题: Embodied/WBC; Agent RL; 优先级: 核心
- [来源](#)

论文: Multi-Loco: Unifying Multi-Embodiment Legged Locomotion via Reinforcement Learning Augmented Diffusion

快速理解: Generalizing locomotion policies across diverse legged robots with varying morphologies is a key challenge due to differences in observation/action dimensions and system dynamics. In this work, we propose textit{Multi-Loco}, a novel unified framework combining a morphology-agnostic generative diffusion model with a lightweight residual policy optimized via reinforcement learning (RL).

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.3.77 QuaDreamer: Controllable Panoramic Video Generation for Quadruped Robots

- 作者: Wu, Sheng; Teng, Fei; Shi, Hao; Jiang, Qi; Luo, Kai; Wang, Kaiwei; Yang, Kailun
- 年份: 2025; 主题: Embodied/WBC; Supporting Perception; 优先级: 核心
- [来源](#)

论文: QuaDreamer: Controllable Panoramic Video Generation for Quadruped Robots

快速理解: Panoramic cameras, capturing comprehensive 360-degree environmental data, are suitable for quadruped robots in surrounding perception and interaction with complex environments. However, the scarcity of high-quality panoramic training data —caused by inherent kinematic constraints and complex sensor calibration challenges —fundamentally limits the development of robust perception systems tailored to these embodied platforms.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.3.78 ReWiND: Language-Guided Rewards Teach Robot Policies without New Demonstrations

- 作者: Zhang, Jiahui; Luo, Yusen; Anwar, Abrar; Sontakke, Sumedh Anand; Lim, Joseph J; Thomason, Jesse; Biyik, Erdem; Zhang, Jesse
- 年份: 2025; 主题: Embodied/WBC; Agent RL; Robot Learning; 优先级: 核心
- [来源](#)

论文: ReWiND: Language-Guided Rewards Teach Robot Policies without New Demonstrations

快速理解: We introduce ReWiND, a framework for learning robot manipulation tasks solely from language instructions without per-task demonstrations. Standard reinforcement learning (RL) and imitation learning methods require expert supervision through human-designed reward functions or demonstrations for every new task.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；强化学习、模仿学习、规划、奖励或多智能体决策；机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：索引摘要。

1.3.3.79 Reflective Planning: Vision-Language Models for Multi-Stage Long-Horizon Robotic Manipulation

- 作者：Feng, Yunhai; Han, Jiaming; Yang, Zhuoran; Yue, Xiangyu; Levine, Sergey; Luo, Jianlan
- 年份：2025；主题：Embodied/WBC; Supporting Perception；优先级：核心
- [来源](#)

论文：Reflective Planning: Vision-Language Models for Multi-Stage Long-Horizon Robotic Manipulation

快速理解：Solving complex long-horizon robotic manipulation problems requires sophisticated high-level planning capabilities, the ability to reason about the physical world, and reactively choose appropriate motor skills. Vision-language models (VLMs) pretrained on Internet data could in principle offer a framework for tackling such problems.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.3.80 Robust Dexterous Grasping of General Objects

- 作者：Zhang, Hui; Wu, Zijian; Huang, Linyi; Christen, Sammy; Song, Jie
- 年份：2025；主题：Embodied/WBC; Robot Learning；优先级：核心
- [来源](#)

论文：Robust Dexterous Grasping of General Objects

快速理解：The ability to robustly grasp a variety of objects is essential for dexterous robots. In this paper, we present a framework for zero-shot dynamic dexterous grasping using single-view visual inputs, designed to be resilient to various disturbances.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：索引摘要。

1.3.3.81 Self-supervised perception for tactile skin covered dexterous hands

- 作者：Sharma, Akash; Higuera, Carolina; Bodduluri, Chaithanya Krishna; Liu, Zixi; Fan, Taosha; Hellebrekers, Tess; Lambeta, Mike; Boots, Byron; Kaess, Michael; Wu, Tingfan; Hogan, Francois Robert; Mukadam, Mustafa
- 年份：2025；主题：Embodied/WBC; Robot Learning；优先级：核心
- [来源](#)

论文：Self-supervised perception for tactile skin covered dexterous hands

快速理解：We present PercepSkin, a pre-trained encoder for magnetic skin sensors distributed across the fingertips, phalanges, and palm of a dexterous robot hand. Magnetic tactile skins offer a flexible form factor for hand-wide coverage with fast response times, in contrast to vision-based tactile sensors that are restricted to the fingertips and limited by bandwidth.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：索引摘要。

1.3.3.82 Steerable Scene Generation with Post Training and Inference-Time Search

- 作者：Pfaff, Nicholas Ezra; Dai, Hongkai; Zakharov, Sergey; Iwase, Shun; Tedrake, Russ
- 年份：2025; 主题：Embodied/WBC; Agent RL; Supporting Perception; 优先级：核心
- [来源](#)

论文：Steerable Scene Generation with Post Training and Inference-Time Search

快速理解：Training robots in simulation requires diverse 3D scenes that reflect the specific challenges of downstream tasks. However, scenes that satisfy strict task requirements, such as high-clutter environments with plausible spatial arrangement, are rare and costly to curate manually.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；强化学习、模仿学习、规划、奖励或多智能体决策；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.3.83 The Sound of Simulation: Learning Multimodal Sim-to-Real Robot Policies with Generative Audio

- 作者：Wang, Renhao; Geng, Haoran; Li, Tingle; Wu, Philipp; Wang, Feishi; Anumanchipalli, Gopala; Darrell, Trevor; Li, Boyi; Abbeel, Pieter; Malik, Jitendra; Efros, Alexei A
- 年份：2025; 主题：Robot Learning; Supporting Perception; 优先级：核心
- [来源](#)

论文：The Sound of Simulation: Learning Multimodal Sim-to-Real Robot Policies with Generative Audio

快速理解：Robots must integrate multiple sensory modalities to act effectively in the real world. Yet, learning such multimodal policies at scale remains challenging.

关注点：机器人数据、策略学习、操作、触觉或 sim-to-real; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.3.84 Tool-as-Interface: Learning Robot Policies from Observing Human Tool Use

- 作者：Chen, Haonan; Zhu, Cheng; Liu, Shuijing; Li, Yunzhu; Driggs-Campbell, Katherine Rose
- 年份：2025; 主题：Teleoperation; Robot Learning; Supporting Perception; 优先级：核心
- [来源](#)

论文：Tool-as-Interface: Learning Robot Policies from Observing Human Tool Use

快速理解：Tool use is essential for enabling robots to perform complex real-world tasks, but learning such skills requires extensive datasets. While teleoperation is widely used, it is slow, delay-sensitive, and poorly suited for dynamic tasks.

关注点：人类示范、远程操作、共享自治或动作重定向；机器人数据、策略学习、操作、触觉或 sim-to-real; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.3.85 Door(s): Junction State Estimation for Efficient Exploration in Reinforcement Learning

- 作者: Fele, Benjamin; Babic, Jan
- 年份: 2025; 主题: Embodied/WBC; Agent RL; 优先级: 核心
- [来源](#)

论文: Door(s): Junction State Estimation for Efficient Exploration in Reinforcement Learning

快速理解: Exploration is one of the important bottlenecks for efficient learning in reinforcement learning, especially in the presence of sparse rewards. One way to traverse the environment faster is by passing through junctions, or metaphorical doors, in the state space.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.3.86 Agreement Volatility: A Second-Order Metric for Uncertainty Quantification in Surgical Robot Learning

- 作者: Thompson, Jordan; Jordan, Britton; Brown, Daniel S.; Kuntz, Alan
- 年份: 2025; 主题: Robot Learning; Systems; 优先级: 核心
- [来源](#)

论文: Agreement Volatility: A Second-Order Metric for Uncertainty Quantification in Surgical Robot Learning

快速理解: Autonomous surgical robots are a promising solution to the increasing demand for surgery amid a shortage of surgeons. Recent work has proposed learning-based approaches for the autonomous manipulation of soft tissue.

关注点: 机器人数据、策略学习、操作、触觉或 sim-to-real; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.3.87 AimBot: A Simple Auxiliary Visual Cue to Enhance Spatial Awareness of Visuomotor Policies

- 作者: Dai, Yinpei; Lee, Jayjun; Zhang, Yichi; Ma, Ziqiao; Yang, Jianing; Zadeh, Amir; Li, Chuan; Fazeli, Nima; Chai, Joyce
- 年份: 2025; 主题: Embodied/WBC; Robot Learning; 优先级: 核心
- [来源](#)

论文: AimBot: A Simple Auxiliary Visual Cue to Enhance Spatial Awareness of Visuomotor Policies

快速理解: In this paper, we propose AimBot, a lightweight visual augmentation technique that provides explicit spatial cues to improve visuomotor policy learning in robotic manipulation. AimBot overlays shooting lines and scope reticles onto multi-view RGB images, offering auxiliary visual guidance that encodes the end-effector's state.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 索引摘要。

1.3.3.88 AnyPlace: Learning Generalizable Object Placement for Robot Manipulation

- 作者: Zhao, Yuchi; Bogdanovic, Miroslav; Luo, Chengyuan; Tohme, Steven; Darvish, Kourosh; Aspuru-Guzik, Alan; Shkurti, Florian; Garg, Animesh
- 年份: 2025; 主题: Embodied/WBC; Supporting Perception; 优先级: 核心

- [来源](#)

论文: AnyPlace: Learning Generalizable Object Placement for Robot Manipulation

快速理解: Object placement in robotic tasks is inherently challenging due to the diversity of object geometries and placement configurations. We address this with AnyPlace, a two-stage method trained entirely on synthetic data, capable of predicting a wide range of feasible placement poses for real-world tasks.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.3.89 CLAMP: Crowdsourcing a LARge-scale in-the-wild haptic dataset with an open-source device for Multimodal robot Perception

- 作者: Thakkar, Pranav N.; Sinha, Shubhangi; Baijal, Karan; Bian, Yuhan (Anjelica); Lackey, Leah; Dodson, Ben; Kong, Heisen; Kwon, Jueun; Li, Amber; Hu, Yifei; rekoutis, alexios; Silver, Tom; Bhattacharjee, Tapomayukh
- 年份: 2025; 主题: Embodied/WBC; Supporting Perception; 优先级: 核心
- [来源](#)

论文: CLAMP: Crowdsourcing a LARge-scale in-the-wild haptic dataset with an open-source device for Multimodal robot Perception

快速理解: Robust robot manipulation in unstructured environments often requires understanding object properties that extend beyond geometry, such as material or compliance—properties that can be challenging to infer using vision alone. Multimodal haptic sensing provides a promising avenue for inferring such properties, yet progress has been constrained by the lack of large, diverse, and realistic haptic datasets.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.3.90 CLASS: Contrastive Learning via Action Sequence Supervision for Robot Manipulation

- 作者: Lee, Sung-Wook; Kang, Xuhui; Yang, Brandon Y.; Kuo, Yen-Ling
- 年份: 2025; 主题: Embodied/WBC; Agent RL; 优先级: 核心
- [来源](#)

论文: CLASS: Contrastive Learning via Action Sequence Supervision for Robot Manipulation

快速理解: Recent advances in Behavior Cloning (BC) have led to strong performance in robotic manipulation, driven by expressive models, sequence modeling of actions, and large-scale demonstration data. However, BC faces significant challenges when applied to heterogeneous datasets, such as visual shift with different camera poses or object appearances, where performance degrades despite the benefits of learning at scale.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.3.91 Constraint-Preserving Data Generation for One-Shot Visuomotor Policy Generalization

- 作者: Lin, Kevin; Ragunath, Varun; McAlinden, Andrew; Prasad, Aaditya; Wu, Jimmy; Zhu, Yuke; Bohg, Jeannette
- 年份: 2025; 主题: Embodied/WBC; Robot Learning; 优先级: 核心
- [来源](#)

论文: Constraint-Preserving Data Generation for One-Shot Visuomotor Policy Generalization

快速理解: Large-scale demonstration data has powered key breakthroughs in robot manipulation, but collecting that data remains costly and time-consuming. To this end, we present Constraint-Preserving Data Generation (CP-Gen), a method that uses a single expert trajectory to generate robot demonstrations containing novel object geometries and poses.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 索引摘要。

1.3.3.92 Decentralized Aerial Manipulation of a Cable-Suspended Load Using Multi-Agent Reinforcement Learning

- 作者: Zeng, Jack; Gimenez, Andreu Matoses; Vinitsky, Eugene; Alonso-Mora, Javier; Sun, Sihao
- 年份: 2025; 主题: Agent RL; Robot Learning; 优先级: 核心
- [来源](#)

论文: Decentralized Aerial Manipulation of a Cable-Suspended Load Using Multi-Agent Reinforcement Learning

快速理解: This paper presents the first decentralized method to enable real-world 6-DoF manipulation of a cable-suspended load using a team of Micro-Aerial Vehicles (MAVs). Our method leverages multi-agent reinforcement learning (MARL) to train an outer-loop control policy for each MAV.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 索引摘要。

1.3.3.93 Enter the Mind Palace: Reasoning and Planning for Long-term Active Embodied Question Answering

- 作者: Ginting, Muhammad Fadhil; Kim, Dong-Ki; Meng, Xiangyun; Reinke, Andrzej Marek; Krishna, Bandi Jai; Kayhani, Navid; Peltzer, Oriana; Fan, David; Shaban, Amirreza; Kim, Sung-Kyun; Kochenderfer, Mykel; Agha-mohammadi, Ali-akbar; Omidshafiei, Shayegan
- 年份: 2025; 主题: Embodied/WBC; Systems; 优先级: 核心
- [来源](#)

论文: Enter the Mind Palace: Reasoning and Planning for Long-term Active Embodied Question Answering

快速理解: As robots become increasingly capable of operating over extended periods—spanning days, weeks, and even months—they are expected to accumulate knowledge of their environments and leverage this experience to assist humans more effectively. This paper studies the problem of Long-term Active Embodied Question Answering (LA-EQA), a new task in which a robot must both recall past experiences and actively explore its environment to answer complex, temporally-grounded

questions.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.3.94 FACET: Force-Adaptive Control via Impedance Reference Tracking for Legged Robots

- 作者: Xu, Botian; Weng, Haoyang; Lu, Qingzhou; Gao, Yang; Xu, Huazhe
- 年份: 2025; 主题: Embodied/WBC; Agent RL; 优先级: 核心
- [来源](#)

论文: FACET: Force-Adaptive Control via Impedance Reference Tracking for Legged Robots

快速理解: Reinforcement learning (RL) has made significant strides in legged robot control, enabling locomotion across diverse terrains and complex loco-manipulation capabilities. However, the commonly used position or velocity tracking-based objectives are agnostic to forces experienced by the robot, leading to stiff and potentially dangerous behaviors and poor control during forceful interactions.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.3.95 Fast Flow-based Visuomotor Policies via Conditional Optimal Transport Couplings

- 作者: Sochopoulos, Andreas; Malkin, Nikolay; Tsagkas, Nikolaos; Moura, Joao; Gienger, Michael; Vijayakumar, Sethu
- 年份: 2025; 主题: Robot Learning; Supporting Perception; 优先级: 核心
- [来源](#)

论文: Fast Flow-based Visuomotor Policies via Conditional Optimal Transport Couplings

快速理解: Diffusion and flow matching policies have recently demonstrated remarkable performance in robotic applications by accurately capturing multimodal robot trajectory distributions. However, their computationally expensive inference, due to the numerical integration of an ODE or SDE, limits their applicability as real-time controllers for robots.

关注点：机器人数据、策略学习、操作、触觉或 sim-to-real; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.3.96 Geometric Red-Teaming for Robotic Manipulation

- 作者: Goel, Divyam; Wang, Yufei; Wu, Tiancheng; Qiao, Guixiu; Pilipchak, Pavel; Held, David; Erickson, Zackory
- 年份: 2025; 主题: Embodied/WBC; Robot Learning; 优先级: 核心
- [来源](#)

论文: Geometric Red-Teaming for Robotic Manipulation

快速理解: Standard evaluation protocols in robotic manipulation typically assess policy performance over curated, in-distribution test sets, offering limited insight into how systems fail under plausible variation. We introduce a red-teaming framework that probes robustness through object-centric ge-

ometric perturbations, automatically generating CrashShapes—structurally valid, user-constrained mesh deformations that trigger catastrophic failures in pre-trained manipulation policies.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：索引摘要。

1.3.3.97 GraphEQA: Using 3D Semantic Scene Graphs for Real-time Embodied Question Answering

- 作者: Saxena, Saumya; Buchanan, Blake; Paxton, Chris; Liu, Peiqi; Chen, Bingqing; Vaskevicius, Narunas; Palmieri, Luigi; Francis, Jonathan; Kroemer, Oliver
- 年份: 2025; 主题: Embodied/WBC; Supporting Perception; 优先级: 核心
- [来源](#)

论文: GraphEQA: Using 3D Semantic Scene Graphs for Real-time Embodied Question Answering

快速理解: In Embodied Question Answering (EQA), agents must explore and develop a semantic understanding of an unseen environment in order to answer a situated question with confidence. This remains a challenging problem in robotics, due to the difficulties in obtaining useful semantic representations, updating these representations online, and leveraging prior world knowledge for efficient exploration and planning.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.3.98 GraspMolmo: Generalizable Task-Oriented Grasping via Large-Scale Synthetic Data Generation

- 作者: Deshpande, Abhay; Deng, Yuquan; Salvador, Jordi; Ray, Arijit; Han, Winson; Duan, Jiafei; Hendrix, Rose; Zhu, Yuke; Krishna, Ranjay
- 年份: 2025; 主题: Embodied/WBC; Robot Learning; 优先级: 核心
- [来源](#)

论文: GraspMolmo: Generalizable Task-Oriented Grasping via Large-Scale Synthetic Data Generation

快速理解: We present GraspMolmo, a generalizable open-vocabulary task-oriented grasping (TOG) model. GraspMolmo predicts semantically appropriate, stable grasps conditioned on a natural language instruction and a single RGB-D frame.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：索引摘要。

1.3.3.99 Imitation Learning Based on Disentangled Representation Learning of Behavioral Characteristics

- 作者: Oishi, Ryoga; Sakaino, Sho; Tsuji, Toshiaki
- 年份: 2025; 主题: Agent RL; Robot Learning; 优先级: 核心
- [来源](#)

论文: Imitation Learning Based on Disentangled Representation Learning of Behavioral Characteristics

快速理解: In the field of robot learning, it is becoming possible to coordinate robot action through language instructions. On the other hand, it is still a difficult task to adjust the action based on human instructions because human instructions are often qualitative, and there are cases where there is no one-to-one correspondence between the behavior and the instructions.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 索引摘要。

1.3.3.100 LaVA-Man: Learning Visual Action Representations for Robot Manipulation

- 作者: Zhu, Chaoran; Wang, Hengyi; Pang, Yik Lung; Oh, Changjae
- 年份: 2025; 主题: Embodied/WBC; Supporting Perception; 优先级: 核心
- [来源](#)

论文: LaVA-Man: Learning Visual Action Representations for Robot Manipulation

快速理解: Visual-textual understanding is essential for language-guided robot manipulation. Recent works leverage pre-trained vision-language models to measure the similarity between encoded visual observations and textual instructions, and then train a model to map this similarity to robot actions.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.3.101 Learning Deployable Locomotion Control via Differentiable Simulation

- 作者: Schwarke, Clemens; Klemm, Victor; Bagajo, Joshua; Sleiman, Jean Pierre; Georgiev, Ignat; Torres, Jesus Tordesillas; Hutter, Marco
- 年份: 2025; 主题: Embodied/WBC; Robot Learning; 优先级: 核心
- [来源](#)

论文: Learning Deployable Locomotion Control via Differentiable Simulation

快速理解: Differentiable simulators promise to improve sample efficiency in robot learning by providing analytic gradients of the system dynamics. Yet, their application to contact-rich tasks like locomotion is complicated by the inherently non-smooth nature of contact, impeding effective gradient-based optimization.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 索引摘要。

1.3.3.102 LodeStar: Long-horizon Dexterity via Synthetic Data Augmentation from Human Demonstrations

- 作者: Wan, Weikang; Fu, Jiawei; Yuan, Xiaodi; Zhu, Yifeng; Su, Hao
- 年份: 2025; 主题: Embodied/WBC; Agent RL; 优先级: 核心
- [来源](#)

论文: LodeStar: Long-horizon Dexterity via Synthetic Data Augmentation from Human Demonstrations

快速理解: Developing robotic systems capable of robustly executing long-horizon manipulation tasks with human-level dexterity is challenging, as such tasks require both physical dexterity and seamless

sequencing of manipulation skills while robustly handling environment variations. While imitation learning offers a promising approach, acquiring comprehensive datasets is resource-intensive.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.3.103 Mastering Multi-Drone Volleyball through Hierarchical Co-Self-Play Reinforcement Learning

- 作者：Zhang, Ruize; Xiang, Sirui; Xu, Zelai; Gao, Feng; Ji, Shilong; Tang, Wenhao; Ding, Wenbo; Yu, Chao; Wang, Yu
- 年份：2025；主题：Embodied/WBC; Agent RL; 优先级：核心
- [来源](#)

论文：Mastering Multi-Drone Volleyball through Hierarchical Co-Self-Play Reinforcement Learning

快速理解：In this paper, we tackle the problem of learning to play 3v3 multi-drone volleyball, a new embodied competitive task that requires both high-level strategic coordination and low-level agile control. The task is turn-based, multi-agent, and physically grounded, posing significant challenges due to its long-horizon dependencies, tight inter-agent coupling, and the underactuated dynamics of quadrotors.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.3.104 Morphologically Symmetric Reinforcement Learning for Ambidextrous Bimanual Manipulation

- 作者：Li, Zechu; Jin, Yufeng; Ordonez-Apraez, Daniel; Semini, Claudio; Liu, Puze; Chalvatzaki, Georgia
- 年份：2025；主题：Embodied/WBC; Agent RL; 优先级：核心
- [来源](#)

论文：Morphologically Symmetric Reinforcement Learning for Ambidextrous Bimanual Manipulation

快速理解：Humans naturally exhibit bilateral symmetry in their gross manipulation skills, effortlessly mirroring simple actions between left and right hands. Bimanual robots—which also feature bilateral symmetry—should similarly exploit this property to perform tasks with either hand.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.3.105 Motion Priors Reimagined: Adapting Flat-Terrain Skills for Complex Quadruped Mobility

- 作者：Zhang, Zewei; Li, Chenhao; Miki, Takahiro; Hutter, Marco
- 年份：2025；主题：Embodied/WBC; Agent RL; 优先级：核心
- [来源](#)

论文：Motion Priors Reimagined: Adapting Flat-Terrain Skills for Complex Quadruped Mobility

快速理解：Reinforcement learning (RL)-based legged locomotion controllers often require meticulous reward tuning to track velocities or goal positions while preserving smooth motion on various ter-

rains. Motion imitation methods via RL using demonstration data reduce reward engineering but fail to generalize to novel environments.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.3.106 Multi-critic Learning for Whole-body End-effector Twist Tracking

- 作者：Vijayan, Aravind Elanjimattathil; Cramariuc, Andrei; Risiglione, Mattia; Gehring, Christian; Hutter, Marco
- 年份：2025；主题：Embodied/WBC; Agent RL；优先级：核心
- [来源](#)

论文：Multi-critic Learning for Whole-body End-effector Twist Tracking

快速理解：Learning whole-body control for locomotion and arm motions in a single policy has challenges, as the two tasks have conflicting goals. For instance, efficient locomotion typically favors a horizontal base orientation, while end-effector tracking may benefit from base tilting to extend reachability.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.3.107 Non-conflicting Energy Minimization in Reinforcement Learning based Robot Control

- 作者：Peri, Skand; Perincherry, Akhil; Pandit, Bikram; Lee, Stefan
- 年份：2025；主题：Embodied/WBC; Agent RL；优先级：核心
- [来源](#)

论文：Non-conflicting Energy Minimization in Reinforcement Learning based Robot Control

快速理解：Efficient robot locomotion often requires balancing task performance with energy expenditure. A common approach in reinforcement learning (RL) is to penalize energy use directly in the reward function.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.3.108 O³Afford: One-Shot 3D Object-to-Object Affordance Grounding for Generalizable Robotic Manipulation

- 作者：Tian, Tongxuan; Kang, Xuhui; Kuo, Yen-Ling
- 年份：2025；主题：Embodied/WBC; Systems；优先级：核心
- [来源](#)

论文：O³Afford: One-Shot 3D Object-to-Object Affordance Grounding for Generalizable Robotic Manipulation

快速理解：Grounding object affordance is fundamental to robotic manipulation as it establishes the critical link between perception and action among interacting objects. However, prior works predominantly focus on predicting single-object affordance, overlooking the fact that most real-world interactions involve relationships between pairs of objects.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.3.109 One Demo is Worth a Thousand Trajectories: Action-View Augmentation for Visuomotor Policies

- 作者：Pan, Chuer; Liang, Litian; Bauer, Dominik; Cousineau, Eric; Burchfiel, Benjamin; Feng, Siyuan; Song, Shuran
- 年份：2025；主题：Robot Learning; Supporting Perception; 优先级：核心
- [来源](#)

论文：One Demo is Worth a Thousand Trajectories: Action-View Augmentation for Visuomotor Policies

快速理解：Visuomotor policies for manipulation have demonstrated remarkable potential in modeling complex robotic behaviors, yet minor alterations in the robot's initial configuration and unseen obstacles easily lead to out-of-distribution observations. Without extensive data collection effort, these result in catastrophic execution failures.

关注点：机器人数据、策略学习、操作、触觉或 sim-to-real; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.3.110 Planning from Point Clouds over Continuous Actions for Multi-object Rearrangement

- 作者：Saha, Kallol; Li, Amber; Rodriguez-Izquierdo, Angela; Yu, Lifan; Eisner, Ben; Likhachev, Maxim; Held, David
- 年份：2025；主题：Supporting Perception; Systems; 优先级：核心
- [来源](#)

论文：Planning from Point Clouds over Continuous Actions for Multi-object Rearrangement

快速理解：Multi-object rearrangement is a challenging task that requires robots to reason about a physical 3D scene and the effects of a sequence of actions. While traditional task planning methods are shown to be effective for long-horizon manipulation, they require discretizing the continuous state and action space into symbolic descriptions of objects, object relationships, and actions.

关注点：3D、视频、多模态视觉或具身感知支撑能力；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.3.111 Point Policy: Unifying Observations and Actions with Key Points for Robot Manipulation

- 作者：Haldar, Siddhant; Pinto, Lerrel
- 年份：2025；主题：Embodied/WBC; Teleoperation; 优先级：核心
- [来源](#)

论文：Point Policy: Unifying Observations and Actions with Key Points for Robot Manipulation

快速理解：Building robotic agents capable of operating across diverse environments and object types remains a significant challenge, often requiring extensive data collection. This is particularly restrictive in robotics, where each data point must be physically executed in the real world.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；人类示范、远程操作、共享自治或动作重定向。

证据层级：索引摘要。

1.3.3.112 Robot Learning from Any Images

- 作者: Zhao, Siheng; Mao, Jiageng; Chow, Wei; Shangguan, Zeyu; Shi, Tianheng; Xue, Rong; Zheng, Yuxi; Weng, Yijia; You, Yang; Seita, Daniel; Guibas, Leonidas; Zakharov, Sergey; Guizilini, Vitor Campagnolo; Wang, Yue
- 年份: 2025; 主题: Embodied/WBC; Robot Learning; 优先级: 核心
- [来源](#)

论文: Robot Learning from Any Images

快速理解: We introduce RoLA, a framework that transforms any in-the-wild image into an interactive, physics-enabled robotic environment. Unlike previous methods, RoLA operates directly on a single image without requiring additional hardware or digital assets.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：索引摘要。

1.3.3.113 SDS -See it, Do it, Sorted: Quadruped Skill Synthesis from Single Video Demonstration

- 作者: Stamatopoulou, Maria; Li, Jeffrey; Kanoulas, Dimitrios
- 年份: 2025; 主题: Embodied/WBC; Robot Learning; 优先级: 核心
- [来源](#)

论文: SDS -See it, Do it, Sorted: Quadruped Skill Synthesis from Single Video Demonstration

快速理解: Imagine a robot learning locomotion skills from any single video, without labels or reward engineering. We introduce SDS ("See it.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：索引摘要。

1.3.3.114 Sampling-based System Identification with Active Exploration for Legged Sim2Real Learning

- 作者: Sobanbabu, Nikhil; He, Guanqi; He, Tairan; Yang, Yuxiang; Shi, Guanya
- 年份: 2025; 主题: Embodied/WBC; Robot Learning; 优先级: 核心
- [来源](#)

论文: Sampling-based System Identification with Active Exploration for Legged Sim2Real Learning

快速理解: Sim-to-real discrepancies hinder learning-based policies from achieving high-precision tasks in the real world. While Domain Randomization (DR) is commonly used to bridge this gap, it often relies on heuristics and can lead to overly conservative policies with degrading performance when not properly tuned.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：索引摘要。

1.3.3.115 Toward Real-World Cooperative and Competitive Soccer with Quadrupedal Robot Teams

- 作者: Su, Zhi; Gao, Yuman; Lukas, Emily; Li, Yunfei; Cai, Jiaze; Talubah, Faris; Gao, Fei; Yu, Chao; Li, Zhongyu; Wu, Yi; Sreenath, Koushil
- 年份: 2025; 主题: Embodied/WBC; Agent RL; 优先级: 核心
- [来源](#)

论文: Toward Real-World Cooperative and Competitive Soccer with Quadrupedal Robot Teams

快速理解: Achieving coordinated teamwork among legged robots requires both fine-grained locomotion control and long-horizon strategic decision-making. Robot soccer offers a compelling testbed for this challenge, combining dynamic, competitive, and multi-agent interactions.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.3.116 Visual Imitation Enables Contextual Humanoid Control

- 作者: Allshire, Arthur; Choi, Hongsuk; Zhang, Junyi; McAllister, David; Zhang, Anthony; Kim, Chung Min; Darrell, Trevor; Abbeel, Pieter; Malik, Jitendra; Kanazawa, Angjoo
- 年份: 2025; 主题: Embodied/WBC; Robot Learning; 优先级: 核心
- [来源](#)

论文: Visual Imitation Enables Contextual Humanoid Control

快速理解: How can we teach humanoids to climb staircases and sit on chairs using the surrounding environment context? Arguably the simplest way is to __just show them__—casually capture a human motion video and feed it to humanoids.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 索引摘要。

1.3.3.117 exUMI: Extensible Robot Teaching System with Action-aware Task-agnostic Tactile Representation

- 作者: Xu, Yue; Wei, Litao; An, Pengyu; Zhang, Qingyu; Li, Yong-Lu
- 年份: 2025; 主题: Agent RL; Robot Learning; 优先级: 核心
- [来源](#)

论文: exUMI: Extensible Robot Teaching System with Action-aware Task-agnostic Tactile Representation

快速理解: Tactile-aware robot learning faces critical challenges in data collection and representation due to data scarcity and sparsity, and the absence of force feedback in existing systems. To address these limitations, we introduce a tactile robot learning system with both hardware and algorithm innovations.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 索引摘要。

1.3.3.118 ATK: Automatic Task-driven Keypoint Selection for Robust Policy Learning

- 作者: Zhang, Yunchu; Mittal, Shubham; Zhang, Zhengyu; Ke, Liyiming; Srinivasa, Siddhartha; Gupta, Abhishek
- 年份: 2025; 主题: Agent RL; Robot Learning; 优先级: 重要
- [来源](#)

论文: ATK: Automatic Task-driven Keypoint Selection for Robust Policy Learning

快速理解: Learning visuomotor policy through imitation learning often suffers from perceptual challenges, where visual differences between training and evaluation environments degrade policy performance. Policies relying on state estimations like 6D pose, require task-specific tracking and are difficult to scale, while raw sensor-based policies may lack robustness to small visual disturbances.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 索引摘要。

1.3.3.119 Action-Free Reasoning for Policy Generalization

- 作者: Clark, Jaden; Mirchandani, Suvir; Sadigh, Dorsa; Belkhale, Suneel
- 年份: 2025; 主题: Agent RL; Robot Learning; 优先级: 重要
- [来源](#)

论文: Action-Free Reasoning for Policy Generalization

快速理解: End-to-end imitation learning offers a promising approach for training robot policies. However, generalizing to new settings—such as unseen scenes, tasks, and object instances—remains a significant challenge.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 索引摘要。

1.3.3.120 BEVCalib: LiDAR-Camera Calibration via Geometry-Guided Bird's-Eye View Representation

- 作者: Yuan, Weiduo; Li, Jerry; Yue, Justin; Shah, Divyank; Karydis, Konstantinos; Qiu, Hang
- 年份: 2025; 主题: AI Infra; Supporting Perception; 优先级: 重要
- [来源](#)

论文: BEVCalib: LiDAR-Camera Calibration via Geometry-Guided Bird's-Eye View Representation

快速理解: Accurate LiDAR-camera calibration is the foundation of accurate multimodal fusion environmental perception for autonomous driving and robotic systems. Traditional calibration methods require extensive data collection in controlled environments and cannot compensate for the transformation changes during the vehicle/robot movement.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.3.121 CaRL: Learning Scalable Planning Policies with Simple Rewards

- 作者: Jaeger, Bernhard; Dauner, Daniel; Bei{ss}wenger, Jens; Gerstenecker, Simon; Chitta, Kashyap; Geiger, Andreas
- 年份: 2025; 主题: Agent RL; AI Infra; 优先级: 重要
- [来源](#)

论文: CaRL: Learning Scalable Planning Policies with Simple Rewards

快速理解: We investigate reinforcement learning (RL) for privileged planning in autonomous driving. State-of-the-art approaches for this task are rule-based, but these methods do not scale to the long tail.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.3.122 Co-Design of Soft Gripper with Neural Physics

- 作者: Yi, Sha; Bai, Xueqian; Singh, Adabhav; Ye, Jianglong; Tolley, Michael T.; Wang, Xiaolong
- 年份: 2025; 主题: Embodied/WBC; Robot Learning; 优先级: 重要
- [来源](#)

论文: Co-Design of Soft Gripper with Neural Physics

快速理解: For robot manipulation, both the controller and end-effector design are crucial. Compared with rigid grippers, soft grippers are more generalizable by deforming to different geometries, but designing such a gripper and finding its grasp pose remains challenging.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 索引摘要。

1.3.3.123 Eye, Robot: Learning to Look to Act with a BC-RL Perception-Action Loop

- 作者: Kerr, Justin; Hari, Kush; Weber, Ethan; Kim, Chung Min; Yi, Brent; bonnen, tyler; Goldberg, Ken; Kanazawa, Angjoo
- 年份: 2025; 主题: Teleoperation; Agent RL; 优先级: 重要
- [来源](#)

论文: Eye, Robot: Learning to Look to Act with a BC-RL Perception-Action Loop

快速理解: Humans do not passively observe the visual world—we actively look in order to act. Motivated by this principle, we introduce EyeRobot, a robotic system with gaze behavior that emerges from the need to complete real-world tasks.

关注点: 人类示范、远程操作、共享自治或动作重定向; 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.3.124 Fabrica: Dual-Arm Assembly of General Multi-Part Objects via Integrated Planning and Learning

- 作者: Tian, Yunsheng; Jacob, Joshua; Huang, Yijiang; Zhao, Jialiang; Gu, Edward Li; Ma, Pingchuan; Zhang, Annan; Javid, Farhad; Romero, Branden; Chitta, Sachin; Sueda, Shinjiro; Li, Hui; Matusik, Wojciech
- 年份: 2025; 主题: Agent RL; Robot Learning; 优先级: 重要
- [来源](#)

论文: Fabrica: Dual-Arm Assembly of General Multi-Part Objects via Integrated Planning and Learning

快速理解: Multi-part assembly poses significant challenges for robotic systems to execute long-horizon,

contact-rich manipulation with generalization across complex geometries. We present a dual-arm robotic system capable of end-to-end planning and control for autonomous assembly of general multi-part objects.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：索引摘要。

1.3.3.125 FetchBot: Learning Generalizable Object Fetching in Cluttered Scenes via Zero-Shot Sim2Real

- 作者：Liu, Weiheng; Wan, Yuxuan; Wang, Jilong; Kuang, Yuxuan; Shi, Xuesong; Li, Haoran; Zhao, Dongbin; Zhang, Zhizheng; Wang, He
- 年份：2025；主题：Embodied/WBC; Robot Learning；优先级：重要
- [来源](#)

论文：FetchBot: Learning Generalizable Object Fetching in Cluttered Scenes via Zero-Shot Sim2Real

快速理解：Generalizable object fetching in cluttered scenes remains a fundamental and application-critical challenge in embodied AI. Closely packed objects cause inevitable occlusions, making safe action generation particularly difficult.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：索引摘要。

1.3.3.126 From Tabula Rasa to Emergent Abilities: Discovering Robot Skills via Real-World Unsupervised Quality-Diversity

- 作者：Grillotti, Luca; Coiffard, Lisa; Pang, Oscar; Faldor, Maxence; Cully, Antoine
- 年份：2025；主题：Embodied/WBC; Robot Learning；优先级：重要
- [来源](#)

论文：From Tabula Rasa to Emergent Abilities: Discovering Robot Skills via Real-World Unsupervised Quality-Diversity

快速理解：Autonomous skill discovery aims to enable robots to acquire diverse behaviors without explicit supervision. Learning such behaviors directly on physical hardware remains challenging due to safety and data efficiency constraints.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：索引摘要。

1.3.3.127 Joint Model-based Model-free Diffusion for Planning with Constraints

- 作者：Jung, Wonsuhk; Mishra, Utkarsh Aashu; Arachchige, Nadun Ranawaka; Chen, Yongxin; Xu, Danfei; Kousik, Shreyas
- 年份：2025；主题：Embodied/WBC; Agent RL；优先级：重要
- [来源](#)

论文：Joint Model-based Model-free Diffusion for Planning with Constraints

快速理解：Model-free diffusion planners have shown great promise for robot motion planning, but practical robotic systems often require combining them with model-based optimization modules to

enforce constraints, such as safety. Naïvely integrating these modules presents compatibility challenges when diffusion's multi-modal outputs behave adversarially to optimization-based modules.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.3.128 KineSoft: Learning Proprioceptive Manipulation Policies with Soft Robot Hands

- 作者: Yoo, Uksang; Francis, Jonathan; Oh, Jean; Ichnowski, Jeffrey
- 年份: 2025; 主题: Embodied/WBC; Agent RL; 优先级: 重要
- [来源](#)

论文: KineSoft: Learning Proprioceptive Manipulation Policies with Soft Robot Hands

快速理解: Underactuated soft robot hands offer inherent safety and adaptability advantages over rigid systems, but developing dexterous manipulation skills remains challenging. While imitation learning shows promise for complex manipulation tasks, traditional approaches struggle with soft systems due to demonstration collection challenges and ineffective state representations.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.3.129 Learning Long-Context Diffusion Policies via Past-Token Prediction

- 作者: Villasevil, Marcel Torne; Tang, Andy; Liu, Yuejiang; Finn, Chelsea
- 年份: 2025; 主题: Agent RL; Systems; 优先级: 重要
- [来源](#)

论文: Learning Long-Context Diffusion Policies via Past-Token Prediction

快速理解: Reasoning over long sequences of observations and actions is essential for many robotic tasks. Yet, learning effective long-context policies from demonstrations remains challenging.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.3.130 Learning from 10 Demos: Generalisable and Sample-Efficient Policy Learning with Oriented Affordance Frames

- 作者: Rana, Krishan; Abou-Chakra, Jad; Garg, Sourav; Lee, Robert; Reid, Ian; Suenderhauf, Niko
- 年份: 2025; 主题: Embodied/WBC; Agent RL; 优先级: 重要
- [来源](#)

论文: Learning from 10 Demos: Generalisable and Sample-Efficient Policy Learning with Oriented Affordance Frames

快速理解: Imitation learning has unlocked the potential for robots to exhibit highly dexterous behaviours. However, it still struggles with long-horizon, multi-object tasks due to poor sample efficiency and limited generalisation.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.3.131 MEReQ: Max-Ent Residual-Q Inverse RL for Sample-Efficient Alignment from Intervention

- 作者: Chen, Yuxin; Tang, Chen; Wei, Jianglan; Li, Chenran; Tian, Thomas; Zhang, Xiang; Zhan, Wei; Stone, Peter; Tomizuka, Masayoshi
- 年份: 2025; 主题: Embodied/WBC; Agent RL; 优先级: 重要
- [来源](#)

论文: MEReQ: Max-Ent Residual-Q Inverse RL for Sample-Efficient Alignment from Intervention

快速理解: Aligning robot behavior with human preferences is crucial for deploying embodied AI agents in human-centered environments. A promising solution is interactive imitation learning from human intervention, where a human expert observes the policy's execution and provides interventions as feedback.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.3.132 MimicFunc: Imitating Tool Manipulation from a Single Human Video via Functional Correspondence

- 作者: Tang, Chao; Xiao, Anxing; Deng, Yuhong; Hu, Tianrun; Dong, Wenlong; Zhang, Hanbo; Hsu, David; Zhang, Hong
- 年份: 2025; 主题: Teleoperation; Robot Learning; 优先级: 重要
- [来源](#)

论文: MimicFunc: Imitating Tool Manipulation from a Single Human Video via Functional Correspondence

快速理解: Imitating tool manipulation from human videos offers an intuitive approach to teaching robots, while also providing a promising and scalable alternative to labor-intensive teleoperation data collection for visuomotor policy learning. While humans can mimic tool manipulation behavior by observing others perform a task just once and effortlessly transfer the skill to diverse tools for functionally equivalent tasks, current robots struggle to achieve this level of generalization.

关注点: 人类示范、远程操作、共享自治或动作重定向; 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 索引摘要。

1.3.3.133 One View, Many Worlds: Single-Image to 3D object Meets Generative Domain Randomization for One-Shot 6D Pose Estimation

- 作者: Geng, Zheng; Wang, Nan; Xu, Shaocong; Ye, Chongjie; Li, Bohan; Chen, Zhaoxi; Peng, Sida; Zhao, Hao
- 年份: 2025; 主题: Embodied/WBC; Robot Learning; 优先级: 重要
- [来源](#)

论文: One View, Many Worlds: Single-Image to 3D object Meets Generative Domain Randomization for One-Shot 6D Pose Estimation

快速理解: Estimating the 6D pose of arbitrary objects from a single reference image is a critical yet challenging task in robotics, especially considering the long-tail distribution of real-world instances. While category-level and model-based approaches have achieved notable progress, they remain lim-

ited in generalizing to unseen objects under one-shot settings.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：索引摘要。

1.3.3.134 Phantom: Training Robots Without Robots Using Only Human Videos

- 作者：Lepert, Marion; Fang, Jiaying; Bohg, Jeannette
- 年份：2025；主题：Teleoperation; Robot Learning; 优先级：重要
- [来源](#)

论文：Phantom: Training Robots Without Robots Using Only Human Videos

快速理解：Training general-purpose robots requires learning from large and diverse data sources. Current approaches rely heavily on teleoperated demonstrations which are difficult to scale.

关注点：人类示范、远程操作、共享自治或动作重定向；机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：索引摘要。

1.3.3.135 Predictive Red Teaming: Breaking Policies Without Breaking Robots

- 作者：Majumdar, Anirudha; Sharma, Mohit; Kalashnikov, Dmitry; Singh, Sumeet; Sermanet, Pierre; Sindhvani, Vikas
- 年份：2025；主题：Agent RL; Robot Learning; 优先级：重要
- [来源](#)

论文：Predictive Red Teaming: Breaking Policies Without Breaking Robots

快速理解：Visuomotor policies trained via imitation learning are capable of performing challenging manipulation tasks, but are often extremely brittle to lighting, visual distractors, and object locations. These vulnerabilities can depend unpredictably on the specifics of training, and are challenging to expose without time-consuming and expensive hardware evaluations.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：索引摘要。

1.3.3.136 ReasonPlan: Unified Scene Prediction and Decision Reasoning for Closed-loop Autonomous Driving

- 作者：Liu, Xueyi; Zhong, Zuodong; Zhang, Qichao; Guo, Yuxin; Zheng, Yupeng; Wang, Junli; Zhao, Dongbin; Liu, Yun-Fu; Su, Zhiguo; Gao, Yinfeng; Lin, Qiao; Huiyong, Chen
- 年份：2025；主题：Agent RL; Supporting Perception; 优先级：重要
- [来源](#)

论文：ReasonPlan: Unified Scene Prediction and Decision Reasoning for Closed-loop Autonomous Driving

快速理解：Due to the powerful vision-language reasoning and generalization abilities, multimodal large language models (MLLMs) have garnered significant attention in the field of end-to-end (E2E) autonomous driving. However, their application to closed-loop systems remains underexplored, and current MLLM-based methods have not shown clear superiority to mainstream E2E imitation learning approaches.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.3.137 RoboChemist: Long-Horizon and Safety-Compliant Robotic Chemical Experimentation

- 作者: Zhang, Zongzheng; Yue, Chenghao; Xu, Haobo; Liao, Minwen; Qi, Xianglin; Gao, Huan-ang; Wang, Ziwei; Zhao, Hao
- 年份: 2025; 主题: VLA; Supporting Perception; 优先级: 重要
- [来源](#)

论文: RoboChemist: Long-Horizon and Safety-Compliant Robotic Chemical Experimentation

快速理解: Robotic chemists promise to both liberate human experts from repetitive tasks and accelerate scientific discovery, yet remain in their infancy. Chemical experiments involve long-horizon procedures over hazardous and deformable substances, where success requires not only task completion but also strict compliance with experimental norms.

关注点：视觉、语言条件与机器人动作策略之间的对齐；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.3.138 TReF-6: Inferring Task-Relevant Frames from a Single Demonstration for One-Shot Skill Generalization

- 作者: Ding, Yuxuan; Wang, Shuangge; Fitzgerald, Tesca
- 年份: 2025; 主题: Agent RL; Supporting Perception; 优先级: 重要
- [来源](#)

论文: TReF-6: Inferring Task-Relevant Frames from a Single Demonstration for One-Shot Skill Generalization

快速理解: Robots often struggle to generalize from a single demonstration due to the lack of a transferable and interpretable spatial representation. In this work, we introduce TReF-6, a method that infers a simplified, abstracted 6DoF Task-Relevant Frame from a single trajectory.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.3.139 TopoCut: Learning Multi-Step Cutting with Spectral Rewards and Discrete Diffusion Policies

- 作者: Wang, Liquan; Bian, Jiangjie; Heiden, Eric; Garg, Animesh
- 年份: 2025; 主题: Embodied/WBC; Agent RL; 优先级: 重要
- [来源](#)

论文: TopoCut: Learning Multi-Step Cutting with Spectral Rewards and Discrete Diffusion Policies

快速理解: Robotic manipulation tasks involving cutting deformable objects remain challenging due to complex topological behaviors, difficulties in perceiving dense object states, and the lack of efficient evaluation methods for cutting outcomes. In this paper, we introduce TopoCut, a comprehensive benchmark for multi-step robotic cutting tasks that integrates a cutting environment and generalized policy learning.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.3.140 UnPose: Uncertainty-Guided Diffusion Priors for Zero-Shot Pose Estimation

- 作者：Jiang, Zhaodong; Sinha, Ashish; Cao, Tongtong; Ren, Yuan; Liu, Bingbing; Xu, Binbin
- 年份：2025；主题：Embodied/WBC; Supporting Perception; 优先级：重要
- [来源](#)

论文：UnPose: Uncertainty-Guided Diffusion Priors for Zero-Shot Pose Estimation

快速理解：Estimating the 6D pose of novel objects is a fundamental yet challenging problem in robotics, often relying on access to object CAD models. However, acquiring such models can be costly and impractical.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.3.141 Versatile Loco-Manipulation through Flexible Interlimb Coordination

- 作者：Zhu, Xinghao; Chen, Yuxin; Sun, Lingfeng; Niroui, Farzad; Cleac'h, Simon Le; Wang, Jiuguang; Fang, Kuan
- 年份：2025；主题：Embodied/WBC; Agent RL; 优先级：重要
- [来源](#)

论文：Versatile Loco-Manipulation through Flexible Interlimb Coordination

快速理解：The ability to flexibly leverage limbs for loco-manipulation is essential for enabling autonomous robots to operate in unstructured environments. Yet, prior work on loco-manipulation is often constrained to specific tasks or predetermined limb configurations.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.3.142 Wheeled Lab: Modern Sim2Real for Low-cost, Open-source Wheeled Robotics

- 作者：Han, Tyler; Shah, Preet; Rajagopal, Sidharth; Bao, Yanda; Jung, Sanghun; Talia, Sidharth; Guo, Gabriel; Xu, Bryan; Mehta, Bhaumik; Romig, Emma; Scalise, Rosario; Boots, Byron
- 年份：2025；主题：Agent RL; Supporting Perception; 优先级：重要
- [来源](#)

论文：Wheeled Lab: Modern Sim2Real for Low-cost, Open-source Wheeled Robotics

快速理解：Simulation has been pivotal in recent robotics milestones and is poised to play a prominent role in the field's future. However, recent robotic advances often rely on expensive and high-maintenance platforms, limiting access to broader robotics audiences.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.3.143 Bipedal Balance Control with Whole-body Musculoskeletal Standing and Falling Simulations

- 作者: Ma, Chengtian; Wei, Yunyue; Zuo, Chenhui; Zhang, Chen; Sui, Yanan
- 年份: 2025; 主题: Embodied/WBC; 优先级: 重要
- [来源](#)

论文: Bipedal Balance Control with Whole-body Musculoskeletal Standing and Falling Simulations

快速理解: Balance control is important for human and bipedal robotic systems. While dynamic balance during locomotion has received considerable attention, quantitative understanding of static balance and falling remains limited.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制。

证据层级: 索引摘要。

1.3.3.144 BranchOut: Capturing Realistic Multimodality in Autonomous Driving Decisions

- 作者: Kim, Hee Jae; Yin, Zekai; Lai, Lei; Lee, Jason; Ohn-Bar, Eshed
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: BranchOut: Capturing Realistic Multimodality in Autonomous Driving Decisions

快速理解: Modeling the nuanced, multimodal nature of human driving remains a core challenge for autonomous systems, as existing methods often fail to capture the diversity of plausible behaviors in complex real-world scenarios. In this work, we introduce a novel benchmark and end-to-end planner for modeling realistic multimodality in autonomous driving decisions.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.3.145 CARE: Enhancing Safety of Visual Navigation through Collision Avoidance via Repulsive Estimation

- 作者: Kim, Joonkyung; Sim, Joonyeol; Kim, Woojun; Sycara, Katia P.; Nam, Changjoo
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: CARE: Enhancing Safety of Visual Navigation through Collision Avoidance via Repulsive Estimation

快速理解: We propose CARE (Collision Avoidance via Repulsive Estimation) for improving the robustness of learning-based visual navigation methods. Recently, visual navigation models, particularly foundation models, have demonstrated promising performance by generating viable trajectories using only RGB images.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.3.146 CDP: Towards Robust Autoregressive Visuomotor Policy Learning via Causal Diffusion

- 作者: Ma, Jiahua; Qin, Yiran; Li, Yixiong; Liao, Xuanqi; Guo, Yulan; Zhang, Ruimao
- 年份: 2025; 主题: Robot Learning; 优先级: 重要

- [来源](#)

论文: CDP: Towards Robust Autoregressive Visuomotor Policy Learning via Causal Diffusion

快速理解: Diffusion Policy (DP) enables robots to learn complex behaviors by imitating expert demonstrations through action diffusion. However, in practical applications, hardware limitations often degrade data quality, while real-time constraints restrict model inference to instantaneous state and scene observations.

关注点: 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 索引摘要。

1.3.3.147 COMBO-Grasp: Learning Constraint-Based Manipulation for Bimanual Occluded Grasping

- 作者: Yamada, Jun; Mitchell, Alexander Luis; Collins, Jack; Posner, Ingmar
- 年份: 2025; 主题: Robot Learning; 优先级: 重要
- [来源](#)

论文: COMBO-Grasp: Learning Constraint-Based Manipulation for Bimanual Occluded Grasping

快速理解: This paper addresses the challenge of occluded robot grasping, i.e. grasping in situations where the desired grasp poses are kinematically infeasible due to environmental constraints such as surface collisions. Existing RL methods struggle with task complexity, and collecting expert demonstrations is often impractical.

关注点: 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 索引摘要。

1.3.3.148 D-Cubed: Latent Diffusion Trajectory Optimisation for Dexterous Deformable Manipulation

- 作者: Yamada, Jun; Zhong, Shaohong; Collins, Jack; Posner, Ingmar
- 年份: 2025; 主题: Embodied/WBC; 优先级: 重要
- [来源](#)

论文: D-Cubed: Latent Diffusion Trajectory Optimisation for Dexterous Deformable Manipulation

快速理解: Mastering deformable object manipulation often necessitates the use of anthropomorphic, high-degree-of-freedom robot hands capable of precise, contact-rich control. However, current trajectory optimisation methods often struggle in these settings due to the large search space and the sparse task information available from shape-matching cost functions, particularly when contact is absent.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制。

证据层级: 索引摘要。

1.3.3.149 Data Retrieval with Importance Weights for Few-Shot Imitation Learning

- 作者: Xie, Amber; Chand, Rahul; Sadigh, Dorsa; Hejna, Joey
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Data Retrieval with Importance Weights for Few-Shot Imitation Learning

快速理解: While large-scale robot datasets have propelled recent progress in imitation learning, learning from smaller task specific datasets remains critical for deployment in new environments and unseen tasks. One such approach to few-shot imitation learning is retrieval-based imitation learning, which extracts relevant samples from large, widely available prior datasets to augment a limited demonstration dataset.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.3.150 DexUMI: Using Human Hand as the Universal Manipulation Interface for Dexterous Manipulation

- 作者: Xu, Mengda; Zhang, Han; Hou, Yifan; Xu, Zhenjia; Fan, Linxi; Veloso, Manuela; Song, Shuran
- 年份: 2025; 主题: Embodied/WBC; 优先级: 重要
- [来源](#)

论文: DexUMI: Using Human Hand as the Universal Manipulation Interface for Dexterous Manipulation

快速理解: We present DexUMI - a data collection and policy learning framework that uses the human hand as the natural interface to transfer dexterous manipulation skills to various robot hands. DexUMI incorporates hardware and software adaptations to minimize the embodiment gap between the human hand and various robot hands.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制。

证据层级: 索引摘要。

1.3.3.151 Diffusion Dynamics Models with Generative State Estimation for Cloth Manipulation

- 作者: Tian, Tongxuan; Li, Haoyang; Ai, Bo; Yuan, Xiaodi; Huang, Zhiao; Su, Hao
- 年份: 2025; 主题: World Model; 优先级: 重要
- [来源](#)

论文: Diffusion Dynamics Models with Generative State Estimation for Cloth Manipulation

快速理解: Cloth manipulation is challenging due to its highly complex dynamics, near-infinite degrees of freedom, and frequent self-occlusions, which complicate both state estimation and dynamics modeling. Inspired by recent advances in generative models, we hypothesize that these expressive models can effectively capture intricate cloth configurations and deformation patterns from data.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策。

证据层级: 索引摘要。

1.3.3.152 Few-Shot Neuro-Symbolic Imitation Learning for Long-Horizon Planning and Acting

- 作者: Lorang, Pierrick; Lu, Hong; Huemer, Johannes; Zips, Patrik; Scheutz, Matthias
- 年份: 2025; 主题: Agent RL; 优先级: 重要

- [来源](#)

论文: Few-Shot Neuro-Symbolic Imitation Learning for Long-Horizon Planning and Acting

快速理解: Imitation learning enables intelligent systems to acquire complex behaviors with minimal supervision. However, existing methods often focus on short-horizon skills, require large datasets, and struggle to solve long-horizon tasks or generalize across task variations and distribution shifts.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.3.153 FlashBack: Consistency Model-Accelerated Shared Autonomy

- 作者: Sun, Luzhe; Ji, Jingtian; Tan, Xiangshan; Walter, Matthew
- 年份: 2025; 主题: Teleoperation; 优先级: 重要
- [来源](#)

论文: FlashBack: Consistency Model-Accelerated Shared Autonomy

快速理解: Abstract: Shared autonomy is an enabling technology that provides users with control authority over robots that would otherwise be difficult if not impossible to directly control. Yet, standard methods make assumptions that limit their adoption in practice—for example, prior knowledge of the user's goals or the objective (i.e., reward) function that they wish to optimize, knowledge of the user's policy, or query-level access to the user during training.

关注点: 人类示范、远程操作、共享自治或动作重定向。

证据层级: 索引摘要。

1.3.3.154 GLOVER++: Unleashing the Potential of Affordance Learning from Human Behaviors for Robotic Manipulation

- 作者: Ma, Teli; Zheng, Jia; Wang, Zifan; Gao, Ziyao; Zhou, Jiaming; Liang, Junwei
- 年份: 2025; 主题: Embodied/WBC; 优先级: 重要
- [来源](#)

论文: GLOVER++: Unleashing the Potential of Affordance Learning from Human Behaviors for Robotic Manipulation

快速理解: Learning manipulation skills from human demonstration videos offers a promising path toward generalizable and interpretable robotic intelligence—particularly through the lens of *actionable affordances*. However, transferring such knowledge remains challenging due to: 1) a lack of large-scale datasets with precise affordance annotations, and 2) insufficient exploration of affordances in diverse manipulation contexts.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制。

证据层级: 索引摘要。

1.3.3.155 Hand-Eye Autonomous Delivery: Learning Humanoid Navigation, Locomotion and Reaching

- 作者: Chen, Sirui; Ye, Yufei; Cao, Zi-ang; Xu, Pei; Lew, Jennifer; Liu, Karen
- 年份: 2025; 主题: Embodied/WBC; 优先级: 重要
- [来源](#)

论文: Hand-Eye Autonomous Delivery: Learning Humanoid Navigation, Locomotion and Reaching

快速理解: We propose Hand-Eye Autonomous Delivery (HEAD), a framework that learns navigation, locomotion, and reaching skills for humanoids, directly from human motion and vision perception data. We take a modular approach where the high-level planner commands the target position and orientation of the hands and eyes of the humanoid, delivered by the low-level policy that controls the whole-body movements.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制。

证据层级: 索引摘要。

1.3.3.156 Hold My Beer: Learning Gentle Humanoid Locomotion and End-Effector Stabilization Control

- 作者: Li, Yitang; Zhang, Yuanhang; Xiao, Wenli; Pan, Chaoyi; Weng, Haoyang; He, Guanqi; He, Tairan; Shi, Guanya
- 年份: 2025; 主题: Embodied/WBC; 优先级: 重要
- [来源](#)

论文: Hold My Beer: Learning Gentle Humanoid Locomotion and End-Effector Stabilization Control

快速理解: Can your humanoid walk up and hand you a full cup of beer—without spilling a drop? While humanoids are increasingly featured in flashy demos—dancing, delivering packages, traversing rough terrain—fine-grained control during locomotion remains a significant challenge.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制。

证据层级: 索引摘要。

1.3.3.157 Learning Smooth State-Dependent Traversability from Dense Point Clouds

- 作者: Dong, Zihao; Papalia, Alan; Jung, Leonard; Spiro, Alenna; Osteen, Philip R; Robison, Christa S.; Everett, Michael
- 年份: 2025; 主题: Systems; 优先级: 重要
- [来源](#)

论文: Learning Smooth State-Dependent Traversability from Dense Point Clouds

快速理解: A key open challenge in off-road autonomy is that the traversability of terrain often depends on the vehicle's state. In particular, some obstacles are only traversable from some orientations.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.3.158 LocoFormer: Generalist Locomotion via Long-context Adaptation

- 作者: Liu, Min; Pathak, Deepak; Agarwal, Ananye
- 年份: 2025; 主题: Embodied/WBC; 优先级: 重要
- [来源](#)

论文: LocoFormer: Generalist Locomotion via Long-context Adaptation

快速理解: Humans and animals exhibit flexible locomotion strategies, such as learning to walk within minutes, and efficient adaptation to changes in morphology. In contrast, modern locomotion con-

trollers are manually tuned for specific embodiments.

关注点：具身任务中的感知、动作生成、全身协调或运动控制。

证据层级：索引摘要。

1.3.3.159 MirrorDuo: Reflection-Consistent Visuomotor Learning from Mirrored Demonstration Pairs

- 作者：Zhuang, Zheyu; Wang, Ruiyu; Marchetti, Giovanni Luca; Pokorny, Florian T.; Kragic, Danica
- 年份：2025; 主题：Robot Learning; 优先级：重要
- [来源](#)

论文：MirrorDuo: Reflection-Consistent Visuomotor Learning from Mirrored Demonstration Pairs

快速理解：Image-based behaviour cloning leverages demonstrations captured from ubiquitous RGB cameras, enabling impressive visuomotor performance. However, it remains constrained by the cost of collecting sufficiently diverse demonstrations, especially for generalizing across workspace variations.

关注点：机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：索引摘要。

1.3.3.160 Mobi-pi: Mobilizing Your Robot Learning Policy

- 作者：Yang, Jingyun; Huang, Isabella; Vu, Brandon; Bajracharya, Max; Antonova, Rika; Bohg, Jeannette
- 年份：2025; 主题：Robot Learning; 优先级：重要
- [来源](#)

论文：Mobi-pi: Mobilizing Your Robot Learning Policy

快速理解：Learned visuomotor policies are capable of performing increasingly complex manipulation tasks. However, most of these policies are trained on data collected from limited robot positions and camera viewpoints.

关注点：机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：索引摘要。

1.3.3.161 Multimodal Fused Learning for Solving the Generalized Traveling Salesman Problem in Robotic Task Planning

- 作者：Cheng, Jiaqi; Fan, Mingfeng; Zhang, Xuefeng; Liang, Jingsong; Cao, Yuhong; Wu, Guohua; Sartoretti, Guillaume Adrien
- 年份：2025; 主题：Supporting Perception; 优先级：重要
- [来源](#)

论文：Multimodal Fused Learning for Solving the Generalized Traveling Salesman Problem in Robotic Task Planning

快速理解：Effective and efficient task planning is essential for mobile robots, especially in applications like warehouse retrieval and environmental monitoring. These tasks often involve selecting one location from each of several target clusters, forming a Generalized Traveling Salesman Problem

(GTSP) that remains challenging to solve both accurately and efficiently.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.3.162 ObjectReact: Learning Object-Relative Control for Visual Navigation

- 作者: Garg, Sourav; Craggs, Dustin; Bhat, Vineeth; Mares, Lachlan; Podgorski, Stefan; Krishna, Madhava; Dayoub, Feras; Reid, Ian
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: ObjectReact: Learning Object-Relative Control for Visual Navigation

快速理解: Visual navigation using only a single camera and a topological map has recently become an appealing alternative to methods that require additional sensors and 3D maps. This is typically achieved through an “image-relative” approach to estimating control from a given pair of current observation and subgoal image.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.3.163 Off Policy Lyapunov Stability in Reinforcement Learning

- 作者: Gill, Sarvan; Constantinescu, Daniela
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Off Policy Lyapunov Stability in Reinforcement Learning

快速理解: Traditional reinforcement learning lacks the ability to provide stability guarantees. More recent algorithms learn Lyapunov functions alongside the control policies to ensure stable learning.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.3.164 ReCoDe: Reinforcement Learning-based Dynamic Constraint Design for Multi-Agent Coordination

- 作者: Amir, Michael; Yang, Guang; Gao, Zhan; Okumura, Keisuke; Woo, Heedo; Prorok, Amanda
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: ReCoDe: Reinforcement Learning-based Dynamic Constraint Design for Multi-Agent Coordination

快速理解: Constraint-based optimization is a cornerstone of robotics, enabling the design of controllers that reliably encode task and safety requirements such as collision avoidance or formation adherence. However, handcrafted constraints can fail in multi-agent settings that demand complex coordination.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.3.165 Reactive In-Air Clothing Manipulation with Confidence-Aware Dense Correspondence and Visuotactile Affordance

- 作者: Sunil, Neha; Tippur, Megha; Portillo, Arnau Saumell; Adelson, Edward H; Garcia, Alberto Rodriguez
- 年份: 2025; 主题: Robot Learning; 优先级: 重要
- [来源](#)

论文: Reactive In-Air Clothing Manipulation with Confidence-Aware Dense Correspondence and Visuotactile Affordance

快速理解: Manipulating clothing is challenging due to their complex, variable configurations and frequent self-occlusion. While prior systems often rely on flattening garments, humans routinely identify keypoints in highly crumpled and suspended states.

关注点: 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 索引摘要。

1.3.3.166 RoboArena: Distributed Real-World Evaluation of Generalist Robot Policies

- 作者: Atreya, Pranav; Pertsch, Karl; Lee, Tony; Kim, Moo Jin; Jain, Arhan; Kuramshin, Artur; Neary, Cyrus; Hu, Edward S.; Arora, Kanav; Ellis, Kirsty; Macesanu, Luca; Leonard, Matthew; Cho, Meedeum; Aslan, Ozgur; Dass, Shivin; Wang, Tony; Yuan, Xingfang; Gupta, Abhishek; Jayaraman, Dinesh; Berseth, Glen; Daniilidis, Kostas; Mart' {i}n-Mart' {i}n, Roberto; Lee, Youngwoon; Liang, Percy; Finn, Chelsea; Levine, Sergey
- 年份: 2025; 主题: VLA; 优先级: 重要
- [来源](#)

论文: RoboArena: Distributed Real-World Evaluation of Generalist Robot Policies

快速理解: Comprehensive, unbiased, and comparable evaluation of modern generalist policies is uniquely challenging: existing approaches for robot benchmarking typically rely on heavy standardization, either by specifying fixed evaluation tasks and environments, or by hosting centralized “robot challenges” , and do not readily scale to evaluating generalist policies across a broad range of tasks and environments. In this work, we propose RoboArena, a new approach for scalable evaluation of generalist robot policies in the real world.

关注点: 视觉、语言条件与机器人动作策略之间的对齐。

证据层级: 索引摘要。

1.3.3.167 Shortcut Learning in Generalist Robot Policies: The Role of Dataset Diversity and Fragmentation

- 作者: Xing, Youguang; Luo, Xu; Xie, Junlin; Gao, Lianli; Shen, Heng Tao; Song, Jingkuan
- 年份: 2025; 主题: VLA; 优先级: 重要
- [来源](#)

论文: Shortcut Learning in Generalist Robot Policies: The Role of Dataset Diversity and Fragmentation

快速理解: Generalist robot policies trained on large-scale datasets such as Open X-Embodiment

(OXE) demonstrate strong performance across a wide range of tasks. However, they often struggle to generalize beyond the distribution of their training data.

关注点：视觉、语言条件与机器人动作策略之间的对齐。

证据层级：索引摘要。

1.3.3.168 SimShear: Sim-to-Real Shear-based Tactile Servoing

- 作者：Freud, Kipp; Lin, Yijiong; Lepora, Nathan F.
- 年份：2025；主题：Robot Learning；优先级：重要
- [来源](#)

论文：SimShear: Sim-to-Real Shear-based Tactile Servoing

快速理解：We present SimShear: a sim-to-real pipeline for tactile control that allows use of shear information without explicitly modeling shear dynamics in simulation. Shear, which arises from lateral movements across contact surfaces, are critical for tasks involving dynamic object interactions but are challenging to simulate.

关注点：机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：索引摘要。

1.3.3.169 SocialNav-SUB: Benchmarking VLMs for Scene Understanding in Social Robot Navigation

- 作者：Munje, Michael Joseph; Tang, Chen; Liu, Shuijing; Hu, Zichao; Zhu, Yifeng; Cui, Jiaxun; Warnell, Garrett; Biswas, Joydeep; Stone, Peter
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：SocialNav-SUB: Benchmarking VLMs for Scene Understanding in Social Robot Navigation

快速理解：Robot navigation in dynamic, human-centered environments requires socially-compliant decisions grounded in robust scene understanding, including spatiotemporal awareness and the ability to interpret human intentions. Recent Vision-Language Models (VLMs) show exhibit promising capabilities such as object recognition, common-sense reasoning, and contextual understanding—that align with the nuanced requirements of social robot navigation.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.3.170 Steering Your Diffusion Policy with Latent Space Reinforcement Learning

- 作者：Wagenmaker, Andrew; Zhang, Yunchu; Nakamoto, Mitsuhiko; Park, Seohong; Yagoub, Waleed; Nagabandi, Anusha; Gupta, Abhishek; Levine, Sergey
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Steering Your Diffusion Policy with Latent Space Reinforcement Learning

快速理解：Robotic control policies learned from human demonstrations have achieved impressive results in many real-world applications. However, in scenarios where initial performance is not satisfactory, as is often the case in novel open-world settings, such behavioral cloning (BC)-learned

policies typically require collecting additional human demonstrations to further improve their behavior—an expensive and time-consuming process.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.3.171 Uncertainty-Aware Scene Understanding via Efficient Sampling-Free Confidence Estimation

- 作者：Miandashti, Hanieh Shojaei; Zou, Qianqian; Brenner, Claus
- 年份：2025; 主题：Supporting Perception; 优先级：重要
- [来源](#)

论文：Uncertainty-Aware Scene Understanding via Efficient Sampling-Free Confidence Estimation

快速理解：Reliable scene understanding requires not only accurate predictions but also well-calibrated confidence estimates to ensure calibrated uncertainty estimation, especially in safety-critical domains like autonomous driving. In this context, semantic segmentation of LiDAR points supports real-time 3D scene understanding, where reliable uncertainty estimates help identify potentially erroneous predictions.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.3.172 UniTac2Pose: A Unified Approach Learned in Simulation for Category-level Visuotactile In-hand Pose Estimation

- 作者：Wu, Mingdong; Yang, Long; Liu, Jin; Huang, Weiyao; Wu, Lehong; Chen, Zelin; Ma, Daolin; Dong, Hao
- 年份：2025; 主题：Robot Learning; 优先级：重要
- [来源](#)

论文：UniTac2Pose: A Unified Approach Learned in Simulation for Category-level Visuotactile In-hand Pose Estimation

快速理解：Accurate estimation of the in-hand pose of an object based on its CAD model is crucial in both industrial applications and everyday tasks—ranging from positioning workpieces and assembling components to seamlessly inserting devices like USB connectors. While existing methods often rely on regression, feature matching, or registration techniques, achieving high precision and generalizability to unseen CAD models remains a significant challenge.

关注点：机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：索引摘要。

1.3.3.173 Unsupervised Skill Discovery as Exploration for Learning Agile Locomotion

- 作者：Rho, Seungeun; Garg, Kartik; Byrd, Morgan; Ha, Sehoon
- 年份：2025; 主题：Embodied/WBC; 优先级：重要
- [来源](#)

论文：Unsupervised Skill Discovery as Exploration for Learning Agile Locomotion

快速理解：Exploration is crucial for legged robots to learn agile locomotion behaviors capable of

overcoming diverse obstacles. For example, a robot may need to try different contact patterns and momentum profiles to successfully jump over an obstacle—but encouraging such diverse exploration is inherently challenging.

关注点：具身任务中的感知、动作生成、全身协调或运动控制。

证据层级：索引摘要。

1.3.3.174 VLM-AD: End-to-End Autonomous Driving through Vision-Language Model Supervision

- 作者: Xu, Yi; Hu, Yuxin; Zhang, Zaiwei; Meyer, Gregory P.; Mustikovela, Siva Karthik; Srinivasa, Siddhartha; Wolff, Eric M.; Huang, Xin
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: VLM-AD: End-to-End Autonomous Driving through Vision-Language Model Supervision

快速理解: Human drivers rely on commonsense reasoning to navigate diverse and dynamic real-world scenarios. Existing end-to-end (E2E) autonomous driving (AD) models are typically optimized to mimic driving patterns observed in data, without capturing the underlying reasoning processes.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.3.175 ZipMPC: Compressed Context-Dependent MPC Cost via Imitation Learning

- 作者: Rickenbach, Rahel; Lahoud, Alan; Schaffernicht, Erik; Zeilinger, Melanie; Stork, Johannes A.
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: ZipMPC: Compressed Context-Dependent MPC Cost via Imitation Learning

快速理解: The computational burden of model predictive control (MPC) limits its application on real-time systems, such as robots, and often requires the use of short prediction horizons. This not only affects the control performance, but also increases the difficulty of designing MPC cost functions that reflect the desired long-term objective.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.3.176 textttSPIN: distilling textttSkill-RRT for long-horizon prehensile and non-prehensile manipulation

- 作者: Jung, Haewon; Lee, Donguk; Park, Haecheol; Hyeop, Kim Jun; Kim, Beomjoon
- 年份: 2025; 主题: Agent RL; 优先级: 扩展
- [来源](#)

论文: textttSPIN: distilling textttSkill-RRT for long-horizon prehensile and non-prehensile manipulation

快速理解: Current robots struggle with long-horizon manipulation tasks requiring sequences of pre-

hensile and non-prehensile skills, contact-rich interactions, and long-term reasoning. We present `texttt{SPIN}` (`textbf{S}`kill `textbf{P}`lanning to `textbf{IN}`ference), a framework that distills a computationally intensive planning algorithm into a policy via imitation learning.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.3.177 AT-Drone: Benchmarking Adaptive Teaming in Multi-Drone Pursuit

- 作者：Li, Yang; Chen, Junfan; Xue, Feng; Qiu, Jiabin; Li, Wenbin; Zhang, Qingrui; Wen, Ying; Pan, Wei
- 年份：2025；主题：AI Infra；优先级：扩展
- [来源](#)

论文：AT-Drone: Benchmarking Adaptive Teaming in Multi-Drone Pursuit

快速理解：Adaptive teaming—the capability of agents to effectively collaborate with unfamiliar teammates without prior coordination—is widely explored in virtual video games but overlooked in real-world multi-robot contexts. Yet, such adaptive collaboration is crucial for real-world applications, including border surveillance, search-and-rescue, and counter-terrorism operations.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.3.178 Articulate AnyMesh: Open-vocabulary 3D Articulated Objects Modeling

- 作者：Qiu, Xiaowen; Yang, Jincheng; Wang, Yian; Chen, Zhehuan; Wang, Yufei; Wang, Tsun-Hsuan; Xian, Zhou; Gan, Chuang
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Articulate AnyMesh: Open-vocabulary 3D Articulated Objects Modeling

快速理解：3D articulated objects modeling has long been a challenging problem, since it requires to capture both accurate surface geometries and semantically meaningful and spatially precise structures, parts, and joints. Existing methods heavily depend on training data from a limited set of handcrafted articulated object categories (e.g., cabinets and drawers), which restricts their ability to model a wide range of articulated objects in an open-vocabulary context.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.3.179 Articulated Object Estimation in the Wild

- 作者：Werby, Abdelrhman; B" {u}chner, Martin; R" {o}fer, Adrian; Huang, Chenguang; Burgard, Wolfram; Valada, Abhinav
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Articulated Object Estimation in the Wild

快速理解：Understanding the 3D motion of articulated objects is essential in robotic scene understanding, mobile manipulation, and motion planning. Prior methods for articulation estimation

have primarily focused on controlled settings, assuming either fixed camera viewpoints or direct observations of various object states, which tend to fail in more realistic, unconstrained environments.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.3.180 Beyond Constant Parameters: Hyper Prediction Models and HyperMPC

- 作者：Wk{e}grzynowski, Jan; Kicki, Piotr; Czechmanowski, Grzegorz; Krupka, Maciej Piotr; Walas, Krzysztof
- 年份：2025；主题：World Model；优先级：扩展
- [来源](#)

论文：Beyond Constant Parameters: Hyper Prediction Models and HyperMPC

快速理解：Model Predictive Control (MPC) is among the most widely adopted and reliable methods for robot control, relying critically on an accurate dynamics model. However, existing dynamics models used in the gradient-based MPC are limited by computational complexity and state representation.

关注点：环境动力学、视频预测、潜变量状态或模型式决策。

证据层级：索引摘要。

1.3.3.181 COLLAGE: Adaptive Fusion-based Retrieval for Augmented Policy Learning

- 作者：Kumar, Sateesh; Dass, Shivin; Pavlakos, Georgios; Mart' {i}n-Mart' {i}n, Roberto
- 年份：2025；主题：Agent RL；优先级：扩展
- [来源](#)

论文：COLLAGE: Adaptive Fusion-based Retrieval for Augmented Policy Learning

快速理解：In this work, we study the problem of data retrieval for few-shot imitation learning: select data from a large dataset to train a performant policy for a specific task, given only a few target demonstrations. Prior methods retrieve data using a single-feature distance heuristic, assuming that the best demonstrations are those that most closely resemble the target examples in visual, semantic, or motion space.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.3.182 CUPID: Curating Data your Robot Loves with Influence Functions

- 作者：Agia, Christopher; Sinha, Rohan; Yang, Jingyun; Antonova, Rika; Pavone, Marco; Nishimura, Haruki; Itkina, Masha; Bohg, Jeannette
- 年份：2025；主题：Agent RL；优先级：扩展
- [来源](#)

论文：CUPID: Curating Data your Robot Loves with Influence Functions

快速理解：In robot imitation learning, policy performance is tightly coupled with the quality and composition of the demonstration data. Yet, developing a precise understanding of how individual demonstrations contribute to downstream outcomes—such as closed-loop task success or failure—remains a persistent challenge.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.3.183 Capability-Aware Shared Hypernetworks for Flexible Heterogeneous Multi-Robot Coordination

- 作者：Fu, Kevin; Jain, Shalin; Howell, Pierce; Ravichandar, Harish
- 年份：2025; 主题：Agent RL; 优先级：扩展
- [来源](#)

论文：Capability-Aware Shared Hypernetworks for Flexible Heterogeneous Multi-Robot Coordination

快速理解：Recent advances have enabled heterogeneous multi-robot teams to learn complex and effective coordination skills. However, existing neural architectures that support heterogeneous teaming tend to force a trade-off between expressivity and efficiency.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.3.184 CoRI: Communication of Robot Intent for Physical Human-Robot Interaction

- 作者：Wang, Junxiang; K" {u}c{c} " {u}ktabak, Emek Bar{i}c{s}; Zarrin, Rana Soltani; Erickson, Zackory
- 年份：2025; 主题：Supporting Perception; 优先级：扩展
- [来源](#)

论文：CoRI: Communication of Robot Intent for Physical Human-Robot Interaction

快速理解：Clear communication of robot intent fosters transparency and interpretability in physical human-robot interaction (pHRI), particularly during assistive tasks involving direct human-robot contact. We introduce CoRI, a pipeline that automatically generates natural language communication of a robot' s upcoming actions directly from its motion plan and visual perception.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.3.185 ComposableNav: Instruction-Following Navigation in Dynamic Environments via Composable Diffusion

- 作者：Hu, Zichao; Tang, Chen; Munje, Michael Joseph; Zhu, Yifeng; Liu, Alex; Liu, Shuijing; Warnell, Garrett; Stone, Peter; Biswas, Joydeep
- 年份：2025; 主题：Agent RL; 优先级：扩展
- [来源](#)

论文：ComposableNav: Instruction-Following Navigation in Dynamic Environments via Composable Diffusion

快速理解：This paper considers the problem of enabling robots to navigate dynamic environments while following instructions. The challenge lies in the combinatorial nature of instruction specifications: each instruction can include multiple specifications, and the number of possible specification combinations grows exponentially as the robot' s skill set expands.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.3.186 Cost-aware Discovery of Contextual Failures using Bayesian Active Learning

- 作者：Parashar, Anjali; Zhang, Joseph; Li, Yingke; Fan, Chuchu
- 年份：2025；主题：Embodied/WBC；优先级：扩展
- [来源](#)

论文：Cost-aware Discovery of Contextual Failures using Bayesian Active Learning

快速理解：Ensuring the robustness of robotic systems is crucial for their deployment in safety-critical domains. Failure discovery, or falsification, is a widely used approach for evaluating robustness, with recent advancements focusing on improving sample efficiency and generalization through probabilistic sampling techniques and learning-theoretic approaches.

关注点：具身任务中的感知、动作生成、全身协调或运动控制。

证据层级：索引摘要。

1.3.3.187 Cross-Sensor Touch Generation

- 作者：Rodriguez, Samanta; Dou, Yiming; Oller, Miquel; Owens, Andrew; Fazeli, Nima
- 年份：2025；主题：Robot Learning；优先级：扩展
- [来源](#)

论文：Cross-Sensor Touch Generation

快速理解：Today' s visuo-tactile sensors come in many shapes and sizes, making it challenging to develop general-purpose tactile representations. This is because most models are tied to a specific sensor design.

关注点：机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：索引摘要。

1.3.3.188 D-CODA: Diffusion for Coordinated Dual-Arm Data Augmentation

- 作者：Liu, I-Chun Arthur; Chen, Jason; Sukhatme, Gaurav S.; Seita, Daniel
- 年份：2025；主题：Agent RL；优先级：扩展
- [来源](#)

论文：D-CODA: Diffusion for Coordinated Dual-Arm Data Augmentation

快速理解：Learning bimanual manipulation is challenging due to its high dimensionality and tight coordination required between two arms. Eye-in-hand imitation learning, which uses wrist-mounted cameras, simplifies perception by focusing on task-relevant views.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.3.189 Deep Reactive Policy: Learning Reactive Manipulator Motion Planning for Dynamic Environments

- 作者: Yang, Jiahui; Liu, Jason Jingzhou; Li, Yulong; Khaky, Youssef; Shaw, Kenneth; Pathak, Deepak
- 年份: 2025; 主题: Systems; 优先级: 扩展
- [来源](#)

论文: Deep Reactive Policy: Learning Reactive Manipulator Motion Planning for Dynamic Environments

快速理解: Generating collision-free motion in dynamic, partially observable environments is a fundamental challenge for robotic manipulators. Classical motion planners can compute globally optimal trajectories but require full environment knowledge and are typically too slow for dynamic scenes.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.3.190 Diffusion-Guided Multi-Arm Motion Planning

- 作者: Parimi, Viraj; Williams, Brian C.
- 年份: 2025; 主题: Agent RL; 优先级: 扩展
- [来源](#)

论文: Diffusion-Guided Multi-Arm Motion Planning

快速理解: Multi-arm motion planning is fundamental for enabling arms to complete collaborative tasks in shared spaces but current methods struggle with scalability due to exponential state-space growth and reliance on large training datasets for learned models. Inspired by Multi-Agent Path Finding (MAPF), which decomposes planning into single-agent problems coupled with collision resolution, we propose a novel diffusion-guided multi-arm planner (DG-MAP) that enhances scalability of learning-based models while reducing their reliance on massive multi-arm datasets.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.3.191 Distilling On-device Language Models for Robot Planning with Minimal Human Intervention

- 作者: Ravichandran, Zachary; Hounie, Ignacio; Cladera, Fernando; Ribeiro, Alejandro; Pappas, George J.; Kumar, Vijay
- 年份: 2025; 主题: Systems; 优先级: 扩展
- [来源](#)

论文: Distilling On-device Language Models for Robot Planning with Minimal Human Intervention

快速理解: Large language models (LLMs) provide robots with powerful contextual reasoning abilities and a natural human interface. Yet, current LLM-enabled robots typically depend on cloud-hosted models, limiting their usability in environments with unreliable communication infrastructure, such as outdoor or industrial settings.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.3.192 Distributed Upload and Active Labeling for Resource-Constrained Fleet Learning

- 作者: Akcin, Oguzhan; Goel, Harsh; Zhao, Ruihan; Chinchali, Sandeep P.
- 年份: 2025; 主题: Systems; 优先级: 扩展
- [来源](#)

论文: Distributed Upload and Active Labeling for Resource-Constrained Fleet Learning

快速理解: In multi-robot systems, fleets are often deployed to collect data that improves the performance of machine learning models for downstream perception and planning. However, real-world robotic deployments generate vast amounts of data across diverse conditions, while only a small portion can be transmitted or labeled due to limited bandwidth, constrained onboard storage, and high annotation costs.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.3.193 Divide, Discover, Deploy: Factorized Skill Learning with Symmetry and Style Priors

- 作者: Cathomen, Rafael; Mittal, Mayank; Vlastelica, Marin; Hutter, Marco
- 年份: 2025; 主题: Embodied/WBC; 优先级: 扩展
- [来源](#)

论文: Divide, Discover, Deploy: Factorized Skill Learning with Symmetry and Style Priors

快速理解: Unsupervised Skill Discovery (USD) allows agents to autonomously learn diverse behaviors without task-specific rewards. While recent USD methods have shown promise, their application to real-world robotics remains underexplored.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制。

证据层级: 索引摘要。

1.3.3.194 Enabling Long(er) Horizon Imitation for Manipulation Tasks by Modeling Subgoal Transitions

- 作者: Jain, Shivam; Sachdeva, Sachit; Paul, Rohan
- 年份: 2025; 主题: Robot Learning; 优先级: 扩展
- [来源](#)

论文: Enabling Long(er) Horizon Imitation for Manipulation Tasks by Modeling Subgoal Transitions

快速理解: Imitation-based policy training for long-horizon manipulation tasks involving multi-step object interactions is often susceptible to compounding action errors. Contemporary methods discover semantic subgoals embedded within the overall task, decomposing the overall task into tractable shorter-horizon goal-conditioned policy learning.

关注点: 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 索引摘要。

1.3.3.195 Extracting Visual Plans from Unlabeled Videos via Symbolic Guidance

- 作者: Yang, Wenyan; Tikna, Ahmet; Zhao, Yi; Zhang, Yuying; Palopoli, Luigi; Roveri, Marco; Pajarinen, Joni
- 年份: 2025; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Extracting Visual Plans from Unlabeled Videos via Symbolic Guidance

快速理解: Visual planning, by offering a sequence of intermediate visual subgoals to a goal-conditioned low-level policy, achieves promising performance on long-horizon manipulation tasks. To obtain the subgoals, existing methods typically resort to video generation models but suffer from model hallucination and computational cost.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.3.196 FOMO-3D: Using Vision Foundation Models for Long-Tailed 3D Object Detection

- 作者: Yang, Anqi Joyce; Tu, James; Dvornik, Nikita; Li, Enxu; Urtasun, Raquel
- 年份: 2025; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: FOMO-3D: Using Vision Foundation Models for Long-Tailed 3D Object Detection

快速理解: In order to navigate complex traffic environments, self-driving vehicles must recognize many semantic classes pertaining to vulnerable road users or traffic control devices. However, many safety-critical objects (e.g., construction worker) appear infrequently in nominal traffic conditions, leading to a severe shortage of training examples from driving data alone.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.3.197 From Real World to Logic and Back: Learning Generalizable Relational Concepts For Long Horizon Robot Planning

- 作者: Shah, Naman; Nagpal, Jayesh; Srivastava, Siddharth
- 年份: 2025; 主题: World Model; 优先级: 扩展
- [来源](#)

论文: From Real World to Logic and Back: Learning Generalizable Relational Concepts For Long Horizon Robot Planning

快速理解: Humans efficiently generalize from limited demonstrations, but robots still struggle to transfer learned knowledge to complex, unseen tasks with longer horizons and increased complexity. We propose the first known method enabling robots to autonomously invent relational concepts directly from small sets of unannotated, unsegmented demonstrations.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策。

证据层级: 索引摘要。

1.3.3.198 Generative Visual Foresight Meets Task-Agnostic Pose Estimation in Robotic Table-top Manipulation

- 作者: Zhang, Chuye; Zhang, Xiaoxiong; Zheng, Linfang; Pan, Wei; Zhang, Wei
- 年份: 2025; 主题: Embodied/WBC; 优先级: 扩展
- [来源](#)

论文: Generative Visual Foresight Meets Task-Agnostic Pose Estimation in Robotic Table-top Manipulation

快速理解: Robotic manipulation in unstructured environments requires systems that can generalize across diverse tasks while maintaining robust and reliable performance. We introduce GVF-TAPE, a closed-loop framework that combines generative visual foresight with task-agnostic pose estimation to enable scalable robotic manipulation.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制。

证据层级: 索引摘要。

1.3.3.199 Granular loco-manipulation: Repositioning rocks through strategic sand avalanche

- 作者: Hu, Haodi; Wu, Yue; Seita, Daniel; Qian, Feifei
- 年份: 2025; 主题: Embodied/WBC; 优先级: 扩展
- [来源](#)

论文: Granular loco-manipulation: Repositioning rocks through strategic sand avalanche

快速理解: Legged robots have the potential to leverage obstacles to climb steep sand slopes. However, efficiently repositioning these obstacles to desired locations is challenging.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制。

证据层级: 索引摘要。

1.3.3.200 HALO : Human Preference Aligned Offline Reward Learning for Robot Navigation

- 作者: Seneviratne, Gershom; An, Jianyu; Ellahy, Sahire; Weerakoon, Kasun; Elnoor, Mohamed Bashir; Kannan, Jonathan Deepak; Sunil, Amogha Thalihalla; Manocha, Dinesh
- 年份: 2025; 主题: Agent RL; 优先级: 扩展
- [来源](#)

论文: HALO : Human Preference Aligned Offline Reward Learning for Robot Navigation

快速理解: In this paper, we introduce HALO, a novel Offline Reward Learning algorithm that quantifies human intuition in navigation into a vision-based reward function for robot navigation. HALO learns a reward model from offline data, leveraging expert trajectories collected from mobile robots.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.3.201 Human-like Navigation in a World Built for Humans

- 作者: Chandaka, Bhargav; Wang, Gloria Xinyue; Chen, Haozhe; Che, Henry; Zhai, Albert J.; Wang, Shenlong
- 年份: 2025; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Human-like Navigation in a World Built for Humans

快速理解: When navigating in a man-made environment they haven't visited before—like an office building—humans employ behaviors such as reading signs and asking others for directions. These behaviors help humans reach their destinations efficiently by reducing the need to search through large areas.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.3.202 HyperTASR: Hypernetwork-Driven Task-Aware Scene Representations for Robust Manipulation

- 作者: Sun, Li; Wu, Jiefeng; Chen, Feng; Liu, Ruizhe; Yang, Yanchao
- 年份: 2025; 主题: Embodied/WBC; 优先级: 扩展
- [来源](#)

论文: HyperTASR: Hypernetwork-Driven Task-Aware Scene Representations for Robust Manipulation

快速理解: Effective policy learning for robotic manipulation requires scene representations that selectively capture task-relevant environmental features. Current approaches typically employ task-agnostic representation extraction, failing to emulate the dynamic perceptual adaptation observed in human cognition.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制。

证据层级: 索引摘要。

1.3.3.203 IRIS: An Immersive Robot Interaction System

- 作者: Jiang, Xinkai; Yuan, Qihao; Dincer, Enes Ulas; Zhou, Hongyi; Li, Ge; Li, Xueyin; Jia, Xiaogang; Schnizer, Timo; Schreiber, Nicolas; Liao, Weiran; Haag, Julius; Li, Kailai; Neumann, Gerhard; Lioutikov, Rudolf
- 年份: 2025; 主题: Systems; 优先级: 扩展
- [来源](#)

论文: IRIS: An Immersive Robot Interaction System

快速理解: This paper introduces IRIS, an Immersive Robot Interaction System leveraging Extended Reality (XR). Existing XR-based systems enable efficient data collection but are often challenging to reproduce and reuse due to their specificity to particular robots, objects, simulators, and environments.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.3.204 ImLPR: Image-based LiDAR Place Recognition using Vision Foundation Models

- 作者: Jung, Minwoo; Fu, Lanke Frank Tarimo; Fallon, Maurice; Kim, Ayong
- 年份: 2025; 主题: Systems; 优先级: 扩展
- [来源](#)

论文: ImLPR: Image-based LiDAR Place Recognition using Vision Foundation Models

快速理解: LiDAR Place Recognition (LPR) is a key component in robotic localization, enabling robots to align current scans with prior maps of their environment. While Visual Place Recognition (VPR) has embraced Vision Foundation Models (VFMs) to enhance descriptor robustness, LPR has relied on task-specific models with limited use of pre-trained foundation-level knowledge.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.3.205 KoopMotion: Learning Almost Divergence Free Koopman Flow Fields for Motion Planning

- 作者: Li, Alice Kate; Silva, Thales C.; Edwards, Victoria; Kumar, Vijay; Hsieh, M. Ani
- 年份: 2025; 主题: World Model; 优先级: 扩展
- [来源](#)

论文: KoopMotion: Learning Almost Divergence Free Koopman Flow Fields for Motion Planning

快速理解: In this work, we propose a novel flow field-based motion planning method that drives a robot from any initial state to a desired reference trajectory such that it converges to the trajectory's end point. Despite demonstrated efficacy in using Koopman operator theory for modeling dynamical systems, Koopman does not inherently enforce convergence to desired trajectories nor to specified goals, a requirement when learning from demonstrations (LfD).

关注点: 环境动力学、视频预测、潜变量状态或模型式决策。

证据层级: 索引摘要。

1.3.3.206 LLM-Guided Probabilistic Program Induction for POMDP Model Estimation

- 作者: Curtis, Aidan; Tang, Hao; Veloso, Thiago; Ellis, Kevin; Tenenbaum, Joshua B.; Lozano-Perez, Tom; Kaelbling, Leslie Pack
- 年份: 2025; 主题: Agent RL; 优先级: 扩展
- [来源](#)

论文: LLM-Guided Probabilistic Program Induction for POMDP Model Estimation

快速理解: Partially Observable Markov Decision Processes (POMDPs) model decision making under uncertainty. While there are many approaches to approximately solving POMDPs, we aim to address the problem of learning such models.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.3.207 Latent Adaptive Planner for Dynamic Manipulation

- 作者: Noh, Donghun; Kong, Deqian; Zhao, Minglu; Lizarraga, Andrew; Xie, Jianwen; Wu, Ying Nian; Hong, Dennis
- 年份: 2025; 主题: Robot Learning; 优先级: 扩展
- [来源](#)

论文: Latent Adaptive Planner for Dynamic Manipulation

快速理解: This paper presents Latent Adaptive Planner (LAP), a novel approach for dynamic non-prehensile manipulation tasks that formulates planning as latent space inference, effectively learned from human demonstration videos. Our method addresses key challenges in visuomotor policy learning through a principled variational replanning framework that maintains temporal consistency while efficiently adapting to environmental changes.

关注点: 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 索引摘要。

1.3.3.208 Latent Theory of Mind: A Decentralized Diffusion Architecture for Cooperative Manipulation

- 作者: He, Chengyang; Camps, Gadiel Mark Sznaiier; Liu, Xu; Schwager, Mac; Sartoretti, Guillaume Adrien
- 年份: 2025; 主题: Embodied/WBC; 优先级: 扩展
- [来源](#)

论文: Latent Theory of Mind: A Decentralized Diffusion Architecture for Cooperative Manipulation

快速理解: We present Latent Theory of Mind (LatentToM), a decentralized diffusion policy architecture for collaborative robot manipulation. Our policy allows multiple manipulators with their own perception and computation to collaborate with each other towards a common task goal with or without explicit communication.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制。

证据层级: 索引摘要。

1.3.3.209 Leveraging Correlation Across Test Platforms for Variance-Reduced Metric Estimation

- 作者: Luo, Rachel; Yang, Heng; Watson, Michael; Sharma, Apoorva; Veer, Sushant; Schmerling, Edward; Pavone, Marco
- 年份: 2025; 主题: Embodied/WBC; 优先级: 扩展
- [来源](#)

论文: Leveraging Correlation Across Test Platforms for Variance-Reduced Metric Estimation

快速理解: Learning-based robotic systems demand rigorous validation to assure reliable performance, but extensive real-world testing is often prohibitively expensive and if conducted may still yield insufficient data for high-confidence guarantees. In this work, we introduce a general estimation framework that leverages *paired* data across test platforms, e.g., paired simulation and real-world observations, to achieve better estimates of real-world metrics via the method of control variates.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制。

证据层级: 索引摘要。

1.3.3.210 MoTo: A Zero-shot Plug-in Interaction-aware Navigation for General Mobile Manipulation

- 作者: Wu, Zhenyu; Ma, Angyuan; Xu, Xiuwei; Yin, Hang; Liang, Yinan; Wang, Ziwei; Lu, Jiwen; Yan, Haibin
- 年份: 2025; 主题: Supporting Perception; 优先级: 扩展

- [来源](#)

论文: MoTo: A Zero-shot Plug-in Interaction-aware Navigation for General Mobile Manipulation

快速理解: Mobile manipulation is the fundamental challenge for robotics in assisting humans with diverse tasks and environments in everyday life. Conventional mobile manipulation approaches often struggle to generalize across different tasks and environments due to the lack of large-scale training.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.3.211 Motion Blender Gaussian Splatting for Dynamic Reconstruction

- 作者: Zhang, Xinyu; Chang, Haonan; Liu, Yuhao; Boularias, Abdeslam
- 年份: 2025; 主题: Robot Learning; 优先级: 扩展
- [来源](#)

论文: Motion Blender Gaussian Splatting for Dynamic Reconstruction

快速理解: Gaussian splatting has emerged as a powerful tool for high-fidelity reconstruction of dynamic scenes. However, existing methods primarily rely on implicit motion representations, such as encoding motions into neural networks or per-Gaussian parameters, which makes it difficult to further manipulate the reconstructed motions.

关注点: 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 索引摘要。

1.3.3.212 Neural Robot Dynamics

- 作者: Xu, Jie; Heiden, Eric; Akinola, Iretiayo; Fox, Dieter; Macklin, Miles; Narang, Yashraj
- 年份: 2025; 主题: World Model; 优先级: 扩展
- [来源](#)

论文: Neural Robot Dynamics

快速理解: Accurate and efficient simulation of modern robots remains challenging due to their high degrees of freedom and intricate mechanisms. Neural simulators have emerged as a promising alternative to traditional analytical simulators, capable of efficiently predicting complex dynamics and adapting to real-world data; however, existing neural simulators typically require application-specific training and fail to generalize to novel tasks and/or environments, primarily due to inadequate representations of the global state.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策。

证据层级: 索引摘要。

1.3.3.213 NeuralSVCD for Efficient Swept Volume Collision Detection

- 作者: Jung, Hojin; Son, Dongwon; Kim, Beomjoon
- 年份: 2025; 主题: Embodied/WBC; 优先级: 扩展
- [来源](#)

论文: NeuralSVCD for Efficient Swept Volume Collision Detection

快速理解: Robot manipulation in unstructured environments requires efficient and reliable Swept

Volume Collision Detection (SVCD) for safe motion planning. Traditional discrete methods potentially miss collisions between these points, whereas SVCD continuously checks for collisions along the entire trajectory.

关注点：具身任务中的感知、动作生成、全身协调或运动控制。

证据层级：索引摘要。

1.3.3.214 OPAL: Visibility-aware LiDAR-to-OpenStreetMap Place Recognition via Adaptive Radial Fusion

- 作者：Kang, Shuhao; Liao, Youqi; Xia, Yan; Wysocki, Olaf; Jutzi, Boris; Cremers, Daniel
- 年份：2025；主题：Systems；优先级：扩展
- [来源](#)

论文：OPAL: Visibility-aware LiDAR-to-OpenStreetMap Place Recognition via Adaptive Radial Fusion

快速理解：LiDAR place recognition is a critical capability for autonomous navigation and cross-modal localization in large-scale outdoor environments. Existing approaches predominantly depend on pre-built 3D dense maps or aerial imagery, which impose significant storage overhead and lack real-time adaptability.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.3.215 Pointing3D: A Benchmark for 3D Object Referral via Pointing Gestures

- 作者：Arslanoglu, Mert; Yilmaz, Kadir; “{O}zaltan, Cemhan Kaan; Linder, Timm; Leibe, Bastian
- 年份：2025；主题：Systems；优先级：扩展
- [来源](#)

论文：Pointing3D: A Benchmark for 3D Object Referral via Pointing Gestures

快速理解：Pointing gestures provide a natural and efficient way to communicate spatial information in human-machine interaction, yet their potential for 3D object referral remains largely under-explored. To fill this gap, we introduce the task of pointing-based 3D segmentation.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.3.216 Poke and Strike: Learning Task-Informed Exploration Policies

- 作者：Aoyama, Marina Y.; Moura, Joao; Ferrandis, Juan Del Aguila; Vijayakumar, Sethu
- 年份：2025；主题：Agent RL；优先级：扩展
- [来源](#)

论文：Poke and Strike: Learning Task-Informed Exploration Policies

快速理解：In many dynamic robotic tasks, such as striking pucks into a goal outside the reachable workspace, the robot must first identify the relevant physical properties of the object for successful task execution, as it is unable to recover from failure or retry without human intervention. To address this challenge, we propose a task-informed exploration approach, based on reinforcement learning, that trains an exploration policy using rewards automatically generated from the sensitivity of a

privileged task policy to errors in estimated properties.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.3.217 Rapid Mismatch Estimation via Neural Network Informed Variational Inference

- 作者：Jaszczyk, Mateusz; Figueroa, Nadia
- 年份：2025；主题：World Model；优先级：扩展
- [来源](#)

论文：Rapid Mismatch Estimation via Neural Network Informed Variational Inference

快速理解：With robots increasingly operating in human-centric environments, ensuring soft and safe physical interactions, whether with humans, surroundings, or other machines, is essential. While compliant hardware can facilitate such interactions, this work focuses on impedance controllers that allow torque-controlled robots to safely and passively respond to contact while accurately executing tasks.

关注点：环境动力学、视频预测、潜变量状态或模型式决策。

证据层级：索引摘要。

1.3.3.218 Robot Operating Home Appliances by Reading User Manuals

- 作者：Zhang, Jian; Zhang, Hanbo; Xiao, Anxing; Hsu, David
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Robot Operating Home Appliances by Reading User Manuals

快速理解：Operating home appliances, among the most common tools in every household, is a critical capability for assistive home robots. This paper presents ApBot, a robot system that operates novel household appliances by “reading” their user manuals.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.3.219 SIREN: Semantic, Initialization-Free Registration of Multi-Robot Gaussian Splatting Maps

- 作者：Shorinwa, Olao; Sun, Jiankai; Schwager, Mac; Majumdar, Anirudha
- 年份：2025；主题：Embodied/WBC；优先级：扩展
- [来源](#)

论文：SIREN: Semantic, Initialization-Free Registration of Multi-Robot Gaussian Splatting Maps

快速理解：We present SIREN for registration of multi-robot Gaussian Splatting (GSplat) maps, with zero access to camera poses, images, and inter-map transforms for initialization or fusion of local submaps. To realize these capabilities, SIREN harnesses the versatility and robustness of semantics in three critical ways to derive a rigorous registration pipeline for multi-robot GSplat maps.

关注点：具身任务中的感知、动作生成、全身协调或运动控制。

证据层级：索引摘要。

1.3.3.220 SafeBimanual: Diffusion-based trajectory optimization for safe bimanual manipulation

- 作者：Deng, Haoyuan; Guo, Wenkai; Wang, Qianzhun; Wu, Zhenyu; Wang, Ziwei
- 年份：2025; 主题：Supporting Perception; 优先级：扩展
- [来源](#)

论文：SafeBimanual: Diffusion-based trajectory optimization for safe bimanual manipulation

快速理解：Bimanual manipulation has been widely applied in household services and manufacturing, which enables the complex task completion with coordination requirements. Recent diffusion-based policy learning approaches have achieved promising performance in modeling action distributions for bimanual manipulation.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.3.221 Search-TTA: A Multi-Modal Test-Time Adaptation Framework for Visual Search in the Wild

- 作者：Tan, Derek Ming Siang; Shailesh, Shailes; Liu, Boyang; Raj, Alok; Ang, Qi Xuan; Dai, Weiheng; Duhan, Tanishq; Chiun, Jimmy; Cao, Yuhong; Shkurti, Florian; Sartoretti, Guillaume Adrien
- 年份：2025; 主题：Supporting Perception; 优先级：扩展
- [来源](#)

论文：Search-TTA: A Multi-Modal Test-Time Adaptation Framework for Visual Search in the Wild

快速理解：To perform autonomous visual search for environmental monitoring, a robot may leverage satellite imagery as a prior map. This can help inform coarse, high level search and exploration strategies, even when such images lack sufficient resolution to allow fine-grained, explicit visual recognition of targets.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.3.222 See, Point, Fly: A Learning-Free VLM Framework for Universal Unmanned Aerial Navigation

- 作者：Hu, Chih Yao; Lin, Yang-Sen; Lee, Yuna; Su, Chih-Hai; Lee, Jie-Ying; Tsai, Shr-Ruei; Lin, Chin-Yang; Chen, Kuan-Wen; Ke, Tsung-Wei; Liu, Yu-Lun
- 年份：2025; 主题：Supporting Perception; 优先级：扩展
- [来源](#)

论文：See, Point, Fly: A Learning-Free VLM Framework for Universal Unmanned Aerial Navigation

快速理解：We present See, Point, Fly (SPF), a training-free aerial vision-and-language navigation (AVLN) framework built atop vision-language models (VLMs). SPF is capable of navigating to any goal based on any type of free-form instructions in any kind of environment.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.3.223 Self-supervised Learning Of Visual Pose Estimation Without Pose Labels By Classifying LED States

- 作者：Carlotti, Nicholas; Nava, Mirko; Giusti, Alessandro
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Self-supervised Learning Of Visual Pose Estimation Without Pose Labels By Classifying LED States

快速理解：We introduce a model for monocular RGB relative pose estimation of a ground robot that trains from scratch without pose labels nor prior knowledge about the robot's shape or appearance. At training time, we assume: (i) a robot fitted with multiple LEDs, whose states are independent and known at each frame; (ii) knowledge of the approximate viewing direction of each LED; and (iii) availability of a calibration image with a known target distance, to address the ambiguity of monocular depth estimation.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.3.224 Streaming Flow Policy: Simplifying diffusion/flow-matching policies by treating action trajectories as flow trajectories

- 作者：Jiang, Sunshine; Fang, Xiaolin; Roy, Nicholas; Lozano-Perez, Tomás; Kaelbling, Leslie Pack; Ancha, Siddharth
- 年份：2025；主题：Agent RL；优先级：扩展
- [来源](#)

论文：Streaming Flow Policy: Simplifying diffusion/flow-matching policies by treating action trajectories as flow trajectories

快速理解：Recent advances in diffusion/flow-matching policies have enabled imitation learning of complex, multi-modal action trajectories. However, they are computationally expensive because they sample a *trajectory of trajectories*—a diffusion/flow trajectory of action trajectories.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.3.225 Towards Generalizable Safety in Crowd Navigation via Conformal Uncertainty Handling

- 作者：Yao, Jianpeng; Zhang, Xiaopan; Xia, Yu; Wang, Zejin; Roy-Chowdhury, Amit; Li, Jiachen
- 年份：2025；主题：Agent RL；优先级：扩展
- [来源](#)

论文：Towards Generalizable Safety in Crowd Navigation via Conformal Uncertainty Handling

快速理解：Mobile robots navigating in crowds trained using reinforcement learning are known to suffer performance degradation when faced with out-of-distribution scenarios. We propose that by properly accounting for the uncertainties of pedestrians, a robot can learn safe navigation policies

that are robust to distribution shifts.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.3.226 Uncertainty-aware Latent Safety Filters for Avoiding Out-of-Distribution Failures

- 作者：Seo, Junwon; Nakamura, Kensuke; Bajcsy, Andrea
- 年份：2025；主题：World Model；优先级：扩展
- [来源](#)

论文：Uncertainty-aware Latent Safety Filters for Avoiding Out-of-Distribution Failures

快速理解：Recent advances in generative world models have enabled classical safe control methods, such as Hamilton-Jacobi (HJ) reachability, to generalize to complex robotic systems operating directly from high-dimensional sensor observations. However, obtaining comprehensive coverage of all safety-critical scenarios during world model training is extremely challenging.

关注点：环境动力学、视频预测、潜变量状态或模型式决策。

证据层级：索引摘要。

1.3.3.227 “Stack It Up!”: 3D Stable Structure Generation from 2D Hand-drawn Sketch

- 作者：Xu, Yiqing; Li, Linfeng; Yu, Cunjun; Hsu, David
- 年份：2025；主题：Embodied/WBC；优先级：扩展
- [来源](#)

论文：“Stack It Up!” : 3D Stable Structure Generation from 2D Hand-drawn Sketch

快速理解：Imagine a child sketching the Eiffel Tower and asking a robot to bring it to life. Today’s robot manipulation systems can’t act on such sketches directly—they require precise 3D block poses as goals, which in turn demand structural analysis and expert tools like CAD.

关注点：具身任务中的感知、动作生成、全身协调或运动控制。

证据层级：索引摘要。

1.3.4 AI · EMNLP

本次会议纳入 **12** 篇。自动主题统计为：Supporting Perception (8)、Agent RL (6)、Systems (4)、VLA (3)、Embodied/WBC (2)、AI Infra (1)。这些统计用于组织阅读顺序，不能单独证明论文结论或实验优越性。

优先阅读候选：- 3MDBench: Medical Multimodal Multi-agent Dialogue Benchmark (优先级分数 60.25) - COLA: Collaborative Multi-Agent Framework with Dynamic Task Scheduling for GUI Automation (优先级分数 60.25)- Cache-Efficient Posterior Sampling for Reinforcement Learning with LLM-Derived Priors Across Discrete and Continuous Domains (优先级分数 60.25) - Cache-of-Thought: Master-Apprentice Framework for Cost-Effective Vision Language Model Reasoning (优先级分数 60.25) - ChatVLA: Unified Multimodal Understanding and Robot Control with Vision-Language-Action Model (优先级分数 60.25)

1.3.4.1 3MDBench: Medical Multimodal Multi-agent Dialogue Benchmark

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL; Supporting Perception；优先级：重要

- [来源](#)

论文: 3MDBench: Medical Multimodal Multi-agent Dialogue Benchmark

快速理解: Though Large Vision-Language Models (LVLMs) are being actively explored in medicine, their ability to conduct complex realworld telemedicine consultations combining accurate diagnosis with professional dialogue remains underexplored. This paper presents 3MDBench (Medical Multimodal Multi-agent Dialogue Benchmark), an open-source framework for simulating and evaluating LVLM-driven telemedical consultations.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.4.2 COLA: Collaborative Multi-Agent Framework with Dynamic Task Scheduling for GUI Automation

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; Systems; 优先级: 重要
- [来源](#)

论文: COLA: Collaborative Multi-Agent Framework with Dynamic Task Scheduling for GUI Automation

快速理解: With the rapid advancements in Large Language Models (LLMs), an increasing number of studies have leveraged LLMs as the cognitive core of agents to address complex task decision-making challenges. Specially, recent research has demonstrated the potential of LLM-based agents on automating GUI operations.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.4.3 Cache-Efficient Posterior Sampling for Reinforcement Learning with LLM-Derived Priors Across Discrete and Continuous Domains

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; Systems; 优先级: 重要
- [来源](#)

论文: Cache-Efficient Posterior Sampling for Reinforcement Learning with LLM-Derived Priors Across Discrete and Continuous Domains

快速理解: Integrating large language models (LLMs) as action proposers in reinforcement learning (RL) boosts performance in text-based environments but incurs high computational costs. We introduce a cache-efficient framework for Bayesian RL with LLM-derived action suggestions, reducing costs while maintaining near-optimal performance.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.4.4 Cache-of-Thought: Master-Apprentice Framework for Cost-Effective Vision Language Model Reasoning

- 作者: 作者信息未提供

- 年份: 2025; 主题: Supporting Perception; Systems; 优先级: 重要
- [来源](#)

论文: Cache-of-Thought: Master-Apprentice Framework for Cost-Effective Vision Language Model Reasoning

快速理解: Vision Language Models (VLMs) have achieved remarkable success in a wide range of vision applications of increasing complexity and scales, yet choosing the right VLM model size involves a trade-off between response quality and cost. While smaller VLMs are cheaper to run, they typically produce responses only marginally better than random guessing on benchmarks such as MMMU.

关注点: 3D、视频、多模态视觉或具身感知支撑能力; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.4.5 ChatVLA: Unified Multimodal Understanding and Robot Control with Vision-Language-Action Model

- 作者: 作者信息未提供
- 年份: 2025; 主题: VLA; Supporting Perception; 优先级: 重要
- [来源](#)

论文: ChatVLA: Unified Multimodal Understanding and Robot Control with Vision-Language-Action Model

快速理解: Humans possess a unified cognitive ability to perceive, comprehend, and interact with the physical world. Why can't large language models replicate this holistic understanding?

关注点: 视觉、语言条件与机器人动作策略之间的对齐; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.4.6 EQA-RM: A Generative Embodied Reward Model with Test-time Scaling

- 作者: 作者信息未提供
- 年份: 2025; 主题: Embodied/WBC; Agent RL; 优先级: 重要
- [来源](#)

论文: EQA-RM: A Generative Embodied Reward Model with Test-time Scaling

快速理解: Reward Models (RMs), vital for large model alignment, are underexplored for complex embodied tasks like Embodied Question Answering (EQA) where nuanced evaluation of agents' spatial, temporal, and logical understanding is critical yet not considered by generic approaches. We introduce EQA-RM, a novel generative multimodal reward model specifically architected for EQA, trained via our innovative Contrastive Group Relative Policy Optimization (C-GRPO) strategy to learn finegrained behavioral distinctions.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.4.7 MAKAR: a Multi-Agent framework based Knowledge-Augmented Reasoning for Grounded Multimodal Named Entity Recognition

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; Supporting Perception; 优先级: 重要

- [来源](#)

论文: MAKAR: a Multi-Agent framework based Knowledge-Augmented Reasoning for Grounded Multimodal Named Entity Recognition

快速理解: Grounded Multimodal Named Entity Recognition (GMNER), which aims to extract textual entities, their types, and corresponding visual regions from image-text data, has become a critical task in multimodal information extraction. However, existing methods face two major challenges.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.4.8 Spec-VLA: Speculative Decoding for Vision-Language-Action Models with Relaxed Acceptance

- 作者: 作者信息未提供
- 年份: 2025; 主题: VLA; Supporting Perception; 优先级: 重要
- [来源](#)

论文: Spec-VLA: Speculative Decoding for Vision-Language-Action Models with Relaxed Acceptance

快速理解: Vision-Language-Action (VLA) models have made substantial progress by leveraging the robust capabilities of Visual Language Models (VLMs). However, VLMs' significant parameter size and autoregressive (AR) decoding nature impose considerable computational demands on VLA models.

关注点: 视觉、语言条件与机器人动作策略之间的对齐; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.4.9 VLA-Mark: A cross modal watermark for large vision-language alignment models

- 作者: 作者信息未提供
- 年份: 2025; 主题: VLA; Supporting Perception; 优先级: 重要
- [来源](#)

论文: VLA-Mark: A cross modal watermark for large vision-language alignment models

快速理解: Vision-language models demand watermarking solutions that protect intellectual property without compromising multimodal coherence. Existing text watermarking methods disrupt visual-textual alignment through biased token selection and static strategies, leaving semanticcritical concepts vulnerable.

关注点: 视觉、语言条件与机器人动作策略之间的对齐; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.4.10 VLP: Vision-Language Preference Learning for Embodied Manipulation

- 作者: 作者信息未提供
- 年份: 2025; 主题: Embodied/WBC; Supporting Perception; 优先级: 重要
- [来源](#)

论文: VLP: Vision-Language Preference Learning for Embodied Manipulation

快速理解: Reward engineering is one of the key challenges in Reinforcement Learning (RL). Preference-based RL effectively addresses this issue by learning from human feedback.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.4.11 ViLBench: A Suite for Vision-Language Process Reward Modeling

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; Supporting Perception; 优先级: 重要
- [来源](#)

论文: ViLBench: A Suite for Vision-Language Process Reward Modeling

快速理解: Process-supervised reward models serve as a fine-grained function that provides detailed step-wise feedback to model responses, facilitating effective selection of reasoning trajectories for complex tasks. Despite its advantages, evaluation on PRMs remains less explored, especially in the multimodal domain.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.4.12 XQuant: Achieving Ultra-Low Bit KV Cache Quantization with Cross-Layer Compression

- 作者: 作者信息未提供
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: XQuant: Achieving Ultra-Low Bit KV Cache Quantization with Cross-Layer Compression

快速理解: Large Language Models (LLMs) have demonstrated remarkable capabilities across diverse natural language processing tasks. However, their extensive memory requirements, particularly due to KV cache growth during long-text understanding and generation, present significant challenges for deployment in resourceconstrained environments.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.5 AI · ICCV

本会议纳入 **178** 篇。自动主题统计为: Supporting Perception (120)、Systems (24)、Embodied/WBC (21)、AI Infra (14)、Agent RL (14)、World Model (12)、Robot Learning (6)、VLA (5)。这些统计用于组织阅读顺序, 不能单独证明论文结论或实验优越性。

优先阅读候选: - FedVLA: Federated Vision-Language-Action Learning with Dual Gating Mixture-of-Experts for Robotic Manipulation (优先级分数 119.25) - DexVLG: Dexterous Vision-Language-Grasp Model at Scale (优先级分数 92.25) - QuantCache: Adaptive Importance-Guided Quantization with Hierarchical Latent and Layer Caching for Video Generation (优先级分数 92.25) - AffordDexGrasp: Open-set Language-guided Dexterous Grasp with Generalizable-Instructive Affordance (优先级分数 65.25) - AirCache: Activating Inter-modal Relevancy KV Cache Compression for Efficient Large Vision-Language Model Inference (优先级分数 65.25)

1.3.5.1 FedVLA: Federated Vision-Language-Action Learning with Dual Gating Mixture-of-Experts for Robotic Manipulation

- 作者：作者信息未提供
- 年份：2025；主题：Embodied/WBC；VLA；AI Infra；Supporting Perception；优先级：核心
- [来源](#)

论文：FedVLA: Federated Vision-Language-Action Learning with Dual Gating Mixture-of-Experts for Robotic Manipulation

快速理解：Vision-language-action (VLA) models have significantly advanced robotic manipulation by enabling robots to interpret language instructions for task execution. However, training these models often relies on large-scale user-specific data, raising concerns about privacy and security, which in turn limits their broader adoption.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；视觉、语言条件与机器人动作策略之间的对齐；训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.5.2 DexVLG: Dexterous Vision-Language-Grasp Model at Scale

- 作者：作者信息未提供
- 年份：2025；主题：Embodied/WBC；Robot Learning；Supporting Perception；优先级：核心
- [来源](#)

论文：DexVLG: Dexterous Vision-Language-Grasp Model at Scale

快速理解：As large models gain traction, vision-language models are enabling robots to tackle increasingly complex tasks. However, limited by the difficulty of data collection, progress has mainly focused on controlling simple gripper end-effectors.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；机器人数据、策略学习、操作、触觉或 sim-to-real；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.3 QuantCache: Adaptive Importance-Guided Quantization with Hierarchical Latent and Layer Caching for Video Generation

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra；Supporting Perception；Systems；优先级：核心
- [来源](#)

论文：QuantCache: Adaptive Importance-Guided Quantization with Hierarchical Latent and Layer Caching for Video Generation

快速理解：Recently, Diffusion Transformers (DiTs) have emerged as a dominant architecture in video generation, surpassing U-Net-based models in terms of performance. However, the enhanced capabilities of DiTs come with significant drawbacks, including increased computational and memory costs, which hinder their deployment on resource-constrained devices. † Corresponding author: Yulun Zhang, yulun100@gmail.com

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；3D、视频、多模态视觉或具身感知支撑能力；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.5.4 AffordDexGrasp: Open-set Language-guided Dexterous Grasp with Generalizable-Instructive Affordance

- 作者：作者信息未提供
- 年份：2025；主题：Embodied/WBC; Robot Learning; 优先级：核心
- [来源](#)

论文：AffordDexGrasp: Open-set Language-guided Dexterous Grasp with Generalizable-Instructive Affordance

快速理解：Language-guided robot dexterous generation enables robots to grasp and manipulate objects based on human commands. However, previous data-driven methods are hard to understand intention and execute grasping with unseen categories in the open set.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：本地 PDF 前页。

1.3.5.5 AirCache: Activating Inter-modal Relevancy KV Cache Compression for Efficient Large Vision-Language Model Inference

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception; Systems; 优先级：核心
- [来源](#)

论文：AirCache: Activating Inter-modal Relevancy KV Cache Compression for Efficient Large Vision-Language Model Inference

快速理解：Recent advancements in Large Visual Language Models (LVLMs) have gained significant attention due to their remarkable reasoning capabilities and proficiency in generalization. However, processing a large number of visual tokens and generating long-context outputs impose substantial computational overhead, leading to excessive demands for key-value (KV) cache.

关注点：3D、视频、多模态视觉或具身感知支撑能力；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.5.6 Benchmarking Multimodal CoT Reward Model Stepwise by Visual Program

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL; Supporting Perception; 优先级：核心
- [来源](#)

论文：Benchmarking Multimodal CoT Reward Model Stepwise by Visual Program

快速理解：Recent advancements in reward signal usage for Large Language Models (LLMs) are remarkable. However, significant challenges exist when transitioning reward signal to the multimodal domain, including labor-intensive annotations, over-reliance on one-step rewards, and inadequate evaluation.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.7 CoA-VLA: Improving Vision-Language-Action Models via Visual-Text Chain-of-Affordance

- 作者：作者信息未提供
- 年份：2025；主题：VLA; Supporting Perception；优先级：核心
- [来源](#)

论文：CoA-VLA: Improving Vision-Language-Action Models via Visual-Text Chain-of-Affordance

快速理解：Robot foundation models, particularly Vision-Language- Action (VLA) models, have garnered significant attention for their ability to enhance robot policy learning, greatly improving robot's generalization and robustness. OpenAI's recent model, O1, showcased impressive capabilities in solving complex problems by utilizing extensive reasoning chains.

关注点：视觉、语言条件与机器人动作策略之间的对齐；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.8 CombatVLA: An Efficient Vision-Language-Action Model for Combat Tasks in 3D Action Role-Playing Games

- 作者：作者信息未提供
- 年份：2025；主题：VLA; Supporting Perception；优先级：核心
- [来源](#)

论文：CombatVLA: An Efficient Vision-Language-Action Model for Combat Tasks in 3D Action Role-Playing Games

快速理解：Recent advances in Vision-Language-Action models (VLAs) have expanded the capabilities of embodied intelligence. However, significant challenges remain in real-time decision-making in complex 3D environments, which demand second-level responses, high-resolution perception, and tactical reasoning under dynamic conditions.

关注点：视觉、语言条件与机器人动作策略之间的对齐；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.9 Disentangled World Models: Learning to Transfer Semantic Knowledge from Distracting Videos for Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：World Model; Agent RL；优先级：核心
- [来源](#)

论文：Disentangled World Models: Learning to Transfer Semantic Knowledge from Distracting Videos for Reinforcement Learning

快速理解：Training visual reinforcement learning (RL) in practical scenarios presents a significant challenge, i.e., RL agents suffer from low sample efficiency in environments with variations. While various approaches have attempted to alleviate this issue by disentangled representation learning, these methods usually start learning from scratch without prior knowledge of the world.

关注点：环境动力学、视频预测、潜变量状态或模型式决策；强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.5.10 DocThinker: Explainable Multimodal Large Language Models with Rule-based Reinforcement Learning for Document Understanding

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL; Supporting Perception; 优先级：核心
- [来源](#)

论文：DocThinker: Explainable Multimodal Large Language Models with Rule-based Reinforcement Learning for Document Understanding

快速理解：Multimodal Large Language Models (MLLMs) have demonstrated remarkable capabilities in document understanding. However, their reasoning processes remain largely black-box, making it difficult to ensure reliability and trustworthiness, especially in high-stakes domains such as legal, financial, and medical document analysis.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.11 Egocentric Action-aware Inertial Localization in Point Clouds with Vision-Language Guidance

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception; Systems; 优先级：核心
- [来源](#)

论文：Egocentric Action-aware Inertial Localization in Point Clouds with Vision-Language Guidance

快速理解：This paper presents a novel inertial localization framework named Egocentric Action-aware Inertial Localization (EAIL), which leverages egocentric action cues from headmounted IMU signals to localize the target individual within a 3D point cloud. Human inertial localization is challenging due to IMU sensor noise that causes trajectory drift over time.

关注点：3D、视频、多模态视觉或具身感知支撑能力；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.5.12 Embodied VideoAgent: Persistent Memory from Egocentric Videos and Embodied Sensors Enables Dynamic Scene Understanding

- 作者：作者信息未提供
- 年份：2025；主题：Embodied/WBC; Supporting Perception; 优先级：核心
- [来源](#)

论文：Embodied VideoAgent: Persistent Memory from Egocentric Videos and Embodied Sensors Enables Dynamic Scene Understanding

快速理解：This paper investigates the problem of understanding dynamic 3D scenes from egocentric observations, a key challenge in robotics and embodied AI. Unlike prior studies that explored this as long-form video understanding and utilized egocentric video only, we instead propose an LLMbased agent, Embodied VideoAgent, which constructs scene memory from both egocentric video and embodied sensory inputs (e.g. depth and pose sensing).

关注点：具身任务中的感知、动作生成、全身协调或运动控制；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.13 EmbodiedOcc: Embodied 3D Occupancy Prediction for Vision-based Online Scene Understanding

- 作者：作者信息未提供
- 年份：2025；主题：Embodied/WBC; Supporting Perception; 优先级：核心
- [来源](#)

论文: EmbodiedOcc: Embodied 3D Occupancy Prediction for Vision-based Online Scene Understanding

快速理解: 3D occupancy prediction provides a comprehensive description of the surrounding scenes and has become an essential task for 3D perception. Most existing methods focus on offline perception from one or a few views and cannot be applied to embodied agents that demand to gradually perceive the scene through progressive embodied exploration.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制；3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.5.14 Exploiting Vision Language Model for Training-Free 3D Point Cloud OOD Detection via Graph Score Propagation

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception; Systems; 优先级：核心
- [来源](#)

论文: Exploiting Vision Language Model for Training-Free 3D Point Cloud OOD Detection via Graph Score Propagation

快速理解: Out-of-distribution (OOD) detection in 3D point cloud data remains a challenge, particularly in applications where safe and robust perception is critical. While existing OOD detection methods have shown progress for 2D image data, extending these to 3D environments involves unique obstacles.

关注点: 3D、视频、多模态视觉或具身感知支撑能力；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.5.15 Free-Form Motion Control: Controlling the 6D Poses of Camera and Objects in Video Generation

- 作者：作者信息未提供
- 年份：2025；主题：Embodied/WBC; Supporting Perception; 优先级：核心
- [来源](#)

论文: Free-Form Motion Control: Controlling the 6D Poses of Camera and Objects in Video Generation

快速理解: Controlling the movements of dynamic objects and the camera within generated videos is a meaningful yet challenging task. Due to the lack of datasets with comprehensive 6D pose annotations, existing text-to-video methods can not simultaneously control the motions of both camera and Henghui Ding (henghui.ding@gmail.com) is the corresponding author with the Institute of Big Data, College of Computer Science and Artificial Intelligence, Fudan University, Shanghai, China. objects in 3D-aware manner, resulting in limited controllability over generated contents.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.16 G-DexGrasp: Generalizable Dexterous Grasping Synthesis Via Part-Aware Prior Retrieval and Prior-Assisted Generation

- 作者：作者信息未提供
- 年份：2025；主题：Embodied/WBC; Robot Learning; 优先级：核心
- [来源](#)

论文：G-DexGrasp: Generalizable Dexterous Grasping Synthesis Via Part-Aware Prior Retrieval and Prior-Assisted Generation

快速理解：Recent advances in dexterous grasping synthesis have demonstrated significant progress in producing reasonable and plausible grasps for many task purposes. But it remains challenging to generalize to unseen object categories and diverse task instructions.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：本地 PDF 前页。

1.3.5.17 GWM: Towards Scalable Gaussian World Models for Robotic Manipulation

- 作者：作者信息未提供
- 年份：2025；主题：Embodied/WBC; World Model; 优先级：核心
- [来源](#)

论文：GWM: Towards Scalable Gaussian World Models for Robotic Manipulation

快速理解：Training robot policies within a learned world model is trending due to the inefficiency of real-world interactions. The established image-based world models and policies have shown prior success, but lack robust geometric information that requires consistent spatial and physical understanding of the three-dimensional world, even pretrained on internet-scale video sources.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；环境动力学、视频预测、潜变量状态或模型式决策。

证据层级：本地 PDF 前页。

1.3.5.18 HERMES: A Unified Self-Driving World Model for Simultaneous 3D Scene Understanding and Generation

- 作者：作者信息未提供
- 年份：2025；主题：World Model; Supporting Perception; 优先级：核心
- [来源](#)

论文：HERMES: A Unified Self-Driving World Model for Simultaneous 3D Scene Understanding and Generation

快速理解：Driving World Models (DWMs) have become essential for autonomous driving by enabling future scene prediction. However, existing DWMs are limited to scene generation and fail to incorporate scene understanding, which involves interpreting and reasoning about the driving environment.

关注点：环境动力学、视频预测、潜变量状态或模型式决策；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.19 IRASim: A Fine-Grained World Model for Robot Manipulation

- 作者：作者信息未提供
- 年份：2025；主题：Embodied/WBC; World Model; 优先级：核心
- [来源](#)

论文：IRASim: A Fine-Grained World Model for Robot Manipulation

快速理解：World models allow autonomous agents to plan and explore by predicting the visual outcomes of different actions. However, for robot manipulation, it is challenging to accurately model the fine-grained robot-object interaction within the visual space using existing methods which overlooks precise alignment between each action and the corresponding frame.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；环境动力学、视频预测、潜变量状态或模型式决策。

证据层级：本地 PDF 前页。

1.3.5.20 InstaDrive: Instance-Aware Driving World Models for Realistic and Consistent Video Generation

- 作者：作者信息未提供
- 年份：2025；主题：World Model; Supporting Perception; 优先级：核心
- [来源](#)

论文：InstaDrive: Instance-Aware Driving World Models for Realistic and Consistent Video Generation

快速理解：Autonomous driving relies on robust models trained on high-quality, large-scale multi-view driving videos. While world models offer a cost-effective solution for generating realistic driving videos, they struggle to maintain instance-level temporal consistency and spatial geometric fidelity.

关注点：环境动力学、视频预测、潜变量状态或模型式决策；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.21 Learning 4D Embodied World Models

- 作者：作者信息未提供
- 年份：2025；主题：Embodied/WBC; World Model; 优先级：核心
- [来源](#)

论文：Learning 4D Embodied World Models

快速理解：This paper presents an effective approach for learning novel 4D embodied world models, TesserAct, which predict the dynamic evolution of 3D scenes over time in response to an embodied agent's actions, providing both spatial and temporal consistency. We propose to learn a 4D world model by training on RGB-DN (RGB, Depth, and Normal) videos.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；环境动力学、视频预测、潜变量状态或模型式决策。

证据层级：本地 PDF 前页。

1.3.5.22 MOERL: When Mixture-of-Experts Meet Reinforcement Learning for Adverse Weather Image Restoration

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL; AI Infra; 优先级：核心
- [来源](#)

论文: MOERL: When Mixture-of-Experts Meet Reinforcement Learning for Adverse Weather Image Restoration

快速理解: Adverse weather conditions, such as rain, snow, and haze, introduce complex degradations that present substantial challenges for effective image restoration. Existing all-in-one models often rely on fixed network structures, limiting their ability to adapt to the varying characteristics of different weather conditions.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 本地 PDF 前页。

1.3.5.23 Multi-Cache Enhanced Prototype Learning for Test-Time Generalization of Vision-Language Models

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception; Systems; 优先级：核心
- [来源](#)

论文: Multi-Cache Enhanced Prototype Learning for Test-Time Generalization of Vision-Language Models

快速理解: In zero-shot setting, test-time adaptation adjusts pre-trained models using unlabeled data from the test phase to enhance performance on unknown test distributions. Existing cache-enhanced TTA methods rely on a low-entropy criterion to select samples for prototype construction, assuming intra-class compactness.

关注点: 3D、视频、多模态视觉或具身感知支撑能力; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.5.24 Multimodal LLMs as Customized Reward Models for Text-to-Image Generation

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL; Supporting Perception; 优先级：核心
- [来源](#)

论文: Multimodal LLMs as Customized Reward Models for Text-to-Image Generation

快速理解: We introduce LLaVA-Reward1 , an efficient reward model designed to automatically evaluate text-to-image (T2I) generations across multiple perspectives, leveraging pretrained multimodal large language models (MLLMs). Existing MLLM-based approaches require instruction-following data for supervised fine-tuning and evaluate generation quality on analyzing text response, which is time-consuming and difficult to train.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.5.25 Omni-scene Perception-oriented Point Cloud Geometry Enhancement for Coordinate Quantization

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra; Systems；优先级：核心
- [来源](#)

论文：Omni-scene Perception-oriented Point Cloud Geometry Enhancement for Coordinate Quantization

快速理解：Information quantization has been widely adopted in multimedia content, such as images, videos, and point clouds. The goal of information quantization is to achieve efficient storage and transmission by reducing data precision or redundancy.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.5.26 R1-VL: Learning to Reason with Multimodal Large Language Models via Step-wise Group Relative Policy Optimization

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL; Supporting Perception；优先级：核心
- [来源](#)

论文：R1-VL: Learning to Reason with Multimodal Large Language Models via Step-wise Group Relative Policy Optimization

快速理解：Recent studies generally enhance MLLMs' reasoning capabilities via supervised fine-tuning on high-quality chain-of-thought reasoning data, which often leads models to merely imitate successful reasoning paths without understanding what the wrong reasoning paths are. In this work, we aim to enhance the MLLMs' reasoning ability beyond passively imitating positive reasoning paths.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.27 RoboTron-Mani: All-in-One Multimodal Large Model for Robotic Manipulation

- 作者：作者信息未提供
- 年份：2025；主题：Embodied/WBC; Supporting Perception；优先级：核心
- [来源](#)

论文：RoboTron-Mani: All-in-One Multimodal Large Model for Robotic Manipulation

快速理解：Recently, robotics has advanced significantly through the integration of larger models and large-scale datasets. However, challenges remain in applying these models to 3D spatial interactions and managing data collection costs.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.28 SAME: Learning Generic Language-Guided Visual Navigation with State-Adaptive Mixture of Experts

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra; Supporting Perception；优先级：核心
- [来源](#)

论文：SAME: Learning Generic Language-Guided Visual Navigation with State-Adaptive Mixture of Experts

快速理解：The academic field of learning instruction-guided visual navigation can be generally categorized into high-level category-specific search and low-level language-guided navigation, depending on the granularity of language instruction, in which the former emphasizes the exploration process, while the latter concentrates on following detailed textual commands. Despite the differing focuses of these tasks, the underlying requirements of interpreting instructions, comprehending the surroundings, and inferring action decisions remain consistent.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.29 Scheduling Weight Transitions for Quantization-Aware Training

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra; Systems；优先级：核心
- [来源](#)

论文：Scheduling Weight Transitions for Quantization-Aware Training

快速理解：Quantization-aware training (QAT) simulates a quantization process during training to lower bit-precision of weights/activations. It learns quantized weights indirectly by updating latent weights, i.e., full-precision inputs to a quantizer, using gradient-based optimizers.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.5.30 Structured Policy Optimization: Enhance Large Vision-Language Model via Self-referenced Dialogue

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL; Supporting Perception；优先级：核心
- [来源](#)

论文：Structured Policy Optimization: Enhance Large Vision-Language Model via Self-referenced Dialogue

快速理解：Preference optimization algorithms typically enhance LLM response quality by leveraging human feedback on multiple answers given a fixed instruction. However, these methods often lack capturing the dynamic nature of conversational exchanges.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.31 Towards Long-Horizon Vision-Language-Action System: Reasoning, Acting and Memory

- 作者：作者信息未提供

- 年份: 2025; 主题: VLA; Supporting Perception; 优先级: 核心
- [来源](#)

论文: Towards Long-Horizon Vision-Language-Action System: Reasoning, Acting and Memory

快速理解: Vision-Language-Action (VLA) is crucial for autonomous decision-making in embodied systems. While current methods have advanced single-skill abilities, their short-horizon capability limits applicability in real-world scenarios.

关注点: 视觉、语言条件与机器人动作策略之间的对齐; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.5.32 VLR-Driver: Large Vision-Language-Reasoning Models for Embodied Autonomous Driving

- 作者: 作者信息未提供
- 年份: 2025; 主题: Embodied/WBC; Supporting Perception; 优先级: 核心
- [来源](#)

论文: VLR-Driver: Large Vision-Language-Reasoning Models for Embodied Autonomous Driving

快速理解: This ICCV paper is the Open Access version, provided by the Computer Vision Foundation. Except for this watermark, it is identical to the accepted version; the final published version of the proceedings is available on IEEE Xplore.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.5.33 VLRGBench: A Comprehensive and Challenging Benchmark for Vision-Language Reward Models

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; Supporting Perception; 优先级: 核心
- [来源](#)

论文: VLRGBench: A Comprehensive and Challenging Benchmark for Vision-Language Reward Models

快速理解: Although large visual-language models (LVLMS) have demonstrated strong performance in multimodal tasks, errors may occasionally arise due to biases during the reasoning process. Recently, reward models (RMs) have become increasingly pivotal in the reasoning process.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.5.34 VQ-VLA: Improving Vision-Language-Action Models via Scaling Vector-Quantized Action Tokenizers

- 作者: 作者信息未提供
- 年份: 2025; 主题: VLA; Supporting Perception; 优先级: 核心
- [来源](#)

论文: VQ-VLA: Improving Vision-Language-Action Models via Scaling Vector-Quantized Action

Tokenizers

快速理解: In this paper, we introduce an innovative vector quantization based action tokenizer built upon the largest-scale action trajectory dataset to date, leveraging over 100 times more data than previous approaches. This extensive dataset enables our tokenizer to capture rich spatiotemporal dynamics, resulting in a model that not only accelerates inference but also generates smoother and more coherent action outputs.

关注点: 视觉、语言条件与机器人动作策略之间的对齐; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.5.35 2.5 Years in Class: A Multimodal Textbook for Vision-Language Pretraining

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: 2.5 Years in Class: A Multimodal Textbook for Vision-Language Pretraining

快速理解: Compared to image-text pair data, interleaved corpora enable Vision-Language Models (VLMs) to understand the world more naturally like humans. However, such existing datasets are crawled from webpage, facing challenges like low knowledge density, loose image-text relations, and poor logical coherence between images.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.5.36 3DGraphLLM: Combining Semantic Graphs and Large Language Models for 3D Scene Understanding

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: 3DGraphLLM: Combining Semantic Graphs and Large Language Models for 3D Scene Understanding

快速理解: A 3D scene graph represents a compact scene model by capturing both the objects present and the semantic relationships between them, making it a promising structure for robotic applications. To effectively interact with users, an embodied intelligent agent should be able to answer a wide range of natural language queries about the surrounding 3D environment.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.5.37 4D Visual Pre-training for Robot Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Robot Learning; 优先级: 扩展
- [来源](#)

论文: 4D Visual Pre-training for Robot Learning

快速理解: General visual representations learned from web-scale datasets for robotics have achieved great success in recent years, enabling data-efficient robot learning on manipulation tasks; yet these pre-trained representations are mostly on 2D images, neglecting the inherent 3D nature of the world. However, due to the scarcity of large-scale 3D data, it is still hard to extract a universal 3D representation from web datasets.

关注点: 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 本地 PDF 前页。

1.3.5.38 A Constrained Optimization Approach for Gaussian Splatting from Coarsely-posed Images and Noisy Lidar Point Clouds

- 作者: 作者信息未提供
- 年份: 2025; 主题: Systems; 优先级: 扩展
- [来源](#)

论文: A Constrained Optimization Approach for Gaussian Splatting from Coarsely-posed Images and Noisy Lidar Point Clouds

快速理解: 3D Gaussian Splatting (3DGS) is a powerful reconstruction technique; however, it requires initialization from accurate camera poses and high-fidelity point clouds. Typically, the initialization is taken from Structure-from-Motion (SfM) algorithms; however, SfM is time-consuming and restricts the application of 3DGS in real-world scenarios and largescale scene reconstruction.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.5.39 A0: An Affordance-Aware Hierarchical Model for General Robotic Manipulation

- 作者: 作者信息未提供
- 年份: 2025; 主题: Embodied/WBC; 优先级: 扩展
- [来源](#)

论文: A0: An Affordance-Aware Hierarchical Model for General Robotic Manipulation

快速理解: Robotic manipulation faces critical challenges in understanding spatial affordances—the “where” and “how” of object interactions—essential for complex manipulation tasks like wiping a board or stacking objects. Existing methods, including modular-based and end-to-end approaches, often lack robust spatial reasoning capabilities.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制。

证据层级: 本地 PDF 前页。

1.3.5.40 AG2aussian: Anchor-Graph Structured Gaussian Splatting for Instance-Level 3D Scene Understanding and Editing

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: AG2aussian: Anchor-Graph Structured Gaussian Splatting for Instance-Level 3D Scene Un-

derstanding and Editing

快速理解: 3D Gaussian Splatting (3DGS) has witnessed exponential adoption across diverse applications, driving a critical need for semantic-aware 3D Gaussian representations to enable scene understanding and editing tasks. Existing approaches typically attach semantic features to a collection of free Gaussians and distill the features via differentiable rendering, leading to noisy segmentation and a messy selection of Gaussians.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.5.41 AHCPTQ: Accurate and Hardware-Compatible Post-Training Quantization for Segment Anything Model

- 作者: 作者信息未提供
- 年份: 2025; 主题: AI Infra; 优先级: 扩展
- [来源](#)

论文: AHCPTQ: Accurate and Hardware-Compatible Post-Training Quantization for Segment Anything Model

快速理解: The Segment Anything Model (SAM) has demonstrated strong versatility across various visual tasks. However, its large storage requirements and high computational cost pose challenges for practical deployment.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 本地 PDF 前页。

1.3.5.42 AID: Adapting Image2Video Diffusion Models for Instruction-guided Video Prediction

- 作者: 作者信息未提供
- 年份: 2025; 主题: World Model; 优先级: 扩展
- [来源](#)

论文: AID: Adapting Image2Video Diffusion Models for Instruction-guided Video Prediction

快速理解: Text-guided video prediction (TVP) involves predicting the motion of future frames from the initial frame according to an instruction, which has wide applications in virtual reality, robotics, and content creation. Previous TVP methods make significant breakthroughs by adapting Stable Diffusion for this task.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策。

证据层级: 本地 PDF 前页。

1.3.5.43 AIGI-Holmes: Towards Explainable and Generalizable AI-Generated Image Detection via Multimodal Large Language Models

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: AIGI-Holmes: Towards Explainable and Generalizable AI-Generated Image Detection via Mul-

timodal Large Language Models

快速理解: The rapid development of AI-generated content (AIGC) technology has led to the misuse of highly realistic AI-generated images (AIGI) in spreading misinformation, posing a threat to public information security. Although existing AIGI detection techniques are generally effective, they face two issues: 1) a lack of human-verifiable explanations, and 2) a lack of generalization in the latest generation technology.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.5.44 AR-VRM: Imitating Human Motions for Visual Robot Manipulation with Analogical Reasoning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Embodied/WBC; 优先级: 扩展
- [来源](#)

论文: AR-VRM: Imitating Human Motions for Visual Robot Manipulation with Analogical Reasoning

快速理解: Visual Robot Manipulation (VRM) aims to enable a robot to follow natural language instructions based on robot states and visual observations, and therefore requires costly multimodal data. To compensate for the deficiency of robot data, existing approaches have employed vision-language pretraining with large-scale data.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制。

证据层级: 本地 PDF 前页。

1.3.5.45 ATCTrack: Aligning Target-Context Cues with Dynamic Target States for Robust Vision-Language Tracking

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: ATCTrack: Aligning Target-Context Cues with Dynamic Target States for Robust Vision-Language Tracking

快速理解: Vision-language tracking aims to locate the target object in the video sequence using a template patch and a language description provided in the initial frame. To achieve robust tracking, especially in complex long-term scenarios that reflect real-world conditions as recently highlighted by MGIT, it is essential not only to characterize the target features but also to utilize the context features related to the target.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.5.46 AV-Link: Temporally-Aligned Diffusion Features for Cross-Modal Audio-Video Generation

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 扩展

- [来源](#)

论文: AV-Link: Temporally-Aligned Diffusion Features for Cross-Modal Audio-Video Generation

快速理解: We propose AV-Link, a unified framework for Video-to-Audio (A2V) and Audio-to-Video (A2V) generation that leverages the activations of frozen video and audio diffusion models for temporally-aligned cross-modal conditioning. The key to our framework is a Fusion Block that facilitates bidirectional information exchange between video and audio diffusion models through temporally-aligned self attention operations.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.5.47 AVAM: a Universal Training-free Adaptive Visual Anchoring Embedded into Multimodal Large Language Model for Multi-image Question Answering

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: AVAM: a Universal Training-free Adaptive Visual Anchoring Embedded into Multimodal Large Language Model for Multi-image Question Answering

快速理解: The advancement of Multimodal Large Language Models (MLLMs) has driven significant progress in Visual Question Answering (VQA), evolving from Single to Multi Image VQA (MVQA). However, the increased number of images in MVQA inevitably introduces substantial visual redundancy that is irrelevant to question answering, negatively impacting both accuracy and efficiency.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.5.48 Accelerating Diffusion Transformer via Gradient-Optimized Cache

- 作者: 作者信息未提供
- 年份: 2025; 主题: Systems; 优先级: 扩展
- [来源](#)

论文: Accelerating Diffusion Transformer via Gradient-Optimized Cache

快速理解: Feature caching has emerged as an effective strategy to accelerate diffusion transformer (DiT) sampling through temporal feature reuse. It is a challenging problem since (1) Progressive error accumulation from cached blocks significantly degrades generation quality, particularly when over 50% of blocks are cached; (2) Current error compensation approaches neglect dynamic perturbation patterns during the caching process, leading to suboptimal error correction.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.5.49 Adaptive Caching for Faster Video Generation with Diffusion Transformers

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Adaptive Caching for Faster Video Generation with Diffusion Transformers

快速理解: Generating temporally-consistent high-fidelity videos can be computationally expensive, especially over longer temporal spans. More-recent Diffusion Transformers (DiTs)—despite making significant headway in this context—have only heightened such challenges as they rely on larger models and heavier attention mechanisms, resulting in slower inference speeds.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.5.50 Adaptive Routing of Text-to-Image Generation Requests Between Large Cloud Model and Light-Weight Edge Model

- 作者: 作者信息未提供
- 年份: 2025; 主题: Systems; 优先级: 扩展
- [来源](#)

论文: Adaptive Routing of Text-to-Image Generation Requests Between Large Cloud Model and Light-Weight Edge Model

快速理解: Large text-to-image models demonstrate impressive generation capabilities; however, their substantial size necessitates expensive cloud servers for deployment. Conversely, light-weight models can be deployed on edge devices at lower cost but often with inferior generation quality for complex user prompts.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.5.51 AdvDreamer Unveils: Are Vision-Language Models Truly Ready for Real-World 3D Variations?

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: AdvDreamer Unveils: Are Vision-Language Models Truly Ready for Real-World 3D Variations?

快速理解: Vision Language Models (VLMs) have exhibited remarkable generalization capabilities, yet their robustness in dynamic real-world scenarios remains largely unexplored. To systematically evaluate VLMs' robustness to real-world 3D variations, we propose AdvDreamer, the first framework capable of generating physically reproducible Adversarial 3D Transformation (Adv-3DT) samples from single-view observations.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.5.52 Aether: Geometric-Aware Unified World Modeling

- 作者: 作者信息未提供
- 年份: 2025; 主题: World Model; 优先级: 扩展
- [来源](#)

论文: Aether: Geometric-Aware Unified World Modeling

快速理解: The integration of geometric reconstruction and generative modeling remains a critical challenge in developing AI systems capable of human-like spatial reasoning. This paper proposes AETHER, a unified framework that enables geometry-aware reasoning in world models by jointly optimizing three core capabilities: (1) 4D dynamic reconstruction, (2) action-conditioned video prediction, and (3) goal-conditioned visual planning.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策。

证据层级: 本地 PDF 前页。

1.3.5.53 Amodal Depth Anything: Amodal Depth Estimation in the Wild

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Amodal Depth Anything: Amodal Depth Estimation in the Wild

快速理解: Amodal depth estimation aims to predict the depth of occluded (invisible) parts of objects in a scene. This task addresses the question of whether models can effectively perceive the geometry of occluded regions based on visible cues.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.5.54 Amodal3R: Amodal 3D Reconstruction from Occluded 2D Images

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Amodal3R: Amodal 3D Reconstruction from Occluded 2D Images

快速理解: Most existing image-to-3D models assume that objects are fully visible, ignoring occlusions that commonly occur in real-world scenarios. In this paper, we introduce Amodal3R, a conditional image-to-3D model designed to reconstruct plausible 3D geometry and appearance from partial observations.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.5.55 Analyzing Finetuning Representation Shift for Multimodal LLMs Steering

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Analyzing Finetuning Representation Shift for Multimodal LLMs Steering

快速理解: Multimodal LLMs (MLLMs) have reached remarkable levels of proficiency in understanding multimodal inputs. However, understanding and interpreting the behavior of such complex models is a challenging task, not to mention the dynamic shifts that may occur during fine-tuning, or due to covariate shift between datasets.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.56 AnyI2V: Animating Any Conditional Image with Motion Control

- 作者：作者信息未提供
- 年份：2025；主题：Embodied/WBC；优先级：扩展
- [来源](#)

论文：AnyI2V: Animating Any Conditional Image with Motion Control

快速理解：Recent advancements in video generation, particularly in diffusion models, have driven notable progress in text-to-video (T2V) and image-to-video (I2V) synthesis. However, challenges remain in effectively integrating dynamic motion signals and flexible spatial constraints.

关注点：具身任务中的感知、动作生成、全身协调或运动控制。

证据层级：本地 PDF 前页。

1.3.5.57 Are They the Same? Exploring Visual Correspondence Shortcomings of Multimodal LLMs

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Are They the Same? Exploring Visual Correspondence Shortcomings of Multimodal LLMs

快速理解：Recent advancements in multimodal large language models (MLLM) have shown a strong ability in visual perception, reasoning abilities, and vision-language understanding. However, the visual matching ability of MLLMs is rarely studied, despite finding the visual correspondence of objects is essential in computer vision.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.58 Articulate3D: Holistic Understanding of 3D Scenes as Universal Scene Description

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Articulate3D: Holistic Understanding of 3D Scenes as Universal Scene Description

快速理解：3D scene understanding is a long-standing challenge in computer vision and a key component in enabling mixed reality, wearable computing, and embodied AI. Providing a solution to these applications requires a multifaceted approach that covers scene-centric, object-centric, as well as interaction-centric capabilities.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.59 Authentic 4D Driving Simulation with a Video Generation Model

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Authentic 4D Driving Simulation with a Video Generation Model

快速理解：Simulating driving environments in 4D is crucial for developing accurate and immersive autonomous driving systems. Despite progress in generating driving scenes, challenges in transforming views and modeling the dynamics of space and time remain.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.60 AutoComPose: Automatic Generation of Pose Transition Descriptions for Composed Pose Retrieval Using Multimodal LLMs

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：AutoComPose: Automatic Generation of Pose Transition Descriptions for Composed Pose Retrieval Using Multimodal LLMs

快速理解：Composed pose retrieval (CPR) enables users to search for human poses by specifying a reference pose and a transition description, but progress in this field is hindered by the scarcity and inconsistency of annotated pose transitions. Existing CPR datasets rely on costly human annotations or heuristic-based rule generation, both of which limit scalability and diversity.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.61 Automated Red Teaming for Text-to-Image Models through Feedback-Guided Prompt Iteration with Vision-Language Models

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Automated Red Teaming for Text-to-Image Models through Feedback-Guided Prompt Iteration with Vision-Language Models

快速理解：Text-to-image models have achieved remarkable progress in generating high-quality images from textual prompts, yet their potential for misuse like generating unsafe content remains a critical concern. Existing safety mechanisms, such as filtering and fine-tuning, remain insufficient in preventing vulnerabilities exposed by adversarial prompts.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.62 BASIC: Boosting Visual Alignment with Intrinsic Refined Embeddings in Multimodal Large Language Models

- 作者：作者信息未提供

- 年份: 2025; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: BASIC: Boosting Visual Alignment with Intrinsic Refined Embeddings in Multimodal Large Language Models

快速理解: Mainstream Multimodal Large Language Models (MLLMs) achieve visual understanding by using a vision projector to bridge well-pretrained vision encoders and large language models (LLMs). The inherent gap between visual and textual modalities makes the embeddings from the vision projector critical for visual comprehension.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.5.63 BUFFER-X: Towards Zero-Shot Point Cloud Registration in Diverse Scenes

- 作者: 作者信息未提供
- 年份: 2025; 主题: Systems; 优先级: 扩展
- [来源](#)

论文: BUFFER-X: Towards Zero-Shot Point Cloud Registration in Diverse Scenes

快速理解: Recent advances in deep learning-based point cloud registration have improved generalization, yet most methods still require retraining or manual parameter tuning for each new environment. In this paper, we identify three key factors limiting generalization: (a) reliance on environment-specific voxel size and search radius, (b) poor out-of-domain robustness of learning-based keypoint detectors, and (c) raw coordinate usage, which exacerbates scale discrepancies.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.5.64 BadVideo: Stealthy Backdoor Attack against Text-to-Video Generation

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: BadVideo: Stealthy Backdoor Attack against Text-to-Video Generation

快速理解: Text-to-video (T2V) generative models have rapidly advanced and found widespread applications across fields like entertainment, education, and marketing. However, the adversarial vulnerabilities of these models remain rarely explored.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.5.65 Benchmarking Multimodal Large Language Models Against Image Corruptions

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Benchmarking Multimodal Large Language Models Against Image Corruptions

快速理解: Multimodal Large Language Models (MLLMs) have made significant strides in visual and language tasks. However, despite their impressive performance on standard datasets, these models encounter considerable robustness challenges when processing corrupted images, raising concerns about their reliability in safety-critical applications.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.5.66 Beyond the Destination: A Novel Benchmark for Exploration-Aware Embodied Question Answering

- 作者: 作者信息未提供
- 年份: 2025; 主题: Embodied/WBC; 优先级: 扩展
- [来源](#)

论文: Beyond the Destination: A Novel Benchmark for Exploration-Aware Embodied Question Answering

快速理解: Embodied Question Answering (EQA) is a challenging task in embodied intelligence that requires agents to dynamically explore 3D environments, actively gather visual information, and perform multi-step reasoning to answer questions. However, current EQA approaches suffer from critical limitations in exploration efficiency, dataset design, and evaluation metrics.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制。

证据层级: 本地 PDF 前页。

1.3.5.67 Bolt3D: Generating 3D Scenes in Seconds

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Bolt3D: Generating 3D Scenes in Seconds

快速理解: We present a latent diffusion model for fast feed-forward 3D scene generation. Given one or more images, our model Bolt3D directly samples a 3D scene representation in less than seven seconds on a single GPU.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.5.68 Boost 3D Reconstruction using Diffusion-based Monocular Camera Calibration

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Boost 3D Reconstruction using Diffusion-based Monocular Camera Calibration

快速理解: In this paper, we present DM-Calib, a diffusion-based approach for estimating pinhole

camera intrinsic parameters from a single input image. Monocular camera calibration is essential for many 3D vision tasks.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.69 Boosting Multimodal Learning via Disentangled Gradient Learning

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Boosting Multimodal Learning via Disentangled Gradient Learning

快速理解：Multimodal learning often encounters the under-optimized problem and may have worse performance than unimodal learning. Existing methods attribute this problem to the imbalanced learning between modalities and rebalance them through gradient modulation.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.70 Boosting Vision Semantic Density with Anatomy Normality Modeling for Medical Vision-language Pre-training

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Boosting Vision Semantic Density with Anatomy Normality Modeling for Medical Vision-language Pre-training

快速理解：Vision-language pre-training (VLP) has great potential for developing multifunctional and general medical diagnostic capabilities. However, aligning medical images with a low signal-to-noise ratio (SNR) to reports with a high SNR presents a semantic density gap, leading to visual alignment bias.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.71 Bootstrapping Grounded Chain-of-Thought in Multimodal LLMs for Data-Efficient Model Adaptation

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Bootstrapping Grounded Chain-of-Thought in Multimodal LLMs for Data-Efficient Model Adaptation

快速理解：Multimodal Large Language Models (MLLMs) have demonstrated remarkable capabilities in interpreting images using natural language. However, without using largescale datasets for retraining, these models are difficult to adapt to specialized vision tasks, e.g., chart understanding.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.72 Bridging Domain Generalization to Multimodal Domain Generalization via Unified Representations

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Bridging Domain Generalization to Multimodal Domain Generalization via Unified Representations

快速理解：Domain Generalization (DG) aims to enhance model robustness in unseen or distributionally shifted target domains through training exclusively on source domains. Although existing DG techniques, such as data manipulation, learning strategies, and representation learning, have shown significant progress, they predominantly address single-modal data.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.73 CAD-Recode: Reverse Engineering CAD Code from Point Clouds

- 作者：作者信息未提供
- 年份：2025；主题：Systems；优先级：扩展
- [来源](#)

论文：CAD-Recode: Reverse Engineering CAD Code from Point Clouds

快速理解：Computer-Aided Design (CAD) models are typically constructed by sequentially drawing parametric sketches and applying CAD operations to obtain a 3D model. The problem of 3D CAD reverse engineering consists of reconstructing the sketch and CAD operation sequences from 3D representations such as point clouds.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.5.74 CAPTURE: Evaluating Spatial Reasoning in Vision Language Models via Occluded Object Counting

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：CAPTURE: Evaluating Spatial Reasoning in Vision Language Models via Occluded Object Counting

快速理解：Recognizing and reasoning about occluded (partially or fully hidden) objects is vital to understanding visual scenes, as occlusions frequently occur in real-world environments and act as obstacles for spatial comprehension. To test models' ability to reason about multiple occluded objects, we introduce a novel task, Counting Amodally for Patterns Through Unseen REgions (CAPTURE), which requires a model to count objects arranged in a pattern by inferring how the pattern continues

behind an occluder (an object which blocks parts of the scene).

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.5.75 CC-OCR: A Comprehensive and Challenging OCR Benchmark for Evaluating Large Multimodal Models in Literacy

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: CC-OCR: A Comprehensive and Challenging OCR Benchmark for Evaluating Large Multimodal Models in Literacy

快速理解: Large Multimodal Models (LMMs) have demonstrated impressive performance in recognizing document images with natural language instructions. However, it remains unclear to what extent capabilities in literacy with rich structure and fine-grained visual challenges.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.5.76 CHORDS: Diffusion Sampling Accelerator with Multi-core Hierarchical ODE Solvers

- 作者: 作者信息未提供
- 年份: 2025; 主题: AI Infra; 优先级: 扩展
- [来源](#)

论文: CHORDS: Diffusion Sampling Accelerator with Multi-core Hierarchical ODE Solvers

快速理解: Diffusion-based generative models have become dominant generators of high-fidelity images and videos but remain limited by their computationally expensive inference procedures. Existing acceleration techniques either require extensive model retraining or compromise significantly on sample quality.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 本地 PDF 前页。

1.3.5.77 CLIP-GS: Unifying Vision-Language Representation with 3D Gaussian Splatting

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: CLIP-GS: Unifying Vision-Language Representation with 3D Gaussian Splatting

快速理解: Recent works in 3D multimodal learning have made remarkable progress. However, typically 3D multimodal models are only capable of handling point clouds.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.5.78 CMAD: Correlation-Aware and Modalities-Aware Distillation for Multimodal Sentiment Analysis with Missing Modalities

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：CMAD: Correlation-Aware and Modalities-Aware Distillation for Multimodal Sentiment Analysis with Missing Modalities

快速理解：Multimodal Sentiment Analysis (MSA) enhances emotion recognition by integrating information from multiple modalities. However, multimodal learning with missing modalities suffers from representation inconsistency and optimization instability, leading to suboptimal performance.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.79 CMT: A Cascade MAR with Topology Predictor for Multimodal Conditional CAD Generation

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：CMT: A Cascade MAR with Topology Predictor for Multimodal Conditional CAD Generation

快速理解：While accurate and user-friendly Computer-Aided Design (CAD) is crucial for industrial design and manufacturing, existing methods still struggle to achieve this due to their over-simplified representations or architectures incapable of supporting multimodal design requirements. In this paper, we attempt to tackle this problem from both methods and datasets aspects.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.80 Calibrating MLLM-as-a-judge via Multimodal Bayesian Prompt Ensembles

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Calibrating MLLM-as-a-judge via Multimodal Bayesian Prompt Ensembles

快速理解：Multimodal large language models (MLLMs) are increasingly used to evaluate text-to-image (TTI) generation systems, providing automated judgments based on visual and textual context. However, these “judge” models often suffer from biases, overconfidence, and inconsistent performance across diverse image domains.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.81 CalliReader: Contextualizing Chinese Calligraphy via an Embedding-Aligned Vision-Language Model

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文: CalliReader: Contextualizing Chinese Calligraphy via an Embedding-Aligned Vision-Language Model

快速理解: Chinese calligraphy, a UNESCO Heritage, remains computationally challenging due to visual ambiguity and cultural complexity. Existing AI systems fail to contextualize their intricate scripts, because of limited annotated data and poor visual-semantic alignment.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.5.82 Can Knowledge be Transferred from Unimodal to Multimodal? Investigating the Transitivity of Multimodal Knowledge Editing

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文: Can Knowledge be Transferred from Unimodal to Multimodal? Investigating the Transitivity of Multimodal Knowledge Editing

快速理解: Multimodal Large Language Models (MLLMs) contain a substantial amount of factual knowledge, which may become outdated or inaccurate over time. Consequently, various knowledge editing techniques have been proposed to update the knowledge encoded within these models.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.5.83 CapeLLM: Support-Free Category-Agnostic Pose Estimation with Multimodal Large Language Models

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文: CapeLLM: Support-Free Category-Agnostic Pose Estimation with Multimodal Large Language Models

快速理解: Category-agnostic pose estimation (CAPE) has traditionally relied on support images with annotated keypoints, a process that is often cumbersome and may fail to fully capture the necessary correspondences across diverse object categories. Recent efforts have explored the use of text queries, leveraging their enhanced stability and generalization capabilities.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.5.84 Causality-guided Prompt Learning for Vision-language Models via Visual Granulation

- 作者：作者信息未提供

- 年份: 2025; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Causality-guided Prompt Learning for Vision-language Models via Visual Granulation

快速理解: Prompt learning has recently attracted much attention for adapting pre-trained vision-language models (e.g., CLIP) to downstream recognition tasks. However, most of the existing CLIP-based prompt learning methods only show a limited ability for handling fine-grained datasets.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.5.85 ChatReID: Open-ended Interactive Person Retrieval via Hierarchical Progressive Tuning for Vision Language Models

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: ChatReID: Open-ended Interactive Person Retrieval via Hierarchical Progressive Tuning for Vision Language Models

快速理解: Person re-identification (Re-ID) is a fundamental task in computer vision that aims to match individuals across non-overlapping camera views. Although recent visionlanguage models (VLMs) have shown strong capabilities in logical reasoning and general-purpose learning, their performance in Re-ID remains suboptimal.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.5.86 CompCap: Improving Multimodal Large Language Models with Composite Captions

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: CompCap: Improving Multimodal Large Language Models with Composite Captions

快速理解: How well can Multimodal Large Language Models (MLLMs) understand composite images? Composite images (CIs) are synthetic visuals created by merging multiple visual elements, such as charts, posters, or screenshots, rather than being captured directly by a camera.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.5.87 Constraint-Aware Feature Learning for Parametric Point Cloud

- 作者: 作者信息未提供
- 年份: 2025; 主题: Systems; 优先级: 扩展
- [来源](#)

论文: Constraint-Aware Feature Learning for Parametric Point Cloud

快速理解: Parametric point clouds are sampled from CAD shapes and are becoming increasingly common in industrial manufacturing. Most CAD-specific deep learning methods focus on geometric features, while overlooking constraints inherent in CAD shapes.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.5.88 Controlling Multimodal LLMs via Reward-guided Decoding

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Controlling Multimodal LLMs via Reward-guided Decoding

快速理解: As Multimodal Large Language Models (MLLMs) gain widespread applicability, it is becoming increasingly desirable to adapt them for diverse user needs. In this paper, we study the adaptation of MLLMs through controlled decoding.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.5.89 Corvid: Improving Multimodal Large Language Models Towards Chain-of-Thought Reasoning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Corvid: Improving Multimodal Large Language Models Towards Chain-of-Thought Reasoning

快速理解: Recent advancements in multimodal large language models (MLLMs) have demonstrated exceptional performance in multimodal perception and understanding. However, leading open-source MLLMs exhibit significant limitations in complex and structured reasoning, particularly in tasks requiring deep reasoning for decision-making and problem-solving.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.5.90 CounterPC: Counterfactual Feature Realignment for Unsupervised Domain Adaptation on Point Clouds

- 作者: 作者信息未提供
- 年份: 2025; 主题: Systems; 优先级: 扩展
- [来源](#)

论文: CounterPC: Counterfactual Feature Realignment for Unsupervised Domain Adaptation on Point Clouds

快速理解: Understanding real-world 3D point clouds is challenging due to domain shifts. The key challenge is disentangling domain-invariant semantics from domain-specific geometric variations, as point clouds exhibit local inconsistency and global redundancy, making direct alignment ineffective.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.5.91 CreatiLayout: Siamese Multimodal Diffusion Transformer for Creative Layout-to-Image Generation

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：CreatiLayout: Siamese Multimodal Diffusion Transformer for Creative Layout-to-Image Generation

快速理解：Diffusion models have been recognized for their ability to generate images that are not only visually appealing but also $\dagger$ Corresponding author. of high artistic quality. As a result, Layout-to-Image (L2I) generation has been proposed to leverage region-specific positions and descriptions to enable more precise and controllable generation.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.92 D3QE: Learning Discrete Distribution Discrepancy-aware Quantization Error for Autoregressive-Generated Image Detection

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra；优先级：扩展
- [来源](#)

论文：D3QE: Learning Discrete Distribution Discrepancy-aware Quantization Error for Autoregressive-Generated Image Detection

快速理解：The emergence of visual autoregressive (AR) models has revolutionized image generation while presenting new challenges for synthetic image detection. Unlike previous GAN or diffusion-based methods, AR models generate images through discrete token prediction, exhibiting both marked improvements in image synthesis quality and unique characteristics in their vector-quantized representations.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.5.93 DAP-MAE: Domain-Adaptive Point Cloud Masked Autoencoder for Effective Cross-Domain Learning

- 作者：作者信息未提供
- 年份：2025；主题：Systems；优先级：扩展
- [来源](#)

论文：DAP-MAE: Domain-Adaptive Point Cloud Masked Autoencoder for Effective Cross-Domain Learning

快速理解：Compared to 2D data, the scale of point cloud data in different domains available for training, is quite limited. Researchers have been trying to combine these data of different domains

for masked autoencoder (MAE) pre-training to leverage such a data scarcity issue.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.5.94 DH-FaceVid-1K: A Large-Scale High-Quality Dataset for Face Video Generation

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：DH-FaceVid-1K: A Large-Scale High-Quality Dataset for Face Video Generation

快速理解：Human-centric generative models are becoming increasingly popular, giving rise to various innovative tools and applications, such as talking face videos conditioned on text or audio prompts. The core of these capabilities lies in powerful pre-trained foundation models, trained on large-scale, high-quality datasets.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.95 DLFR-Gen: Diffusion-based Video Generation with Dynamic Latent Frame Rate

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：DLFR-Gen: Diffusion-based Video Generation with Dynamic Latent Frame Rate

快速理解：Diffusion Transformer(DiT)-based generation models have achieved remarkable success in video generation. However, their inherent computational demands pose significant efficiency challenges.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.96 DMQ: Dissecting Outliers of Diffusion Models for Post-Training Quantization

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra；优先级：扩展
- [来源](#)

论文：DMQ: Dissecting Outliers of Diffusion Models for Post-Training Quantization

快速理解：Diffusion models have achieved remarkable success in image generation but come with significant computational costs, posing challenges for deployment in resourceconstrained environments. Recent post-training quantization (PTQ) methods have attempted to mitigate this issue by focusing on the iterative nature of diffusion models.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.5.97 DOLLAR: Few-Step Video Generation via Distillation and Latent Reward Optimization

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：DOLLAR: Few-Step Video Generation via Distillation and Latent Reward Optimization

快速理解：Diffusion probabilistic models have shown significant progress in video generation; however, their computational efficiency is limited by the large number of sampling steps required. Reducing sampling steps often compromises video quality or generation diversity.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.98 DPoser-X: Diffusion Model as Robust 3D Whole-body Human Pose Prior

- 作者：作者信息未提供
- 年份：2025；主题：Embodied/WBC；优先级：扩展
- [来源](#)

论文：DPoser-X: Diffusion Model as Robust 3D Whole-body Human Pose Prior

快速理解：We present DPoser-X, a diffusion-based prior model for 3D whole-body human poses. Building a versatile and robust full-body human pose prior remains challenging due to the inherent complexity of articulated human poses and the scarcity of high-quality whole-body pose datasets.

关注点：具身任务中的感知、动作生成、全身协调或运动控制。

证据层级：本地 PDF 前页。

1.3.5.99 Dataset Distillation via Vision-Language Category Prototype

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Dataset Distillation via Vision-Language Category Prototype

快速理解：Dataset distillation (DD) condenses large datasets into compact yet informative substitutes, preserving performance comparable to the original dataset while reducing storage, transmission costs, and computational consumption. However, previous DD methods mainly focus on distilling information from images, often overlooking the semantic information inherent in the data.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.100 DeGauss: Dynamic-Static Decomposition with Gaussian Splatting for Distractor-free 3D Reconstruction

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文: DeGauss: Dynamic-Static Decomposition with Gaussian Splatting for Distractor-free 3D Reconstruction

快速理解: Reconstructing clean, distractor-free 3D scenes from realworld captures remains a significant challenge, particularly in highly dynamic and cluttered settings such as egocentric videos. To tackle this problem, we introduce DeGauss, a simple and robust self-supervised framework for dynamic scene reconstruction based on a decoupled dynamic-static Gaussian Splatting design.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.5.101 DeSPITE: Exploring Contrastive Deep Skeleton-Pointcloud-IMU-Text Embeddings for Advanced Point Cloud Human Activity Understanding

- 作者: 作者信息未提供
- 年份: 2025; 主题: Systems; 优先级: 扩展
- [来源](#)

论文: DeSPITE: Exploring Contrastive Deep Skeleton-Pointcloud-IMU-Text Embeddings for Advanced Point Cloud Human Activity Understanding

快速理解: Despite LiDAR (Light Detection and Ranging) being an effective privacy-preserving alternative to RGB cameras to perceive human activities, it remains largely underexplored in the context of multi-modal contrastive pre-training for human activity understanding tasks, such as human activity recognition (HAR), retrieval, or person re-identification (RE-ID). To close this gap, our work explores learning the correspondence between LiDAR point clouds, human skeleton poses, IMU data, and text in a joint embedding space.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.5.102 Deciphering Cross-Modal Alignment in Large Vision-Language Models via Modality Integration Rate

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Deciphering Cross-Modal Alignment in Large Vision-Language Models via Modality Integration Rate

快速理解: The early stage of multi-modal pre-training plays a pivotal role in aligning two modalities for Large Vision- Language Models (LVLMs), while evaluating its training quality usually requires the costly supervised fine-tuning (SFT) stage to verify the downstream benchmark scores. Loss, perplexity, and in-context evaluation results are commonly used pre-training metrics for Large Language Models (LLMs), while we observed that these metrics are less indicative when quantifying the pre-trained LVLMs.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.5.103 DeepMesh: Auto-Regressive Artist-mesh Creation with Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：扩展
- [来源](#)

论文：DeepMesh: Auto-Regressive Artist-mesh Creation with Reinforcement Learning

快速理解：Triangle meshes play a crucial role in 3D applications for efficient manipulation and rendering. While autoregressive methods generate structured meshes by predicting discrete vertex tokens, they are often constrained by limited face counts and mesh incompleteness.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.5.104 Democratizing High-Fidelity Co-Speech Gesture Video Generation

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Democratizing High-Fidelity Co-Speech Gesture Video Generation

快速理解：Co-speech gesture video generation aims to synthesize realistic, audio-aligned videos of speakers, complete with synchronized facial expressions and body gestures. This task presents challenges due to the significant one-to-many mapping between audio and visual content, further complicated by the scarcity of large-scale public datasets and high computational demands.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.105 Depth Any Event Stream: Enhancing Event-based Monocular Depth Estimation via Dense-to-Sparse Distillation

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Depth Any Event Stream: Enhancing Event-based Monocular Depth Estimation via Dense-to-Sparse Distillation

快速理解：With the superior sensitivity of event cameras to high-speed motion and extreme lighting conditions, event-based monocular depth estimation has gained popularity to predict structural information about surrounding scenes in challenging environments. However, the scarcity of labeled event data constrains prior supervised learning methods.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.106 Depth AnyEvent: A Cross-Modal Distillation Paradigm for Event-Based Monocular Depth Estimation

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Depth AnyEvent: A Cross-Modal Distillation Paradigm for Event-Based Monocular Depth Estimation

快速理解：Event cameras capture sparse, high-temporal-resolution visual information, making them particularly suitable for challenging environments with high-speed motion and strongly varying lighting conditions. However, the lack of large datasets with dense ground-truth depth annotations hinders learning-based monocular depth estimation from event data.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.107 Derm1M: A Million-scale Vision-Language Dataset Aligned with Clinical Ontology Knowledge for Dermatology

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Derm1M: A Million-scale Vision-Language Dataset Aligned with Clinical Ontology Knowledge for Dermatology

快速理解：The emergence of vision-language models has transformed medical AI, enabling unprecedented advances in diagnostic capability and clinical applications. However, progress in dermatology has lagged behind other medical domains due to the lack of standard image-text pairs.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.108 DiffPCI: Large Motion Point Cloud frame Interpolation with Diffusion Model

- 作者：作者信息未提供
- 年份：2025；主题：Systems；优先级：扩展
- [来源](#)

论文：DiffPCI: Large Motion Point Cloud frame Interpolation with Diffusion Model

快速理解：Point cloud interpolation aims to recover intermediate frames for temporally smoothing a point cloud sequence. However, real-world challenges, such as uneven or large scene motions, cause existing methods to struggle with limited interpolation precision.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.5.109 DiffRefine: Diffusion-based Proposal Specific Point Cloud Densification for Cross-Domain Object Detection

- 作者：作者信息未提供
- 年份：2025；主题：Systems；优先级：扩展

- [来源](#)

论文: DiffRefine: Diffusion-based Proposal Specific Point Cloud Densification for Cross-Domain Object Detection

快速理解: The robustness of 3D object detection in large-scale outdoor point clouds degrades significantly when deployed in an unseen environment due to domain shifts. To minimize the domain gap, existing works on domain adaptive detection focuses on several factors, including point density, object shape and sizes, to reduce the false negative detections.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.5.110 Differential-informed Sample Selection Accelerates Multimodal Contrastive Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Differential-informed Sample Selection Accelerates Multimodal Contrastive Learning

快速理解: The remarkable success of contrastive-learning-based multimodal models has been greatly driven by training on ever-larger datasets with expensive compute consumption. Sample selection as an alternative efficient paradigm plays an important direction to accelerate the training process.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.5.111 Diffusion Guided Adaptive Augmentation for Generalization in Visual Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 扩展
- [来源](#)

论文: Diffusion Guided Adaptive Augmentation for Generalization in Visual Reinforcement Learning

快速理解: Reinforcement learning (RL) has proven its potential in complex decision-making tasks. Yet, many RL systems rely on manually crafted state representations, requiring effort in feature engineering.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.5.112 Does Your Vision-Language Model Get Lost in the Long Video Sampling Dilemma?

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Does Your Vision-Language Model Get Lost in the Long Video Sampling Dilemma?

快速理解: The rise of Large Vision-Language Models (LVLMs) has significantly advanced video understanding. However, efficiently processing long videos remains a challenge due to the “Sampling Dilemma”: low-density sampling risks missing critical information, while high-density sampling introduces redundancy.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.5.113 DrivingGPT: Unifying Driving World Modeling and Planning with Multi-modal Autoregressive Transformers

- 作者: 作者信息未提供
- 年份: 2025; 主题: World Model; 优先级: 扩展
- [来源](#)

论文: DrivingGPT: Unifying Driving World Modeling and Planning with Multi-modal Autoregressive Transformers

快速理解: World model-based searching and planning are widely recognized as a promising path toward human-level physical intelligence. However, current driving world models primarily rely on video diffusion models, which specialize in visual generation but lack the flexibility to incorporate other modalities like action.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策。

证据层级: 本地 PDF 前页。

1.3.5.114 DropletVideo: A Dataset and Approach to Explore Integral Spatio-Temporal Consistent Video Generation

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: DropletVideo: A Dataset and Approach to Explore Integral Spatio-Temporal Consistent Video Generation

快速理解: Spatio-temporal consistency is a critical topic in video generation. A qualified generated video segment must ensure plot plausibility and coherence while maintaining visual consistency of objects and scenes across varying viewpoints.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.5.115 Dual-Expert Consistency Model for Efficient and High-Quality Video Generation

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Dual-Expert Consistency Model for Efficient and High-Quality Video Generation

快速理解: Diffusion Models have achieved remarkable results in video synthesis but require iterative

denoising steps, leading to substantial computational overhead. Consistency Models have made significant progress in accelerating diffusion models.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.116 Dynamic Multimodal Prototype Learning in Vision-Language Models

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Dynamic Multimodal Prototype Learning in Vision-Language Models

快速理解：With the increasing attention to pre-trained visionlanguage models (VLMs), e.g., CLIP, substantial efforts have been devoted to many downstream tasks, especially in testtime adaptation (TTA). However, previous works focus on learning prototypes only in the textual modality while overlooking the ambiguous semantics in class names.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.117 Dynamic Point Maps: A Versatile Representation for Dynamic 3D Reconstruction

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Dynamic Point Maps: A Versatile Representation for Dynamic 3D Reconstruction

快速理解：DUST3R has recently demonstrated that many tasks in multiview geometry, including estimating camera intrinsics and extrinsics, reconstructing 3D scenes, and establishing image correspondences, can be reduced to predicting a pair of viewpoint-invariant point maps, i.e., pixel-aligned point clouds defined in a common reference frame. While this formulation is elegant and powerful, it is limited to static scenes.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.118 EC-Flow: Enabling Versatile Robotic Manipulation from Action-Unlabeled Videos via Embodiment-Centric Flow

- 作者：作者信息未提供
- 年份：2025；主题：Embodied/WBC；优先级：扩展
- [来源](#)

论文：EC-Flow: Enabling Versatile Robotic Manipulation from Action-Unlabeled Videos via Embodiment-Centric Flow

快速理解：Current language-guided robotic manipulation systems often require low-level action-labeled datasets for imitation learning. While object-centric flow prediction methods mitigate this issue, they remain limited to scenarios involving rigid objects with clear displacement and minimal occlusion.

关注点：具身任务中的感知、动作生成、全身协调或运动控制。

证据层级：本地 PDF 前页。

1.3.5.119 EVEv2: Improved Baselines for Encoder-Free Vision-Language Models

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：EVEv2: Improved Baselines for Encoder-Free Vision-Language Models

快速理解：Existing encoder-free vision-language models (VLMs) are rapidly narrowing the performance gap with their encoderbased counterparts, highlighting the promising potential for unified multimodal systems with structural simplicity and efficient deployment. We systematically clarify the performance gap between VLMs using pre-trained vision encoders, discrete tokenizers, and minimalist visual layers from scratch, deeply excavating the under-examined characteristics of encoder-free VLMs.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.120 Efficient Spiking Point Mamba for Point Cloud Analysis

- 作者：作者信息未提供
- 年份：2025；主题：Systems；优先级：扩展
- [来源](#)

论文：Efficient Spiking Point Mamba for Point Cloud Analysis

快速理解：Bio-inspired Spiking Neural Networks (SNNs) provide an energy-efficient way to extract 3D spatio-temporal features. However, existing 3D SNNs have struggled with long-range dependencies until the recent emergence of Mamba, which offers superior computational efficiency and sequence modeling capability.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.5.121 Efficient Visual Place Recognition Through Multimodal Semantic Knowledge Integration

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Efficient Visual Place Recognition Through Multimodal Semantic Knowledge Integration

快速理解：Visual place recognition is crucial for autonomous navigation and robotic mapping. Current methods struggle with perceptual aliasing and computational inefficiency.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.122 EgoM2P: Egocentric Multimodal Multitask Pretraining

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：EgoM2P: Egocentric Multimodal Multitask Pretraining

快速理解：Understanding multimodal signals in egocentric vision, such as RGB video, depth, camera poses, and gaze, is essential for applications in augmented reality, robotics, and human-computer interaction, enabling systems to better interpret the camera wearer' s actions, intentions, and surrounding environment. However, building large-scale egocentric multimodal and multitask models presents unique challenges.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.123 Embodied Image Captioning: Self-supervised Learning Agents for Spatially Coherent Image Descriptions

- 作者：作者信息未提供
- 年份：2025；主题：Embodied/WBC；优先级：扩展
- [来源](#)

论文：Embodied Image Captioning: Self-supervised Learning Agents for Spatially Coherent Image Descriptions

快速理解：We present a self-supervised method to improve an agent' s abilities in describing arbitrary objects while actively exploring a generic environment. This is a challenging problem, as current models struggle to obtain coherent image captions due to different camera viewpoints and clutter.

关注点：具身任务中的感知、动作生成、全身协调或运动控制。

证据层级：本地 PDF 前页。

1.3.5.124 Embodied Navigation with Auxiliary Task of Action Description Prediction

- 作者：作者信息未提供
- 年份：2025；主题：Embodied/WBC；优先级：扩展
- [来源](#)

论文：Embodied Navigation with Auxiliary Task of Action Description Prediction

快速理解：The field of multimodal robot navigation in indoor environments has garnered significant attention in recent years. However, as tasks and methods become more advanced, the action decision systems tend to become more complex and operate as black-boxes.

关注点：具身任务中的感知、动作生成、全身协调或运动控制。

证据层级：本地 PDF 前页。

1.3.5.125 Embodied Representation Alignment with Mirror Neurons

- 作者：作者信息未提供
- 年份：2025；主题：Embodied/WBC；优先级：扩展
- [来源](#)

论文: Embodied Representation Alignment with Mirror Neurons

快速理解: Mirror neurons are a class of neurons that activate both when an individual observes an action and when they perform the same action. This mechanism reveals a fundamental interplay between action understanding and embodied execution, suggesting that these two abilities are inherently connected.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制。

证据层级: 本地 PDF 前页。

1.3.5.126 EmbodiedSplat: Personalized Real-to-Sim-to-Real Navigation with Gaussian Splats from a Mobile Device

- 作者: 作者信息未提供
- 年份: 2025; 主题: Robot Learning; 优先级: 扩展
- [来源](#)

论文: EmbodiedSplat: Personalized Real-to-Sim-to-Real Navigation with Gaussian Splats from a Mobile Device

快速理解: The field of Embodied AI predominantly relies on simulation for training and evaluation, often using either fully synthetic environments that lack photorealism or high-fidelity real-world reconstructions captured with expensive hardware. As a result, sim-to-real transfer remains a major challenge.

关注点: 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 本地 PDF 前页。

1.3.5.127 End-to-End Driving with Online Trajectory Evaluation via BEV World Model

- 作者: 作者信息未提供
- 年份: 2025; 主题: World Model; 优先级: 扩展
- [来源](#)

论文: End-to-End Driving with Online Trajectory Evaluation via BEV World Model

快速理解: End-to-end autonomous driving has achieved remarkable progress by integrating perception, prediction, and planning into a fully differentiable framework. Yet, to fully realize its potential, an effective online trajectory evaluation is indispensable to ensure safety.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策。

证据层级: 本地 PDF 前页。

1.3.5.128 Enhancing Few-Shot Vision-Language Classification with Large Multimodal Model Features

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Enhancing Few-Shot Vision-Language Classification with Large Multimodal Model Features

快速理解: Generative Large Multimodal Models (LMMs) like LLaVA and Qwen-VL excel at a wide variety of vision-language (VL) tasks. Despite strong performance, LMMs' generative outputs are not specialized for vision-language classification tasks (i.e., tasks with vision-language inputs and discrete labels) such as image classification and multiplechoice VQA.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.5.129 Enhancing Reward Models for High-quality Image Generation: Beyond Text-Image Alignment

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 扩展
- [来源](#)

论文: Enhancing Reward Models for High-quality Image Generation: Beyond Text-Image Alignment

快速理解: Contemporary image generation systems have achieved high fidelity and superior aesthetic quality beyond basic text-image alignment. However, existing evaluation frameworks have failed to evolve in parallel.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.5.130 Enhancing Spatial Reasoning in Multimodal Large Language Models through Reasoning-based Segmentation

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Enhancing Spatial Reasoning in Multimodal Large Language Models through Reasoning-based Segmentation

快速理解: Recent advances in point cloud perception have demonstrated remarkable progress in scene understanding through vision-language alignment leveraging large language models (LLMs). However, existing methods may still encounter challenges in handling complex instructions that require accurate spatial reasoning, even if the 3D point cloud data provides detailed spatial cues such as size and position for identifying the targets.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.5.131 Enrich and Detect: Video Temporal Grounding with Multimodal LLMs

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Enrich and Detect: Video Temporal Grounding with Multimodal LLMs

快速理解: We introduce ED-VTG, a method for fine-grained video temporal grounding utilizing multi-modal large language models. Our approach harnesses the capabilities of multimodal LLMs

to jointly process text and video, in order to effectively localize natural language queries in videos through a two-stage process.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.132 Entropy-Adaptive Diffusion Policy Optimization with Dynamic Step Alignment

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：扩展
- [来源](#)

论文：Entropy-Adaptive Diffusion Policy Optimization with Dynamic Step Alignment

快速理解：While fine-tuning diffusion models with reinforcement learning (RL) has demonstrated effectiveness in directly optimizing downstream objectives, existing RL frameworks are prone to overfitting the rewards, leading to outputs that deviate from the true data distribution and exhibit reduced diversity. To address this issue, we introduce entropy as a quantitative measure to enhance the exploratory capacity of diffusion models' denoising policies.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.5.133 Epona: Autoregressive Diffusion World Model for Autonomous Driving

- 作者：作者信息未提供
- 年份：2025；主题：World Model；优先级：扩展
- [来源](#)

论文：Epona: Autoregressive Diffusion World Model for Autonomous Driving

快速理解：Diffusion models have demonstrated exceptional visual quality in video generation, making them promising for autonomous driving world modeling. However, existing video diffusion-based world models struggle with flexible-length, long-horizon predictions and integrating trajectory planning.

关注点：环境动力学、视频预测、潜变量状态或模型式决策。

证据层级：本地 PDF 前页。

1.3.5.134 Equipping Vision Foundation Model with Mixture of Experts for Out-of-Distribution Detection

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra；优先级：扩展
- [来源](#)

论文：Equipping Vision Foundation Model with Mixture of Experts for Out-of-Distribution Detection

快速理解：Pre-trained vision foundation models have transformed many computer vision tasks. Despite their strong ability to learn discriminative and generalizable features crucial for out-of-distribution (OOD) detection, their impact on this task remains underexplored.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.5.135 Event-Driven Storytelling with Multiple Lifelike Humans in a 3D Scene

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Event-Driven Storytelling with Multiple Lifelike Humans in a 3D Scene

快速理解：In this work, we propose a framework that creates a lively virtual dynamic scene with contextual motions of multiple humans. Generating multi-human contextual motion requires holistic reasoning over dynamic relationships among human-human and human-scene interactions.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.136 EvolvingGrasp: Evolutionary Grasp Generation via Efficient Preference Alignment

- 作者：作者信息未提供
- 年份：2025；主题：Robot Learning；优先级：扩展
- [来源](#)

论文：EvolvingGrasp: Evolutionary Grasp Generation via Efficient Preference Alignment

快速理解：Dexterous robotic hands often struggle to generalize effectively in complex environments due to models trained on low-diversity data. However, the real world presents an inherently unbounded range of scenarios.

关注点：机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：本地 PDF 前页。

1.3.5.137 ExCap3D: Expressive 3D Scene Understanding via Object Captioning with Varying Detail

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：ExCap3D: Expressive 3D Scene Understanding via Object Captioning with Varying Detail

快速理解：Generating text descriptions of objects in 3D indoor scenes is an important building block of embodied understanding. Existing methods describe objects at a single level of detail and do not capture fine-grained details of the parts of objects.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.138 Explaining Human Preferences via Metrics for Structured 3D Reconstruction

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Explaining Human Preferences via Metrics for Structured 3D Reconstruction

快速理解：“What cannot be measured cannot be improved” while likely never uttered by Lord Kelvin, summarizes effectively the driving force behind this work. This paper presents a detailed discussion of automated metrics for evaluating structured 3D reconstructions.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.139 Exploiting Frequency Dynamics for Enhanced Multimodal Event-based Action Recognition

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Exploiting Frequency Dynamics for Enhanced Multimodal Event-based Action Recognition

快速理解：While event cameras excel in capturing microsecond temporal dynamics, they suffer from sparse spatial representations compared to traditional RGB data. Thus, multimodal event-based action recognition approaches aim to synergize complementary strengths by independently extracting and integrating paired RGB-Event features.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.140 ExploreGS: Explorable 3D Scene Reconstruction with Virtual Camera Samplings and Diffusion Priors

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：ExploreGS: Explorable 3D Scene Reconstruction with Virtual Camera Samplings and Diffusion Priors

快速理解：Recent advances in novel view synthesis (NVS) have enabled real-time rendering with 3D Gaussian Splatting (3DGS). However, existing methods struggle with artifacts and missing regions when rendering from viewpoints that deviate from the training trajectory, limiting seamless scene exploration.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.141 Exploring Multimodal Diffusion Transformers for Enhanced Prompt-based Image Editing

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展

- [来源](#)

论文: Exploring Multimodal Diffusion Transformers for Enhanced Prompt-based Image Editing

快速理解: Transformer-based diffusion models have recently superseded traditional U-Net architectures, with multimodal diffusion transformers (MM-DiT) emerging as the dominant approach in state-of-the-art models like Stable Diffusion 3 and Flux.1. Previous approaches have relied on unidirectional cross-attention mechanisms, with information flowing from text embeddings to image latents.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.5.142 FA: Forced Prompt Learning of Vision-Language Models for Out-of-Distribution Detection

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: FA: Forced Prompt Learning of Vision-Language Models for Out-of-Distribution Detection

快速理解: Pre-trained vision-language models (VLMs) have advanced out-of-distribution (OOD) detection recently. However, existing CLIP-based methods often focus on learning OOD-related knowledge to improve OOD detection, showing limited generalization or reliance on external largescale auxiliary datasets.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.5.143 FALCON: Resolving Visual Redundancy and Fragmentation in High-resolution Multimodal Large Language Models via Visual Registers

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: FALCON: Resolving Visual Redundancy and Fragmentation in High-resolution Multimodal Large Language Models via Visual Registers

快速理解: The incorporation of high-resolution visual input equips multimodal large language models (MLLMs) with enhanced visual perception capabilities for real-world tasks. However, most existing high-resolution MLLMs rely on a cropping-based approach to process images, which leads to fragmented visual encoding and a sharp increase in redundant tokens.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.5.144 FastVAR: Linear Visual Autoregressive Modeling via Cached Token Pruning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Systems; 优先级: 扩展
- [来源](#)

论文: FastVAR: Linear Visual Autoregressive Modeling via Cached Token Pruning

快速理解: Visual Autoregressive (VAR) modeling has gained popularity for its shift towards next-scale prediction. However, existing VAR paradigms process the entire token map at each scale step, leading to the complexity and runtime scaling dramatically with image resolution.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.5.145 Feather the Throttle: Revisiting Visual Token Pruning for Vision-Language Model Acceleration

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: Feather the Throttle: Revisiting Visual Token Pruning for Vision-Language Model Acceleration

快速理解: Recent works on accelerating Vision-Language Models achieve strong performance across a variety of visionlanguage tasks despite highly compressing visual information. In this work, we examine the popular acceleration approach of early pruning of visual tokens inside the language model.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.5.146 Feature Extraction and Representation of Pre-training Point Cloud Based on Diffusion Models

- 作者: 作者信息未提供
- 年份: 2025; 主题: Systems; 优先级: 扩展
- [来源](#)

论文: Feature Extraction and Representation of Pre-training Point Cloud Based on Diffusion Models

快速理解: The pretrain-finetune paradigm of pre-training a model on large amounts of image and text data and then fine-tuning the model for a specific task has led to significant progress in many 2D image and natural language processing tasks. Similarly, the use of pre-training methods in point cloud data can also enhance the working performance and generalization ability of the model.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.5.147 FedMVP: Federated Multimodal Visual Prompt Tuning for Vision-Language Models

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: FedMVP: Federated Multimodal Visual Prompt Tuning for Vision-Language Models

快速理解: In federated learning, textual prompt tuning adapts Vision-Language Models (e.g., CLIP)

by tuning lightweight input tokens (or prompts) on local client data, while keeping network weights frozen. After training, only the prompts are shared by the clients with the central server for aggregation.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.148 FedWSQ: Efficient Federated Learning with Weight Standardization and Distribution-Aware Non-Uniform Quantization

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra；优先级：扩展
- [来源](#)

论文：FedWSQ: Efficient Federated Learning with Weight Standardization and Distribution-Aware Non-Uniform Quantization

快速理解：Federated learning (FL) often suffers from performance degradation due to key challenges such as data heterogeneity and communication constraints. To address these limitations, we present a novel FL framework called FedWSQ, which integrates weight standardization (WS) and the proposed distribution-aware non-uniform quantization (DANUQ).

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.5.149 Federated Prompt-Tuning with Heterogeneous and Incomplete Multimodal Client Data

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Federated Prompt-Tuning with Heterogeneous and Incomplete Multimodal Client Data

快速理解：This paper introduces a generalized federated prompttuning framework for practical scenarios where local datasets are multi-modal and exhibit different distributional patterns of missing features at the input level. The proposed framework bridges the gap between federated learning and multi-modal prompt-tuning which have traditionally focused on either uni-modal or centralized data.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.150 Few-Shot Image Quality Assessment via Adaptation of Vision-Language Models

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Few-Shot Image Quality Assessment via Adaptation of Vision-Language Models

快速理解：Image Quality Assessment (IQA) remains an unresolved challenge in computer vision due

to complex distortions, diverse image content, and limited data availability. Existing Blind IQA (BIQA) methods largely rely on extensive human annotations, which are labor-intensive and costly due to the demanding nature of creating IQA datasets.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.151 FiffDepth: Feed-forward Transformation of Diffusion-Based Generators for Detailed Depth Estimation

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：FiffDepth: Feed-forward Transformation of Diffusion-Based Generators for Detailed Depth Estimation

快速理解：Monocular Depth Estimation (MDE) is a fundamental 3D vision problem with numerous applications such as 3D scene reconstruction, autonomous navigation, and AI content creation. However, robust and generalizable MDE remains challenging due to limited real-world labeled data and distribution gaps between synthetic datasets and real data.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.152 FinMMR: Make Financial Numerical Reasoning More Multimodal, Comprehensive, and Challenging

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：FinMMR: Make Financial Numerical Reasoning More Multimodal, Comprehensive, and Challenging

快速理解：We present FinMMR, a novel bilingual multimodal benchmark tailored to evaluate the reasoning capabilities of multimodal large language models (MLLMs) in financial numerical reasoning tasks. Compared to existing benchmarks, our work introduces three significant advancements. (1) Multimodality: We meticulously transform existing financial reasoning benchmarks, and construct novel questions from the latest Chinese financial research reports.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.153 Fine-Grained Evaluation of Large Vision-Language Models in Autonomous Driving

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Fine-Grained Evaluation of Large Vision-Language Models in Autonomous Driving

快速理解: Existing benchmarks for Vision-Language Model (VLM) in autonomous driving (AD) primarily assess interpretability through open-form visual question answering (QA) within coarse-grained tasks, which remain insufficient to assess capabilities in complex driving scenarios. To this end, we introduce VLADBench, a challenging and finegrained benchmark featuring close-form QAs that progress from static foundational knowledge and elements to advanced reasoning for dynamic on-road situations.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.5.154 FlashDepth: Real-time Streaming Video Depth Estimation at 2K Resolution

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: FlashDepth: Real-time Streaming Video Depth Estimation at 2K Resolution

快速理解: A versatile video depth estimation model should (1) be accurate and consistent across frames, (2) produce highresolution depth maps, and (3) support real-time streaming. We propose FlashDepth, a method that satisfies all three requirements, performing depth estimation on a 2044→1148 streaming video at 24 FPS.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.5.155 FlowR: Flowing from Sparse to Dense 3D Reconstructions

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: FlowR: Flowing from Sparse to Dense 3D Reconstructions

快速理解: 3D Gaussian splatting enables high-quality novel view synthesis (NVS) at real-time frame rates. However, its quality drops sharply as we depart from the training views.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.5.156 Forensic-MoE: Exploring Comprehensive Synthetic Image Detection Traces with Mixture of Experts

- 作者: 作者信息未提供
- 年份: 2025; 主题: AI Infra; 优先级: 扩展
- [来源](#)

论文: Forensic-MoE: Exploring Comprehensive Synthetic Image Detection Traces with Mixture of Experts

快速理解: Recently, synthetic images have evolved incredibly realistic with the development of generative techniques. To avoid the spread of misinformation and identify synthetic content, research on synthetic image detection becomes urgent.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.5.157 FrameFusion: Combining Similarity and Importance for Video Token Reduction on Large Vision Language Models

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：FrameFusion: Combining Similarity and Importance for Video Token Reduction on Large Vision Language Models

快速理解：The increasing demand to process long and high-resolution videos significantly burdens Large Vision-Language Models (LVLMs) due to the enormous number of visual tokens. Existing token reduction methods primarily prune tokens based on importance metrics, such as cumulative attention scores.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.158 FreeSplatter: Pose-free Gaussian Splatting for Sparse-view 3D Reconstruction

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：FreeSplatter: Pose-free Gaussian Splatting for Sparse-view 3D Reconstruction

快速理解：Sparse-view reconstruction models typically require precise camera poses, yet obtaining these parameters from sparse-view images remains challenging. We introduce FreeSplatter, a scalable feed-forward framework that generates high-quality 3D Gaussians from uncalibrated sparse-view images while estimating camera parameters within seconds.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.159 From Gaze to Movement: Predicting Visual Attention for Autonomous Driving Human-Machine Interaction based on Programmatic Imitation Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：扩展
- [来源](#)

论文：From Gaze to Movement: Predicting Visual Attention for Autonomous Driving Human-Machine Interaction based on Programmatic Imitation Learning

快速理解：Human-machine interaction technology requires not only the distribution of human visual attention but also the prediction of the gaze point trajectory. We introduce PILOT, a programmatic imitation learning approach that predicts a driver's eye movements based on a set of rule-based conditions.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.5.160 FullDiT: Video Generative Foundation Models with Multimodal Control via Full Attention

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：FullDiT: Video Generative Foundation Models with Multimodal Control via Full Attention

快速理解：Current video generative foundation models primarily focus on text-to-video tasks, providing limited control for fine-grained video content creation. Although adapterbased approaches (e.g., ControlNet) enable additional controls with minimal fine-tuning, they encounter challenges when integrating multiple conditions, including: branch conflicts between independently trained adapters, parameter redundancy leading to increased computational cost, and suboptimal performance compared to full fine-tuning.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.161 Fusion Meets Diverse Conditions: A High-diversity Benchmark and Baseline for UAV-based Multimodal Object Detection with Condition Cues

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Fusion Meets Diverse Conditions: A High-diversity Benchmark and Baseline for UAV-based Multimodal Object Detection with Condition Cues

快速理解：Unmanned aerial vehicles (UAV)-based object detection with visible (RGB) and infrared (IR) images facilitates robust around-the-clock detection, driven by advancements in deep learning techniques and the availability of highquality dataset. However, the existing dataset struggles to fully capture real-world complexity for limited imaging conditions.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.162 Fuzzy Contrastive Decoding to Alleviate Object Hallucination in Large Vision-Language Models

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Fuzzy Contrastive Decoding to Alleviate Object Hallucination in Large Vision-Language Models

快速理解：Large vision-language models (LVLMs) often exhibit object hallucination, a phenomenon where models generate descriptions of non-existent objects within images. Prior methods have sought to mitigate this issue by adjusting model logits to reduce linguistic bias, but they often lack precise

control over visual uncertainty, sometimes exacerbating hallucinations instead of mitigating them.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.163 G2D: Boosting Multimodal Learning with Gradient-Guided Distillation

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：G2D: Boosting Multimodal Learning with Gradient-Guided Distillation

快速理解：Multimodal learning aims to leverage information from diverse data modalities to achieve more comprehensive performance. However, conventional multimodal models often suffer from modality imbalance, where one or a few modalities dominate model optimization, leading to suboptimal feature representation and underutilization of weak modalities.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.164 G2SF: Geometry-Guided Score Fusion for Multimodal Industrial Anomaly Detection

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：G2SF: Geometry-Guided Score Fusion for Multimodal Industrial Anomaly Detection

快速理解：Industrial quality inspection plays a critical role in modern manufacturing by identifying defective products during production. While single-modality approaches using either 3D point clouds or 2D RGB images suffer from information incompleteness, multimodal anomaly detection offers promise through the complementary fusion of crossmodal data.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.165 GAP: Gaussianize Any Point Clouds with Text Guidance

- 作者：作者信息未提供
- 年份：2025；主题：Systems；优先级：扩展
- [来源](#)

论文：GAP: Gaussianize Any Point Clouds with Text Guidance

快速理解：3D Gaussian Splatting (3DGS) has demonstrated its advantages in achieving fast and high-quality rendering. As point clouds serve as a widely-used and easily accessible form of 3D representation, bridging the gap between point clouds and Gaussians becomes increasingly important.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.5.166 GEOBench-VLM: Benchmarking Vision-Language Models for Geospatial Tasks

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：GEOBench-VLM: Benchmarking Vision-Language Models for Geospatial Tasks

快速理解：While numerous recent benchmarks focus on evaluating generic Vision-Language Models (VLMs), they do not effectively address the specific challenges of geospatial applications. Generic VLM benchmarks are not designed to handle the complexities of geospatial data, an essential component for applications such as environmental monitoring, urban planning, and disaster management.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.167 GM-MoE: Low-Light Enhancement with Gated-Mechanism Mixture-of-Experts

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra；优先级：扩展
- [来源](#)

论文：GM-MoE: Low-Light Enhancement with Gated-Mechanism Mixture-of-Experts

快速理解：Low-light enhancement has wide applications in autonomous driving, 3D reconstruction, remote sensing, surveillance, and so on, which can significantly improve information utilization. However, most existing methods lack generalization and are limited to specific tasks such as image recovery.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.5.168 GRAB: A Challenging GRaph Analysis Benchmark for Large Multimodal Models

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：GRAB: A Challenging GRaph Analysis Benchmark for Large Multimodal Models

快速理解：Large multimodal models (LMMs) have exhibited proficiencies across many visual tasks. Although numerous wellknown benchmarks exist to evaluate model performance, they increasingly have insufficient headroom.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.169 GReg: Geometry-Aware Region Refinement for Sign Language Video Generation

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：GReg: Geometry-Aware Region Refinement for Sign Language Video Generation

快速理解：Sign Language Video Generation (SLVG) aims to transform sign language sequences into natural and fluent sign language videos. Existing SLVG methods lack geometric modeling of human anatomical structures, leading to anatomically implausible and temporally inconsistent generation.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.170 GSOT3D: Towards Generic 3D Single Object Tracking in the Wild

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：GSOT3D: Towards Generic 3D Single Object Tracking in the Wild

快速理解：In this paper, we present a novel benchmark, GSOT3D, that aims at facilitating development of generic 3D single object tracking (SOT) in the wild. Specifically, GSOT3D offers 620 sequences with 123K frames, and covers a wide selection of 54 object categories.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.171 Gaussian-based World Model: Gaussian Priors for Voxel-Based Occupancy Prediction and Future Motion Prediction

- 作者：作者信息未提供
- 年份：2025；主题：World Model；优先级：扩展
- [来源](#)

论文：Gaussian-based World Model: Gaussian Priors for Voxel-Based Occupancy Prediction and Future Motion Prediction

快速理解：In autonomous driving, accurately predicting occupancy and motion is crucial for safe navigation within dynamic environments. However, existing methods often suffer from difficulties in handling complex scenes and uncertainty arising from sensor data.

关注点：环境动力学、视频预测、潜变量状态或模型式决策。

证据层级：本地 PDF 前页。

1.3.5.172 GenFlow3D: Generative Scene Flow Estimation and Prediction on Point Cloud Sequences

- 作者：作者信息未提供
- 年份：2025；主题：Systems；优先级：扩展
- [来源](#)

论文：GenFlow3D: Generative Scene Flow Estimation and Prediction on Point Cloud Sequences

快速理解: Scene flow provides the fundamental information of the scene dynamics. Existing scene flow estimation methods typically rely on the correlation between only a consecutive point cloud pair, which makes them limited to the instantaneous state of the scene and face challenges in real-world scenarios with factors like occlusion, noise, and diverse motion of background and foreground.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.5.173 GenFlowRL: Shaping Rewards with Generative Object-Centric Flow in Visual Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 扩展
- [来源](#)

论文: GenFlowRL: Shaping Rewards with Generative Object-Centric Flow in Visual Reinforcement Learning

快速理解: Recent advances have shown that video generation models can enhance robot learning by deriving effective robot actions through inverse dynamics. However, these methods heavily depend on the quality of generated data and struggle with fine-grained manipulation due to the lack of environment feedback.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.5.174 General Compression Framework for Efficient Transformer Object Tracking

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 扩展
- [来源](#)

论文: General Compression Framework for Efficient Transformer Object Tracking

快速理解: Previous works have attempted to improve tracking efficiency through lightweight architecture design or knowledge distillation from teacher models to compact student trackers. However, these solutions often sacrifice accuracy for speed to a great extent, and also have the problems of complex training process and structural limitations.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.5.175 Generalized Few-Shot Point Cloud Segmentation via LLM-Assisted Hyper-Relation Matching

- 作者: 作者信息未提供
- 年份: 2025; 主题: Systems; 优先级: 扩展
- [来源](#)

论文: Generalized Few-Shot Point Cloud Segmentation via LLM-Assisted Hyper-Relation Matching

快速理解: Generalized few-shot point cloud segmentation (GFS- 3DSeg) aims to segment objects of both base and novel classes using abundant base class samples and limited novel class samples.

Existing GFS-3DSeg methods encounter bottlenecks due to the scarcity of novel class data and inter-class confusion.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.5.176 Generating, Fast and Slow: Scalable Parallel Video Generation with Video Interface Networks

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Generating, Fast and Slow: Scalable Parallel Video Generation with Video Interface Networks

快速理解：Diffusion Transformers (DiTs) can generate short photorealistic videos, yet directly training and sampling longer videos with full attention across the video remains computationally challenging. Alternative methods break long videos down into sequential generation of short video segments, requiring multiple sampling chain iterations and specialized consistency modules.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.177 Generative Gaussian Splatting: Generating 3D Scenes with Video Diffusion Priors

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：Generative Gaussian Splatting: Generating 3D Scenes with Video Diffusion Priors

快速理解：Synthesizing consistent and photorealistic 3D scenes is an open problem in computer vision. Video diffusion models generate impressive videos but cannot directly synthesize 3D representations, i.e., lack 3D consistency in the generated sequences.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.5.178 GenieBlue: Integrating both Linguistic and Multimodal Capabilities for Large Language Models on Mobile Devices

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：扩展
- [来源](#)

论文：GenieBlue: Integrating both Linguistic and Multimodal Capabilities for Large Language Models on Mobile Devices

快速理解：Recent advancements in Multimodal Large Language Models (MLLMs) have enabled their deployment on mobile devices. However, challenges persist in maintaining strong language capabilities and ensuring hardware compatibility, both of which are crucial for user experience and practical deployment efficiency.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.6 AI · ICML

本会议纳入 **486** 篇。自动主题统计为：Agent RL (231)、Supporting Perception (155)、AI Infra (116)、Systems (62)、World Model (36)、Embodied/WBC (33)、Robot Learning (7)、VLA (6)。这些统计用于组织阅读顺序，不能单独证明论文结论或实验优越性。

优先阅读候选：- Latent Diffusion Planning for Imitation Learning (优先级分数 116.25) - Video Prediction Policy: A Generalist Robot Policy with Predictive Visual Representations (优先级分数 116.25) - CommVQ: Commutative Vector Quantization for KV Cache Compression (优先级分数 89.25) - Empowering World Models with Reflection for Embodied Video Prediction (优先级分数 89.25) - MENTOR: Mixture-of-Experts Network with Task-Oriented Perturbation for Visual Reinforcement Learning (优先级分数 89.25)

1.3.6.1 Latent Diffusion Planning for Imitation Learning

- 作者：Xie, Amber; Rybkin, Oleh; Sadigh, Dorsa; Finn, Chelsea
- 年份：2025；主题：Embodied/WBC; World Model; Agent RL; Robot Learning; Supporting Perception; 优先级：核心
- [来源](#)

论文：Latent Diffusion Planning for Imitation Learning

快速理解：Recent progress in imitation learning has been enabled by policy architectures that scale to complex visuomotor tasks, multimodal distributions, and large datasets. However, these methods often rely on learning from large amount of expert demonstrations.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；环境动力学、视频预测、潜变量状态或模型式决策；强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.2 Video Prediction Policy: A Generalist Robot Policy with Predictive Visual Representations

- 作者：Hu, Yucheng; Guo, Yanjiang; Wang, Pengchao; Chen, Xiaoyu; Wang, Yen-Jen; Zhang, Jianke; Sreenath, Koushil; Lu, Chaochao; Chen, Jianyu
- 年份：2025；主题：Embodied/WBC; VLA; World Model; Robot Learning; 优先级：核心
- [来源](#)

论文：Video Prediction Policy: A Generalist Robot Policy with Predictive Visual Representations

快速理解：Visual representations play a crucial role in developing generalist robotic policies. Previous vision encoders, typically pre-trained with single-image reconstruction or two-image contrastive learning, tend to capture static information, often neglecting the dynamic aspects vital for embodied tasks.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；视觉、语言条件与机器人动作策略之间的对齐；环境动力学、视频预测、潜变量状态或模型式决策。

证据层级：索引摘要。

1.3.6.3 CommVQ: Commutative Vector Quantization for KV Cache Compression

- 作者: Li, Junyan; Zhang, Yang; Hassan, Muhammad Yusuf; Chafekar, Talha; Cai, Tianle; Ren, Zhile; Guo, Pengsheng; Karimzadeh, Foroozan; Reed, Colorado; Wang, Chong; Gan, Chuang
- 年份: 2025; 主题: Embodied/WBC; AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: CommVQ: Commutative Vector Quantization for KV Cache Compression

快速理解: Large Language Models (LLMs) are increasingly used in applications requiring long context lengths, but the key-value (KV) cache often becomes a memory bottleneck on GPUs as context grows. To address this, we propose Commutative Vector Quantization (CommVQ) to significantly reduce memory usage for long-context LLM inference.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.6.4 Empowering World Models with Reflection for Embodied Video Prediction

- 作者: Chi, Xiaowei; Fan, Chun-Kai; Zhang, Hengyuan; Qi, Xingqun; Zhang, Rongyu; Chen, Anthony; Chan, Chi-Min; Xue, Wei; Liu, Qifeng; Zhang, Shanghang; Guo, Yike
- 年份: 2025; 主题: Embodied/WBC; World Model; Supporting Perception; 优先级: 核心
- [来源](#)

论文: Empowering World Models with Reflection for Embodied Video Prediction

快速理解: Video generation models have made significant progress in simulating future states, showcasing their potential as world simulators in embodied scenarios. However, existing models often lack robust understanding, limiting their ability to perform multi-step predictions or handle Out-of-Distribution (OOD) scenarios.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 环境动力学、视频预测、潜变量状态或模型式决策; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.5 MENTOR: Mixture-of-Experts Network with Task-Oriented Perturbation for Visual Reinforcement Learning

- 作者: Huang, Suning; Zhang, Zheyu Aqa; Liang, Tianhai; Xu, Yihan; Kou, Zhehao; Lu, Chenhao; Xu, Guowei; Xue, Zhengrong; Xu, Huazhe
- 年份: 2025; 主题: Embodied/WBC; Agent RL; AI Infra; 优先级: 核心
- [来源](#)

论文: MENTOR: Mixture-of-Experts Network with Task-Oriented Perturbation for Visual Reinforcement Learning

快速理解: Visual deep reinforcement learning (RL) enables robots to acquire skills from visual input for unstructured tasks. However, current algorithms suffer from low sample efficiency, limiting their practical applicability.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 强化学习、模仿学习、规划、奖励或多智能体决策; 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.6.6 OTTER: A Vision-Language-Action Model with Text-Aware Visual Feature Extraction

- 作者: Huang, Huang; Liu, Fangchen; Fu, Letian; Wu, Tingfan; Mukadam, Mustafa; Malik, Jitendra; Goldberg, Ken; Abbeel, Pieter
- 年份: 2025; 主题: VLA; AI Infra; Supporting Perception; 优先级: 核心
- [来源](#)

论文: OTTER: A Vision-Language-Action Model with Text-Aware Visual Feature Extraction

快速理解: Vision-Language-Action (VLA) models aim to predict robotic actions based on visual observations and language instructions. Existing approaches require fine-tuning pre-trained vision-language models (VLMs) as visual and language features are independently fed into downstream policies, degrading the pre-trained semantic alignments.

关注点: 视觉、语言条件与机器人动作策略之间的对齐; 训练、推理、服务、编译、调度、通信或 AI 加速; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.7 UP-VLA: A Unified Understanding and Prediction Model for Embodied Agent

- 作者: Zhang, Jianke; Guo, Yanjiang; Hu, Yucheng; Chen, Xiaoyu; Zhu, Xiang; Chen, Jianyu
- 年份: 2025; 主题: Embodied/WBC; VLA; Supporting Perception; 优先级: 核心
- [来源](#)

论文: UP-VLA: A Unified Understanding and Prediction Model for Embodied Agent

快速理解: Recent advancements in Vision-Language-Action (VLA) models have leveraged pre-trained Vision-Language Models (VLMs) to improve the generalization capabilities. VLMs, typically pre-trained on vision-language understanding tasks, provide rich semantic knowledge and reasoning abilities.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 视觉、语言条件与机器人动作策略之间的对齐; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.8 DiffusionVLA: Scaling Robot Foundation Models via Unified Diffusion and Autoregression

- 作者: Wen, Junjie; Zhu, Yichen; Zhu, Minjie; Tang, Zhibin; Li, Jinming; Zhou, Zhongyi; Liu, Xiaoyu; Shen, Chaomin; Peng, Yaxin; Feng, Feifei
- 年份: 2025; 主题: VLA; AI Infra; Supporting Perception; 优先级: 核心
- [来源](#)

论文: DiffusionVLA: Scaling Robot Foundation Models via Unified Diffusion and Autoregression

快速理解: In this paper, we present DiffusionVLA, a novel framework that integrates autoregressive reasoning with diffusion policies to address the limitations of existing methods: while autoregressive Vision-Language-Action (VLA) models lack precise and robust action generation, diffusion-based policies inherently lack reasoning capabilities. Central to our approach is autoregressive reasoning—a task decomposition and explanation process enabled by a pre-trained VLM—to guide diffusion-based action policies.

关注点：视觉、语言条件与机器人动作策略之间的对齐；训练、推理、服务、编译、调度、通信或 AI 加速；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.6.9 Goal-Oriented Skill Abstraction for Offline Multi-Task Reinforcement Learning

- 作者：He, Jinmin; Li, Kai; Zang, Yifan; Fu, Haobo; Fu, Qiang; Xing, Junliang; Cheng, Jian
- 年份：2025；主题：Embodied/WBC; Agent RL; AI Infra; 优先级：核心
- [来源](#)

论文：Goal-Oriented Skill Abstraction for Offline Multi-Task Reinforcement Learning

快速理解：Offline multi-task reinforcement learning aims to learn a unified policy capable of solving multiple tasks using only pre-collected task-mixed datasets, without requiring any online interaction with the environment. However, it faces significant challenges in effectively sharing knowledge across tasks.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；强化学习、模仿学习、规划、奖励或多智能体决策；训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.6.10 Multi-Stage Manipulation with Demonstration-Augmented Reward, Policy, and World Model Learning

- 作者：Escoriza, Adri{a} Lopez; Hansen, Nicklas; Tao, Stone; Mu, Tongzhou; Su, Hao
- 年份：2025；主题：Embodied/WBC; World Model; Agent RL; 优先级：核心
- [来源](#)

论文：Multi-Stage Manipulation with Demonstration-Augmented Reward, Policy, and World Model Learning

快速理解：Long-horizon tasks in robotic manipulation present significant challenges in reinforcement learning (RL) due to the difficulty of designing dense reward functions and effectively exploring the expansive state-action space. However, despite a lack of dense rewards, these tasks often have a multi-stage structure, which can be leveraged to decompose the overall objective into manageable sub-goals.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；环境动力学、视频预测、潜变量状态或模型式决策；强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.11 Q-VDiT: Towards Accurate Quantization and Distillation of Video-Generation Diffusion Transformers

- 作者：Feng, Weilun; Yang, Chuanguang; Qin, Haotong; Li, Xiangqi; Wang, Yu; An, Zhulin; Huang, Libo; Diao, Boyu; Zhao, Zixiang; Xu, Yongjun; Magno, Michele
- 年份：2025；主题：AI Infra; Supporting Perception; Systems; 优先级：核心
- [来源](#)

论文：Q-VDiT: Towards Accurate Quantization and Distillation of Video-Generation Diffusion Trans-

formers

快速理解: Diffusion transformers (DiT) have demonstrated exceptional performance in video generation. However, their large number of parameters and high computational complexity limit their deployment on edge devices.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 3D、视频、多模态视觉或具身感知支撑能力; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.6.12 ReinboT: Amplifying Robot Visual-Language Manipulation with Reinforcement Learning

- 作者: Zhang, Hongyin; Zhuang, Zifeng; Zhao, Han; Ding, Pengxiang; Lu, Hongchao; Wang, Donglin
- 年份: 2025; 主题: VLA; Agent RL; Supporting Perception; 优先级: 核心
- [来源](#)

论文: ReinboT: Amplifying Robot Visual-Language Manipulation with Reinforcement Learning

快速理解: Vision-Language-Action (VLA) models have shown great potential in general robotic decision-making tasks via imitation learning. However, the variable quality of training data often constrains the performance of these models.

关注点: 视觉、语言条件与机器人动作策略之间的对齐; 强化学习、模仿学习、规划、奖励或多智能体决策; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.13 SENSEI: Semantic Exploration Guided by Foundation Models to Learn Versatile World Models

- 作者: Sancaktar, Cansu; Gumbsch, Christian; Zadaianchuk, Andrii; Kolev, Pavel; Martius, Georg
- 年份: 2025; 主题: World Model; Agent RL; Supporting Perception; 优先级: 核心
- [来源](#)

论文: SENSEI: Semantic Exploration Guided by Foundation Models to Learn Versatile World Models

快速理解: Exploration is a cornerstone of reinforcement learning (RL). Intrinsic motivation attempts to decouple exploration from external, task-based rewards.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策; 强化学习、模仿学习、规划、奖励或多智能体决策; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.14 SOLD: Slot Object-Centric Latent Dynamics Models for Relational Manipulation Learning from Pixels

- 作者: Mosbach, Malte; Ewertz, Jan Niklas; Villar-Corrales, Angel; Behnke, Sven
- 年份: 2025; 主题: Embodied/WBC; World Model; Agent RL; 优先级: 核心
- [来源](#)

论文: SOLD: Slot Object-Centric Latent Dynamics Models for Relational Manipulation Learning from Pixels

快速理解: Learning a latent dynamics model provides a task-agnostic representation of an agent's understanding of its environment. Leveraging this knowledge for model-based reinforcement learning (RL) holds the potential to improve sample efficiency over model-free methods by learning from imagined rollouts.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 环境动力学、视频预测、潜变量状态或模型式决策; 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.15 Scaling Laws for Pre-training Agents and World Models

- 作者: Pearce, Tim; Rashid, Tabish; Bignell, David; Georgescu, Raluca; Devlin, Sam; Hofmann, Katja
- 年份: 2025; 主题: Embodied/WBC; World Model; Agent RL; 优先级: 核心
- [来源](#)

论文: Scaling Laws for Pre-training Agents and World Models

快速理解: The performance of embodied agents has been shown to improve by increasing model parameters, dataset size, and compute. This has been demonstrated in domains from robotics to video games, when generative learning objectives on offline datasets (pre-training) are used to model an agent's behavior (imitation learning) or their environment (world modeling).

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 环境动力学、视频预测、潜变量状态或模型式决策; 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.16 Towards Efficient Online Tuning of VLM Agents via Counterfactual Soft Reinforcement Learning

- 作者: Feng, Lang; Tan, Weihao; Lyu, Zhiyi; Zheng, Longtao; Xu, Haiyang; Yan, Ming; Huang, Fei; An, Bo
- 年份: 2025; 主题: Embodied/WBC; Agent RL; Supporting Perception; 优先级: 核心
- [来源](#)

论文: Towards Efficient Online Tuning of VLM Agents via Counterfactual Soft Reinforcement Learning

快速理解: Online fine-tuning vision-language model (VLM) agents with reinforcement learning (RL) has shown promise for equipping agents with multi-step, goal-oriented capabilities in dynamic environments. However, their open-ended textual action space and non-end-to-end nature of action generation present significant challenges to effective online exploration in RL, e.g., explosion of the exploration space.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 强化学习、模仿学习、规划、奖励或多智能体决策; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.17 Video-Enhanced Offline Reinforcement Learning: A Model-Based Approach

- 作者: Pan, Minting; Zheng, Yitao; Li, Jiajian; Wang, Yunbo; Yang, Xiaokang
- 年份: 2025; 主题: Embodied/WBC; World Model; Agent RL; 优先级: 核心

- [来源](#)

论文: Video-Enhanced Offline Reinforcement Learning: A Model-Based Approach

快速理解: Offline reinforcement learning (RL) enables policy optimization using static datasets, avoiding the risks and costs of extensive real-world exploration. However, it struggles with suboptimal offline behaviors and inaccurate value estimation due to the lack of environmental interaction.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 环境动力学、视频预测、潜变量状态或模型式决策; 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.18 textttI²MoE: Interpretable Multimodal Interaction-aware Mixture-of-Experts

- 作者: Xin, Jiayi; Yun, Sukwon; Peng, Jie; Choi, Inyoung; Ballard, Jenna L.; Chen, Tianlong; Long, Qi
- 年份: 2025; 主题: AI Infra; Supporting Perception; 优先级: 核心
- [来源](#)

论文: textttI²MoE: Interpretable Multimodal Interaction-aware Mixture-of-Experts

快速理解: Modality fusion is a cornerstone of multimodal learning, enabling information integration from diverse data sources. However, existing approaches are limited by textbf{(a)} their focus on modality correspondences, which neglects heterogeneous interactions between modalities, and textbf{(b)} the fact that they output a single multimodal prediction without offering interpretable insights into the multimodal interactions present in the data.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.19 Cache Me If You Must: Adaptive Key-Value Quantization for Large Language Models

- 作者: Shutova, Alina; Malinovskii, Vladimir; Egiazarian, Vage; Kuznedelev, Denis; Mazur, Denis; Nikita, Surkov; Ermakov, Ivan; Alistarh, Dan
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: Cache Me If You Must: Adaptive Key-Value Quantization for Large Language Models

快速理解: Efficient real-world deployments of large language models (LLMs) rely on Key-Value (KV) caching for processing and generating long outputs, reducing the need for repetitive computation. For large contexts, Key-Value caches can take up tens of gigabytes of device memory, as they store vector representations for each token and layer.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.6.20 Continual Reinforcement Learning by Planning with Online World Models

- 作者: Liu, Zichen; Fu, Guoji; Du, Chao; Lee, Wee Sun; Lin, Min
- 年份: 2025; 主题: World Model; Agent RL; 优先级: 核心

- [来源](#)

论文: Continual Reinforcement Learning by Planning with Online World Models

快速理解: Continual reinforcement learning (CRL) refers to a naturalistic setting where an agent needs to endlessly evolve, by trial and error, to solve multiple tasks that are presented sequentially. One of the largest obstacles to CRL is that the agent may forget how to solve previous tasks when learning a new task, known as catastrophic forgetting.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策; 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.21 Enhancing Rating-Based Reinforcement Learning to Effectively Leverage Feedback from Large Vision-Language Models

- 作者: Luu, Tung Minh; Lee, Younghwan; Lee, Donghoon; Kim, Sunho; Kim, Min Jun; Yoo, Chang D.
- 年份: 2025; 主题: Agent RL; Supporting Perception; 优先级: 核心
- [来源](#)

论文: Enhancing Rating-Based Reinforcement Learning to Effectively Leverage Feedback from Large Vision-Language Models

快速理解: Designing effective reward functions remains a fundamental challenge in reinforcement learning (RL), as it often requires extensive human effort and domain expertise. While RL from human feedback has been successful in aligning agents with human intent, acquiring high-quality feedback is costly and labor-intensive, limiting its scalability.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.22 FOUNDER: Grounding Foundation Models in World Models for Open-Ended Embodied Decision Making

- 作者: Wang, Yucen; Yu, Rui; Wan, Shenghua; Gan, Le; Zhan, De-Chuan
- 年份: 2025; 主题: Embodied/WBC; World Model; 优先级: 核心
- [来源](#)

论文: FOUNDER: Grounding Foundation Models in World Models for Open-Ended Embodied Decision Making

快速理解: Foundation Models (FMs) and World Models (WMs) offer complementary strengths in task generalization at different levels. In this work, we propose FOUNDER, a framework that integrates the generalizable knowledge embedded in FMs with the dynamic modeling capabilities of WMs to enable open-ended task solving in embodied environments in a reward-free manner.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 环境动力学、视频预测、潜变量状态或模型式决策。

证据层级: 索引摘要。

1.3.6.23 GoIRL: Graph-Oriented Inverse Reinforcement Learning for Multimodal Trajectory Prediction

- 作者: Pei, Muleilan; Shi, Shaoshuai; Zhang, Lu; Li, Peiliang; Shen, Shaojie

- 年份: 2025; 主题: Agent RL; Supporting Perception; 优先级: 核心
- [来源](#)

论文: GoIRL: Graph-Oriented Inverse Reinforcement Learning for Multimodal Trajectory Prediction

快速理解: Trajectory prediction for surrounding agents is a challenging task in autonomous driving due to its inherent uncertainty and underlying multimodality. Unlike prevailing data-driven methods that primarily rely on supervised learning, in this paper, we introduce a novel Graph-oriented Inverse Reinforcement Learning (GoIRL) framework, which is an IRL-based predictor equipped with vectorized context representations.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.24 Hi Robot: Open-Ended Instruction Following with Hierarchical Vision-Language-Action Models

- 作者: Shi, Lucy Xiaoyang; Ichter, Brian; Equi, Michael Robert; Ke, Liyiming; Pertsch, Karl; Vuong, Quan; Tanner, James; Walling, Anna; Wang, Haohuan; Fusai, Niccolo; Li-Bell, Adrian; Driess, Danny; Groom, Lachy; Levine, Sergey; Finn, Chelsea
- 年份: 2025; 主题: VLA; Supporting Perception; 优先级: 核心
- [来源](#)

论文: Hi Robot: Open-Ended Instruction Following with Hierarchical Vision-Language-Action Models

快速理解: Generalist robots that can perform a range of different tasks in open-world settings must be able to not only reason about the steps needed to accomplish their goals, but also process complex instructions, prompts, and even feedback during task execution. Intricate instructions (e.g., “Could you make me a vegetarian sandwich?” or “I don’t like that one”) require not just the ability to physically perform the individual steps, but the ability to situate complex commands and feedback in the physical world.

关注点: 视觉、语言条件与机器人动作策略之间的对齐; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.25 How Far Is Video Generation from World Model: A Physical Law Perspective

- 作者: Kang, Bingyi; Yue, Yang; Lu, Rui; Lin, Zhijie; Zhao, Yang; Wang, Kaixin; Huang, Gao; Feng, Jiashi
- 年份: 2025; 主题: World Model; Supporting Perception; 优先级: 核心
- [来源](#)

论文: How Far Is Video Generation from World Model: A Physical Law Perspective

快速理解: Scaling video generation models is believed to be promising in building world models that adhere to fundamental physical laws. However, whether these models can discover physical laws purely from vision can be questioned.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.26 KVTuner: Sensitivity-Aware Layer-Wise Mixed-Precision KV Cache Quantization for Efficient and Nearly Lossless LLM Inference

- 作者: Li, Xing; Xing, Zeyu; Li, Yiming; Qu, Linping; Zhen, Hui-Ling; Yao, Yiwu; Liu, Wulong; Pan, Sinno Jialin; Yuan, Mingxuan
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: KVTuner: Sensitivity-Aware Layer-Wise Mixed-Precision KV Cache Quantization for Efficient and Nearly Lossless LLM Inference

快速理解: KV cache quantization can improve Large Language Models (LLMs) inference throughput and latency in long contexts and large batch-size scenarios while preserving LLMs effectiveness. However, current methods have three unsolved issues: overlooking layer-wise sensitivity to KV cache quantization, high overhead of online fine-grained decision-making, and low flexibility to different LLMs and constraints.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.6.27 LMAct: A Benchmark for In-Context Imitation Learning with Long Multi-modal Demonstrations

- 作者: Ruoss, Anian; Pardo, Fabio; Chan, Harris; Li, Bonnie; Mnih, Volodymyr; Genewein, Tim
- 年份: 2025; 主题: Agent RL; Supporting Perception; 优先级: 核心
- [来源](#)

论文: LMAct: A Benchmark for In-Context Imitation Learning with Long Multimodal Demonstrations

快速理解: In this paper, we present a benchmark to pressure-test today's frontier models' multi-modal decision-making capabilities in the very long-context regime (up to one million tokens) and investigate whether these models can learn from large numbers of expert demonstrations in their context. We evaluate the performance of Claude 3.5 Sonnet, Gemini 1.5 Flash, Gemini 1.5 Pro, Gemini 2.0 Flash Experimental, GPT-4o, o1-mini, o1-preview, and o1 as policies across a battery of simple interactive decision-making tasks: playing tic-tac-toe, chess, and Atari, navigating grid worlds, solving crosswords, and controlling a simulated cheetah.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.28 Novelty Detection in Reinforcement Learning with World Models

- 作者: Zollicoffer, Geigh; Eaton, Kenneth; Balloch, Jonathan C; Kim, Julia; Zhou, Wei; Wright, Robert; Riedl, Mark
- 年份: 2025; 主题: World Model; Agent RL; 优先级: 核心
- [来源](#)

论文: Novelty Detection in Reinforcement Learning with World Models

快速理解: Reinforcement learning (RL) using world models has found significant recent successes. However, when a sudden change to world mechanics or properties occurs then agent performance

and reliability can dramatically decline.

关注点：环境动力学、视频预测、潜变量状态或模型式决策；强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.29 PIGDreamer: Privileged Information Guided World Models for Safe Partially Observable Reinforcement Learning

- 作者：Huang, Dongchi; Wang, Jiaqi; Li, Yang; Xia, Chunhe; Zhang, Tianle; Zhang, Kaige
- 年份：2025；主题：World Model; Agent RL; 优先级：核心
- [来源](#)

论文：PIGDreamer: Privileged Information Guided World Models for Safe Partially Observable Reinforcement Learning

快速理解：Partial observability presents a significant challenge for Safe Reinforcement Learning (Safe RL), as it impedes the identification of potential risks and rewards. Leveraging specific types of privileged information during training to mitigate the effects of partial observability has yielded notable empirical successes.

关注点：环境动力学、视频预测、潜变量状态或模型式决策；强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.30 PISA Experiments: Exploring Physics Post-Training for Video Diffusion Models by Watching Stuff Drop

- 作者：Li, Chenyu; Michel, Oscar; Pan, Xichen; Liu, Sainan; Roberts, Mike; Xie, Saining
- 年份：2025；主题：World Model; Agent RL; Supporting Perception; 优先级：核心
- [来源](#)

论文：PISA Experiments: Exploring Physics Post-Training for Video Diffusion Models by Watching Stuff Drop

快速理解：Large-scale pre-trained video generation models excel in content creation but are not reliable as physically accurate world simulators out of the box. This work studies the process of post-training these models for accurate world modeling through the lens of the simple, yet fundamental, physics task of modeling object freefall.

关注点：环境动力学、视频预测、潜变量状态或模型式决策；强化学习、模仿学习、规划、奖励或多智能体决策；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.6.31 R2-T2: Re-Routing in Test-Time for Multimodal Mixture-of-Experts

- 作者：Li, Zhongyang; Li, Ziyue; Zhou, Tianyi
- 年份：2025；主题：AI Infra; Supporting Perception; 优先级：核心
- [来源](#)

论文：R2-T2: Re-Routing in Test-Time for Multimodal Mixture-of-Experts

快速理解：In large multimodal models (LMMs), the perception of non-language modalities (e.g., visual representations) is usually not on par with the large language models (LLMs)' powerful reasoning capabilities, deterring LMMs' performance on challenging downstream tasks. This weakness has been

recently mitigated by replacing the vision encoder with a mixture-of-experts (MoE), which provides rich, multi-granularity, and diverse representations required by different downstream tasks.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.6.32 R3DM: Enabling Role Discovery and Diversity Through Dynamics Models in Multi-agent Reinforcement Learning

- 作者：Goel, Harsh; Omama, Mohammad; Chalaki, Behdad; Tadiparthi, Vaishnav; Moradi Pari, Ehsan; Chinchali, Sandeep P.
- 年份：2025；主题：World Model; Agent RL; 优先级：核心
- [来源](#)

论文：R3DM: Enabling Role Discovery and Diversity Through Dynamics Models in Multi-agent Reinforcement Learning

快速理解：Multi-agent reinforcement learning (MARL) has achieved significant progress in large-scale traffic control, autonomous vehicles, and robotics. Drawing inspiration from biological systems where roles naturally emerge to enable coordination, role-based MARL methods have been proposed to enhance cooperation learning for complex tasks.

关注点：环境动力学、视频预测、潜变量状态或模型式决策；强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.33 Rethinking Latent Redundancy in Behavior Cloning: An Information Bottleneck Approach for Robot Manipulation

- 作者：Bai, Shuanghao; Zhou, Wanqi; Ding, Pengxiang; Zhao, Wei; Wang, Donglin; Chen, Badong
- 年份：2025；主题：Embodied/WBC; Agent RL; 优先级：核心
- [来源](#)

论文：Rethinking Latent Redundancy in Behavior Cloning: An Information Bottleneck Approach for Robot Manipulation

快速理解：Behavior Cloning (BC) is a widely adopted visual imitation learning method in robot manipulation. Current BC approaches often enhance generalization by leveraging large datasets and incorporating additional visual and textual modalities to capture more diverse information.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.34 Reward-free World Models for Online Imitation Learning

- 作者：Li, Shangzhe; Huang, Zhiao; Su, Hao
- 年份：2025；主题：World Model; Agent RL; 优先级：核心
- [来源](#)

论文：Reward-free World Models for Online Imitation Learning

快速理解：Imitation learning (IL) enables agents to acquire skills directly from expert demonstrations, providing a compelling alternative to reinforcement learning. However, prior online IL approaches struggle with complex tasks characterized by high-dimensional inputs and complex dynamics.

关注点：环境动力学、视频预测、潜变量状态或模型式决策；强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.35 RocketKV: Accelerating Long-Context LLM Inference via Two-Stage KV Cache Compression

- 作者: Behnam, Payman; Fu, Yaosheng; Zhao, Ritchie; Tsai, Po-An; Yu, Zhiding; Tumanov, Alexey
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: RocketKV: Accelerating Long-Context LLM Inference via Two-Stage KV Cache Compression

快速理解: Transformer-based Large Language Models rely critically on the KV cache to efficiently handle extended contexts during the decode phase. Yet, the size of the KV cache grows proportionally with the input length, burdening both memory bandwidth and capacity as decoding progresses.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.6.36 ShadowKV: KV Cache in Shadows for High-Throughput Long-Context LLM Inference

- 作者: Sun, Hanshi; Chang, Li-Wen; Bao, Wenlei; Zheng, Size; Zheng, Ningxin; Liu, Xin; Dong, Harry; Chi, Yuejie; Chen, Beidi
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: ShadowKV: KV Cache in Shadows for High-Throughput Long-Context LLM Inference

快速理解: With the widespread deployment of long-context large language models (LLMs), there has been a growing demand for efficient support of high-throughput inference. However, as the key-value (KV) cache expands with the sequence length, the increasing memory footprint and the need to access it for decoding both result in low throughput when serving long-context LLMs.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.6.37 Speeding up Policy Simulation in Supply Chain RL

- 作者: Farias, Vivek; Gijsbrechts, Joren; Khojandi, Aryan I.; Peng, Tianyi; Zheng, Andrew
- 年份: 2025; 主题: Agent RL; AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: Speeding up Policy Simulation in Supply Chain RL

快速理解: Simulating a single trajectory of a dynamical system under some state-dependent policy is a core bottleneck in policy optimization (PO) algorithms. The many inherently serial policy evaluations that must be performed in a single simulation constitute the bulk of this bottleneck.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.6.38 Test-Time Adaptation for Online Vision-Language Navigation with Feedback-based Reinforcement Learning

- 作者: Kim, Sungjune; Oh, Gyeongrok; Ko, Heeju; Ji, Daehyun; Lee, Dongwook; Lee, Byung-Jun; Jang, Sujin; Kim, Sangpil
- 年份: 2025; 主题: Agent RL; Supporting Perception; 优先级: 核心
- [来源](#)

论文: Test-Time Adaptation for Online Vision-Language Navigation with Feedback-based Reinforcement Learning

快速理解: Navigating in an unfamiliar environment during deployment poses a critical challenge for a vision-language navigation (VLN) agent. Yet, test-time adaptation (TTA) remains relatively underexplored in robotic navigation, leading us to the fundamental question: what are the key properties of TTA for online VLN?

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.39 The Devil Is in the Details: Tackling Unimodal Spurious Correlations for Generalizable Multimodal Reward Models

- 作者: Li, Zichao; Wen, Xueru; Lou, Jie; Ji, Yuqiu; Lu, Yaojie; Han, Xianpei; Zhang, Debing; Sun, Le
- 年份: 2025; 主题: Agent RL; Supporting Perception; 优先级: 核心
- [来源](#)

论文: The Devil Is in the Details: Tackling Unimodal Spurious Correlations for Generalizable Multimodal Reward Models

快速理解: Multimodal Reward Models (MM-RMs) are crucial for aligning Large Language Models (LLMs) with human preferences, particularly as LLMs increasingly interact with multimodal data. However, we find that MM-RMs trained on existing datasets often struggle to generalize to out-of-distribution data due to their reliance on unimodal spurious correlations, primarily text-only shortcuts within the training distribution, which prevents them from leveraging true multimodal reward functions.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.40 Windows Agent Arena: Evaluating Multi-Modal OS Agents at Scale

- 作者: Bonatti, Rogerio; Zhao, Dan; Bonacci, Francesco; Dupont, Dillon; Abdali, Sara; Li, Yinheng; Lu, Yadong; Wagle, Justin; Koishida, Kazuhito; Buckner, Arthur; Jang, Lawrence Keunho; Hui, Zheng
- 年份: 2025; 主题: Agent RL; Supporting Perception; Systems; 优先级: 核心
- [来源](#)

论文: Windows Agent Arena: Evaluating Multi-Modal OS Agents at Scale

快速理解: Large language models (LLMs) show potential as computer agents, enhancing productivity and software accessibility in multi-modal tasks. However, measuring agent performance in sufficiently

realistic and complex environments becomes increasingly challenging as: (i) most benchmarks are limited to specific modalities/domains (e.g., text-only, web navigation, Q&A) and (ii) full benchmark evaluations are slow (on order of magnitude of multiple hours/days) given the multi-step sequential nature of tasks.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；3D、视频、多模态视觉或具身感知支撑能力；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.6.41 World Model Implanting for Test-time Adaptation of Embodied Agents

- 作者：Yoo, Minjong; Jang, Jinwoo; Yoon, Sihyung; Woo, Honguk
- 年份：2025；主题：Embodied/WBC; World Model; 优先级：核心
- [来源](#)

论文：World Model Implanting for Test-time Adaptation of Embodied Agents

快速理解：In embodied AI, a persistent challenge is enabling agents to robustly adapt to novel domains without requiring extensive data collection or retraining. To address this, we present a world model implanting framework (WorMI) that combines the reasoning capabilities of large language models (LLMs) with independently learned, domain-specific world models through test-time composition.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；环境动力学、视频预测、潜变量状态或模型式决策。

证据层级：索引摘要。

1.3.6.42 3D Question Answering via only 2D Vision-Language Models

- 作者：Wang, Fengyun; Yu, Sicheng; Wu, Jiawei; Tang, Jinhui; Zhang, Hanwang; Sun, Qianru
- 年份：2025；主题：Supporting Perception; Systems; 优先级：重要
- [来源](#)

论文：3D Question Answering via only 2D Vision-Language Models

快速理解：Large vision-language models (LVLMs) have significantly advanced numerous fields. In this work, we explore how to harness their potential to address 3D scene understanding tasks, using 3D question answering (3D-QA) as a representative example.

关注点：3D、视频、多模态视觉或具身感知支撑能力；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.6.43 ABNet: Adaptive explicit-Barrier Net for Safe and Scalable Robot Learning

- 作者：Xiao, Wei; Wang, Tsun-Hsuan; Gan, Chuang; Rus, Daniela
- 年份：2025；主题：Embodied/WBC; Robot Learning; 优先级：重要
- [来源](#)

论文：ABNet: Adaptive explicit-Barrier Net for Safe and Scalable Robot Learning

快速理解：Safe learning is central to AI-enabled robots where a single failure may lead to catastrophic results. Existing safe learning methods are not scalable, inefficient and hard to train, and tend to generate unstable signals under noisy inputs that are challenging to be deployed for robots.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：索引摘要。

1.3.6.44 Agent Reviewers: Domain-specific Multimodal Agents with Shared Memory for Paper Review

- 作者：Lu, Kai; Xu, Shixiong; Li, Jinqiu; Ding, Kun; Meng, Gaofeng
- 年份：2025；主题：Agent RL; Supporting Perception；优先级：重要
- [来源](#)

论文：Agent Reviewers: Domain-specific Multimodal Agents with Shared Memory for Paper Review

快速理解：Feedback from peer review is essential to improve the quality of scientific articles. However, at present, many manuscripts do not receive sufficient external feedback for refinement before or during submission.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.6.45 Aggregation of Dependent Expert Distributions in Multimodal Variational Autoencoders

- 作者：A. Mancisidor, Rogelio; Jenssen, Robert; Yu, Shujian; Kampffmeyer, Michael
- 年份：2025；主题：AI Infra; Supporting Perception；优先级：重要
- [来源](#)

论文：Aggregation of Dependent Expert Distributions in Multimodal Variational Autoencoders

快速理解：Multimodal learning with variational autoencoders (VAEs) requires estimating joint distributions to evaluate the evidence lower bound (ELBO). Current methods, the product and mixture of experts, aggregate single-modality distributions assuming independence for simplicity, which is an overoptimistic assumption.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.6.46 Breaking the Curse of Multiagency in Robust Multi-Agent Reinforcement Learning

- 作者：Shi, Laixi; Gai, Jingchu; Mazumdar, Eric; Chi, Yuejie; Wierman, Adam
- 年份：2025；主题：Agent RL; Robot Learning；优先级：重要
- [来源](#)

论文：Breaking the Curse of Multiagency in Robust Multi-Agent Reinforcement Learning

快速理解：Standard multi-agent reinforcement learning (MARL) algorithms are vulnerable to sim-to-real gaps. To address this, distributionally robust Markov games (RMGs) have been proposed to enhance robustness in MARL by optimizing the worst-case performance when game dynamics shift within a prescribed uncertainty set.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：索引摘要。

1.3.6.47 Ca2-VDM: Efficient Autoregressive Video Diffusion Model with Causal Generation and Cache Sharing

- 作者: Gao, Kaifeng; Shi, Jiaxin; Zhang, Hanwang; Wang, Chunping; Xiao, Jun; Chen, Long
- 年份: 2025; 主题: Supporting Perception; Systems; 优先级: 重要
- [来源](#)

论文: Ca2-VDM: Efficient Autoregressive Video Diffusion Model with Causal Generation and Cache Sharing

快速理解: With the advance of diffusion models, today' s video generation has achieved impressive quality. To extend the generation length and facilitate real-world applications, a majority of video diffusion models (VDMs) generate videos in an autoregressive manner, i.e., generating subsequent clips conditioned on the last frame(s) of the previous clip.

关注点: 3D、视频、多模态视觉或具身感知支撑能力; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.6.48 Can Compressed LLMs Truly Act? An Empirical Evaluation of Agentic Capabilities in LLM Compression

- 作者: Dong, Peijie; Tang, Zhenheng; Liu, Xiang; Li, Lujun; Chu, Xiaowen; Li, Bo
- 年份: 2025; 主题: Agent RL; AI Infra; 优先级: 重要
- [来源](#)

论文: Can Compressed LLMs Truly Act? An Empirical Evaluation of Agentic Capabilities in LLM Compression

快速理解: Post-training compression reduces the computational and memory costs of large language models (LLMs), enabling resource-efficient deployment. However, existing compression benchmarks focus narrowly on language modeling (e.g., perplexity) and natural language understanding tasks (e.g., GLUE accuracy), ignoring the agentic capabilities—workflow, tool use/function call, long-context understanding and real-world application.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.6.49 CateKV: On Sequential Consistency for Long-Context LLM Inference Acceleration

- 作者: Jiang, Haoyun; Li, Haolin; Zhang, Jianwei; Huang, Fei; Hu, Qiang; Sun, Minmin; Xiao, Shuai; Li, Yong; Lin, Junyang; Yao, Jiangchao
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: CateKV: On Sequential Consistency for Long-Context LLM Inference Acceleration

快速理解: Large language models (LLMs) have demonstrated strong capabilities in handling long-context tasks, but processing such long contexts remains challenging due to the substantial memory requirements and inference latency. In this work, we discover that certain attention heads exhibit sequential consistency in their attention patterns, which can be persistently identified using a coefficient-of-variation-based algorithm.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.6.50 Compute or Load KV Cache? Why Not Both?

- 作者：Jin, Shuowei; Liu, Xueshen; Zhang, Qingzhao; Mao, Zhuoqing
- 年份：2025；主题：AI Infra; Systems；优先级：重要
- [来源](#)

论文：Compute or Load KV Cache? Why Not Both?

快速理解：Large Language Models (LLMs) are increasingly deployed in large-scale online services, enabling sophisticated applications. However, the computational overhead of generating key-value (KV) caches in the prefill stage presents a major bottleneck, particularly for long-context inputs.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.6.51 Cowpox: Towards the Immunity of VLM-based Multi-Agent Systems

- 作者：Wu, Yutong; Zhang, Jie; Li, Yiming; Zhang, Chao; Guo, Qing; Qiu, Han; Lukas, Nils; Zhang, Tianwei
- 年份：2025；主题：Agent RL; Supporting Perception；优先级：重要
- [来源](#)

论文：Cowpox: Towards the Immunity of VLM-based Multi-Agent Systems

快速理解：Vision Language Model (VLM) Agents are stateful, autonomous entities capable of perceiving and interacting with their environments through vision and language. Multi-agent systems comprise specialized agents who collaborate to solve a (complex) task.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.6.52 DINO-WM: World Models on Pre-trained Visual Features enable Zero-shot Planning

- 作者：Zhou, Gaoyue; Pan, Hengkai; Lecun, Yann; Pinto, Lerrel
- 年份：2025；主题：World Model; Agent RL；优先级：重要
- [来源](#)

论文：DINO-WM: World Models on Pre-trained Visual Features enable Zero-shot Planning

快速理解：The ability to predict future outcomes given control actions is fundamental for physical reasoning. However, such predictive models, often called world models, remain challenging to learn and are typically developed for task-specific solutions with online policy learning.

关注点：环境动力学、视频预测、潜变量状态或模型式决策；强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.53 Demystifying Cost-Efficiency in LLM Serving over Heterogeneous GPUs

- 作者: Jiang, Youhe; Fu, Fangcheng; Yao, Xiaozhe; He, Guoliang; Miao, Xupeng; Klimovic, Ana; Cui, Bin; Yuan, Binhang; Yoneki, Eiko
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Demystifying Cost-Efficiency in LLM Serving over Heterogeneous GPUs

快速理解: Recent advancements in Large Language Models (LLMs) have led to increasingly diverse requests, accompanied with varying resource (compute and memory) demands to serve them. However, this in turn degrades the cost-efficiency of LLM serving as common practices primarily rely on homogeneous GPU resources.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.6.54 Diving into Self-Evolving Training for Multimodal Reasoning

- 作者: Liu, Wei; Li, Junlong; Zhang, Xiwen; Zhou, Fan; Cheng, Yu; He, Junxian
- 年份: 2025; 主题: Agent RL; Supporting Perception; 优先级: 重要
- [来源](#)

论文: Diving into Self-Evolving Training for Multimodal Reasoning

快速理解: Self-evolving training—where models iteratively learn from their own outputs—has emerged as a key approach for complex reasoning tasks, addressing the scarcity of high-quality chain-of-thought data. However, its effectiveness in multimodal reasoning, a domain more intricate than text-only reasoning, remains underexplored, and the understanding of critical factors in this training paradigm remains limited.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.55 ELEMENTAL: Interactive Learning from Demonstrations and Vision-Language Models for Reward Design in Robotics

- 作者: Chen, Letian; Moorman, Nina Marie; Gombolay, Matthew Craig
- 年份: 2025; 主题: Agent RL; Supporting Perception; 优先级: 重要
- [来源](#)

论文: ELEMENTAL: Interactive Learning from Demonstrations and Vision-Language Models for Reward Design in Robotics

快速理解: Reinforcement learning (RL) has demonstrated compelling performance in robotic tasks, but its success often hinges on the design of complex, ad hoc reward functions. Researchers have explored how Large Language Models (LLMs) could enable non-expert users to specify reward functions more easily.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.56 Efficient Online Reinforcement Learning for Diffusion Policy

- 作者: Ma, Haitong; Chen, Tianyi; Wang, Kai; Li, Na; Dai, Bo

- 年份: 2025; 主题: Embodied/WBC; Agent RL; 优先级: 重要
- [来源](#)

论文: Efficient Online Reinforcement Learning for Diffusion Policy

快速理解: Diffusion policies have achieved superior performance in imitation learning and offline reinforcement learning (RL) due to their rich expressiveness. However, the conventional diffusion training procedure requires samples from target distribution, which is impossible in online RL since we cannot sample from the optimal policy.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.57 Efficiently Serving Large Multimodal Models Using EPD Disaggregation

- 作者: Singh, Gursimran; Wang, Xinglu; Hu, Yifan; Yu, Timothy Tin Long; Xing, Linzi; Jiang, Wei; Wang, Zhefeng; Xiaolong, Bai; Li, Yi; Xiong, Ying; Zhang, Yong; Fan, Zhenan
- 年份: 2025; 主题: Supporting Perception; Systems; 优先级: 重要
- [来源](#)

论文: Efficiently Serving Large Multimodal Models Using EPD Disaggregation

快速理解: Large Multimodal Models (LMMs) extend Large Language Models (LLMs) by handling diverse inputs such as images, audio, and video, but at the cost of adding a multimodal encoding stage that increases both computational and memory overhead. This step negatively affects key Service Level Objectives (SLOs), such as time to first token (TTFT) and time per output token (TPOT).

关注点: 3D、视频、多模态视觉或具身感知支撑能力; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.6.58 EncryptedLLM: Privacy-Preserving Large Language Model Inference via GPU-Accelerated Fully Homomorphic Encryption

- 作者: De Castro, Leo; Escudero, Daniel; Agrawal, Adya; Polychroniadou, Antigoni; Veloso, Manuela
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: EncryptedLLM: Privacy-Preserving Large Language Model Inference via GPU-Accelerated Fully Homomorphic Encryption

快速理解: As large language models (LLMs) become more powerful, the computation required to run these models is increasingly outsourced to a third-party cloud. While this saves clients' computation, it risks leaking the clients' LLM queries to the cloud provider.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.6.59 Explainable Concept Generation through Vision-Language Preference Learning for Understanding Neural Networks' Internal Representations

- 作者: Taparia, Aditya; Sagar, Som; Senanayake, Ransalu

- 年份: 2025; 主题: Agent RL; Supporting Perception; 优先级: 重要
- [来源](#)

论文: Explainable Concept Generation through Vision-Language Preference Learning for Understanding Neural Networks' Internal Representations

快速理解: Understanding the inner representation of a neural network helps users improve models. Concept-based methods have become a popular choice for explaining deep neural networks post-hoc because, unlike most other explainable AI techniques, they can be used to test high-level visual "concepts" that are not directly related to feature attributes.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.60 Falcon: Fast Visuomotor Policies via Partial Denoising

- 作者: Chen, Haojun; Liu, Minghao; Ma, Chengdong; Ma, Xiaojian; Ma, Zailin; Wu, Huimin; Chen, Yuanpei; Zhong, Yifan; Wang, Mingzhi; Li, Qing; Yang, Yaodong
- 年份: 2025; 主题: Robot Learning; Supporting Perception; 优先级: 重要
- [来源](#)

论文: Falcon: Fast Visuomotor Policies via Partial Denoising

快速理解: Diffusion policies are widely adopted in complex visuomotor tasks for their ability to capture multimodal action distributions. However, the multiple sampling steps required for action generation significantly harm real-time inference efficiency, which limits their applicability in real-time decision-making scenarios.

关注点: 机器人数据、策略学习、操作、触觉或 sim-to-real; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.61 Gradual Transition from Bellman Optimality Operator to Bellman Operator in Online Reinforcement Learning

- 作者: Omura, Motoki; Ota, Kazuki; Osa, Takayuki; Mukuta, Yusuke; Harada, Tatsuya
- 年份: 2025; 主题: Embodied/WBC; Agent RL; 优先级: 重要
- [来源](#)

论文: Gradual Transition from Bellman Optimality Operator to Bellman Operator in Online Reinforcement Learning

快速理解: For continuous action spaces, actor-critic methods are widely used in online reinforcement learning (RL). However, unlike RL algorithms for discrete actions, which generally model the optimal value function using the Bellman optimality operator, RL algorithms for continuous actions typically model Q-values for the current policy using the Bellman operator.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.62 Graph World Model

- 作者: Feng, Tao; Wu, Yexin; Lin, Guanyu; You, Jiaxuan
- 年份: 2025; 主题: World Model; Agent RL; 优先级: 重要

- [来源](#)

论文: Graph World Model

快速理解: World models (WMs) demonstrate strong capabilities in prediction, generation, and planning tasks. Existing WMs primarily focus on unstructured data while cannot leverage the ubiquitous structured data, often represented as graphs, in the digital world.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策; 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.63 Graph-Assisted Stitching for Offline Hierarchical Reinforcement Learning

- 作者: Baek, Seungho; Park, Taegeon; Park, Jongchan; Oh, Seungjun; Kim, Yusung
- 年份: 2025; 主题: Embodied/WBC; Agent RL; 优先级: 重要
- [来源](#)

论文: Graph-Assisted Stitching for Offline Hierarchical Reinforcement Learning

快速理解: Existing offline hierarchical reinforcement learning methods rely on high-level policy learning to generate subgoal sequences. However, their efficiency degrades as task horizons increase, and they lack effective strategies for stitching useful state transitions across different trajectories.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.64 I Think, Therefore I Diffuse: Enabling Multimodal In-Context Reasoning in Diffusion Models

- 作者: Mi, Zhenxing; Wang, Kuan-Chieh; Qian, Guocheng; Ye, Hanrong; Liu, Runtao; Tulyakov, Sergey; Aberman, Kfir; Xu, Dan
- 年份: 2025; 主题: AI Infra; Supporting Perception; 优先级: 重要
- [来源](#)

论文: I Think, Therefore I Diffuse: Enabling Multimodal In-Context Reasoning in Diffusion Models

快速理解: This paper presents ThinkDiff, a novel alignment paradigm that empowers text-to-image diffusion models with multimodal in-context understanding and reasoning capabilities by integrating the strengths of vision-language models (VLMs). Existing multimodal diffusion finetuning methods largely focus on pixel-level reconstruction rather than in-context reasoning, and are constrained by the complexity and limited availability of reasoning-based datasets.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.65 Imitation Learning from a Single Temporally Misaligned Video

- 作者: Huey, William; Wang, Huaxiaoyue; Wu, Anne; Artzi, Yoav; Choudhury, Sanjiban
- 年份: 2025; 主题: Embodied/WBC; Agent RL; 优先级: 重要
- [来源](#)

论文: Imitation Learning from a Single Temporally Misaligned Video

快速理解: We examine the problem of learning sequential tasks from a single visual demonstration.

A key challenge arises when demonstrations are temporally misaligned due to variations in timing, differences in embodiment, or inconsistencies in execution.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.66 Knowledge Retention in Continual Model-Based Reinforcement Learning

- 作者：Fu, Haotian; Sun, Yixiang; Littman, Michael; Konidaris, George
- 年份：2025；主题：World Model; Agent RL；优先级：重要
- [来源](#)

论文：Knowledge Retention in Continual Model-Based Reinforcement Learning

快速理解：We propose DRAGO, a novel approach for continual model-based reinforcement learning aimed at improving the incremental development of world models across a sequence of tasks that differ in their reward functions but not the state space or dynamics. DRAGO comprises two key components: Synthetic Experience Rehearsal, which leverages generative models to create synthetic experiences from past tasks, allowing the agent to reinforce previously learned dynamics without storing data, and Regaining Memories Through Exploration, which introduces an intrinsic reward mechanism to guide the agent toward revisiting relevant states from prior tasks.

关注点：环境动力学、视频预测、潜变量状态或模型式决策；强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.67 LARM: Large Auto-Regressive Model for Long-Horizon Embodied Intelligence

- 作者：Li, Zhuoling; Xu, Xiaogang; Xu, Zhenhua; Lim, Ser-Nam; Zhao, Hengshuang
- 年份：2025；主题：Embodied/WBC; Agent RL；优先级：重要
- [来源](#)

论文：LARM: Large Auto-Regressive Model for Long-Horizon Embodied Intelligence

快速理解：Recent embodied agents are primarily built based on reinforcement learning (RL) or large language models (LLMs). Among them, RL agents are efficient for deployment but only perform very few tasks.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.68 LV-XAttn: Distributed Cross-Attention for Long Visual Inputs in Multimodal Large Language Models

- 作者：Chang, Tzu-Tao; Venkataraman, Shivaram
- 年份：2025；主题：AI Infra; Supporting Perception；优先级：重要
- [来源](#)

论文：LV-XAttn: Distributed Cross-Attention for Long Visual Inputs in Multimodal Large Language Models

快速理解：Cross-attention is commonly adopted in multimodal large language models (MLLMs) for integrating visual information into the language backbone. However, in applications with large visual

inputs, such as video understanding, processing a large number of visual tokens in cross-attention layers leads to high memory demands and often necessitates distributed computation across multiple GPUs.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.6.69 Layer-wise Alignment: Examining Safety Alignment Across Image Encoder Layers in Vision Language Models

- 作者：Bachu, Saketh; Shayegani, Erfan; Lal, Rohit; Chakraborty, Trishna; Dutta, Arindam; Song, Chengyu; Dong, Yue; Abu-Ghazaleh, Nael B.; Roy-Chowdhury, Amit
- 年份：2025；主题：Agent RL; Supporting Perception；优先级：重要
- [来源](#)

论文：Layer-wise Alignment: Examining Safety Alignment Across Image Encoder Layers in Vision Language Models

快速理解：Vision-language models (VLMs) have improved significantly in their capabilities, but their complex architecture makes their safety alignment challenging. In this paper, we reveal an uneven distribution of harmful information across the intermediate layers of the image encoder and show that skipping a certain set of layers and exiting early can increase the chance of the VLM generating harmful responses.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.6.70 Lexico: Extreme KV Cache Compression via Sparse Coding over Universal Dictionaries

- 作者：Kim, Junhyuck; Park, Jongho; Cho, Jaewoong; Papailiopoulos, Dimitris
- 年份：2025；主题：AI Infra; Systems；优先级：重要
- [来源](#)

论文：Lexico: Extreme KV Cache Compression via Sparse Coding over Universal Dictionaries

快速理解：We introduce Lexico, a novel KV cache compression method that leverages sparse coding with a universal dictionary. Our key finding is that key-value cache in modern LLMs can be accurately approximated using sparse linear combination from a small, input-agnostic dictionary of 4k atoms, enabling efficient compression across different input prompts, tasks and models.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.6.71 M3-JEPA: Multimodal Alignment via Multi-gate MoE based on the Joint-Embedding Predictive Architecture

- 作者：Lei, Hongyang; Cheng, Xiaolong; Qin, Qi; Wang, Dan; Huang, Huazhen; Gu, Qingqing; Wu, Yetao; Ji, Luo
- 年份：2025；主题：AI Infra; Supporting Perception；优先级：重要
- [来源](#)

论文: M3-JEPA: Multimodal Alignment via Multi-gate MoE based on the Joint-Embedding Predictive Architecture

快速理解: Current multimodal learning strategies primarily optimize in the original token space. Such a framework is easy to incorporate with the backbone of pretrained language model, but might result in modality collapse.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.72 MAS-GPT: Training LLMs to Build LLM-based Multi-Agent Systems

- 作者: Ye, Rui; Tang, Shuo; Ge, Rui; Du, Yaxin; Yin, Zhenfei; Chen, Siheng; Shao, Jing
- 年份: 2025; 主题: Agent RL; AI Infra; 优先级: 重要
- [来源](#)

论文: MAS-GPT: Training LLMs to Build LLM-based Multi-Agent Systems

快速理解: LLM-based multi-agent systems (MAS) have shown significant potential in tackling diverse tasks. However, to design effective MAS, existing approaches heavily rely on manual configurations or multiple calls of advanced LLMs, resulting in inadaptability and high inference costs.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.6.73 MM-RLHF: The Next Step Forward in Multimodal LLM Alignment

- 作者: Zhang, Yifan; Yu, Tao; Tian, Haochen; Fu, Chaoyou; Li, Peiyan; Zeng, Jianshu; Xie, Wulin; Shi, Yang; Zhang, Huanyu; Wu, Junkang; Wang, Xue; Hu, Yibo; Wen, Bin; Gao, Tingting; Zhang, Zhang; Yang, Fan; Zhang, Di; Wang, Liang; Jin, Rong
- 年份: 2025; 主题: Agent RL; Supporting Perception; 优先级: 重要
- [来源](#)

论文: MM-RLHF: The Next Step Forward in Multimodal LLM Alignment

快速理解: Existing efforts to align multimodal large language models (MLLMs) with human preferences have only achieved progress in narrow areas, such as hallucination reduction, but remain limited in practical applicability and generalizability. To this end, we introduce MM-RLHF, a dataset containing 120k fine-grained, human-annotated preference comparison pairs.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.74 Mastering Massive Multi-Task Reinforcement Learning via Mixture-of-Expert Decision Transformer

- 作者: Kong, Yilun; Ma, Guozheng; Zhao, Qi; Wang, Haoyu; Shen, Li; Wang, Xueqian; Tao, Dacheng
- 年份: 2025; 主题: Agent RL; AI Infra; 优先级: 重要
- [来源](#)

论文: Mastering Massive Multi-Task Reinforcement Learning via Mixture-of-Expert Decision Transformer

快速理解: Despite recent advancements in offline multi-task reinforcement learning (MTRL) have harnessed the powerful capabilities of the Transformer architecture, most approaches focus on a limited number of tasks, with scaling to extremely massive tasks remaining a formidable challenge. In this paper, we first revisit the key impact of task numbers on current MTRL method, and further reveal that naively expanding the parameters proves insufficient to counteract the performance degradation as the number of tasks escalates.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策；训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.6.75 Maximum Entropy Reinforcement Learning with Diffusion Policy

- 作者: Dong, Xiaoyi; Cheng, Jian; Zhang, Xi Sheryl
- 年份: 2025; 主题: Agent RL; Supporting Perception; 优先级: 重要
- [来源](#)

论文: Maximum Entropy Reinforcement Learning with Diffusion Policy

快速理解: The Soft Actor-Critic (SAC) algorithm with a Gaussian policy has become a mainstream implementation for realizing the Maximum Entropy Reinforcement Learning (MaxEnt RL) objective, which incorporates entropy maximization to encourage exploration and enhance policy robustness. While the Gaussian policy performs well on simpler tasks, its exploration capacity and potential performance in complex multi-goal RL environments are limited by its inherent unimodality.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策；3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.76 Multi-Timescale Dynamics Model Bayesian Optimization for Plasma Stabilization in Tokamaks

- 作者: Sonker, Rohit; Capone, Alexandre; Rothstein, Andrew; Kaga, Hiro Josep Farre; Kolemen, Egemen; Schneider, Jeff
- 年份: 2025; 主题: World Model; Agent RL; 优先级: 重要
- [来源](#)

论文: Multi-Timescale Dynamics Model Bayesian Optimization for Plasma Stabilization in Tokamaks

快速理解: Machine learning algorithms often struggle to control complex real-world systems. In the case of nuclear fusion, these challenges are exacerbated, as the dynamics are notoriously complex, data is poor, hardware is subject to failures, and experiments often affect dynamics beyond the experiment's duration.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策；强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.77 NestQuant: nested lattice quantization for matrix products and LLMs

- 作者: Savkin, Semyon; Porat, Eitan; Ordentlich, Or; Polyanskiy, Yury
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: NestQuant: nested lattice quantization for matrix products and LLMs

快速理解: Post-training quantization (PTQ) has emerged as a critical technique for efficient deployment of large language models (LLMs). This work proposes NestQuant, a novel PTQ scheme for weights and activations that is based on self-similar nested lattices.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.6.78 Neurosymbolic World Models for Sequential Decision Making

- 作者: Hernandez Cano, Leonardo; Perroni-Scharf, Maxine; Dhir, Neil; Ramamurthy, Arun; Solar-Lezama, Armando
- 年份: 2025; 主题: World Model; Agent RL; 优先级: 重要
- [来源](#)

论文: Neurosymbolic World Models for Sequential Decision Making

快速理解: We present Structured World Modeling for Policy Optimization (SWMPO), a framework for unsupervised learning of neurosymbolic Finite State Machines (FSM) that capture environmental structure for policy optimization. Traditional unsupervised world modeling methods rely on unstructured representations, such as neural networks, that do not explicitly represent high-level patterns within the system (e.g., patterns in the dynamics of regions such as water and land).

关注点: 环境动力学、视频预测、潜变量状态或模型式决策; 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.79 OmniBal: Towards Fast Instruction-Tuning for Vision-Language Models via Omniverse Computation Balance

- 作者: Yao, Yongqiang; Tan, Jingru; Zhang, Feizhao; Hu, Jiahao; Niu, Yazhe; Xin, Jin; Li, Bo; Liu, Pengfei; Gong, Ruihao; Lin, Dahua; Xu, Ningyi
- 年份: 2025; 主题: AI Infra; Supporting Perception; 优先级: 重要
- [来源](#)

论文: OmniBal: Towards Fast Instruction-Tuning for Vision-Language Models via Omniverse Computation Balance

快速理解: Vision-language instruction-tuning models have recently achieved significant performance improvements. In this work, we discover that large-scale 3D parallel training on those models leads to an imbalanced computation load across different devices.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.80 One-Step Diffusion Policy: Fast Visuomotor Policies via Diffusion Distillation

- 作者: Wang, Zhendong; Li, Max; Mandlekar, Ajay; Xu, Zhenjia; Fan, Jiaojiao; Narang, Yashraj; Fan, Linxi; Zhu, Yuke; Balaji, Yogesh; Zhou, Mingyuan; Liu, Ming-Yu; Zeng, Yu
- 年份: 2025; 主题: Agent RL; Robot Learning; 优先级: 重要
- [来源](#)

论文: One-Step Diffusion Policy: Fast Visuomotor Policies via Diffusion Distillation

快速理解: Diffusion models, praised for their success in generative tasks, are increasingly being applied

to robotics, demonstrating exceptional performance in behavior cloning. However, their slow generation process stemming from iterative denoising steps poses a challenge for real-time applications in resource-constrained robotics setups and dynamically changing environments.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：索引摘要。

1.3.6.81 Overcoming the Curse of Dimensionality in Reinforcement Learning Through Approximate Factorization

- 作者：Lu, Chenbei; Shi, Laixi; Chen, Zaiwei; Wu, Chenye; Wierman, Adam
- 年份：2025；主题：Agent RL; Systems；优先级：重要
- [来源](#)

论文：Overcoming the Curse of Dimensionality in Reinforcement Learning Through Approximate Factorization

快速理解：Factored Markov Decision Processes (FMDPs) offer a promising framework for overcoming the curse of dimensionality in reinforcement learning (RL) by decomposing high-dimensional MDPs into smaller and independently evolving components. Despite their potential, existing studies on FMDPs face three key limitations: reliance on perfectly factorizable models, suboptimal sample complexity guarantees for model-based algorithms, and the absence of model-free algorithms.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.6.82 Quamba2: A Robust and Scalable Post-training Quantization Framework for Selective State Space Models

- 作者：Chiang, Hung-Yueh; Chang, Chi-Chih; Frumkin, Natalia; Wu, Kai-Chiang; Abdelfattah, Mohamed S.; Marculescu, Diana
- 年份：2025；主题：AI Infra; Systems；优先级：重要
- [来源](#)

论文：Quamba2: A Robust and Scalable Post-training Quantization Framework for Selective State Space Models

快速理解：State Space Models (SSMs) are gaining attention as an efficient alternative to Transformers due to their constant memory complexity and comparable performance. Yet, deploying large-scale SSMs on cloud-based services or resource-constrained devices faces challenges.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.6.83 QuantSpec: Self-Speculative Decoding with Hierarchical Quantized KV Cache

- 作者：Tiwari, Rishabh; Xi, Haocheng; Tomar, Aditya; Hooper, Coleman Richard Charles; Kim, Sehoon; Horton, Maxwell; Najibi, Mahyar; Mahoney, Michael W.; Keutzer, Kurt; Gholami, Amir
- 年份：2025；主题：AI Infra; Systems；优先级：重要
- [来源](#)

论文: QuantSpec: Self-Speculative Decoding with Hierarchical Quantized KV Cache

快速理解: Large Language Models (LLMs) are increasingly being deployed on edge devices for long-context settings, creating a growing need for fast and efficient long-context inference. In these scenarios, the Key-Value (KV) cache is the primary bottleneck in terms of both GPU memory and latency, as the full KV cache must be loaded for each decoding step.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.6.84 ResQ: Mixed-Precision Quantization of Large Language Models with Low-Rank Residuals

- 作者: Saxena, Utkarsh; Sharify, Sayeh; Roy, Kaushik; Wang, Xin
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: ResQ: Mixed-Precision Quantization of Large Language Models with Low-Rank Residuals

快速理解: Post-training quantization (PTQ) of large language models (LLMs) holds the promise in reducing the prohibitive computational cost at inference time. Quantization of all weight, activation and key-value (KV) cache tensors to 4-bit without significantly degrading generalizability is challenging, due to the high quantization error caused by extreme outliers in activations.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.6.85 RoSTE: An Efficient Quantization-Aware Supervised Fine-Tuning Approach for Large Language Models

- 作者: Wei, Quan; Yau, Chung-Yiu; Wai, Hoi To; Zhao, Yang; Kang, Dongyeop; Park, Youngsuk; Hong, Mingyi
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: RoSTE: An Efficient Quantization-Aware Supervised Fine-Tuning Approach for Large Language Models

快速理解: Supervised fine-tuning is a standard method for adapting pre-trained large language models (LLMs) to downstream tasks. Quantization has been recently studied as a post-training technique for efficient LLM deployment.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.6.86 Robust Multimodal Large Language Models Against Modality Conflict

- 作者: Zhang, Zongmeng; Zhou, Wengang; Zhao, Jie; Li, Houqiang
- 年份: 2025; 主题: Agent RL; Supporting Perception; 优先级: 重要
- [来源](#)

论文: Robust Multimodal Large Language Models Against Modality Conflict

快速理解: Despite the impressive capabilities of multimodal large language models (MLLMs) in vision-language tasks, they are prone to hallucinations in real-world scenarios. This paper investigates the hallucination phenomenon in MLLMs from the perspective of modality conflict.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.87 SKIM: Any-bit Quantization Pushing The Limits of Post-Training Quantization

- 作者: Bai, Runsheng; Liu, Bo; Liu, Qiang
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: SKIM: Any-bit Quantization Pushing The Limits of Post-Training Quantization

快速理解: Large Language Models (LLMs) exhibit impressive performance across various tasks, but deploying them for inference poses challenges. Their high resource demands often necessitate complex, costly multi-GPU pipelines, or the use of smaller, less capable models.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.6.88 STAR: Learning Diverse Robot Skill Abstractions through Rotation-Augmented Vector Quantization

- 作者: Li, Hao; Lv, Qi; Shao, Rui; Deng, Xiang; Li, Yinchuan; Hao, Jianye; Nie, Liqiang
- 年份: 2025; 主题: Embodied/WBC; AI Infra; 优先级: 重要
- [来源](#)

论文: STAR: Learning Diverse Robot Skill Abstractions through Rotation-Augmented Vector Quantization

快速理解: Transforming complex actions into discrete skill abstractions has demonstrated strong potential for robotic manipulation. Existing approaches mainly leverage latent variable models, e.g., VQ-VAE, to learn skill abstractions through learned vectors (codebooks), while they suffer from codebook collapse and modeling the causal relationship between learned skills. To address these limitations, we present Skill Training with Augmented Rotation (STAR), a framework that advances both skill learning and composition to complete complex behaviors.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.6.89 SageAttention2: Efficient Attention with Thorough Outlier Smoothing and Per-thread INT4 Quantization

- 作者: Zhang, Jintao; Huang, Haofeng; Zhang, Pengle; Wei, Jia; Zhu, Jun; Chen, Jianfei
- 年份: 2025; 主题: AI Infra; Supporting Perception; 优先级: 重要
- [来源](#)

论文: SageAttention2: Efficient Attention with Thorough Outlier Smoothing and Per-thread INT4 Quantization

快速理解: Although quantization for linear layers has been widely used, its application to accelerate the attention process remains limited. To further enhance the efficiency of attention computation compared to SageAttention while maintaining precision, we propose SageAttention2, which utilizes significantly faster 4-bit matrix multiplication (Matmul) alongside additional precision-enhancing techniques.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.90 SpeCache: Speculative Key-Value Caching for Efficient Generation of LLMs

- 作者: Jie, Shibo; Tang, Yehui; Han, Kai; Deng, Zhi-Hong; Han, Jing
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: SpeCache: Speculative Key-Value Caching for Efficient Generation of LLMs

快速理解: Transformer-based large language models (LLMs) have already achieved remarkable results on long-text tasks, but the limited GPU memory (VRAM) resources struggle to accommodate the linearly growing demand for key-value (KV) cache as the sequence length increases, which has become a bottleneck for the application of LLMs on long sequences. Existing KV cache compression methods include eviction, merging, or quantization of the KV cache to reduce its size.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.6.91 Star Attention: Efficient LLM Inference over Long Sequences

- 作者: Acharya, Shantanu; Jia, Fei; Ginsburg, Boris
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Star Attention: Efficient LLM Inference over Long Sequences

快速理解: Inference with Transformer-based Large Language Models (LLMs) on long sequences is both costly and slow due to the quadratic complexity of the self-attention mechanism. We introduce Star Attention, a two-phase block-sparse approximation that improves computational efficiency by sharding attention across multiple hosts while minimizing communication overhead.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.6.92 Topological Signatures of Adversaries in Multimodal Alignments

- 作者: Vu, Minh N.; Zollicoffer, Geigh; Mai, Huy; Nebgen, Ben; Alexandrov, Boian; Bhattarai, Manish
- 年份: 2025; 主题: AI Infra; Supporting Perception; 优先级: 重要
- [来源](#)

论文: Topological Signatures of Adversaries in Multimodal Alignments

快速理解: Multimodal Machine Learning systems, particularly those aligning text and image data like CLIP/BLIP models, have become increasingly prevalent, yet remain susceptible to adversar-

ial attacks. While substantial research has addressed adversarial robustness in unimodal contexts, defense strategies for multimodal systems are underexplored.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.6.93 Unifying 2D and 3D Vision-Language Understanding

- 作者：Jain, Ayush; Swerdlow, Alexander; Wang, Yuzhou; Arnaud, Sergio; Martin, Ada; Sax, Alexander; Meier, Franziska; Fragkiadaki, Katerina
- 年份：2025；主题：Embodied/WBC; Supporting Perception；优先级：重要
- [来源](#)

论文：Unifying 2D and 3D Vision-Language Understanding

快速理解：Progress in 3D vision-language learning has been hindered by the scarcity of large-scale 3D datasets. We introduce UniVLG, a unified architecture for 2D and 3D vision-language understanding that bridges the gap between existing 2D-centric models and the rich 3D sensory data available in embodied systems.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.6.94 WorldSimBench: Towards Video Generation Models as World Simulators

- 作者：Qin, Yiran; Shi, Zhelun; Yu, Jiwen; Wang, Xijun; Zhou, Enshen; Li, Lijun; Yin, Zhenfei; Liu, Xihui; Sheng, Lu; Shao, Jing; Bai, Lei; Zhang, Ruimao
- 年份：2025；主题：Embodied/WBC; Supporting Perception；优先级：重要
- [来源](#)

论文：WorldSimBench: Towards Video Generation Models as World Simulators

快速理解：Recent advancements in predictive models have demonstrated exceptional capabilities in predicting the future state of objects and scenes. However, the lack of categorization based on inherent characteristics continues to hinder the progress of predictive model development.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.6.95 Zero-Shot Offline Imitation Learning via Optimal Transport

- 作者：Rupf, Thomas; Bagatella, Marco; G" {u}rtler, Nico; Frey, Jonas; Martius, Georg
- 年份：2025；主题：World Model; Agent RL；优先级：重要
- [来源](#)

论文：Zero-Shot Offline Imitation Learning via Optimal Transport

快速理解：Zero-shot imitation learning algorithms hold the promise of reproducing unseen behavior from as little as a single demonstration at test time. Existing practical approaches view the expert demonstration as a sequence of goals, enabling imitation with a high-level goal selector, and a low-level goal-conditioned policy.

关注点：环境动力学、视频预测、潜变量状态或模型式决策；强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.96 Activation by Interval-wise Dropout: A Simple Way to Prevent Neural Networks from Plasticity Loss

- 作者：Park, Sangyeon; Han, Isaac; Oh, Seungwon; Kim, Kyungjoong
- 年份：2025；主题：Agent RL; AI Infra; 优先级：重要
- [来源](#)

论文：Activation by Interval-wise Dropout: A Simple Way to Prevent Neural Networks from Plasticity Loss

快速理解：Plasticity loss, a critical challenge in neural network training, limits a model's ability to adapt to new tasks or shifts in data distribution. While widely used techniques like L2 regularization and Layer Normalization have proven effective in mitigating this issue, Dropout remains notably ineffective.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.6.97 Active Fine-Tuning of Multi-Task Policies

- 作者：Bagatella, Marco; H" {u}botter, Jonas; Martius, Georg; Krause, Andreas
- 年份：2025；主题：Agent RL; Robot Learning; 优先级：重要
- [来源](#)

论文：Active Fine-Tuning of Multi-Task Policies

快速理解：Pre-trained generalist policies are rapidly gaining relevance in robot learning due to their promise of fast adaptation to novel, in-domain tasks. This adaptation often relies on collecting new demonstrations for a specific task of interest and applying imitation learning algorithms, such as behavioral cloning.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：索引摘要。

1.3.6.98 BOPO: Neural Combinatorial Optimization via Best-anchored and Objective-guided Preference Optimization

- 作者：Liao, Zijun; Chen, Jinbiao; Wang, Debing; Zhang, Zizhen; Wang, Jiahai
- 年份：2025；主题：Agent RL; Systems; 优先级：重要
- [来源](#)

论文：BOPO: Neural Combinatorial Optimization via Best-anchored and Objective-guided Preference Optimization

快速理解：Neural Combinatorial Optimization (NCO) has emerged as a promising approach for NP-hard problems. However, prevailing RL-based methods suffer from low sample efficiency due to sparse rewards and underused solutions.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.6.99 Behavioral Exploration: Learning to Explore via In-Context Adaptation

- 作者: Wagenmaker, Andrew; Zhou, Zhiyuan; Levine, Sergey
- 年份: 2025; 主题: Embodied/WBC; Agent RL; 优先级: 重要
- [来源](#)

论文: Behavioral Exploration: Learning to Explore via In-Context Adaptation

快速理解: Developing autonomous agents that quickly explore an environment and adapt their behavior online is a canonical challenge in robotics and machine learning. While humans are able to achieve such fast online exploration and adaptation, often acquiring new information and skills in only a handful of interactions, existing algorithmic approaches tend to rely on random exploration and slow, gradient-based behavior updates.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.100 CSV-Occ: Fusing Multi-frame Alignment for Occupancy Prediction with Temporal Cross State Space Model and Central Voting Mechanism

- 作者: Zhu, Ziming; Zhu, Yu; Chen, Jiahao; Ling, Xiaofeng; Chen, Huanlei; Sun, Lihua
- 年份: 2025; 主题: Supporting Perception; Systems; 优先级: 重要
- [来源](#)

论文: CSV-Occ: Fusing Multi-frame Alignment for Occupancy Prediction with Temporal Cross State Space Model and Central Voting Mechanism

快速理解: Recently, image-based 3D semantic occupancy prediction has become a hot topic in 3D scene understanding for autonomous driving. Compared with the bounding box form of 3D object detection, the ability to describe the fine-grained contours of any obstacles in the scene is the key insight of voxel occupancy representation, which facilitates subsequent tasks of autonomous driving.

关注点: 3D、视频、多模态视觉或具身感知支撑能力; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.6.101 Closed-Loop Long-Horizon Robotic Planning via Equilibrium Sequence Modeling

- 作者: Li, Jinghan; Sun, Zhicheng; Mu, Yadong
- 年份: 2025; 主题: World Model; Agent RL; 优先级: 重要
- [来源](#)

论文: Closed-Loop Long-Horizon Robotic Planning via Equilibrium Sequence Modeling

快速理解: In the endeavor to make autonomous robots take actions, task planning is a major challenge that requires translating high-level task descriptions to long-horizon action sequences. Despite recent advances in language model agents, they remain prone to planning errors and limited in their ability to plan ahead.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策; 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.102 Controlling Large Language Model with Latent Action

- 作者: Jia, Chengxing; Li, Ziniu; Wang, Pengyuan; Li, Yi-Chen; Hou, Zhenyu; Dong, Yuxiao; Yu, Yang
- 年份: 2025; 主题: World Model; Agent RL; 优先级: 重要
- [来源](#)

论文: Controlling Large Language Model with Latent Action

快速理解: Adapting Large Language Models (LLMs) to downstream tasks using Reinforcement Learning (RL) has proven to be an effective approach. However, LLMs do not inherently define the structure of an agent for RL training, particularly in terms of specifying the action space.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策; 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.103 Delta Decompression for MoE-based LLMs Compression

- 作者: Gu, Hao; Li, Wei; Li, Lujun; Qiyuan, Zhu; Lee, Mark G.; Sun, Shengjie; Xue, Wei; Guo, Yike
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Delta Decompression for MoE-based LLMs Compression

快速理解: Mixture-of-Experts (MoE) architectures in large language models (LLMs) achieve exceptional performance, but face prohibitive storage and memory requirements. To address these challenges, we present D²-MoE, a new delta decompression compressor for reducing the parameters of MoE LLMs.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.6.104 Determining Layer-wise Sparsity for Large Language Models Through a Theoretical Perspective

- 作者: Huang, Weizhong; Zhang, Yuxin; Zheng, Xiawu; Chao, Fei; Ji, Rongrong
- 年份: 2025; 主题: AI Infra; Supporting Perception; 优先级: 重要
- [来源](#)

论文: Determining Layer-wise Sparsity for Large Language Models Through a Theoretical Perspective

快速理解: In this paper, we address the challenge of determining the layer-wise sparsity rates of large language models (LLMs) through a theoretical perspective. Specifically, we identify a critical issue of “reconstruction error explosion” in existing LLMs sparsification methods.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.105 Diffusion Instruction Tuning

- 作者: Jin, Chen; Tanno, Ryutaro; Saseendran, Amrutha; Diethe, Tom; Teare, Philip Alexander
- 年份: 2025; 主题: AI Infra; Supporting Perception; 优先级: 重要
- [来源](#)

论文: Diffusion Instruction Tuning

快速理解: We introduce Lavender, a simple supervised fine-tuning (SFT) method that boosts the performance of advanced vision-language models (VLMs) by leveraging state-of-the-art image generation models such as Stable Diffusion. Specifically, Lavender aligns the text-vision attention in the VLM transformer with the equivalent used by Stable Diffusion during SFT, instead of adapting separate encoders.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.106 Diffusion Sampling Correction via Approximately 10 Parameters

- 作者: Wang, Guangyi; Peng, Wei; Li, Lijiang; Chen, Wenyu; Cai, Yuren; Su, Song-Zhi
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Diffusion Sampling Correction via Approximately 10 Parameters

快速理解: While powerful for generation, Diffusion Probabilistic Models (DPMs) face slow sampling challenges, for which various distillation-based methods have been proposed. However, they typically require significant additional training costs and model parameter storage, limiting their practicality.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.6.107 Diversifying Policy Behaviors with Extrinsic Behavioral Curiosity

- 作者: Wan, Zhenglin; Yu, Xingrui; Bossens, David Mark; Lyu, Yueming; Guo, Qing; Fan, Flint Xiaofeng; Ong, Yew-Soon; Tsang, Ivor
- 年份: 2025; 主题: Embodied/WBC; Agent RL; 优先级: 重要
- [来源](#)

论文: Diversifying Policy Behaviors with Extrinsic Behavioral Curiosity

快速理解: Imitation learning (IL) has shown promise in various applications (e.g. robot locomotion) but is often limited to learning a single expert policy, constraining behavior diversity and robustness in unpredictable real-world scenarios. To address this, we introduce Quality Diversity Inverse Reinforcement Learning (QD-IRL), a novel framework that integrates quality-diversity optimization with IRL methods, enabling agents to learn diverse behaviors from limited demonstrations.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.108 Efficient Long Context Fine-tuning with Chunk Flow

- 作者: Yuan, Xiulong; Xu, Hongtao; Shen, Wenting; Wang, Ang; Qiu, Xiafei; Zhang, Jie; Liu, Yuqiong; Yu, Bowen; Lin, Junyang; Li, Mingzhen; Jia, Weile; Li, Yong; Lin, Wei
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Efficient Long Context Fine-tuning with Chunk Flow

快速理解: Long context fine-tuning of large language models(LLMs) involves training on datasets that are predominantly composed of short sequences and a small proportion of longer sequences. However, existing approaches overlook this long-tail distribution and employ training strategies designed specifically for long sequences.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.6.109 FlexiReID: Adaptive Mixture of Expert for Multi-Modal Person Re-Identification

- 作者: Sun, Zhen; Tan, Lei; Shen, Yunhang; Cai, Chengmao; Sun, Xing; Dai, Pingyang; Cao, Liujuan; Ji, Rongrong
- 年份: 2025; 主题: AI Infra; Supporting Perception; 优先级: 重要
- [来源](#)

论文: FlexiReID: Adaptive Mixture of Expert for Multi-Modal Person Re-Identification

快速理解: Multimodal person re-identification (Re-ID) aims to match pedestrian images across different modalities. However, most existing methods focus on limited cross-modal settings and fail to support arbitrary query-retrieval combinations, hindering practical deployment.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.110 Flow of Reasoning: Training LLMs for Divergent Reasoning with Minimal Examples

- 作者: Yu, Fangxu; Jiang, Lai; Kang, Haoqiang; Hao, Shibo; Qin, Lianhui
- 年份: 2025; 主题: Embodied/WBC; Agent RL; 优先级: 重要
- [来源](#)

论文: Flow of Reasoning: Training LLMs for Divergent Reasoning with Minimal Examples

快速理解: The ability to generate diverse solutions to a given problem is a hallmark of human creativity. This divergent reasoning is also crucial for machines, enhancing their robustness and enabling them to assist humans in many applications such as scientific discovery.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.111 From Thousands to Billions: 3D Visual Language Grounding via Render-Supervised Distillation from 2D VLMs

- 作者: Cao, Ang; Arnaud, Sergio; Maksymets, Oleksandr; Yang, Jianing; Jain, Ayush; Martin, Ada; Berges, Vincent-Pierre; Mcvay, Paul; Partsey, Ruslan; Rajeswaran, Aravind; Meier, Franziska; Johnson, Justin; Park, Jeong Joon; Sax, Alexander
- 年份: 2025; 主题: Supporting Perception; Systems; 优先级: 重要
- [来源](#)

论文: From Thousands to Billions: 3D Visual Language Grounding via Render-Supervised Distillation from 2D VLMs

快速理解: 3D vision-language grounding faces a fundamental data bottleneck: while 2D models train on billions of images, 3D models have access to only thousands of labeled scenes—a six-order-of-magnitude gap that severely limits performance. We introduce LIFT-GS, a practical distillation technique that overcomes this limitation by using differentiable rendering to bridge 3D and 2D supervision.

关注点: 3D、视频、多模态视觉或具身感知支撑能力; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.6.112 Geometric Feature Embedding for Effective 3D Few-Shot Class Incremental Learning

- 作者: Li, Xiangqi; Huang, Libo; An, Zhulin; Feng, Weilun; Yang, Chuanguang; Diao, Boyu; Wang, Fei; Xu, Yongjun
- 年份: 2025; 主题: Supporting Perception; Systems; 优先级: 重要
- [来源](#)

论文: Geometric Feature Embedding for Effective 3D Few-Shot Class Incremental Learning

快速理解: 3D few-shot class incremental learning (FSCIL) aims to learn new point cloud categories from limited samples while preventing the forgetting of previously learned categories. This research area significantly enhances the capabilities of self-driving vehicles and computer vision systems.

关注点: 3D、视频、多模态视觉或具身感知支撑能力; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.6.113 Hardware and Software Platform Inference

- 作者: Zhang, Cheng; Foerster, Hanna; Mullins, Robert D.; Zhao, Yiren; Shumailov, Ilia
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Hardware and Software Platform Inference

快速理解: It is now a common business practice to buy access to large language model (LLM) inference rather than self-host, because of significant upfront hardware infrastructure and energy costs. However, as a buyer, there is no mechanism to verify the authenticity of the advertised service including the serving hardware platform, e.g. that it is actually being served using an NVIDIA H100.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.6.114 High Dynamic Range Novel View Synthesis with Single Exposure

- 作者: Zhang, Kaixuan; Wang, Hu; Li, Minxian; Ren, Mingwu; Ye, Mao; Zhu, Xiatian
- 年份: 2025; 主题: Supporting Perception; Systems; 优先级: 重要
- [来源](#)

论文: High Dynamic Range Novel View Synthesis with Single Exposure

快速理解: High Dynamic Range Novel View Synthesis (HDR-NVS) aims to establish a 3D scene HDR model from Low Dynamic Range (LDR) imagery. Typically, multiple-exposure LDR images are employed to capture a wider range of brightness levels in a scene, as a single LDR image cannot

represent both the brightest and darkest regions simultaneously.

关注点：3D、视频、多模态视觉或具身感知支撑能力；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.6.115 Improving LLM Video Understanding with 16 Frames Per Second

- 作者：Li, Yixuan; Tang, Changli; Zhuang, Jimin; Yang, Yudong; Sun, Guangzhi; Li, Wei; Ma, Zejun; Zhang, Chao
- 年份：2025；主题：AI Infra; Supporting Perception; 优先级：重要
- [来源](#)

论文：Improving LLM Video Understanding with 16 Frames Per Second

快速理解：Human vision is dynamic and continuous. However, in video understanding with multimodal large language models (LLMs), existing methods primarily rely on static features extracted from images sampled at a fixed low frame rate of frame-per-second (FPS) ≤ 2 , leading to critical visual information loss.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.6.116 LADA: Scalable Label-Specific CLIP Adapter for Continual Learning

- 作者：Luo, Mao-Lin; Zhou, Zi-Hao; Wei, Tong; Zhang, Min-Ling
- 年份：2025；主题：AI Infra; Supporting Perception; 优先级：重要
- [来源](#)

论文：LADA: Scalable Label-Specific CLIP Adapter for Continual Learning

快速理解：Continual learning with vision-language models like CLIP offers a pathway toward scalable machine learning systems by leveraging its transferable representations. Existing CLIP-based methods adapt the pre-trained image encoder by adding multiple sets of learnable parameters, with each task using a partial set of parameters.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.6.117 LBI-FL: Low-Bit Integerized Federated Learning with Temporally Dynamic Bit-Width Allocation

- 作者：Ding, Li; Zhang, Hao; Dai, Wenrui; Li, Chenglin; Lu, Weijia; Yang, Zhifei; Zhang, Xiaodong; Ma, Xiaofeng; Zou, Junni; Xiong, Hongkai
- 年份：2025；主题：Agent RL; AI Infra; 优先级：重要
- [来源](#)

论文：LBI-FL: Low-Bit Integerized Federated Learning with Temporally Dynamic Bit-Width Allocation

快速理解：Federated learning (FL) is greatly challenged by the communication bottleneck and computation limitation on clients. Existing methods based on quantization for FL cannot simultaneously reduce the uplink and downlink communication cost and mitigate the computation burden on clients.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.6.118 LOCATE 3D: Real-World Object Localization via Self-Supervised Learning in 3D

- 作者：Mcvay, Paul; Arnaud, Sergio; Martin, Ada; Majumdar, Arjun; Jatavallabhula, Krishna Murthy; Thomas, Phillip; Partsey, Ruslan; Dugas, Daniel; Gejji, Abha; Sax, Alexander; Berges, Vincent-Pierre; Henaff, Mikael; Jain, Ayush; Cao, Ang; Prasad, Ishita; Kalakrishnan, Mrinal; Rabbat, Michael; Ballas, Nicolas; Assran, Mido; Maksymets, Oleksandr; Rajeswaran, Aravind; Meier, Franziska
- 年份：2025；主题：Supporting Perception; Systems；优先级：重要
- [来源](#)

论文：LOCATE 3D: Real-World Object Localization via Self-Supervised Learning in 3D

快速理解：We present LOCATE 3D, a model for localizing objects in 3D scenes from referring expressions like “the small coffee table between the sofa and the lamp.” LOCATE 3D sets a new state-of-the-art on standard referential grounding benchmarks and showcases robust generalization capabilities. Notably, LOCATE 3D operates directly on sensor observation streams (posed RGB-D frames), enabling real-world deployment on robots and AR devices.

关注点：3D、视频、多模态视觉或具身感知支撑能力；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.6.119 LineFlow: A Framework to Learn Active Control of Production Lines

- 作者：Müller, Kai; Wenzel, Martin; Windisch, Tobias
- 年份：2025；主题：Agent RL; Systems；优先级：重要
- [来源](#)

论文：LineFlow: A Framework to Learn Active Control of Production Lines

快速理解：Many production lines require active control mechanisms, such as adaptive routing, worker reallocation, and rescheduling, to maintain optimal performance. However, designing these control systems is challenging for various reasons, and while reinforcement learning (RL) has shown promise in addressing these challenges, a standardized and general framework is still lacking.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.6.120 LipsNet++: Unifying Filter and Controller into a Policy Network

- 作者：Song, Xujie; Chen, Liangfa; Liu, Tong; Wang, Wenxuan; Wang, Yinuo; Qin, Shentao; Ma, Yinsong; Duan, Jingliang; Li, Shengbo Eben
- 年份：2025；主题：Embodied/WBC; Agent RL；优先级：重要
- [来源](#)

论文：LipsNet++: Unifying Filter and Controller into a Policy Network

快速理解：Deep reinforcement learning (RL) is effective for decision-making and control tasks like autonomous driving and embodied AI. However, RL policies often suffer from the action fluctuation

problem in real-world applications, resulting in severe actuator wear, safety risk, and performance degradation.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.121 MMInference: Accelerating Pre-filling for Long-Context Visual Language Models via Modality-Aware Permutation Sparse Attention

- 作者: Li, Yucheng; Jiang, Huiqiang; Zhang, Chengruidong; Wu, Qianhui; Luo, Xufang; Ahn, Surin; Abdi, Amir H.; Li, Dongsheng; Gao, Jianfeng; Yang, Yuqing; Qiu, Lili
- 年份: 2025; 主题: AI Infra; Supporting Perception; 优先级: 重要
- [来源](#)

论文: MMInference: Accelerating Pre-filling for Long-Context Visual Language Models via Modality-Aware Permutation Sparse Attention

快速理解: The integration of long-context capabilities with visual understanding unlocks unprecedented potential for Vision Language Models (VLMs). However, the quadratic attention complexity during the pre-filling phase remains a significant obstacle to real-world deployment.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.6.122 Masked Generative Nested Transformers with Decode Time Scaling

- 作者: Goyal, Sahil; Tula, Debapriya; Jain, Gagan; Shenoy, Pradeep; Jain, Prateek; Paul, Sujoy
- 年份: 2025; 主题: Supporting Perception; Systems; 优先级: 重要
- [来源](#)

论文: Masked Generative Nested Transformers with Decode Time Scaling

快速理解: Recent advances in visual generation have made significant strides in producing content of exceptional quality. However, most methods suffer from a fundamental problem - a bottleneck of inference computational efficiency.

关注点：3D、视频、多模态视觉或具身感知支撑能力；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.6.123 Mixture of Lookup Experts

- 作者: Jie, Shibo; Tang, Yehui; Han, Kai; Li, Yitong; Tang, Duyu; Deng, Zhi-Hong; Wang, Yunhe
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Mixture of Lookup Experts

快速理解: Mixture-of-Experts (MoE) activates only a subset of experts during inference, allowing the model to maintain low inference FLOPs and latency even as the parameter count scales up. However, since MoE dynamically selects the experts, all the experts need to be loaded into VRAM.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.6.124 Model Swarms: Collaborative Search to Adapt LLM Experts via Swarm Intelligence

- 作者: Feng, Shangbin; Wang, Zifeng; Wang, Yike; Ebrahimi, Sayna; Palangi, Hamid; Miculicich, Lesly; Kulshrestha, Achin; Rauschmayr, Nathalie; Choi, Yejin; Tsvetkov, Yulia; Lee, Chen-Yu; Pfister, Tomas
- 年份: 2025; 主题: Agent RL; AI Infra; 优先级: 重要
- [来源](#)

论文: Model Swarms: Collaborative Search to Adapt LLM Experts via Swarm Intelligence

快速理解: We propose Model Swarms, a collaborative search algorithm to adapt LLMs via swarm intelligence, the collective behavior guiding individual systems. Specifically, Model Swarms starts with a pool of LLM experts and a utility function.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.6.125 Open-Det: An Efficient Learning Framework for Open-Ended Detection

- 作者: Cao, Guiping; Wang, Tao; Huang, Wenjian; Lan, Xiangyuan; Zhang, Jianguo; Jiang, Dongmei
- 年份: 2025; 主题: AI Infra; Supporting Perception; 优先级: 重要
- [来源](#)

论文: Open-Det: An Efficient Learning Framework for Open-Ended Detection

快速理解: Open-Ended object Detection (OED) is a novel and challenging task that detects objects and generates their category names in a free-form manner, without requiring additional vocabularies during inference. However, the existing OED models, such as GenerateU, require large-scale datasets for training, suffer from slow convergence, and exhibit limited performance.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.126 Orthus: Autoregressive Interleaved Image-Text Generation with Modality-Specific Heads

- 作者: Kou, Siqi; Jin, Jiachun; Liu, Zhihong; Liu, Chang; Ma, Ye; Jia, Jian; Chen, Quan; Jiang, Peng; Deng, Zhijie
- 年份: 2025; 主题: AI Infra; Supporting Perception; 优先级: 重要
- [来源](#)

论文: Orthus: Autoregressive Interleaved Image-Text Generation with Modality-Specific Heads

快速理解: We introduce Orthus, a unified multimodal model that excels in generating interleaved images and text from mixed-modality inputs by simultaneously handling discrete text tokens and continuous image features under the AR modeling principle. The continuous treatment of visual signals minimizes the information loss while the fully AR formulation renders the characterization of the correlation between modalities straightforward.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.127 ParallelComp: Parallel Long-Context Compressor for Length Extrapolation

- 作者: Xiong, Jing; Shen, Jianghan; Zheng, Chuanyang; Wan, Zhongwei; Zhao, Chenyang; Yang, Chiwan; Ye, Fanghua; Yang, Hongxia; Kong, Lingpeng; Wong, Ngai
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: ParallelComp: Parallel Long-Context Compressor for Length Extrapolation

快速理解: Extrapolating ultra-long contexts (text length >128K) remains a major challenge for large language models (LLMs), as most training-free extrapolation methods are not only severely limited by memory bottlenecks, but also suffer from the attention sink, which restricts their scalability and effectiveness in practice. In this work, we propose ParallelComp, a parallel long-context compression method that effectively overcomes the memory bottleneck, enabling 8B-parameter LLMs to extrapolate from 8K to 128K tokens on a single A100 80GB GPU in a training-free setting.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.6.128 Parrot: Multilingual Visual Instruction Tuning

- 作者: Sun, Hai-Long; Zhou, Da-Wei; Li, Yang; Lu, Shiyin; Yi, Chao; Chen, Qing-Guo; Xu, Zhao; Luo, Weihua; Zhang, Kaifu; Zhan, De-Chuan; Ye, Han-Jia
- 年份: 2025; 主题: AI Infra; Supporting Perception; 优先级: 重要
- [来源](#)

论文: Parrot: Multilingual Visual Instruction Tuning

快速理解: The rapid development of Multimodal Large Language Models (MLLMs), such as GPT-4, marks a significant step toward artificial general intelligence. Existing methods typically align vision encoders with LLMs via supervised fine-tuning (SFT), but this often deteriorates their ability to handle multiple languages as training progresses.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.129 Preference Adaptive and Sequential Text-to-Image Generation

- 作者: Nabati, Ofir; Tennenholtz, Guy; Hsu, Chihwei; Ryu, Moonkyung; Ramachandran, Deepak; Chow, Yinlam; Li, Xiang; Boutilier, Craig
- 年份: 2025; 主题: Agent RL; Supporting Perception; 优先级: 重要
- [来源](#)

论文: Preference Adaptive and Sequential Text-to-Image Generation

快速理解: We address the problem of interactive text-to-image (T2I) generation, designing a reinforcement learning (RL) agent which iteratively improves a set of generated images for a user through a sequence of prompt expansions. Using human raters, we create a novel dataset of sequential preferences, which we leverage, together with large-scale open-source (non-sequential) datasets.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.130 Probabilistic Interactive 3D Segmentation with Hierarchical Neural Processes

- 作者: Liu, Jie; Zhou, Pan; Xiao, Zehao; Shen, Jiayi; Yin, Wenzhe; Sonke, Jan-Jakob; Gavves, Efstratios
- 年份: 2025; 主题: Supporting Perception; Systems; 优先级: 重要
- [来源](#)

论文: Probabilistic Interactive 3D Segmentation with Hierarchical Neural Processes

快速理解: Interactive 3D segmentation has emerged as a promising solution for generating accurate object masks in complex 3D scenes by incorporating user-provided clicks. However, two critical challenges remain underexplored: (1) effectively generalizing from sparse user clicks to produce accurate segmentations and (2) quantifying predictive uncertainty to help users identify unreliable regions.

关注点: 3D、视频、多模态视觉或具身感知支撑能力; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.6.131 Robust Noise Attenuation via Adaptive Pooling of Transformer Outputs

- 作者: Brothers, Greyson
- 年份: 2025; 主题: Agent RL; AI Infra; 优先级: 重要
- [来源](#)

论文: Robust Noise Attenuation via Adaptive Pooling of Transformer Outputs

快速理解: We investigate the design of pooling methods used to summarize the outputs of transformer embedding models, primarily motivated by reinforcement learning and vision applications. This work considers problems where a subset of the input vectors contains requisite information for a downstream task (signal) while the rest are distractors (noise).

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.6.132 Safe-EF: Error Feedback for Non-smooth Constrained Optimization

- 作者: Islamov, Rustem; As, Yarden; Fatkhullin, Ilyas
- 年份: 2025; 主题: Embodied/WBC; Agent RL; 优先级: 重要
- [来源](#)

论文: Safe-EF: Error Feedback for Non-smooth Constrained Optimization

快速理解: Federated learning faces severe communication bottlenecks due to the high dimensionality of model updates. Communication compression with contractive compressors (e.g., Top-K) is often preferable in practice but can degrade performance without proper handling.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.133 Scaling Video-Language Models to 10K Frames via Hierarchical Differential Distillation

- 作者: Cheng, Chuanqi; Guan, Jian; Wu, Wei; Yan, Rui
- 年份: 2025; 主题: AI Infra; Supporting Perception; 优先级: 重要
- [来源](#)

论文: Scaling Video-Language Models to 10K Frames via Hierarchical Differential Distillation

快速理解: Long-form video processing fundamentally challenges vision-language models (VLMs) due to the high computational costs of handling extended temporal sequences. Existing token pruning and feature merging methods often sacrifice critical temporal dependencies or dilute semantic information.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.134 Sparse Video-Gen: Accelerating Video Diffusion Transformers with Spatial-Temporal Sparsity

- 作者: Xi, Haocheng; Yang, Shuo; Zhao, Yilong; Xu, Chenfeng; Li, MUYANG; Li, Xiuyu; Lin, Yujun; Cai, Han; Zhang, Jintao; Li, Dacheng; Chen, Jianfei; Stoica, Ion; Keutzer, Kurt; Han, Song
- 年份: 2025; 主题: AI Infra; Supporting Perception; 优先级: 重要
- [来源](#)

论文: Sparse Video-Gen: Accelerating Video Diffusion Transformers with Spatial-Temporal Sparsity

快速理解: Diffusion Transformers (DiTs) dominate video generation but their high computational cost severely limits real-world applicability, usually requiring tens of minutes to generate a few seconds of video even on high-performance GPUs. This inefficiency primarily arises from the quadratic computational complexity of 3D full attention with respect to the context length.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.135 TOPLOC: A Locality Sensitive Hashing Scheme for Trustless Verifiable Inference

- 作者: Ong, Jack Min; Ferrante, Matthew Di; Pazdera, Aaron; Garner, Ryan; Jaghouar, Sami; Basra, Manveer; Ryabinin, Max; Hagemann, Johannes
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: TOPLOC: A Locality Sensitive Hashing Scheme for Trustless Verifiable Inference

快速理解: Large language models (LLMs) have proven to be very capable, but access to frontier models currently relies on inference providers. This introduces trust challenges: how can we be sure that the provider is using the model configuration they claim?

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.6.136 VTGaussian-SLAM: RGBD SLAM for Large Scale Scenes with Splatting View-Tied 3D Gaussians

- 作者: Hu, Pengchong; Han, Zhizhong

- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: VTGaussian-SLAM: RGBD SLAM for Large Scale Scenes with Splatting View-Tied 3D Gaussians

快速理解: Jointly estimating camera poses and mapping scenes from RGBD images is a fundamental task in simultaneous localization and mapping (SLAM). State-of-the-art methods employ 3D Gaussians to represent a scene, and render these Gaussians through splatting for higher efficiency and better rendering.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.6.137 video-SALMONN-o1: Reasoning-enhanced Audio-visual Large Language Model

- 作者: Sun, Guangzhi; Yang, Yudong; Zhuang, Jimin; Tang, Changli; Li, Yixuan; Li, Wei; Ma, Zejun; Zhang, Chao
- 年份: 2025; 主题: Agent RL; Supporting Perception; 优先级: 重要
- [来源](#)

论文: video-SALMONN-o1: Reasoning-enhanced Audio-visual Large Language Model

快速理解: While recent advancements in reasoning optimization have significantly enhanced the capabilities of large language models (LLMs), existing efforts to improve reasoning have been limited to solving mathematical problems and focusing on visual graphical inputs, neglecting broader applications in general video understanding. This paper proposes video-SALMONN-o1, the first open-source reasoning-enhanced audio-visual LLM designed for general video understanding tasks.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.138 A Bayesian Model Selection Criterion for Selecting Pretraining Checkpoints

- 作者: Munn, Michael; Wei, Susan
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: A Bayesian Model Selection Criterion for Selecting Pretraining Checkpoints

快速理解: Recent advances in artificial intelligence have been fueled by the development of foundation models such as BERT, GPT, T5, and Vision Transformers. These models are first pretrained on vast and diverse datasets and then adapted to specific downstream tasks, often with significantly less data.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.6.139 A Causal World Model Underlying Next Token Prediction: Exploring GPT in a Controlled Environment

- 作者: Yehezkel Rohekar, Raanan; Gurwicz, Yaniv; Yu, Sungduk; Aflalo, Estelle; Lal, Vasudev

- 年份: 2025; 主题: World Model; 优先级: 重要
- [来源](#)

论文: A Causal World Model Underlying Next Token Prediction: Exploring GPT in a Controlled Environment

快速理解: Are generative pre-trained transformer (GPT) models, trained only to predict the next token, implicitly learning a world model from which sequences are generated one token at a time? We address this question by deriving a causal interpretation of the attention mechanism in GPT and presenting a causal world model that arises from this interpretation.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策。

证据层级: 索引摘要。

1.3.6.140 A Closer Look at Multimodal Representation Collapse

- 作者: Chaudhuri, Abhra; Dutta, Anjan; Bui, Tu; Georgescu, Serban
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: A Closer Look at Multimodal Representation Collapse

快速理解: We aim to develop a fundamental understanding of modality collapse, a recently observed empirical phenomenon wherein models trained for multimodal fusion tend to rely only on a subset of the modalities, ignoring the rest. We show that modality collapse happens when noisy features from one modality are entangled, via a shared set of neurons in the fusion head, with predictive features from another, effectively masking out positive contributions from the predictive features of the former modality and leading to its collapse.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.141 A Forget-and-Grow Strategy for Deep Reinforcement Learning Scaling in Continuous Control

- 作者: Kang, Zilin; Hu, Chenyuan; Luo, Yu; Yuan, Zhecheng; Zheng, Ruijie; Xu, Huazhe
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: A Forget-and-Grow Strategy for Deep Reinforcement Learning Scaling in Continuous Control

快速理解: Deep reinforcement learning for continuous control has recently achieved impressive progress. However, existing methods often suffer from primacy bias—a tendency to overfit early experiences stored in the replay buffer—which limits an RL agent's sample efficiency and generalizability.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.142 A Reduction Framework for Distributionally Robust Reinforcement Learning under Average Reward

- 作者: Roch, Zachary Andrew; Atia, George K.; Wang, Yue
- 年份: 2025; 主题: Agent RL; 优先级: 重要

- [来源](#)

论文: A Reduction Framework for Distributionally Robust Reinforcement Learning under Average Reward

快速理解: Robust reinforcement learning (RL) under the average reward criterion, which seeks to optimize long-term system performance in uncertain environments, remains a largely unexplored area. To address this challenge, we propose a reduction-based framework that transforms robust average reward optimization into the more extensively studied robust discounted reward optimization by employing a specific discount factor.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.143 A Reductions Approach to Risk-Sensitive Reinforcement Learning with Optimized Certainty Equivalents

- 作者: Wang, Kaiwen; Liang, Dawen; Kallus, Nathan; Sun, Wen
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: A Reductions Approach to Risk-Sensitive Reinforcement Learning with Optimized Certainty Equivalents

快速理解: We study risk-sensitive RL where the goal is learn a history-dependent policy that optimizes some risk measure of cumulative rewards. We consider a family of risks called the optimized certainty equivalents (OCE), which captures important risk measures such as conditional value-at-risk (CVaR), entropic risk and Markowitz' s mean-variance.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.144 A Sharper Global Convergence Analysis for Average Reward Reinforcement Learning via an Actor-Critic Approach

- 作者: Ganesh, Swetha; Mondal, Washim Uddin; Aggarwal, Vaneet
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: A Sharper Global Convergence Analysis for Average Reward Reinforcement Learning via an Actor-Critic Approach

快速理解: This work examines average-reward reinforcement learning with general policy parametrization. Existing state-of-the-art (SOTA) guarantees for this problem are either suboptimal or hindered by several challenges, including poor scalability with respect to the size of the state-action space, high iteration complexity, and a significant dependence on knowledge of mixing times and hitting times.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.145 AEQA-NAT : Adaptive End-to-end Quantization Alignment Training Framework for Non-autoregressive Machine Translation

- 作者: Qu, Xiangyu; Liu, Guojing; Li, Liang
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: AEQA-NAT : Adaptive End-to-end Quantization Alignment Training Framework for Non-autoregressive Machine Translation

快速理解: Non-autoregressive Transformers (NATs) have garnered significant attention due to their efficient decoding compared to autoregressive methods. However, existing conditional dependency modeling schemes based on masked language modeling introduce a training-inference gap in NATs.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.6.146 AGAV-Rater: Adapting Large Multimodal Model for AI-Generated Audio-Visual Quality Assessment

- 作者: Cao, Yuqin; Min, Xiongkuo; Gao, Yixuan; Sun, Wei; Zhai, Guangtao
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: AGAV-Rater: Adapting Large Multimodal Model for AI-Generated Audio-Visual Quality Assessment

快速理解: Many video-to-audio (VTA) methods have been proposed for dubbing silent AI-generated videos. An efficient quality assessment method for AI-generated audio-visual content (AGAV) is crucial for ensuring audio-visual quality.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.147 ARS: Adaptive Reward Scaling for Multi-Task Reinforcement Learning

- 作者: Cho, Myungsik; Park, Jongeui; Kim, Jeonghye; Sung, Youngchul
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: ARS: Adaptive Reward Scaling for Multi-Task Reinforcement Learning

快速理解: Multi-task reinforcement learning (RL) encounters significant challenges due to varying task complexities and their reward distributions from the environment. To address these issues, in this paper, we propose Adaptive Reward Scaling (ARS), a novel framework that dynamically adjusts reward magnitudes and leverages a periodic network reset mechanism.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.148 AUTOCIRCUIT-RL: Reinforcement Learning-Driven LLM for Automated Circuit Topology Generation

- 作者: Vijayaraghavan, Prashanth; Shi, Luyao; Degan, Ehsan; Mukherjee, Vandana; Zhang, Xin
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: AUTOCIRCUIT-RL: Reinforcement Learning-Driven LLM for Automated Circuit Topology Generation

快速理解: Analog circuit topology synthesis is integral to Electronic Design Automation (EDA), enabling the automated creation of circuit structures tailored to specific design requirements. However, the vast design search space and strict constraint adherence make efficient synthesis challenging.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.149 Accelerating LLM Inference with Lossless Speculative Decoding Algorithms for Heterogeneous Vocabularies

- 作者: Timor, Nadav; Mamou, Jonathan; Korat, Daniel; Berchansky, Moshe; Jain, Gaurav; Pereg, Oren; Wasserblat, Moshe; Harel, David
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Accelerating LLM Inference with Lossless Speculative Decoding Algorithms for Heterogeneous Vocabularies

快速理解: Accelerating the inference of large language models (LLMs) is a critical challenge in generative AI. Speculative decoding (SD) methods offer substantial efficiency gains by generating multiple tokens using a single target forward pass.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.6.150 Accelerating Quantum Reinforcement Learning with a Quantum Natural Policy Gradient Based Approach

- 作者: Xu, Yang; Aggarwal, Vaneet
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Accelerating Quantum Reinforcement Learning with a Quantum Natural Policy Gradient Based Approach

快速理解: We address the problem of quantum reinforcement learning (QRL) under model-free settings with quantum oracle access to the Markov Decision Process (MDP). This paper introduces a Quantum Natural Policy Gradient (QNPG) algorithm, which replaces the random sampling used in classical Natural Policy Gradient (NPG) estimators with a deterministic gradient estimation approach, enabling seamless integration into quantum systems.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.151 Accurate and Efficient World Modeling with Masked Latent Transformers

- 作者: Burchi, Maxime; Timofte, Radu
- 年份: 2025; 主题: World Model; 优先级: 重要
- [来源](#)

论文: Accurate and Efficient World Modeling with Masked Latent Transformers

快速理解: The Dreamer algorithm has recently obtained remarkable performance across diverse environment domains by training powerful agents with simulated trajectories. However, the compressed nature of its world model's latent space can result in the loss of crucial information, negatively affecting the agent's performance.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策。

证据层级: 索引摘要。

1.3.6.152 Action-Constrained Imitation Learning

- 作者: Yeh, Chia-Han; Nan, Tse-Sheng; Vuorio, Risto; Hung, Wei; Wu, Hung Yen; Sun, Shao-Hua; Hsieh, Ping-Chun
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Action-Constrained Imitation Learning

快速理解: Policy learning under action constraints plays a central role in ensuring safe behaviors in various robot control and resource allocation applications. In this paper, we study a new problem setting termed Action-Constrained Imitation Learning (ACIL), where an action-constrained imitator aims to learn from a demonstrative expert with larger action space.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.153 Active Reward Modeling: Adaptive Preference Labeling for Large Language Model Alignment

- 作者: Shen, Yunyi; Sun, Hao; Ton, Jean-Francois
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Active Reward Modeling: Adaptive Preference Labeling for Large Language Model Alignment

快速理解: Building neural reward models from human preferences is a pivotal component in reinforcement learning from human feedback (RLHF) and large language model alignment research. Given the scarcity and high cost of human annotation, how to select the most informative pairs to annotate is an essential yet challenging open problem.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.154 AdaWorld: Learning Adaptable World Models with Latent Actions

- 作者: Gao, Shenyan; Zhou, Siyuan; Du, Yilun; Zhang, Jun; Gan, Chuang
- 年份: 2025; 主题: World Model; 优先级: 重要
- [来源](#)

论文: AdaWorld: Learning Adaptable World Models with Latent Actions

快速理解: World models aim to learn action-controlled future prediction and have proven essential for the development of intelligent agents. However, most existing world models rely heavily on substantial action-labeled data and costly training, making it challenging to adapt to novel environments with heterogeneous actions through limited interactions.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策。

证据层级: 索引摘要。

1.3.6.155 Adaptive Sample Sharing for Multi Agent Linear Bandits

- 作者: Cherkaoui, Hamza; Barlier, Merwan; Colin, Igor
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Adaptive Sample Sharing for Multi Agent Linear Bandits

快速理解: The multi-agent linear bandit setting is a well-known setting for which designing efficient collaboration between agents remains challenging. This paper studies the impact of data sharing among agents on regret minimization.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.156 Adaptive Self-improvement LLM Agentic System for ML Library Development

- 作者: Zhang, Genghan; Liang, Weixin; Hsu, Olivia; Olukotun, Kunle
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Adaptive Self-improvement LLM Agentic System for ML Library Development

快速理解: ML libraries, often written in architecture-specific programming languages (ASPLs) that target domain-specific architectures, are key to efficient ML systems. However, writing these high-performance ML libraries is challenging because it requires expert knowledge of both ML algorithms and the ASPL.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.157 Adversarial Inception Backdoor Attacks against Reinforcement Learning

- 作者: Rathbun, Ethan; Oprea, Alina; Amato, Christopher
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Adversarial Inception Backdoor Attacks against Reinforcement Learning

快速理解: Recent works have demonstrated the vulnerability of Deep Reinforcement Learning (DRL) algorithms against training-time, backdoor poisoning attacks. The objectives of these attacks are twofold: induce pre-determined, adversarial behavior in the agent upon observing a fixed trigger

during deployment while allowing the agent to solve its intended task during training.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.158 AffectGPT: A New Dataset, Model, and Benchmark for Emotion Understanding with Multimodal Large Language Models

- 作者: Lian, Zheng; Chen, Haoyu; Chen, Lan; Sun, Haiyang; Sun, Licai; Ren, Yong; Cheng, Zebang; Liu, Bin; Liu, Rui; Peng, Xiaojiang; Yi, Jiangyan; Tao, Jianhua
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: AffectGPT: A New Dataset, Model, and Benchmark for Emotion Understanding with Multimodal Large Language Models

快速理解: The emergence of multimodal large language models (MLLMs) advances multimodal emotion recognition (MER) to the next level—from naive discriminative tasks to complex emotion understanding with advanced video understanding abilities and natural language description. However, the current community suffers from a lack of large-scale datasets with intensive, descriptive emotion annotations, as well as a multimodal-centric framework to maximize the potential of MLLMs for emotion understanding.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.6.159 Agent-Centric Actor-Critic for Asynchronous Multi-Agent Reinforcement Learning

- 作者: Jung, Whiyoung; Hong, Sunghoon; Yoon, Deunsol; Lee, Kanghoon; Lim, Woohyung
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Agent-Centric Actor-Critic for Asynchronous Multi-Agent Reinforcement Learning

快速理解: Multi-Agent Reinforcement Learning (MARL) struggles with coordination in sparse reward environments. Macro-actions —sequences of actions executed as single decisions—facilitate long-term planning but introduce asynchrony, complicating Centralized Training with Decentralized Execution (CTDE).

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.160 Aligning Multimodal Representations through an Information Bottleneck

- 作者: Almud' {e}var, Antonio; Hern' {a}ndez-Lobato, Jos' {e} Miguel; Khurana, Sameer; Marxer, Ricard; Ortega, Alfonso
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Aligning Multimodal Representations through an Information Bottleneck

快速理解: Contrastive losses have been extensively used as a tool for multimodal representation

learning. However, it has been empirically observed that their use is not effective to learn an aligned representation space.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.6.161 AlphaQCM: Alpha Discovery in Finance with Distributional Reinforcement Learning

- 作者：Zhu, Zhoufan; Zhu, Ke
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：AlphaQCM: Alpha Discovery in Finance with Distributional Reinforcement Learning

快速理解：For researchers and practitioners in finance, finding synergistic formulaic alphas is very important but challenging. In this paper, we reconsider the discovery of synergistic formulaic alphas from the viewpoint of sequential decision-making, and conceptualize the entire alpha discovery process as a non-stationary and reward-sparse Markov decision process.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.162 Arrow: Accelerator for Time Series Causal Discovery with Time Weaving

- 作者：Yao, Yuanyuan; Dong, Yuan; Chen, Lu; Kuang, Kun; Fang, Ziquan; Long, Cheng; Gao, Yunjun; Li, Tianyi
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：Arrow: Accelerator for Time Series Causal Discovery with Time Weaving

快速理解：Current causal discovery methods for time series data can effectively address a variety of scenarios; however, they remain constrained by inefficiencies. The significant inefficiencies arise primarily from the high computational costs associated with binning, the uncertainty in selecting appropriate time lags, and the extensive sets of candidate variables.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.6.163 Assessing Safety Risks and Quantization-aware Safety Patching for Quantized Large Language Models

- 作者：Chen, Kejia; Zhang, Jiawen; Hu, Jiacong; Wang, Yu; Lou, Jian; Feng, Zunlei; Song, Mingli
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：Assessing Safety Risks and Quantization-aware Safety Patching for Quantized Large Language Models

快速理解：Quantized large language models (LLMs) have gained increasing attention and significance for enabling deployment in resource-constrained environments. However, emerging studies on a few calibration dataset-free quantization methods suggest that quantization may compromise the safety

capabilities of LLMs, underscoring the urgent need for systematic safety evaluations and effective mitigation strategies.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.6.164 Auto-reconfiguration for Latency Minimization in CPU-based DNN Serving

- 作者：Bhardwaj, Ankit; Phanishayee, Amar; Narayanan, Deepak; Stutsman, Ryan
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：Auto-reconfiguration for Latency Minimization in CPU-based DNN Serving

快速理解：In this paper, we investigate how to push the performance limits of serving Deep Neural Network (DNN) models on CPU-based servers. Specifically, we observe that while intra-operator parallelism across multiple threads is an effective way to reduce inference latency, it provides diminishing returns.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.6.165 AutoML-Agent: A Multi-Agent LLM Framework for Full-Pipeline AutoML

- 作者：Trirat, Patara; Jeong, Wonyong; Hwang, Sung Ju
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：AutoML-Agent: A Multi-Agent LLM Framework for Full-Pipeline AutoML

快速理解：Automated machine learning (AutoML) accelerates AI development by automating tasks in the development pipeline, such as optimal model search and hyperparameter tuning. Existing AutoML systems often require technical expertise to set up complex tools, which is in general time-consuming and requires a large amount of human effort.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.166 Automated Hypothesis Validation with Agentic Sequential Falsifications

- 作者：Huang, Kexin; Jin, Ying; Li, Ryan; Li, Michael Y.; Candes, Emmanuel; Leskovec, Jure
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Automated Hypothesis Validation with Agentic Sequential Falsifications

快速理解：Hypotheses are central to information acquisition, decision-making, and discovery. However, many real-world hypotheses are abstract, high-level statements that are difficult to validate directly.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.167 BRIDGE: Bootstrapping Text to Control Time-Series Generation via Multi-Agent Iterative Optimization and Diffusion Modeling

- 作者: Li, Hao; Huang, Yu-Hao; Xu, Chang; Schlegel, Viktor; Jiang, Renhe; Batista-Navarro, Riza; Nenadic, Goran; Bian, Jiang
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: BRIDGE: Bootstrapping Text to Control Time-Series Generation via Multi-Agent Iterative Optimization and Diffusion Modeling

快速理解: Time-series Generation (TSG) is a prominent research area with broad applications in simulations, data augmentation, and counterfactual analysis. While existing methods have shown promise in unconditional single-domain TSG, real-world applications demand for cross-domain approaches capable of controlled generation tailored to domain-specific constraints and instance-level requirements.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.168 Behavior-Regularized Diffusion Policy Optimization for Offline Reinforcement Learning

- 作者: Gao, Chen-Xiao; Wu, Chenyang; Cao, Mingjun; Xiao, Chenjun; Yu, Yang; Zhang, Zongzhang
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Behavior-Regularized Diffusion Policy Optimization for Offline Reinforcement Learning

快速理解: Behavior regularization, which constrains the policy to stay close to some behavior policy, is widely used in offline reinforcement learning (RL) to manage the risk of hazardous exploitation of unseen actions. Nevertheless, existing literature on behavior-regularized RL primarily focuses on explicit policy parameterizations, such as Gaussian policies.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.169 Behavior-agnostic Task Inference for Robust Offline In-context Reinforcement Learning

- 作者: Ma, Long; Zhong, Fangwei; Wang, Yizhou
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Behavior-agnostic Task Inference for Robust Offline In-context Reinforcement Learning

快速理解: The ability to adapt to new environments with noisy dynamics and unseen objectives is crucial for AI agents. In-context reinforcement learning (ICRL) has emerged as a paradigm to build adaptive policies, employing a context trajectory of the test-time interactions to infer the true task and the corresponding optimal policy efficiently without gradient updates.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.170 Bellman Unbiasedness: Toward Provably Efficient Distributional Reinforcement Learning with General Value Function Approximation

- 作者: Cho, Taehyun; Han, Seungyub; Ju, Seokhun; Kim, Dohyeong; Lee, Kyungjae; Lee, Jungwoo
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Bellman Unbiasedness: Toward Provably Efficient Distributional Reinforcement Learning with General Value Function Approximation

快速理解: Distributional reinforcement learning improves performance by capturing environmental stochasticity, but a comprehensive theoretical understanding of its effectiveness remains elusive. In addition, the intractable element of the infinite dimensionality of distributions has been overlooked.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.171 Benchmarking Quantum Reinforcement Learning

- 作者: Meyer, Nico; Ufrecht, Christian; Yammine, George; Kontes, Georgios; Mutschler, Christopher; Scherer, Daniel
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Benchmarking Quantum Reinforcement Learning

快速理解: Benchmarking and establishing proper statistical validation metrics for reinforcement learning (RL) remain ongoing challenges, where no consensus has been established yet. The emergence of quantum computing and its potential applications in quantum reinforcement learning (QRL) further complicate benchmarking efforts.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.172 Beyond CVaR: Leveraging Static Spectral Risk Measures for Enhanced Decision-Making in Distributional Reinforcement Learning

- 作者: Moghimi, Mehrdad; Ku, Hyejin
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Beyond CVaR: Leveraging Static Spectral Risk Measures for Enhanced Decision-Making in Distributional Reinforcement Learning

快速理解: In domains such as finance, healthcare, and robotics, managing worst-case scenarios is critical, as failure to do so can lead to catastrophic outcomes. Distributional Reinforcement Learning (DRL) provides a natural framework to incorporate risk sensitivity into decision-making processes.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.173 Beyond The Rainbow: High Performance Deep Reinforcement Learning on a Desktop PC

- 作者: Clark, Tyler; Towers, Mark; Evers, Christine; Hare, Jonathon
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Beyond The Rainbow: High Performance Deep Reinforcement Learning on a Desktop PC

快速理解: Rainbow Deep Q-Network (DQN) demonstrated combining multiple independent enhancements could significantly boost a reinforcement learning (RL) agent's performance. In this paper, we present "Beyond The Rainbow" (BTR), a novel algorithm that integrates six improvements from across the RL literature to Rainbow DQN, establishing a new state-of-the-art for RL using a desktop PC, with a human-normalized interquartile mean (IQM) of 7.6 on Atari-60.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.174 BlockDialect: Block-wise Fine-grained Mixed Format Quantization for Energy-Efficient LLM Inference

- 作者: Jang, Wonsuk; Tambe, Thierry
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: BlockDialect: Block-wise Fine-grained Mixed Format Quantization for Energy-Efficient LLM Inference

快速理解: The rapidly increasing size of large language models (LLMs) presents significant challenges in memory usage and computational costs. Quantizing both weights and activations can address these issues, with hardware-supported fine-grained scaling emerging as a promising solution to mitigate outliers.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.6.175 BoA: Attention-aware Post-training Quantization without Backpropagation

- 作者: Kim, Junhan; Kim, Ho-Young; Cho, Eulrang; Lee, Chungman; Kim, Joonyoung; Jeon, Yongkweon
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: BoA: Attention-aware Post-training Quantization without Backpropagation

快速理解: Post-training quantization (PTQ) is a promising solution for deploying large language models (LLMs) on resource-constrained devices. Early methods developed for small-scale networks, such as ResNet, rely on gradient-based optimization, which becomes impractical for hyper-scale LLMs with billions of parameters.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.6.176 Boosting Virtual Agent Learning and Reasoning: A Step-Wise, Multi-Dimensional, and Generalist Reward Model with Benchmark

- 作者: Miao, Bingchen; Wu, Yang; Gao, Minghe; Yu, Qifan; Bu, Wendong; Zhang, Wenqiao; Li, Yunfei; Tang, Siliang; Chua, Tat-Seng; Li, Juncheng
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Boosting Virtual Agent Learning and Reasoning: A Step-Wise, Multi-Dimensional, and Generalist Reward Model with Benchmark

快速理解: The development of Generalist Virtual Agents (GVAs) has shown significant promise in autonomous task execution. However, current training paradigms face critical limitations, including reliance on outcome supervision and labor-intensive human annotations.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.177 C2IQL: Constraint-Conditioned Implicit Q-learning for Safe Offline Reinforcement Learning

- 作者: Liu, Zifan; Li, Xinran; Zhang, Jun
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: C2IQL: Constraint-Conditioned Implicit Q-learning for Safe Offline Reinforcement Learning

快速理解: Safe offline reinforcement learning aims to develop policies that maximize cumulative rewards while satisfying safety constraints without the need for risky online interaction. However, existing methods often struggle with the out-of-distribution (OOD) problem, leading to potentially unsafe and suboptimal policies.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.178 CERTAIN: Context Uncertainty-aware One-Shot Adaptation for Context-based Offline Meta Reinforcement Learning

- 作者: Zhou, Hongtu; Yang, Ruiling; Zhu, Yakun; Zhao, Haoqi; Zhang, Hai; Zhang, Di; Zhao, Junqiao; Ye, Chen; Jiang, Changjun
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: CERTAIN: Context Uncertainty-aware One-Shot Adaptation for Context-based Offline Meta Reinforcement Learning

快速理解: Existing context-based offline meta-reinforcement learning (COMRL) methods primarily focus on task representation learning and given-context adaptation performance. They often assume that the adaptation context is collected using task-specific behavior policies or through multiple rounds of collection.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.179 CLARIFY: Contrastive Preference Reinforcement Learning for Untangling Ambiguous Queries

- 作者: Mu, Ni; Hu, Hao; Hu, Xiao; Yang, Yiqin; Xu, Bo; Jia, Qing-Shan
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: CLARIFY: Contrastive Preference Reinforcement Learning for Untangling Ambiguous Queries

快速理解: Preference-based reinforcement learning (PbRL) bypasses explicit reward engineering by inferring reward functions from human preference comparisons, enabling better alignment with human intentions. However, humans often struggle to label a clear preference between similar segments, reducing label efficiency and limiting PbRL's real-world applicability.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.180 CLIMB: Data Foundations for Large Scale Multimodal Clinical Foundation Models

- 作者: Dai, Wei; Chen, Peilin; Lu, Malinda; Li, Daniel A; Wei, Haowen; Cui, Hejie; Liang, Paul Pu
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: CLIMB: Data Foundations for Large Scale Multimodal Clinical Foundation Models

快速理解: Recent advances in clinical AI have enabled remarkable progress across many clinical domains. However, existing benchmarks and models are primarily limited to a small set of modalities and tasks, which hinders the development of large-scale multimodal methods that can make holistic assessments of patient health and well-being.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.181 CPCF: A Cross-Prompt Contrastive Framework for Referring Multimodal Large Language Models

- 作者: Zhu, Lanyun; Ji, Deyi; Chen, Tianrun; Wu, Haiyang; Soh, De Wen; Liu, Jun
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: CPCF: A Cross-Prompt Contrastive Framework for Referring Multimodal Large Language Models

快速理解: Referring MLLMs extend conventional multimodal large language models by allowing them to receive referring visual prompts and generate responses tailored to the indicated regions. However, these models often suffer from suboptimal performance due to incorrect responses tailored to misleading areas adjacent to or similar to the target region.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.182 Can Large Language Models Understand Intermediate Representations in Compilers?

- 作者: Jiang, Hailong; Zhu, Jianfeng; Wan, Yao; Fang, Bo; Zhang, Hongyu; Jin, Ruoming; Guan, Qiang
- 年份: 2025; 主题: Systems; 优先级: 重要
- [来源](#)

论文: Can Large Language Models Understand Intermediate Representations in Compilers?

快速理解: Intermediate Representations (IRs) play a critical role in compiler design and program analysis, yet their comprehension by Large Language Models (LLMs) remains underexplored. In this paper, we present an explorative empirical study evaluating the capabilities of six state-of-the-art LLMs—GPT-4, GPT-3, DeepSeek, Gemma 2, Llama 3, and Code Llama—in understanding IRs.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.6.183 Can MLLMs Reason in Multimodality? EMMA: An Enhanced MultiModal ReAsoning Benchmark

- 作者: Hao, Yunzhuo; Gu, Jiawei; Wang, Huichen Will; Li, Linjie; Yang, Zhengyuan; Wang, Lijuan; Cheng, Yu
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Can MLLMs Reason in Multimodality? EMMA: An Enhanced MultiModal ReAsoning Benchmark

快速理解: The ability to organically reason over and with both text and images is a pillar of human intelligence, yet the ability of Multimodal Large Language Models (MLLMs) to perform such multimodal reasoning remains under-explored. Existing benchmarks often emphasize text-dominant reasoning or rely on shallow visual cues, failing to adequately assess integrated visual and textual reasoning.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.184 Can RLHF be More Efficient with Imperfect Reward Models? A Policy Coverage Perspective

- 作者: Huang, Jiawei; Li, Bingcong; Dann, Christoph; He, Niao
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Can RLHF be More Efficient with Imperfect Reward Models? A Policy Coverage Perspective

快速理解: Sample efficiency is critical for online Reinforcement Learning from Human Feedback (RLHF). While existing works investigate sample-efficient online exploration strategies, the potential of utilizing misspecified yet relevant reward models to accelerate learning remains underexplored.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.185 Can We Predict Performance of Large Models across Vision-Language Tasks?

- 作者：Zhao, Qinyu; Xu, Ming; Gupta, Kartik; Asthana, Akshay; Zheng, Liang; Gould, Stephen
- 年份：2025; 主题：Supporting Perception; 优先级：重要
- [来源](#)

论文：Can We Predict Performance of Large Models across Vision-Language Tasks?

快速理解：Evaluating large vision-language models (LVLMs) is very expensive, due to high computational cost and the wide variety of tasks. The good news is that if we already have some observed performance scores, we may be able to infer unknown ones.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.6.186 Catch Your Emotion: Sharpening Emotion Perception in Multimodal Large Language Models

- 作者：Fang, Yiyang; Liang, Jian; Huang, Wenke; Li, He; Su, Kehua; Ye, Mang
- 年份：2025; 主题：Supporting Perception; 优先级：重要
- [来源](#)

论文：Catch Your Emotion: Sharpening Emotion Perception in Multimodal Large Language Models

快速理解：Multimodal large language models (MLLMs) have achieved impressive progress in tasks such as visual question answering and visual understanding, but they still face significant challenges in emotional reasoning. Current methods to enhance emotional understanding typically rely on fine-tuning or manual annotations, which are resource-intensive and limit scalability.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.6.187 Categorical Distributional Reinforcement Learning with Kullback-Leibler Divergence: Convergence and Asymptotics

- 作者：Kastner, Tyler; Rowland, Mark; Tang, Yunhao; Erdogdu, Murat A; Farahmand, Amir-Massoud
- 年份：2025; 主题：Agent RL; 优先级：重要
- [来源](#)

论文：Categorical Distributional Reinforcement Learning with Kullback-Leibler Divergence: Convergence and Asymptotics

快速理解：We study the problem of distributional reinforcement learning using categorical parametrisations and a KL divergence loss. Previous work analyzing categorical distributional RL has done so using a Cramér distance-based loss, simplifying the analysis but creating a theory-practice gap.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.188 Combinatorial Reinforcement Learning with Preference Feedback

- 作者: Lee, Joongkyu; Oh, Min-Hwan
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Combinatorial Reinforcement Learning with Preference Feedback

快速理解: In this paper, we consider combinatorial reinforcement learning with preference feedback, where a learning agent sequentially offers an action—an assortment of multiple items—to a user, whose preference feedback follows a multinomial logistic (MNL) model. This framework allows us to model real-world scenarios, particularly those involving long-term user engagement, such as in recommender systems and online advertising.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.189 Compositional Scene Understanding through Inverse Generative Modeling

- 作者: Wang, Yanbo; Dauwels, Justin; Du, Yilun
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Compositional Scene Understanding through Inverse Generative Modeling

快速理解: Generative models have demonstrated remarkable abilities in generating high-fidelity visual content. In this work, we explore how generative models can further be used not only to synthesize visual content but also to understand the properties of a scene given a natural image.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.190 Compression via Pre-trained Transformers: A Study on Byte-Level Multimodal Data

- 作者: Heurtel-Depeiges, David; Ruoss, Anian; Veness, Joel; Genewein, Tim
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Compression via Pre-trained Transformers: A Study on Byte-Level Multimodal Data

快速理解: Foundation models are strong data compressors, but when accounting for their parameter size, their compression ratios are inferior to standard compression algorithms. Naively reducing the parameter count does not necessarily help as it deteriorates predictions and, accordingly, compression.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.191 Concurrent Reinforcement Learning with Aggregated States via Randomized Least Squares Value Iteration

- 作者: Chen, Yan; Bai, Qinxun; Zhang, Yiteng; Dimakopoulou, Maria; Dong, Shi; Sun, Qi; Zhou,

Zhengyuan

- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Concurrent Reinforcement Learning with Aggregated States via Randomized Least Squares Value Iteration

快速理解: Designing learning agents that explore efficiently in a complex environment has been widely recognized as a fundamental challenge in reinforcement learning. While a number of works have demonstrated the effectiveness of techniques based on randomized value functions on a single agent, it remains unclear, from a theoretical point of view, whether injecting randomization can help a society of agents concurrently explore an environment.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.192 Constrained Exploitability Descent: An Offline Reinforcement Learning Method for Finding Mixed-Strategy Nash Equilibrium

- 作者: Lu, Runyu; Zhu, Yuanheng; Zhao, Dongbin
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Constrained Exploitability Descent: An Offline Reinforcement Learning Method for Finding Mixed-Strategy Nash Equilibrium

快速理解: This paper proposes Constrained Exploitability Descent (CED), a model-free offline reinforcement learning (RL) algorithm for solving adversarial Markov games (MGs). CED combines the game-theoretical approach of Exploitability Descent (ED) with policy constraint methods from offline RL.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.193 Controlling Underestimation Bias in Constrained Reinforcement Learning for Safe Exploration

- 作者: Gao, Shiqing; Ding, Jiabin; Fu, Luoyi; Wang, Xinbing
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Controlling Underestimation Bias in Constrained Reinforcement Learning for Safe Exploration

快速理解: Constrained Reinforcement Learning (CRL) aims to maximize cumulative rewards while satisfying constraints. However, existing CRL algorithms often encounter significant constraint violations during training, limiting their applicability in safety-critical scenarios.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.194 Convex Markov Games: A New Frontier for Multi-Agent Reinforcement Learning

- 作者: Gemp, Ian; Haupt, Andreas Alexander; Marris, Luke; Liu, Siqu; Piliouras, Georgios
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Convex Markov Games: A New Frontier for Multi-Agent Reinforcement Learning

快速理解: Behavioral diversity, expert imitation, fairness, safety goals and others give rise to preferences in sequential decision making domains that do not decompose additively across time. We introduce the class of convex Markov games that allow general convex preferences over occupancy measures.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.195 CoreMatching: A Co-adaptive Sparse Inference Framework with Token and Neuron Pruning for Comprehensive Acceleration of Vision-Language Models

- 作者: Wang, Qinsi; Ye, Hancheng; Chung, Ming-Yu; Liu, Yudong; Lin, Yueqian; Kuo, Martin; Ma, Mingyuan; Zhang, Jianyi; Chen, Yiran
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: CoreMatching: A Co-adaptive Sparse Inference Framework with Token and Neuron Pruning for Comprehensive Acceleration of Vision-Language Models

快速理解: Vision-Language Models (VLMs) excel across diverse tasks but suffer from high inference costs in time and memory. Token sparsity mitigates inefficiencies in token usage, while neuron sparsity reduces high-dimensional computations, both offering promising solutions to enhance efficiency.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.196 Cost-efficient Collaboration between On-device and Cloud Language Models

- 作者: Narayan, Avanika; Biderman, Dan; Eyuboglu, Sabri; May, Avner; Linderman, Scott; Zou, James; Re, Christopher
- 年份: 2025; 主题: Systems; 优先级: 重要
- [来源](#)

论文: Cost-efficient Collaboration between On-device and Cloud Language Models

快速理解: We investigate an emerging setup in which a small, on-device language model (LM) with access to local data collaborates with a frontier, cloud-hosted LM to solve real-world tasks involving financial, medical, and scientific reasoning over long documents. Can a local-remote collaboration reduce cloud inference costs while preserving quality? First, we consider a naïve collaboration protocol, coined MINION, where the local and remote models simply chat back and forth. Because only the local model ingests the full context, this protocol reduces cloud costs by 30.4x, but recovers only 87% of the performance of the frontier model.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.6.197 Counterfactual Effect Decomposition in Multi-Agent Sequential Decision Making

- 作者: Triantafyllou, Stelios; Sukovic, Aleksa; Zolfimoselo, Yasaman; Radanovic, Goran
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Counterfactual Effect Decomposition in Multi-Agent Sequential Decision Making

快速理解: We address the challenge of explaining counterfactual outcomes in multi-agent Markov decision processes. In particular, we aim to explain the total counterfactual effect of an agent's action on the outcome of a realized scenario through its influence on the environment dynamics and the agents' behavior.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.198 Craftium: Bridging Flexibility and Efficiency for Rich 3D Single- and Multi-Agent Environments

- 作者: Malag' {o}n, Mikel; Ceberio, Josu; Lozano, Jose A.
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Craftium: Bridging Flexibility and Efficiency for Rich 3D Single- and Multi-Agent Environments

快速理解: Advances in large models, reinforcement learning, and open-endedness have accelerated progress toward autonomous agents that can learn and interact in the real world. To achieve this, flexible tools are needed to create rich, yet computationally efficient, environments.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.199 Cross-environment Cooperation Enables Zero-shot Multi-agent Coordination

- 作者: Jha, Kunal; Carvalho, Wilka; Liang, Yancheng; Du, Simon Shaolei; Kleiman-Weiner, Max; Jaques, Natasha
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Cross-environment Cooperation Enables Zero-shot Multi-agent Coordination

快速理解: Zero-shot coordination (ZSC), the ability to adapt to a new partner in a cooperative task, is a critical component of human-compatible AI. While prior work has focused on training agents to cooperate on a single task, these specialized models do not generalize to new tasks, even if they are highly similar.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.200 CtrlSynth: Controllable Image Text Synthesis for Data-Efficient Multimodal Learning

- 作者: Cao, Qingqing; Najibi, Mahyar; Mehta, Sachin
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: CtrlSynth: Controllable Image Text Synthesis for Data-Efficient Multimodal Learning

快速理解: Pretraining robust vision or multimodal foundation models (e.g., CLIP) relies on large-scale datasets that may be noisy, potentially misaligned, and have long-tail distributions. Previous works have shown promising results in augmenting datasets by generating synthetic samples.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.201 DEFAME: Dynamic Evidence-based FAct-checking with Multimodal Experts

- 作者: Braun, Tobias; Rothermel, Mark; Rohrbach, Marcus; Rohrbach, Anna
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: DEFAME: Dynamic Evidence-based FAct-checking with Multimodal Experts

快速理解: The proliferation of disinformation demands reliable and scalable fact-checking solutions. We present Dynamic Evidence-based FAct-checking with Multimodal Experts (DEFAME), a modular, zero-shot MLLM pipeline for open-domain, text-image claim verification.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.202 DIME: Diffusion-Based Maximum Entropy Reinforcement Learning

- 作者: Celik, Onur; Li, Zechu; Blessing, Denis; Li, Ge; Palenicek, Daniel; Peters, Jan; Chalvatzaki, Georgia; Neumann, Gerhard
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: DIME: Diffusion-Based Maximum Entropy Reinforcement Learning

快速理解: Maximum entropy reinforcement learning (MaxEnt-RL) has become the standard approach to RL due to its beneficial exploration properties. Traditionally, policies are parameterized using Gaussian distributions, which significantly limits their representational capacity.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.203 DS-VLM: Diffusion Supervision Vision Language Model

- 作者: Sun, Zhen; Shen, Yunhang; Li, Jie; Sun, Xing; Dai, Pingyang; Cao, Liujuan; Ji, Rongrong
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: DS-VLM: Diffusion Supervision Vision Language Model

快速理解: Vision-Language Models (VLMs) face two critical limitations in visual representation learning: degraded supervision due to information loss during gradient propagation, and the inherent semantic sparsity of textual supervision compared to visual data. We propose the Diffusion Supervision Vision-Language Model (DS-VLM), a plug-and-play framework that introduces diffusion-based direct supervision for vision-language alignment.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.204 Data-Juicer Sandbox: A Feedback-Driven Suite for Multimodal Data-Model Co-development

- 作者: Chen, Daoyuan; Wang, Haibin; Huang, Yilun; Ge, Ce; Li, Yaliang; Ding, Bolin; Zhou, Jingren
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Data-Juicer Sandbox: A Feedback-Driven Suite for Multimodal Data-Model Co-development

快速理解: The emergence of multimodal large models has advanced artificial intelligence, introducing unprecedented levels of performance and functionality. However, optimizing these models remains challenging due to historically isolated paths of model-centric and data-centric developments, leading to suboptimal outcomes and inefficient resource utilization.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.205 Decomposition of Graphic Design with Unified Multimodal Model

- 作者: Nie, Hui; Zhang, Zhao; Cheng, Yutao; Yang, Maoke; Shi, Gonglei; Xie, Qingsong; Shao, Jie; Wu, Xinglong
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Decomposition of Graphic Design with Unified Multimodal Model

快速理解: We propose Layer Decomposition of Graphic Designs (LDGD), a novel vision task that converts composite graphic design (e.g., posters) into structured representations comprising ordered RGB-A layers and metadata. By transforming visual content into structured data, LDGD facilitates precise image editing and offers significant advantages for digital content creation, management, and reuse.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.206 Deep Reinforcement Learning from Hierarchical Preference Design

- 作者: Bukharin, Alexander; Li, Yixiao; He, Pengcheng; Zhao, Tuo
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Deep Reinforcement Learning from Hierarchical Preference Design

快速理解: Reward design is a fundamental, yet challenging aspect of reinforcement learning (RL). Researchers typically utilize feedback signals from the environment to handcraft a reward function, but this process is not always effective due to the varying scale and intricate dependencies of the feedback signals.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.207 Deliberation in Latent Space via Differentiable Cache Augmentation

- 作者: Liu, Luyang; Pfeiffer, Jonas; Wu, Jiaxing; Xie, Jun; Szlam, Arthur
- 年份: 2025; 主题: Systems; 优先级: 重要
- [来源](#)

论文: Deliberation in Latent Space via Differentiable Cache Augmentation

快速理解: Techniques enabling large language models (LLMs) to “think more” by generating and attending to intermediate reasoning steps have shown promise in solving complex problems. However, the standard approaches generate sequences of discrete tokens immediately before responding, and so they can incur significant latency costs and be challenging to optimize.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.6.208 Demonstration Selection for In-Context Learning via Reinforcement Learning

- 作者: Wang, Xubin; Wu, Jianfei; Yichen, Yuan; Cai, Deyu; Li, Mingzhe; Jia, Weijia
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Demonstration Selection for In-Context Learning via Reinforcement Learning

快速理解: Diversity in demonstration selection is critical for enhancing model generalization by enabling broader coverage of structures and concepts. Constructing appropriate demonstration sets remains a key research challenge.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.209 Dialogue Without Limits: Constant-Sized KV Caches for Extended Response in LLMs

- 作者: Ghadia, Ravi; Kumar, Avinash; Jain, Gaurav; Nair, Prashant J.; Das, Poulami
- 年份: 2025; 主题: Systems; 优先级: 重要
- [来源](#)

论文: Dialogue Without Limits: Constant-Sized KV Caches for Extended Response in LLMs

快速理解: Autoregressive Transformers rely on Key-Value (KV) caching to accelerate inference. However, the linear growth of the KV cache with context length leads to excessive memory consumption

and bandwidth constraints.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.6.210 Diffuse Everything: Multimodal Diffusion Models on Arbitrary State Spaces

- 作者: Rojas, Kevin; Zhu, Yuchen; Zhu, Sichen; Ye, Felix X-F.; Tao, Molei
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Diffuse Everything: Multimodal Diffusion Models on Arbitrary State Spaces

快速理解: Diffusion models have demonstrated remarkable performance in generating unimodal data across various tasks, including image, video, and text generation. On the contrary, the joint generation of multimodal data through diffusion models is still in the early stages of exploration.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.211 Diffusion Adversarial Post-Training for One-Step Video Generation

- 作者: Lin, Shanchuan; Xia, Xin; Ren, Yuxi; Yang, Ceyuan; Xiao, Xuefeng; Jiang, Lu
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Diffusion Adversarial Post-Training for One-Step Video Generation

快速理解: The diffusion models are widely used for image and video generation, but their iterative generation process is slow and expansive. While existing distillation approaches have demonstrated the potential for one-step generation in the image domain, they still suffer from significant quality degradation.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.212 Directly Forecasting Belief for Reinforcement Learning with Delays

- 作者: Wu, Qingyuan; Wang, Yuhui; Zhan, Simon Sinong; Wang, Yixuan; Lin, Chung-Wei; Lv, Chen; Zhu, Qi; Schmidhuber, J" {u}rgen; Huang, Chao
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Directly Forecasting Belief for Reinforcement Learning with Delays

快速理解: Reinforcement learning (RL) with delays is challenging as sensory perceptions lag behind the actual events: the RL agent needs to estimate the real state of its environment based on past observations. State-of-the-art (SOTA) methods typically employ recursive, step-by-step forecasting of states.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.213 Discriminative Finetuning of Generative Large Language Models without Reward Models and Human Preference Data

- 作者: Guo, Siqi; Hong, Ilgee; Balmaseda, Vicente; Yu, Changlong; Qiu, Liang; Liu, Xin; Jiang, Haoming; Zhao, Tuo; Yang, Tianbao
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Discriminative Finetuning of Generative Large Language Models without Reward Models and Human Preference Data

快速理解: Supervised fine-tuning (SFT) has become a crucial step for aligning pretrained large language models (LLMs) using supervised datasets of input-output pairs. However, despite being supervised, SFT is inherently limited by its generative training objective.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.214 Discriminative Policy Optimization for Token-Level Reward Models

- 作者: Chen, Hongzhan; Yang, Tao; Gao, Shiping; Chen, Ruijun; Quan, Xiaojun; Tian, Hongtao; Yao, Ting
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Discriminative Policy Optimization for Token-Level Reward Models

快速理解: Process reward models (PRMs) provide more nuanced supervision compared to outcome reward models (ORMs) for optimizing policy models, positioning them as a promising approach to enhancing the capabilities of LLMs in complex reasoning tasks. Recent efforts have advanced PRMs from step-level to token-level granularity by integrating reward modeling into the training of generative models, with reward scores derived from token generation probabilities.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.215 Distributionally Robust Multi-Agent Reinforcement Learning for Dynamic Chute Mapping

- 作者: Liu, Guangyi; Iloglu, Suzan; Caldara, Michael; Durham, Joseph W; Zavlanos, Michael M.
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Distributionally Robust Multi-Agent Reinforcement Learning for Dynamic Chute Mapping

快速理解: In Amazon robotic warehouses, the destination-to-chute mapping problem is crucial for efficient package sorting. Often, however, this problem is complicated by uncertain and dynamic package induction rates, which can lead to increased package recirculation.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.216 Divide and Conquer: Grounding LLMs as Efficient Decision-Making Agents via Offline Hierarchical Reinforcement Learning

- 作者: Hu, Zican; Liu, Wei; Qu, Xiaoye; Yue, Xiangyu; Chen, Chunlin; Wang, Zhi; Cheng, Yu
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Divide and Conquer: Grounding LLMs as Efficient Decision-Making Agents via Offline Hierarchical Reinforcement Learning

快速理解: While showing sophisticated reasoning abilities, large language models (LLMs) still struggle with long-horizon decision-making tasks due to deficient exploration and long-term credit assignment, especially in sparse-reward scenarios. Inspired by the divide-and-conquer principle, we propose an innovative framework GLIDER (Grounding Language Models as Efficient Decision-Making Agents via Offline HiErarchical Reinforcement Learning) that introduces a parameter-efficient and generally applicable hierarchy to LLM policies.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.217 Do Vision-Language Models Really Understand Visual Language?

- 作者: Hou, Yifan; Giledereli, Buse; Tu, Yilei; Sachan, Mrinmaya
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Do Vision-Language Models Really Understand Visual Language?

快速理解: Visual language is a system of communication that conveys information through symbols, shapes, and spatial arrangements. Diagrams are a typical example of a visual language depicting complex concepts and their relationships in the form of an image.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.218 Double-Filter: Efficient Fine-tuning of Pre-trained Vision-Language Models via Patch&Layer Filtering

- 作者: He, Yaoqin; Fu, Junchen; Zheng, Kaiwen; Xu, Songpei; Chen, Fuhai; Li, Jie; Jose, Joemon M.; Ge, Xuri
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Double-Filter: Efficient Fine-tuning of Pre-trained Vision-Language Models via Patch&Layer Filtering

快速理解: In this paper, we present a novel approach, termed Double-Filter, to “slim down” the fine-tuning process of vision-language pre-trained (VLP) models via filtering redundancies in feature inputs and architectural components. We enhance the fine-tuning process using two approaches.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.219 DyPolySeg: Taylor Series-Inspired Dynamic Polynomial Fitting Network for Few-shot Point Cloud Semantic Segmentation

- 作者: Wang, Changshuo; Fang, Xiang; Tiwari, Prayag
- 年份: 2025; 主题: Systems; 优先级: 重要
- [来源](#)

论文: DyPolySeg: Taylor Series-Inspired Dynamic Polynomial Fitting Network for Few-shot Point Cloud Semantic Segmentation

快速理解: Few-shot point cloud semantic segmentation effectively addresses data scarcity by identifying unlabeled query samples through semantic prototypes generated from a small set of labeled support samples. However, pre-training-based methods suffer from domain shifts and increased training time.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.6.220 DynaMind: Reasoning over Abstract Video Dynamics for Embodied Decision-Making

- 作者: Wang, Zirui; Wang, Mengmeng; Dai, Jade; Ma, Teli; Qi, Guo-Jun; Liu, Yong; Dai, Guang; Wang, Jingdong
- 年份: 2025; 主题: Embodied/WBC; 优先级: 重要
- [来源](#)

论文: DynaMind: Reasoning over Abstract Video Dynamics for Embodied Decision-Making

快速理解: Integrating natural language instructions and visual perception with decision-making is a critical challenge for embodied agents. Existing methods often struggle to balance the conciseness of language commands with the richness of video content.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制。

证据层级: 索引摘要。

1.3.6.221 Dynamic Mixture of Curriculum LoRA Experts for Continual Multimodal Instruction Tuning

- 作者: Ge, Chendi; Wang, Xin; Zhang, Zeyang; Chen, Hong; Fan, Jiawei; Huang, Longtao; Xue, Hui; Zhu, Wenwu
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Dynamic Mixture of Curriculum LoRA Experts for Continual Multimodal Instruction Tuning

快速理解: Continual multimodal instruction tuning is crucial for adapting Multimodal Large Language Models (MLLMs) to evolving tasks. However, most existing methods adopt a fixed architecture, struggling with adapting to new tasks due to static model capacity.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.222 EARL-BO: Reinforcement Learning for Multi-Step Lookahead, High-Dimensional Bayesian Optimization

- 作者: Cheon, Mujin; Lee, Jay H; Koh, Dong-Yeun; Tsay, Calvin
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: EARL-BO: Reinforcement Learning for Multi-Step Lookahead, High-Dimensional Bayesian Optimization

快速理解: To avoid myopic behavior, multi-step lookahead Bayesian optimization (BO) algorithms consider the sequential nature of BO and have demonstrated promising results in recent years. However, owing to the curse of dimensionality, most of these methods make significant approximations or suffer scalability issues.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.223 ELMO : Efficiency via Low-precision and Peak Memory Optimization in Large Output Spaces

- 作者: Zhang, Jinbin; Ullah, Nasib; Schultheis, Erik; Babbar, Rohit
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: ELMO : Efficiency via Low-precision and Peak Memory Optimization in Large Output Spaces

快速理解: Large output spaces, also referred to as Extreme multilabel classification (XMC), is a setting that arises, e.g., in large-scale tagging and product-to-product recommendation, and is characterized by the number of labels ranging from hundreds of thousands to millions. This means that the linear classification head, usually only a tiny fraction of the overall model, turns into the main driver for compute and memory demand.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.6.224 ERICT: Enhancing Robustness by Identifying Concept Tokens in Zero-Shot Vision Language Models

- 作者: Dong, Xinpeng; Zhang, Min; Zhu, Didi; Jian, Ye Jun; Keli, Zhang; Zhou, Aimin; Wu, Fei; Kuang, Kun
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: ERICT: Enhancing Robustness by Identifying Concept Tokens in Zero-Shot Vision Language Models

快速理解: Pre-trained vision-language models (VLMs) have revolutionized the field of machine learning, demonstrating exceptional performance across a wide range of tasks. However, their robustness remains vulnerable to the spurious-correlation problem.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.225 EasyRef: Omni-Generalized Group Image Reference for Diffusion Models via Multimodal LLM

- 作者: Zong, Zhuofan; Jiang, Dongzhi; Ma, Bingqi; Song, Guanglu; Shao, Hao; Shen, Dazhong; Liu, Yu; Li, Hongsheng
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: EasyRef: Omni-Generalized Group Image Reference for Diffusion Models via Multimodal LLM

快速理解: Significant achievements in personalization of diffusion models have been witnessed. Conventional tuning-free methods mostly encode multiple reference images by averaging or concatenating their image embeddings as the injection condition, but such an image-independent operation cannot perform interaction among images to capture consistent visual elements within multiple references.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.226 Efficient Generative Modeling with Residual Vector Quantization-Based Tokens

- 作者: Kim, Jaehyeon; Moon, Taehong; Lee, Keon; Cho, Jaewoong
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Efficient Generative Modeling with Residual Vector Quantization-Based Tokens

快速理解: We introduce ResGen, an efficient Residual Vector Quantization (RVQ)-based generative model for high-fidelity generation with fast sampling. RVQ improves data fidelity by increasing the number of quantization steps, referred to as depth, but deeper quantization typically increases inference steps in generative models.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.6.227 Efficient Parallel Training Methods for Spiking Neural Networks with Constant Time Complexity

- 作者: Feng, Wanjin; Gao, Xingyu; Du, Wenqian; Shi, Hailong; Zhao, Peilin; Wu, Pengcheng; Miao, Chunyan
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Efficient Parallel Training Methods for Spiking Neural Networks with Constant Time Complexity

快速理解: Spiking Neural Networks (SNNs) often suffer from high time complexity $O(T)$ due to the sequential processing of T spikes, making training computationally expensive. In this paper, we propose a novel Fixed-point Parallel Training (FPT) method to accelerate SNN training without modifying the network architecture or introducing additional assumptions.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.6.228 Efficient Quantification of Multimodal Interaction at Sample Level

- 作者: Yang, Zequn; Wang, Hongfa; Hu, Di
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Efficient Quantification of Multimodal Interaction at Sample Level

快速理解: Interactions between modalities—redundancy, uniqueness, and synergy—collectively determine the composition of multimodal information. Understanding these interactions is crucial for analyzing information dynamics in multimodal systems, yet their accurate sample-level quantification presents significant theoretical and computational challenges.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.229 Efficiently Vectorized MCMC on Modern Accelerators

- 作者: Dance, Hugh; Glaser, Pierre; Orbanz, Peter; Adams, Ryan P
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Efficiently Vectorized MCMC on Modern Accelerators

快速理解: With the advent of automatic vectorization tools (e.g., JAX' s vmap), writing multi-chain MCMC algorithms is often now as simple as invoking those tools on single-chain code. Whilst convenient, for various MCMC algorithms this results in a synchronization problem—loosely speaking, at each iteration all chains running in parallel must wait until the last chain has finished drawing its sample.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.6.230 Elucidating the Design Space of Multimodal Protein Language Models

- 作者: Hsieh, Cheng-Yen; Wang, Xinyou; Zhang, Daiheng; Xue, Dongyu; Ye, Fei; Huang, Shujian; Zheng, Zaixiang; Gu, Quanquan
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Elucidating the Design Space of Multimodal Protein Language Models

快速理解: Multimodal protein language models (PLMs) integrate sequence and token-based structural information, serving as a powerful foundation for protein modeling, generation, and design. However, the reliance on tokenizing 3D structures into discrete tokens causes substantial loss of fidelity about fine-grained structural details and correlations.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.231 EmbodiedBench: Comprehensive Benchmarking Multi-modal Large Language Models for Vision-Driven Embodied Agents

- 作者: Yang, Rui; Chen, Hanyang; Zhang, Junyu; Zhao, Mark; Qian, Cheng; Wang, Kangrui; Wang, Qineng; Koripella, Teja Venkat; Movahedi, Marziyeh; Li, Manling; Ji, Heng; Zhang, Huan; Zhang, Tong
- 年份: 2025; 主题: Embodied/WBC; 优先级: 重要
- [来源](#)

论文: EmbodiedBench: Comprehensive Benchmarking Multi-modal Large Language Models for Vision-Driven Embodied Agents

快速理解: Leveraging Multi-modal Large Language Models (MLLMs) to create embodied agents offers a promising avenue for tackling real-world tasks. While language-centric embodied agents have garnered substantial attention, MLLM-based embodied agents remain underexplored due to the lack of comprehensive evaluation frameworks.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制。

证据层级: 索引摘要。

1.3.6.232 Enhancing Cooperative Multi-Agent Reinforcement Learning with State Modelling and Adversarial Exploration

- 作者: Kontogiannis, Andreas; Papathanasiou, Konstantinos; Shen, Yi; Stamou, Giorgos; Zavlanos, Michael M.; Vouros, George
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Enhancing Cooperative Multi-Agent Reinforcement Learning with State Modelling and Adversarial Exploration

快速理解: Learning to cooperate in distributed partially observable environments with no communication abilities poses significant challenges for multi-agent deep reinforcement learning (MARL). This paper addresses key concerns in this domain, focusing on inferring state representations from individual agent observations and leveraging these representations to enhance agents' exploration and collaborative task execution policies.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.233 Expert Race: A Flexible Routing Strategy for Scaling Diffusion Transformer with Mixture of Experts

- 作者: Yuan, Yike; Wang, Ziyu; Huang, Zihao; Zhu, Defa; Zhou, Xun; Yu, Jingyi; Min, Qiyang
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Expert Race: A Flexible Routing Strategy for Scaling Diffusion Transformer with Mixture of Experts

快速理解: Diffusion models have emerged as mainstream framework in visual generation. Building upon this success, the integration of Mixture of Experts (MoE) methods has shown promise in enhancing model scalability and performance.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.6.234 Explicit Exploration for High-Welfare Equilibria in Game-Theoretic Multi-agent Reinforcement Learning

- 作者：Nguyen, Austin A.; Gu, Anri; Wellman, Michael P.
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Explicit Exploration for High-Welfare Equilibria in Game-Theoretic Multiagent Reinforcement Learning

快速理解：Iterative extension of empirical game models through deep reinforcement learning (RL) has proved an effective approach for finding equilibria in complex games. When multiple equilibria exist, we may also be interested in finding solutions with particular characteristics.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.235 Explicit Preference Optimization: No Need for an Implicit Reward Model

- 作者：Hu, Xiangkun; Kong, Lemin; He, Tong; Wipf, David
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Explicit Preference Optimization: No Need for an Implicit Reward Model

快速理解：The generated responses of large language models (LLMs) are often fine-tuned to human preferences through a process called reinforcement learning from human feedback (RLHF). As RLHF relies on a challenging training sequence, whereby a separate reward model is independently learned and then later applied to LLM policy updates, ongoing research effort has targeted more straightforward alternatives.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.236 Exploring Vision Semantic Prompt for Efficient Point Cloud Understanding

- 作者：Zha, Yixin; Wang, Chuxin; Yang, Wenfei; Zhang, Tianzhu; Wu, Feng
- 年份：2025；主题：Systems；优先级：重要
- [来源](#)

论文：Exploring Vision Semantic Prompt for Efficient Point Cloud Understanding

快速理解：A series of pre-trained models have demonstrated promising results in point cloud understanding tasks and are widely applied to downstream tasks through fine-tuning. However, full fine-tuning leads to the forgetting of pretrained knowledge and substantial storage costs on edge devices.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.6.237 Extreme Value Policy Optimization for Safe Reinforcement Learning

- 作者: Gao, Shiqing; Zhou, Yihang; Shao, Shuai; Luo, Haoyu; Bing, Yiheng; Ding, Jiaxin; Fu, Luoyi; Wang, Xinbing
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Extreme Value Policy Optimization for Safe Reinforcement Learning

快速理解: Ensuring safety is a critical challenge in applying Reinforcement Learning (RL) to real-world scenarios. Constrained Reinforcement Learning (CRL) addresses this by maximizing returns under predefined constraints, typically formulated as the expected cumulative cost.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.238 FOCOOp: Enhancing Out-of-Distribution Robustness in Federated Prompt Learning for Vision-Language Models

- 作者: Liao, Xinting; Liu, Weiming; Qian, Jiaming; Zhou, Pengyang; Xu, Jiahe; Wang, Wenjie; Chen, Chaochao; Zheng, Xiaolin; Chua, Tat-Seng
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: FOCOOp: Enhancing Out-of-Distribution Robustness in Federated Prompt Learning for Vision-Language Models

快速理解: Federated prompt learning (FPL) for vision-language models is a powerful approach to collaboratively adapt models across distributed clients while preserving data privacy. However, existing FPL approaches suffer from a trade-off between performance and robustness, particularly in out-of-distribution (OOD) shifts, limiting their reliability in real-world scenarios.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.239 Fast Video Generation with Sliding Tile Attention

- 作者: Zhang, Peiyuan; Chen, Yongqi; Su, Runlong; Ding, Hangliang; Stoica, Ion; Liu, Zhengzhong; Zhang, Hao
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Fast Video Generation with Sliding Tile Attention

快速理解: Diffusion Transformers (DiTs) with 3D full attention power state-of-the-art video generation, but suffer from prohibitive compute cost -when generating just a 5-second 720P video, attention alone takes 800 out of 950 seconds of total inference time. This paper introduces sliding tile attention (STA) to address this challenge.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.240 Finite-Sample Convergence Bounds for Trust Region Policy Optimization in Mean Field Games

- 作者: Ocello, Antonio; Tiapkin, Daniil; Mancini, Lorenzo; Lauriere, Mathieu; Moulines, Eric
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Finite-Sample Convergence Bounds for Trust Region Policy Optimization in Mean Field Games

快速理解: We introduce Mean Field Trust Region Policy Optimization (MF-TRPO), a novel algorithm designed to compute approximate Nash equilibria for ergodic Mean Field Games (MFGs) in finite state-action spaces. Building on the well-established performance of TRPO in the reinforcement learning (RL) setting, we extend its methodology to the MFG framework, leveraging its stability and robustness in policy optimization.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.241 Finite-Time Global Optimality Convergence in Deep Neural Actor-Critic Methods for Decentralized Multi-Agent Reinforcement Learning

- 作者: Zhang, Zhiyao; Oh, Myeung Suk; Hairi, Fnu; Luo, Ziyue; Velasquez, Alvaro; Liu, Jia
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Finite-Time Global Optimality Convergence in Deep Neural Actor-Critic Methods for Decentralized Multi-Agent Reinforcement Learning

快速理解: Actor-critic methods for decentralized multi-agent reinforcement learning (MARL) facilitate collaborative optimal decision making without centralized coordination, thus enabling a wide range of applications in practice. To date, however, most theoretical convergence studies for existing actor-critic decentralized MARL methods are limited to the guarantee of a stationary solution under the linear function approximation.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.242 FlatQuant: Flatness Matters for LLM Quantization

- 作者: Sun, Yuxuan; Liu, Ruikang; Bai, Haoli; Bao, Han; Zhao, Kang; Li, Yuening; Hu, Jiaxin; Yu, Xianzhi; Hou, Lu; Yuan, Chun; Jiang, Xin; Liu, Wulong; Yao, Jun
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: FlatQuant: Flatness Matters for LLM Quantization

快速理解: Recently, quantization has been widely used for the compression and acceleration of large language models (LLMs). Due to the outliers in LLMs, it is crucial to flatten weights and activations to minimize quantization error with equally spaced quantization points.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.6.243 FloE: On-the-Fly MoE Inference on Memory-constrained GPU

- 作者: Zhou, Yuxin; Li, Zheng; Zhang, Jun; Wang, Jue; Wang, Yiping; Xie, Zhongle; Chen, Ke;

Shou, Lidan

- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: FloE: On-the-Fly MoE Inference on Memory-constrained GPU

快速理解: With the widespread adoption of Mixture-of-Experts (MoE) models, there is a growing demand for efficient inference on memory-constrained devices. While offloading expert parameters to CPU memory and loading activated experts on demand has emerged as a potential solution, the large size of activated experts overburdens the limited PCIe bandwidth, hindering the effectiveness in latency-sensitive scenarios.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.6.244 Flow Matching for Few-Trial Neural Adaptation with Stable Latent Dynamics

- 作者: Wang, Puli; Qi, Yu; Wang, Yueming; Pan, Gang
- 年份: 2025; 主题: World Model; 优先级: 重要
- [来源](#)

论文: Flow Matching for Few-Trial Neural Adaptation with Stable Latent Dynamics

快速理解: The primary goal of brain-computer interfaces (BCIs) is to establish a direct linkage between neural activities and behavioral actions via neural decoders. Due to the nonstationary property of neural signals, BCIs trained on one day usually obtain degraded performance on other days, hindering the user experience.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策。

证据层级: 索引摘要。

1.3.6.245 FrameBridge: Improving Image-to-Video Generation with Bridge Models

- 作者: Wang, Yuji; Chen, Zehua; Xiaoyu, Chen; Wei, Yixiang; Zhu, Jun; Chen, Jianfei
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: FrameBridge: Improving Image-to-Video Generation with Bridge Models

快速理解: Diffusion models have achieved remarkable progress on image-to-video (I2V) generation, while their noise-to-data generation process is inherently mismatched with this task, which may lead to suboptimal synthesis quality. In this work, we present FrameBridge.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.246 From Black Boxes to Transparent Minds: Evaluating and Enhancing the Theory of Mind in Multimodal Large Language Models

- 作者: Li, Xinyang; Liu, Siqi; Zou, Bochao; Chen, Jiansheng; Ma, Huimin
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: From Black Boxes to Transparent Minds: Evaluating and Enhancing the Theory of Mind in Multimodal Large Language Models

快速理解: As large language models evolve, there is growing anticipation that they will emulate human-like Theory of Mind (ToM) to assist with routine tasks. However, existing methods for evaluating machine ToM focus primarily on unimodal models and largely treat these models as black boxes, lacking an interpretative exploration of their internal mechanisms.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.247 From Debate to Equilibrium: Belief-Driven Multi-Agent LLM Reasoning via Bayesian Nash Equilibrium

- 作者: Yi, Xie; Zhou, Zhanke; Cao, Chentao; Niu, Qiyu; Liu, Tongliang; Han, Bo
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: From Debate to Equilibrium: Belief-Driven Multi-Agent LLM Reasoning via Bayesian Nash Equilibrium

快速理解: Multi-agent frameworks can substantially boost the reasoning power of large language models (LLMs), but they typically incur heavy computational costs and lack convergence guarantees. To overcome these challenges, we recast multi-LLM coordination as an incomplete-information game and seek a Bayesian Nash equilibrium (BNE), in which each agent optimally responds to its probabilistic beliefs about the strategies of others.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.248 From Local Details to Global Context: Advancing Vision-Language Models with Attention-Based Selection

- 作者: Cai, Lincan; Kang, Jingxuan; Li, Shuang; Ma, Wenxuan; Xie, Binhui; Qin, Zhida; Liang, Jian
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: From Local Details to Global Context: Advancing Vision-Language Models with Attention-Based Selection

快速理解: Pretrained vision-language models (VLMs), e.g., CLIP, demonstrate impressive zero-shot capabilities on downstream tasks. Prior research highlights the crucial role of visual augmentation techniques, like random cropping, in alignment with fine-grained class descriptions generated by large language models (LLMs), significantly enhancing zero-shot performance by incorporating multi-view information.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.249 G-Designer: Architecting Multi-agent Communication Topologies via Graph Neural Networks

- 作者: Zhang, Guibin; Yue, Yanwei; Sun, Xiangguo; Wan, Guancheng; Yu, Miao; Fang, Junfeng; Wang, Kun; Chen, Tianlong; Cheng, Dawei
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: G-Designer: Architecting Multi-agent Communication Topologies via Graph Neural Networks

快速理解: Recent advancements in large language model (LLM)-based agents have demonstrated that collective intelligence can significantly surpass the capabilities of individual agents, primarily due to well-crafted inter-agent communication topologies. Despite the diverse and high-performing designs available, practitioners often face confusion when selecting the most effective pipeline for their specific task: Which topology is the best choice for my task, avoiding unnecessary communication token overhead while ensuring high-quality solution? In response to this dilemma, we introduce G-Designer, an adaptive, efficient, and robust solution for multi-agent deployment, which dynamically designs task-aware, customized communication topologies.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.250 GANQ: GPU-Adaptive Non-Uniform Quantization for Large Language Models

- 作者: Zhao, Pengxiang; Yuan, Xiaoming
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: GANQ: GPU-Adaptive Non-Uniform Quantization for Large Language Models

快速理解: Large Language Models (LLMs) face significant deployment challenges due to their substantial resource requirements. While low-bit quantized weights can reduce memory usage and improve inference efficiency, current hardware lacks native support for mixed-precision General Matrix Multiplication (mpGEMM), resulting in inefficient dequantization-based implementations.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.6.251 GAPrompt: Geometry-Aware Point Cloud Prompt for 3D Vision Model

- 作者: Ai, Zixiang; Liu, Zichen; Lei, Yuanhang; Cui, Zhenyu; Zou, Xu; Zhou, Jiahuan
- 年份: 2025; 主题: Systems; 优先级: 重要
- [来源](#)

论文: GAPrompt: Geometry-Aware Point Cloud Prompt for 3D Vision Model

快速理解: Pre-trained 3D vision models have gained significant attention for their promising performance on point cloud data. However, fully fine-tuning these models for downstream tasks is computationally expensive and storage-intensive.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.6.252 GPTAQ: Efficient Finetuning-Free Quantization for Asymmetric Calibration

- 作者: Li, Yuhang; Yin, Ruokai; Lee, Donghyun; Xiao, Shiting; Panda, Priyadarshini
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: GPTAQ: Efficient Finetuning-Free Quantization for Asymmetric Calibration

快速理解: We introduce GPTAQ, a novel finetuning-free quantization method for compressing large-scale transformer architectures. Unlike the previous GPTQ method, which independently calibrates each layer, we always match the quantized layer's output to the exact output in the full-precision model, resulting in a scheme that we call asymmetric calibration.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.6.253 GRADEO: Towards Human-Like Evaluation for Text-to-Video Generation via Multi-Step Reasoning

- 作者: Mou, Zhun; Xia, Bin; Huang, Zhengchao; Yang, Wenming; Jia, Jiaya
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: GRADEO: Towards Human-Like Evaluation for Text-to-Video Generation via Multi-Step Reasoning

快速理解: Recent great advances in video generation models have demonstrated their potential to produce high-quality videos, bringing challenges to effective evaluation. Unlike human evaluation, existing automated evaluation metrics lack high-level semantic understanding and reasoning capabilities for video, thus making them infeasible and unexplainable.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.254 GRAM: A Generative Foundation Reward Model for Reward Generalization

- 作者: Wang, Chenglong; Gan, Yang; Huo, Yifu; Mu, Yongyu; He, Qiaozhi; Yang, Murun; Li, Bei; Xiao, Tong; Zhang, Chunliang; Liu, Tongran; Zhu, Jingbo
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: GRAM: A Generative Foundation Reward Model for Reward Generalization

快速理解: In aligning large language models (LLMs), reward models have played an important role, but are standardly trained as discriminative models and rely only on labeled human preference data. In this paper, we explore methods that train reward models using both unlabeled and labeled data.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.255 GS-Bias: Global-Spatial Bias Learner for Single-Image Test-Time Adaptation of Vision-Language Models

- 作者: Huang, Zhaohong; Zhang, Yuxin; Xie, Jingjing; Chao, Fei; Ji, Rongrong
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: GS-Bias: Global-Spatial Bias Learner for Single-Image Test-Time Adaptation of Vision-Language Models

快速理解: Recent advances in test-time adaptation (TTA) for Vision-Language Models (VLMs) have garnered increasing attention, particularly through the use of multiple augmented views of a single image to boost zero-shot generalization. Unfortunately, existing methods fail to strike a satisfactory balance between performance and efficiency, either due to excessive overhead of tuning text prompts or unstable benefits from handcrafted, training-free visual feature enhancement.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.256 General agents need world models

- 作者: Richens, Jonathan; Everitt, Tom; Abel, David
- 年份: 2025; 主题: World Model; 优先级: 重要
- [来源](#)

论文: General agents need world models

快速理解: Are world models a necessary ingredient for flexible, goal-directed behaviour, or is model-free learning sufficient? We provide a formal answer to this question, showing that any agent capable of generalizing to multi-step goal-directed tasks must have learned a predictive model of its environment.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策。

证据层级: 索引摘要。

1.3.6.257 Generative Point Cloud Registration

- 作者: Jiang, Haobo; Xie, Jin; Yang, Jian; Yu, Liang; Zheng, Jianmin
- 年份: 2025; 主题: Systems; 优先级: 重要
- [来源](#)

论文: Generative Point Cloud Registration

快速理解: In this paper, we propose a novel 3D registration paradigm, Generative Point Cloud Registration, which bridges advanced 2D generative models with 3D matching tasks to enhance registration performance. Our key idea is to generate cross-view consistent image pairs that are well-aligned with the source and target point clouds, enabling geometric-color feature fusion to facilitate robust matching.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.6.258 GeoPixel: Pixel Grounding Large Multimodal Model in Remote Sensing

- 作者: Shabbir, Akashah; Zumri, Mohammed; Bennamoun, Mohammed; Khan, Fahad Shahbaz; Khan, Salman
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: GeoPixel: Pixel Grounding Large Multimodal Model in Remote Sensing

快速理解: Recent advances in large multimodal models (LMMs) have recognized fine-grained grounding as an imperative factor of visual understanding and dialogue. However, the benefits of such representation in LMMs are limited to the natural image domain, and these models perform poorly for remote sensing (RS).

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.259 GradPS: Resolving Futile Neurons in Parameter Sharing Network for Multi-Agent Reinforcement Learning

- 作者: Qin, Haoyuan; Liu, Zhengzhu; Lin, Chenxing; Ma, Chennan; Mei, Songzhu; Shen, Siqu; Wang, Cheng
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: GradPS: Resolving Futile Neurons in Parameter Sharing Network for Multi-Agent Reinforcement Learning

快速理解: Parameter-sharing (PS) techniques have been widely adopted in cooperative Multi-Agent Reinforcement Learning (MARL). In PS, all the agents share a policy network with identical parameters, which enjoys good sample efficiency.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.260 Gradient Boosting Reinforcement Learning

- 作者: Fuhrer, Benjamin; Tessler, Chen; Dalal, Gal
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Gradient Boosting Reinforcement Learning

快速理解: We present Gradient Boosting Reinforcement Learning (GBRL), a framework that adapts the strengths of Gradient Boosting Trees (GBT) to reinforcement learning (RL) tasks. While neural networks (NNs) have become the de facto choice for RL, they face significant challenges with structured and categorical features and tend to generalize poorly to out-of-distribution samples.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.261 Gradient Inversion of Multimodal Models

- 作者: Hemo, Omri Ben; Zolfi, Alon; Yehezkel, Oryan; Hofman, Omer; Vainshtein, Roman; Kojima, Hisashi; Elovici, Yuval; Shabtai, Asaf

- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Gradient Inversion of Multimodal Models

快速理解: Federated learning (FL) enables privacy-preserving distributed machine learning by sharing gradients instead of raw data. However, FL remains vulnerable to gradient inversion attacks, in which shared gradients can reveal sensitive training data.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.262 Graph Diffusion for Robust Multi-Agent Coordination

- 作者: Zeng, Xianghua; Su, Hang; Wang, Zhengyi; Lin, Zhiyuan
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Graph Diffusion for Robust Multi-Agent Coordination

快速理解: Offline multi-agent reinforcement learning (MARL) struggles to estimate out-of-distribution states and actions due to the absence of real-time environmental feedback. While diffusion models show promise in addressing these challenges, their application primarily focuses on independently diffusing the historical trajectories of individual agents, neglecting crucial multi-agent coordination dynamics and reducing policy robustness in dynamic environments.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.263 Graph4MM: Weaving Multimodal Learning with Structural Information

- 作者: Ning, Xuying; Fu, Dongqi; Wei, Tianxin; Xu, Wujiang; He, Jingrui
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Graph4MM: Weaving Multimodal Learning with Structural Information

快速理解: Real-world multimodal data usually exhibit complex structural relationships beyond traditional one-to-one mappings like image-caption pairs. Entities across modalities interact in intricate ways, with images and text forming diverse interconnections through contextual dependencies and co-references.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.264 GuidedQuant: Large Language Model Quantization via Exploiting End Loss Guidance

- 作者: Kim, Jinuk; El Halabi, Marwa; Park, Wonpyo; Schaefer, Clemens Js; Lee, Deokjae; Park, Yeonhong; Lee, Jae W.; Song, Hyun Oh
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: GuidedQuant: Large Language Model Quantization via Exploiting End Loss Guidance

快速理解: Post-training quantization is a key technique for reducing the memory and inference latency of large language models by quantizing weights and activations without requiring retraining. However, existing methods either (1) fail to account for the varying importance of hidden features to the end loss or, when incorporating end loss, (2) neglect the critical interactions between model weights.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.6.265 HYGMA: Hypergraph Coordination Networks with Dynamic Grouping for Multi-Agent Reinforcement Learning

- 作者: Liu, Chiqiang; Li, Dazi
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: HYGMA: Hypergraph Coordination Networks with Dynamic Grouping for Multi-Agent Reinforcement Learning

快速理解: Cooperative multi-agent reinforcement learning faces significant challenges in effectively organizing agent relationships and facilitating information exchange, particularly when agents need to adapt their coordination patterns dynamically. This paper presents a novel framework that integrates dynamic spectral clustering with hypergraph neural networks to enable adaptive group formation and efficient information processing in multi-agent systems.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.266 Handling Imbalanced Pseudolabels for Vision-Language Models with Concept Alignment and Confusion-Aware Calibrated Margin

- 作者: Wang, Yuchen; Bai, Xuefeng; Li, Xiucheng; Guan, Weili; Nie, Liqiang; Chen, Xinyang
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Handling Imbalanced Pseudolabels for Vision-Language Models with Concept Alignment and Confusion-Aware Calibrated Margin

快速理解: Adapting vision-language models (VLMs) to downstream tasks with pseudolabels has gained increasing attention. A major obstacle is that the pseudolabels generated by VLMs tend to be imbalanced, leading to inferior performance.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.267 HealthGPT: A Medical Large Vision-Language Model for Unifying Comprehension and Generation via Heterogeneous Knowledge Adaptation

- 作者: Lin, Tianwei; Zhang, Wenqiao; Li, Sijing; Yuan, Yuqian; Yu, Binhe; Li, Haoyuan; He, Wanggui; Jiang, Hao; Li, Mengze; Xiaohui, Song; Tang, Siliang; Xiao, Jun; Lin, Hui; Zhuang,

Yueting; Ooi, Beng Chin

- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: HealthGPT: A Medical Large Vision-Language Model for Unifying Comprehension and Generation via Heterogeneous Knowledge Adaptation

快速理解: We present HealthGPT, a powerful Medical Large Vision-Language Model (Med-LVLM) that integrates medical visual comprehension and generation capabilities within a unified autoregressive paradigm. Our bootstrapping philosophy is to progressively adapt heterogeneous comprehension and generation knowledge to pre-trained Large Language Models (LLMs).

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.268 Hidden No More: Attacking and Defending Private Third-Party LLM Inference

- 作者: Thomas, Rahul Krishna; Zahran, Louai; Choi, Erica; Potti, Akilesh; Goldblum, Micah; Pal, Arka
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Hidden No More: Attacking and Defending Private Third-Party LLM Inference

快速理解: Recent advances in Large Language Models (LLMs) have led to widespread adoption of third-party inference services, raising critical privacy concerns. In this work, we introduce a novel reconstruction technique that can recover original prompts from hidden states with nearly perfect accuracy across multiple state-of-the-art LLMs in the increasingly important open-weights setting.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.6.269 Hierarchical Reinforcement Learning with Targeted Causal Interventions

- 作者: Khorasani, Mohammadsadegh; Salehkaleybar, Saber; Kiyavash, Negar; Grossglauser, Matthias
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Hierarchical Reinforcement Learning with Targeted Causal Interventions

快速理解: Hierarchical reinforcement learning (HRL) improves the efficiency of long-horizon reinforcement-learning tasks with sparse rewards by decomposing the task into a hierarchy of subgoals. The main challenge of HRL is efficient discovery of the hierarchical structure among subgoals and utilizing this structure to achieve the final goal.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.270 Hierarchical Reinforcement Learning with Uncertainty-Guided Diffusional Subgoals

- 作者: Wang, Vivienne Huiling; Wang, Tinghuai; Pajarinen, Joni
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Hierarchical Reinforcement Learning with Uncertainty-Guided Diffusional Subgoals

快速理解: Hierarchical reinforcement learning (HRL) learns to make decisions on multiple levels of temporal abstraction. A key challenge in HRL is that the low-level policy changes over time, making it difficult for the high-level policy to generate effective subgoals.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.271 Hybrid Quantum-Classical Multi-Agent Pathfinding

- 作者: Gerlach, Thore; Lee, Loong Kuan; Barbaresco, Frederic; Piatkowski, Nico
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Hybrid Quantum-Classical Multi-Agent Pathfinding

快速理解: Multi-Agent Path Finding (MAPF) focuses on determining conflict-free paths for multiple agents navigating through a shared space to reach specified goal locations. This problem becomes computationally challenging, particularly when handling large numbers of agents, as frequently encountered in practical applications like coordinating autonomous vehicles.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.272 HybridGS: High-Efficiency Gaussian Splatting Data Compression using Dual-Channel Sparse Representation and Point Cloud Encoder

- 作者: Yang, Qi; Yang, Le; Van Der Auwera, Geert; Li, Zhu
- 年份: 2025; 主题: Systems; 优先级: 重要
- [来源](#)

论文: HybridGS: High-Efficiency Gaussian Splatting Data Compression using Dual-Channel Sparse Representation and Point Cloud Encoder

快速理解: Most existing 3D Gaussian Splatting (3DGS) compression schemes focus on producing compact 3DGS representation via implicit data embedding. They have long encoding and decoding times and highly customized data format, making it difficult for widespread deployment.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.6.273 Hyper: Hyperparameter Robust Efficient Exploration in Reinforcement Learning

- 作者: Wang, Yiran; Liu, Chenshu; Li, Yunfan; Amani, Sanae; Zhou, Bolei; Yang, Lin
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Hyper: Hyperparameter Robust Efficient Exploration in Reinforcement Learning

快速理解: The exploration & exploitation dilemma poses significant challenges in reinforcement learning (RL). Recently, curiosity-based exploration methods achieved great success in tackling hard-exploration problems.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.274 Hyperspherical Normalization for Scalable Deep Reinforcement Learning

- 作者: Lee, Hojoon; Lee, Youngdo; Seno, Takuma; Kim, Donghu; Stone, Peter; Choo, Jaegul
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Hyperspherical Normalization for Scalable Deep Reinforcement Learning

快速理解: Scaling up the model size and computation has brought consistent performance improvements in supervised learning. However, this lesson often fails to apply to reinforcement learning (RL) because training the model on non-stationary data easily leads to overfitting and unstable optimization.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.275 IL-SOAR : Imitation Learning with Soft Optimistic Actor cRitic

- 作者: Viel, Stefano; Viano, Luca; Cevher, Volkan
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: IL-SOAR : Imitation Learning with Soft Optimistic Actor cRitic

快速理解: This paper introduces the SOAR framework for imitation learning. SOAR is an algorithmic template which learns a policy from expert demonstrations with a primal-dual style algorithm which alternates cost and policy updates.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.276 Imagine While Reasoning in Space: Multimodal Visualization-of-Thought

- 作者: Li, Chengzu; Wu, Wenshan; Zhang, Huanyu; Xia, Yan; Mao, Shaoguang; Dong, Li; Vuli' {c}, Ivan; Wei, Furu
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Imagine While Reasoning in Space: Multimodal Visualization-of-Thought

快速理解: Chain-of-Thought (CoT) prompting has proven highly effective for enhancing complex reasoning in Large Language Models (LLMs) and Multimodal Large Language Models (MLLMs). Yet, it struggles in complex spatial reasoning tasks.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.6.277 Improved Off-policy Reinforcement Learning in Biological Sequence Design

- 作者：Kim, Hyeonah; Kim, Minsu; Yun, Taeyoung; Choi, Sanghyeok; Bengio, Emmanuel; Hernandez-Garcia, Alex; Park, Jinkyoo
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Improved Off-policy Reinforcement Learning in Biological Sequence Design

快速理解：Designing biological sequences with desired properties is challenging due to vast search spaces and limited evaluation budgets. Although reinforcement learning methods use proxy models for rapid reward evaluation, insufficient training data can cause proxy misspecification on out-of-distribution inputs.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.278 Improving Multimodal Learning Balance and Sufficiency through Data Remixing

- 作者：Ma, Xiaoyu; Chen, Hao; Deng, Yongjian
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Improving Multimodal Learning Balance and Sufficiency through Data Remixing

快速理解：Different modalities hold considerable gaps in optimization trajectories, including speeds and paths, which lead to modality laziness and modality clash when jointly training multimodal models, resulting in insufficient and imbalanced multimodal learning. Existing methods focus on enforcing the weak modality by adding modality-specific optimization objectives, aligning their optimization speeds, or decomposing multimodal learning to enhance unimodal learning.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.6.279 Improving Reward Model Generalization from Adversarial Process Enhanced Preferences

- 作者：Zhang, Zhilong; Xu, Tian; Du, Xinghao; Cao, Xingchen; Sun, Yihao; Yu, Yang
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Improving Reward Model Generalization from Adversarial Process Enhanced Preferences

快速理解：In sequential decision-making, the reward function serves as the primary supervision signal, guiding agents to acquire the desired behaviors. Traditional reward modeling methods rely heavily on human expertise, limiting their scalability.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.280 Improving Transformer World Models for Data-Efficient RL

- 作者: Dedieu, Antoine; Ortiz, Joseph; Lou, Xinghua; Wendelken, Carter; Guntupalli, J Swaroop; Lehrach, Wolfgang; Lazaro-Gredilla, Miguel; Murphy, Kevin Patrick
- 年份: 2025; 主题: World Model; 优先级: 重要
- [来源](#)

论文: Improving Transformer World Models for Data-Efficient RL

快速理解: We present an approach to model-based RL that achieves a new state of the art performance on the challenging Craftax-classic benchmark, an open-world 2D survival game that requires agents to exhibit a wide range of general abilities—such as strong generalization, deep exploration, and long-term reasoning. With a series of careful design choices aimed at improving sample efficiency, our MBRL algorithm achieves a reward of 69.66% after only 1M environment steps, significantly outperforming DreamerV3, which achieves 53.2%, and, for the first time, exceeds human performance of 65.0%.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策。

证据层级: 索引摘要。

1.3.6.281 Improving Zero-Shot Adversarial Robustness in Vision-Language Models by Closed-form Alignment of Adversarial Path Simplices

- 作者: Dong, Junhao; Koniusz, Piotr; Zhang, Yifei; Zhu, Hao; Liu, Weiming; Qu, Xinghua; Ong, Yew-Soon
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Improving Zero-Shot Adversarial Robustness in Vision-Language Models by Closed-form Alignment of Adversarial Path Simplices

快速理解: Vision-Language Models (VLMs) such as CLIP excel at zero-shot classification due to large-scale pre-training but are vulnerable to adversarial examples. Adversarial fine-tuning robustifies zero-shot models by aligning prediction scores of individual adversaries with their clean counterparts, which typically overlooks intermediate adversarial samples along the adversarial trajectory crossing the decision boundary.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.282 Improving the Continuity of Goal-Achievement Ability via Policy Self-Regularization for Goal-Conditioned Reinforcement Learning

- 作者: Gong, Xudong; Yang, Sen; Dawei, Feng; Xu, Kele; Ding, Bo; Wang, Huaimin; Dou, Yong
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Improving the Continuity of Goal-Achievement Ability via Policy Self-Regularization for Goal-Conditioned Reinforcement Learning

快速理解: This paper addresses the challenge of discontinuity in goal-achievement capabilities observed in Goal-conditioned Reinforcement Learning (GCRL) algorithms. Through a theoretical analysis,

we identify that the reuse of successful trajectories or policies during training can aid in achieving adjacent goals of achievable goals.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.283 In-Context Reinforcement Learning From Suboptimal Historical Data

- 作者：Dong, Juncheng; Guo, Moyang; Fang, Ethan X; Yang, Zhuoran; Tarokh, Vahid
- 年份：2025; 主题：Agent RL; 优先级：重要
- [来源](#)

论文：In-Context Reinforcement Learning From Suboptimal Historical Data

快速理解：Transformer models have achieved remarkable empirical successes, largely due to their in-context learning capabilities. Inspired by this, we explore training an autoregressive transformer for in-context reinforcement learning (ICRL).

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.284 Incentivize without Bonus: Provably Efficient Model-based Online Multi-agent RL for Markov Games

- 作者：Yang, Tong; Dai, Bo; Xiao, Lin; Chi, Yuejie
- 年份：2025; 主题：Agent RL; 优先级：重要
- [来源](#)

论文：Incentivize without Bonus: Provably Efficient Model-based Online Multi-agent RL for Markov Games

快速理解：Multi-agent reinforcement learning (MARL) lies at the heart of a plethora of applications involving the interaction of a group of agents in a shared unknown environment. A prominent framework for studying MARL is Markov games, with the goal of finding various notions of equilibria in a sample-efficient manner, such as the Nash equilibrium (NE) and the coarse correlated equilibrium (CCE).

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.285 Inducing, Detecting and Characterising Neural Modules: A Pipeline for Functional Interpretability in Reinforcement Learning

- 作者：Soligo, Anna; Ferraro, Pietro; Boyle, David
- 年份：2025; 主题：Agent RL; 优先级：重要
- [来源](#)

论文：Inducing, Detecting and Characterising Neural Modules: A Pipeline for Functional Interpretability in Reinforcement Learning

快速理解：Interpretability is crucial for ensuring RL systems align with human values. However, it remains challenging to achieve in complex decision making domains.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.286 InfoCons: Identifying Interpretable Critical Concepts in Point Clouds via Information Theory

- 作者: Li, Feifei; Zhang, Mi; Wang, Zhaoxiang; Yang, Min
- 年份: 2025; 主题: Systems; 优先级: 重要
- [来源](#)

论文: InfoCons: Identifying Interpretable Critical Concepts in Point Clouds via Information Theory

快速理解: Interpretability of point cloud (PC) models becomes imperative given their deployment in safety-critical scenarios such as autonomous vehicles. We focus on attributing PC model outputs to interpretable critical concepts, defined as meaningful subsets of the input point cloud.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.6.287 Intersectional Fairness in Reinforcement Learning with Large State and Constraint Spaces

- 作者: Eaton, Eric; Hussing, Marcel; Kearns, Michael; Roth, Aaron; Sengupta, Sikata Bela; Sorrell, Jessica
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Intersectional Fairness in Reinforcement Learning with Large State and Constraint Spaces

快速理解: In traditional reinforcement learning (RL), the learner aims to solve a single objective optimization problem: find the policy that maximizes expected reward. However, in many real-world settings, it is important to optimize over multiple objectives simultaneously.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.288 Inverse Reinforcement Learning with Switching Rewards and History Dependency for Characterizing Animal Behaviors

- 作者: Ke, Jingyang; Wu, Feiyang; Wang, Jiyi; Markowitz, Jeffrey; Wu, Anqi
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Inverse Reinforcement Learning with Switching Rewards and History Dependency for Characterizing Animal Behaviors

快速理解: Traditional approaches to studying decision-making in neuroscience focus on simplified behavioral tasks where animals perform repetitive, stereotyped actions to receive explicit rewards. While informative, these methods constrain our understanding of decision-making to short timescale behaviors driven by explicit goals.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.289 Joint MoE Scaling Laws: Mixture of Experts Can Be Memory Efficient

- 作者: Ludziewski, Jan; Pi{ro}, Maciej; Krajewski, Jakub; Stefaniak, Maciej; Krutul, Micha{l}; Ma{l}a{snicki}, Jan; Cygan, Marek; Sankowski, Piotr; Adamczewski, Kamil; Mi{o}{s}, Piotr; Jaszczur, Sebastian
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Joint MoE Scaling Laws: Mixture of Experts Can Be Memory Efficient

快速理解: Mixture of Experts (MoE) architectures have significantly increased computational efficiency in both research and real-world applications of large-scale machine learning models. However, their scalability and efficiency under memory constraints remain relatively underexplored.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.6.290 KABB: Knowledge-Aware Bayesian Bandits for Dynamic Expert Coordination in Multi-Agent Systems

- 作者: Zhang, Jusheng; Huang, Zimeng; Fan, Yijia; Liu, Ningyuan; Li, Mingyan; Yang, Zhuojie; Yao, Jiawei; Wang, Jian; Wang, Keze
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: KABB: Knowledge-Aware Bayesian Bandits for Dynamic Expert Coordination in Multi-Agent Systems

快速理解: As scaling large language models faces prohibitive costs, multi-agent systems emerge as a promising alternative, though challenged by static knowledge assumptions and coordination inefficiencies. We introduce Knowledge-Aware Bayesian Bandits (KABB), a novel framework that enhances multi-agent system coordination through semantic understanding and dynamic adaptation.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.291 KBQA-o1: Agentic Knowledge Base Question Answering with Monte Carlo Tree Search

- 作者: Luo, Haoran; E, Haihong; Guo, Yikai; Lin, Qika; Wu, Xiaobao; Mu, Xinyu; Liu, Wenhao; Song, Meina; Zhu, Yifan; Luu, Anh Tuan
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: KBQA-o1: Agentic Knowledge Base Question Answering with Monte Carlo Tree Search

快速理解: Knowledge Base Question Answering (KBQA) aims to answer natural language questions with a large-scale structured knowledge base (KB). Despite advancements with large language models (LLMs), KBQA still faces challenges in weak KB awareness, imbalance between effectiveness

and efficiency, and high reliance on annotated data.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.292 Kernel-based Unsupervised Embedding Alignment for Enhanced Visual Representation in Vision-language Models

- 作者：Gong, Shizhan; Jiang, Yankai; Dou, Qi; Farnia, Farzan
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Kernel-based Unsupervised Embedding Alignment for Enhanced Visual Representation in Vision-language Models

快速理解：Vision-language models, such as CLIP, have achieved significant success in aligning visual and textual representations, becoming essential components of many multi-modal large language models (MLLMs) like LLaVA and OpenFlamingo. However, numerous studies have identified CLIP's limited fine-grained perception as a critical drawback, leading to substantial failures in downstream MLLMs.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.6.293 KernelBench: Can LLMs Write Efficient GPU Kernels?

- 作者：Ouyang, Anne; Guo, Simon; Arora, Simran; Zhang, Alex L; Hu, William; Re, Christopher; Mirhoseini, Azalia
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：KernelBench: Can LLMs Write Efficient GPU Kernels?

快速理解：Efficient GPU kernels are crucial for building performant machine learning architectures, but writing them is a time-consuming challenge that requires significant expertise; therefore, we explore using language models (LMs) to automate kernel generation. We introduce KernelBench, an open-source framework for evaluating LMs' ability to write fast and correct kernels on a suite of 250 carefully selected PyTorch ML workloads.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.6.294 LEMoN: Label Error Detection using Multimodal Neighbors

- 作者：Zhang, Haoran; Balagopalan, Aparna; Oufattole, Nassim; Jeong, Hyewon; Wu, Yan; Zhu, Jiacheng; Ghassemi, Marzyeh
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：LEMoN: Label Error Detection using Multimodal Neighbors

快速理解：Large repositories of image-caption pairs are essential for the development of vision-language models. However, these datasets are often extracted from noisy data scraped from the web, and

contain many mislabeled instances.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.295 LLM-Assisted Semantically Diverse Teammate Generation for Efficient Multi-agent Coordination

- 作者: Li, Lihe; Yuan, Lei; Liu, Pengsen; Jiang, Tao; Yu, Yang
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: LLM-Assisted Semantically Diverse Teammate Generation for Efficient Multi-agent Coordination

快速理解: Training with diverse teammates is the key for learning generalizable agents. Typical approaches aim to generate diverse teammates by utilizing techniques like randomization, designing regularization terms, or reducing policy compatibility, etc.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.296 LMRL Gym: Benchmarks for Multi-Turn Reinforcement Learning with Language Models

- 作者: Abdulhai, Marwa; White, Isadora; Snell, Charlie Victor; Sun, Charles; Hong, Joey; Zhai, Yuexiang; Xu, Kelvin; Levine, Sergey
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: LMRL Gym: Benchmarks for Multi-Turn Reinforcement Learning with Language Models

快速理解: Large language models (LLMs) provide excellent text-generation capabilities, but standard prompting and generation methods generally do not lead to intentional or goal-directed agents and might necessitate considerable prompt tuning. Even the best current LLMs rarely ask clarifying questions, engage in explicit information gathering, or take actions that lead to better decisions after multiple turns.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.297 LRA-QViT: Integrating Low-Rank Approximation and Quantization for Robust and Efficient Vision Transformers

- 作者: Kang, Beom Jin; Kim, Namjoon; Kim, Hyun
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: LRA-QViT: Integrating Low-Rank Approximation and Quantization for Robust and Efficient Vision Transformers

快速理解: Recently, transformer-based models have demonstrated state-of-the-art performance across various computer vision tasks, including image classification, detection, and segmentation. However,

their substantial parameter count poses significant challenges for deployment in resource-constrained environments such as edge or mobile devices.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.6.298 LaCache: Ladder-Shaped KV Caching for Efficient Long-Context Modeling of Large Language Models

- 作者：Shi, Dachuan; Fu, Yonggan; Yuan, Xiangchi; Yu, Zhongzhi; You, Haoran; Li, Sixu; Dong, Xin; Kautz, Jan; Molchanov, Pavlo; Lin, Yingyan Celine
- 年份：2025；主题：Systems；优先级：重要
- [来源](#)

论文：LaCache: Ladder-Shaped KV Caching for Efficient Long-Context Modeling of Large Language Models

快速理解：Recent advancements in Large Language Models (LLMs) have spurred interest in numerous applications requiring robust long-range capabilities, essential for processing extensive input contexts and continuously generating extended outputs. As sequence lengths increase, the number of Key-Value (KV) pairs in LLMs escalates, creating a significant efficiency bottleneck.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.6.299 LangTime: A Language-Guided Unified Model for Time Series Forecasting with Proximal Policy Optimization

- 作者：Niu, Wenzhe; Xie, Zongxia; Sun, Yanru; He, Wei; Xu, Man; Hao, Chao
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：LangTime: A Language-Guided Unified Model for Time Series Forecasting with Proximal Policy Optimization

快速理解：Recent research has shown an increasing interest in utilizing pre-trained large language models (LLMs) for a variety of time series applications. However, there are three main challenges when using LLMs as foundational models for time series forecasting: (1) Cross-domain generalization. (2) Cross-modality alignment. (3) Error accumulation in autoregressive frameworks.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.300 Layer-wise Quantization for Quantized Optimistic Dual Averaging

- 作者：Nguyen, Anh Duc; Markov, Ilia; Wu, Zhengqing; Ramezani-Kebrya, Ali; Antonakopoulos, Kimon; Alistarh, Dan; Cevher, Volkan
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：Layer-wise Quantization for Quantized Optimistic Dual Averaging

快速理解：Modern deep neural networks exhibit heterogeneity across numerous layers of various

types such as residuals, multi-head attention, etc., due to varying structures (dimensions, activation functions, etc.), distinct representation characteristics, which impact predictions. We develop a general layer-wise quantization framework with tight variance and code-length bounds, adapting to the heterogeneities over the course of training.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.6.301 Learn from Downstream and Be Yourself in Multimodal Large Language Models Fine-Tuning

- 作者：Huang, Wenke; Liang, Jian; Shi, Zekun; Zhu, Didi; Wan, Guancheng; Li, He; Du, Bo; Tao, Dacheng; Ye, Mang
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Learn from Downstream and Be Yourself in Multimodal Large Language Models Fine-Tuning

快速理解：Multimodal Large Language Model (MLLM) has demonstrated strong generalization capabilities across diverse distributions and tasks, largely due to extensive pre-training datasets. Fine-tuning MLLM has become a common practice to improve performance on specific downstream tasks.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.6.302 Learning Invariant Causal Mechanism from Vision-Language Models

- 作者：Song, Zeen; Zhao, Siyu; Zhang, Xingyu; Li, Jiangmeng; Zheng, Changwen; Qiang, Wenwen
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Learning Invariant Causal Mechanism from Vision-Language Models

快速理解：Contrastive Language-Image Pretraining (CLIP) has achieved remarkable success, but its performance can degrade when fine-tuned in out-of-distribution (OOD) scenarios. We model the prediction process using a Structural Causal Model (SCM) and show that the causal mechanism involving both invariant and variant factors in training environments differs from that in test environments.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.6.303 Learning Optimal Multimodal Information Bottleneck Representations

- 作者：Wu, Qilong; Shao, Yiyang; Wang, Jun; Sun, Xiaobo
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Learning Optimal Multimodal Information Bottleneck Representations

快速理解：Leveraging high-quality joint representations from multimodal data can greatly enhance model performance in various machine-learning based applications. Recent multimodal learning

methods, based on the multimodal information bottleneck (MIB) principle, aim to generate optimal MIB with maximal task-relevant information and minimal superfluous information via regularization.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.304 Learning Progress Driven Multi-Agent Curriculum

- 作者: Zhao, Wenshuai; Li, Zhiyuan; Pajarinen, Joni
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Learning Progress Driven Multi-Agent Curriculum

快速理解: The number of agents can be an effective curriculum variable for controlling the difficulty of multi-agent reinforcement learning (MARL) tasks. Existing work typically uses manually defined curricula such as linear schemes.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.305 Learning Strategic Language Agents in the Werewolf Game with Iterative Latent Space Policy Optimization

- 作者: Xu, Zelai; Gu, Wanjun; Yu, Chao; Wu, Yi; Wang, Yu
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Learning Strategic Language Agents in the Werewolf Game with Iterative Latent Space Policy Optimization

快速理解: Large language model (LLM) agents have recently demonstrated impressive capabilities in various domains like open-ended conversation and multi-step decision-making. However, it remains challenging for these agents to solve strategic language games, such as Werewolf, which demand both strategic decision-making and free-form language interactions.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.306 Learning from Loss Landscape: Generalizable Mixed-Precision Quantization via Adaptive Sharpness-Aware Gradient Aligning

- 作者: Ma, Lianbo; Ma, Jianlun; Zhou, Yuee; Xie, Guoyang; He, Qiang; Lu, Zhichao
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Learning from Loss Landscape: Generalizable Mixed-Precision Quantization via Adaptive Sharpness-Aware Gradient Aligning

快速理解: Mixed Precision Quantization (MPQ) has become an essential technique for optimizing neural network by determining the optimal bitwidth per layer. Existing MPQ methods, however, face a major hurdle: they require a computationally expensive search for quantization strategies on

large-scale datasets.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.6.307 Learning to Trust Bellman Updates: Selective State-Adaptive Regularization for Offline RL

- 作者：Luo, Qin-Wen; Xie, Ming-Kun; Wang, Ye-Wen; Huang, Sheng-Jun
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Learning to Trust Bellman Updates: Selective State-Adaptive Regularization for Offline RL

快速理解：Offline reinforcement learning (RL) aims to learn an effective policy from a static dataset. To alleviate extrapolation errors, existing studies often uniformly regularize the value function or policy updates across all states.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.308 Logarithmic Regret for Online KL-Regularized Reinforcement Learning

- 作者：Zhao, Heyang; Ye, Chenlu; Xiong, Wei; Gu, Quanquan; Zhang, Tong
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Logarithmic Regret for Online KL-Regularized Reinforcement Learning

快速理解：Recent advances in Reinforcement Learning from Human Feedback (RLHF) have shown that KL-regularization plays a pivotal role in improving the efficiency of RL fine-tuning for large language models (LLMs). Despite its empirical advantage, the theoretical difference between KL-regularized RL and standard RL remains largely under-explored.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.309 Look Twice Before You Answer: Memory-Space Visual Retracing for Hallucination Mitigation in Multimodal Large Language Models

- 作者：Zou, Xin; Wang, Yizhou; Yan, Yibo; Lyu, Yuanhuiyi; Zheng, Kening; Huang, Sirui; Chen, Junkai; Jiang, Peijie; Liu, Jia; Tang, Chang; Hu, Xuming
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Look Twice Before You Answer: Memory-Space Visual Retracing for Hallucination Mitigation in Multimodal Large Language Models

快速理解：Despite their impressive capabilities, Multimodal Large Language Models (MLLMs) are prone to hallucinations, i.e., the generated content that is nonsensical or unfaithful to input sources. Unlike in LLMs, hallucinations in MLLMs often stem from the sensitivity of text decoder to visual tokens, leading to a phenomenon akin to “amnesia” about visual information.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.6.310 Low-distortion and GPU-compatible Tree Embeddings in Hyperbolic Space

- 作者：Van Spengler, Max; Mettes, Pascal
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：Low-distortion and GPU-compatible Tree Embeddings in Hyperbolic Space

快速理解：Embedding tree-like data, from hierarchies to ontologies and taxonomies, forms a well-studied problem for representing knowledge across many domains. Hyperbolic geometry provides a natural solution for embedding trees, with vastly superior performance over Euclidean embeddings.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.6.311 M³HF: Multi-agent Reinforcement Learning from Multi-phase Human Feedback of Mixed Quality

- 作者：Wang, Ziyang; Zhang, Zhicheng; Fang, Fei; Du, Yali
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：M³HF: Multi-agent Reinforcement Learning from Multi-phase Human Feedback of Mixed Quality

快速理解：Designing effective reward functions in multi-agent reinforcement learning (MARL) is a significant challenge, often leading to suboptimal or misaligned behaviors in complex, coordinated environments. We introduce Multi-agent Reinforcement Learning from Multi-phase Human Feedback of Mixed Quality (text{M}^3text{HF}), a novel framework that integrates multi-phase human feedback of mixed quality into the MARL training process.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.312 MERGE³: Efficient Evolutionary Merging on Consumer-grade GPUs

- 作者：Mencattini, Tommaso; Minut, Robert Adrian; Crisostomi, Donato; Santilli, Andrea; Rodol'f, Emanuele
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：MERGE³: Efficient Evolutionary Merging on Consumer-grade GPUs

快速理解：Evolutionary model merging enables the creation of high-performing multi-task models but remains computationally prohibitive for consumer hardware. We introduce MERGE³, an efficient framework that makes evolutionary merging of Large Language Models (LLMs) feasible on a single GPU by reducing fitness computation costs 50times while retaining a large fraction of the original performance.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.6.313 MME-CoT: Benchmarking Chain-of-Thought in Large Multimodal Models for Reasoning Quality, Robustness, and Efficiency

- 作者: Jiang, Dongzhi; Zhang, Renrui; Guo, Ziyu; Li, Yanwei; Qi, Yu; Chen, Xinyan; Wang, Liuhui; Jin, Jianhan; Guo, Claire; Yan, Shen; Zhang, Bo; Fu, Chaoyou; Gao, Peng; Li, Hongsheng
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: MME-CoT: Benchmarking Chain-of-Thought in Large Multimodal Models for Reasoning Quality, Robustness, and Efficiency

快速理解: Answering questions with Chain-of-Thought (CoT) has significantly enhanced the reasoning capabilities of Large Language Models (LLMs), yet its impact on Large Multimodal Models (LMMs) still lacks a systematic assessment and in-depth investigation. In this paper, we introduce MME-CoT, a specialized benchmark evaluating the CoT reasoning performance of LMMs, spanning six domains: math, science, OCR, logic, space-time, and general scenes.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.6.314 MMedPO: Aligning Medical Vision-Language Models with Clinical-Aware Multimodal Preference Optimization

- 作者: Zhu, Kangyu; Xia, Peng; Li, Yun; Zhu, Hongtu; Wang, Sheng; Yao, Huaxiu
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: MMedPO: Aligning Medical Vision-Language Models with Clinical-Aware Multimodal Preference Optimization

快速理解: The advancement of Large Vision-Language Models (LVLMs) has propelled their application in the medical field. However, Medical LVLMs (Med-LVLMs) encounter factuality challenges due to modality misalignment, where the models prioritize textual knowledge over visual input, leading to hallucinations that contradict information in medical images.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.6.315 MODA: MODular Duplex Attention for Multimodal Perception, Cognition, and Emotion Understanding

- 作者: Zhang, Zhicheng; Xia, Wuyou; Zhao, Chenxi; Yan, Zhou; Liu, Xiaoqiang; Zhu, Yongjie; Qin, Wenyu; Wan, Pengfei; Zhang, Di; Yang, Jufeng
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: MODA: MODular Duplex Attention for Multimodal Perception, Cognition, and Emotion Understanding

快速理解: Multimodal large language models (MLLMs) recently showed strong capacity in integrat-

ing data among multiple modalities, empowered by generalizable attention architecture. Advanced methods predominantly focus on language-centric tuning while less exploring multimodal tokens mixed through attention, posing challenges in high-level tasks that require fine-grained cognition and emotion understanding.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.6.316 MODULI: Unlocking Preference Generalization via Diffusion Models for Offline Multi-Objective Reinforcement Learning

- 作者：Yuan, Yifu; Zheng, Zhenrui; Dong, Zibin; Hao, Jianye
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文:MODULI: Unlocking Preference Generalization via Diffusion Models for Offline Multi-Objective Reinforcement Learning

快速理解: Multi-objective Reinforcement Learning (MORL) seeks to develop policies that simultaneously optimize multiple conflicting objectives, but it requires extensive online interactions. Offline MORL provides a promising solution by training on pre-collected datasets to generalize to any preference upon deployment.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.317 MP-Nav: Enhancing Data Poisoning Attacks against Multimodal Learning

- 作者：Zhang, Jingfeng; Krishnamurthy, Prashanth; Patel, Naman; Tzes, Anthony; Khorrami, Farshad
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文: MP-Nav: Enhancing Data Poisoning Attacks against Multimodal Learning

快速理解: Despite the success of current multimodal learning at scale, its susceptibility to data poisoning attacks poses security concerns in critical applications. Attacker can manipulate model behavior by injecting maliciously crafted yet minute instances into the training set, stealthily mismatching distinct concepts.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.6.318 MTSTRec: Multimodal Time-Aligned Shared Token Recommender

- 作者：Hong, Ming-Yi; Hsu, Yen-Jung; Chiang, Miao-Chen; Lin, Che
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文: MTSTRec: Multimodal Time-Aligned Shared Token Recommender

快速理解: Sequential recommendation in e-commerce utilizes users' anonymous browsing histories to personalize product suggestions without relying on private information. Existing item ID-based

methods and multimodal models often overlook the temporal alignment of modalities like textual descriptions, visual content, and prices in user browsing sequences.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.6.319 Make LoRA Great Again: Boosting LoRA with Adaptive Singular Values and Mixture-of-Experts Optimization Alignment

- 作者：Fan, Chenghao; Lu, Zhenyi; Liu, Sichen; Gu, Chengfeng; Qu, Xiaoye; Wei, Wei; Cheng, Yu
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：Make LoRA Great Again: Boosting LoRA with Adaptive Singular Values and Mixture-of-Experts Optimization Alignment

快速理解：While Low-Rank Adaptation (LoRA) enables parameter-efficient fine-tuning for Large Language Models (LLMs), its performance often falls short of Full Fine-Tuning (Full FT). Current methods optimize LoRA by initializing with static singular value decomposition (SVD) subsets, leading to suboptimal leveraging of pre-trained knowledge.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.6.320 Matryoshka Quantization

- 作者：Nair, Pranav Ajit; Datta, Puranjay; Dean, Jeff; Jain, Prateek; Kusupati, Aditya
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：Matryoshka Quantization

快速理解：Quantizing model weights is critical for reducing the communication and inference costs of large models. However, quantizing models -especially to low precisions like int4 or int2 -requires a trade-off in model quality; int2, in particular, is known to severely degrade model quality.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.6.321 Maximizing Intermediate Checkpoint Value in LLM Pretraining with Bayesian Optimization

- 作者：Liu, Deyuan; Wang, Zecheng; Wang, Bingning; Chen, Weipeng; Li, Chunshan; Tu, Zhiying; Chu, Dianhui; Sui, Dianbo
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：Maximizing Intermediate Checkpoint Value in LLM Pretraining with Bayesian Optimization

快速理解：The rapid proliferation of large language models (LLMs), such as GPT-4 and Gemini, underscores the intense demand for resources during their training processes, posing significant challenges due to substantial computational and environmental costs. In this paper, we introduce a novel checkpoint merging strategy aimed at making efficient use of intermediate checkpoints during

LLM pretraining.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.6.322 Maximum Total Correlation Reinforcement Learning

- 作者：You, Bang; Liu, Puze; Liu, Huaping; Peters, Jan; Arenz, Oleg
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Maximum Total Correlation Reinforcement Learning

快速理解：Simplicity is a powerful inductive bias. In reinforcement learning, regularization is used for simpler policies, data augmentation for simpler representations, and sparse reward functions for simpler objectives, all that, with the underlying motivation to increase generalizability and robustness by focusing on the essentials.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.323 Merge-Friendly Post-Training Quantization for Multi-Target Domain Adaptation

- 作者：Shin, Juncheol; Seok, Minsang; Kim, Seonggon; Park, Eunhyeok
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：Merge-Friendly Post-Training Quantization for Multi-Target Domain Adaptation

快速理解：Model merging has emerged as a powerful technique for combining task-specific weights, achieving superior performance in multi-target domain adaptation. However, when applied to practical scenarios, such as quantized models, new challenges arise.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.6.324 Meta-Reinforcement Learning with Adaptation from Human Feedback via Preference-Order-Preserving Task Embedding

- 作者：Xu, Siyuan; Zhu, Minghui
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Meta-Reinforcement Learning with Adaptation from Human Feedback via Preference-Order-Preserving Task Embedding

快速理解：This paper studies meta-reinforcement learning with adaptation from human feedback. It aims to pre-train a meta-model that can achieve few-shot adaptation for new tasks from human preference queries without relying on reward signals.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.325 MetaAgent: Automatically Constructing Multi-Agent Systems Based on Finite State Machines

- 作者: Zhang, Yaolun; Liu, Xiaogeng; Xiao, Chaowei
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: MetaAgent: Automatically Constructing Multi-Agent Systems Based on Finite State Machines

快速理解: Large Language Models (LLMs) have demonstrated the ability to solve a wide range of practical tasks within multi-agent systems. However, existing human-designed multi-agent frameworks are typically limited to a small set of pre-defined scenarios, while current automated design methods suffer from several limitations, such as the lack of tool integration, dependence on external training data, and rigid communication structures.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.326 MimicMotion: High-Quality Human Motion Video Generation with Confidence-aware Pose Guidance

- 作者: Zhang, Yuang; Gu, Jiayi; Wang, Li-Wen; Wang, Han; Cheng, Junqi; Zhu, Yuefeng; Zou, Fangyuan
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: MimicMotion: High-Quality Human Motion Video Generation with Confidence-aware Pose Guidance

快速理解: In recent years, while generative AI has advanced significantly in image generation, video generation continues to face challenges in controllability, length, and detail quality, which hinder its application. We present MimicMotion, a framework for generating high-quality human videos of arbitrary length using motion guidance.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.327 Mind the Gap: A Practical Attack on GGUF Quantization

- 作者: Egashira, Kazuki; Staab, Robin; Vero, Mark; He, Jingxuan; Vechev, Martin
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Mind the Gap: A Practical Attack on GGUF Quantization

快速理解: With the increasing size of frontier LLMs, post-training quantization has become the standard for memory-efficient deployment. Recent work has shown that basic rounding-based quantization schemes pose security risks, as they can be exploited to inject malicious behaviors into quantized models that remain hidden in full precision.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.6.328 Minimax Optimal Regret Bound for Reinforcement Learning with Trajectory Feedback

- 作者: Zhang, Zihan; Chen, Yuxin; Lee, Jason D.; Du, Simon Shaolei; Wang, Ruosong
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Minimax Optimal Regret Bound for Reinforcement Learning with Trajectory Feedback

快速理解: In this work, we study reinforcement learning (RL) with trajectory feedback. Compared to the standard RL setting, in RL with trajectory feedback, the agent only observes the accumulative reward along the trajectory, and therefore, this model is particularly suitable for scenarios where querying the reward in each single step incurs prohibitive cost.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.329 Mitigating Object Hallucination in Large Vision-Language Models via Image-Grounded Guidance

- 作者: Zhao, Linxi; Deng, Yihe; Zhang, Weitong; Gu, Quanquan
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Mitigating Object Hallucination in Large Vision-Language Models via Image-Grounded Guidance

快速理解: The advancement of Large Vision-Language Models (LVLMs) has increasingly highlighted the critical issue of their tendency to hallucinate non-existing objects in the images. To address this issue, previous works focused on using specially curated datasets or powerful LLMs to rectify the outputs of LVLMs.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.330 Mitigating Plasticity Loss in Continual Reinforcement Learning by Reducing Churn

- 作者: Tang, Hongyao; Obando-Ceron, Johan; Castro, Pablo Samuel; Courville, Aaron; Berseth, Glen
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Mitigating Plasticity Loss in Continual Reinforcement Learning by Reducing Churn

快速理解: Plasticity, or the ability of an agent to adapt to new tasks, environments, or distributions, is crucial for continual learning. In this paper, we study the loss of plasticity in deep continual RL from the lens of churn: network output variability induced by the data in each training batch.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.331 Mixture of Experts Made Intrinsically Interpretable

- 作者: Yang, Xingyi; Venhoff, Constantin; Khakzar, Ashkan; Schroeder De Witt, Christian; Doka-
nia, Puneet K.; Bibi, Adel; Torr, Philip
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Mixture of Experts Made Intrinsically Interpretable

快速理解: Neurons in large language models often exhibit polysemanticity, simultaneously encoding multiple unrelated concepts and obscuring interpretability. Instead of relying on post-hoc methods, we present MoE-X, a mixture-of-experts (MoE) language model designed to be intrinsically interpretable.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.6.332 Mixture of Experts Provably Detect and Learn the Latent Cluster Structure in Gradient-Based Learning

- 作者: Kawata, Ryotaro; Matsutani, Kohsei; Kinoshita, Yuri; Nishikawa, Naoki; Suzuki, Taiji
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Mixture of Experts Provably Detect and Learn the Latent Cluster Structure in Gradient-Based Learning

快速理解: Mixture of Experts (MoE), an ensemble of specialized models equipped with a router that dynamically distributes each input to appropriate experts, has achieved successful results in the field of machine learning. However, theoretical understanding of this architecture is falling behind due to its inherent complexity.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.6.333 MoE-SVD: Structured Mixture-of-Experts LLMs Compression via Singular Value Decomposition

- 作者: Li, Wei; Li, Lujun; Gu, Hao; Huang, You-Liang; Lee, Mark G.; Sun, Shengjie; Xue, Wei;
Guo, Yike
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: MoE-SVD: Structured Mixture-of-Experts LLMs Compression via Singular Value Decomposition

快速理解: Mixture of Experts (MoE) architecture improves Large Language Models (LLMs) with better scaling, but its higher parameter counts and memory demands create challenges for deployment. In this paper, we present MoE-SVD, a new decomposition-based compression framework tailored for MoE LLMs without any extra training.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.6.334 MoEQuant: Enhancing Quantization for Mixture-of-Experts Large Language Models via Expert-Balanced Sampling and Affinity Guidance

- 作者: Chen, Zhixuan; Hu, Xing; Yang, Dawei; Xu, Zukang; Chen, Xu; Yuan, Zhihang; Zhou, Sifan; Yu, Jiangyong
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: MoEQuant: Enhancing Quantization for Mixture-of-Experts Large Language Models via Expert-Balanced Sampling and Affinity Guidance

快速理解: Mixture-of-Experts (MoE) large language models (LLMs), which leverage dynamic routing and sparse activation to enhance efficiency and scalability, have achieved higher performance while reducing computational costs. However, these models face significant memory overheads, limiting their practical deployment and broader adoption.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.6.335 Modularized Self-Reflected Video Reasoner for Multimodal LLM with Application to Video Question Answering

- 作者: Song, Zihan; Wang, Xin; Qian, Zi; Chen, Hong; Huang, Longtao; Xue, Hui; Zhu, Wenwu
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Modularized Self-Reflected Video Reasoner for Multimodal LLM with Application to Video Question Answering

快速理解: Multimodal Large Language Models (Multimodal LLMs) have shown their strength in Video Question Answering (VideoQA). However, due to the black-box nature of end-to-end training strategies, existing approaches based on Multimodal LLMs suffer from the lack of interpretability for VideoQA: they can neither present reasoning paths nor indicate where the answers are derived from the video.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.336 Modulated Diffusion: Accelerating Generative Modeling with Modulated Quantization

- 作者: Gao, Weizhi; Hou, Zhichao; Yin, Junqi; Wang, Feiyi; Peng, Linyu; Liu, Xiaorui
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Modulated Diffusion: Accelerating Generative Modeling with Modulated Quantization

快速理解: Diffusion models have emerged as powerful generative models, but their high computational cost in iterative sampling remains a significant bottleneck. In this work, we present an in-depth and insightful study of state-of-the-art acceleration techniques for diffusion models, including caching and quantization, and reveal their limitations in computation error and generation quality.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.6.337 Moirai-MoE: Empowering Time Series Foundation Models with Sparse Mixture of Experts

- 作者：Liu, Xu; Liu, Juncheng; Woo, Gerald; Aksu, Taha; Liang, Yuxuan; Zimmermann, Roger; Liu, Chenghao; Li, Junnan; Savarese, Silvio; Xiong, Caiming; Sahoo, Doyen
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：Moirai-MoE: Empowering Time Series Foundation Models with Sparse Mixture of Experts

快速理解：Achieving effective unified pretraining on large time series corpora remains an open challenge in developing time series foundation models. Existing methods, such as Moirai, introduce multiple projection layers for time series of different frequencies to account for high data heterogeneity.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.6.338 More Than Meets the Eye: Enhancing Multi-Object Tracking Even with Prolonged Occlusions

- 作者：Galoaa, Bishoy; Amraee, Somaieh; Ostadabbas, Sarah
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：More Than Meets the Eye: Enhancing Multi-Object Tracking Even with Prolonged Occlusions

快速理解：This paper introduces MOTE (MOre Than meets the Eye), a novel multi-object tracking (MOT) algorithm designed to address the challenges of tracking occluded objects. By integrating deformable detection transformers with a custom disocclusion matrix, MOTE significantly enhances the ability to track objects even when they are temporarily hidden from view.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.6.339 Multi-Modal Object Re-identification via Sparse Mixture-of-Experts

- 作者：Feng, Yingying; Li, Jie; Xie, Chi; Tan, Lei; Ji, Jiayi
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：Multi-Modal Object Re-identification via Sparse Mixture-of-Experts

快速理解：We present MFRNet, a novel network for multi-modal object re-identification that integrates multi-modal data features to effectively retrieve specific objects across different modalities. Current methods suffer from two principal limitations: (1) insufficient interaction between pixel-level semantic features across modalities, and (2) difficulty in balancing modality-shared and modality-specific features within a unified architecture.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.6.340 Multi-agent Architecture Search via Agentic Supernet

- 作者: Zhang, Guibin; Niu, Luyang; Fang, Junfeng; Wang, Kun; Bai, Lei; Wang, Xiang
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Multi-agent Architecture Search via Agentic Supernet

快速理解: Large Language Model (LLM)-empowered multi-agent systems extend the cognitive boundaries of individual agents through disciplined collaboration and interaction, while constructing these systems often requires labor-intensive manual designs. Despite the availability of methods to automate the design of agentic workflows, they typically seek to identify a static, complex, one-size-fits-all system, which, however, fails to dynamically allocate inference resources based on the difficulty and domain of each query.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.341 Multi-objective Linear Reinforcement Learning with Lexicographic Rewards

- 作者: Xue, Bo; Bu, Dake; Cheng, Ji; Wan, Yuanyu; Zhang, Qingfu
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Multi-objective Linear Reinforcement Learning with Lexicographic Rewards

快速理解: Reinforcement Learning (RL) with linear transition kernels and reward functions has recently attracted growing attention due to its computational efficiency and theoretical advancements. However, prior theoretical research in RL has primarily focused on single-objective problems, resulting in limited theoretical development for multi-objective reinforcement learning (MORL).

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.342 Multimodal Medical Code Tokenizer

- 作者: Su, Xiaorui; Messica, Shvat; Huang, Yepeng; Johnson, Ruth; Fesser, Lukas; Gao, Shanghua; Sahneh, Faryad; Zitnik, Marinka
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Multimodal Medical Code Tokenizer

快速理解: Foundation models trained on patient electronic health records (EHRs) require tokenizing medical data into sequences of discrete vocabulary items. Existing tokenizers treat medical codes from EHRs as isolated textual tokens.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.343 MxMoE: Mixed-precision Quantization for MoE with Accuracy and Performance Co-Design

- 作者: Duanmu, Haojie; Li, Xiuhong; Yuan, Zhihang; Zheng, Size; Duan, Jiangfei; Zhang, Xingcheng; Lin, Dahua
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: MxMoE: Mixed-precision Quantization for MoE with Accuracy and Performance Co-Design

快速理解: Mixture-of-Experts (MoE) models face deployment challenges due to their large parameter counts and computational demands. We explore quantization for MoE models and highlight two key insights: 1) linear blocks exhibit varying quantization sensitivity, and 2) divergent expert activation frequencies create heterogeneous computational characteristics.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.6.344 Navigating the Social Welfare Frontier: Portfolios for Multi-objective Reinforcement Learning

- 作者: Kim, Cheol Woo; Moondra, Jai; Verma, Shresth; Pollack, Madeleine; Kong, Ling kai; Tambe, Milind; Gupta, Swati
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Navigating the Social Welfare Frontier: Portfolios for Multi-objective Reinforcement Learning

快速理解: In many real-world applications of Reinforcement Learning (RL), deployed policies have varied impacts on different stakeholders, creating challenges in reaching consensus on how to effectively aggregate their preferences. Generalized p-means form a widely used class of social welfare functions for this purpose, with broad applications in fair resource allocation, AI alignment, and decision-making.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.345 Near-optimal Regret Using Policy Optimization in Online MDPs with Aggregate Bandit Feedback

- 作者: Lancewicki, Tal; Mansour, Yishay
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Near-optimal Regret Using Policy Optimization in Online MDPs with Aggregate Bandit Feedback

快速理解: We study online finite-horizon Markov Decision Processes with adversarially changing loss and aggregate bandit feedback (a.k.a full-bandit). Under this type of feedback, the agent observes only the total loss incurred over the entire trajectory, rather than the individual losses at each intermediate step within the trajectory.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.346 Network Sparsity Unlocks the Scaling Potential of Deep Reinforcement Learning

- 作者: Ma, Guozheng; Li, Lu; Wang, Zilin; Shen, Li; Bacon, Pierre-Luc; Tao, Dacheng
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Network Sparsity Unlocks the Scaling Potential of Deep Reinforcement Learning

快速理解: Effectively scaling up deep reinforcement learning models has proven notoriously difficult due to network pathologies during training, motivating various targeted interventions such as periodic reset and architectural advances such as layer normalization. Instead of pursuing more complex modifications, we show that introducing static network sparsity alone can unlock further scaling potential beyond their dense counterparts with state-of-the-art architectures.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.347 Neural Event-Triggered Control with Optimal Scheduling

- 作者: Yang, Luan; Zhang, Jingdong; Zhu, Qunxi; Lin, Wei
- 年份: 2025; 主题: Systems; 优先级: 重要
- [来源](#)

论文: Neural Event-Triggered Control with Optimal Scheduling

快速理解: Learning-enabled controllers with stability certificate functions have demonstrated impressive empirical performance in addressing control problems in recent years. Nevertheless, directly deploying the neural controllers onto actual digital platforms requires impractically excessive communication resources due to a continuously updating demand from the closed-loop feedback controller.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.6.348 O-MAPL: Offline Multi-agent Preference Learning

- 作者: Bui, The Viet; Mai, Tien Anh; Nguyen, Thanh Hong
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: O-MAPL: Offline Multi-agent Preference Learning

快速理解: Inferring reward functions from demonstrations is a key challenge in reinforcement learning (RL), particularly in multi-agent RL (MARL). The large joint state-action spaces and intricate inter-agent interactions in MARL make inferring the joint reward function especially challenging.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.349 OV-MER: Towards Open-Vocabulary Multimodal Emotion Recognition

- 作者: Lian, Zheng; Sun, Haiyang; Sun, Licai; Chen, Haoyu; Chen, Lan; Gu, Hao; Wen, Zhuofan;

Chen, Shun; Siyuan, Zhang; Yao, Hailiang; Liu, Bin; Liu, Rui; Liang, Shan; Li, Ya; Yi, Jiangyan; Tao, Jianhua

- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: OV-MER: Towards Open-Vocabulary Multimodal Emotion Recognition

快速理解: Multimodal Emotion Recognition (MER) is a critical research area that seeks to decode human emotions from diverse data modalities. However, existing machine learning methods predominantly rely on predefined emotion taxonomies, which fail to capture the inherent complexity, subtlety, and multi-appraisal nature of human emotional experiences, as demonstrated by studies in psychology and cognitive science.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.350 Offline-to-Online Reinforcement Learning with Classifier-Free Diffusion Generation

- 作者: Huang, Xiao; Liu, Xu; Zhang, Enze; Yu, Tong; Li, Shuai
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Offline-to-Online Reinforcement Learning with Classifier-Free Diffusion Generation

快速理解: Offline-to-online Reinforcement Learning (O2O RL) aims to perform online fine-tuning on an offline pre-trained policy to minimize costly online interactions. Existing work used offline datasets to generate data that conform to the online data distribution for data augmentation.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.351 On Path to Multimodal Generalist: General-Level and General-Bench

- 作者: Fei, Hao; Zhou, Yuan; Li, Juncheng; Li, Xiangtai; Xu, Qingshan; Li, Bobo; Wu, Shengqiong; Wang, Yaoting; Zhou, Junbao; Meng, Jiahao; Shi, Qingyu; Zhou, Zhiyuan; Shi, Liangtao; Gao, Minghe; Zhang, Daoan; Ge, Zhiqi; Tang, Siliang; Pan, Kaihang; Ye, Yaobo; Yuan, Haobo; Zhang, Tao; Wu, Weiming; Ju, Tianjie; Meng, Zixiang; Xu, Shilin; Jia, Liyu; Hu, Wentao; Luo, Meng; Luo, Jiebo; Chua, Tat-Seng; Yan, Shuicheng; Zhang, Hanwang
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: On Path to Multimodal Generalist: General-Level and General-Bench

快速理解: The Multimodal Large Language Model (MLLM) is currently experiencing rapid growth, driven by the advanced capabilities of language-based LLMs. Unlike their specialist predecessors, existing MLLMs are evolving towards a Multimodal Generalist paradigm.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.352 On the Resilience of LLM-Based Multi-Agent Collaboration with Faulty Agents

- 作者: Huang, Jen-Tse; Zhou, Jiaxu; Jin, Tailin; Zhou, Xuhui; Chen, Zixi; Wang, Wenxuan; Yuan, Youliang; Lyu, Michael; Sap, Maarten
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: On the Resilience of LLM-Based Multi-Agent Collaboration with Faulty Agents

快速理解: Large language model-based multi-agent systems have shown great abilities across various tasks due to the collaboration of expert agents, each focusing on a specific domain. However, the impact of clumsy or even malicious agents—those who frequently make errors in their tasks—on the overall performance of the system remains underexplored.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.353 On the Robustness of Reward Models for Language Model Alignment

- 作者: Hong, Jiwoo; Lee, Noah; Kim, Eunki; Son, Guijin; Chung, Woojin; Gupta, Aman; Tang, Shao; Thorne, James
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: On the Robustness of Reward Models for Language Model Alignment

快速理解: The Bradley-Terry (BT) model is widely practiced in reward modeling for reinforcement learning with human feedback (RLHF). Despite its effectiveness, reward models (RMs) trained with BT model loss as one-way classifiers are prone to over-optimization, losing generalizability to unseen inputs.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.354 Online Episodic Convex Reinforcement Learning

- 作者: Moreno, Bianca Marin; Eldowa, Khaled; Gaillard, Pierre; Br' {e}g' {e}re, Margaux; Oudjane, Nadia
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Online Episodic Convex Reinforcement Learning

快速理解: We study online learning in episodic finite-horizon Markov decision processes (MDPs) with convex objective functions, known as the concave utility reinforcement learning (CURL) problem. This setting generalizes RL from linear to convex losses on the state-action distribution induced by the agent' s policy.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.355 Online Laplacian-Based Representation Learning in Reinforcement Learning

- 作者: Ahmed, Maheed H.; Bhargav, Jayanth; Ghasemi, Mahsa
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Online Laplacian-Based Representation Learning in Reinforcement Learning

快速理解: Representation learning plays a crucial role in reinforcement learning, especially in complex environments with high-dimensional and unstructured states. Effective representations can enhance the efficiency of learning algorithms by improving sample efficiency and generalization across tasks.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.356 Online Pre-Training for Offline-to-Online Reinforcement Learning

- 作者: Shin, Yongjae; Kim, Jeonghye; Jung, Whiyoung; Hong, Sunghoon; Yoon, Deunsol; Jang, Youngsoo; Kim, Geon-Hyeong; Chae, Jongseong; Sung, Youngchul; Lee, Kanghoon; Lim, Woohyung
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Online Pre-Training for Offline-to-Online Reinforcement Learning

快速理解: Offline-to-online reinforcement learning (RL) aims to integrate the complementary strengths of offline and online RL by pre-training an agent offline and subsequently fine-tuning it through online interactions. However, recent studies reveal that offline pre-trained agents often underperform during online fine-tuning due to inaccurate value estimation caused by distribution shift, with random initialization proving more effective in certain cases.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.357 Online Robust Reinforcement Learning Through Monte-Carlo Planning

- 作者: Dam, Tuan Quang; Panaganti, Kishan; Driss, Brahim; Wierman, Adam
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Online Robust Reinforcement Learning Through Monte-Carlo Planning

快速理解: Monte Carlo Tree Search (MCTS) is a powerful framework for solving complex decision-making problems, yet it often relies on the assumption that the simulator and the real-world dynamics are identical. Although this assumption helps achieve the success of MCTS in games like Chess, Go, and Shogi, the real-world scenarios incur ambiguity due to their modeling mismatches in low-fidelity simulators.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.358 Optimal Sensor Scheduling and Selection for Continuous-Discrete Kalman Filtering with Auxiliary Dynamics

- 作者: Ahdab, Mohamad Al; Leth, John; Tan, Zheng-Hua
- 年份: 2025; 主题: Systems; 优先级: 重要
- [来源](#)

论文: Optimal Sensor Scheduling and Selection for Continuous-Discrete Kalman Filtering with Auxiliary Dynamics

快速理解: We study the Continuous-Discrete Kalman Filter (CD-KF) for State-Space Models (SSMs) where continuous-time dynamics are observed via multiple sensors with discrete, irregularly timed measurements. Our focus extends to scenarios in which the measurement process is coupled with the states of an auxiliary SSM.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.6.359 Optimizing Language Models for Inference Time Objectives using Reinforcement Learning

- 作者: Tang, Yunhao; Zheng, Kunhao; Synnaeve, Gabriel; Munos, Remi
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Optimizing Language Models for Inference Time Objectives using Reinforcement Learning

快速理解: In this work, we investigate the merits of explicitly optimizing for inference time algorithmic performance during model training. We show how optimizing for inference time performance can improve overall model efficacy.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.360 Optimizing Large Language Model Training Using FP4 Quantization

- 作者: Wang, Ruizhe; Gong, Yeyun; Liu, Xiao; Zhao, Guoshuai; Yang, Ziyue; Guo, Baining; Zha, Zheng-Jun; Cheng, Peng
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Optimizing Large Language Model Training Using FP4 Quantization

快速理解: The growing computational demands of training large language models (LLMs) necessitate more efficient methods. Quantized training presents a promising solution by enabling low-bit arithmetic operations to reduce these costs.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.6.361 Optimizing Robustness and Accuracy in Mixture of Experts: A Dual-Model Approach

- 作者: Zhang, Xu; Xu, Kaidi; Hu, Ziqing; Wang, Ren
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Optimizing Robustness and Accuracy in Mixture of Experts: A Dual-Model Approach

快速理解: Mixture of Experts (MoE) have shown remarkable success in leveraging specialized expert networks for complex machine learning tasks. However, their susceptibility to adversarial attacks presents a critical challenge for deployment in robust applications.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.6.362 OrthoRank: Token Selection via Sink Token Orthogonality for Efficient LLM inference

- 作者: Shin, Seungjun; Oh, Jaehoon; Oh, Dokwan
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: OrthoRank: Token Selection via Sink Token Orthogonality for Efficient LLM inference

快速理解: Attention mechanisms are central to the success of large language models (LLMs), enabling them to capture intricate token dependencies and implicitly assign importance to each token. Recent studies have revealed the sink token, which receives disproportionately high attention despite their limited semantic role.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.6.363 Outlier-Aware Post-Training Quantization for Discrete Graph Diffusion Models

- 作者: Gong, Zheng; Sun, Ying
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Outlier-Aware Post-Training Quantization for Discrete Graph Diffusion Models

快速理解: Discrete Graph Diffusion Models (DGDMs) mark a pivotal advancement in graph generation, effectively preserving sparsity and structural integrity, thereby enhancing the learning of graph data distributions for diverse generative applications. Despite their potential, DGDMs are computationally intensive due to the numerous low-parameter yet high-computation operations, thereby increasing the need of inference acceleration.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.6.364 Overcoming Multi-step Complexity in Multimodal Theory-of-Mind Reasoning: A Scalable Bayesian Planner

- 作者: Zhang, Chunhui; Ouyang, Zhongyu; Lee, Kwonjoon; Agarwal, Nakul; Houlihan, Sean Dae; Vosoughi, Soroush; Lo, Shao-Yuan

- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Overcoming Multi-step Complexity in Multimodal Theory-of-Mind Reasoning: A Scalable Bayesian Planner

快速理解: Theory-of-mind (ToM) enables humans to infer mental states—such as beliefs, desires, and intentions—forming the foundation of social cognition. Existing computational ToM methods rely on structured workflows with ToM-specific priors or deep model fine-tuning but struggle with scalability in multimodal environments.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.365 PARM: Multi-Objective Test-Time Alignment via Preference-Aware Autoregressive Reward Model

- 作者: Lin, Baijiong; Jiang, Weisen; Xu, Yuancheng; Chen, Hao; Chen, Ying-Cong
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: PARM: Multi-Objective Test-Time Alignment via Preference-Aware Autoregressive Reward Model

快速理解: Multi-objective test-time alignment aims to adapt large language models (LLMs) to diverse multi-dimensional user preferences during inference while keeping LLMs frozen. Recently, GenARM (Xu et al., 2025) first independently trains Autoregressive Reward Models (ARMs) for each preference dimension without awareness of each other, then combines their outputs based on user-specific preference vectors during inference to achieve multi-objective test-time alignment, leading to two key limitations: the need for multiple ARMs increases the inference cost, and the separate training of ARMs causes the misalignment between the guided generation and the user preferences.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.366 PARQ: Piecewise-Affine Regularized Quantization

- 作者: Jin, Lisa; Ma, Jianhao; Liu, Zechun; Gromov, Andrey; Defazio, Aaron; Xiao, Lin
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: PARQ: Piecewise-Affine Regularized Quantization

快速理解: We develop a novel optimization method for quantization-aware training (QAT). Specifically, we show that convex, piecewise-affine regularization (PAR) can effectively induce neural network weights to cluster towards discrete values.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.6.367 PILAF: Optimal Human Preference Sampling for Reward Modeling

- 作者: Feng, Yunzhen; Kwiatkowski, Ariel; Zheng, Kunhao; Kempe, Julia; Duan, Yaqi

- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: PILAF: Optimal Human Preference Sampling for Reward Modeling

快速理解: As large language models increasingly drive real-world applications, aligning them with human values becomes paramount. Reinforcement Learning from Human Feedback (RLHF) has emerged as a key technique, translating preference data into reward models when oracle human values remain inaccessible.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.368 Parameters vs FLOPs: Scaling Laws for Optimal Sparsity for Mixture-of-Experts Language Models

- 作者: Abnar, Samira; Shah, Harshay; Busbridge, Dan; El-Nouby, Alaaeldin; Susskind, Joshua M.; Thilak, Vimal
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Parameters vs FLOPs: Scaling Laws for Optimal Sparsity for Mixture-of-Experts Language Models

快速理解: Scaling the capacity of language models has consistently proven to be a reliable approach for improving performance and unlocking new capabilities. Capacity can be primarily defined by two dimensions: the number of model parameters and the compute per example.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.6.369 Partially Observable Reinforcement Learning with Memory Traces

- 作者: Eberhard, Onno; Muehlebach, Michael; Vernade, Claire
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Partially Observable Reinforcement Learning with Memory Traces

快速理解: Partially observable environments present a considerable computational challenge in reinforcement learning due to the need to consider long histories. Learning with a finite window of observations quickly becomes intractable as the window length grows.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.370 Penalizing Infeasible Actions and Reward Scaling in Reinforcement Learning with Offline Data

- 作者: Kim, Jeonghye; Shin, Yongjae; Jung, Whiyoung; Hong, Sunghoon; Yoon, Deunsol; Sung, Youngchul; Lee, Kanghoon; Lim, Woohyung
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Penalizing Infeasible Actions and Reward Scaling in Reinforcement Learning with Offline Data

快速理解: Reinforcement learning with offline data suffers from Q-value extrapolation errors. To address this issue, we first demonstrate that linear extrapolation of the Q-function beyond the data range is particularly problematic.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.371 Pessimism Principle Can Be Effective: Towards a Framework for Zero-Shot Transfer Reinforcement Learning

- 作者: Zhang, Chi; Jia, Ziyang; Atia, George K.; He, Sihong; Wang, Yue
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Pessimism Principle Can Be Effective: Towards a Framework for Zero-Shot Transfer Reinforcement Learning

快速理解: Transfer reinforcement learning aims to derive a near-optimal policy for a target environment with limited data by leveraging abundant data from related source domains. However, it faces two key challenges: the lack of performance guarantees for the transferred policy, which can lead to undesired actions, and the risk of negative transfer when multiple source domains are involved.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.372 PipeOffload: Improving Scalability of Pipeline Parallelism with Memory Optimization

- 作者: Wan, Xinyi; Qi, Penghui; Huang, Guangxing; Lin, Min; Li, Jialin
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: PipeOffload: Improving Scalability of Pipeline Parallelism with Memory Optimization

快速理解: Pipeline parallelism (PP) is widely used for training large language models (LLMs), yet its scalability is often constrained by high activation memory consumption as the number of in-flight microbatches grows with the degree of PP. In this paper, we focus on addressing this challenge by leveraging the under-explored memory offload strategy in PP.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.6.373 PlaySlot: Learning Inverse Latent Dynamics for Controllable Object-Centric Video Prediction and Planning

- 作者: Villar-Corrales, Angel; Behnke, Sven
- 年份: 2025; 主题: World Model; 优先级: 重要
- [来源](#)

论文: PlaySlot: Learning Inverse Latent Dynamics for Controllable Object-Centric Video Prediction and Planning

快速理解: Predicting future scene representations is a crucial task for enabling robots to understand and interact with the environment. However, most existing methods rely on videos and simulations with precise action annotations, limiting their ability to leverage the large amount of available unlabeled video data.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策。

证据层级: 索引摘要。

1.3.6.374 Point Cloud Dataset Distillation

- 作者: Bo, Deyu; Wang, Xinchao
- 年份: 2025; 主题: Systems; 优先级: 重要
- [来源](#)

论文: Point Cloud Dataset Distillation

快速理解: This study introduces dataset distillation (DD) tailored for 3D data, particularly point clouds. DD aims to substitute large-scale real datasets with a small set of synthetic samples while preserving model performance.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.6.375 Point-Level Topological Representation Learning on Point Clouds

- 作者: Grande, Vincent Peter; Schaub, Michael T
- 年份: 2025; 主题: Systems; 优先级: 重要
- [来源](#)

论文: Point-Level Topological Representation Learning on Point Clouds

快速理解: Topological Data Analysis (TDA) allows us to extract powerful topological and higher-order information on the global shape of a data set or point cloud. Tools like Persistent Homology give a single complex description of the global structure of the point cloud.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.6.376 PoisonedEye: Knowledge Poisoning Attack on Retrieval-Augmented Generation based Large Vision-Language Models

- 作者: Zhang, Chenyang; Zhang, Xiaoyu; Lou, Jian; Wu, Kai; Wang, Zilong; Chen, Xiaofeng
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: PoisonedEye: Knowledge Poisoning Attack on Retrieval-Augmented Generation based Large Vision-Language Models

快速理解: Vision-Language Retrieval-Augmented Generation (VLRAG) systems have been widely applied to Large Vision-Language Models (LVLMs) to enhance their generation ability. However, the reliance on external multimodal knowledge databases renders VLRAG systems vulnerable to malicious poisoning attacks.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.6.377 Policy Filtration for RLHF to Mitigate Noise in Reward Models

- 作者：Zhang, Chuheng; Shen, Wei; Zhao, Li; Zhang, Xuyun; Xu, Xiaolong; Dou, Wanchun; Bian, Jiang
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Policy Filtration for RLHF to Mitigate Noise in Reward Models

快速理解：While direct policy optimization methods exist, pioneering LLMs are fine-tuned with reinforcement learning from human feedback (RLHF) to generate better responses under the supervision of a reward model learned from preference data. One major challenge of RLHF is the inaccuracy of the intermediate reward model, especially in the tasks that requires complex reasoning for the reward model to score a response.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.378 Policy Optimization for CMDPs with Bandit Feedback: Learning Stochastic and Adversarial Constraints

- 作者：Stradi, Francesco Emanuele; Lunghi, Anna; Castiglioni, Matteo; Marchesi, Alberto; Gatti, Nicola
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Policy Optimization for CMDPs with Bandit Feedback: Learning Stochastic and Adversarial Constraints

快速理解：We study online learning in constrained Markov decision processes (CMDPs) in which rewards and constraints may be either stochastic or adversarial. In such settings, stradi et al. (2024) proposed the first best-of-both-worlds algorithm able to seamlessly handle stochastic and adversarial constraints, achieving optimal regret and constraint violation bounds in both cases.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.379 Policy Regularization on Globally Accessible States in Cross-Dynamics Reinforcement Learning

- 作者：Xue, Zhenghai; Feng, Lang; Xu, Jiacheng; Kang, Kang; Wen, Xiang; An, Bo; Yan, Shuicheng
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Policy Regularization on Globally Accessible States in Cross-Dynamics Reinforcement Learning

快速理解：To learn from data collected in diverse dynamics, Imitation from Observation (IfO) methods leverage expert state trajectories based on the premise that recovering expert state distributions in other dynamics facilitates policy learning in the current one. However, Imitation Learning inherently

imposes a performance upper bound of learned policies.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.380 Polynomial-Time Approximability of Constrained Reinforcement Learning

- 作者：Mcmahan, Jeremy
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Polynomial-Time Approximability of Constrained Reinforcement Learning

快速理解：We study the computational complexity of approximating general constrained Markov decision processes. Our primary contribution is the design of a polynomial time $(0, \epsilon)$ -additive bicriteria approximation algorithm for finding optimal constrained policies across a broad class of recursively computable constraints, including almost-sure, chance, expectation, and their anytime variants.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.381 Position: Constants are Critical in Regret Bounds for Reinforcement Learning

- 作者：Drago, Simone; Mussi, Marco; Metelli, Alberto Maria
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Position: Constants are Critical in Regret Bounds for Reinforcement Learning

快速理解：Mainstream research in theoretical RL is currently focused on designing online learning algorithms with regret bounds that match the corresponding regret lower bound up to multiplicative constants (and, sometimes, logarithmic terms). In this position paper, we constructively question this trend, arguing that algorithms should be designed to at least minimize the amount of unnecessary exploration, and we highlight the significant role constants play in algorithms' actual performances.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.382 Position: Lifetime tuning is incompatible with continual reinforcement learning

- 作者：Mesbahi, Golnaz; Mohammad Panahi, Parham; Mastikhina, Olya; Tang, Steven; White, Martha; White, Adam
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Position: Lifetime tuning is incompatible with continual reinforcement learning

快速理解：In continual RL we want agents capable of never-ending learning, and yet our evaluation methodologies do not reflect this. The standard practice in RL is to assume unfettered access to

the deployment environment for the full lifetime of the agent.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.383 Prediction-Aware Learning in Multi-Agent Systems

- 作者: Capitaine, Aymeric; Boursier, Etienne; Moulines, Eric; Jordan, Michael I.; Oliviero Durmus, Alain
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Prediction-Aware Learning in Multi-Agent Systems

快速理解: The framework of uncoupled online learning in multiplayer games has made significant progress in recent years. In particular, the development of time-varying games has considerably expanded its modeling capabilities.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.384 Preference Controllable Reinforcement Learning with Advanced Multi-Objective Optimization

- 作者: Yang, Yucheng; Zhou, Tianyi; Pechenizkiy, Mykola; Fang, Meng
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Preference Controllable Reinforcement Learning with Advanced Multi-Objective Optimization

快速理解: Practical reinforcement learning (RL) usually requires agents to be optimized for multiple potentially conflicting criteria, e.g. speed vs. safety. Although Multi-Objective RL (MORL) algorithms have been studied in previous works, their trained agents often cover limited Pareto optimal solutions and they lack precise controllability of the delicate trade-off among multiple objectives.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.385 Privacy-Shielded Image Compression: Defending Against Exploitation from Vision-Language Pretrained Models

- 作者: Shen, Xuelin; Xu, Jiayin; Yin, Kangsheng; Yang, Wenhan
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Privacy-Shielded Image Compression: Defending Against Exploitation from Vision-Language Pretrained Models

快速理解: The improved semantic understanding of vision-language pretrained (VLP) models has made it increasingly difficult to protect publicly posted images from being exploited by search engines and other similar tools. In this context, this paper seeks to protect users' privacy by implementing defenses at the image compression stage to prevent exploitation.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.6.386 Provable Zero-Shot Generalization in Offline Reinforcement Learning

- 作者：Wang, Zhiyong; Yang, Chen; Lui, John C.S.; Zhou, Dongruo
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Provable Zero-Shot Generalization in Offline Reinforcement Learning

快速理解：In this work, we study offline reinforcement learning (RL) with zero-shot generalization property (ZSG), where the agent has access to an offline dataset including experiences from different environments, and the goal of the agent is to train a policy over the training environments which performs well on test environments without further interaction. Existing work showed that classical offline RL fails to generalize to new, unseen environments.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.387 Provably Efficient Exploration in Inverse Constrained Reinforcement Learning

- 作者：Yue, Bo; Li, Jian; Liu, Guiliang
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Provably Efficient Exploration in Inverse Constrained Reinforcement Learning

快速理解：Optimizing objective functions subject to constraints is fundamental in many real-world applications. However, these constraints are often not readily defined and must be inferred from expert agent behaviors, a problem known as Inverse Constraint Inference.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.388 PyTDC: A multimodal machine learning training, evaluation, and inference platform for biomedical foundation models

- 作者：Velez-Arce, Alejandro; Zitnik, Marinka
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：PyTDC: A multimodal machine learning training, evaluation, and inference platform for biomedical foundation models

快速理解：Existing biomedical benchmarks do not provide end-to-end infrastructure for training, evaluation, and inference of models that integrate multimodal biological data and a broad range of machine learning tasks in therapeutics. We present PyTDC, an open-source machine-learning platform providing streamlined training, evaluation, and inference software for multimodal biological AI models.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.6.389 Q-Supervised Contrastive Representation: A State Decoupling Framework for Safe Offline Reinforcement Learning

- 作者: Yang, Zhihe; Xu, Yunjian; Zhang, Yang
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Q-Supervised Contrastive Representation: A State Decoupling Framework for Safe Offline Reinforcement Learning

快速理解: Safe offline reinforcement learning (RL), which aims to learn the safety-guaranteed policy without risky online interaction with environments, has attracted growing recent attention for safety-critical scenarios. However, existing approaches encounter out-of-distribution problems during the testing phase, which can result in potentially unsafe outcomes.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.390 QT-DoG: Quantization-Aware Training for Domain Generalization

- 作者: Javed, Saqib; Le, Hieu; Salzmänn, Mathieu
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: QT-DoG: Quantization-Aware Training for Domain Generalization

快速理解: A key challenge in Domain Generalization (DG) is preventing overfitting to source domains, which can be mitigated by finding flatter minima in the loss landscape. In this work, we propose Quantization-aware Training for Domain Generalization (QT-DoG) and demonstrate that weight quantization effectively leads to flatter minima in the loss landscape, thereby enhancing domain generalization.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.6.391 RATE: Causal Explainability of Reward Models with Imperfect Counterfactuals

- 作者: Reber, David; Richardson, Sean M; Nief, Todd; Garbacea, Cristina; Veitch, Victor
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: RATE: Causal Explainability of Reward Models with Imperfect Counterfactuals

快速理解: Reward models are widely used as proxies for human preferences when aligning or evaluating LLMs. However, reward models are black boxes, and it is often unclear what, exactly, they are actually rewarding.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.392 RLEF: Grounding Code LLMs in Execution Feedback with Reinforcement Learning

- 作者: Gehring, Jonas; Zheng, Kunhao; Copet, Jade; Mella, Vegard; Cohen, Taco; Synnaeve, Gabriel
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: RLEF: Grounding Code LLMs in Execution Feedback with Reinforcement Learning

快速理解: Large language models (LLMs) deployed as agents solve user-specified tasks over multiple steps while keeping the required manual engagement to a minimum. Crucially, such LLMs need to ground their generations in any feedback obtained to reliably achieve the desired outcomes.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.393 RWKVQuant: Quantizing the RWKV Family with Proxy Guided Hybrid of Scalar and Vector Quantization

- 作者: Xu, Chen; Yue, Yuxuan; Xu, Zukang; Hu, Xing; Yu, Jiangyong; Chen, Zhixuan; Zhou, Sifan; Yuan, Zhihang; Yang, Dawei
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: RWKVQuant: Quantizing the RWKV Family with Proxy Guided Hybrid of Scalar and Vector Quantization

快速理解: RWKV is a modern RNN architecture with comparable performance to Transformer, but still faces challenges when deployed to resource-constrained devices. Post Training Quantization (PTQ), which is an essential technique to reduce model size and inference latency, has been widely used in Transformer models.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.6.394 Ranked from Within: Ranking Large Multimodal Models Without Labels

- 作者: Tu, Weijie; Deng, Weijian; Campbell, Dylan; Yao, Yu; Zheng, Jiyang; Gedeon, Tom; Liu, Tongliang
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Ranked from Within: Ranking Large Multimodal Models Without Labels

快速理解: Can the relative performance of a pre-trained large multimodal model (LMM) be predicted without access to labels? As LMMs proliferate, it becomes increasingly important to develop efficient ways to choose between them when faced with new data or tasks.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.395 Re-ranking Reasoning Context with Tree Search Makes Large Vision-Language Models Stronger

- 作者: Yang, Qi; Zhang, Chenghao; Fan, Lubin; Ding, Kun; Ye, Jieping; Xiang, Shiming
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Re-ranking Reasoning Context with Tree Search Makes Large Vision-Language Models Stronger

快速理解: Recent advancements in Large Vision Language Models (LVLMs) have significantly improved performance in Visual Question Answering (VQA) tasks through multimodal Retrieval-Augmented Generation (RAG). However, existing methods still face challenges, such as the scarcity of knowledge with reasoning examples and erratic responses from retrieved knowledge.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.396 Reasoning Limitations of Multimodal Large Language Models. A case study of Bongard Problems

- 作者: Ma^{l}ki^{n}ski, Miko^{l}aj; Pawlonka, Szymon; Ma^{n}'dziuk, Jacek
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Reasoning Limitations of Multimodal Large Language Models. A case study of Bongard Problems

快速理解: Abstract visual reasoning (AVR) involves discovering shared concepts across images through analogy, akin to solving IQ test problems. Bongard Problems (BPs) remain a key challenge in AVR, requiring both visual reasoning and verbal description.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.397 Regret-Free Reinforcement Learning for Temporal Logic Specifications

- 作者: Majumdar, R; Salamati, Mahmoud; Soudjani, Sadegh
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Regret-Free Reinforcement Learning for Temporal Logic Specifications

快速理解: Learning to control an unknown dynamical system with respect to high-level temporal specifications is an important problem in control theory. We present the first regret-free online algorithm for learning a controller for linear temporal logic (LTL) specifications for systems with unknown dynamics.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.398 Reidentify: Context-Aware Identity Generation for Contextual Multi-Agent Reinforcement Learning

- 作者: Xu, Zhiwei; Hu, Kun; Xin, Xin; Meng, Weiliang; Shi, Yiwei; Mao, Hangyu; Zhang, Bin; Li, Dapeng; Yin, Jiangjin
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Reidentify: Context-Aware Identity Generation for Contextual Multi-Agent Reinforcement Learning

快速理解: Generalizing multi-agent reinforcement learning (MARL) to accommodate variations in problem configurations remains a critical challenge in real-world applications, where even subtle differences in task setups can cause pre-trained policies to fail. To address this, we propose Context-Aware Identity Generation (CAID), a novel framework to enhance MARL performance under the Contextual MARL (CMARL) setting.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.399 Reinforce LLM Reasoning through Multi-Agent Reflection

- 作者: Yuan, Yurun; Xie, Tengyang
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Reinforce LLM Reasoning through Multi-Agent Reflection

快速理解: Leveraging more test-time computation has proven to be an effective way to boost the reasoning capabilities of large language models (LLMs). Among various methods, the verify-and-improve paradigm stands out for enabling dynamic solution exploration and feedback incorporation.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.400 Reinforcement Learning Control of a Physical Robot Device for Assisted Human Walking without a Simulator

- 作者: Zhong, Junmin; Yumbla, Emiliano Quinones; Soltanian, Seyed Yousef; Wu, Ruofan; Zhang, Wenlong; Si, Jennie
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Reinforcement Learning Control of a Physical Robot Device for Assisted Human Walking without a Simulator

快速理解: This study presents an innovative reinforcement learning (RL) control approach to facilitate soft exosuit-assisted human walking. Our goal is to address the ongoing challenges in developing reliable RL-based methods for controlling physical devices.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.401 Reinforcement Learning for Quantum Control under Physical Constraints

- 作者: Ernst, Jan Ole; Chatterjee, Aniket; Franzmeyer, Tim; Kuhn, Axel

- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Reinforcement Learning for Quantum Control under Physical Constraints

快速理解: Quantum control is concerned with the realisation of desired dynamics in quantum systems, serving as a linchpin for advancing quantum technologies and fundamental research. Analytic approaches and standard optimisation algorithms do not yield satisfactory solutions for more complex quantum systems, and especially not for real world quantum systems which are open and noisy.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.402 Reinforcement Learning with Adaptive Reward Modeling for Expensive-to-Evaluate Systems

- 作者: Su, Hongyuan; Zheng, Yu; Yuan, Yuan; Lin, Yuming; Jin, Depeng; Li, Yong
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Reinforcement Learning with Adaptive Reward Modeling for Expensive-to-Evaluate Systems

快速理解: Training reinforcement learning (RL) agents requires extensive trials and errors, which becomes prohibitively time-consuming in systems with costly reward evaluations. To address this challenge, we propose adaptive reward modeling (AdaReMo) which accelerates RL training by decomposing the complicated reward function into multiple localized fast reward models approximating direct reward evaluation with neural networks.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.403 Reinforcement Learning with Random Time Horizons

- 作者: Ribera Borrell, Enric; Richter, Lorenz; Schuette, Christof
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Reinforcement Learning with Random Time Horizons

快速理解: We extend the standard reinforcement learning framework to random time horizons. While the classical setting typically assumes finite and deterministic or infinite runtimes of trajectories, we argue that multiple real-world applications naturally exhibit random (potentially trajectory-dependent) stopping times.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.404 Reinforcement Learning with Segment Feedback

- 作者: Du, Yihan; Winnicki, Anna; Dalal, Gal; Mannor, Shie; Srikant, R.
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Reinforcement Learning with Segment Feedback

快速理解: Standard reinforcement learning (RL) assumes that an agent can observe a reward for each state-action pair. However, in practical applications, it is often difficult and costly to collect a reward for each state-action pair.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.405 RepLoRA: Reparameterizing Low-rank Adaptation via the Perspective of Mixture of Experts

- 作者: Truong, Tuan; Nguyen, Chau; Nguyen, Huy; Le, Minh; Le, Trung; Ho, Nhat
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: RepLoRA: Reparameterizing Low-rank Adaptation via the Perspective of Mixture of Experts

快速理解: Low-rank Adaptation (LoRA) has emerged as a powerful and efficient method for fine-tuning large-scale foundation models. Despite its popularity, the theoretical understanding of LoRA has remained underexplored.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.6.406 Rethinking Point Cloud Data Augmentation: Topologically Consistent Deformation

- 作者: Bi, Jian; Wu, Qianliang; Li, Xiang; Chen, Shuo; Qian, Jianjun; Luo, Lei; Yang, Jian
- 年份: 2025; 主题: Systems; 优先级: 重要
- [来源](#)

论文: Rethinking Point Cloud Data Augmentation: Topologically Consistent Deformation

快速理解: Data augmentation has been widely used in machine learning. Its main goal is to transform and expand the original data using various techniques, creating a more diverse and enriched training dataset.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.6.407 Revisiting Cooperative Off-Policy Multi-Agent Reinforcement Learning

- 作者: Li, Yueheng; Xie, Guangming; Lu, Zongqing
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Revisiting Cooperative Off-Policy Multi-Agent Reinforcement Learning

快速理解: Cooperative Multi-Agent Reinforcement Learning (MARL) has become a critical tool for addressing complex real-world problems. However, off-policy MARL methods, which rely on joint Q-functions, face significant scalability challenges due to the exponentially growing joint action space.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.408 Reward Modeling with Ordinal Feedback: Wisdom of the Crowd

- 作者：Liu, Shang; Pan, Yu; Chen, Guanting; Li, Xiaocheng
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Reward Modeling with Ordinal Feedback: Wisdom of the Crowd

快速理解：The canonical setup of learning a reward model (RM) from human preferences with binary feedback discards potentially useful samples (such as “tied” between the two responses) and loses fine-grained information (such as “slightly better”). This paper proposes a framework for learning RMs under ordinal feedback, generalizing the binary feedback to arbitrary granularity.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.409 Reward-Guided Prompt Evolving in Reinforcement Learning for LLMs

- 作者：Ye, Ziyu; Agarwal, Rishabh; Liu, Tianqi; Joshi, Rishabh; Velury, Sarmishta; Le, Quoc V; Tan, Qijun; Liu, Yuan
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Reward-Guided Prompt Evolving in Reinforcement Learning for LLMs

快速理解：Existing reinforcement learning (RL) methods for large language models (LLMs) rely on static prompt sets, where prompts are curated a priori, and sampled in a fixed schedule for training, regardless of their usefulness to the RL process. We design eva, the first method that allows LLMs to prioritize and adaptively create useful prompts during RL training by reward signals.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.410 Robot-Gated Interactive Imitation Learning with Adaptive Intervention Mechanism

- 作者：Cai, Haoyuan; Peng, Zhenghao; Zhou, Bolei
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Robot-Gated Interactive Imitation Learning with Adaptive Intervention Mechanism

快速理解：Interactive Imitation Learning (IIL) allows agents to acquire desired behaviors through human interventions, but current methods impose high cognitive demands on human supervisors. We propose the Adaptive Intervention Mechanism (AIM), a novel robot-gated IIL algorithm that learns an adaptive criterion for requesting human demonstrations.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.411 Robust Multi-Agent Reinforcement Learning with Stochastic Adversary

- 作者: Zhou, Ziyuan; Liu, Guanjin; Zhou, Mengchu; Guo, Weiran
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Robust Multi-Agent Reinforcement Learning with Stochastic Adversary

快速理解: The performance of models trained by Multi-Agent Reinforcement Learning (MARL) is sensitive to perturbations in observations, lowering their trustworthiness in complex environments. Adversarial training is a valuable approach to enhance their performance robustness.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.412 Robust Offline Reinforcement Learning with Linearly Structured f-Divergence Regularization

- 作者: Tang, Cheng; Liu, Zhishuai; Xu, Pan
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Robust Offline Reinforcement Learning with Linearly Structured f-Divergence Regularization

快速理解: The Robust Regularized Markov Decision Process (RRMDP) is proposed to learn policies robust to dynamics shifts by adding regularization to the transition dynamics in the value function. Existing methods mostly use unstructured regularization, potentially leading to conservative policies under unrealistic transitions.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.413 RobustLight: Improving Robustness via Diffusion Reinforcement Learning for Traffic Signal Control

- 作者: Li, Mingyuan; Wang, Jiahao; Yu, Guangsheng; Wang, Xu; Chen, Qianrun; Ni, Wei; Li, Lixiang; Peng, Haipeng
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: RobustLight: Improving Robustness via Diffusion Reinforcement Learning for Traffic Signal Control

快速理解: Reinforcement Learning (RL) optimizes Traffic Signal Control (TSC) to reduce congestion and emissions, but real-world TSC systems face challenges like adversarial attacks and missing data, leading to incorrect signal decisions and increased congestion. Existing methods, limited to offline data predictions, address only one issue and fail to meet TSC's dynamic, real-time needs.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.414 RobustZero: Enhancing MuZero Reinforcement Learning Robustness to State Perturbations

- 作者: Li, Yushuai; Liu, Hengyu; Pedersen, Torben Bach; He, Yuqiang; Larsen, Kim Guldstrand; Chen, Lu; Jensen, Christian S.; Xu, Jiachen; Li, Tianyi
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: RobustZero: Enhancing MuZero Reinforcement Learning Robustness to State Perturbations

快速理解: The MuZero reinforcement learning method has achieved superhuman performance at games, and advances that enable MuZero to contend with complex actions now enable use of MuZero-class methods in real-world decision-making applications. However, some real-world applications are susceptible to state perturbations caused by malicious attacks and noisy sensors.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.415 RollingQ: Reviving the Cooperation Dynamics in Multimodal Transformer

- 作者: Ni, Haotian; Wei, Yake; Liu, Hang; Chen, Gong; Peng, Chong; Lin, Hao; Hu, Di
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: RollingQ: Reviving the Cooperation Dynamics in Multimodal Transformer

快速理解: Multimodal learning faces challenges in effectively fusing information from diverse modalities, especially when modality quality varies across samples. Dynamic fusion strategies, such as attention mechanism in Transformers, aim to address such challenge by adaptively emphasizing modalities based on the characteristics of input data.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.416 RuleAdapter: Dynamic Rules for training Safety Reward Models in RLHF

- 作者: Li, Xiaomin; Gao, Mingye; Zhang, Zhiwei; Fan, Jingxuan; Li, Weiyu
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: RuleAdapter: Dynamic Rules for training Safety Reward Models in RLHF

快速理解: Reinforcement Learning from Human Feedback (RLHF) is widely used to align models with human preferences, particularly to enhance the safety of responses generated by LLMs. This method traditionally relies on choosing preferred responses from response pairs.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.417 S2-Track: A Simple yet Strong Approach for End-to-End 3D Multi-Object Tracking

- 作者: Tang, Tao; Zhou, Lijun; Hao, Pengkun; He, Zihang; Ho, Kalok; Gu, Shuo; Hao, Zhihui; Sun, Haiyang; Zhan, Kun; Jia, Peng; Lang, Xianpeng; Liang, Xiaodan
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: S2-Track: A Simple yet Strong Approach for End-to-End 3D Multi-Object Tracking

快速理解: 3D multiple object tracking (MOT) plays a crucial role in autonomous driving perception. Recent end-to-end query-based trackers simultaneously detect and track objects, which have shown promising potential for the 3D MOT task.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.418 SAE-V: Interpreting Multimodal Models for Enhanced Alignment

- 作者: Lou, Hantao; Li, Changye; Ji, Jiaming; Yang, Yaodong
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: SAE-V: Interpreting Multimodal Models for Enhanced Alignment

快速理解: With the integration of image modality, the semantic space of multimodal large language models (MLLMs) is more complex than text-only models, making their interpretability more challenging and their alignment less stable, particularly susceptible to low-quality data, which can lead to inconsistencies between modalities, hallucinations, and biased outputs. As a result, developing interpretability methods for MLLMs is crucial for improving alignment quality and efficiency.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.419 SAFER: A Calibrated Risk-Aware Multimodal Recommendation Model for Dynamic Treatment Regimes

- 作者: Shen, Yishan; Ye, Yuyang; Xiong, Hui; Chen, Yong
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: SAFER: A Calibrated Risk-Aware Multimodal Recommendation Model for Dynamic Treatment Regimes

快速理解: Dynamic treatment regimes (DTRs) are critical to precision medicine, optimizing long-term outcomes through personalized, real-time decision-making in evolving clinical contexts, but require careful supervision for unsafe treatment risks. Existing efforts rely primarily on clinician-prescribed gold standards despite the absence of a known optimal strategy, and predominantly using structured EHR data without extracting valuable insights from clinical notes, limiting their reliability for treatment recommendations.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.420 SAM2Act: Integrating Visual Foundation Model with A Memory Architecture for Robotic Manipulation

- 作者: Fang, Haoquan; Grotz, Markus; Pumacay, Wilbert; Wang, Yi Ru; Fox, Dieter; Krishna, Ranjay; Duan, Jiafei
- 年份: 2025; 主题: Embodied/WBC; 优先级: 重要

- [来源](#)

论文: SAM2Act: Integrating Visual Foundation Model with A Memory Architecture for Robotic Manipulation

快速理解: Robotic manipulation systems operating in diverse, dynamic environments must exhibit three critical abilities: multitask interaction, generalization to unseen scenarios, and spatial memory. While significant progress has been made in robotic manipulation, existing approaches often fall short in generalization to complex environmental variations and addressing memory-dependent tasks.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制。

证据层级: 索引摘要。

1.3.6.421 SECOND: Mitigating Perceptual Hallucination in Vision-Language Models via Selective and Contrastive Decoding

- 作者: Park, Woohyeon; Kim, Woojin; Kim, Jaeik; Do, Jaeyoung
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: SECOND: Mitigating Perceptual Hallucination in Vision-Language Models via Selective and Contrastive Decoding

快速理解: Despite significant advancements in Vision-Language Models (VLMs), the performance of existing VLMs remains hindered by object hallucination, a critical challenge to achieving accurate visual understanding. To address this issue, we propose SECOND: Selective and Contrastive Decoding, a novel approach that enables VLMs to effectively leverage multi-scale visual information with an object-centric manner, closely aligning with human visual perception.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.422 SEFE: Superficial and Essential Forgetting Eliminator for Multimodal Continual Instruction Tuning

- 作者: Chen, Jinpeng; Cong, Runmin; Zhao, Yuzhi; Yang, Hongzheng; Hu, Guangneng; Ip, Horace; Kwong, Sam
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: SEFE: Superficial and Essential Forgetting Eliminator for Multimodal Continual Instruction Tuning

快速理解: Multimodal Continual Instruction Tuning (MCIT) aims to enable Multimodal Large Language Models (MLLMs) to incrementally learn new tasks without catastrophic forgetting, thus adapting to evolving requirements. In this paper, we explore the forgetting caused by such incremental training, categorizing it into superficial forgetting and essential forgetting.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.423 SK-VQA: Synthetic Knowledge Generation at Scale for Training Context-Augmented Multimodal LLMs

- 作者: Su, Xin; Luo, Man; Pan, Kris W; Chou, Tien Pei; Lal, Vasudev; Howard, Phillip
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: SK-VQA: Synthetic Knowledge Generation at Scale for Training Context-Augmented Multimodal LLMs

快速理解: Multimodal retrieval-augmented generation (RAG) plays a crucial role in domains such as knowledge-based visual question answering (KB-VQA), where models should effectively integrate additional knowledge to generate a response. However, existing vision and language models (VLMs) are not inherently designed for context-augmented generation, limiting their effectiveness in such tasks.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.424 SLiM: One-shot Quantization and Sparsity with Low-rank Approximation for LLM Weight Compression

- 作者: Mozaffari, Mohammad; Yazdanbakhsh, Amir; Mehri Dehnavi, Maryam
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: SLiM: One-shot Quantization and Sparsity with Low-rank Approximation for LLM Weight Compression

快速理解: Conventional model compression techniques for LLMs address high memory consumption and slow inference challenges but typically require computationally expensive retraining to preserve accuracy. In contrast, one-shot compression methods eliminate retraining cost, but struggle to achieve accuracy comparable to dense models.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.6.425 SMART-PC: Skeletal Model Adaptation for Robust Test-Time Training in Point Clouds

- 作者: Bahri, Ali; Yazdanpanah, Moslem; Dastani, Sahar; Noori, Mehrdad; Vargas Hakim, Gustavo Adolfo; Osowiechi, David; Beizae, Farzad; Ayed, Ismail Ben; Desrosiers, Christian
- 年份: 2025; 主题: Systems; 优先级: 重要
- [来源](#)

论文: SMART-PC: Skeletal Model Adaptation for Robust Test-Time Training in Point Clouds

快速理解: Test-Time Training has emerged as a promising solution to address distribution shifts in 3D point cloud classification. However, existing methods often rely on computationally expensive backpropagation during adaptation, limiting their applicability in real-world, time-sensitive scenarios.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.6.426 Sable: a Performant, Efficient and Scalable Sequence Model for MARL

- 作者: Mahjoub, Omayma; Abramowitz, Sasha; De Kock, Ruan John; Khlifi, Wiem; Toit, Simon Verster Du; Daniel, Jemma; Nessir, Louay Ben; Beyers, Louise; Formanek, Juan Claude; Clark, Liam; Pretorius, Arnu
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Sable: a Performant, Efficient and Scalable Sequence Model for MARL

快速理解: As multi-agent reinforcement learning (MARL) progresses towards solving larger and more complex problems, it becomes increasingly important that algorithms exhibit the key properties of (1) strong performance, (2) memory efficiency, and (3) scalability. In this work, we introduce Sable, a performant, memory-efficient, and scalable sequence modelling approach to MARL.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.427 SafeAuto: Knowledge-Enhanced Safe Autonomous Driving with Multimodal Foundation Models

- 作者: Zhang, Jiawei; Yang, Xuan; Wang, Taiqi; Yao, Yu; Petiushko, Aleksandr; Li, Bo
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: SafeAuto: Knowledge-Enhanced Safe Autonomous Driving with Multimodal Foundation Models

快速理解: Traditional autonomous driving systems often struggle to connect high-level reasoning with low-level control, leading to suboptimal and sometimes unsafe behaviors. Recent advances in multimodal large language models (MLLMs), which process both visual and textual data, offer an opportunity to unify perception and reasoning.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.6.428 Safety-Polarized and Prioritized Reinforcement Learning

- 作者: Fan, Ke; Zhang, Jinpeng; Zhang, Xuefeng; Wu, Yunze; Cao, Jingyu; Zhou, Yuan; Ma, Jianzhu
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Safety-Polarized and Prioritized Reinforcement Learning

快速理解: Motivated by the first priority of safety in many real-world applications, we propose MaxSafe, a chance-constrained bi-level optimization framework for safe reinforcement learning. MaxSafe first minimizes the unsafe probability and then maximizes the return among the safest policies.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.429 Sample Complexity of Distributionally Robust Off-Dynamics Reinforcement Learning with Online Interaction

- 作者: He, Yiting; Liu, Zhishuai; Wang, Weixin; Xu, Pan
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Sample Complexity of Distributionally Robust Off-Dynamics Reinforcement Learning with Online Interaction

快速理解: Off-dynamics reinforcement learning (RL), where training and deployment transition dynamics are different, can be formulated as learning in a robust Markov decision process (RMDP) where uncertainties in transition dynamics are imposed. Existing literature mostly assumes access to generative models allowing arbitrary state-action queries or pre-collected datasets with a good state coverage of the deployment environment, bypassing the challenge of exploration.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.430 Satori: Reinforcement Learning with Chain-of-Action-Thought Enhances LLM Reasoning via Autoregressive Search

- 作者: Shen, Maohao; Zeng, Guangtao; Qi, Zhenting; Hong, Zhang-Wei; Chen, Zhenfang; Lu, Wei; Wornell, Gregory W.; Das, Subhro; Cox, David Daniel; Gan, Chuang
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Satori: Reinforcement Learning with Chain-of-Action-Thought Enhances LLM Reasoning via Autoregressive Search

快速理解: Large language models (LLMs) have demonstrated remarkable reasoning capabilities across diverse domains. Recent studies have shown that increasing test-time computation enhances LLMs' reasoning capabilities.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.431 Scaling Laws for Floating-Point Quantization Training

- 作者: Sun, Xingwu; Li, Shuaipeng; Xie, Ruobing; Han, Weidong; Wu, Kan; Yang, Zhen; Li, Yixing; Wang, An; Li, Shuai; Xue, Jinbao; Cheng, Yu; Tao, Yangyu; Kang, Zhanhui; Xu, Cheng-Zhong; Wang, Di; Jiang, Jie
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Scaling Laws for Floating-Point Quantization Training

快速理解: Low-precision training is considered an effective strategy for reducing both training and downstream inference costs. Previous scaling laws for precision mainly focus on integer quantization, which pay less attention to the constituents in floating-point (FP) quantization, and thus cannot well fit the LLM losses in this scenario.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.6.432 Scaling Laws for Upcycling Mixture-of-Experts Language Models

- 作者：Liew, Seng Pei; Kato, Takuya; Takase, Sho
- 年份：2025; 主题：AI Infra; 优先级：重要
- [来源](#)

论文：Scaling Laws for Upcycling Mixture-of-Experts Language Models

快速理解：Pretraining large language models (LLMs) is resource-intensive, often requiring months of training time even with high-end GPU clusters. There are two approaches of mitigating such computational demands: reusing smaller models to train larger ones (upcycling), and training computationally efficient models like mixture-of-experts (MoE).

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.6.433 Score as Action: Fine Tuning Diffusion Generative Models by Continuous-time Reinforcement Learning

- 作者：Zhao, Hanyang; Chen, Haoxian; Zhang, Ji; Yao, David; Tang, Wenpin
- 年份：2025; 主题：Agent RL; 优先级：重要
- [来源](#)

论文：Score as Action: Fine Tuning Diffusion Generative Models by Continuous-time Reinforcement Learning

快速理解：Reinforcement learning from human feedback (RLHF), which aligns a diffusion model with input prompt, has become a crucial step in building reliable generative AI models. Most works in this area uses a discrete-time formulation, which is prone to induced errors, and often not applicable to models with higher-order/black-box solvers.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.434 Score-Based Diffusion Policy Compatible with Reinforcement Learning via Optimal Transport

- 作者：Sun, Mingyang; Ding, Pengxiang; Zhang, Weinan; Wang, Donglin
- 年份：2025; 主题：Agent RL; 优先级：重要
- [来源](#)

论文：Score-Based Diffusion Policy Compatible with Reinforcement Learning via Optimal Transport

快速理解：Diffusion policies have shown promise in learning complex behaviors from demonstrations, particularly for tasks requiring precise control and long-term planning. However, they face challenges in robustness when encountering distribution shifts.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.435 Settling the Maximin Share Fairness for Scheduling among Groups of Machines

- 作者: Li, Bo; Wang, Fangxiao; Shiji, Xing
- 年份: 2025; 主题: Systems; 优先级: 重要
- [来源](#)

论文: Settling the Maximin Share Fairness for Scheduling among Groups of Machines

快速理解: We study the fair scheduling of jobs among groups of (unrelated) machines and focus on the maximin share (MMS) fairness at the group level. The problem was first introduced by Li et al. [NeurIPS 2023], where each group consists of a number of identical machines (or identical up to different speeds), and the cost of a group is determined by the minimum makespan on completing all jobs assigned to it.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.6.436 Shortcut-connected Expert Parallelism for Accelerating Mixture of Experts

- 作者: Cai, Weilin; Jiang, Juyong; Qin, Le; Cui, Junwei; Kim, Sunghun; Huang, Jiayi
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Shortcut-connected Expert Parallelism for Accelerating Mixture of Experts

快速理解: Expert parallelism has emerged as a key strategy for distributing the computational workload of sparsely-gated mixture-of-experts (MoE) models across multiple devices, enabling the processing of increasingly large-scale models. However, the All-to-All communication inherent to expert parallelism poses a significant bottleneck, limiting the efficiency of MoE models.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.6.437 Simple Policy Optimization

- 作者: Xie, Zhengpeng; Zhang, Qiang; Yang, Fan; Hutter, Marco; Xu, Renjing
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Simple Policy Optimization

快速理解: Model-free reinforcement learning algorithms have seen remarkable progress, but key challenges remain. Trust Region Policy Optimization (TRPO) is known for ensuring monotonic policy improvement through conservative updates within a trust region, backed by strong theoretical guarantees.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.438 Sleeping Reinforcement Learning

- 作者: Drago, Simone; Mussi, Marco; Metelli, Alberto Maria

- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Sleeping Reinforcement Learning

快速理解: In the standard Reinforcement Learning (RL) paradigm, the action space is assumed to be fixed and immutable throughout the learning process. However, in many real-world scenarios, not all actions are available at every decision stage.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.439 SLiM-LLM: Salience-Driven Mixed-Precision Quantization for Large Language Models

- 作者: Huang, Wei; Qin, Haotong; Liu, Yangdong; Li, Yawei; Liu, Qinshuo; Liu, Xianglong; Benini, Luca; Magno, Michele; Zhang, Shiming; Qi, Xiaojuan
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: SLiM-LLM: Salience-Driven Mixed-Precision Quantization for Large Language Models

快速理解: Post-training quantization (PTQ) is an effective technique for compressing large language models (LLMs). However, while uniform-precision quantization is computationally efficient, it often compromises model performance.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.6.440 Sliding Puzzles Gym: A Scalable Benchmark for State Representation in Visual Reinforcement Learning

- 作者: De Oliveira, Bryan Lincoln Marques; Martins, Luana Guedes Barros; Brandão, Bruno; Luz, Murilo Lopes Da; De Lima Soares, Telma Woerle; Carvalho Melo, Luckeciano
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Sliding Puzzles Gym: A Scalable Benchmark for State Representation in Visual Reinforcement Learning

快速理解: Effective visual representation learning is crucial for reinforcement learning (RL) agents to extract task-relevant information from raw sensory inputs and generalize across diverse environments. However, existing RL benchmarks lack the ability to systematically evaluate representation learning capabilities in isolation from other learning challenges.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.441 SparseVLM: Visual Token Sparsification for Efficient Vision-Language Model Inference

- 作者: Zhang, Yuan; Fan, Chun-Kai; Ma, Junpeng; Zheng, Wenzhao; Huang, Tao; Cheng, Kuan; Gudovskiy, Denis A; Okuno, Tomoyuki; Nakata, Yohei; Keutzer, Kurt; Zhang, Shanghang

- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: SparseVLM: Visual Token Sparsification for Efficient Vision-Language Model Inference

快速理解: In vision-language models (VLMs), visual tokens usually consume a significant amount of computational overhead, despite their sparser information density compared to text tokens. To address this, most existing methods learn a network to prune redundant visual tokens and require additional training data.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.442 Stay Hungry, Keep Learning: Sustainable Plasticity for Deep Reinforcement Learning

- 作者: Zhou, Huaicheng; Zhuang, Zifeng; Wang, Donglin
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Stay Hungry, Keep Learning: Sustainable Plasticity for Deep Reinforcement Learning

快速理解: The integration of Deep Neural Networks in Reinforcement Learning (RL) systems has led to remarkable progress in solving complex tasks but also introduced challenges like primacy bias and dead neurons. Primacy bias skews learning towards early experiences, while dead neurons diminish the network's capacity to acquire new knowledge.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.443 Stealing That Free Lunch: Exposing the Limits of Dyna-Style Reinforcement Learning

- 作者: Barkley, Brett; Fridovich-Keil, David
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Stealing That Free Lunch: Exposing the Limits of Dyna-Style Reinforcement Learning

快速理解: Dyna-style off-policy model-based reinforcement learning (DMBRL) algorithms are a family of techniques for generating synthetic state transition data and thereby enhancing the sample efficiency of off-policy RL algorithms. This paper identifies and investigates a surprising performance gap observed when applying DMBRL algorithms across different benchmark environments with proprioceptive observations.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.444 Surrogate Prompt Learning: Towards Efficient and Diverse Prompt Learning for Vision-Language Models

- 作者: Liu, Liangchen; Wang, Nannan; Yang, Xi; Gao, Xinbo; Liu, Tongliang
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要

- [来源](#)

论文: Surrogate Prompt Learning: Towards Efficient and Diverse Prompt Learning for Vision-Language Models

快速理解: Prompt learning is a cutting-edge parameter-efficient fine-tuning technique for pre-trained vision-language models (VLMs). Instead of learning a single text prompt, recent works have revealed that learning diverse text prompts can effectively boost the performances on downstream tasks, as the diverse prompted text features can comprehensively depict the visual concepts from different perspectives.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.445 Symmetric Reinforcement Learning Loss for Robust Learning on Diverse Tasks and Model Scales

- 作者: Byun, Ju-Seung; Perrault, Andrew
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Symmetric Reinforcement Learning Loss for Robust Learning on Diverse Tasks and Model Scales

快速理解: Reinforcement learning (RL) training is inherently unstable due to factors such as moving targets and high gradient variance. Reinforcement Learning from Human Feedback (RLHF) and Reinforcement Learning from AI Feedback (RLAIF) introduce additional challenges.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.446 T1: Advancing Language Model Reasoning through Reinforcement Learning and Inference Scaling

- 作者: Hou, Zhenyu; Lv, Xin; Lu, Rui; Zhang, Jiajie; Li, Yujiang; Yao, Zijun; Li, Juanzi; Tang, Jie; Dong, Yuxiao
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: T1: Advancing Language Model Reasoning through Reinforcement Learning and Inference Scaling

快速理解: Large language models (LLMs) have demonstrated remarkable capabilities in complex reasoning tasks. However, existing approaches mainly rely on imitation learning and struggle to achieve effective test-time scaling.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.447 TRUST-VLM: Thorough Red-Teaming for Uncovering Safety Threats in Vision-Language Models

- 作者: Chen, Kangjie; Muyang, Li; Li, Guanlin; Zhang, Shudong; Guo, Shangwei; Zhang, Tianwei

- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: TRUST-VLM: Thorough Red-Teaming for Uncovering Safety Threats in Vision-Language Models

快速理解: Vision-Language Models (VLMs) have become a cornerstone in multi-modal artificial intelligence, enabling seamless integration of visual and textual information for tasks such as image captioning, visual question answering, and cross-modal retrieval. Despite their impressive capabilities, these models often exhibit inherent vulnerabilities that can lead to safety failures in critical applications.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.448 Task-Aware Virtual Training: Enhancing Generalization in Meta-Reinforcement Learning for Out-of-Distribution Tasks

- 作者: Kim, Jeongmo; Park, Yisak; Kim, Minung; Han, Seungyul
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Task-Aware Virtual Training: Enhancing Generalization in Meta-Reinforcement Learning for Out-of-Distribution Tasks

快速理解: Meta reinforcement learning aims to develop policies that generalize to unseen tasks sampled from a task distribution. While context-based meta-RL methods improve task representation using task latents, they often struggle with out-of-distribution (OOD) tasks.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.449 Teaching Language Models to Critique via Reinforcement Learning

- 作者: Xie, Zhihui; Chen, Jie; Chen, Liyu; Mao, Weichao; Xu, Jingjing; Kong, Lingpeng
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Teaching Language Models to Critique via Reinforcement Learning

快速理解: Teaching large language models (LLMs) to critique and refine their outputs is crucial for building systems that can iteratively improve, yet it is fundamentally limited by the ability to provide accurate judgments and actionable suggestions. In this work, we study LLM critics for code generation and propose `texttt{CTRL}`, a framework for `texttt{C}`ritic `texttt{T}`raining via `texttt{R}`einforcement `texttt{L}`earning, which trains a critic model to generate feedback that maximizes correction performance for a fixed generator model without human supervision.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.450 Temporal Distance-aware Transition Augmentation for Offline Model-based Reinforcement Learning

- 作者: Lee, Dongsu; Kwon, Minhae
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Temporal Distance-aware Transition Augmentation for Offline Model-based Reinforcement Learning

快速理解: The goal of offline reinforcement learning (RL) is to extract the best possible policy from the previously collected dataset considering the out-of-distribution (OOD) sample issue. Offline model-based RL (MBRL) is a captivating solution capable of alleviating such issues through a state-action transition augmentation with a learned dynamic model.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.451 Test-Time Multimodal Backdoor Detection by Contrastive Prompting

- 作者: Niu, Yuwei; He, Shuo; Wei, Qi; Wu, Zongyu; Liu, Feng; Feng, Lei
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Test-Time Multimodal Backdoor Detection by Contrastive Prompting

快速理解: While multimodal contrastive learning methods (e.g., CLIP) can achieve impressive zero-shot classification performance, recent research has revealed that these methods are vulnerable to backdoor attacks. To defend against backdoor attacks on CLIP, existing defense methods focus on either the pre-training stage or the fine-tuning stage, which would unfortunately cause high computational costs due to numerous parameter updates and are not applicable in black-box settings.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.452 Test-time Adapted Reinforcement Learning with Action Entropy Regularization

- 作者: Xu, Shoukai; Lian, Zihao; Tan, Mingkui; Liu, Liu; Zhang, Zhong; Zhao, Peilin
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Test-time Adapted Reinforcement Learning with Action Entropy Regularization

快速理解: Offline reinforcement learning is widely applied in multiple fields due to its advantages in efficiency and risk control. However, a major problem it faces is the distribution shift between offline datasets and online environments.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.453 Testing the Limits of Fine-Tuning for Improving Visual Cognition in Vision Language Models

- 作者: Schulze Buschoff, Luca M.; Voudouris, Konstantinos; Akata, Elif; Bethge, Matthias; Tenenbaum, Joshua B.; Schulz, Eric

- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Testing the Limits of Fine-Tuning for Improving Visual Cognition in Vision Language Models

快速理解: Pre-trained vision language models still fall short of human visual cognition. In an effort to improve visual cognition and align models with human behavior, we introduce visual stimuli and human judgments on visual cognition tasks, allowing us to systematically evaluate performance across cognitive domains under a consistent environment.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.454 Textural or Textual: How Vision-Language Models Read Text in Images

- 作者: Wang, Hanzhang; Ma, Qingyuan
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Textural or Textual: How Vision-Language Models Read Text in Images

快速理解: Typographic attacks are often attributed to the ability of multimodal pre-trained models to fuse textual semantics into visual representations, yet the mechanisms and locus of such interference remain unclear. We examine whether such models genuinely encode textual semantics or primarily rely on texture-based visual features.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.455 The Berkeley Function Calling Leaderboard (BFCL): From Tool Use to Agentic Evaluation of Large Language Models

- 作者: Patil, Shishir G; Mao, Huanzhi; Yan, Fanjia; Ji, Charlie Cheng-Jie; Suresh, Vishnu; Stoica, Ion; Gonzalez, Joseph E.
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: The Berkeley Function Calling Leaderboard (BFCL): From Tool Use to Agentic Evaluation of Large Language Models

快速理解: Function calling, also called tool use, refers to an LLM's ability to invoke external functions, APIs, or user-defined tools in response to user queries—an essential capability for agentic LLM applications. Despite its prominence, there did not exist a standard benchmark to evaluate function calling abilities, due to two reasons –the challenging nature of evaluating when a function call is valid, and the challenge of acquiring diverse, real-world functions.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.456 The Butterfly Effect: Neural Network Training Trajectories Are Highly Sensitive to Initial Conditions

- 作者: Alt{ i }ntac{ s }, G" { u }l Sena; Kwok, Devin; Raffel, Colin; Rolnick, David

- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: The Butterfly Effect: Neural Network Training Trajectories Are Highly Sensitive to Initial Conditions

快速理解: Neural network training is inherently sensitive to initialization and the randomness induced by stochastic gradient descent. However, it is unclear to what extent such effects lead to meaningfully different networks, either in terms of the models' weights or the underlying functions that were learned.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.6.457 The Courage to Stop: Overcoming Sunk Cost Fallacy in Deep Reinforcement Learning

- 作者: Liu, Jiashun; Obando-Ceron, Johan; Castro, Pablo Samuel; Courville, Aaron; Pan, Ling
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: The Courage to Stop: Overcoming Sunk Cost Fallacy in Deep Reinforcement Learning

快速理解: Off-policy deep reinforcement learning (RL) agents typically leverage replay buffers for reusing past experiences during learning. This can help sample efficiency when the collected data is informative and aligned with the learning objectives; when that is not the case, it has the effect of “polluting” the replay buffer with data that can exacerbate optimization challenges in addition to wasting environment interactions due to redundant sampling.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.458 The Hidden Life of Tokens: Reducing Hallucination of Large Vision-Language Models Via Visual Information Steering

- 作者: Li, Zhuowei; Shi, Haizhou; Gao, Yunhe; Liu, Di; Wang, Zhenting; Chen, Yuxiao; Liu, Ting; Zhao, Long; Wang, Hao; Metaxas, Dimitris N.
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: The Hidden Life of Tokens: Reducing Hallucination of Large Vision-Language Models Via Visual Information Steering

快速理解: Large Vision-Language Models (LVLMs) can reason effectively over both textual and visual inputs, but they tend to hallucinate syntactically coherent yet visually ungrounded contents. In this paper, we investigate the internal dynamics of hallucination by examining the tokens logits rankings throughout the generation process, revealing three key patterns in how LVLMs process information: (1) gradual visual information loss –visually grounded tokens gradually become less favored throughout generation, and (2) early excitation –semantically meaningful tokens achieve peak activation in the layers earlier than the final layer. (3) hidden genuine information –visually grounded tokens though not being eventually decided still retain relatively high rankings at inference.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.6.459 The Impact of On-Policy Parallelized Data Collection on Deep Reinforcement Learning Networks

- 作者：Mayor, Walter; Obando-Ceron, Johan; Courville, Aaron; Castro, Pablo Samuel
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：The Impact of On-Policy Parallelized Data Collection on Deep Reinforcement Learning Networks

快速理解：The use of parallel actors for data collection has been an effective technique used in reinforcement learning (RL) algorithms. The manner in which data is collected in these algorithms, controlled via the number of parallel environments and the rollout length, induces a form of bias-variance trade-off; the number of training passes over the collected data, on the other hand, must strike a balance between sample efficiency and overfitting.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.460 The Surprising Agreement Between Convex Optimization Theory and Learning-Rate Scheduling for Large Model Training

- 作者：Schaipp, Fabian; H" {a}gele, Alexander; Taylor, Adrien; Simsekli, Umut; Bach, Francis
- 年份：2025；主题：Systems；优先级：重要
- [来源](#)

论文：The Surprising Agreement Between Convex Optimization Theory and Learning-Rate Scheduling for Large Model Training

快速理解：We show that learning-rate schedules for large model training behave surprisingly similar to a performance bound from non-smooth convex optimization theory. We provide a bound for the constant schedule with linear cooldown; in particular, the practical benefit of cooldown is reflected in the bound due to the absence of logarithmic terms.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.6.461 Time-Aware World Model for Adaptive Prediction and Control

- 作者：Nhu, Anh N; Son, Sanghyun; Lin, Ming
- 年份：2025；主题：World Model；优先级：重要
- [来源](#)

论文：Time-Aware World Model for Adaptive Prediction and Control

快速理解：In this work, we introduce the Time-Aware World Model (TAWM), a model-based approach that explicitly incorporates temporal dynamics. By conditioning on the time-step size, Δt , and training over a diverse range of Δt values –rather than sampling at a fixed time-step –TAWM learns both high- and low-frequency task dynamics across diverse control problems.

关注点：环境动力学、视频预测、潜变量状态或模型式决策。

证据层级：索引摘要。

1.3.6.462 Time-VLM: Exploring Multimodal Vision-Language Models for Augmented Time Series Forecasting

- 作者: Zhong, Siru; Ruan, Weilin; Jin, Ming; Li, Huan; Wen, Qingsong; Liang, Yuxuan
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Time-VLM: Exploring Multimodal Vision-Language Models for Augmented Time Series Forecasting

快速理解: Recent advancements in time series forecasting have explored augmenting models with text or vision modalities to improve accuracy. While text provides contextual understanding, it often lacks fine-grained temporal details.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.6.463 Towards World Simulator: Crafting Physical Commonsense-Based Benchmark for Video Generation

- 作者: Meng, Fanqing; Liao, Jiaqi; Tan, Xinyu; Lu, Quanfeng; Shao, Wenqi; Zhang, Kaipeng; Cheng, Yu; Li, Dianqi; Luo, Ping
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Towards World Simulator: Crafting Physical Commonsense-Based Benchmark for Video Generation

快速理解: Text-to-video (T2V) models like Sora have made significant strides in visualizing complex prompts, which is increasingly viewed as a promising path towards constructing the universal world simulator. Cognitive psychologists believe that the foundation for achieving this goal is the ability to understand intuitive physics.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.6.464 Trajectory World Models for Heterogeneous Environments

- 作者: Yin, Shaofeng; Wu, Jialong; Huang, Siqiao; Su, Xingjian; He, Xu; Hao, Jianye; Long, Ming-sheng
- 年份: 2025; 主题: World Model; 优先级: 重要
- [来源](#)

论文: Trajectory World Models for Heterogeneous Environments

快速理解: Heterogeneity in sensors and actuators across environments poses a significant challenge to building large-scale pre-trained world models on top of this low-dimensional sensor information. In this work, we explore pre-training world models for heterogeneous environments by addressing key transfer barriers in both data diversity and model flexibility.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策。

证据层级：索引摘要。

1.3.6.465 Understanding Multimodal LLMs Under Distribution Shifts: An Information-Theoretic Approach

- 作者：Oh, Changdae; Fang, Zhen; Im, Shawn; Du, Xuefeng; Li, Yixuan
- 年份：2025; 主题：Supporting Perception; 优先级：重要
- [来源](#)

论文：Understanding Multimodal LLMs Under Distribution Shifts: An Information-Theoretic Approach

快速理解：Multimodal large language models (MLLMs) have shown promising capabilities but struggle under distribution shifts, where evaluation data differ from instruction tuning distributions. Although previous works have provided empirical evaluations, we argue that establishing a formal framework that can characterize and quantify the risk of MLLMs is necessary to ensure the safe and reliable application of MLLMs in the real world.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.6.466 Understanding and Mitigating Miscalibration in Prompt Tuning for Vision-Language Models

- 作者：Wang, Shuoyuan; Li, Yixuan; Wei, Hongxin
- 年份：2025; 主题：Supporting Perception; 优先级：重要
- [来源](#)

论文：Understanding and Mitigating Miscalibration in Prompt Tuning for Vision-Language Models

快速理解：Confidence calibration is critical for the safe deployment of machine learning models in the real world. However, such issue in vision-language models like CLIP, particularly after fine-tuning, has not been fully addressed.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.6.467 Understanding the Emergence of Multimodal Representation Alignment

- 作者：Tjandrasuwita, Megan; Ekbote, Chanakya; Ziyin, Liu; Liang, Paul Pu
- 年份：2025; 主题：Supporting Perception; 优先级：重要
- [来源](#)

论文：Understanding the Emergence of Multimodal Representation Alignment

快速理解：Multimodal representation learning is fundamentally about transforming incomparable modalities into comparable representations. While prior research has primarily focused on explicitly aligning these representations through targeted learning objectives and model architectures, a recent line of work has found that independently trained unimodal models of increasing scale and performance can become implicitly aligned with each other.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.6.468 Understanding the Unfairness in Network Quantization

- 作者: Liu, Bing; Miao, Wenjun; Zhang, Boyu; Zhang, Qiankun; Yuan, Bin; Wang, Jing; Liu, Shenghao; Deng, Xianjun
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Understanding the Unfairness in Network Quantization

快速理解: Network quantization, one of the most widely studied model compression methods, effectively quantizes a floating-point model to obtain a fixed-point one with negligible accuracy loss. Although great success was achieved in reducing the model size, it may exacerbate the unfairness in model accuracy across different groups of datasets.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.6.469 Unlocking the Capabilities of Large Vision-Language Models for Generalizable and Explainable Deepfake Detection

- 作者: Yu, Peipeng; Fei, Jianwei; Gao, Hui; Feng, Xuan; Xia, Zhihua; Chang, Chip Hong
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Unlocking the Capabilities of Large Vision-Language Models for Generalizable and Explainable Deepfake Detection

快速理解: Current Large Vision-Language Models (LVLMs) have demonstrated remarkable capabilities in understanding multimodal data, but their potential remains underexplored for deepfake detection due to the misalignment of their knowledge and forensics patterns. To this end, we present a novel framework that unlocks LVLMs' potential capabilities for deepfake detection.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.470 Unpaired Point Cloud Completion via Unbalanced Optimal Transport

- 作者: Lee, Taekyung; Choi, Jaemoo; Choi, Jaewoong; Kang, Myungjoo
- 年份: 2025; 主题: Systems; 优先级: 重要
- [来源](#)

论文: Unpaired Point Cloud Completion via Unbalanced Optimal Transport

快速理解: Unpaired point cloud completion is crucial for real-world applications, where ground-truth data for complete point clouds are often unavailable. By learning a completion map from unpaired incomplete and complete point cloud data, this task avoids the reliance on paired datasets.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.6.471 Unveiling Markov heads in Pretrained Language Models for Offline Reinforcement Learning

- 作者: Zhao, Wenhao; Xu, Qiushui; Xu, Linjie; Song, Lei; Wang, Jinyu; Zhou, Chunlai; Bian, Jiang
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Unveiling Markov heads in Pretrained Language Models for Offline Reinforcement Learning

快速理解: Recently, incorporating knowledge from pretrained language models (PLMs) into decision transformers (DTs) has generated significant attention in offline reinforcement learning (RL). These PLMs perform well in RL tasks, raising an intriguing question: what kind of knowledge from PLMs has been transferred to RL to achieve such good results?

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.472 VIP: Vision Instructed Pre-training for Robotic Manipulation

- 作者: Li, Zhuoling; Ren, Liangliang; Yang, Jinrong; Zhao, Yong; Wu, Xiaoyang; Xu, Zhenhua; Bai, Xiang; Zhao, Hengshuang
- 年份: 2025; 主题: Embodied/WBC; 优先级: 重要
- [来源](#)

论文: VIP: Vision Instructed Pre-training for Robotic Manipulation

快速理解: The effectiveness of scaling up training data in robotic manipulation is still limited. A primary challenge in manipulation is the tasks are diverse, and the trained policy would be confused if the task targets are not specified clearly.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制。

证据层级: 索引摘要。

1.3.6.473 VersaPRM: Multi-Domain Process Reward Model via Synthetic Reasoning Data

- 作者: Zeng, Thomas; Zhang, Shuibai; Wu, Shutong; Classen, Christian; Chae, Daewon; Ewer, Ethan; Lee, Minjae; Kim, Heeju; Kang, Wonjun; Kunde, Jackson; Fan, Ying; Kim, Jungtaek; Koo, Hyung Il; Ramchandran, Kannan; Papailiopoulos, Dimitris; Lee, Kangwook
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: VersaPRM: Multi-Domain Process Reward Model via Synthetic Reasoning Data

快速理解: Process Reward Models (PRMs) have proven effective at enhancing mathematical reasoning for Large Language Models (LLMs) by leveraging increased inference-time computation. However, they are predominantly trained on mathematical data and their generalizability to non-mathematical domains has not been rigorously studied.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.474 Vintix: Action Model via In-Context Reinforcement Learning

- 作者: Polubarov, Andrei; Nikita, Lyubaykin; Derevyagin, Alexander; Zisman, Ilya; Tarasov, Denis; Nikulin, Alexander; Kurenkov, Vladislav

- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Vintix: Action Model via In-Context Reinforcement Learning

快速理解: In-Context Reinforcement Learning (ICRL) represents a promising paradigm for developing generalist agents that learn at inference time through trial-and-error interactions, analogous to how large language models adapt contextually, but with a focus on reward maximization. However, the scalability of ICRL beyond toy tasks and single-domain settings remains an open challenge.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.475 Vision-Language Model Selection and Reuse for Downstream Adaptation

- 作者: Tan, Hao-Zhe; Zhou, Zhi; Li, Yu-Feng; Guo, Lan-Zhe
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Vision-Language Model Selection and Reuse for Downstream Adaptation

快速理解: Pre-trained Vision-Language Models (VLMs) are becoming increasingly popular across various visual tasks, and several open-sourced VLM variants have been released. However, selecting the best-performing pre-trained VLM for a specific downstream task is challenging since no single VLM can achieve promising performance on all downstream tasks, and evaluating all available VLMs is impossible due to time and data limitations.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.476 Vision-Language Models Create Cross-Modal Task Representations

- 作者: Luo, Grace; Darrell, Trevor; Bar, Amir
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Vision-Language Models Create Cross-Modal Task Representations

快速理解: Autoregressive vision-language models (VLMs) can handle many tasks within a single model, yet the representations that enable this capability remain opaque. We find that VLMs align conceptually equivalent inputs into a shared task vector, which is invariant to modality (text, image) and format (examples, instruction), and may simplify VLM processing.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.477 Visual Attention Never Fades: Selective Progressive Attention ReCalibration for Detailed Image Captioning in Multimodal Large Language Models

- 作者: Jung, Mingi; Lee, Saehyung; Kim, Eunji; Yoon, Sungroh
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Visual Attention Never Fades: Selective Progressive Attention ReCalibration for Detailed Image Captioning in Multimodal Large Language Models

快速理解: Detailed image captioning is essential for tasks like data generation and aiding visually impaired individuals. High-quality captions require a balance between precision and recall, which remains challenging for current multimodal large language models (MLLMs).

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.478 Visual Graph Arena: Evaluating Visual Conceptualization of Vision and Multimodal Large Language Models

- 作者: Babaiee, Zahra; Kiasari, Peyman; Rus, Daniela; Grosu, Radu
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Visual Graph Arena: Evaluating Visual Conceptualization of Vision and Multimodal Large Language Models

快速理解: Recent advancements in multimodal large language models have driven breakthroughs in visual question answering. Yet, a critical gap persists, ‘conceptualization’ —the ability to recognize and reason about the same concept despite variations in visual form, a basic ability of human reasoning.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.479 WMarkGPT: Watermarked Image Understanding via Multimodal Large Language Models

- 作者: Tan, Songbai; Qiu, Xuerui; Shu, Yao; Xu, Gang; Xu, Linrui; Xu, Xiangyu; Zhuang, Huiping; Li, Ming; Yu, Fei
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: WMarkGPT: Watermarked Image Understanding via Multimodal Large Language Models

快速理解: Invisible watermarking is widely used to protect digital images from unauthorized use. Accurate assessment of watermarking efficacy is crucial for advancing algorithmic development.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.480 Wasserstein Policy Optimization

- 作者: Pfau, David; Davies, Ian; Borsa, Diana L; Ara' {u}jo, Jo~{a}o Guilherme Madeira; Tracey, Brendan Daniel; Van Hasselt, Hado
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Wasserstein Policy Optimization

快速理解: We introduce Wasserstein Policy Optimization (WPO), an actor-critic algorithm for reinforcement learning in continuous action spaces. WPO can be derived as an approximation to Wasserstein gradient flow over the space of all policies projected into a finite-dimensional parameter space (e.g., the weights of a neural network), leading to a simple and completely general closed-form update.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 索引摘要。

1.3.6.481 What Has a Foundation Model Found? Using Inductive Bias to Probe for World Models

- 作者: Vafa, Keyon; Chang, Peter G.; Rambachan, Ashesh; Mullainathan, Sendhil
- 年份: 2025; 主题: World Model; 优先级: 重要
- [来源](#)

论文: What Has a Foundation Model Found? Using Inductive Bias to Probe for World Models

快速理解: Foundation models are premised on the idea that sequence prediction can uncover deeper domain understanding, much like how Kepler's predictions of planetary motion later led to the discovery of Newtonian mechanics. However, evaluating whether these models truly capture deeper structure remains a challenge.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策。

证据层级: 索引摘要。

1.3.6.482 When Every Millisecond Counts: Real-Time Anomaly Detection via the Multimodal Asynchronous Hybrid Network

- 作者: Xiao, Dong; Chen, Guangyao; Peng, Peixi; Huang, Yangru; Zhao, Yifan; Dai, Yongxing; Tian, Yonghong
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: When Every Millisecond Counts: Real-Time Anomaly Detection via the Multimodal Asynchronous Hybrid Network

快速理解: Anomaly detection is essential for the safety and reliability of autonomous driving systems. Current methods often focus on detection accuracy but neglect response time, which is critical in time-sensitive driving scenarios.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.6.483 When Maximum Entropy Misleads Policy Optimization

- 作者: Zhang, Ruipeng; Chang, Ya-Chien; Gao, Sicun
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: When Maximum Entropy Misleads Policy Optimization

快速理解: The Maximum Entropy Reinforcement Learning (MaxEnt RL) framework is a leading

approach for achieving efficient learning and robust performance across many RL tasks. However, MaxEnt methods have also been shown to struggle with performance-critical control problems in practice, where non-MaxEnt algorithms can successfully learn.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.484 When do neural networks learn world models?

- 作者：Zhang, Tianren; Chen, Guanyu; Chen, Feng
- 年份：2025；主题：World Model；优先级：重要
- [来源](#)

论文：When do neural networks learn world models?

快速理解：Humans develop world models that capture the underlying generation process of data. Whether neural networks can learn similar world models remains an open problem.

关注点：环境动力学、视频预测、潜变量状态或模型式决策。

证据层级：索引摘要。

1.3.6.485 Which Agent Causes Task Failures and When? On Automated Failure Attribution of LLM Multi-Agent Systems

- 作者：Zhang, Shaokun; Yin, Ming; Zhang, Jieyu; Liu, Jiale; Han, Zhiguang; Zhang, Jingyang; Li, Beibin; Wang, Chi; Wang, Huazheng; Chen, Yiran; Wu, Qingyun
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Which Agent Causes Task Failures and When? On Automated Failure Attribution of LLM Multi-Agent Systems

快速理解：Failure attribution in LLM multi-agent systems—identifying the agent and step responsible for task failures—provides crucial clues for systems debugging but remains underexplored and labor-intensive. In this paper, we propose and formulate a new research area: automated failure attribution for LLM multi-agent systems.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.6.486 Wolfpack Adversarial Attack for Robust Multi-Agent Reinforcement Learning

- 作者：Lee, Sunwoo; Hwang, Jaebak; Jo, Yonghyeon; Han, Seungyul
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Wolfpack Adversarial Attack for Robust Multi-Agent Reinforcement Learning

快速理解：Traditional robust methods in multi-agent reinforcement learning (MARL) often struggle against coordinated adversarial attacks in cooperative scenarios. To address this limitation, we propose the Wolfpack Adversarial Attack framework, inspired by wolf hunting strategies, which targets an initial agent and its assisting agents to disrupt cooperation.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：索引摘要。

1.3.7 AI · NeurIPS

本会议纳入 **875** 篇。自动主题统计为：Agent RL (353)、Supporting Perception (332)、AI Infra (81)、Systems (71)、Embodied/WBC (44)、World Model (39)、VLA (18)、Robot Learning (16)。这些统计用于组织阅读顺序，不能单独证明论文结论或实验优越性。

优先阅读候选：- VLA-Cache: Efficient Vision-Language-Action Manipulation via Adaptive Token Caching (优先级分数 98.25) - 4D-VLA: Spatiotemporal Vision-Language-Action Pretraining with Cross-Scene Calibration (优先级分数 71.25) - Accurate KV Cache Eviction via Anchor Direction Projection for Efficient LLM Inference (优先级分数 71.25) - Ada-KV: Optimizing KV Cache Eviction by Adaptive Budget Allocation for Efficient LLM Inference (优先级分数 71.25) - AffordBot: 3D Fine-grained Embodied Reasoning via Multimodal Large Language Models (优先级分数 71.25)

1.3.7.1 VLA-Cache: Efficient Vision-Language-Action Manipulation via Adaptive Token Caching

- 作者：作者信息未提供
- 年份：2025；主题：VLA; Supporting Perception; Systems；优先级：核心
- [来源](#)

论文：VLA-Cache: Efficient Vision-Language-Action Manipulation via Adaptive Token Caching

快速理解：Vision-Language-Action (VLA) models have demonstrated strong multi-modal reasoning capabilities, enabling direct action generation from visual perception and language instructions in an end-to-end manner. However, their substantial computational cost poses a challenge for real-time robotic control, where rapid decision-making is essential.

关注点：视觉、语言条件与机器人动作策略之间的对齐；3D、视频、多模态视觉或具身感知支撑能力；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.7.2 4D-VLA: Spatiotemporal Vision-Language-Action Pretraining with Cross-Scene Calibration

- 作者：作者信息未提供
- 年份：2025；主题：VLA; Supporting Perception；优先级：核心
- [来源](#)

论文：4D-VLA: Spatiotemporal Vision-Language-Action Pretraining with Cross-Scene Calibration

快速理解：Leveraging diverse robotic data for pretraining remains a critical challenge. Existing methods typically model the dataset's action distribution using simple observations as inputs.

关注点：视觉、语言条件与机器人动作策略之间的对齐；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.3 Accurate KV Cache Eviction via Anchor Direction Projection for Efficient LLM Inference

- 作者：作者信息未提供

- 年份: 2025; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: Accurate KV Cache Eviction via Anchor Direction Projection for Efficient LLM Inference

快速理解: Key-Value (KV) cache eviction—which retains the KV pairs of the most important tokens while discarding less important ones—is a critical technique for optimizing both memory usage and inference latency in large language models (LLMs). However, existing approaches often rely on simple heuristics—such as attention weights—to measure token importance, overlooking the spatial relationships between token value states in the vector space.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.7.4 Ada-KV: Optimizing KV Cache Eviction by Adaptive Budget Allocation for Efficient LLM Inference

- 作者: 作者信息未提供
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: Ada-KV: Optimizing KV Cache Eviction by Adaptive Budget Allocation for Efficient LLM Inference

快速理解: Large Language Models have excelled in various domains but face efficiency challenges due to the growing Key-Value (KV) cache required for long-sequence inference. Recent efforts aim to reduce KV cache size by evicting vast non-critical cache elements during runtime while preserving generation quality.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.7.5 AffordBot: 3D Fine-grained Embodied Reasoning via Multimodal Large Language Models

- 作者: 作者信息未提供
- 年份: 2025; 主题: Embodied/WBC; Supporting Perception; 优先级: 核心
- [来源](#)

论文: AffordBot: 3D Fine-grained Embodied Reasoning via Multimodal Large Language Models

快速理解: Effective human-agent collaboration in physical environments requires understanding not only what to act upon, but also where the actionable elements are and how to interact with them. Existing approaches often operate at the object level or disjointedly handle fine-grained affordance reasoning, lacking coherent, instruction-driven grounding and reasoning.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.6 AutoVLA: A Vision-Language-Action Model for End-to-End Autonomous Driving with Adaptive Reasoning and Reinforcement Fine-Tuning

- 作者: 作者信息未提供

- 年份: 2025; 主题: VLA; Supporting Perception; 优先级: 核心
- [来源](#)

论文: AutoVLA: A Vision-Language-Action Model for End-to-End Autonomous Driving with Adaptive Reasoning and Reinforcement Fine-Tuning

快速理解: Recent advancements in Vision-Language-Action (VLA) models have shown promise for end-to-end autonomous driving by leveraging world knowledge and reasoning capabilities. However, current VLA models often struggle with physically infeasible action outputs, complex model structures, or unnecessarily long reasoning.

关注点: 视觉、语言条件与机器人动作策略之间的对齐; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.7 BadVLA: Towards Backdoor Attacks on Vision-Language-Action Models via Objective-Decoupled Optimization

- 作者: 作者信息未提供
- 年份: 2025; 主题: VLA; Supporting Perception; 优先级: 核心
- [来源](#)

论文: BadVLA: Towards Backdoor Attacks on Vision-Language-Action Models via Objective-Decoupled Optimization

快速理解: Vision-Language-Action (VLA) models have advanced robotic control by enabling end-to-end decision-making directly from multimodal inputs. However, their tightly coupled architectures expose novel security vulnerabilities.

关注点: 视觉、语言条件与机器人动作策略之间的对齐; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.8 CADGrasp: Learning Contact and Collision Aware General Dexterous Grasping in Cluttered Scenes

- 作者: 作者信息未提供
- 年份: 2025; 主题: Embodied/WBC; Robot Learning; 优先级: 核心
- [来源](#)

论文: CADGrasp: Learning Contact and Collision Aware General Dexterous Grasping in Cluttered Scenes

快速理解: Dexterous grasping in cluttered environments presents substantial challenges due to the high degrees of freedom of dexterous hands, occlusion, and potential collisions arising from diverse object geometries and complex layouts. To address these challenges, we propose CADGrasp, a two-stage algorithm for general dexterous grasping using single-view point cloud inputs.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 本地 PDF 前页。

1.3.7.9 ChatVLA-2: Vision-Language-Action Model with Open-World Reasoning

- 作者: 作者信息未提供
- 年份: 2025; 主题: VLA; Supporting Perception; 优先级: 核心

- [来源](#)

论文: ChatVLA-2: Vision-Language-Action Model with Open-World Reasoning

快速理解: Vision-language-action (VLA) models have emerged as the next generation of models in robotics. However, despite leveraging powerful pre-trained Vision-Language Models (VLMs), existing end-to-end VLA systems often lose key capabilities during fine-tuning as the model adapts to specific robotic tasks.

关注点: 视觉、语言条件与机器人动作策略之间的对齐; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.10 ChunkKV: Semantic-Preserving KV Cache Compression for Efficient Long-Context LLM Inference

- 作者: 作者信息未提供
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: ChunkKV: Semantic-Preserving KV Cache Compression for Efficient Long-Context LLM Inference

快速理解: Large Language Models (LLMs) require significant GPU memory when processing long texts, with the key value (KV) cache consuming up to 70% of total memory during inference. Although existing compression methods reduce memory by evaluating the importance of individual tokens, they overlook critical semantic relationships between tokens, resulting in fragmented context and degraded performance.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.7.11 Co-Reinforcement Learning for Unified Multimodal Understanding and Generation

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; Supporting Perception; 优先级: 核心
- [来源](#)

论文: Co-Reinforcement Learning for Unified Multimodal Understanding and Generation

快速理解: This paper presents a pioneering exploration of reinforcement learning (RL) via group relative policy optimization for unified multimodal large language models (ULMs), aimed at simultaneously reinforcing generation and understanding capabilities. Through systematic pilot studies, we uncover the significant potential of ULMs to enable the synergistic co-evolution of dual capabilities within a shared policy optimization framework.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.12 CogVLA: Cognition-Aligned Vision-Language-Action Models via Instruction-Driven Routing & Sparsification

- 作者: 作者信息未提供

- 年份: 2025; 主题: VLA; Supporting Perception; 优先级: 核心
- [来源](#)

论文: CogVLA: Cognition-Aligned Vision-Language-Action Models via Instruction-Driven Routing & Sparsification

快速理解: Recent Vision-Language-Action (VLA) models built on pre-trained Vision- Language Models (VLMs) require extensive post-training, resulting in high computational overhead that limits scalability and deployment. Existing sparsification strategies—such as Mixture-of-Depths, layer skipping, and early exit—fall short by neglecting the semantic coupling across vision-language-action modalities, *Corresponding author 39th Conference on Neural Information Processing Systems (NeurIPS 2025). and focusing narrowly on intra-LLM computation while overlooking end-to-end coherence from perception to control.

关注点: 视觉、语言条件与机器人动作策略之间的对齐; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.13 Compiler-R1: Towards Agentic Compiler Auto-tuning with Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; Systems; 优先级: 核心
- [来源](#)

论文: Compiler-R1: Towards Agentic Compiler Auto-tuning with Reinforcement Learning

快速理解: Compiler auto-tuning optimizes pass sequences to improve performance metrics such as Intermediate Representation (IR) instruction count. Although recent advances leveraging Large Language Models (LLMs) have shown promise in automating compiler tuning, two significant challenges still remain: the absence of highquality reasoning datasets for agents training, and limited effective interactions with the compilation environment.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.7.14 Contact Map Transfer with Conditional Diffusion Model for Generalizable Dexterous Grasp Generation

- 作者: 作者信息未提供
- 年份: 2025; 主题: Embodied/WBC; Robot Learning; 优先级: 核心
- [来源](#)

论文: Contact Map Transfer with Conditional Diffusion Model for Generalizable Dexterous Grasp Generation

快速理解: Dexterous grasp generation is a fundamental challenge in robotics, requiring both grasp stability and adaptability across diverse objects and tasks. Analytical methods ensure stable grasps but are inefficient and lack task adaptability, while generative approaches improve efficiency and task integration but generalize poorly to unseen objects and tasks due to data limitations.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 本地 PDF 前页。

1.3.7.15 Cost-Efficient LLM Training with Lifetime-Aware Tensor Offloading via GPUDirect Storage

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra; Systems；优先级：核心
- [来源](#)

论文：Cost-Efficient LLM Training with Lifetime-Aware Tensor Offloading via GPUDirect Storage

快速理解：We present the design and implementation of a new lifetime-aware tensor offloading framework for GPU memory expansion using low-cost PCIe-based solid-state drives (SSDs). Our framework, TERAIO, is developed explicitly for large language model (LLM) training with multiple GPUs and multiple SSDs.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.7.16 DreamPRM: Domain-reweighted Process Reward Model for Multimodal Reasoning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL; Supporting Perception；优先级：核心
- [来源](#)

论文：DreamPRM: Domain-reweighted Process Reward Model for Multimodal Reasoning

快速理解：Reasoning has substantially improved the performance of large language models (LLMs) on complicated tasks. Central to the current reasoning studies, Process Reward Models (PRMs) offer a fine-grained evaluation of intermediate reasoning steps and guide the reasoning process.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.17 DreamVLA: A Vision-Language-Action Model Dreamed with Comprehensive World Knowledge

- 作者：作者信息未提供
- 年份：2025；主题：VLA; Supporting Perception；优先级：核心
- [来源](#)

论文：DreamVLA: A Vision-Language-Action Model Dreamed with Comprehensive World Knowledge

快速理解：Recent advances in vision-language-action (VLA) models have shown promise in integrating image generation with action prediction to improve generalization and reasoning in robot manipulation. However, existing methods are limited to challenging image-based forecasting, which suffers from redundant information and lacks comprehensive and critical world knowledge, including dynamic, spatial and semantic information.

关注点：视觉、语言条件与机器人动作策略之间的对齐；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.18 DynamicVerse: A Physically-Aware Multimodal Framework for 4D World Modeling

- 作者：作者信息未提供
- 年份：2025；主题：World Model; Supporting Perception；优先级：核心
- [来源](#)

论文：DynamicVerse: A Physically-Aware Multimodal Framework for 4D World Modeling

快速理解：Understanding the dynamic physical world, characterized by its evolving 3D structure, real-world motion, and semantic content with textual descriptions, is crucial for human-agent interaction and enables embodied agents to perceive and act within real environments with human-like capabilities. However, existing datasets are often derived from limited simulators or utilize traditional Structurefrom-Motion for up-to-scale annotation and offer limited descriptive captioning, which restricts the capacity of foundation models to accurately interpret real-world dynamics from monocular videos, commonly sourced from the internet.

关注点：环境动力学、视频预测、潜变量状态或模型式决策；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.19 EfficientVLA: Training-Free Acceleration and Compression for Vision-Language-Action Models

- 作者：作者信息未提供
- 年份：2025；主题：VLA; Supporting Perception；优先级：核心
- [来源](#)

论文：EfficientVLA: Training-Free Acceleration and Compression for Vision-Language-Action Models

快速理解：Vision-Language-Action (VLA) models, particularly diffusion-based architectures, demonstrate transformative potential for embodied intelligence but are severely hampered by high computational and memory demands stemming from extensive inherent and inference-time redundancies. While existing acceleration efforts often target isolated inefficiencies, such piecemeal solutions typically fail to holistically address the varied computational and memory bottlenecks across the entire VLA pipeline, thereby limiting practical deployability.

关注点：视觉、语言条件与机器人动作策略之间的对齐；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.20 Enhancing Tactile-based Reinforcement Learning for Robotic Control

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL; Robot Learning；优先级：核心
- [来源](#)

论文：Enhancing Tactile-based Reinforcement Learning for Robotic Control

快速理解：Achieving safe, reliable real-world robotic manipulation requires agents to evolve beyond vision and incorporate tactile sensing to overcome sensory deficits and reliance on idealised state information. Despite its potential, the efficacy of tactile sensing in reinforcement learning (RL) remains inconsistent.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：本地 PDF 前页。

1.3.7.21 Exploring the Limits of Vision-Language-Action Manipulation in Cross-task Generalization

- 作者：作者信息未提供
- 年份：2025；主题：VLA; Supporting Perception；优先级：核心
- [来源](#)

论文：Exploring the Limits of Vision-Language-Action Manipulation in Cross-task Generalization

快速理解：The generalization capabilities of vision-language-action (VLA) models to unseen tasks are crucial to achieving general-purpose robotic manipulation in open-world settings. However, the cross-task generalization capabilities of existing VLA models remain significantly underexplored.

关注点：视觉、语言条件与机器人动作策略之间的对齐；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.22 FlowMoE: A Scalable Pipeline Scheduling Framework for Distributed Mixture-of-Experts Training

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra; Systems；优先级：核心
- [来源](#)

论文：FlowMoE: A Scalable Pipeline Scheduling Framework for Distributed Mixture-of-Experts Training

快速理解：The parameter size of modern large language models (LLMs) can be scaled up via the sparsely-activated Mixture-of-Experts (MoE) technique to avoid excessive increase of the computational costs. To further improve training efficiency, pipelining computation and communication has become a promising solution for distributed MoE training.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.7.23 Grasp2Grasp: Vision-Based Dexterous Grasp Translation via Schrödinger Bridges

- 作者：作者信息未提供
- 年份：2025；主题：Embodied/WBC; Robot Learning；优先级：核心
- [来源](#)

论文：Grasp2Grasp: Vision-Based Dexterous Grasp Translation via Schrödinger Bridges

快速理解：We propose a new approach to vision-based dexterous grasp translation, which aims to transfer grasp intent across robotic hands with differing morphologies. Given a visual observation of a source hand grasping an object, our goal is to synthesize a functionally equivalent grasp for a target hand without requiring paired demonstrations or hand-specific simulations.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：本地 PDF 前页。

1.3.7.24 HiFC: High-efficiency Flash-based KV Cache Swapping for Scaling LLM Inference

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra; Systems；优先级：核心
- [来源](#)

论文：HiFC: High-efficiency Flash-based KV Cache Swapping for Scaling LLM Inference

快速理解：Large-language-model inference with long contexts often produces key-value (KV) caches whose footprint exceeds the capacity of high-bandwidth memory on a GPU. Prior LLM inference frameworks such as vLLM mitigate this pressure by swapping KV cache pages to host DRAM.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.7.25 KVCOMM: Online Cross-context KV-cache Communication for Efficient LLM-based Multi-agent Systems

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL; Systems；优先级：核心
- [来源](#)

论文：KVCOMM: Online Cross-context KV-cache Communication for Efficient LLM-based Multi-agent Systems

快速理解：Multi-agent large language model (LLM) systems are increasingly adopted for complex language processing tasks that require communication and coordination among agents. However, these systems often suffer substantial overhead from repeated reprocessing of overlapping contexts across agents.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.7.26 KeyDiff: Key Similarity-Based KV Cache Eviction for Long-Context LLM Inference in Resource-Constrained Environments

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra; Systems；优先级：核心
- [来源](#)

论文：KeyDiff: Key Similarity-Based KV Cache Eviction for Long-Context LLM Inference in Resource-Constrained Environments

快速理解：We demonstrate that geometrically distinctive keys during LLM inference tend to have high attention scores. Based on the phenomenon we propose KEYDIFF, a training-free KV cache eviction method based solely on key similarity.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.7.27 Knowledge Insulating Vision-Language-Action Models: Train Fast, Run Fast, Generalize Better

- 作者：作者信息未提供
- 年份：2025；主题：VLA; Supporting Perception；优先级：核心

- [来源](#)

论文: Knowledge Insulating Vision-Language-Action Models: Train Fast, Run Fast, Generalize Better

快速理解: Vision-language-action (VLA) models provide a powerful approach to training control policies for physical systems, such as robots, by combining end-to-end learning with transfer of semantic knowledge from web-scale vision-language model (VLM) training. However, the constraints of real-time control are often at odds with the design of VLMs: the most powerful VLMs have tens or hundreds of billions of parameters, presenting an obstacle to real-time inference, and operate on discrete tokens rather than the continuous-valued outputs that are required for controlling robots.

关注点: 视觉、语言条件与机器人动作策略之间的对齐; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.28 Learning 3D Persistent Embodied World Models

- 作者: 作者信息未提供
- 年份: 2025; 主题: Embodied/WBC; World Model; 优先级: 核心
- [来源](#)

论文: Learning 3D Persistent Embodied World Models

快速理解: The ability to simulate the effects of future actions on the world is a crucial ability of intelligent embodied agents, enabling agents to anticipate the effects of their actions and make plans accordingly. While a large body of existing work has explored how to construct such world models using video models, they are often myopic in nature, without any memory of a scene not captured by currently observed images, preventing agents from making consistent long-horizon plans in complex environments where many parts of the scene are partially observed.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 环境动力学、视频预测、潜变量状态或模型式决策。

证据层级: 本地 PDF 前页。

1.3.7.29 Learning Interactive World Model for Object-Centric Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: World Model; Agent RL; 优先级: 核心
- [来源](#)

论文: Learning Interactive World Model for Object-Centric Reinforcement Learning

快速理解: Agents that understand objects and their interactions can learn policies that are more robust and transferable. However, most object-centric RL methods factor state by individual objects while leaving interactions implicit.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策; 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.30 Learning World Models for Interactive Video Generation

- 作者: 作者信息未提供
- 年份: 2025; 主题: World Model; Supporting Perception; 优先级: 核心
- [来源](#)

论文: Learning World Models for Interactive Video Generation

快速理解: Foundational world models must be both interactive and preserve spatiotemporal coherence for effective future planning with action choices. However, present models for long video generation have limited inherent world modeling capabilities due to two main challenges: compounding errors and insufficient memory mechanisms.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.31 Learning and Planning Multi-Agent Tasks via an MoE-based World Model

- 作者: 作者信息未提供
- 年份: 2025; 主题: World Model; Agent RL; 优先级: 核心
- [来源](#)

论文: Learning and Planning Multi-Agent Tasks via an MoE-based World Model

快速理解: Multi-task multi-agent reinforcement learning (MT-MARL) aims to develop a single model capable of solving a diverse set of tasks. However, existing methods often fall short due to the substantial variation in optimal policies across tasks, making it challenging for a single policy model to generalize effectively.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策; 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.32 MPCache: MPC-Friendly KV Cache Eviction for Efficient Private LLM Inference

- 作者: 作者信息未提供
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: MPCache: MPC-Friendly KV Cache Eviction for Efficient Private LLM Inference

快速理解: Private large language model (LLM) inference based on secure multi-party computation (MPC) achieves formal data privacy protection but suffers from significant latency overhead, especially for long input sequences. While key-value (KV) cache eviction and sparse attention algorithms have been proposed for efficient LLM inference in plaintext, they are not designed for MPC and cannot benefit private LLM inference directly.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.7.33 MUSTAFAR: Promoting Unstructured Sparsity for KV Cache Pruning in LLM Inference

- 作者: 作者信息未提供
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: MUSTAFAR: Promoting Unstructured Sparsity for KV Cache Pruning in LLM Inference

快速理解: We demonstrate that unstructured sparsity significantly improves KV cache compression for LLMs, enabling sparsity levels up to 70% without compromising accuracy or requiring fine-tuning. We conduct a systematic exploration of pruning strategies and find per-token magnitude-based pruning as highly effective for both Key and Value caches under unstructured sparsity, surpassing prior structured pruning schemes.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.7.34 MagCache: Fast Video Generation with Magnitude-Aware Cache

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; Systems; 优先级: 核心
- [来源](#)

论文: MagCache: Fast Video Generation with Magnitude-Aware Cache

快速理解: Existing acceleration techniques for video diffusion models often rely on uniform heuristics or time-embedding variants to skip timesteps and reuse cached features. These approaches typically require extensive calibration with curated prompts and risk inconsistent outputs due to prompt-specific overfitting.

关注点: 3D、视频、多模态视觉或具身感知支撑能力; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.7.35 Memo: Training Memory-Efficient Embodied Agents with Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Embodied/WBC; Agent RL; 优先级: 核心
- [来源](#)

论文: Memo: Training Memory-Efficient Embodied Agents with Reinforcement Learning

快速理解: To enable embodied agents to operate effectively over extended timeframes, it is crucial to develop models that form and access memories to stay contextualized in their environment. In the current paradigm of training transformer-based policies for embodied sequential decision-making tasks, visual inputs often overwhelm the context limits of transformers, while humans can maintain and utilize a lifetime of experience compressed as memories.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.36 Mixture-of-Experts Meets In-Context Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; AI Infra; 优先级: 核心
- [来源](#)

论文: Mixture-of-Experts Meets In-Context Reinforcement Learning

快速理解: In-context reinforcement learning (ICRL) has emerged as a promising paradigm for adapting RL agents to downstream tasks through prompt conditioning. However, two notable challenges

remain in fully harnessing in-context learning within RL domains: the intrinsic multi-modality of the state-action-reward data and the diverse, heterogeneous nature of decision tasks.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.7.37 NSNQuant: A Double Normalization Approach for Calibration-Free Low-Bit Vector Quantization of KV Cache

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra; Systems; 优先级：核心
- [来源](#)

论文：NSNQuant: A Double Normalization Approach for Calibration-Free Low-Bit Vector Quantization of KV Cache

快速理解：Large Language Model (LLM) inference is typically memory-intensive, especially when processing large batch sizes and long sequences, due to the large size of key-value (KV) cache. Vector Quantization (VQ) is recently adopted to alleviate this issue, but we find that the existing approach is susceptible to distribution shift due to its reliance on calibration datasets.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.7.38 NavBench: Probing Multimodal Large Language Models for Embodied Navigation

- 作者：作者信息未提供
- 年份：2025；主题：Embodied/WBC; Supporting Perception; 优先级：核心
- [来源](#)

论文：NavBench: Probing Multimodal Large Language Models for Embodied Navigation

快速理解：Multimodal Large Language Models (MLLMs) have demonstrated strong generalization in vision-language tasks, yet their ability to understand and act within embodied environments remains underexplored. We present NavBench, a benchmark to evaluate the embodied navigation capabilities of MLLMs under zero-shot settings.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.39 PRIME: Preference-based Reinforcement Learning with Multimodal Feedback and Trajectory Synthesis from Foundation Models

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL; Supporting Perception; 优先级：核心
- [来源](#)

论文：PRIME: Preference-based Reinforcement Learning with Multimodal Feedback and Trajectory Synthesis from Foundation Models

快速理解：Preference-based reinforcement learning (PbRL) has emerged as a promising paradigm for teaching robots complex behaviors without reward engineering. However, its effectiveness is often

limited by two critical challenges: the reliance on extensive human input and the inherent difficulties in resolving query ambiguity and credit assignment during reward learning.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.40 PeRL: Permutation-Enhanced Reinforcement Learning for Interleaved Vision-Language Reasoning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL; Supporting Perception；优先级：核心
- [来源](#)

论文：PeRL: Permutation-Enhanced Reinforcement Learning for Interleaved Vision-Language Reasoning

快速理解：Inspired by the impressive reasoning capabilities demonstrated by reinforcement learning approaches like DeepSeek-R1, recent emerging research has begun exploring the use of reinforcement learning (RL) to enhance vision-language models (VLMs) for multimodal reasoning tasks. However, most existing multimodal reinforcement learning approaches remain limited to spatial reasoning within single-image contexts, yet still struggle to generalize to more complex and realworld scenarios involving multi-image positional reasoning, where understanding the relationships across images is crucial.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.41 Point4Bit: Post Training 4-bit Quantization for Point Cloud 3D Detection

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra; Systems；优先级：核心
- [来源](#)

论文：Point4Bit: Post Training 4-bit Quantization for Point Cloud 3D Detection

快速理解：Voxel-based 3D object detectors have achieved remarkable performance in point cloud perception, yet their high computational and memory demands pose significant challenges for deployment on resource-constrained edge devices. Posttraining quantization (PTQ) provides a practical means to compress models and accelerate inference; however, existing PTQ methods for point cloud detection are typically limited to INT8 and lack support for lower-bit formats such as INT4, which restricts their deployment potential.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.7.42 PointMapPolicy: Structured Point Cloud Processing for Multi-Modal Imitation Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL; Systems；优先级：核心
- [来源](#)

论文: PointMapPolicy: Structured Point Cloud Processing for Multi-Modal Imitation Learning

快速理解: Robotic manipulation systems benefit from complementary sensing modalities, where each provides unique environmental information. Point clouds capture detailed geometric structure, while RGB images provide rich semantic context.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.7.43 PolarQuant: Leveraging Polar Transformation for Key Cache Quantization and Decoding Acceleration

- 作者: 作者信息未提供
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: PolarQuant: Leveraging Polar Transformation for Key Cache Quantization and Decoding Acceleration

快速理解: The increasing demand for long-context generation has made the KV cache in large language models a bottleneck in memory consumption. Quantizing the cache to lower bit widths is an effective way to reduce memory costs; however, previous methods struggle with key cache quantization due to outliers, resulting in suboptimal performance.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.7.44 Provable Ordering and Continuity in Vision-Language Pretraining for Generalizable Embodied Agents

- 作者: 作者信息未提供
- 年份: 2025; 主题: Embodied/WBC; Supporting Perception; 优先级: 核心
- [来源](#)

论文: Provable Ordering and Continuity in Vision-Language Pretraining for Generalizable Embodied Agents

快速理解: Pre-training vision-language representations on human action videos has emerged as a promising approach to reduce reliance on large-scale expert demonstrations for training embodied agents. However, prior methods often employ time contrastive learning based on goal-reaching heuristics, progressively aligning language instructions from the initial to the final frame.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.45 QSCA: Quantization with Self-Compensating Auxiliary for Monocular Depth Estimation

- 作者: 作者信息未提供
- 年份: 2025; 主题: AI Infra; Supporting Perception; 优先级: 核心
- [来源](#)

论文: QSCA: Quantization with Self-Compensating Auxiliary for Monocular Depth Estimation

快速理解: Monocular depth estimation has advanced significantly with foundation models like Depth Anything, leveraging large-scale transformer architectures for the superior generalization. However, the deployment on resource-constrained devices remains challenging due to the high computation and memory requirement.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.46 REASONING COMPILER: LLM-Guided Optimizations for Efficient Model Serving

- 作者: 作者信息未提供
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: REASONING COMPILER: LLM-Guided Optimizations for Efficient Model Serving

快速理解: While model serving has unlocked unprecedented capabilities, the high cost of serving large-scale models continues to be a significant barrier to widespread accessibility and rapid innovation. Compiler optimizations have long driven substantial performance improvements, but existing compilers struggle with neural workloads due to the exponentially large and highly interdependent space of possible transformations.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.7.47 RLGF: Reinforcement Learning with Geometric Feedback for Autonomous Driving Video Generation

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; Supporting Perception; 优先级: 核心
- [来源](#)

论文: RLGF: Reinforcement Learning with Geometric Feedback for Autonomous Driving Video Generation

快速理解: Synthetic data is crucial for advancing autonomous driving (AD) systems, yet current state-of-the-art video generation models, despite their visual realism, suffer from subtle geometric distortions that limit their utility for downstream perception tasks. We identify and quantify this critical issue, demonstrating a significant performance gap in 3D object detection when using synthetic versus real data.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.48 RLVR-World: Training World Models with Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: World Model; Agent RL; 优先级: 核心
- [来源](#)

论文: RLVR-World: Training World Models with Reinforcement Learning

快速理解: World models predict state transitions in response to actions and are increasingly developed across diverse modalities. However, standard training objectives such as maximum likelihood estimation (MLE) often misalign with task-specific goals of world models, i.e., transition prediction metrics like accuracy or perceptual quality.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策; 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.49 ROVER: Recursive Reasoning Over Videos with Vision-Language Models for Embodied Tasks

- 作者: 作者信息未提供
- 年份: 2025; 主题: Embodied/WBC; Supporting Perception; 优先级: 核心
- [来源](#)

论文: ROVER: Recursive Reasoning Over Videos with Vision-Language Models for Embodied Tasks

快速理解: Vision-language models (VLMs) have exhibited impressive capabilities across diverse image understanding tasks, but still struggle in settings that require reasoning over extended sequences of camera frames from a video. This limits their utility in embodied settings, which require reasoning over long frame sequences from a continuous stream of visual input at each moment of a task attempt.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.50 Raw2Drive: Reinforcement Learning with Aligned World Models for End-to-End Autonomous Driving (in CARLA v2)

- 作者: 作者信息未提供
- 年份: 2025; 主题: World Model; Agent RL; 优先级: 核心
- [来源](#)

论文: Raw2Drive: Reinforcement Learning with Aligned World Models for End-to-End Autonomous Driving (in CARLA v2)

快速理解: Reinforcement Learning (RL) can mitigate the causal confusion and distribution shift inherent in imitation learning (IL). However, applying RL to end-to-end autonomous driving (E2E-AD) remains an open problem for its training difficulty, and IL is still the mainstream paradigm in both academia and industry.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策; 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.51 Reinforcement learning for one-shot DAG scheduling with comparability identification and dense reward

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; Systems; 优先级: 核心
- [来源](#)

论文: Reinforcement learning for one-shot DAG scheduling with comparability identification and

dense reward

快速理解: In recent years, many studies proposed to generate solutions for Directed Acyclic Graph (DAG) scheduling problem in one shot by combining reinforcement learning and list scheduling heuristic. However, these existing methods suffer from biased estimation of sampling probabilities and inefficient guidance in training, due to redundant comparisons among node priorities and the sparse reward challenge.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.7.52 Revisiting Multi-Agent World Modeling from a Diffusion-Inspired Perspective

- 作者: 作者信息未提供
- 年份: 2025; 主题: World Model; Agent RL; 优先级: 核心
- [来源](#)

论文: Revisiting Multi-Agent World Modeling from a Diffusion-Inspired Perspective

快速理解: World models have recently attracted growing interest in Multi-Agent Reinforcement Learning (MARL) due to their ability to improve sample efficiency for policy learning. However, accurately modeling environments in MARL is challenging due to the exponentially large joint action space and highly uncertain dynamics inherent in multi-agent systems.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策; 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.53 RoboScape: Physics-informed Embodied World Model

- 作者: 作者信息未提供
- 年份: 2025; 主题: Embodied/WBC; World Model; 优先级: 核心
- [来源](#)

论文: RoboScape: Physics-informed Embodied World Model

快速理解: World models have become indispensable tools for embodied intelligence, serving as powerful simulators capable of generating realistic robotic videos while addressing critical data scarcity challenges. However, current embodied world models exhibit limited physical awareness, particularly in modeling 3D geometry and motion dynamics, resulting in unrealistic video generation for contact-rich robotic scenarios.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 环境动力学、视频预测、潜变量状态或模型式决策。

证据层级: 本地 PDF 前页。

1.3.7.54 Robot-R1: Reinforcement Learning for Enhanced Embodied Reasoning in Robotics

- 作者: 作者信息未提供
- 年份: 2025; 主题: Embodied/WBC; Agent RL; 优先级: 核心
- [来源](#)

论文: Robot-R1: Reinforcement Learning for Enhanced Embodied Reasoning in Robotics

快速理解: Large Vision-Language Models (LVLMs) have recently shown great promise in advancing robotics by combining embodied reasoning with robot control. A common approach involves training on embodied reasoning tasks related to robot control using Supervised Fine-Tuning (SFT).

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.55 SAFE: Multitask Failure Detection for Vision-Language-Action Models

- 作者: 作者信息未提供
- 年份: 2025; 主题: VLA; Supporting Perception; 优先级: 核心
- [来源](#)

论文: SAFE: Multitask Failure Detection for Vision-Language-Action Models

快速理解: While vision-language-action models (VLAs) have shown promising robotic behaviors across a diverse set of manipulation tasks, they achieve limited success rates when deployed on novel tasks out of the box. To allow these policies to safely interact with their environments, we need a failure detector that gives a timely alert such that the robot can stop, backtrack, or ask for help.

关注点: 视觉、语言条件与机器人动作策略之间的对齐; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.56 SAM-R1: Leveraging SAM for Reward Feedback in Multimodal Segmentation via Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; Supporting Perception; 优先级: 核心
- [来源](#)

论文: SAM-R1: Leveraging SAM for Reward Feedback in Multimodal Segmentation via Reinforcement Learning

快速理解: Leveraging multimodal large models for image segmentation has become a prominent research direction. However, existing approaches typically rely heavily on manually annotated datasets that include explicit reasoning processes, which are costly and time-consuming to produce.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.57 SRPO: Enhancing Multimodal LLM Reasoning via Reflection-Aware Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; Supporting Perception; 优先级: 核心
- [来源](#)

论文: SRPO: Enhancing Multimodal LLM Reasoning via Reflection-Aware Reinforcement Learning

快速理解: Multimodal large language models (MLLMs) have shown promising capabilities in reasoning tasks, yet still struggle significantly with complex problems requiring explicit self-reflection and self-correction, especially compared to their unimodal text-based counterparts. Existing reflection methods are simplistic and struggle to generate meaningful, instructive feedback, as the reasoning

ability and knowledge limits of pre-trained models are largely fixed during initial training.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.58 STRATUS: A Multi-agent System for Autonomous Reliability Engineering of Modern Clouds

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL; Systems；优先级：核心
- [来源](#)

论文：STRATUS: A Multi-agent System for Autonomous Reliability Engineering of Modern Clouds

快速理解：In cloud-scale systems, failures are the norm. A distributed computing cluster exhibits hundreds of machine failures and thousands of disk failures; software bugs and misconfigurations are reported to be more frequent.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.7.59 SafeVLA: Towards Safety Alignment of Vision-Language-Action Model via Constrained Learning

- 作者：作者信息未提供
- 年份：2025；主题：VLA; Supporting Perception；优先级：核心
- [来源](#)

论文：SafeVLA: Towards Safety Alignment of Vision-Language-Action Model via Constrained Learning

快速理解：Vision-language-action models (VLAs) show potential as generalist robot policies. However, these models pose extreme safety challenges during real-world deployment, including the risk of harm to the environment, the robot itself, and humans.

关注点：视觉、语言条件与机器人动作策略之间的对齐；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.60 Scaffolding Dexterous Manipulation with Vision-Language Models

- 作者：作者信息未提供
- 年份：2025；主题：Embodied/WBC; Supporting Perception；优先级：核心
- [来源](#)

论文：Scaffolding Dexterous Manipulation with Vision-Language Models

快速理解：Dexterous robotic hands are essential for performing complex manipulation tasks, yet remain difficult to train due to the challenges of demonstration collection and high-dimensional control. While reinforcement learning (RL) can alleviate the data bottleneck by generating experience in simulation, it typically relies on carefully designed, task-specific reward functions, which hinder scalability and generalization.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.61 Semi-off-Policy Reinforcement Learning for Vision-Language Slow-Thinking Reasoning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL; Supporting Perception；优先级：核心
- [来源](#)

论文：Semi-off-Policy Reinforcement Learning for Vision-Language Slow-Thinking Reasoning

快速理解：Enhancing large vision-language models (LVLMs) with visual slow-thinking reasoning is crucial for solving complex multimodal tasks. However, since LVLMs are mainly trained with vision-language alignment, it is difficult to adopt on-policy reinforcement learning (RL) to develop the slow thinking ability because the rollout space is restricted by its initial abilities.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.62 SmallKV: Small Model Assisted Compensation of KV Cache Compression for Efficient LLM Inference

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra; Systems；优先级：核心
- [来源](#)

论文：SmallKV: Small Model Assisted Compensation of KV Cache Compression for Efficient LLM Inference

快速理解：KV cache eviction has emerged as an effective solution to alleviate resource constraints faced by LLMs in long-context scenarios. However, existing token-level eviction methods often overlook two critical aspects: (1) their irreversible eviction strategy fails to adapt to dynamic attention patterns during decoding (the saliency shift problem), and (2) they treat both marginally important tokens and truly unimportant tokens equally, despite the collective significance of marginal tokens to model performance (the marginal information over-compression problem).

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.7.63 SmartCache: Context-aware Semantic Cache for Efficient Multi-turn LLM Inference

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra; Systems；优先级：核心
- [来源](#)

论文：SmartCache: Context-aware Semantic Cache for Efficient Multi-turn LLM Inference

快速理解：Large Language Models (LLMs) for multi-turn conversations suffer from inefficiency: semantically similar queries across different user sessions trigger redundant computation and duplicate memory-intensive Key-Value (KV) caches. Existing optimizations such as prefix caching overlook semantic similarities, while typical semantic caches either ignore conversational context or are not

integrated with low-level KV cache management.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.7.64 Taccel: Scaling Up Vision-based Tactile Robotics via High-performance GPU Simulation

- 作者：作者信息未提供
- 年份：2025；主题：Robot Learning; AI Infra; 优先级：核心
- [来源](#)

论文：Taccel: Scaling Up Vision-based Tactile Robotics via High-performance GPU Simulation

快速理解：Tactile sensing is crucial for achieving human-level robotic capabilities in manipulation tasks [54]. As a promising solution, Vision-based tactile sensors (VBTs) [61, 37] offer high spatial resolution and cost-effectiveness, but present unique challenges in robotics for their complex physical characteristics and visual signal processing requirements.

关注点：机器人数据、策略学习、操作、触觉或 sim-to-real；训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.7.65 Think or Not? Selective Reasoning via Reinforcement Learning for Vision-Language Models

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL; Supporting Perception; 优先级：核心
- [来源](#)

论文：Think or Not? Selective Reasoning via Reinforcement Learning for Vision-Language Models

快速理解：Reinforcement Learning (RL) has proven to be an effective post-training strategy for enhancing reasoning in vision-language models (VLMs). Group Relative Policy Optimization (GRPO) is a recent prominent method that encourages models to generate complete reasoning traces before answering, leading to increased token usage and computational cost.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.66 ThinkAct: Vision-Language-Action Reasoning via Reinforced Visual Latent Planning

- 作者：作者信息未提供
- 年份：2025；主题：VLA; Supporting Perception; 优先级：核心
- [来源](#)

论文：ThinkAct: Vision-Language-Action Reasoning via Reinforced Visual Latent Planning

快速理解：Vision-language-action (VLA) reasoning tasks require agents to interpret multimodal instructions, perform long-horizon planning, and act adaptively in dynamic environments. Existing approaches typically train VLA models in an end-to-end fashion, directly mapping inputs to actions without explicit reasoning, which hinders their ability to plan over multiple steps or adapt to complex task variations.

关注点：视觉、语言条件与机器人动作策略之间的对齐；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.67 Time Reversal Symmetry for Efficient Robotic Manipulations in Deep Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Embodied/WBC；Agent RL；优先级：核心
- [来源](#)

论文：Time Reversal Symmetry for Efficient Robotic Manipulations in Deep Reinforcement Learning

快速理解：Symmetry is pervasive in robotics and has been widely exploited to improve sample efficiency in deep reinforcement learning (DRL). However, existing approaches primarily focus on spatial symmetries—such as reflection, rotation, and translation—while largely neglecting temporal symmetries.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.68 Towards Generalizable Multi-Policy Optimization with Self-Evolution for Job Scheduling

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；Systems；优先级：核心
- [来源](#)

论文：Towards Generalizable Multi-Policy Optimization with Self-Evolution for Job Scheduling

快速理解：Reinforcement Learning (RL) has shown promising results in solving Job Scheduling Problems (JSPs), automatically deriving powerful dispatching rules from data without relying on expert knowledge. However, most RL-based methods train only a single decision-maker, which limits exploration capability and leaves significant room for performance improvement.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.7.69 Towards Unified Multimodal Interleaved Generation via Group Relative Policy Optimization

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；Supporting Perception；优先级：核心
- [来源](#)

论文：Towards Unified Multimodal Interleaved Generation via Group Relative Policy Optimization

快速理解：Unified vision-language models have made significant progress in multimodal understanding and generation, yet they largely fall short in producing multimodal interleaved outputs, which is a crucial capability for tasks like visual storytelling and step-by-step visual reasoning. In this work, we propose a reinforcement learningbased post-training strategy to unlock this capability in existing unified models, without relying on large-scale multimodal interleaved datasets.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.70 Unified Multimodal Chain-of-Thought Reward Model through Reinforcement Fine-Tuning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL; Supporting Perception；优先级：核心
- [来源](#)

论文：Unified Multimodal Chain-of-Thought Reward Model through Reinforcement Fine-Tuning

快速理解：Recent advances in multimodal Reward Models (RMs) have shown significant promise in delivering reward signals to align vision models with human preferences. However, current RMs are generally restricted to providing direct responses or engaging in shallow reasoning processes with limited depth, often leading to inaccurate reward signals.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.71 Unified Reinforcement and Imitation Learning for Vision-Language Models

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL; Supporting Perception；优先级：核心
- [来源](#)

论文：Unified Reinforcement and Imitation Learning for Vision-Language Models

快速理解：Vision-Language Models (VLMs) have achieved remarkable progress, yet their large scale often renders them impractical for resource-constrained environments. This paper introduces Unified Reinforcement and Imitation Learning (RIL), a novel and efficient training algorithm designed to create powerful, lightweight VLMs.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.72 Universal Visuo-Tactile Video Understanding for Embodied Interaction

- 作者：作者信息未提供
- 年份：2025；主题：Embodied/WBC; Robot Learning；优先级：核心
- [来源](#)

论文：Universal Visuo-Tactile Video Understanding for Embodied Interaction

快速理解：Tactile perception is essential for embodied agents to understand physical attributes of objects that cannot be determined through visual inspection alone. While existing approaches have made progress in visual and language modalities for physical understanding, they fail to effectively incorporate tactile information that provides crucial haptic feedback for real-world interaction.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：本地 PDF 前页。

1.3.7.73 Unlocking Multimodal Mathematical Reasoning via Process Reward Model

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL; Supporting Perception；优先级：核心
- [来源](#)

论文：Unlocking Multimodal Mathematical Reasoning via Process Reward Model

快速理解：Process Reward Models (PRMs) have shown promise in enhancing the mathematical reasoning capabilities of Large Language Models (LLMs) through Test-Time Scaling (TTS). However, their integration into multimodal reasoning remains largely unexplored.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.74 VL-Rethinker: Incentivizing Self-Reflection of Vision-Language Models with Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL; Supporting Perception；优先级：核心
- [来源](#)

论文：VL-Rethinker: Incentivizing Self-Reflection of Vision-Language Models with Reinforcement Learning

快速理解：Recently, slow-thinking systems like GPT-o1 and DeepSeek-R1 have demonstrated great potential in solving challenging problems through explicit reflection. They significantly outperform the best fast-thinking models, such as GPT-4o, on various math and science benchmarks.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.75 VLA-OS: Structuring and Dissecting Planning Representations and Paradigms in Vision-Language-Action Models

- 作者：作者信息未提供
- 年份：2025；主题：VLA; Supporting Perception；优先级：核心
- [来源](#)

论文：VLA-OS: Structuring and Dissecting Planning Representations and Paradigms in Vision-Language-Action Models

快速理解：Recent studies on Vision-Language-Action (VLA) models have shifted from the end-to-end action-generation paradigm toward a pipeline involving task planning followed by action generation, demonstrating improved performance on various complex, long-horizon manipulation tasks. However, existing approaches vary significantly in terms of network architectures, planning paradigms, representations, and training data sources, making it challenging for researchers to identify the precise sources of performance gains and components to be further improved.

关注点：视觉、语言条件与机器人动作策略之间的对齐；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.76 Vector Quantization in the Brain: Grid-like Codes in World Models

- 作者：作者信息未提供

- 年份: 2025; 主题: World Model; AI Infra; 优先级: 核心
- [来源](#)

论文: Vector Quantization in the Brain: Grid-like Codes in World Models

快速理解: We propose Grid-like Code Quantization (GCQ), a brain-inspired method for compressing observation-action sequences into discrete representations using grid-like patterns in attractor dynamics. Unlike conventional vector quantization approaches that operate on static inputs, GCQ performs spatiotemporal compression through an action-conditioned codebook, where codewords are derived from continuous attractor neural networks and dynamically selected based on actions.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策; 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 本地 PDF 前页。

1.3.7.77 VisionThink: Smart and Efficient Vision Language Model via Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; Supporting Perception; 优先级: 核心
- [来源](#)

论文: VisionThink: Smart and Efficient Vision Language Model via Reinforcement Learning

快速理解: Recent advancements in vision-language models (VLMs) have improved performance by increasing the number of visual tokens, which are often significantly longer than text tokens. However, we observe that most real-world scenarios do not require such an extensive number of visual tokens.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.78 Whole-Body Conditioned Egocentric Video Prediction

- 作者: 作者信息未提供
- 年份: 2025; 主题: Embodied/WBC; World Model; 优先级: 核心
- [来源](#)

论文: Whole-Body Conditioned Egocentric Video Prediction

快速理解: We train models to Predict Ego-centric Video from human Actions (PEVA), given the past video and an action represented by the relative 3D body pose. By conditioning on kinematic pose trajectories, structured by the joint hierarchy of the body, our model learns to simulate how physical human actions shape the environment from a first-person point of view.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 环境动力学、视频预测、潜变量状态或模型式决策。

证据层级: 本地 PDF 前页。

1.3.7.79 ΔEnergy : Optimizing Energy Change During Vision-Language Alignment Improves both OOD Detection and OOD Generalization

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: ΔEnergy : Optimizing Energy Change During Vision-Language Alignment Improves both OOD Detection and OOD Generalization

快速理解: Recent approaches for vision-language models (VLMs) have shown remarkable success in achieving fast downstream adaptation. When applied to real-world downstream tasks, VLMs inevitably encounter both the in-distribution (ID) data and out-of-distribution (OOD) data.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.80 Hyper-GoalNet : Goal-Conditioned Manipulation Policy Learning with HyperNetworks

- 作者: 作者信息未提供
- 年份: 2025; 主题: Robot Learning; 优先级: 重要
- [来源](#)

论文: Hyper-GoalNet : Goal-Conditioned Manipulation Policy Learning with HyperNetworks

快速理解: Goal-conditioned policy learning for robotic manipulation presents significant challenges in maintaining performance across diverse objectives and environments. We introduce Hyper-GoalNet, a framework that generates task-specific policy network parameters from goal specifications using hypernetworks.

关注点: 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 本地 PDF 前页。

1.3.7.81 $G1$: Teaching LLMs to Reason on Graphs with Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: $G1$: Teaching LLMs to Reason on Graphs with Reinforcement Learning

快速理解: Although Large Language Models (LLMs) have demonstrated remarkable progress, their proficiency in graph-related tasks remains notably limited, hindering the development of truly general-purpose models. Previous attempts, including pretraining graph foundation models or employing supervised fine-tuning, often face challenges such as the scarcity of large-scale, universally represented graph data.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.82 3D Equivariant Visuomotor Policy Learning via Spherical Projection

- 作者: 作者信息未提供
- 年份: 2025; 主题: Robot Learning; 优先级: 重要
- [来源](#)

论文: 3D Equivariant Visuomotor Policy Learning via Spherical Projection

快速理解: Equivariant models have recently been shown to improve the data efficiency of diffusion policy by a significant margin. However, prior work that explored this direction focused primarily on point cloud inputs generated by multiple cameras fixed in the workspace.

关注点: 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 本地 PDF 前页。

1.3.7.83 3D Visual Illusion Depth Estimation

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: 3D Visual Illusion Depth Estimation

快速理解: 3D visual illusion is a perceptual phenomenon where a two-dimensional plane is manipulated to simulate three-dimensional spatial relationships, making a flat artwork or object look three-dimensional in the human visual system. In this paper, we reveal that the machine visual system is also seriously fooled by 3D visual illusions, including monocular and binocular depth estimation.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.84 3D-Agent: A Tri-Modal Multi-Agent Responsive Framework for Comprehensive 3D Object Annotation

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: 3D-Agent: A Tri-Modal Multi-Agent Responsive Framework for Comprehensive 3D Object Annotation

快速理解: Driven by the applications in autonomous driving, robotics, and augmented reality, 3D object annotation is a critical task compared to 2D annotation, such as spatial complexity, occlusion, and viewpoint inconsistency. The existing methods relying on single models often struggle with these issues.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.85 3DLLM-Mem: Long-Term Spatial-Temporal Memory for Embodied 3D Large Language Model

- 作者: 作者信息未提供
- 年份: 2025; 主题: Embodied/WBC; 优先级: 重要
- [来源](#)

论文: 3DLLM-Mem: Long-Term Spatial-Temporal Memory for Embodied 3D Large Language Model

快速理解: Humans excel at performing complex tasks by leveraging long-term memory across temporal and spatial experiences. In contrast, current Large Language Models (LLMs) struggle to effectively plan and act in dynamic, multi-room 3D environments.

关注点：具身任务中的感知、动作生成、全身协调或运动控制。

证据层级：本地 PDF 前页。

1.3.7.86 3DRS: MLLMs Need 3D-Aware Representation Supervision for Scene Understanding

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：3DRS: MLLMs Need 3D-Aware Representation Supervision for Scene Understanding

快速理解：Recent advances in scene understanding have leveraged multimodal large language models (MLLMs) for 3D reasoning by capitalizing on their strong 2D pretraining. However, the lack of explicit 3D data during MLLM pretraining limits 3D representation capability.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.87 4KAgent: Agentic Any Image to 4K Super-Resolution

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：4KAgent: Agentic Any Image to 4K Super-Resolution

快速理解：We present 4KAgent, a unified agentic super-resolution generalist system designed to universally upscale any image to 4K resolution (and even higher, if applied iteratively). Our system can transform images from extremely low resolutions with severe degradations, for example, highly distorted inputs at 256×256 , into crystal-clear, photorealistic 4K outputs.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.88 70% Size, 100% Accuracy: Lossless LLM Compression for Efficient GPU Inference via Dynamic-Length Float (DFloat11)

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：70% Size, 100% Accuracy: Lossless LLM Compression for Efficient GPU Inference via Dynamic-Length Float (DFloat11)

快速理解：Large-scale AI models, such as Large Language Models (LLMs) and Diffusion Models (DMs), have grown rapidly in size, creating significant challenges for efficient deployment on resource-constrained hardware. In this paper, we introduce Dynamic-Length Float (DFloat11), a lossless compression framework that reduces LLM and DM size by 30% while preserving outputs that are bit-for-bit identical to the original model.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.7.89 A Bayesian Fast-Slow Framework to Mitigate Interference in Non-Stationary Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：A Bayesian Fast-Slow Framework to Mitigate Interference in Non-Stationary Reinforcement Learning

快速理解：A Bayesian Fast-Slow Framework to Mitigate Interference in Non-Stationary Reinforcement Learning Yihuan Mao Institute for Interdisciplinary Information Sciences Tsinghua University maoyh1024@gmail.com Chongjie Zhang Department of Computer Science & Engineering Washington University in St. Louis chongjie@wustl.edu Abstract Given the ever-changing nature of the world and its inhabitants, agents must possess the ability to adapt and evolve over time.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.90 A Clean Slate for Offline Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：A Clean Slate for Offline Reinforcement Learning

快速理解：Progress in offline reinforcement learning (RL) has been impeded by ambiguous problem definitions and entangled algorithmic designs, resulting in inconsistent implementations, insufficient ablations, and unfair evaluations. Although offline RL explicitly avoids environment interaction, prior methods frequently employ extensive, undocumented online evaluation for hyperparameter tuning, complicating method comparisons.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.91 A Differential and Pointwise Control Approach to Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：A Differential and Pointwise Control Approach to Reinforcement Learning

快速理解：Reinforcement learning (RL) in continuous state-action spaces remains challenging in scientific computing due to poor sample efficiency and lack of pathwise physical consistency. We introduce Differential Reinforcement Learning (Differential RL), a novel framework that reformulates RL from a continuous-time control perspective via a differential dual formulation.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.92 A Multimodal BiMamba Network with Test-Time Adaptation for Emotion Recognition Based on Physiological Signals

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：A Multimodal BiMamba Network with Test-Time Adaptation for Emotion Recognition Based on Physiological Signals

快速理解：Emotion recognition based on physiological signals plays a vital role in psychological health and human-computer interaction, particularly with the substantial advances in multimodal emotion recognition techniques. However, two key challenges remain unresolved: 1) how to effectively model the intra-modal long-range dependencies and inter-modal correlations in multimodal physiological emotion signals, and 2) how to address the performance limitations resulting from missing multimodal data.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.93 A Principle of Targeted Intervention for Multi-Agent Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：A Principle of Targeted Intervention for Multi-Agent Reinforcement Learning

快速理解：Steering cooperative multi-agent reinforcement learning (MARL) towards desired outcomes is challenging, particularly when the global guidance from a human on the whole multi-agent system is impractical in a large-scale MARL. On the other hand, designing external mechanisms (e.g., intrinsic rewards and human feedback) to coordinate agents mostly relies on empirical studies, lacking a easy-to-use research tool.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.94 A Provable Approach for End-to-End Safe Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：A Provable Approach for End-to-End Safe Reinforcement Learning

快速理解：A longstanding goal in safe reinforcement learning (RL) is a method to ensure the safety of a policy throughout the entire process, from learning to operation. However, existing safe RL paradigms inherently struggle to achieve this objective.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.95 A Reinforcement Learning-based Bidding Strategy for Data Consumers in Auction-based Federated Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：A Reinforcement Learning-based Bidding Strategy for Data Consumers in Auction-based Federated Learning

快速理解：Auction-based Federated Learning (AFL) fosters collaboration among selfinterested data consumers (DCs) and data owners (DOs). A major challenge in AFL pertains to how DCs select and bid for DOs.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.96 A Snapshot of Influence: A Local Data Attribution Framework for Online Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：A Snapshot of Influence: A Local Data Attribution Framework for Online Reinforcement Learning

快速理解：Online reinforcement learning (RL) excels in complex, safety-critical domains but suffers from sample inefficiency, training instability, and limited interpretability. Data attribution provides a principled way to trace model behavior back to training samples, yet existing methods assume fixed datasets, which is violated in online RL where each experience both updates the policy and shapes future data collection.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.97 A Stable Whitening Optimizer for Efficient Neural Network Training

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：A Stable Whitening Optimizer for Efficient Neural Network Training

快速理解：In this work, we take an experimentally grounded look at neural network optimization. Building on the Shampoo family of algorithms, we identify and alleviate three key issues, resulting in the proposed SPlus method.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.7.98 A TRIANGLE Enables Multimodal Alignment Beyond Cosine Similarity

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：A TRIANGLE Enables Multimodal Alignment Beyond Cosine Similarity

快速理解：Multimodal learning plays a pivotal role in advancing artificial intelligence systems by incorporating information from multiple modalities to build a more comprehensive representation. Despite its importance, current state-of-the-art models still suffer from severe limitations that prevent the successful development of a fully multimodal model.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.99 A-Mem: Agentic Memory for LLM Agents

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：A-Mem: Agentic Memory for LLM Agents

快速理解：While large language model (LLM) agents can effectively use external tools for complex real-world tasks, they require memory systems to leverage historical experiences. Current memory systems enable basic storage and retrieval but lack sophisticated memory organization, despite recent attempts to incorporate graph databases.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.100 ACT as Human: Multimodal Large Language Model Data Annotation with Critical Thinking

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：ACT as Human: Multimodal Large Language Model Data Annotation with Critical Thinking

快速理解：Supervised learning relies on high-quality labeled data, but obtaining such data through human annotation is both expensive and time-consuming. Recent work explores using large language models (LLMs) for annotation, but LLM-generated labels still fall short of human-level quality.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.101 ADG: Ambient Diffusion-Guided Dataset Recovery for Corruption-Robust Offline Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文: ADG: Ambient Diffusion-Guided Dataset Recovery for Corruption-Robust Offline Reinforcement Learning

快速理解: Real-world datasets collected from sensors or human inputs are prone to noise and errors, posing significant challenges for applying offline reinforcement learning (RL). While existing methods have made progress in addressing corrupted actions and rewards, they remain insufficient for handling corruption in high-dimensional state spaces and for cases where multiple elements in the dataset are corrupted simultaneously.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.102 ADMN: A Layer-Wise Adaptive Multimodal Network for Dynamic Input Noise and Compute Resources

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: ADMN: A Layer-Wise Adaptive Multimodal Network for Dynamic Input Noise and Compute Resources

快速理解: Multimodal deep learning systems are deployed in dynamic scenarios due to the robustness afforded by multiple sensing modalities. Nevertheless, they struggle with varying compute resource availability (due to multi-tenancy, device heterogeneity, etc.) and fluctuating quality of inputs (from sensor feed corruption, environmental noise, etc.).

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.103 AOR: Anatomical Ontology-Guided Reasoning for Medical Large Multimodal Model in Chest X-Ray Interpretation

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: AOR: Anatomical Ontology-Guided Reasoning for Medical Large Multimodal Model in Chest X-Ray Interpretation

快速理解: Chest X-rays (CXRs) are the most frequently performed imaging examinations in clinical settings. Recent advancements in Medical Large Multimodal Models (MLMMs) have enabled automated CXR interpretation, improving diagnostic accuracy and efficiency.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.104 APML: Adaptive Probabilistic Matching Loss for Robust 3D Point Cloud Reconstruction

- 作者: 作者信息未提供
- 年份: 2025; 主题: Systems; 优先级: 重要

- [来源](#)

论文: APML: Adaptive Probabilistic Matching Loss for Robust 3D Point Cloud Reconstruction

快速理解: Training deep learning models for point cloud prediction tasks such as shape completion and generation depends critically on loss functions that measure discrepancies between predicted and ground-truth point sets. Commonly used functions such as Chamfer Distance (CD), HyperCD, InfoCD and Density-aware CD rely on nearest-neighbor assignments, which often induce many-to-one correspondences, leading to point congestion in dense regions and poor coverage in sparse regions.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.7.105 AREAL: A Large-Scale Asynchronous Reinforcement Learning System for Language Reasoning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: AREAL: A Large-Scale Asynchronous Reinforcement Learning System for Language Reasoning

快速理解: Reinforcement learning (RL) has become a trending paradigm for training large language models (LLMs), particularly for reasoning tasks. Effective RL for LLMs requires massive parallelization and poses an urgent need for efficient training systems.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.106 ASDSV: Multimodal Generation Made Efficient with Approximate Speculative Diffusion and Speculative Verification

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: ASDSV: Multimodal Generation Made Efficient with Approximate Speculative Diffusion and Speculative Verification

快速理解: Diffusion in transformer is central to advances in high-quality multimodal generation but suffer from high inference latency due to their iterative nature. Inspired by speculative decoding's success in accelerating large language models, we propose Approximate Speculative Diffusion with Speculative Verification (ASDSV), a novel method to enhance the efficiency of diffusion models.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.107 Accelerating Multimodal Large Language Models via Dynamic Visual-Token Exit and the Empirical Findings

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要

- [来源](#)

论文: Accelerating Multimodal Large Language Models via Dynamic Visual-Token Exit and the Empirical Findings

快速理解: In this paper, we study the visual redundancy problem of multimodal large language models (MLLMs) from the perspective of attention behaviors. Via extensive empirical experiments, we observe and conclude three main inference stages of MLLMs: (i) Early fusion between tokens is first accomplished quickly. (ii) Intra-modality modeling then comes to play. (iii) Multimodal reasoning resumes and lasts until the end of inference.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.108 Accelerating Parallel Diffusion Model Serving with Residual Compression

- 作者: 作者信息未提供
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Accelerating Parallel Diffusion Model Serving with Residual Compression

快速理解: Diffusion models produce realistic images and videos but require substantial computational resources, necessitating multi-accelerator parallelism for real-time deployment. However, parallel inference introduces significant communication overhead from exchanging large activations between devices, limiting efficiency and scalability.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 本地 PDF 前页。

1.3.7.109 AceReason-Nemotron: Advancing Math and Code Reasoning through Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: AceReason-Nemotron: Advancing Math and Code Reasoning through Reinforcement Learning

快速理解: Despite recent progress in large-scale reinforcement learning (RL) for reasoning, the training recipe for building high-performing reasoning models remains elusive. Key implementation details of frontier models, such as DeepSeek-R1, including data curation strategies and RL training recipe, are often omitted.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.110 Act Only When It Pays: Efficient Reinforcement Learning for LLM Reasoning via Selective Rollouts

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Act Only When It Pays: Efficient Reinforcement Learning for LLM Reasoning via Selective Rollouts

快速理解: Reinforcement learning, such as PPO and GRPO, has powered recent breakthroughs in LLM reasoning. Scaling rollout to sample more prompts enables models to selectively use higher-quality data for training, which can stabilize RL training and improve model performance, but at the cost of significant computational overhead.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.111 Actial: Activate Spatial Reasoning Ability of Multimodal Large Language Models

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Actial: Activate Spatial Reasoning Ability of Multimodal Large Language Models

快速理解: Recent advances in Multimodal Large Language Models (MLLMs) have significantly improved 2D visual understanding, prompting interest in their application to complex 3D reasoning tasks. However, it remains unclear whether these models can effectively capture the detailed spatial information required for robust real-world performance, especially cross-view consistency, a key requirement for accurate 3D reasoning.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.112 Active Test-time Vision-Language Navigation

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Active Test-time Vision-Language Navigation

快速理解: Vision-Language Navigation (VLN) policies trained on offline datasets often exhibit degraded task performance when deployed in unfamiliar navigation environments at test time, where agents are typically evaluated without access to external interaction or feedback. Entropy minimization has emerged as a practical solution for reducing prediction uncertainty at test time; however, it can suffer from accumulated errors, as agents may become overconfident in incorrect actions without sufficient contextual grounding.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.113 ActiveVOO: Value of Observation Guided Active Knowledge Acquisition for Open-World Embodied Lifted Regression Planning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Embodied/WBC; 优先级: 重要

- [来源](#)

论文: ActiveVOO: Value of Observation Guided Active Knowledge Acquisition for Open-World Embodied Lifted Regression Planning

快速理解: The ability to actively acquire information is essential for open-world planning under partial observability and incomplete knowledge. However, most existing embodied AI systems either assume a known object category or rely on passive perception strategies that exhaustively gather object and relational information from the environment.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制。

证据层级: 本地 PDF 前页。

1.3.7.114 Adaptive 3D Reconstruction via Diffusion Priors and Forward Curvature--Matching Likelihood Updates

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Adaptive 3D Reconstruction via Diffusion Priors and Forward Curvature-Matching Likelihood Updates

快速理解: Reconstructing high-quality point clouds from images remains challenging in computer vision. Existing generative models, particularly diffusion models, based approaches that directly learn the posterior may suffer from inflexibility—they require conditioning signals during training, support only a fixed number of input views, and need complete retraining for different measurements.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.115 Adaptive Batch-Wise Sample Scheduling for Direct Preference Optimization

- 作者: 作者信息未提供
- 年份: 2025; 主题: Systems; 优先级: 重要
- [来源](#)

论文: Adaptive Batch-Wise Sample Scheduling for Direct Preference Optimization

快速理解: Adaptive Batch-Wise Sample Scheduling for Direct Preference Optimization Zixuan Huang¹ Yikun Ban^{1*} Lean Fu² Xiaojie Li¹ Zhongxiang Dai³ Jianxin Li¹ Deqing Wang^{1*} ¹Beihang University ²Bytedance Inc ³The Chinese University of Hong Kong, Shenzhen {huang_zx, yikunb, li_xiaojie, dqwang}@buaa.edu.cn lijx@act.buaa.edu.cn fulean@bytedance.com daizhongxiang@cuhk.edu.cn Abstract Direct Preference Optimization (DPO) has emerged as an effective approach for aligning large language models (LLMs) with human preferences. However, its performance is highly dependent on the quality of the underlying human preference data.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.7.116 Adaptive Context Length Optimization with Low-Frequency Truncation for Multi-Agent Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Adaptive Context Length Optimization with Low-Frequency Truncation for Multi-Agent Reinforcement Learning

快速理解：Recently, deep multi-agent reinforcement learning (MARL) has demonstrated promising performance for solving challenging tasks, such as long-term dependencies and non-Markovian environments. Its success is partly attributed to conditioning policies on large fixed context length.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.117 Adaptive Defense against Harmful Fine-Tuning for Large Language Models via Bayesian Data Scheduler

- 作者：作者信息未提供
- 年份：2025；主题：Systems；优先级：重要
- [来源](#)

论文：Adaptive Defense against Harmful Fine-Tuning for Large Language Models via Bayesian Data Scheduler

快速理解：Harmful fine-tuning poses critical safety risks to fine-tuning-as-a-service for large language models. Existing defense strategies preemptively build robustness via attack simulation but suffer from fundamental limitations: (i) the infeasibility of extending attack simulations beyond bounded threat models due to the inherent difficulty of anticipating unknown attacks, and (ii) limited adaptability to varying attack settings, as simulation fails to capture their variability and complexity.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.7.118 Adaptive Divergence Regularized Policy Optimization for Fine-tuning Generative Models

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Adaptive Divergence Regularized Policy Optimization for Fine-tuning Generative Models

快速理解：Balancing exploration and exploitation during reinforcement learning fine-tuning of generative models presents a critical challenge, as existing approaches rely on fixed divergence regularization that creates an inherent dilemma: strong regularization preserves model capabilities but limits reward optimization, while weak regularization enables greater alignment but risks instability or reward hacking. We introduce Adaptive Divergence Regularized Policy Optimization (ADRPO), which automatically adjusts regularization strength based on advantage estimates—reducing regularization for high-value samples while applying stronger regularization to poor samples, enabling policies to navigate between exploration and aggressive exploitation according to data quality.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.119 Adaptive Neighborhood-Constrained Q Learning for Offline Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Adaptive Neighborhood-Constrained Q Learning for Offline Reinforcement Learning

快速理解：Offline reinforcement learning (RL) suffers from extrapolation errors induced by out-of-distribution (OOD) actions. To address this, offline RL algorithms typically impose constraints on action selection, which can be systematically categorized into density, support, and sample constraints.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.120 Adaptive Quantization in Generative Flow Networks for Probabilistic Sequential Prediction

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：Adaptive Quantization in Generative Flow Networks for Probabilistic Sequential Prediction

快速理解：Probabilistic time series forecasting, essential in domains like healthcare and neuroscience, requires models capable of capturing uncertainty and intricate temporal dependencies. While deep learning has advanced forecasting, generating calibrated probability distributions over continuous future values remains challenging.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.7.121 Adaptive Re-calibration Learning for Balanced Multimodal Intention Recognition

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Adaptive Re-calibration Learning for Balanced Multimodal Intention Recognition

快速理解：Multimodal Intention Recognition (MIR) plays a critical role in applications such as intelligent assistants, service robots, and autonomous systems. However, in realworld scenarios, different modalities often vary significantly in informativeness, reliability, and noise levels.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.122 Adaptive Surrogate Gradients for Sequential Reinforcement Learning in Spiking Neural Networks

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Adaptive Surrogate Gradients for Sequential Reinforcement Learning in Spiking Neural Networks

快速理解：Neuromorphic computing systems are set to revolutionize energy-constrained robotics by achieving orders-of-magnitude efficiency gains, while enabling native temporal processing. Spiking Neural Networks (SNNs) represent a promising algorithmic approach for these systems, yet their application to complex control tasks faces two critical challenges: (1) the non-differentiable nature of spiking neurons necessitates surrogate gradients with unclear optimization properties, and (2) the stateful dynamics of SNNs require training on sequences, which in reinforcement learning (RL) is hindered by limited sequence lengths during early training, preventing the network from bridging its warm-up period.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.123 AdvEDM: Fine-grained Adversarial Attack against VLM-based Embodied Agents

- 作者：作者信息未提供
- 年份：2025；主题：Embodied/WBC；优先级：重要
- [来源](#)

论文：AdvEDM: Fine-grained Adversarial Attack against VLM-based Embodied Agents

快速理解：Vision-Language Models (VLMs), with their strong reasoning and planning capabilities, are widely used in embodied decision-making (EDM) tasks in embodied agents, such as autonomous driving and robotic manipulation. Recent research has increasingly explored adversarial attacks on VLMs to reveal their vulnerabilities.

关注点：具身任务中的感知、动作生成、全身协调或运动控制。

证据层级：本地 PDF 前页。

1.3.7.124 Advanced Sign Language Video Generation with Compressed and Quantized Multi-Condition Tokenization

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Advanced Sign Language Video Generation with Compressed and Quantized Multi-Condition Tokenization

快速理解：Sign Language Video Generation (SLVG) seeks to generate identity-preserving sign language videos from spoken language texts. Existing methods primarily rely on the single coarse condition (e.g., skeleton sequences) as the intermediary to bridge the translation model and the video

generation model, which limits both the naturalness and expressiveness of the generated videos.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.125 Adversarial Diffusion for Robust Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Adversarial Diffusion for Robust Reinforcement Learning

快速理解: Robustness to modeling errors and uncertainties remains a central challenge in reinforcement learning (RL). In this work, we address this challenge by leveraging diffusion models to train robust RL policies.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.126 Adversarial Locomotion and Motion Imitation for Humanoid Policy Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Embodied/WBC; 优先级: 重要
- [来源](#)

论文: Adversarial Locomotion and Motion Imitation for Humanoid Policy Learning

快速理解: Humans exhibit diverse and expressive whole-body movements. However, attaining human-like whole-body coordination in humanoid robots remains challenging, as conventional approaches that mimic whole-body motions often neglect the distinct roles of upper and lower body.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制。

证据层级: 本地 PDF 前页。

1.3.7.127 Afterburner: Reinforcement Learning Facilitates Self-Improving Code Efficiency Optimization

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Afterburner: Reinforcement Learning Facilitates Self-Improving Code Efficiency Optimization

快速理解: Large Language Models (LLMs) generate functionally correct solutions but often fall short in code efficiency, a critical bottleneck for real-world deployment. In this paper, we introduce a novel test-time iterative optimization framework to address this, employing a closed-loop system where LLMs iteratively refine code based on empirical performance feedback from an execution sandbox.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.128 AgentBreeder: Mitigating the AI Safety Risks of Multi-Agent Scaffolds via Self-Improvement

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：AgentBreeder: Mitigating the AI Safety Risks of Multi-Agent Scaffolds via Self-Improvement

快速理解：Scaffolding Large Language Models (LLMs) into multi-agent systems often improves performance on complex tasks, but the safety impact of such scaffolds has not been thoroughly explored. We introduce AGENTBREEDER, a framework for multi-objective self-improving evolutionary search over scaffolds.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.129 AgentNet: Decentralized Evolutionary Coordination for LLM-based Multi-Agent Systems

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：AgentNet: Decentralized Evolutionary Coordination for LLM-based Multi-Agent Systems

快速理解：The rapid advancement of large language models (LLMs) has enabled the development of multi-agent systems where multiple LLM-based agents collaborate on complex tasks. However, existing systems often rely on centralized coordination, leading to scalability bottlenecks, reduced adaptability, and single points of failure.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.130 Agentic Plan Caching: Test-Time Memory for Fast and Cost-Efficient LLM Agents

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Agentic Plan Caching: Test-Time Memory for Fast and Cost-Efficient LLM Agents

快速理解：LLM-based agent applications have shown increasingly remarkable capabilities in complex workflows but incur substantial costs and latency due to extensive planning and reasoning requirements. Existing LLM caching techniques (like context caching and semantic caching), primarily designed for serving chatbots, are insufficient for agent applications where outputs depend on external data and environmental contexts.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.131 Agentic RL Scaling Law: Spontaneous Code Execution for Mathematical Problem Solving

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Agentic RL Scaling Law: Spontaneous Code Execution for Mathematical Problem Solving

快速理解：Large Language Models (LLMs) often struggle with mathematical reasoning tasks requiring precise, verifiable computation. While Reinforcement Learning (RL) from outcome-based rewards enhances text-based reasoning, understanding how agents autonomously learn to leverage external tools like code execution remains crucial.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.132 Agents Robust to Distribution Shifts Learn Causal World Models Even Under Mediation

- 作者：作者信息未提供
- 年份：2025；主题：World Model；优先级：重要
- [来源](#)

论文：Agents Robust to Distribution Shifts Learn Causal World Models Even Under Mediation

快速理解：In this work, we prove that agents capable of adapting to distribution shifts must have learned the causal model of their environment even in the presence of mediation. This term describes situations where an agent's actions affect its environment, a dynamic common to most real-world settings.

关注点：环境动力学、视频预测、潜变量状态或模型式决策。

证据层级：本地 PDF 前页。

1.3.7.133 AlignVLM: Bridging Vision and Language Latent Spaces for Multimodal Document Understanding

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：AlignVLM: Bridging Vision and Language Latent Spaces for Multimodal Document Understanding

快速理解：Aligning visual features with language embeddings is a key challenge in visionlanguage models (VLMs). The performance of such models hinges on having a good connector that maps visual features generated by a vision encoder to a shared embedding space with the LLM while preserving semantic similarity.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.134 Aligning What Matters: Masked Latent Adaptation for Text-to-Audio-Video Generation

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Aligning What Matters: Masked Latent Adaptation for Text-to-Audio-Video Generation

快速理解：Text-to-Audio-Video (T2AV) generation aims to produce temporally and semantically aligned visual and auditory content from natural language descriptions. While recent progress in text-to-audio and text-to-video models has improved generation quality within each modality, jointly modeling them remains challenging due to incomplete and asymmetric correspondence: audio often reflects only a subset of the visual scene, and vice versa.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.135 Aligning by Misaligning: Boundary-aware Curriculum Learning for Multimodal Alignment

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Aligning by Misaligning: Boundary-aware Curriculum Learning for Multimodal Alignment

快速理解：Most multimodal models treat every negative pair alike, ignoring the ambiguous negatives that differ from the positive by only a small detail. We propose Boundary-Aware Curriculum with Local Attention (BACL), a lightweight add-on that turns these borderline cases into a curriculum signal.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.136 Among Us: A Sandbox for Measuring and Detecting Agentic Deception

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Among Us: A Sandbox for Measuring and Detecting Agentic Deception

快速理解：Prior studies on deception in language-based AI agents typically assess whether the agent produces a false statement about a topic, or makes a binary choice prompted by a goal, rather than allowing open-ended deceptive behavior to emerge in pursuit of a longer-term goal. To fix this, we introduce Among Us1, a sandbox social deception game where LLM-agents exhibit long-term, open-ended deception as a consequence of the game objectives.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.137 Amplifying Prominent Representations in Multimodal Learning via Variational Dirichlet Process

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Amplifying Prominent Representations in Multimodal Learning via Variational Dirichlet Process

快速理解：Developing effective multimodal fusion approaches has become increasingly essential in many real-world scenarios, such as health care and finance. The key challenge is how to preserve the feature expressiveness in each modality while learning cross-modal interactions.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.138 An Information-theoretical Framework for Understanding Out-of-distribution Detection with Pretrained Vision-Language Models

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：An Information-theoretical Framework for Understanding Out-of-distribution Detection with Pretrained Vision-Language Models

快速理解：Out-of-distribution (OOD) detection, recognized for its ability to identify samples of unknown classes, provides solid advantages in ensuring the reliability of machine learning models. Among existing OOD detection methods, pre-trained visionlanguage models have emerged as powerful post-hoc OOD detectors by leveraging textual and visual information.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.139 Analytic Energy-Guided Policy Optimization for Offline Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Analytic Energy-Guided Policy Optimization for Offline Reinforcement Learning

快速理解：Conditional decision generation with diffusion models has shown powerful competitiveness in reinforcement learning (RL). Recent studies reveal the relation between energy-function-guidance diffusion models and constrained RL problems.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.140 Analyzing Fine-Grained Alignment and Enhancing Vision Understanding in Multimodal Language Models

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Analyzing Fine-Grained Alignment and Enhancing Vision Understanding in Multimodal Language Models

快速理解：Achieving better alignment between vision embeddings and Large Language Models (LLMs) is crucial for enhancing the abilities of Multimodal LLMs (MLLMs), particularly for recent models that rely on powerful pretrained vision encoders and LLMs. A common approach to connect the pretrained vision encoder and LLM is through a projector applied after the vision encoder.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.141 Approximate Domain Unlearning for Vision-Language Models

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Approximate Domain Unlearning for Vision-Language Models

快速理解：Pre-trained Vision-Language Models (VLMs) exhibit strong generalization capabilities, enabling them to recognize a wide range of objects across diverse domains without additional training. However, they often retain irrelevant information beyond the requirements of specific target downstream tasks, raising concerns about computational efficiency and potential information leakage.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.142 Approximating Shapley Explanations in Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Approximating Shapley Explanations in Reinforcement Learning

快速理解：Reinforcement learning has achieved remarkable success in complex decisionmaking environments, yet its lack of transparency limits its deployment in practice, especially in safety-critical settings. Shapley values from cooperative game theory provide a principled framework for explaining reinforcement learning; however, the computational cost of Shapley explanations is an obstacle for their use.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.143 Ask a Strong LLM Judge when Your Reward Model is Uncertain

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文: Ask a Strong LLM Judge when Your Reward Model is Uncertain

快速理解: Reward model (RM) plays a pivotal role in reinforcement learning with human feedback (RLHF) for aligning large language models (LLMs). However, classical RMs trained on human preferences are vulnerable to reward hacking and generalize poorly to out-of-distribution (OOD) inputs.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.144 Asymmetric REINFORCE for off-Policy Reinforcement Learning: Balancing positive and negative rewards

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Asymmetric REINFORCE for off-Policy Reinforcement Learning: Balancing positive and negative rewards

快速理解: Reinforcement learning (RL) is increasingly used to align large language models (LLMs). Off-policy methods offer greater implementation simplicity and data efficiency than on-policy techniques, but often result in suboptimal performance.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.145 Attention! Your Vision Language Model Could Be Maliciously Manipulated

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Attention! Your Vision Language Model Could Be Maliciously Manipulated

快速理解: Large Vision-Language Models (VLMs) have achieved remarkable success in understanding complex real-world scenarios and supporting data-driven decisionmaking processes. However, VLMs exhibit significant vulnerability against adversarial examples, either text or image, which can lead to various adversarial outcomes, e.g., jailbreaking, hijacking, and hallucination, etc.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.146 AttentionPredictor: Temporal Patterns Matter for KV Cache Compression

- 作者: 作者信息未提供
- 年份: 2025; 主题: Systems; 优先级: 重要
- [来源](#)

论文: AttentionPredictor: Temporal Patterns Matter for KV Cache Compression

快速理解: With the development of large language models (LLMs), efficient inference through Key-Value (KV) cache compression has attracted considerable attention, especially for long-context

generation. To compress the KV cache, recent methods identify critical KV tokens through static modeling of attention scores.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.7.147 Audio-Sync Video Generation with Multi-Stream Temporal Control

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Audio-Sync Video Generation with Multi-Stream Temporal Control

快速理解：Audio is inherently temporal and closely synchronized with the visual world, making it a naturally aligned and expressive control signal for controllable video generation (e.g., movies). Beyond control, directly translating audio into video is essential for understanding and visualizing rich audio narratives (e.g., Podcasts or historical recordings).

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.148 AutoData: A Multi-Agent System for Open Web Data Collection

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：AutoData: A Multi-Agent System for Open Web Data Collection

快速理解：The exponential growth of data-driven systems and AI technologies has intensified the demand for high-quality web-sourced datasets. While existing datasets have proven valuable, conventional web data collection approaches face significant limitations in terms of human effort and scalability.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.149 Automated Composition of Agents: A Knapsack Approach for Agentic Component Selection

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Automated Composition of Agents: A Knapsack Approach for Agentic Component Selection

快速理解：Designing effective agentic systems requires the seamless composition and integration of agents, tools, and models within dynamic and uncertain environments. Most existing methods rely on static, semantic retrieval approaches for tool or agent discovery.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.150 Autoregressive Adversarial Post-Training for Real-Time Interactive Video Generation

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Autoregressive Adversarial Post-Training for Real-Time Interactive Video Generation

快速理解：Autoregressive Adversarial Post-Training for Real-Time Interactive Video Generation Shanchuan Lin* Ceyuan Yang Hao He† Jianwen Jiang‡ Yuxi Ren Xin Xia Yang Zhao Xuefeng Xiao Lu Jiang ByteDance Seed <https://seaweed-apt.com/2> Abstract Existing large-scale video generation models are computationally intensive, preventing adoption in real-time and interactive applications. In this work, we propose autoregressive adversarial post-training (AAPT) to transform a pre-trained latent video diffusion model into a real-time, interactive video generator.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.151 Aux-Think: Exploring Reasoning Strategies for Data-Efficient Vision-Language Navigation

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Aux-Think: Exploring Reasoning Strategies for Data-Efficient Vision-Language Navigation

快速理解：Vision-Language Navigation (VLN) is a critical task for developing embodied agents that can follow natural language instructions to navigate in complex realworld environments. Recent advances driven by large pretrained models have significantly improved generalization and instruction grounding compared to traditional approaches.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.152 BEAST: Efficient Tokenization of B-Splines Encoded Action Sequences for Imitation Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：BEAST: Efficient Tokenization of B-Splines Encoded Action Sequences for Imitation Learning

快速理解：We present the B-spline Encoded Action Sequence Tokenizer (BEAST), a novel action tokenizer that encodes action sequences into compact discrete or continuous tokens using B-spline. In contrast to existing action tokenizers based on vector quantization or byte pair encoding, BEAST requires no separate tokenizer training and consistently produces tokens of uniform length, enabling fast action sequence generation via parallel decoding.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.153 Balanced Token Pruning: Accelerating Vision Language Models Beyond Local Optimization

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Balanced Token Pruning: Accelerating Vision Language Models Beyond Local Optimization

快速理解：Large Vision-Language Models (LVLMs) have shown impressive performance across multi-modal tasks by encoding images into thousands of tokens. However, the large number of image tokens results in significant computational overhead, and the use of dynamic high-resolution inputs further increases this burden.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.154 Balancing Multimodal Training Through Game-Theoretic Regularization

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Balancing Multimodal Training Through Game-Theoretic Regularization

快速理解：Multimodal learning holds promise for richer information extraction by capturing dependencies across data sources. Yet, current training methods often underperform due to modality competition, a phenomenon where modalities contend for training resources leaving some underoptimized.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.155 Bayesian Ego-graph Inference for Networked Multi-Agent Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Bayesian Ego-graph Inference for Networked Multi-Agent Reinforcement Learning

快速理解：In networked multi-agent reinforcement learning (Networked-MARL), decentralized agents must act autonomously under local observability and constrained communication over fixed physical graphs. Existing methods often assume static neighborhoods, limiting adaptability to dynamic or heterogeneous environments.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.156 Behavior Injection: Preparing Language Models for Reinforcement Learning

- 作者：作者信息未提供

- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Behavior Injection: Preparing Language Models for Reinforcement Learning

快速理解: Reinforcement learning (RL) has emerged as a powerful post-training technique to incentivize the reasoning ability of large language models (LLMs). However, LLMs can respond very inconsistently to RL finetuning: some show substantial performance gains, while others plateau or even degrade.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.157 Belief-Calibrated Multi-Agent Consensus Seeking for Complex NLP Tasks

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Belief-Calibrated Multi-Agent Consensus Seeking for Complex NLP Tasks

快速理解: A multi-agent system (MAS) enhances its capacity to solve complex natural language processing (NLP) tasks through collaboration among multiple agents, where consensus-seeking serves as a fundamental mechanism. However, existing consensus-seeking approaches typically rely on voting mechanisms to judge consensus, overlooking contradictions in system-internal beliefs that destabilize the consensus.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.158 Better Tokens for Better 3D: Advancing Vision-Language Modeling in 3D Medical Imaging

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Better Tokens for Better 3D: Advancing Vision-Language Modeling in 3D Medical Imaging

快速理解: Recent progress in vision-language modeling for 3D medical imaging has been fueled by large-scale computed tomography (CT) corpora with paired free-text reports, stronger architectures, and powerful pretrained models. This has enabled applications such as automated report generation and text-conditioned 3D image synthesis.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.159 Beyond Accuracy: Dissecting Mathematical Reasoning for LLMs Under Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Beyond Accuracy: Dissecting Mathematical Reasoning for LLMs Under Reinforcement Learning

快速理解: Reinforcement learning (RL) has become the dominant paradigm for improving the performance of language models on complex reasoning tasks. Despite the substantial empirical gains demonstrated by RL-based training methods like GRPO, a granular understanding of why and how RL enhances performance is still lacking.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.160 Beyond Average Value Function in Precision Medicine: Maximum Probability-Driven Reinforcement Learning for Survival Analysis

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Beyond Average Value Function in Precision Medicine: Maximum Probability-Driven Reinforcement Learning for Survival Analysis

快速理解: Constructing multistage optimal decisions for alternating recurrent event data is critically important in medical and healthcare research. Current reinforcement learning (RL) algorithms have only been applied to time-to-event data, with the objective of maximizing expected survival time.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.161 Beyond Modality Collapse: Representation Blending for Multimodal Dataset Distillation

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Beyond Modality Collapse: Representation Blending for Multimodal Dataset Distillation

快速理解: Multimodal Dataset Distillation (MDD) seeks to condense large-scale image-text datasets into compact surrogates while retaining their effectiveness for cross-modal learning. Despite recent progress, existing MDD approaches often suffer from Modality Collapse, characterized by over-concentrated intra-modal representations and enlarged distributional gap across modalities.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.162 Beyond Verifiable Rewards: Scaling Reinforcement Learning in Language Models to Unverifiable Data

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Beyond Verifiable Rewards: Scaling Reinforcement Learning in Language Models to Unverifiable Data

快速理解: We propose to scale RL to unverifiable data with a novel algorithm JEPO (Jensen's Evidence lower bound for Policy Optimization). While most prior effort on scaling RL for LLMs focuses on verifiable data where ground truth answers are typically short-form and can be matched easily, we investigate the case where such assumptions are less valid (e.g., when answers are long-form such as mathematical proofs).

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.163 Beyond the 80/20 Rule: High-Entropy Minority Tokens Drive Effective Reinforcement Learning for LLM Reasoning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Beyond the 80/20 Rule: High-Entropy Minority Tokens Drive Effective Reinforcement Learning for LLM Reasoning

快速理解: Reinforcement Learning with Verifiable Rewards (RLVR) has emerged as a powerful approach to enhancing the reasoning capabilities of Large Language Models (LLMs), while its mechanisms are not yet well understood. In this work, we undertake a pioneering exploration of RLVR through the novel perspective of token entropy patterns, comprehensively analyzing how different tokens influence reasoning performance.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.164 Bi-Level Knowledge Transfer for Multi-Task Multi-Agent Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Bi-Level Knowledge Transfer for Multi-Task Multi-Agent Reinforcement Learning

快速理解: Multi-Agent Reinforcement Learning (MARL) has achieved remarkable success in various real-world scenarios, but its high cost of online training makes it impractical to learn each task from scratch. To enable effective policy reuse, we consider the problem of zero-shot generalization from offline data across multiple tasks.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.165 Bifrost-1: Bridging Multimodal LLMs and Diffusion Models with Patch-level CLIP Latents

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Bifrost-1: Bridging Multimodal LLMs and Diffusion Models with Patch-level CLIP Latents

快速理解: There is growing interest in integrating high-fidelity visual synthesis capabilities into large language models (LLMs) without compromising their strong reasoning capabilities. Existing methods that directly train LLMs or bridge LLMs and diffusion models usually suffer from costly training since the backbone LLMs have not seen image representations during pretraining.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.166 Binary Quadratic Quantization: Beyond First-Order Quantization for Real-Valued Matrix Compression

- 作者: 作者信息未提供
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Binary Quadratic Quantization: Beyond First-Order Quantization for Real-Valued Matrix Compression

快速理解: This paper proposes a novel matrix quantization method, Binary Quadratic Quantization (BQQ). In contrast to conventional first-order quantization approaches—such as uniform quantization and binary coding quantization—that approximate real-valued matrices via linear combinations of binary bases, BQQ leverages the expressive power of binary quadratic expressions while maintaining an extremely compact data format.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 本地 PDF 前页。

1.3.7.167 BioReason: Incentivizing Multimodal Biological Reasoning within a DNA-LLM Model

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: BioReason: Incentivizing Multimodal Biological Reasoning within a DNA-LLM Model

快速理解: Unlocking deep, interpretable biological reasoning from complex genomic data is a major AI challenge hindering scientific discovery. Current DNA foundation models, despite strong sequence representation, struggle with multi-step reasoning and lack inherent transparent, biologically intuitive explanations.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.168 Blending Complementary Memory Systems in Hybrid Quadratic-Linear Transformers

- 作者: 作者信息未提供
- 年份: 2025; 主题: Systems; 优先级: 重要
- [来源](#)

论文: Blending Complementary Memory Systems in Hybrid Quadratic-Linear Transformers

快速理解: We develop hybrid memory architectures for general-purpose sequence processing neural networks, that combine key-value memory using softmax attention (KV-memory) with fast weight memory through dynamic synaptic modulation (FWmemory)—the core principles of quadratic and linear transformers, respectively. These two memory systems have complementary but individually limited properties: KV-memory offers precise retrieval but is constrained by quadratic complexity in sequence length, while FW-memory supports arbitrarily long sequences and enables more expressive computation but sacrifices precise recall.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.7.169 Blindfolded Experts Generalize Better: Insights from Robotic Manipulation and Videogames

- 作者: 作者信息未提供
- 年份: 2025; 主题: Embodied/WBC; 优先级: 重要
- [来源](#)

论文: Blindfolded Experts Generalize Better: Insights from Robotic Manipulation and Videogames

快速理解: Behavioral cloning is a simple yet effective technique for learning sequential decision-making from demonstrations. Recently, it has gained prominence as the core of foundation models for the physical world, where achieving generalization requires countless demonstrations of a multitude of tasks.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制。

证据层级: 本地 PDF 前页。

1.3.7.170 Boosting Knowledge Utilization in Multimodal Large Language Models via Adaptive Logits Fusion and Attention Reallocation

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Boosting Knowledge Utilization in Multimodal Large Language Models via Adaptive Logits Fusion and Attention Reallocation

快速理解: Despite their recent progress, Multimodal Large Language Models (MLLMs) often struggle in knowledge-intensive tasks due to the limited and outdated parametric knowledge acquired during training. Multimodal Retrieval Augmented Generation addresses this issue by retrieving contextual knowledge from external databases, thereby enhancing MLLMs with expanded knowledge sources.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.171 Bootstrap Off-policy with World Model

- 作者: 作者信息未提供
- 年份: 2025; 主题: World Model; 优先级: 重要

- [来源](#)

论文: Bootstrap Off-policy with World Model

快速理解: Online planning has proven effective in reinforcement learning (RL) for improving sample efficiency and final performance. However, using planning for environment interaction inevitably introduces a divergence between the collected data and the policy's actual behaviors, degrading both model learning and policy improvement.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策。

证据层级: 本地 PDF 前页。

1.3.7.172 Boundary-to-Region Supervision for Offline Safe Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Boundary-to-Region Supervision for Offline Safe Reinforcement Learning

快速理解: Offline safe reinforcement learning aims to learn policies that satisfy predefined safety constraints from static datasets. Existing sequence-model-based methods condition action generation on symmetric input tokens for return-to-go and cost-to-go, neglecting their intrinsic asymmetry: return-to-go (RTG) serves as a flexible performance target, while cost-to-go (CTG) should represent a rigid safety boundary.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.173 BraVE: Offline Reinforcement Learning for Discrete Combinatorial Action Spaces

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: BraVE: Offline Reinforcement Learning for Discrete Combinatorial Action Spaces

快速理解: Offline reinforcement learning in high-dimensional, discrete action spaces is challenging due to the exponential scaling of the joint action space with the number of sub-actions and the complexity of modeling sub-action dependencies. Existing methods either exhaustively evaluate the action space, making them computationally infeasible, or factorize Q-values, failing to represent joint sub-action effects.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.174 Brain Harmony: A Multimodal Foundation Model Unifying Morphology and Function into 1D Tokens

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Brain Harmony: A Multimodal Foundation Model Unifying Morphology and Function into 1D Tokens

快速理解: We present Brain Harmony (BrainHarmonix), the first multimodal brain foundation model that unifies structural morphology and functional dynamics into compact 1D token representations. The model was pretrained on two of the largest neuroimaging datasets to date, encompassing 64,594 T1-weighted structural MRI 3D volumes (~14 million images) and 70,933 functional MRI (fMRI) time series.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.175 Breaking the Performance Ceiling in Reinforcement Learning requires Inference Strategies

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Breaking the Performance Ceiling in Reinforcement Learning requires Inference Strategies

快速理解: Reinforcement learning (RL) systems have countless applications, from energygrid management to protein design. However, such real-world scenarios are often extremely difficult, combinatorial in nature, and require complex coordination between multiple agents.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.176 BridgeVLA: Input-Output Alignment for Efficient 3D Manipulation Learning with Vision-Language Models

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: BridgeVLA: Input-Output Alignment for Efficient 3D Manipulation Learning with Vision-Language Models

快速理解: Recently, leveraging pre-trained vision-language models (VLMs) for building vision-language-action (VLA) models has emerged as a promising approach to effective robot manipulation learning. However, only few methods incorporate 3D signals into VLMs for action prediction, and they do not fully leverage the spatial structure inherent in 3D data, leading to low data efficiency.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.177 CAM: A Constructivist View of Agentic Memory for LLM-Based Reading Comprehension

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: CAM: A Constructivist View of Agentic Memory for LLM-Based Reading Comprehension

快速理解: Current Large Language Models (LLMs) are confronted with overwhelming information volume when comprehending long-form documents. This challenge raises the imperative of a cohesive memory module, which can elevate vanilla LLMs into autonomous reading agents.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.178 CAML: Collaborative Auxiliary Modality Learning for Multi-Agent Systems

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: CAML: Collaborative Auxiliary Modality Learning for Multi-Agent Systems

快速理解: Multi-modal learning has emerged as a key technique for improving performance across domains such as autonomous driving, robotics, and reasoning. However, in certain scenarios, particularly in resource-constrained environments, some modalities available during training may be absent during inference.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.179 CF-VLM:CounterFactual Vision-Language Fine-tuning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: CF-VLM:CounterFactual Vision-Language Fine-tuning

快速理解: Recent advances in vision-language models (VLMs) have greatly improved crossmodal semantic understanding, yet significant limitations remain in fine-grained discrimination and deep causal reasoning tasks. Existing VLMs often rely on superficial statistical correlations, lacking the ability to capture the underlying causal logic between visual and textual content.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.180 CHPO: Constrained Hybrid-action Policy Optimization for Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: CHPO: Constrained Hybrid-action Policy Optimization for Reinforcement Learning

快速理解: Constrained hybrid-action reinforcement learning (RL) promises to learn a safe policy within a parameterized action space, which is particularly valuable for safety-critical applications involving discrete-continuous hybrid action spaces. However, existing hybrid-action RL algorithms

primarily focus on reward maximization, which faces significant challenges for tasks involving both cost constraints and hybrid action spaces.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.181 CLIPGaussian: Universal and Multimodal Style Transfer Based on Gaussian Splatting

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：CLIPGaussian: Universal and Multimodal Style Transfer Based on Gaussian Splatting

快速理解：Gaussian Splatting (GS) has recently emerged as an efficient representation for rendering 3D scenes from 2D images and has been extended to images, videos, and dynamic 4D content. However, applying style transfer to GS-based representations, especially beyond simple color changes, remains challenging.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.182 CLIPTTA: Robust Contrastive Vision-Language Test-Time Adaptation

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：CLIPTTA: Robust Contrastive Vision-Language Test-Time Adaptation

快速理解：Vision-language models (VLMs) like CLIP exhibit strong zero-shot capabilities but often fail to generalize under distribution shifts. Test-time adaptation (TTA) allows models to update at inference time without labeled data, typically via entropy minimization.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.183 CMoB: Modality Valuation via Causal Effect for Balanced Multimodal Learning

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：CMoB: Modality Valuation via Causal Effect for Balanced Multimodal Learning

快速理解：Existing early and late fusion frameworks in multimodal learning are confronted with the fundamental challenge of modality imbalance, wherein disparities in representational capacities induce inter-modal competition during training. Current research methodologies primarily rely on modality-level contribution assessments to measure gaps in representational capabilities and enhance poorly learned modalities, overlooking the dynamic variations of modality contributions across individual samples.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.184 COLA: Towards Efficient Multi-Objective Reinforcement Learning with Conflict Objective Regularization in Latent Space

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：COLA: Towards Efficient Multi-Objective Reinforcement Learning with Conflict Objective Regularization in Latent Space

快速理解：Many real-world control problems require continual policy adjustments to balance multiple objectives, which requires the acquisition of high-quality policies to cover diverse preferences. Multi-Objective Reinforcement Learning (MORL) provides a general framework to solve such problems.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.185 COME: Adding Scene-Centric Forecasting Control to Occupancy World Model

- 作者：作者信息未提供
- 年份：2025；主题：World Model；优先级：重要
- [来源](#)

论文：COME: Adding Scene-Centric Forecasting Control to Occupancy World Model

快速理解：World models are critical for autonomous driving to simulate environmental dynamics and generate synthetic data. Existing methods struggle to disentangle ego-vehicle motion (perspective shifts) from scene evolution (agent interactions), leading to suboptimal predictions.

关注点：环境动力学、视频预测、潜变量状态或模型式决策。

证据层级：本地 PDF 前页。

1.3.7.186 CORE: Collaborative Optimization with Reinforcement Learning and Evolutionary Algorithm for Floorplanning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：CORE: Collaborative Optimization with Reinforcement Learning and Evolutionary Algorithm for Floorplanning

快速理解：Floorplanning is the initial step in the physical design process of Electronic Design Automation (EDA), directly influencing subsequent placement, routing, and final power of the chip. However, the solution space in floorplanning is vast, and current algorithms often struggle to explore it sufficiently, making them prone to getting trapped in local optima.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.187 CORE: Reducing UI Exposure in Mobile Agents via Collaboration Between Cloud and Local LLMs

- 作者：作者信息未提供
- 年份：2025；主题：Systems；优先级：重要
- [来源](#)

论文：CORE: Reducing UI Exposure in Mobile Agents via Collaboration Between Cloud and Local LLMs

快速理解：Mobile agents rely on Large Language Models (LLMs) to plan and execute tasks on smartphone user interfaces (UIs). While cloud-based LLMs achieve high task accuracy, they require uploading the full UI state at every step, exposing unnecessary and often irrelevant information.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.7.188 CPPO: Accelerating the Training of Group Relative Policy Optimization-Based Reasoning Models

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：CPPO: Accelerating the Training of Group Relative Policy Optimization-Based Reasoning Models

快速理解：This paper introduces Completion Pruning Policy Optimization (CPPO) to accelerate the training of reasoning models based on Group Relative Policy Optimization (GRPO). GRPO, while effective, incurs high training costs due to the need to sample multiple completions for each question.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.189 CREA: A Collaborative Multi-Agent Framework for Creative Image Editing and Generation

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：CREA: A Collaborative Multi-Agent Framework for Creative Image Editing and Generation

快速理解：Creativity in AI imagery remains a fundamental challenge, requiring not only the generation of visually compelling content but also the capacity to add novel, expressive, and artistically rich transformations to images. Unlike conventional editing tasks that rely on direct prompt-based modifications, creative image editing requires an autonomous, iterative approach that balances originality, coherence, and artistic intent.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.190 CURV: Coherent Uncertainty-Aware Reasoning in Vision-Language Models for X-Ray Report Generation

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：CURV: Coherent Uncertainty-Aware Reasoning in Vision-Language Models for X-Ray Report Generation

快速理解：Vision-language models have been explored for radiology report generation with promising results. Yet, uncertainty elaborated in findings and the reasoning process for reaching clinical impressions are seldom explicitly modeled, reducing the clinical accuracy and trustworthiness of the generated reports.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.191 Centralized Reward Agent for Knowledge Sharing and Transfer in Multi-Task Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Centralized Reward Agent for Knowledge Sharing and Transfer in Multi-Task Reinforcement Learning

快速理解：Reward shaping is effective in addressing the sparse-reward challenge in reinforcement learning (RL) by providing immediate feedback through auxiliary, informative rewards. Based on the reward shaping strategy, we propose a novel multi-task reinforcement learning framework that integrates a centralized reward agent (CRA) and multiple distributed policy agents.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.192 Certifying Stability of Reinforcement Learning Policies using Generalized Lyapunov Functions

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Certifying Stability of Reinforcement Learning Policies using Generalized Lyapunov Functions

快速理解：Establishing stability certificates for closed-loop systems under reinforcement learning (RL) policies is essential to move beyond empirical performance and offer guarantees of system behavior. Classical Lyapunov methods require a strict stepwise decrease in the Lyapunov function but such certificates are difficult to construct for learned policies.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.193 Chain-of-Action: Trajectory Autoregressive Modeling for Robotic Manipulation

- 作者：作者信息未提供
- 年份：2025；主题：Embodied/WBC；优先级：重要
- [来源](#)

论文：Chain-of-Action: Trajectory Autoregressive Modeling for Robotic Manipulation

快速理解：We present Chain-of-Action (CoA), a novel visuomotor policy paradigm built upon Trajectory Autoregressive Modeling. Unlike conventional approaches that predict next step action(s) forward, CoA generates an entire trajectory by explicit backward reasoning with task-specific goals through an action-level Chain-of- Thought (CoT) process.

关注点：具身任务中的感知、动作生成、全身协调或运动控制。

证据层级：本地 PDF 前页。

1.3.7.194 ChartSketcher: Reasoning with Multimodal Feedback and Reflection for Chart Understanding

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：ChartSketcher: Reasoning with Multimodal Feedback and Reflection for Chart Understanding

快速理解：Charts are high-density visualization carriers for complex data, serving as a crucial medium for information extraction and analysis. Automated chart understanding poses significant challenges to existing multimodal large language models (MLLMs) due to the need for precise and complex visual reasoning.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.195 Checklists Are Better Than Reward Models For Aligning Language Models

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Checklists Are Better Than Reward Models For Aligning Language Models

快速理解：Language models must be adapted to understand and follow user instructions. Reinforcement learning is widely used to facilitate this -typically using fixed criteria such as “helpfulness” and “harmfulness” .

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.196 Chiron-o1: Igniting Multimodal Large Language Models towards Generalizable Medical Reasoning via Mentor-Intern Collaborative Search

- 作者：作者信息未提供

- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Chiron-o1: Igniting Multimodal Large Language Models towards Generalizable Medical Reasoning via Mentor-Intern Collaborative Search

快速理解: Multimodal large language models (MLLMs) have begun to demonstrate robust reasoning capabilities on general tasks, yet their application in the medical domain remains in its early stages. Constructing chain-of-thought (CoT) training data is essential for bolstering the reasoning abilities of medical MLLMs.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.197 Cloud4D: Estimating Cloud Properties at a High Spatial and Temporal Resolution

- 作者: 作者信息未提供
- 年份: 2025; 主题: Systems; 优先级: 重要
- [来源](#)

论文: Cloud4D: Estimating Cloud Properties at a High Spatial and Temporal Resolution

快速理解: There has been great progress in improving numerical weather prediction and climate models using machine learning. However, most global models act at a kilometer-scale, making it challenging to model individual clouds and factors such as extreme precipitation, wind gusts, turbulence, and surface irradiance.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.7.198 ClusterFusion: Expanding Operator Fusion Scope for LLM Inference via Cluster-Level Collective Primitive

- 作者: 作者信息未提供
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: ClusterFusion: Expanding Operator Fusion Scope for LLM Inference via Cluster-Level Collective Primitive

快速理解: Large language model (LLM) decoding suffers from high latency due to fragmented execution across operators and heavy reliance on off-chip memory for data exchange and reduction. This execution model limits opportunities for fusion and incurs significant memory traffic and kernel launch overhead.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 本地 PDF 前页。

1.3.7.199 Co-Evolving LLM Coder and Unit Tester via Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要

- [来源](#)

论文: Co-Evolving LLM Coder and Unit Tester via Reinforcement Learning

快速理解: Mathematical reasoning in large language models has been successfully incentivized through reinforcement learning with verifiable rewards, leading to improved one-shot precision. In this work, we turn our focus to the coding domain.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.200 CoC-VLA: Delving into Adversarial Domain Transfer for Explainable Autonomous Driving via Chain-of-Causality Visual-Language-Action Model

- 作者: 作者信息未提供
- 年份: 2025; 主题: VLA; 优先级: 重要
- [来源](#)

论文: CoC-VLA: Delving into Adversarial Domain Transfer for Explainable Autonomous Driving via Chain-of-Causality Visual-Language-Action Model

快速理解: Autonomous driving represents a prominent application of artificial intelligence. Recent approaches have shifted from focusing solely on common scenarios to addressing complex, long-tail situations such as subtle human behaviors, traffic accidents, and non-compliant driving patterns.

关注点: 视觉、语言条件与机器人动作策略之间的对齐。

证据层级: 本地 PDF 前页。

1.3.7.201 CoDA: Coordinated Diffusion Noise Optimization for Whole-Body Manipulation of Articulated Objects

- 作者: 作者信息未提供
- 年份: 2025; 主题: Embodied/WBC; 优先级: 重要
- [来源](#)

论文: CoDA: Coordinated Diffusion Noise Optimization for Whole-Body Manipulation of Articulated Objects

快速理解: Synthesizing whole-body manipulation of articulated objects, including body motion, hand motion, and object motion, is a critical yet challenging task with broad applications in virtual humans and robotics. The core challenges are twofold.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制。

证据层级: 本地 PDF 前页。

1.3.7.202 CoP: Agentic Red-teaming for Large Language Models using Composition of Principles

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: CoP: Agentic Red-teaming for Large Language Models using Composition of Principles

快速理解: Recent advances in Large Language Models (LLMs) have spurred transformative applications in various domains, ranging from open-source to proprietary LLMs. However, jailbreak attacks, which aim to break safety alignment and user compliance by tricking the target LLMs into answering harmful and risky responses, are becoming an urgent concern.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.203 Coarse-to-fine Q-Network with Action Sequence for Data-Efficient Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Coarse-to-fine Q-Network with Action Sequence for Data-Efficient Reinforcement Learning

快速理解: Predicting a sequence of actions has been crucial in the success of recent behavior cloning algorithms in robotics. Can similar ideas improve reinforcement learning (RL)?

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.204 Cognitive Predictive Processing: A Human-inspired Framework for Adaptive Exploration in Open-World Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Cognitive Predictive Processing: A Human-inspired Framework for Adaptive Exploration in Open-World Reinforcement Learning

快速理解: Open-world reinforcement learning challenges agents to develop intelligent behavior in vast exploration spaces. Recent approaches like LS-Imagine have advanced the field by extending imagination horizons through jumpy state transitions, yet remain limited by fixed exploration mechanisms and static jump thresholds that cannot adapt across changing task phases, resulting in inefficient exploration and lower completion rates.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.205 Complete Structure Guided Point Cloud Completion via Cluster- and Instance-Level Contrastive Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Systems; 优先级: 重要
- [来源](#)

论文: Complete Structure Guided Point Cloud Completion via Cluster- and Instance-Level Contrastive Learning

快速理解: Point cloud completion, aiming to reconstruct missing part from incomplete point clouds, is

a pivotal task in 3D computer vision. Traditional supervised approaches often necessitate complete point clouds for training supervision, which are not readily accessible in real-world applications.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.7.206 Composite Flow Matching for Reinforcement Learning with Shifted-Dynamics Data

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Composite Flow Matching for Reinforcement Learning with Shifted-Dynamics Data

快速理解：Incorporating pre-collected offline data can substantially improve the sample efficiency of reinforcement learning (RL), but its benefits can break down when the transition dynamics in the offline dataset differ from those encountered online. Existing approaches typically mitigate this issue by penalizing or filtering offline transitions in regions with large dynamics gap.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.207 Compress & Cache: Vision token compression for efficient generation and retrieval

- 作者：作者信息未提供
- 年份：2025；主题：Systems；优先级：重要
- [来源](#)

论文：Compress & Cache: Vision token compression for efficient generation and retrieval

快速理解：This work aims to compress the vision tokens of an LVLM into a representation that is simultaneously suitable for (a) generative and (b) discriminative tasks, (c) is nearly lossless, and (d) storage-efficient. To this end, we propose C&C, a novel compression method that leverages the LVLM itself for task-agnostic visual token compression.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.7.208 Computational Hardness of Reinforcement Learning with Partial q^{π} -Realizability

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Computational Hardness of Reinforcement Learning with Partial q^{π} -Realizability

快速理解：This paper investigates the computational complexity of reinforcement learning within a novel linear function approximation regime, termed partial q -realizability. In this framework, the objective is to learn an ϵ -optimal policy with respect to a predefined policy set Π , under the assumption that all value functions corresponding to policies in Π are linearly realizable.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.209 Confounding Robust Deep Reinforcement Learning: A Causal Approach

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Confounding Robust Deep Reinforcement Learning: A Causal Approach

快速理解：A key task in Artificial Intelligence is learning effective policies for controlling agents in unknown environments to optimize performance measures. Off-policy learning methods, like Q-learning, allow learners to make optimal decisions based on past experiences.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.210 Consistent Paths Lead to Truth: Self-Rewarding Reinforcement Learning for LLM Reasoning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Consistent Paths Lead to Truth: Self-Rewarding Reinforcement Learning for LLM Reasoning

快速理解：Recent advances of Reinforcement Learning (RL) have highlighted its potential in complex reasoning tasks, yet effective training often relies on external supervision, which limits the broader applicability. In this work, we propose a novel self-rewarding reinforcement learning framework to enhance Large Language Model (LLM) reasoning by leveraging the consistency of intermediate reasoning states across different reasoning trajectories.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.211 Consistently Simulating Human Personas with Multi-Turn Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Consistently Simulating Human Personas with Multi-Turn Reinforcement Learning

快速理解：Large Language Models (LLMs) are increasingly used to simulate human users in interactive settings such as therapy, education, and social role-play. While these simulations enable scalable training and evaluation of AI agents, off-the-shelf LLMs often drift from their assigned personas, contradict earlier statements, or abandon role-appropriate behavior.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.212 Contextual Integrity in LLMs via Reasoning and Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Contextual Integrity in LLMs via Reasoning and Reinforcement Learning

快速理解：As the era of autonomous agents making decisions on behalf of users unfolds, ensuring contextual integrity (CI) –what is the appropriate information to share while carrying out a certain task –becomes a central question to the field. We posit that CI demands a form of reasoning where the agent needs to reason about the context in which it is operating.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.213 Continual Knowledge Adaptation for Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Continual Knowledge Adaptation for Reinforcement Learning

快速理解：Reinforcement Learning enables agents to learn optimal behaviors through interactions with environments. However, real-world environments are typically non-stationary, requiring agents to continuously adapt to new tasks and changing conditions.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.214 Continual Multimodal Contrastive Learning

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Continual Multimodal Contrastive Learning

快速理解：Multimodal Contrastive Learning (MCL) advances in aligning different modalities and generating multimodal representations in a joint space. By leveraging contrastive learning across diverse modalities, large-scale multimodal data enhances representational quality.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.215 Continuous Q-Score Matching: Diffusion Guided Reinforcement Learning for Continuous-Time Control

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Continuous Q-Score Matching: Diffusion Guided Reinforcement Learning for Continuous-Time

Control

快速理解: Reinforcement learning (RL) has achieved significant success across a wide range of domains, however, most existing methods are formulated in discrete time. In this work, we introduce a novel RL method for continuous-time control, where stochastic differential equations govern state-action dynamics.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.216 Convergence Theorems for Entropy-Regularized and Distributional Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Convergence Theorems for Entropy-Regularized and Distributional Reinforcement Learning

快速理解: In the pursuit of finding an optimal policy, reinforcement learning (RL) methods generally ignore the properties of learned policies apart from their expected return. Thus, even when successful, it is difficult to characterize which policies will be learned and what they will do.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.217 Counteractive RL: Rethinking Core Principles for Efficient and Scalable Deep Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Counteractive RL: Rethinking Core Principles for Efficient and Scalable Deep Reinforcement Learning

快速理解: Following the pivotal success of learning strategies to win at tasks, solely by interacting with an environment without any supervision, agents have gained the ability to make sequential decisions in complex MDPs. Yet, reinforcement learning policies face exponentially growing state spaces in high dimensional MDPs resulting in a dichotomy between computational complexity and policy success.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.218 Counterfactual Evolution of Multimodal Datasets via Visual Programming

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Counterfactual Evolution of Multimodal Datasets via Visual Programming

快速理解: The rapid development of Multimodal Large Language Models (MLLMs) poses increasing demands on the diversity and complexity of multimodal datasets. Yet manual annotation pipelines can no longer keep pace.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.219 CovMatch: Cross-Covariance Guided Multimodal Dataset Distillation with Trainable Text Encoder

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: CovMatch: Cross-Covariance Guided Multimodal Dataset Distillation with Trainable Text Encoder

快速理解: Multimodal dataset distillation aims to synthesize a small set of image-text pairs that enables efficient training of large-scale vision-language models. While dataset distillation has shown promise in unimodal tasks, extending it to multimodal contrastive learning presents key challenges: learning cross-modal alignment and managing the high computational cost of large encoders.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.220 CroPe: Cross-Modal Semantic Compensation Adaptation for All Adverse Scene Understanding

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: CroPe: Cross-Modal Semantic Compensation Adaptation for All Adverse Scene Understanding

快速理解: Scene understanding in adverse conditions, such as fog, snow, and night, is challenging due to the visual appearance degeneration. In this context, we propose a Cross-modal Semantic Compensation Adaptation method (CroPe) for scene understanding.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.221 CryptoMoE: Privacy-Preserving and Scalable Mixture of Experts Inference via Balanced Expert Routing

- 作者: 作者信息未提供
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: CryptoMoE: Privacy-Preserving and Scalable Mixture of Experts Inference via Balanced Expert Routing

快速理解: Private large language model (LLM) inference based on cryptographic primitives offers a promising path towards privacy-preserving deep learning. However, existing frameworks only sup-

port dense LLMs like LLaMA-1 and struggle to scale to mixture-of-experts (MoE) architectures.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.7.222 CyIN: Cyclic Informative Latent Space for Bridging Complete and Incomplete Multimodal Learning

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：CyIN: Cyclic Informative Latent Space for Bridging Complete and Incomplete Multimodal Learning

快速理解：Multimodal machine learning, mimicking the human brain's ability to integrate various modalities has seen rapid growth. Most previous multimodal models are trained on perfectly paired multimodal input to reach optimal performance.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.223 Cypher-RI: Reinforcement Learning for Integrating Schema Selection into Cypher Generation

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Cypher-RI: Reinforcement Learning for Integrating Schema Selection into Cypher Generation

快速理解：The increasing utilization of graph databases across various fields stems from their capacity to represent intricate interconnections. Nonetheless, exploiting the full capabilities of graph databases continues to be a significant hurdle, largely because of the inherent difficulty in translating natural language into Cypher.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.224 DAIL: Beyond Task Ambiguity for Language-Conditioned Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：DAIL: Beyond Task Ambiguity for Language-Conditioned Reinforcement Learning

快速理解：Comprehending natural language and following human instructions are critical capabilities for intelligent agents. However, the flexibility of linguistic instructions induces substantial ambiguity across language-conditioned tasks, severely degrading algorithmic performance.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.225 DAPO : Improving Multi-Step Reasoning Abilities of Large Language Models with Direct Advantage-Based Policy Optimization

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：DAPO : Improving Multi-Step Reasoning Abilities of Large Language Models with Direct Advantage-Based Policy Optimization

快速理解：The role of reinforcement learning (RL) in enhancing the reasoning of large language models (LLMs) is becoming increasingly significant. Despite the success of RL in many scenarios, there are still many challenges in improving the reasoning of LLMs.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.226 DAPO: An Open-Source LLM Reinforcement Learning System at Scale

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：DAPO: An Open-Source LLM Reinforcement Learning System at Scale

快速理解：Inference scaling empowers LLMs with unprecedented reasoning ability, with reinforcement learning as the core technique to elicit complex reasoning. However, key technical details of state-of-the-art reasoning LLMs are concealed (such as in OpenAI o1 blog and DeepSeek R1 technical report), thus the community still struggles to reproduce their RL training results.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.227 DEAL: Diffusion Evolution Adversarial Learning for Sim-to-Real Transfer

- 作者：作者信息未提供
- 年份：2025；主题：Robot Learning；优先级：重要
- [来源](#)

论文：DEAL: Diffusion Evolution Adversarial Learning for Sim-to-Real Transfer

快速理解：Training Reinforcement Learning (RL) controllers in simulation offers costefficiency and safety advantages. However, the resultant policies often suffer significant performance degradation during real-world deployment due to the reality gap.

关注点：机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：本地 PDF 前页。

1.3.7.228 DEXTER: Diffusion-Guided EXplanations with TExtual Reasoning for Vision Models

- 作者：作者信息未提供
- 年份：2025；主题：Embodied/WBC；优先级：重要
- [来源](#)

论文：DEXTER: Diffusion-Guided EXplanations with TExtual Reasoning for Vision Models

快速理解：Understanding and explaining the behavior of machine learning models is essential for building transparent and trustworthy AI systems. We introduce DEX-TER, a data-free framework that employs diffusion models and large language models to generate global, textual explanations of visual classifiers.

关注点：具身任务中的感知、动作生成、全身协调或运动控制。

证据层级：本地 PDF 前页。

1.3.7.229 DISC: Dynamic Decomposition Improves LLM Inference Scaling

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：DISC: Dynamic Decomposition Improves LLM Inference Scaling

快速理解：Inference scaling methods for LLMs often rely on decomposing problems into steps (or groups of tokens), followed by sampling and selecting the best next steps. However, these steps and their sizes are often predetermined or manually designed based on domain knowledge.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.7.230 DISCOVER: Automated Curricula for Sparse-Reward Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：DISCOVER: Automated Curricula for Sparse-Reward Reinforcement Learning

快速理解：Sparse-reward reinforcement learning (RL) can model a wide range of highly complex tasks. Solving sparse-reward tasks is RL's core premise—requiring efficient exploration coupled with long-horizon credit assignment—and overcoming these challenges is key for building self-improving agents with superhuman ability.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.231 DMWM: Dual-Mind World Model with Long-Term Imagination

- 作者：作者信息未提供
- 年份：2025；主题：World Model；优先级：重要
- [来源](#)

论文：DMWM: Dual-Mind World Model with Long-Term Imagination

快速理解: Imagination in world models is crucial for enabling agents to learn long-horizon policies in a sample-efficient manner. Existing recurrent state-space model (RSSM)- based world models depend on single-step statistical inference to capture the environment dynamics, and, hence, they cannot effectively perform long-term imagination tasks due to the accumulation of prediction errors.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策。

证据层级: 本地 PDF 前页。

1.3.7.232 DOTA: Distributional Test-time Adaptation of Vision-Language Models

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: DOTA: Distributional Test-time Adaptation of Vision-Language Models

快速理解: Vision-language foundation models (VLMs), such as CLIP, exhibit remarkable performance across a wide range of tasks. However, deploying these models can be unreliable when significant distribution gaps exist between training and test data, while fine-tuning for diverse scenarios is often costly.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.233 DPAIL: Training Diffusion Policy for Adversarial Imitation Learning without Policy Optimization

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: DPAIL: Training Diffusion Policy for Adversarial Imitation Learning without Policy Optimization

快速理解: Human experts employ diverse strategies to complete a task, producing to multimodal demonstration data. Although traditional Adversarial Imitation Learning (AIL) methods have achieved notable success, they often collapse theses multimodal behaviors into a single strategy, failing to replicate expert behaviors.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.234 DREAM: Drafting with Refined Target Features and Entropy-Adaptive Cross-Attention Fusion for Multimodal Speculative Decoding

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: DREAM: Drafting with Refined Target Features and Entropy-Adaptive Cross-Attention Fusion for Multimodal Speculative Decoding

快速理解: Speculative decoding (SD) has emerged as a powerful method for accelerating autoregres-

sive generation in large language models (LLMs), yet its integration into vision-language models (VLMs) remains underexplored. We introduce DREAM, a novel speculative decoding framework tailored for VLMs that combines three key innovations: (1) a cross-attention-based mechanism to inject intermediate features from the target model into the draft model for improved alignment, (2) adaptive intermediate feature selection based on attention entropy to guide efficient draft model training, and (3) visual token compression to reduce draft model latency.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.235 DartQuant: Efficient Rotational Distribution Calibration for LLM Quantization

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：DartQuant: Efficient Rotational Distribution Calibration for LLM Quantization

快速理解：Quantization plays a crucial role in accelerating the inference of large-scale models, and rotational matrices have been shown to effectively improve quantization performance by smoothing outliers. However, end-to-end fine-tuning of rotational optimization algorithms incurs high computational costs and is prone to overfitting.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.7.236 Data Selection Matters: Towards Robust Instruction Tuning of Large Multimodal Models

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Data Selection Matters: Towards Robust Instruction Tuning of Large Multimodal Models

快速理解：Selecting a compact subset of visual instruction-following data has emerged as an effective way to align large multimodal models with human intentions while avoiding the high cost of full-dataset training. Yet we observe that both full-data training and existing state-of-the-art data selection methods tend to inherit underlying dataset biases such as position bias and spurious correlations, leading to biased model behaviors.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.237 Dataset Distillation of 3D Point Clouds via Distribution Matching

- 作者：作者信息未提供
- 年份：2025；主题：Systems；优先级：重要
- [来源](#)

论文：Dataset Distillation of 3D Point Clouds via Distribution Matching

快速理解: Large-scale datasets are usually required to train deep neural networks; however, they increase computational complexity, hindering practical applications. Recently, dataset distillation for images and texts has attracted considerable attention, as it reduces the original dataset to a small synthetic one to alleviate the computational burden of training while preserving essential task-relevant information.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.7.238 Debate or Vote: Which Yields Better Decisions in Multi-Agent Large Language Models?

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Debate or Vote: Which Yields Better Decisions in Multi-Agent Large Language Models?

快速理解: Multi-Agent Debate (MAD) has emerged as a promising paradigm for improving the performance of large language models through collaborative reasoning. Despite recent advances, the key factors driving MAD's effectiveness remain unclear.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.239 Decoupling Contrastive Decoding: Robust Hallucination Mitigation in Multimodal Large Language Models

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Decoupling Contrastive Decoding: Robust Hallucination Mitigation in Multimodal Large Language Models

快速理解: Although multimodal large language models (MLLMs) exhibit remarkable reasoning capabilities on complex multimodal understanding tasks, they still suffer from the notorious “hallucination” issue: generating outputs misaligned with obvious visual or factual evidence. Currently, training-based solutions, like direct preference optimization (DPO), leverage paired preference data to suppress hallucinations.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.240 Deep Video Discovery: Agentic Search with Tool Use for Long-form Video Understanding

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Deep Video Discovery: Agentic Search with Tool Use for Long-form Video Understanding

快速理解: Long-form video understanding presents significant challenges due to extensive temporal-spatial complexity and the difficulty of question answering under such extended contexts. While Large Language Models (LLMs) have demonstrated considerable advancements in video analysis capabilities and long context handling, they continue to exhibit limitations when processing information-dense hour-long videos.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.241 DeepDiver: Adaptive Web-Search Intensity Scaling via Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: DeepDiver: Adaptive Web-Search Intensity Scaling via Reinforcement Learning

快速理解: Information seeking demands iterative evidence gathering and reflective reasoning, yet large language models (LLMs) still struggle with it in open-web question answering. Existing prompting and supervised fine-tuning (SFT) methods remain fixed by prompt rules or training corpora, and are usually benchmarked only on wellstructured wiki sources, limiting real-world adaptability.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.242 Defending Multimodal Backdoored Models by Repulsive Visual Prompt Tuning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Defending Multimodal Backdoored Models by Repulsive Visual Prompt Tuning

快速理解: Multimodal contrastive learning models (e.g., CLIP) can learn high-quality representations from large-scale image-text datasets, while they exhibit significant vulnerabilities to backdoor attacks, raising serious safety concerns. In this paper, we reveal that CLIP's vulnerabilities primarily stem from its tendency to encode features beyond in-dataset predictive patterns, compromising its visual feature resistivity to input perturbations.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.243 Dense Backpropagation Improves Training for Sparse Mixture-of-Experts

- 作者: 作者信息未提供
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Dense Backpropagation Improves Training for Sparse Mixture-of-Experts

快速理解: Mixture of Experts (MoE) pretraining is more scalable than dense Transformer pretraining,

because MoEs learn to route inputs to a sparse set of their feedforward parameters. However, this means that MoEs only receive a sparse backward update, leading to training instability and suboptimal performance.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.7.244 Dense Metric Depth Estimation via Event-based Differential Focus Volume Prompting

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Dense Metric Depth Estimation via Event-based Differential Focus Volume Prompting

快速理解：Dense metric depth estimation has witnessed great developments in recent years. While single-image-based methods have demonstrated commendable performance in certain circumstances, they may encounter challenges regarding scale ambiguities and visual illusions in real world.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.245 DexFlyWheel: A Scalable and Self-improving Data Generation Framework for Dexterous Manipulation

- 作者：作者信息未提供
- 年份：2025；主题：Embodied/WBC；优先级：重要
- [来源](#)

论文：DexFlyWheel: A Scalable and Self-improving Data Generation Framework for Dexterous Manipulation

快速理解：Dexterous manipulation is critical for advancing robot capabilities in real-world applications, yet diverse and high-quality datasets remain scarce. Existing data collection methods either rely on human teleoperation or require significant human engineering, or generate data with limited diversity, which restricts their scalability and generalization.

关注点：具身任务中的感知、动作生成、全身协调或运动控制。

证据层级：本地 PDF 前页。

1.3.7.246 DexGarmentLab: Dexterous Garment Manipulation Environment with Generalizable Policy

- 作者：作者信息未提供
- 年份：2025；主题：Embodied/WBC；优先级：重要
- [来源](#)

论文：DexGarmentLab: Dexterous Garment Manipulation Environment with Generalizable Policy

快速理解：Garment manipulation is a critical challenge due to the diversity in garment categories, geometries, and deformations. Despite this, humans can effortlessly handle garments, thanks to the dexterity of our hands.

关注点：具身任务中的感知、动作生成、全身协调或运动控制。

证据层级：本地 PDF 前页。

1.3.7.247 DiEP: Adaptive Mixture-of-Experts Compression through Differentiable Expert Pruning

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：DiEP: Adaptive Mixture-of-Experts Compression through Differentiable Expert Pruning

快速理解：DiEP: Adaptive Mixture-of-Experts Compression through Differentiable Expert Pruning Sikai Bai HKUST Hong Kong, China sbaiae@connect.ust.hk Haoxi Li HKUST Hong Kong, China hligb@connect.ust.hk Jie Zhang * HKUST Hong Kong, China csejzhang@ust.hk Zicong Hong HKUST Hong Kong, China congcong@ust.hk Song Guo * HKUST Hong Kong, China song-guo@cse.ust.hk Abstract Despite the significant breakthrough of Mixture-of-Experts (MoE), the increasing scale of these MoE models presents huge memory and storage challenges. Existing MoE pruning methods, which involve reducing parameter size with a uniform sparsity across all layers, often lead to suboptimal outcomes and performance degradation due to varying expert redundancy in different MoE layers.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.7.248 Diffusion Guided Adversarial State Perturbations in Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Diffusion Guided Adversarial State Perturbations in Reinforcement Learning

快速理解：Reinforcement learning (RL) systems, while achieving remarkable success across various domains, are vulnerable to adversarial attacks. This is especially a concern in vision-based environments where minor manipulations of high-dimensional image inputs can easily mislead the agent's behavior.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.249 Diffusion Model as a Noise-Aware Latent Reward Model for Step-Level Preference Optimization

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Diffusion Model as a Noise-Aware Latent Reward Model for Step-Level Preference Optimization

快速理解：Preference optimization for diffusion models aims to align them with human preferences

for images. Previous methods typically use Vision-Language Models (VLMs) as pixel-level reward models to approximate human preferences.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.250 Discovering Important Experts for Mixture-of-Experts Models Pruning Through a Theoretical Perspective

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：Discovering Important Experts for Mixture-of-Experts Models Pruning Through a Theoretical Perspective

快速理解：Mixture-of-Experts (MoE) architectures enable efficient scaling of large language models but face prohibitive memory demands due to massive parameterization. Existing pruning methods rely on heuristic metrics or impractical enumeration of expert subsets, leading to suboptimal performance or scalability.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.7.251 Distil-E2D: Distilling Image-to-Depth Priors for Event-Based Monocular Depth Estimation

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Distil-E2D: Distilling Image-to-Depth Priors for Event-Based Monocular Depth Estimation

快速理解：Event cameras are neuromorphic vision sensors that asynchronously capture pixellevel intensity changes with high temporal resolution and dynamic range. These make them well suited for monocular depth estimation under challenging lighting conditions.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.252 Distributed Multi-Agent Bandits Over Erdős-Rényi Random Networks

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Distributed Multi-Agent Bandits Over Erdős-Rényi Random Networks

快速理解：We study the distributed multi-agent multi-armed bandit problem with heterogeneous rewards over random communication graphs. Uniquely, at each time step t agents communicate over a time-varying random graph G_t generated by applying the Erdős-Rényi model to a fixed connected base graph G (for classical Erdős-Rényi graphs, G is a complete graph), where each potential edge in G is randomly and independently present with the link probability p .

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.253 Diversity-Aware Policy Optimization for Large Language Model Reasoning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Diversity-Aware Policy Optimization for Large Language Model Reasoning

快速理解：The reasoning capabilities of large language models (LLMs) have advanced rapidly, particularly following the release of DeepSeek-R1, which has inspired a surge of research into data quality and reinforcement learning (RL) algorithms. Despite the pivotal role diversity plays in RL, its influence on LLM reasoning remains largely underexplored.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.254 Does Reinforcement Learning Really Incentivize Reasoning Capacity in LLMs Beyond the Base Model?

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Does Reinforcement Learning Really Incentivize Reasoning Capacity in LLMs Beyond the Base Model?

快速理解：Reinforcement Learning with Verifiable Rewards (RLVR) has recently demonstrated notable success in enhancing the reasoning performance of large language models (LLMs), particularly in mathematics and programming tasks. It is widely believed that, similar to how traditional RL helps agents to explore and learn new strategies, RLVR enables LLMs to continuously self-improve, thus acquiring novel reasoning abilities that exceed the capacity of the corresponding base models.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.255 Domain-Specific Pruning of Large Mixture-of-Experts Models with Few-shot Demonstrations

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：Domain-Specific Pruning of Large Mixture-of-Experts Models with Few-shot Demonstrations

快速理解：Mixture-of-Experts (MoE) models achieve a favorable trade-off between performance and inference efficiency by activating only a subset of experts. However, the memory overhead of storing all experts remains a major limitation, especially in large-scale MoE models such as DeepSeek-R1 (671B).

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.7.256 Don' t Trade Off Safety: Diffusion Regularization for Constrained Offline RL

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Don' t Trade Off Safety: Diffusion Regularization for Constrained Offline RL

快速理解：Constrained reinforcement learning (RL) seeks high-performance policies under safety constraints. We focus on an offline setting where the agent learns from a fixed dataset—a common requirement in realistic tasks to prevent unsafe exploration.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.257 Dual-Path Temporal Decoder for End-to-End Multi-Object Tracking

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Dual-Path Temporal Decoder for End-to-End Multi-Object Tracking

快速理解：We present a novel end-to-end transformer-based framework for Multiple Object Tracking (MOT) that advances temporal modeling and identity preservation. Despite recent progress in transformer-based MOT, existing methods still struggle to maintain consistent object identities across frames, especially under occlusions, appearance changes, or detection failures.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.258 DualCnst: Enhancing Zero-Shot Out-of-Distribution Detection via Text-Image Consistency in Vision-Language Models

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：DualCnst: Enhancing Zero-Shot Out-of-Distribution Detection via Text-Image Consistency in Vision-Language Models

快速理解：Pretrained vision-language models (VLMs), such as CLIP, have shown promising zero-shot out-of-distribution (OOD) detection capabilities by leveraging semantic similarities between input images and textual labels. However, most existing approaches focus solely on expanding the label space in the text domain, ignoring complementary visual cues that can further enhance discriminative power.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.259 DyFlow: Dynamic Workflow Framework for Agentic Reasoning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：DyFlow: Dynamic Workflow Framework for Agentic Reasoning

快速理解：Agent systems based on large language models (LLMs) have shown great potential in complex reasoning tasks, but building efficient and generalizable workflows remains a major challenge. Most existing approaches rely on manually designed processes, which limits their adaptability across different tasks.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.260 DyMoDreamer: World Modeling with Dynamic Modulation

- 作者：作者信息未提供
- 年份：2025；主题：World Model；优先级：重要
- [来源](#)

论文：DyMoDreamer: World Modeling with Dynamic Modulation

快速理解：A critical bottleneck in deep reinforcement learning (DRL) is sample inefficiency, as training high-performance agents often demands extensive environmental interactions. Model-based reinforcement learning (MBRL) mitigates this by building world models that simulate environmental dynamics and generate synthetic experience, improving sample efficiency.

关注点：环境动力学、视频预测、潜变量状态或模型式决策。

证据层级：本地 PDF 前页。

1.3.7.261 Dyn-O: Building Structured World Models with Object-Centric Representations

- 作者：作者信息未提供
- 年份：2025；主题：World Model；优先级：重要
- [来源](#)

论文：Dyn-O: Building Structured World Models with Object-Centric Representations

快速理解：World models aim to capture the dynamics of the environment, enabling agents to predict and plan for future states. In most scenarios of interest, the dynamics are highly centered on interactions among objects within the environment.

关注点：环境动力学、视频预测、潜变量状态或模型式决策。

证据层级：本地 PDF 前页。

1.3.7.262 DynaNav: Dynamic Feature and Layer Selection for Efficient Visual Navigation

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要

- [来源](#)

论文: DynaNav: Dynamic Feature and Layer Selection for Efficient Visual Navigation

快速理解: Visual navigation is essential for robotics and embodied AI. However, existing foundation models, particularly those with transformer decoders, suffer from high computational overhead and lack interpretability, limiting their deployment in resource-tight scenarios.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.263 DynaRend: Learning 3D Dynamics via Masked Future Rendering for Robotic Manipulation

- 作者: 作者信息未提供
- 年份: 2025; 主题: Embodied/WBC; 优先级: 重要
- [来源](#)

论文: DynaRend: Learning 3D Dynamics via Masked Future Rendering for Robotic Manipulation

快速理解: Learning generalizable robotic manipulation policies remains a key challenge due to the scarcity of diverse real-world training data. While recent approaches have attempted to mitigate this through self-supervised representation learning, most either rely on 2D vision pretraining paradigms such as masked image modeling, which primarily focus on static semantics or scene geometry, or utilize large-scale video prediction models that emphasize 2D dynamics, thus failing to jointly learn the geometry, semantics, and dynamics required for effective manipulation.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制。

证据层级: 本地 PDF 前页。

1.3.7.264 Dynamic Configuration for Cutting Plane Separators via Reinforcement Learning on Incremental Graph

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Dynamic Configuration for Cutting Plane Separators via Reinforcement Learning on Incremental Graph

快速理解: Cutting planes (cuts) are essential for solving mixed-integer linear programming (MILP) problems, as they tighten the feasible solution space and accelerate the solving process. Modern MILP solvers offer diverse cutting plane separators to generate cuts, enabling users to leverage their potential complementary strengths to tackle problems with different structures.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.265 Dynamic Focused Masking for Autoregressive Embodied Occupancy Prediction

- 作者: 作者信息未提供
- 年份: 2025; 主题: Embodied/WBC; 优先级: 重要

- [来源](#)

论文: Dynamic Focused Masking for Autoregressive Embodied Occupancy Prediction

快速理解: Visual autoregressive modeling has recently demonstrated potential in image tasks by enabling coarse-to-fine, next-level prediction. Most indoor 3D occupancy prediction methods, however, continue to rely on dense voxel grids and convolution-heavy backbones, which incur high computational costs when applying such coarse-to-fine frameworks.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制。

证据层级: 本地 PDF 前页。

1.3.7.266 Dynamics-Aligned Latent Imagination in Contextual World Models for Zero-Shot Generalization

- 作者: 作者信息未提供
- 年份: 2025; 主题: World Model; 优先级: 重要
- [来源](#)

论文: Dynamics-Aligned Latent Imagination in Contextual World Models for Zero-Shot Generalization

快速理解: Real-world reinforcement learning demands adaptation to unseen environmental conditions without costly retraining. Contextual Markov Decision Processes (cMDP) model this challenge, but existing methods often require explicit context variables (e.g., friction, gravity), limiting their use when contexts are latent or hard to measure.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策。

证据层级: 本地 PDF 前页。

1.3.7.267 EDELINE: Enhancing Memory in Diffusion-based World Models via Linear-Time Sequence Modeling

- 作者: 作者信息未提供
- 年份: 2025; 主题: World Model; 优先级: 重要
- [来源](#)

论文: EDELINE: Enhancing Memory in Diffusion-based World Models via Linear-Time Sequence Modeling

快速理解: World models represent a promising approach for training reinforcement learning agents with significantly improved sample efficiency. While most world model methods primarily rely on sequences of discrete latent variables to model environment dynamics, this compression often neglects critical visual details essential for reinforcement learning.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策。

证据层级: 本地 PDF 前页。

1.3.7.268 ESCA: Contextualizing Embodied Agents via Scene-Graph Generation

- 作者: 作者信息未提供
- 年份: 2025; 主题: Embodied/WBC; 优先级: 重要
- [来源](#)

论文: ESCA: Contextualizing Embodied Agents via Scene-Graph Generation

快速理解: Multi-modal large language models (MLLMs) are making rapid progress toward general-purpose embodied agents. However, existing MLLMs do not reliably capture fine-grained links between low-level visual features and high-level textual semantics, leading to weak grounding and inaccurate perception.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制。

证据层级: 本地 PDF 前页。

1.3.7.269 Eagle 2.5: Boosting Long-Context Post-Training for Frontier Vision-Language Models

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Eagle 2.5: Boosting Long-Context Post-Training for Frontier Vision-Language Models

快速理解: We introduce Eagle2.5, a frontier vision-language model (VLM) for long-context multimodal learning. Our work addresses the challenges in long video comprehension and high-resolution image understanding, introducing a generalist framework for both tasks.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.270 EasySpec: Layer-Parallel Speculative Decoding for Efficient Multi-GPU Utilization

- 作者: 作者信息未提供
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: EasySpec: Layer-Parallel Speculative Decoding for Efficient Multi-GPU Utilization

快速理解: Speculative decoding is an effective and lossless method for Large Language Model (LLM) inference acceleration. It employs a smaller model to generate a draft token sequence, which is then verified by the original base model.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 本地 PDF 前页。

1.3.7.271 EchoShot: Multi-Shot Portrait Video Generation

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: EchoShot: Multi-Shot Portrait Video Generation

快速理解: Video diffusion models substantially boost the productivity of artistic workflows with high-quality portrait video generative capacity. However, prevailing pipelines are primarily constrained to single-shot creation, while real-world applications urge multiple shots with identity consistency

and flexible content controllability.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.272 Effective Policy Learning for Multi-Agent Online Coordination Beyond Submodular Objectives

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Effective Policy Learning for Multi-Agent Online Coordination Beyond Submodular Objectives

快速理解：In this paper, we present two effective policy learning algorithms for multi-agent online coordination(MA-OC) problem. The first one, MA-SPL, not only can achieve the optimal $(1 - c e)$ -approximation guarantee for the MA-OC problem with submodular objectives but also can handle the unexplored γ -weakly DR-submodular and (γ, β) -weakly submodular scenarios, where c is the curvature of the investigated submodular functions, γ denotes the diminishing-return(DR) ratio and the tuple (γ, β) represents the submodularity ratios.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.273 Efficient Multi-bit Quantization Network Training via Weight Bias Correction and Bit-wise Coreset Sampling

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：Efficient Multi-bit Quantization Network Training via Weight Bias Correction and Bit-wise Coreset Sampling

快速理解：Multi-bit quantization networks enable flexible deployment of deep neural networks by supporting multiple precision levels within a single model. However, existing approaches suffer from significant training overhead as full-dataset updates are repeated for each supported bit-width, resulting in a cost that scales linearly with the number of precisions.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.7.274 Efficient Multimodal Dataset Distillation via Generative Models

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Efficient Multimodal Dataset Distillation via Generative Models

快速理解：Dataset distillation aims to synthesize a small dataset from a large dataset, enabling the model trained on it to perform well on the original dataset. With the blooming of large language models and multimodal large language models, the importance of multimodal datasets, particularly

image-text datasets, has grown significantly.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.275 Efficient Policy Optimization in Robust Constrained MDPs with Iteration Complexity Guarantees

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Efficient Policy Optimization in Robust Constrained MDPs with Iteration Complexity Guarantees

快速理解: Constrained decision-making is essential for designing safe policies in real-world control systems, yet simulated environments often fail to capture real-world adversities. We consider the problem of learning a policy that will maximize the cumulative reward while satisfying a constraint, even when there is a mismatch between the real model and an accessible simulator/nominal model.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.276 Efficient Pre-Training of LLMs via Topology-Aware Communication Alignment on More Than 9600 GPUs

- 作者: 作者信息未提供
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Efficient Pre-Training of LLMs via Topology-Aware Communication Alignment on More Than 9600 GPUs

快速理解: The scaling law for large language models (LLMs) depicts that the path towards machine intelligence necessitates training at large scale. Thus, companies continuously build large-scale GPU clusters, and launch training jobs that span over thousands of computing nodes.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 本地 PDF 前页。

1.3.7.277 Efficient Preference-Based Reinforcement Learning: Randomized Exploration meets Experimental Design

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Efficient Preference-Based Reinforcement Learning: Randomized Exploration meets Experimental Design

快速理解: We study reinforcement learning from human feedback in general Markov decision processes, where agents learn from trajectory-level preference comparisons. A central challenge in this setting is to design algorithms that select informative preference queries to identify the underlying reward

while ensuring theoretical guarantees.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.278 Efficient Safe Meta-Reinforcement Learning: Provable Near-Optimality and Anytime Safety

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Efficient Safe Meta-Reinforcement Learning: Provable Near-Optimality and Anytime Safety

快速理解：This paper studies the problem of safe meta-reinforcement learning (safe meta- RL), where an agent efficiently adapts to unseen tasks while satisfying safety constraints at all times during adaptation. We propose a framework consisting of two complementary modules: safe policy adaptation and safe meta-policy training.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.279 Efficient and Generalizable Mixed-Precision Quantization via Topological Entropy

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：Efficient and Generalizable Mixed-Precision Quantization via Topological Entropy

快速理解：Network quantization effectively reduces both memory footprints and inference time of deep neural networks, enabling their deployment on resource-constrained devices. To fully utilize the multiple bit-width arithmetic operations of the hardware, mixed-precision quantization (MPQ) is developed to assign different bit-widths to each layer.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.7.280 ElasticMM: Efficient Multimodal LLMs Serving with Elastic Multimodal Parallelism

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：ElasticMM: Efficient Multimodal LLMs Serving with Elastic Multimodal Parallelism

快速理解：Multimodal large language models (MLLMs) extend LLMs to handle images, videos, and audio by incorporating feature extractors and projection modules. However, these additional components—combined with complex inference pipelines and heterogeneous workloads—introduce significant inference overhead.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.281 Elevating Visual Perception in Multimodal LLMs with Visual Embedding Distillation

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Elevating Visual Perception in Multimodal LLMs with Visual Embedding Distillation

快速理解：In recent times, the standard practice for developing MLLMs is to feed features from vision encoder(s) into the LLM and train with natural language supervision. This approach often causes models to lean towards language comprehension and undermine the rich visual perception signals present in the data, which are critical for tasks involving spatial reasoning in the domain of embodied AI and robotics.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.282 Embodied Cognition Augmented End2End Autonomous Driving

- 作者：作者信息未提供
- 年份：2025；主题：Embodied/WBC；优先级：重要
- [来源](#)

论文：Embodied Cognition Augmented End2End Autonomous Driving

快速理解：In recent years, vision-based end-to-end autonomous driving has emerged as a new paradigm. However, popular end-to-end approaches typically rely on visual feature extraction networks trained under label supervision.

关注点：具身任务中的感知、动作生成、全身协调或运动控制。

证据层级：本地 PDF 前页。

1.3.7.283 Embodied Crowd Counting

- 作者：作者信息未提供
- 年份：2025；主题：Embodied/WBC；优先级：重要
- [来源](#)

论文：Embodied Crowd Counting

快速理解：Occlusion is one of the fundamental challenges in crowd counting. In the community, various data-driven approaches have been developed to address this issue, yet their effectiveness is limited.

关注点：具身任务中的感知、动作生成、全身协调或运动控制。

证据层级：本地 PDF 前页。

1.3.7.284 Empirical Study on Robustness and Resilience in Cooperative Multi-Agent Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文: Empirical Study on Robustness and Resilience in Cooperative Multi-Agent Reinforcement Learning

快速理解: In cooperative Multi-Agent Reinforcement Learning (MARL), it is a common practice to tune hyperparameters in ideal simulated environments to maximize cooperative performance. However, policies tuned for cooperation often fail to maintain robustness and resilience under real-world uncertainties.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.285 EnerVerse: Envisioning Embodied Future Space for Robotics Manipulation

- 作者：作者信息未提供
- 年份：2025；主题：Embodied/WBC；优先级：重要
- [来源](#)

论文: EnerVerse: Envisioning Embodied Future Space for Robotics Manipulation

快速理解: We introduce ENERVERSE, a generative robotics foundation model that constructs and interprets embodied spaces. ENERVERSE employs a chunk-wise autoregressive video diffusion framework to predict future embodied spaces from instructions, enhanced by a sparse context memory for long-term reasoning.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制。

证据层级: 本地 PDF 前页。

1.3.7.286 Enhanced Expert Merging for Mixture-of-Experts in Graph Foundation Models

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文: Enhanced Expert Merging for Mixture-of-Experts in Graph Foundation Models

快速理解: Graph foundation models (GFMs) have emerged as a promising paradigm for learning transferable knowledge across diverse graph-structured data. The inherent heterogeneity in features and graph structures poses significant challenges for building scalable and generalizable GFMs.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 本地 PDF 前页。

1.3.7.287 Enhanced Self-Distillation Framework for Efficient Spiking Neural Network Training

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文: Enhanced Self-Distillation Framework for Efficient Spiking Neural Network Training

快速理解: Enhanced Self-Distillation Framework for Efficient Spiking Neural Network Training Xiaochen Zhao^{1†} Chengting Yu^{1,2†} Kairong Yu^{1,2} Lei Liu¹ Aili Wang^{1,2*} ¹ ZJU-UIUC Institute, Zhejiang University ² College of Information Science and Electronic Engineering, Zhejiang University {xiaochen.24,chengting.21,aili.wang}@intl.zju.edu.cn Abstract Spiking Neural Networks (SNNs) exhibit exceptional energy efficiency on neuromorphic hardware due to their sparse activation patterns. However, conventional training methods based on surrogate gradients and Backpropagation Through Time (BPTT) not only lag behind Artificial Neural Networks (ANNs) in performance, but also incur significant computational and memory overheads that grow linearly with the temporal dimension.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 本地 PDF 前页。

1.3.7.288 Enhancing 3D Reconstruction for Dynamic Scenes

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Enhancing 3D Reconstruction for Dynamic Scenes

快速理解: In this work, we address the task of 3D reconstruction in dynamic scenes, where object motions frequently degrade the quality of previous 3D pointmap regression methods, such as DUST3R, that are originally designed for static 3D scene reconstruction. Although these methods provide an elegant and powerful solution in static settings, they struggle in the presence of dynamic motions that disrupt alignment based solely on camera poses.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.289 Enhancing Interpretability in Deep Reinforcement Learning through Semantic Clustering

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Enhancing Interpretability in Deep Reinforcement Learning through Semantic Clustering

快速理解: In this paper, we explore semantic clustering properties of deep reinforcement learning (DRL) to improve its interpretability and deepen our understanding of its internal semantic organization. In this context, semantic clustering refers to the ability of neural networks to cluster inputs based on their semantic similarity in the feature space.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.290 Enhancing Privacy in Multimodal Federated Learning with Information Theory

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Enhancing Privacy in Multimodal Federated Learning with Information Theory

快速理解：Multimodal federated learning (MMFL) has gained increasing popularity due to its ability to leverage the correlation between various modalities, meanwhile preserving data privacy for different clients. However, recent studies show that correlation between modalities increase the vulnerability of federated learning against Gradient Inversion Attack (GIA).

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.291 Enhancing Safety in Reinforcement Learning with Human Feedback via Rectified Policy Optimization

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Enhancing Safety in Reinforcement Learning with Human Feedback via Rectified Policy Optimization

快速理解：Balancing helpfulness and safety (harmlessness) is a critical challenge in aligning large language models (LLMs). Current approaches often decouple these two objectives, training separate preference models for helpfulness and safety, while framing safety as a constraint within a constrained Markov Decision Process (CMDP) framework.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.292 Enhancing Vision-Language Model Reliability with Uncertainty-Guided Dropout Decoding

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Enhancing Vision-Language Model Reliability with Uncertainty-Guided Dropout Decoding

快速理解：Large vision-language models (LVLMs) excel at multimodal tasks but are prone to misinterpreting visual inputs, often resulting in hallucinations and unreliable outputs. We present DROPOUT DECODING, a novel inference-time approach that quantifies the uncertainty of visual tokens and selectively masks uncertain tokens to improve decoding.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.293 Entropic Time Schedulers for Generative Diffusion Models

- 作者：作者信息未提供
- 年份：2025；主题：Systems；优先级：重要

- [来源](#)

论文: Entropic Time Schedulers for Generative Diffusion Models

快速理解: The practical performance of generative diffusion models depends on the appropriate choice of the noise scheduling function, which can also be equivalently expressed as a time reparameterization. In this paper, we present a time scheduler that selects sampling points based on entropy rather than uniform time spacing, ensuring that each point contributes an equal amount of information to the final generation.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.7.294 Equilibrium Policy Generalization: A Reinforcement Learning Framework for Cross-Graph Zero-Shot Generalization in Pursuit-Evasion Games

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Equilibrium Policy Generalization: A Reinforcement Learning Framework for Cross-Graph Zero-Shot Generalization in Pursuit-Evasion Games

快速理解: Equilibrium learning in adversarial games is an important topic widely examined in the fields of game theory and reinforcement learning (RL). Pursuit-evasion game (PEG), as an important class of real-world games from the fields of robotics and security, requires exponential time to be accurately solved.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.295 Evaluating Robustness of Monocular Depth Estimation with Procedural Scene Perturbations

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Evaluating Robustness of Monocular Depth Estimation with Procedural Scene Perturbations

快速理解: Recent years have witnessed substantial progress on monocular depth estimation, particularly as measured by the success of large models on standard benchmarks. However, performance on standard benchmarks does not offer a complete assessment, because most evaluate accuracy but not robustness.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.296 ExGra-Med: Extended Context Graph Alignment for Medical Vision-Language Models

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要

- [来源](#)

论文: ExGra-Med: Extended Context Graph Alignment for Medical Vision-Language Models

快速理解: State-of-the-art medical multi-modal LLMs (med-MLLMs), such as LLAVA-MED and BIOMEDGPT, primarily depend on scaling model size and data volume, with training driven largely by autoregressive objectives. However, we reveal that this approach can lead to weak vision-language alignment, making these models overly dependent on costly instruction-following data.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.297 ExPO: Unlocking Hard Reasoning with Self-Explanation-Guided Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: ExPO: Unlocking Hard Reasoning with Self-Explanation-Guided Reinforcement Learning

快速理解: Self-improvement via RL often fails on complex reasoning tasks because GRPOstyle post-training methods rely on the model's initial ability to generate positive samples. Without guided exploration, these approaches merely reinforce what the model already knows (distribution-sharpening) rather than enabling the model to solve problems where it initially generates no correct solutions.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.298 Explainable Reinforcement Learning from Human Feedback to Improve Alignment

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Explainable Reinforcement Learning from Human Feedback to Improve Alignment

快速理解: A common and effective strategy for humans to improve an unsatisfactory outcome in daily life is to find a cause of this outcome and correct the cause. In this paper, we investigate whether this human improvement strategy can be applied to improving reinforcement learning from human feedback (RLHF) for alignment of language models (LMs).

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.299 Explainably Safe Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Explainably Safe Reinforcement Learning

快速理解: Trust in a decision-making system requires both safety guarantees and the ability to interpret and understand its behavior. This is particularly important for learned systems, whose decision-making processes are often highly opaque.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.300 Explaining Similarity in Vision-Language Encoders with Weighted Banzhaf Interactions

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Explaining Similarity in Vision-Language Encoders with Weighted Banzhaf Interactions

快速理解: Language-image pre-training (LIP) enables the development of vision-language models capable of zero-shot classification, localization, multimodal retrieval, and semantic understanding. Various explanation methods have been proposed to visualize the importance of input image-text pairs on the model's similarity outputs.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.301 Exploring Data Scaling Trends and Effects in Reinforcement Learning from Human Feedback

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Exploring Data Scaling Trends and Effects in Reinforcement Learning from Human Feedback

快速理解: Reinforcement Learning from Human Feedback (RLHF) is essential for aligning large language models (LLMs) with human preferences and values. While recent research has primarily focused on algorithmic advancements—such as reducing computational overhead or strengthening reward models to mitigate reward hacking—the critical role of prompt-data construction and its scalability has received comparatively less attention.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.302 Extremely Simple Multimodal Outlier Synthesis for Out-of-Distribution Detection and Segmentation

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Extremely Simple Multimodal Outlier Synthesis for Out-of-Distribution Detection and Segmentation

快速理解: Out-of-distribution (OOD) detection and segmentation are crucial for deploying machine

learning models in safety-critical applications such as autonomous driving and robot-assisted surgery. While prior research has primarily focused on unimodal image data, real-world applications are inherently multimodal, requiring the integration of multiple modalities for improved OOD detection.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.303 FANS: A Flatness-Aware Network Structure for Generalization in Offline Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：FANS: A Flatness-Aware Network Structure for Generalization in Offline Reinforcement Learning

快速理解：Offline reinforcement learning (RL) aims to learn optimal policies from static datasets while enhancing generalization to out-of-distribution (OOD) data. To mitigate overfitting to suboptimal behaviors in offline datasets, existing methods often relax constraints on policy and data or extract informative patterns through data-driven techniques.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.304 FOCUS: Unified Vision-Language Modeling for Interactive Editing Driven by Referential Segmentation

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：FOCUS: Unified Vision-Language Modeling for Interactive Editing Driven by Referential Segmentation

快速理解：Recent Large Vision Language Models (LVLMs) demonstrate promising capabilities in unifying visual understanding and generative modeling, enabling both accurate content understanding and flexible editing. However, current approaches treat “what to see” and “how to edit” separately: they either perform isolated object segmentation or utilize segmentation masks merely as conditional prompts for local edit generation tasks, often relying on multiple disjointed models.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.305 FairDICE: Fairness-Driven Offline Multi-Objective Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：FairDICE: Fairness-Driven Offline Multi-Objective Reinforcement Learning

快速理解：Multi-objective reinforcement learning (MORL) aims to optimize policies in the presence of

conflicting objectives, where linear scalarization is commonly used to reduce vector-valued returns into scalar signals. While effective for certain preferences, this approach cannot capture fairness-oriented goals such as Nash social welfare or max-min fairness, which require nonlinear and non-additive trade-offs.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.306 Faithful Dynamic Imitation Learning from Human Intervention with Dynamic Regret Minimization

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Faithful Dynamic Imitation Learning from Human Intervention with Dynamic Regret Minimization

快速理解：Human-in-the-loop (HIL) imitation learning enables agents to learn complex behaviors safely through real-time human intervention. However, existing methods struggle to efficiently leverage agent-generated data due to dynamically evolving trajectory distributions and imperfections caused by human intervention delays, often failing to faithfully imitate the human expert policy.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.307 Fast-Slow Thinking GRPO for Large Vision-Language Model Reasoning

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Fast-Slow Thinking GRPO for Large Vision-Language Model Reasoning

快速理解：When applying reinforcement learning—typically through GRPO—to large visionlanguage model reasoning struggles to effectively scale reasoning length or generates verbose outputs across all tasks with only marginal gains in accuracy. To address this issue, we present FAST-GRPO, a variant of GRPO that dynamically adapts reasoning depth based on question characteristics.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.308 Fast-in-Slow: A Dual-System VLA Model Unifying Fast Manipulation within Slow Reasoning

- 作者：作者信息未提供
- 年份：2025；主题：VLA；优先级：重要
- [来源](#)

论文：Fast-in-Slow: A Dual-System VLA Model Unifying Fast Manipulation within Slow Reasoning

快速理解：Generalized policy and execution efficiency constitute the two critical challenges in robotic manipulation. While recent foundation policies benefit from the commonsense reasoning capabilities

of internet-scale pretrained vision-language models (VLMs), they often suffer from low execution frequency.

关注点：视觉、语言条件与机器人动作策略之间的对齐。

证据层级：本地 PDF 前页。

1.3.7.309 Few-Shot Learning from Gigapixel Images via Hierarchical Vision-Language Alignment and Modeling

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Few-Shot Learning from Gigapixel Images via Hierarchical Vision-Language Alignment and Modeling

快速理解：Vision-language models (VLMs) have recently been integrated into multiple instance learning (MIL) frameworks to address the challenge of few-shot, weakly supervised classification of whole slide images (WSIs). A key trend involves leveraging multi-scale information to better represent hierarchical tissue structures.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.310 Fin3R: Fine-tuning Feed-forward 3D Reconstruction Models via Monocular Knowledge Distillation

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Fin3R: Fine-tuning Feed-forward 3D Reconstruction Models via Monocular Knowledge Distillation

快速理解：We present Fin3R, a simple, effective, and general fine-tuning method for feedforward 3D reconstruction models. The family of feed-forward reconstruction model regresses pointmap of all input images to a reference frame coordinate system, along with other auxiliary outputs, in a single forward pass.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.311 FineRS: Fine-grained Reasoning and Segmentation of Small Objects with Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：FineRS: Fine-grained Reasoning and Segmentation of Small Objects with Reinforcement Learning

快速理解：Multi-modal Large Language Models (MLLMs) have shown remarkable capabilities across

a wide range of vision-language tasks. However, due to the restricted input resolutions, MLLMs face significant challenges in precisely understanding and localizing visual details in high-resolution images—particularly when dealing with extra-small objects embedded in cluttered contexts.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.312 Finite-Sample Analysis of Policy Evaluation for Robust Average Reward Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Finite-Sample Analysis of Policy Evaluation for Robust Average Reward Reinforcement Learning

快速理解：We present the first finite-sample analysis of policy evaluation in robust averagereward Markov Decision Processes (MDPs). Prior work in this setting have established only asymptotic convergence guarantees, leaving open the question of sample complexity.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.313 First Attentions Last: Better Exploiting First Attentions for Efficient Parallel Training

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：First Attentions Last: Better Exploiting First Attentions for Efficient Parallel Training

快速理解：As training billion-scale transformers becomes increasingly common, employing multiple distributed GPUs along with parallel training methods has become a standard practice. However, existing transformer designs suffer from significant communication overhead, especially in Tensor Parallelism (TP), where each block's MHA-MLP connection requires an all-reduce communication.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.7.314 Fit the Distribution: Cross-Image/Prompt Adversarial Attacks on Multimodal Large Language Models

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Fit the Distribution: Cross-Image/Prompt Adversarial Attacks on Multimodal Large Language Models

快速理解：Although Multimodal Large Language Models (MLLMs) have demonstrated remarkable achievements in recent years, they remain vulnerable to adversarial examples that result in harmful

responses. Existing attacks typically focus on optimizing adversarial perturbations for a certain multimodal image-prompt pair or fixed training dataset, which often leads to overfitting.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.315 Flex-Judge: Text-Only Reasoning Unleashes Zero-Shot Multimodal Evaluators

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Flex-Judge: Text-Only Reasoning Unleashes Zero-Shot Multimodal Evaluators

快速理解：Human-generated reward signals are critical for aligning generative models with human preferences, guiding both training and inference-time evaluations. While large language models (LLMs) employed as proxy evaluators, i.e., LLM-as-a-Judge, significantly reduce the costs associated with manual annotations, they typically require extensive modality-specific training data and fail to generalize well across diverse multimodal tasks.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.316 FlexAC: Towards Flexible Control of Associative Reasoning in Multimodal Large Language Models

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：FlexAC: Towards Flexible Control of Associative Reasoning in Multimodal Large Language Models

快速理解：Multimodal large language models (MLLMs) face an inherent trade-off between faithfulness and creativity, as different tasks require varying degrees of associative reasoning. However, existing methods lack the flexibility to modulate this reasoning strength, limiting MLLMs' adaptability across factual and creative scenarios.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.317 FlexWorld: Progressively Expanding 3D Scenes for Flexible-View Exploration

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：FlexWorld: Progressively Expanding 3D Scenes for Flexible-View Exploration

快速理解：Generating flexible-view 3D scenes, including 360° rotation and zooming, from single images is challenging due to a lack of 3D data. To this end, we introduce FlexWorld, a novel framework

that progressively constructs a persistent 3D Gaussian splatting representation by synthesizing and integrating new 3D content.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.318 Flow-Based Policy for Online Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Flow-Based Policy for Online Reinforcement Learning

快速理解：We present FlowRL, a novel framework for online reinforcement learning that integrates flow-based policy representation with Wasserstein-2-regularized optimization. We argue that in addition to training signals, enhancing the expressiveness of the policy class is crucial for the performance gains in RL.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.319 FlowCut: Rethinking Redundancy via Information Flow for Efficient Vision-Language Models

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：FlowCut: Rethinking Redundancy via Information Flow for Efficient Vision-Language Models

快速理解：Large vision-language models (LVLMs) excel at multimodal understanding but suffer from high computational costs due to redundant vision tokens. Existing pruning methods typically rely on single-layer attention scores to rank and prune redundant visual tokens to solve this inefficiency.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.320 FlowMo: Variance-Based Flow Guidance for Coherent Motion in Video Generation

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：FlowMo: Variance-Based Flow Guidance for Coherent Motion in Video Generation

快速理解：Text-to-video diffusion models are notoriously limited in their ability to model temporal aspects such as motion, physics, and dynamic interactions. Existing approaches address this limitation by retraining the model or introducing external conditioning signals to enforce temporal consistency.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.321 FlyLoRA: Boosting Task Decoupling and Parameter Efficiency via Implicit Rank-Wise Mixture-of-Experts

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文: FlyLoRA: Boosting Task Decoupling and Parameter Efficiency via Implicit Rank-Wise Mixture-of-Experts

快速理解: Low-Rank Adaptation (LoRA) is a widely used parameter-efficient fine-tuning method for foundation models, but it suffers from parameter interference, resulting in suboptimal performance. Although Mixture-of-Experts (MoE)-based LoRA variants show promise in mitigating intra-task correlations in single-task instruction tuning, they introduce additional router parameters and remain ineffective in multi-task model merging where inter-task interference arises.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.7.322 Force Prompting: Video Generation Models Can Learn And Generalize Physics-based Control Signals

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文: Force Prompting: Video Generation Models Can Learn And Generalize Physics-based Control Signals

快速理解: Recent advances in video generation models have sparked interest in world models capable of simulating realistic environments. While navigation has been wellexplored, physically meaningful interactions that mimic real-world forces remain largely understudied.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.323 ForceVLA: Enhancing VLA Models with a Force-aware MoE for Contact-rich Manipulation

- 作者：作者信息未提供
- 年份：2025；主题：VLA；优先级：重要
- [来源](#)

论文: ForceVLA: Enhancing VLA Models with a Force-aware MoE for Contact-rich Manipulation

快速理解: Vision-Language-Action (VLA) models have advanced general-purpose robotic manipulation by leveraging pretrained visual and linguistic representations. However, they struggle with contact-rich tasks that require fine-grained control involving force, especially under visual occlusion or dynamic uncertainty.

关注点：视觉、语言条件与机器人动作策略之间的对齐。

证据层级：本地 PDF 前页。

1.3.7.324 Forecasting in Offline Reinforcement Learning for Non-stationary Environments

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Forecasting in Offline Reinforcement Learning for Non-stationary Environments

快速理解：Offline Reinforcement Learning (RL) provides a promising avenue for training policies from pre-collected datasets when gathering additional interaction data is infeasible. However, existing offline RL methods often assume stationarity or only consider synthetic perturbations at test time, assumptions that often fail in real-world scenarios characterized by abrupt, time-varying offsets.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.325 Foresight: Adaptive Layer Reuse for Accelerated and High-Quality Text-to-Video Generation

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Foresight: Adaptive Layer Reuse for Accelerated and High-Quality Text-to-Video Generation

快速理解：Diffusion Transformers (DiTs) achieve state-of-the-art results in text-to-image, text-to-video generation, and editing. However, their large model size and the quadratic cost of spatial-temporal attention over multiple denoising steps make video generation computationally expensive.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.326 ForgerySleuth: Empowering Multimodal Large Language Models for Image Manipulation Detection

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：ForgerySleuth: Empowering Multimodal Large Language Models for Image Manipulation Detection

快速理解：Multimodal large language models have unlocked new possibilities for various multimodal tasks. However, their potential in image manipulation detection remains unexplored.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.327 Fourier Clouds: Fast Bias Correction for Imbalanced Semi-Supervised Learning

- 作者：作者信息未提供
- 年份：2025；主题：Systems；优先级：重要
- [来源](#)

论文：Fourier Clouds: Fast Bias Correction for Imbalanced Semi-Supervised Learning

快速理解：Pseudo-label-based Semi-Supervised Learning (SSL) often suffers from classifier bias, particularly under class imbalance, as inaccurate pseudo-labels tend to exacerbate existing biases towards majority classes. Existing methods, such as CD- MAD[30], utilize simplistic reference inputs—typically uniform or blank-colored images—to estimate and correct this bias.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.7.328 Frame In-N-Out: Unbounded Controllable Image-to-Video Generation

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Frame In-N-Out: Unbounded Controllable Image-to-Video Generation

快速理解：Controllability, temporal coherence, and detail synthesis remain the most critical challenges in video generation. In this paper, we focus on a commonly used yet underexplored cinematic technique known as Frame In and Frame Out.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.329 FreqPolicy: Efficient Flow-based Visuomotor Policy via Frequency Consistency

- 作者：作者信息未提供
- 年份：2025；主题：Robot Learning；优先级：重要
- [来源](#)

论文：FreqPolicy: Efficient Flow-based Visuomotor Policy via Frequency Consistency

快速理解：Generative modeling-based visuomotor policies have been widely adopted in robotic manipulation, attributed to their ability to model multimodal action distributions. However, the high inference cost of multi-step sampling limits its applicability in real-time robotic systems.

关注点：机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：本地 PDF 前页。

1.3.7.330 FreqPolicy: Frequency Autoregressive Visuomotor Policy with Continuous Tokens

- 作者：作者信息未提供
- 年份：2025；主题：Robot Learning；优先级：重要
- [来源](#)

论文：FreqPolicy: Frequency Autoregressive Visuomotor Policy with Continuous Tokens

快速理解: Learning effective visuomotor policies for robotic manipulation is challenging, as it requires generating precise actions while maintaining computational efficiency. Existing methods remain unsatisfactory due to inherent limitations in the essential action representation and the basic network architectures.

关注点: 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 本地 PDF 前页。

1.3.7.331 From Experts to a Generalist: Toward General Whole-Body Control for Humanoid Robots

- 作者: 作者信息未提供
- 年份: 2025; 主题: Embodied/WBC; 优先级: 重要
- [来源](#)

论文: From Experts to a Generalist: Toward General Whole-Body Control for Humanoid Robots

快速理解: Achieving general agile whole-body control on humanoid robots remains a major challenge due to diverse motion demands and data conflicts. While existing frameworks excel in training single motion-specific policies, they struggle to generalize across highly varied behaviors due to conflicting control requirements and mismatched data distributions.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制。

证据层级: 本地 PDF 前页。

1.3.7.332 From Forecasting to Planning: Policy World Model for Collaborative State-Action Prediction

- 作者: 作者信息未提供
- 年份: 2025; 主题: World Model; 优先级: 重要
- [来源](#)

论文: From Forecasting to Planning: Policy World Model for Collaborative State-Action Prediction

快速理解: Despite remarkable progress in driving world models, their potential for autonomous systems remains largely untapped: the world models are mostly learned for world simulation and decoupled from trajectory planning. While recent efforts aim to unify world modeling and planning in a single framework, the synergistic facilitation mechanism of world modeling for planning still requires further exploration.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策。

证据层级: 本地 PDF 前页。

1.3.7.333 From Human Attention to Diagnosis: Semantic Patch-Level Integration of Vision-Language Models in Medical Imaging

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: From Human Attention to Diagnosis: Semantic Patch-Level Integration of Vision-Language Models in Medical Imaging

快速理解: Predicting human eye movements during goal-directed visual search is critical for enhancing interactive AI systems. In medical imaging, such prediction can support radiologists in interpreting complex data, such as chest X-rays.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.334 From Pose to Muscle: Multimodal Learning for Piano Hand Muscle Electromyography

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: From Pose to Muscle: Multimodal Learning for Piano Hand Muscle Electromyography

快速理解: Muscle coordination is fundamental when humans interact with the world. Reliable estimation of hand muscle engagement can serve as a source of internal feedback, supporting the development of embodied intelligence and the acquisition of dexterous skills.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.335 Fully Spiking Neural Networks for Unified Frame-Event Object Tracking

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Fully Spiking Neural Networks for Unified Frame-Event Object Tracking

快速理解: The integration of image and event streams offers a promising approach for achieving robust visual object tracking in complex environments. However, current fusion methods achieve high performance at the cost of significant computational overhead and struggle to efficiently extract the sparse, asynchronous information from event streams, failing to leverage the energy-efficient advantages of event-driven spiking paradigms.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.336 Fuz-RL: A Fuzzy-Guided Robust Framework for Safe Reinforcement Learning under Uncertainty

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Fuz-RL: A Fuzzy-Guided Robust Framework for Safe Reinforcement Learning under Uncertainty

快速理解: Safe Reinforcement Learning (RL) is crucial for achieving high performance while ensuring safety in real-world applications. However, the complex interplay of multiple uncertainty sources in

real environments poses significant challenges for interpretable risk assessment and robust decision-making.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.337 G-Memory: Tracing Hierarchical Memory for Multi-Agent Systems

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：G-Memory: Tracing Hierarchical Memory for Multi-Agent Systems

快速理解：Large language model (LLM)-powered multi-agent systems (MAS) have demonstrated cognitive and execution capabilities that far exceed those of single LLM agents, yet their capacity for self-evolution remains hampered by underdeveloped memory architectures. Upon close inspection, we are alarmed to discover that prevailing MAS memory mechanisms (1) are overly simplistic, completely disregarding the nuanced inter-agent collaboration trajectories, and (2) lack cross-trial and agent-specific customization, in stark contrast to the expressive memory developed for single agents.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.338 GRE Suite: Geo-localization Inference via Fine-Tuned Vision-Language Models and Enhanced Reasoning Chains

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：GRE Suite: Geo-localization Inference via Fine-Tuned Vision-Language Models and Enhanced Reasoning Chains

快速理解：Recent advances in Visual Language Models (VLMs) have demonstrated exceptional performance in visual reasoning tasks. However, geo-localization presents unique challenges, requiring the extraction of multigranular visual cues from images and their integration with external world knowledge for systematic reasoning.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.339 GUARDIAN: Safeguarding LLM Multi-Agent Collaborations with Temporal Graph Modeling

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：GUARDIAN: Safeguarding LLM Multi-Agent Collaborations with Temporal Graph Modeling

快速理解：The emergence of large language models (LLMs) enables the development of intelligent

agents capable of engaging in complex and multi-turn dialogues. However, multi-agent collaboration faces critical safety challenges, such as hallucination amplification and error injection and propagation.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.340 GUI Exploration Lab: Enhancing Screen Navigation in Agents via Multi-Turn Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：GUI Exploration Lab: Enhancing Screen Navigation in Agents via Multi-Turn Reinforcement Learning

快速理解：With the rapid development of Large Vision Language Models, the focus of Graphical User Interface (GUI) agent tasks shifts from single-screen tasks to complex screen navigation challenges. However, real-world GUI environments, such as PC software and mobile Apps, are often complex and proprietary, making it difficult to obtain the comprehensive environment information needed for agent training and evaluation.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.341 GUI-Reflection: Empowering Multimodal GUI Models with Self-Reflection Behavior

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：GUI-Reflection: Empowering Multimodal GUI Models with Self-Reflection Behavior

快速理解：Multimodal Large Language Models (MLLMs) have shown great potential in revolutionizing Graphical User Interface (GUI) automation. However, existing GUI models mostly rely on learning from nearly error-free offline trajectories, thus lacking reflection and error recovery capabilities.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.342 GVPO: Group Variance Policy Optimization for Large Language Model Post-Training

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：GVPO: Group Variance Policy Optimization for Large Language Model Post-Training

快速理解：Post-training plays a crucial role in refining and aligning large language models to meet specific tasks and human preferences. While recent advancements in posttraining techniques, such

as Group Relative Policy Optimization (GRPO), leverage increased sampling with relative reward scoring to achieve superior performance, these methods often suffer from training instability that limits their practical adoption.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.343 GauDP: Reinventing Multi-Agent Collaboration through Gaussian-Image Synergy in Diffusion Policies

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：GauDP: Reinventing Multi-Agent Collaboration through Gaussian-Image Synergy in Diffusion Policies

快速理解：Recently, effective coordination in embodied multi-agent systems remains a fundamental challenge—particularly in scenarios where agents must balance individual perspectives with global environmental awareness. Existing approaches often struggle to balance fine-grained local control with comprehensive scene understanding, resulting in limited scalability and compromised collaboration quality.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.344 GeGS-PCR: Fast and Robust Color 3D Point Cloud Registration with Two-Stage Geometric-3DGS Fusion

- 作者：作者信息未提供
- 年份：2025；主题：Systems；优先级：重要
- [来源](#)

论文：GeGS-PCR: Fast and Robust Color 3D Point Cloud Registration with Two-Stage Geometric-3DGS Fusion

快速理解：We address the challenge of point cloud registration using color information, where traditional methods relying solely on geometric features often struggle in lowoverlap and incomplete scenarios. To overcome these limitations, we propose GeGS-PCR, a novel two-stage method that combines geometric, color, and Gaussian information for robust registration.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.7.345 GenPO: Generative Diffusion Models Meet On-Policy Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：GenPO: Generative Diffusion Models Meet On-Policy Reinforcement Learning

快速理解: Recent advances in reinforcement learning (RL) have demonstrated the powerful exploration capabilities and multimodality of generative diffusion-based policies. While substantial progress has been made in offline RL and off-policy RL settings, integrating diffusion policies into on-policy frameworks like PPO remains underexplored.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.346 Generalized Contrastive Learning for Universal Multimodal Retrieval

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Generalized Contrastive Learning for Universal Multimodal Retrieval

快速理解: Despite their consistent performance improvements, cross-modal retrieval models (e.g., CLIP) show degraded performances with retrieving keys composed of fused image-text modality (e.g., Wikipedia pages with both images and text). To address this critical challenge, multimodal retrieval has been recently explored to develop a unified single retrieval model capable of retrieving keys across diverse modality combinations.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.347 Generate, but Verify: Reducing Hallucination in Vision-Language Models with Retrospective Resampling

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Generate, but Verify: Reducing Hallucination in Vision-Language Models with Retrospective Resampling

快速理解: Vision-Language Models (VLMs) excel at visual understanding but often suffer from visual hallucinations, where they generate descriptions of nonexistent objects, actions, or concepts, posing significant risks in safety-critical applications. Existing hallucination mitigation methods typically follow one of two paradigms: generation adjustment, which modifies decoding behavior to align text with visual inputs, and post-hoc verification, where external models assess and correct outputs.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.348 Generating and Checking DNN Verification Proofs

- 作者: 作者信息未提供
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Generating and Checking DNN Verification Proofs

快速理解: Deep Neural Networks (DNN) have emerged as an effective approach to implementing

challenging subproblems. They are increasingly being used as components in critical transportation, medical, and military systems.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.7.349 Genesis: Multimodal Driving Scene Generation with Spatio-Temporal and Cross-Modal Consistency

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Genesis: Multimodal Driving Scene Generation with Spatio-Temporal and Cross-Modal Consistency

快速理解：We present Genesis, a unified world model for joint generation of multi-view driving videos and LiDAR sequences with spatio-temporal and cross-modal consistency. Genesis employs a two-stage architecture that integrates a DiT-based video diffusion model with 3D-VAE encoding, and a BEV-represented LiDAR generator with NeRF-based rendering and adaptive sampling.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.350 GeoLLaVA-8K: Scaling Remote-Sensing Multimodal Large Language Models to 8K Resolution

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：GeoLLaVA-8K: Scaling Remote-Sensing Multimodal Large Language Models to 8K Resolution

快速理解：Ultra-high-resolution (UHR) remote sensing (RS) imagery offers valuable data for Earth observation but pose challenges for existing multimodal foundation models due to two key bottlenecks: (1) limited availability of UHR training data, and (2) token explosion caused by the large image size. To address data scarcity, we introduce SuperRS-VQA (avg.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.351 GeoVideo: Introducing Geometric Regularization into Video Generation Model

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：GeoVideo: Introducing Geometric Regularization into Video Generation Model

快速理解：Recent advances in video generation have enabled the synthesis of high-quality and visually realistic clips using diffusion transformer models. However, most existing approaches operate purely in the 2D pixel space and lack explicit mechanisms for modeling 3D structures, often resulting in

temporally inconsistent geometries, implausible motions, and structural artifacts.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.352 Glance2Gaze: Efficient Vision-Language Models from Glance Fusion to Gaze Compression

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Glance2Gaze: Efficient Vision-Language Models from Glance Fusion to Gaze Compression

快速理解：Vision-language models heavily rely on visual representations, yet ensuring its efficiency remains a critical challenge. Most existing approaches focus on reducing visual tokens either at the visual encoder phase or during the LLM decoder stage.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.353 Globally Optimal Policy Gradient Algorithms for Reinforcement Learning with PID Control Policies

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Globally Optimal Policy Gradient Algorithms for Reinforcement Learning with PID Control Policies

快速理解：We develop policy gradient algorithms with global optimality and convergence guarantees for reinforcement learning (RL) with proportional-integral-derivative (PID) parameterized control policies. RL enables learning control policies through direct interaction with a system, without explicit model knowledge that is typically assumed in classical control.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.354 GoalLadder: Incremental Goal Discovery with Vision-Language Models

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：GoalLadder: Incremental Goal Discovery with Vision-Language Models

快速理解：Natural language can offer a concise and human-interpretable means of specifying reinforcement learning (RL) tasks. The ability to extract rewards from a language instruction can enable the development of robotic systems that can learn from human guidance; however, it remains a challenging problem, especially in visual environments.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.355 Graphs Help Graphs: Multi-Agent Graph Socialized Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Graphs Help Graphs: Multi-Agent Graph Socialized Learning

快速理解：Graphs in the real world are fragmented and dynamic, lacking collaboration akin to that observed in human societies. Existing paradigms present collaborative information collapse and forgetting, making collaborative relationships poorly autonomous and interactive information insufficient.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.356 Ground-Compose-Reinforce: Grounding Language in Agentic Behaviours using Limited Data

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Ground-Compose-Reinforce: Grounding Language in Agentic Behaviours using Limited Data

快速理解：Grounding language in perception and action is a key challenge when building situated agents that can interact with humans, or other agents, via language. In the past, addressing this challenge has required manually designing the language grounding or curating massive datasets that associate language with the environment.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.357 Grounded Reinforcement Learning for Visual Reasoning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Grounded Reinforcement Learning for Visual Reasoning

快速理解：While reinforcement learning (RL) over chains of thought has significantly advanced language models in tasks such as mathematics and coding, visual reasoning introduces added complexity by requiring models to direct visual attention, interpret perceptual inputs, and ground abstract reasoning in spatial evidence. We introduce ViGoRL (Visually Grounded Reinforcement Learning), a vision-language model trained with RL to explicitly anchor each reasoning step to specific visual coordinates.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.358 Group-in-Group Policy Optimization for LLM Agent Training

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Group-in-Group Policy Optimization for LLM Agent Training

快速理解：Recent advances in group-based reinforcement learning (RL) have driven frontier large language models (LLMs) in single-turn tasks like mathematical reasoning. However, their scalability to multi-turn LLM agent training remains limited.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.359 Guiding LLM Decision-Making with Fairness Reward Models

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Guiding LLM Decision-Making with Fairness Reward Models

快速理解：Large language models are increasingly used to support high-stakes decisions, potentially influencing who is granted bail or receives a loan. Naive chain-of-thought sampling can improve average decision accuracy, but has also been shown to amplify unfair bias.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.360 HBLLM: Wavelet-Enhanced High-Fidelity 1-Bit Quantization for LLMs

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：HBLLM: Wavelet-Enhanced High-Fidelity 1-Bit Quantization for LLMs

快速理解：We introduce HBLLM, a wavelet-enhanced high-fidelity 1-bit post-training quantization method for Large Language Models (LLMs). By leveraging Haar wavelet transforms to enhance expressive capacity through frequency decomposition, HBLLM significantly improves quantization fidelity while maintaining minimal overhead.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.7.361 HCRMP: An LLM-Hinted Contextual Reinforcement Learning Framework for Autonomous Driving

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：HCRMP: An LLM-Hinted Contextual Reinforcement Learning Framework for Autonomous

Driving

快速理解: Integrating the understanding and reasoning capabilities of Large Language Models (LLM) with the self-learning capabilities of Reinforcement Learning (RL) enables more reliable driving performance under complex driving conditions. There has been a lot of work exploring LLM-Dominated RL methods in the field of autonomous driving motion planning.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.362 HMARL-CBF -Hierarchical Multi-Agent Reinforcement Learning with Control Barrier Functions for Safety-Critical Autonomous Systems

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: HMARL-CBF -Hierarchical Multi-Agent Reinforcement Learning with Control Barrier Functions for Safety-Critical Autonomous Systems

快速理解: We address the problem of safe policy learning in multi-agent safety-critical autonomous systems. In such systems, it is necessary for each agent to meet the safety requirements at all times while also cooperating with other agents to accomplish the task.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.363 H MVLM:Human Motion-Vision-Language Model via MoE LoRA

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: H MVLM:Human Motion-Vision-Language Model via MoE LoRA

快速理解: The expansion of instruction-tuning data has enabled foundation language models to exhibit improved instruction adherence and superior performance across diverse downstream tasks. Semantically-rich 3D human motion is being progressively integrated with these foundation models to enhance multimodal understanding and cross-modal generation capabilities.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.364 HQA-VLAttack: Towards High Quality Adversarial Attack on Vision-Language Pre-Trained Models

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: HQA-VLAttack: Towards High Quality Adversarial Attack on Vision-Language Pre-Trained Models

快速理解: Black-box adversarial attack on vision-language pre-trained models is a practical and challenging task, as text and image perturbations need to be considered simultaneously, and only the predicted results are accessible. Research on this problem is in its infancy, and only a handful of methods are available.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.365 Hallucination at a Glance: Controlled Visual Edits and Fine-Grained Multimodal Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Hallucination at a Glance: Controlled Visual Edits and Fine-Grained Multimodal Learning

快速理解: Multimodal large language models (MLLMs) have achieved strong performance on vision-language tasks but still struggle with fine-grained visual differences, leading to hallucinations or missed semantic shifts. We attribute this to limitations in both training data and learning objectives.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.366 Hawaii: Hierarchical Visual Knowledge Transfer for Efficient Vision-Language Models

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Hawaii: Hierarchical Visual Knowledge Transfer for Efficient Vision-Language Models

快速理解: Improving the visual understanding ability of vision-language models (VLMs) is crucial for enhancing their performance across various tasks. While using multiple pretrained visual experts has shown great promise, it often incurs significant computational costs during training and inference.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.367 Head Pursuit: Probing Attention Specialization in Multimodal Transformers

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Head Pursuit: Probing Attention Specialization in Multimodal Transformers

快速理解: Language and vision-language models have shown impressive performance across a wide range of tasks, but their internal mechanisms remain only partly understood. In this work, we study how individual attention heads in text-generative models specialize in specific semantic or visual attributes.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.368 HermesFlow: Seamlessly Closing the Gap in Multimodal Understanding and Generation

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：HermesFlow: Seamlessly Closing the Gap in Multimodal Understanding and Generation

快速理解：The remarkable success of the autoregressive paradigm has made significant advancement in Multimodal Large Language Models (MLLMs), with powerful models like Show-o, Transfusion and Emu3 achieving notable progress in unified image understanding and generation. For the first time, we uncover a common phenomenon: the understanding capabilities of MLLMs are typically stronger than their generative capabilities, with a significant gap between the two.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.369 Heterogeneous Graph Transformers for Simultaneous Mobile Multi-Robot Task Allocation and Scheduling under Temporal Constraints

- 作者：作者信息未提供
- 年份：2025；主题：Systems；优先级：重要
- [来源](#)

论文：Heterogeneous Graph Transformers for Simultaneous Mobile Multi-Robot Task Allocation and Scheduling under Temporal Constraints

快速理解：Coordinating large teams of heterogeneous mobile agents to perform complex tasks efficiently has scalability bottlenecks in feasible and optimal task scheduling, with critical applications in logistics, manufacturing, and disaster response. Existing task allocation and scheduling methods, including heuristics and optimizationbased solvers, often fail to scale and overlook inter-task dependencies and agent heterogeneity.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.7.370 HiMoLE: Towards OOD-Robust LoRA via Hierarchical Mixture of Experts

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：HiMoLE: Towards OOD-Robust LoRA via Hierarchical Mixture of Experts

快速理解：Parameter-efficient fine-tuning (PEFT) methods, such as LoRA, have enabled the efficient adaptation of large language models (LLMs) by updating only a small subset of parameters. However, their robustness under out-of-distribution (OOD) conditions remains insufficiently studied.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.7.371 HiPoNet: A Multi-View Simplicial Complex Network for High Dimensional Point-Cloud and Single-Cell data

- 作者：作者信息未提供
- 年份：2025；主题：Systems；优先级：重要
- [来源](#)

论文：HiPoNet: A Multi-View Simplicial Complex Network for High Dimensional Point-Cloud and Single-Cell data

快速理解：In this paper, we propose HiPoNet, an end-to-end differentiable neural network for regression, classification, and representation learning on high-dimensional point clouds. Our work is motivated by single-cell data which can have very high-dimensionality –exceeding the capabilities of existing methods for point clouds which are mostly tailored for 3D data.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.7.372 Hierarchical Fine-grained Preference Optimization for Physically Plausible Video Generation

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Hierarchical Fine-grained Preference Optimization for Physically Plausible Video Generation

快速理解：Recent advancements in video generation have enabled the creation of high-quality, visually compelling videos. However, generating videos that adhere to the laws of physics remains a critical challenge for applications requiring realism and accuracy.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.373 Hierarchical Information Aggregation for Incomplete Multimodal Alzheimer' s Disease Diagnosis

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Hierarchical Information Aggregation for Incomplete Multimodal Alzheimer's Disease Diagnosis

快速理解：Alzheimer' s Disease (AD) poses a significant health threat to the aging population, underscoring the critical need for early diagnosis to delay disease progression and improve patient quality of life. Recent advances in heterogeneous multimodal artificial intelligence (AI) have facilitated comprehensive joint diagnosis, yet practical clinical scenarios frequently encounter incomplete modalities due to factors like high acquisition costs or radiation risks.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.374 Hierarchical Semantic-Augmented Navigation: Optimal Transport and Graph-Driven Reasoning for Vision-Language Navigation

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Hierarchical Semantic-Augmented Navigation: Optimal Transport and Graph-Driven Reasoning for Vision-Language Navigation

快速理解：Vision-Language Navigation in Continuous Environments (VLN-CE) poses a formidable challenge for autonomous agents, requiring seamless integration of natural language instructions and visual observations to navigate complex 3D indoor spaces. Existing approaches often falter in long-horizon tasks due to limited scene understanding, inefficient planning, and lack of robust decision-making frameworks.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.375 High-dimensional neuronal activity from low-dimensional latent dynamics: a solvable model

- 作者：作者信息未提供
- 年份：2025；主题：World Model；优先级：重要
- [来源](#)

论文：High-dimensional neuronal activity from low-dimensional latent dynamics: a solvable model

快速理解：Computation in recurrent networks of neurons has been hypothesized to occur at the level of low-dimensional latent dynamics, both in artificial systems and in the brain. This hypothesis seems at odds with evidence from large-scale neuronal recordings in mice showing that neuronal population activity is high-dimensional.

关注点：环境动力学、视频预测、潜变量状态或模型式决策。

证据层级：本地 PDF 前页。

1.3.7.376 High-order Interactions Modeling for Interpretable Multi-Agent Q-Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：High-order Interactions Modeling for Interpretable Multi-Agent Q-Learning

快速理解：The ability to model interactions among agents is crucial for effective coordination and understanding their cooperation mechanisms in multi-agent reinforcement learning (MARL). However, previous efforts to model high-order interactions have been primarily hindered by the combinatorial explosion or the opaque nature of their black-box network structures.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.377 HoPE: Hybrid of Position Embedding for Long Context Vision-Language Models

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：HoPE: Hybrid of Position Embedding for Long Context Vision-Language Models

快速理解：Vision-Language Models (VLMs) have made significant progress in multimodal tasks. However, their performance often deteriorates in long-context scenarios, particularly long videos.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.378 HoliGS: Holistic Gaussian Splatting for Embodied View Synthesis

- 作者：作者信息未提供
- 年份：2025；主题：Embodied/WBC；优先级：重要
- [来源](#)

论文：HoliGS: Holistic Gaussian Splatting for Embodied View Synthesis

快速理解：We propose HoliGS, a novel deformable Gaussian splatting framework that addresses embodied view synthesis from long monocular RGB videos. Unlike prior 4D Gaussian splatting and dynamic NeRF pipelines, which struggle with training overhead in minute-long captures, our method leverages invertible Gaussian Splatting deformation networks to reconstruct large-scale, dynamic environments accurately.

关注点：具身任务中的感知、动作生成、全身协调或运动控制。

证据层级：本地 PDF 前页。

1.3.7.379 Homogeneous Keys, Heterogeneous Values: Exploiting Local KV Cache Asymmetry for Long-Context LLMs

- 作者：作者信息未提供
- 年份：2025；主题：Systems；优先级：重要
- [来源](#)

论文：Homogeneous Keys, Heterogeneous Values: Exploiting Local KV Cache Asymmetry for Long-Context LLMs

快速理解：Recent advances in Large Language Models (LLMs) have highlighted the critical importance of extending context length, yet the quadratic complexity of attention mechanisms poses significant challenges for efficient long-context modeling. KV cache compression has emerged as a key approach to address this challenge.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.7.380 How Ensembles of Distilled Policies Improve Generalisation in Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：How Ensembles of Distilled Policies Improve Generalisation in Reinforcement Learning

快速理解：In the zero-shot policy transfer setting in reinforcement learning, the goal is to train an agent on a fixed set of training environments so that it can generalise to similar, but unseen, testing environments. Previous work has shown that policy distillation after training can sometimes produce a policy that outperforms the original in the testing environments.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.381 How Many Tokens Do 3D Point Cloud Transformer Architectures Really Need?

- 作者：作者信息未提供
- 年份：2025；主题：Systems；优先级：重要
- [来源](#)

论文：How Many Tokens Do 3D Point Cloud Transformer Architectures Really Need?

快速理解：Recent advances in 3D point cloud transformers have led to state-of-the-art results in tasks such as semantic segmentation and reconstruction. However, these models typically rely on dense token representations, incurring high computational and memory costs during training and inference.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.7.382 HumanoidGen: Data Generation for Bimanual Dexterous Manipulation via LLM Reasoning

- 作者：作者信息未提供
- 年份：2025；主题：Embodied/WBC；优先级：重要
- [来源](#)

论文：HumanoidGen: Data Generation for Bimanual Dexterous Manipulation via LLM Reasoning

快速理解：For robotic manipulation, existing robotics datasets and simulation benchmarks predominantly cater to robot-arm platforms. However, for humanoid robots equipped with dual arms and dexterous hands, simulation tasks and high-quality demonstrations are notably lacking.

关注点：具身任务中的感知、动作生成、全身协调或运动控制。

证据层级：本地 PDF 前页。

1.3.7.383 Hybrid Latent Reasoning via Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文: Hybrid Latent Reasoning via Reinforcement Learning

快速理解: Recent advances in large language models (LLMs) have introduced latent reasoning as a promising alternative to autoregressive reasoning. By performing internal computation with hidden states from previous steps, latent reasoning benefit from more informative features rather than sampling a discrete chain-of-thought (CoT) path.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.384 HypRL: Reinforcement Learning of Control Policies for Hyperproperties

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: HypRL: Reinforcement Learning of Control Policies for Hyperproperties

快速理解: Reward shaping in multi-agent reinforcement learning (MARL) for complex tasks remains a significant challenge. Existing approaches often fail to find optimal solutions or cannot efficiently handle such tasks.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.385 Hyper-Modality Enhancement for Multimodal Sentiment Analysis with Missing Modalities

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Hyper-Modality Enhancement for Multimodal Sentiment Analysis with Missing Modalities

快速理解: Multimodal Sentiment Analysis (MSA) aims to infer human emotions by integrating complementary signals from diverse modalities. However, in real-world scenarios, missing modalities are common due to data corruption, sensor failure, or privacy concerns, which can significantly degrade model performance.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.386 HyperMARL: Adaptive Hypernetworks for Multi-Agent RL

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: HyperMARL: Adaptive Hypernetworks for Multi-Agent RL

快速理解: Adaptive cooperation in multi-agent reinforcement learning (MARL) requires policies to express homogeneous, specialised, or mixed behaviours, yet achieving this adaptivity remains a critical challenge. While parameter sharing (PS) is standard for efficient learning, it notoriously

suppresses the behavioural diversity required for specialisation.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.387 IOSTOM: Offline Imitation Learning from Observations via State Transition Occupancy Matching

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文:IOSTOM: Offline Imitation Learning from Observations via State Transition Occupancy Matching

快速理解：Offline Learning from Observation (LfO) focuses on enabling agents to imitate expert behavior using datasets that contain only expert state trajectories and separate transition data with suboptimal actions. This setting is both practical and critical in real-world scenarios where direct environment interaction or access to expert action labels is costly, risky, or infeasible.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.388 IllumiCraft: Unified Geometry and Illumination Diffusion for Controllable Video Generation

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文: IllumiCraft: Unified Geometry and Illumination Diffusion for Controllable Video Generation

快速理解：Although diffusion-based models can generate high-quality and high-resolution video sequences from textual or image inputs, they lack explicit integration of geometric cues when controlling scene lighting and visual appearance across frames. To address this limitation, we propose IllumiCraft, an end-to-end diffusion framework accepting three complementary inputs: (1) high-dynamic-range (HDR) video 39th Conference on Neural Information Processing Systems (NeurIPS 2025). maps for detailed lighting control; (2) synthetically relit frames with randomized illumination changes (optionally paired with a static background reference image) to provide appearance cues; and (3) 3D point tracks that capture precise 3D geometry information.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.389 Image Token Matters: Mitigating Hallucination in Discrete Tokenizer-based Large Vision-Language Models via Latent Editing

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文: Image Token Matters: Mitigating Hallucination in Discrete Tokenizer-based Large Vision-

Language Models via Latent Editing

快速理解: Large Vision-Language Models (LVLMs) with discrete image tokenizers unify multimodal representations by encoding visual inputs into a finite set of tokens. Despite their effectiveness, we find that these models still hallucinate non-existent objects.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.390 Image as a World: Generating Interactive World from Single Image via Panoramic Video Generation

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Image as a World: Generating Interactive World from Single Image via Panoramic Video Generation

快速理解: Generating an interactive visual world from a single image is both challenging and practically valuable, as single-view inputs are easy to acquire and align well with prompt-driven applications such as gaming and virtual reality. This paper introduces a novel unified framework, Image as a World (IaaW), which synthesizes high-quality 360-degree videos from a single image that are both controllable and temporally continuable.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.391 Imagine Beyond ! Distributionally Robust Autoencoding for State Space Coverage in Online Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Imagine Beyond ! Distributionally Robust Autoencoding for State Space Coverage in Online Reinforcement Learning

快速理解: Goal-Conditioned Reinforcement Learning (GCRL) enables agents to autonomously acquire diverse behaviors, but faces major challenges in visual environments due to high-dimensional, semantically sparse observations. In the online setting, where agents learn representations while exploring, the latent space evolves with the agent's policy, to capture newly discovered areas of the environment.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.392 Imagine360: Immersive 360 Video Generation from Perspective Anchor

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Imagine360: Immersive 360 Video Generation from Perspective Anchor

快速理解: 360° videos offer a hyper-immersive experience that allows the viewers to explore a dynamic scene from full 360 degrees. To achieve more accessible and personalized content creation in 360° video format, we seek to lift standard perspective videos into 360° equirectangular videos.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.393 Imitation Learning with Temporal Logic Constraints

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Imitation Learning with Temporal Logic Constraints

快速理解: Designing reinforcement learning agents to satisfy complex temporal objectives expressed in Linear Temporal Logic (LTL), presents significant challenges, particularly in ensuring sample efficiency and task alignment over infinite horizons. Recent works have shown that by leveraging the corresponding Limit Deterministic Büchi Automaton (LDBA) representation, LTL formulas can be translated into variable discounting schemes over LDBA-accepting states to maximize a lower bound on the probability of formula satisfaction.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.394 Improve Temporal Reasoning in Multimodal Large Language Models via Video Contrastive Decoding

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Improve Temporal Reasoning in Multimodal Large Language Models via Video Contrastive Decoding

快速理解: A major distinction between video and image understanding is that the former requires reasoning over time. Existing Video Large Language Models (VLLMs) demonstrate promising performance in general video understanding, such as brief captioning or object recognition within individual frames.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.395 Improving Generative Behavior Cloning via Self-Guidance and Adaptive Chunking

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Improving Generative Behavior Cloning via Self-Guidance and Adaptive Chunking

快速理解: Generative Behavior Cloning (GBC) is a simple yet effective framework for robot learning, particularly in multi-task settings. Recent GBC methods often employ diffusion policies with open-loop (OL) control, where actions are generated via a diffusion process and executed in multi-step chunks without replanning.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.396 Improving Model Representation and Reducing KV Cache via Skip Connections with First Value Heads

- 作者: 作者信息未提供
- 年份: 2025; 主题: Systems; 优先级: 重要
- [来源](#)

论文: Improving Model Representation and Reducing KV Cache via Skip Connections with First Value Heads

快速理解: Transformer models have driven breakthroughs across various language tasks by their strong capability to learn rich contextual representations. Scaling them to improve representation, however, often demands substantial memory and compute costs, such as the Key-Value (KV) cache used during auto-regressive decoding.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.7.397 Improving Model-Based Reinforcement Learning by Converging to Flatter Minima

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Improving Model-Based Reinforcement Learning by Converging to Flatter Minima

快速理解: Model-based reinforcement learning (MBRL) hinges on a learned dynamics model whose errors can compound along imagined rollouts. We study how encouraging flatness in the model's training loss affects downstream control, and show that steering optimization toward flatter minima yields a better policy.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.398 Improving Retrieval-Augmented Generation through Multi-Agent Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Improving Retrieval-Augmented Generation through Multi-Agent Reinforcement Learning

快速理解: Retrieval-augmented generation (RAG) is widely utilized to incorporate external knowledge

into large language models, thereby enhancing factuality and reducing hallucinations in question-answering (QA) tasks. A standard RAG pipeline consists of several components, such as query rewriting, document retrieval, document filtering, and answer generation.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.399 Improving Reward Models with Proximal Policy Exploration for Preference-Based Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Improving Reward Models with Proximal Policy Exploration for Preference-Based Reinforcement Learning

快速理解：Reinforcement learning (RL) heavily depends on well-designed reward functions, which are often biased and difficult to design for complex behaviors. Preferencebased RL (PbRL) addresses this by learning reward models from human feedback, but its practicality is constrained by a critical dilemma: while existing methods reduce human effort through query optimization, they neglect the preference buffer's restricted coverage—a factor that fundamentally determines the reliability of reward model.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.400 Improving Task-Specific Multimodal Sentiment Analysis with General MLLMs via Prompting

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Improving Task-Specific Multimodal Sentiment Analysis with General MLLMs via Prompting

快速理解：Multimodal Sentiment Analysis (MSA) aims to predict sentiment from diverse data types, such as video, audio, and language. Recent progress in Multimodal Large Language Models (MLLMs) have demonstrated impressive performance across various tasks.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.401 Improving Video Generation with Human Feedback

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Improving Video Generation with Human Feedback

快速理解：Video generation has achieved significant advances through rectified flow techniques, but issues like unsmooth motion and misalignment between videos and prompts persist. In this work,

we develop a systematic pipeline that harnesses human feedback to mitigate these problems and refine the video generation model.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.402 In-Context Fully Decentralized Cooperative Multi-Agent Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：In-Context Fully Decentralized Cooperative Multi-Agent Reinforcement Learning

快速理解：In this paper, we consider fully decentralized cooperative multi-agent reinforcement learning, where each agent has access only to the states, its local actions, and the shared rewards. The absence of information about other agents' actions typically leads to the non-stationarity problem during per-agent value function updates, and the relative overgeneralization issue during value function estimation.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.403 InfMasking: Unleashing Synergistic Information by Contrastive Multimodal Interactions

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：InfMasking: Unleashing Synergistic Information by Contrastive Multimodal Interactions

快速理解：In multimodal representation learning, synergistic interactions between modalities not only provide complementary information but also create unique outcomes through specific interaction patterns that no single modality could achieve alone. Existing methods may struggle to effectively capture the full spectrum of synergistic information, leading to suboptimal performance in tasks where such interactions are critical.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.404 InfantAgent-Next: A Multimodal Generalist Agent for Automated Computer Interaction

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：InfantAgent-Next: A Multimodal Generalist Agent for Automated Computer Interaction

快速理解：This paper introduces INFANTAGENT-NEXT, a generalist agent capable of interacting with computers in a multimodal manner, encompassing text, images, audio, and video. Unlike

existing approaches that either build intricate workflows around a single large model or only provide workflow modularity, our agent integrates tool-based and pure vision agents within a highly modular architecture, enabling different models to collaboratively solve decoupled tasks in a step-by-step manner.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.405 Inference of Whole Brain Electrophysiological Networks Through Multimodal Integration of Simultaneous Scalp and Intracranial EEG

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Inference of Whole Brain Electrophysiological Networks Through Multimodal Integration of Simultaneous Scalp and Intracranial EEG

快速理解：Brain imaging research has transitioned over the past decades from identifying isolated regions of task-evoked activation to characterizing the spatiotemporal dynamics of large-scale brain networks. Electrophysiological signals are the direct manifestation of brain activity; thus, characterizing whole-brain electrophysiological networks (WBEN) can serve as a fundamental tool for neuroscience studies and clinical applications.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.406 Inference-Time Hyper-Scaling with KV Cache Compression

- 作者：作者信息未提供
- 年份：2025；主题：Systems；优先级：重要
- [来源](#)

论文：Inference-Time Hyper-Scaling with KV Cache Compression

快速理解：Inference-time scaling trades efficiency for increased reasoning accuracy by generating longer or more parallel sequences. However, in Transformer LLMs, generation cost is bottlenecked by the size of the key-value (KV) cache, rather than the number of generated tokens.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.7.407 InfiniPot-V: Memory-Constrained KV Cache Compression for Streaming Video Understanding

- 作者：作者信息未提供
- 年份：2025；主题：Systems；优先级：重要
- [来源](#)

论文：InfiniPot-V: Memory-Constrained KV Cache Compression for Streaming Video Understanding

快速理解：Modern multimodal large language models (MLLMs) can reason over hour-long video, yet their key-value (KV) cache grows linearly with time—quickly exceeding the fixed memory of phones,

AR glasses, and edge robots. Prior compression schemes either assume the whole video and user query are available offline or must first build the full cache, so memory still scales with stream length.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.7.408 Interactive Cross-modal Learning for Text-3D Scene Retrieval

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Interactive Cross-modal Learning for Text-3D Scene Retrieval

快速理解：Text-3D Scene Retrieval (T3SR) aims to retrieve relevant scenes using linguistic queries. Although traditional T3SR methods have made significant progress in capturing fine-grained associations, they implicitly assume that query descriptions are information-complete.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.409 Interactive and Hybrid Imitation Learning: Provably Beating Behavior Cloning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Interactive and Hybrid Imitation Learning: Provably Beating Behavior Cloning

快速理解：Imitation learning (IL) is a paradigm for learning sequential decision-making policies from experts, leveraging offline demonstrations, interactive annotations, or both. Recent advances show that when annotation cost is tallied per trajectory, Behavior Cloning (BC)—which relies solely on offline demonstrations—cannot be improved in general, leaving limited conditions for interactive methods such as DAgger to help.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.410 Intrinsic Benefits of Categorical Distributional Loss: Uncertainty-aware Regularized Exploration in Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Intrinsic Benefits of Categorical Distributional Loss: Uncertainty-aware Regularized Exploration in Reinforcement Learning

快速理解：The remarkable empirical performance of distributional reinforcement learning (RL) has garnered increasing attention to understanding its theoretical advantages over classical RL. By decomposing the categorical distributional loss commonly employed in distributional RL, we find

that the potential superiority of distributional RL can be attributed to a derived distribution-matching entropy regularization.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.411 Intrinsic Goals for Autonomous Agents: Model-Based Exploration in Virtual Zebrafish Predicts Ethological Behavior and Whole-Brain Dynamics

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文: Intrinsic Goals for Autonomous Agents: Model-Based Exploration in Virtual Zebrafish Predicts Ethological Behavior and Whole-Brain Dynamics

快速理解: Autonomy is a hallmark of animal intelligence, enabling adaptive and intelligent behavior in complex environments without relying on external reward or task structure. Existing reinforcement learning approaches to exploration in rewardfree environments, including a class of methods known as model-based intrinsic motivation, exhibit inconsistent exploration patterns and do not converge to an exploratory policy, thus failing to capture robust autonomous behaviors observed in animals.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.412 Inverse Q-Learning Done Right: Offline Imitation Learning in $\hat{Q}^{\pi}$ -Realizable MDPs

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文: Inverse Q-Learning Done Right: Offline Imitation Learning in $\hat{Q}^{\pi}$ -Realizable MDPs

快速理解: We study the problem of offline imitation learning in Markov decision processes (MDPs), where the goal is to learn a well-performing policy given a dataset of state-action pairs generated by an expert policy. Complementing a recent line of work on this topic that assumes the expert belongs to a tractable class of known policies, we approach this problem from a new angle and leverage a different type of structural assumption about the environment.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.413 Iterative Self-Incentivization Empowers Large Language Models as Agentic Searchers

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文: Iterative Self-Incentivization Empowers Large Language Models as Agentic Searchers

快速理解: Large language models have been widely integrated into information retrieval to advance

traditional techniques. However, effectively enabling LLMs to seek accurate knowledge in complex tasks remains a challenge due to the complexity of multi-hop queries as well as the irrelevant retrieved content.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.414 Iterative Tool Usage Exploration for Multimodal Agents via Step-wise Preference Tuning

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Iterative Tool Usage Exploration for Multimodal Agents via Step-wise Preference Tuning

快速理解：Multimodal agents, which integrate a controller (e.g., a vision language model) with external tools, have demonstrated remarkable capabilities in tackling complex multimodal tasks. Existing approaches for training these agents, both supervised fine-tuning and reinforcement learning, depend on extensive human-annotated task-answer pairs and tool trajectories.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.415 JailBound: Jailbreaking Internal Safety Boundaries of Vision-Language Models

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：JailBound: Jailbreaking Internal Safety Boundaries of Vision-Language Models

快速理解：Vision-Language Models (VLMs) exhibit impressive performance, yet the integration of powerful vision encoders has significantly broadened their attack surface, rendering them increasingly susceptible to jailbreak attacks. However, lacking well-defined attack objectives, existing jailbreak methods often struggle with gradient-based strategies prone to local optima and lacking precise directional guidance, and typically decouple visual and textual modalities, thereby limiting their effectiveness by neglecting crucial cross-modal interactions.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.416 Janus-Pro-R1: Advancing Collaborative Visual Comprehension and Generation via Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Janus-Pro-R1: Advancing Collaborative Visual Comprehension and Generation via Reinforcement Learning

快速理解: Recent endeavors in Multimodal Large Language Models (MLLMs) aim to unify visual comprehension and generation. However, these two capabilities remain largely independent, as if they are two separate functions encapsulated within the same model.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.417 Jasmine: Harnessing Diffusion Prior for Self-supervised Depth Estimation

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Jasmine: Harnessing Diffusion Prior for Self-supervised Depth Estimation

快速理解: In this paper, we propose Jasmine, the first Stable Diffusion (SD)-based selfsupervised framework for monocular depth estimation, which effectively harnesses SD's visual priors to enhance the sharpness and generalization of unsupervised prediction. Previous SD-based methods are all supervised since adapting diffusion models for dense prediction requires high-precision supervision.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.418 Joint Velocity-Growth Flow Matching for Single-Cell Dynamics Modeling

- 作者: 作者信息未提供
- 年份: 2025; 主题: World Model; 优先级: 重要
- [来源](#)

论文: Joint Velocity-Growth Flow Matching for Single-Cell Dynamics Modeling

快速理解: Learning the underlying dynamics of single cells from snapshot data has gained increasing attention in scientific and machine learning research. The destructive measurement technique and cell proliferation/death result in unpaired and unbalanced data between snapshots, making the learning of the underlying dynamics challenging.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策。

证据层级: 本地 PDF 前页。

1.3.7.419 KARMA: Leveraging Multi-Agent LLMs for Automated Knowledge Graph Enrichment

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: KARMA: Leveraging Multi-Agent LLMs for Automated Knowledge Graph Enrichment

快速理解: Maintaining comprehensive and up-to-date knowledge graphs (KGs) is critical for modern AI systems, but manual curation struggles to scale with the rapid growth of scientific literature. This paper presents KARMA, a novel framework employing multi-agent large language models (LLMs) to automate KG enrichment through structured analysis of unstructured text.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.420 KVFlow: Efficient Prefix Caching for Accelerating LLM-Based Multi-Agent Workflows

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：KVFlow: Efficient Prefix Caching for Accelerating LLM-Based Multi-Agent Workflows

快速理解：Large language model (LLM) based agentic workflows have become a popular paradigm for coordinating multiple specialized agents to solve complex tasks. To improve serving efficiency, existing LLM systems employ prefix caching to reuse key-value (KV) tensors corresponding to agents' fixed prompts, thereby avoiding redundant computation across repeated invocations.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.421 KVLink: Accelerating Large Language Models via Efficient KV Cache Reuse

- 作者：作者信息未提供
- 年份：2025；主题：Systems；优先级：重要
- [来源](#)

论文：KVLink: Accelerating Large Language Models via Efficient KV Cache Reuse

快速理解：We describe KVLINK, an approach for efficient key-value (KV) cache reuse in large language models (LLMs). In many LLM applications, different inputs can share overlapping context, such as the same retrieved document appearing in multiple queries.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.7.422 KVzip: Query-Agnostic KV Cache Compression with Context Reconstruction

- 作者：作者信息未提供
- 年份：2025；主题：Systems；优先级：重要
- [来源](#)

论文：KVzip: Query-Agnostic KV Cache Compression with Context Reconstruction

快速理解：Transformer-based large language models (LLMs) cache context as key-value (KV) pairs during inference. As context length grows, KV cache sizes expand, leading to substantial memory overhead and increased attention latency.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.7.423 Know What You Don' t Know: Uncertainty Calibration of Process Reward Models

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Know What You Don' t Know: Uncertainty Calibration of Process Reward Models

快速理解：Process reward models (PRMs) play a central role in guiding inference-time scaling algorithms for large language models (LLMs). However, we observe that even stateof-the-art PRMs can be poorly calibrated.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.424 Knowledge Starts with Practice: Knowledge-Aware Exercise Generative Recommendation with Adaptive Multi-Agent Cooperation

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Knowledge Starts with Practice: Knowledge-Aware Exercise Generative Recommendation with Adaptive Multi-Agent Cooperation

快速理解：Adaptive learning, which requires the in-depth understanding of students' learning processes and rational planning of learning resources, plays a crucial role in intelligent education. However, how to effectively model these two processes and seamlessly integrate them poses significant implementation challenges for adaptive learning.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.425 Knowledge-based Visual Question Answer with Multimodal Processing, Retrieval and Filtering

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Knowledge-based Visual Question Answer with Multimodal Processing, Retrieval and Filtering

快速理解：Knowledge-based visual question answering (KB-VQA) requires visual language models (VLMs) to integrate visual understanding with external knowledge retrieval. Although retrieval-augmented generation (RAG) achieves significant advances in this task by combining knowledge-base querying, it still struggles with the quality of multimodal queries and the relevance of retrieved results.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.426 KungfuBot: Physics-Based Humanoid Whole-Body Control for Learning Highly-Dynamic Skills

- 作者：作者信息未提供

- 年份: 2025; 主题: Embodied/WBC; 优先级: 重要
- [来源](#)

论文: KungfuBot: Physics-Based Humanoid Whole-Body Control for Learning Highly-Dynamic Skills

快速理解: Humanoid robots are promising to acquire various skills by imitating human behaviors. However, existing algorithms are only capable of tracking smooth, low-speed human motions, even with delicate reward and curriculum design.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制。

证据层级: 本地 PDF 前页。

1.3.7.427 LAsER: Learning to Adaptively Select Reward Models with Multi-Arm Bandits

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: LAsER: Learning to Adaptively Select Reward Models with Multi-Arm Bandits

快速理解: Reward Models (RMs) are crucial to aligning large language models (LLMs), but the degree to which an RM specialized to one task (e.g. writing) generalizes to new tasks (e.g. math) is often not known a priori, often making using only one fixed RM to train LLMs suboptimal. However, optimizing LLMs with multiple RMs simultaneously can incur a prohibitively high computational cost and lead to conflicting signals from different RMs that may degrade performance.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.428 LBMKGC: Large Model-Driven Balanced Multimodal Knowledge Graph Completion

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: LBMKGC: Large Model-Driven Balanced Multimodal Knowledge Graph Completion

快速理解: Multi-modal Knowledge Graph Completion (MMKGC) aims to predict missing entities, relations, or attributes in knowledge graphs by collaboratively modeling the triple structure and multimodal information (e.g., text, images, videos) associated with entities. This approach facilitates the automatic discovery of previously unobserved factual knowledge.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.429 LLM Query Scheduling with Prefix Reuse and Latency Constraints

- 作者: 作者信息未提供
- 年份: 2025; 主题: Systems; 优先级: 重要
- [来源](#)

论文: LLM Query Scheduling with Prefix Reuse and Latency Constraints

快速理解: The efficient deployment of large language models (LLMs) in online settings requires optimizing inference performance under stringent latency constraints, particularly the time-to-first-token (TTFT) and time-per-output-token (TPOT). This paper focuses on the query scheduling problem for LLM inference with prefix reuse, a technique that leverages shared prefixes across queries to reduce computational overhead.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.7.430 LLM Strategic Reasoning: Agentic Study through Behavioral Game Theory

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: LLM Strategic Reasoning: Agentic Study through Behavioral Game Theory

快速理解: What does it truly mean for a language model to “reason” strategically, and can scaling up alone guarantee intelligent, context-aware decisions? Strategic decisionmaking requires adaptive reasoning, where agents anticipate and respond to others’ actions under uncertainty.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.431 LLM-Explorer: A Plug-in Reinforcement Learning Policy Exploration Enhancement Driven by Large Language Models

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: LLM-Explorer: A Plug-in Reinforcement Learning Policy Exploration Enhancement Driven by Large Language Models

快速理解: Policy exploration is critical in reinforcement learning (RL), where existing approaches include ϵ -greedy, Gaussian process, etc. However, these approaches utilize preset stochastic processes and are indiscriminately applied in all kinds of RL tasks without considering task-specific features that influence policy exploration.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.432 LMFusion: Adapting Pretrained Language Models for Multimodal Generation

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: LMFusion: Adapting Pretrained Language Models for Multimodal Generation

快速理解: We present LMFusion, a framework for empowering pretrained text-only large language models (LLMs) with multimodal generative capabilities, enabling them to understand and generate both text and images in arbitrary sequences. LMFusion leverages existing Llama-3' s weights for processing texts autoregressively while introducing additional and parallel transformer modules for processing images with diffusion.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.433 LOMIA: Label-Only Membership Inference Attacks against Pre-trained Large Vision-Language Models

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: LOMIA: Label-Only Membership Inference Attacks against Pre-trained Large Vision-Language Models

快速理解: Large vision-language models (VLLMs) have driven significant progress in multimodal systems, enabling a wide range of applications across domains such as healthcare, education, and content generation. Despite the success, the large-scale datasets used to train these models often contain sensitive or personally identifiable information, raising serious privacy concerns.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.434 LaRes: Evolutionary Reinforcement Learning with LLM-based Adaptive Reward Search

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: LaRes: Evolutionary Reinforcement Learning with LLM-based Adaptive Reward Search

快速理解: The integration of evolutionary algorithms (EAs) with reinforcement learning (RL) has shown superior performance compared to standalone methods. However, previous research focuses on exploration in policy parameter space, while overlooking the reward function search.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.435 LaViDa: A Large Diffusion Language Model for Multimodal Understanding

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: LaViDa: A Large Diffusion Language Model for Multimodal Understanding

快速理解: Modern Vision-Language Models (VLMs) can solve a wide range of tasks requiring visual reasoning. In real-world scenarios, desirable properties for VLMs include fast inference and

controllable generation (e.g., constraining outputs to adhere to a desired format).

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.436 Latent Policy Barrier: Learning Robust Visuomotor Policies by Staying In-Distribution

- 作者: 作者信息未提供
- 年份: 2025; 主题: Robot Learning; 优先级: 重要
- [来源](#)

论文: Latent Policy Barrier: Learning Robust Visuomotor Policies by Staying In-Distribution

快速理解: Visuomotor policies trained via behavior cloning are vulnerable to covariate shift, where small deviations from expert trajectories can compound into failure. Common strategies to mitigate this issue involve expanding the training distribution through human-in-the-loop corrections or synthetic data augmentation.

关注点: 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 本地 PDF 前页。

1.3.7.437 Layer-Wise Modality Decomposition for Interpretable Multimodal Sensor Fusion

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Layer-Wise Modality Decomposition for Interpretable Multimodal Sensor Fusion

快速理解: In autonomous driving, transparency in the decision-making of perception models is critical, as even a single misperception can be catastrophic. Yet with multi-sensor inputs, it is difficult to determine how each modality contributes to a prediction because sensor information becomes entangled within the fusion network.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.438 LeMiCa: Lexicographic Minimax Path Caching for Efficient Diffusion-Based Video Generation

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: LeMiCa: Lexicographic Minimax Path Caching for Efficient Diffusion-Based Video Generation

快速理解: We present LeMiCa, a training-free and efficient acceleration framework for diffusion-based video generation. While existing caching strategies primarily focus on reducing local heuristic errors, they often overlook the accumulation of global errors, leading to noticeable content degradation between accelerated and original videos.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.439 Learned Prefix Caching for Efficient LLM Inference

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：Learned Prefix Caching for Efficient LLM Inference

快速理解：Prefix caching is a key technique for reducing Large Language Model (LLM) inference costs. However, the prevalent least-recently-used (LRU) eviction algorithm has a large gap to the optimal algorithm.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.7.440 Learning Equilibria from Data: Provably Efficient Multi-Agent Imitation Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Learning Equilibria from Data: Provably Efficient Multi-Agent Imitation Learning

快速理解：This paper provides the first expert sample complexity characterization for learning a Nash equilibrium from expert data in Markov Games. We show that a new quantity named the all policy deviation concentrability coefficient is unavoidable in the non-interactive imitation learning setting, and we provide an upper bound for behavioral cloning (BC) featuring such coefficient.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.441 Learning Human-Like RL Agents Through Trajectory Optimization With Action Quantization

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：Learning Human-Like RL Agents Through Trajectory Optimization With Action Quantization

快速理解：Human-like agents have long been one of the goals in pursuing artificial intelligence. Although reinforcement learning (RL) has achieved superhuman performance in many domains, relatively little attention has been focused on designing human-like RL agents.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.7.442 Learning Intractable Multimodal Policies with Reparameterization and Diversity Regularization

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Learning Intractable Multimodal Policies with Reparameterization and Diversity Regularization

快速理解：Traditional continuous deep reinforcement learning (RL) algorithms employ deterministic or unimodal Gaussian actors, which cannot express complex multimodal decision distributions. This limitation can hinder their performance in diversitycritical scenarios.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.443 Learning Memory-Enhanced Improvement Heuristics for Flexible Job Shop Scheduling

- 作者：作者信息未提供
- 年份：2025；主题：Systems；优先级：重要
- [来源](#)

论文：Learning Memory-Enhanced Improvement Heuristics for Flexible Job Shop Scheduling

快速理解：The rise of smart manufacturing under Industry 4.0 introduces mass customization and dynamic production, demanding more advanced and flexible scheduling techniques. The flexible job-shop scheduling problem (FJSP) has attracted significant attention due to its complex constraints and strong alignment with real-world production scenarios.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.7.444 Learning Personalized Ad Impact via Contextual Reinforcement Learning under Delayed Rewards

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Learning Personalized Ad Impact via Contextual Reinforcement Learning under Delayed Rewards

快速理解：Online advertising platforms use automated auctions to connect advertisers with potential customers, requiring effective bidding strategies to maximize profits. Accurate ad impact estimation requires considering three key factors: delayed and long-term effects, cumulative ad impacts such as reinforcement or fatigue, and customer heterogeneity.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.445 Learning Reconfigurable Representations for Multimodal Federated Learning with Missing Data

- 作者：作者信息未提供

- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Learning Reconfigurable Representations for Multimodal Federated Learning with Missing Data

快速理解: Multimodal federated learning in real-world settings often encounters incomplete and heterogeneous data across clients. This results in misaligned local feature representations that limit the effectiveness of model aggregation.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.446 Learning Robust Vision-Language Models from Natural Latent Spaces

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Learning Robust Vision-Language Models from Natural Latent Spaces

快速理解: Pre-trained vision-language models (VLMs) exhibit significant vulnerability to imperceptible adversarial perturbations. Current advanced defense strategies typically employ adversarial prompt tuning to improve the adversarial robustness of VLMs, which struggle to simultaneously maintain generalization across both natural and adversarial examples under different benchmarks and downstream tasks.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.447 Learning Sparse Approximate Inverse Preconditioners for Conjugate Gradient Solvers on GPUs

- 作者: 作者信息未提供
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Learning Sparse Approximate Inverse Preconditioners for Conjugate Gradient Solvers on GPUs

快速理解: The conjugate gradient solver (CG) is a prevalent method for solving symmetric and positive definite linear systems $Ax = b$, where effective preconditioners are crucial for fast convergence. Traditional preconditioners rely on prescribed algorithms to offer rigorous theoretical guarantees, while limiting their ability to exploit optimization from data.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 本地 PDF 前页。

1.3.7.448 Learning from Reward-Free Offline Data: A Case for Planning with Latent Dynamics Models

- 作者: 作者信息未提供
- 年份: 2025; 主题: World Model; 优先级: 重要
- [来源](#)

论文: Learning from Reward-Free Offline Data: A Case for Planning with Latent Dynamics Models

快速理解: A long-standing goal in AI is to develop agents capable of solving diverse tasks across a range of environments, including those never seen during training. Two dominant paradigms address this challenge: (i) reinforcement learning (RL), which learns policies via trial and error, and (ii) optimal control, which plans actions using a known or learned dynamics model.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策。

证据层级: 本地 PDF 前页。

1.3.7.449 Learning to Clean: Reinforcement Learning for Noisy Label Correction

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Learning to Clean: Reinforcement Learning for Noisy Label Correction

快速理解: The challenge of learning with noisy labels is significant in machine learning, as it can severely degrade the performance of prediction models if not addressed properly. This paper introduces a novel framework that conceptualizes noisy label correction as a reinforcement learning (RL) problem.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.450 Learning to Route: Per-Sample Adaptive Routing for Multimodal Multitask Prediction

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Learning to Route: Per-Sample Adaptive Routing for Multimodal Multitask Prediction

快速理解: We propose a unified framework for adaptive routing in multitask, multimodal prediction settings where data heterogeneity and task interactions vary across samples. We introduce a routing-based architecture that dynamically selects modality processing pathways and task-sharing strategies on a per-sample basis.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.451 Learning to Steer: Input-dependent Steering for Multimodal LLMs

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Learning to Steer: Input-dependent Steering for Multimodal LLMs

快速理解: Steering has emerged as a practical approach to enable post-hoc guidance of LLMs towards enforcing a specific behavior. However, it remains largely underexplored for multimodal LLMs

(MLLMs); furthermore, existing steering techniques, such as mean steering, rely on a single steering vector, applied independently of the input query.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.452 Less is More: an Attention-free Sequence Prediction Modeling for Offline Embodied Learning

- 作者：作者信息未提供
- 年份：2025；主题：Embodied/WBC；优先级：重要
- [来源](#)

论文：Less is More: an Attention-free Sequence Prediction Modeling for Offline Embodied Learning

快速理解：Offline reinforcement learning (offline RL) is increasingly approached as a sequence modeling task, with methods leveraging advanced architectures like Transformers to capture trajectory dependencies. Despite significant progress, the mechanisms underlying their effectiveness and limitations remain insufficiently understood.

关注点：具身任务中的感知、动作生成、全身协调或运动控制。

证据层级：本地 PDF 前页。

1.3.7.453 Lessons Learned: A Multi-Agent Framework for Code LLMs to Learn and Improve

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Lessons Learned: A Multi-Agent Framework for Code LLMs to Learn and Improve

快速理解：Recent studies show that LLMs possess different skills and specialize in different tasks. In fact, we observe that their varied performance occur in several levels of granularity.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.454 Let Them Talk: Audio-Driven Multi-Person Conversational Video Generation

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Let Them Talk: Audio-Driven Multi-Person Conversational Video Generation

快速理解：Audio-driven human animation methods, such as talking head and talking body generation, have made remarkable progress in generating synchronized facial movements and appealing visual quality videos. However, existing methods primarily focus on single human animation and struggle with multi-stream audio inputs, facing incorrect binding problems between audio and persons.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.455 Leveraging Conditional Dependence for Efficient World Model Denoising

- 作者：作者信息未提供
- 年份：2025；主题：World Model；优先级：重要
- [来源](#)

论文：Leveraging Conditional Dependence for Efficient World Model Denoising

快速理解：Effective denoising is critical for managing complex visual inputs contaminated with noisy distractors in model-based reinforcement learning (RL). Current methods often oversimplify the decomposition of observations by neglecting the conditional dependence between task-relevant and task-irrelevant components given an observation.

关注点：环境动力学、视频预测、潜变量状态或模型式决策。

证据层级：本地 PDF 前页。

1.3.7.456 Limited Preference Data? Learning Better Reward Model with Latent Space Synthesis

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Limited Preference Data? Learning Better Reward Model with Latent Space Synthesis

快速理解：Reward modeling, crucial for aligning large language models (LLMs) with human preferences, is often bottlenecked by the high cost of preference data. Existing textual data synthesis methods are computationally expensive.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.457 LiteReality: Graphics-Ready 3D Scene Reconstruction from RGB-D Scans

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：LiteReality: Graphics-Ready 3D Scene Reconstruction from RGB-D Scans

快速理解：We propose LiteReality, a novel pipeline that converts RGB-D scans of indoor environments into compact, realistic, and interactive 3D virtual replicas. LiteReality not only reconstructs scenes that visually resemble reality but also supports key features essential for graphics pipelines—such as object individuality, articulation, high-quality physically based rendering materials.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.458 LittleBit: Ultra Low-Bit Quantization via Latent Factorization

- 作者：作者信息未提供

- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: LittleBit: Ultra Low-Bit Quantization via Latent Factorization

快速理解: The deployment of large language models (LLMs) is frequently hindered by prohibitive memory and computational requirements. While quantization mitigates these bottlenecks, maintaining model fidelity in the sub-1-bit regime remains a persistent challenge.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 本地 PDF 前页。

1.3.7.459 LoTA-QAF: Lossless Ternary Adaptation for Quantization-Aware Fine-Tuning

- 作者: 作者信息未提供
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: LoTA-QAF: Lossless Ternary Adaptation for Quantization-Aware Fine-Tuning

快速理解: Quantization and fine-tuning are crucial for deploying large language models (LLMs) on resource-constrained edge devices. However, fine-tuning quantized models presents significant challenges, primarily stemming from: First, the mismatch in data types between the low-precision quantized weights (e.g., 4-bit) and the high-precision adaptation weights (e.g., 16-bit).

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 本地 PDF 前页。

1.3.7.460 Luminance-Aware Statistical Quantization: Unsupervised Hierarchical Learning for Illumination Enhancement

- 作者: 作者信息未提供
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Luminance-Aware Statistical Quantization: Unsupervised Hierarchical Learning for Illumination Enhancement

快速理解: Low-light image enhancement (LLIE) faces persistent challenges in balancing reconstruction fidelity with cross-scenario generalization. While existing methods predominantly focus on deterministic pixel-level mappings between paired low/normal-light images, they often neglect the continuous physical process of luminance transitions in real-world environments, leading to performance drop when normal-light references are unavailable.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 本地 PDF 前页。

1.3.7.461 Lyapunov-Stable Adaptive Control for Multimodal Concept Drift

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Lyapunov-Stable Adaptive Control for Multimodal Concept Drift

快速理解: Multimodal learning systems often struggle in non-stationary environments due to concept drift, where changing data distributions can degrade performance. Modality-specific drifts and the lack of mechanisms for continuous, stable adaptation compound this challenge.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.462 MACS: Multi-Agent Reinforcement Learning for Optimization of Crystal Structures

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: MACS: Multi-Agent Reinforcement Learning for Optimization of Crystal Structures

快速理解: Geometry optimization of atomic structures is a common and crucial task in computational chemistry and materials design. Following the learning to optimize paradigm, we propose a new multi-agent reinforcement learning method called Multi-Agent Crystal Structure optimization (MACS) to address periodic crystal structure optimization.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.463 MAESTRO : Adaptive Sparse Attention and Robust Learning for Multimodal Dynamic Time Series

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: MAESTRO : Adaptive Sparse Attention and Robust Learning for Multimodal Dynamic Time Series

快速理解: From clinical healthcare to daily living, continuous sensor monitoring across multiple modalities has shown great promise for real-world intelligent decision-making but also faces various challenges. In this work, we introduce MAESTRO, a novel framework that overcomes key limitations of existing multimodal learning approaches: (1) reliance on a single primary modality for alignment, (2) pairwise modeling of modalities, and (3) assumption of complete modality observations.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.464 MAGNET: A Multi-agent Framework for Finding Audio-Visual Needles by Reasoning over Multi-Video Haystacks

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: MAGNET: A Multi-agent Framework for Finding Audio-Visual Needles by Reasoning over

Multi-Video Haystacks

快速理解: Large multimodal models (LMMs) have shown remarkable progress in audiovisual understanding, yet they struggle with real-world scenarios that require complex reasoning across extensive video collections. Existing benchmarks for video question answering remain limited in scope, typically involving one clip per query, which falls short of representing the challenges of large-scale, audiovisual retrieval and reasoning encountered in practical applications.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.465 MALinZero: Efficient Low-Dimensional Search for Mastering Complex Multi-Agent Planning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: MALinZero: Efficient Low-Dimensional Search for Mastering Complex Multi-Agent Planning

快速理解: Monte Carlo Tree Search (MCTS), which leverages Upper Confidence Bound for Trees (UCTs) to balance exploration and exploitation through randomized sampling, is instrumental to solving complex planning problems. However, for multi-agent planning, MCTS is confronted with a large combinatorial action space that often grows exponentially with the number of agents.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.466 MANGO: Multimodal Attention-based Normalizing Flow Approach to Fusion Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: MANGO: Multimodal Attention-based Normalizing Flow Approach to Fusion Learning

快速理解: Multimodal learning has gained much success in recent years. However, current multimodal fusion methods adopt the attention mechanism of Transformers to implicitly learn the underlying correlation of multimodal features.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.467 MASTER: Enhancing Large Language Model via Multi-Agent Simulated Teaching

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: MASTER: Enhancing Large Language Model via Multi-Agent Simulated Teaching

快速理解: Instruction fine-tuning is crucial in NLP tasks, enhancing pretrained models' instruction-following capabilities and task-specific performance. However, obtaining high-quality fine-tuning data for large models is challenging due to data collection difficulties and high production costs.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.468 MAT-Agent: Adaptive Multi-Agent Training Optimization

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: MAT-Agent: Adaptive Multi-Agent Training Optimization

快速理解: Multi-label image classification demands adaptive training strategies to navigate complex, evolving visual-semantic landscapes, yet conventional methods rely on static configurations that falter in dynamic settings. We propose MAT-Agent, a novel multi-agent framework that reimagines training as a collaborative, real-time optimization process.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.469 MEGADance: Mixture-of-Experts Architecture for Genre-Aware 3D Dance Generation

- 作者: 作者信息未提供
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: MEGADance: Mixture-of-Experts Architecture for Genre-Aware 3D Dance Generation

快速理解: Music-driven 3D dance generation has attracted increasing attention in recent years, with promising applications in choreography, virtual reality, and creative content creation. Previous research has generated promising realistic dance movement from audio signals.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 本地 PDF 前页。

1.3.7.470 MEgoHand: Multimodal Egocentric Hand-Object Interaction Motion Generation

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: MEgoHand: Multimodal Egocentric Hand-Object Interaction Motion Generation

快速理解: Egocentric hand-object motion generation is crucial for immersive AR/VR and robotic imitation but remains challenging due to unstable viewpoints, selfocclusions, perspective distortion, and noisy ego-motion. Existing methods rely on predefined 3D object priors, limiting generalization to novel objects, which restricts their generalizability to novel objects.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.471 MIDAS: Misalignment-based Data Augmentation Strategy for Imbalanced Multimodal Learning

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：MIDAS: Misalignment-based Data Augmentation Strategy for Imbalanced Multimodal Learning

快速理解：Multimodal models often over-rely on dominant modalities, failing to achieve optimal performance. While prior work focuses on modifying training objectives or optimization procedures, data-centric solutions remain underexplored.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.472 MIP against Agent: Malicious Image Patches Hijacking Multimodal OS Agents

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：MIP against Agent: Malicious Image Patches Hijacking Multimodal OS Agents

快速理解：Recent advances in operating system (OS) agents have enabled vision-language models (VLMs) to directly control a user's computer. Unlike conventional VLMs that passively output text, OS agents autonomously perform computer-based tasks in response to a single user prompt.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.473 MIRAGE: Assessing Hallucination in Multimodal Reasoning Chains of MLLM

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：MIRAGE: Assessing Hallucination in Multimodal Reasoning Chains of MLLM

快速理解：Multimodal hallucination in multimodal large language models (MLLMs) restricts the correctness of MLLMs. However, multimodal hallucinations are multi-sourced and arise from diverse causes.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.474 MJ-Video: Benchmarking and Rewarding Video Generation with Fine-Grained Video Preference

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：MJ-Video: Benchmarking and Rewarding Video Generation with Fine-Grained Video Preference

快速理解：Recent advancements in video generation have significantly improved the ability to synthesize videos from text instructions. However, existing models still struggle with key challenges such as instruction misalignment, content hallucination, safety concerns, and generation bias.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.475 MLLM-For3D: Adapting Multimodal Large Language Model for 3D Reasoning Segmentation

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：MLLM-For3D: Adapting Multimodal Large Language Model for 3D Reasoning Segmentation

快速理解：Reasoning segmentation aims to segment target objects in complex scenes based on human intent and spatial reasoning. While recent multimodal large language models (MLLMs) have demonstrated impressive 2D image reasoning segmentation, adapting these capabilities to 3D scenes remains underexplored.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.476 MLZero: A Multi-Agent System for End-to-end Machine Learning Automation

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：MLZero: A Multi-Agent System for End-to-end Machine Learning Automation

快速理解：Existing AutoML systems have advanced the automation of machine learning (ML); however, they still require substantial manual configuration and expert input, particularly when handling multimodal data. We introduce MLZero, a novel multi-agent framework powered by Large Language Models (LLMs) that enables end-to-end ML automation across diverse data modalities with minimal human intervention.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.477 MMaDA: Multimodal Large Diffusion Language Models

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：MMaDA: Multimodal Large Diffusion Language Models

快速理解：We introduce MMaDA, a novel class of multimodal diffusion foundation models designed to achieve superior performance across diverse domains such as textual reasoning, multimodal understanding, and text-to-image generation. The approach is distinguished by three key innovations: (i) MMaDA adopts a unified diffusion architecture with a shared probabilistic formulation and a modality-agnostic design, eliminating the need for modality-specific components.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.478 MOSDT: Self-Distillation-Based Decision Transformer for Multi-Agent Offline Safe Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：MOSDT: Self-Distillation-Based Decision Transformer for Multi-Agent Offline Safe Reinforcement Learning

快速理解：We introduce MOSDT, the first algorithm designed for multi-agent offline safe reinforcement learning (MOSRL), alongside MOSDB, the first dataset and benchmark for this domain. Different from most existing knowledge distillation-based multiagent RL methods, we propose policy self-distillation (PSD) with a new global information reconstruction scheme by fusing the observation features of all agents, streamlining training and improving parameter efficiency.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.479 MTRec: Learning to Align with User Preferences via Mental Reward Models

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：MTRec: Learning to Align with User Preferences via Mental Reward Models

快速理解：Recommendation models are predominantly trained using implicit user feedback, since explicit feedback is often costly to obtain. However, implicit feedback, such as clicks, does not always reflect users' real preferences.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.480 MURKA: Multi-Reward Reinforcement Learning with Knowledge Alignment for Optimization Tasks

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文: MURKA: Multi-Reward Reinforcement Learning with Knowledge Alignment for Optimization Tasks

快速理解: Optimization plays a central role in Operations Research (OR) and numerous industrial applications, yet automating the end-to-end process of translating natural language descriptions into executable optimization programs remains a formidable challenge. While recent efforts have applied Large Language Models (LLMs) to this task, existing approaches are hindered by high inference costs, limited robustness across domains, and weak verification mechanisms.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.481 Mamba Goes HoME: Hierarchical Soft Mixture-of-Experts for 3D Medical Image Segmentation

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文: Mamba Goes HoME: Hierarchical Soft Mixture-of-Experts for 3D Medical Image Segmentation

快速理解: In recent years, artificial intelligence has significantly advanced medical image segmentation. Nonetheless, challenges remain, including efficient 3D medical image processing across diverse modalities and handling data variability.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 本地 PDF 前页。

1.3.7.482 Mean-Field Sampling for Cooperative Multi-Agent Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文: Mean-Field Sampling for Cooperative Multi-Agent Reinforcement Learning

快速理解: Designing efficient algorithms for multi-agent reinforcement learning (MARL) is fundamentally challenging because the size of the joint state and action spaces grows exponentially in the number of agents. These difficulties are exacerbated when balancing sequential global decision-making with local agent interactions.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.483 Measure gradients, not activations! Enhancing neuronal activity in deep reinforcement learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要

- [来源](#)

论文: Measure gradients, not activations! Enhancing neuronal activity in deep reinforcement learning

快速理解: Deep reinforcement learning (RL) agents frequently suffer from neuronal activity loss, which impairs their ability to adapt to new data and learn continually. A common method to quantify and address this issue is the -dormant neuron ratio, which uses activation statistics to measure the expressive ability of neurons.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.484 Mechanism Design for LLM Fine-tuning with Multiple Reward Models

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Mechanism Design for LLM Fine-tuning with Multiple Reward Models

快速理解: Fine-tuning large language models (LLMs) to aggregate multiple preferences has attracted considerable research attention. With aggregation algorithms advancing, a potential economic scenario arises where fine-tuning services are provided to agents with different preferences.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.485 Memory-Efficient Visual Autoregressive Modeling with Scale-Aware KV Cache Compression

- 作者: 作者信息未提供
- 年份: 2025; 主题: Systems; 优先级: 重要
- [来源](#)

论文: Memory-Efficient Visual Autoregressive Modeling with Scale-Aware KV Cache Compression

快速理解: Visual Autoregressive (VAR) modeling has garnered significant attention for its innovative next-scale prediction approach, which yields substantial improvements in efficiency, scalability, and zero-shot generalization. Nevertheless, the coarse-to-fine methodology inherent in VAR results in exponential growth of the KV cache during inference, causing considerable memory consumption and computational redundancy.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.7.486 MeshCoder: LLM-Powered Structured Mesh Code Generation from Point Clouds

- 作者: 作者信息未提供
- 年份: 2025; 主题: Systems; 优先级: 重要
- [来源](#)

论文: MeshCoder: LLM-Powered Structured Mesh Code Generation from Point Clouds

快速理解: Reconstructing 3D objects into editable programs is pivotal for applications like reverse engineering and shape editing. However, existing methods often rely on limited domain-specific languages (DSLs) and small-scale datasets, restricting their ability to model complex geometries and structures.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.7.487 MetaGS: A Meta-Learned Gaussian-Phong Model for Out-of-Distribution 3D Scene Relighting

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: MetaGS: A Meta-Learned Gaussian-Phong Model for Out-of-Distribution 3D Scene Relighting

快速理解: Out-of-distribution (OOD) 3D relighting requires novel view synthesis under unseen lighting conditions that differ significantly from the observed images. Existing relighting methods, which assume consistent light source distributions between training and testing, often degrade in OOD scenarios.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.488 MetaMind: Modeling Human Social Thoughts with Metacognitive Multi-Agent Systems

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: MetaMind: Modeling Human Social Thoughts with Metacognitive Multi-Agent Systems

快速理解: Human social interactions depend on the ability to infer others' unspoken intentions, emotions, and beliefs—a cognitive skill grounded in the psychological concept of Theory of Mind (ToM). While large language models (LLMs) excel in semantic understanding tasks, they struggle with the ambiguity and contextual nuance inherent in human communication.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.489 Mind the GAP! The Challenges of Scale in Pixel-based Deep Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Mind the GAP! The Challenges of Scale in Pixel-based Deep Reinforcement Learning

快速理解: Scaling deep reinforcement learning in pixel-based environments presents a significant challenge, often resulting in diminished performance. While recent works have proposed algorithmic

and architectural approaches to address this, the underlying cause of the performance drop remains unclear.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.490 MindForge: Empowering Embodied Agents with Theory of Mind for Lifelong Cultural Learning

- 作者：作者信息未提供
- 年份：2025；主题：Embodied/WBC；优先级：重要
- [来源](#)

论文：MindForge: Empowering Embodied Agents with Theory of Mind for Lifelong Cultural Learning

快速理解：Embodied agents powered by large language models (LLMs), such as Voyager, promise open-ended competence in worlds such as Minecraft. However, when powered by open-weight LLMs they still falter on elementary tasks after domainspecific fine-tuning.

关注点：具身任务中的感知、动作生成、全身协调或运动控制。

证据层级：本地 PDF 前页。

1.3.7.491 MindJourney: Test-Time Scaling with World Models for Spatial Reasoning

- 作者：作者信息未提供
- 年份：2025；主题：World Model；优先级：重要
- [来源](#)

论文：MindJourney: Test-Time Scaling with World Models for Spatial Reasoning

快速理解：Spatial reasoning in 3D space is central to human cognition and indispensable for embodied tasks such as navigation and manipulation. However, state-of-the-art vision-language models (VLMs) struggle frequently with tasks as simple as anticipating how a scene will look after an egocentric motion: they perceive 2D images but lack an internal model of 3D dynamics.

关注点：环境动力学、视频预测、潜变量状态或模型式决策。

证据层级：本地 PDF 前页。

1.3.7.492 MindOmni: Unleashing Reasoning Generation in Vision Language Models with RGPO

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：MindOmni: Unleashing Reasoning Generation in Vision Language Models with RGPO

快速理解：Recent text-to-image systems face limitations in handling multimodal inputs and complex reasoning tasks. We introduce MindOmni, a unified multimodal large language model that addresses these challenges by incorporating reasoning generation through reinforcement learning.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.493 Mint: A Simple Test-Time Adaptation of Vision-Language Models against Common Corruptions

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Mint: A Simple Test-Time Adaptation of Vision-Language Models against Common Corruptions

快速理解：Pretrained vision-language models such as CLIP achieve strong zero-shot generalization but remain vulnerable to distribution shifts caused by input corruptions. In this work, we investigate how corruptions affect CLIP's image embeddings and uncover a consistent phenomenon we term as embedding variance collapse, where both intra-class and inter-class variances shrink as corruption severity increases.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.494 MisoDICE: Multi-Agent Imitation from Mixed-Quality Demonstrations

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：MisoDICE: Multi-Agent Imitation from Mixed-Quality Demonstrations

快速理解：We study offline imitation learning (IL) in cooperative multi-agent settings, where demonstrations have unlabeled mixed quality —containing both expert and suboptimal trajectories. Our proposed solution is structured in two stages: trajectory labeling and multi-agent imitation learning, designed jointly to enable effective learning from heterogeneous, unlabeled data.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.495 Mitigating Hallucination Through Theory-Consistent Symmetric Multimodal Preference Optimization

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Mitigating Hallucination Through Theory-Consistent Symmetric Multimodal Preference Optimization

快速理解：Direct Preference Optimization (DPO) has emerged as an effective approach for mitigating hallucination in Multimodal Large Language Models (MLLMs). Although existing methods have achieved significant progress by utilizing vision-oriented contrastive objectives for enhancing MLLMs' attention to visual inputs and hence reducing hallucination, they suffer from non-rigorous optimization objective function and indirect preference supervision.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.496 Mitigating Intra- and Inter-modal Forgetting in Continual Learning of Unified Multimodal Models

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Mitigating Intra- and Inter-modal Forgetting in Continual Learning of Unified Multimodal Models

快速理解：Unified Multimodal Generative Models (UMGMs) unify visual understanding and image generation within a single autoregressive framework. However, their ability to continually learn new tasks is severely hindered by catastrophic forgetting, both within a modality (intra-modal) and across modalities (inter-modal).

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.497 MixPrompt: Efficient Mixed Prompting for Multimodal Semantic Segmentation

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：MixPrompt: Efficient Mixed Prompting for Multimodal Semantic Segmentation

快速理解：Recent advances in multimodal semantic segmentation show that incorporating auxiliary inputs—such as depth or thermal images—can significantly improve performance over single-modality (RGB-only) approaches. However, most existing solutions rely on parallel backbone networks and complex fusion modules, greatly increasing model size and computational demands.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.498 Mixture-of-Experts Operator Transformer for Large-Scale PDE Pre-Training

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：Mixture-of-Experts Operator Transformer for Large-Scale PDE Pre-Training

快速理解：Pre-training has proven effective in addressing data scarcity and performance limitations in solving PDE problems with neural operators. However, challenges remain due to the heterogeneity of PDE datasets in equation types, which leads to high errors in mixed training.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.7.499 Mixtures of Subspaces for Bandwidth Efficient Context Parallel Training

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：Mixtures of Subspaces for Bandwidth Efficient Context Parallel Training

快速理解：Pretraining language models with extended context windows enhances their ability to leverage rich information during generation. Existing methods split input sequences into chunks, broadcast them across multiple devices, and compute attention block by block which incurs significant communication overhead.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.7.500 MoEMeta: Mixture-of-Experts Meta Learning for Few-Shot Relational Learning

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：MoEMeta: Mixture-of-Experts Meta Learning for Few-Shot Relational Learning

快速理解：Few-shot knowledge graph relational learning seeks to perform reasoning over relations given only a limited number of training examples. While existing approaches largely adopt a meta-learning framework for enabling fast adaptation to new relations, they suffer from two key pitfalls.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.7.501 MoRE-Brain: Routed Mixture of Experts for Interpretable and Generalizable Cross-Subject fMRI Visual Decoding

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：MoRE-Brain: Routed Mixture of Experts for Interpretable and Generalizable Cross-Subject fMRI Visual Decoding

快速理解：Decoding visual experiences from fMRI offers a powerful avenue to understand human perception and develop advanced brain-computer interfaces. However, current progress often prioritizes maximizing reconstruction fidelity while overlooking interpretability, an essential aspect for deriving neuroscientific insight.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.7.502 Modality-Aware SAM: Sharpness-Aware-Minimization Driven Gradient Modulation for Harmonized Multimodal Learning

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要

- [来源](#)

论文: Modality-Aware SAM: Sharpness-Aware-Minimization Driven Gradient Modulation for Harmonized Multimodal Learning

快速理解: In multimodal learning, dominant modalities often overshadow others, limiting generalization. We propose Modality-Aware Sharpness-Aware Minimization (M-SAM), a model-agnostic framework that applies to many modalities and supports early and late fusion scenarios.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.503 Modelling the control of offline processing with reinforcement learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Modelling the control of offline processing with reinforcement learning

快速理解: Brains reorganise knowledge offline to improve future behaviour, with ‘replay’ involved in consolidating memories, abstracting patterns from experience, and simulating new scenarios. However, there are few models of how the brain might orchestrate these processes, and of when different types of replay might be useful.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.504 MokA: Multimodal Low-Rank Adaptation for MLLMs

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: MokA: Multimodal Low-Rank Adaptation for MLLMs

快速理解: In this paper, we reveal that most current efficient multimodal fine-tuning methods are hindered by a key limitation: they are directly borrowed from LLMs, often neglecting the intrinsic differences of multimodal scenarios and even affecting the full utilization of all modalities. Inspired by our empirical observation, we argue that unimodal adaptation and cross-modal adaptation are two essential parts for the effective fine-tuning of MLLMs.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.505 MoodAngels: A Retrieval-augmented Multi-agent Framework for Psychiatry Diagnosis

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: MoodAngels: A Retrieval-augmented Multi-agent Framework for Psychiatry Diagnosis

快速理解: The application of AI in psychiatric diagnosis faces significant challenges, including the subjective nature of mental health assessments, symptom overlap across disorders, and privacy constraints limiting data availability. To address these issues, we present MoodAngels, the first specialized multi-agent framework for mood disorder diagnosis.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.506 More Than Generation: Unifying Generation and Depth Estimation via Text-to-Image Diffusion Models

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: More Than Generation: Unifying Generation and Depth Estimation via Text-to-Image Diffusion Models

快速理解: Generative depth estimation methods leverage the rich visual priors stored in pre-trained text-to-image diffusion models, demonstrating astonishing zero-shot capability. However, parameter updates during training lead to catastrophic degradation in the image generation capability of the pre-trained model.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.507 More Thinking, Less Seeing? Assessing Amplified Hallucination in Multimodal Reasoning Models

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: More Thinking, Less Seeing? Assessing Amplified Hallucination in Multimodal Reasoning Models

快速理解: Test-time compute has empowered multimodal large language models to generate extended reasoning chains, yielding strong performance on tasks such as multimodal math reasoning. However, we observe that this improved reasoning ability often comes with increased hallucination: as generations become longer, models tend to drift away from image-grounded content and rely more on language priors.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.508 Motion4D: Learning 3D-Consistent Motion and Semantics for 4D Scene Understanding

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Motion4D: Learning 3D-Consistent Motion and Semantics for 4D Scene Understanding

快速理解: Recent advancements in foundation models for 2D vision have substantially improved the analysis of dynamic scenes from monocular videos. However, despite their strong generalization capabilities, these models often lack 3D consistency, a fundamental requirement for understanding scene geometry and motion, thereby causing severe spatial misalignment and temporal flickering in complex 3D environments.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.509 MotionRAG: Motion Retrieval-Augmented Image-to-Video Generation

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: MotionRAG: Motion Retrieval-Augmented Image-to-Video Generation

快速理解: Image-to-video generation has made remarkable progress with the advancements in diffusion models, yet generating videos with realistic motion remains highly challenging. This difficulty arises from the complexity of accurately modeling motion, which involves capturing physical constraints, object interactions, and domain-specific dynamics that are not easily generalized across diverse scenarios.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.510 MuSLR: Multimodal Symbolic Logical Reasoning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: MuSLR: Multimodal Symbolic Logical Reasoning

快速理解: Multimodal symbolic logical reasoning, which aims to deduce new facts from multimodal input via formal logic, is critical in high-stakes applications such as autonomous driving and medical diagnosis, as its rigorous, deterministic reasoning helps prevent serious consequences. To evaluate such capabilities of current state-of-the-art vision language models (VLMs), we introduce MuSLR, the first multimodal symbolic logical reasoning grounded in formal logical rules.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.511 Multi-Agent Collaboration via Evolving Orchestration

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Multi-Agent Collaboration via Evolving Orchestration

快速理解: Large language models (LLMs) have achieved remarkable results across diverse downstream tasks, but their monolithic nature restricts scalability and efficiency in complex problem-solving. While recent research explores multi-agent collaboration among LLMs, most approaches rely on static organizational structures that struggle to adapt as task complexity and agent numbers grow, resulting in coordination overhead and inefficiencies.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.512 Multi-Agent Debate for LLM Judges with Adaptive Stability Detection

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Multi-Agent Debate for LLM Judges with Adaptive Stability Detection

快速理解: With the advancing reasoning capabilities of Large Language Models (LLMs), they are increasingly employed for complex evaluation tasks, such as grading student responses, verifying factual claims, and comparing competing answers. Leveraging multiple LLMs as automated judges can enhance robustness and accuracy by aggregating diverse perspectives, yet existing approaches often rely on static and simple aggregation methods, such as majority voting, which may produce incorrect judgments despite correct individual assessments.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.513 Multi-Agent Imitation by Learning and Sampling from Factorized Soft Q-Function

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Multi-Agent Imitation by Learning and Sampling from Factorized Soft Q-Function

快速理解: Learning from multi-agent expert demonstrations, known as Multi-Agent Imitation Learning (MAIL), provides a promising approach to sequential decision-making. However, existing MAIL methods including Behavior Cloning (BC) and Adversarial Imitation Learning (AIL) face significant challenges: BC suffers from the compounding error issue, while the very nature of adversarial optimization makes AIL prone to instability.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.514 Multi-Agent Learning under Uncertainty: Recurrence vs. Concentration

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Multi-Agent Learning under Uncertainty: Recurrence vs. Concentration

快速理解: In this paper, we examine the convergence landscape of multi-agent learning under uncertainty. Specifically, we analyze two stochastic models of regularized learning in continuous games—one in continuous and one in discrete time—with the aim of characterizing the long-run behavior of the induced sequence of play.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.515 Multi-Agent Reinforcement Learning with Communication-Constrained Priors

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Multi-Agent Reinforcement Learning with Communication-Constrained Priors

快速理解: Communication is one of the effective means to improve the learning of cooperative policy in multi-agent systems. However, in most real-world scenarios, lossy communication is a prevalent issue.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.516 Multi-Objective Reinforcement Learning with Max-Min Criterion: A Game-Theoretic Approach

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Multi-Objective Reinforcement Learning with Max-Min Criterion: A Game-Theoretic Approach

快速理解: In this paper, we propose a provably convergent and practical framework for multi-objective reinforcement learning with max-min criterion. From a game-theoretic perspective, we reformulate max-min multi-objective reinforcement learning as a two-player zero-sum regularized continuous game and introduce an efficient algorithm based on mirror descent.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.517 Multi-agent KTO: Enhancing Strategic Interactions of Large Language Model in Language Game

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Multi-agent KTO: Enhancing Strategic Interactions of Large Language Model in Language Game

快速理解: Achieving Artificial General Intelligence (AGI) requires AI agents that can not only make strategic decisions but also engage in flexible and meaningful communication. Inspired by Wittgen-

stein's language game theory, we propose that language agents can learn through in-context interaction rather than traditional multi-stage frameworks that separate decision-making from language expression.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.518 Multi-agent Markov Entanglement

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Multi-agent Markov Entanglement

快速理解：Value decomposition has long been a fundamental technique in multi-agent reinforcement learning and dynamic programming. Specifically, the value function of a global state $(s_1, s_2, \dots, s_N)$ is often approximated as the sum of local functions: $V(s_1, s_2, \dots, s_N) \approx \sum_{i=1}^N V_i(s_i)$.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.519 Multi-scale Temporal Prediction via Incremental Generation and Multi-agent Collaboration

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Multi-scale Temporal Prediction via Incremental Generation and Multi-agent Collaboration

快速理解：Accurate temporal prediction is the bridge between comprehensive scene understanding and embodied artificial intelligence. However, predicting multiple finegrained states of scene at multiple temporal scales is difficult for vision-language models.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.520 Multimodal 3D Genome Pre-training

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Multimodal 3D Genome Pre-training

快速理解：Deep learning techniques have driven significant progress in various analytical tasks within 3D genomics in computational biology. However, a holistic understanding of 3D genomics knowledge remains underexplored.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.521 Multimodal Bandits: Regret Lower Bounds and Optimal Algorithms

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Multimodal Bandits: Regret Lower Bounds and Optimal Algorithms

快速理解：We consider a stochastic multi-armed bandit problem with i.i.d. rewards where the expected reward function is multimodal with at most m modes. We propose the first known computationally tractable algorithm for computing the solution to the Graves-Lai optimization problem, which in turn enables the implementation of asymptotically optimal algorithms for this bandit problem.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.522 Multimodal Causal Reasoning for UAV Object Detection

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Multimodal Causal Reasoning for UAV Object Detection

快速理解：Unmanned Aerial Vehicle (UAV) object detection faces significant challenges due to complex environmental conditions and different imaging conditions. These factors introduce significant changes in scale and appearance, particularly for small objects that occupy limited pixels and exhibit limited information, complicating detection tasks.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.523 Multimodal Disease Progression Modeling via Spatiotemporal Disentanglement and Multiscale Alignment

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Multimodal Disease Progression Modeling via Spatiotemporal Disentanglement and Multiscale Alignment

快速理解：Longitudinal multimodal data, including electronic health records (EHR) and sequential chest X-rays (CXRs), is critical for modeling disease progression, yet remains underutilized due to two key challenges: (1) redundancy in consecutive CXR sequences, where static anatomical regions dominate over clinically-meaningful dynamics, and (2) temporal misalignment between sparse, irregular imaging and continuous EHR data. We introduce DiPro, a novel framework that addresses these challenges through region-aware disentanglement and multi-timescale alignment.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.524 Multimodal LiDAR-Camera Novel View Synthesis with Unified Pose-free Neural Fields

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Multimodal LiDAR-Camera Novel View Synthesis with Unified Pose-free Neural Fields

快速理解：Pose-free Neural Radiance Field (NeRF) aims at novel view synthesis (NVS) without relying on accurate poses, exhibiting significant practical value. Image and LiDAR point cloud are two pivotal modalities in autonomous driving scenarios.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.525 Multimodal Negative Learning

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Multimodal Negative Learning

快速理解：Multimodal learning systems often encounter challenges related to modality imbalance, where a dominant modality may overshadow others, thereby hindering the learning of weak modalities. Conventional approaches often force weak modalities to align with dominant ones in “Learning to be (the same)” (Positive Learning), which risks suppressing the unique information inherent in the weak modalities.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.526 Multimodal Tabular Reasoning with Privileged Structured Information

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Multimodal Tabular Reasoning with Privileged Structured Information

快速理解：Tabular reasoning requires complex, multi-step information extraction and logical inference, such as aggregation, comparison, or calculation over tabular data. While recent advances have leveraged large language models (LLMs) for reasoning over structured text tables, such high-quality textual representations are often unavailable in real-world settings, where tables typically appear as images.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.527 NAUTILUS: A Large Multimodal Model for Underwater Scene Understanding

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：NAUTILUS: A Large Multimodal Model for Underwater Scene Understanding

快速理解：Underwater exploration offers critical insights into our planet and attracts increasing attention for its broader applications in resource exploration, national security, etc. We study the underwater scene understanding methods, which aim to achieve automated underwater exploration.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.528 NaViL: Rethinking Scaling Properties of Native Multimodal Large Language Models under Data Constraints

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：NaViL: Rethinking Scaling Properties of Native Multimodal Large Language Models under Data Constraints

快速理解：Compositional training has been the de-facto paradigm in existing Multimodal Large Language Models (MLLMs), where pre-trained visual encoders are connected with pre-trained LLMs through continuous multimodal pre-training. However, the multimodal scaling property of this paradigm remains difficult to explore due to the separated training.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.529 NeSyPr: Neurosymbolic Proceduralization For Efficient Embodied Reasoning

- 作者：作者信息未提供
- 年份：2025；主题：Embodied/WBC；优先级：重要
- [来源](#)

论文：NeSyPr: Neurosymbolic Proceduralization For Efficient Embodied Reasoning

快速理解：We address the challenge of adopting language models (LMs) for embodied tasks in dynamic environments, where online access to large-scale inference engines or symbolic planners is constrained due to latency, connectivity, and resource limitations. To this end, we present NESYPR, a novel embodied reasoning framework that compiles knowledge via neurosymbolic proceduralization, thereby equipping LM-based agents with structured, adaptive, and timely reasoning capabilities.

关注点：具身任务中的感知、动作生成、全身协调或运动控制。

证据层级：本地 PDF 前页。

1.3.7.530 NoisyGRPO: Incentivizing Multimodal CoT Reasoning via Noise Injection and Bayesian Estimation

- 作者：作者信息未提供

- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: NoisyGRPO: Incentivizing Multimodal CoT Reasoning via Noise Injection and Bayesian Estimation

快速理解: Reinforcement learning (RL) has shown promise in enhancing the general Chain-of-Thought (CoT) reasoning capabilities of multimodal large language models (MLLMs). However, when applied to improve general CoT reasoning, existing RL frameworks often struggle to generalize beyond the training distribution.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.531 Non-Clairvoyant Scheduling with Progress Bars

- 作者: 作者信息未提供
- 年份: 2025; 主题: Systems; 优先级: 重要
- [来源](#)

论文: Non-Clairvoyant Scheduling with Progress Bars

快速理解: In non-clairvoyant scheduling, the goal is to minimize the total job completion time without prior knowledge of individual job processing times. This classical online optimization problem has recently gained attention through the framework of learning-augmented algorithms.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.7.532 Non-Line-of-Sight 3D Reconstruction with Radar

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Non-Line-of-Sight 3D Reconstruction with Radar

快速理解: Seeing hidden structures and objects around corners is critical for robots operating in complex, cluttered environments. Existing methods, however, are limited to detecting and tracking hidden objects rather than reconstructing the occluded full scene.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.533 Novel Class Discovery for Point Cloud Segmentation via Joint Learning of Causal Representation and Reasoning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Systems; 优先级: 重要
- [来源](#)

论文: Novel Class Discovery for Point Cloud Segmentation via Joint Learning of Causal Representation and Reasoning

快速理解: In this paper, we focus on Novel Class Discovery for Point Cloud Segmentation (3D-NCD), aiming to learn a model that can segment unlabeled (novel) 3D classes using only the supervision from labeled (base) 3D classes. The key to this task is to setup the exact correlations between the point representations and their base class labels, as well as the representation correlations between the points from base and novel classes.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.7.534 OOD-Barrier: Build a Middle-Barrier for Open-Set Single-Image Test Time Adaptation via Vision Language Models

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: OOD-Barrier: Build a Middle-Barrier for Open-Set Single-Image Test Time Adaptation via Vision Language Models

快速理解: In real-world environments, a well-designed model must be capable of handling dynamically evolving distributions, where both in-distribution (ID) and out-of-distribution (OOD) samples appear unpredictably and individually, making realtime adaptation particularly challenging. While open-set test-time adaptation has demonstrated effectiveness in adjusting to distribution shifts, existing methods often rely on batch processing and struggle to manage single-sample data stream in open-set environments.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.535 OPHR: Mastering Volatility Trading with Multi-Agent Deep Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: OPHR: Mastering Volatility Trading with Multi-Agent Deep Reinforcement Learning

快速理解: Options markets represent one of the most sophisticated segments of the financial ecosystem, with prices that directly reflect market uncertainty. In this paper, we introduce the first reinforcement learning (RL) framework specifically designed for volatility trading through options, focusing on profit from the difference between implied volatility and realized volatility.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.536 ORIGAMISPACE: Benchmarking Multimodal LLMs in Multi-Step Spatial Reasoning with Mathematical Constraints

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: ORIGAMISPACE: Benchmarking Multimodal LLMs in Multi-Step Spatial Reasoning with Mathematical Constraints

快速理解: Spatial reasoning is a key capability in the field of artificial intelligence, especially crucial in areas such as robotics, computer vision, and natural language understanding. However, evaluating the ability of multimodal large language models (MLLMs) in complex spatial reasoning still faces challenges, particularly in scenarios requiring multi-step reasoning and precise mathematical constraints.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.537 OWL: Optimized Workforce Learning for General Multi-Agent Assistance in Real-World Task Automation

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: OWL: Optimized Workforce Learning for General Multi-Agent Assistance in Real-World Task Automation

快速理解: Large Language Model (LLM)-based multi-agent systems show promise for automating real-world tasks but struggle to transfer across domains due to their domain-specific nature. Current approaches face two critical shortcomings: they require complete architectural redesign and full retraining of all components when applied to new domains.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.538 OWMM-Agent: Open World Mobile Manipulation With Multi-modal Agentic Data Synthesis

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: OWMM-Agent: Open World Mobile Manipulation With Multi-modal Agentic Data Synthesis

快速理解: The rapid progress of navigation, manipulation, and vision models has made mobile manipulators capable in many specialized tasks. However, the open-world mobile manipulation (OWMM) task remains a challenge due to the need for generalization to open-ended instructions and environments, as well as the systematic complexity to integrate high-level decision making with low-level robot control based on both global scene understanding and current agent state.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.539 ObCLIP: Oblivious CCloud-Device Hybrid Image Generation with Privacy Preservation

- 作者: 作者信息未提供

- 年份: 2025; 主题: Systems; 优先级: 重要
- [来源](#)

论文: ObCLIP: Oblivious CCloud-Device Hybrid Image Generation with Privacy Preservation

快速理解: Diffusion Models have gained significant popularity due to their remarkable capabilities in image generation, albeit at the cost of intensive computation requirement. Meanwhile, despite their widespread deployment in inference services such as Midjourney, concerns about the potential leakage of sensitive information in uploaded user prompts have arisen.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.7.540 Object-centric 3D Motion Field for Robot Learning from Human Videos

- 作者: 作者信息未提供
- 年份: 2025; 主题: Robot Learning; 优先级: 重要
- [来源](#)

论文: Object-centric 3D Motion Field for Robot Learning from Human Videos

快速理解: Learning robot control policies from human videos is a promising direction for scaling up robot learning. However, how to extract action knowledge (or action representations) from videos for policy learning remains a key challenge.

关注点: 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 本地 PDF 前页。

1.3.7.541 Off-policy Reinforcement Learning with Model-based Exploration Augmentation

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Off-policy Reinforcement Learning with Model-based Exploration Augmentation

快速理解: Exploration is fundamental to reinforcement learning (RL), as it determines how effectively an agent discovers and exploits the underlying structure of its environment to achieve optimal performance. Existing exploration methods generally fall into two categories: active exploration and passive exploration.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.542 Offline Goal-conditioned Reinforcement Learning with Quasimetric Representations

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Offline Goal-conditioned Reinforcement Learning with Quasimetric Representations

快速理解: Approaches for goal-conditioned reinforcement learning (GCRL) often use learned state representations to extract goal-reaching policies. Two frameworks for representation structure have yielded particularly effective GCRL algorithms: (1) contrastive representations, in which methods learn “successor features” with a contrastive objective that performs inference over future outcomes, and (2) temporal distances, which link the (quasimetric) distance in representation space to the transit time from states to goals.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.543 Offline Guarded Safe Reinforcement Learning for Medical Treatment Optimization Strategies

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Offline Guarded Safe Reinforcement Learning for Medical Treatment Optimization Strategies

快速理解: When applying offline reinforcement learning (RL) in healthcare scenarios, the out-of-distribution (OOD) issues pose significant risks, as inappropriate generalization beyond clinical expertise can result in potentially harmful recommendations. While existing methods like conservative Q-learning (CQL) attempt to address the OOD issue, their effectiveness is limited by only constraining action selection by suppressing uncertain actions.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.544 Offline RL by Reward-Weighted Fine-Tuning for Conversation Optimization

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Offline RL by Reward-Weighted Fine-Tuning for Conversation Optimization

快速理解: Offline reinforcement learning (RL) is a variant of RL where the policy is learned from a previously collected dataset of trajectories and rewards. In our work, we propose a practical approach to offline RL with large language models (LLMs).

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.545 Omni-R1: Reinforcement Learning for Multimodal Reasoning via Two-System Collaboration

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Omni-R1: Reinforcement Learning for Multimodal Reasoning via Two-System Collaboration

快速理解: Long-horizon video-audio reasoning and fine-grained pixel understanding impose conflicting

requirements on omnimodal models: dense temporal coverage demands many low-resolution frames, whereas precise grounding calls for high-resolution inputs. We tackle this trade-off with a two-system architecture: a Global Reasoning System selects informative keyframes and rewrites the task at low spatial cost, while a Detail Understanding System performs pixel-level grounding on the selected high-resolution snippets.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.546 OmniResponse: Online Multimodal Conversational Response Generation in Dyadic Interactions

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：OmniResponse: Online Multimodal Conversational Response Generation in Dyadic Interactions

快速理解：In this paper, we introduce Online Multimodal Conversational Response Generation (OMCRG), a novel task designed to produce synchronized verbal and non-verbal listener feedback online, based on the speaker's multimodal inputs. OMCRG captures natural dyadic interactions and introduces new challenges in aligning generated audio with listeners' facial responses.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.547 OmniTalker: One-shot Real-time Text-Driven Talking Audio-Video Generation With Multimodal Style Mimicking

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：OmniTalker: One-shot Real-time Text-Driven Talking Audio-Video Generation With Multimodal Style Mimicking

快速理解：Although significant progress has been made in audio-driven talking head generation, text-driven methods remain underexplored. In this work, we present OmniTalker, a unified framework that jointly generates synchronized talking audiovideo content from input text while emulating the target identity's speaking and facial movement styles, including speech characteristics, head motion, and facial dynamics.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.548 OmniVCus: Feedforward Subject-driven Video Customization with Multimodal Control Conditions

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文: OmniVCus: Feedforward Subject-driven Video Customization with Multimodal Control Conditions

快速理解: Existing feedforward subject-driven video customization methods mainly study single-subject scenarios due to the difficulty of constructing multi-subject training data pairs. Another challenging problem that how to use the signals such as depth, mask, camera, and text prompts to control and edit the subject in the customized video is still less explored.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.549 Omnidirectional 3D Scene Reconstruction from Single Image

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Omnidirectional 3D Scene Reconstruction from Single Image

快速理解: Reconstruction of 3D scenes from a single image is a crucial step towards enabling next-generation AI-powered immersive experiences. However, existing diffusion-based methods often struggle with reconstructing omnidirectional scenes due to geometric distortions and inconsistencies across the generated novel views, hindering accurate 3D recovery.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.550 On Epistemic Uncertainty of Visual Tokens for Object Hallucinations in Large Vision-Language Models

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: On Epistemic Uncertainty of Visual Tokens for Object Hallucinations in Large Vision-Language Models

快速理解: Large vision-language models (LVLMs), which integrate a vision encoder (VE) with a large language model, have achieved remarkable success across various tasks. However, there are still crucial challenges in LVLMs such as object hallucination, generating descriptions of objects that are not in the input image.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.551 On Fairness of Unified Multimodal Large Language Model for Image Generation

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: On Fairness of Unified Multimodal Large Language Model for Image Generation

快速理解: Unified multimodal large language models (U-MLLMs) have demonstrated impressive performance in end-to-end visual understanding and generation tasks. However, compared to generation-only systems (e.g., Stable Diffusion), the unified architecture of U-MLLMs introduces new risks of propagating demographic stereotypes.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.552 On Feasible Rewards in Multi-Agent Inverse Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: On Feasible Rewards in Multi-Agent Inverse Reinforcement Learning

快速理解: Multi-agent Inverse Reinforcement Learning (MAIRL) aims to recover agent reward functions from expert demonstrations. We characterize the feasible reward set in Markov games, identifying all reward functions that rationalize a given equilibrium.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.553 On Geometry-Enhanced Parameter-Efficient Fine-Tuning for 3D Scene Segmentation

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: On Geometry-Enhanced Parameter-Efficient Fine-Tuning for 3D Scene Segmentation

快速理解: The emergence of large-scale pre-trained point cloud models has significantly advanced 3D scene understanding, but adapting these models to specific downstream tasks typically demands full fine-tuning, incurring high computational and storage costs. Parameter-efficient fine-tuning (PEFT) techniques, successful in natural language processing and 2D vision tasks, would underperform when naively applied to 3D point cloud models due to significant geometric and spatial distribution shifts.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.554 On Linear Mode Connectivity of Mixture-of-Experts Architectures

- 作者: 作者信息未提供
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: On Linear Mode Connectivity of Mixture-of-Experts Architectures

快速理解: Linear Mode Connectivity (LMC) is a notable phenomenon in the loss landscapes of neural networks, wherein independently trained models have been observed to be connected-up to permutation symmetries-by linear paths in parameter space along which the loss remains consistently low. This observation challenges classical views of non-convex optimization and has implications for

model ensembling, generalization, and our understanding of neural loss geometry.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.7.555 On Minimax Estimation of Parameters in Softmax-Contaminated Mixture of Experts

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：On Minimax Estimation of Parameters in Softmax-Contaminated Mixture of Experts

快速理解：The softmax-contaminated mixture of experts (MoE) model is deployed when a large-scale pre-trained model, which plays the role of a fixed expert, is fine-tuned for learning downstream tasks by including a new contamination part, or prompt, functioning as a new, trainable expert. Despite its popularity and relevance, the theoretical properties of the softmax-contaminated MoE have remained unexplored in the literature.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.7.556 On the Expressive Power of Mixture-of-Experts for Structured Complex Tasks

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：On the Expressive Power of Mixture-of-Experts for Structured Complex Tasks

快速理解：Mixture-of-experts networks (MoEs) have demonstrated remarkable efficiency in modern deep learning. Despite their empirical success, the theoretical foundations underlying their ability to model complex tasks remain poorly understood.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.7.557 On the Global Optimality of Policy Gradient Methods in General Utility Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：On the Global Optimality of Policy Gradient Methods in General Utility Reinforcement Learning

快速理解：Reinforcement learning with general utilities (RLGU) offers a unifying framework to capture several problems beyond standard expected returns, including imitation learning, pure exploration, and safe RL. Despite recent fundamental advances in the theoretical analysis of policy gradient (PG) methods for standard RL and recent efforts in RLGU, the understanding of these PG algorithms

and their scope of application in RLGU still remain limited.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.558 On the Sample Complexity Bounds of Bilevel Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：On the Sample Complexity Bounds of Bilevel Reinforcement Learning

快速理解：Bilevel reinforcement learning (BRL) has emerged as a powerful framework for aligning generative models, yet its theoretical foundations, especially sample complexity bounds, remain relatively underexplored. In this work, we present the first sample complexity bound for BRL, establishing a rate of $\tilde{O}(-3)$ in continuous state-action spaces.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.559 On the Sample Complexity of Differentially Private Policy Optimization

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：On the Sample Complexity of Differentially Private Policy Optimization

快速理解：Policy optimization (PO) is a cornerstone of modern reinforcement learning (RL), with diverse applications spanning robotics, healthcare, and large language model training. The increasing deployment of PO in sensitive domains, however, raises significant privacy concerns.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.560 On the Value of Cross-Modal Misalignment in Multimodal Representation Learning

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：On the Value of Cross-Modal Misalignment in Multimodal Representation Learning

快速理解：Multimodal representation learning, exemplified by multimodal contrastive learning (MMCL) using image-text pairs, aims to learn powerful representations by aligning cues across modalities. This approach relies on the core assumption that the exemplar image-text pairs constitute two representations of an identical concept.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.561 On-Policy Optimization with Group Equivalent Preference for Multi-Programming Language Understanding

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：On-Policy Optimization with Group Equivalent Preference for Multi-Programming Language Understanding

快速理解：Large language models (LLMs) achieve remarkable performance in code generation tasks. However, a significant performance disparity persists between popular programming languages (e.g., Python, C++) and others.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.562 One Subgoal at a Time: Zero-Shot Generalization to Arbitrary Linear Temporal Logic Requirements in Multi-Task Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：One Subgoal at a Time: Zero-Shot Generalization to Arbitrary Linear Temporal Logic Requirements in Multi-Task Reinforcement Learning

快速理解：Generalizing to complex and temporally extended task objectives and safety constraints remains a critical challenge in reinforcement learning (RL). Linear temporal logic (LTL) offers a unified formalism to specify such requirements, yet existing methods are limited in their abilities to handle nested long-horizon tasks and safety constraints, and cannot identify situations when a subgoal is not satisfiable and an alternative should be sought.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.563 Online Mixture of Experts: No-Regret Learning for Optimal Collective Decision-Making

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：Online Mixture of Experts: No-Regret Learning for Optimal Collective Decision-Making

快速理解：We explore the use of expert-guided bandit learning, which we refer to as online mixture-of-experts (OMoE). In this setting, given a context, a candidate committee of experts must determine how to aggregate their outputs to achieve optimal results in terms of aggregate accuracy.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.7.564 Online Optimization for Offline Safe Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Online Optimization for Offline Safe Reinforcement Learning

快速理解：We study the problem of Offline Safe Reinforcement Learning (OSRL), where the goal is to learn a reward-maximizing policy from fixed data under a cumulative cost constraint. We propose a novel OSRL approach that frames the problem as a minimax objective and solves it by combining offline RL with online optimization algorithms.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.565 OnlineSplatter: Pose-Free Online 3D Reconstruction for Free-Moving Objects

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：OnlineSplatter: Pose-Free Online 3D Reconstruction for Free-Moving Objects

快速理解：Free-moving object reconstruction from monocular video remains challenging, particularly without reliable pose or depth cues and under arbitrary object motion. We introduce OnlineSplatter, a novel online feed-forward framework generating highquality, object-centric 3D Gaussians directly from RGB frames without requiring camera pose, depth priors, or bundle optimization.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.566 Open-Reasoner-Zero: An Open Source Approach to Scaling Up Reinforcement Learning on the Base Model

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Open-Reasoner-Zero: An Open Source Approach to Scaling Up Reinforcement Learning on the Base Model

快速理解：We introduce Open-Reasoner-Zero, the first open source implementation of largescale reasoning-oriented RL training on the base model focusing on scalability, simplicity and accessibility. Through extensive experiments, we demonstrate that a minimalist approach, vanilla PPO with GAE ($\gamma = 1$, $\beta = 1$) and straightforward rule-based rewards, without any KL regularization, is sufficient to scale up both benchmark performance and response length, replicating the scaling phenomenon observed in DeepSeek-R1-Zero.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.567 OpenHOI: Open-World Hand-Object Interaction Synthesis with Multimodal Large Language Model

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：OpenHOI: Open-World Hand-Object Interaction Synthesis with Multimodal Large Language Model

快速理解：Understanding and synthesizing realistic 3D hand-object interactions (HOI) is critical for applications ranging from immersive AR/VR to dexterous robotics. Existing methods struggle with generalization, performing well on closed-set objects and predefined tasks but failing to handle unseen objects or open-vocabulary instructions.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.568 OpenOmni: Advancing Open-Source Omnimodal Large Language Models with Progressive Multimodal Alignment and Real-time Emotional Speech Synthesis

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：OpenOmni: Advancing Open-Source Omnimodal Large Language Models with Progressive Multimodal Alignment and Real-time Emotional Speech Synthesis

快速理解：Recent advancements in omnimodal learning have significantly improved understanding and generation across images, text, and speech, yet these developments remain predominantly confined to proprietary models. The lack of high-quality omnimodal datasets and the challenges of real-time emotional speech synthesis have notably hindered progress in open-source research.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.569 OpenVLThinker: Complex Vision-Language Reasoning via Iterative SFT-RL Cycles

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：OpenVLThinker: Complex Vision-Language Reasoning via Iterative SFT-RL Cycles

快速理解：We introduce OpenVLThinker, one of the first open-source large vision-language models (LVLMs) to exhibit sophisticated chain-of-thought reasoning, achieving notable performance gains on challenging visual reasoning tasks. While textbased reasoning models (e.g., Deepseek R1) show promising results in text-only tasks, distilling their reasoning into LVLMs via supervised fine-tuning (SFT) often results in performance degradation due to imprecise visual grounding.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.570 Optimal Dynamic Regret by Transformers for Non-Stationary Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Optimal Dynamic Regret by Transformers for Non-Stationary Reinforcement Learning

快速理解：Transformers have demonstrated exceptional performance across a wide range of domains. While their ability to perform reinforcement learning in-context has been established both theoretically and empirically, their behavior in nonstationary environments remains less understood.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.571 Optimal Regret Bounds via Low-Rank Structured Variation in Non-Stationary Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Optimal Regret Bounds via Low-Rank Structured Variation in Non-Stationary Reinforcement Learning

快速理解：We study reinforcement learning in non-stationary communicating MDPs whose transition drift admits a low-rank plus sparse structure. We propose SVUCRL (Structured Variation UCRL) and prove the dynamic-regret bound $e O \sqrt{SA T} + D_{\max} S \sqrt{A T} + L_{\max} B_r + D_{\max} L_{\max} B_p + D_{\max} S p A B_p + D_{\max} B B_p + D_{\max} \sqrt{K T}$, (up to the additional planning-tolerance term $PT t=1 (m(t))$). where S is the number of states, A the number of actions, T the horizon, $D_{\max}$ the MDP diameter, B_r/B_p the total reward/transition variation budgets, and $K \leq SA$ the rank of the structured drift, $L_{\max}$ is the maximum episode length.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.572 Optimal Single-Policy Sample Complexity and Transient Coverage for Average-Reward Offline RL

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Optimal Single-Policy Sample Complexity and Transient Coverage for Average-Reward Offline RL

快速理解：We study offline reinforcement learning in average-reward MDPs, which presents increased challenges from the perspectives of distribution shift and non-uniform coverage, and has been relatively underexamined from a theoretical perspective. While previous work obtains performance guarantees under single-policy data coverage assumptions, such guarantees utilize additional complexity measures which are uniform over all policies, such as the uniform mixing time.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.573 Optimizing Anytime Reasoning via Budget Relative Policy Optimization

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Optimizing Anytime Reasoning via Budget Relative Policy Optimization

快速理解：Scaling test-time compute is crucial for enhancing the reasoning capabilities of large language models (LLMs). Existing approaches typically employ reinforcement learning (RL) to maximize a verifiable reward obtained at the end of reasoning traces.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.574 Optimizing Retrieval for RAG via Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Optimizing Retrieval for RAG via Reinforcement Learning

快速理解：As retrieval-augmented generation (RAG) becomes more widespread, the role of retrieval is shifting from retrieving information for human browsing to retrieving context for AI reasoning. This shift creates more complex search environments, where relevance is difficult to pre-define.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.575 Optimizing the Unknown: Black Box Bayesian Optimization with Energy-Based Model and Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Optimizing the Unknown: Black Box Bayesian Optimization with Energy-Based Model and Reinforcement Learning

快速理解：Existing Bayesian Optimization (BO) methods typically balance exploration and exploitation to optimize costly objective functions. However, these methods often suffer from a significant one-step bias, which may lead to convergence towards local optima and poor performance in complex or high-dimensional tasks.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.576 Option-aware Temporally Abstracted Value for Offline Goal-Conditioned Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Option-aware Temporally Abstracted Value for Offline Goal-Conditioned Reinforcement Learning

快速理解：Offline goal-conditioned reinforcement learning (GCRL) offers a practical learning paradigm in which goal-reaching policies are trained from abundant state-action trajectory datasets without additional environment interaction. However, offline GCRL still struggles with long-horizon tasks, even with recent advances that employ hierarchical policy structures, such as HIQL [33].

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.577 OrbitZoo: Real Orbital Systems Challenges for Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：OrbitZoo: Real Orbital Systems Challenges for Reinforcement Learning

快速理解：The increasing number of satellites and orbital debris has made space congestion a critical issue, threatening satellite safety and sustainability. Challenges such as collision avoidance, station-keeping, and orbital maneuvering require advanced techniques to handle dynamic uncertainties and multi-agent interactions.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.578 Oryx: a Scalable Sequence Model for Many-Agent Coordination in Offline MARL

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Oryx: a Scalable Sequence Model for Many-Agent Coordination in Offline MARL

快速理解：A key challenge in offline multi-agent reinforcement learning (MARL) is achieving effective many-agent multi-step coordination in complex environments. In this work, we propose Oryx, a novel algorithm for offline cooperative MARL to directly address this challenge.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.579 Outcome-Based Online Reinforcement Learning: Algorithms and Fundamental Limits

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文: Outcome-Based Online Reinforcement Learning: Algorithms and Fundamental Limits

快速理解: Reinforcement learning with outcome-based feedback faces a fundamental challenge: when rewards are only observed at trajectory endpoints, how do we assign credit to the right actions? This paper provides the first comprehensive analysis of this problem in online RL with general function approximation.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.580 Overcoming Challenges of Long-Horizon Prediction in Driving World Models

- 作者: 作者信息未提供
- 年份: 2025; 主题: World Model; 优先级: 重要
- [来源](#)

论文: Overcoming Challenges of Long-Horizon Prediction in Driving World Models

快速理解: Existing world models for autonomous driving struggle with long-horizon generation and generalization to challenging scenarios. In this work, we develop a model using simple design choices, and without additional supervision or sensors, such as maps, depth, or multiple cameras.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策。

证据层级: 本地 PDF 前页。

1.3.7.581 PANDA: Towards Generalist Video Anomaly Detection via Agentic AI Engineer

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: PANDA: Towards Generalist Video Anomaly Detection via Agentic AI Engineer

快速理解: Video anomaly detection (VAD) is a critical yet challenging task due to the complex and diverse nature of real-world scenarios. Previous methods typically rely on domain-specific training data and manual adjustments when applying to new scenarios and unseen anomaly types, suffering from high labor costs and limited generalization.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.582 PARCO: Parallel AutoRegressive Models for Multi-Agent Combinatorial Optimization

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: PARCO: Parallel AutoRegressive Models for Multi-Agent Combinatorial Optimization

快速理解: Combinatorial optimization problems involving multiple agents are notoriously challenging

due to their NP-hard nature and the necessity for effective agent coordination. Despite advancements in learning-based methods, existing approaches often face critical limitations, including suboptimal agent coordination, poor generalization, and high computational latency.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.583 PARTONOMY: Large Multimodal Models with Part-Level Visual Understanding

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：PARTONOMY: Large Multimodal Models with Part-Level Visual Understanding

快速理解：Real-world objects are composed of distinctive, object-specific parts. Identifying these parts is key to performing fine-grained, compositional reasoning—yet, large multimodal models (LMMs) struggle to perform this seemingly straightforward task.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.584 PLANA3R: Zero-shot Metric Planar 3D Reconstruction via Feed-forward Planar Splatting

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：PLANA3R: Zero-shot Metric Planar 3D Reconstruction via Feed-forward Planar Splatting

快速理解：This paper addresses metric 3D reconstruction of indoor scenes by exploiting their inherent geometric regularities with compact representations. Using planar 3D primitives –a well-suited representation for man-made environments –we introduce PLANA3R, a pose-free framework for metric Planar 3D Reconstruction from unposed two-view images.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.585 PMQ-VE: Progressive Multi-Frame Quantization for Video Enhancement

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：PMQ-VE: Progressive Multi-Frame Quantization for Video Enhancement

快速理解：Multi-frame video enhancement tasks aim to improve the spatial and temporal resolution and quality of video sequences by leveraging temporal information from multiple frames, which are widely used in streaming video processing, surveillance, and generation. Although numerous Transformer-based enhancement methods have achieved impressive performance, their computational and memory demands hinder deployment on edge devices.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.7.586 PT-MoE: An Efficient Finetuning Framework for Integrating Mixture-of-Experts into Prompt Tuning

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：PT-MoE: An Efficient Finetuning Framework for Integrating Mixture-of-Experts into Prompt Tuning

快速理解：Parameter-efficient fine-tuning (PEFT) methods have shown promise in adapting large language models, yet existing approaches exhibit counter-intuitive phenomena: integrating either matrix decomposition or mixture-of-experts (MoE) individually decreases performance across tasks, though decomposition improves results on specific domains despite reducing parameters, while MoE increases parameter count without corresponding decrease in training efficiency. Motivated by these observations and the modular nature of PT, we propose PT-MoE, a novel framework that integrates matrix decomposition with MoE routing for efficient PT.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.7.587 PanoWan: Lifting Diffusion Video Generation Models to 360° with Latitude/Longitude-aware Mechanisms

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：PanoWan: Lifting Diffusion Video Generation Models to 360° with Latitude/Longitude-aware Mechanisms

快速理解：Panoramic video generation enables immersive 360° content creation, valuable in applications that demand scene-consistent world exploration. However, existing panoramic video generation models struggle to leverage pre-trained generative priors from conventional text-to-video models for high-quality and diverse panoramic videos generation, due to limited dataset scale and the gap in spatial feature representations.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.588 ParetoQ: Improving Scaling Laws in Extremely Low-bit LLM Quantization

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：ParetoQ: Improving Scaling Laws in Extremely Low-bit LLM Quantization

快速理解：The optimal bit-width for achieving the best trade-off between quantized model size and

accuracy has been a subject of ongoing debate. While some advocate for 4-bit quantization, others propose that 1.58-bit offers superior results.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.7.589 Parsimonious Predictions for Strategyproof Scheduling

- 作者：作者信息未提供
- 年份：2025；主题：Systems；优先级：重要
- [来源](#)

论文：Parsimonious Predictions for Strategyproof Scheduling

快速理解：We consider the problem of scheduling m jobs on n unrelated strategic machines to minimize the maximum load of any machine. As the machines are strategic they may misreport processing times to minimize their own load.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.7.590 Pass@K Policy Optimization: Solving Harder Reinforcement Learning Problems

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Pass@K Policy Optimization: Solving Harder Reinforcement Learning Problems

快速理解：Reinforcement Learning algorithms commonly sample multiple ($n > 1$) solution attempts for each problem and reward them independently. This optimizes for pass@1 performance and prioritizes individual sample performance over the diversity and collective utility of a set of samples.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.591 PathVQ: Reforming Computational Pathology Foundation Model for Whole Slide Image Analysis via Vector Quantization

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：PathVQ: Reforming Computational Pathology Foundation Model for Whole Slide Image Analysis via Vector Quantization

快速理解：Pathology whole slide image (WSI) analysis is vital for disease diagnosis and understanding. While foundation models (FMs) have driven recent advances, their scalability in pathology remains a key challenge.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.7.592 Perception-R1: Pioneering Perception Policy with Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Perception-R1: Pioneering Perception Policy with Reinforcement Learning

快速理解：Inspired by the success of DeepSeek-R1, we explore the potential of rule-based reinforcement learning (RL) in MLLM post-training for perception policy learning. While promising, our initial experiments reveal that incorporating a thinking process through RL does not consistently lead to performance gains across all visual perception tasks.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.593 PhysCtrl: Generative Physics for Controllable and Physics-Grounded Video Generation

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：PhysCtrl: Generative Physics for Controllable and Physics-Grounded Video Generation

快速理解：Existing video generation models excel at producing photo-realistic videos from text or images, but often lack physical plausibility and 3D controllability. To overcome these limitations, we introduce PhysCtrl, a novel framework for physicsgrounded image-to-video generation with physical parameters and force control.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.594 PhysVLM-AVR: Active Visual Reasoning for Multimodal Large Language Models in Physical Environments

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：PhysVLM-AVR: Active Visual Reasoning for Multimodal Large Language Models in Physical Environments

快速理解：Visual reasoning in multimodal large language models (MLLMs) has primarily been studied in static, fully observable settings, limiting their effectiveness in real-world environments where information is often incomplete due to occlusion or limited field of view. Humans, in contrast, actively explore and interact with their environment—moving, examining, and manipulating objects—to gather information through a closed-loop process integrating perception, reasoning, and action.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.595 Physics-informed Value Learner for Offline Goal-Conditioned Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Physics-informed Value Learner for Offline Goal-Conditioned Reinforcement Learning

快速理解：Offline Goal-Conditioned Reinforcement Learning (GCRL) holds great promise for domains such as autonomous navigation and locomotion, where collecting interactive data is costly and unsafe. However, it remains challenging in practice due to the need to learn from datasets with limited coverage of the state-action space and to generalize across long-horizon tasks.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.596 Pixel Reasoner: Incentivizing Pixel Space Reasoning via Curiosity-Driven Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Pixel Reasoner: Incentivizing Pixel Space Reasoning via Curiosity-Driven Reinforcement Learning

快速理解：Chain-of-thought reasoning has significantly improved the performance of Large Language Models (LLMs) across various domains. However, this reasoning process has been confined exclusively to textual space, limiting its effectiveness in visually intensive tasks.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.597 PlanarGS: High-Fidelity Indoor 3D Gaussian Splatting Guided by Vision-Language Planar Priors

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：PlanarGS: High-Fidelity Indoor 3D Gaussian Splatting Guided by Vision-Language Planar Priors

快速理解：Three-dimensional Gaussian Splatting (3DGS) has recently emerged as an efficient representation for novel-view synthesis, achieving impressive visual quality. However, in scenes dominated by large and low-texture regions, common in indoor environments, the photometric loss used to optimize 3DGS yields ambiguous geometry and fails to recover high-fidelity 3D surfaces.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.598 Plenodium: Underwater 3D Scene Reconstruction with Plenoptic Medium Representation

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Plenodium: Underwater 3D Scene Reconstruction with Plenoptic Medium Representation

快速理解：We present Plenodium (plenoptic medium), an effective and efficient 3D representation framework capable of jointly modeling both objects and the participating medium. In contrast to existing medium representations that rely solely on viewdependent modeling, our novel plenoptic medium representation incorporates both directional and positional information through spherical harmonics encoding, enabling highly accurate underwater scene reconstruction.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.599 Plug-and-play Feature Causality Decomposition for Multimodal Representation Learning

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Plug-and-play Feature Causality Decomposition for Multimodal Representation Learning

快速理解：Multimodal representation learning is critical for a wide range of applications, such as multimodal sentiment analysis. Current multimodal representation learning methods mainly focus on the multimodal alignment or fusion strategies, such that the complementary and consistent information among heterogeneous modalities can be fully explored.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.600 PoE-World: Compositional World Modeling with Products of Programmatic Experts

- 作者：作者信息未提供
- 年份：2025；主题：World Model；优先级：重要
- [来源](#)

论文：PoE-World: Compositional World Modeling with Products of Programmatic Experts

快速理解：Learning how the world works is central to building AI agents that can adapt to complex environments. Traditional world models based on deep learning demand vast amounts of training data, and do not flexibly update their knowledge from sparse observations.

关注点：环境动力学、视频预测、潜变量状态或模型式决策。

证据层级：本地 PDF 前页。

1.3.7.601 Point Cloud Synthesis Using Inner Product Transforms

- 作者：作者信息未提供
- 年份：2025；主题：Systems；优先级：重要
- [来源](#)

论文：Point Cloud Synthesis Using Inner Product Transforms

快速理解：Point cloud synthesis, i.e. the generation of novel point clouds from an input distribution, remains a challenging task, for which numerous complex machine learning models have been devised. We develop a novel method that encodes geometrical-topological characteristics of point clouds using inner products, leading to a highly-efficient point cloud representation with provable expressivity properties.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.7.602 Point-MaDi: Masked Autoencoding with Diffusion for Point Cloud Pre-training

- 作者：作者信息未提供
- 年份：2025；主题：Systems；优先级：重要
- [来源](#)

论文：Point-MaDi: Masked Autoencoding with Diffusion for Point Cloud Pre-training

快速理解：Self-supervised pre-training is essential for 3D point cloud representation learning, as annotating their irregular, topology-free structures is costly and labor-intensive. Masked autoencoders (MAEs) offer a promising framework but rely on explicit positional embeddings, such as patch center coordinates, which leak geometric information and limit data-driven structural learning.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.7.603 Point-RFT: Improving Multimodal Reasoning with Visually Grounded Reinforcement Finetuning

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Point-RFT: Improving Multimodal Reasoning with Visually Grounded Reinforcement Finetuning

快速理解：Recent advances in large language models have significantly improved textual reasoning through the effective use of Chain-of-Thought (CoT) and reinforcement learning. However, extending these successes to vision-language tasks remains challenging due to inherent limitations in text-only CoT, such as visual hallucinations and insufficient multimodal integration.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.604 Point3R: Streaming 3D Reconstruction with Explicit Spatial Pointer Memory

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Point3R: Streaming 3D Reconstruction with Explicit Spatial Pointer Memory

快速理解：Dense 3D scene reconstruction from an ordered sequence or unordered image collections is a critical step when bringing research in computer vision into practical scenarios. Following the paradigm introduced by DUS3R, which unifies an image pair densely into a shared coordinate system, subsequent methods maintain an implicit memory to achieve dense 3D reconstruction from more images.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.605 PointMAC: Meta-Learned Adaptation for Robust Test-Time Point Cloud Completion

- 作者：作者信息未提供
- 年份：2025；主题：Systems；优先级：重要
- [来源](#)

论文：PointMAC: Meta-Learned Adaptation for Robust Test-Time Point Cloud Completion

快速理解：Point cloud completion is essential for robust 3D perception in safety-critical applications such as robotics and augmented reality. However, existing models perform static inference and rely heavily on inductive biases learned during training, limiting their ability to adapt to novel structural patterns and sensor-induced distortions at test time.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.7.606 PointTruss: K-Truss for Point Cloud Registration

- 作者：作者信息未提供
- 年份：2025；主题：Systems；优先级：重要
- [来源](#)

论文：PointTruss: K-Truss for Point Cloud Registration

快速理解：Point cloud registration is a fundamental task in 3D computer vision. Recent advances have shown that graph-based methods are effective for outlier rejection in this context.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.7.607 PolyVivid: Vivid Multi-Subject Video Generation with Cross-Modal Interaction and Enhancement

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文: PolyVivid: Vivid Multi-Subject Video Generation with Cross-Modal Interaction and Enhancement

快速理解: Despite recent advances in video generation, existing models still lack finegrained controllability, especially for multi-subject customization with consistent identity and interaction. In this paper, we propose PolyVivid, a multi-subject video customization framework that enables flexible and identity-consistent generation.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.608 Praxis-VLM: Vision-Grounded Decision Making via Text-Driven Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Praxis-VLM: Vision-Grounded Decision Making via Text-Driven Reinforcement Learning

快速理解: Vision Language Models exhibit impressive performance for various tasks, yet they often lack the sophisticated situational reasoning required for complex decisionmaking. This paper shows that VLMs can achieve surprisingly strong decisionmaking performance when visual scenes are replaced by textual descriptions, suggesting foundational reasoning can be effectively learned from language.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.609 Pre-Trained Policy Discriminators are General Reward Models

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Pre-Trained Policy Discriminators are General Reward Models

快速理解: We offer a novel perspective on reward modeling by formulating it as a policy discriminator, which quantifies the difference between two policies to generate a reward signal, guiding the training policy towards a target policy with desired behaviors. Based on this conceptual insight, we propose a scalable pre-training method named POLicy DiscriminAtive LeaRning (POLAR), which trains a reward model (RM) to discern identical policies and discriminate different ones.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.610 Preference Distillation via Value based Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Preference Distillation via Value based Reinforcement Learning

快速理解: Direct Preference Optimization (DPO) is a powerful paradigm to align language models with human preferences using pairwise comparisons. However, its binary win-or-loss supervision often proves insufficient for training small models with limited capacity.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.611 Preference-based Reinforcement Learning beyond Pairwise Comparisons: Benefits of Multiple Options

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Preference-based Reinforcement Learning beyond Pairwise Comparisons: Benefits of Multiple Options

快速理解: We study online preference-based reinforcement learning (PbRL) with the goal of improving sample efficiency. While a growing body of theoretical work has emerged—motivated by PbRL’s recent empirical success, particularly in aligning large language models (LLMs)—most existing studies focus only on pairwise comparisons.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.612 PrefixKV: Adaptive Prefix KV Cache is What Vision Instruction-Following Models Need for Efficient Generation

- 作者: 作者信息未提供
- 年份: 2025; 主题: Systems; 优先级: 重要
- [来源](#)

论文: PrefixKV: Adaptive Prefix KV Cache is What Vision Instruction-Following Models Need for Efficient Generation

快速理解: Recently, large vision-language models (LVLMs) have rapidly gained popularity for their strong generation and reasoning capabilities given diverse multimodal inputs. However, these models incur significant computational and memory overhead during inference, which greatly hinders the efficient deployment in practical scenarios.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.7.613 Pretraining a Shared Q-Network for Data-Efficient Offline Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Pretraining a Shared Q-Network for Data-Efficient Offline Reinforcement Learning

快速理解: Offline reinforcement learning (RL) aims to learn a policy from a fixed dataset without

additional environment interaction. However, effective offline policy learning often requires a large and diverse dataset to mitigate epistemic uncertainty.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.614 Prior-Guided Diffusion Planning for Offline Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Prior-Guided Diffusion Planning for Offline Reinforcement Learning

快速理解：Diffusion models have recently gained prominence in offline reinforcement learning due to their ability to effectively learn high-performing, generalizable policies from static datasets. Diffusion-based planners facilitate long-horizon decisionmaking by generating high-quality trajectories through iterative denoising, guided by return-maximizing objectives.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.615 ProRL: Prolonged Reinforcement Learning Expands Reasoning Boundaries in Large Language Models

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：ProRL: Prolonged Reinforcement Learning Expands Reasoning Boundaries in Large Language Models

快速理解：Recent advances in reasoning-centric language models have highlighted reinforcement learning (RL) as a promising method for aligning models with verifiable rewards. However, it remains contentious whether RL truly expands a model's reasoning capabilities or merely amplifies high-reward outputs already latent in the base model's distribution, and whether continually scaling up RL compute reliably leads to improved reasoning performance.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.616 Process vs. Outcome Reward: Which is Better for Agentic RAG Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Process vs. Outcome Reward: Which is Better for Agentic RAG Reinforcement Learning

快速理解：Retrieval-augmented generation (RAG) enhances the text generation capabilities of large language models (LLMs) by integrating external knowledge and up-to-date information. However, traditional RAG systems are limited by static workflows and lack the adaptability required for

multistep reasoning and complex task management.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.617 ProfiX: Improving Profile-Guided Optimization in Compilers with Graph Neural Networks

- 作者：作者信息未提供
- 年份：2025；主题：Systems；优先级：重要
- [来源](#)

论文：ProfiX: Improving Profile-Guided Optimization in Compilers with Graph Neural Networks

快速理解：Profile-guided optimization (PGO) advances the frontiers of compiler optimization by leveraging dynamic runtime information to generate highly optimized binaries. Traditional instrumentation-based profiling collects accurate profile data but often suffers from heavy runtime overhead.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.7.618 Progress Reward Model for Reinforcement Learning via Large Language Models

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Progress Reward Model for Reinforcement Learning via Large Language Models

快速理解：Traditional reinforcement learning (RL) algorithms face significant limitations in handling long-term tasks with sparse rewards. Recent advancements have leveraged large language models (LLMs) to enhance RL by utilizing their world knowledge for task planning and reward generation.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.619 Prompted Policy Search: Reinforcement Learning through Linguistic and Numerical Reasoning in LLMs

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Prompted Policy Search: Reinforcement Learning through Linguistic and Numerical Reasoning in LLMs

快速理解：Reinforcement Learning (RL) traditionally relies on scalar reward signals, limiting its ability to leverage the rich semantic knowledge often available in real-world tasks. In contrast, humans learn efficiently by combining numerical feedback with language, prior knowledge, and common sense.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.620 Prot2Text-V2: Protein Function Prediction with Multimodal Contrastive Alignment

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Prot2Text-V2: Protein Function Prediction with Multimodal Contrastive Alignment

快速理解：Predicting protein function from sequence is a central challenge in computational biology. While existing methods rely heavily on structured ontologies or similaritybased techniques, they often lack the flexibility to express structure-free functional descriptions and novel biological functions.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.621 Provably Efficient Online RLHF with One-Pass Reward Modeling

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Provably Efficient Online RLHF with One-Pass Reward Modeling

快速理解：Reinforcement Learning from Human Feedback (RLHF) has shown remarkable success in aligning Large Language Models (LLMs) with human preferences. Traditional RLHF methods rely on a fixed dataset, which often suffers from limited coverage.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.622 Puzzles: Unbounded Video-Depth Augmentation for Scalable End-to-End 3D Reconstruction

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Puzzles: Unbounded Video-Depth Augmentation for Scalable End-to-End 3D Reconstruction

快速理解：Multi-view 3D reconstruction remains a core challenge in computer vision. Recent methods, such as DUS3R and its successors, directly regress pointmaps from image pairs without relying on known scene geometry or camera parameters.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.623 Q-Insight: Understanding Image Quality via Visual Reinforcement Learning

- 作者：作者信息未提供

- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Q-Insight: Understanding Image Quality via Visual Reinforcement Learning

快速理解: Image quality assessment (IQA) focuses on the perceptual visual quality of images, playing a crucial role in downstream tasks such as image reconstruction, compression, and generation. The rapid advancement of multi-modal large language models (MLLMs) has significantly broadened the scope of IQA, moving toward comprehensive image quality understanding that incorporates content analysis, degradation perception, and comparison reasoning beyond mere numerical scoring.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.624 QBasicVSR: Temporal Awareness Adaptation Quantization for Video Super-Resolution

- 作者: 作者信息未提供
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: QBasicVSR: Temporal Awareness Adaptation Quantization for Video Super-Resolution

快速理解: While model quantization has become pivotal for deploying super-resolution (SR) networks on mobile devices, existing works focus on quantization methods only for image super-resolution. Different from image SR quantization, the temporal error propagation, shared temporal parameterization, and temporal metric mismatch significantly degrade the quantization performance of a video SR model.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 本地 PDF 前页。

1.3.7.625 QSVD: Efficient Low-rank Approximation for Unified Query-Key-Value Weight Compression in Low-Precision Vision-Language Models

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: QSVD: Efficient Low-rank Approximation for Unified Query-Key-Value Weight Compression in Low-Precision Vision-Language Models

快速理解: Vision-Language Models (VLMs) are integral to tasks such as image captioning and visual question answering, but their high computational cost, driven by large memory footprints and processing time, limits their scalability and real-time applicability. In this work, we propose leveraging Singular-Value Decomposition (SVD) over the joint query (Q), key (K), and value (V) weight matrices to reduce KV cache size and computational overhead.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.626 QoQ-Med: Building Multimodal Clinical Foundation Models with Domain-Aware GRPO Training

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：QoQ-Med: Building Multimodal Clinical Foundation Models with Domain-Aware GRPO Training

快速理解：Clinical decision-making routinely demands reasoning over heterogeneous data, yet existing multimodal language models (MLLMs) remain largely vision-centric and fail to generalize across clinical specialties. To bridge this gap, we introduce QoQ-Med-7B/32B, the first open generalist clinical foundation model that jointly reasons across medical images, time-series signals, and text reports.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.627 Quality-Driven Curation of Remote Sensing Vision-Language Data via Learned Scoring Models

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Quality-Driven Curation of Remote Sensing Vision-Language Data via Learned Scoring Models

快速理解：Vision-Language Models (VLMs) have demonstrated great potential in interpreting remote sensing (RS) images through language-guided semantic. However, the effectiveness of these VLMs critically depends on high-quality image-text training data that captures rich semantic relationships between visual content and language descriptions.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.628 Quantifying Cross-Modality Memorization in Vision-Language Models

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Quantifying Cross-Modality Memorization in Vision-Language Models

快速理解：Understanding what and how neural networks memorize during training is crucial, both from the perspective of unintentional memorization of potentially sensitive information and from the standpoint of effective knowledge acquisition for realworld, knowledge-intensive tasks. While previous studies primarily investigate memorization within a single modality, such as text memorization in large language models or image memorization in diffusion models, unified multimodal models are becoming increasingly prevalent in practical applications.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.629 Quantile Reward Policy Optimization: Alignment with Pointwise Regression and Exact Partition Functions

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Quantile Reward Policy Optimization: Alignment with Pointwise Regression and Exact Partition Functions

快速理解：Aligning large language models with pointwise absolute rewards has so far required online, on-policy algorithms such as PPO and GRPO. In contrast, simpler methods that can leverage offline or off-policy data, such as DPO and REBEL, are limited to learning from preference pairs or relative signals.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.630 Quantization Error Propagation: Revisiting Layer-Wise Post-Training Quantization

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：Quantization Error Propagation: Revisiting Layer-Wise Post-Training Quantization

快速理解：Layer-wise PTQ is a promising technique for compressing large language models (LLMs), due to its simplicity and effectiveness without requiring retraining. However, recent progress in this area is saturating, underscoring the need to revisit its core limitations and explore further improvements.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.7.631 Quantization-Free Autoregressive Action Transformer

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：Quantization-Free Autoregressive Action Transformer

快速理解：Current transformer-based imitation learning approaches introduce discrete action representations and train an autoregressive transformer decoder on the resulting latent code. However, the initial quantization breaks the continuous structure of the action space thereby limiting the capabilities of the generative model.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.7.632 R-KV: Redundancy-aware KV Cache Compression for Reasoning Models

- 作者：作者信息未提供
- 年份：2025；主题：Systems；优先级：重要
- [来源](#)

论文：R-KV: Redundancy-aware KV Cache Compression for Reasoning Models

快速理解：Reasoning models have demonstrated impressive performance in self-reflection and chain-of-thought reasoning. However, they often produce excessively long outputs, leading to prohibitively large key-value (KV) caches during inference.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.7.633 R1-ShareVL: Incentivizing Reasoning Capabilities of Multimodal Large Language Models via Share-GRPO

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：R1-ShareVL: Incentivizing Reasoning Capabilities of Multimodal Large Language Models via Share-GRPO

快速理解：In this work, we aim to incentivize the reasoning ability of Multimodal Large Language Models (MLLMs) via reinforcement learning (RL) and develop an effective approach that mitigates the sparse reward and advantage vanishing issues during RL. To this end, we propose Share-GRPO, a novel RL approach that tackle these issues by exploring and sharing diverse reasoning trajectories over expanded question space.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.634 RAD: Training an End-to-End Driving Policy via Large-Scale 3DGS-based Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：RAD: Training an End-to-End Driving Policy via Large-Scale 3DGS-based Reinforcement Learning

快速理解：Existing end-to-end autonomous driving (AD) algorithms typically follow the Imitation Learning (IL) paradigm, which faces challenges such as causal confusion and an open-loop gap. In this work, we propose RAD, a 3DGS-based closed-loop Reinforcement Learning (RL) framework for end-to-end Autonomous Driving.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.635 RAPID Hand: Robust, Affordable, Perception-Integrated, Dexterous Manipulation Platform for Embodied Intelligence

- 作者：作者信息未提供
- 年份：2025；主题：Embodied/WBC；优先级：重要
- [来源](#)

论文：RAPID Hand: Robust, Affordable, Perception-Integrated, Dexterous Manipulation Platform for Embodied Intelligence

快速理解：This paper addresses the scarcity of low-cost but high-dexterity platforms for collecting real-world multi-fingered robot manipulation data towards generalist robot autonomy. To achieve it, we propose the RAPID Hand, a co-optimized hardware and software platform where the compact 20-DoF hand, robust whole-hand perception, and high-DoF teleoperation interface are jointly designed.

关注点：具身任务中的感知、动作生成、全身协调或运动控制。

证据层级：本地 PDF 前页。

1.3.7.636 REGen: Multimodal Retrieval-Embedded Generation for Long-to-Short Video Editing

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：REGen: Multimodal Retrieval-Embedded Generation for Long-to-Short Video Editing

快速理解：Short videos are an effective tool for promoting contents and improving knowledge accessibility. While existing extractive video summarization methods struggle to produce a coherent narrative, existing abstractive methods cannot ‘quote’ from the input videos, i.e., inserting short video clips in their outputs.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.637 RPG360: Robust 360 Depth Estimation with Perspective Foundation Models and Graph Optimization

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：RPG360: Robust 360 Depth Estimation with Perspective Foundation Models and Graph Optimization

快速理解：The increasing use of 360° images across various domains has emphasized the need for robust depth estimation techniques tailored for omnidirectional images. However, obtaining large-scale labeled datasets for 360° depth estimation remains a significant challenge.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.638 RSAVQ: Riemannian Sensitivity-Aware Vector Quantization for Large Language Models

- 作者：作者信息未提供

- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: RSAVQ: Riemannian Sensitivity-Aware Vector Quantization for Large Language Models

快速理解: Large language models (LLMs) have demonstrated remarkable performance across a wide range of natural language processing tasks. However, their exponentially increasing parameters pose significant challenges for deployment on resourceconstrained devices.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 本地 PDF 前页。

1.3.7.639 RadZero: Similarity-Based Cross-Attention for Explainable Vision-Language Alignment in Chest X-ray with Zero-Shot Multi-Task Capability

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: RadZero: Similarity-Based Cross-Attention for Explainable Vision-Language Alignment in Chest X-ray with Zero-Shot Multi-Task Capability

快速理解: Recent advancements in multimodal models have significantly improved visionlanguage (VL) alignment in radiology. However, existing approaches struggle to effectively utilize complex radiology reports for learning and offer limited interpretability through attention probability visualizations.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.640 Radial Attention: $\mathcal{O}(n \log n)$ Sparse Attention with Energy Decay for Long Video Generation

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Radial Attention: $\mathcal{O}(n \log n)$ Sparse Attention with Energy Decay for Long Video Generation

快速理解: Recent advances in diffusion models have enabled high-quality video generation, but the additional temporal dimension significantly increases computational costs, making training and inference on long videos prohibitively expensive. In this paper, we identify a phenomenon we term Spatiotemporal Energy Decay in video diffusion models: post-softmax attention scores diminish as spatial and temporal distance between tokens increase, akin to the physical decay of signal or waves over space and time in nature.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.641 Rainbow Delay Compensation: A Multi-Agent Reinforcement Learning Framework for Mitigating Observation Delays

- 作者: 作者信息未提供

- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Rainbow Delay Compensation: A Multi-Agent Reinforcement Learning Framework for Mitigating Observation Delays

快速理解: In real-world multi-agent systems (MASs), observation delays are ubiquitous, preventing agents from making decisions based on the environment's true state. An individual agent's local observation typically comprises multiple components from other agents or dynamic entities within the environment.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.642 ReAgent-V: A Reward-Driven Multi-Agent Framework for Video Understanding

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: ReAgent-V: A Reward-Driven Multi-Agent Framework for Video Understanding

快速理解: Video understanding is fundamental to tasks such as action recognition, video reasoning, and robotic control. Early video understanding methods based on large vision-language models (LVLMs) typically adopt a single-pass reasoning paradigm without dynamic feedback, limiting the model's capacity to self-correct and adapt in complex scenarios.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.643 ReDit: Reward Dithering for Improved LLM Policy Optimization

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: ReDit: Reward Dithering for Improved LLM Policy Optimization

快速理解: DeepSeek-R1 has successfully enhanced Large Language Models (LLMs) reasoning capabilities through its rule-based reward system. While it's a "perfect" reward system that effectively mitigates reward hacking, such reward functions are often discrete.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.644 ReMA: Learning to Meta-Think for LLMs with Multi-agent Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: ReMA: Learning to Meta-Think for LLMs with Multi-agent Reinforcement Learning

快速理解: Recent research on Reasoning of Large Language Models (LLMs) has sought to further enhance their performance by integrating meta-thinking—enabling models to monitor, evaluate, and control their reasoning processes for more adaptive and effective problem-solving. However, current single-agent work lacks a specialized design for acquiring meta-thinking, resulting in low efficacy.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.645 ReSearch: Learning to Reason with Search for LLMs via Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: ReSearch: Learning to Reason with Search for LLMs via Reinforcement Learning

快速理解: Large Language Models (LLMs) have shown remarkable capabilities in reasoning, exemplified by the success of OpenAI-o1 and DeepSeek-R1. However, integrating reasoning with external search processes remains challenging, especially for complex multi-hop questions requiring multiple retrieval steps.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.646 Real-World Adverse Weather Image Restoration via Dual-Level Reinforcement Learning with High-Quality Cold Start

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Real-World Adverse Weather Image Restoration via Dual-Level Reinforcement Learning with High-Quality Cold Start

快速理解: Adverse weather severely impairs real-world visual perception, while existing vision models trained on synthetic data with fixed parameters struggle to generalize to complex degradations. To address this, we first construct HFLS-Weather, a physics-driven, high-fidelity dataset that simulates diverse weather phenomena, and then design a dual-level reinforcement learning framework initialized with HFLS-Weather for cold-start training.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.647 Real-World Reinforcement Learning of Active Perception Behaviors

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Real-World Reinforcement Learning of Active Perception Behaviors

快速理解: A robot's instantaneous sensory observations do not always reveal task-relevant state information. Under such partial observability, optimal behavior typically involves explicitly acting to gain the missing information.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.648 Reason-RFT: Reinforcement Fine-Tuning for Visual Reasoning of Vision Language Models

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Reason-RFT: Reinforcement Fine-Tuning for Visual Reasoning of Vision Language Models

快速理解: Visual reasoning abilities play a crucial role in understanding complex multimodal data, advancing both domain-specific applications and artificial general intelligence (AGI). Existing methods improve Vision-Language Models (VLMs) reasoning via Chain-of-Thought (CoT) supervised fine-tuning, using meticulously annotated training data to enhance visual reasoning capabilities.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.649 Reasoning Models Hallucinate More: Factuality-Aware Reinforcement Learning for Large Reasoning Models

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Reasoning Models Hallucinate More: Factuality-Aware Reinforcement Learning for Large Reasoning Models

快速理解: Large language models (LLMs) have significantly advanced in reasoning tasks through reinforcement learning (RL) optimization, achieving impressive capabilities across various challenging benchmarks. However, our empirical analysis reveals a critical drawback: reasoning-oriented RL fine-tuning significantly increases the prevalence of hallucinations.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.650 Rebalancing Return Coverage for Conditional Sequence Modeling in Offline Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Rebalancing Return Coverage for Conditional Sequence Modeling in Offline Reinforcement Learning

快速理解: Recent advancements in offline reinforcement learning (RL) have underscored the capa-

bilities of conditional sequence modeling (CSM), a paradigm that models the action distribution conditioned on both historical trajectories and target returns associated with each state. However, due to the imbalanced return distribution caused by suboptimal datasets, CSM is grappling with a serious distributional shift problem when conditioning on high returns.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.651 Recognition through Reasoning: Reinforcing Image Geo-localization with Large Vision-Language Models

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Recognition through Reasoning: Reinforcing Image Geo-localization with Large Vision-Language Models

快速理解：Previous methods for image geo-localization have typically treated the task as either classification or retrieval, often relying on black-box decisions that lack interpretability. The rise of large vision-language models (LVLMs) has enabled a rethinking of geo-localization as a reasoning-driven task grounded in visual cues.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.652 Rectified Point Flow: Generic Point Cloud Pose Estimation

- 作者：作者信息未提供
- 年份：2025；主题：Systems；优先级：重要
- [来源](#)

论文：Rectified Point Flow: Generic Point Cloud Pose Estimation

快速理解：We present Rectified Point Flow, a unified parameterization that formulates pairwise point cloud registration and multi-part shape assembly as a single conditional generative problem. Given unposed point clouds, our method learns a continuous point-wise velocity field that transports noisy points toward their target positions, from which part poses are recovered.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.7.653 Recursive Inference Scaling: A Winning Path to Scalable Inference in Language and Multimodal Systems

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Recursive Inference Scaling: A Winning Path to Scalable Inference in Language and Multimodal Systems

快速理解：Inspired by recent findings on the fractal geometry of language, we introduce Recursive

Inference Scaling (RINS) as a complementary, plug-in recipe for scaling inference time in language and multimodal systems. RINS is a particular form of recursive depth that significantly outperforms +55 other variants, including the recent “repeat-all-over” (RAO) strategy in Mobile LLM (Liu et al., 2024) and latent recurrent thinking (Geiping et al., 2025).

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.654 Regret Lower Bounds for Decentralized Multi-Agent Stochastic Shortest Path Problems

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Regret Lower Bounds for Decentralized Multi-Agent Stochastic Shortest Path Problems

快速理解：Multi-agent systems (MAS) are central to applications such as swarm robotics and traffic routing, where agents must coordinate in a decentralized manner to achieve a common objective. Stochastic Shortest Path (SSP) problems provide a natural framework for modeling decentralized control in such settings.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.655 Regret-Optimal Q-Learning with Low Cost for Single-Agent and Federated Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Regret-Optimal Q-Learning with Low Cost for Single-Agent and Federated Reinforcement Learning

快速理解：Motivated by real-world settings where data collection and policy deployment—whether for a single agent or across multiple agents—are costly, we study the problem of on-policy single-agent reinforcement learning (RL) and federated RL (FRL) with a focus on minimizing burn-in costs (the sample sizes needed to reach near-optimal regret) and policy switching or communication costs. In parallel finite-horizon episodic Markov Decision Processes (MDPs) with S states and A actions, existing methods either require superlinear burn-in costs in S and A or fail to achieve logarithmic switching or communication costs.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.656 ReinFlow: Fine-tuning Flow Matching Policy with Online Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文: ReinFlow: Fine-tuning Flow Matching Policy with Online Reinforcement Learning

快速理解: We propose ReinFlow, a simple yet effective online reinforcement learning (RL) framework that fine-tunes a family of flow matching policies for continuous robotic control. Derived from rigorous RL theory, ReinFlow injects learnable noise into a flow policy's deterministic path, converting the flow into a discrete-time Markov Process for exact and straightforward likelihood computation.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.657 Reinforcement Learning Finetunes Small Subnetworks in Large Language Models

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Reinforcement Learning Finetunes Small Subnetworks in Large Language Models

快速理解: Reinforcement learning (RL) yields substantial improvements in large language models' (LLMs) downstream task performance and alignment with human values. Surprisingly, such large gains result from updating only a small subnetwork comprising just 5%-30% of the parameters, with the rest effectively unchanged.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.658 Reinforcement Learning Meets Masked Generative Models: Mask-GRPO for Text-to-Image Generation

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Reinforcement Learning Meets Masked Generative Models: Mask-GRPO for Text-to-Image Generation

快速理解: Reinforcement learning (RL) has garnered increasing attention in text-to-image (T2I) generation. However, most existing RL approaches are tailored to either diffusion models or autoregressive models, overlooking an important alternative: masked generative models.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.659 Reinforcement Learning Teachers of Test Time Scaling

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Reinforcement Learning Teachers of Test Time Scaling

快速理解: Training reasoning language models (LMs) with reinforcement learning (RL) for one-hot

correctness inherently relies on the LM being able to explore and solve its task with some chance at initialization. Furthermore, a key use case of reasoning LMs is to act as teachers for distilling new students and cold-starting future RL iterations rather than being deployed themselves.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.660 Reinforcement Learning for Out-of-Distribution Reasoning in LLMs: An Empirical Study on Diagnosis-Related Group Coding

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Reinforcement Learning for Out-of-Distribution Reasoning in LLMs: An Empirical Study on Diagnosis-Related Group Coding

快速理解：Diagnosis-Related Group (DRG) codes are essential for hospital reimbursement and operations but require labor-intensive assignment. Large Language Models (LLMs) struggle with DRG coding due to the out-of-distribution (OOD) nature of the task: pretraining corpora rarely contain private clinical or billing data.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.661 Reinforcement Learning for Reasoning in Large Language Models with One Training Example

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Reinforcement Learning for Reasoning in Large Language Models with One Training Example

快速理解：We show that reinforcement learning with verifiable reward using one training example (1-shot RLVR) is effective in incentivizing the mathematical reasoning capabilities of large language models (LLMs). Applying RLVR to the base model Qwen2.5-Math-1.5B, we identify a single example that elevates model performance on MATH500 from 36.0% to 73.6% (8.6% improvement beyond format correction), and improves the average performance across six common mathematical reasoning benchmarks from 17.6% to 35.7% (7.0% non-format gain).

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.662 Reinforcement Learning with Action Chunking

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Reinforcement Learning with Action Chunking

快速理解：We present Q-chunking, a simple yet effective recipe for improving reinforcement learning

(RL) algorithms for long-horizon, sparse-reward tasks. Our recipe is designed for the offline-to-online RL setting, where the goal is to leverage an offline prior dataset to maximize the sample-efficiency of online learning.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.663 Reinforcement Learning with Backtracking Feedback

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Reinforcement Learning with Backtracking Feedback

快速理解：Addressing the critical need for robust safety in Large Language Models (LLMs), particularly against adversarial attacks and in-distribution errors, we introduce Reinforcement Learning with Backtracking Feedback (RLBF). This framework advances upon prior methods, such as BSAFE, by primarily leveraging a Reinforcement Learning (RL) stage where models learn to dynamically correct their own generation errors.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.664 Reinforcement Learning with Imperfect Transition Predictions: A Bellman--Jensen Approach

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Reinforcement Learning with Imperfect Transition Predictions: A Bellman-Jensen Approach

快速理解：Traditional reinforcement learning (RL) assumes the agents make decisions based on Markov decision processes (MDPs) with one-step transition models. In many real-world applications, such as energy management and stock investment, agents can access multi-step predictions of future states, which provide additional advantages for decision making.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.665 Reinforcing Spatial Reasoning in Vision-Language Models with Interwoven Thinking and Visual Drawing

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Reinforcing Spatial Reasoning in Vision-Language Models with Interwoven Thinking and Visual Drawing

快速理解：As textual reasoning with large language models (LLMs) has advanced significantly, there has been growing interest in enhancing the multimodal reasoning capabilities of large vision-language

models (LVLMs). However, existing methods primarily approach multimodal reasoning in a straightforward, text-centric manner, where both reasoning and answer derivation are conducted purely through text, with the only difference being the presence of multimodal input.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.666 Reliable Lifelong Multimodal Editing: Conflict-Aware Retrieval Meets Multi-Level Guidance

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Reliable Lifelong Multimodal Editing: Conflict-Aware Retrieval Meets Multi-Level Guidance

快速理解: The dynamic nature of real-world information demands efficient knowledge editing in multimodal large language models (MLLMs) to ensure continuous knowledge updates. However, existing methods often struggle with precise matching in largescale knowledge retrieval and lack multi-level guidance for coordinated editing, leading to less reliable outcomes.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.667 Rendering-Aware Reinforcement Learning for Vector Graphics Generation

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Rendering-Aware Reinforcement Learning for Vector Graphics Generation

快速理解: Scalable Vector Graphics (SVG) offer a powerful format for representing visual designs as interpretable code. Recent advances in vision-language models (VLMs) have enabled high-quality SVG generation by framing the problem as a code generation task and leveraging large-scale pre-training.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.668 Repurposing Marigold for Zero-Shot Metric Depth Estimation via Defocus Blur Cues

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Repurposing Marigold for Zero-Shot Metric Depth Estimation via Defocus Blur Cues

快速理解: Recent monocular metric depth estimation (MMDE) methods have made notable progress towards zero-shot generalization. However, they still exhibit a significant performance drop on out-of-distribution datasets.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.669 ResponseRank: Data-Efficient Reward Modeling through Preference Strength Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：ResponseRank: Data-Efficient Reward Modeling through Preference Strength Learning

快速理解：Binary choices, as often used for reinforcement learning from human feedback (RLHF), convey only the direction of a preference. A person may choose apples over oranges and bananas over grapes, but which preference is stronger?

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.670 Restage4D: Reanimating Deformable 3D Reconstruction from a Single Video

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Restage4D: Reanimating Deformable 3D Reconstruction from a Single Video

快速理解：Creating deformable 3D content has gained increasing attention with the rise of text-to-image and image-to-video generative models. While these models provide rich semantic priors for appearance, they struggle to capture the physical realism and motion dynamics needed for authentic 4D scene synthesis.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.671 Rethinking Multimodal Learning from the Perspective of Mitigating Classification Ability Disproportion

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Rethinking Multimodal Learning from the Perspective of Mitigating Classification Ability Disproportion

快速理解：Multimodal learning (MML) is significantly constrained by modality imbalance, leading to suboptimal performance in practice. While existing approaches primarily focus on balancing the learning of different modalities to address this issue, they fundamentally overlook the inherent disproportion in model classification ability, which serves as the primary cause of this phenomenon.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.672 RetrievalAttention: Accelerating Long-Context LLM Inference via Vector Retrieval

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：RetrievalAttention: Accelerating Long-Context LLM Inference via Vector Retrieval

快速理解：Transformer-based Large Language Models (LLMs) have become increasingly important. However, scaling LLMs to longer contexts incurs slow inference speed and high GPU memory consumption for caching key-value (KV) vectors.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.7.673 Retro-R1: LLM-based Agentic Retrosynthesis

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Retro-R1: LLM-based Agentic Retrosynthesis

快速理解：Retrosynthetic planning is a fundamental task in chemical discovery. Due to the vast combinatorial search space, identifying viable synthetic routes remains a significant challenge—even for expert chemists.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.674 Retrv-R1: A Reasoning-Driven MLLM Framework for Universal and Efficient Multimodal Retrieval

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Retrv-R1: A Reasoning-Driven MLLM Framework for Universal and Efficient Multimodal Retrieval

快速理解：The success of DeepSeek-R1 demonstrates the immense potential of using reinforcement learning (RL) to enhance LLMs' reasoning capabilities. This paper introduces Retrv-R1, the first R1-style MLLM specifically designed for multimodal universal retrieval, achieving higher performance by employing step-by-step reasoning to produce more accurate retrieval results.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.675 Revealing Multimodal Causality with Large Language Models

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Revealing Multimodal Causality with Large Language Models

快速理解：Uncovering cause-and-effect mechanisms from data is fundamental to scientific progress. While large language models (LLMs) show promise for enhancing causal discovery (CD) from unstructured data, their application to the increasingly prevalent multimodal setting remains a critical challenge.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.676 Reverse Engineering Human Preferences with Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Reverse Engineering Human Preferences with Reinforcement Learning

快速理解：The capabilities of Large Language Models (LLMs) are routinely evaluated by other LLMs trained to predict human preferences. This framework—known as LLM-as-a-judge—is highly scalable and relatively low cost.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.677 Reward-Aware Proto-Representations in Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Reward-Aware Proto-Representations in Reinforcement Learning

快速理解：In recent years, the successor representation (SR) has attracted increasing attention in reinforcement learning (RL), and it has been used to address some of its key challenges, such as exploration, credit assignment, and generalization. The SR can be seen as representing the underlying credit assignment structure of the environment by implicitly encoding its induced transition dynamics.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.678 RiOSWorld: Benchmarking the Risk of Multimodal Computer-Use Agents

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：RiOSWorld: Benchmarking the Risk of Multimodal Computer-Use Agents

快速理解: With the rapid development of multimodal large language models (MLLMs), they are increasingly deployed as autonomous computer-use agents capable of accomplishing complex computer tasks. However, a pressing issue arises: Can the safety risk principles designed and aligned for general MLLMs in dialogue scenarios be effectively transferred to real-world computer-use scenarios?

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.679 Rig3R: Rig-Aware Conditioning and Discovery for 3D Reconstruction

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Rig3R: Rig-Aware Conditioning and Discovery for 3D Reconstruction

快速理解: Estimating agent pose and 3D scene structure from multi-camera rigs is a central task in embodied AI applications such as autonomous driving. Recent learned approaches such as DUST3R have shown impressive performance in multiview settings.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.680 Risk-Averse Constrained Reinforcement Learning with Optimized Certainty Equivalents

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Risk-Averse Constrained Reinforcement Learning with Optimized Certainty Equivalents

快速理解: Constrained optimization provides a common framework for dealing with conflicting objectives in reinforcement learning (RL). In most of these settings, the objectives (and constraints) are expressed through the expected accumulated reward.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.681 Risk-Averse Total-Reward Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Risk-Averse Total-Reward Reinforcement Learning

快速理解: Risk-averse total-reward Markov Decision Processes (MDPs) offer a promising framework for modeling and solving undiscounted infinite-horizon objectives. Existing model-based algorithms for risk measures like the entropic risk measure (ERM) and entropic value-at-risk (EVaR) are effective in small problems, but require full access to transition probabilities.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.682 RoME: Domain-Robust Mixture-of-Experts for MILP Solution Prediction across Domains

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：RoME: Domain-Robust Mixture-of-Experts for MILP Solution Prediction across Domains

快速理解：Mixed-Integer Linear Programming (MILP) is a fundamental and powerful framework for modeling complex optimization problems across diverse domains. Recently, learning-based methods have shown great promise in accelerating MILP solvers by predicting high-quality solutions.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.7.683 RoboRefer: Towards Spatial Referring with Reasoning in Vision-Language Models for Robotics

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：RoboRefer: Towards Spatial Referring with Reasoning in Vision-Language Models for Robotics

快速理解：Spatial referring is a fundamental capability of embodied robots to interact with the 3D physical world. However, even with the powerful pretrained vision language models (VLMs), recent approaches are still not qualified to accurately understand the complex 3D scenes and dynamically reason about the instructionindicated locations for interaction.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.684 Robust Reinforcement Learning in Finance: Modeling Market Impact with Elliptic Uncertainty Sets

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Robust Reinforcement Learning in Finance: Modeling Market Impact with Elliptic Uncertainty Sets

快速理解：In financial applications, reinforcement learning (RL) agents are commonly trained on historical data, where their actions do not influence prices. However, during deployment, these agents trade in live markets where their own transactions can shift asset prices, a phenomenon known as market impact.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.685 Robust SuperAlignment: Weak-to-Strong Robustness Generalization for Vision-Language Models

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文: Robust SuperAlignment: Weak-to-Strong Robustness Generalization for Vision-Language Models

快速理解: Numerous well-established studies have demonstrated the superhuman capabilities of modern Vision-Language Models (VLMs) across a wide range of tasks. However, growing is the doubt about the continuing availability of reliable high-quality labeling (supervision) from human annotators, leading to stagnation of the model's performance.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.686 Robust and Diverse Multi-Agent Learning via Rational Policy Gradient

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文: Robust and Diverse Multi-Agent Learning via Rational Policy Gradient

快速理解: Adversarial optimization algorithms that explicitly search for flaws in agents' policies have been successfully applied to finding robust and diverse policies in multi-agent settings. However, the success of adversarial optimization has been largely limited to zero-sum settings because its naive application in cooperative settings leads to a critical failure mode: agents are irrationally incentivized to selfsabotage, blocking the completion of tasks and halting further learning.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.687 Robust and Scalable Autonomous Reinforcement Learning in Irreversible Environments

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文: Robust and Scalable Autonomous Reinforcement Learning in Irreversible Environments

快速理解: Reinforcement learning (RL) typically assumes repetitive resets to provide an agent with diverse and unbiased experiences. These resets require significant human intervention and result in poor training efficiency in real-world settings.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.688 Role-aware Multi-agent Reinforcement Learning for Coordinated Emergency Traffic Control

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Role-aware Multi-agent Reinforcement Learning for Coordinated Emergency Traffic Control

快速理解：Emergency traffic control presents an increasingly critical challenge, requiring seamless coordination among emergency vehicles, regular vehicles, and traffic lights to ensure efficient passage for all vehicles. Existing models primarily only focus on traffic light control, leaving emergency and regular vehicles prone to delay due to the lack of navigation strategies.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.689 Router-R1: Teaching LLMs Multi-Round Routing and Aggregation via Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Router-R1: Teaching LLMs Multi-Round Routing and Aggregation via Reinforcement Learning

快速理解：The rapid emergence of diverse large language models (LLMs) has spurred the development of LLM routers that assign user queries to the most suitable model. However, existing LLM routers typically perform a single-round, one-to-one mapping (i.e., assigning each query to a single model in isolation), which limits their capability to tackle complex tasks that demand the complementary strengths of multiple LLMs.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.690 Routing Mamba: Scaling State Space Models with Mixture-of-Experts Projection

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：Routing Mamba: Scaling State Space Models with Mixture-of-Experts Projection

快速理解：State Space Models (SSMs) offer remarkable performance gains in efficient sequence modeling, with constant per-step inference-time computation and memory complexity. Recent advances, such as Mamba, further enhance SSMs with inputdependent gating and hardware-aware implementations, positioning them as strong alternatives to Transformers for long sequence modeling.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.7.691 S-GRPO: Early Exit via Reinforcement Learning in Reasoning Models

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要

- [来源](#)

论文: S-GRPO: Early Exit via Reinforcement Learning in Reasoning Models

快速理解: As Test-Time Scaling emerges as an active research focus in the large language model community, advanced post-training methods increasingly emphasize extending chain-of-thought (CoT) generation length, thereby enhancing reasoning capabilities to approach Deepseek R1-like reasoning models. However, recent studies reveal that reasoning models (even Qwen3) consistently exhibit excessive thought redundancy in CoT generation.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.692 SALS: Sparse Attention in Latent Space for KV Cache Compression

- 作者: 作者信息未提供
- 年份: 2025; 主题: Systems; 优先级: 重要
- [来源](#)

论文: SALS: Sparse Attention in Latent Space for KV Cache Compression

快速理解: Large Language Models (LLMs) capable of handling extended contexts are in high demand, yet their inference remains challenging due to substantial Key-Value (KV) cache size and high memory bandwidth requirements. Previous research has demonstrated that KV cache exhibits low-rank characteristics within the hidden dimension, suggesting the potential for effective compression.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.7.693 SAMPO: Scale-wise Autoregression with Motion Prompt for Generative World Models

- 作者: 作者信息未提供
- 年份: 2025; 主题: World Model; 优先级: 重要
- [来源](#)

论文: SAMPO: Scale-wise Autoregression with Motion Prompt for Generative World Models

快速理解: World models allow agents to simulate the consequences of actions in imagined environments for planning, control, and long-horizon decision-making. However, existing autoregressive world models struggle with visually coherent predictions due to disrupted spatial structure, inefficient decoding, and inadequate motion modeling.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策。

证据层级: 本地 PDF 前页。

1.3.7.694 SATURN: SAT-based Reinforcement Learning to Unleash LLMs Reasoning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: SATURN: SAT-based Reinforcement Learning to Unleash LLMs Reasoning

快速理解: How to design reinforcement learning (RL) tasks that effectively unleash the reasoning capability of large language models (LLMs) remains an open question. Existing RL tasks (e.g., math, programming, and constructing reasoning tasks) face three key limitations: Scalability.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.695 SD-VLM: Spatial Measuring and Understanding with Depth-Encoded Vision-Language Models

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: SD-VLM: Spatial Measuring and Understanding with Depth-Encoded Vision-Language Models

快速理解: While vision language models (VLMs) excel in 2D semantic visual understanding, their ability to quantitatively reason about 3D spatial relationships remains underexplored due to the deficiency of spatial representation ability of 2D images. In this paper, we analyze the problem hindering VLMs' spatial understanding abilities and propose SD-VLM, a novel framework that significantly enhances fundamental spatial perception abilities of VLMs through two key contributions: (1) propose Massive Spatial Measuring and Understanding (MSMU) dataset with precise spatial annotations, and (2) introduce a simple depth positional encoding method strengthening VLMs' spatial awareness.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.696 SE-GUI: Enhancing Visual Grounding for GUI Agents via Self-Evolutionary Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: SE-GUI: Enhancing Visual Grounding for GUI Agents via Self-Evolutionary Reinforcement Learning

快速理解: Graphical User Interface (GUI) agents have made substantial strides in understanding and executing user instructions across diverse platforms. Yet, grounding these instructions to precise interface elements remains challenging—especially in complex, high-resolution, professional environments.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.697 SEEA-R1: Tree-Structured Reinforcement Fine-Tuning for Self-Evolving Embodied Agents

- 作者: 作者信息未提供
- 年份: 2025; 主题: Embodied/WBC; 优先级: 重要
- [来源](#)

论文: SEEA-R1: Tree-Structured Reinforcement Fine-Tuning for Self-Evolving Embodied Agents

快速理解: Self-evolution, the ability of agents to autonomously improve their reasoning and behavior, is essential for the embodied domain with long-horizon, real-world tasks. Despite current advancements in reinforcement fine-tuning (RFT) showing strong performance in enhancing reasoning in LLMs, its potential to enable self-evolving embodied intelligence with multi-modal interactions remains largely unexplored.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制。

证据层级: 本地 PDF 前页。

1.3.7.698 SIU3R: Simultaneous Scene Understanding and 3D Reconstruction Beyond Feature Alignment

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: SIU3R: Simultaneous Scene Understanding and 3D Reconstruction Beyond Feature Alignment

快速理解: Simultaneous understanding and 3D reconstruction plays an important role in developing end-to-end embodied intelligent systems. To achieve this, recent approaches resort to 2D-to-3D feature alignment paradigm, which leads to limited 3D understanding capability and potential semantic information loss.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.699 SPARTAN: A Sparse Transformer World Model Attending to What Matters

- 作者: 作者信息未提供
- 年份: 2025; 主题: World Model; 优先级: 重要
- [来源](#)

论文: SPARTAN: A Sparse Transformer World Model Attending to What Matters

快速理解: Capturing the interactions between entities in a structured way plays a central role in world models that flexibly adapt to changes in the environment. Recent works motivate the benefits of models that explicitly represent the structure of interactions and formulate the problem as discovering local causal structures.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策。

证据层级: 本地 PDF 前页。

1.3.7.700 SPOT: Scalable Policy Optimization with Trees for Markov Decision Processes

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: SPOT: Scalable Policy Optimization with Trees for Markov Decision Processes

快速理解: Interpretable reinforcement learning policies are essential for high-stakes decisionmaking, yet optimizing decision tree policies in Markov Decision Processes (MDPs) remains challenging. We propose SPOT, a novel method for computing decision tree policies, which formulates the optimization problem as a mixedinteger linear program (MILP).

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.701 SPiDR: A Simple Approach for Zero-Shot Safety in Sim-to-Real Transfer

- 作者: 作者信息未提供
- 年份: 2025; 主题: Robot Learning; 优先级: 重要
- [来源](#)

论文: SPiDR: A Simple Approach for Zero-Shot Safety in Sim-to-Real Transfer

快速理解: Deploying reinforcement learning (RL) safely in the real world is challenging, as policies trained in simulators must face the inevitable ‘sim-to-real gap’. Robust safe RL techniques are provably safe however difficult to scale, while domain randomization is more practical yet prone to unsafe behaviors.

关注点: 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 本地 PDF 前页。

1.3.7.702 SQL-R1: Training Natural Language to SQL Reasoning Model By Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: SQL-R1: Training Natural Language to SQL Reasoning Model By Reinforcement Learning

快速理解: Natural Language to SQL (NL2SQL) enables intuitive interactions with databases by transforming natural language queries into structured SQL statements. Despite recent advancements in enhancing human-computer interaction within database applications, significant challenges persist, particularly regarding the reasoning performance in complex scenarios involving multi-table joins and nested queries.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.703 SSR: Enhancing Depth Perception in Vision-Language Models via Rationale-Guided Spatial Reasoning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: SSR: Enhancing Depth Perception in Vision-Language Models via Rationale-Guided Spatial Reasoning

快速理解: Despite impressive advancements in Visual-Language Models (VLMs) for multimodal tasks,

their reliance on RGB inputs limits precise spatial understanding. Existing methods for integrating spatial cues, such as point clouds or depth, either require specialized sensors or fail to effectively exploit depth information for higher-order reasoning.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.704 ST²360D: Spatial-to-Temporal Consistency for Training-free 360 Monocular Depth Estimation

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文:ST²360D: Spatial-to-Temporal Consistency for Training-free 360 Monocular Depth Estimation

快速理解：360° monocular depth estimation plays a crucial role in scene understanding owing to its 180° × 360° field-of-view (FoV). To mitigate the distortions brought by equirectangular projection, existing methods typically divide 360° images into distortion-less perspective patches.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.705 STAIR: Addressing Stage Misalignment through Temporal-Aligned Preference Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文: STAIR: Addressing Stage Misalignment through Temporal-Aligned Preference Reinforcement Learning

快速理解：Preference-based reinforcement learning (PbRL) bypasses complex reward engineering by learning rewards directly from human preferences, enabling better alignment with human intentions. However, its effectiveness in multi-stage tasks, where agents sequentially perform sub-tasks (e.g., navigation, grasping), is limited by stage misalignment: Comparing segments from mismatched stages, such as movement versus manipulation, results in uninformative feedback, thus hindering policy learning.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.706 STAR: Efficient Preference-based Reinforcement Learning via Dual Regularization

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文: STAR: Efficient Preference-based Reinforcement Learning via Dual Regularization

快速理解：Preference-based reinforcement learning (PbRL) bypasses complex reward engineering by

learning from human feedback. However, due to the high cost of obtaining feedback, PbRL typically relies on a limited set of preference-labeled samples.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.707 STAR: Spatial-Temporal Tracklet Matching for Multi-Object Tracking

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：STAR: Spatial-Temporal Tracklet Matching for Multi-Object Tracking

快速理解：Existing tracking-by-detection Multi-Object Tracking methods mainly rely on associating objects with tracklets using motion and appearance features. However, variations in viewpoint and occlusions can result in discrepancies between the features of current objects and those of historical tracklets.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.708 SWE-RL: Advancing LLM Reasoning via Reinforcement Learning on Open Software Evolution

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：SWE-RL: Advancing LLM Reasoning via Reinforcement Learning on Open Software Evolution

快速理解：The recent DeepSeek-R1 release has demonstrated the immense potential of reinforcement learning (RL) in enhancing the general reasoning capabilities of large language models (LLMs). While DeepSeek-R1 and other follow-up work primarily focus on applying RL to competitive coding and math problems, this paper introduces SWE-RL, the first approach to scale RL-based LLM reasoning for real-world software engineering.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.709 SYMPHONY: Synergistic Multi-agent Planning with Heterogeneous Language Model Assembly

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：SYMPHONY: Synergistic Multi-agent Planning with Heterogeneous Language Model Assembly

快速理解：Recent advancements have increasingly focused on leveraging large language models (LLMs) to construct autonomous agents for complex problem-solving tasks. However, existing approaches predominantly employ a single-agent framework to generate search branches and estimate rewards during Monte Carlo Tree Search (MCTS) planning.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.710 Safe + Safe = Unsafe? Exploring How Safe Images Can Be Exploited to Jailbreak Large Vision-Language Models

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Safe + Safe = Unsafe? Exploring How Safe Images Can Be Exploited to Jailbreak Large Vision-Language Models

快速理解：Recent advances in Large Vision-Language Models (LVLMs) have showcased strong reasoning abilities across multiple modalities, achieving significant breakthroughs in various real-world applications. Despite this great success, the safety guardrail of LVLMs may not cover the unforeseen domains introduced by the visual modality.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.711 Safe RLHF-V: Safe Reinforcement Learning from Multi-modal Human Feedback

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Safe RLHF-V: Safe Reinforcement Learning from Multi-modal Human Feedback

快速理解：Multimodal large language models (MLLMs) are essential for building generalpurpose AI assistants; however, they pose increasing safety risks. How can we ensure safety alignment of MLLMs to prevent undesired behaviors?

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.712 Safe-Sora: Safe Text-to-Video Generation via Graphical Watermarking

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Safe-Sora: Safe Text-to-Video Generation via Graphical Watermarking

快速理解：The explosive growth of generative video models has amplified the demand for reliable copyright preservation of AI-generated content. Despite its popularity in image synthesis, invisible generative watermarking remains largely underexplored in video generation.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.713 SafePTR: Token-Level Jailbreak Defense in Multimodal LLMs via Prune-then-Restore Mechanism

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：SafePTR: Token-Level Jailbreak Defense in Multimodal LLMs via Prune-then-Restore Mechanism

快速理解：Content Warning: This paper contains a few harmful images and texts! By incorporating visual inputs, Multimodal Large Language Models (MLLMs) extend LLMs to support visual reasoning.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.714 Sample Complexity of Distributionally Robust Average-Reward Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Sample Complexity of Distributionally Robust Average-Reward Reinforcement Learning

快速理解：Motivated by practical applications where stable long-term performance is critical—such as robotics, operations research, and healthcare—we study the problem of distributionally robust (DR) average-reward reinforcement learning. We propose two algorithms that achieve near-optimal sample complexity.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.715 Sample-Efficient Tabular Self-Play for Offline Robust Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Sample-Efficient Tabular Self-Play for Offline Robust Reinforcement Learning

快速理解：Multi-agent reinforcement learning (MARL), as a thriving field, explores how multiple agents independently make decisions in a shared dynamic environment. Due to environmental uncertainties, policies in MARL must remain robust to tackle the sim-to-real gap.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.716 Scaling Off-Policy Reinforcement Learning with Batch and Weight Normalization

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Scaling Off-Policy Reinforcement Learning with Batch and Weight Normalization

快速理解：Reinforcement learning has achieved significant milestones, but sample efficiency remains a bottleneck for real-world applications. Recently, CrossQ has demonstrated state-of-the-art sample efficiency with a low update-to-data (UTD) ratio of 1.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.717 Scaling Offline RL via Efficient and Expressive Shortcut Models

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Scaling Offline RL via Efficient and Expressive Shortcut Models

快速理解：Diffusion and flow models have emerged as powerful generative approaches capable of modeling diverse and multimodal behavior. However, applying these models to offline reinforcement learning (RL) remains challenging due to the iterative nature of their noise sampling processes, making policy optimization difficult.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.718 SceneWeaver: All-in-One 3D Scene Synthesis with an Extensible and Self-Reflective Agent

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：SceneWeaver: All-in-One 3D Scene Synthesis with an Extensible and Self-Reflective Agent

快速理解：Indoor scene synthesis has become increasingly important with the rise of Embodied AI, which requires 3D environments that are not only visually realistic but also physically plausible and functionally diverse. While recent approaches have advanced visual fidelity, they often remain constrained to fixed scene categories, lack sufficient object-level detail and physical consistency, and struggle to align with complex user instructions.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.719 SeRL: Self-play Reinforcement Learning for Large Language Models with Limited Data

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文: SeRL: Self-play Reinforcement Learning for Large Language Models with Limited Data

快速理解: Recent advances have demonstrated the effectiveness of Reinforcement Learning (RL) in improving the reasoning capabilities of Large Language Models (LLMs). However, existing works inevitably rely on high-quality instructions and verifiable rewards for effective training, both of which are often difficult to obtain in specialized domains.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.720 See&Trek: Training-Free Spatial Prompting for Multimodal Large Language Model

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: See&Trek: Training-Free Spatial Prompting for Multimodal Large Language Model

快速理解: We introduce SEE&TREK, the first training-free prompting framework tailored to enhance the spatial understanding of Multimodal Large Language Models (MLLMs) under vision-only constraints. While prior efforts have incorporated modalities like depth or point clouds to improve spatial reasoning, purely visuospatial understanding remains underexplored.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.721 Seeing is Believing? Mitigating OCR Hallucinations in Multimodal Large Language Models

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Seeing is Believing? Mitigating OCR Hallucinations in Multimodal Large Language Models

快速理解: Recent advancements in multimodal large language models (MLLMs) have enhanced document understanding by integrating textual and visual information. However, existing models exhibit incompleteness within their paradigm in real-world scenarios, particularly under visual degradation (e.g., blur, occlusion, low contrast).

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.722 Seeing the Arrow of Time in Large Multimodal Models

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Seeing the Arrow of Time in Large Multimodal Models

快速理解: The Arrow of Time (AoT)—time’s irreversible flow shaping physical events—is fundamental

to video comprehension, yet remains a significant challenge for modern large multimodal models (LMMs). Current LMMs struggle to perceive and utilize temporal directionality in video when responding to language queries, obstructing deeper temporal understanding.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.723 Seeing through Uncertainty: Robust Task-Oriented Optimization in Visual Navigation

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Seeing through Uncertainty: Robust Task-Oriented Optimization in Visual Navigation

快速理解：Visual navigation is a fundamental problem in embodied AI, yet practical deployments demand long-horizon planning capabilities to address multi-objective tasks. A major bottleneck is data scarcity: policies learned from limited data often overfit and fail to generalize OOD.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.724 Segment Policy Optimization: Effective Segment-Level Credit Assignment in RL for Large Language Models

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Segment Policy Optimization: Effective Segment-Level Credit Assignment in RL for Large Language Models

快速理解：Enhancing the reasoning capabilities of large language models effectively using reinforcement learning (RL) remains a crucial challenge. Existing approaches primarily adopt two contrasting advantage estimation granularities: token-level methods (e.g., PPO) aim to provide fine-grained advantage signals but suffer from inaccurate estimation due to difficulties in training an accurate critic model.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.725 Self-Improving Embodied Foundation Models

- 作者：作者信息未提供
- 年份：2025；主题：Embodied/WBC；优先级：重要
- [来源](#)

论文：Self-Improving Embodied Foundation Models

快速理解：Foundation models trained on web-scale data have revolutionized robotics, but their application to low-level control remains largely limited to behavioral cloning. Drawing inspiration from the success of the reinforcement learning stage in finetuning large language models, we propose a

two-stage post-training approach for robotics.

关注点：具身任务中的感知、动作生成、全身协调或运动控制。

证据层级：本地 PDF 前页。

1.3.7.726 Selftok-Zero: Reinforcement Learning for Visual Generation via Discrete and Autoregressive Visual Tokens

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Selftok-Zero: Reinforcement Learning for Visual Generation via Discrete and Autoregressive Visual Tokens

快速理解：Reinforcement learning (RL) has become an indispensable post-training step for unlocking the full potential of Large Language Models (LLMs). Its core motivation is to incentivize the model's inference trajectory via a reward model, effectively balancing the exploration-exploitation trade-off in scenarios where collecting exhaustive input-output ground-truth pairs is infeasible.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.727 Sequential Monte Carlo for Policy Optimization in Continuous POMDPs

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Sequential Monte Carlo for Policy Optimization in Continuous POMDPs

快速理解：Optimal decision-making under partial observability requires agents to balance reducing uncertainty (exploration) against pursuing immediate objectives (exploitation). In this paper, we introduce a novel policy optimization framework for continuous partially observable Markov decision processes (POMDPs) that explicitly addresses this challenge.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.728 Sequential Multi-Agent Dynamic Algorithm Configuration

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Sequential Multi-Agent Dynamic Algorithm Configuration

快速理解：The performance of an algorithm often critically depends on its hyperparameter configuration. Dynamic algorithm configuration (DAC) is a recent trend in automated machine learning, which can dynamically adjust the algorithm's configuration during the execution process and relieve users from tedious trial-and-error tuning tasks.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.729 ShapeLLM-Omni: A Native Multimodal LLM for 3D Generation and Understanding

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：ShapeLLM-Omni: A Native Multimodal LLM for 3D Generation and Understanding

快速理解：Recently, the powerful text-to-image capabilities of GPT-4o have led to growing appreciation for native multimodal large language models. However, its multimodal capabilities remain confined to images and text.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.730 SharpZO: Hybrid Sharpness-Aware Vision Language Model Prompt Tuning via Forward-Only Passes

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：SharpZO: Hybrid Sharpness-Aware Vision Language Model Prompt Tuning via Forward-Only Passes

快速理解：Fine-tuning vision language models (VLMs) has achieved remarkable performance across various downstream tasks; yet, it requires access to model gradients through backpropagation (BP), making them unsuitable for memory-constrained, inference-only edge devices. To address this limitation, previous work has explored various BP-free fine-tuning methods.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.731 Sherlock: Self-Correcting Reasoning in Vision-Language Models

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Sherlock: Self-Correcting Reasoning in Vision-Language Models

快速理解：Reasoning Vision-Language Models (VLMs) have shown promising performance on complex multimodal tasks. However, they still face significant challenges: they are highly sensitive to reasoning errors, require large volumes of annotated data or accurate verifiers, and struggle to generalize beyond specific domains.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.732 Shift Before You Learn: Enabling Low-Rank Representations in Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Shift Before You Learn: Enabling Low-Rank Representations in Reinforcement Learning

快速理解：Low-rank structure is a common implicit assumption in many modern reinforcement learning (RL) algorithms. For instance, reward-free and goal-conditioned RL methods often presume that the successor measure admits a low-rank representation.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.733 ShotBench: Expert-Level Cinematic Understanding in Vision-Language Models

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：ShotBench: Expert-Level Cinematic Understanding in Vision-Language Models

快速理解：Cinematography, the fundamental visual language of film, is essential for conveying narrative, emotion, and aesthetic quality. While recent Vision-Language Models (VLMs) demonstrate strong general visual understanding, their proficiency in comprehending the nuanced cinematic grammar embedded within individual shots remains largely unexplored and lacks robust evaluation.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.734 Show-o2: Improved Native Unified Multimodal Models

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Show-o2: Improved Native Unified Multimodal Models

快速理解：This paper presents improved native unified multimodal models, i.e., Show-o2, that leverage autoregressive modeling and flow matching. Built upon a 3D causal variational autoencoder space, unified visual representations are constructed through a dual-path of spatial (-temporal) fusion, enabling scalability across image and video modalities while ensuring effective multimodal understanding and generation.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.735 Sim-LLM: Optimizing LLM Inference at the Edge through Inter-Task KV Reuse

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：Sim-LLM: Optimizing LLM Inference at the Edge through Inter-Task KV Reuse

快速理解：KV cache technology, by storing key-value pairs, helps reduce the computational overhead incurred by large language models (LLMs). It facilitates their deployment on resource-constrained edge computing nodes like edge servers.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.7.736 SimWorld-Robotics: Synthesizing Photorealistic and Dynamic Urban Environments for Multimodal Robot Navigation and Collaboration

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：SimWorld-Robotics: Synthesizing Photorealistic and Dynamic Urban Environments for Multimodal Robot Navigation and Collaboration

快速理解：Recent advances in foundation models have shown promising results in developing generalist robotics that can perform diverse tasks in open-ended scenarios given multimodal inputs. However, current work has been mainly focused on indoor, household scenarios.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.737 Simultaneous Statistical Inference for Off-Policy Evaluation in Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Simultaneous Statistical Inference for Off-Policy Evaluation in Reinforcement Learning

快速理解：This work presents the first theoretically justified simultaneous inference framework for off-policy evaluation (OPE). In contrast to existing methods that focus on point estimates or pointwise confidence intervals (CIs), the new framework quantifies global uncertainty across an infinite or continuous initial state space, offering valid inference over the entire state space.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.738 Single GPU Task Adaptation of Pathology Foundation Models for Whole Slide Image Analysis

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文: Single GPU Task Adaptation of Pathology Foundation Models for Whole Slide Image Analysis

快速理解: Pathology foundation models (PFMs) have emerged as powerful tools for analyzing whole slide images (WSIs). However, adapting these pretrained PFMs for specific clinical tasks presents considerable challenges, primarily due to the availability of only weak (WSI-level) labels for gigapixel images, necessitating multiple instance learning (MIL) paradigm for effective WSI analysis.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 本地 PDF 前页。

1.3.7.739 SiriuS: Self-improving Multi-agent Systems via Bootstrapped Reasoning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: SiriuS: Self-improving Multi-agent Systems via Bootstrapped Reasoning

快速理解: Multi-agent AI systems powered by large language models (LLMs) are increasingly applied to solve complex tasks. However, these systems often rely on fragile, manually designed prompts and heuristics, making optimization difficult.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.740 SketchMind: A Multi-Agent Cognitive Framework for Assessing Student--Drawn Scientific Sketches

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: SketchMind: A Multi-Agent Cognitive Framework for Assessing Student-Drawn Scientific Sketches

快速理解: Scientific sketches (e.g., models) offer a powerful lens into students' conceptual understanding, yet AI-powered automated assessment of such free-form, visually diverse artifacts remains a critical challenge. Existing solutions often treat sketch evaluation as either an image classification task or monolithic vision-language models, which lack interpretability, pedagogical alignment, and adaptability across cognitive levels.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.741 Skrull: Towards Efficient Long Context Fine-tuning through Dynamic Data Scheduling

- 作者: 作者信息未提供
- 年份: 2025; 主题: Systems; 优先级: 重要
- [来源](#)

论文: Skrull: Towards Efficient Long Context Fine-tuning through Dynamic Data Scheduling

快速理解: Long-context supervised fine-tuning (Long-SFT) plays a vital role in enhancing the performance of large language models (LLMs) on long-context tasks. To smoothly adapt LLMs to long-context scenarios, this process typically entails training on mixed datasets containing both long and short sequences.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.7.742 SkyLadder: Better and Faster Pretraining via Context Window Scheduling

- 作者: 作者信息未提供
- 年份: 2025; 主题: Systems; 优先级: 重要
- [来源](#)

论文: SkyLadder: Better and Faster Pretraining via Context Window Scheduling

快速理解: Recent advancements in LLM pretraining have featured ever-expanding context windows to process longer sequences. However, our controlled study reveals that models pretrained with shorter context windows consistently outperform their long-context counterparts under a fixed token budget.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.7.743 Social World Model-Augmented Mechanism Design Policy Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: World Model; 优先级: 重要
- [来源](#)

论文: Social World Model-Augmented Mechanism Design Policy Learning

快速理解: Designing adaptive mechanisms to align individual and collective interests remains a central challenge in artificial social intelligence. Existing methods often struggle with modeling heterogeneous agents possessing persistent latent traits (e.g., skills, preferences) and dealing with complex multi-agent system dynamics.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策。

证据层级: 本地 PDF 前页。

1.3.7.744 Solving Continuous Mean Field Games: Deep Reinforcement Learning for Non-Stationary Dynamics

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Solving Continuous Mean Field Games: Deep Reinforcement Learning for Non-Stationary Dynamics

快速理解: Mean field games (MFGs) have emerged as a powerful framework for modeling interactions in large-scale multi-agent systems. Despite recent advancements in reinforcement learning (RL) for MFGs, existing methods are typically limited to finite spaces or stationary models, hindering their applicability to real-world problems.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.745 SpaceServe: Spatial Multiplexing of Complementary Encoders and Decoders for Multimodal LLMs

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：SpaceServe: Spatial Multiplexing of Complementary Encoders and Decoders for Multimodal LLMs

快速理解：Recent multimodal large language models (MLLMs) marry modality-specific vision or audio encoders with a shared text decoder. While the encoder is computeintensive but memory-light, the decoder is the opposite, yet state-of-the-art serving stacks still time-multiplex these complementary kernels, idling SMs or HBM in turn.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.746 Sparse Autoencoders Learn Monosemantic Features in Vision-Language Models

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Sparse Autoencoders Learn Monosemantic Features in Vision-Language Models

快速理解：Sparse Autoencoders (SAEs) have recently gained attention as a means to improve the interpretability and steerability of Large Language Models (LLMs), both of which are essential for AI safety. In this work, we extend the application of SAEs to Vision-Language Models (VLMs), such as CLIP, and introduce a comprehensive framework for evaluating monosemanticity at the neuron-level in visual representations.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.747 Sparse VideoGen2: Accelerate Video Generation with Sparse Attention via Semantic-Aware Permutation

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Sparse VideoGen2: Accelerate Video Generation with Sparse Attention via Semantic-Aware Permutation

快速理解：Diffusion Transformers (DiTs) are essential for video generation but suffer from significant latency due to the quadratic complexity of attention. By computing only critical tokens, sparse attention reduces computational costs and offers a promising acceleration approach.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.748 Spatial-Aware Decision-Making with Ring Attractors in Reinforcement Learning Systems

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Spatial-Aware Decision-Making with Ring Attractors in Reinforcement Learning Systems

快速理解：Ring attractors, mathematical models inspired by neural circuit dynamics, provide a biologically plausible mechanism to improve learning speed and accuracy in Reinforcement Learning (RL). Serving as specialized brain-inspired structures that encode spatial information and uncertainty, ring attractors explicitly encode the action space, facilitate the organization of neural activity, and enable the distribution of spatial representations across the neural network in the context of Deep Reinforcement Learning (DRL).

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.749 SpecMAS: A Multi-Agent System for Self-Verifying System Generation via Formal Model Checking

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：SpecMAS: A Multi-Agent System for Self-Verifying System Generation via Formal Model Checking

快速理解：We present SpecMAS, a novel multi-agent system that autonomously constructs and formally verifies executable system models from natural language specifications. Given a Standard Operating Procedure (SOP) describing a target system, SpecMAS parses the specification, identifies relevant operational modes, variables, transitions, and properties, and generates a formal model in NuSMV code syntax, an industry-standard symbolic model checker.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.750 Spot the Fake: Large Multimodal Model-Based Synthetic Image Detection with Artifact Explanation

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Spot the Fake: Large Multimodal Model-Based Synthetic Image Detection with Artifact Explanation

快速理解：With the rapid advancement of Artificial Intelligence Generated Content (AIGC) technolo-

gies, synthetic images have become increasingly prevalent in everyday life, posing new challenges for authenticity assessment and detection. Despite the effectiveness of existing methods in evaluating image authenticity and locating forgeries, these approaches often lack human interpretability and do not fully address the growing complexity of synthetic data.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.751 Spotlight Attention: Towards Efficient LLM Generation via Non-linear Hashing-based KV Cache Retrieval

- 作者：作者信息未提供
- 年份：2025；主题：Systems；优先级：重要
- [来源](#)

论文：Spotlight Attention: Towards Efficient LLM Generation via Non-linear Hashing-based KV Cache Retrieval

快速理解：Reducing the key-value (KV) cache burden in Large Language Models (LLMs) significantly accelerates inference. Dynamically selecting critical KV caches during decoding helps maintain performance.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.7.752 Stable Cinemetrics : Structured Taxonomy and Evaluation for Professional Video Generation

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Stable Cinemetrics : Structured Taxonomy and Evaluation for Professional Video Generation

快速理解：Recent advances in video generation have enabled high-fidelity video synthesis from user provided prompts. However, existing models and benchmarks fail to capture the complexity and requirements of professional video generation.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.753 Stable Gradients for Stable Learning at Scale in Deep Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Stable Gradients for Stable Learning at Scale in Deep Reinforcement Learning

快速理解：Scaling deep reinforcement learning networks is challenging and often results in degraded performance, yet the root causes of this failure mode remain poorly understood. Several recent works have proposed mechanisms to address this, but they are often complex and fail to highlight

the causes underlying this difficulty.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.754 Stable Part Diffusion 4D: Multi-View RGB and Kinematic Parts Video Generation

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Stable Part Diffusion 4D: Multi-View RGB and Kinematic Parts Video Generation

快速理解：We present Stable Part Diffusion 4D (SP4D), a framework for generating paired RGB and kinematic part videos from monocular inputs. Unlike conventional part segmentation methods that rely on appearance-based semantic cues, SP4D learns to produce kinematic parts —structural components aligned with object articulation and consistent across views and time.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.755 Staggered Environment Resets Improve Massively Parallel On-Policy Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Staggered Environment Resets Improve Massively Parallel On-Policy Reinforcement Learning

快速理解：Massively parallel GPU simulation environments have accelerated reinforcement learning (RL) research by enabling fast data collection for on-policy RL algorithms like Proximal Policy Optimization (PPO). To maximize throughput, it is common to use short rollouts per policy update, increasing the update-to-data (UTD) ratio.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.756 State Entropy Regularization for Robust Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：State Entropy Regularization for Robust Reinforcement Learning

快速理解：State entropy regularization has empirically shown better exploration and sample complexity in reinforcement learning (RL). However, its theoretical guarantees have not been studied.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.757 StateSpaceDiffuser: Bringing Long Context to Diffusion World Models

- 作者：作者信息未提供
- 年份：2025；主题：World Model；优先级：重要
- [来源](#)

论文：StateSpaceDiffuser: Bringing Long Context to Diffusion World Models

快速理解：World models have recently gained prominence for action-conditioned visual prediction in complex environments. However, relying on only a few recent observations causes them to lose long-term context.

关注点：环境动力学、视频预测、潜变量状态或模型式决策。

证据层级：本地 PDF 前页。

1.3.7.758 Statistics Caching Test-Time Adaptation for Vision-Language Models

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Statistics Caching Test-Time Adaptation for Vision-Language Models

快速理解：Test-time adaptation (TTA) for Vision-Language Models (VLMs) aims to enhance performance on unseen test data. However, existing methods struggle to achieve robust and continuous knowledge accumulation during test time.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.759 Stop Summation: Min-Form Credit Assignment Is All Process Reward Model Needs for Reasoning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Stop Summation: Min-Form Credit Assignment Is All Process Reward Model Needs for Reasoning

快速理解：Process reward models (PRMs) have proven effective for test-time scaling of Large Language Models (LLMs) on challenging reasoning tasks. However, reward hacking issues with PRMs limit their successful application in reinforcement finetuning.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.760 Strategyproof Reinforcement Learning from Human Feedback

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Strategyproof Reinforcement Learning from Human Feedback

快速理解: We study Reinforcement Learning from Human Feedback (RLHF) in settings where multiple labelers may strategically misreport feedback to steer the learned policy toward their own preferences. We show that existing RLHF algorithms, including recent pluralistic methods, are not strategyproof, and that even a single strategic labeler can cause arbitrarily large misalignment with social welfare.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.761 Structural Information-based Hierarchical Diffusion for Offline Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Structural Information-based Hierarchical Diffusion for Offline Reinforcement Learning

快速理解: Diffusion-based generative methods have shown promising potential for modeling trajectories from offline reinforcement learning (RL) datasets, and hierarchical diffusion has been introduced to mitigate variance accumulation and computational challenges in long-horizon planning tasks. However, existing approaches typically assume a fixed two-layer diffusion hierarchy with a single predefined temporal scale, which limits adaptability to diverse downstream tasks and reduces flexibility in decision making.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.762 Structure-Aware Cooperative Ensemble Evolutionary Optimization on Combinatorial Problems with Multimodal Large Language Models

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Structure-Aware Cooperative Ensemble Evolutionary Optimization on Combinatorial Problems with Multimodal Large Language Models

快速理解: Evolutionary algorithms (EAs) have proven effective in exploring the vast solution spaces typical of graph-structured combinatorial problems. However, traditional encoding schemes, such as binary or numerical representations, often fail to straightforwardly capture the intricate structural properties of networks.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.763 Structure-Aware Fusion with Progressive Injection for Multimodal Molecular Representation Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Structure-Aware Fusion with Progressive Injection for Multimodal Molecular Representation Learning

快速理解: Multimodal molecular models often suffer from 3D conformer unreliability and modality collapse, limiting their robustness and generalization. We propose MuMo, a structured multimodal fusion framework that addresses these challenges in molecular representation through two key strategies.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.764 Structured Reinforcement Learning for Combinatorial Decision-Making

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Structured Reinforcement Learning for Combinatorial Decision-Making

快速理解: Reinforcement learning (RL) is increasingly applied to real-world problems involving complex and structured decisions, such as routing, scheduling, and assortment planning. These settings challenge standard RL algorithms, which struggle to scale, generalize, and exploit structure in the presence of combinatorial action spaces.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.765 Structured Spectral Reasoning for Frequency-Adaptive Multimodal Recommendation

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Structured Spectral Reasoning for Frequency-Adaptive Multimodal Recommendation

快速理解: Multimodal recommendation aims to integrate collaborative signals with heterogeneous content such as visual and textual information, but remains challenged by modality-specific noise, semantic inconsistency, and unstable propagation over user-item graphs. These issues are often exacerbated by naive fusion or shallow modeling strategies, leading to degraded generalization and poor robustness.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.766 Succeed or Learn Slowly: Sample Efficient Off-Policy Reinforcement Learning for Mobile App Control

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Succeed or Learn Slowly: Sample Efficient Off-Policy Reinforcement Learning for Mobile App

Control

快速理解: Reinforcement learning (RL) using foundation models for policy approximations in multi-turn tasks remains challenging. We identify two main limitations related to sparse reward settings and policy gradient updates, based on which we formulate a key insight: updates from positive samples with high returns typically do not require policy regularisation, whereas updates from negative samples, reflecting undesirable behaviour, can harm model performance.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.767 SwS: Self-aware Weakness-driven Problem Synthesis in Reinforcement Learning for LLM Reasoning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: SwS: Self-aware Weakness-driven Problem Synthesis in Reinforcement Learning for LLM Reasoning

快速理解: Reinforcement Learning with Verifiable Rewards (RLVR) has proven effective for training large language models (LLMs) on complex reasoning tasks, such as mathematical problem solving. A prerequisite for the scalability of RLVR is a high-quality problem set with precise and verifiable answers.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.768 Switchable Token-Specific Codebook Quantization For Face Image Compression

- 作者: 作者信息未提供
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Switchable Token-Specific Codebook Quantization For Face Image Compression

快速理解: With the ever-increasing volume of visual data, the efficient and lossless transmission, along with its subsequent interpretation and understanding, has become a critical bottleneck in modern information systems. The emerged codebook-based solution utilize a globally shared codebook to quantize and dequantize each token, controlling the bpp by adjusting the number of tokens or the codebook size.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 本地 PDF 前页。

1.3.7.769 TRACE: Grounding Time Series in Context for Multimodal Embedding and Retrieval

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要

- [来源](#)

论文: TRACE: Grounding Time Series in Context for Multimodal Embedding and Retrieval

快速理解: The ubiquity of dynamic data in domains such as weather, healthcare, and energy underscores a growing need for effective interpretation and retrieval of time-series data. These data are inherently tied to domain-specific contexts, such as clinical notes or weather narratives, making cross-modal retrieval essential not only for downstream tasks but also for developing robust time-series foundation models by retrieval-augmented generation (RAG).

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.770 TRAP: Targeted Redirecting of Agentic Preferences

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: TRAP: Targeted Redirecting of Agentic Preferences

快速理解: Autonomous agentic AI systems powered by vision-language models (VLMs) are rapidly advancing toward real-world deployment, yet their cross-modal reasoning capabilities introduce new attack surfaces for adversarial manipulation that exploit semantic reasoning across modalities. Existing adversarial attacks typically rely on visible pixel perturbations or require privileged model or environment access, making them impractical for stealthy, real-world exploitation.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.771 TRoVe: Discovering Error-Inducing Static Feature Biases in Temporal Vision-Language Models

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: TRoVe: Discovering Error-Inducing Static Feature Biases in Temporal Vision-Language Models

快速理解: Vision-language models (VLMs) have made great strides in addressing temporal understanding tasks, which involve characterizing visual changes across a sequence of images. However, recent works have suggested that when making predictions, VLMs may rely on static feature biases, such as background or object features, rather than dynamic visual changes.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.772 TTRL: Test-Time Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: TTRL: Test-Time Reinforcement Learning

快速理解: This paper investigates Reinforcement Learning (RL) on data without explicit labels for reasoning tasks in Large Language Models (LLMs). The core challenge of the problem is reward estimation during inference while not having access to ground-truth information.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.773 Table2LaTeX-RL: High-Fidelity LaTeX Code Generation from Table Images via Reinforced Multimodal Language Models

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Table2LaTeX-RL: High-Fidelity LaTeX Code Generation from Table Images via Reinforced Multimodal Language Models

快速理解: In this work, we address the task of table image to LaTeX code generation, with the goal of automating the reconstruction of high-quality, publication-ready tables from visual inputs. A central challenge of this task lies in accurately handling complex tables—those with large sizes, deeply nested structures, and semantically rich or irregular cell content—where existing methods often fail.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.774 Tackling Continual Offline RL through Selective Weights Activation on Aligned Spaces

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Tackling Continual Offline RL through Selective Weights Activation on Aligned Spaces

快速理解: Continual offline reinforcement learning (CORL) has shown impressive ability in diffusion-based continual learning systems by modeling the joint distributions of trajectories. However, most research only focuses on limited continual task settings where the tasks have the same observation and action space, which deviates from the realistic demands of training agents in various environments.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.775 Tail-Optimized Caching for LLM Inference

- 作者: 作者信息未提供
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Tail-Optimized Caching for LLM Inference

快速理解: Prompt caching is critical for reducing latency and cost in LLM inference—OpenAI and Anthropic report up to 50-90% cost savings through prompt reuse. Despite its widespread success, little is known about what constitutes an optimal prompt caching policy, particularly when optimizing tail latency—a metric of central importance to practitioners.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 本地 PDF 前页。

1.3.7.776 Tapered Off-Policy REINFORCE - Stable and efficient reinforcement learning for large language models

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Tapered Off-Policy REINFORCE - Stable and efficient reinforcement learning for large language models

快速理解: We propose a new algorithm for fine-tuning large language models using reinforcement learning. Tapered Off-Policy REINFORCE (TOPR) uses an asymmetric, tapered variant of importance sampling to speed up learning while maintaining stable learning dynamics, even without the use of KL regularization.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.777 Task-Optimized Convolutional Recurrent Networks Align with Tactile Processing in the Rodent Brain

- 作者: 作者信息未提供
- 年份: 2025; 主题: Robot Learning; 优先级: 重要
- [来源](#)

论文: Task-Optimized Convolutional Recurrent Networks Align with Tactile Processing in the Rodent Brain

快速理解: Tactile sensing remains far less understood in neuroscience and less effective in artificial systems compared to more mature modalities such as vision and language. We bridge these gaps by introducing an Encoder-Attender-Decoder (EAD) framework to systematically explore the space of task-optimized temporal neural networks trained on realistic tactile input sequences from a customized rodent whisker-array simulator.

关注点: 机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级: 本地 PDF 前页。

1.3.7.778 Temporal Logic-Based Multi-Vehicle Backdoor Attacks against Offline RL Agents in End-to-end Autonomous Driving

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Temporal Logic-Based Multi-Vehicle Backdoor Attacks against Offline RL Agents in End-to-end Autonomous Driving

快速理解: Assessing the safety of autonomous driving (AD) systems against security threats, particularly backdoor attacks, is a stepping stone for real-world deployment. However, existing works mainly focus on pixel-level triggers that are impractical to deploy in the real world.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.779 TensorRL-QAS: Reinforcement learning with tensor networks for improved quantum architecture search

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: TensorRL-QAS: Reinforcement learning with tensor networks for improved quantum architecture search

快速理解: Variational quantum algorithms hold the promise to address meaningful quantum problems already on noisy intermediate-scale quantum hardware. In spite of the promise, they face the challenge of designing quantum circuits that both solve the target problem and comply with device limitations.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.780 Test-Time Adaptation of Vision-Language Models for Open-Vocabulary Semantic Segmentation

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Test-Time Adaptation of Vision-Language Models for Open-Vocabulary Semantic Segmentation

快速理解: Recently, test-time adaptation has attracted wide interest in the context of visionlanguage models for image classification. However, to the best of our knowledge, the problem is completely overlooked in dense prediction tasks such as Open- Vocabulary Semantic Segmentation (OVSS).

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.781 Test-Time Spectrum-Aware Latent Steering for Zero-Shot Generalization in Vision-Language Models

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Test-Time Spectrum-Aware Latent Steering for Zero-Shot Generalization in Vision-Language Models

快速理解: Vision-Language Models (VLMs) excel at zero-shot inference but often degrade under test-time domain shifts. For this reason, episodic test-time adaptation strategies have recently emerged as powerful techniques for adapting VLMs to a single unlabeled image.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.782 Text-to-Decision Agent: Offline Meta-Reinforcement Learning from Natural Language Supervision

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Text-to-Decision Agent: Offline Meta-Reinforcement Learning from Natural Language Supervision

快速理解: Offline meta-RL usually tackles generalization by inferring task beliefs from highquality samples or warmup explorations. The restricted form limits their generality and usability since these supervision signals are expensive and even infeasible to acquire in advance for unseen tasks.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.783 The Indra Representation Hypothesis for Multimodal Alignment

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: The Indra Representation Hypothesis for Multimodal Alignment

快速理解: Recent studies have uncovered an interesting phenomenon: unimodal foundation models tend to learn convergent representations, regardless of differences in architecture, training objectives, or data modalities. However, these representations are essentially internal abstractions of samples that characterize samples independently, leading to limited expressiveness.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.784 The Narrow Gate: Localized Image-Text Communication in Native Multimodal Models

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: The Narrow Gate: Localized Image-Text Communication in Native Multimodal Models

快速理解: Recent advances in multimodal training have significantly improved the integration of image understanding and generation within a unified model. This study investigates how vision-language models (VLMs) handle image-understanding tasks, focusing on how visual information is processed and transferred to the textual domain.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.785 The Omni-Expert: A Computationally Efficient Approach to Achieve a Mixture of Experts in a Single Expert Model

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：The Omni-Expert: A Computationally Efficient Approach to Achieve a Mixture of Experts in a Single Expert Model

快速理解：Mixture-of-Experts (MoE) models have become popular in machine learning, boosting performance by partitioning tasks across multiple experts. However, the need for several experts often results in high computational costs, limiting their application on resource-constrained devices with stringent real-time requirements, such as cochlear implants (CIs).

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.7.786 The Rise of Parameter Specialization for Knowledge Storage in Large Language Models

- 作者：作者信息未提供
- 年份：2025；主题：Systems；优先级：重要
- [来源](#)

论文：The Rise of Parameter Specialization for Knowledge Storage in Large Language Models

快速理解：Over time, a growing wave of large language models from various series has been introduced to the community. Researchers are striving to maximize the performance of language models with constrained parameter sizes.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.7.787 Think-RM: Enabling Long-Horizon Reasoning in Generative Reward Models

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Think-RM: Enabling Long-Horizon Reasoning in Generative Reward Models

快速理解：Reinforcement learning from human feedback (RLHF) has become a powerful posttraining paradigm for aligning large language models with human preferences. A core challenge in RLHF is constructing accurate reward signals, where the conventional Bradley-Terry reward models (BT RMs) often suffer from sensitivity to data size and coverage, as well as vulnerability to reward hacking.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.788 ThinkSound: Chain-of-Thought Reasoning in Multimodal LLMs for Audio Generation and Editing

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：ThinkSound: Chain-of-Thought Reasoning in Multimodal LLMs for Audio Generation and Editing

快速理解：While end-to-end video-to-audio generation has greatly improved, producing highfidelity audio that authentically captures the nuances of visual content remains challenging. Like professionals in the creative industries, this generation requires sophisticated reasoning about items such as visual dynamics, acoustic environments, and temporal relationships.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.789 Tighter CMI-Based Generalization Bounds via Stochastic Projection and Quantization

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：Tighter CMI-Based Generalization Bounds via Stochastic Projection and Quantization

快速理解：In this paper, we leverage stochastic projection and lossy compression to establish new conditional mutual information (CMI) bounds on the generalization error of statistical learning algorithms. It is shown that these bounds are generally tighter than the existing ones.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.7.790 Time-R1: Post-Training Large Vision Language Model for Temporal Video Grounding

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Time-R1: Post-Training Large Vision Language Model for Temporal Video Grounding

快速理解：Temporal Video Grounding (TVG), the task of locating specific video segments based on language queries, is a core challenge in long-form video understanding. While recent Large Vision-Language Models (LVLMs) have shown early promise in tackling TVG through supervised fine-tuning (SFT), their ability to generalize remains limited.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.791 Too Late to Recall: Explaining the Two-Hop Problem in Multimodal Knowledge Retrieval

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Too Late to Recall: Explaining the Two-Hop Problem in Multimodal Knowledge Retrieval

快速理解：Training vision language models (VLMs) aims to align visual representations from a vision encoder with the textual representations of a pretrained large language model (LLM). However, many VLMs exhibit reduced factual recall performance compared to their LLM backbones, raising the question of how effective multimodal fine-tuning is at extending existing mechanisms within the LLM to visual inputs.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.792 Touch in the Wild: Learning Fine-Grained Manipulation with a Portable Visuo-Tactile Gripper

- 作者：作者信息未提供
- 年份：2025；主题：Robot Learning；优先级：重要
- [来源](#)

论文：Touch in the Wild: Learning Fine-Grained Manipulation with a Portable Visuo-Tactile Gripper

快速理解：Handheld grippers are increasingly used to collect human demonstrations due to their ease of deployment and versatility. However, most existing designs lack tactile sensing, despite the critical role of tactile feedback in precise manipulation.

关注点：机器人数据、策略学习、操作、触觉或 sim-to-real。

证据层级：本地 PDF 前页。

1.3.7.793 Toward Efficient Inference Attacks: Shadow Model Sharing via Mixture-of-Experts

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：Toward Efficient Inference Attacks: Shadow Model Sharing via Mixture-of-Experts

快速理解：Machine learning models are often vulnerable to inference attacks that expose sensitive information from their training data. Shadow model technique is commonly employed in such attacks, such as membership inference.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.7.794 Towards Building Model/Prompt-Transferable Attackers against Large Vision-Language Models

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Towards Building Model/Prompt-Transferable Attackers against Large Vision-Language Models

快速理解：Although Large Vision-Language Models (LVLMs) exhibit impressive multimodal capabilities, their vulnerability to adversarial examples has raised serious security concerns. Existing LVLM attackers simply optimize adversarial images that easily overfit a certain model/prompt, making them ineffective once they are transferred to attack a different model/prompt.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.795 Towards Comprehensive Scene Understanding: Integrating First and Third-Person Views for LVLMs

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Towards Comprehensive Scene Understanding: Integrating First and Third-Person Views for LVLMs

快速理解：Large vision-language models (LVLMs) are increasingly deployed in interactive applications such as virtual and augmented reality, where a first-person (egocentric) view captured by head-mounted cameras serves as key input. While this view offers fine-grained cues about user attention and hand-object interactions, its narrow field of view and lack of global context often lead to failures on spatially or contextually demanding queries.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.796 Towards Dynamic 3D Reconstruction of Hand-Instrument Interaction in Ophthalmic Surgery

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Towards Dynamic 3D Reconstruction of Hand-Instrument Interaction in Ophthalmic Surgery

快速理解：Accurate 3D reconstruction of hands and instruments is critical for vision-based analysis of ophthalmic microsurgery, yet progress has been hampered by the lack of realistic, large-scale datasets and reliable annotation tools. In this work, we introduce OphNet-3D, the first extensive RGB-D dynamic 3D reconstruction dataset for ophthalmic surgery, comprising 41 sequences from 40 surgeons and totaling 7.1 million frames, with fine-grained annotations of 12 surgical phases, 10 instrument categories, dense MANO hand meshes, and full 6-DoF instrument poses.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.797 Towards General Continuous Memory for Vision-Language Models

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Towards General Continuous Memory for Vision-Language Models

快速理解：Language models (LMs) and their extension, vision-language models (VLMs), have achieved remarkable performance across various tasks. However, they still struggle with complex reasoning tasks that require multimodal or multilingual real-world knowledge.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.798 Towards Large-Scale In-Context Reinforcement Learning by Meta-Training in Randomized Worlds

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Towards Large-Scale In-Context Reinforcement Learning by Meta-Training in Randomized Worlds

快速理解：In-Context Reinforcement Learning (ICRL) enables agents to learn automatically and on-the-fly from their interactive experiences. However, a major challenge in scaling up ICRL is the lack of scalable task collections.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.799 Towards Physical Understanding in Video Generation: A 3D Point Regularization Approach

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Towards Physical Understanding in Video Generation: A 3D Point Regularization Approach

快速理解：We present a novel video generation framework that integrates 3-dimensional geometry and dynamic awareness. To achieve this, we augment 2D videos with 3D point trajectories and align them in pixel space.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.800 Towards Principled Unsupervised Multi-Agent Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文: Towards Principled Unsupervised Multi-Agent Reinforcement Learning

快速理解: In reinforcement learning, we typically refer to unsupervised pre-training when we aim to pre-train a policy without a priori access to the task specification, i.e., rewards, to be later employed for efficient learning of downstream tasks. In singleagent settings, the problem has been extensively studied and mostly understood.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.801 Towards Provable Emergence of In-Context Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Towards Provable Emergence of In-Context Reinforcement Learning

快速理解: Typically, a modern reinforcement learning (RL) agent solves a task by updating its neural network parameters to adapt its policy to the task. Recently, it has been observed that some RL agents can solve a wide range of new out-of-distribution tasks without parameter updates after pretraining on some task distribution.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.802 Towards Realistic Earth-Observation Constellation Scheduling: Benchmark and Methodology

- 作者: 作者信息未提供
- 年份: 2025; 主题: Systems; 优先级: 重要
- [来源](#)

论文: Towards Realistic Earth-Observation Constellation Scheduling: Benchmark and Methodology

快速理解: Agile Earth Observation Satellites (AEOSs) constellations offer unprecedented flexibility for monitoring the Earth's surface, but their scheduling remains challenging under large-scale scenarios, dynamic environments, and stringent constraints. Existing methods often simplify these complexities, limiting their realworld performance.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.7.803 Towards Reliable Code-as-Policies: A Neuro-Symbolic Framework for Embodied Task Planning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Embodied/WBC; 优先级: 重要
- [来源](#)

论文: Towards Reliable Code-as-Policies: A Neuro-Symbolic Framework for Embodied Task Planning

快速理解: Recent advances in large language models (LLMs) have enabled the automatic generation

of executable code for task planning and control in embodied agents such as robots, demonstrating the potential of LLM-based embodied intelligence. However, these LLM-based code-as-policies approaches often suffer from limited environmental grounding, particularly in dynamic or partially observable settings, leading to suboptimal task success rates due to incorrect or incomplete code generation.

关注点：具身任务中的感知、动作生成、全身协调或运动控制。

证据层级：本地 PDF 前页。

1.3.7.804 Towards Robust Zero-Shot Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Towards Robust Zero-Shot Reinforcement Learning

快速理解：The recent development of zero-shot reinforcement learning (RL) has opened a new avenue for learning pre-trained generalist policies that can adapt to arbitrary new tasks in a zero-shot manner. While the popular Forward-Backward representations (FB) and related methods have shown promise in zero-shot RL, we empirically found that their modeling lacks expressivity and that extrapolation errors caused by out-of-distribution (OOD) actions during offline learning sometimes lead to biased representations, ultimately resulting in suboptimal performance.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.805 Towards Self-Refinement of Vision-Language Models with Triangular Consistency

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Towards Self-Refinement of Vision-Language Models with Triangular Consistency

快速理解：Vision-Language Models (VLMs) integrate visual knowledge with the analytical capabilities of Large Language Models (LLMs) through supervised visual instruction tuning, using image-question-answer triplets. However, the potential of VLMs trained without supervised instruction remains largely unexplored.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.806 Towards foundational LiDAR world models with efficient latent flow matching

- 作者：作者信息未提供
- 年份：2025；主题：World Model；优先级：重要
- [来源](#)

论文：Towards foundational LiDAR world models with efficient latent flow matching

快速理解: LiDAR-based world models offer more structured and geometry-aware representations than their image-based counterparts. However, existing LiDAR world models are narrowly trained; each model excels only in the domain for which it was built.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策。

证据层级: 本地 PDF 前页。

1.3.7.807 Training-Free Efficient Video Generation via Dynamic Token Carving

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Training-Free Efficient Video Generation via Dynamic Token Carving

快速理解: Despite the remarkable generation quality of video Diffusion Transformer (DiT) models, their practical deployment is severely hindered by extensive computational requirements. This inefficiency stems from two key challenges: the quadratic complexity of self-attention with respect to token length and the multi-step nature of diffusion models.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.808 Training-Free Test-Time Adaptation via Shape and Style Guidance for Vision-Language Models

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Training-Free Test-Time Adaptation via Shape and Style Guidance for Vision-Language Models

快速理解: Test-time adaptation with pre-trained vision-language models shows impressive zero-shot classification abilities, and training-free methods further improve the performance without any optimization burden. However, existing training-free test-time adaptation methods typically rely on entropy criteria to select the visual features and update the visual caches, while ignoring the generalizable factors, such as shape-sensitive and style-insensitive factors.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.809 Trajectory Graph Learning: Aligning with Long Trajectories in Reinforcement Learning Without Reward Design

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Trajectory Graph Learning: Aligning with Long Trajectories in Reinforcement Learning Without Reward Design

快速理解: Reinforcement learning (RL) often relies on manually designed reward functions, which are difficult to specify and can lead to issues such as reward hacking and suboptimal behavior.

Alternatives like inverse RL and preference-based RL attempt to infer surrogate rewards from demonstrations or preferences but suffer from ambiguity and distribution mismatch.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.810 Trust, But Verify: A Self-Verification Approach to Reinforcement Learning with Verifiable Rewards

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Trust, But Verify: A Self-Verification Approach to Reinforcement Learning with Verifiable Rewards

快速理解：Large Language Models (LLMs) show great promise in complex reasoning, with Reinforcement Learning with Verifiable Rewards (RLVR) being a key enhancement strategy. However, a prevalent issue is “superficial self-reflection”, where models fail to robustly verify their own outputs.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.811 Two-Steps Diffusion Policy for Robotic Manipulation via Genetic Denoising

- 作者：作者信息未提供
- 年份：2025；主题：Embodied/WBC；优先级：重要
- [来源](#)

论文：Two-Steps Diffusion Policy for Robotic Manipulation via Genetic Denoising

快速理解：Diffusion models, such as diffusion policy, have achieved state-of-the-art results in robotic manipulation by imitating expert demonstrations. While diffusion models were originally developed for vision tasks like image and video generation, many of their inference strategies have been directly transferred to control domains without adaptation.

关注点：具身任务中的感知、动作生成、全身协调或运动控制。

证据层级：本地 PDF 前页。

1.3.7.812 U-CAN: Unsupervised Point Cloud Denoising with Consistency-Aware Noise2Noise Matching

- 作者：作者信息未提供
- 年份：2025；主题：Systems；优先级：重要
- [来源](#)

论文：U-CAN: Unsupervised Point Cloud Denoising with Consistency-Aware Noise2Noise Matching

快速理解：Point clouds captured by scanning sensors are often perturbed by noise, which have a highly negative impact on downstream tasks (e.g. surface reconstruction and shape understanding). Previous works mostly focus on training neural networks with noisy-clean point cloud pairs for learning denoising priors, which requires extensively manual efforts.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.7.813 UFO-RL: Uncertainty-Focused Optimization for Efficient Reinforcement Learning Data Selection

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：UFO-RL: Uncertainty-Focused Optimization for Efficient Reinforcement Learning Data Selection

快速理解：A primary impediment to scaling reinforcement learning (RL) for large language model (LLM) training is the substantial computational cost, predominantly arising from the necessity of multi-sampling for policy optimization and evaluation. This underscores the critical yet challenging nature of efficient training data selection.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.814 URDF-Anything: Constructing Articulated Objects with 3D Multimodal Language Model

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：URDF-Anything: Constructing Articulated Objects with 3D Multimodal Language Model

快速理解：Constructing accurate digital twins of articulated objects is essential for robotic simulation training and embodied AI world model building, yet historically requires painstaking manual modeling or multi-stage pipelines. In this work, we propose URDF-Anything, an end-to-end automatic reconstruction framework based on a 3D multimodal large language model (MLLM).

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.815 Uncertainty-Aware Multi-Objective Reinforcement Learning-Guided Diffusion Models for 3D De Novo Molecular Design

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Uncertainty-Aware Multi-Objective Reinforcement Learning-Guided Diffusion Models for 3D De Novo Molecular Design

快速理解：Designing de novo 3D molecules with desirable properties remains a fundamental challenge in drug discovery and molecular engineering. While diffusion models have demonstrated remarkable capabilities in generating high-quality 3D molecular structures, they often struggle to effectively control complex multi-objective constraints critical for real-world applications.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.816 Uncertainty-Based Smooth Policy Regularisation for Reinforcement Learning with Few Demonstrations

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Uncertainty-Based Smooth Policy Regularisation for Reinforcement Learning with Few Demonstrations

快速理解：In reinforcement learning with sparse rewards, demonstrations can accelerate learning, but determining when to imitate them remains challenging. We propose Smooth Policy Regularisation from Demonstrations (SPReD), a framework that addresses the fundamental question: when should an agent imitate a demonstration versus follow its own policy?

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.817 Understanding and Mitigating Numerical Sources of Nondeterminism in LLM Inference

- 作者：作者信息未提供
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：Understanding and Mitigating Numerical Sources of Nondeterminism in LLM Inference

快速理解：Large Language Models (LLMs) are now integral across various domains and have demonstrated impressive performance. Progress, however, rests on the premise that benchmark scores are both accurate and reproducible.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.7.818 Understanding protein function with a multimodal retrieval-augmented foundation model

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Understanding protein function with a multimodal retrieval-augmented foundation model

快速理解：Protein language models (PLMs) learn probability distributions over natural protein sequences. By learning from hundreds of millions of natural protein sequences, protein understanding and design capabilities emerge.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.819 Understanding the Gain from Data Filtering in Multimodal Contrastive Learning

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Understanding the Gain from Data Filtering in Multimodal Contrastive Learning

快速理解：The success of modern multimodal representation learning relies on internet-scale datasets. Due to the low quality of a large fraction of raw web data, data curation has become a critical step in the training pipeline.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.820 Uni-MuMER: Unified Multi-Task Fine-Tuning of Vision-Language Model for Handwritten Mathematical Expression Recognition

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Uni-MuMER: Unified Multi-Task Fine-Tuning of Vision-Language Model for Handwritten Mathematical Expression Recognition

快速理解：Handwritten Mathematical Expression Recognition (HMER) remains a persistent challenge in Optical Character Recognition (OCR) due to the inherent freedom of symbol layouts and variability in handwriting styles. Prior methods have faced performance bottlenecks by proposing isolated architectural modifications, making them difficult to integrate coherently into a unified framework.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.821 Uni-RL: Unifying Online and Offline RL via Implicit Value Regularization

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Uni-RL: Unifying Online and Offline RL via Implicit Value Regularization

快速理解：The practical implementations of reinforcement learning (RL) often face diverse settings, such as online, offline, and offline-to-online learning. Instead of developing separate algorithms for each setting, we propose Uni-RL, a unified model-free RL framework that addresses all these scenarios within a single formulation.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.822 UniGen: Enhanced Training & Test-Time Strategies for Unified Multimodal Understanding and Generation

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文: UniGen: Enhanced Training & Test-Time Strategies for Unified Multimodal Understanding and Generation

快速理解: We introduce UniGen, a unified multimodal large language model (MLLM) capable of image understanding and generation. We study the full training pipeline of Uni- Gen from a data-centric perspective, including multi-stage pre-training, supervised fine-tuning, and direct preference optimization.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.823 Unlabeled Data Improves Fine-Grained Image Zero-shot Classification with Multimodal LLMs

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文: Unlabeled Data Improves Fine-Grained Image Zero-shot Classification with Multimodal LLMs

快速理解: Despite Multimodal Large Language Models (MLLMs) showing promising results on general zero-shot image classification tasks, fine-grained image classification remains challenging. It demands precise attention to subtle visual details to distinguish between visually similar subcategories—details that MLLMs may easily overlook without explicit guidance.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.824 Unleashing the Potential of Multimodal LLMs for Zero-Shot Spatio-Temporal Video Grounding

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文: Unleashing the Potential of Multimodal LLMs for Zero-Shot Spatio-Temporal Video Grounding

快速理解: Spatio-temporal video grounding (STVG) aims at localizing the spatio-temporal tube of a video, as specified by the input text query. In this paper, we utilize multimodal large language models (MLLMs) to explore a zero-shot solution in STVG.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.825 Unveiling Chain of Step Reasoning for Vision-Language Models with Fine-grained Rewards

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要

- [来源](#)

论文: Unveiling Chain of Step Reasoning for Vision-Language Models with Fine-grained Rewards

快速理解: Chain of thought reasoning has demonstrated remarkable success in large language models, yet its adaptation to vision-language reasoning remains an open challenge with unclear best practices. Existing attempts typically employ reasoning chains at a coarse-grained level, which struggles to perform fine-grained structured reasoning and, more importantly, are difficult to evaluate the reward and quality of intermediate reasoning.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.826 Unveiling the Compositional Ability Gap in Vision-Language Reasoning Model

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Unveiling the Compositional Ability Gap in Vision-Language Reasoning Model

快速理解: While large language models (LLMs) demonstrate strong reasoning capabilities utilizing reinforcement learning (RL) with verifiable reward, whether large visionlanguage models (VLMs) can directly inherit such capabilities through similar posttraining strategies remains underexplored. In this work, we conduct a systematic compositional probing study to evaluate whether current VLMs trained with RL or other post-training strategies can compose capabilities across modalities or tasks under out-of-distribution conditions.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.827 Unveiling the Uncertainty in Embodied and Operational Carbon of Large AI Models through a Probabilistic Carbon Accounting Model

- 作者: 作者信息未提供
- 年份: 2025; 主题: Embodied/WBC; 优先级: 重要
- [来源](#)

论文: Unveiling the Uncertainty in Embodied and Operational Carbon of Large AI Models through a Probabilistic Carbon Accounting Model

快速理解: The rapid growth of large AI models has raised significant environmental concerns due to their substantial carbon footprint. Existing carbon accounting methods for AI models are fundamentally deterministic and fail to account for inherent uncertainties in embodied and operational carbon emissions.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制。

证据层级: 本地 PDF 前页。

1.3.7.828 VAGEN: Reinforcing World Model Reasoning for Multi-Turn VLM Agents

- 作者: 作者信息未提供

- 年份: 2025; 主题: World Model; 优先级: 重要
- [来源](#)

论文: VAGEN: Reinforcing World Model Reasoning for Multi-Turn VLM Agents

快速理解: A major challenge in training VLM agents, compared to LLM agents, is that states shift from simple texts to complex visual observations, which introduces partial observability and demands robust world modeling. We ask: can VLM agents build internal world models through explicit visual state reasoning?

关注点: 环境动力学、视频预测、潜变量状态或模型式决策。

证据层级: 本地 PDF 前页。

1.3.7.829 VETA-DiT: Variance-Equalized and Temporally Adaptive Quantization for Efficient 4-bit Diffusion Transformers

- 作者: 作者信息未提供
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: VETA-DiT: Variance-Equalized and Temporally Adaptive Quantization for Efficient 4-bit Diffusion Transformers

快速理解: Diffusion Transformers (DiTs) have recently demonstrated remarkable performance in visual generation tasks, surpassing traditional U-Net-based diffusion models by significantly improving image and video generation quality and scalability. However, the large model size and iterative denoising process introduce substantial computational and memory overhead, limiting their deployment in realworld applications.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 本地 PDF 前页。

1.3.7.830 VL-SAE: Interpreting and Enhancing Vision-Language Alignment with a Unified Concept Set

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: VL-SAE: Interpreting and Enhancing Vision-Language Alignment with a Unified Concept Set

快速理解: The alignment of vision-language representations endows current Vision-Language Models (VLMs) with strong multi-modal reasoning capabilities. However, the interpretability of the alignment component remains uninvestigated due to the difficulty in mapping the semantics of multi-modal representations into a unified concept set.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.831 VLForgery Face Triad: Detection, Localization and Attribution via Multi-modal Large Language Models

- 作者: 作者信息未提供

- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: VLForgery Face Triad: Detection, Localization and Attribution via Multimodal Large Language Models

快速理解: Faces synthesized by diffusion models (DMs) with high-quality and controllable attributes pose a significant challenge for Deepfake detection. Most state-of-the-art detectors only yield a binary decision, incapable of forgery localization, attribution of forgery methods, and providing analysis on the cause of forgeries.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.832 VLM in a flash: I/O-Efficient Sparsification of Vision-Language Model via Neuron Chunking

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: VLM in a flash: I/O-Efficient Sparsification of Vision-Language Model via Neuron Chunking

快速理解: Edge deployment of large Vision-Language Models (VLMs) increasingly relies on flash-based weight offloading, where activation sparsification is used to reduce I/O overhead. However, conventional sparsification remains model-centric, selecting neurons solely by activation magnitude and neglecting how access patterns influence flash performance.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.833 VLM-R³: Region Recognition, Reasoning, and Refinement for Enhanced Multimodal Chain-of-Thought

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: VLM-R³: Region Recognition, Reasoning, and Refinement for Enhanced Multimodal Chain-of-Thought

快速理解: Recently, reasoning-based MLLMs have achieved a degree of success in generating long-form textual reasoning chains. However, they still struggle with complex tasks that necessitate dynamic and iterative focusing on and revisiting of visual regions to achieve precise grounding of textual reasoning in visual evidence.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.834 VLMLight: Safety-Critical Traffic Signal Control via Vision-Language Meta-Control and Dual-Branch Reasoning Architecture

- 作者: 作者信息未提供

- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: VLMLight: Safety-Critical Traffic Signal Control via Vision-Language Meta-Control and Dual-Branch Reasoning Architecture

快速理解: Traffic signal control (TSC) is a core challenge in urban mobility, where real-time decisions must balance efficiency and safety. Existing methods—ranging from rulebased heuristics to reinforcement learning (RL)—often struggle to generalize to complex, dynamic, and safety-critical scenarios.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.835 VRAG-RL: Empower Vision-Perception-Based RAG for Visually Rich Information Understanding via Iterative Reasoning with Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: VRAG-RL: Empower Vision-Perception-Based RAG for Visually Rich Information Understanding via Iterative Reasoning with Reinforcement Learning

快速理解: Effectively retrieving, reasoning and understanding visually rich information remains a challenge for traditional Retrieval-Augmented Generation (RAG) methods. On the one hand, traditional text-based methods cannot handle visual-related information.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.836 VaMP: Variational Multi-Modal Prompt Learning for Vision-Language Models

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: VaMP: Variational Multi-Modal Prompt Learning for Vision-Language Models

快速理解: Vision-language models (VLMs), such as CLIP, have shown strong generalization under zero-shot settings, yet adapting them to downstream tasks with limited supervision remains a significant challenge. Existing multi-modal prompt learning methods typically rely on fixed, shared prompts and deterministic parameters, which limits their ability to capture instance-level variation or model uncertainty across diverse tasks and domains.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.837 Value Diffusion Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要

- [来源](#)

论文: Value Diffusion Reinforcement Learning

快速理解: Model-free reinforcement learning (RL) combined with diffusion models has achieved significant progress in addressing complex continuous control tasks. However, a persistent challenge in RL remains the accurate estimation of Q-values, which critically governs the efficacy of policy optimization.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.838 Value-Guided Decision Transformer: A Unified Reinforcement Learning Framework for Online and Offline Settings

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Value-Guided Decision Transformer: A Unified Reinforcement Learning Framework for Online and Offline Settings

快速理解: The Conditional Sequence Modeling (CSM) paradigm, benefiting from the transformer's powerful distribution modeling capabilities, has demonstrated considerable promise in Reinforcement Learning (RL) tasks. However, much of the work has focused on applying CSM to single online or offline settings, with the general architecture rarely explored.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.839 ViCrit: A Verifiable Reinforcement Learning Proxy Task for Visual Perception in VLMs

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: ViCrit: A Verifiable Reinforcement Learning Proxy Task for Visual Perception in VLMs

快速理解: Reinforcement learning (RL) has shown great effectiveness for fine-tuning large language models (LLMs) using tasks that are challenging yet easily verifiable, such as math reasoning or code generation. However, extending this success to visual perception in vision-language models (VLMs) has been impeded by the scarcity of vision-centric tasks that are simultaneously challenging and unambiguously verifiable.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.840 ViSpec: Accelerating Vision-Language Models with Vision-Aware Speculative Decoding

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要

- [来源](#)

论文: ViSpec: Accelerating Vision-Language Models with Vision-Aware Speculative Decoding

快速理解: Speculative decoding is a widely adopted technique for accelerating inference in large language models (LLMs), yet its application to vision-language models (VLMs) remains underexplored, with existing methods achieving only modest speedups ($< 1.5\times$). This gap is increasingly significant as multimodal capabilities become central to large-scale models.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.841 Video Perception Models for 3D Scene Synthesis

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: Video Perception Models for 3D Scene Synthesis

快速理解: Automating the expert-dependent and labor-intensive task of 3D scene synthesis would significantly benefit fields such as architectural design, robotics simulation, and virtual reality. Recent approaches to 3D scene synthesis often rely on the commonsense reasoning of large language models (LLMs) or strong visual priors from image generation models.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.842 Video World Models with Long-term Spatial Memory

- 作者: 作者信息未提供
- 年份: 2025; 主题: World Model; 优先级: 重要
- [来源](#)

论文: Video World Models with Long-term Spatial Memory

快速理解: Emerging world models autoregressively generate video frames in response to actions, such as camera movements and text prompts, among other control signals. Due to limited temporal context window sizes, these models often struggle to maintain scene consistency during revisits, leading to severe forgetting of previously generated environments.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策。

证据层级: 本地 PDF 前页。

1.3.7.843 VideoChat-R1.5: Visual Test-Time Scaling to Reinforce Multimodal Reasoning by Iterative Perception

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: VideoChat-R1.5: Visual Test-Time Scaling to Reinforce Multimodal Reasoning by Iterative Perception

快速理解: Inducing reasoning in multimodal large language models (MLLMs) is critical for achieving human-level perception and understanding. Existing methods mainly leverage LLM reasoning to analyze parsed visuals, often limited by static perception stages.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.844 VideoMAR: Autoregressive Video Generation with Continuous Tokens

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: VideoMAR: Autoregressive Video Generation with Continuous Tokens

快速理解: Mask-based autoregressive models have demonstrated promising image generation capability in continuous space. However, their potential for video generation remains under-explored.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.845 VideoREPA: Learning Physics for Video Generation through Relational Alignment with Foundation Models

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: VideoREPA: Learning Physics for Video Generation through Relational Alignment with Foundation Models

快速理解: Recent advancements in text-to-video (T2V) diffusion models have enabled highfidelity and realistic video synthesis. However, current T2V models often struggle to generate physically plausible content due to their limited inherent ability to accurately understand physics.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.846 VideoTitans: Scalable Video Prediction with Integrated Short- and Long-term Memory

- 作者: 作者信息未提供
- 年份: 2025; 主题: World Model; 优先级: 重要
- [来源](#)

论文: VideoTitans: Scalable Video Prediction with Integrated Short- and Long-term Memory

快速理解: Accurate video forecasting enables autonomous vehicles to anticipate hazards, robotics and surveillance systems to predict human intent, and environmental models to issue timely warnings for extreme weather events. However, existing methods remain limited: transformers rely on global attention with quadratic complexity, making them impractical for high-resolution, long-horizon video prediction, while convolutional and recurrent networks suffer from short-range receptive fields and vanishing gradients, losing key information over extended sequences.

关注点：环境动力学、视频预测、潜变量状态或模型式决策。

证据层级：本地 PDF 前页。

1.3.7.847 Videos are Sample-Efficient Supervisions: Behavior Cloning from Videos via Latent Representations

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Videos are Sample-Efficient Supervisions: Behavior Cloning from Videos via Latent Representations

快速理解：Humans can efficiently extract knowledge and learn skills from the videos within only a few trials and errors. However, it poses a big challenge to replicate this learning process for autonomous agents, due to the complexity of visual input, the absence of action or reward signals, and the limitations of interaction steps.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.848 ViewPoint: Panoramic Video Generation with Pretrained Diffusion Models

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：ViewPoint: Panoramic Video Generation with Pretrained Diffusion Models

快速理解：Panoramic video generation aims to synthesize 360-degree immersive videos, holding significant importance in the fields of VR, world models, and spatial intelligence. Existing works fail to synthesize high-quality panoramic videos due to the inherent modality gap between panoramic data and perspective data, which constitutes the majority of the training data for modern diffusion models.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.849 Vinci: Deep Thinking in Text-to-Image Generation using Unified Model with Reinforcement Learning

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Vinci: Deep Thinking in Text-to-Image Generation using Unified Model with Reinforcement Learning

快速理解：With the continuous development of large language models and reasoning chain technologies, the potential of deep reasoning based on reinforcement learning has shown remarkable promise in multi-task scenarios. However, existing unified models have yet to achieve end-to-end integration in image generation and understanding tasks, limiting the model's self-reflection ability and the

realization of cross-modal reasoning chains.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.850 Vision Function Layer in Multimodal LLMs

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Vision Function Layer in Multimodal LLMs

快速理解：This study identifies that visual-related functional decoding is distributed across different decoder layers in Multimodal Large Language Models (MLLMs). Typically, each function, such as counting, grounding, or OCR recognition, narrows down to two or three layers, which we define as Vision Function Layers (VFL).

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.851 Visual Thoughts: A Unified Perspective of Understanding Multimodal Chain-of-Thought

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Visual Thoughts: A Unified Perspective of Understanding Multimodal Chain-of-Thought

快速理解：Large Vision-Language Models (LVLMs) have achieved significant success in multimodal tasks, with multimodal chain-of-thought (MCoT) further enhancing performance and interpretability. Recent MCoT methods fall into two categories: (i) Textual-MCoT (T-MCoT), which takes multimodal input and produces textual output; and (ii) Interleaved-MCoT (I-MCoT), which generates interleaved imagetext outputs.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.852 VisualQuality-R1: Reasoning-Induced Image Quality Assessment via Reinforcement Learning to Rank

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：VisualQuality-R1: Reasoning-Induced Image Quality Assessment via Reinforcement Learning to Rank

快速理解：DeepSeek-R1 has demonstrated remarkable effectiveness in incentivizing reasoning and generalization capabilities of large language models (LLMs) through reinforcement learning. Nevertheless, the potential of reasoning-induced computation has not been thoroughly explored in the context of image quality assessment (IQA), a task depending critically on visual reasoning.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.853 WALL-E: World Alignment by NeuroSymbolic Learning improves World Model-based LLM Agents

- 作者：作者信息未提供
- 年份：2025；主题：World Model；优先级：重要
- [来源](#)

论文：WALL-E: World Alignment by NeuroSymbolic Learning improves World Model-based LLM Agents

快速理解：Can we build accurate world models out of large language models (LLMs)? How can world models benefit LLM agents?

关注点：环境动力学、视频预测、潜变量状态或模型式决策。

证据层级：本地 PDF 前页。

1.3.7.854 WHAT MAKES MATH PROBLEMS HARD FOR REINFORCEMENT LEARNING: A CASE STUDY

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：WHAT MAKES MATH PROBLEMS HARD FOR REINFORCEMENT LEARNING: A CASE STUDY

快速理解：Using a long-standing conjecture from combinatorial group theory, we explore, from multiple perspectives, the challenges of finding rare instances carrying disproportionately high rewards. Based on lessons learned in the context defined by the Andrews-Curtis conjecture, we analyze how reinforcement learning agents handle problems of varying hardness.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.855 WISA: World simulator assistant for physics-aware text-to-video generation

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：WISA: World simulator assistant for physics-aware text-to-video generation

快速理解：Recent advances in text-to-video (T2V) generation, exemplified by models such as Sora and Kling, have demonstrated strong potential for constructing world simulators. However, existing T2V models still struggle to understand abstract physical principles and to generate videos that faithfully obey physical laws.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.856 Wan-Move: Motion-controllable Video Generation via Latent Trajectory Guidance

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Wan-Move: Motion-controllable Video Generation via Latent Trajectory Guidance

快速理解：We present Wan-Move, a simple and scalable framework that brings motion control to video generative models. Existing motion-controllable methods typically suffer from coarse control granularity and limited scalability, leaving their outputs insufficient for practical use.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.857 Watch and Listen: Understanding Audio-Visual-Speech Moments with Multimodal LLM

- 作者：作者信息未提供
- 年份：2025；主题：Supporting Perception；优先级：重要
- [来源](#)

论文：Watch and Listen: Understanding Audio-Visual-Speech Moments with Multimodal LLM

快速理解：Humans naturally understand moments in a video by integrating visual and auditory cues. For example, localizing a scene in the video like “A scientist passionately speaks on wildlife conservation as dramatic orchestral music plays, with the audience nodding and applauding” requires simultaneous processing of visual, audio, and speech signals.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.858 What Can RL Bring to VLA Generalization? An Empirical Study

- 作者：作者信息未提供
- 年份：2025；主题：VLA；优先级：重要
- [来源](#)

论文：What Can RL Bring to VLA Generalization? An Empirical Study

快速理解：Large Vision-Language Action (VLA) models have shown significant potential for embodied AI. However, their predominant training via supervised fine-tuning (SFT) limits generalization due to susceptibility to compounding errors under distribution shifts.

关注点：视觉、语言条件与机器人动作策略之间的对齐。

证据层级：本地 PDF 前页。

1.3.7.859 What Makes a Reward Model a Good Teacher? An Optimization Perspective

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要

- [来源](#)

论文: What Makes a Reward Model a Good Teacher? An Optimization Perspective

快速理解: The success of Reinforcement Learning from Human Feedback (RLHF) critically depends on the quality of the reward model. However, while this quality is primarily evaluated through accuracy, it remains unclear whether accuracy fully captures what makes a reward model an effective teacher.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.860 What We Miss Matters: Learning from the Overlooked in Point Cloud Transformers

- 作者: 作者信息未提供
- 年份: 2025; 主题: Systems; 优先级: 重要
- [来源](#)

论文: What We Miss Matters: Learning from the Overlooked in Point Cloud Transformers

快速理解: Point Cloud Transformers have become a cornerstone in 3D representation for their ability to model long-range dependencies via self-attention. However, these models tend to overemphasize salient regions while neglecting other informative regions, which limits feature diversity and compromises robustness.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.7.861 When Can Model-Free Reinforcement Learning be Enough for Thinking?

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: When Can Model-Free Reinforcement Learning be Enough for Thinking?

快速理解: Recent work on large language models has demonstrated the use of model-free reinforcement learning (RL) to train reasoning-like capabilities. The emergence of “thinking” through model-free RL is interesting as thinking actions neither produce reward nor change the external world state to one where the agent is more likely to get reward.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.862 When Semantics Mislead Vision: Mitigating Large Multimodal Models Hallucinations in Scene Text Spotting and Understanding

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: When Semantics Mislead Vision: Mitigating Large Multimodal Models Hallucinations in Scene Text Spotting and Understanding

快速理解: Large Multimodal Models (LMMs) have achieved impressive progress in visual perception and reasoning. However, when confronted with visually ambiguous or non-semantic scene text, they often struggle to accurately spot and understand the content, frequently generating semantically plausible yet visually incorrect answers, which we refer to as semantic hallucination.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.863 Who Speaks for the Trigger? Dynamic Expert Routing in Backdoored Mixture-of-Experts Transformers

- 作者: 作者信息未提供
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Who Speaks for the Trigger? Dynamic Expert Routing in Backdoored Mixture-of-Experts Transformers

快速理解: Large language models (LLMs) with Mixture-of-Experts (MoE) architectures achieve impressive performance and efficiency by dynamically routing inputs to specialized subnetworks, known as experts. However, this sparse routing mechanism inherently exhibits task preferences due to expert specialization, introducing a new and underexplored vulnerability to backdoor attacks.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 本地 PDF 前页。

1.3.7.864 Wider or Deeper? Scaling LLM Inference-Time Compute with Adaptive Branching Tree Search

- 作者: 作者信息未提供
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Wider or Deeper? Scaling LLM Inference-Time Compute with Adaptive Branching Tree Search

快速理解: Recent advances demonstrate that increasing inference-time computation can significantly boost the reasoning capabilities of large language models (LLMs). Although repeated sampling (i.e., generating multiple candidate outputs) is a highly effective strategy, it does not leverage external feedback signals for refinement, which are often available in real tasks like coding.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 本地 PDF 前页。

1.3.7.865 With Limited Data for Multimodal Alignment, Let the STRUCTURE Guide You

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: With Limited Data for Multimodal Alignment, Let the STRUCTURE Guide You

快速理解: Multimodal models have demonstrated powerful capabilities in complex tasks requiring

multimodal alignment, including zero-shot classification and cross-modal retrieval. However, existing models typically rely on millions of paired multimodal samples, which are prohibitively expensive or infeasible to obtain in many domains.

关注点: 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 本地 PDF 前页。

1.3.7.866 Wonder Wins Ways: Curiosity-Driven Exploration through Multi-Agent Contextual Calibration

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: Wonder Wins Ways: Curiosity-Driven Exploration through Multi-Agent Contextual Calibration

快速理解: Autonomous exploration in complex multi-agent reinforcement learning (MARL) with sparse rewards critically depends on providing agents with effective intrinsic motivation. While artificial curiosity offers a powerful self-supervised signal, it often confuses environmental stochasticity with meaningful novelty.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.867 World Models as Reference Trajectories for Rapid Motor Adaptation

- 作者: 作者信息未提供
- 年份: 2025; 主题: World Model; 优先级: 重要
- [来源](#)

论文: World Models as Reference Trajectories for Rapid Motor Adaptation

快速理解: Learned control policies often fail when deployed in real-world environments with changing dynamics. When system dynamics shift unexpectedly, performance degrades until models are retrained on new data.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策。

证据层级: 本地 PDF 前页。

1.3.7.868 World-aware Planning Narratives Enhance Large Vision-Language Model Planner

- 作者: 作者信息未提供
- 年份: 2025; 主题: Supporting Perception; 优先级: 重要
- [来源](#)

论文: World-aware Planning Narratives Enhance Large Vision-Language Model Planner

快速理解: Large Vision-Language Models (LVLMs) show promise for embodied planning tasks but struggle with complex scenarios involving unfamiliar environments and multi-step goals. Current approaches rely on environment-agnostic imitation learning that disconnects instructions from environmental contexts, causing models to struggle with context-sensitive instructions and rely on supplementary cues rather than visual reasoning during long-horizon interactions.

关注点：3D、视频、多模态视觉或具身感知支撑能力。

证据层级：本地 PDF 前页。

1.3.7.869 Zero-Shot Context Generalization in Reinforcement Learning from Few Training Contexts

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Zero-Shot Context Generalization in Reinforcement Learning from Few Training Contexts

快速理解：Deep reinforcement learning (DRL) has achieved remarkable success across multiple domains, including competitive games, natural language processing, and robotics. Despite these advancements, policies trained via DRL often struggle to generalize to evaluation environments with different parameters.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.870 Zero-Shot Detection of LLM-Generated Text via Implicit Reward Model

- 作者：作者信息未提供
- 年份：2025；主题：Agent RL；优先级：重要
- [来源](#)

论文：Zero-Shot Detection of LLM-Generated Text via Implicit Reward Model

快速理解：Large language models (LLMs) have demonstrated remarkable capabilities across various tasks. However, their ability to generate human-like text has raised concerns about potential misuse.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级：本地 PDF 前页。

1.3.7.871 Zero-shot World Models via Search in Memory

- 作者：作者信息未提供
- 年份：2025；主题：World Model；优先级：重要
- [来源](#)

论文：Zero-shot World Models via Search in Memory

快速理解：World Models have vastly permeated the field of Reinforcement Learning. Their ability to model the transition dynamics of an environment have greatly improved sample efficiency in online RL.

关注点：环境动力学、视频预测、潜变量状态或模型式决策。

证据层级：本地 PDF 前页。

1.3.7.872 ZigzagPointMamba: Spatial-Semantic Mamba for Point Cloud Understanding

- 作者：作者信息未提供

- 年份: 2025; 主题: Systems; 优先级: 重要
- [来源](#)

论文: ZigzagPointMamba: Spatial-Semantic Mamba for Point Cloud Understanding

快速理解: State Space models (SSMs) like PointMamba provide efficient feature extraction for point cloud self-supervised learning with linear complexity, surpassing Transformers in computational efficiency. However, existing PointMamba-based methods rely on complex token ordering and random masking, disrupting spatial continuity and local semantic correlations.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.7.873 d1: Scaling Reasoning in Diffusion Large Language Models via Reinforcement Learning

- 作者: 作者信息未提供
- 年份: 2025; 主题: Agent RL; 优先级: 重要
- [来源](#)

论文: d1: Scaling Reasoning in Diffusion Large Language Models via Reinforcement Learning

快速理解: Recent large language models (LLMs) have demonstrated strong reasoning capabilities that benefits from online reinforcement learning (RL). These capabilities have primarily been demonstrated within the left-to-right autoregressive (AR) generation paradigm.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策。

证据层级: 本地 PDF 前页。

1.3.7.874 dKV-Cache: The Cache for Diffusion Language Models

- 作者: 作者信息未提供
- 年份: 2025; 主题: Systems; 优先级: 重要
- [来源](#)

论文: dKV-Cache: The Cache for Diffusion Language Models

快速理解: Diffusion Language Models (DLMs) have been seen as a promising competitor for autoregressive language models (ARs). However, diffusion language models have long been constrained by slow inference.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.7.875 seq-JEPA: Autoregressive Predictive Learning of Invariant-Equivariant World Models

- 作者: 作者信息未提供
- 年份: 2025; 主题: World Model; 优先级: 重要
- [来源](#)

论文: seq-JEPA: Autoregressive Predictive Learning of Invariant-Equivariant World Models

快速理解: Joint-embedding self-supervised learning (SSL) commonly relies on transformations such

as data augmentation and masking to learn visual representations, a task achieved by enforcing invariance or equivariance with respect to these transformations applied to two views of an image. This dominant two-view paradigm in SSL often limits the flexibility of learned representations for downstream adaptation by creating performance trade-offs between high-level invariance-demanding tasks such as image classification and more fine-grained equivariance-related tasks.

关注点：环境动力学、视频预测、潜变量状态或模型式决策。

证据层级：本地 PDF 前页。

1.3.8 SYSTEMS · ASPLOS

本会议纳入 **55** 篇。自动主题统计为：Systems (54)、AI Infra (53)、Agent RL (4)、World Model (2)、Supporting Perception (1)、Teleoperation (1)。这些统计用于组织阅读顺序，不能单独证明论文结论或实验优越性。

优先阅读候选：- TelaMalloc: Efficient On-Chip Memory Allocation for Production Machine Learning Accelerators (优先级分数 80.709) - ARC: Warp-level Adaptive Atomic Reduction in GPUs to Accelerate Differentiable Rendering (优先级分数 80.572) - History Doesn't Repeat Itself but Rollouts Rhyme: Accelerating Reinforcement Learning with RhymeRL (优先级分数 79.085) - PyTorch 2: Faster Machine Learning Through Dynamic Python Bytecode Transformation and Graph Compilation (优先级分数 77.0) - TAPAS: Thermal- and Power-Aware Scheduling for LLM Inference in Cloud Platforms (优先级分数 73.676)

1.3.8.1 TelaMalloc: Efficient On-Chip Memory Allocation for Production Machine Learning Accelerators

- 作者：Martin Maas; Ulysse Beaunon; A. Chauhan; Berkin Ilbeyi
- 年份：2022; 主题：World Model; AI Infra; Systems; 优先级：核心
- [来源](#)

论文：TelaMalloc: Efficient On-Chip Memory Allocation for Production Machine Learning Accelerators

快速理解：Memory buffer allocation for on-chip memories is a major challenge in modern machine learning systems that target ML accelerators. In interactive systems such as mobile phones, it is on the critical path of launching ML-enabled applications.

关注点：环境动力学、视频预测、潜变量状态或模型式决策；训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.8.2 ARC: Warp-level Adaptive Atomic Reduction in GPUs to Accelerate Differentiable Rendering

- 作者：Sankeerth Durvasula; Adrian Zhao; Fan Chen; Ruofan Liang; Pawan Kumar Sanjaya; Yushi Guan; Christina Giannoula; Nandita Vijaykumar
- 年份：2025; 主题：AI Infra; Supporting Perception; Systems; 优先级：核心
- [来源](#)

论文：ARC: Warp-level Adaptive Atomic Reduction in GPUs to Accelerate Differentiable Rendering

快速理解：Differentiable rendering is widely used in emerging applications that represent any 3D scene as a model trained using gradient descent from 2D images. Recent works (e.g., 3D Gaussian Splatting) use rasterization to enable rendering photo-realistic imagery at high speeds from these

learned 3D models.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；3D、视频、多模态视觉或具身感知支撑能力；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.8.3 History Doesn't Repeat Itself but Rollouts Rhyme: Accelerating Reinforcement Learning with RhymeRL

- 作者：Jingkai He; Tianjian Li; Erhu Feng; Dong Du; Qian Liu; Tao Liu; Yu-Bin Xia; Haibo Chen
- 年份：2026；主题：Agent RL; AI Infra; Systems；优先级：核心
- [来源](#)

论文：History Doesn't Repeat Itself but Rollouts Rhyme: Accelerating Reinforcement Learning with RhymeRL

快速理解：With the rapid advancement of large language models (LLMs), reinforcement learning (RL) has emerged as a pivotal methodology for enhancing the reasoning capabilities of LLMs. Unlike traditional pre-training approaches, RL encompasses multiple stages: rollout, reward, and training, which necessitates collaboration among various worker types.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.8.4 PyTorch 2: Faster Machine Learning Through Dynamic Python Bytecode Transformation and Graph Compilation

- 作者：Jason Ansel; Edward Yang; Horace He; N. Gimsheine; Animesh Jain; Michael Voznesensky; Bin Bao; Peter Bell; D. Berard; Evgeni Burovski; Geeta Chauhan; Anjali Chourdia; W. Constable; Alban Desmaison; Zachary DeVito; Elias Ellison; W. Feng; Jiong Gong; Michael Gschwind; B. Hirsh; Sherlock Huang; Kshiteej Kalambarkar; Laurent Kirsch; Michael Lazos; M. Lezcano; Yanbo Liang; J. Liang; Yinghai Lu; C. Luk; Bertrand A. Maher; Yunjie Pan; Christian Puhersch; Matthias Reso; Mark-Albert Saroufim; Marcos Yukio Siraichi; Helen Suk; Shunting Zhang; Michael Suo; P. Tillet; Xu Zhao; Eikan Wang; Keren Zhou; Richard Zou; Xiaodong Wang; Ajit Mathews; W. Wen; Gregory Chanan; Peng Wu; Soumith Chintala
- 年份：2024；主题：World Model; AI Infra; Systems；优先级：核心
- [来源](#)

论文：PyTorch 2: Faster Machine Learning Through Dynamic Python Bytecode Transformation and Graph Compilation

快速理解：This paper introduces two extensions to the popular PyTorch machine learning framework, TorchDynamo and TorchInductor, which implement the torch.compile feature released in PyTorch 2. TorchDynamo is a Python-level just-in-time (JIT) compiler that enables graph compilation in PyTorch programs without sacrificing the flexibility of Python.

关注点：环境动力学、视频预测、潜变量状态或模型式决策；训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.8.5 TAPAS: Thermal- and Power-Aware Scheduling for LLM Inference in Cloud Platforms

- 作者: Jovan Stojkovic; Chaojie Zhang; Íñigo Goiri; Esha Choukse; Haoran Qiu; Rodrigo Fonseca; J. Torrellas; Ricardo Bianchini
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: TAPAS: Thermal- and Power-Aware Scheduling for LLM Inference in Cloud Platforms

快速理解: The rising demand for generative large language models (LLMs) poses challenges for thermal and power management in cloud datacenters. Traditional techniques are often inadequate for LLM inference due to the fine-grained, millisecond-scale execution phases, each with distinct performance, thermal, and power profiles.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.8.6 ExeGPT: Constraint-Aware Resource Scheduling for LLM Inference

- 作者: Hyungjun Oh; Kihong Kim; Jaemin Kim; Sungkyun Kim; Junyeol Lee; Du-seong Chang; Jiwon Seo
- 年份: 2024; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: ExeGPT: Constraint-Aware Resource Scheduling for LLM Inference

快速理解: This paper presents ExeGPT, a distributed system designed for constraint-aware LLM inference. ExeGPT finds and runs with an optimal execution schedule to maximize inference throughput while satisfying a given latency constraint.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.8.7 IANUS: Integrated Accelerator based on NPU-PIM Unified Memory System

- 作者: Minseok Seo; X. Nguyen; Seokha Hwang; Yongkee Kwon; Guhyun Kim; C. Park; Il-Hyu Kim; Jaehan Park; Jeongbin Kim; Woojae Shin; Jong-Hak Won; Haerang Choi; Kyuyoung Kim; Daehan Kwon; Chunseok Jeong; Sangheon Lee; Yongseok Choi; Wooseok Byun; Seungcheol Baek; Hyuk-Jae Lee; John Kim
- 年份: 2024; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: IANUS: Integrated Accelerator based on NPU-PIM Unified Memory System

快速理解: Accelerating end-to-end inference of transformer-based large language models (LLMs) is a critical component of AI services in datacenters. However, the diverse compute characteristics of LLMs' end-to-end inference present challenges as previously proposed accelerators only address certain operations or stages (e.g., self-attention, generation stage, etc.).

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.8.8 Medusa: Accelerating Serverless LLM Inference with Materialization

- 作者: Shaoxun Zeng; Min-Hui Xie; Shiwei Gao; Youmin Chen; You-You Lu
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: Medusa: Accelerating Serverless LLM Inference with Materialization

快速理解: Serverless is a promising paradigm to provide scalable, cost-efficient, and easy-to-use model inference services. However, the cold start of model inference functions requires loading models to the devices, which incurs high latencies and undermines the benefits of serverless computing.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.8.9 Dilu: Enabling GPU Resourcing-on-Demand for Serverless DL Serving via Introspective Elasticity

- 作者: Cunchi Lv; Xiao Shi; Zhengyu Lei; Jinyue Huang; Wenting Tan; Xiaohui Zheng; Xiaofang Zhao
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: Dilu: Enabling GPU Resourcing-on-Demand for Serverless DL Serving via Introspective Elasticity

快速理解: Serverless computing, with its ease of management, auto-scaling, and cost-effectiveness, is widely adopted by deep learning (DL) applications. DL workloads, especially with large language models, require substantial GPU resources to ensure QoS.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.8.10 Spindle: Efficient Distributed Training of Multi-Task Large Models via Wavefront Scheduling

- 作者: Yujie Wang; Shenhan Zhu; Fangcheng Fu; Xupeng Miao; Jie Zhang; Juan Zhu; Fan Hong; Yong Li; Bin Cui
- 年份: 2024; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: Spindle: Efficient Distributed Training of Multi-Task Large Models via Wavefront Scheduling

快速理解: Recent foundation models are capable of handling multiple tasks and multiple data modalities with the unified base model structure and several specialized model components. However, efficient training of such multi-task (MT) multi-modal (MM) models poses significant system challenges due to the sophisticated model architecture and the heterogeneous workloads of different tasks and modalities.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.8.11 C4CAM: A Compiler for CAM-based In-memory Accelerators

- 作者: H. Farzaneh; J. P. C. de Lima; Meng-Yuan Li; Asif Ali Khan; Xiaobo Sharon Hu; J. Castrillón
- 年份: 2023; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: C4CAM: A Compiler for CAM-based In-memory Accelerators

快速理解: Machine learning and data analytics applications increasingly suffer from the high latency and energy consumption of conventional von Neumann architectures. Recently, several in-memory and near-memory systems have been proposed to overcome this von Neumann bottleneck.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.8.12 OS2G: A High-Performance DPU Offloading Architecture for GPU-based Deep Learning with Object Storage

- 作者: Zhengxu Jin; Yiquan Chen; Ming Liang; Yijing Wang; Guoju Fang; Ao Zhou; Keyao Zhang; Jiexiong Xu; Wenhai Lin; Yiquan Lin; Shushu Zhao; Wenkai Shi; Zhenhua He; Shishun Cai; Wenzhi Chen
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: OS2G: A High-Performance DPU Offloading Architecture for GPU-based Deep Learning with Object Storage

快速理解: Object storage is increasingly attractive for deep learning (DL) applications due to its cost-effectiveness and high scalability. However, it exacerbates CPU burdens in DL clusters due to intensive object storage processing and multiple data movements.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.8.13 FastTTS: Accelerating Test-Time Scaling for Edge LLM Reasoning

- 作者: H. Chen; Zhiwen Mo; Guanxi Lu; Shuang Liang; Lingxiao Ma; Wayne Luk; Hongxiang Fan
- 年份: 2025; 主题: Agent RL; AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: FastTTS: Accelerating Test-Time Scaling for Edge LLM Reasoning

快速理解: Recent advances in reasoning Large Language Models (LLMs) are driving the emergence of agentic AI systems. Edge deployment of LLM agents near end users is increasingly necessary to protect data privacy, enable offline use, and provide responsive interaction with local context.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.8.14 Characterizing Power Management Opportunities for LLMs in the Cloud

- 作者: P. Patel; Esha Choukse; Chaojie Zhang; Íñigo Goiri; Brijesh Warriar; Nithish Mahalingam; Ricardo Bianchini
- 年份: 2024; 主题: AI Infra; Systems; 优先级: 核心

- [来源](#)

论文: Characterizing Power Management Opportunities for LLMs in the Cloud

快速理解: Recent innovation in large language models (LLMs), and their myriad use cases have rapidly driven up the compute demand for datacenter GPUs. Several cloud providers and other enterprises plan to substantially grow their datacenter capacity to support these new workloads.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.8.15 Helix: Serving Large Language Models over Heterogeneous GPUs and Network via Max-Flow

- 作者: Yixuan Mei; Yonghao Zhuang; Xupeng Miao; Juncheng Yang; Zhihao Jia; Rashmi Vinayak
- 年份: 2024; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Helix: Serving Large Language Models over Heterogeneous GPUs and Network via Max-Flow

快速理解: This paper introduces Helix, a distributed system for high-throughput, low-latency large language model (LLM) serving in heterogeneous GPU clusters. The key idea behind Helix is to formulate inference computation of LLMs over heterogeneous GPUs and network connections as a max-flow problem on directed, weighted graphs, whose nodes represent GPU instances and edges capture both GPU and network heterogeneity through their capacities.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.8.16 Centauri: Enabling Efficient Scheduling for Communication-Computation Overlap in Large Model Training via Communication Partitioning

- 作者: Chang Chen; Xiuhong Li; Qianchao Zhu; Jiangfei Duan; Peng Sun; Xingcheng Zhang; Chao Yang
- 年份: 2024; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Centauri: Enabling Efficient Scheduling for Communication-Computation Overlap in Large Model Training via Communication Partitioning

快速理解: Efficiently training large language models (LLMs) necessitates the adoption of hybrid parallel methods, integrating multiple communications collectives within distributed partitioned graphs. Overcoming communication bottlenecks is crucial and is often achieved through communication and computation overlaps.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.8.17 NNSmith: Generating Diverse and Valid Test Cases for Deep Learning Compilers

- 作者: Jiawei Liu; Jinkun Lin; Fabian Ruffy; Cheng Tan; Jinyang Li; Aurojit Panda; Lingming Zhang

- 年份: 2022; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: NNSmith: Generating Diverse and Valid Test Cases for Deep Learning Compilers

快速理解: Deep-learning (DL) compilers such as TVM and TensorRT are increasingly being used to optimize deep neural network (DNN) models to meet performance, resource utilization and other requirements. Bugs in these compilers can result in models whose semantics differ from the original ones, producing incorrect results that corrupt the correctness of downstream applications.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.8.18 PIM Is All You Need: A CXL-Enabled GPU-Free System for Large Language Model Inference

- 作者: Yu-Feng Gu; Alireza Khadem; S. Umesh; Ning Liang; Xavier Servot; O. Mutlu; Ravi Iyer; Reetuparna Das
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: PIM Is All You Need: A CXL-Enabled GPU-Free System for Large Language Model Inference

快速理解: Large Language Model (LLM) inference uses an autoregressive manner to generate one token at a time, which exhibits notably lower operational intensity compared to earlier Machine Learning (ML) models such as encoder-only transformers and Convolutional Neural Networks. At the same time, LLMs possess large parameter sizes and use key-value caches to store context information.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.8.19 Mind mappings: enabling efficient algorithm-accelerator mapping space search

- 作者: Kartik Hegde; Po-An Tsai; Sitao Huang; V. Chandra; A. Parashar; Christopher W. Fletcher
- 年份: 2021; 主题: Agent RL; AI Infra; 优先级: 重要
- [来源](#)

论文: Mind mappings: enabling efficient algorithm-accelerator mapping space search

快速理解: Modern day computing increasingly relies on specialization to satiate growing performance and efficiency requirements. A core challenge in designing such specialized hardware architectures is how to perform mapping space search, i.e., search for an optimal mapping from algorithm to hardware.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.8.20 ElasticFlow: An Elastic Serverless Training Platform for Distributed Deep Learning

- 作者: Diandian Gu; Yihao Zhao; Yinmin Zhong; Yifan Xiong; Zhenhua Han; Peng Cheng; Fan Yang; Gang Huang; Xin Jin; Xuanzhe Liu

- 年份: 2023; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: ElasticFlow: An Elastic Serverless Training Platform for Distributed Deep Learning

快速理解: This paper proposes ElasticFlow, an elastic serverless training platform for distributed deep learning. ElasticFlow provides a serverless interface with two distinct features: (i) users specify only the deep neural network (DNN) model and hyperparameters for a job, but not the number of GPUs; (ii) users specify the deadline for a job, but not the amount of time to occupy GPUs.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.8.21 Forecasting GPU Performance for Deep Learning Training and Inference

- 作者: Seonho Lee; Amar Phanishayee; Divya Mahajan
- 年份: 2024; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Forecasting GPU Performance for Deep Learning Training and Inference

快速理解: Deep learning kernels exhibit a high level of predictable memory accesses and compute patterns, making GPU's architecture well-suited for their execution. Moreover, software and runtime system for GPUs further enable optimizations that aim to better utilize the stream multiprocessors, on-chip bandwidth, multiple levels of cache hierarchy, and off-chip high-bandwidth memory.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.8.22 Proteus: A High-Throughput Inference-Serving System with Accuracy Scaling

- 作者: Sohaib Ahmad; Hui Guan; Brian D. Friedman; Thomas Williams; R. Sitaraman; Thomas Woo
- 年份: 2024; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Proteus: A High-Throughput Inference-Serving System with Accuracy Scaling

快速理解: Existing machine learning inference-serving systems largely rely on hardware scaling by adding more devices or using more powerful accelerators to handle increasing query demands. However, hardware scaling might not be feasible for fixed-size edge clusters or private clouds due to their limited hardware resources.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.8.23 MSCCLang: Microsoft Collective Communication Language

- 作者: M. Cowan; Saeed Maleki; Madan Musuvathi; Olli Saarikivi; Yifan Xiong
- 年份: 2023; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: MSCCLang: Microsoft Collective Communication Language

快速理解: Machine learning models with millions or billions of parameters are increasingly trained and served on large multi-GPU systems. As models grow in size and execute on more GPUs, collective communication becomes a bottleneck.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.8.24 In-Network Aggregation with Transport Transparency for Distributed Training

- 作者: Shuo Liu; Qiaoling Wang; Junyi Zhang; Wenfei Wu; Qinliang Lin; Yao Liu; Meng Xu; Marco Canini; R. Cheung; Jianfei He
- 年份: 2023; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: In-Network Aggregation with Transport Transparency for Distributed Training

快速理解: Recent In-Network Aggregation (INA) solutions offload the all-reduce operation onto network switches to accelerate and scale distributed training (DT). On end hosts, these solutions build custom network stacks to replace the transport layer.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.8.25 FleetIO: Managing Multi-Tenant Cloud Storage with Multi-Agent Reinforcement Learning

- 作者: Jinghan Sun; Benjamin Reidys; Daixuan Li; Jichuan Chang; Marc Snir; Jian Huang
- 年份: 2025; 主题: Agent RL; Systems; 优先级: 重要
- [来源](#)

论文: FleetIO: Managing Multi-Tenant Cloud Storage with Multi-Agent Reinforcement Learning

快速理解: Cloud platforms have been virtualizing storage devices like flash-based solid-state drives (SSDs) to make effective use of storage resources. They enable either software-isolated instance or hardware-isolated instance for facilitating the storage sharing between multi-tenant applications.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.8.26 Hydrice: A Retargetable and Extensible Synthesis-based Compiler for Modern Hardware Architectures

- 作者: Akash Kothari; Abdul Rafae Noor; Muchen Xu; Hassam Uddin; Dhruv Baronia; Stefanos Baziotis; V. Adve; Charith Mendis; S. Sengupta
- 年份: 2024; 主题: Teleoperation; Systems; 优先级: 重要
- [来源](#)

论文: Hydrice: A Retargetable and Extensible Synthesis-based Compiler for Modern Hardware Architectures

快速理解: As modern hardware architectures evolve to support increasingly diverse, complex instruction sets for meeting the performance demands of modern workloads in image processing, deep learning, etc., it has become ever more crucial for compilers to provide robust support for evolution of their internal abstractions and retargetable code generation support to keep pace with emerging instruction sets. We propose Hydrice, a novel approach to compiling for complex, emerging hardware architectures.

关注点: 人类示范、远程操作、共享自治或动作重定向; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.8.27 Graphene: An IR for Optimized Tensor Computations on GPUs

- 作者: Bastian Hagedorn; Bin Fan; Han-Feng Chen; C. Cecka; Michael Garland; Vinod Grover
- 年份: 2023; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Graphene: An IR for Optimized Tensor Computations on GPUs

快速理解: Modern GPUs accelerate computations and data movements of multi-dimensional tensors in hardware. However, expressing optimized tensor computations in software is extremely challenging even for experts.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.8.28 A full-stack search technique for domain optimized deep learning accelerators

- 作者: Dan Zhang; Safeen Huda; Ebrahim M. Songhori; Kartik Prabhu; Quoc V. Le; Anna Goldie; Azalia Mirhoseini
- 年份: 2021; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: A full-stack search technique for domain optimized deep learning accelerators

快速理解: The rapidly-changing deep learning landscape presents a unique opportunity for building inference accelerators optimized for specific datacenter-scale workloads. We propose Full-stack Accelerator Search Technique (FAST), a hardware accelerator search framework that defines a broad optimization environment covering key design decisions within the hardware-software stack, including hardware datapath, software scheduling, and compiler passes such as operation fusion and tensor padding.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.8.29 FSMoE: A Flexible and Scalable Training System for Sparse Mixture-of-Experts Models

- 作者: Xinglin Pan; Wen-Jing Lin; Lin Zhang; S. Shi; Zhenheng Tang; Rui Wang; Bo Li; Xiaowen Chu
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: FSMoE: A Flexible and Scalable Training System for Sparse Mixture-of-Experts Models

快速理解: Recent large language models (LLMs) have tended to leverage sparsity to reduce computations, employing the sparsely activated mixture-of-experts (MoE) technique. MoE introduces four modules, including token routing, token communication, expert computation, and expert parallelism, that impact model quality and training efficiency.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.8.30 GraphPipe: Improving Performance and Scalability of DNN Training with Graph Pipeline Parallelism

- 作者: Byungsoo Jeon; Mengdi Wu; Shiyi Cao; Sunghyun Kim; Sunghyun Park; Neeraj Aggarwal; Colin Unger; Daiyaan Arfeen; Peiyuan Liao; Xupeng Miao; Mohammad Alizadeh; G. R. Ganger; Tianqi Chen; Zhihao Jia
- 年份: 2024; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: GraphPipe: Improving Performance and Scalability of DNN Training with Graph Pipeline Parallelism

快速理解: Deep neural networks (DNNs) continue to grow rapidly in size, making them infeasible to train on a single device (e.g. GPU).

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.8.31 Aqua: Network-Accelerated Memory Offloading for LLMs in Scale-Up GPU Domains

- 作者: Abhishek Vijaya Kumar; Gianni Antichi; Rachee Singh
- 年份: 2024; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Aqua: Network-Accelerated Memory Offloading for LLMs in Scale-Up GPU Domains

快速理解: Inference on large-language models (LLMs) is constrained by GPU memory capacity. A sudden increase in the number of inference requests to a cloud-hosted LLM can deplete GPU memory, leading to contention between multiple prompts for limited resources.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.8.32 Skybox: Open-Source Graphic Rendering on Programmable RISC-V GPUs

- 作者: Blaise Tine; Varun Saxena; Santosh Srivatsan; Joshua R. Simpson; Fadi Alzammar; L. Cooper; Hyesoon Kim
- 年份: 2023; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Skybox: Open-Source Graphic Rendering on Programmable RISC-V GPUs

快速理解: Graphics rendering remains one of the most compute intensive and memory bound applications of GPUs and has been driving their push for performance and energy efficiency since its inception. Early GPU architectures focused only on accelerating graphics rendering and implemented dedicated fixed- function rasterizer hardware to speed-up their rendering pipeline.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.8.33 DREAM: A Dynamic Scheduler for Dynamic Real-time Multi-model ML Workloads

- 作者: Seah Kim; Hyoukjun Kwon; Jinook Song; Jihyuck Jo; Yu-Hsin Chen; Liangzhen Lai; Vikas Chandra
- 年份: 2022; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: DREAM: A Dynamic Scheduler for Dynamic Real-time Multi-model ML Workloads

快速理解: Emerging real-time multi-model ML (RTMM) workloads such as AR/VR and drone control involve dynamic behaviors in various granularity; task, model, and layers within a model. Such dynamic behaviors introduce new challenges to the system software in an ML system since the overall system load is not completely predictable, unlike traditional ML workloads.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.8.34 PCcheck: Persistent Concurrent Checkpointing for ML

- 作者: F. Strati; Michal Friedman; A. Klimovic
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: PCcheck: Persistent Concurrent Checkpointing for ML

快速理解: Training large-scale machine learning (ML) models is expensive and time-intensive, consuming many hardware accelerators for days or weeks. As the scale of hardware deployments and training time continue to grow, the probability of failures also increases.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.8.35 HIR: An MLIR-based Intermediate Representation for Hardware Accelerator Description

- 作者: Kingshuk Majumder; Uday Bondhugula
- 年份: 2021; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: HIR: An MLIR-based Intermediate Representation for Hardware Accelerator Description

快速理解: The emergence of machine learning, image and audio processing on edge devices has motivated research towards power-efficient custom hardware accelerators. Though FPGAs are an ideal target for custom accelerators, the difficulty of hardware design and the lack of vendor agnostic,

standardized hardware compilation infrastructure has hindered their adoption.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.8.36 ZipServ: Fast and Memory-Efficient LLM Inference with Hardware-Aware Lossless Compression

- 作者: Ruibo Fan; Xiangrui Yu; Xinglin Pan; Zeyu Li; Weile Luo; Qiang Wang; Wei Wang; Xiao-Xia Chu
- 年份: 2026; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: ZipServ: Fast and Memory-Efficient LLM Inference with Hardware-Aware Lossless Compression

快速理解: Lossless model compression holds tremendous promise for alleviating the memory and bandwidth bottlenecks in bit-exact Large Language Model (LLM) serving. However, existing approaches often result in substantial inference slowdowns due to fundamental design mismatches with GPU architectures: at the kernel level, variable-length bitstreams produced by traditional entropy codecs break SIMT parallelism; at the system level, decoupled pipelines lead to redundant memory traffic.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.8.37 ACES: Accelerating Sparse Matrix Multiplication with Adaptive Execution Flow and Concurrency-Aware Cache Optimizations

- 作者: Xiaoyang Lu; Boyu Long; Xiaoming Chen; Yinhe Han; Xian-He Sun
- 年份: 2024; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: ACES: Accelerating Sparse Matrix Multiplication with Adaptive Execution Flow and Concurrency-Aware Cache Optimizations

快速理解: Sparse matrix-matrix multiplication (SpMM) is a critical computational kernel in numerous scientific and machine learning applications. SpMM involves massive irregular memory accesses and poses great challenges to conventional cache-based computer architectures.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.8.38 CIM-MLC: A Multi-level Compilation Stack for Computing-In-Memory Accelerators

- 作者: Songyun Qu; Shixin Zhao; Bing Li; Yintao He; Xuyi Cai; Lei Zhang; Ying Wang
- 年份: 2024; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: CIM-MLC: A Multi-level Compilation Stack for Computing-In-Memory Accelerators

快速理解: In recent years, various computing-in-memory (CIM) processors have been presented, showing superior performance over traditional architectures. To unleash the potential of various CIM architectures, such as device precision, crossbar size, and crossbar number, it is necessary to develop

compilation tools that are fully aware of the CIM architectural details and implementation diversity.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.8.39 RECom: A Compiler Approach to Accelerating Recommendation Model Inference with Massive Embedding Columns

- 作者：Zaifeng Pan; Zhen Zheng; Feng Zhang; Ruofan Wu; Hao Liang; Dalin Wang; Xiafei Qiu; Jun-Jie Bai; Wei Lin; Xiaoyong Du
- 年份：2023; 主题：AI Infra; Systems; 优先级：重要
- [来源](#)

论文：RECom: A Compiler Approach to Accelerating Recommendation Model Inference with Massive Embedding Columns

快速理解：Embedding columns are important for deep recommendation models to achieve high accuracy, but they can be very time-consuming during inference. Machine learning (ML) compilers are used broadly in real businesses to optimize ML models automatically.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.8.40 QoServe: Breaking the Silos of LLM Inference Serving

- 作者：Kanishk Goel; Jayashree Mohan; Nipun Kwatra; R. S. Anupindi; R. Ramjee
- 年份：2026; 主题：AI Infra; Systems; 优先级：重要
- [来源](#)

论文：QoServe: Breaking the Silos of LLM Inference Serving

快速理解：The widespread adoption of Large Language Models (LLMs) has enabled diverse applications with very different latency requirements. Existing LLM serving frameworks rely on siloed infrastructure with coarse-grained workload segregation –interactive and batch –leading to inefficient resource utilization and limited support for fine-grained Quality-of-Service (QoS) differentiation.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.8.41 BiSon-e: a lightweight and high-performance accelerator for narrow integer linear algebra computing on the edge

- 作者：Enrico Reggiani; Cristóbal Ramírez Lazo; R. F. Bague; A. Cristal; M. Olivieri; O. Unsal
- 年份：2022; 主题：AI Infra; Systems; 优先级：重要
- [来源](#)

论文：BiSon-e: a lightweight and high-performance accelerator for narrow integer linear algebra computing on the edge

快速理解：Linear algebra computational kernels based on byte and sub-byte integer data formats are at the base of many classes of applications, ranging from Deep Learning to Pattern Matching. Porting the computation of these applications from cloud to edge and mobile devices would enable significant improvements in terms of security, safety, and energy efficiency.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.8.42 LAER-MoE: Load-Adaptive Expert Re-layout for Efficient Mixture-of-Experts Training

- 作者：Xinyi Liu; Yujie Wang; Fangcheng Fu; Xuefeng Xiao; Huixia Li; Jiashi Li; Bin Cui
- 年份：2026；主题：AI Infra; Systems；优先级：重要
- [来源](#)

论文：LAER-MoE: Load-Adaptive Expert Re-layout for Efficient Mixture-of-Experts Training

快速理解：Expert parallelism is vital for effectively training Mixture-of-Experts (MoE) models, enabling different devices to host distinct experts, with each device processing different input data. However, during expert parallel training, dynamic routing results in significant load imbalance among experts: a handful of overloaded experts hinder overall iteration, emerging as a training bottleneck.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.8.43 Neuralink: Fast on-Device LLM Inference with Neuron Co-Activation Linking

- 作者：Tuowei Wang; Ruwen Fan; Minxing Huang; Zixu Hao; Kun Li; Ting Cao; You-You Lu; Yao-Xue Zhang; Ju Ren
- 年份：2024；主题：AI Infra; Systems；优先级：重要
- [来源](#)

论文：Neuralink: Fast on-Device LLM Inference with Neuron Co-Activation Linking

快速理解：Large Language Models (LLMs) have achieved remarkable success across various domains, yet deploying them on mobile devices remains an arduous challenge due to their extensive computational and memory demands. While lightweight LLMs have been developed to fit mobile environments, they suffer from degraded model accuracy.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.8.44 A Cost-Effective Near-Storage Processing Solution for Offline Inference of Long-Context LLMs

- 作者：Hongsun Jang; Jaeyong Song; Chang-Min Shin; Si Ung Noh; Jaewon Jung; Jisung Park; Jinho Lee
- 年份：2025；主题：AI Infra; Systems；优先级：重要
- [来源](#)

论文：A Cost-Effective Near-Storage Processing Solution for Offline Inference of Long-Context LLMs

快速理解：The computational and memory demands of large language models for generative inference present significant challenges for practical deployment. One promising solution targeting offline inference is offloading-based batched inference, which extends the GPU's memory hierarchy with host memory and storage.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.8.45 Shift Parallelism: Low-Latency, High-Throughput LLM Inference for Dynamic Workloads

- 作者：Mert Hidayetoğlu; Aurick Qiao; Michael Wyatt; Jeff Rasley; Yuxiong He; Samyam Rajbhandari
- 年份：2025；主题：AI Infra; Systems；优先级：重要
- [来源](#)

论文：Shift Parallelism: Low-Latency, High-Throughput LLM Inference for Dynamic Workloads

快速理解：Efficient parallelism is necessary for achieving low-latency, high-throughput inference with large language models (LLMs). Tensor parallelism (TP) is the state-of-the-art method for reducing LLM response latency, however GPU communications reduces combined token throughput.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.8.46 Vela: A Virtualized LLM Training System with GPU Direct RoCE

- 作者：Apoorve Mohan; Robert Walkup; B. Karaçali; Ming-hung Chen; Apo Kayi; L. Schour; Shweta Salaria; Sophia Wen; I-hsin Chung; Abdul Alim; C. Evangelinos; Lixiang Luo; Marc Dombrowa; Laurent Schares; Ali Sydney; P. Maniotis; Sandhya Koteshwara; Brent Tang; Joel Belog; Rei Odaira; Vasily Tarasov; Eran Gampel; Drew Thorstensen; Talia Gershon; S. Seelam
- 年份：2025；主题：AI Infra; Systems；优先级：重要
- [来源](#)

论文：Vela: A Virtualized LLM Training System with GPU Direct RoCE

快速理解：Vela is a cloud-native system designed for LLM training workloads built using off-the-shelf hardware, Linux KVM-based virtualization, and a virtualized RDMA over Converged Ethernet (RoCE) network. Vela virtual machines (VMs) support peer-to-peer DMA between the GPUs and SRIOV-based network interface.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.8.47 Gigaflow: Pipeline-Aware Sub-Traversal Caching for Modern SmartNICs

- 作者：Annus Zulfiqar; Ali Imran; Venkat Kunaparaju; Ben Pfaff; Gianni Antichi; Muhammad Shahbaz
- 年份：2025；主题：AI Infra; Systems；优先级：重要
- [来源](#)

论文：Gigaflow: Pipeline-Aware Sub-Traversal Caching for Modern SmartNICs

快速理解：The success of modern public/edge clouds hinges heavily on the performance of their end-host network stacks if they are to support the emerging and diverse tenants' workloads (e.g., distributed training in the cloud to fast inference at the edge). Virtual Switches (vSwitches) are vital components of this stack, providing a unified interface to enforce high-level policies on incoming

packets and route them to physical interfaces, containers, or virtual machines.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.8.48 Hector: An Efficient Programming and Compilation Framework for Implementing Relational Graph Neural Networks in GPU Architectures

- 作者：Kun Wu; Mert Hidayetoğlu; Xiang Song; Sitao Huang; Da Zheng; Israt Nisa; W. Hwu
- 年份：2023；主题：AI Infra; Systems；优先级：重要
- [来源](#)

论文：Hector: An Efficient Programming and Compilation Framework for Implementing Relational Graph Neural Networks in GPU Architectures

快速理解：Relational graph neural networks (RGNNs) are graph neural networks with dedicated structures for modeling the different types of nodes and edges in heterogeneous graphs. While RGNNs have been increasingly adopted in many real-world applications due to their versatility and accuracy, they pose performance and system design challenges: inherent memory-intensive computation patterns, the gap between the programming interface and kernel APIs, and heavy programming effort required to optimize kernels caused by their coupling with data layout and heterogeneity.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.8.49 Mosaic: Exploiting Instruction-Level Parallelism on Deep Learning Accelerators with iTex Tessellation

- 作者：Jianxing Xu; Yuanbo Wen; Zikang Liu; Ruibai Xu; Tingfeng Ruan; Jun Bi; Rui Zhang; Di Huang; Xinkai Song; Yifan Hao; Xin-Gui Hu; Zidong Du; Chongqing Zhao; Jiang Jie; Qi Guo
- 年份：2025；主题：AI Infra; Systems；优先级：重要
- [来源](#)

论文：Mosaic: Exploiting Instruction-Level Parallelism on Deep Learning Accelerators with iTex Tessellation

快速理解：Deep learning has achieved great success in numerous application areas at the cost of high computational complexity. To meet the ever-increasing computational demand, commodity hardware platforms (e.g., CPUs and GPUs) offer abundant computing resources including scalar, vector, and tensor units for deep learning that could execute in parallel.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.8.50 DFVG: A Heterogeneous Architecture for Speculative Decoding with Draft-on-FPGA and Verify-on-GPU

- 作者：Shaoqiang Lu; Yangbo Wei; Junhong Qian; Dongge Qin; Shiji Gao; Yizhi Ding; Qifan Wang; Chen Wu; Xiao Shi; Lei He
- 年份：2026；主题：AI Infra; Systems；优先级：重要
- [来源](#)

论文: DFVG: A Heterogeneous Architecture for Speculative Decoding with Draft-on-FPGA and Verify-on-GPU

快速理解: Speculative decoding is a promising paradigm that accelerates LLM inference by generating drafts and performing verification. However, such systems still face three major challenges: (1) The imbalance in resource requirements between draft and verification models result in low utilization and energy inefficiency when deployed together. (2) Fixed-pattern token trees produce many candidates but few valid paths, resulting in redundant drafts due to the lack of full leverage of the inherent confidence in dynamic generation. (3) Asynchronous execution with frequent alternation between the two stages suffers from idle waiting and rollback overhead.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.8.51 REPA: Reconfigurable PIM for the Joint Acceleration of KV Cache Offloading and Processing

- 作者: Yang Hong; Jun-Long Yang; Bo Peng; Jian-Guo Yao
- 年份: 2026; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: REPA: Reconfigurable PIM for the Joint Acceleration of KV Cache Offloading and Processing

快速理解: The use of KV cache in LLM inference leads to large memory footprint and sub-optimal decoding performance. Prior studies typically address one of these two limitations by either offloading or stage-split inference.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.8.52 SwiftSpec: Disaggregated Speculative Decoding and Fused Kernels for Low-Latency LLM Inference

- 作者: Ziyi Zhang; Ziheng Jiang; Chengquan Jiang; Menghan Yu; Size Zheng; Haibin Lin; Xin Liu; Henry Hoffmann
- 年份: 2026; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: SwiftSpec: Disaggregated Speculative Decoding and Fused Kernels for Low-Latency LLM Inference

快速理解: Low-latency, single-request decoding of large language models is critical for interactive systems with tight SLA demands. Prior work reduces latency through speculative decoding (combining a small draft model with a larger target model), but the draft model remains on the critical path, and communication overhead limits scaling across GPUs due to the small batch size associated with single-request decoding.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.8.53 NDPipe: Exploiting Near-data Processing for Scalable Inference and Continuous Training in Photo Storage

- 作者: Jungwoo Kim; Seonggyun Oh; Jaeha Kung; Yeseong Kim; Sungjin Lee
- 年份: 2024; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: NDPipe: Exploiting Near-data Processing for Scalable Inference and Continuous Training in Photo Storage

快速理解: This paper proposes a novel photo storage system called NDPipe, which accelerates the performance of training and inference for image data by leveraging near-data processing in photo storage servers. NDPipe distributes storage servers with inexpensive commodity GPUs in a data center and uses their collective intelligence to perform inference and training near image data.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.8.54 BitRed: Taming Non-Uniform Bit-Level Sparsity with a Programmable RISC-V ISA for DNN Acceleration

- 作者: Yan-Huan Liu; Wenming Li; Kunming Zhang; Yu-Qun Liu; S. Wen; Lexin Wang; Tian-Yu Liu; Haibin Wu; Zhihua Fan; Xiaochun Ye; Dongrui Fan; Xuejun An
- 年份: 2026; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: BitRed: Taming Non-Uniform Bit-Level Sparsity with a Programmable RISC-V ISA for DNN Acceleration

快速理解: The non-uniform and dynamic nature of Bit-Level Sparsity (BLS) poses a critical load-imbalance challenge for parallel hardware accelerators. While the Bit-Interleaving paradigm, represented by state-of-the-art accelerators like Bitlet, shows promise, it is fundamentally constrained by a rigid datapath and severe inter-channel load imbalance.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.8.55 RedFuser: An Automatic Operator Fusion Framework for Cascaded Reductions on AI Accelerators

- 作者: Xin-Sheng Tang; Yang-Cheng Li; Nan Wang; Zhiyi Shu; Xingyu Ling; Junna Xing; Peng Zhou; Qiang Liu
- 年份: 2026; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: RedFuser: An Automatic Operator Fusion Framework for Cascaded Reductions on AI Accelerators

快速理解: Operator fusion, as a key performance optimization technique in the deployment of AI models, significantly improves execution efficiency and has been widely adopted in modern AI compilers. However, for cascaded reduction operations involving multiple loops with inter-loop data dependencies, such as the safe softmax followed by GEMM within attention mechanisms, existing compilers lack effective automated fusion and kernel generation capabilities.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.9 SYSTEMS · EuroSys

本次会议纳入 30 篇。自动主题统计为：AI Infra (29)、Systems (28)、Supporting Perception (2)、Agent RL (2)。这些统计用于组织阅读顺序，不能单独证明论文结论或实验优越性。

优先阅读候选：- Efficient Multimodal Serving via Module Multiplexing (优先级分数 76.5) - ScheMoE: An Extensible Mixture-of-Experts Distributed Training System with Tasks Scheduling (优先级分数 72.883) - CacheBlend: Fast Large Language Model Serving for RAG with Cached Knowledge Fusion (优先级分数 72.676) - Fleche: an efficient GPU embedding cache for personalized recommendations (优先级分数 71.709) - MEPipe: Democratizing LLM Training with Memory-Efficient Slice-Level Pipeline Scheduling on Cost-Effective Accelerators (优先级分数 70.835)

1.3.9.1 Efficient Multimodal Serving via Module Multiplexing

- 作者：Zicong Hong; Yuyan Chen; Hao-Yue Zhang; Peng Li; Wuhui Chen; Song Guo; Xiao-Wei Shen
- 年份：2026; 主题：AI Infra; Supporting Perception; Systems; 优先级：核心
- [来源](#)

论文：Efficient Multimodal Serving via Module Multiplexing

快速理解：Multimodal learning enables models to process and reason over diverse information sources, unlocking human-like perceptual and cognitive capabilities. As such models gain adoption, efficiently serving them on GPUs has become increasingly important.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；3D、视频、多模态视觉或具身感知支撑能力；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.9.2 ScheMoE: An Extensible Mixture-of-Experts Distributed Training System with Tasks Scheduling

- 作者：S. Shi; Xinglin Pan; Qiang Wang; Chengjian Liu; Xiaozhe Ren; Zhongzhe Hu; Yu Yang; Bo Li; Xiaowen Chu
- 年份：2024; 主题：AI Infra; Systems; 优先级：核心
- [来源](#)

论文：ScheMoE: An Extensible Mixture-of-Experts Distributed Training System with Tasks Scheduling

快速理解：In recent years, large-scale models can be easily scaled to trillions of parameters with sparsely activated mixture-of-experts (MoE), which significantly improves the model quality while only requiring a sub-linear increase in computational costs. However, MoE layers require the input data to be dynamically routed to a particular GPU for computing during distributed training.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.9.3 CacheBlend: Fast Large Language Model Serving for RAG with Cached Knowledge Fusion

- 作者: Jiayi Yao; Hanchen Li; Yuhao Liu; Siddhant Ray; Yihua Cheng; Qizheng Zhang; Kuntai Du; Shan Lu; Junchen Jiang
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: CacheBlend: Fast Large Language Model Serving for RAG with Cached Knowledge Fusion

快速理解: Large language models (LLMs) often incorporate multiple text chunks in their inputs to provide the necessary contexts. To speed up the prefill of the long LLM inputs, one can pre-compute the KV cache of a text and re-use the KV cache when the context is reused as the prefix of another LLM input.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.9.4 Fleche: an efficient GPU embedding cache for personalized recommendations

- 作者: Min-Hui Xie; Youyou Lu; Jiazhen Lin; Qing Wang; Jian Gao; Kai Ren; Jiwu Shu
- 年份: 2022; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: Fleche: an efficient GPU embedding cache for personalized recommendations

快速理解: Deep learning based models have dominated current production recommendation systems. However, the gap between CPU-side DRAM data accessing and GPU processing still impedes their inference performance.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.9.5 MEPipe: Democratizing LLM Training with Memory-Efficient Slice-Level Pipeline Scheduling on Cost-Effective Accelerators

- 作者: Zhenbo Sun; Shengqi Chen; Yuanwei Wang; Jian Sha; Guan-Yu Feng; Wenguang Chen
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: MEPipe: Democratizing LLM Training with Memory-Efficient Slice-Level Pipeline Scheduling on Cost-Effective Accelerators

快速理解: The training of large language models (LLMs) typically needs costly GPUs, such as NVIDIA A100 or H100. They possess substantial high-bandwidth on-chip memory and rapid interconnects like NVLinks.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.9.6 Untangling GPU Power Consumption: Job-Level Inference in Cloud Shared Settings

- 作者: P. Jacquet; Maxime Agusti; Eddy Caron; Camille Coti; M. D. de Assunção; Laurent Lefèvre; Anne-Cécile Orgerie
- 年份: 2026; 主题: AI Infra; Systems; 优先级: 核心

- [来源](#)

论文: Untangling GPU Power Consumption: Job-Level Inference in Cloud Shared Settings

快速理解: As the demand for AI-driven workloads increases, the energy consumption of Graphics Processing Units (GPUs) devices has come under intense scrutiny, particularly in hyperscale data centers where large numbers of accelerators are centralized and leased to diverse clients. In the context of cloud hyperscalers, GPUs power monitoring presents several challenges that vary depending on the product offered.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.9.7 Efficient Data Passing for Serverless Inference Workflows: A GPU-Centric Approach

- 作者: Hao Wu; Yaochen Liu; Min-Chen Yu; Qizhen Weng; Jun Deng; Yue Yu; Hao Fan; Song Wu; Wei Wang; Hai Jin
- 年份: 2026; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: Efficient Data Passing for Serverless Inference Workflows: A GPU-Centric Approach

快速理解: Serverless computing offers a compelling paradigm for deploying machine learning inference workflows composed of heterogeneous CPU and GPU functions. However, existing data-passing solutions in serverless systems primarily rely on host memory for data exchange (host-centric), leading to substantial data movement and salient I/O overhead.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.9.8 Multipath Collective Communication Beyond Scale-up Networks in GPU Clouds

- 作者: Yuchen Xu; Jianglong Nie; Baojia Li; Mingzhuo Chen; Hao Lu; Guanyu Qu; Zhenchuan Liu; Shuangshuang Yin; Xiaojie Huang; Chunzhi He; Yinben Xia; Quan Wen; Xiang Li; Zekun He; Yachen Wang; Xianneng Zou; Congcong Miao; Wenfei Wu
- 年份: 2026; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: Multipath Collective Communication Beyond Scale-up Networks in GPU Clouds

快速理解: Hardware vendors introduce scale-up networks interconnecting accelerators to speed up communication in distributed training. We argue that in the context of GPU clouds, the scale-out network can be a good complement to the collective communication that is originally performed on scale-up networks.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.9.9 RoPeerTo: A Datacenter-Scale Architecture for Peer-To-Peer DMA between GPUs and FPGAs

- 作者: Marco Venere; Giuseppe Sorrentino; Benjamin Ramhorst; M. J. Heer; Lucian Petrica; Dario Korolija; Marco D. Santambrogio; Davide Conficconi; Gustavo Alonso; Kenneth O' Brien
- 年份: 2026; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: RoPeerTo: A Datacenter-Scale Architecture for Peer-To-Peer DMA between GPUs and FPGAs

快速理解: Modern datacenters integrate heterogeneous accelerators, such as GPUs and FPGAs, to speed up different stages of compute-intensive pipelines. GPUs are best suited for massively parallel workloads (e.g., deep learning), while FPGAs excel at task-level parallelism, stream-oriented processing, and in-network acceleration.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.9.10 AdaGen: Workload-Adaptive Cluster Scheduler for Latency-Optimal LLM Inference Serving

- 作者: Sudipta Saha Shubha; Ayush Goel; Diman Zad Tootaghaj; Khaled Diab; Hardik Soni; K. Ramakrishnan; Puneet Sharma; Haiying Shen
- 年份: 2026; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: AdaGen: Workload-Adaptive Cluster Scheduler for Latency-Optimal LLM Inference Serving

快速理解: The inference workloads of Large Language Models (LLMs) pose significant latency and cost challenges due to increasing model sizes and demand for real-time responses. Existing cluster schedulers for multi-instance LLM serving primarily focus on load balancing to optimize memory usage, which is insufficient for workloads with diverse request characteristics.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.9.11 Automated End-to-End Model Serving with Cooperative Compilation and Scheduling

- 作者: Yi-Kang Zhang; Junlong Chen; Wei Wang; Jia Liu; Nan Hu; Hai-Peng Dai
- 年份: 2026; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: Automated End-to-End Model Serving with Cooperative Compilation and Scheduling

快速理解: Model serving systems are critical for deep learning inference, managing GPU infrastructure to deliver end-to-end services. Current frameworks typically treat operators as basic compilation and scheduling units, which often fail to maximize GPU utilization due to hardware-unfriendly kernels and coarse-grained scheduling.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.9.12 Bridging the GPU Utilization Gap: Predictive Multi-Dimensional Resource Scheduling for AI Workloads

- 作者: Yilei Lu; Dongbiao He; Teng Ma; Zhe Liu; Letian Ruan; Jinlei Jiang; Yong-Wei Wu
- 年份: 2026; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: Bridging the GPU Utilization Gap: Predictive Multi-Dimensional Resource Scheduling for AI Workloads

快速理解: Modern AI data centers face a critical paradox: while machine learning workloads dominate infrastructure demands, actual GPU utilization remains consistently low. Existing schedulers fail to coordinate heterogeneous resources effectively, lack predictive capabilities for dynamic workloads, and cannot balance isolation requirements with sharing optimization in multi-tenant clusters.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.9.13 PointShuffler: Accelerating Point Cloud Neural Networks on General-Purpose GPUs

- 作者: Yangfan Li; Zhen Jin; Yueyi Tian; Mengquan Li; Fengxiao Tang; Ming Zhao; Cen Chen
- 年份: 2026; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: PointShuffler: Accelerating Point Cloud Neural Networks on General-Purpose GPUs

快速理解: Point Cloud Neural Networks (PCNNs) have emerged as a vital tool for latency-sensitive 3D perception applications, such as autonomous driving and AR/VR. However, their inherent computational redundancy—arising from excessive global sampling/search operations and repeated feature updates/aggregations caused by shared neighbors—severely constrains execution efficiency.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.9.14 GNNLab: a factored system for sample-based GNN training over GPUs

- 作者: Jianbang Yang; Dahai Tang; Xiaoni Song; Lei Wang; Qiang Yin; Rong Chen; Wenyan Yu; Jingren Zhou
- 年份: 2022; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: GNNLab: a factored system for sample-based GNN training over GPUs

快速理解: We propose GNNLab, a sample-based GNN training system in a single machine multi-GPU setup. GNNLab adopts a factored design for multiple GPUs, where each GPU is dedicated to the task of graph sampling or model training.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.9.15 Lyra: Elastic Scheduling for Deep Learning Clusters

- 作者: Jiamin Li; Hong-Yu Xu; Yibo Zhu; Zherui Liu; Chuanxiong Guo; Cong Wang
- 年份: 2022; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Lyra: Elastic Scheduling for Deep Learning Clusters

快速理解: Organizations often build separate training and inference clusters for deep learning, and use separate schedulers to manage them. This leads to problems for both: inference clusters have low utilization when the traffic load is low; training jobs often experience long queuing due to a lack of resources.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.9.16 Egeria: Efficient DNN Training with Knowledge-Guided Layer Freezing

- 作者: Yiding Wang; Decang Sun; Kai Chen; Fan Lai; Mosharaf Chowdhury
- 年份: 2022; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Egeria: Efficient DNN Training with Knowledge-Guided Layer Freezing

快速理解: Training deep neural networks (DNNs) is time-consuming. While most existing solutions try to overlap/schedule computation and communication for efficient training, this paper goes one step further by skipping computing and communication through DNN layer freezing.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.9.17 Fast State Restoration in LLM Serving with HCache

- 作者: Shiwei Gao; Youmin Chen; Ji-Wu Shu
- 年份: 2024; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Fast State Restoration in LLM Serving with HCache

快速理解: The growing complexity of LLM usage today, e.g., multi-round conversation and retrieval-augmented generation (RAG), makes contextual states (i.e., KV cache) reusable across user requests. Given the capacity constraints of GPU memory, only a limited number of contexts can be cached on GPU for reusing.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.9.18 SpInfer: Leveraging Low-Level Sparsity for Efficient Large Language Model Inference on GPUs

- 作者: Ruibo Fan; Xiangrui Yu; Peijie Dong; Zeyu Li; G. Gong; Qiang Wang; Wei Wang; Xiao-Xia Chu
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: SpInfer: Leveraging Low-Level Sparsity for Efficient Large Language Model Inference on GPUs

快速理解: Large Language Models (LLMs) have demonstrated remarkable capabilities, but their immense scale poses significant challenges in terms of both memory and computational costs. While

unstructured pruning offers promising solutions by introducing sparsity to reduce resource requirements, realizing its benefits in LLM inference remains elusive.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.9.19 Towards VM Rescheduling Optimization Through Deep Reinforcement Learning

- 作者: Xianzhong Ding; Yunkai Zhang; Binbin Chen; Donghao Ying; Tieying Zhang; Jianjun Chen; Lei Zhang; Alberto E. Cerpa; Wan Du
- 年份: 2025; 主题: Agent RL; Systems; 优先级: 重要
- [来源](#)

论文: Towards VM Rescheduling Optimization Through Deep Reinforcement Learning

快速理解: Modern industry-scale data centers need to manage a large number of virtual machines (VMs). Due to the continual creation and release of VMs, many small resource fragments are scattered across physical machines (PMs).

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.9.20 SiloD: A Co-design of Caching and Scheduling for Deep Learning Clusters

- 作者: Hanyu Zhao; Zhenhua Han; Zhi Yang; Quanlu Zhang; Mingxia Li; Fan Yang; Qianxi Zhang; Binyang Li; Yuqing Yang; Lili Qiu; Lintao Zhang; Lidong Zhou
- 年份: 2023; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: SiloD: A Co-design of Caching and Scheduling for Deep Learning Clusters

快速理解: Deep learning training on cloud platforms usually follows the tradition of the separation of storage and computing. The training executes on a compute cluster equipped with GPUs/TPUs while reading data from a separate cluster hosting the storage service.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.9.21 Multiplexing Dynamic Deep Learning Workloads with SLO-awareness in GPU Clusters

- 作者: Wenyan Chen; Chengzhi Lu; Huan-Le Xu; Ke Ye; Cheng-zhong Xu
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Multiplexing Dynamic Deep Learning Workloads with SLO-awareness in GPU Clusters

快速理解: Deep learning (DL) inference services are widely recognized as crucial workloads in large-scale cloud clusters. However, due to the stringent latency requirements, cloud providers often over-provision GPU resources, resulting in underutilization of the available GPU potential.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.9.22 Out-of-order backprop: an effective scheduling technique for deep learning

- 作者: Hyungjun Oh; Junyeol Lee; Hyeongju Kim; Jiwon Seo
- 年份: 2022; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Out-of-order backprop: an effective scheduling technique for deep learning

快速理解: Neural network training requires a large amount of computation and thus GPUs are often used for the acceleration. While they improve the performance, GPUs are underutilized during the training.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.9.23 Mist: Efficient Distributed Training of Large Language Models via Memory--Parallelism Co-Optimization

- 作者: Zhanda Zhu; Christina Giannoula; Muralidhar Andoorveedu; Qidong Su; K. Mangalam; Bojian Zheng; Gennady Pekhimenko
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Mist: Efficient Distributed Training of Large Language Models via Memory-Parallelism Co-Optimization

快速理解: Various parallelism, such as data, tensor, and pipeline parallelism, along with memory optimizations like activation checkpointing, redundancy elimination, and offloading, have been proposed to accelerate distributed training for Large Language Models. To find the best combination of these techniques, automatic distributed training systems are proposed.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.9.24 SkyWalker: A Locality-Aware Cross-Region Load Balancer for LLM Inference

- 作者: Tian Xia; Zi-Ming Mao; Jamison Kerney; Ethan J. Jackson; Zhifei Li; Jiarong Xing; S. Shenker; Ion Stoica
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: SkyWalker: A Locality-Aware Cross-Region Load Balancer for LLM Inference

快速理解: Serving Large Language Models (LLMs) efficiently in multi-region setups remains a challenge. Due to cost and GPU availability concerns, providers typically deploy LLMs in multiple regions using instance with long-term commitments, like reserved instances or on-premise clusters, which are often underutilized due to their region-local traffic handling and diurnal traffic variance.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.9.25 Blox: A Modular Toolkit for Deep Learning Schedulers

- 作者: Saurabh Agarwal; Amar Phanishayee; Shivaram Venkataraman
- 年份: 2023; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Blox: A Modular Toolkit for Deep Learning Schedulers

快速理解: Deep Learning (DL) workloads have rapidly increased in popularity in enterprise clusters and several new cluster schedulers have been proposed in recent years to support these workloads. With rapidly evolving DL workloads, it is challenging to quickly prototype and compare scheduling policies across workloads.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.9.26 Comprehensive Deadlock Prevention for GPU Collective Communication

- 作者: Lichen Pan; Juncheng Liu; Yong-Quan Fu; Jinhui Yuan; Rongkai Zhang; Pengze Li; Zhen Xiao
- 年份: 2023; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Comprehensive Deadlock Prevention for GPU Collective Communication

快速理解: Distributed deep neural network training necessitates efficient GPU collective communications, which are inherently susceptible to deadlocks. GPU collective deadlocks arise easily in distributed deep learning applications when multiple collectives circularly wait for each other.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.9.27 Flex: Fast, Accurate DNN Inference on Low-Cost Edges Using Heterogeneous Accelerator Execution

- 作者: Tanmoy Sen; Hai-Ying Shen; Anand Iyer
- 年份: 2025; 主题: Agent RL; AI Infra; 优先级: 重要
- [来源](#)

论文: Flex: Fast, Accurate DNN Inference on Low-Cost Edges Using Heterogeneous Accelerator Execution

快速理解: Significant breakthroughs in machine learning (ML) and the advantages of on-device processing have led to edge devices increasingly incorporating accelerators like GPUs, NPUs, and DSPs. However, these accelerators consume energy, prompting users to limit their floating-point precision.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.9.28 HetAuto: Cross-Cluster Auto-Parallelism for Heterogeneous Distributed Training

- 作者: Guicheng Qi; Junwei Su; Liqiang Yang; Tao Li; Tingwen Xie; Yerui Sun; Yuchen Xie; Chuan Wu
- 年份: 2026; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: HetAuto: Cross-Cluster Auto-Parallelism for Heterogeneous Distributed Training

快速理解: As large neural network models (e.g., LLMs) grow in scale, single-cluster resources become insufficient, making cross-cluster distributed training essential. Cross-cluster training is challenging: hardware heterogeneity complicates load balancing and parallelization strategy and introduces hardware compatibility issues in implementation; cross-cluster communication bottlenecks severely impact training throughput.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.9.29 MegaScale-Omni: A Hyper-Scale, Workload-Resilient System for MultiModal LLM Training in Production

- 作者: Chunyu Xue; Yangrui Chen; Jianyu Jiang; Ningxin Zheng; Junda Feng; Jingji Chen; Shixiong Zhao; Shen Yan; Yi Lin; Lei Shi; Zhanbo Wang; Lishu Luo; Faming Wu; Haibin Lin; Yanghua Peng; Xin Liu; Quan Chen
- 年份: 2026; 主题: AI Infra; Supporting Perception; 优先级: 重要
- [来源](#)

论文: MegaScale-Omni: A Hyper-Scale, Workload-Resilient System for MultiModal LLM Training in Production

快速理解: As the foundational component of versatile AI applications, training an multimodal large language model (MLLM) relies on multimodal datasets with dynamic modality mixture proportions and sample length distributions. However, existing MLLM systems remain inefficient under dynamic workloads, due to statically coupled decisions of resource allocation and model parallelization between encoders and the LLM backbone.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.9.30 FlashPS: Efficient Generative Image Editing with Mask-aware Caching and Scheduling

- 作者: Xiaoxiao Jiang; Suyi Li; Ling-Yun Yang; Tianyu Feng; Zhipeng Di; Wei-Yi Lu; Guoxuan Zhu; Xiu Lin; Kan Liu; Yinghao Yu; Tao Lan; Guodong Yang; Lin Qu; Liping Zhang; Wei Wang
- 年份: 2026; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: FlashPS: Efficient Generative Image Editing with Mask-aware Caching and Scheduling

快速理解: Generative image editing using diffusion models has become a prevalent application in today's AI cloud services. In production environments, image editing typically involves a mask that specifies the regions of an image template to be edited.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.10 SYSTEMS · FAST

本次会议纳入 **13** 篇。自动主题统计为：Systems (9)、AI Infra (5)。这些统计用于组织阅读顺序，不能单独证明论文结论或实验优越性。

优先阅读候选：- Accelerating Model Loading in LLM Inference by Programmable Page Cache (优先级分数 67.5) - Hardware/Software Co-Programmable Framework for Computational SSDs to Accelerate Deep Learning Service on Large-Scale Graphs (优先级分数 53.544) - Preparation Meets Opportunity: Enhancing Data Preprocessing for ML Training With Seneca (优先级分数 51.25) - Behemoth: A Flash-centric Training Accelerator for Extreme-scale DNNs (优先级分数 44.709) - AdaCheck: An Adaptive Checkpointing System for Efficient LLM Training with Redundancy Utilization (优先级分数 42.5)

1.3.10.1 Accelerating Model Loading in LLM Inference by Programmable Page Cache

- 作者: Yubo Liu; Hong-Bo Li; Xiaojia Huang; Yongfeng Wang; Hanjun Guo; Huimin Chen; Yu-Xin Ren; Ning Jia
- 年份: 2026; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: Accelerating Model Loading in LLM Inference by Programmable Page Cache

快速理解: 摘要未随索引或本地 PDF 前页提供; 根据标题与主题标签, 本文关注 AI Infra、Systems 相关问题。以下判断仅用于快速阅读排序, 不能替代全文核对。

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 标题/主题推断。

1.3.10.2 Hardware/Software Co-Programmable Framework for Computational SSDs to Accelerate Deep Learning Service on Large-Scale Graphs

- 作者: M. Kwon; Donghyun Gouk; Sangwon Lee; Myoungsoo Jung
- 年份: 2022; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Hardware/Software Co-Programmable Framework for Computational SSDs to Accelerate Deep Learning Service on Large-Scale Graphs

快速理解: Graph neural networks (GNNs) process large-scale graphs consisting of a hundred billion edges. In contrast to traditional deep learning, unique behaviors of the emerging GNNs are engaged with a large set of graphs and embedding data on storage, which exhibits complex and irregular preprocessing.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.10.3 Preparation Meets Opportunity: Enhancing Data Preprocessing for ML Training With Seneca

- 作者: Omkar Desai; Zi-Yang Jiao; Shuyi Pei; J. Bhimani; B. Kim
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Preparation Meets Opportunity: Enhancing Data Preprocessing for ML Training With Seneca

快速理解: Input data preprocessing is a common bottleneck when concurrently training multimedia machine learning (ML) models in modern systems. To alleviate these bottlenecks and reduce the training time for concurrent jobs, we present Seneca, a data loading system that optimizes cache partitioning and data sampling for the data storage and ingestion (DSI) pipeline.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.10.4 Behemoth: A Flash-centric Training Accelerator for Extreme-scale DNNs

- 作者: Shine Kim; Yunho Jin; G. Sohn; Jonghyun Bae; Tae Jun Ham; Jae W. Lee
- 年份: 2021; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Behemoth: A Flash-centric Training Accelerator for Extreme-scale DNNs

快速理解: 摘要未随索引或本地 PDF 前页提供; 根据标题与主题标签, 本文关注 AI Infra 相关问题。以下判断仅用于快速阅读排序, 不能替代全文核对。

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 标题/主题推断。

1.3.10.5 AdaCheck: An Adaptive Checkpointing System for Efficient LLM Training with Redundancy Utilization

- 作者: Weijie Liu; Shengwei Li; Zhiquan Lai; Ke-shi Ge; Qiaoling Chen; Peng Sun; Dongsheng Li; Kai Lu
- 年份: 2026; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: AdaCheck: An Adaptive Checkpointing System for Efficient LLM Training with Redundancy Utilization

快速理解: 摘要未随索引或本地 PDF 前页提供; 根据标题与主题标签, 本文关注 AI Infra 相关问题。以下判断仅用于快速阅读排序, 不能替代全文核对。

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 标题/主题推断。

1.3.10.6 Learning Cache Replacement with CACHEUS

- 作者: Liana V. Rodriguez; Farzana Beente Yusuf; Steven Lyons; Eysler Paz; R. Rangaswami; Jason Liu; Ming Zhao; G. Narasimhan
- 年份: 2021; 主题: Systems; 优先级: 扩展
- [来源](#)

论文: Learning Cache Replacement with CACHEUS

快速理解: 摘要未随索引或本地 PDF 前页提供; 根据标题与主题标签, 本文关注 Systems 相关问题。以下判断仅用于快速阅读排序, 不能替代全文核对。

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：标题/主题推断。

1.3.10.7 GL-Cache: Group-level learning for efficient and high-performance caching

- 作者：Juncheng Yang; Ziming Mao; Yao Yue; K. V. Rashmi
- 年份：2023; 主题：Systems; 优先级：扩展
- [来源](#)

论文：GL-Cache: Group-level learning for efficient and high-performance caching

快速理解：摘要未随索引或本地 PDF 前页提供；根据标题与主题标签，本文关注 Systems 相关问题。以下判断仅用于快速阅读排序，不能替代全文核对。

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：标题/主题推断。

1.3.10.8 IMPRESS: An Importance-Informed Multi-Tier Prefix KV Storage System for Large Language Model Inference

- 作者：Wei-Jian Chen; Shui-Bing He; Haoyang Qu; Ruidong Zhang; Si-Ling Yang; Ping Chen; Yi Zheng; Baoxing Huai; Gang Chen
- 年份：2025; 主题：Systems; 优先级：扩展
- [来源](#)

论文：IMPRESS: An Importance-Informed Multi-Tier Prefix KV Storage System for Large Language Model Inference

快速理解：摘要未随索引或本地 PDF 前页提供；根据标题与主题标签，本文关注 Systems 相关问题。以下判断仅用于快速阅读排序，不能替代全文核对。

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：标题/主题推断。

1.3.10.9 Separating Data via Block Invalidation Time Inference for Write Amplification Reduction in Log-Structured Storage

- 作者：Qiuping Wang; Jinhong Li; P. Lee; Tao Ouyang; Chao Shi; Lilong Huang
- 年份：2021; 主题：Systems; 优先级：扩展
- [来源](#)

论文：Separating Data via Block Invalidation Time Inference for Write Amplification Reduction in Log-Structured Storage

快速理解：Log-structured storage has been widely deployed in various domains of storage systems, yet its garbage collection incurs write amplification (WA) due to the rewrites of live data. We show that there exists an optimal data placement scheme that minimizes WA using the future knowledge of block invalidation time (BIT) of each written block, yet it is infeasible to realize in practice.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.10.10 SHADE: Enable Fundamental Cacheability for Distributed Deep Learning Training

- 作者: R. Khan; Ahmad Hossein Yazdani; Yuqi Fu; Arnab K. Paul; Bo Ji; Xun Jian; Yue Cheng; A. Butt
- 年份: 2023; 主题: Systems; 优先级: 扩展
- [来源](#)

论文: SHADE: Enable Fundamental Cacheability for Distributed Deep Learning Training

快速理解: 摘要未随索引或本地 PDF 前页提供; 根据标题与主题标签, 本文关注 Systems 相关问题。以下判断仅用于快速阅读排序, 不能替代全文核对。

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 标题/主题推断。

1.3.10.11 DeepSketch: A New Machine Learning-Based Reference Search Technique for Post-Deduplication Delta Compression

- 作者: Jisung Park; Jeoggyun Kim; Yeseong Kim; Sungjin Lee; Onur Mutlu
- 年份: 2022; 主题: Systems; 优先级: 扩展
- [来源](#)

论文: DeepSketch: A New Machine Learning-Based Reference Search Technique for Post-Deduplication Delta Compression

快速理解: Data reduction in storage systems is becoming increasingly important as an effective solution to minimize the management cost of a data center. To maximize data-reduction efficiency, existing post-deduplication delta-compression techniques perform delta compression along with traditional data deduplication and lossless compression.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.10.12 FlashNeuron: SSD-Enabled Large-Batch Training of Very Deep Neural Networks

- 作者: Jonghyun Bae; Jongsung Lee; Yunho Jin; Sam Son; Shine Kim; Hakbeom Jang; Tae Jun Ham; Jae W. Lee
- 年份: 2021; 主题: 未标注主题; 优先级: 扩展
- [来源](#)

论文: FlashNeuron: SSD-Enabled Large-Batch Training of Very Deep Neural Networks

快速理解: 摘要未随索引或本地 PDF 前页提供; 根据标题与主题标签, 本文关注未标注主题相关问题。以下判断仅用于快速阅读排序, 不能替代全文核对。

关注点: 需打开全文确认研究问题、方法和实验边界。

证据层级: 标题/主题推断。

1.3.10.13 LeapGNN: Accelerating Distributed GNN Training Leveraging Feature-Centric Model Migration

- 作者: Wei-Jian Chen; Shui-Bing He; Haoyang Qu; Xuechen Zhang
- 年份: 2025; 主题: 未标注主题; 优先级: 扩展
- [来源](#)

论文: LeapGNN: Accelerating Distributed GNN Training Leveraging Feature-Centric Model Migration

快速理解: 摘要未随索引或本地 PDF 前页提供; 根据标题与主题标签, 本文关注未标注主题相关问题。以下判断仅用于快速阅读排序, 不能替代全文核对。

关注点: 需打开全文确认研究问题、方法和实验边界。

证据层级: 标题/主题推断。

1.3.11 SYSTEMS · HPCA

本会议纳入 50 篇。自动主题统计为: Systems (49)、AI Infra (48)、Supporting Perception (2)、Agent RL (2)。这些统计用于组织阅读顺序, 不能单独证明论文结论或实验优越性。

优先阅读候选: - V-Rex: Real-Time Streaming Video LLM Acceleration via Dynamic KV Cache Retrieval (优先级分数 79.057) - InstAttention: In-Storage Attention Offloading for Cost-Effective Long-Context LLM Inference (优先级分数 72.709) - iCache: An Importance-Sampling-Informed Cache for Accelerating I/O-Bound DNN Model Training (优先级分数 71.45) - LightTrader: A Standalone High-Frequency Trading System with Deep Learning Inference Accelerators and Proactive Scheduler (优先级分数 70.75) - FHENDI: A Near-DRAM Accelerator for Compiler-Generated Fully Homomorphic Encryption Applications (优先级分数 69.835)

1.3.11.1 V-Rex: Real-Time Streaming Video LLM Acceleration via Dynamic KV Cache Retrieval

- 作者: Donghyuk Kim; Sejeong Yang; Wonjin Shin; Joo-Young Kim
- 年份: 2025; 主题: AI Infra; Supporting Perception; Systems; 优先级: 核心
- [来源](#)

论文: V-Rex: Real-Time Streaming Video LLM Acceleration via Dynamic KV Cache Retrieval

快速理解: Streaming video large language models (LLMs) are increasingly used for real-time multi-modal tasks such as video captioning, question answering, conversational agents, and augmented reality. However, these models face fundamental memory and computational challenges because their key-value (KV) caches grow substantially with continuous streaming video input.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 3D、视频、多模态视觉或具身感知支撑能力; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.11.2 InstAttention: In-Storage Attention Offloading for Cost-Effective Long-Context LLM Inference

- 作者: Xiurui Pan; Endian Li; Qiao Li; Shengwen Liang; Yizhou Shan; Ke Zhou; Yingwei Luo; Xiao-Lin Wang; Jie Zhang
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: InstAttention: In-Storage Attention Offloading for Cost-Effective Long-Context LLM Inference

快速理解: The widespread of Large Language Models (LLMs) marks a significant milestone in generative AI. Nevertheless, the increasing context length and batch size in offline LLM inference escalate the memory requirement of the key-value (KV) cache, which imposes a huge burden on the GPU

VRAM, especially for resource-constrained scenarios (e.g., edge computing).

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.11.3 iCache: An Importance-Sampling-Informed Cache for Accelerating I/O-Bound DNN Model Training

- 作者: Weijian Chen; Shuibing He; Yaowen Xu; Xuechen Zhang; Si-Ling Yang; Shuang Hu; Xian-He Sun; Gang Chen
- 年份: 2023; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: iCache: An Importance-Sampling-Informed Cache for Accelerating I/O-Bound DNN Model Training

快速理解: Fetching a large amount of DNN training data from storage systems incurs long I/O latency and fetch stalls of GPUs. Importance sampling in DNN training can reduce the amount of data computing on GPUs while maintaining a similar model accuracy.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.11.4 LightTrader: A Standalone High-Frequency Trading System with Deep Learning Inference Accelerators and Proactive Scheduler

- 作者: Sungyeob Yoo; Hyuncheol Kim; Jinseok Kim; Sunghyun Park; Joo-Young Kim; Jinwook Oh
- 年份: 2023; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: LightTrader: A Standalone High-Frequency Trading System with Deep Learning Inference Accelerators and Proactive Scheduler

快速理解: Recent research shows that artificial intelligence (AI) algorithms can dramatically improve the profitability of high-frequency trading (HFT) with accurate market prediction, overcoming the limitation of conventional latency-oriented approaches. However, it is challenging to integrate the computationally intensive AI algorithm into the existing trading pipeline due to its excessively long latency and insufficient throughput, necessitating a breakthrough in hardware.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.11.5 FHENDI: A Near-DRAM Accelerator for Compiler-Generated Fully Homomorphic Encryption Applications

- 作者: Yongmo Park; Aporva Amarnath; Subhankar Pal; Karthik Swaminathan; A. Buyuktosunoglu; Hayim Shaul; E. Aharoni; Nir Drucker; Wei D. Lu; Omri Soceanu; P. Bose
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: FHENDI: A Near-DRAM Accelerator for Compiler-Generated Fully Homomorphic Encryption

Applications

快速理解: Fully homomorphic encryption (FHE) is a powerful cryptographic technique that enables computation on encrypted data without needing to decrypt it. It has broad applications in scenarios where sensitive data needs to be processed in the cloud or in other untrusted environments.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.11.6 Towards Resource-Efficient Serverless LLM Inference with SLINFER

- 作者: Chuhao Xu; Zijun Li; Quan Chen; Han Zhao; Xueyan Tang; Minyi Guo
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: Towards Resource-Efficient Serverless LLM Inference with SLINFER

快速理解: The rise of LLMs has driven demand for private serverless deployments, characterized by moderate-sized models and infrequent requests. While existing serverless solutions follow exclusive GPU allocation, we take a step back to explore modern platforms and find that: Emerging CPU architectures with built-in accelerators are capable of serving LLMs but remain underutilized, and both CPUs and GPUs can accommodate multiple LLMs simultaneously.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.11.7 High Performance and Power Efficient Accelerator for Cloud Inference

- 作者: Jianguo Yao; Hao Zhou; Yalin Zhang; Ying Li; Chuang Feng; Shi Chen; Jiaoyan Chen; Yongdong Wang; Qiaojuan Hu
- 年份: 2023; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: High Performance and Power Efficient Accelerator for Cloud Inference

快速理解: Facing the growing complexity of Deep Neural Networks (DNNs), high-performance and power-efficient AI accelerators are desired to provide effective and affordable cloud inference services. We introduce our flagship product, i.e., the Cloudblazer i20 accelerator, which integrates the innovated Deep Thinking Unit (DTU 2.0).

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.11.8 Characterizing Cloud-Native LLM Inference at Bytedance and Exposing Optimization Challenges and Opportunities for Future AI Accelerators

- 作者: Jing-Wei Cai; De-Hao Kong; Hantao Huang; Zishang Jiang; Zixuan Ma; Qingyu Guo; Zhenxing Zhang; Guiming Shi; Mingyu Gao; Kai-Sheng Ma; Minghui Yu
- 年份: 2026; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: Characterizing Cloud-Native LLM Inference at Bytedance and Exposing Optimization Challenges and Opportunities for Future AI Accelerators

快速理解: As a major provider of LLM inference services, ByteDance has continuously explored diverse accelerator options to meet the rapidly growing inference demands of various heterogeneous LLM scenarios with higher cost-effectiveness, thereby enabling LLMs to serve more people worldwide. However, during this process, we have found that the complexity and opacity of cloud scenarios and corresponding cloud accelerators make it difficult for academia and many innovative chip startups to fully understand the real demands and challenges of these scenarios, which in turn severely restricts innovation and application potential in this field.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.11.9 FAB: An FPGA-based Accelerator for Bootstrappable Fully Homomorphic Encryption

- 作者: R. Agrawal; Leo de Castro; Guo-Wei Yang; C. Juvekar; R. Yazicigil; A. Chandrakasan; V. Vaikuntanathan; A. Joshi
- 年份: 2022; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: FAB: An FPGA-based Accelerator for Bootstrappable Fully Homomorphic Encryption

快速理解: Fully Homomorphic Encryption (FHE) offers protection to private data on third-party cloud servers by allowing computations on the data in encrypted form. To support general-purpose encrypted computations, all existing FHE schemes require an expensive operation known as “bootstrapping” .

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.11.10 TensorFHE: Achieving Practical Computation on Encrypted Data Using GPGPU

- 作者: Shengyu Fan; Zhi-Wei Wang; Weizhi Xu; Rui Hou; Dan Meng; M. Zhang
- 年份: 2022; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: TensorFHE: Achieving Practical Computation on Encrypted Data Using GPGPU

快速理解: In the cloud computing era, privacy protection is becoming pervasive in a broad range of applications (e.g., machine learning, data mining, etc). Fully Homomorphic Encryption (FHE) is considered the perfect solution as it enables privacy-preserved computation on untrusted servers.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.11.11 Gemini: Mapping and Architecture Co-exploration for Large-scale DNN Chiplet Accelerators

- 作者: Jing-Wei Cai; Zuotong Wu; Sen Peng; Yuchen Wei; Zhanhong Tan; Guiming Shi; Mingyu Gao; Kai-Sheng Ma
- 年份: 2023; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Gemini: Mapping and Architecture Co-exploration for Large-scale DNN Chiplet Accelerators

快速理解: Chiplet technology enables the integration of an increasing number of transistors on a single accelerator with higher yield in the post-Moore era, addressing the immense computational demands arising from rapid AI advancements. However, it also introduces more expensive packaging costs and costly Die-to-Die (D2D) interfaces, which require more area, consume higher power, and offer lower bandwidth than onchip interconnects.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.11.12 Sentinel: Efficient Tensor Migration and Allocation on Heterogeneous Memory Systems for Deep Learning

- 作者: Jie Ren; Jiaolin Luo; Kai Wu; Minjia Zhang; Hyeran Jeon
- 年份: 2021; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Sentinel: Efficient Tensor Migration and Allocation on Heterogeneous Memory Systems for Deep Learning

快速理解: Memory capacity is a major bottleneck for training deep neural networks (DNN). Heterogeneous memory (HM) combining fast and slow memories provides a promising direction to increase memory capacity.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.11.13 CHROME: Concurrency-Aware Holistic Cache Management Framework with Online Reinforcement Learning

- 作者: Xiaoyang Lu; Hamed Najafi; Jason Liu; Xian-He Sun
- 年份: 2024; 主题: Agent RL; Systems; 优先级: 重要
- [来源](#)

论文: CHROME: Concurrency-Aware Holistic Cache Management Framework with Online Reinforcement Learning

快速理解: 摘要未随索引或本地 PDF 前页提供; 根据标题与主题标签, 本文关注 Agent RL、Systems 相关问题。以下判断仅用于快速阅读排序, 不能替代全文核对。

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 标题/主题推断。

1.3.11.14 AutoCAT: Reinforcement Learning for Automated Exploration of Cache-Timing Attacks

- 作者: Mulong Luo; Wenjie Xiong; Geunbae Lee; Yueying Li; Xiaomeng Yang; Amy Zhang; Yuan-dong Tian; Hsien-Hsin S. Lee; G. Suh
- 年份: 2022; 主题: Agent RL; Systems; 优先级: 重要
- [来源](#)

论文: AutoCAT: Reinforcement Learning for Automated Exploration of Cache-Timing Attacks

快速理解: The aggressive performance optimizations in modern microprocessors can result in security vulnerabilities. For example, timing-based attacks in processor caches can steal secret keys or break randomization.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.11.15 MLPerf Power: Benchmarking the Energy Efficiency of Machine Learning Systems from Watts to MWatts for Sustainable AI

- 作者: Arya Tschand; Arun Tejusve Raghunath Rajan; Sachin Idgunji; Anirban Ghosh; Jeremy Holleman; Csaba Király; Pawan Ambalkar; Ritika Borkar; Ramesh Chukka; Trevor Cockrell; Oliver Curtis; G. Fursin; Miro Hodak; H. Kassa; Anton Lokhmotov; Dejan Miskovic; Yuechao Pan; Manu Prasad Manmathan; Liz Raymond; Tom St. John; Arjun Suresh; Rowan Taubitz; Sean Zhan; Scott Wasson; David Kanter; V. Reddi
- 年份: 2024; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: MLPerf Power: Benchmarking the Energy Efficiency of Machine Learning Systems from Watts to MWatts for Sustainable AI

快速理解: Rapid adoption of machine learning (ML) technologies has led to a surge in power consumption across diverse systems, from tiny IoT devices to massive datacenter clusters. Benchmarking the energy efficiency of these systems is crucial for optimization, but presents novel challenges due to the variety of hardware platforms, workload characteristics, and system-level interactions.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.11.16 Smart-Infinity: Fast Large Language Model Training using Near-Storage Processing on a Real System

- 作者: Hongsun Jang; Jaeyong Song; Jaewon Jung; Jaeyoung Park; Youngsok Kim; Jinho Lee
- 年份: 2024; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Smart-Infinity: Fast Large Language Model Training using Near-Storage Processing on a Real System

快速理解: The recent huge advance of Large Language Models (LLMs) is mainly driven by the increase in the number of parameters. This has led to substantial memory capacity requirements, necessitating the use of dozens of GPUs just to meet the capacity.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.11.17 Need for Speed: Experiences Building a Trustworthy System-Level GPU Simulator

- 作者: Oreste Villa; Daniel Lustig; Zi Yan; Evgeny Bolotin; Yaosheng Fu; Niladrish Chatterjee; Nan Jiang; D. Nellans
- 年份: 2021; 主题: AI Infra; Systems; 优先级: 重要

- [来源](#)

论文: Need for Speed: Experiences Building a Trustworthy System-Level GPU Simulator

快速理解: The demands of high-performance computing (HPC) and machine learning (ML) workloads have resulted in the rapid architectural evolution of GPUs over the last decade. The growing memory footprint and diversity of data types in these workloads has required GPUs to embrace micro-architectural heterogeneity and increased memory system sophistication to scale performance.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.11.18 Atomic Dataflow based Graph-Level Workload Orchestration for Scalable DNN Accelerators

- 作者: Shixuan Zheng; Xianjue Zhang; Leibo Liu; Shaojun Wei; Shouyi Yin
- 年份: 2022; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Atomic Dataflow based Graph-Level Workload Orchestration for Scalable DNN Accelerators

快速理解: To efficiently deploy state-of-the-art deep neural network (DNN) workloads with growing computational intensity and structural complexity, scalable DNN accelerators have been proposed in recent years, which are featured by multi-tensor engines and distributed on-chip buffers. Such spatial architectures have significantly expanded scheduling space in terms of parallelism and data reuse potentials, which demands for delicate workload orchestration.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.11.19 throttLL' eM: Predictive GPU Throttling for Energy Efficient LLM Inference Serving

- 作者: A. Kakolyris; D. Masouros; Petros Vavaroutsos; Sotirios Xydis; Dimitrios Soudris
- 年份: 2024; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: throttLL' eM: Predictive GPU Throttling for Energy Efficient LLM Inference Serving

快速理解: As Large Language Models (LLMs) gain traction, their reliance on power-hungry GPUs places ever-increasing energy demands, raising environmental and monetary concerns. Inference dominates LLM workloads, presenting a critical challenge for providers: minimizing energy costs under Service-Level Objectives (SLOs) that ensure optimal user experience.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.11.20 Layerweaver: Maximizing Resource Utilization of Neural Processing Units via Layer-Wise Scheduling

- 作者: Young H. Oh; Seonghak Kim; Yunho Jin; Sam Son; Jonghyun Bae; Jongsung Lee; Yeonhong Park; Dong Uk Kim; Tae Jun Ham; Jae W. Lee
- 年份: 2021; 主题: AI Infra; Systems; 优先级: 重要

- [来源](#)

论文: Layerweaver: Maximizing Resource Utilization of Neural Processing Units via Layer-Wise Scheduling

快速理解: To meet surging demands for deep learning inference services, many cloud computing vendors employ high-performance specialized accelerators, called neural processing units (NPUs). One important challenge for effective use of NPUs is to achieve high resource utilization over a wide spectrum of deep neural network (DNN) models with diverse arithmetic intensities.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.11.21 Tacker: Tensor-CUDA Core Kernel Fusion for Improving the GPU Utilization while Ensuring QoS

- 作者: Han Zhao; Weihao Cui; Quan Chen; Youtao Zhang; Yanchao Lu; C. Li; Jingwen Leng; Minyi Guo
- 年份: 2022; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Tacker: Tensor-CUDA Core Kernel Fusion for Improving the GPU Utilization while Ensuring QoS

快速理解: The proliferation of machine learning applications has promoted both CUDA Cores and Tensor Cores' integration to meet their acceleration demands. While studies have shown that co-locating multiple tasks on the same GPU can effectively improve system throughput and resource utilization, existing schemes focus on scheduling the resources of traditional CUDA Cores and thus lack the ability to exploit the parallelism between Tensor Cores and CUDA Cores. In this paper, we propose Tacker, a static kernel fusion and scheduling approach to improve GPU utilization of both types of cores while ensuring the QoS (Quality-of-Service) of co-located tasks.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.11.22 Make LLM Inference Affordable to Everyone: Augmenting GPU Memory with NDP-DIMM

- 作者: Lian Liu; Shixin Zhao; Bing Li; Hai-Meng Ren; Zhaohui Xu; Mengdi Wang; Xiaowei Li; Yi Han; Ying Wang
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Make LLM Inference Affordable to Everyone: Augmenting GPU Memory with NDP-DIMM

快速理解: The billion-scale Large Language Models (LLMs) necessitate deployment on expensive server-grade GPUs with large-storage HBMs and abundant computation capability. As LLM-assisted services become popular, achieving cost-effective LLM inference on budget-friendly hardware becomes the current trend.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.11.23 VQ-LLM: High-performance Code Generation for Vector Quantization Augmented LLM Inference

- 作者: Zihan Liu; Xinhao Luo; Junxian Guo; Wentao Ni; Yangjie Zhou; Yue Guan; Cong Guo; Weihao Cui; Yu Feng; Minyi Guo; Yuhao Zhu; Minjia Zhang; Jingwen Leng; Chen Jin
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: VQ-LLM: High-performance Code Generation for Vector Quantization Augmented LLM Inference

快速理解: Vector quantization (VQ), which treats a vector as a compression unit, gains increasing research interests for its potential to accelerate large language models (LLMs). Compared to conventional element-wise quantization methods, VQ algorithms can compress weight and KV cache tensors in LLMs with a greater ratio while maintaining the high model accuracy.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.11.24 Morphling: A Throughput-Maximized TFHE-based Accelerator using Transform-domain Reuse

- 作者: Prasetyo; Adiwena Putra; Joo-Young Kim
- 年份: 2024; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Morphling: A Throughput-Maximized TFHE-based Accelerator using Transform-domain Reuse

快速理解: Fully Homomorphic Encryption (FHE) has become an increasingly important aspect in modern computing, particularly in preserving privacy in cloud computing by enabling computation directly on encrypted data. Despite its potential, FHE generally poses major computational challenges, including huge computational and memory requirements.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.11.25 OptimStore: In-Storage Optimization of Large Scale DNNs with On-Die Processing

- 作者: Junkyum Kim; Myeonggu Kang; Yunki Han; Yanggon Kim; Lee-Sup Kim
- 年份: 2023; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: OptimStore: In-Storage Optimization of Large Scale DNNs with On-Die Processing

快速理解: Training deep neural network (DNN) models is a resource-intensive, iterative process. For this reason, nowadays, complex optimizers like Adam are widely adopted as it increases the speed and efficiency of training.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.11.26 Lincoln: Real-Time 50~100B LLM Inference on Consumer Devices with LPDDR-Interfaced, Compute-Enabled Flash Memory

- 作者: Wei-Yi Sun; Ming-Yu Gao; Zhaoshi Li; Aoyang Zhang; Iris Ying Chou; Jianfeng Zhu; Shaojun Wei; Lei-Bo Liu
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Lincoln: Real-Time 50~100B LLM Inference on Consumer Devices with LPDDR-Interfaced, Compute-Enabled Flash Memory

快速理解: With the widespread use of large language models (LLMs), and with the privacy and cost concerns on cloud-based services, vendors are now pushing LLM inference to consumer devices. However, current attempts only enable real-time inference of low-quality small-sized LLMs.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.11.27 MEGA: A Memory-Efficient GNN Accelerator Exploiting Degree-Aware Mixed-Precision Quantization

- 作者: Zeyu Zhu; Fanrong Li; Gang Li; Zejian Liu; Zitao Mo; Qinghao Hu; Xiaoyao Liang; Jian Cheng
- 年份: 2023; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: MEGA: A Memory-Efficient GNN Accelerator Exploiting Degree-Aware Mixed-Precision Quantization

快速理解: Graph Neural Networks (GNNs) are becoming a promising technique in various domains due to their excellent capabilities in modeling non-Euclidean data. Although a spectrum of accelerators has been proposed to accelerate the inference of GNNs, our analysis demonstrates that the latency and energy consumption induced by DRAM access still significantly impedes the improvement of performance and energy efficiency.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.11.28 Rambda: RDMA-driven Acceleration Framework for Memory-intensive μ s-scale Datacenter Applications

- 作者: Yifan Yuan; Jinghan Huang; Y. Sun; Tianchen Wang; Jacob Nelson; Dan R. K. Ports; Yipeng Wang; Ren Wang; Charlie Tai; Nam Sung Kim
- 年份: 2023; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Rambda: RDMA-driven Acceleration Framework for Memory-intensive μ s-scale Datacenter Applications

快速理解: Responding to the “datacenter tax” and “killer microseconds” problems for memory-intensive datacenter applications, diverse solutions including Smart NIC-based ones have been proposed. Nonetheless, they often suffer from high overhead of communications over network and/or PCIe

links.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.11.29 Anda: Unlocking Efficient LLM Inference with a Variable-Length Grouped Activation Data Format

- 作者：Chao Fang; Man Shi; Robin Geens; Arne Symons; Zhongfeng Wang; Marian Verhelst
- 年份：2024; 主题：AI Infra; Systems; 优先级：重要
- [来源](#)

论文：Anda: Unlocking Efficient LLM Inference with a Variable-Length Grouped Activation Data Format

快速理解：The widely-used, weight-only quantized large language models (LLMs), which leverage low-bit integer (INT) weights and retain floating-point (FP) activations, reduce storage requirements while maintaining accuracy. However, this shifts the energy and latency bottlenecks towards the FP activations that are associated with costly memory accesses and computations.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.11.30 Common Counters: Compressed Encryption Counters for Secure GPU Memory

- 作者：Seonjin Na; Sun-Ho Lee; Yeonjae Kim; Jongse Park; Jaehyuk Huh
- 年份：2021; 主题：AI Infra; Systems; 优先级：重要
- [来源](#)

论文：Common Counters: Compressed Encryption Counters for Secure GPU Memory

快速理解：Hardware-based trusted execution has opened a promising new opportunity for enabling secure cloud computing. Nevertheless, the current trusted execution environments are limited to the traditional CPU-based computation, while more and more critical workloads such as machine learning require to use GPUs.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.11.31 PAISE: PIM-Accelerated Inference Scheduling Engine for Transformer-based LLM

- 作者：Hyojung Lee; Daehyeon Baek; Jimyoung Son; Jieun Choi; Kihyo Moon; Minsung Jang
- 年份：2025; 主题：AI Infra; Systems; 优先级：重要
- [来源](#)

论文：PAISE: PIM-Accelerated Inference Scheduling Engine for Transformer-based LLM

快速理解：Transformer-based Large Language Models (LLMs) demand significant computational and memory resources due to the autoregressive token generation in decoder blocks. In particular, the attention layer in LLM models has low arithmetic intensity but high memory traffic, thus requiring frequent updates to the KV matrices with each decoder iteration.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.11.32 Know Your Enemy To Save Cloud Energy: Energy-Performance Characterization of Machine Learning Serving

- 作者：Junyeol Yu; Jongseok Kim; Euseong Seo
- 年份：2023; 主题：AI Infra; Systems; 优先级：重要
- [来源](#)

论文：Know Your Enemy To Save Cloud Energy: Energy-Performance Characterization of Machine Learning Serving

快速理解：The proportion of machine learning (ML) inference in modern cloud workloads is rapidly increasing, and graphic processing units (GPUs) are the most preferred computational accelerators for it. The massively parallel computing capability of GPUs is well-suited to the inference workloads but consumes more power than conventional CPUs.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.11.33 WASP: Exploiting GPU Pipeline Parallelism with Hardware-Accelerated Automatic Warp Specialization

- 作者：N. Crago; S. Damani; K. Sankaralingam; Stephen W. Keckler
- 年份：2024; 主题：AI Infra; Systems; 优先级：重要
- [来源](#)

论文：WASP: Exploiting GPU Pipeline Parallelism with Hardware-Accelerated Automatic Warp Specialization

快速理解：Graphics processing units (GPUs) are an important class of parallel processors that offer high compute throughput and memory bandwidth. GPUs are used in a variety of important computing domains, such as machine learning, high performance computing, sparse linear algebra, autonomous vehicles, and robotics.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.11.34 FlashGNN: An In-SSD Accelerator for GNN Training

- 作者：Fuping Niu; Jianhui Yue; Jiangqiu Shen; Xiaofei Liao; Hai Jin
- 年份：2024; 主题：AI Infra; Systems; 优先级：重要
- [来源](#)

论文：FlashGNN: An In-SSD Accelerator for GNN Training

快速理解：Recently, Graph Neural Networks (GNNs) have emerged as powerful tools for data analysis, surpassing traditional algorithms in various applications. However, the growing size of real-world datasets has outpaced the capabilities of centralized CPU or GPU-based systems.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.11.35 TinyTS: Memory-Efficient TinyML Model Compiler Framework on Microcontrollers

- 作者：Yu-Yuan Liu; Hong-Sheng Zheng; Yu-Fang Hu; Chen-Fong Hsu; Tsung-Tai Yeh
- 年份：2024; 主题：AI Infra; Systems; 优先级：重要
- [来源](#)

论文：TinyTS: Memory-Efficient TinyML Model Compiler Framework on Microcontrollers

快速理解：Deploying deep neural network (DNN) models on Microcontroller Units (MCUs) is typically limited by the tightness of the SRAM memory budget. Previously, machine learning system frameworks often allocated tensor memory layer-wise, but this will result in out-of-memory exceptions when a DNN model includes a large tensor.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.11.36 EXMA: A Genomics Accelerator for Exact-Matching

- 作者：Lei Jiang; Farzaneh Zokaee
- 年份：2021; 主题：AI Infra; Systems; 优先级：重要
- [来源](#)

论文：EXMA: A Genomics Accelerator for Exact-Matching

快速理解：Genomics is the foundation of precision medicine, global food security and virus surveillance. Exact-match is one of the most essential operations widely used in almost every step of genomics such as alignment, assembly, annotation, and compression.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.11.37 NeuroMeter: An Integrated Power, Area, and Timing Modeling Framework for Machine Learning Accelerators Industry Track Paper

- 作者：Tianqi Tang; Sheng Li; Lifeng Nai; N. Jouppi; Yuan Xie
- 年份：2021; 主题：AI Infra; Systems; 优先级：重要
- [来源](#)

论文：NeuroMeter: An Integrated Power, Area, and Timing Modeling Framework for Machine Learning Accelerators Industry Track Paper

快速理解：As Machine Learning (ML) becomes pervasive in the era of artificial intelligence, ML specific tools and frameworks are required for architectural research. This paper introduces NeuroMeter, an integrated power, area, and timing modeling framework for ML accelerators.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.11.38 Data Motion Acceleration: Chaining Cross-Domain Multi Accelerators

- 作者: Shu-Ting Wang; Hanyang Xu; Amin Mamandipoor; R. Mahapatra; Byung Hoon Ahn; Soroush Ghodrati; K. Kailas; Mohammad Alian; H. Esmaeilzadeh
- 年份: 2024; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Data Motion Acceleration: Chaining Cross-Domain Multi Accelerators

快速理解: There has been an arms race for devising accelerators for deep learning in recent years. However, real-world applications are not only neural networks but often span across multiple domains, e.g., database queries, compression, encryption, video coding, signal processing, and traditional machine learning, which may or may not contain deep learning.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.11.39 Deadline-Aware Offloading for High-Throughput Accelerators

- 作者: Tsung-Tai Yeh; Matthew D. Sinclair; Bradford M. Beckmann; Timothy G. Rogers
- 年份: 2021; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Deadline-Aware Offloading for High-Throughput Accelerators

快速理解: Contemporary GPUs are widely used for throughput-oriented data-parallel workloads and increasingly are being considered for latency-sensitive applications in datacenters. Examples include recurrent neural network (RNN) inference, network packet processing, and intelligent personal assistants.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.11.40 LAD: Efficient Accelerator for Generative Inference of LLM with Locality Aware Decoding

- 作者: Hao-Ran Wang; Yuming Li; Haobo Xu; Ying Wang; Liqi Liu; Jun Yang; Yi Han
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: LAD: Efficient Accelerator for Generative Inference of LLM with Locality Aware Decoding

快速理解: Large Language Models (LLMs) have emerged as the cornerstone of content generation applications due to their ability to capture relations between newly generated token and the full preceding context. However, this ability stems from the attention mechanism for decoding that retains the entire generation history as key value cache (KV cache).

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.11.41 Hydra: Scale-out FHE Accelerator Architecture for Secure Deep Learning on FPGA

- 作者: Ying-Hao Yang; Xi-Cheng Xu; Haibin Zhang; Jie Song; Xin Tang; Hang Lu; Xiaowei Li
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要

- [来源](#)

论文: Hydra: Scale-out FHE Accelerator Architecture for Secure Deep Learning on FPGA

快速理解: Deep learning, including Convolutional Neural Network (CNN) and Large Language Model (LLM), under Fully Homomorphic Encryption (FHE) is very computationally intensive because of the burdensome computations like ciphertext convolution and matrix multiplication, non-linear layers, and bootstrapping. Existing FHE accelerators focus on the high throughput computational units, stacking parallelized clusters to maximize ciphertext inference performance.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.11.42 PIMphony: Overcoming Bandwidth and Capacity Inefficiency in PIM-Based Long-Context LLM Inference System

- 作者: Hyucksung Kwon; K. Koo; Janghyeon Kim; Woongkyu Lee; Minjae Lee; Gyeonggeun Jung; Hyungdeok Lee; Yousub Jung; Jaehan Park; Yosub Song; Byeongsu Yang; Haerang Choi; Guhyun Kim; Jong-Hak Won; Woojae Shin; Changhyun Kim; Gyeongcheol Shin; Yongkee Kwon; Il-Hyu Kim; Euicheol Lim; John Kim; Jungwook Choi
- 年份: 2024; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: PIMphony: Overcoming Bandwidth and Capacity Inefficiency in PIM-Based Long-Context LLM Inference System

快速理解: The expansion of long-context Large Language Models (LLMs) creates significant memory system challenges. While Processing-in-Memory (PIM) is a promising accelerator, we identify that it suffers from critical inefficiencies when scaled to long contexts: severe channel underutilization, performance-limiting I/O bottlenecks, and massive memory waste from static KV cache management.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.11.43 Enabling Efficient Large-Scale Deep Learning Training with Cache Coherent Disaggregated Memory Systems

- 作者: Zixuan Wang; Joonseop Sim; Euicheol Lim; Jishen Zhao
- 年份: 2022; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Enabling Efficient Large-Scale Deep Learning Training with Cache Coherent Disaggregated Memory Systems

快速理解: Modern deep learning (DL) training is memory-consuming, constrained by the memory capacity of each computation component and cross-device communication bandwidth. In response to such constraints, current approaches include increasing parallelism in distributed training and optimizing inter-device communication.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.11.44 BitDecoding: Unlocking Tensor Cores for Long-Context LLMs with Low-Bit KV Cache

- 作者: D. Du; Shijie Cao; Jianyi Cheng; Luo Mai; Ting Cao; Mao Yang
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: BitDecoding: Unlocking Tensor Cores for Long-Context LLMs with Low-Bit KV Cache

快速理解: The rise of long-context Large Language Models (LLMs) amplifies memory and bandwidth demands during autoregressive decoding, as the Key-Value (KV) cache grows with each generated token. Low-bit KV-cache quantization (e.g., 4-bit or 2-bit) can reduce memory footprint while preserving accuracy, but existing systems suffer from slow decoding due to their exclusive reliance on CUDA cores, neglecting Tensor Cores—the primary source of compute on modern GPUs.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.11.45 EDA: Energy-Efficient Inter-Layer Model Compilation for Edge DNN Inference Acceleration

- 作者: Bo-Ren Pao; I-Chia Chen; E. Chang; Tsung-Tai Yeh
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: EDA: Energy-Efficient Inter-Layer Model Compilation for Edge DNN Inference Acceleration

快速理解: Modern handheld devices often employ neural processing units (NPUs) to accelerate deep neural network (DNN) inference applications. Unlike the AI accelerator of a data center, the NPU of an edge device has strict price and energy budgets.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.11.46 CROSS: Compiler-Driven Optimization of Sparse DNNs Using Sparse/Dense Computation Kernels

- 作者: Fangxin Liu; Shiyuan Huang; Ning Yang; Zongwu Wang; Haomin Li; Li Jiang
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: CROSS: Compiler-Driven Optimization of Sparse DNNs Using Sparse/Dense Computation Kernels

快速理解: As deep learning models continue to grow larger and more complex, exploiting sparsity is becoming one of the most critical areas for enhancing efficiency and scalability. Several methods for leveraging sparsity have been proposed to more effectively balance the trade-off between compression ratio and accuracy.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.11.47 AQPIM: Breaking the PIM Capacity Wall for LLMs with in-Memory Activation Quantization

- 作者: Kosuke Matsushima; Yasuyuki Okoshi; Masato Motomura; Daichi Fujiki
- 年份: 2026; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: AQPIM: Breaking the PIM Capacity Wall for LLMs with in-Memory Activation Quantization

快速理解: Processing-in-Memory (PIM) architectures offer a promising solution to the memory bottlenecks in data-intensive machine learning, yet often overlook the growing challenge of activation memory footprint. Conventional PIM approaches struggle with massive KV cache sizes generated in long-context scenarios by Transformer-based models, frequently exceeding PIM's limited memory capacity, while techniques like sparse attention can conflict with PIM's need for data locality.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.11.48 Adyna: Accelerating Dynamic Neural Networks with Adaptive Scheduling

- 作者: Zhiyao Li; Bohan Yang; Jiaxiang Li; Taijie Chen; Xintong Li; Mingyu Gao
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Adyna: Accelerating Dynamic Neural Networks with Adaptive Scheduling

快速理解: Dynamic architecture neural networks (DynNNs) are an emerging type of deep learning models that can leverage different processing difficulties of data samples to dynamically reduce computation demands at runtime. However, current GPUs and specialized accelerators lack the necessary architecture and dataflow support to achieve the promised theoretical efficiency improvements due to the high dynamism in DynNN execution.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.11.49 Focus: A Streaming Concentration Architecture for Efficient Vision-Language Models

- 作者: Chi-Yue Wei; Cong Guo; Jun-Yao Zhang; Haoxuan Shan; Yifan Xu; Ziyue Zhang; Yudong Liu; Qinsi Wang; Changchun Zhou; Hai Li; Yi-Ran Chen
- 年份: 2025; 主题: AI Infra; Supporting Perception; 优先级: 重要
- [来源](#)

论文: Focus: A Streaming Concentration Architecture for Efficient Vision-Language Models

快速理解: Vision-Language Models (VLMs) have demonstrated strong performance on tasks such as video captioning and visual question answering. However, their growing scale and video-level inputs lead to significant computational and memory overhead, posing challenges for real-time deployment on hardware accelerators.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.11.50 MERCURY: Accelerating DNN Training By Exploiting Input Similarity

- 作者: Vahid Janfaza; Kevin Weston; Moein Razavi; Shantanu Mandal; Farabi Mahmud; Alex Hilty; Abdullah Muzahid
- 年份: 2021; 主题: AI Infra; Systems; 优先级: 重要
- 来源

论文: MERCURY: Accelerating DNN Training By Exploiting Input Similarity

快速理解: Deep Neural Networks (DNN) are computationally intensive to train. It consists of a large number of multidimensional dot products between many weights and input vectors.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.12 SYSTEMS · ISCA

本次会议纳入 45 篇。自动主题统计为: AI Infra (44)、Systems (40)、Supporting Perception (5)、Agent RL (4)、Embodied/WBC (2)、Teleoperation (2)、VLA (1)、Robot Learning (1)。这些统计用于组织阅读顺序, 不能单独证明论文结论或实验优越性。

优先阅读候选: - DiTPA: A DiT-Based Action Planner Accelerator Exploiting Action-Denoising-Multimodality Redundancy for Embodied Artificial Intelligence (优先级分数 130.5) - Combating the Memory Walls: Optimization Pathways for Long-Context Agentic Llm Inference (优先级分数 89.057) - EdgePC: Efficient Deep Learning Analytics for Point Clouds on Edge Devices (优先级分数 80.75) - PowerWeave: Unlocking Energy-Efficient ML on GPUs with OS-Level Spatial Power Management (优先级分数 77.5) - Inter-layer Scheduling Space Definition and Exploration for Tiled Accelerators (优先级分数 73.365)

1.3.12.1 DiTPA: A DiT-Based Action Planner Accelerator Exploiting Action-Denoising-Multimodality Redundancy for Embodied Artificial Intelligence

- 作者: Xin Zhao; Longke Yan; Jiancong Li; Yongkun Wu; Fengbin Tu
- 年份: 2026; 主题: Embodied/WBC; VLA; AI Infra; Supporting Perception; Systems; 优先级: 核心
- 来源

论文: DiTPA: A DiT-Based Action Planner Accelerator Exploiting Action-Denoising-Multimodality Redundancy for Embodied Artificial Intelligence

快速理解: Recent advances in multimodal vision-languageaction (VLA) models have endowed embodied artificial intelligence (embodied AI) systems with remarkable perception, reasoning, and planning capabilities. Among these VLA models, diffusion transformers (DiTs) have become the backbone for action planning due to their strong and continuous generation capability.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 视觉、语言条件与机器人动作策略之间的对齐; 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.12.2 Combating the Memory Walls: Optimization Pathways for Long-Context Agentic Llm Inference

- 作者: Haoran Wu; Can Xiao; Jiayi Nie; Xuan Guo; Binglei Lou; Jeffrey T. H. Wong; Zhiwen Mo; Cheng Zhang; Przemyslaw Forys; Cheng-Yang Ai; T. Adeniran; Wayne Luk; Hongxiang Fan;

Jian-Yi Cheng; Timothy M. Jones; R. Antonova; Robert Mullins; Aaron Zhao

- 年份: 2025; 主题: Agent RL; AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: Combating the Memory Walls: Optimization Pathways for Long-Context Agentic Llm Inference

快速理解: Large Language Models (LLMs) serve as the core components of AI agents used across a wide range of applications, including enterprise workflow automation, software engineering, web automation, computer use, and research. These agentic LLM inference tasks are fundamentally different from traditional chatbot-focused inference—they often have much larger context lengths to capture complex, prolonged inputs, such as an entire webpage DOM or complicated tool call trajectories.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.12.3 EdgePC: Efficient Deep Learning Analytics for Point Clouds on Edge Devices

- 作者: Ziyu Ying; Sandeepa Bhuyan; Yan Kang; Yingtian Zhang; M. Kandemir; C. Das
- 年份: 2023; 主题: AI Infra; Supporting Perception; Systems; 优先级: 核心
- [来源](#)

论文: EdgePC: Efficient Deep Learning Analytics for Point Clouds on Edge Devices

快速理解: Recently, point cloud (PC) has gained popularity in modeling various 3D objects (including both synthetic and real-life) and has been extensively utilized in a wide range of applications such as AR/VR, 3D reconstruction, and autonomous driving. For such applications, it is critical to analyze/understand the surrounding scenes properly.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 3D、视频、多模态视觉或具身感知支撑能力; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.12.4 PowerWeave: Unlocking Energy-Efficient ML on GPUs with OS-Level Spatial Power Management

- 作者: Vasilis Kypriotis; Eric Dubberstein; Patrick H. Coppock; Eliot H. Solomon; Rayyan Zamir; T. Srimani; Dimitrios Skarlatos
- 年份: 2026; 主题: Agent RL; AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: PowerWeave: Unlocking Energy-Efficient ML on GPUs with OS-Level Spatial Power Management

快速理解: GPU power consumption has become a central systems challenge as modern AI datacenters increasingly rely on accelerators whose power footprints reach unprecedented scales. A single NVIDIA B200 already draws around 1000 W, making large multi-GPU nodes among the most power-hungry computing platforms ever deployed.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.12.5 Inter-layer Scheduling Space Definition and Exploration for Tiled Accelerators

- 作者：Jing-Wei Cai; Yuchen Wei; Zuotong Wu; Sen Peng; Kaisheng Ma
- 年份：2023；主题：AI Infra; Systems；优先级：核心
- [来源](#)

论文：Inter-layer Scheduling Space Definition and Exploration for Tiled Accelerators

快速理解：With the continuous expansion of the DNN accelerator scale, inter-layer scheduling, which studies the allocation of computing resources to each layer and the computing order of all layers in a DNN, plays an increasingly important role in maintaining a high utilization rate and energy efficiency of DNN inference accelerators. However, current inter-layer scheduling is mainly conducted based on some heuristic patterns.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.12.6 Gen-NeRF: Efficient and Generalizable Neural Radiance Fields via Algorithm-Hardware Co-Design

- 作者：Y. Fu; Zhi-Fan Ye; Jiayi Yuan; Shun Yao Zhang; Sixu Li; Haoran You; Yingyan Lin
- 年份：2023；主题：AI Infra; Supporting Perception; Systems；优先级：核心
- [来源](#)

论文：Gen-NeRF: Efficient and Generalizable Neural Radiance Fields via Algorithm-Hardware Co-Design

快速理解：Novel view synthesis is an essential functionality for enabling immersive experiences in various Augmented- and Virtual-Reality (AR/VR) applications, for which Neural Radiance Field (NeRF) has emerged as the state-of-the-art (SOTA) technique. In particular, generalizable NeRFs have gained increasing popularity thanks to their cross-scene generalization capability, which enables NeRFs to be instantly serviceable for new scenes without per-scene training.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；3D、视频、多模态视觉或具身感知支撑能力；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.12.7 Themis: a network bandwidth-aware collective scheduling policy for distributed training of DL models

- 作者：Saeed Rashidi; William Won; S. Srinivasan; Srinivas Sridharan; Tushar Krishna
- 年份：2021；主题：AI Infra; Systems；优先级：核心
- [来源](#)

论文：Themis: a network bandwidth-aware collective scheduling policy for distributed training of DL models

快速理解：Distributed training is a solution to reduce DNN training time by splitting the task across multiple NPUs (e.g., GPU/TPU). However, distributed training adds communication overhead

between the NPUs in order to synchronize the gradients and/or activation, depending on the parallelization strategy.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.12.8 REDUCT: Keep it Close, Keep it Cool! : Efficient Scaling of DNN Inference on Multi-core CPUs with Near-Cache Compute

- 作者：Anant V. Nori; Rahul Bera; S. Balachandran; Joydeep Rakshit; O. J. Omer; Avishai Abuhatzera; B. Kuttanna; S. Subramoney
- 年份：2021；主题：AI Infra; Systems；优先级：核心
- [来源](#)

论文：REDUCT: Keep it Close, Keep it Cool! : Efficient Scaling of DNN Inference on Multi-core CPUs with Near-Cache Compute

快速理解：Deep Neural Networks (DNN) are used in a variety of applications and services. With the evolving nature of DNNs, the race to build optimal hardware (both in datacenter and edge) continues.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.12.9 HPVM-HDC: A Heterogeneous Programming System for Accelerating Hyperdimensional Computing

- 作者：Russel Arbore; Xavier Routh; Abdul Rafae Noor; Akash Kothari; Haichao Yang; Weihong Xu; Sumukh Pinge; Minxuan Zhou; T. Rosing; V. Adve
- 年份：2024；主题：Teleoperation; AI Infra; Systems；优先级：核心
- [来源](#)

论文：HPVM-HDC: A Heterogeneous Programming System for Accelerating Hyperdimensional Computing

快速理解：Hyperdimensional Computing (HDC), a technique inspired by cognitive models of computation, has been proposed as an efficient and robust alternative basis for machine learning. HDC programs are often manually written in low-level and target specific languages targeting CPUs, GPUs, and FPGAs—these codes cannot be easily retargeted onto HDC-specific accelerators.

关注点：人类示范、远程操作、共享自治或动作重定向；训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.12.10 Disdp: Disaggregating Compute, Network, and Storage for Model-Sharded Data-Parallel Training

- 作者：Mo Sun; Zihan Yang; Changyue Liao; Ying-Tao Li; Jie Zhang; Kaiqi Chen; Fei Wu; Zeke Wang
- 年份：2024；主题：AI Infra; Systems；优先级：核心
- [来源](#)

论文: Disdp: Disaggregating Compute, Network, and Storage for Model-Sharded Data-Parallel Training

快速理解: Model-sharded data parallelism (MSDP), e.g., ZeRO, evenly shards the model states across all GPUs, and thus has been widely adopted by LLM pre-training, such as Llama and DeepSeek, due to its low GPU memory capacity requirement. However, MSDP introduces severe overhead from additional network communication collectives (i.e., AllGather and ReduceScatter).

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.12.11 PowerGrad: Hierarchical Power Management for Power-Limited ML Inference Clusters

- 作者: Hyounghook Nam; Raghavendra Pradyumna Pothukuchi; A. Buyuktosunoglu; Aporva Amarnath; P. Bose; J. Torrellas
- 年份: 2026; 主题: Teleoperation; AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: PowerGrad: Hierarchical Power Management for Power-Limited ML Inference Clusters

快速理解: As machine learning (ML) workloads demand more power and datacenters integrate renewable energy, workloads have to deal with situations where power demands exceed supply. In such situations, intelligently allocating the power among the nodes is key to maximizing efficiency.

关注点: 人类示范、远程操作、共享自治或动作重定向; 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.12.12 Splitwise: Efficient Generative LLM Inference Using Phase Splitting

- 作者: P. Patel; Esha Choukse; Chaojie Zhang; Aashaka Shah; Í. Goiri; Saeed Maleki; Ricardo Bianchini
- 年份: 2023; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: Splitwise: Efficient Generative LLM Inference Using Phase Splitting

快速理解: Generative large language model (LLM) applications are growing rapidly, leading to large-scale deployments of expensive and power-hungry GPUs. Our characterization of LLM inference shows that each inference request undergoes two phases: a compute-intensive prompt computation phase and a memory intensive token generation phase, each with distinct latency, throughput, memory, and power characteristics.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.12.13 KernelEvolve: Scaling Agentic Kernel Coding for Heterogeneous AI Accelerators at Meta

- 作者: Gang Liao; Hongsen Qin; Ying Wang; Alicia Golden; Michael Kuchnik; Yavuz Yetim; Rui Xiao; Jia Jiunn Ang; Chunli Fu; Yihan He; Samuel Hsia; Zewei Jiang; Roman Levenstein; Dian

Li; Li-Yuan Li; Ajit Mathews; Varna Puvvada; F. Shi; Na Yan; Xiayu Yu; Uladzimir Pashkevich; Matt Steiner; Carole-Jean Wu; Gaoxiang Liu

- 年份: 2026; 主题: Agent RL; AI Infra; 优先级: 核心
- [来源](#)

论文: KernelEvolve: Scaling Agentic Kernel Coding for Heterogeneous AI Accelerators at Meta

快速理解: Making deep learning recommendation model (DLRM) inference fast and efficient is important. However, this presents three key system challenges - model architecture diversity, kernel primitive diversity, and hardware generation and architecture heterogeneity.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.12.14 Omni-LUT: Energy-Efficient LUT-Based Accelerator with Hardware-Aware KV Cache Quantization

- 作者: Cheng-Han Tsai; Kuan-Chen Chou; Yu-Hsin Wang; Chieh-Dun Wen; Tsung-Tai Yeh
- 年份: 2026; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: Omni-LUT: Energy-Efficient LUT-Based Accelerator with Hardware-Aware KV Cache Quantization

快速理解: Large language model (LLM) inference incurs substantial computation and energy consumption. Lookup-table (LUT)-based general matrix multiplication (GEMM) accelerators reduce this burden by replacing costly multiplications with table lookups.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.12.15 CraterLake: a hardware accelerator for efficient unbounded computation on encrypted data

- 作者: Nikola Samardžić; Axel Feldmann; Aleksandar Krastev; Nathan Manohar; Nicholas Genise; S. Devadas; Karim Eldefrawy; Chris Peikert; Daniel Sánchez
- 年份: 2022; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: CraterLake: a hardware accelerator for efficient unbounded computation on encrypted data

快速理解: Fully Homomorphic Encryption (FHE) enables offloading computation to untrusted servers with cryptographic privacy. Despite its attractive security, FHE is not yet widely adopted due to its prohibitive overheads, about 10,000X over unencrypted computation.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.12.16 Sibyl: adaptive and extensible data placement in hybrid storage systems using online reinforcement learning

- 作者: Gagandeep Singh; Rakesh Nadig; Jisung Park; Rahul Bera; Nastaran Hajinazar; David Novo; Juan Gómez-Luna; S. Stuijk; H. Corporaal; Onur Mutlu

- 年份: 2022; 主题: Agent RL; Systems; 优先级: 核心
- [来源](#)

论文: Sibyl: adaptive and extensible data placement in hybrid storage systems using online reinforcement learning

快速理解: Hybrid storage systems (HSS) use multiple different storage devices to provide high and scalable storage capacity at high performance. Data placement across different devices is critical to maximize the benefits of such a hybrid system.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.12.17 Instant-3D: Instant Neural Radiance Field Training Towards On-Device AR/VR 3D Reconstruction

- 作者: Sixu Li; Chaojian Li; Wenbo Zhu; Bo Yu; Yang Zhao; Cheng Wan; Haoran You; Huihong Shi; Yingyan Lin
- 年份: 2023; 主题: AI Infra; Supporting Perception; 优先级: 重要
- [来源](#)

论文: Instant-3D: Instant Neural Radiance Field Training Towards On-Device AR/VR 3D Reconstruction

快速理解: Neural Radiance Field (NeRF) based 3D reconstruction is highly desirable for immersive Augmented and Virtual Reality (AR/VR) applications, but achieving instant (i.e., < 5 seconds) on-device NeRF training remains a challenge. In this work, we first identify the inefficiency bottleneck: the need to interpolate NeRF embeddings up to 200,000 times from a 3D embedding grid during each training iteration.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.12.18 Understanding data storage and ingestion for large-scale deep recommendation model training: industrial product

- 作者: Mark Zhao; Niket Agarwal; Aarti Basant; B. Gedik; S. Pan; Muhammet Mustafa Ozdal; Rakesh Komuravelli; Jerry Y. Pan; Tianshu Bao; Haowei Lu; Sundaram Narayanan; Jack Langman; Kevin Wilfong; Harsha Rastogi; Carole-Jean Wu; Christos Kozyrakis; Parikshit Pol
- 年份: 2021; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Understanding data storage and ingestion for large-scale deep recommendation model training: industrial product

快速理解: Datacenter-scale AI training clusters consisting of thousands of domain-specific accelerators (DSA) are used to train increasingly-complex deep learning models. These clusters rely on a data storage and ingestion (DSI) pipeline, responsible for storing exabytes of training data and serving it at tens of terabytes per second.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.12.19 AMOS: enabling automatic mapping for tensor computations on spatial accelerators with hardware abstraction

- 作者: Size Zheng; Renze Chen; Anjiang Wei; Yicheng Jin; Qin Han; Liqiang Lu; Bingyang Wu; Xiuhong Li; Shengen Yan; Yun Liang
- 年份: 2022; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: AMOS: enabling automatic mapping for tensor computations on spatial accelerators with hardware abstraction

快速理解: Hardware specialization is a promising trend to sustain performance growth. Spatial hardware accelerators that employ specialized and hierarchical computation and memory resources have recently shown high performance gains for tensor applications such as deep learning, scientific computing, and data mining.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.12.20 Spy in the GPU-box: Covert and Side Channel Attacks on Multi-GPU Systems

- 作者: S. B. Dutta; Hoda Naghibijouybari; Arjun Gupta; Nael B. Abu-Ghazaleh; A. Márquez; K. Barker
- 年份: 2022; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Spy in the GPU-box: Covert and Side Channel Attacks on Multi-GPU Systems

快速理解: The deep learning revolution has been enabled in large part by GPUs, and more recently accelerators, which make it possible to carry out computationally demanding training and inference in acceptable times. As the size of machine learning networks and workloads continues to increase, multi-GPU machines have emerged as an important platform offered on High Performance Computing and cloud data centers.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.12.21 Crescent: taming memory irregularities for accelerating deep point cloud analytics

- 作者: Yu Feng; Gunnar Hammonds; Yiming Gan; Yuhao Zhu
- 年份: 2022; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Crescent: taming memory irregularities for accelerating deep point cloud analytics

快速理解: 3D perception in point clouds is transforming the perception ability of future intelligent machines. Point cloud algorithms, however, are plagued by irregular memory accesses, leading to massive inefficiencies in the memory sub-system, which bottlenecks the overall efficiency.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.12.22 LUT Tensor Core: A Software-Hardware Co-Design for LUT-Based Low-Bit LLM Inference

- 作者: Zhiwen Mo; Lei Wang; Jianyu Wei; Zhichen Zeng; Shijie Cao; Lingxiao Ma; Nai-Feng Jing; Ting Cao; Jilong Xue; Fan Yang; Mao Yang
- 年份: 2024; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: LUT Tensor Core: A Software-Hardware Co-Design for LUT-Based Low-Bit LLM Inference

快速理解: Large Language Model (LLM) inference becomes resource-intensive, prompting a shift toward low-bit model weights to reduce the memory footprint and improve efficiency. Such low-bit LLMs necessitate the mixed-precision matrix multiplication (mpGEMM), an important yet underexplored operation involving the multiplication of lower-precision weights with higher-precision activations.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.12.23 MECLA: Memory-Compute-Efficient LLM Accelerator with Scaling Sub-matrix Partition

- 作者: Yu-Bin Qin; Yang Wang; Zhi-Ren Zhao; Xiao-Long Yang; Yang Zhou; Shaojun Wei; Yang Hu; Shou-Yi Yin
- 年份: 2024; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: MECLA: Memory-Compute-Efficient LLM Accelerator with Scaling Sub-matrix Partition

快速理解: Large language models (LLMs) have been showing surprising performance in processing language tasks, bringing a new prevalence to deploy LLM from cloud to edge. However, being a scaling auto-regressive Transformer with a huge parameter amount and generating output one by one, LLM introduces overwhelming memory footprints and computation during its inference, especially from its linear layers.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.12.24 SARA: Scaling a Reconfigurable Dataflow Accelerator

- 作者: Yaqi Zhang; Nathan Zhang; Tian Zhao; Matthew Vili; M. Shahbaz; K. Olukotun
- 年份: 2021; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: SARA: Scaling a Reconfigurable Dataflow Accelerator

快速理解: The need for speed in modern data-intensive work-loads and the rise of “dark silicon” in the semiconductor industry are pushing for larger, faster, and more energy and area-efficient architectures, such as Reconfigurable Dataflow Accelerators (RDAs). Nevertheless, challenges remain in developing mechanisms to effectively utilize the compute power of these large-scale RDAs.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.12.25 Dadu-Corki: Algorithm-Architecture Co-Design for Embodied AI-powered Robotic Manipulation

- 作者: Yiyang Huang; Yu-Hsiang Hao; Bo Yu; Feng Yan; Yuxin Yang; Feng Min; Yinhe Han; Lin Ma; Shaoshan Liu; Qiang Liu; Yiming Gan
- 年份: 2024; 主题: Embodied/WBC; AI Infra; 优先级: 重要
- [来源](#)

论文: Dadu-Corki: Algorithm-Architecture Co-Design for Embodied AI-powered Robotic Manipulation

快速理解: Embodied AI robots have the potential to fundamentally improve the way human beings live and manufacture. Continued progress in the burgeoning field of using large language models to control robots depends critically on an efficient computing substrate, and this trend is strongly evident in manipulation tasks.

关注点: 具身任务中的感知、动作生成、全身协调或运动控制; 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.12.26 REIS: A High-Performance and Energy-Efficient Retrieval System with In-Storage Processing

- 作者: Kangqi Chen; A. Kakolyris; Rakesh Nadig; Manos Frouzakis; Nika Mansouri-Ghiasi; Yu Liang; Haiyu Mao; Jisung Park; Mohammad Sadrosadati; Onur Mutlu
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: REIS: A High-Performance and Energy-Efficient Retrieval System with In-Storage Processing

快速理解: Large Language Models (LLMs) face an inherent challenge: their knowledge is confined to the data that they have been trained on. This limitation, combined with the significant cost of retraining renders them incapable of providing up-to-date responses.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.12.27 Training personalized recommendation systems from (GPU) scratch: look forward not backwards

- 作者: Youngeun Kwon; Minsoo Rhu
- 年份: 2022; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Training personalized recommendation systems from (GPU) scratch: look forward not backwards

快速理解: Personalized recommendation models (RecSys) are one of the most popular machine learning workload serviced by hyperscalers. A critical challenge of training RecSys is its high memory capacity requirements, reaching hundreds of GBs to TBs of model size.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.12.28 ECSSD: Hardware/Data Layout Co-Designed In-Storage-Computing Architecture for Extreme Classification

- 作者: Si-Qi Li; Fengbin Tu; Liu Liu; Jilan Lin; Zheng Wang; Yangwook Kang; Yufei Ding; Yuan Xie
- 年份: 2023; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: ECSSD: Hardware/Data Layout Co-Designed In-Storage-Computing Architecture for Extreme Classification

快速理解: With the rapid growth of classification scale in deep learning systems, the final classification layer becomes extreme classification with a memory footprint exceeding the main memory capacity of the CPU or GPU. The emerging in-storage-computing technique offers an opportunity on account of the fact that SSD has enough storage capacity for the parameters of extreme classification.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.12.29 MAD-Max Beyond Single-Node: Enabling Large Machine Learning Model Acceleration on Distributed Systems

- 作者: Samuel Hsia; Alicia Golden; Bilge Acun; Newsha Ardalani; Zachary DeVito; Gu-Yeon Wei; David Brooks; Carole-Jean Wu
- 年份: 2023; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: MAD-Max Beyond Single-Node: Enabling Large Machine Learning Model Acceleration on Distributed Systems

快速理解: Training and deploying large-scale machine learning models is time-consuming, requires significant distributed computing infrastructures, and incurs high operational costs. Our analysis, grounded in real-world large model training on datacenter-scale infrastructures, reveals that 14~32% of all GPU hours are spent on communication with no overlapping computation.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.12.30 AI accelerator on IBM Telum processor: industrial product

- 作者: C. Lichtenau; A. Buyuktosunoglu; Ramon Bertran; Peter Figuli; C. Jacobi; N. Papandreou; H. Pozidis; A. Saporito; Andrew Sica; Elpida Tzortzatos
- 年份: 2022; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: AI accelerator on IBM Telum processor: industrial product

快速理解: IBM Telum is the next generation processor chip for IBM Z and LinuxONE systems. The Telum design is focused on enterprise class workloads and it achieves over 40% per socket performance growth compared to IBM z15.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.12.31 Hybe: GPU-NPU Hybrid System for Efficient LLM Inference with Million-Token Context Window

- 作者: Seungjae Moon; Junseo Cha; Hyunjun Park; Joo-Young Kim
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Hybe: GPU-NPU Hybrid System for Efficient LLM Inference with Million-Token Context Window

快速理解: The growth of context window size in large language model (LLM) inference poses a very distinct computational challenge of hardware inefficiency. The inefficiency arises from the computational imbalance during LLM inference between the compute-intensive prefill stage, and memory-intensive decode stage.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.12.32 AMALI: An Analytical Model for Accurately Modeling LLM Inference on Modern GPUs

- 作者: Shiheng Cao; Junmin Wu; Junshi Chen; Hong An; Zhibin Yu
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: AMALI: An Analytical Model for Accurately Modeling LLM Inference on Modern GPUs

快速理解: Large language model (LLM) inference applications are surging in recent years, which largely relies on modern GPUs. On the other hand, GPU analytical model is a commonly used tool for architects to precisely identify bottlenecks quickly with deep insights.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.12.33 PreSto: An In-Storage Data Preprocessing System for Training Recommendation Models

- 作者: Yunjae Lee; Hyeseong Kim; Minsoo Rhu
- 年份: 2024; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: PreSto: An In-Storage Data Preprocessing System for Training Recommendation Models

快速理解: Training recommendation systems (RecSys) faces several challenges as it requires the “data preprocessing” stage to preprocess an ample amount of raw data and feed them to the GPU for training in a seamless manner. To sustain high training throughput, state-of-the-art solutions reserve a large fleet of CPU servers for preprocessing which incurs substantial deployment cost and power consumption.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.12.34 P^3 -LLM: An Integrated NPU-PIM Accelerator for Edge LLM Inference Using Hybrid Numerical Formats

- 作者: Yuzong Chen; Chao Fang; Xilai Dai; Yuheng Wu; Thierry Tambe; Marian Verhelst; Mohamed S. Abdelfattah
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: P^3 -LLM: An Integrated NPU-PIM Accelerator for Edge LLM Inference Using Hybrid Numerical Formats

快速理解: The substantial memory bandwidth and computational demands of large language models (LLMs) present critical challenges for efficient inference. To tackle this, the literature has explored heterogeneous systems that combine neural processing units (NPUs) with DRAM-based processing-in-memory (PIM) for LLM acceleration.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.12.35 Lit Silicon: A Case Where Thermal Imbalance Couples Concurrent Execution in Multiple GPUs

- 作者: Marco Kurzynski; Shaizeen Aga; Di Wu
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Lit Silicon: A Case Where Thermal Imbalance Couples Concurrent Execution in Multiple GPUs

快速理解: GPU systems are increasingly powering modern datacenters at scale. Despite being highly performant, GPU systems can exhibit performance variation at the node and cluster levels.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.12.36 TrioSim: A Lightweight Simulator for Large-Scale DNN Workloads on Multi-GPU Systems

- 作者: Ying Li; Yuhui Bao; Gongyu Wang; Xinxin Mei; Pranav Vaid; Anandaroop Ghosh; Adwait Jog; Darius Bunandar; Ajay Joshi; Yi-Fan Sun
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: TrioSim: A Lightweight Simulator for Large-Scale DNN Workloads on Multi-GPU Systems

快速理解: Deep Neural Networks (DNNs) have become increasingly capable of performing tasks ranging from image recognition to content generation. The training and inference of DNNs heavily rely on GPUs, as GPUs' massively parallel architecture delivers extremely high computing capability.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.12.37 HiPER: Hierarchically-Composed Processing for Efficient Robot Learning-Based Control

- 作者: Justin Ting; Minsik Kim; Junkang Zhu; Haotian Sheng; Zheng-Ya Zhang
- 年份: 2025; 主题: Robot Learning; AI Infra; 优先级: 重要
- [来源](#)

论文: HiPER: Hierarchically-Composed Processing for Efficient Robot Learning-Based Control

快速理解: Learning-Based Model Predictive Control (LMPC) is a class of algorithms that enhances Model Predictive Control (MPC) by including machine learning methods, improving robot navigation in complex environments. However, the combination of machine learning and MPC computation in LMPC creates a unique workload that cannot be efficiently handled by a simple GPU and CPU integration.

关注点: 机器人数据、策略学习、操作、触觉或 sim-to-real; 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.12.38 NASGuard: A Novel Accelerator Architecture for Robust Neural Architecture Search (NAS) Networks

- 作者: Xing-Bin Wang; Boyan Zhao; Rui Hou; Amro Awad; Zhihong Tian; Dan Meng
- 年份: 2021; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: NASGuard: A Novel Accelerator Architecture for Robust Neural Architecture Search (NAS) Networks

快速理解: Due to the wide deployment of deep learning applications in safety-critical systems, robust and secure execution of deep learning workloads is imperative. Adversarial examples, where the inputs are carefully designed to mislead the machine learning model is among the most challenging attacks to detect and defeat.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.12.39 Early Silicon of Raptor: The First 3D-DRAM Accelerator for Generative Inference

- 作者: Prashant J. Nair; Ramyad Hadidi; Subramani Ganesh; Sangamesh Kodge; Shubhankit Rathore; Neil Thanawala; Nikitha Reddy; Gyanesh Saharia; Vinayak Patankar; Arun Tiruvur; Nithesh Kurella; Sudeep Bhoja
- 年份: 2026; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Early Silicon of Raptor: The First 3D-DRAM Accelerator for Generative Inference

快速理解: Generative Inference is largely memory-bound. Autoregressive decoding dominates inference runtime and thus drives memory bandwidth and capacity demands.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.12.40 LightML: A Photonic Accelerator for Efficient General Purpose Machine Learning

- 作者: Liang Liu; S. R. Kari; Xin Xin; N. Youngblood; You-Tao Zhang; Jun Yang
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: LightML: A Photonic Accelerator for Efficient General Purpose Machine Learning

快速理解: The rapid integration of AI technologies into everyday life across sectors such as health-care, autonomous driving, and smart home applications requires extensive computational resources, placing strain on server infrastructure and incurring significant costs. We present LightML, the first system-level photonic crossbar design, optimized for high-performance machine learning applications.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.12.41 The Sparsity-Aware LazyGPU Architecture

- 作者: Changxi Liu; Miao Yu; Yi-Fan Sun; Trevor E. Carlson
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: The Sparsity-Aware LazyGPU Architecture

快速理解: General-Purpose Graphics Processing Units (GPUs) are essential accelerators in data-parallel applications, including machine learning, and physical simulations. Although GPUs utilize fast wavefront context switching to hide memory access latency, memory continues to be a significant bottleneck, limiting overall performance for many important workloads.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.12.42 AQuant: Repurposing CODEC for VLM Acceleration via Adaptive Quantization

- 作者: Zhuoran Song; Chunyu Qi; Jian Weng; Xiaoyao Liang; Haibing Guan
- 年份: 2026; 主题: AI Infra; Supporting Perception; 优先级: 重要
- [来源](#)

论文: AQuant: Repurposing CODEC for VLM Acceleration via Adaptive Quantization

快速理解: Vision-Language Models (VLMs) have reached the forefront of accuracy in various vision understanding tasks. Despite their remarkable success, the computing costs of VLMs scale significantly with the high image resolutions or the increasing number of video frames that need to be processed, posing substantial challenges for deployment to real-time applications.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.12.43 ATX: Accelerator Task Extensions

- 作者: Gerasimos Gerogiannis; Stijn Eyerman; J. Torrellas; Wim Heirman
- 年份: 2026; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: ATX: Accelerator Task Extensions

快速理解: Integrating accelerators in CPU multicores combines the benefits of accelerated computation with the flexibility and programmability of CPUs. CPU-integrated accelerators can be classified into In-Core Accelerators (ICAs), which typically reside inside the core's pipeline, and Out-of-Core Accelerators (OCAs), which are typically attached at the core's cache subsystem.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.12.44 DarkStream: Exploiting Internal Throughput Contention in Data Streaming Accelerator for Timing Attacks

- 作者: Hyosang Kim; Ki-Dong Kang; Gyeongseo Park; Sungju Kim; Daehoon Kim
- 年份: 2026; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: DarkStream: Exploiting Internal Throughput Contention in Data Streaming Accelerator for Timing Attacks

快速理解: Modern Intel processors integrate a variety of off-core accelerators to enhance data movement efficiency for memory-intensive workloads in data centers. Among them, the Data Streaming Accelerator (DSA) offloads repetitive memory operations such as copying and integrity checking to improve performance and energy efficiency.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.12.45 PipeComm: Maximizing Link Utilization Through Pipeline-Aware Collective Communication Synthesis

- 作者: Ruifan Xu; Yu-Ze Luo; Yu-Hao Meng; Size Zheng; Meng Li; Y. Liang
- 年份: 2026; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: PipeComm: Maximizing Link Utilization Through Pipeline-Aware Collective Communication Synthesis

快速理解: Efficient collective communication is crucial for distributed training. While recent topology-aware synthesis approaches attempt to optimize communication based on the network topology, they struggle with heterogeneous environments where links exhibit distinct bandwidths and asymmetric connections.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.13 SYSTEMS · MICRO

本次会议纳入 34 篇。自动主题统计为: Systems (34)、AI Infra (33)、Supporting Perception (2)、Agent RL (2)、Robot Learning (1)、Teleoperation (1)。这些统计用于组织阅读顺序, 不能单独证明论文结论或实验优越性。

优先阅读候选：- Fusion-3D: Integrated Acceleration for Instant 3D Reconstruction and Real-Time Rendering (优先级分数 80.17) - Cohmeleon: Learning-Based Orchestration of Accelerator Coherence in Heterogeneous SoCs (优先级分数 80.108) - Cambricon-R: A Fully Fused Accelerator for Real-Time Learning of Neural Scene Representation (优先级分数 79.998) - Pushing the Performance Envelope of DNN-based Recommendation Systems Inference on GPUs (优先级分数 78.807) - TAIDL: Tensor Accelerator ISA Definition Language with Auto-generation of Scalable Test Oracles(优先级分数 78.572)

1.3.13.1 Fusion-3D: Integrated Acceleration for Instant 3D Reconstruction and Real-Time Rendering

- 作者: Sixu Li; Yang Zhao; Chaojian Li; Bo-Wei Guo; Jingqun Zhang; Wenbo Zhu; Zhi-Fan Ye; Cheng Wan; Y. Lin
- 年份: 2024; 主题: AI Infra; Supporting Perception; Systems; 优先级: 核心
- [来源](#)

论文: Fusion-3D: Integrated Acceleration for Instant 3D Reconstruction and Real-Time Rendering

快速理解: Recent breakthroughs in Neural Radiance Field (NeRF) based 3D reconstruction and rendering have spurred the possibility of immersive experiences in augmented and virtual reality (AR/VR). However, current NeRF acceleration techniques are still inadequate for real-world AR/VR applications due to: 1) the lack of end-to-end pipeline acceleration support, which causes impractical off-chip bandwidth demands for edge devices, and 2) limited scalability in handling large-scale scenes.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 3D、视频、多模态视觉或具身感知支撑能力; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.13.2 Cohmeleon: Learning-Based Orchestration of Accelerator Coherence in Heterogeneous SoCs

- 作者: Joseph Zuckerman; Davide Giri; Jihye Kwon; Paolo Mantovani; L. Carloni
- 年份: 2021; 主题: Agent RL; AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: Cohmeleon: Learning-Based Orchestration of Accelerator Coherence in Heterogeneous SoCs

快速理解: One of the most critical aspects of integrating loosely-coupled accelerators in heterogeneous SoC architectures is orchestrating their interactions with the memory hierarchy, especially in terms of navigating the various cache-coherence options: from accelerators accessing off-chip memory directly, bypassing the cache hierarchy, to accelerators having their own private cache. By running real-size applications on FPGA-based prototypes of many-accelerator multi-core SoCs, we show that the best cache-coherence mode for a given accelerator varies at runtime, depending on the accelerator's characteristics, the workload size, and the overall SoC status.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.13.3 Cambricon-R: A Fully Fused Accelerator for Real-Time Learning of Neural Scene Representation

- 作者: Xinkai Song; Yuanbo Wen; Xin-Gui Hu; Tianbo Liu; Haoxuan Zhou; Husheng Han; Tian Zhi; Zidong Du; Wei Li; Rui Zhang; Chen Zhang; Lin Gao; Qi Guo; Tianshi Chen
- 年份: 2023; 主题: AI Infra; Supporting Perception; Systems; 优先级: 核心
- [来源](#)

论文: Cambricon-R: A Fully Fused Accelerator for Real-Time Learning of Neural Scene Representation

快速理解: Neural scene representation (NSR) initiates a new methodology of encoding a 3D scene with neural networks by learning from dozens of photos taken from different camera positions. NSR not only achieves significant improvement in the quality of novel view synthesis and 3D reconstruction but also reduces the camera cost from the expensive laser cameras to the cheap color cameras on the shelf.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 3D、视频、多模态视觉或具身感知支撑能力; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.13.4 Pushing the Performance Envelope of DNN-based Recommendation Systems Inference on GPUs

- 作者: Rishabh Jain; Vivek M. Bhasi; Adwait Jog; A. Sivasubramaniam; M. Kandemir; Chita R. Das
- 年份: 2024; 主题: Robot Learning; AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: Pushing the Performance Envelope of DNN-based Recommendation Systems Inference on GPUs

快速理解: Personalized recommendation is a ubiquitous application on the internet, with many industries and hyperscalers extensively leveraging Deep Learning Recommendation Models (DLRMs) for their personalization needs (like ad serving or movie suggestions). With growing model and dataset sizes pushing computation and memory requirements, GPUs are being increasingly preferred for executing DLRM inference.

关注点: 机器人数据、策略学习、操作、触觉或 sim-to-real; 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.13.5 TAIDL: Tensor Accelerator ISA Definition Language with Auto-generation of Scalable Test Oracles

- 作者: Devansh Jain; Marco Frigo; Jai Arora; Akash Pardeshi; Zhihao Wang; Krut Patel; Charith Mendis
- 年份: 2025; 主题: Teleoperation; AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: TAIDL: Tensor Accelerator ISA Definition Language with Auto-generation of Scalable Test Oracles

快速理解: With the increasing importance of deep learning workloads, many hardware accelerators have been proposed in both academia and industry. However, software tooling for the vast majority

of them does not exist compared to the software ecosystem and innovations proposed for established platforms such as CPUs and GPUs.

关注点：人类示范、远程操作、共享自治或动作重定向；训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.13.6 PointAcc: Efficient Point Cloud Accelerator

- 作者：Yujun Lin; Zhekai Zhang; Haotian Tang; Hanrui Wang; Song Han
- 年份：2021；主题：AI Infra; Systems；优先级：核心
- [来源](#)

论文：PointAcc: Efficient Point Cloud Accelerator

快速理解：Deep learning on point clouds plays a vital role in a wide range of applications such as autonomous driving and AR/VR. These applications interact with people in real time on edge devices and thus require low latency and low energy.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.13.7 G10: Enabling An Efficient Unified GPU Memory and Storage Architecture with Smart Tensor Migrations

- 作者：Haoyang Zhang; Y. Zhou; Yu Xue; Yiqi Liu; Jian Huang
- 年份：2023；主题：AI Infra; Systems；优先级：核心
- [来源](#)

论文：G10: Enabling An Efficient Unified GPU Memory and Storage Architecture with Smart Tensor Migrations

快速理解：To break the GPU memory wall for scaling deep learning workloads, a variety of architecture and system techniques have been proposed recently. Their typical approaches include memory extension with flash memory and direct storage access.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.13.8 Chameleon: Adaptive Caching and Scheduling for Many-Adapter LLM Inference Environments

- 作者：Nikoleta Iliakopoulou; Jovan Stojkovic; Chloe Alverti; Tianyin Xu; Hubertus Franke; J. Torrellas
- 年份：2024；主题：AI Infra; Systems；优先级：核心
- [来源](#)

论文：Chameleon: Adaptive Caching and Scheduling for Many-Adapter LLM Inference Environments

快速理解：The effectiveness of LLMs has triggered an exponential rise in their deployment, imposing substantial demands on inference clusters. Such clusters often handle numerous concurrent queries for different LLM downstream tasks.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.13.9 MAICC : A Lightweight Many-core Architecture with In-Cache Computing for Multi-DNN Parallel Inference

- 作者: Renhao Fan; Yikai Cui; Qilin Chen; Mingyu Wang; Youhui Zhang; Wei-Min Zheng; Zhao-Lin Li
- 年份: 2023; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: MAICC : A Lightweight Many-core Architecture with In-Cache Computing for Multi-DNN Parallel Inference

快速理解: The growing complexity and diversity of neural networks in the fields of autonomous driving and intelligent robots have facilitated the research of many-core architectures, which can offer sufficient programming flexibility to simultaneously support multi-DNN parallel inference with different network structures and sizes compared to domain-specific architectures. However, due to the tight constraints of area and power consumption, many-core architectures typically use lightweight scalar cores without vector units and are almost unable to meet the high-performance computing needs of multi-DNN parallel inference.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.13.10 Crane: Inter-Layer Scheduling Framework for DNN Inference and Training Co-Support on Tiled Architecture

- 作者: Yu Gong; Lingyi Huang; Haodong Chang; Rongjian Liang; Cheng Yang; Zhexiang Tang; Jiang Hu; Bo Yuan
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: Crane: Inter-Layer Scheduling Framework for DNN Inference and Training Co-Support on Tiled Architecture

快速理解: Tiled architectures have emerged as a compelling platform for scaling deep neural network (DNN) execution, offering both compute density and communication efficiency. To harness their full potential, effective inter-layer scheduling is crucial for managing operation order, memory behavior, and compute resource coordination.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.13.11 F1: A Fast and Programmable Accelerator for Fully Homomorphic Encryption

- 作者: Axel S. Feldmann; Nikola Samardžić; Aleksandar Krastev; S. Devadas; Ron Dreslinski; Karim Eldefrawy; Nicholas Genise; Chris Peikert; Daniel Sánchez
- 年份: 2021; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: F1: A Fast and Programmable Accelerator for Fully Homomorphic Encryption

快速理解: Fully Homomorphic Encryption (FHE) allows computing on encrypted data, enabling secure offloading of computation to untrusted servers. Though it provides ideal security, FHE is expensive when executed in software, 4 to 5 orders of magnitude slower than computing on unencrypted data.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.13.12 Micro-Armed Bandit: Lightweight & Reusable Reinforcement Learning for Microarchitecture Decision-Making

- 作者: Gerasimos Gerogiannis; J. Torrellas
- 年份: 2023; 主题: Agent RL; Systems; 优先级: 重要
- [来源](#)

论文: Micro-Armed Bandit: Lightweight & Reusable Reinforcement Learning for Microarchitecture Decision-Making

快速理解: Online Reinforcement Learning (RL) has been adopted as an effective mechanism in various decision-making problems in microarchitecture. Its high adaptability and the ability to learn at runtime are attractive characteristics in microarchitecture settings.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.13.13 TorchSparse++: Efficient Training and Inference Framework for Sparse Convolution on GPUs

- 作者: Haotian Tang; Shang Yang; Zhijian Liu; Ke Hong; Zhongming Yu; Xiuyu Li; Guohao Dai; Yu Wang; Song Han
- 年份: 2023; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: TorchSparse++: Efficient Training and Inference Framework for Sparse Convolution on GPUs

快速理解: Sparse convolution plays a pivotal role in emerging workloads, including point cloud processing in AR/VR, autonomous driving, and graph understanding in recommendation systems. Since the computation pattern is sparse and irregular, specialized high-performance kernels are required.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.13.14 Point-X: A Spatial-Locality-Aware Architecture for Energy-Efficient Graph-Based Point-Cloud Deep Learning

- 作者: Jie-Fang Zhang; Zhengya Zhang
- 年份: 2021; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Point-X: A Spatial-Locality-Aware Architecture for Energy-Efficient Graph-Based Point-Cloud Deep Learning

快速理解: Deep learning on point clouds has attracted increasing attention in the fields of 3D computer vision and robotics. In particular, graph-based point-cloud deep neural networks (DNNs) have demonstrated promising performance in 3D object classification and scene segmentation tasks.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.13.15 Eureka: Efficient Tensor Cores for One-sided Unstructured Sparsity in DNN Inference

- 作者: Ashish Gondimalla; M. Thottethodi; T. Vijaykumar
- 年份: 2023; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Eureka: Efficient Tensor Cores for One-sided Unstructured Sparsity in DNN Inference

快速理解: Deep neural networks (DNNs), while enormously popular, continue to place ever higher compute demand for which GPUs provide specialized matrix multipliers called tensor cores. To reduce the compute demand via sparsity, Nvidia Ampere' s tensor cores support 2:4 structured sparsity in the filters (i.e., two non-zeros out of four values) which provides uniform 50% sparsity without any load imbalance issues.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.13.16 Coruscant: Co-Designing GPU Kernel and Sparse Tensor Core to Advocate Unstructured Sparsity in Efficient LLM Inference

- 作者: Donghyeon Joo; Helya Hosseini; Ramyad Hadidi; Bahar Asgari
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Coruscant: Co-Designing GPU Kernel and Sparse Tensor Core to Advocate Unstructured Sparsity in Efficient LLM Inference

快速理解: In the era of large language models (LLMs) and long-context generation, model compression techniques such as pruning, quantization, and distillation offer effective ways to reduce memory usage. Among them, pruning is constrained by the difficulty of exploiting unstructured sparsity on modern hardware.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.13.17 ESCALATE: Boosting the Efficiency of Sparse CNN Accelerator with Kernel Decomposition

- 作者: Shiyu Li; Edward Hanson; Xuehai Qian; H. Li; Yiran Chen
- 年份: 2021; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: ESCALATE: Boosting the Efficiency of Sparse CNN Accelerator with Kernel Decomposition

快速理解: The ever-growing parameter size and computation cost of Convolutional Neural Network

(CNN) models hinder their deployment onto resource-constrained platforms. Network pruning techniques are proposed to remove the redundancy in CNN parameters and produce a sparse model.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.13.18 MX+: Pushing the Limits of Microscaling Formats for Efficient Large Language Model Serving

- 作者：Jungi Lee; Junyong Park; S. Cha; Jaehoon Cho; Jaewoong Sim
- 年份：2025；主题：AI Infra; Systems；优先级：重要
- [来源](#)

论文：MX+: Pushing the Limits of Microscaling Formats for Efficient Large Language Model Serving

快速理解：Reduced-precision data formats are crucial for cost-effective serving of large language models (LLMs). While numerous reduced-precision formats have been introduced thus far, they often require intrusive modifications to the software frameworks or are rather unconventional for widespread adoption across hardware vendors.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.13.19 Point Cloud Acceleration by Exploiting Geometric Similarity

- 作者：Cen Chen; Xiaofeng Zou; Hongen Shao; Yangfan Li; Kenli Li
- 年份：2023；主题：AI Infra; Systems；优先级：重要
- [来源](#)

论文：Point Cloud Acceleration by Exploiting Geometric Similarity

快速理解：Deep learning on point clouds has attracted increasing attention for various emerging 3D computer vision applications, such as autonomous driving, robotics, and virtual reality. These applications interact with people in real-time on edge devices and thus require low latency and low energy.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.13.20 SMART: A Heterogeneous Scratchpad Memory Architecture for Superconductor SFQ-based Systolic CNN Accelerators

- 作者：Farzaneh Zokaee; Lei Jiang
- 年份：2021；主题：AI Infra; Systems；优先级：重要
- [来源](#)

论文：SMART: A Heterogeneous Scratchpad Memory Architecture for Superconductor SFQ-based Systolic CNN Accelerators

快速理解：Ultra-fast & low-power superconductor single-flux-quantum (SFQ)-based CNN systolic accelerators are built to enhance the CNN inference throughput. However, shift-register (SHIFT)-based scratchpad memory (SPM) arrays prevent a SFQ CNN accelerator from exceeding 40% of its peak throughput, due to the lack of random access capability.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.13.21 Photon: A Fine-grained Sampled Simulation Methodology for GPU Workloads

- 作者：Changxi Liu; Yi-Fan Sun; Trevor E. Carlson
- 年份：2023；主题：AI Infra; Systems；优先级：重要
- [来源](#)

论文：Photon: A Fine-grained Sampled Simulation Methodology for GPU Workloads

快速理解：GPUs, due to their massively-parallel computing architectures, provide high performance for data-parallel applications. However, existing GPU simulators are too slow to enable architects to quickly evaluate their hardware designs and software analysis studies.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.13.22 MCBP: A Memory-Compute Efficient LLM Inference Accelerator Leveraging Bit-Slice-enabled Sparsity and Repetitiveness

- 作者：Hui-Zheng Wang; Zichuan Wang; Zhi-Heng Yue; Yousheng Long; Taiquan Wei; Jian-Xun Yang; Yang Wang; Chao Li; Shao-Jun Wei; Yang Hu; Shou-Yi Yin
- 年份：2025；主题：AI Infra; Systems；优先级：重要
- [来源](#)

论文：MCBP: A Memory-Compute Efficient LLM Inference Accelerator Leveraging Bit-Slice-enabled Sparsity and Repetitiveness

快速理解：Large language models (LLMs) face significant inference latency due to inefficiencies in GEMM operations, weight access, and KV cache access, especially in real-time scenarios. This highlights the need for a versatile compute-memory efficient accelerator.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.13.23 SparseAdapt: Runtime Control for Sparse Linear Algebra on a Reconfigurable Accelerator

- 作者：S. Pal; Aporva Amarnath; Siying Feng; M. O' Boyle; R. Dreslinski; Christophe Dubach
- 年份：2021；主题：AI Infra; Systems；优先级：重要
- [来源](#)

论文：SparseAdapt: Runtime Control for Sparse Linear Algebra on a Reconfigurable Accelerator

快速理解：Dynamic adaptation is a post-silicon optimization technique that adapts the hardware to workload phases. However, current adaptive approaches are oblivious to implicit phases that arise from operating on irregular data, such as sparse linear algebra operations.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.13.24 Hardware-Assisted Virtualization of Neural Processing Units for Cloud Platforms

- 作者: Yu Xue; Yiqi Liu; Lifeng Nai; Jian Huang
- 年份: 2024; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Hardware-Assisted Virtualization of Neural Processing Units for Cloud Platforms

快速理解: Cloud platforms today have been deploying hardware accelerators like neural processing units (NPUs) for powering machine learning (ML) inference services. To maximize the resource utilization while ensuring reasonable quality of service, a natural approach is to virtualize NPUs for efficient resource sharing for multi-tenant ML services.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.13.25 Optimizing All-to-All Collective Communication with Fault Tolerance on Torus Networks

- 作者: Le Qin; Junwei Cui; Weilin Cai; Meng Niu; Yan Yang; Jiayi Huang
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Optimizing All-to-All Collective Communication with Fault Tolerance on Torus Networks

快速理解: Large-scale distributed processing is extensively employed for large model inference and training, such as Deep Learning Recommendation Models (DLRMs) and Mixture-of-Experts (MoE) models. However, the All-to-All collective, with its complex point-to-point communication patterns and blocking nature, has become a major performance bottleneck in distributed DLRM and MoE accelerations.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.13.26 StreamTensor: Make Tensors Stream in Dataflow Accelerators for LLMs

- 作者: Hanchen Ye; Deming Chen
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: StreamTensor: Make Tensors Stream in Dataflow Accelerators for LLMs

快速理解: Efficient execution of deep learning workloads on dataflow architectures is crucial for overcoming memory bottlenecks and maximizing performance. While streaming intermediate results between computation kernels can significantly improve efficiency, existing approaches struggle with inter-kernel correlations, external memory access management, and buffer optimization.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.13.27 ORCHES: Orchestrated Test-Time-Compute-based LLM Reasoning on Collaborative GPU-PIM HETerogeneous System

- 作者: Sixu Li; Yuzhou Chen; Chaojian Li; Yonggan Fu; Zhengyi Wang; Zhongzhi Yu; Hao-Ran You; Zhi-Fan Ye; Wei Zhou; Yongan Zhang; Y. Lin
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: ORCHES: Orchestrated Test-Time-Compute-based LLM Reasoning on Collaborative GPU-PIM HEterogeneous System

快速理解: Recent breakthroughs in AI reasoning, enabled by test-time compute (TTC) on compact large language models (LLMs), offer great potential for edge devices to effectively execute complex reasoning tasks. However, the intricate inference pipelines associated with TTC pose new efficiency bottlenecks, limiting achievable latency and hindering widespread adoption.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.13.28 TT-GNN: Efficient On-Chip Graph Neural Network Training via Embedding Reformation and Hardware Optimization

- 作者: Zheng Qu; Dimin Niu; Shuangchen Li; Hong-Zhong Zheng; Yuan Xie
- 年份: 2023; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: TT-GNN: Efficient On-Chip Graph Neural Network Training via Embedding Reformation and Hardware Optimization

快速理解: Training Graph Neural Networks on large graphs is challenging due to the need to store graph data and move them along the memory hierarchy. In this work, we tackle this by effectively compressing graph embedding matrix such that the model training can be fully enabled with on-chip compute and memory resources.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.13.29 Equinox: Training (for Free) on a Custom Inference Accelerator

- 作者: M. Drumond; Louis Coulon; Arash Pourhabibi; Ahmet Caner Yüzügüler; B. Falsafi; Martin Jaggi
- 年份: 2021; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Equinox: Training (for Free) on a Custom Inference Accelerator

快速理解: DNN inference accelerators executing online services exhibit low average loads because of service demand variability, leading to poor resource utilization. Unfortunately, reclaiming idle inference cycles is difficult as other workloads can not execute on a custom accelerator.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.13.30 Grape: Practical and Efficient Graph-based Executions for Dynamic Deep Neural Networks on GPUs

- 作者: Bojian Zheng; C. Yu; Jie Wang; Yaoyao Ding; Yizhi Liu; Yida Wang; Gennady Pekhimenko
- 年份: 2023; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Grape: Practical and Efficient Graph-based Executions for Dynamic Deep Neural Networks on GPUs

快速理解: Achieving high performance in machine learning workloads is a crucial yet difficult task. To achieve high runtime performance on hardware platforms such as GPUs, graph-based executions such as CUDA graphs are often used to eliminate CPU runtime overheads by submitting jobs in the granularity of multiple kernels.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.13.31 MAD MAcce: Supporting Multiply-Add Operations for Democratizing Matrix-Multiplication Accelerators

- 作者: Seunghwan Sung; Sujin Hur; Sungwoo Kim; Dongho Ha; Yunho Oh; Won Woo Ro
- 年份: 2023; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: MAD MAcce: Supporting Multiply-Add Operations for Democratizing Matrix-Multiplication Accelerators

快速理解: Modern GPUs commonly employ specialized matrix multiplication units (MXUs) to accelerate matrix multiplication, the core computation of deep learning workloads. However, it is challenging to exploit the MXUs for GPGPU applications whose fundamental algorithms do not rely on matrix multiplication.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.13.32 HgPCN: A Heterogeneous Architecture for E2E Embedded Point Cloud Inference

- 作者: Yiming Gao; Chao Jiang; Wesley Piard; Xiangrui Chen; Bhavesh Patel; Herman Lam
- 年份: 2024; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: HgPCN: A Heterogeneous Architecture for E2E Embedded Point Cloud Inference

快速理解: Point cloud is an important type of geometric data structure for many embedded applications such as autonomous driving and augmented reality. Current Point Cloud Networks (PCNs) have proven to achieve great success in using inference to perform point cloud analysis, including object part segmentation, shape classification, and so on.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.13.33 ADA-GP: Accelerating DNN Training By Adaptive Gradient Prediction

- 作者: Vahid Janfaza; Shantanu Mandal; Farabi Mahmud; Abdullah Muzahid

- 年份: 2023; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: ADA-GP: Accelerating DNN Training By Adaptive Gradient Prediction

快速理解: Neural network training is inherently sequential where the layers finish the forward propagation in succession, followed by the calculation and back-propagation of gradients (based on a loss function) starting from the last layer. The sequential computations significantly slow down neural network training, especially the deeper ones.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.13.34 PolymorPIC: Embedding Polymorphic Processing-in-Cache in RISC-V based Processor for Full-stack Efficient AI Inference

- 作者: Chen Zou; Ziling Wei; Jun-Yan Lee; Chen Nie; Kang You; Zhezhi He
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: PolymorPIC: Embedding Polymorphic Processing-in-Cache in RISC-V based Processor for Full-stack Efficient AI Inference

快速理解: The growing demand for neural network (NN) driven applications in AIoT devices necessitates efficient matrix multiplication (MM) acceleration. While domain-specific accelerators (DSAs) for NN are widely used, their large area overhead of dedicated buffers and low reusability limit cost-effectiveness.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.14 SYSTEMS · MLSys

本次会议纳入 **100** 篇。自动主题统计为: AI Infra (97)、Systems (38)、Agent RL (6)、Embodied/WBC (1)、World Model (1)。这些统计用于组织阅读顺序, 不能单独证明论文结论或实验优越性。

优先阅读候选: - AgenticCache: Cache-Driven Asynchronous Planning for Embodied AI Agents (优先级分数 87.5) - NEO: Saving GPU Memory Crisis with CPU Offloading for Online LLM Inference (优先级分数 81.858) - RL-Scope: Cross-Stack Profiling for Deep Reinforcement Learning Workloads (优先级分数 79.572) - Q-Hitter: A Better Token Oracle for Efficient LLM Inference via Sparse-Quantized KV Cache (优先级分数 72.524) - The CoRa Tensor Compiler: Compilation for Ragged Tensors with Minimal Padding (优先级分数 71.572)

1.3.14.1 AgenticCache: Cache-Driven Asynchronous Planning for Embodied AI Agents

- 作者: 作者信息未提供
- 年份: 2026; 主题: Embodied/WBC; Agent RL; Systems; 优先级: 核心
- [来源](#)

论文: AgenticCache: Cache-Driven Asynchronous Planning for Embodied AI Agents

快速理解: Embodied AI agents increasingly rely on large language models (LLMs) for planning, yet per-step LLM calls impose severe latency and cost. In this paper, we show that embodied tasks

exhibit strong plan locality, where the next plan is largely predictable from the current one.

关注点：具身任务中的感知、动作生成、全身协调或运动控制；强化学习、模仿学习、规划、奖励或多智能体决策；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.14.2 NEO: Saving GPU Memory Crisis with CPU Offloading for Online LLM Inference

- 作者：Xu Jiang; Yang Zhou; Shiyi Cao; Ion Stoica; Minlan Yu
- 年份：2024; 主题：Agent RL; AI Infra; Systems; 优先级：核心
- [来源](#)

论文：NEO: Saving GPU Memory Crisis with CPU Offloading for Online LLM Inference

快速理解：Online LLM inference powers many exciting applications such as intelligent chatbots and autonomous agents. Modern LLM inference engines widely rely on request batching to improve inference throughput, aiming to make it cost-efficient when running on expensive GPU accelerators.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.14.3 RL-Scope: Cross-Stack Profiling for Deep Reinforcement Learning Workloads

- 作者：James Gleeson; Srivatsan Krishnan; Moshe Gabel; V. Reddi; E. de Lara; Gennady Pekhimenko
- 年份：2021; 主题：Agent RL; AI Infra; Systems; 优先级：核心
- [来源](#)

论文：RL-Scope: Cross-Stack Profiling for Deep Reinforcement Learning Workloads

快速理解：Deep reinforcement learning (RL) has made groundbreaking advancements in robotics, data center management and other applications. Unfortunately, system-level bottlenecks in RL workloads are poorly understood; we observe fundamental structural differences in RL workloads that make them inherently less GPU-bound than supervised learning (SL).

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.14.4 Q-Hitter: A Better Token Oracle for Efficient LLM Inference via Sparse-Quantized KV Cache

- 作者：Zhenyu (Allen) Zhang; Shiwei Liu; Runjin Chen; B. Kailkhura; Beidi Chen; Atlas Wang
- 年份：2024; 主题：AI Infra; Systems; 优先级：核心
- [来源](#)

论文：Q-Hitter: A Better Token Oracle for Efficient LLM Inference via Sparse-Quantized KV Cache

快速理解：摘要未随索引或本地 PDF 前页提供；根据标题与主题标签，本文关注 AI Infra、Systems 相关问题。以下判断仅用于快速阅读排序，不能替代全文核对。

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：标题/主题推断。

1.3.14.5 The CoRa Tensor Compiler: Compilation for Ragged Tensors with Minimal Padding

- 作者：Pratik Fegade; Tianqi Chen; Phillip B. Gibbons; T. Mowry
- 年份：2021；主题：AI Infra; Systems；优先级：核心
- [来源](#)

论文：The CoRa Tensor Compiler: Compilation for Ragged Tensors with Minimal Padding

快速理解：There is often variation in the shape and size of input data used for deep learning. In many cases, such data can be represented using tensors with non-uniform shapes, or ragged tensors.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.14.6 Wavelet: Efficient DNN Training with Tick-Tock Scheduling

- 作者：Guanhua Wang; Kehan Wang; Kenan Jiang; Xiangjun Li; Ion Stoica
- 年份：2021；主题：AI Infra; Systems；优先级：核心
- [来源](#)

论文：Wavelet: Efficient DNN Training with Tick-Tock Scheduling

快速理解：摘要未随索引或本地 PDF 前页提供；根据标题与主题标签，本文关注 AI Infra、Systems 相关问题。以下判断仅用于快速阅读排序，不能替代全文核对。

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证据层级：标题/主题推断。

1.3.14.7 ALCOP: Automatic Load-Compute Pipelining in Deep Learning Compiler for AI-GPUs

- 作者：Guyue Huang; Yang Bai; L. Liu; Yuke Wang; Bei Yu; Yufei Ding; Yuan Xie
- 年份：2022；主题：AI Infra; Systems；优先级：核心
- [来源](#)

论文：ALCOP: Automatic Load-Compute Pipelining in Deep Learning Compiler for AI-GPUs

快速理解：Pipelining between data loading and computation is a critical tensor program optimization for GPUs. In order to unleash the high performance of latest GPUs, we must perform a synergetic optimization of multi-stage pipelining across the multi-level buffer hierarchy of GPU.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.14.8 Efficient LLM Inference using Dynamic Input Pruning and Cache-Aware Masking

- 作者：Marco Federici; Davide Belli; M. V. Baalen; Amir Jalalirad; Andrii Skliar; B. Major; Markus Nagel; Paul N. Whatmough

- 年份: 2024; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: Efficient LLM Inference using Dynamic Input Pruning and Cache-Aware Masking

快速理解: While mobile devices provide ever more compute power, improvements in DRAM bandwidth are much slower. This is unfortunate for large language model (LLM) token generation, which is heavily memory-bound.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.14.9 SLA-Driven ML Inference Framework for Clouds with Heterogeneous Accelerators

- 作者: Jung-Yon Cho; D. Z. Tootaghaj; Lianjie Cao; Puneet Sharma
- 年份: 2022; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: SLA-Driven ML Inference Framework for Clouds with Heterogeneous Accelerators

快速理解: 摘要未随索引或本地 PDF 前页提供; 根据标题与主题标签, 本文关注 AI Infra、Systems 相关问题。以下判断仅用于快速阅读排序, 不能替代全文核对。

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证据层级: 标题/主题推断。

1.3.14.10 Cupcake: A Compression Scheduler for Scalable Communication-Efficient Distributed Training

- 作者: Zhuang Wang; X. Wu; Zhaozhuo Xu; T. S. Eugene Ng
- 年份: 2023; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: Cupcake: A Compression Scheduler for Scalable Communication-Efficient Distributed Training

快速理解: 摘要未随索引或本地 PDF 前页提供; 根据标题与主题标签, 本文关注 AI Infra、Systems 相关问题。以下判断仅用于快速阅读排序, 不能替代全文核对。

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证据层级: 标题/主题推断。

1.3.14.11 Hydrozoa: Dynamic Hybrid-Parallel DNN Training on Serverless Containers

- 作者: R. Guo; Victor J. W. Guo; Antonio Kim; Joshua Hildred; K. Daudjee
- 年份: 2022; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: Hydrozoa: Dynamic Hybrid-Parallel DNN Training on Serverless Containers

快速理解: 摘要未随索引或本地 PDF 前页提供; 根据标题与主题标签, 本文关注 AI Infra、Systems 相关问题。以下判断仅用于快速阅读排序, 不能替代全文核对。

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证据层级：标题/主题推断。

1.3.14.12 AutoScratch: ML-Optimized Cache Management for Inference-Oriented GPUs

- 作者: Yaosheng Fu; Evgeny Bolotin; A. Jaleel; Gal Dalal; Shie Mannor; Jacob Subag; Noam Korem; M. Behar; D. Nellans
- 年份: 2023; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: AutoScratch: ML-Optimized Cache Management for Inference-Oriented GPUs

快速理解: 摘要未随索引或本地 PDF 前页提供; 根据标题与主题标签, 本文关注 AI Infra、Systems 相关问题。以下判断仅用于快速阅读排序, 不能替代全文核对。

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 标题/主题推断。

1.3.14.13 A Transferable Approach for Partitioning Machine Learning Models on Multi-Chip-Modules

- 作者: Xinfeng Xie; Prakash Prabhu; Ulysse Beaugnon; P. Phothilimthana; Sudip Roy; Azalia Mirhoseini; E. Brevdo; J. Laudon; Yanqi Zhou
- 年份: 2021; 主题: Agent RL; AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: A Transferable Approach for Partitioning Machine Learning Models on Multi-Chip-Modules

快速理解: Multi-Chip-Modules (MCMs) reduce the design and fabrication cost of machine learning (ML) accelerators while delivering performance and energy efficiency on par with a monolithic large chip. However, ML compilers targeting MCMs need to solve complex optimization problems optimally and efficiently to achieve this high performance.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.14.14 AWQ: Activation-aware Weight Quantization for On-Device LLM Compression and Acceleration

- 作者: Ji Lin; Jiaming Tang; Haotian Tang; Shang Yang; Xingyu Dang; Song Han
- 年份: 2023; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: AWQ: Activation-aware Weight Quantization for On-Device LLM Compression and Acceleration

快速理解: Large language models (LLMs) have transformed numerous AI applications. On-device LLM is becoming increasingly important: running LLMs locally on edge devices can reduce cloud computing costs and protect users' privacy.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.14.15 AccelOpt: A Self-Improving LLM Agentic System for AI Accelerator Kernel Optimization

- 作者：作者信息未提供
- 年份：2026；主题：Agent RL; AI Infra; 优先级：核心
- [来源](#)

论文：AccelOpt: A Self-Improving LLM Agentic System for AI Accelerator Kernel Optimization

快速理解：We present AccelOpt, a self-improving large language model (LLM) agentic system that autonomously optimizes kernels for emerging AI accelerators, eliminating the need for expert-provided hardware-specific optimization knowledge. AccelOpt explores the kernel optimization space through iterative generation, informed by an optimization memory that curates experiences and insights from previously encountered slow-fast kernel pairs.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.14.16 Kitty: Accurate and Efficient 2-bit KV Cache Quantization with Dynamic Channel-wise Precision Boost

- 作者：作者信息未提供
- 年份：2026；主题：AI Infra; Systems; 优先级：核心
- [来源](#)

论文：Kitty: Accurate and Efficient 2-bit KV Cache Quantization with Dynamic Channel-wise Precision Boost

快速理解：The KV cache is a dominant memory bottleneck for LLM inference. While 4-bit KV quantization preserves accuracy, 2-bit often degrades it, especially on long-context reasoning.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.14.17 Locality-Aware Beam Scheduling for Efficient Test-Time Compute with a Consumer-grade GPU

- 作者：作者信息未提供
- 年份：2026；主题：AI Infra; Systems; 优先级：核心
- [来源](#)

论文：Locality-Aware Beam Scheduling for Efficient Test-Time Compute with a Consumer-grade GPU

快速理解：Large Language Models (LLMs) are central to modern NLP applications, yet their deployment on consumer-grade GPUs is constrained by limited memory capacity and bandwidth. In typical single-batch inference on local devices, the key-value (KV) cache occupies only a small fraction of total memory, so prior studies have largely focused on model weights.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：本地 PDF 前页。

1.3.14.18 SuperInfer: SLO-Aware Rotary Scheduling and Memory Management for LLM Inference on Superchips

- 作者: 作者信息未提供
- 年份: 2026; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: SuperInfer: SLO-Aware Rotary Scheduling and Memory Management for LLM Inference on Superchips

快速理解: Large Language Model (LLM) serving faces a fundamental tension between stringent latency Service Level Objectives (SLOs) and limited GPU memory capacity. When high request rates exhaust the KV cache budget, existing LLM inference systems often suffer severe head-of-line (HOL) blocking.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 本地 PDF 前页。

1.3.14.19 FlashInfer: Efficient and Customizable Attention Engine for LLM Inference Serving

- 作者: Zihao Ye; Lequn Chen; Ruihang Lai; Wuwei Lin; Yineng Zhang; Stephanie Wang; Tian-Qi Chen; Baris Kasikci; Vinod Grover; Arvind Krishnamurthy; Luis Ceze
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: FlashInfer: Efficient and Customizable Attention Engine for LLM Inference Serving

快速理解: Transformers, driven by attention mechanisms, form the foundation of large language models (LLMs). As these models scale up, efficient GPU attention kernels become essential for high-throughput and low-latency inference.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.14.20 Prompt Cache: Modular Attention Reuse for Low-Latency Inference

- 作者: In Gim; Guojun Chen; Seung-seob Lee; Nikhil Sarda; Anurag Khandelwal; Lin Zhong
- 年份: 2023; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: Prompt Cache: Modular Attention Reuse for Low-Latency Inference

快速理解: We present Prompt Cache, an approach for accelerating inference for large language models (LLM) by reusing attention states across different LLM prompts. Many input prompts have overlapping text segments, such as system messages, prompt templates, and documents provided for context.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.14.21 QServe: W4A8KV4 Quantization and System Co-design for Efficient LLM Serving

- 作者: Yujun Lin; Haotian Tang; Shang Yang; Zhekai Zhang; Guangxuan Xiao; Chuang Gan; Song Han
- 年份: 2024; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: QServe: W4A8KV4 Quantization and System Co-design for Efficient LLM Serving

快速理解: Quantization can accelerate large language model (LLM) inference. Going beyond INT8 quantization, the research community is actively exploring even lower precision, such as INT4.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.14.22 Tutel: Adaptive Mixture-of-Experts at Scale

- 作者: Changho Hwang; Wei Cui; Yifan Xiong; Ziyue Yang; Ze Liu; Han Hu; Zilong Wang; Rafael Salas; Jithin Jose; Prabhat Ram; J. Chau; Peng Cheng; Fan Yang; Mao Yang; Yongqiang Xiong
- 年份: 2022; 主题: World Model; AI Infra; 优先级: 核心
- [来源](#)

论文: Tutel: Adaptive Mixture-of-Experts at Scale

快速理解: Sparsely-gated mixture-of-experts (MoE) has been widely adopted to scale deep learning models to trillion-plus parameters with fixed computational cost. The algorithmic performance of MoE relies on its token routing mechanism that forwards each input token to the right sub-models or experts.

关注点: 环境动力学、视频预测、潜变量状态或模型式决策; 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.14.23 Vidur: A Large-Scale Simulation Framework For LLM Inference

- 作者: Amey Agrawal; Nitin Kedia; Jayashree Mohan; Ashish Panwar; Nipun Kwatra; Bhargav S. Gulavani; R. Ramjee; Alexey Tumanov
- 年份: 2024; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: Vidur: A Large-Scale Simulation Framework For LLM Inference

快速理解: Optimizing the deployment of Large language models (LLMs) is expensive today since it requires experimentally running an application workload against an LLM implementation while exploring large configuration space formed by system knobs such as parallelization strategies, batching techniques, and scheduling policies. To address this challenge, we present Vidur - a large-scale, high-fidelity, easily-extensible simulation framework for LLM inference performance.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.14.24 TorchSparse: Efficient Point Cloud Inference Engine

- 作者: Haotian Tang; Zhijian Liu; Xiuyu Li; Yujun Lin; Song Han
- 年份: 2022; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: TorchSparse: Efficient Point Cloud Inference Engine

快速理解: Deep learning on point clouds has received increased attention thanks to its wide applications in AR/VR and autonomous driving. These applications require low latency and high accuracy to provide real-time user experience and ensure user safety.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.14.25 Lancet: Accelerating Mixture-of-Experts Training via Whole Graph Computation-Communication Overlapping

- 作者: Chenyu Jiang; Ye Tian; Zhen Jia; Shuai Zheng; Chuan Wu; Yida Wang
- 年份: 2024; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Lancet: Accelerating Mixture-of-Experts Training via Whole Graph Computation-Communication Overlapping

快速理解: The Mixture-of-Expert (MoE) technique plays a crucial role in expanding the size of DNN model parameters. However, it faces the challenge of extended all-to-all communication latency during the training process.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.14.26 Seesaw: High-throughput LLM Inference via Model Re-sharding

- 作者: Qidong Su; Wei Zhao; X. Li; Muralidhar Andoorvedu; Che Jiang; Zhanda Zhu; Kevin Song; Christina Giannoula; Gennady Pekhimenko
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Seesaw: High-throughput LLM Inference via Model Re-sharding

快速理解: To improve the efficiency of distributed large language model (LLM) inference, various parallelization strategies, such as tensor and pipeline parallelism, have been proposed. However, the distinct computational characteristics inherent in the two stages of LLM inference-prefilling and decoding-render a single static parallelization strategy insufficient for the effective optimization of both stages.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.14.27 dPRO: A Generic Performance Diagnosis and Optimization Toolkit for Expediting Distributed DNN Training

- 作者: Han Hu; Chenyu Jiang; Yuchen Zhong; Yanghua Peng; Chuan Wu; Yibo Zhu; Haibin Lin; Chuanxiong Guo
- 年份: 2022; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: dPRO: A Generic Performance Diagnosis and Optimization Toolkit for Expediting Distributed

DNN Training

快速理解: Distributed training using multiple devices (e.g., GPUs) has been widely adopted for learning DNN models over large datasets. However, the performance of large-scale distributed training tends to be far from linear speed-up in practice.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.14.28 Rubick: Exploiting Job Reconfigurability for Deep Learning Cluster Scheduling

- 作者: Xinyi Zhang; Hanyu Zhao; Wencong Xiao; Xianyan Jia; Fei Xu; Yong Li; Wei Lin; Fangming Liu
- 年份: 2024; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Rubick: Exploiting Job Reconfigurability for Deep Learning Cluster Scheduling

快速理解: The era of large deep learning models has given rise to advanced training strategies such as 3D parallelism and the ZeRO series. These strategies enable various (re-)configurable execution plans for a training job, which exhibit remarkably different requirements of multiple resource types.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.14.29 Context Parallelism for Scalable Million-Token Inference

- 作者: Amy Yang; Jing-Yi Yang; Aya Ibrahim; Xinfeng Xie; Bangsheng Tang; Grigory Sizov; J. Reizenstein; Jongsoo Park; Jianyu Huang
- 年份: 2024; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Context Parallelism for Scalable Million-Token Inference

快速理解: We present context parallelism for long-context large language model inference, which achieves near-linear scaling for long-context prefill latency with up to 128 H100 GPUs across 16 nodes. Particularly, our method achieves 1M context prefill with Llama3 405B model in 77s (93% parallelization efficiency, 63% FLOPS utilization) and 128K context prefill in 3.8s.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.14.30 TileLink: Generating Efficient Compute-Communication Overlapping Kernels using Tile-Centric Primitives

- 作者: Size Zheng; Jin Fang; Xuegui Zheng; Qi Hou; Wenlei Bao; Ningxin Zheng; Ziheng Jiang; Dongyang Wang; Jia-jun Ye; Haibin Lin; Li-Wen Chang; Xin Liu
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: TileLink: Generating Efficient Compute-Communication Overlapping Kernels using Tile-Centric Primitives

快速理解: Large deep learning models have achieved state-of-the-art performance in a wide range of tasks. These models often necessitate distributed systems for efficient training and inference.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.14.31 ReaL: Efficient RLHF Training of Large Language Models with Parameter Reallocation

- 作者: Zhiyu Mei; Wei Fu; Kaiwei Li; Guangju Wang; Huanchen Zhang; Yi Wu
- 年份: 2024; 主题: Agent RL; AI Infra; 优先级: 重要
- [来源](#)

论文: ReaL: Efficient RLHF Training of Large Language Models with Parameter Reallocation

快速理解: Reinforcement Learning from Human Feedback (RLHF) is a pivotal technique for empowering large language model (LLM) applications. Compared with the supervised training process of LLMs, the RLHF training process is much more sophisticated, requiring a diverse range of computation workloads with intricate dependencies between multiple LLM instances.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.14.32 Optimizing LLM Queries in Relational Data Analytics Workloads

- 作者: Shu Liu; Asim Biswal; Amog Kamsetty; Audrey Cheng; L. G. Schroeder; Liana Patel; Shiyi Cao; Xiangxi Mo; Ion Stoica; Joseph E. Gonzalez; M. Zaharia
- 年份: 2024; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Optimizing LLM Queries in Relational Data Analytics Workloads

快速理解: Batch data analytics is a growing application for Large Language Models (LLMs). LLMs enable users to perform a wide range of natural language tasks, such as classification, entity extraction, and translation, over large datasets.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.14.33 Communication-Efficient Graph Neural Networks with Probabilistic Neighborhood Expansion Analysis and Caching

- 作者: Tim Kaler; Alexandros-Stavros Iliopoulos; P. Murzynowski; T. Schardl; C. Leiserson; Jie Chen
- 年份: 2023; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Communication-Efficient Graph Neural Networks with Probabilistic Neighborhood Expansion Analysis and Caching

快速理解: Training and inference with graph neural networks (GNNs) on massive graphs has been actively studied since the inception of GNNs, owing to the widespread use and success of GNNs in applications such as recommendation systems and financial forensics. This paper is concerned

with minibatch training and inference with GNNs that employ node-wise sampling in distributed settings, where the necessary partitioning of vertex features across distributed storage causes feature communication to become a major bottleneck that hampers scalability.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.14.34 DiffusionPipe: Training Large Diffusion Models with Efficient Pipelines

- 作者：Ye Tian; Zhen Jia; Ziyue Luo; Yida Wang; Chuan Wu
- 年份：2024；主题：AI Infra; Systems；优先级：重要
- [来源](#)

论文：DiffusionPipe: Training Large Diffusion Models with Efficient Pipelines

快速理解：Diffusion models have emerged as dominant performers for image generation. To support training large diffusion models, this paper studies pipeline parallel training of diffusion models and proposes DiffusionPipe, a synchronous pipeline training system that advocates innovative pipeline bubble filling technique, catering to structural characteristics of diffusion models.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.14.35 Swift for TensorFlow: A portable, flexible platform for deep learning

- 作者：Brennan Saeta; Denys Shabalin; M. Rasi; Brad Larson; Xihui Wu; P. Schuh; M. Casbon; Daniel Zheng; Saleem Abdulrasool; A. Efremov; Dave Abrahams; C. Lattner; Richard Wei
- 年份：2021；主题：AI Infra; Systems；优先级：重要
- [来源](#)

论文：Swift for TensorFlow: A portable, flexible platform for deep learning

快速理解：Swift for TensorFlow is a deep learning platform that scales from mobile devices to clusters of hardware accelerators in data centers. It combines a language-integrated automatic differentiation system and multiple Tensor implementations within a modern ahead-of-time compiled language oriented around mutable value semantics.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.14.36 Subgraph Stationary Hardware-Software Inference Co-Design

- 作者：Payman Behnam; Jianming Tong; Alind Khare; Yangyu Chen; Yue Pan; Pranav Gadikar; A. Bambhaniya; Tushar Krishna; Alexey Tumanov
- 年份：2023；主题：AI Infra; Systems；优先级：重要
- [来源](#)

论文：Subgraph Stationary Hardware-Software Inference Co-Design

快速理解：A growing number of applications depend on Machine Learning (ML) functionality and benefits from both higher quality ML predictions and better timeliness (latency) at the same time. A growing body of research in computer architecture, ML, and systems software literature focuses on reaching better latency-accuracy tradeoffs for ML models.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.14.37 ML-EXray: Visibility into ML Deployment on the Edge

- 作者：Hang Qiu; Ioanna Vavelidou; Jian Li; Evgenya Pergament; Pete Warden; Sandeep P. Chinchali; Zain Asgar; S. Katti
- 年份：2021；主题：AI Infra; Systems；优先级：重要
- [来源](#)

论文：ML-EXray: Visibility into ML Deployment on the Edge

快速理解：Benefiting from expanding cloud infrastructure, deep neural networks (DNNs) today have increasingly high performance when trained in the cloud. Researchers spend months of effort competing for an extra few percentage points of model accuracy.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.14.38 ACroBat: Optimizing Auto-batching of Dynamic Deep Learning at Compile Time

- 作者：Pratik Fegade; Tianqi Chen; Phillip B. Gibbons; T. Mowry
- 年份：2023；主题：AI Infra; Systems；优先级：重要
- [来源](#)

论文：ACroBat: Optimizing Auto-batching of Dynamic Deep Learning at Compile Time

快速理解：Dynamic control flow is an important technique often used to design expressive and efficient deep learning computations for applications such as text parsing, machine translation, exiting early out of deep models and so on. The control flow divergence resulting from dynamic control flow makes batching, an important optimization enabling high throughput and hardware utilization, difficult to perform manually.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.14.39 MAS-Attention: Memory-Aware Stream Processing for Attention Acceleration on Resource-Constrained Edge Devices

- 作者：Mohammadali Shakerdargah; Shan Lu; Chao Gao; Di Niu
- 年份：2024；主题：AI Infra; Systems；优先级：重要
- [来源](#)

论文：MAS-Attention: Memory-Aware Stream Processing for Attention Acceleration on Resource-Constrained Edge Devices

快速理解：The advent of foundation models have revolutionized various fields, enabling unprecedented task accuracy and flexibility in computational linguistics, computer vision and other domains. Attention mechanism has become an essential component of foundation models, due to their superb capability of capturing correlations in a sequence.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.14.40 MegaBlocks: Efficient Sparse Training with Mixture-of-Experts

- 作者: Trevor Gale; D. Narayanan; C. Young; M. Zaharia
- 年份: 2022; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: MegaBlocks: Efficient Sparse Training with Mixture-of-Experts

快速理解: We present MegaBlocks, a system for efficient Mixture-of-Experts (MoE) training on GPUs. Our system is motivated by the limitations of current frameworks, which restrict the dynamic routing in MoE layers to satisfy the constraints of existing software and hardware.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.14.41 VS-Quant: Per-vector Scaled Quantization for Accurate Low-Precision Neural Network Inference

- 作者: Steve Dai; Rangharajan Venkatesan; Haoxing Ren; Brian Zimmer; W. Dally; Brucek Khailany
- 年份: 2021; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: VS-Quant: Per-vector Scaled Quantization for Accurate Low-Precision Neural Network Inference

快速理解: Quantization enables efficient acceleration of deep neural networks by reducing model memory footprint and exploiting low-cost integer math hardware units. Quantization maps floating-point weights and activations in a trained model to low-bitwidth integer values using scale factors.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.14.42 An Efficient Statistical-based Gradient Compression Technique for Distributed Training Systems

- 作者: A. Abdelmoniem; Ahmed Elzanaty; Mohamed-Slim Alouini; Marco Canini
- 年份: 2021; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: An Efficient Statistical-based Gradient Compression Technique for Distributed Training Systems

快速理解: The recent many-fold increase in the size of deep neural networks makes efficient distributed training challenging. Many proposals exploit the compressibility of the gradients and propose lossy compression techniques to speed up the communication stage of distributed training.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.14.43 Randomness In Neural Network Training: Characterizing The Impact of Tooling

- 作者: Donglin Zhuang; Xingyao Zhang; S. Song; Sara Hooker
- 年份: 2021; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Randomness In Neural Network Training: Characterizing The Impact of Tooling

快速理解: The quest for determinism in machine learning has disproportionately focused on characterizing the impact of noise introduced by algorithmic design choices. In this work, we address a less well understood and studied question: how does our choice of tooling introduce randomness to deep neural network training.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.14.44 SampleAttention: Near-Lossless Acceleration of Long Context LLM Inference with Adaptive Structured Sparse Attention

- 作者: Qianchao Zhu; Jiangfei Duan; Chang Chen; Si-Ran Liu; Xiuhong Li; Guanyu Feng; Xin Lv; Huanqi Cao; Chuanfu Xiao; Xingcheng Zhang; Dahua Lin; Chao Yang
- 年份: 2024; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: SampleAttention: Near-Lossless Acceleration of Long Context LLM Inference with Adaptive Structured Sparse Attention

快速理解: Large language models (LLMs) now support extremely long context windows, but the quadratic complexity of vanilla attention results in significantly long Time-to-First-Token (TTFT) latency. Existing approaches to address this complexity require additional pretraining or finetuning, and often sacrifice model accuracy.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.14.45 SiDA: Sparsity-Inspired Data-Aware Serving for Efficient and Scalable Large Mixture-of-Experts Models

- 作者: Zhi-Xu Du; Shiyu Li; Yuhao Wu; Xiangyu Jiang; Jingwei Sun; Qilin Zheng; Yong-Kai Wu; Ang Li; Hai Li; Yiran Chen
- 年份: 2023; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: SiDA: Sparsity-Inspired Data-Aware Serving for Efficient and Scalable Large Mixture-of-Experts Models

快速理解: Mixture-of-Experts (MoE) has emerged as a favorable architecture in the era of large models due to its inherent advantage, i.e., enlarging model capacity without incurring notable computational overhead. Yet, the realization of such benefits often results in ineffective GPU memory utilization, as large portions of the model parameters remain dormant during inference.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.14.46 Efficient Post-training Quantization with FP8 Formats

- 作者: Haihao Shen; Naveen Mellempudi; Xin He; Qun Gao; Chang-Bao Wang; Meng-Ni Wang
- 年份: 2023; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Efficient Post-training Quantization with FP8 Formats

快速理解: Recent advances in deep learning methods such as LLMs and Diffusion models have created a need for improved quantization methods that can meet the computational demands of these modern architectures while maintaining accuracy. Towards this goal, we study the advantages of FP8 data formats for post-training quantization across 75 unique network architectures covering a wide range of tasks, including machine translation, language modeling, text generation, image classification, generation, and segmentation.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.14.47 On Optimizing the Communication of Model Parallelism

- 作者: Yonghao Zhuang; Hexu Zhao; Lianmin Zheng; Zhuohan Li; Eric P. Xing; Qirong Ho; Joseph E. Gonzalez; Ion Stoica; Haotong Zhang
- 年份: 2022; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: On Optimizing the Communication of Model Parallelism

快速理解: We study a novel and important communication pattern in large-scale model-parallel deep learning (DL), which we call cross-mesh resharding. This pattern emerges when the two paradigms of model parallelism - intra-operator and inter-operator parallelism - are combined to support large models on large clusters.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.14.48 On the Utility of Gradient Compression in Distributed Training Systems

- 作者: Saurabh Agarwal; Hongyi Wang; S. Venkataraman; Dimitris Papailiopoulos
- 年份: 2021; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: On the Utility of Gradient Compression in Distributed Training Systems

快速理解: A rich body of prior work has highlighted the existence of communication bottlenecks in synchronous data-parallel training. To alleviate these bottlenecks, a long line of recent work proposes gradient and model compression methods.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.14.49 Adaptive Message Quantization and Parallelization for Distributed Full-graph GNN Training

- 作者: Borui Wan; Jun Zhao; Chuan Wu
- 年份: 2023; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Adaptive Message Quantization and Parallelization for Distributed Full-graph GNN Training

快速理解: Distributed full-graph training of Graph Neural Networks (GNNs) over large graphs is bandwidth-demanding and time-consuming. Frequent exchanges of node features, embeddings and embedding gradients (all referred to as messages) across devices bring significant communication overhead for nodes with remote neighbors on other devices (marginal nodes) and unnecessary waiting time for nodes without remote neighbors (central nodes) in the training graph.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.14.50 FastTree: Optimizing Attention Kernel and Runtime for Tree-Structured LLM Inference

- 作者: Zaifeng Pan; Yi-Tong Ding; Yue Guan; Zheng Wang; Zhongkai Yu; Xulong Tang; Yida Wang; Yu-Fei Ding
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: FastTree: Optimizing Attention Kernel and Runtime for Tree-Structured LLM Inference

快速理解: 摘要未随索引或本地 PDF 前页提供; 根据标题与主题标签, 本文关注 AI Infra 相关问题。以下判断仅用于快速阅读排序, 不能替代全文核对。

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 标题/主题推断。

1.3.14.51 Accelerating SLIDE Deep Learning on Modern CPUs: Vectorization, Quantizations, Memory Optimizations, and More

- 作者: Shabnam Daghighi; Nicholas Meisburger; Mengnan Zhao; Yong Wu; Sameh Gobriel; Charlie Tai; Anshumali Shrivastava
- 年份: 2021; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Accelerating SLIDE Deep Learning on Modern CPUs: Vectorization, Quantizations, Memory Optimizations, and More

快速理解: Deep learning implementations on CPUs (Central Processing Units) are gaining more traction. Enhanced AI capabilities on commodity x86 architectures are commercially appealing due to the reuse of existing hardware and virtualization ease.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.14.52 Does compressing activations help model parallel training?

- 作者: S. Bian; Da-Cheng Li; Hongyi Wang; Eric P. Xing; S. Venkataraman
- 年份: 2023; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Does compressing activations help model parallel training?

快速理解: Large-scale Transformer models are known for their exceptional performance in a range of tasks, but training them can be difficult due to the requirement for communication-intensive model parallelism. One way to improve training speed is to compress the message size in communication.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.14.53 FlexInfer: Flexible LLM Inference with CPU Computations

- 作者: Seonjin Na; Geonhwa Jeong; Byung Hoon Ahn; A. Jezghani; Jeffrey Young; C. Hughes; Tushar Krishna; Hyesoon Kim
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: FlexInfer: Flexible LLM Inference with CPU Computations

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关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 标题/主题推断。

1.3.14.54 Just-in-time Quantization with Processing-In-Memory for Efficient ML Training

- 作者: M. Ibrahim; Shaizeen Aga; Ada Li; Suchita Pati; Mahzabeen Islam
- 年份: 2023; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Just-in-time Quantization with Processing-In-Memory for Efficient ML Training

快速理解: Data format innovations have been critical for machine learning (ML) scaling, which in turn fuels ground-breaking ML capabilities. However, even in the presence of low-precision formats, model weights are often stored in both high-precision and low-precision during training.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.14.55 Practical Edge Kernels for Integer-Only Vision Transformers Under Post-training Quantization

- 作者: Zi-Ning Zhang; Bingsheng He; Zhen-Jie Zhang
- 年份: 2023; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Practical Edge Kernels for Integer-Only Vision Transformers Under Post-training Quantization

快速理解: 摘要未随索引或本地 PDF 前页提供; 根据标题与主题标签, 本文关注 AI Infra 相关问题。以下判断仅用于快

速阅读排序，不能替代全文核对。

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：标题/主题推断。

1.3.14.56 PipeFill: Using GPUs During Bubbles in Pipeline-parallel LLM Training

- 作者：Daiyaan Arfeen; Zhen Zhang; Xinwei Fu; G. R. Ganger; Yida Wang
- 年份：2024; 主题：AI Infra; 优先级：重要
- [来源](#)

论文：PipeFill: Using GPUs During Bubbles in Pipeline-parallel LLM Training

快速理解：Training Deep Neural Networks (DNNs) with billions of parameters generally involves pipeline-parallel (PP) execution. Unfortunately, PP model training can use GPUs inefficiently, especially at large scale, due to idle GPU time caused by pipeline bubbles, which are often 15-30% and can exceed 60% of the training job's GPU allocation.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.14.57 Rethinking Floating Point Overheads for Mixed Precision DNN Accelerators

- 作者：Hamzah Abdel-Aziz; A. Shafiee; Jong-Hoon Shin; A. Pedram; Joseph Hassoun
- 年份：2021; 主题：AI Infra; 优先级：重要
- [来源](#)

论文：Rethinking Floating Point Overheads for Mixed Precision DNN Accelerators

快速理解：In this paper, we propose a mixed-precision convolution unit architecture which supports different integer and floating point (FP) precisions. The proposed architecture is based on low-bit inner product units and realizes higher precision based on temporal decomposition.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.14.58 Distributed Matrix-Based Sampling for Graph Neural Network Training

- 作者：A. Tripathy; K. Yelick; A. Buluç
- 年份：2023; 主题：AI Infra; 优先级：重要
- [来源](#)

论文：Distributed Matrix-Based Sampling for Graph Neural Network Training

快速理解：Graph Neural Networks (GNNs) offer a compact and computationally efficient way to learn embeddings and classifications on graph data. GNN models are frequently large, making distributed minibatch training necessary.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.14.59 Efficient Strong Scaling Through Burst Parallel Training

- 作者: S. Park; Joshua Fried; Sunghyun Kim; Mohammad Alizadeh; Adam Belay
- 年份: 2021; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Efficient Strong Scaling Through Burst Parallel Training

快速理解: As emerging deep neural network (DNN) models continue to grow in size, using large GPU clusters to train DNNs is becoming an essential requirement to achieving acceptable training times. In this paper, we consider the case where future increases in cluster size will cause the global batch size that can be used to train models to reach a fundamental limit: beyond a certain point, larger global batch sizes cause sample efficiency to degrade, increasing overall time to accuracy.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.14.60 Hotline Profiler: Automatic Annotation and A Multi-Scale Timeline for Visualizing Time-Use in DNN Training

- 作者: Daniel Snider; Fanny Chevalier; Gennady Pekhimenko
- 年份: 2023; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Hotline Profiler: Automatic Annotation and A Multi-Scale Timeline for Visualizing Time-Use in DNN Training

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关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 标题/主题推断。

1.3.14.61 vMCU: Coordinated Memory Management and Kernel Optimization for DNN Inference on MCUs

- 作者: Size Zheng; Renze Chen; Meng Li; Zihao Ye; Luis Ceze; Yun Liang
- 年份: 2024; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: vMCU: Coordinated Memory Management and Kernel Optimization for DNN Inference on MCUs

快速理解: IoT devices based on microcontroller units (MCU) provide ultra-low power consumption and ubiquitous computation for near-sensor deep learning models (DNN). However, the memory of MCU is usually 2-3 orders of magnitude smaller than mobile devices, which makes it challenging to map DNNs onto MCUs.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.14.62 Torch2Chip: An End-to-end Customizable Deep Neural Network Compression and Deployment Toolkit for Prototype Hardware Accelerator Design

- 作者: Jian Meng; Yuan Liao; Anupreetham Anupreetham; Ahmed Hassan; Shixing Yu; Han-Sok

Suh; Xiaofeng Hu; Jae-sun Seo

- 年份: 2024; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Torch2Chip: An End-to-end Customizable Deep Neural Network Compression and Deployment Toolkit for Prototype Hardware Accelerator Design

快速理解: The development of model compression is continuously motivated by the evolution of various neural network accelerators with ASIC or FPGA. On the algorithm side, the ultimate goal of quantization or pruning is accelerating the expensive DNN computations on low-power hardware.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.14.63 GPU Semiring Primitives for Sparse Neighborhood Methods

- 作者: Corey Nolet; Divye Gala; Edward Raff; Joe Eaton; Brad Rees; John Zedlewski; T. Oates
- 年份: 2021; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: GPU Semiring Primitives for Sparse Neighborhood Methods

快速理解: High-performance primitives for mathematical operations on sparse vectors must deal with the challenges of skewed degree distributions and limits on memory consumption that are typically not issues in dense operations. We demonstrate that a sparse semiring primitive can be flexible enough to support a wide range of critical distance measures while maintaining performance and memory efficiency on the GPU.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.14.64 -TWO: 3× Faster Multi-Model Training with Orchestration and Memory Optimization

- 作者: S. Purandare; A. Wasay; Animesh Jain; Stratos Idreos
- 年份: 2023; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: -TWO: 3× Faster Multi-Model Training with Orchestration and Memory Optimization

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关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 标题/主题推断。

1.3.14.65 Spa: Scaling Graph Neural Network Training on Large graphs via Probabilistic splitting

- 作者: Sandeep Polisetty; Juelin Liu; Yi-Ren Fung; S. Lim; Hui Guan; Marco Serafini
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Spa: Scaling Graph Neural Network Training on Large graphs via Probabilistic splitting

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关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 标题/主题推断。

1.3.14.66 The Hidden Bloat in Machine Learning Systems

- 作者: Huaifeng Zhang; Ahmed Ali-Eldin
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: The Hidden Bloat in Machine Learning Systems

快速理解: Software bloat refers to code and features that is not used by a software during runtime. For Machine Learning (ML) systems, bloat is a major contributor to their technical debt leading to decreased performance and resource wastage.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.14.67 A Lightweight High-Throughput Collective-Capable NoC for Large-Scale ML Accelerators

- 作者: 作者信息未提供
- 年份: 2026; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: A Lightweight High-Throughput Collective-Capable NoC for Large-Scale ML Accelerators

快速理解: The exponential increase in Machine Learning (ML) model size and complexity has driven unprecedented demand for high-performance acceleration systems. As technology scaling enables the integration of thousands of computing elements onto a single die, the boundary between distributed and on-chip systems has blurred, making efficient on-chip collective communication increasingly critical.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 本地 PDF 前页。

1.3.14.68 BEAM: Joint Resource-Power Optimization for Energy-Efficient LLM Inference under SLO constraints

- 作者: 作者信息未提供
- 年份: 2026; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: BEAM: Joint Resource-Power Optimization for Energy-Efficient LLM Inference under SLO constraints

快速理解: Large Language Model (LLM) serving is rapidly becoming one of the most power-intensive workloads in modern datacenters. Unlike training, where throughput dominates, inference must

satisfy strict per-request latency targets such as Time-to-First-Token (TTFT) and Time-Between-Tokens (TBT).

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.14.69 BOute: Cost-Efficient LLM Serving with Heterogeneous LLMs and GPUs via Multi-Objective Bayesian Optimization

- 作者：作者信息未提供
- 年份：2026；主题：AI Infra；优先级：重要
- [来源](#)

论文：BOute: Cost-Efficient LLM Serving with Heterogeneous LLMs and GPUs via Multi-Objective Bayesian Optimization

快速理解：The rapid growth of large language model (LLM) deployments has made cost-efficient serving systems essential. Recent efforts to enhance system cost-efficiency adopt two main perspectives: (i) An algorithmic perspective that exploits heterogeneous model capabilities to route simpler queries to lower-cost models and complex queries to higher-cost models (i.e., heterogeneous model routing); and (ii) a systems perspective that utilizes heterogeneous GPU resources as cost-effective alternatives to homogeneous high-end GPUs (i.e., heterogeneous model deployment).

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.14.70 BatchLLM: Optimizing Large Batched LLM Inference with Global Prefix Sharing and Throughput-oriented Token Batching

- 作者：作者信息未提供
- 年份：2026；主题：AI Infra；优先级：重要
- [来源](#)

论文：BatchLLM: Optimizing Large Batched LLM Inference with Global Prefix Sharing and Throughput-oriented Token Batching

快速理解：Large language models (LLMs) increasingly play an important role in a wide range of information processing and management tasks in industry. Many of these tasks are performed in large batches or even offline, and the performance indicator for which is throughput.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.14.71 Blueprint, Bootstrap, and Bridge: A Security Look at NVIDIA GPU Confidential Computing

- 作者：作者信息未提供
- 年份：2026；主题：AI Infra；优先级：重要
- [来源](#)

论文：Blueprint, Bootstrap, and Bridge: A Security Look at NVIDIA GPU Confidential Computing

快速理解：NVIDIA GPU Confidential Computing (GPU-CC) aims to provide secure execution for AI

workloads. For end users, enabling GPU-CC is seamless and requires no modifications to existing applications.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.14.72 CAGE: Curvature-Aware Gradient Estimation For Accurate Quantization-Aware Training

- 作者：作者信息未提供
- 年份：2026；主题：AI Infra；优先级：重要
- [来源](#)

论文：CAGE: Curvature-Aware Gradient Estimation For Accurate Quantization-Aware Training

快速理解：Despite significant work on low-bit quantization-aware training (QAT), there is still an accuracy gap between such techniques and native training. To address this, we introduce CAGE (Curvature-Aware Gradient Estimation), a new QAT method that augments the straight-through estimator (STE) gradient with a curvature-aware correction designed to counteract the loss increase induced by quantization.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.14.73 CRAFT: Fine-Grained Cost-Aware Expert Replication For Efficient Mixture-of-Experts Serving

- 作者：作者信息未提供
- 年份：2026；主题：AI Infra；优先级：重要
- [来源](#)

论文：CRAFT: Fine-Grained Cost-Aware Expert Replication For Efficient Mixture-of-Experts Serving

快速理解：Mixture-of-Experts (MoE) has recently emerged as the mainstream architecture for efficiently scaling large language models while maintaining near-constant computational cost. Expert parallelism distributes parameters by partitioning experts across devices, but this introduces token-level load imbalance during inference.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.14.74 Demystifying the Mixture of Experts Serving Tax

- 作者：作者信息未提供
- 年份：2026；主题：AI Infra；优先级：重要
- [来源](#)

论文：Demystifying the Mixture of Experts Serving Tax

快速理解：Mixture-of-Experts (MoEs) enable massive model sizes but incur higher serving overheads than dense models at the same per-token compute cost. This MoE tax varies with the model architecture, inference phase, and parallelism strategy.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.14.75 DriftBench: Measuring and Predicting Infrastructure Drift in LLM Serving Systems

- 作者：作者信息未提供
- 年份：2026；主题：AI Infra；优先级：重要
- [来源](#)

论文：DriftBench: Measuring and Predicting Infrastructure Drift in LLM Serving Systems

快速理解：Production LLM deployments lack systematic methods to assess output consistency risks when infrastructure changes. We present DriftBench, a measurement and prediction framework comprising 236,985 prompt-response pairs across 105 configurations spanning 5 models, 4 GPU platforms, 3 frameworks, 3 precisions.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.14.76 FP8-Flow-MoE: A Casting-Free FP8 Recipe without Double Quantization Error

- 作者：作者信息未提供
- 年份：2026；主题：AI Infra；优先级：重要
- [来源](#)

论文：FP8-Flow-MoE: A Casting-Free FP8 Recipe without Double Quantization Error

快速理解：Training large Mixture-of-Experts (MoE) models remains computationally prohibitive due to their extreme compute and memory demands. Although low-precision training promises to accelerate computation and reduce memory footprint, existing implementations still rely on BF16-dominated dataflows with frequent quantizedequantize (Q/DQ) conversions.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.14.77 FarSkip-Collective: Unhobbling Blocking Communication in Mixture of Experts Models

- 作者：作者信息未提供
- 年份：2026；主题：AI Infra；优先级：重要
- [来源](#)

论文：FarSkip-Collective: Unhobbling Blocking Communication in Mixture of Experts Models

快速理解：Blocking communication presents a major hurdle in running MoEs efficiently in distributed settings. To address this, we present FarSkip-Collective which modifies the architecture of modern models to enable overlapping of their computation with communication.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.14.78 FlexTrain: Scalable Hybrid-Parallel Training with Elastic Resource Utilization and Consistent Accuracy

- 作者：作者信息未提供
- 年份：2026；主题：AI Infra；优先级：重要
- [来源](#)

论文：FlexTrain: Scalable Hybrid-Parallel Training with Elastic Resource Utilization and Consistent Accuracy

快速理解：Large language model (LLM) training has become a critical workload in shared GPU clusters. However, our observations reveal that these clusters suffer from significant underutilization.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.14.79 FreeScale: Distributed Training for Sequence Recommendation Models with Minimal Scaling Cost

- 作者：作者信息未提供
- 年份：2026；主题：AI Infra；优先级：重要
- [来源](#)

论文：FreeScale: Distributed Training for Sequence Recommendation Models with Minimal Scaling Cost

快速理解：Modern industrial Deep Learning Recommendation Models typically extract user preferences through the analysis of sequential interaction histories, subsequently generating predictions based on these derived interests. The inherent heterogeneity in data characteristics frequently result in substantial under-utilization of computational resources during large-scale training, primarily due to computational bubbles caused by severe stragglers and slow blocking communications.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.14.80 G-HEMP: FAST MULTI-GPU PRIVATE INFERENCE FOR LARGE-SCALE GCNS WITH HOMOMORPHIC ENCRYPTION

- 作者：作者信息未提供
- 年份：2026；主题：AI Infra；优先级：重要
- [来源](#)

论文：G-HEMP: FAST MULTI-GPU PRIVATE INFERENCE FOR LARGE-SCALE GCNS WITH HOMOMORPHIC ENCRYPTION

快速理解：Homomorphic Encryption (HE) offers a promising solution for privacy-preserving Graph Convolutional Network (GCN) inference in untrusted cloud environments by enabling computation directly on encrypted data. This capability is particularly valuable in domains such as recommendation systems, financial analysis, and bioinformatics, where data confidentiality is paramount.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.14.81 GhostServe: A Lightweight Checkpointing System in the Shadow for Fault-Tolerant LLM Serving

- 作者：作者信息未提供
- 年份：2026；主题：AI Infra；优先级：重要
- [来源](#)

论文：GhostServe: A Lightweight Checkpointing System in the Shadow for Fault-Tolerant LLM Serving

快速理解：The rise of million-token, agent-based applications has placed unprecedented demands on large language model (LLM) inference services. The long-running nature of these tasks increases their susceptibility to hardware and software faults, leading to costly job failures, wasted resources, and degraded user experience.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.14.82 HELIOS : Adaptive Model And Early-Exit Selection for Efficient LLM Inference Serving

- 作者：作者信息未提供
- 年份：2026；主题：AI Infra；优先级：重要
- [来源](#)

论文：HELIOS : Adaptive Model And Early-Exit Selection for Efficient LLM Inference Serving

快速理解：Early-Exit Large Language Models (EE-LLMs) enable high throughput inference by allowing tokens to exit early at intermediate layers. However, their throughput is limited by the computational and memory savings.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.14.83 Hawkeye: Reproducing GPU-Level Non-Determinism

- 作者：作者信息未提供
- 年份：2026；主题：AI Infra；优先级：重要
- [来源](#)

论文：Hawkeye: Reproducing GPU-Level Non-Determinism

快速理解：We present Hawkeye, a system for analyzing and reproducing GPU-level arithmetic operations. Using our framework, anyone can re-execute on a CPU the exact matrix multiplication operations underlying a machine learning model training or inference workflow that was executed on an NVIDIA GPU, without any precision loss.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：本地 PDF 前页。

1.3.14.84 Massive-Scale Out-Of-Core UMAP on the GPU

- 作者：作者信息未提供

- 年份: 2026; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Massive-Scale Out-Of-Core UMAP on the GPU

快速理解: The Uniform Manifold Approximation and Projection (UMAP) algorithm has become a widely popular technique to reduce the dimensionality of a set of vectors, both for visualization and as a pre-processing step for follow-on machine learning tasks. UMAP is often an integral part of iterative and exploratory workflows, but the heavy amount of compute and memory required makes scaling to tens or even hundreds of gigabytes of vectors intractable on the CPU, often taking several hours to days to complete.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 本地 PDF 前页。

1.3.14.85 MixLLM: LLM Quantization with Global Mixed-precision between Output-features and Highly-efficient System Design

- 作者: 作者信息未提供
- 年份: 2026; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: MixLLM: LLM Quantization with Global Mixed-precision between Output-features and Highly-efficient System Design

快速理解: Quantization has become one of the most effective methodologies to compress LLMs into smaller size. However, the existing quantization solutions still show limitations of either non-negligible accuracy drop or low system efficiency.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 本地 PDF 前页。

1.3.14.86 MoEBlaze: Breaking the Memory Wall for Efficient MoE Training on Modern GPUs

- 作者: 作者信息未提供
- 年份: 2026; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: MoEBlaze: Breaking the Memory Wall for Efficient MoE Training on Modern GPUs

快速理解: The pervasive “memory wall” bottleneck is significantly amplified in modern large-scale Mixture-of-Experts (MoE) architectures. MoE’s inherent architectural sparsity leads to sparse arithmetic compute and also introduces substantial activation memory overheads—driven by large token routing buffers and the need to materialize and buffer intermediate tensors.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 本地 PDF 前页。

1.3.14.87 Optimizing Deployment Configurations for LLM Inference

- 作者: 作者信息未提供
- 年份: 2026; 主题: AI Infra; 优先级: 重要

- [来源](#)

论文: Optimizing Deployment Configurations for LLM Inference

快速理解: Meta's Large Language Models (LLMs)—the Llama model family—serve nearly one billion monthly active users. Deploying these models for inference involves navigating a complex design space that spans diverse hardware options (e.g., H100, H200, MI300X), multiple parallelism strategies (tensor, pipeline, expert, context, and data parallelism), and nuanced runtime choices (e.g., continuous batching versus prefill-decode disaggregation)—all while leveraging workload-specific characteristics and meeting stringent service level objectives (SLOs).

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 本地 PDF 前页。

1.3.14.88 PLA-Serve: A Prefill-Length-Aware LLM Serving System

- 作者: 作者信息未提供
- 年份: 2026; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: PLA-Serve: A Prefill-Length-Aware LLM Serving System

快速理解: Length-Aware Prefill Serving (LAPS) identifies and disaggregates requests with different prompt lengths in LLM serving to reduce TTFT latency. While recent systems have decoupled the prefill and decode stages to improve throughput, they still rely on unified scheduling policies that fail to adapt to heterogeneous workload characteristics.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 本地 PDF 前页。

1.3.14.89 ParallelKittens: Systematic and Practical Simplification of Multi-GPU AI Kernels

- 作者: 作者信息未提供
- 年份: 2026; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: ParallelKittens: Systematic and Practical Simplification of Multi-GPU AI Kernels

快速理解: Inter-GPU communication has become a major bottleneck for modern AI workloads as models scale and improvements in hardware compute throughput outpace improvements in interconnect bandwidth. Existing systems mitigate this through compute-communication overlap but often fail to meet theoretical peak performance across heterogeneous workloads and new accelerators.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 本地 PDF 前页。

1.3.14.90 ProfInfer: An eBPF-based Fine-Grained LLM Inference Profiler

- 作者: 作者信息未提供
- 年份: 2026; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: ProfInfer: An eBPF-based Fine-Grained LLM Inference Profiler

快速理解: As large language models (LLMs) move from research to production, understanding how inference engines behave in real time has become both essential and elusive. Unlike general-purpose engines such as ONNX Runtime, today's LLM inference systems offer little operator-level visibility, leaving developers blind to where time and resources go.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 本地 PDF 前页。

1.3.14.91 Scaling Up Large Language Models Serving Systems for Semantic Job Search

- 作者: 作者信息未提供
- 年份: 2026; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Scaling Up Large Language Models Serving Systems for Semantic Job Search

快速理解: Large Language Models (LLMs) have demonstrated impressive quality when applied to predictive tasks such as relevance ranking and semantic search. However, deployment of such LLMs remains prohibitively expensive for industry applications with strict latency and throughput requirements.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 本地 PDF 前页。

1.3.14.92 SwiftGS: Algorithm and System Co-Optimization for Fast 3D Gaussian Splatting on GPUs

- 作者: 作者信息未提供
- 年份: 2026; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: SwiftGS: Algorithm and System Co-Optimization for Fast 3D Gaussian Splatting on GPUs

快速理解: Recent advances in 3D Gaussian Splatting (3DGS) have enabled high-quality and efficient novel view synthesis, demonstrating great potential in real-world applications such as robotic perception and digital-twin construction. However, 3DGS requires processing up to millions of Gaussians in parallel, imposing significant computational and memory demands that limit its deployment on resource-constrained platforms.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 本地 PDF 前页。

1.3.14.93 TokenWeave: Efficient Compute-Communication Overlap for Distributed LLM Inference

- 作者: 作者信息未提供
- 年份: 2026; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: TokenWeave: Efficient Compute-Communication Overlap for Distributed LLM Inference

快速理解: Distributed inference of large language models (LLMs) using tensor parallelism can introduce communication overheads of 20% even over GPUs connected via NVLink, a high-speed GPU interconnect. Several techniques have been proposed to mitigate these overheads by decomposing computations into smaller tasks and overlapping communication with these subtasks.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 本地 PDF 前页。

1.3.14.94 When Enough is Enough: Rank-Aware Early Termination for Vector Search

- 作者: 作者信息未提供
- 年份: 2026; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: When Enough is Enough: Rank-Aware Early Termination for Vector Search

快速理解: Graph-based vector search underpins modern LLM applications such as retrieval-augmented generation (RAG), but its efficiency is increasingly constrained by disk I/O. Existing systems continue searching long after discovering the higher-ranked (i.e., more valuable) results for downstream applications.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 本地 PDF 前页。

1.3.14.95 Zorse: Optimizing LLM Training Efficiency on Heterogeneous GPU Clusters

- 作者: 作者信息未提供
- 年份: 2026; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Zorse: Optimizing LLM Training Efficiency on Heterogeneous GPU Clusters

快速理解: Large language models (LLMs) require vast amounts of GPU compute to train, but limited availability and high costs of GPUs make homogeneous clusters impractical for many organizations. Instead, assembling heterogeneous clusters by pooling together GPUs of different generations allows them to achieve higher aggregate compute and make use of all available GPUs.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 本地 PDF 前页。

1.3.14.96 Keyformer: KV Cache Reduction through Key Tokens Selection for Efficient Generative Inference

- 作者: Muhammad Adnan; Akhil Arunkumar; Gaurav Jain; Prashant J. Nair; Ilya Soloveychik; Purushotham Kamath
- 年份: 2024; 主题: Systems; 优先级: 重要
- [来源](#)

论文: Keyformer: KV Cache Reduction through Key Tokens Selection for Efficient Generative Inference

快速理解: Transformers have emerged as the underpinning architecture for Large Language Models (LLMs). In generative language models, the inference process involves two primary phases: prompt

processing and token generation.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.14.97 Efficiently Scaling Transformer Inference

- 作者：Reiner Pope; Sholto Douglas; A. Chowdhery; Jacob Devlin; James Bradbury; Anselm Levskaya; J. Heek; Kefan Xiao; Shivani Agrawal; J. Dean
- 年份：2022; 主题：AI Infra; 优先级：重要
- [来源](#)

论文：Efficiently Scaling Transformer Inference

快速理解：We study the problem of efficient generative inference for Transformer models, in one of its most challenging settings: large deep models, with tight latency targets and long sequence lengths. Better understanding of the engineering tradeoffs for inference for large Transformer-based models is important as use cases of these models are growing rapidly throughout application areas.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.14.98 Reducing Activation Recomputation in Large Transformer Models

- 作者：V. Korthikanti; J. Casper; Sangkug Lym; Lawrence C. McAfee; M. Andersch; M. Shoeybi; Bryan Catanzaro
- 年份：2022; 主题：AI Infra; 优先级：扩展
- [来源](#)

论文：Reducing Activation Recomputation in Large Transformer Models

快速理解：Training large transformer models is one of the most important computational challenges of modern AI. In this paper, we show how to significantly accelerate training of large transformer models by reducing activation recomputation.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.14.99 Gyro Dropout: Maximizing Ensemble Effect in Neural Network Training

- 作者：Jiwon Seo
- 年份：2022; 主题：AI Infra; 优先级：扩展
- [来源](#)

论文：Gyro Dropout: Maximizing Ensemble Effect in Neural Network Training

快速理解：摘要未随索引或本地 PDF 前页提供；根据标题与主题标签，本文关注 AI Infra 相关问题。以下判断仅用于快速阅读排序，不能替代全文核对。

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：标题/主题推断。

1.3.14.100 Learning on Distributed Traces for Data Center Storage Systems

- 作者: Giulio Zhou; Martin Maas
- 年份: 2021; 主题: Systems; 优先级: 扩展
- [来源](#)

论文: Learning on Distributed Traces for Data Center Storage Systems

快速理解: 摘要未随索引或本地 PDF 前页提供; 根据标题与主题标签, 本文关注 Systems 相关问题。以下判断仅用于快速阅读排序, 不能替代全文核对。

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 标题/主题推断。

1.3.15 SYSTEMS · NSDI

本会议纳入 30 篇。自动主题统计为: AI Infra (27)、Systems (12)、Agent RL (4)、Supporting Perception (1)。这些统计用于组织阅读顺序, 不能单独证明论文结论或实验优越性。

优先阅读候选: - ServeGen: Workload Characterization and Generation of Large Language Model Serving in Production(优先级分数 81.95)- FAST: An Efficient Scheduler for All-to-All GPU Communication (优先级分数 70.572) - Characterization of Large Language Model Development in the Datacenter (优先级分数 65.547) - Mutant: Learning Congestion Control from Existing Protocols via Online Reinforcement Learning (优先级分数 64.894) - BGL: GPU-Efficient GNN Training by Optimizing Graph Data I/O and Preprocessing (优先级分数 64.193)

1.3.15.1 ServeGen: Workload Characterization and Generation of Large Language Model Serving in Production

- 作者: Yuxing Xiang; Xue Li; Kun Qian; Yan Zhang; Wen-Yuan Yu; Ennan Zhai; Xin Jin; Jing-Ren Zhou
- 年份: 2025; 主题: AI Infra; Supporting Perception; Systems; 优先级: 核心
- [来源](#)

论文: ServeGen: Workload Characterization and Generation of Large Language Model Serving in Production

快速理解: With the widespread adoption of Large Language Models (LLMs), serving LLM inference requests has become an increasingly important task, attracting active research advancements. Practical workloads play an essential role in this process: they are critical for motivating and benchmarking serving techniques and systems.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 3D、视频、多模态视觉或具身感知支撑能力; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.15.2 FAST: An Efficient Scheduler for All-to-All GPU Communication

- 作者: Yiran Lei; Dongjoo Lee; Liangyu Zhao; D. Kurniawan; Chanmyeong Kim; Heetaek Jeong; Changsu Kim; Hyeonseon Choi; Liangcheng Yu; Arvind Krishnamurthy; Justine Sherry; Eriko Nurvitadhi
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: FAST: An Efficient Scheduler for All-to-All GPU Communication

快速理解: All-to-All(v) communication is a critical primitive in modern machine learning workloads, particularly mixture-of-experts (MoE) models. Unfortunately, efficient scheduling is challenging due to workload skew, heterogeneous two-tier fabrics, and incast congestion, compounded by the dynamic nature of MoE workloads, where traffic shifts every few hundred milliseconds.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.15.3 Characterization of Large Language Model Development in the Datacenter

- 作者: Qi Hu; Zhisheng Ye; Zerui Wang; Guoteng Wang; Meng Zhang; Qiaoling Chen; Peng Sun; Dahua Lin; Xiaolin Wang; Yingwei Luo; Yonggang Wen; Tianwei Zhang
- 年份: 2024; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: Characterization of Large Language Model Development in the Datacenter

快速理解: Large Language Models (LLMs) have presented impressive performance across several transformative tasks. However, it is non-trivial to efficiently utilize large-scale cluster resources to develop LLMs, often riddled with numerous challenges such as frequent hardware failures, intricate parallelization strategies, and imbalanced resource utilization.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.15.4 Mutant: Learning Congestion Control from Existing Protocols via Online Reinforcement Learning

- 作者: Lorenzo Pappone; Alessio Sacco; Flavio Esposito
- 年份: 2025; 主题: Agent RL; Systems; 优先级: 重要
- [来源](#)

论文: Mutant: Learning Congestion Control from Existing Protocols via Online Reinforcement Learning

快速理解: 摘要未随索引或本地 PDF 前页提供; 根据标题与主题标签, 本文关注 Agent RL、Systems 相关问题。以下判断仅用于快速阅读排序, 不能替代全文核对。

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 标题/主题推断。

1.3.15.5 BGL: GPU-Efficient GNN Training by Optimizing Graph Data I/O and Preprocessing

- 作者: Tianfeng Liu; Yangrui Chen; Dan Li; Chuan Wu; Yibo Zhu; Jun He; Yanghua Peng; Hong-Zheng Chen; Hong-Zhi Chen; Chuanxiong Guo
- 年份: 2021; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: BGL: GPU-Efficient GNN Training by Optimizing Graph Data I/O and Preprocessing

快速理解: Graph neural networks (GNNs) have extended the success of deep neural networks (DNNs) to non-Euclidean graph data, achieving ground-breaking performance on various tasks such as node

classification and graph property prediction. Nonetheless, existing systems are inefficient to train large graphs with billions of nodes and edges with GPUs.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.15.6 Parcae: Proactive, Liveput-Optimized DNN Training on Preemptible Instances

- 作者: Jiangfei Duan; Ziang Song; Xupeng Miao; Xiaoli Xi; Dahua Lin; Harry Xu; Minjia Zhang; Zhihao Jia
- 年份: 2024; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Parcae: Proactive, Liveput-Optimized DNN Training on Preemptible Instances

快速理解: Deep neural networks (DNNs) are becoming progressively large and costly to train. This paper aims to reduce DNN training costs by leveraging preemptible instances on modern clouds, which can be allocated at a much lower price when idle but may be preempted by the cloud provider at any time.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.15.7 ByteCheckpoint: A Unified Checkpointing System for Large Foundation Model Development

- 作者: Borui Wan; Mingji Han; Yiyao Sheng; Yanghua Peng; Haibin Lin; Mofan Zhang; Zhichao Lai; Menghan Yu; Jun-Da Zhang; Zuquan Song; Xin Liu; Chuan Wu
- 年份: 2024; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: ByteCheckpoint: A Unified Checkpointing System for Large Foundation Model Development

快速理解: Checkpointing to preserve training states is crucial during the development of Large Foundation Models (LFMs), for training resumption upon various failures or changes in GPU resources and parallelism configurations. In addition, saved checkpoints are dispatched to evaluation tasks or transferred across different training stages (e.g., from pre-training to post-training).

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.15.8 Cloud-LoRa: Enabling Cloud Radio Access LoRa Networks Using Reinforcement Learning Based Bandwidth-Adaptive Compression

- 作者: Muhammad Osama Shahid; D. Koch; Jayaram Raghuram; Bhuvana Krishnaswamy; Krishna Chintalapudi; Suman Banerjee
- 年份: 2024; 主题: Agent RL; Systems; 优先级: 重要
- [来源](#)

论文: Cloud-LoRa: Enabling Cloud Radio Access LoRa Networks Using Reinforcement Learning Based Bandwidth-Adaptive Compression

快速理解：摘要未随索引或本地 PDF 前页提供；根据标题与主题标签，本文关注 Agent RL、Systems 相关问题。以下判断仅用于快速阅读排序，不能替代全文核对。

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：标题/主题推断。

1.3.15.9 FlexLLM: Token-Level Co-Serving of LLM Inference and Finetuning with SLO Guarantees

- 作者：Gabriele Oliaro; Xupeng Miao; Xinhao Cheng; Vineeth Kada; Mengdi Wu; Ruo-Han Gao; Yingyi Huang; Rémi Delacourt; April Yang; Yingcheng Wang; Colin Unger; Zhihao Jia
- 年份：2024；主题：AI Infra; Systems；优先级：重要
- [来源](#)

论文：FlexLLM: Token-Level Co-Serving of LLM Inference and Finetuning with SLO Guarantees

快速理解：Finetuning large language models (LLMs) is essential for task adaptation, yet today's serving stacks isolate inference and finetuning on separate GPU clusters -wasting resources and under-utilizing hardware. We introduce FlexLLM, the first system to co-serve LLM inference and PEFT-based finetuning on shared GPUs by fusing computation at the token level.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.15.10 RLBoost: Harvesting Preemptible Cloud Resources for Cost-Efficient Reinforcement Learning on LLMs

- 作者：Yongji Wu; Xueshen Liu; Haizhong Zheng; Juncheng Gu; Beidi Chen; Z. Mao; Arvind Krishnamurthy; Ion Stoica
- 年份：2026；主题：Agent RL; Systems；优先级：重要
- [来源](#)

论文：RLBoost: Harvesting Preemptible Cloud Resources for Cost-Efficient Reinforcement Learning on LLMs

快速理解：摘要未随索引或本地 PDF 前页提供；根据标题与主题标签，本文关注 Agent RL、Systems 相关问题。以下判断仅用于快速阅读排序，不能替代全文核对。

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证据层级：标题/主题推断。

1.3.15.11 Bamboo: Making Preemptible Instances Resilient for Affordable Training of Large DNNs

- 作者：John Thorpe; Pengzhan Zhao; Jon Eyolfson; Yifan Qiao; Zhihao Jia; Minjia Zhang; Ravi Netravali; Guoqing Harry Xu
- 年份：2022；主题：AI Infra; Systems；优先级：重要
- [来源](#)

论文：Bamboo: Making Preemptible Instances Resilient for Affordable Training of Large DNNs

快速理解：DNN models across many domains continue to grow in size, resulting in high resource requirements for effective training, and unpalatable (and often unaffordable) costs for organizations

and research labs across scales. This paper aims to significantly reduce training costs with effective use of preemptible instances, i.e., those that can be obtained at a much cheaper price while idle, but may be preempted whenever requested by priority users.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.15.12 Optimizing RLHF Training for Large Language Models with Stage Fusion

- 作者：Yinmin Zhong; Zili Zhang; Bing-Ya Wu; Sheng-Yu Liu; Yukun Chen; Changyi Wan; Hanpeng Hu; Lei Xia; Ranchen Ming; Yi-Bo Zhu; Xin Jin
- 年份：2024；主题：Agent RL; AI Infra; 优先级：重要
- [来源](#)

论文：Optimizing RLHF Training for Large Language Models with Stage Fusion

快速理解：We present RLHFuse, an efficient training system with stage fusion for Reinforcement Learning from Human Feedback (RLHF). Due to the intrinsic nature of RLHF training, i.e., the data skewness in the generation stage and the pipeline bubbles in the training stage, existing RLHF systems suffer from low GPU utilization.

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证据层级：索引摘要。

1.3.15.13 EROICA: Online Performance Troubleshooting for Large-scale Model Training

- 作者：Yu Guan; Zhiyu Yin; Haoyu Chen; Shengxun Cheng; Chaojie Yang; Kun Qian; Tianyin Xu; Pengcheng Zhang; Yang Zhang; Hanyu Zhao; Yong Li; Dennis Cai; Ennan Zhai
- 年份：2025；主题：AI Infra; Systems; 优先级：重要
- [来源](#)

论文：EROICA: Online Performance Troubleshooting for Large-scale Model Training

快速理解：Troubleshooting performance problems of large model training (LMT) is immensely challenging, due to unprecedented scales of modern GPU clusters, the complexity of software-hardware interactions, and the data intensity of the training process. Existing troubleshooting approaches designed for traditional distributed systems or datacenter networks fall short and can hardly apply to real-world training systems.

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证据层级：索引摘要。

1.3.15.14 MegaScale: Scaling Large Language Model Training to More Than 10, 000 GPUs

- 作者：Ziheng Jiang; Haibin Lin; Yinmin Zhong; Qi Huang; Yangrui Chen; Zhi Zhang; Yanghua Peng; Xiang Li; Cong Xie; Shibiao Nong; Yulu Jia; Sunan He; Hongmin Chen; Zhihao Bai; Qi Hou; Shipeng Yan; Ding Zhou; Yiyao Sheng; Zhuo Jiang; Haohan Xu; Hao-Ran Wei; Zhang Zhang; Peng-Fei Nie; Leqi Zou; Sida Zhao; L. Xiang; Zherui Liu; Zhe Li; Xiaoying Jia; Jia-jun Ye; Xin Jin; Xin Liu
- 年份：2024；主题：AI Infra; 优先级：重要

- [来源](#)

论文: MegaScale: Scaling Large Language Model Training to More Than 10, 000 GPUs

快速理解: We present the design, implementation and engineering experience in building and deploying MegaScale, a production system for training large language models (LLMs) at the scale of more than 10,000 GPUs. Training LLMs at this scale brings unprecedented challenges to training efficiency and stability.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.15.15 Zeus: Understanding and Optimizing GPU Energy Consumption of DNN Training

- 作者: Jie You; Jaehoon Chung; Mosharaf Chowdhury
- 年份: 2022; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Zeus: Understanding and Optimizing GPU Energy Consumption of DNN Training

快速理解: Training deep neural networks (DNNs) is becoming increasingly more resource- and energy-intensive every year. Unfortunately, existing works primarily focus on optimizing DNN training for faster completion, often without considering the impact on energy efficiency.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.15.16 TopoOpt: Co-optimizing Network Topology and Parallelization Strategy for Distributed Training Jobs

- 作者: Weiyang Wang; Moein Khazraee; Zhizhen Zhong; M. Ghobadi; Zhihao Jia; Dheevatsa Mudigere; Ying Zhang; A. Kewitsch
- 年份: 2022; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: TopoOpt: Co-optimizing Network Topology and Parallelization Strategy for Distributed Training Jobs

快速理解: We propose TopoOpt, a novel direct-connect fabric for deep neural network (DNN) training workloads. TopoOpt co-optimizes the distributed training process across three dimensions: computation, communication, and network topology.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.15.17 Towards Domain-Specific Network Transport for Distributed DNN Training

- 作者: Hao Wang; Han Tian; Jingrong Chen; Xinchun Wan; Jiacheng Xia; Gaoxiong Zeng; Wei Bai; Junchen Jiang; Yong Wang; Kai Chen
- 年份: 2024; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Towards Domain-Specific Network Transport for Distributed DNN Training

快速理解: 摘要未随索引或本地 PDF 前页提供; 根据标题与主题标签, 本文关注 AI Infra 相关问题。以下判断仅用于快速阅读排序, 不能替代全文核对。

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证据层级: 标题/主题推断。

1.3.15.18 Mistify: Automating DNN Model Porting for On-Device Inference at the Edge

- 作者: Peizhen Guo; Bo Hu; Wenjun Hu
- 年份: 2021; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Mistify: Automating DNN Model Porting for On-Device Inference at the Edge

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证据层级: 标题/主题推断。

1.3.15.19 AutoCCL: Automated Collective Communication Tuning for Accelerating Distributed and Parallel DNN Training

- 作者: Guanbin Xu; Zhi-Hao Le; Yinhe Chen; Zhiqi Lin; Ze-Wen Jin; Youshan Miao; Cheng Li
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: AutoCCL: Automated Collective Communication Tuning for Accelerating Distributed and Parallel DNN Training

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关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 标题/主题推断。

1.3.15.20 ModelKeeper: Accelerating DNN Training via Automated Training Warmup

- 作者: Fan Lai; Yinwei Dai; H. Madhyastha; Mosharaf Chowdhury
- 年份: 2023; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: ModelKeeper: Accelerating DNN Training via Automated Training Warmup

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证据层级: 标题/主题推断。

1.3.15.21 GPU-Disaggregated Serving for Deep Learning Recommendation Models at Scale

- 作者: Ling-Yun Yang; Yongchen Wang; Yinghao Yu; Qizhen Weng; Jianbo Dong; Kan Liu; Chi Zhang; Yanyi Zi; Hao Li; Zechao Zhang; Nan Wang; Yuyi Dong; Menglei Zheng; Lan-Lan Xi; Xiao-Wei Lu; Liang Ye; Guodong Yang; Binzhang Fu; Tao Lan; Liping Zhang; Lin Qu; Wei Wang
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: GPU-Disaggregated Serving for Deep Learning Recommendation Models at Scale

快速理解: 摘要未随索引或本地 PDF 前页提供; 根据标题与主题标签, 本文关注 AI Infra 相关问题。以下判断仅用于快速阅读排序, 不能替代全文核对。

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证据层级: 标题/主题推断。

1.3.15.22 Holmes: Localizing Irregularities in LLM Training with Mega-scale GPU Clusters

- 作者: Zhiyi Yao; Peng-Bo Hu; Congcong Miao; Xuya Jia; Zuning Liang; Yuedong Xu; Chunzhi He; Hao Lu; Mingzhuo Chen; Xiang Li; Zekun He; Yachen Wang; Xianneng Zou; Junchen Jiang
- 年份: 2025; 主题: AI Infra; 优先级: 重要
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论文: Holmes: Localizing Irregularities in LLM Training with Mega-scale GPU Clusters

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证据层级: 标题/主题推断。

1.3.15.23 Accelerating Collective Communication in Data Parallel Training across Deep Learning Frameworks

- 作者: J. Romero; Junqi Yin; N. Laanait; Bing Xie; M. Young; Sean Treichler; V. Starchenko; A. Borisevich; Alexander Sergeev; Michael A. Matheson
- 年份: 2022; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Accelerating Collective Communication in Data Parallel Training across Deep Learning Frameworks

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证据层级: 标题/主题推断。

1.3.15.24 ARK: GPU-driven Code Execution for Distributed Deep Learning

- 作者: Changho Hwang; Kyoungsoo Park; Ran Shu; Xinyuan Qu; Peng Cheng; Yongqiang Xiong

- 年份: 2023; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: ARK: GPU-driven Code Execution for Distributed Deep Learning

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证据层级: 标题/主题推断。

1.3.15.25 ForestColl: Throughput-Optimal Collective Communications on Heterogeneous Network Fabrics

- 作者: Liangyu Zhao; Saeed Maleki; Yuanhong Wang; Zezhou Wang; Ziyue Yang; Hossein Pourreza; Arvind Krishnamurthy
- 年份: 2024; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: ForestColl: Throughput-Optimal Collective Communications on Heterogeneous Network Fabrics

快速理解: As modern DNN models grow ever larger, collective communications between the accelerators (allreduce, etc.) emerge as a significant performance bottleneck. Designing efficient communication schedules is challenging, given today's heterogeneous and diverse network fabrics.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.15.26 FLARE: Anomaly Diagnostics for Divergent LLM Training in GPU Clusters of Thousand-Plus Scale

- 作者: Wei Cui; Ji Zhang; Han Zhao; Chao Liu; Jian Sha; Bo Sang; Bingsheng He; M. Guo; Quan Chen
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: FLARE: Anomaly Diagnostics for Divergent LLM Training in GPU Clusters of Thousand-Plus Scale

快速理解: The rapid proliferation of large language models has driven the need for efficient GPU training clusters. However, it is challenging due to the frequent occurrence of training anomalies.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.15.27 Phantora: Maximizing Code Reuse in Simulation-based Machine Learning System Performance Estimation

- 作者: Jianxing Qin; Jingrong Chen; Xinhao Kong; Yongji Wu; Tianjun Yuan; Liang Luo; Zhaodong Wang; Ying Zhang; Tingjun Chen; A. Lebeck; Danyang Zhuo
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Phantora: Maximizing Code Reuse in Simulation-based Machine Learning System Performance Estimation

快速理解: Modern machine learning (ML) training workloads place substantial demands on both computational and communication resources. Consequently, accurate performance estimation has become increasingly critical for guiding system design decisions, such as the selection of parallelization strategies, cluster configurations, and hardware provisioning.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.15.28 Sparse Checkpointing for Fast and Reliable MoE Training

- 作者: Swapnil Gandhi; Christoforos E. Kozyrakis
- 年份: 2024; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Sparse Checkpointing for Fast and Reliable MoE Training

快速理解: As large language models scale, training them requires thousands of GPUs over extended durations-making frequent failures an inevitable reality. While checkpointing remains the primary fault-tolerance mechanism, existing methods fall short when applied to Mixture-of-Experts (MoE) models.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.15.29 SYMI: Efficient Mixture-of-Experts Training via Model and Optimizer State Decoupling

- 作者: Athinagoras Skiadopoulos; Mark Zhao; Swapnil Gandhi; Thomas Norrie; Shrijeet Mukherjee; Christoforos E. Kozyrakis
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: SYMI: Efficient Mixture-of-Experts Training via Model and Optimizer State Decoupling

快速理解: Mixture-of-Experts (MoE) models have become a widely-adopted solution to continue scaling model sizes without a corresponding linear increase in compute. During MoE model training, each input token is dynamically routed to a subset of experts -sparsely-activated feed-forward networks -within each transformer layer.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.15.30 Checkmate: Zero Performance Overhead Model Checkpointing via Network Gradient Replication

- 作者: Ankit Bhardwaj; Weiyang Wang; Jeremy Carin; Adam Belay; M. Ghobadi
- 年份: 2026; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Checkmate: Zero Performance Overhead Model Checkpointing via Network Gradient Replica-

tion

快速理解：摘要未随索引或本地 PDF 前页提供；根据标题与主题标签，本文关注 AI Infra 相关问题。以下判断仅用于快速阅读排序，不能替代全文核对。

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：标题/主题推断。

1.3.16 SYSTEMS · OSDI

本次会议纳入 36 篇。自动主题统计为：AI Infra (23)、Systems (14)。这些统计用于组织阅读顺序，不能单独证明论文结论或实验优越性。

优先阅读候选：- Lllumnix: Dynamic Scheduling for Large Language Model Serving (优先级分数 75.224) - Taming Throughput-Latency Tradeoff in LLM Inference with Sarathi-Serve (优先级分数 67.662) - InfiniGen: Efficient Generative Inference of Large Language Models with Dynamic KV Cache Management (优先级分数 66.371) - ServerlessLLM: Low-Latency Serverless Inference for Large Language Models (优先级分数 65.637) - Dorylus: Affordable, Scalable, and Accurate GNN Training with Distributed CPU Servers and Serverless Threads (优先级分数 64.781)

1.3.16.1 Lllumnix: Dynamic Scheduling for Large Language Model Serving

- 作者：Biao Sun; Ziming Huang; Hanyu Zhao; Wencong Xiao; Xinyi Zhang; Yong Li; Wei Lin
- 年份：2024；主题：AI Infra; Systems；优先级：核心
- [来源](#)

论文：Lllumnix: Dynamic Scheduling for Large Language Model Serving

快速理解：Inference serving for large language models (LLMs) is the key to unleashing their potential in people's daily lives. However, efficient LLM serving remains challenging today because the requests are inherently heterogeneous and unpredictable in terms of resource and latency requirements, as a result of the diverse applications and the dynamic execution nature of LLMs.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.16.2 Taming Throughput-Latency Tradeoff in LLM Inference with Sarathi-Serve

- 作者：Amey Agrawal; Nitin Kedia; Ashish Panwar; Jayashree Mohan; Nipun Kwatra; Bhargav S. Gulavani; Alexey Tumanov; R. Ramjee
- 年份：2024；主题：AI Infra; Systems；优先级：核心
- [来源](#)

论文：Taming Throughput-Latency Tradeoff in LLM Inference with Sarathi-Serve

快速理解：Each LLM serving request goes through two phases. The first is prefill which processes the entire input prompt and produces the first output token and the second is decode which generates the rest of output tokens, one-at-a-time.

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证据层级：索引摘要。

1.3.16.3 InfiniGen: Efficient Generative Inference of Large Language Models with Dynamic KV Cache Management

- 作者: Wonbeom Lee; Jungi Lee; Junghwa Seo; Jaewoong Sim
- 年份: 2024; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: InfiniGen: Efficient Generative Inference of Large Language Models with Dynamic KV Cache Management

快速理解: Transformer-based large language models (LLMs) demonstrate impressive performance across various natural language processing tasks. Serving LLM inference for generating long contents, however, poses a challenge due to the enormous memory footprint of the transient state, known as the key-value (KV) cache, which scales with the sequence length and batch size.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.16.4 ServerlessLLM: Low-Latency Serverless Inference for Large Language Models

- 作者: Yao Fu; Leyang Xue; Yeqi Huang; Andrei-Octavian Brabete; D. Ustiugov; Yuvraj Patel; Luo Mai
- 年份: 2024; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: ServerlessLLM: Low-Latency Serverless Inference for Large Language Models

快速理解: This paper presents ServerlessLLM, a distributed system designed to support low-latency serverless inference for Large Language Models (LLMs). By harnessing the substantial near-GPU storage and memory capacities of inference servers, ServerlessLLM achieves effective local checkpoint storage, minimizing the need for remote checkpoint downloads and ensuring efficient checkpoint loading.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.16.5 Dorylus: Affordable, Scalable, and Accurate GNN Training with Distributed CPU Servers and Serverless Threads

- 作者: John Thorpe; Yifan Qiao; Jon Eyolfson; Shen Teng; Guanzhou Hu; Zhihao Jia; Jinliang Wei; Keval Vora; Ravi Netravali; Miryung Kim; G. Xu
- 年份: 2021; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Dorylus: Affordable, Scalable, and Accurate GNN Training with Distributed CPU Servers and Serverless Threads

快速理解: A graph neural network (GNN) enables deep learning on structured graph data. There are two major GNN training obstacles: 1) it relies on high-end servers with many GPUs which are expensive to purchase and maintain, and 2) limited memory on GPUs cannot scale to today's billion-edge graphs.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.16.6 Privacy Budget Scheduling

- 作者: Tao Luo; Mingen Pan; Pierre Tholoniati; Asaf Cidon; Roxana Geambasu; M. L'ecuyer
- 年份: 2021; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Privacy Budget Scheduling

快速理解: Machine learning (ML) models trained on personal data have been shown to leak information about users. Differential privacy (DP) enables model training with a guaranteed bound on this leakage.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.16.7 QiMeng-Xpiler: Transcompiling Tensor Programs for Deep Learning Systems with a Neural-Symbolic Approach

- 作者: Shouyang Dong; Yuanbo Wen; Jun Bi; Di Huang; Jiaming Guo; Jianxing Xu; Ruibai Xu; Xinkai Song; Yifan Hao; Xuehai Zhou; Tianshi Chen; Qi Guo; Yunji Chen
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: QiMeng-Xpiler: Transcompiling Tensor Programs for Deep Learning Systems with a Neural-Symbolic Approach

快速理解: Heterogeneous deep learning systems (DLS) such as GPUs and ASICs have been widely deployed in industrial data centers, which requires to develop multiple low-level tensor programs for different platforms. An attractive solution to relieve the programming burden is to transcompile the legacy code of one platform to others.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.16.8 Fairness in Serving Large Language Models

- 作者: Ying Sheng; Shiyi Cao; Da-Cheng Li; Banghua Zhu; Zhuohan Li; Danyang Zhuo; Joseph E. Gonzalez; Ion Stoica
- 年份: 2023; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Fairness in Serving Large Language Models

快速理解: High-demand LLM inference services (e.g., ChatGPT and BARD) support a wide range of requests from short chat conversations to long document reading. To ensure that all client requests are processed fairly, most major LLM inference services have request rate limits, to ensure that no client can dominate the request queue.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.16.9 Caravan: Practical Online Learning of In-Network ML Models with Labeling Agents

- 作者: Qizheng Zhang; Ali Imran; Enkeleda Bardhi; Tushar Swamy; Nathan Zhang; Muhammad Shahbaz; K. Olukotun
- 年份: 2024; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Caravan: Practical Online Learning of In-Network ML Models with Labeling Agents

快速理解: Recent work on in-network machine learning (ML) anticipates offline models to operate well in modern networking environments. However, upon deployment, these models struggle to cope with fluctuating traffic patterns and network conditions and, therefore, must be validated and updated frequently in an online fashion.

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证据层级: 索引摘要。

1.3.16.10 ACCL+: an FPGA-Based Collective Engine for Distributed Applications

- 作者: Zhenhao He; Dario Korolija; Yu Zhu; Benjamin Ramhorst; Tristan Laan; Lucian Petrica; M. Blott; Gustavo Alonso
- 年份: 2023; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: ACCL+: an FPGA-Based Collective Engine for Distributed Applications

快速理解: FPGAs are increasingly prevalent in cloud deployments, serving as Smart NICs or network-attached accelerators. Despite their potential, developing distributed FPGA-accelerated applications remains cumbersome due to the lack of appropriate infrastructure and communication abstractions.

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证据层级: 索引摘要。

1.3.16.11 Orca: A Distributed Serving System for Transformer-Based Generative Models

- 作者: Gyeong-In Yu; Joo Seong Jeong
- 年份: 2022; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Orca: A Distributed Serving System for Transformer-Based Generative Models

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证据层级: 标题/主题推断。

1.3.16.12 AlpaServe: Statistical Multiplexing with Model Parallelism for Deep Learning Serving

- 作者: Zhuohan Li; Lianmin Zheng; Yinmin Zhong; Vincent Liu; Ying Sheng; Xin Jin; Yanping Huang; Z. Chen; Hao Zhang; Joseph E. Gonzalez; Ion Stoica
- 年份: 2023; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: AlpaServe: Statistical Multiplexing with Model Parallelism for Deep Learning Serving

快速理解: Model parallelism is conventionally viewed as a method to scale a single large deep learning model beyond the memory limits of a single device. In this paper, we demonstrate that model parallelism can be additionally used for the statistical multiplexing of multiple devices when serving multiple models, even when a single model can fit into a single device.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.16.13 Unity: Accelerating DNN Training Through Joint Optimization of Algebraic Transformations and Parallelization

- 作者: Colin Unger; Zhihao Jia; Wei Wu; Sina Lin; Mandeep Baines; C. Narvaez; Vinay Ramakrishnaiah; Nirmal Prajapati; Patrick S. McCormick; Jamal Mohd-Yusof; Xi Luo; Dheevatsa Mudigere; Jongsoo Park; Misha Smelyanskiy; Alexander Aiken
- 年份: 2022; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Unity: Accelerating DNN Training Through Joint Optimization of Algebraic Transformations and Parallelization

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证据层级: 标题/主题推断。

1.3.16.14 Quake: Adaptive Indexing for Vector Search

- 作者: J. Mohoney; Devesh Sarda; Meng-Ze Tang; S. R. Chowdhury; Anil Pacaci; Ihab F. Ilyas; Theodoros Rekatsinas; Shivaram Venkataraman
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Quake: Adaptive Indexing for Vector Search

快速理解: Vector search, the task of finding the k-nearest neighbors of a query vector against a database of high-dimensional vectors, underpins many machine learning applications, including retrieval-augmented generation, recommendation systems, and information retrieval. However, existing approximate nearest neighbor (ANN) methods perform poorly under dynamic and skewed workloads where data distributions evolve.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.16.15 Enabling Efficient GPU Communication over Multiple NICs with FuseLink

- 作者: Zhenghang Ren; Yuxuan Li; Zilong Wang; Xinyang Huang; Wenxue Li; Kaiqiang Xu; Xudong Liao; Yijun Sun; Bowen Liu; Ying Zhang; Kai Chen; Han Tian; Junxue Zhang; Mingfei Wang; Zhizhen Zhong; Guyue Liu
- 年份: 2025; 主题: AI Infra; 优先级: 重要

- [来源](#)

论文: Enabling Efficient GPU Communication over Multiple NICs with FuseLink

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证据层级: 标题/主题推断。

1.3.16.16 MonoNN: Enabling a New Monolithic Optimization Space for Neural Network Inference Tasks on Modern GPU-Centric Architectures

- 作者: Donglin Zhuang; Zhen Zheng; Haojun Xia; Xiafei Qiu; Jun-Jie Bai; Wei Lin; S. Song
- 年份: 2024; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: MonoNN: Enabling a New Monolithic Optimization Space for Neural Network Inference Tasks on Modern GPU-Centric Architectures

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证据层级: 标题/主题推断。

1.3.16.17 DecDEC: A Systems Approach to Advancing Low-Bit LLM Quantization

- 作者: Yeonhong Park; Jake Hyun; Hojoon Kim; Jae W. Lee
- 年份: 2024; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: DecDEC: A Systems Approach to Advancing Low-Bit LLM Quantization

快速理解: Quantization of Large Language Models (LLMs) has recently gained popularity, particularly for on-device settings with limited hardware resources. While efficient, quantization inevitably degrades model quality, especially in aggressive low-bit settings such as 3-bit and 4-bit precision.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.16.18 ZEN: Empowering Distributed Training with Sparsity-driven Data Synchronization

- 作者: Zhuang Wang; Zhaozhuo Xu; Jingyi Xi; Yuke Wang; Anshumali Shrivastava; T. S. Eugene Ng
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: ZEN: Empowering Distributed Training with Sparsity-driven Data Synchronization

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证据层级：标题/主题推断。

1.3.16.19 Performance Interfaces for Hardware Accelerators

- 作者: Jiacheng Ma; Rishabh R. Iyer; Sahand Kashani; Mahyar Emami; Thomas Bourgeat; George Candea
- 年份: 2024; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Performance Interfaces for Hardware Accelerators

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证据层级: 标题/主题推断。

1.3.16.20 Welder: Scheduling Deep Learning Memory Access via Tile-graph

- 作者: Yining Shi; Zhi Yang; Jilong Xue; Lingxiao Ma; Yuqing Xia; Ziming Miao; Yuxiao Guo; Fan Yang; Lidong Zhou
- 年份: 2023; 主题: Systems; 优先级: 扩展
- [来源](#)

论文: Welder: Scheduling Deep Learning Memory Access via Tile-graph

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关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 标题/主题推断。

1.3.16.21 Alpa: Automating Inter- and Intra-Operator Parallelism for Distributed Deep Learning

- 作者: Lianmin Zheng; Zhuohan Li; Hao Zhang; Yonghao Zhuang; Zhifeng Chen; Yan-Ping Huang; Yi-Da Wang; Yuan-Zhong Xu; Danyang Zhuo; Eric P. Xing; Joseph E. Gonzalez; Ion Stoica
- 年份: 2022; 主题: AI Infra; 优先级: 扩展
- [来源](#)

论文: Alpa: Automating Inter- and Intra-Operator Parallelism for Distributed Deep Learning

快速理解: Alpa automates model-parallel training of large deep learning (DL) models by generating execution plans that unify data, operator, and pipeline parallelism. Existing model-parallel training systems either require users to manually create a parallelization plan or automatically generate one from a limited space of model parallelism configurations.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.16.22 MAST: Global Scheduling of ML Training across Geo-Distributed Datacenters at Hyperscale

- 作者: Arnab Choudhury; Yang Wang; Tuomas Pelkonen; Kutta Srinivasan; Abha Jain; Shenghao Lin; Delia David; Siavash Soleimanifard; Michael Chen; Abhishek Yadav; Ritesh Tijoriwala; Denis Samoylov; Chunqiang Tang
- 年份: 2024; 主题: Systems; 优先级: 扩展
- [来源](#)

论文: MAST: Global Scheduling of ML Training across Geo-Distributed Datacenters at Hyperscale

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证据层级: 标题/主题推断。

1.3.16.23 Walle: An End-to-End, General-Purpose, and Large-Scale Production System for Device-Cloud Collaborative Machine Learning

- 作者: Chengfei Lv; Chaoyue Niu; Renjie Gu; Xiaotang Jiang; Zhaode Wang; B. Liu; Zi-Qi Wu; Qiulin Yao; Congyu Huang; Panos Huang; Tao Huang; Hui Shu; Jinde Song; Bingzhang Zou; Peng Lan; Guohuan Xu; Fei Wu; Shaojie Tang; Fan Wu; Guihai Chen
- 年份: 2022; 主题: Systems; 优先级: 扩展
- [来源](#)

论文: Walle: An End-to-End, General-Purpose, and Large-Scale Production System for Device-Cloud Collaborative Machine Learning

快速理解: To break the bottlenecks of mainstream cloud-based machine learning (ML) paradigm, we adopt device-cloud collaborative ML and build the first end-to-end and general-purpose system, called Walle, as the foundation. Walle consists of a deployment platform, distributing ML tasks to billion-scale devices in time; a data pipeline, efficiently preparing task input; and a compute container, providing a cross-platform and high-performance execution environment, while facilitating daily task iteration.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.16.24 BlockFlex: Enabling Storage Harvesting with Software-Defined Flash in Modern Cloud Platforms

- 作者: Benjamin Reidys; Jinghan Sun; Anirudh Badam; S. Noghabi; Jian Huang
- 年份: 2022; 主题: Systems; 优先级: 扩展
- [来源](#)

论文: BlockFlex: Enabling Storage Harvesting with Software-Defined Flash in Modern Cloud Platforms

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证据层级: 标题/主题推断。

1.3.16.25 Learning Massive Graph Embeddings on a Single Machine

- 作者: J. Mohoney; R. Waleffe; Yiheng Xu; Theodoros Rekatsinas; S. Venkataraman
- 年份: 2021; 主题: AI Infra; 优先级: 扩展
- [来源](#)

论文: Learning Massive Graph Embeddings on a Single Machine

快速理解: We propose a new framework for computing the embeddings of large-scale graphs on a single machine. A graph embedding is a fixed length vector representation for each node (and/or edge-type) in a graph and has emerged as the de-facto approach to apply modern machine learning on graphs.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.16.26 Understanding Stragglers in Large Model Training Using What-if Analysis

- 作者: Jinkun Lin; Ziheng Jiang; Zuquan Song; Sida Zhao; Menghan Yu; Zhanghan Wang; Chen-Yuan Wang; Zuocheng Shi; Xiang Shi; Wei Jia; Zherui Liu; Shuguang Wang; Haibin Lin; Xin Liu; Aurojit Panda; Jinyang Li
- 年份: 2025; 主题: AI Infra; 优先级: 扩展
- [来源](#)

论文: Understanding Stragglers in Large Model Training Using What-if Analysis

快速理解: Large language model (LLM) training is one of the most demanding distributed computations today, often requiring thousands of GPUs with frequent synchronization across machines. Such a workload pattern makes it susceptible to stragglers, where the training can be stalled by few slow workers.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.16.27 WaferLLM: Large Language Model Inference at Wafer Scale

- 作者: Congjie He; Yeqi Huang; Pei Mu; Ziming Miao; Jilong Xue; Lingxiao Ma; Fan Yang; Luo Mai
- 年份: 2025; 主题: AI Infra; 优先级: 扩展
- [来源](#)

论文: WaferLLM: Large Language Model Inference at Wafer Scale

快速理解: Emerging AI accelerators increasingly adopt wafer-scale manufacturing technologies, integrating hundreds of thousands of AI cores in a mesh architecture with large distributed on-chip memory (tens of GB in total) and ultra-high on-chip memory bandwidth (tens of PB/s). However, current LLM inference systems, optimized for shared memory architectures like GPUs, fail to exploit these accelerators fully.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.16.28 DistAI: Data-Driven Automated Invariant Learning for Distributed Protocols

- 作者: Jianan Yao; Runzhou Tao; Ronghui Gu; Jason Nieh; S. Jana; Gabriel Ryan
- 年份: 2021; 主题: 未标注主题; 优先级: 扩展
- [来源](#)

论文: DistAI: Data-Driven Automated Invariant Learning for Distributed Protocols

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关注点: 需打开全文确认研究问题、方法和实验边界。

证据层级: 标题/主题推断。

1.3.16.29 nnScaler: Constraint-Guided Parallelization Plan Generation for Deep Learning Training

- 作者: Zhiqi Lin; Youshan Miao; Quanlu Zhang; Fan Yang; Yi Zhu; Cheng Li; Saeed Maleki; Xu Cao; Ning Shang; Yilei Yang; Weijiang Xu; Mao Yang; Lintao Zhang; Lidong Zhou
- 年份: 2024; 主题: 未标注主题; 优先级: 扩展
- [来源](#)

论文: nnScaler: Constraint-Guided Parallelization Plan Generation for Deep Learning Training

快速理解: 摘要未随索引或本地 PDF 前页提供; 根据标题与主题标签, 本文关注未标注主题相关问题。以下判断仅用于快速阅读排序, 不能替代全文核对。

关注点: 需打开全文确认研究问题、方法和实验边界。

证据层级: 标题/主题推断。

1.3.16.30 USHER: Holistic Interference Avoidance for Resource Optimized ML Inference

- 作者: Sudipta Saha Shubha; Hai-Ying Shen; Anand Iyer
- 年份: 2024; 主题: 未标注主题; 优先级: 扩展
- [来源](#)

论文: USHER: Holistic Interference Avoidance for Resource Optimized ML Inference

快速理解: 摘要未随索引或本地 PDF 前页提供; 根据标题与主题标签, 本文关注未标注主题相关问题。以下判断仅用于快速阅读排序, 不能替代全文核对。

关注点: 需打开全文确认研究问题、方法和实验边界。

证据层级: 标题/主题推断。

1.3.16.31 WLB-LLM: Workload-Balanced 4D Parallelism for Large Language Model Training

- 作者: Zheng Wang; Anna Cai; Xinfeng Xie; Zaifeng Pan; Yue Guan; Weiwei Chu; Jie Wang; Shikai Li; Jianyu Huang; Chris Cai; Yuchen Hao; Yu-Fei Ding
- 年份: 2025; 主题: 未标注主题; 优先级: 扩展
- [来源](#)

论文: WLB-LLM: Workload-Balanced 4D Parallelism for Large Language Model Training

快速理解: In this work, we present WLB-LLM, a workLoad-balanced 4D parallelism for large language model training. We first thoroughly analyze the workload imbalance issue in LLM training and identify two primary sources of imbalance at the pipeline parallelism and context parallelism levels.

关注点: 需打开全文确认研究问题、方法和实验边界。

证据层级: 索引摘要。

1.3.16.32 DuoAI: Fast, Automated Inference of Inductive Invariants for Verifying Distributed Protocols

- 作者: Jianan Yao; Runzhou Tao; Ronghui Gu; Jason Nieh
- 年份: 2022; 主题: 未标注主题; 优先级: 扩展
- [来源](#)

论文: DuoAI: Fast, Automated Inference of Inductive Invariants for Verifying Distributed Protocols

快速理解: 摘要未随索引或本地 PDF 前页提供; 根据标题与主题标签, 本文关注未标注主题相关问题。以下判断仅用于快速阅读排序, 不能替代全文核对。

关注点: 需打开全文确认研究问题、方法和实验边界。

证据层级: 标题/主题推断。

1.3.16.33 Polyjuice: High-Performance Transactions via Learned Concurrency Control

- 作者: Jia-Chen Wang; Ding Ding; Huan Wang; Conrad Christensen; Zhaoguo Wang; Haibo Chen; Jinyang Li
- 年份: 2021; 主题: 未标注主题; 优先级: 扩展
- [来源](#)

论文: Polyjuice: High-Performance Transactions via Learned Concurrency Control

快速理解: Concurrency control algorithms are key determinants of the performance of in-memory databases. Existing algorithms are designed to work well for certain workloads.

关注点: 需打开全文确认研究问题、方法和实验边界。

证据层级: 索引摘要。

1.3.16.34 Ekko: A Large-Scale Deep Learning Recommender System with Low-Latency Model Update

- 作者: Chijun Sima; Yao Fu; Man-Kit Sit; Liyi Guo; X. Gong; Feng Lin; Junyu Wu; Yongsheng Li; Haidong Rong; Pierre-Louis Aublin; Luo Mai
- 年份: 2022; 主题: 未标注主题; 优先级: 扩展
- [来源](#)

论文: Ekko: A Large-Scale Deep Learning Recommender System with Low-Latency Model Update

快速理解: 摘要未随索引或本地 PDF 前页提供; 根据标题与主题标签, 本文关注未标注主题相关问题。以下判断仅用于快速阅读排序, 不能替代全文核对。

关注点: 需打开全文确认研究问题、方法和实验边界。

证据层级：标题/主题推断。

1.3.16.35 Training with Confidence: Catching Silent Errors in Deep Learning Training with Automated Proactive Checks

- 作者：Yuxuan Jiang; Ziming Zhou; Boyu Xu; Beijie Liu; Runhui Xu; Peng Huang
- 年份：2025；主题：未标注主题；优先级：扩展
- [来源](#)

论文：Training with Confidence: Catching Silent Errors in Deep Learning Training with Automated Proactive Checks

快速理解：Training deep learning (DL) models is a complex process, making it prone to silent errors that are challenging to detect and diagnose. This paper presents TRAINCHECK, a framework that takes a proactive checking approach to address silent training errors.

关注点：需打开全文确认研究问题、方法和实验边界。

证据层级：索引摘要。

1.3.16.36 Inductive Invariants That Spark Joy: Using Invariant Taxonomies to Streamline Distributed Protocol Proofs

- 作者：T. Zhang; Travis Hance; Manos Kapritsos; Tej Chajed; Bryan Parno
- 年份：2024；主题：未标注主题；优先级：扩展
- [来源](#)

论文：Inductive Invariants That Spark Joy: Using Invariant Taxonomies to Streamline Distributed Protocol Proofs

快速理解：摘要未随索引或本地 PDF 前页提供；根据标题与主题标签，本文关注未标注主题相关问题。以下判断仅用于快速阅读排序，不能替代全文核对。

关注点：需打开全文确认研究问题、方法和实验边界。

证据层级：标题/主题推断。

1.3.17 SYSTEMS · PPOPP

本次会议纳入 20 篇。自动主题统计为：AI Infra (20)、Systems (18)、Teleoperation (1)。这些统计用于组织阅读顺序，不能单独证明论文结论或实验优越性。

优先阅读候选：- ApproxTuner: a compiler and runtime system for adaptive approximations (优先级分数 80.294) - JanusQuant: Accurate and Efficient 2-bit KV Cache Quantization for Long-Context Inference(优先级分数 68.5)- TAC: Cache-Based System for Accelerating Billion-Scale GNN Training on Multi-GPU Platform(优先级分数 67.5)- Shared Memory-contention-aware Concurrent DNN Execution for Diversely Heterogeneous System-on-Chips(优先级分数 62.45)- SGDRC: Software-Defined Dynamic Resource Control for Concurrent DNN Inference on NVIDIA GPUs (优先级分数 62.322)

1.3.17.1 ApproxTuner: a compiler and runtime system for adaptive approximations

- 作者：Hashim Sharif; Yifan Zhao; Maria Kotsifakou; Akash Kothari; Ben Schreiber; Elizabeth Wang; Yasmin Sarita; Nathan Zhao; Keyur Joshi; V. Adve; Sasa Misailovic; S. Adve
- 年份：2021；主题：Teleoperation; AI Infra; Systems；优先级：核心
- [来源](#)

论文: ApproxTuner: a compiler and runtime system for adaptive approximations

快速理解: Manually optimizing the tradeoffs between accuracy, performance and energy for resource-intensive applications with flexible accuracy or precision requirements is extremely difficult. We present ApproxTuner, an automatic framework for accuracy-aware optimization of tensor-based applications while requiring only high-level end-to-end quality specifications.

关注点: 人类示范、远程操作、共享自治或动作重定向; 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.17.2 JanusQuant: Accurate and Efficient 2-bit KV Cache Quantization for Long-Context Inference

- 作者: Chengyu Sun; Yaqi Xia; Hulin Wang; Donglin Yang; Xiao-Bo Zhou; Dazhao Cheng
- 年份: 2026; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: JanusQuant: Accurate and Efficient 2-bit KV Cache Quantization for Long-Context Inference

快速理解: Long-context large language models (LLMs) have seen widespread adoption in recent years. However, during inference, the key-value (KV) cache—which stores intermediate activations—consumes significant memory, particularly as sequence lengths grow.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.17.3 TAC: Cache-Based System for Accelerating Billion-Scale GNN Training on Multi-GPU Platform

- 作者: Zhiqiang Liang; Hongyu Gao; Jue Wang; Fang Liu; Xingguo Shi; Junyu Gu; Peng Di; Sian Li; Lei Tang; Chunbao Zhou; Lian Zhao; Yangang Wang; Xuebin Chi
- 年份: 2026; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: TAC: Cache-Based System for Accelerating Billion-Scale GNN Training on Multi-GPU Platform

快速理解: Graph neural networks (GNNs) have been proven to have increasingly widespread applications in the real world. In the mainstream mini-batch training mode, multiple cache-based GNN training acceleration systems have been proposed because of the possibility of selecting the same vertex multiple times during the sampling process.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.17.4 Shared Memory-contention-aware Concurrent DNN Execution for Diversely Heterogeneous System-on-Chips

- 作者: Ismet Dagli; M. E. Belviranli
- 年份: 2023; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Shared Memory-contention-aware Concurrent DNN Execution for Diversely Heterogeneous System-on-Chips

快速理解: Two distinguishing features of state-of-the-art mobile and autonomous systems are: 1) There are often multiple workloads, mainly deep neural network (DNN) inference, running concurrently and continuously. 2) They operate on shared memory System-on-Chips (SoC) that embed heterogeneous accelerators tailored for specific operations.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.17.5 SGDRC: Software-Defined Dynamic Resource Control for Concurrent DNN Inference on NVIDIA GPUs

- 作者: Yongkang Zhang; Haoxuan Yu; Chenxia Han; Cheng Wang; Baotong Lu; Yun-Zhe Li; Zhifeng Jiang; Y. Li; Xiao-Wen Chu; Huaicheng Li
- 年份: 2024; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: SGDRC: Software-Defined Dynamic Resource Control for Concurrent DNN Inference on NVIDIA GPUs

快速理解: Cloud service providers heavily colocate high-priority, latency sensitive (LS), and low-priority, best-effort (BE) DNN inference services on the same GPU to improve resource utilization in data centers. Among the critical shared GPU resources, there has been very limited analysis on the dynamic allocation of compute units and VRAM bandwidth, mainly for two reasons: (1) The native GPU resource management solutions are either hardware-specific, or unable to dynamically allocate resources to different tenants, or both; (2) NVIDIA doesn't expose interfaces for VRAM bandwidth allocation, and the software stack and VRAM channel architectures are black-box, both of which limit the software-level resource management.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.17.6 Mario: Near Zero-cost Activation Checkpointing in Pipeline Parallelism

- 作者: Weijian Liu; Ming-Zhen Li; Guangming Tan; Wei-Le Jia
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Mario: Near Zero-cost Activation Checkpointing in Pipeline Parallelism

快速理解: Large language models have to be trained in parallel due to their large number of parameters and significant memory footprint. Among various parallelism techniques, pipeline parallelism is widely adopted in inter-node scenarios with minimal communication overhead.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.17.7 Efficient All-Reduce for Distributed DNN Training in Optical Interconnect Systems

- 作者: Fei Dai; Yawen Chen; Zhiyi Huang; Haibo Zhang; Fangfang Zhang
- 年份: 2022; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Efficient All-Reduce for Distributed DNN Training in Optical Interconnect Systems

快速理解: All-reduce is the crucial communication primitive to reduce model parameters in distributed Deep Neural Networks (DNN) training. Most existing all-reduce algorithms are designed for traditional electrical interconnect systems, which cannot meet the communication requirements for distributed training of large DNNs due to the low data bandwidth of the electrical interconnect systems.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.17.8 Accelerating Sparse Transformer Inference on GPU

- 作者: Wen-Hao Dai; Hao-Dong Deng; Mengfei Rong; Xin-Yu Yang; Hongyu Liu; Fangxin Liu; Hailong Yang; Qi Cao; Qing-Xiao Sun
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Accelerating Sparse Transformer Inference on GPU

快速理解: Large language models (LLMs) are popular around the world due to their powerful understanding capabilities. As the core component of LLMs, accelerating Transformer through parallelization has gradually become a hot research topic.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.17.9 POSTER: FineCo: Fine-grained Heterogeneous Resource Management for Concurrent DNN Inferences

- 作者: Lixian Ma; Haoruo Chen; En Shao; Leping Wang; Quan Chen; Guang-Ming Tan
- 年份: 2024; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: POSTER: FineCo: Fine-grained Heterogeneous Resource Management for Concurrent DNN Inferences

快速理解: Co-locating multiple DNN servings to share GPU resource is widely used to improve resource utilization while guaranteeing user QoS. Existing GPU sharing mechanism is restricted to model level, and fluctuations in kernel-level resource demands highlight a suboptimal utilization of the current sharing mechanism.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.17.10 MARLIN: Mixed-Precision Auto-Regressive Parallel Inference on Large Language Models

- 作者: Elias Frantar; Roberto L. Castro; Jiale Chen; T. Hoefler; Dan Alistarh

- 年份: 2024; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: MARLIN: Mixed-Precision Auto-Regressive Parallel Inference on Large Language Models

快速理解: As inference on Large Language Models (LLMs) emerges as an important workload in machine learning applications, model weight quantization has become a standard technique for efficient GPU deployment. Quantization not only reduces model size, but has also been shown to yield substantial speedups for single-user inference, due to reduced memory movement, with low accuracy impact.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.17.11 FlashSparse: Minimizing Computation Redundancy for Fast Sparse Matrix Multiplications on Tensor Cores

- 作者: Jin-Liang Shi; Shigang Li; You-Xuan Xu; Rong-Tian Fu; Xue-Ying Wang; Tong Wu
- 年份: 2024; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: FlashSparse: Minimizing Computation Redundancy for Fast Sparse Matrix Multiplications on Tensor Cores

快速理解: Sparse Matrix-matrix Multiplication (SpMM) and Sampled Dense-dense Matrix Multiplication (SDDMM) are important sparse operators in scientific computing and deep learning. Tensor Core Units (TCUs) enhance modern accelerators with superior computing power, which is promising to boost the performance of matrix operators to a higher level.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.17.12 Liger: Interleaving Intra- and Inter-Operator Parallelism for Distributed Large Model Inference

- 作者: Jiangsu Du; Jinhui Wei; Jiazhi Jiang; Shenggan Cheng; Dan-E Huang; Zhi-Guang Chen; Yutong Lu
- 年份: 2024; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Liger: Interleaving Intra- and Inter-Operator Parallelism for Distributed Large Model Inference

快速理解: Distributed large model inference is still in a dilemma where balancing cost and effect. The online scenarios demand intraoperator parallelism to achieve low latency and intensive communications makes it costly.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.17.13 POSTER: Pattern-Aware Sparse Communication for Scalable Recommendation Model Training

- 作者: Jiaao He; Shengqi Chen; Ji-Dong Zhai

- 年份: 2024; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: POSTER: Pattern-Aware Sparse Communication for Scalable Recommendation Model Training

快速理解: Recommendation models are an important category of deep learning models whose size is growing enormous. They consist of a sparse part with TBs of memory footprint and a dense part that demands PFLOPs of computing capability to train.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.17.14 BEEMS: Boosting Machine Vision Efficiency via Computation Graph-Based Memory Smoothing

- 作者: Hanjing Shen; Fangxin Liu; Jian Liu; Li Jiang; Haibing Guan
- 年份: 2026; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: BEEMS: Boosting Machine Vision Efficiency via Computation Graph-Based Memory Smoothing

快速理解: With the rapid advances of deep learning-based computer vision (CV) technology, digital images are increasingly processed not by humans, but by downstream CV algorithms. In particular, the growing popularity of vision foundation models has heightened interest in deploying these models on edge devices.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.17.15 ElasGNN: An Elastic Training Framework for Distributed GNN Training

- 作者: Siqi Wang; Hailong Yang; Pengbo Wang; Hongliang Cao; Yufan Xu; Xuezhu Wang; Zhongzhi Luan; Yi Liu; Depei Qian
- 年份: 2026; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: ElasGNN: An Elastic Training Framework for Distributed GNN Training

快速理解: Graph Neural Networks (GNNs) have emerged as powerful machine learning models for numerous graph-based applications. However, existing GNN training frameworks cannot scale the training process elastically, resulting in poor training throughput and low cluster utilization.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.17.16 Exploiting Efficient Mapping and Pipelined Execution for Accelerating SpMV on Tensor Cores

- 作者: Kaige Zhang; Hailong Yang; Xin You; Tianyu Feng; Yufan Xu; Zhongzhi Luan; Yi Liu; Depei Qian
- 年份: 2026; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Exploiting Efficient Mapping and Pipelined Execution for Accelerating SpMV on Tensor Cores

快速理解: Sparse matrix-vector multiplication (SpMV) is a fundamental operation in scientific computing, machine learning, and graph analytics, demanding efficient execution on modern hardware. Recent advances in hardware accelerators, such as Tensor Cores, have significantly improved the performance of many compute-intensive workloads.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.17.17 FlashAttention-T: Towards Fully Tensorized Attention by Exploiting Tensor-Vector Parallelism

- 作者: Jianxing Xu; Yuanbo Wen; Jun Bi; Ruibai Xu; Guanglin Xu; Rui Zhang; Wei Li; Ling Li; Tianshi Chen; Qi Guo; Yunji Chen
- 年份: 2026; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: FlashAttention-T: Towards Fully Tensorized Attention by Exploiting Tensor-Vector Parallelism

快速理解: The attention mechanism is central to modern deep learning, particularly in large language models (LLMs), but suffers from quadratic computational complexity. To accelerate attention computation on GPUs, fused attention techniques (e.g., FlashAttention) consolidate the matrix multiplication (GEMM) and softmax computations into a single kernel.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.17.18 ParDiff: Efficiently Parallelizing Reverse-Mode Automatic Differentiation with Direct Indexing

- 作者: Shuhong Huang; Shizhi Tang; Yuan Wen; Huan-Qi Cao; Ruibai Tang; Yidong Chen; Jiping Yu; Yang Li; Chao Jiang; Liming Xiao; Ji-Dong Zhai
- 年份: 2026; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: ParDiff: Efficiently Parallelizing Reverse-Mode Automatic Differentiation with Direct Indexing

快速理解: Automatic Differentiation (AD) is a technique that computes the derivatives of numerical programs by systematically applying the chain rule, playing a critical role in domains such as machine learning, simulation, and control systems. However, parallelizing differentiated programs remains a significant challenge due to the conflict between tapes (a data structure for intermediate variable storage) and summations: the differentiation process inherently introduces inter-thread summation patterns, which require prohibitively expensive atomic operations; and traditional tape designs tightly couple data retrieval with the program's control flow, preventing code restructuring needed to eliminate these costly dependencies.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.17.19 TileSpGEMM: a tiled algorithm for parallel sparse general matrix-matrix multiplication on GPUs

- 作者: Yuyao Niu; Zhengyang Lu; H. Ji; Shuhui Song; Zhou Jin; Weifeng Liu
- 年份: 2022; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: TileSpGEMM: a tiled algorithm for parallel sparse general matrix-matrix multiplication on GPUs

快速理解: Sparse general matrix-matrix multiplication (SpGEMM) is one of the most fundamental building blocks in sparse linear solvers, graph processing frameworks and machine learning applications. The existing parallel approaches for shared memory SpGEMM mostly use the row-row style with possibly good parallelism.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.17.20 DSP: Efficient GNN Training with Multiple GPUs

- 作者: Zhenkun Cai; Qihui Zhou; Xiao Yan; Da Zheng; Xiang Song; Chen-Guang Zheng; James Cheng; G. Karypis
- 年份: 2023; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: DSP: Efficient GNN Training with Multiple GPUs

快速理解: Jointly utilizing multiple GPUs to train graph neural networks (GNNs) is crucial for handling large graphs and achieving high efficiency. However, we find that existing systems suffer from high communication costs and low GPU utilization due to improper data layout and training procedures.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.18 SYSTEMS · SC

本会议纳入 20 篇。自动主题统计为: AI Infra (20)、Systems (11)、Agent RL (1)。这些统计用于组织阅读顺序, 不能单独证明论文结论或实验优越性。

优先阅读候选: - HyTiS: Hybrid Tile Scheduling for GPU GEMM with Enhanced Wave Utilization and Cache Locality (优先级分数 70.057) - Phoenix: A Refactored I/O Stack for GPU Direct Storage without Phony Buffers (优先级分数 69.835) - Benchmark-driven Models for Energy Analysis and Attribution of GPU-Accelerated Supercomputing (优先级分数 61.572) - MLP-Offload: Multi-Level, Multi-Path Offloading for LLM Pre-training to Break the GPU Memory Wall (优先级分数 61.25) - LowDiff: Efficient Frequent Checkpointing via Low-Cost Differential for High-Performance Distributed Training Systems (优先级分数 60.835)

1.3.18.1 HyTiS: Hybrid Tile Scheduling for GPU GEMM with Enhanced Wave Utilization and Cache Locality

- 作者: Zheng Zhang; Hulin Wang; Hongming Xu; Donglin Yang; Xiao-Bo Zhou; Dazhao Cheng
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: HyTiS: Hybrid Tile Scheduling for GPU GEMM with Enhanced Wave Utilization and Cache

Locality

快速理解: General matrix-matrix multiplication (GEMM) is a fundamental operation in both deep learning and scientific computing. To accelerate these workloads, GPUs with a large number of streaming multiprocessors (SMs) are widely used.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.18.2 Phoenix: A Refactored I/O Stack for GPU Direct Storage without Phony Buffers

- 作者: Jianqin Yan; Shi Qiu; Yina Lv; Yifan Hu; Hao Chen; Zhi-Rong Shen; Xin Yao; Renhai Chen; Ji-Wu Shu; Gong Zhang; Yi-Ming Zhang
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: Phoenix: A Refactored I/O Stack for GPU Direct Storage without Phony Buffers

快速理解: GPU Direct Storage (GDS) plays a vital role in GPU-based training and inference systems, leveraging Peer-to-Peer Direct Memory Access (P2P-DMA) to establish a direct data transfer path between the GPU and the storage device. The direct I/O path reduces GPU storage access latency and CPU overhead, thus improving the efficiency of data transfer.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.18.3 Benchmark-driven Models for Energy Analysis and Attribution of GPU-Accelerated Supercomputing

- 作者: O. Antepara; Zhengji Zhao; Brian Austin; Nan Ding; L. Oliker; Nicholas J. Wright; Samuel Williams
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Benchmark-driven Models for Energy Analysis and Attribution of GPU-Accelerated Supercomputing

快速理解: As advances in energy-efficiency become the primary limiter to increases in power-constrained supercomputing and machine learning performance, it is imperative developers, architects, and practitioners understand how modern GPUs consume energy when running HPC and ML applications. Rather than opaque coarse-grained metrics, in this paper, we develop an extensible, microbenchmark-parameterized energy model capable of attributing application energy not only by functional unit (FPU, tensor core, integer ALU) and memory level (L1, L2, HBM), but can also differentiate control energy from datapath energy.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.18.4 MLP-Offload: Multi-Level, Multi-Path Offloading for LLM Pre-training to Break the GPU Memory Wall

- 作者: Avinash Maurya; M. M. Rafique; Franck Cappello; Bogdan Nicolae
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: MLP-Offload: Multi-Level, Multi-Path Offloading for LLM Pre-training to Break the GPU Memory Wall

快速理解: Training LLMs larger than the aggregated memory of multiple GPUs is increasingly necessary due to the faster growth of LLM sizes compared to GPU memory. To this end, multi-tier host memory or disk offloading techniques are proposed by state of art.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.18.5 LowDiff: Efficient Frequent Checkpointing via Low-Cost Differential for High-Performance Distributed Training Systems

- 作者: Chenxuan Yao; Feifan Liu; Yuchong Hu; Zhengyu Liu; Xin-Jue Zheng; Wen Zhou
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: LowDiff: Efficient Frequent Checkpointing via Low-Cost Differential for High-Performance Distributed Training Systems

快速理解: Distributed training of large deep-learning models often leads to failures, so checkpointing is commonly employed for recovery. State-of-the-art studies focus on frequent checkpointing for fast recovery from failures.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.18.6 Matrix Is All You Need: Rearchitecting Quantum Chemistry to Scale on AI Accelerators

- 作者: Haozhi Han; Kun Li; Fusong Ju; Qi Li; Hong An; Yifeng Chen; Yun-Quan Zhang; Ting Cao; Mao Yang
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Matrix Is All You Need: Rearchitecting Quantum Chemistry to Scale on AI Accelerators

快速理解: Scientific computing remains fundamentally misaligned with the execution paradigm of modern AI accelerators, which rely on structured, low-precision matrix operations for performance and scalability. Quantum chemistry exemplifies this gap through three core scalability limits: irregular computational patterns, fragmented hardware utilization, and limited scientific reach.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.18.7 BOER: Enhancing Resource Utilization for Deep Learning Inference with Hybrid Spatial GPU Sharing

- 作者: Bowen Zhang; Yu-Hang Wang; Zhuozhao Li

- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: BOER: Enhancing Resource Utilization for Deep Learning Inference with Hybrid Spatial GPU Sharing

快速理解: Many inference systems leverage spatial multiplexing technologies, such as Multi-Process Service (MPS) and Multi-Instance GPU (MIG), to serve deep learning models concurrently on a single GPU. However, existing solutions suffer from interference under MPS and rigid partition sizes in MIG.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.18.8 Moment: Co-optimizing Physical Communication Topology and Data Placement for Multi-GPU Out-of-core GNN Training

- 作者: Zuocheng Shi; Jie Sun; Ziyu Song; Mo Sun; Yang Xiao; Fei Wu; Ze-Ke Wang
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Moment: Co-optimizing Physical Communication Topology and Data Placement for Multi-GPU Out-of-core GNN Training

快速理解: Graph Neural Networks (GNNs) are widely employed in applications like recommendation systems, social network analysis, and fraud detection, but training large-scale GNNs is challenging due to its memory limitations. Existing systems face a trade-off between throughput and monetary cost: Distributed systems require expensive memory scaling, while single-machine out-of-core systems are limited by GPU/PCIe throughput.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.18.9 SDR-RDMA: Software-Defined Reliability Architecture for Planetary Scale RDMA Communication

- 作者: Mikhail Khalilov; Siyuan Shen; Marcin Chrapek; Tiancheng Chen; K. Nakano; Nicola Mazzeo; Peter-Jan Gootzen; S. Di Girolamo; Rami Nudelman; Gil Bloch; Jithin Jose; Abdul Kabani; S. Anantharamu; Jie Zhang; Konstantin Taranov; Zhuo-Long Yu; Scott Moe; Mahmoud Elhaddad; T. Hoefler
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: SDR-RDMA: Software-Defined Reliability Architecture for Planetary Scale RDMA Communication

快速理解: RDMA is vital for efficient distributed training across datacenters, but millisecond-scale latencies complicate the design of its reliability layer. We show that depending on long-haul link characteristics, such as drop rate, distance and bandwidth, the widely used Selective Repeat algorithm can be inefficient, warranting alternatives like Erasure Coding.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.18.10 Diff-MoE: Efficient Batched MoE Inference with Priority-Driven Differential Expert Caching

- 作者: Kexin Li; Wenkan Huang; Qing-Gang Wang; Long Zheng; Xiaofei Liao; Hai Jin; Jingling Xue
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Diff-MoE: Efficient Batched MoE Inference with Priority-Driven Differential Expert Caching

快速理解: The emerging Mixture-of-Experts (MoE) model mitigates the high compute cost of large-scale LLMs by sparsely activating a subset of experts during inference. However, MoE requires storing massive expert parameters, creating a severe memory bottleneck on resource-constrained GPUs.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.18.11 PerfDojo: Automated ML Library Generation for Heterogeneous Architectures

- 作者: Andrei Ivanov; Siyuan Shen; Gioele Gottardo; Marcin Chrapek; Afif Boudaoud; Timo Schneider; Luca Benini; T. Hoefler
- 年份: 2025; 主题: Agent RL; AI Infra; 优先级: 重要
- [来源](#)

论文: PerfDojo: Automated ML Library Generation for Heterogeneous Architectures

快速理解: The increasing complexity of machine learning models and the proliferation of diverse hardware architectures (CPUs, GPUs, accelerators) make achieving optimal performance a significant challenge. Heterogeneity in instruction sets, specialized kernel requirements for different data types and model features (e.g., sparsity, quantization), and architecture-specific optimizations complicate performance tuning.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.18.12 Compile-Time QoS Scheme for Deep Learning Inferences

- 作者: Sungin Hong; Hyunjun Kim; Hwansoo Han
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Compile-Time QoS Scheme for Deep Learning Inferences

快速理解: With the proliferation of deep learning technologies across various service domains, the sharing of accelerators such as GPUs, TPUs, and NPUs for inference processing has become increasingly common. These accelerators must efficiently handle multiple deep learning services operating concurrently.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.18.13 TurboFNO: High-Performance Fourier Neural Operator with Fused FFT-GEMM-iFFT on GPU

- 作者: Shixun Wu; Yujia Zhai; Huangliang Dai; Hairui Zhao; Yue Zhu; Haiyang Hu; Zizhong Chen
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: TurboFNO: High-Performance Fourier Neural Operator with Fused FFT-GEMM-iFFT on GPU

快速理解: Fourier Neural Operators (FNO) are widely used for learning partial differential equation solution operators. However, FNO lacks architecture-aware optimizations, with its Fourier layers executing FFT, filtering, GEMM, zero padding, and iFFT as separate stages, incurring multiple kernel launches and significant global memory traffic.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.18.14 TT-LoRA MoE: Using Parameter-Efficient Fine-Tuning and Sparse Mixture-Of-Experts

- 作者: Pradip Kunwar; Minh N. Vu; Maanak Gupta; Mahmoud Abdelsalam; Manish Bhattarai
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: TT-LoRA MoE: Using Parameter-Efficient Fine-Tuning and Sparse Mixture-Of-Experts

快速理解: We propose Tensor-Trained Low-Rank Adaptation Mixture of Experts (TT-LoRA MoE), a novel computational framework integrating Parameter-Efficient Fine-Tuning (PEFT) with sparse MoE routing to address scalability challenges in large model deployments. Unlike traditional MoE approaches, which face substantial computational overhead as expert counts grow, TT-LoRA MoE decomposes training into two distinct, optimized stages.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.18.15 X-MoE: Enabling Scalable Training for Emerging Mixture-of-Experts Architectures on HPC Platforms

- 作者: Yueming Yuan; Ahan Gupta; Jianping Li; Sajal Dash; Feiyi Wang; Minjia Zhang
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: X-MoE: Enabling Scalable Training for Emerging Mixture-of-Experts Architectures on HPC Platforms

快速理解: Emerging expert-specialized Mixture-of-Experts (MoE) architectures, such as DeepSeek-MoE, deliver strong model quality through fine-grained expert segmentation and large top-k routing. However, their scalability is limited by substantial activation memory overhead and costly all-to-all communication.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.18.16 PGT-I: Scaling Spatiotemporal GNNs with Memory-Efficient Distributed Training

- 作者: Seth Ockerman; Amal Gueroudji; Tanwi Mallick; Yixuan He; L. Pouchard; Robert B. Ross; Shivaram Venkataraman
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: PGT-I: Scaling Spatiotemporal GNNs with Memory-Efficient Distributed Training

快速理解: Spatiotemporal graph neural networks (ST-GNNs) are powerful tools for modeling spatial and temporal data dependencies. However, their applications have been limited primarily to small-scale datasets because of memory constraints.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.18.17 Towards Efficient LLM Inference via Collective and Adaptive Speculative Decoding

- 作者: Siqi Wang; Hailong Yang; Xuezhu Wang; Tongxuan Liu; Pengbo Wang; Yufan Xu; Xuning Liang; Kejie Ma; Tianyu Feng; Xin You; Ruihao Gong; Rui Wang; Zhongzhi Luan; Yi Liu; Depei Qian
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Towards Efficient LLM Inference via Collective and Adaptive Speculative Decoding

快速理解: Large language models (LLMs) have gained considerable attention for their remarkable performance across a wide range of tasks. However, efficient LLM inference remains challenging because of the autoregressive decoding process, which generates only one token at a time.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.18.18 Fine-grained Automated Failure Management for Extreme-Scale GPU Accelerated Systems

- 作者: Yonatan Levitt; Richard Barella; Sam Zeltner; Tom Musta; Lance Cheney; Gustavo Espinosa; O. Franza; Balazs Gerofi
- 年份: 2025; 主题: AI Infra; 优先级: 重要
- [来源](#)

论文: Fine-grained Automated Failure Management for Extreme-Scale GPU Accelerated Systems

快速理解: As high-performance computing (HPC) systems scale in size, system wide hardware failure rates increase. Historical data from previous large-scale HPC installations illustrate this trend, with the mean time between failures (MTBF) decreasing steadily over the past decade.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.18.19 StraGCN: GPU-Accelerated Strassen's Sparse-Dense Matrix Multiplication for Graph Convolutional Network Training

- 作者：Wei-Dong He; Haikun Liu; Zhuohui Duan; Xiaofei Liao; Shuhao Zhang; Fubing Mao; Hai Jin
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：StraGCN: GPU-Accelerated Strassen's Sparse-Dense Matrix Multiplication for Graph Convolutional Network Training

快速理解：Graph Convolutional Networks (GCNs) are a fundamental approach to deep learning on graph-structured data. However, they face a significant challenge in training efficiency due to the high computational cost of Sparse-Dense Matrix Multiplication (SpMM).

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.18.20 Reproducibility Report for SC25 Paper MLP-Offload: Multi-Level, Multi-Path Offloading for LLM Pre-training to Break the GPU Memory Wall

- 作者：Benjamin Brock
- 年份：2025；主题：AI Infra；优先级：重要
- [来源](#)

论文：Reproducibility Report for SC25 Paper MLP-Offload: Multi-Level, Multi-Path Offloading for LLM Pre-training to Break the GPU Memory Wall

快速理解：This reproducibility report provides details about the artifact evaluation done with regards to the Artifact Description and Evaluation appendix of SC25 paper “MLP-Offload: Multi-Level, Multi-Path Offloading for LLM Pre-training to Break the GPU Memory Wall” by Avinash Maurya, et al. The work was done as part of the Reproducibility Initiative of SC25.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.19 SYSTEMS · SIGCOMM

本会议纳入 20 篇。自动主题统计为：AI Infra (20)、Systems (19)、Agent RL (3)、Supporting Perception (1)。这些统计用于组织阅读顺序，不能单独证明论文结论或实验优越性。

优先阅读候选：- DualPath: Accelerating Agentic LLM Inference by Harvesting Disaggregated KV-Cache Storage I/O (优先级分数 94.5) - ByteScale: Communication-Efficient Scaling of LLM Training with a 2048K Context Length on 16384 GPUs (优先级分数 80.774) - Teal: Learning-Accelerated Optimization of WAN Traffic Engineering (优先级分数 73.489) - Crux: GPU-Efficient Communication Scheduling for Deep Learning Training(优先级分数 73.358)- SyCCL: Exploiting Symmetry for Efficient Collective Communication Scheduling (优先级分数 72.108)

1.3.19.1 DualPath: Accelerating Agentic LLM Inference by Harvesting Disaggregated KV-Cache Storage I/O

- 作者: Yong-Tong Wu; Shao-Yuan Chen; Rilin Huang; Yi-Xuan Tan; Yinmin Zhong; Mingxing Zhang; Xin Jin; Pan-Pan Huang
- 年份: 2026; 主题: Agent RL; AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: DualPath: Accelerating Agentic LLM Inference by Harvesting Disaggregated KV-Cache Storage I/O

快速理解: The performance of multi-turn, agentic LLM inference is increasingly dominated by KV-Cache storage I/O rather than computation. In prevalent disaggregated architectures, loading the massive KV-Cache from external storage creates a fundamental imbalance: storage NICs on prefill engines become bandwidth-saturated, while those on decoding engines remain idle.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.19.2 ByteScale: Communication-Efficient Scaling of LLM Training with a 2048K Context Length on 16384 GPUs

- 作者: Hao Ge; Junda Feng; Qi Huang; Fangcheng Fu; Xiaonan Nie; Lei Zuo; Haibin Lin; Bin Cui; Xin Liu
- 年份: 2025; 主题: Agent RL; AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: ByteScale: Communication-Efficient Scaling of LLM Training with a 2048K Context Length on 16384 GPUs

快速理解: Scaling long-context ability is essential for Large Language Models (LLMs). To amortize the memory consumption across multiple devices in long-context training, inter-data partitioning (a.k.a.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.19.3 Teal: Learning-Accelerated Optimization of WAN Traffic Engineering

- 作者: Zhiying Xu; Francis Y. Yan; Rachee Singh; Justin T Chiu; Alexander M. Rush; Minlan Yu
- 年份: 2022; 主题: Agent RL; AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: Teal: Learning-Accelerated Optimization of WAN Traffic Engineering

快速理解: The rapid expansion of global cloud wide-area networks (WANs) has posed a challenge for commercial optimization engines to efficiently solve network traffic engineering (TE) problems at scale. Existing acceleration strategies decompose TE optimization into concurrent subproblems but realize limited parallelism due to an inherent tradeoff between run time and allocation performance.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.19.4 Crux: GPU-Efficient Communication Scheduling for Deep Learning Training

- 作者: Jiamin Cao; Yu Guan; Kun Qian; Jiaqi Gao; Wencong Xiao; Jianbo Dong; Binzhang Fu; Dennis Cai; Ennan Zhai
- 年份: 2024; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: Crux: GPU-Efficient Communication Scheduling for Deep Learning Training

快速理解: Deep learning training (DLT), e.g., large language model (LLM) training, has become one of the most important services in multitenant cloud computing. By deeply studying in-production DLT jobs, we observed that communication contention among different DLT jobs seriously influences the overall GPU computation utilization, resulting in the low efficiency of the training cluster.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.19.5 SyCCL: Exploiting Symmetry for Efficient Collective Communication Scheduling

- 作者: Jiamin Cao; Shangfeng Shi; Jiaqi Gao; Weisen Liu; Yi-Fan Yang; Yichi Xu; Zhilong Zheng; Yu Guan; Kun Qian; Ying Liu; Mingwei Xu; Tianshu Wang; Ning Wang; Jianbo Dong; Binzhang Fu; Dennis Cai; Ennan Zhai
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: SyCCL: Exploiting Symmetry for Efficient Collective Communication Scheduling

快速理解: The performance of collective communication schedules is crucial for the efficiency of machine learning jobs and GPU cluster utilization. Existing open-source collective communication libraries (such as NCCL and RCCL) rely on fixed schedules and cannot adjust to varying topology and model requirements.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.19.6 ResCCL: Resource-Efficient Scheduling for Collective Communication

- 作者: Tongrui Liu; Chenyang Hei; Fu-Liang Li; Chengxi Gao; Jiamin Cao; Tianshu Wang; Ennan Zhai; Xingwei Wang
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: ResCCL: Resource-Efficient Scheduling for Collective Communication

快速理解: As distributed deep learning training (DLT) systems scale, collective communication has become a significant performance bottleneck. While current approaches optimize bandwidth utilization and task completion time, existing communication libraries (CCLs) backends fail to efficiently manage GPU resources during algorithm execution, limiting the performance of advanced algorithms.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.19.7 HACK: Homomorphic Acceleration via Compression of the Key-Value Cache for Disaggregated LLM Inference

- 作者: Zeyu Zhang; Haiying Shen; S. Vargaftik; Ran Ben Basat; M. Mitzenmacher; Minlan Yu
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: HACK: Homomorphic Acceleration via Compression of the Key-Value Cache for Disaggregated LLM Inference

快速理解: Disaggregated Large Language Model (LLM) inference decouples the compute-intensive prefill stage from the memory-intensive decode stage, allowing low-end, compute-focused GPUs for prefill and high-end, memory-rich GPUs for decode, which reduces cost while maintaining high throughput. However, transmitting Key-Value (KV) data between the two stages can be a bottleneck, especially for long prompts.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.19.8 ARK: Avoiding Routing Collisions for KV Cache Transfer in Disaggregated LLM Inference

- 作者: Hung-Chun Lin; Ting-Wei Hsu; Chung-En Ho; Ahmed Saeed
- 年份: 2026; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: ARK: Avoiding Routing Collisions for KV Cache Transfer in Disaggregated LLM Inference

快速理解: Disaggregated LLM inference separates the prefill and decode phases across GPU pools, generating massive KV-cache transfers. Because these transfers last hundreds of milliseconds to seconds, they behave as mega elephant flows that dominate link bandwidth utilization.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.19.9 Efficient Remote KV Cache Reuse with GPU-native Video Codec

- 作者: Liang Mi; Wei-Jun Wang; Jing-Han Chen; Ting Cao; Haipeng Dai; Yunxin Liu
- 年份: 2026; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: Efficient Remote KV Cache Reuse with GPU-native Video Codec

快速理解: Remote KV cache reuse fetches KV cache for identical contexts from remote storage, avoiding recomputation, accelerating LLM inference. While it excels in high-speed networks, its performance degrades significantly in bandwidth-limited scenarios.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.19.10 GeoOrchestra: Orchestrating Heterogeneous Geo-Distributed Training with Network-Aware Scheduling

- 作者: Ting Liu; Qinghua Wu; Jun Zhou; Yuan Sun; Jing-Bin Yang; Ji Pei; Chen Zhang; Heng Pan; Zhen-Yu Li; Yun-jie Liu
- 年份: 2026; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: GeoOrchestra: Orchestrating Heterogeneous Geo-Distributed Training with Network-Aware Scheduling

快速理解: Scaling Large Language Models (LLMs) increasingly mandates the aggregation of geographically dispersed resources. However, cross-datacenter (Cross-DC) training faces significant hurdles due to extreme hardware heterogeneity and limited Wide-Area Network (WAN) bandwidth.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.19.11 Alibaba HPN: A Data Center Network for Large Language Model Training

- 作者: Kun Qian; Yongqing Xi; Jiamin Cao; Jiaqi Gao; Yichi Xu; Yu Guan; Binzhang Fu; Xuemei Shi; Fangbo Zhu; Rui Miao; Chao Wang; Peng Wang; Pengcheng Zhang; Xianlong Zeng; Eddie Ruan; Zhiping Yao; Ennan Zhai; Dennis Cai
- 年份: 2024; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: Alibaba HPN: A Data Center Network for Large Language Model Training

快速理解: This paper presents HPN, Alibaba Cloud's data center network for large language model (LLM) training. Due to the differences between LLMs and general cloud computing (e.g., in terms of traffic patterns and fault tolerance), traditional data center networks are not well-suited for LLM training.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.19.12 Rethinking Machine Learning Collective Communication as a Multi-Commodity Flow Problem

- 作者: Xuting Liu; Behnaz Arzani; S. Kakarla; Lian Zhao; Vincent Liu; M. Castro; Srikanth Kandula; Luke Marshall
- 年份: 2023; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Rethinking Machine Learning Collective Communication as a Multi-Commodity Flow Problem

快速理解: Cloud operators utilize collective communication optimizers to enhance the efficiency of the single-tenant, centrally managed training clusters they manage. However, current optimizers struggle to scale for such use cases and often compromise solution quality for scalability.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.19.13 Janus: A Unified Distributed Training Framework for Sparse Mixture-of-Experts Models

- 作者: Juncai Liu; Jessie Hui Wang; Yimin Jiang
- 年份: 2023; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Janus: A Unified Distributed Training Framework for Sparse Mixture-of-Experts Models

快速理解: Scaling models to large sizes to improve performance has led a trend in deep learning, and sparsely activated Mixture-of-Expert (MoE) is a promising architecture to scale models. However, training MoE models in existing systems is expensive, mainly due to the All-to-All communication between layers.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.19.14 AdaInf: Data Drift Adaptive Scheduling for Accurate and SLO-guaranteed Multiple-Model Inference Serving at Edge Servers

- 作者: Sudipta Saha Shubha; Hai-Ying Shen
- 年份: 2023; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: AdaInf: Data Drift Adaptive Scheduling for Accurate and SLO-guaranteed Multiple-Model Inference Serving at Edge Servers

快速理解: Various audio and video applications rely on multiple deep neural network (DNN) models deployed on edge servers to conduct inference with ms-level latency service-level-objectives (SLOs). To avoid accuracy decreases caused by data drift, continual retraining is necessary.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.19.15 DistTrain: Addressing Model and Data Heterogeneity with Disaggregated Training for Multimodal Large Language Models

- 作者: Zili Zhang; Yinmin Zhong; Yimin Jiang; Hanpeng Hu; Jianjian Sun; Zheng Ge; Yibo Zhu; Daxin Jiang; Xin Jin
- 年份: 2024; 主题: AI Infra; Supporting Perception; 优先级: 重要
- [来源](#)

论文: DistTrain: Addressing Model and Data Heterogeneity with Disaggregated Training for Multimodal Large Language Models

快速理解: Multimodal large language models (LLMs) empower LLMs to ingest inputs and generate outputs in multiple forms, such as text, image, and audio. However, the integration of multiple modalities introduces heterogeneity in both the model and training data, creating unique systems challenges.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 3D、视频、多模态视觉或具身感知支撑能力。

证据层级: 索引摘要。

1.3.19.16 Accelerating Model Training in Multi-cluster Environments with Consumer-grade GPUs

- 作者: Hwijoon Lim; Juncheol Ye; Sangeetha Abdu Jyothi; Dongsu Han
- 年份: 2024; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Accelerating Model Training in Multi-cluster Environments with Consumer-grade GPUs

快速理解: Rapid advances in machine learning necessitate significant computing power and memory for training, which is accessible only to large corporations today. Small-scale players like academics often only have consumer-grade GPU clusters locally and can afford cloud GPU instances to a limited extent.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.19.17 Alibaba Stellar: A New Generation RDMA Network for Cloud AI

- 作者: Jie Lu; Jiaqi Gao; Fei Feng; Zhiqiang He; Menglei Zheng; Kun Liu; Junxian He; Binbin Liao; Suwei Xu; Ke Sun; Yongjia Mo; Qinghua Peng; Jilie Luo; Qingxu Li; Gang Lu; Zishu Wang; Jianbo Dong; Kunling He; Sheng Cheng; Jiamin Cao; Hairong Jiao; Pengcheng Zhang; Shu Ma; Lingjun Zhu; Chao Shi; Yangming Zhang; Yiquan Chen; Wei Wang; Shuhong Zhu; Xingru Li; Qiang Wang; Jiang Liu; Chao Wang; Wei Lin; Ennan Zhai; Jie-Sheng Wu; Qiang Liu; Binzhang Fu; Dennis Cai
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Alibaba Stellar: A New Generation RDMA Network for Cloud AI

快速理解: The rapid adoption of Large Language Models (LLMs) in cloud environments has intensified the demand for high-performance AI training and inference, where Remote Direct Memory Access (RDMA) plays a critical role. However, existing RDMA virtualization solutions, such as Single-Root Input/Output Virtualization (SR-IOV), face significant limitations in scalability, performance, and stability.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.19.18 InfiniteHBD: Building Datacenter-Scale High-Bandwidth Domain for LLM with Optical Circuit Switching Transceivers

- 作者: Chenchen Shou; G. Liu; Hao Nie; Huaiyu Meng; Yu Zhou; Yimin Jiang; Wenqing Lv; Yelong Xu; Yuanwei Lu; Zhang Chen; Yanbo Yu; Yichen Shen; Yibo Zhu; Daxin Jiang
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: InfiniteHBD: Building Datacenter-Scale High-Bandwidth Domain for LLM with Optical Circuit Switching Transceivers

快速理解: Scaling Large Language Model (LLM) training relies on multidimensional parallelism, where High-Bandwidth Domains (HBDs) are critical for communication-intensive parallelism like Tensor Parallelism. However, existing HBD architectures face fundamental limitations in scalability, cost, and fault resiliency: switch-centric HBDs (e.g., NVL-72) incur prohibitive scaling costs, while GPU-centric HBDs (e.g., TPUv3/Dojo) suffer from severe fault propagation.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.19.19 KVServe: Service-Aware KV Cache Compression for Communication-Efficient Disaggregated LLM Serving

- 作者：Ze-Dong Liu; Xinyang Ma; Dejun Luo; Hai-Rui Zhao; Bing Lu; Wenjing Huang; Yida Gu; Xing-Chen Liu; Zheng Wei; Jinyang Liu; Dingwen Tao; Guang-Ming Tan
- 年份：2026；主题：AI Infra; Systems；优先级：重要
- [来源](#)

论文：KVServe: Service-Aware KV Cache Compression for Communication-Efficient Disaggregated LLM Serving

快速理解：LLMs are widely adopted in production, pushing inference systems to their limits. Disaggregated LLM serving (e.g., PD separation and KV state disaggregation) improves scalability and cost efficiency, but it also turns KV into an explicit payload crossing network and storage boundaries, making KV a dominant end-to-end bottle-neck.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.19.20 POSTER:In-network Model Inference for Distributed Systems via Programmable Switches

- 作者：Jianqiang Zhong; Yihong Li; Shuo Liu; Jingpu Duan; Xiaoxi Zhang; Xu Chen
- 年份：2024；主题：AI Infra; Systems；优先级：重要
- [来源](#)

论文：POSTER:In-network Model Inference for Distributed Systems via Programmable Switches

快速理解：Model parallelism is crucial for accelerating distributed DNN inference. As a core component of distributed systems, programmable switches have demonstrated their ability in assisting both communication and partial computation.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.20 SYSTEMS · SOSP

本次会议纳入 36 篇。自动主题统计为：AI Infra (30)、Systems (20)、Teleoperation (1)、Supporting Perception (1)、Agent RL (1)。这些统计用于组织阅读顺序，不能单独证明论文结论或实验优越性。

优先阅读候选：- SilvanForge: A Schedule-Guided Retargetable Compiler for Decision Tree Inference (优先级分数 88.17) - Paella: Low-latency Model Serving with Software-defined GPU Scheduling (优先级分数 72.837) - LithOS: An Operating System for Efficient Machine Learning on GPUs (优先级分数 72.459) - UGACHE: A Unified GPU Cache for Embedding-based Deep Learning (优先级分数 71.394) - GEMINI: Fast Failure Recovery in Distributed Training with In-Memory Checkpoints (优先级分数 65.218)

1.3.20.1 SilvanForge: A Schedule-Guided Retargetable Compiler for Decision Tree Inference

- 作者: Ashwin Prasad; Sampath Rajendra; Kaushik Rajan; R. Govindarajan; Uday Bondhugula
- 年份: 2024; 主题: Teleoperation; AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: SilvanForge: A Schedule-Guided Retargetable Compiler for Decision Tree Inference

快速理解: The proliferation of machine learning together with the rapid evolution of the hardware ecosystem has led to a surge in the demand for model inference on a variety of hardware. Decision tree based models are the most popular models on tabular data.

关注点: 人类示范、远程操作、共享自治或动作重定向; 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.20.2 Paella: Low-latency Model Serving with Software-defined GPU Scheduling

- 作者: Kelvin K. W. Ng; Henri Maxime Demoulin; Vincent Liu
- 年份: 2023; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: Paella: Low-latency Model Serving with Software-defined GPU Scheduling

快速理解: Model serving systems play a critical role in multiplexing machine learning inference jobs across shared GPU infrastructure. These systems have traditionally sat at a high level of abstraction—receiving jobs from clients through a narrow API and relying on black-box GPU scheduling mechanisms when dispatching them.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.20.3 LithOS: An Operating System for Efficient Machine Learning on GPUs

- 作者: Patrick H. Coppock; B. Zhang; Eliot H. Solomon; Vasilis Kypriotis; Leon Yang; Bikash Sharma; Dan Schatzberg; Todd C. Mowry; Dimitrios Skarlatos
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: LithOS: An Operating System for Efficient Machine Learning on GPUs

快速理解: The rapid growth of machine learning (ML) has made GPUs indispensable in datacenters and underscores the urgency of improving their efficiency. However, balancing diverse model demands with high utilization remains a fundamental challenge.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.20.4 UGACHE: A Unified GPU Cache for Embedding-based Deep Learning

- 作者: Xiaoni Song; Yiwen Zhang; Rong Chen; Haibo Chen
- 年份: 2023; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: UGACHE: A Unified GPU Cache for Embedding-based Deep Learning

快速理解: This paper presents UGache, a unified multi-GPU cache system for embedding-based deep learning (EmbDL). UGache is primarily motivated by the unique characteristics of EmbDL applications, namely read-only, batched, skewed, and predictable embedding accesses.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.20.5 GEMINI: Fast Failure Recovery in Distributed Training with In-Memory Checkpoints

- 作者: Zhuang Wang; Zhen Jia; Shuai Zheng; Zhen Zhang; Xinwei Fu; T. S. Eugene Ng; Yida Wang
- 年份: 2023; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: GEMINI: Fast Failure Recovery in Distributed Training with In-Memory Checkpoints

快速理解: Large deep learning models have recently garnered substantial attention from both academia and industry. Nonetheless, frequent failures are observed during large model training due to large-scale resources involved and extended training time.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.20.6 Sia: Heterogeneity-aware, goodput-optimized ML-cluster scheduling

- 作者: Suhas Jayaram Subramanya; Daiyaan Arfeen; Shouxu Lin; Aurick Qiao; Zhihao Jia; G. Ganger
- 年份: 2023; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Sia: Heterogeneity-aware, goodput-optimized ML-cluster scheduling

快速理解: The Sia scheduler efficiently assigns heterogeneous deep learning (DL) cluster resources to elastic resource-adaptive jobs. Although some recent schedulers address one aspect or another (e.g., heterogeneity or resource-adaptivity), none addresses all and most scale poorly to large clusters and/or heavy workloads even without the full complexity of the combined scheduling problem.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.20.7 Aegaeon: Effective GPU Pooling for Concurrent LLM Serving on the Market

- 作者: Yuxing Xiang; Xue Li; Kun Qian; Yufan Yang; Diwen Zhu; Wen-Yuan Yu; Ennan Zhai; Xuazhe Liu; Xin Jin; Jing-Ren Zhou
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Aegaeon: Effective GPU Pooling for Concurrent LLM Serving on the Market

快速理解: Model markets (e.g., Hugging Face) feature a wide variety of models with unique characteristics and varying levels of popularity. Serving sporadic and unpredictable requests in concurrent inference workloads with dedicated GPU instances results in substantial resource waste.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.20.8 KTransformers: Unleashing the Full Potential of CPU/GPU Hybrid Inference for MoE Models

- 作者：Hongtao Chen; Weiyu Xie; Bo Zhang; Jing Tang; Jiahao Wang; Jianwei Dong; Shao-Yuan Chen; Zi-Wei Yuan; Chen Lin; Cheng Qiu; Yuening Zhu; Qingliang Ou; Jia-Qi Liao; Xianglin Chen; Zhiyuan Ai; Yong-Wei Wu; Mingxing Zhang
- 年份：2025；主题：AI Infra; Systems；优先级：重要
- [来源](#)

论文：KTransformers: Unleashing the Full Potential of CPU/GPU Hybrid Inference for MoE Models

快速理解：Due to the sparse nature of Mixture-of-Experts (MoE) models, they are particularly suitable for hybrid CPU/GPU inference, especially in low-concurrency scenarios. This hybrid approach leverages both the large, cost-effective memory capacity of CPU/DRAM and the high bandwidth of GPU/VRAM.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.20.9 Characterizing Mobile SoC for Accelerating Heterogeneous LLM Inference

- 作者：Le Chen; Dahu Feng; Erhu Feng; Ying-Rui Wang; Rong Zhao; Yubin Xia; Pinjie Xu; Haibo Chen
- 年份：2025；主题：AI Infra; Supporting Perception；优先级：重要
- [来源](#)

论文：Characterizing Mobile SoC for Accelerating Heterogeneous LLM Inference

快速理解：With the rapid advancement of artificial intelligence technologies such as ChatGPT, AI agents, and video generation, contemporary mobile systems have begun integrating these AI capabilities on local devices to enhance privacy and reduce response latency. To meet the computational demands of AI tasks, current mobile SoCs are equipped with diverse AI accelerators, including GPUs and Neural Processing Units (NPU).

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；3D、视频、多模态视觉或具身感知支撑能力。

证据层级：索引摘要。

1.3.20.10 Coyote v2: Raising the Level of Abstraction for Data Center FPGAs

- 作者：Benjamin Ramhorst; Dario Korolija; M. J. Heer; Jonas Dann; Luhao Liu; Gustavo Alonso
- 年份：2025；主题：AI Infra; Systems；优先级：重要
- [来源](#)

论文：Coyote v2: Raising the Level of Abstraction for Data Center FPGAs

快速理解：In the trend towards hardware specialization, FPGAs play a dual role as accelerators for offloading, e.g., network virtualization, and as a vehicle for prototyping and exploring hardware designs. While FPGAs offer versatility and performance, integrating them in larger systems remains challenging.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.20.11 Random Walks on Huge Graphs at Cache Efficiency

- 作者：Ke Yang; Xiaosong Ma; Saravanan Thirumuruganathan; Kang Chen; Yongwei Wu
- 年份：2021；主题：AI Infra; Systems；优先级：重要
- [来源](#)

论文：Random Walks on Huge Graphs at Cache Efficiency

快速理解：Data-intensive applications dominated by random accesses to large working sets fail to utilize the computing power of modern processors. Graph random walk, an indispensable workhorse for many important graph processing and learning applications, is one prominent case of such applications.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.20.12 PrefillOnly: An Inference Engine for Prefill-only Workloads in Large Language Model Applications

- 作者：Kuntai Du; Bowen Wang; Chen Zhang; Yi-Ming Cheng; Qing Lan; Hejian Sang; Yihua Cheng; Jiayi Yao; Xiaoxuan Liu; Yifan Qiao; Ion Stoica; Junchen Jiang
- 年份：2025；主题：AI Infra; Systems；优先级：重要
- [来源](#)

论文：PrefillOnly: An Inference Engine for Prefill-only Workloads in Large Language Model Applications

快速理解：Besides typical generative applications, like ChatGPT, GitHub Copilot, and Cursor, we observe an emerging trend that LLMs are increasingly used in traditional discriminative tasks, such as recommendation, credit verification, and data labeling. The key characteristic of these emerging use cases is that the LLM generates only a single output token, rather than an arbitrarily long sequence of tokens.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.20.13 Reducing Energy Bloat in Large Model Training

- 作者：Jae-Won Chung; Yile Gu; Insu Jang; Luoxi Meng; Nikhil Bansal; Mosharaf Chowdhury
- 年份：2023；主题：AI Infra; Systems；优先级：重要
- [来源](#)

论文：Reducing Energy Bloat in Large Model Training

快速理解：Training large AI models on numerous GPUs consumes a massive amount of energy, making power delivery one of the largest limiting factors in building and operating datacenters for AI workloads. However, we observe that not all energy consumed during training directly contributes to end-to-end throughput; a significant portion can be removed without slowing down training.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.20.14 PIT: Optimization of Dynamic Sparse Deep Learning Models via Permutation Invariant Transformation

- 作者：Ningxin Zheng; Huiqiang Jiang; Quan Zhang; Zhenhua Han; Yuqing Yang; Lingxiao Ma; Fan Yang; Lili Qiu; Mao Yang; Lidong Zhou
- 年份：2023; 主题：AI Infra; Systems; 优先级：重要
- [来源](#)

论文：PIT: Optimization of Dynamic Sparse Deep Learning Models via Permutation Invariant Transformation

快速理解：Dynamic sparsity, where the sparsity patterns are unknown until runtime, poses a significant challenge to deep learning. The state-of-the-art sparsity-aware deep learning solutions are restricted to pre-defined, static sparsity patterns due to significant overheads associated with preprocessing.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.20.15 Tenplex: Dynamic Parallelism for Deep Learning using Parallelizable Tensor Collections

- 作者：Marcel Wagenländer; Guo Li; Bo Zhao; Luo Mai; Peter R. Pietzuch
- 年份：2023; 主题：AI Infra; Systems; 优先级：重要
- [来源](#)

论文：Tenplex: Dynamic Parallelism for Deep Learning using Parallelizable Tensor Collections

快速理解：Deep learning (DL) jobs use multi-dimensional parallelism, i.e., combining data, model, and pipeline parallelism, to use large GPU clusters efficiently. Long-running jobs may experience changes to their GPU allocation: (i) resource elasticity during training adds or removes GPUs; (ii) hardware maintenance may require redeployment on different GPUs; and (iii) GPU failures force jobs to run with fewer devices.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.20.16 Scaling Deep Learning Computation over the Inter-Core Connected Intelligence Processor with T10

- 作者：Yiqi Liu; Yu Xue; Yu Cheng; Lingxiao Ma; Ziming Miao; Jilong Xue; Jian Huang
- 年份：2024; 主题：AI Infra; Systems; 优先级：重要
- [来源](#)

论文：Scaling Deep Learning Computation over the Inter-Core Connected Intelligence Processor with T10

快速理解：As AI chips incorporate numerous parallelized cores to scale deep learning (DL) computing, inter-core communication is enabled recently by employing high-bandwidth and low-latency interconnect links on the chip (e.g., Graphcore IPU). It allows each core to directly access the fast scratchpad memory in other cores, which enables new parallel computing paradigms.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.20.17 PowerInfer: Fast Large Language Model Serving with a Consumer-grade GPU

- 作者：Yi-Xin Song; Zeyu Mi; Haotong Xie; Hai-Bo Chen
- 年份：2023; 主题：AI Infra; 优先级：重要
- [来源](#)

论文：PowerInfer: Fast Large Language Model Serving with a Consumer-grade GPU

快速理解：This paper introduces PowerInfer, a high-speed Large Language Model (LLM) inference engine on a personal computer (PC) equipped with a single consumer-grade GPU. The key principle underlying the design of PowerInfer is exploiting the high locality inherent in LLM inference, characterized by a power-law distribution in neuron activation.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.20.18 Oobleck: Resilient Distributed Training of Large Models Using Pipeline Templates

- 作者：Insu Jang; Zhenning Yang; Zhen Zhang; Xin Jin; Mosharaf Chowdhury
- 年份：2023; 主题：AI Infra; 优先级：重要
- [来源](#)

论文：Oobleck: Resilient Distributed Training of Large Models Using Pipeline Templates

快速理解：Oobleck enables resilient distributed training of large DNN models with guaranteed fault tolerance. It takes a planning-execution co-design approach, where it first generates a set of heterogeneous pipeline templates and instantiates at least $f + 1$ logically equivalent pipeline replicas to tolerate any f simultaneous failures.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.20.19 Sailor: Automating Distributed Training over Dynamic, Heterogeneous, and Geo-distributed Clusters

- 作者：F. Strati; Zhendong Zhang; G. Manos; Ixeia Sanchez Perez; Qinghao Hu; Tiancheng Chen; Berk Buzcu; Song Han; Pamela Delgado; A. Klimovic
- 年份：2025; 主题：AI Infra; 优先级：重要
- [来源](#)

论文：Sailor: Automating Distributed Training over Dynamic, Heterogeneous, and Geo-distributed Clusters

快速理解：The high GPU demand of ML training makes it hard to allocate large homogeneous clusters of high-end GPUs in a single availability zone. Leveraging heterogeneous GPUs available within and across zones can improve throughput at a reasonable cost.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.20.20 Automated SmartNIC Offloading Insights for Network Functions

- 作者：Yiming Qiu; Jiarong Xing; Kuo-Feng Hsu; Qiao Kang; Ming G. Liu; S. Narayana; Ang Chen
- 年份：2021; 主题：AI Infra; 优先级：重要
- [来源](#)

论文：Automated SmartNIC Offloading Insights for Network Functions

快速理解：The gap between CPU and networking speeds has motivated the development of SmartNICs for NF (network functions) offloading. However, offloading performance is predicated upon intricate knowledge about SmartNIC hardware and careful hand-tuning of the ported programs.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.20.21 Gradient Compression Supercharged High-Performance Data Parallel DNN Training

- 作者：Youhui Bai; Cheng Li; Quan Zhou; Jun Yi; Ping Gong; Feng Yan; Ruichuan Chen; Yinlong Xu
- 年份：2021; 主题：AI Infra; 优先级：重要
- [来源](#)

论文：Gradient Compression Supercharged High-Performance Data Parallel DNN Training

快速理解：Gradient compression is a promising approach to alleviating the communication bottleneck in data parallel deep neural network (DNN) training by significantly reducing the data volume of gradients for synchronization. While gradient compression is being actively adopted by the industry (e.g., Facebook and AWS), our study reveals that there are two critical but often overlooked challenges: 1) inefficient coordination between compression and communication during gradient synchronization incurs substantial overheads, and 2) developing, optimizing, and integrating gradient compression algorithms into DNN systems imposes heavy burdens on DNN practitioners, and ad-hoc compression implementations often yield surprisingly poor system performance.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.20.22 gSampler: General and Efficient GPU-based Graph Sampling for Graph Learning

- 作者：Ping Gong; Renjie Liu; Zunyao Mao; Zhenkun Cai; Xiao Yan; Cheng Li; Minjie Wang; Zhuozhao Li
- 年份：2023; 主题：AI Infra; 优先级：重要
- [来源](#)

论文：gSampler: General and Efficient GPU-based Graph Sampling for Graph Learning

快速理解：Graph sampling prepares training samples for graph learning and can dominate the training time. Due to the increasing algorithm diversity and complexity, existing sampling frameworks are

insufficient in the generality of expression and the efficiency of execution.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.20.23 Apparate: Rethinking Early Exits to Tame Latency-Throughput Tensions in ML Serving

- 作者：Yinwei Dai; Rui Pan; Anand Iyer; Kai Li; Ravi Netravali
- 年份：2023; 主题：AI Infra; 优先级：重要
- [来源](#)

论文：Apparate: Rethinking Early Exits to Tame Latency-Throughput Tensions in ML Serving

快速理解：Machine learning (ML) inference platforms are tasked with balancing two competing goals: ensuring high throughput given many requests, and delivering low-latency responses to support interactive applications. Unfortunately, existing platform knobs (e.g., batch sizes) fail to ease this fundamental tension, and instead only enable users to harshly trade off one property for the other.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.20.24 Mycroft: Tracing Dependencies in Collective Communication Towards Reliable LLM Training

- 作者：Yangtao Deng; Lei Zhang; Qinlong Wang; Xiaoyun Zhi; Xin-Lei Zhang; Zhuo Jiang; Hao-Han Xu; Lei Wang; Zuquan Song; Gaohong Liu; Yang-Yang Bai; Shuguang Wang; Wencong Xiao; Jia-jun Ye; Min-Lan Yu; Hong Xu
- 年份：2025; 主题：AI Infra; 优先级：重要
- [来源](#)

论文：Mycroft: Tracing Dependencies in Collective Communication Towards Reliable LLM Training

快速理解：Reliability is essential for ensuring efficiency in LLM training. However, many real-world reliability issues remain difficult to resolve, resulting in wasted resources and degraded model performance.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.20.25 Tempo: Compiled Dynamic Deep Learning with Symbolic Dependence Graphs

- 作者：Pedro F. Silvestre; Peter R. Pietzuch
- 年份：2025; 主题：Agent RL; Systems; 优先级：重要
- [来源](#)

论文：Tempo: Compiled Dynamic Deep Learning with Symbolic Dependence Graphs

快速理解：Deep learning (DL) algorithms are often defined in terms of temporal relationships: a tensor at one timestep may depend on tensors from earlier or later timesteps. Such dynamic dependencies (and corresponding dynamic tensor shapes) are difficult to express and optimize: while eager DL systems support such dynamism, they cannot apply compiler-based optimizations; graph-based

systems require static tensor shapes, which forces users to pad tensors or break-up programs into multiple static graphs.

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.20.26 Improving DNN Inference Throughput Using Practical, Per-Input Compute Adaptation

- 作者：Anand Padmanabha Iyer; Mingyu Guan; Yinwei Dai; Rui Pan; Swapnil Gandhi; Ravi Netravali
- 年份：2024；主题：AI Infra；优先级：重要
- [来源](#)

论文：Improving DNN Inference Throughput Using Practical, Per-Input Compute Adaptation

快速理解：Machine learning inference platforms continue to face high request rates and strict latency constraints. Existing solutions largely focus on compressing models to substantially lower compute costs (and time) with mild accuracy degradations.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.20.27 Unlocking True Elasticity for the Cloud-Native Era with Dandelion

- 作者：Tom Kuchler; Pinghe Li; Ya-Zhuo Zhang; Lazar Cvetković; Boris Goranov; Tobias Stocker; Leon Thomm; Simone Kalbermatter; Tim Notter; Andrea Lattuada; A. Klimovic
- 年份：2025；主题：Systems；优先级：扩展
- [来源](#)

论文：Unlocking True Elasticity for the Cloud-Native Era with Dandelion

快速理解：Elasticity is fundamental to cloud computing. An elastic platform can quickly allocate resources to match the demand of each workload as it arrives, rather than pre-provisioning resources to meet performance objectives.

关注点：操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.20.28 ReCycle: Resilient Training of Large DNNs using Pipeline Adaptation

- 作者：Swapnil Gandhi; Mark Zhao; Athinagoras Skiadopoulos; Christos Kozyrakis
- 年份：2024；主题：AI Infra；优先级：扩展
- [来源](#)

论文：ReCycle: Resilient Training of Large DNNs using Pipeline Adaptation

快速理解：Training large Deep Neural Network (DNN) models requires thousands of GPUs over the course of several days or weeks. At this scale, failures are frequent and can have a big impact on training throughput.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级：索引摘要。

1.3.20.29 Robust LLM Training Infrastructure at ByteDance

- 作者: Borui Wan; Gaohong Liu; Zuquan Song; Jun Wang; Yun Zhang; Guangming Sheng; Shuguang Wang; Houmin Wei; Chen-Yuan Wang; Weiqiang Lou; Xi Yang; Mofan Zhang; Kaihua Jiang; Cheng Ren; Xiaoyun Zhi; Menghan Yu; Zhe Nan; Zhuolin Zheng; B. Zhong; Qinlong Wang; Huan Yu; Jinxin Chi; Wang Zhang; Yuhan Li; Zixian Du; Sida Zhao; Yong-Qiang Zhang; Jing-Zhe Tang; Zherui Liu; Chuan Wu; Yanghua Peng; Haibin Lin; Wencong Xiao; Xin Liu; Liang Xiang
- 年份: 2025; 主题: AI Infra; 优先级: 扩展
- [来源](#)

论文: Robust LLM Training Infrastructure at ByteDance

快速理解: The training scale of large language models (LLMs) has reached tens of thousands of GPUs and is still continuously expanding, enabling faster learning of larger models. Accompanying the expansion of the resource scale is the prevalence of failures (CUDA error, NaN values, job hang, etc.), which poses significant challenges to training stability.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.20.30 DCP: Addressing Input Dynamism In Long-Context Training via Dynamic Context Parallelism

- 作者: Chenyu Jiang; Zhenkun Cai; Ye Tian; Zhen Jia; Yida Wang; Chuan Wu
- 年份: 2025; 主题: AI Infra; 优先级: 扩展
- [来源](#)

论文: DCP: Addressing Input Dynamism In Long-Context Training via Dynamic Context Parallelism

快速理解: Context parallelism has emerged as a key technique to support long-context training, a growing trend in generative AI for modern large models. However, existing context parallel methods rely on static parallelization configurations that overlook the dynamic nature of training data, specifically, the variability in sequence lengths and token relationships (i.e., attention patterns) across samples.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.20.31 TrainVerify: Equivalence-Based Verification for Distributed LLM Training

- 作者: Yunchi Lu; Youshan Miao; Cheng Tan; Peng Huang; Yi Zhu; Xian Zhang; Fan Yang
- 年份: 2025; 主题: AI Infra; 优先级: 扩展
- [来源](#)

论文: TrainVerify: Equivalence-Based Verification for Distributed LLM Training

快速理解: Training large language models (LLMs) at scale requires parallel execution across thousands of devices, incurring enormous computational costs. Yet, these costly distributed trainings are prone to correctness bugs, causing silent errors and potentially wasting millions of GPU hours.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.20.32 SAND: A New Programming Abstraction for Video-based Deep Learning

- 作者: Juncheol Ye; Seungkook Lee; Hwijoon Lim; Jihyuk Lee; Uitaek Hong; Youngjin Kwon; Dongsu Han
- 年份: 2025; 主题: AI Infra; 优先级: 扩展
- [来源](#)

论文: SAND: A New Programming Abstraction for Video-based Deep Learning

快速理解: Video-based deep learning (VDL) is increasingly used across diverse applications and has become highly popular, but it faces significant challenges in preprocessing highly compressed video data. Preprocessing pipelines are complex, requiring extensive engineering effort, and introduce computational bottlenecks, with latency exceeding GPU training time.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.20.33 HEALER: Relation Learning Guided Kernel Fuzzing

- 作者: Hao Sun; Yu-Heng Shen; Cong Wang; Jian-Zhong Liu; Yu Jiang; Ting Chen; Aiguo Cui
- 年份: 2021; 主题: Systems; 优先级: 扩展
- [来源](#)

论文: HEALER: Relation Learning Guided Kernel Fuzzing

快速理解: Modern operating system kernels are too complex to be free of bugs. Fuzzing is a promising approach for vulnerability detection and has been applied to kernel testing.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.20.34 Bagpipe: Accelerating Deep Recommendation Model Training

- 作者: Saurabh Agarwal; Chengpo Yan; Ziyi Zhang; S. Venkataraman
- 年份: 2022; 主题: Systems; 优先级: 扩展
- [来源](#)

论文: Bagpipe: Accelerating Deep Recommendation Model Training

快速理解: Deep learning based recommendation models (DLRM) are widely used in several business critical applications. Training such recommendation models efficiently is challenging because they contain billions of embedding-based parameters, leading to significant overheads from embedding access.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.20.35 Enabling Parallelism Hot Switching for Efficient Training of Large Language Models

- 作者: Hao Ge; Fangcheng Fu; Haoyang Li; Xuanyu Wang; Sheng Lin; Yujie Wang; Xiaonan Nie; Hailin Zhang; Xupeng Miao; Bin Cui
- 年份: 2024; 主题: Systems; 优先级: 扩展

- [来源](#)

论文: Enabling Parallelism Hot Switching for Efficient Training of Large Language Models

快速理解: Training of large-scale deep learning models necessitates parallelizing the model and data across numerous devices, and the choice of parallelism strategy substantially depends on the training workloads such as memory consumption, computation cost, and communication cost. Current approaches generally assume uniform training workloads across samples in a given task.

关注点: 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.20.36 Coeus: A System for Oblivious Document Ranking and Retrieval

- 作者: Ishtiyaque Ahmad; Laboni Sarker; D. Agrawal; A. El Abbadi; Trinabh Gupta
- 年份: 2021; 主题: 未标注主题; 优先级: 扩展
- [来源](#)

论文: Coeus: A System for Oblivious Document Ranking and Retrieval

快速理解: Given a private string q and a remote server that holds a set of public documents D , how can one of the K most relevant documents to q in D be selected and viewed without anyone (not even the server) learning anything about q or the document? This is the oblivious document ranking and retrieval problem.

关注点: 需打开全文确认研究问题、方法和实验边界。

证据层级: 索引摘要。

1.3.21 SYSTEMS · USENIX ATC

本次会议纳入 11 篇。自动主题统计为: Systems (10)、AI Infra (9)、Agent RL (3)、Supporting Perception (1)。这些统计用于组织阅读顺序, 不能单独证明论文结论或实验优越性。

优先阅读候选: - Cost-Efficient Large Language Model Serving for Multi-turn Conversations with CachedAttention (优先级分数 74.977) - Beware of Fragmentation: Scheduling GPU-Sharing Workloads with Fragmentation Gradient Descent (优先级分数 73.583) - Optimus: Accelerating Large-Scale Multi-Modal LLM Training by Bubble Exploitation (优先级分数 72.087) - Torpor: GPU-Enabled Serverless Computing for Low-Latency, Resource-Efficient Inference (优先级分数 72.035) - StreamBox: A Lightweight GPU SandBox for Serverless Inference Workflow (优先级分数 71.459)

1.3.21.1 Cost-Efficient Large Language Model Serving for Multi-turn Conversations with CachedAttention

- 作者: Bin Gao; Zhuomin He; Puru Sharma; Qingxuan Kang; Djordje Jevdjic; Junbo Deng; Xingkun Yang; Zhou Yu; Pengfei Zuo
- 年份: 2024; 主题: AI Infra; Systems; 优先级: 核心
- [来源](#)

论文: Cost-Efficient Large Language Model Serving for Multi-turn Conversations with CachedAttention

快速理解: Interacting with humans through multi-turn conversations is a fundamental feature of large language models (LLMs). However, existing LLM serving engines executing multi-turn conversations are inefficient due to the need to repeatedly compute the key-value (KV) caches of historical tokens,

incurring high serving costs.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.21.2 Beware of Fragmentation: Scheduling GPU-Sharing Workloads with Fragmentation Gradient Descent

- 作者：Qizhen Weng; Lingyun Yang; Yinghao Yu; Wen Wang; Xiaochuan Tang; Guodong Yang; Liping Zhang
- 年份：2023；主题：AI Infra; Systems；优先级：核心
- [来源](#)

论文：Beware of Fragmentation: Scheduling GPU-Sharing Workloads with Fragmentation Gradient Descent

快速理解：摘要未随索引或本地 PDF 前页提供；根据标题与主题标签，本文关注 AI Infra、Systems 相关问题。以下判断仅用于快速阅读排序，不能替代全文核对。

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：标题/主题推断。

1.3.21.3 Optimus: Accelerating Large-Scale Multi-Modal LLM Training by Bubble Exploitation

- 作者：Weiqi Feng; Yangrui Chen; Shaoyu Wang; Yanghua Peng; Haibin Lin; Min-Lan Yu
- 年份：2024；主题：AI Infra; Supporting Perception; Systems；优先级：核心
- [来源](#)

论文：Optimus: Accelerating Large-Scale Multi-Modal LLM Training by Bubble Exploitation

快速理解：Multimodal large language models (MLLMs) have extended the success of large language models (LLMs) to multiple data types, such as image, text and audio, achieving significant performance in various domains, including multimodal translation, visual question answering and content generation. Nonetheless, existing systems are inefficient to train MLLMs due to substantial GPU bubbles caused by the heterogeneous modality models and complex data dependencies in 3D parallelism.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；3D、视频、多模态视觉或具身感知支撑能力；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.21.4 Torpor: GPU-Enabled Serverless Computing for Low-Latency, Resource-Efficient Inference

- 作者：Minchen Yu; Ao Wang; Dong-dong Chen; Haoxuan Yu; Xiaonan Luo; Zhuohao Li; W. Wang; Ruichuan Chen; Dapeng Nie; Haoran Yang
- 年份：2023；主题：AI Infra; Systems；优先级：核心
- [来源](#)

论文：Torpor: GPU-Enabled Serverless Computing for Low-Latency, Resource-Efficient Inference

快速理解：Serverless computing offers a compelling cloud model for online inference services. However,

existing serverless platforms lack efficient support for GPUs, hindering their ability to deliver high-performance inference.

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：索引摘要。

1.3.21.5 StreamBox: A Lightweight GPU SandBox for Serverless Inference Workflow

- 作者：Hao Wu; Yue Yu; Jun Deng; Shadi Ibrahim; Song Wu; Haoqiang Fan; Ziyue Cheng; Hai Jin
- 年份：2024；主题：AI Infra; Systems；优先级：核心
- [来源](#)

论文：StreamBox: A Lightweight GPU SandBox for Serverless Inference Workflow

快速理解：摘要未随索引或本地 PDF 前页提供；根据标题与主题标签，本文关注 AI Infra、Systems 相关问题。以下判断仅用于快速阅读排序，不能替代全文核对。

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：标题/主题推断。

1.3.21.6 HyCache: Hybrid Caching for Accelerating DNN Input Preprocessing Pipelines

- 作者：K. Jha; Shweta Pandey; Murali Annavaram; Arkaprava Basu
- 年份：2025；主题：AI Infra; Systems；优先级：核心
- [来源](#)

论文：HyCache: Hybrid Caching for Accelerating DNN Input Preprocessing Pipelines

快速理解：摘要未随索引或本地 PDF 前页提供；根据标题与主题标签，本文关注 AI Infra、Systems 相关问题。以下判断仅用于快速阅读排序，不能替代全文核对。

关注点：训练、推理、服务、编译、调度、通信或 AI 加速；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：标题/主题推断。

1.3.21.7 AWARE: Automate Workload Autoscaling with Reinforcement Learning in Production Cloud Systems

- 作者：Haoran Qiu; Weichao Mao; Chen Wang; H. Franke; A. Youssef; Z. Kalbarczyk; T. Başar; Ravishankar K. Iyer
- 年份：2023；主题：Agent RL; Systems；优先级：核心
- [来源](#)

论文：AWARE: Automate Workload Autoscaling with Reinforcement Learning in Production Cloud Systems

快速理解：摘要未随索引或本地 PDF 前页提供；根据标题与主题标签，本文关注 Agent RL、Systems 相关问题。以下判断仅用于快速阅读排序，不能替代全文核对。

关注点：强化学习、模仿学习、规划、奖励或多智能体决策；操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级：标题/主题推断。

1.3.21.8 AUTO: Adaptive Congestion Control Based on Multi-Objective Reinforcement Learning for the Satellite-Ground Integrated Network

- 作者: Xu Li; Feilong Tang; Jia-Cheng Liu; L. Yang; Luoyi Fu; Long Chen
- 年份: 2021; 主题: Agent RL; Systems; 优先级: 重要
- [来源](#)

论文: AUTO: Adaptive Congestion Control Based on Multi-Objective Reinforcement Learning for the Satellite-Ground Integrated Network

快速理解: 摘要未随索引或本地 PDF 前页提供; 根据标题与主题标签, 本文关注 Agent RL、Systems 相关问题。以下判断仅用于快速阅读排序, 不能替代全文核对。

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 标题/主题推断。

1.3.21.9 Legion: Automatically Pushing the Envelope of Multi-GPU System for Billion-Scale GNN Training

- 作者: Si-Jie Shen; Zi-Hang Yao; Ling-Quan Shi; Lei Wang; Long-Bin Lai; Qian Tao; Li Su; Rong Chen; Wen-Yuan Yu; Haibo Chen; Binyu Zang; Jing-Ren Zhou
- 年份: 2023; 主题: AI Infra; Systems; 优先级: 重要
- [来源](#)

论文: Legion: Automatically Pushing the Envelope of Multi-GPU System for Billion-Scale GNN Training

快速理解: Graph neural network(GNN) has been widely applied in real-world applications, such as product recommendation in e-commerce platforms and risk control in financial management systems. Several cache-based GNN systems have been built to accelerate GNN training in a single machine with multiple GPUs.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.3.21.10 MSRL: Distributed Reinforcement Learning with Dataflow Fragments

- 作者: Huanzhou Zhu; Bo Zhao; Gang Chen; Weifeng Chen; Yi-Jie Chen; Liang Shi; Yaodong Yang; P. Pietzuch; Lei Chen
- 年份: 2022; 主题: Agent RL; AI Infra; 优先级: 重要
- [来源](#)

论文: MSRL: Distributed Reinforcement Learning with Dataflow Fragments

快速理解: Reinforcement learning (RL) trains many agents, which is resource-intensive and must scale to large GPU clusters. Different RL training algorithms offer different opportunities for distributing and parallelising the computation.

关注点: 强化学习、模仿学习、规划、奖励或多智能体决策; 训练、推理、服务、编译、调度、通信或 AI 加速。

证据层级: 索引摘要。

1.3.21.11 PPipe: Efficient Video Analytics Serving on Heterogeneous GPU Clusters via Pool-Based Pipeline Parallelism

- 作者: Z. Kong; Qiang Xu; Y. C. Hu
- 年份: 2025; 主题: AI Infra; Systems; 优先级: 重要
- 来源

论文: PPipe: Efficient Video Analytics Serving on Heterogeneous GPU Clusters via Pool-Based Pipeline Parallelism

快速理解: With the rapid innovation of GPUs, heterogeneous GPU clusters in both public clouds and on-premise data centers have become increasingly commonplace. In this paper, we demonstrate how pipeline parallelism, a technique wellstudied for throughput-oriented deep learning model training, can be used effectively for serving latency-bound model inference, e.g., in video analytics systems, on heterogeneous GPU clusters.

关注点: 训练、推理、服务、编译、调度、通信或 AI 加速; 操作系统、分布式系统、网络、存储、架构或并行运行时。

证据层级: 索引摘要。

1.4 4. 后续精读顺序

1. 先读同时命中 VLA、World Model、Embodied/WBC、Teleoperation 的具身论文。
2. 再读 Agent RL、Robot Learning 与 Supporting Perception 的高优先级论文。
3. 最后按 AI Infra、Systems、分布式训练、推理服务、架构和存储路线阅读系统论文。

1.5 5. 使用说明

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